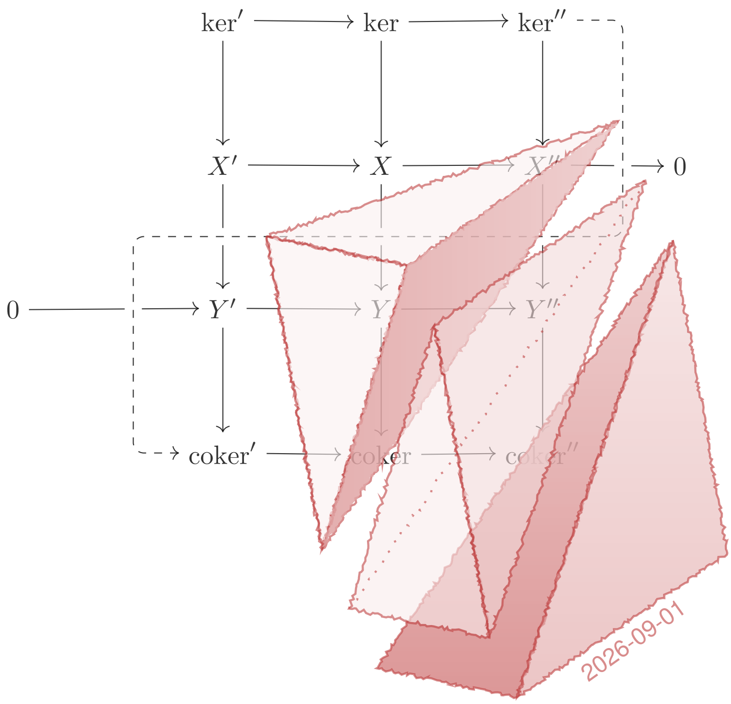
Methods of Algebra

Volume 2: Linear Algebra

Wen-Wei Li author



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This independent edition translates the complete active introduction, Chapters 1–9, Appendices A–B, exercises, hints, headings, and reader interfaces. Formulae, notation, numbering, labels, cross-references, citations, diagrams, and bibliographic identities are preserved. Two clearly identified mastery bridges are independent additions derived from the Indonesian course edition. Wen-Wei Li and Higher Education Press do not endorse this independent translation.

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Introduction

This book is a sequel to [51] (hereafter Volume I). Its purpose is to introduce a range of methods, ideas, and techniques that may reasonably be grouped under the broad heading of linear algebra, assuming familiarity with the basic structures of algebra. These methods pervade contemporary mathematics. To address practical needs as comprehensively as possible, the relevant techniques must be forged into purer and more concise forms; categories and functors are an indispensable language for doing so.

The book is divided into three main parts: the Inner Part, the Outer Part, and the appendices. It is intended chiefly for advanced undergraduates, graduate students, researchers, teachers, and independent learners who work on or are interested in these subjects. The prerequisites include familiarity with algebraic structures such as groups, rings, modules, and fields, as well as category theory. Volume I covers all of this background, though other good textbooks will serve equally well apart from minor differences in notation. The motivation for writing the book was explained in the introduction to Volume I; the original purpose remains unchanged and need not be repeated here.

What Is Linear Algebra?

In the usage of algebraists, "linear" broadly describes properties connected with structures that have addition and its inverse operation, subtraction. Addition and subtraction in turn give rise to multiplication by integers; an immediate generalization is scalar multiplication by elements of a ring R. The prototypical examples are left and right R-modules, while a $\mathbb{Z}$-module is simply an abelian group. For a given R-module (taken to be a left module unless otherwise stated), one may study its submodules and quotient modules, as well as the kernel $\ker(f)$, cokernel $\operatorname{coker}(f)$, and image $\operatorname{im}(f)$ of a module homomorphism f. All are old friends from elementary algebra; at least the special case in which R is a field—that is, the case of vector spaces—is widely familiar.

Since the twentieth century, mathematical practice has gradually shown that extending the study of R-modules and their homomorphisms to complexes of R-modules is not only useful and convenient, but often necessary. Such a complex is a sequence of module homomorphisms satisfying $d^{n+1}d^n = 0$:

$$\cdots \longrightarrow X^{n-1} \xrightarrow{d^{n-1}} X^n \xrightarrow{d^n} X^{n+1} \xrightarrow{d^{n+1}} \cdots .$$

The condition on the homomorphisms is sometimes written $d^2 = 0$, while the full data of the complex are often abbreviated as $(X^n, d^n)_n$, (X, d), or X. Because the notation uses increasing superscripts, this is also called a cochain complex. If instead one uses decreasing subscripts, $\cdots \to X_n \xrightarrow{d_n} X_{n-1} \to \cdots$, the resulting mathematical object is called a chain complex; the distinction is purely formal.

A single R-module M may be regarded as the special complex with $X^0 = M$ and all other terms zero. For a complex X, the identity $d^n d^{n-1} = 0$ allows its nth cohomology to be defined as the quotient module

$$\mathrm{H}^n(X) := \ker\left(d^n\right) / \operatorname{im}\left(d^{n-1}\right).$$

A complex satisfying $\mathrm{H}^n(X) = 0$ for every n is called an exact sequence or an acyclic complex. For a chain complex, correspondingly, one has homology $\mathrm{H}_n(X) := \ker(d_n)/\operatorname{im}(d_{n+1})$. The cohomology of a complex (or the homology of a chain complex) often contains important information about the mathematical object under study.

For modules and other mathematical structures equipped with some kind of linear operation—for example, objects of the abelian categories introduced later—the study of complexes and their cohomology is the classical subject matter of homological algebra. This forms a proper part of "linear algebra" in its genuine sense, yet lies at the heart of this book. A brief account of its origins follows.

A Brief History of Homology

Mathematics need not be grounded in history; nevertheless, looking back to understand what lies ahead can only be beneficial. We shall divide the development of homology theory, somewhat artificially, into several stages.

The Formative Period The first impetus for the development of homological algebra came from topology. In the latter half of the nineteenth century, B. Riemann and E. Betti studied the genus of surfaces and its higher-dimensional generalizations, now called Betti numbers. Around the turn of the century, H. Poincaré developed these ideas into a more rigorous theory. Roughly speaking, Poincaré's original idea was to decompose a space into polygons, polyhedra, or higher-dimensional analogues glued together; compute Betti numbers and their torsion analogues from matrices encoding the gluing; and prove that these quantities are topological invariants.

In a short paper from 1925, E. Noether explained how homology could be defined as an abelian group, with the Betti numbers and their torsion analogues merely numerical invariants derived from it. This already comes close to the modern definition in algebraic topology: for a decomposed space E and an abelian group A, one defines a chain complex $C(E, A)$ that records how the pieces are glued; the homology groups of E with coefficients in A are then $\mathrm{H}_n(E; A) := \mathrm{H}_n(C(E, A))$, while the cohomology groups $\mathrm{H}^n(E; A)$ are defined dually.

From then until about 1950, algebraic topology entered an extraordinarily fertile period. From the algebraic point of view, notable developments included:

- ◇ the universal coefficient theorem for homology (or cohomology), which says that the case of integer coefficients suffices to determine homology (or cohomology) with coefficients in an arbitrary abelian group A; its explicit description involves the Tor (or Ext) functor, to be studied in depth later;
- ◇ the cup product on cohomology, a multiplicative structure carried by cohomology groups;
- ◇ aspherical spaces: connected spaces E satisfying $n > 1 \implies \pi_n(E) = 0$, where $\pi_n(E)$ denotes the nth homotopy group relative to a chosen base point. W. Hurewicz proved that the homology and cohomology groups of an aspherical space are entirely determined by $\pi_1(E)$ and can be described algebraically. This marked the beginning of group cohomology.

Another impetus for homological algebra came from within algebra itself. The classical example is Hilbert's syzygy theorem in invariant theory (1890): in modern language, it states that every module over the polynomial ring $\Bbbk[X_1, \ldots, X_n]$ has a free resolution of length $\leq n$.

The Ext functor likewise arises from algebraic questions. For abelian groups A and B, a short exact sequence of abelian groups $0 \to A \to E \to B \to 0$ is also called a group extension; equivalence classes of such extensions correspond bijectively to the elements of $\mathrm{Ext}^1(B, A)$. Moreover, if B is replaced by an arbitrary group G, the classification of group extensions leads naturally to an algebraic definition of group cohomology $\mathrm{H}^2(G, A)$.

Derivations on rings attracted algebraists' attention early on. In 1942, G. Hochschild defined invariants of a ring R now called Hochschild homology and cohomology; the degree-one cohomology group classifies derivations modulo inner derivations.[1)] These constructions extend to arbitrary (R, R)-bimodules M and continue to play an important role at the frontiers of modern mathematics.

Also worth mentioning is J. Leray's wartime work on sheaves and spectral sequences, motivated initially by fixed-point problems related to partial differential equations. Sheaves are geometric structures that carry local–global information, while spectral sequences provide powerful tools for computing sheaf cohomology. Influenced by Leray's theory and Kiyoshi Oka's work in several complex variables, H. Cartan and others set about rewriting the foundations of algebraic topology after the war; homological algebra thereby took on an entirely new form.

The Cartan–Eilenberg Revolution The monumental and far-reaching work of H. Cartan and S. Eilenberg [8] was long regarded as the canonical reference. It marked a new period in the development of homological algebra. Its main contributions include:

- defining injective and projective modules, and injective and projective resolutions of modules;
- defining right and left derived functors in the framework of module theory, thereby giving general definitions of Ext and Tor, and introducing the injective and projective dimensions of a module—tools that soon proved highly effective in the theory of commutative rings;
- using this machinery to explain earlier constructions, such as group cohomology and Hochschild homology and cohomology;
- systematically establishing the general theory of spectral sequences and their multiplicative structures.

[1)] Translator's note: the source says that the degree-one term "classifies all derivations." This has been corrected here: $\mathrm{H}^1_{\mathrm{Hoch}}(R, R)$ classifies derivations modulo inner derivations; see correction O014-C001.

The term "homological algebra" also originates in [8]. With the emergence of this grand theory, homological algebra entered its youth.

Abelian Categories From a geometric—more precisely, sheaf-theoretic—viewpoint, the module-theoretic framework for homological algebra is too restrictive. One seeks categories $\mathcal{A}$ more general than the category $R\text{-}\mathsf{Mod}$ of left R-modules, while retaining some "linear" character. An early attempt appeared in D. Buchsbaum's 1955 doctoral thesis, also included as an appendix to [8]; he abstracted the notion of an exact sequence from module theory.

The now standard approach comes from A. Grothendieck's paper [16]; it led to what are now called abelian categories. An abelian category is an additive category with kernels and cokernels in which every morphism $f : M \to N$ is strict. Roughly speaking, strictness corresponds to the familiar module-theoretic isomorphism theorem $M/\ker(f) \xrightarrow{\sim} \mathrm{im}(f)$. In the same work, Grothendieck also:

- used long exact sequences to define δ-functors and universal δ-functors, characterized the universal ones, and showed that derived functors are special cases of universal δ-functors;
- gave the Grothendieck spectral sequence for deriving a composite of functors;
- defined a special class of abelian categories, now called Grothendieck categories, and established the existence of injective resolutions in them.

These theories made it possible to study sheaf cohomology on more general spaces and were crucial to the development of geometry.

Derived Categories In his 1963 doctoral thesis, J.-L. Verdier defined the derived category $\mathsf{D}(\mathcal{A})$ of an abelian category $\mathcal{A}$. The motivation came from difficulties Grothendieck encountered while studying duality in algebraic geometry. Those obstacles prompted the search for a framework in which one works with complexes themselves rather than merely with their cohomology, formally inverting the quasi-isomorphisms in the category of complexes $\mathsf{C}(\mathcal{A})$. A quasi-isomorphism is a morphism of complexes $f : X \to Y$ inducing isomorphisms $\mathrm{H}^n(f) : \mathrm{H}^n(X) \xrightarrow{\sim} \mathrm{H}^n(Y)$ on cohomology. The operation of formally inverting these morphisms is called Gabriel–Zisman localization, a categorical generalization of ring localization.

The derived category $\mathsf{D}(\mathcal{A})$ contains more information than cohomology alone, and its formalism is correspondingly different: sequences of morphisms in $\mathsf{D}(\mathcal{A})$ called distinguished triangles,

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

play the role of short exact sequences in $\mathcal{A}$ or in $\mathsf{C}(\mathcal{A})$. Here $X[1]$ is the shift of the complex X:

$$X[1]^n = X^{n+1}, \quad d^n_{X[1]} = -d^n_X.$$

Verdier further abstracted these constructions into structures called triangulated categories. From the topological viewpoint, the triangulated structure of a derived category is equally natural: distinguished triangles are analogous to cofiber sequences in homotopy theory. This relates to the homotopical algebra discussed later. Also motivated by homotopy theory, D. Puppe independently discovered a similar structure in 1962, although his conditions did not include Verdier's octahedral axiom.

Although derived categories are not ideal for every application, their language remained the preferred one among geometers and algebraists into the early twenty-first century. One prominent application is the theory of $\mathscr{D}$-modules, developed chiefly by Masaki Kashiwara, Z. Mebkhout, and others; some of its basic operations admit satisfactory definitions only at the level of derived categories.

The derived category of a ring R, $\mathsf{D}(R) := \mathsf{D}(R\text{-}\mathsf{Mod})$, may also be regarded as an invariant of R. If two rings have equivalent derived categories as triangulated categories, they are called derived equivalent. Questions and methods surrounding derived equivalence form a major current in the representation theory of algebras, pioneered by D. Happel, J. Rickard, and others.

Homotopical Algebra The reader will recall that the topological origin of homological algebra lay in decomposing spaces and then extracting topological invariants from suitable chain or cochain complexes. There are many ways to carry out the initial decomposition; simplicial sets, defined by Eilenberg and Zilber in 1950, are one of them, and they generalize further to simplicial objects in a given category. This language has unusually broad explanatory power: the homotopy theory of topological spaces can be developed using simplicial sets, while the construction called the "nerve" interprets categories as a special kind of simplicial set.

What happens if a little "linearity" is added to this picture? The Dold–Kan correspondence (1958) directly connects simplicial objects in an abelian category with the theory of complexes. For the category Ab of abelian groups, for example, the Dold–Kan correspondence is an explicitly defined adjoint equivalence

$$\mathrm{N} : \mathsf{sAb} \rightleftarrows \mathsf{Ch}_{\geq 0}(\mathsf{Ab}) : \Gamma$$

where sAb is the category of simplicial objects in Ab and $\mathsf{Ch}_{\geq 0}(\mathsf{Ab})$ is the category of nonnegative chain complexes in Ab. Other conventions for complexes can be accommodated by reversing the indexing or passing to the opposite category.

The Dold–Kan correspondence gives topological interpretations of many definitions and constructions concerning chain complexes. Simplicial methods likewise provide a unified account of a class of resolutions collectively called “bar constructions” in homological algebra.

Homotopy is a central idea in simplicial theory. Algebraic methods involving simplicial objects are therefore sometimes called homotopical algebra. Their power is displayed especially clearly in the higher K-theory developed by D. Quillen, though their applications are by no means limited to it.

In summary, simplicial methods in linear algebra may be viewed as a higher-level confluence of topology—especially homotopy theory—and homological algebra. In the early twenty-first century, this confluence inspired another wave: infinity-category theory.

The Purpose of This Book

Organizational Aims The preceding introduction shows that what is called linear algebra divides roughly into two main branches: the theory of abelian categories and the theory of complexes built upon them. This book centers on these two branches and seeks to provide the ideas and techniques needed to understand contemporary mathematics. Its contents include both the algebraic methods used in applications and the theoretical foundations those methods require. Since the methods primarily serve research in pure mathematics and are meant to apply as broadly as possible, the book may also be characterized as “pure applied mathematics.”

The book also attempts to serve both as a textbook and as a reference, which necessarily calls for attention to detail and systematic development. These goals pull against one another, yet the book must state its purpose, develop its themes, and bring them home within a finite space. This creates an enormous organizational challenge, but one that must be accepted.

As to content, derived categories, derived functors, and spectral sequences are essential knowledge for contemporary mathematicians. Yet they pose real difficulties for beginners, chiefly at the level of ideas and definitions. The Inner Part is organized around this objective and lays the foundation for the Outer Part. To keep the exposition flowing and save space, we shall not retrace history step by step, but will more often follow the logical order made visible by hindsight.

Constrained by these aims, what critics of art and literature call an “aura” inevitably recedes from the prose, replaced at times by the heaviness of systematic structure and rigorous proof. The aura opposed to that heaviness is mathematics’ first knock upon the human mind, and perhaps its last. In the author's view, however, a light, agile, direct style belongs either to a theory's formative period or relies upon a complete supporting literature. Such support must be embedded in the native linguistic culture and shared by every reader. The author has chosen to carry the burden, hoping that the appearance of this book will leave future authors free to travel more lightly.

During its preparation, the book drew extensively on related works, among them [46, 33, 23, 37, 48]. The Stacks Project [42] and the nLab website also proved invaluable.

Approach and Trade-offs The book must accommodate many fields and styles while finding a balance between classical and modern approaches, so that the material remains reasonably accessible to beginners. Compromise is therefore the author's guiding principle. Few readers, however, open a book in search of compromise. Compromise brings detours, so the path of this book is a geodesic for no reader; that is unavoidable.

For the same reason, the selection of material cannot satisfy everyone. Geometrically inclined readers will not find sheaves or t-structures, while readers interested in the representation theory of algebras will not find tilting theory. These topics are omitted because the corresponding theories cannot be adequately motivated without first entering those subjects more deeply.

Category theorists may regard the book's approach as no more than semiclassical. Even so, readers may notice that the main text and exercises leave entry points for more advanced viewpoints, such as dg-categories, monad theory, and simplicial objects, though each is touched on only briefly. These topics can be viewed as a warm-up for infinity-category theory and point toward a new world deeper than linear algebra, called advanced or higher algebra, which lies beyond the scope of this book.

One further choice deserves explanation: the book treats general abelian categories and develops their basic properties directly from the axioms, unlike some homological-algebra texts that consider only modules. This generality is demanded both by theoretical completeness and by practical necessity: even if modules were our sole starting point, constructions such as functor categories and Serre quotients would still require it. Because diagram chases in module theory do not extend directly to the general setting, some foundational results about abelian categories require rather indirect proofs. To help with these details, the book expects readers to have some elementary experience with complexes of modules and their cohomology, such as the material of [51, Chapter 6], though this is not logically necessary.

The celebrated Freyd–Mitchell embedding theorem states that every small abelian category admits an exact, fully faithful embedding into $\mathsf{Mod}\text{-}R$ for some ring R. Granting this theorem, many basic facts about abelian categories can be checked in the concrete module category $\mathsf{Mod}\text{-}R$. A proof of the theorem is given in Appendix §B.6. Because the proof is itself nontrivial and depends on basic properties of abelian categories, the theorem does not essentially simplify their theory; its role is chiefly heuristic.

Guide to Using the Book

How the Book May Be Used To serve as a reference work, the book needs a reasonably systematic organization. Its chapters therefore follow a primarily logical order. Dependencies between earlier and later material are especially pronounced in the Inner Part, while the few necessary topics that diverge from the main line or are more technical have been assigned to appendices.

Independent readers should decide what to read according to their own interests and needs. Most of these choices arise in the Outer Part, whereas the Inner Part supplies the common foundational framework, although even there some sections are optional. More abstract or technical details may be omitted on an initial reading, while material involving the appendices is best skipped at first or merely skimmed. Detailed advice on these choices appears at the beginning of each chapter and appendix, especially in the reading guides.

As noted above in the discussion of the book's aims, we have unavoidably wrapped mathematical ideas in layer upon layer of prose. Independent readers must make a determined effort to discern the mathematics itself within those layers, like catching lightning amid roiling clouds.

For readers using the book as a reference, the author has tried to organize the exposition modularly so that its parts can be read separately. This is reflected not only in the division into chapters, but also in the following two features.

- ◇ As far as possible, the book uses the standard notation of contemporary literature, supplemented by reviews and explanations. The Conventions at the end of the introduction and the index at the end of the book provide two further aids to understanding the notation.

- ◇ Statements and proofs contain copious cross-references, allowing readers who already know something of the subject to understand the discussion quickly and build an efficient conceptual map.

In particular, the author hopes that the book will help readers preparing to teach or participating in seminars to assemble the knowledge they need quickly and then recast it in their own language.

As for classroom use, one must frankly admit that teaching schedules are not governed by individual wishes. While writing the book, the author had almost no opportunity to try out this material in the classroom, so its effectiveness there remains unknown. Judging from past experience, even with details omitted in lectures, the book clearly contains enough material for more than one academic year; in a streamlined presentation, the Inner Part might perhaps be compressed into a single semester. An instructor using the book as the primary textbook will first have to decide how to divide the material. The author hopes that the modular design just mentioned will make this task easier.

The lack of classroom testing has two further consequences. First, the book must still contain many undiscovered errors. Second, it lacks the finish of a classic textbook. This is due both to practical circumstances and to the author's own limitations; the author can only ask the reader's indulgence.

On the Exercises Every chapter and appendix ends with exercises arranged in the order of the corresponding material in the main text and sometimes accompanied by hints. Their chief purposes are to supplement the text with further properties, provide examples, and broaden the reader's perspective. Some exercises help readers become familiar with techniques and generally make few technical demands. A few ask readers to fill in details omitted from the text; these details are either trivial, easy, or, more often, both. In short, the author hopes that beginners will attempt as many exercises as possible, but they need not force themselves or feel obliged to complete them all.

Categories and Their Size Most of this book is expressed in categorical language because practice has shown category theory to be an efficient means of understanding and handling algebraic problems. Nevertheless, linear algebra—or what is called homological algebra—is no more a branch of category theory than probability theory is a branch of measure theory.

Unlike some textbooks, both this book and Volume I require that, for a category, the collection of all objects and the collection of all morphisms each be a set; it is not enough for the objects merely to form a "class." The price of excluding proper classes from the definitions is that, strictly speaking, one must introduce Grothendieck universes to measure the size of sets and assume that all sets belong to some universe $\mathcal{U}$; see [51, Assumption 1.5.2, Remark 1.5.6]. When concrete problems are considered case by case, the axiom that there are sufficiently many universes—equivalently, sufficiently many strongly inaccessible cardinals—can often be weakened. Since this issue is not central to the book, we shall not pursue it further.

Overview of the Structure

Readers are advised to begin with the Conventions at the end of the introduction, in order to familiarize themselves with the notation and conventions used throughout the book. Beyond that, beginners are discouraged from reading strictly in sequence unless they are already highly receptive to abstract methods. Each chapter opens with a detailed summary and reading guide.

First comes the Inner Part, which lays the foundations.

▷ Chapter 1: Supplements to Category Theory   Reviews and rounds out the requisite category theory, emphasizing strict morphisms, kernels and cokernels in additive categories, Kan extensions, Gabriel–Zisman localization, and related topics.

▷ Chapter 2: Abelian Categories   Develops the basic theory of abelian categories, supplemented by material on Grothendieck categories and other topics.

▷ Chapter 3: Complexes   Discusses basic notions such as complexes, double complexes, homotopies, and mapping cones in additive categories, as well as cohomology in abelian categories. The second half treats derived functors from the classical viewpoint, including special cases such as Ext, Tor, and $\lim^1$. To handle unbounded complexes, it concludes with K-injective and K-projective resolutions.

▷ Chapter 4: Triangulated and Derived Categories   Introduces the general notions of triangulated categories and triangulated functors, then concentrates on derived categories of abelian categories and derived functors between derived categories. The classical Ext and Tor are upgraded accordingly to RHom and $\overset{\mathrm{L}}{\otimes}$.

▷ Chapter 5: Spectral Sequences   Begins with the general definition of spectral sequences and exact couples, then specializes progressively to the spectral sequences of filtered complexes and double complexes and to their applications in algebra. Exact couples are not absolutely necessary for the spectral sequence of a filtered complex.

Next comes the Outer Part, devoted to applications or further developments.

▷ Chapter 6: Group Homology and Cohomology   Studies group homology and cohomology, together with a variety of related computations and examples. It also treats Tate cohomology, the cohomology of profinite groups, and nonabelian cohomology, each of which plays an important role in applications.

- ▷ Chapter 7: Monad Theory Starts from the general notion of an "algebra" in a monoidal category and first discusses differential graded structures related to complexes. After treating examples that include coalgebras, bialgebras, and Hopf algebras, it turns to Beck's monadicity theorem. The second half considers module theory, introduces Morita theory, and applies monadicity to flat descent for modules and to Galois descent.
- ▷ Chapter 8: Simplicial Methods Covers the general theory of simplicial objects and simplicial sets, the Dold–Kan correspondence, and the bar construction formulated in the language of monads. The final sections concern the generalized Eilenberg–Zilber theorem and a topological interpretation of mapping cones.
- ▷ Chapter 9: Duality Begins with duality in monoidal categories and, after basic examples, studies reconstruction theorems and Tannakian categories. This chapter is not directly concerned with complexes.

The original plan called for the Outer Part to include the foundations of representation theory, which seemed likely to play a natural role in the overall structure. Late in the process, the idea was abandoned because the book was already too long and several mature textbooks were available.

The appendices are as follows.

- ▷ Appendix A: Further Material on Abelian Categories Collects material directly or indirectly related to abelian categories, either cited in the main text or developed as a side topic.
- ▷ Appendix B: An Introduction to Ind- and Pro-objects Starts from the theory of profinite groups, naturally introduces ind- and pro-objects, and studies the Ind-construction for categories and functors. As an application, it then proves the Freyd–Mitchell embedding theorem for abelian categories.

Finally, the author's personal website provides the latest supplementary information, including errata, for this book and the author's other works. Readers are invited to follow the updates there and to write to the author if they find any errors in the book.

Acknowledgments

During the preparation of this book, the author benefited from corrections and suggestions by teachers, colleagues, and friends. The process lasted so long that omissions and errors of memory are inevitable; the following list is incomplete: Huan Zhen, Huang Chenxin, Li Jinghui, Lei Jiale, Li Changyuan, Sun Yuze, Tang Yiming, Xue Hanyu, Yang Enlin, Zhang Haofeng, and Zhao Zelong (in Hanyu Pinyin order).

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Like all the author's works to date, this book was written entirely in an open-source software environment, including the operating system and the fonts used in preparing it for press. The author salutes the many selfless developers who put into practice the universal values of sharing and altruism.

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The author also thanks Zhao Tianfu, an editor at Higher Education Press, for his patience and professionalism. The writing of this book ought to have followed seamlessly from Volume I, but various obstacles delayed its completion until the end of 2022.

Despite the delay, preparing the book consumed even more of the author's mental energy than Volume I. This was especially true of the repeated checking and revision in the final stages, which felt like walking through an endless dark tunnel. Much of this late work was done in Beijing subway cars. Working on the train was hardly comfortable and required making every minute and second count, quite literally. Yet amid that immense, anonymous current, day after day, the author seemed to sense a kind of tacit mutual understanding rarely found in working life, and to draw a glimmer of consolation from it. Was it so, or was it not? The explanation may be fanciful, but the feeling was real. The prelude now draws to a close: homage to the boundless distance, and to the endless multitudes.

Wen-Wei Li

December 2022

Shijingshan

Conventions

The notational conventions in this book are the same as in Volume I. They are summarized here, at the risk of some length.

Basic Conventions The book uses standard logical symbols such as $\forall$, $\exists$, and $\implies$. The symbol $\exists!$ means "there exists exactly one," and $\iff$ means "if and only if." The end of a proof is marked by $\square$. The notation $A := B$ means "A is defined to be B." If the definition of a mathematical object does not depend on choices of auxiliary data, it is called well-defined.

The arrow $\xrightarrow{\sim}$ denotes an isomorphism between objects, sometimes written without direction as $\simeq$. The symbols $\xrightarrow{1:1}$ and $\xleftrightarrow{1:1}$ denote a bijection between sets, that is, a one-to-one correspondence. Notation such as $F(\cdot)$ emphasizes the variable of a map or functor F.

The notation $n \gg m$ means that n is sufficiently larger than m. Thus $n \gg 0$ means that n is sufficiently large, while $n \ll 0$ means that $-n$ is sufficiently large.

The language of commutative diagrams is used extensively. The simplest case is the following.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "h"'] & B \arrow[dl, "g"] \\ C & \end{tikzcd} \quad \text{commutes} \overset{\text{definition}}{\iff} g \circ f = h.$$

Here f, g, and h may be maps, homomorphisms, or morphisms in an arbitrary category. More complicated commutative diagrams are understood in the same way.

Sets and Order Structures The book uses Zermelo–Fraenkel axiomatic set theory and accepts the axiom of choice.

The empty set is denoted by $\varnothing$. The notation $A \subset B$ means that A is a subset of B, with equality allowed; strict inclusion is written $A \subsetneq B$. Set difference is $A \smallsetminus B := \{a \in A : a \notin B\}$, and disjoint union is $A \sqcup B$. For a map $f : A \to B$, the notation $a \mapsto b$ means that $a \in A$ is mapped by f to $b \in B$, and $f|_{A'}$ is the restriction of f to a subset $A' \subset A$. The identity map on A is denoted by $\mathrm{id} = \mathrm{id}_A$. The number of elements of A, also called its cardinality or cardinal, is denoted by $|A|$.

Following [51, Definition 1.2.1], a **partially ordered set** is a set P equipped with a reflexive, transitive, and antisymmetric binary relation $\leq$. Omitting antisymmetry gives a **preordered set**. The notation $x < y$ means $x \lneq y$. A partially ordered set $(P, \leq)$ is often abbreviated to P.

- If $m \in P$ satisfies $\forall x \in P,\ x \geq m \iff x = m$, then m is a maximal element of P. Replacing $\geq$ by $\leq$ defines a minimal element. In general these need not be unique.
- Let S be a subset of P. If $b \in P$ satisfies $\forall x \in S,\ x \leq b$, then b is an upper bound of S in P. Lower bounds are defined similarly.
- If b is an upper bound of S and every other upper bound b' satisfies $b' \geq b$, then b is the supremum of S, denoted $\sup S$. A supremum, if it exists, is unique. The infimum $\inf S$ is defined similarly.

A map $f : P \to Q$ of partially ordered sets is order-preserving if $a \leq b \implies f(a) \leq f(b)$. If f is bijective and both f and f^{-1} are order-preserving, it is an isomorphism of partially ordered sets, or an order-preserving bijection.

If every two elements of a partially ordered set P are comparable, then P is totally ordered. A nonempty totally ordered set in which every nonempty subset has a least element is **well-ordered** . Certain special well-ordered sets are called **ordinals**, and any two ordinals are comparable. Briefly:

- for every ordinal α, one has $\alpha = \{\beta : \text{ordinal},\ \beta < \alpha\}$;
- if an ordinal κ, as a set, is not equinumerous with any ordinal smaller than κ, then it is called a **cardinal**;
- the cardinality $|S|$ of a set S is defined as the least ordinal in its equinumerosity class. This uses the fact that every set can be well-ordered (which requires the axiom of choice) and that every well-ordered set is isomorphic to a unique ordinal;
- in particular, $|\alpha| \leq \alpha$ for every ordinal α.

This is von Neumann's viewpoint; see [51, § 1.2] or [21]. It is slightly more circuitous than the usual definition of a cardinal as an equinumerosity class, but has its conveniences.

Finite ordinals may be identified with nonnegative integers: for each $n \in \mathbb{Z}_{\geq 0}$ there is a corresponding totally ordered set $\mathbf{n}$, recursively defined as a set by $\mathbf{0} := \varnothing$ and $\mathbf{n+1} := \{\mathbf{0}, \ldots, \mathbf{n}\}$, with the evident order. The least infinite ordinal ω is the well-ordered set $\mathbf{0} \leq \mathbf{1} \leq \cdots$, and is also the countable cardinal $\aleph_0$.

We choose a **Grothendieck universe** $\mathcal{U}$ to distinguish sizes of sets. Elements of $\mathcal{U}$ are called $\mathcal{U}$-sets; sets equinumerous with a $\mathcal{U}$-set are called $\mathcal{U}$-small, or simply **small sets**. If a cardinal μ, regarded as a set, is $\mathcal{U}$-small, it is a **small cardinal**. We assume that every set X belongs to some $\mathcal{U}$; see [51, Assumption 1.5.2] and the accompanying discussion.

Categories Our conventions for categories agree with [51, § 1.5, Definition 2.1.3]: the objects and morphisms of a category $\mathcal{C}$ each form a set, denoted by $\mathrm{Ob}(\mathcal{C})$ and $\mathrm{Mor}(\mathcal{C})$. The set of morphisms between two objects is denoted by $\mathrm{Hom}_{\mathcal{C}}(\cdot,\cdot)$, or simply $\mathrm{Hom}(\cdot,\cdot)$. The identity morphism of X is id_X. Monomorphisms are marked by $\hookrightarrow$ and epimorphisms by $\twoheadrightarrow$; a monomorphism is also called an embedding.

- For any morphism $f : X \to Y$ in a category and any object T, f induces pullback f^* and pushforward f_* operations on Hom sets:

$$f^* : \mathrm{Hom}(Y,T) \to \mathrm{Hom}(X,T), \qquad f_* : \mathrm{Hom}(T,X) \to \mathrm{Hom}(T,Y)$$
$$\beta \mapsto \beta f \qquad\qquad \alpha \mapsto f\alpha.$$

- Subcategories are written using subset notation, $\mathcal{C}' \subset \mathcal{C}$. If $\mathrm{Hom}_{\mathcal{C}'}(X,Y) = \mathrm{Hom}_{\mathcal{C}}(X,Y)$, then $\mathcal{C}'$ is a full subcategory.

- The product $\mathcal{C}_1 \times \mathcal{C}_2$ of two categories has object set $\mathrm{Ob}(\mathcal{C}_1) \times \mathrm{Ob}(\mathcal{C}_2)$. A morphism from (X_1, X_2) to (Y_1, Y_2) is an element of $\mathrm{Hom}_{\mathcal{C}_1}(X_1,Y_1) \times \mathrm{Hom}_{\mathcal{C}_2}(X_2,Y_2)$, and composition is componentwise. This immediately defines products $\prod_i \mathcal{C}_i$ of arbitrary families of categories. Likewise, $\mathcal{C}^n := \mathcal{C} \times \cdots \times \mathcal{C}$ (n factors), or more generally $\mathcal{C}^I$ for a set I. See [51, Definition 2.3.1].

For example, every preordered set $(P, \leq)$ determines a category $\mathcal{P}$ with object set P and

$$\forall a,b \in P, \quad \mathrm{Hom}_{\mathcal{P}}(a,b) = \begin{cases} \text{the singleton } \{\star\}, & a \leq b \\ \varnothing, & \text{otherwise.} \end{cases}$$

In particular, $\mathbf{0}$ may be identified with the empty category, having no objects and no morphisms, while $\mathbf{1}$ is identified with the category having exactly one object $\mathbf{0}$ and one morphism $\mathrm{id}_{\mathbf{0}}$.

A category satisfying $\mathrm{Mor}(\mathcal{C}) = \{\mathrm{id}_X : X \in \mathrm{Ob}(\mathcal{C})\}$ is called **discrete**. Discrete categories correspond bijectively to sets under $\mathcal{C} \mapsto \mathrm{Ob}(\mathcal{C})$.

Terminology: Small and Large Categories Fix a Grothendieck universe $\mathcal{U}$, and let $\mathcal{C}$ be a category. If $\mathrm{Mor}(\mathcal{C})$ is $\mathcal{U}$-small, then $\mathcal{C}$ is a $\mathcal{U}$-small category. If only $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ is required to be $\mathcal{U}$-small for every $X, Y \in \mathrm{Ob}(\mathcal{C})$, then $\mathcal{C}$ is a $\mathcal{U}$-category. [2)]

For example, all $\mathcal{U}$-sets and maps between them form the $\mathcal{U}$-category Set, which is not $\mathcal{U}$-small. Without a choice of $\mathcal{U}$, on the other hand, all sets form a proper class and not a category in the sense of this book. Similarly, Grp (or Ab) denotes the category of all groups (or abelian groups) realized on $\mathcal{U}$-sets; these are $\mathcal{U}$-categories.

To see that size is a genuine issue, observe that the Hom functor $\mathrm{Hom}_{\mathcal{C}} : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathsf{Set}$ can be defined only when $\mathcal{C}$ is a $\mathcal{U}$-category.

Unless otherwise stated, categories in this book are $\mathcal{U}$-categories, and $\mathcal{U}$-small categories are simply called **small**. A category not necessarily a $\mathcal{U}$-category is called **large**.

Large categories are hard to avoid in many constructions, especially in theories involving functor categories or localization. Whenever a large category may occur and has substantive consequences, this will be stated explicitly.

Functors and Functor Categories The identity functor of $\mathcal{C}$ is denoted by $\mathrm{id}_{\mathcal{C}}$. A functor $F : \mathcal{C} \to \mathcal{D}$ sends a morphism $f : X \to Y$ in $\mathcal{C}$ to a morphism $Ff : FX \to FY$ in $\mathcal{D}$. If $\mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(FX, FY)$ is injective (respectively bijective) for every X, Y, then F is faithful (respectively fully faithful). If every object of $\mathcal{D}$ is isomorphic to some FX, then F is essentially surjective.

For categories associated with partially ordered sets, an order-preserving map $P \to Q$ is precisely a functor $\mathcal{P} \to \mathcal{Q}$.

In some of the literature, a morphism $\varphi : F \to G$ between functors is also called a natural transformation and is sometimes written with a double arrow $\varphi : F \Rightarrow G$. It consists of a compatible family $(\varphi_X : FX \to GX)_{X \in \mathrm{Ob}(\mathcal{C})}$, and may be represented by the **2-cell** diagram introduced in [51, § 2.2]:

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left=50, "F"] \arrow[r, bend right=50, "G"'] \arrow[r, phantom, "\Downarrow\varphi"] & \mathcal{D}. \end{tikzcd}$$

When Ab-categories (see §1.3) or $\Bbbk$-linear categories (see §1.4) are discussed later, corresponding conditions will also be imposed on functors.

Given functors $\mathcal{B} \xrightarrow{R} \mathcal{C} \overset{F}{\underset{G}{\rightrightarrows}} \mathcal{D} \xrightarrow{L} \mathcal{E}$ and a morphism $\varphi : F \to G$, one naturally obtains $L\varphi : LF \to LG$ and $\varphi R : FR \to GR$. Morphisms $\varphi : F \to G$

[2)] In much of the literature, the objects of a category are understood to form a class, while the Hom between any two objects is a set. Thus their "classes" correspond to this book's sets, their "sets" to this book's $\mathcal{U}$-sets, and their "categories" essentially to this book's $\mathcal{U}$-categories.

and $\psi : G \to H$ can also be composed to form $\psi\varphi : F \to H$. These operations can be expressed as composition of 2-cells, that is, by pasting diagrams. The triangle identities reviewed shortly are a basic example.

All functors from $\mathcal{C}$ to $\mathcal{D}$ form the functor category $\mathcal{D}^{\mathcal{C}}$, also denoted $\mathrm{Fct}(\mathcal{C}, \mathcal{D})$. Unless $\mathcal{C}$ is small, $\mathcal{D}^{\mathcal{C}}$ is often a "large category." As special cases, for $n \in \mathbb{Z}_{\geq 0}$ and the corresponding totally ordered set $\mathbf{n}$, the categories $\mathcal{C}^{\mathbf{n}}$ are described as follows.

- ◇ by definition, $\mathcal{C}^{\mathbf{0}} := \mathbf{1}$;
- ◇ there is a unique functor $\mathcal{C} \to \mathbf{1}$, hence $\mathbf{1}^{\mathcal{C}} = \mathbf{1}$;
- ◇ specifying a functor $\mathbf{1} \to \mathcal{C}$ is equivalent to specifying an object of $\mathcal{C}$, hence $\mathcal{C}^{\mathbf{1}} \simeq \mathcal{C}$;
- ◇ the objects of $\mathcal{C}^{\mathbf{2}}$ are morphisms in $\mathcal{C}$, and its morphisms are commutative squares between them.

Thus $\mathbf{0}$ may be called the initial category, $\mathbf{1} = [\bullet]$ the terminal category, and $\mathbf{2} = [\bullet \to \bullet]$ may be pictured as a walking arrow.

The opposite category $\mathcal{C}^{\mathrm{op}}$ has the same objects as $\mathcal{C}$. Every morphism $f : X \to Y$ may be regarded as a morphism $Y \to X$ in $\mathcal{C}^{\mathrm{op}}$ and is then written f^{op}. Moreover, $\mathrm{Fct}(\mathcal{C}, \mathcal{D})^{\mathrm{op}}$ is naturally identified with $\mathrm{Fct}(\mathcal{C}^{\mathrm{op}}, \mathcal{D}^{\mathrm{op}})$. Passing from $\mathcal{C}$ to $\mathcal{C}^{\mathrm{op}}$ reverses arrows; this mechanism is called duality in category theory.

If functors $F : \mathcal{C} \leftrightarrows \mathcal{D} : G$ satisfy $GF \simeq \mathrm{id}_{\mathcal{C}}$ and $FG \simeq \mathrm{id}_{\mathcal{D}}$, they are mutually quasi-inverse. A functor having a quasi-inverse is an equivalence of categories, a fundamental notion in category theory. If the strict equalities $GF = \mathrm{id}_{\mathcal{C}}$ and $FG = \mathrm{id}_{\mathcal{D}}$ hold, then they are mutually inverse isomorphisms of categories.

Algebraic Structures The identity element of a group is denoted by 1 in multiplicative notation and by 0 for an abelian group in additive notation. The opposite group, with reversed multiplication, is G^{op}. If H is a subgroup of G, its index is $(G : H)$, also equal to the number of cosets $|G/H|$ or $|H\backslash G|$.

Unless otherwise stated, rings are associative and unital, as are algebras over commutative rings. The multiplicative identity of R is 1 or 1_R, and its invertible elements form the multiplicative group $R^{\times}$. The characteristic of a field or integral domain R is $\mathrm{char}(R)$. The opposite ring R^{op} has multiplication in the reverse order.

Let r be a nonzero element of a commutative ring R. If the additive-group homomorphism $R \xrightarrow{\text{multiplication by } r} R$ has a nonzero kernel, then r is a zero divisor. The ideal generated by $x, y, \ldots$ in a commutative ring is denoted by $(x, y, \ldots)$.

The polynomial algebra over a commutative ring R in variables $X_1, X_2, \ldots$ is $R[X_1, X_2, \ldots]$, and the formal power-series ring is $R[\![X_1, X_2, \ldots]\!]$. If M is a monoid, its monoid algebra over R is $R[M]$.

By convention, $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ denote, respectively, the ring of integers, the field of rational numbers, the field of real numbers, and the field of complex numbers. If q

is a prime power, $\mathbb{F}_q$ is the finite field with q elements. If $F \hookrightarrow E$ is an embedding of fields, the corresponding field extension is denoted by $E|F$ and its degree by $[E : F]$. If $E|F$ is Galois, its Galois group is $\mathrm{Gal}(E|F)$.

The common categories associated with some basic algebraic structures are denoted as follows.

Monoids	Mon
Groups	Grp
Abelian groups	$\mathsf{Ab} = \mathbb{Z}\text{-}\mathsf{Mod}$
Left or right R-modules	$R\text{-}\mathsf{Mod}$ or $\mathsf{Mod}\text{-}R$
$\Bbbk$-vector spaces	$\mathsf{Vect}(\Bbbk)$
(R, S)-bimodules	$(R, S)\text{-}\mathsf{Mod}$
$\Bbbk$-algebras	$\Bbbk\text{-}\mathsf{Alg}$
Commutative $\Bbbk$-algebras	$\Bbbk\text{-}\mathsf{CAlg}$

R, S: arbitrary rings,

$\Bbbk$: a field (for vector spaces),

$\Bbbk$: a commutative ring (for algebras),

the module category $\Bbbk\text{-}\mathsf{Mod}$ does not distinguish left from right.

If $\Bbbk$ is fixed and R, S are $\Bbbk$-algebras, then an (R, S)-bimodule is understood by default to arise from a left $R \underset{\Bbbk}{\otimes} S^{\mathrm{op}}$-module structure. In other words, the left and right actions of $\Bbbk$ are required to agree.

The group of homomorphisms between two left R-modules or two right R-modules is denoted by $\mathrm{Hom}_R(X, Y)$. For (R, S)-bimodules, the analogous notation is $\mathrm{Hom}_{(R,S)}(X, Y)$.

For a commutative ring R and a multiplicative subset U, the corresponding localization is $R[U^{-1}]$.

As with the category of sets Set, groups, rings, modules, and the other structures here are understood to be realized on $\mathcal{U}$-sets for the chosen Grothendieck universe $\mathcal{U}$. Every category in the table therefore has a forgetful functor to Set. Other examples used in this book, such as the category Top of topological spaces and the category CHaus of compact Hausdorff spaces, obey the same convention: their spaces are realized on $\mathcal{U}$-sets.

Adjunctions An adjoint pair (F, G, φ) consists of two functors

$$\mathcal{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{C}'$$

together with a family of canonical bijections $\varphi_{X,Y} : \mathrm{Hom}_{\mathcal{C}'}(FX, Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(X, GY)$, that is, an isomorphism of functors

$$\varphi : \mathrm{Hom}_{\mathcal{C}'}(F(\cdot), \cdot) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(\cdot, G(\cdot)) : \mathcal{C}^{\mathrm{op}} \times \mathcal{C}' \to \mathsf{Set}.$$

An adjoint pair is also often displayed as

$$F : \mathcal{C} \rightleftarrows \mathcal{C}' : G$$

meaning that F is left adjoint to G, or G is right adjoint to F. The datum φ is frequently omitted, and the adjoint pair is then written (F, G).

Equivalently, φ can be characterized by a **unit** $\eta : \mathrm{id}_{\mathcal{C}} \to GF$ and a **counit** $\varepsilon : FG \to \mathrm{id}_{\mathcal{C}'}$. The data $(F, G, \eta, \varepsilon)$ form an adjoint pair if and only if they satisfy the **triangle identities** :

$$G\varepsilon \circ \eta G = \mathrm{id}_G, \quad \varepsilon F \circ F\eta = \mathrm{id}_F.$$

See [51, (2.6) + Proposition 2.6.5]. Equivalently, these identities may be written as equalities of composites of 2-cells:

$$\begin{tikzcd} \mathcal{C}' \arrow[r, "G"] \arrow[rr, bend right=60, "\mathrm{id}_{\mathcal{C}'}"'] \arrow[r, phantom, bend right=40, "\Downarrow\varepsilon"] & \mathcal{C} \arrow[r, "F"] \arrow[rr, bend left=60, "\mathrm{id}_{\mathcal{C}}"] & \mathcal{C}' \arrow[r, "G"] \arrow[u, phantom, "\Downarrow\eta"] & \mathcal{C} \end{tikzcd} = \begin{tikzcd} \mathcal{C}' \arrow[r, bend left=50, "G"] \arrow[r, bend right=50, "G"'] \arrow[r, phantom, "\Downarrow \mathrm{id}_G"] & \mathcal{C} \end{tikzcd},$$

$$\begin{tikzcd} \mathcal{C} \arrow[r, "F"] \arrow[rr, bend left=60, "\mathrm{id}_{\mathcal{C}}"] & \mathcal{C}' \arrow[r, "G"] \arrow[u, phantom, "\Downarrow\eta"] \arrow[rr, bend right=60, "\mathrm{id}_{\mathcal{C}'}"'] & \mathcal{C} \arrow[r, "F"] \arrow[d, phantom, "\Downarrow\varepsilon"] & \mathcal{C}' \end{tikzcd} = \begin{tikzcd} \mathcal{C} \arrow[r, bend left=50, "F"] \arrow[r, bend right=50, "F"'] \arrow[r, phantom, "\Downarrow \mathrm{id}_F"] & \mathcal{C}' \end{tikzcd}.$$

For details, see [51, Remark 2.6.6].

Limits Consider a functor $\alpha : I \to \mathcal{C}$.

- ◇ The inductive limit of α, if it exists, is denoted by $\varinjlim \alpha$ or $\varinjlim_{i \in \mathrm{Ob}(I)} \alpha(i)$. It is an object of $\mathcal{C}$ equipped with morphisms $\alpha(i) \to \varinjlim \alpha$ and determined by a universal property.
- ◇ Similarly, the projective limit of α, if it exists, is denoted by $\varprojlim \alpha$ or $\varprojlim_{i \in \mathrm{Ob}(I)} \alpha(i)$. It is equipped with morphisms $\varprojlim \alpha \to \alpha(i)$ and determined by a universal property.

The two kinds of limits are dual: the $\varinjlim$ of $\alpha : I \to \mathcal{C}$ is the $\varprojlim$ of $\alpha^{\mathrm{op}} : I^{\mathrm{op}} \to \mathcal{C}^{\mathrm{op}}$. For this reason, functors involved in $\varprojlim$ are often written in the form $I^{\mathrm{op}} \to \mathcal{C}$.

In much of the literature, $\varinjlim$ is called a colimit and denoted by colim, while $\varprojlim$ is called a limit and denoted by lim. Unless otherwise stated, “limit” in this book includes both $\varinjlim$ and $\varprojlim$.

If I is small, the corresponding $\varinjlim$ or $\varprojlim$ is a **small limit**. Following the standard terminology of [51, § 2.8], a category $\mathcal{C}$ having all small $\varprojlim$ (respectively all small $\varinjlim$) is **complete** (respectively **cocomplete**). This notion depends on the choice of $\mathcal{U}$.

To fix notation, choose a category $\mathcal{C}$ and review several common special cases of limits. Detailed definitions may be found in [51, § 2.7] or other textbooks. In the universal properties below, T is an arbitrary object of $\mathcal{C}$ and I is an arbitrary set.

Limit	Input	Object	Canonical morphism	Universal property
Equalizer	$X \overset{f}{\underset{g}{\rightrightarrows}} Y$	$\ker(f,g)$	$\ker(f,g) \xrightarrow{\iota} X$	$\begin{array}{cc} \mathrm{Hom}(T, \ker(f,g)) & \psi \\ {\scriptstyle 1:1}\downarrow & \downarrow \\ \left\{ \begin{array}{l} \phi \in \mathrm{Hom}(T,X) : \\ f\phi = g\phi \end{array} \right\} & \iota\psi \end{array}$
Coequalizer	(as above)	$\operatorname{coker}(f,g)$	$Y \xrightarrow{p} \operatorname{coker}(f,g)$	$\begin{array}{cc} \mathrm{Hom}(\operatorname{coker}(f,g), T) & \psi \\ {\scriptstyle 1:1}\downarrow & \downarrow \\ \left\{ \begin{array}{l} \phi \in \mathrm{Hom}(Y,T) : \\ \phi f = \phi g \end{array} \right\} & \psi p \end{array}$
Product	$(X_i)_{i \in I}$	$\prod_{i \in I} X_i$	$\prod_{j \in I} X_j \xrightarrow{p_i} X_i$	$\begin{array}{cc} \mathrm{Hom}\left(T, \prod_{i \in I} X_i\right) & \psi \\ {\scriptstyle 1:1}\downarrow & \downarrow \\ \prod_{i \in I} \mathrm{Hom}(T, X_i) & (p_i\psi)_{i \in I} \end{array}$
Coproduct	(as above)	$\coprod_{i \in I} X_i$	$X_i \xrightarrow{\iota_i} \coprod_{j \in I} X_j$	$\begin{array}{cc} \mathrm{Hom}\left(\coprod_{i \in I} X_i, T\right) & \psi \\ {\scriptstyle 1:1}\downarrow & \downarrow \\ \prod_{i \in I} \mathrm{Hom}(X_i, T) & (\psi\iota_i)_{i \in I} \end{array}$

Translator's note. In the coproduct row, the source gives the target of the canonical injection as $\prod_{j\in I} X_j$. It has been corrected to $\coprod_{j\in I} X_j$, in agreement with the preceding object column and the following universal property; see correction O014-C003.

Finite products (respectively coproducts) are also written $X_1 \times X_2 \times \cdots$ (respectively $X_1 \sqcup X_2 \sqcup \cdots$). If $\mathcal{C}$ is additive, the coproduct $\coprod_i$ is often written using the direct-sum symbol $\bigoplus_i$ from module theory.

Equalizers and coequalizers, and products and coproducts, form dual pairs. These universal-property characterizations may also be written as commutative diagrams as in [51, § 2.7], or interpreted through the Yoneda embedding reviewed in §A.1. Suppose that $\coprod_{i\in I} X_i$ and $\prod_{j\in J} Y_j$ exist. Their universal properties give a bijection

$$\operatorname{Hom}\left(\coprod_{i\in I} X_i, \prod_{j\in J} Y_j\right) \xrightarrow{1:1} \prod_{\substack{i\in I\\ j\in J}} \operatorname{Hom}(X_i, Y_j), \quad \psi \mapsto (p_j \psi \iota_i)_{i,j}.$$

- ▷ Terminal object A product corresponding to $I = \varnothing$, if it exists, is a terminal object of $\mathcal{C}$, temporarily denoted by Z. By convention, $\prod_{i\in\varnothing} := \{\varnothing\}$, so the universal property of Z says that $\operatorname{Hom}(T, Z)$ is a singleton for every $T \in \operatorname{Ob}(\mathcal{C})$.

- ▷ Initial object A coproduct corresponding to $I = \varnothing$, if it exists, is an initial object of $\mathcal{C}$, temporarily denoted by S. Its universal property says that $\operatorname{Hom}(S, T)$ is a singleton for every $T \in \operatorname{Ob}(\mathcal{C})$.

- ▷ Zero object and zero morphism If an object 0 of $\mathcal{C}$ is both initial and terminal, it is called a zero object; morphisms to and from it exist and are unique. For $X, Y \in \operatorname{Ob}(\mathcal{C})$, the composite $X \to 0 \to Y$ gives a canonical element of $\operatorname{Hom}(X, Y)$ called the zero morphism.

Because an isomorphism from any object X to a zero object, if it exists, is unique, one customarily says that X is a zero object or writes $X = 0$, rather than saying that X is isomorphic to a zero object.

We next review two important, mutually dual kinds of limit.

Name	Input	Object	Canonical morphism	Universal property
Fiber product	$\left(X_i \xrightarrow{f_i} Z\right)_{i\in I}$	$\prod_{i\in I}(X_i \to Z)$	$\prod_{j\in I}(X_j \to Z) \xrightarrow{p_i} X_i$	$\mathrm{Hom}\left(T, \prod_{i\in I}(X_i \to Z)\right) \xrightarrow[1:1]{\psi\mapsto(p_i\psi)_i} \left\{\begin{array}{l}(\xi_i : T \to X_i)_{i\in I} : \\ \forall i,j \in I, \\ f_i\xi_i = f_j\xi_j\end{array}\right\}$
Fiber coproduct	$\left(Z \xrightarrow{g_i} X_i\right)_{i\in I}$	$\coprod_{i\in I}(Z \to X_i)$	$X_i \xrightarrow{\iota_i} \coprod_{j\in I}(Z \to X_j)$	$\mathrm{Hom}\left(\coprod_{i\in I}(Z \to X_i), T\right) \xrightarrow[1:1]{\psi\mapsto(\psi\iota_i)_i} \left\{\begin{array}{l}(\xi_i : X_i \to T)_{i\in I} : \\ \forall i,j \in I, \\ \xi_i g_i = \xi_j g_j\end{array}\right\}$

Taking $\psi = \mathrm{id}$ in the universal property of the fiber product shows that $f_ip_i = f_jp_j$ for all i, j, yielding a canonical morphism $\prod_{i\in I}(X_i \to Z) \to Z$. Similarly, for the fiber coproduct, taking $\psi = \mathrm{id}$ shows that $\iota_i g_i = \iota_j g_j$ for all i, j, yielding a canonical morphism $Z \to \coprod_{i\in I}(Z \to X_i)$.

Fiber products of finitely many objects $X_1 \underset{Z}{\times} X_2 \underset{Z}{\times} \cdots$, and fiber coproducts $X_1 \underset{Z}{\sqcup} X_2 \underset{Z}{\sqcup} \cdots$, reduce to repeated instances of the two-object case. Consider the following two commutative diagrams.

$$\begin{tikzcd} X \underset{Z}{\times} Y \arrow[r, "p_1"] \arrow[d, "p_2"'] & X \arrow[d] \\ Y \arrow[r] & Z \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} Z \arrow[r] \arrow[d] & X \arrow[d, "\iota_1"] \\ Y \arrow[r, "\iota_2"'] & X \underset{Z}{\sqcup} Y \end{tikzcd}$$

These are called a **pullback** diagram and a **pushout** diagram, respectively. The morphism $X \underset{Z}{\times} Y \to Y$ is called the pullback of $X \to Z$, and $Y \to X \underset{Z}{\sqcup} Y$ the pushout of $Z \to X$; the roles of X and Y are symmetric. More generally, commutative diagrams isomorphic to these are also called pullback and pushout diagrams; see [51, Definition 2.8.5].

In this book, the center of the square in a pullback (respectively pushout) diagram is marked by $\square$ (respectively $\boxplus$).

Translation In the Chinese source edition, the conventions for translation agree with [51]; Chinese terminology is kept compatible with [50] unless that work's translation is clearly inappropriate or erroneous.

In the source index, Chinese terms are accompanied by English equivalents. This helps readers who need to read or write articles in English, provides a reference for readers whose native language is not Chinese, and, finally, reflects the historical fact that knowledge of foreign languages remains necessary for understanding the meaning of many mathematical symbols.

> **Translator's note.** This English edition indexes English headwords and records the Chinese–English terminological choices in a separate production glossary.

Core Part

Chapter 1 Supplements to Category Theory

As its title suggests, this chapter introduces several tools from category theory that will be needed in later chapters, both as a review and as an opportunity to see familiar ideas afresh. Some of this material appears in [51] or similar textbooks; it is nevertheless restated here in order to review the concepts and standardize the notation. For other basic terminology, the reader may consult the introduction to this book.

- ◇ Sections 1.1–1.2 begin with a brief introduction to subobjects and quotient objects, including the general properties of monomorphisms and epimorphisms. They then introduce the image $\mathrm{im}(f)$ and coimage $\mathrm{coim}(f)$ of a morphism f, as well as morphisms for which "image equals coimage," called strict morphisms. These are basic notions that make sense in arbitrary categories.

- ◇ Sections 1.3–1.4 form the linear part of the chapter. They introduce a special kind of category in which every morphism set has the structure of an abelian group, called an Ab-category; a functor that preserves addition of morphisms is called additive. In such categories one can define kernels and cokernels of morphisms, generalizing those of module homomorphisms. An Ab-category with a zero object and finite products is called an additive category. In an additive category one can form finite direct sums of objects, and morphisms between finite direct sums can be manipulated in much the same way as matrices.

More generally, the additive structure can be enhanced with scalar multiplication by a commutative ring $\Bbbk$, leading to the notions of a $\Bbbk$-Mod-category and a $\Bbbk$-linear category. Categories with these linear structures are a principal concern of this book.

◇ Sections 1.5–1.6 cover the foundations of limits. In particular, Definition 1.5.2 introduces the terminology for a functor $F : \mathcal{C} \to \mathcal{D}$ that preserves limits (from $\mathcal{C}$ to $\mathcal{D}$) or creates them (from $\mathcal{D}$ to $\mathcal{C}$). Cofinal functors and filtered inductive limits are also discussed; these are useful and important notions.

◇ Left and right Kan extensions, discussed in Sections 1.7–1.8, are fundamental operations in category theory. Under suitable conditions they can be constructed using $\varinjlim$ or $\varprojlim$. The argument is somewhat involved, but the underlying idea is simple: express the problem of extending a functor as a "best approximation" formulated in terms of limits. We also record the non-obvious Theorem 1.7.7, which explains the connection between absolute left or right Kan extensions and adjoint pairs.

◇ Gabriel–Zisman localization, introduced in Section 1.9 and henceforth called simply **localization**, is the construction that formally adjoins inverses to a family S of morphisms in a category. It is analogous to localization of rings, but the concrete details are considerably more complicated. To retain better control over the resulting category, in practice S is often assumed to be a **multiplicative system** in the sense of Definition 1.9.5. The principal use of localization in this book is to construct derived categories; another is to define Serre quotients of abelian categories in Section 2.9. Section 1.10 explains how to form Kan extensions along a localization of categories. This may be viewed as the localization of a functor and is essential for understanding derived functors.

◇ The adjoint functor theorem discussed in Section 1.11 characterizes, under suitable conditions, when a functor has a left or right adjoint. Its principal application in this book is to the study of Grothendieck categories in Section 2.10. The theorem has uses far beyond this application, so the section is valuable in its own right. The notions of generator and cogenerator discussed there (Definition 1.11.8) are also essential.

This chapter draws extensively on [23, 37].

Reading Guide

The basic theory of abelian categories and complexes requires only Sections 1.1–1.4 of this chapter. Sections 1.5–1.6 provide supporting material and are also cited repeatedly in other chapters. The theory of derived categories uses Sections 1.7–1.10. This material has applications beyond that theory, but readers may judge their own needs and postpone it until just before studying derived categories. The final section, Section 1.11, is used only in Section 2.10; nevertheless, its material on generators and cogenerators is essential and may be consulted as needed.

1.1 Subquotients

We begin by recalling the notions of monomorphism and epimorphism. Fix a category $\mathcal{C}$.

- A morphism $f : X \to Y$ is called a **monomorphism**, also written $f : X \hookrightarrow Y$, if left cancellation holds:
$$\forall T \in \mathrm{Ob}(\mathcal{C}),\ \forall g, h : T \to X, \quad fg = fh \iff g = h;$$
- A morphism $f : X \to Y$ is called an **epimorphism**, also written $f : X \twoheadrightarrow Y$, if right cancellation holds:
$$\forall T \in \mathrm{Ob}(\mathcal{C}),\ \forall g, h : Y \to T, \quad gf = hf \iff g = h.$$

It follows immediately that $f : X \to Y$ is monic if and only if $f_* : \mathrm{Hom}(T, X) \to \mathrm{Hom}(T, Y)$ is injective for every T. Likewise, f is epic if and only if $f^* : \mathrm{Hom}(Y, T) \to \mathrm{Hom}(X, T)$ is injective for every T. Monicity and epicity are dual properties: f is monic if and only if f^{op} is epic.

Example 1.1.1 If the equalizer $\ker(f, g)$ of a pair of morphisms $f, g : X \to Y$ exists, its universal property shows that the canonical morphism $\iota : \ker(f, g) \to X$ is monic. Dually, if the coequalizer $\mathrm{coker}(f, g)$ exists, the canonical morphism $p : Y \to \mathrm{coker}(f, g)$ is epic. Here is another simple observation: if $\mathcal{C}$ has a zero object, denoted by 0, then $0 \to X$ is monic and $X \to 0$ is epic. The reader is invited to verify these details.

Proposition 1.1.2 Consider morphisms $X \xrightarrow{a} Y \xrightarrow{b} Z$. Suppose that b is monic or that a is epic. Then $ba : X \to Z$ is an isomorphism if and only if both a and b are isomorphisms.

Proof One direction is clear. Now suppose that ba is an isomorphism and that b is monic. Set $c := a(ba)^{-1} : Z \to Y$; then $bc = \mathrm{id}_Z$. Moreover, $b(cb) = (bc)b = b$, so left cancellation by b gives $cb = \mathrm{id}_Y$. Thus b is an isomorphism, and consequently so is a. Passing to $\mathcal{C}^{\mathrm{op}}$ gives the case in which a is epic. □

In categories commonly used in algebra, such as **Set**, **Ab**, and R-**Mod**, a morphism is monic or epic precisely when its underlying map is injective or surjective, respectively. In these examples one can also speak of subsets, subgroups, and so forth. This notion extends to arbitrary categories, but one additional step is needed. We formulate it in the language of partially ordered sets.

Let $\mathcal{C}$ be a category and let X be an object of $\mathcal{C}$. Given a pair of monomorphisms $f_i : S_i \hookrightarrow X$, where $i = 1, 2$, write $(S_1, f_1) \subset (S_2, f_2)$, or simply $S_1 \subset S_2$, if there is a morphism $g : S_1 \to S_2$ such that $f_2 g = f_1$. The monicity of f_2 implies that g, if it exists, is unique and is itself monic. The relation $\subset$ defined in this way is not yet a partial order. If $S_1 \subset S_2$ and $S_2 \subset S_1$, we say that S_1 and S_2 are equivalent. This is the same as requiring a commutative diagram

$$\begin{tikzcd} S_1 \arrow[r, "g_1"] \arrow[dr, hook, "f_1"'] & S_2 \arrow[r, "g_2"] \arrow[d, hook, "f_2"] & S_1 \arrow[dl, hook', "f_1"] \\ & X & \end{tikzcd}$$

Uniqueness gives $g_2 g_1 = \mathrm{id}_{S_1}$ and, similarly, $g_1 g_2 = \mathrm{id}_{S_2}$. Thus S_1 and S_2 are equivalent if and only if there is an isomorphism $g : S_1 \xrightarrow{\sim} S_2$ with $f_2 g = f_1$; moreover, this isomorphism is unique. This is, of course, a standard argument in algebra.

Similarly, given a pair of epimorphisms $f_i : X \twoheadrightarrow Q_i$, where $i = 1, 2$, if there is a morphism $g : Q_2 \to Q_1$ such that $g f_2 = f_1$, then g is unique and epic; in this case write $Q_1 \twoheadleftarrow Q_2$. If $Q_1 \twoheadleftarrow Q_2$ and $Q_2 \twoheadleftarrow Q_1$, we say that Q_1 and Q_2 are equivalent. Equivalently, there is a unique isomorphism $g : Q_2 \xrightarrow{\sim} Q_1$ satisfying $g f_2 = f_1$.

The binary relation $\subset$ (or $\twoheadleftarrow$) induces a partial order on the corresponding equivalence classes. We have borrowed the set-inclusion symbol $\subset$ here. Although this may cause some ambiguity, it is consistent with the notation for other algebraic substructures, such as subgroups, submodules, and subspaces.

Definition 1.1.3 Let $\mathcal{C}$ be a category and let X be an object of $\mathcal{C}$. An equivalence class, under the relation above, of monomorphisms $S \hookrightarrow X$ (respectively, of epimorphisms $X \twoheadrightarrow Q$) is called a **subobject** (respectively, a **quotient object**) of X. These equivalence classes form the partially ordered set $(\mathrm{Sub}_X, \subset)$ (respectively, $(\mathrm{Quot}_X, \twoheadleftarrow)$), in which $\mathrm{id}_X : X \to X$ gives the unique maximal element.

Subobjects and quotient objects are dual: $(\mathrm{Sub}_X, \subset)$ as defined in $\mathcal{C}$ is the same as $(\mathrm{Quot}_X, \twoheadleftarrow)$ as defined in $\mathcal{C}^{\mathrm{op}}$. For convenience, this section deals mainly with subobjects. Because the isomorphism g occurring in the equivalence relation is unique, one may choose a representative $f : S \hookrightarrow X$ of an equivalence class without ambiguity, or abbreviate that representative simply as S. Recall that $f_* : \mathrm{Hom}(\cdot, S) \to \mathrm{Hom}(\cdot, X)$ is injective.

Lemma 1.1.4 For any two subobjects S_1, S_2 of X, one has

$$S_1 \subset S_2 \iff \forall T \in \mathrm{Ob}(\mathcal{C}),\ \mathrm{Hom}(T, S_1) \subset \mathrm{Hom}(T, S_2),$$

where $\mathrm{Hom}(T, S_1)$ and $\mathrm{Hom}(T, S_2)$ on the right-hand side are understood as subsets of $\mathrm{Hom}(T, X)$.

Proof The implication $\implies$ is immediate. Conversely, take $T := S_1$. Under $\mathrm{Hom}(S_1, S_1) \subset \mathrm{Hom}(S_1, X)$, the element id_{S_1} corresponds to f_1. The hypothesis supplies $g \in \mathrm{Hom}(S_1, S_2)$ whose image in $\mathrm{Hom}(S_1, X)$ is f_1, that is, $f_2 g = f_1$. This is the required morphism. $\square$

Suppose that $\mathcal{C}$ has a zero object and that $(X_i)_{i \in I}$ is a family of objects. For every $(i, j) \in I \times I$, define $\delta_{ij} \in \mathrm{Hom}(X_i, X_j)$ by $\delta_{ij} = \mathrm{id}$ when $i = j$, and as the zero morphism otherwise. If the product and coproduct of $(X_i)_{i \in I}$ exist, then $(\delta_{ij})_{i,j}$ determines a canonical morphism

$$\delta : \coprod_{i \in I} X_i \to \prod_{i \in I} X_i. \tag{1.1.1}$$

For the additive categories to be reviewed shortly, δ is always an isomorphism when I is finite. This need not hold in an arbitrary category.

Lemma 1.1.5 Let $X \to Z$ be a monomorphism. Its pullback along $Y \to Z$, namely $X \underset{Z}{\times} Y \to Y$ (if it exists), is also a monomorphism. Dually, a pushout of an epimorphism is again an epimorphism.

Proof By duality, it suffices to treat the case of monomorphisms. Let $p_2 : X \underset{Z}{\times} Y \to Y$ be the canonical morphism supplied by the fiber product. The question reduces to showing, for every object T, that

$$\begin{tikzcd}
\mathrm{Hom}(T, X) \underset{\mathrm{Hom}(T,Z)}{\times} \mathrm{Hom}(T, Y) \arrow[d, leftrightarrow, "1:1"', "\text{universal property}"] & \\
\mathrm{Hom}\left(T, X \underset{Z}{\times} Y\right) \arrow[r, "(p_2)_*"'] & \mathrm{Hom}(T, Y)
\end{tikzcd}$$

is injective. Indeed, the monicity of $X \to Z$ makes $\mathrm{Hom}(T, X) \to \mathrm{Hom}(T, Z)$ injective, and the map from the upper left to the lower right is the evident projection from the fiber product. $\square$

Lemma 1.1.6 Let $(f_i : X_i \to Z)_{i \in I}$ be a family of monomorphisms. Then the canonical morphism from $\prod_{i \in I}(X_i \to Z)$ (if it exists) to Z is also a monomorphism. Dually, given a family of epimorphisms $(g_i : Z \to X_i)_{i \in I}$, the canonical morphism from Z to $\coprod_{i \in I}(Z \to X_i)$ (if it exists) is also an epimorphism.

Proof As in Lemma 1.1.5, the question reduces to showing that the bottom row of

$$\begin{tikzcd}
\prod_{i \in I} \left(\mathrm{Hom}(T, X_i) \to \mathrm{Hom}(T, Z)\right) \arrow[d, leftrightarrow, "1:1"', "\text{universal property}"] & \\
\mathrm{Hom}\left(T, \prod_{i \in I}(X_i \to Z)\right) \arrow[r] & \mathrm{Hom}(T, Z)
\end{tikzcd}$$

is injective. This follows from the hypothesis, since every map $\mathrm{Hom}(T, X_i) \to \mathrm{Hom}(T, Z)$ is injective. $\square$

We return to subobjects and quotient objects.

Definition 1.1.7 Let $\mathcal{C}$ be any category and let X, Y be objects of $\mathcal{C}$. If Y can be realized as a quotient object of a subobject of X, then Y is called a **subquotient** of X.

Considering a subobject of a quotient object amounts to discussing a subquotient in $\mathcal{C}^{\mathrm{op}}$. The next result shows that, for a certain class of categories, the two approaches yield the same notion. This class includes the abelian categories studied later; see Corollary 2.1.7.

Proposition 1.1.8 Suppose that $\mathcal{C}$ has finite fiber products and finite fiber coproducts, that pullbacks of epimorphisms remain epimorphisms, and that pushouts of monomorphisms remain monomorphisms. Then a subquotient may equivalently be defined as a subobject of a quotient object.

Proof Consider a subquotient diagram $Y \twoheadleftarrow Z \hookrightarrow X$ in $\mathcal{C}$, and set $W := X \sqcup_Z Y$. The morphism $X \to W$ is the pushout of $Z \to Y$, so it is epic by Lemma 1.1.5; the morphism $Y \to W$ is the pushout of $Z \to X$, so it is monic by hypothesis. Thus $Y \hookrightarrow W \twoheadleftarrow X$, exhibiting a subquotient of X in $\mathcal{C}^{\mathrm{op}}$. Since the assumptions are self-dual, a subquotient in $\mathcal{C}^{\mathrm{op}}$ likewise yields a subquotient in $\mathcal{C}$. $\square$

Once the property "quotient of a subobject = subobject of a quotient" has been established, it follows that a subquotient of a subquotient is again a subquotient.

The definition of a subquotient does not specify how it is to be realized as a quotient object of a subobject. For example, in $\mathcal{C} = \mathsf{Ab}$, $\mathbb{Z}/2\mathbb{Z}$ is a subquotient of $\mathbb{Z}/4\mathbb{Z}$: it can be realized either as a quotient object of $\mathbb{Z}/4\mathbb{Z}$ or as the subobject $2\mathbb{Z}/4\mathbb{Z}$ of $\mathbb{Z}/4\mathbb{Z}$.

1.2 Images, Coimages, and Strict Morphisms

Let $\mathcal{C}$ be a category. For any morphism $f : X \to Y$, the fiber coproduct $Y \underset{X}{\sqcup} Y$, if it exists, comes equipped with a canonical pair $Y \rightrightarrows Y \underset{X}{\sqcup} Y$. Dually, the fiber product $X \underset{Y}{\times} X$, if it exists, comes equipped with a canonical pair $X \underset{Y}{\times} X \rightrightarrows X$, usually called the projection morphisms.

Definition 1.2.1 (Image and Coimage) Let $f : X \to Y$ be a morphism in $\mathcal{C}$. Suppose that $Y \underset{X}{\sqcup} Y$ exists. If the equalizer $\ker\left(Y \rightrightarrows Y \underset{X}{\sqcup} Y\right)$ exists, this equalizer is called the **image** of f and denoted by $\operatorname{im}(f)$; it carries a canonical monomorphism $\operatorname{im}(f) \hookrightarrow Y$.

Dually, suppose that $X \underset{Y}{\times} X$ exists. If the coequalizer $\operatorname{coker}\left(X \underset{Y}{\times} X \rightrightarrows X\right)$ exists, this coequalizer is called the **coimage** of f and denoted by $\operatorname{coim}(f)$; it carries a canonical epimorphism $X \twoheadrightarrow \operatorname{coim}(f)$.

Remark 1.2.2 The duality between ker and coker immediately shows that $\operatorname{im}(f)$ exists in $\mathcal{C}$ if and only if $\operatorname{coim}(f^{\mathrm{op}})$ exists in $\mathcal{C}^{\mathrm{op}}$, and then $\operatorname{im}(f) = \operatorname{coim}(f^{\mathrm{op}})$.

Besides being useful, the following lemma sheds light on the nature of images and coimages.

Lemma 1.2.3 Suppose that $\operatorname{coim}(f)$ exists (respectively, that $\operatorname{im}(f)$ exists) for a morphism $f : X \to Y$. For any monomorphism $j : Y' \hookrightarrow Y$ (respectively epimorphism $q : X \twoheadrightarrow X'$), if $g : X \to Y'$ satisfies $jg = f$ (respectively $g : X' \to Y$ satisfies $gq = f$), then there is a unique $\overline{g}$ making the corresponding diagram commute:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, two heads] \arrow[dr, "g"] & Y \\ \operatorname{coim}(f) \arrow[r, "\overline{g}"'] & Y' \arrow[u, hook, "j"'] \end{tikzcd} \quad \text{or} \quad \begin{tikzcd} X \arrow[r, "f"] \arrow[d, two heads, "q"'] & Y \\ X' \arrow[r, "\overline{g}"'] \arrow[ur, "g"] & \operatorname{im}(f). \arrow[u, hook] \end{tikzcd}$$

Proof The two versions are dual, so it suffices to consider the one involving j. Let p_1, p_2 be the projections $X \underset{Y}{\times} X \rightrightarrows X$. Since $jgp_1 = fp_1 = fp_2 = jgp_2$ and j is monic, $gp_1 = gp_2$. The universal property of $\operatorname{coim}(f)$ as a coequalizer then uniquely determines $\overline{g}$ making the diagram commute. $\square$

Applying Lemma 1.2.3 successively to $\operatorname{coim}(f)$ and $\operatorname{im}(f)$ shows that there is a comparison morphism $\operatorname{coim}(f) \to \operatorname{im}(f)$ making the following diagram commute:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, two heads] & Y \\ \operatorname{coim}(f) \arrow[r] & \operatorname{im}(f) \arrow[u, hook] \end{tikzcd} \tag{1.2.1}$$

This comparison morphism is also unique, by left cancellation with a monomorphism or right cancellation with an epimorphism.

Definition 1.2.4 (Strict morphism) Suppose that $\operatorname{im}(f)$ and $\operatorname{coim}(f)$ exist. If the morphism $\operatorname{coim}(f) \to \operatorname{im}(f)$ in (1.2.1) is an isomorphism, then f is called **strict**.

For example, every morphism in $\mathcal{C} = \mathsf{Set}$ is strict, and both its image and its coimage are canonically identified with its set-theoretic image. Further examples appear in the exercises for this chapter.

Proposition 1.2.5 Given morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$, there are unique morphisms $\alpha : \operatorname{coim}(f) \to \operatorname{im}(gf)$ and $\beta : \operatorname{coim}(gf) \to \operatorname{im}(g)$ making the following diagrams commute, provided the indicated coimages and images exist:

$$\begin{tikzcd} X \arrow[r, two heads] \arrow[dr] & \operatorname{coim}(f) \arrow[d, "\alpha"] \\ & \operatorname{im}(gf) \end{tikzcd} \qquad \begin{tikzcd} Z & \operatorname{im}(g) \arrow[l, hook'] \\ & \operatorname{coim}(gf). \arrow[ul] \arrow[u, "\beta"'] \end{tikzcd}$$

Proof For α, uniqueness follows by right cancellation with the epimorphism $X \twoheadrightarrow \operatorname{coim}(f)$. Existence follows by applying the right-hand version of Lemma 1.2.3 to

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, two heads] & Y \arrow[r, "g"] & Z \\ \operatorname{coim}(f) \arrow[ur] \arrow[urr] & & \operatorname{im}(gf) \arrow[u, hook] \end{tikzcd}$$

where $\operatorname{coim}(f) \to Y$ is the composite $\operatorname{coim}(f) \to \operatorname{im}(f) \hookrightarrow Y$. □

Lemma 1.2.6 For a morphism $f : X \to Y$ in $\mathcal{C}$,

$$f \text{ epic} \iff \operatorname{im}(f) \xrightarrow{\sim} Y, \quad f \text{ monic} \iff X \xrightarrow{\sim} \operatorname{coim}(f).$$

Proof By duality it suffices to prove the first equivalence. If f is epic, the canonical morphisms $i_1, i_2 : Y \rightrightarrows Y \sqcup_X Y$ satisfy $i_1 f = i_2 f$, hence $i_1 = i_2$, and therefore $\operatorname{im}(f) := \ker(i_1, i_2) = Y$.

Conversely, suppose the canonical morphism $\operatorname{im}(f) \to Y$ is an isomorphism. Since composing it with i_1 and i_2 gives the same morphism, again $i_1 = i_2$. If $Z \in \operatorname{Ob}(\mathcal{C})$ and

$g_1, g_2 : Y \rightrightarrows Z$ satisfy $g_1 f = g_2 f$, the universal property yields $g : Y \coprod_X Y \to Z$ with $g_j = g i_j$ for $j = 1, 2$. Hence $g_1 = g_2$, proving that f is epic. □

Proposition 1.2.7 A strict morphism f is an isomorphism if and only if it is both monic and epic.

Proof If f is strict, monic, and epic, factor it as $X \twoheadrightarrow \mathrm{coim}(f) \xrightarrow{\sim} \mathrm{im}(f) \hookrightarrow Y$ and apply Proposition 1.1.2 and Lemma 1.2.6. □

The preceding arguments involve diagrams $X \twoheadrightarrow C \hookrightarrow Y$ whose composite is f. Such a diagram is an **epi–mono factorization** of $f : X \to Y$. In practice one studies not only the factorizations themselves, but also the morphisms between them.

Lemma 1.2.8 Consider the diagram below, whose solid-arrow part commutes,

$$\begin{tikzcd} X \arrow[r, "e"] \arrow[d, "a"'] & C \arrow[r, "m"] \arrow[d, dashed, "\varphi"] & Y \arrow[d, "b"] \\ X' \arrow[r, "e'"'] & C' \arrow[r, "m'"'] & Y' \end{tikzcd}$$

and suppose that e, e' are epic (respectively m, m' are monic).

(i) If a φ as shown by the dashed arrow makes the left half (respectively right half) commute, then the other half also commutes. Such a φ, if it exists, is unique.

(ii) If a is epic (respectively b is monic) and φ exists, then φ is epic (respectively monic).

Proof Suppose that $\varphi : C \to C'$ makes the left half commute and that e, e' are epic. The equation $\varphi e = e'a$, together with the epicity of e, uniquely determines φ. Since the outer rectangle commutes, $m'\varphi e = m'e'a = bme$; cancelling e then gives $m'\varphi = bm$, so the right half commutes. If a is also epic, then $\varphi e = e'a$ is epic, and hence φ is epic.

This proves one half of (i) and (ii). The case where φ makes the right half commute and m, m' are monic is analogous, or equivalently dual. □

Definition 1.2.9 For $f : X \to Y$, form the category $\mathsf{epi.mono}(f)$ whose objects are all epi–mono factorizations of f and whose morphisms are commutative diagrams

$$\begin{tikzcd} & C \arrow[dd, "\varphi"] \arrow[dr, hook, "m"] & \\ X \arrow[ur, two heads, "e"] \arrow[dr, two heads, "e'"'] & & Y. \\ & C' \arrow[ur, hook, "m'"'] & \end{tikzcd}$$

Taking $a = \mathrm{id}_X$ and $b = \mathrm{id}_Y$ in Lemma 1.2.8 shows that the φ above, if it exists, is unique and both epic and monic. Thus $\mathsf{epi.mono}(f)$ is the category associated with a preordered set. This section considers only one special case of epi–mono factorizations.

Proposition 1.2.10 Suppose that every morphism in $\mathcal{C}$ is strict. Then every $f : X \to Y$ has an epi–mono factorization $X \twoheadrightarrow C \hookrightarrow Y$, unique up to isomorphism in $\mathsf{epi.mono}(f)$.

Proof Taking $C = \mathrm{im}(f)$ proves existence. For uniqueness, Proposition 1.2.7 and the preceding discussion show that every morphism in $\mathsf{epi.mono}(f)$ is an isomorphism. It remains to construct a morphism $\varphi : C \to \mathrm{im}(f)$ in that category. Apply the right-hand version of Lemma 1.2.3 to $X \overset{q}{\twoheadrightarrow} C \overset{g}{\hookrightarrow} Y$. It supplies a φ making the right half of

$$\begin{tikzcd} & C \arrow[dr, hook] \arrow[dd, "\varphi"] & \\ X \arrow[ur, two heads] \arrow[dr, two heads] & & Y \\ & \mathrm{im}(f) \arrow[ur, hook] & \end{tikzcd}$$

commute. Lemma 1.2.8 then shows that the whole diagram commutes. □

Inferring that a subdiagram commutes from the commutativity of its outer frame is a standard technique that will be used repeatedly.

1.3 Additive Categories: Kernels and Cokernels

This section uses the notion of a functor preserving $\varinjlim$ or $\varprojlim$. This is basic category theory; see [51, Definition 2.8.8] or the brief review in Section 1.5. We first recall Ab-categories and additive categories; see [51, § 3.4].

- ◇ A category $\mathcal{C}$ is an **Ab-category**, or a **preadditive category**, if every $\mathrm{Hom}(X,Y)$ is an abelian group, written additively (equivalently, a $\mathbb{Z}$-module), and every composition map $\mathrm{Hom}(Y,Z) \times \mathrm{Hom}(X,Y) \to \mathrm{Hom}(X,Z)$ is $\mathbb{Z}$-bilinear. One may therefore speak of the morphism $0 \in \mathrm{Hom}(X,Y)$. An Ab-category may have Ab-subcategories; every full subcategory is automatically one.
- ◇ If $\mathcal{C}_1, \mathcal{C}_2$ are Ab-categories, a functor $F : \mathcal{C}_1 \to \mathcal{C}_2$ is **additive** if the induced maps on Hom sets are group homomorphisms. Equivalences of Ab-categories and their quasi-inverses are always understood to be additive.

We shall need the following elementary but fundamental observation. By convention, a linear map of $\mathbb{Z}$-modules means a module homomorphism.

Proposition 1.3.1 Let $\mathcal{C}$ and $\mathcal{C}'$ be Ab-categories.

1. Given adjoint functors

$$F : \mathcal{C} \rightleftarrows \mathcal{C}' : G ,$$

with F, G additive, the adjunction isomorphism $\varphi : \mathrm{Hom}_{\mathcal{C}'}(F(\cdot), \cdot) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(\cdot, G(\cdot))$ is linear.

2. For any $\alpha : I \to \mathcal{C}$, the isomorphism in the universal property

$$\mathrm{Hom}_{\mathcal{C}}\left(\varinjlim \alpha, T\right) \simeq \varprojlim_{i \in \mathrm{Ob}(I)} \mathrm{Hom}_{\mathcal{C}}(\alpha(i), T), \quad T \in \mathrm{Ob}(\mathcal{C})$$

is linear whenever the colimit exists. The analogous assertion holds for $\varprojlim$.

Proof For (i), the discussion following [51, Definition 2.6.3] says that φ is determined by the unit η and counit ε:

$$\varphi(f) = Gf \circ \eta_X, \quad \varphi^{-1}(g) = \varepsilon_Y \circ Fg, \quad f \in \mathrm{Hom}_{\mathcal{C}'}(FX, Y),\ g \in \mathrm{Hom}_{\mathcal{C}}(X, GY);$$

bilinearity of composition makes φ linear. For (ii), the isomorphism is described by composition with $\iota_i : \alpha(i) \to \varinjlim \alpha$, so it too is linear. □

In an Ab-category, $X_1 \times X_2$ exists if and only if $X_1 \sqcup X_2$ exists, and then there is a canonical isomorphism $X_1 \sqcup X_2 \xrightarrow{\sim} X_1 \times X_2$. This structure is called a **biproduct**; see [51, Definition 3.4.8, Theorem 3.4.9].

More generally, a biproduct of $X_1, \ldots, X_n$ is an object Z with morphisms $X_i \underset{\iota_i}{\overset{p_i}{\rightleftarrows}} Z$, $i = 1, \ldots, n$, satisfying

$$p_i \iota_j = \begin{cases} \mathrm{id}, & i = j \\ 0, & i \neq j \end{cases} \qquad \sum_{i=1}^{n} \iota_i p_i = \mathrm{id}_Z. \tag{1.3.1}$$

Then the families $\iota_i : X_i \to Z$ and $p_i : Z \to X_i$ induce isomorphisms

$$\coprod_{i=1}^{n} X_i \xrightarrow{\sim} Z \xrightarrow{\sim} \prod_{i=1}^{n} X_i.$$

By convention, the biproduct for $n = 0$ is a zero object.

If an Ab-category $\mathcal{C}$ has a zero object, [51, Lemma 3.4.11] ensures that the zero morphism between any $X, Y \in \mathrm{Ob}(\mathcal{C})$ agrees with $0 \in \mathrm{Hom}(X, Y)$. Moreover, the preceding $\coprod_i X_i \xrightarrow{\sim} \prod_i X_i$ is precisely the morphism δ of (1.1.1); this follows directly from (1.3.1).

Convention 1.3.2 Write the n-fold biproduct Z as $X_1 \oplus \cdots \oplus X_n$. If binary biproducts exist, all n-fold biproducts for $n \geq 1$ can be formed iteratively using the canonical isomorphism $X_1 \oplus X_2 \oplus X_3 \simeq (X_1 \oplus X_2) \oplus X_3$. Also set

$$X^{\oplus n} := \underbrace{X \oplus \cdots \oplus X}_{n \text{ terms}}.$$

The product diagonal $\delta_X : X \to X^{\oplus n}$ and the coproduct codiagonal $\check{\delta}_X : X^{\oplus n} \to X$ are characterized by $p_i \delta_X = \mathrm{id}_X = \check{\delta}_X \iota_i$ for $1 \leq i \leq n$. Explicitly, $\delta_X = \sum_{i=1}^n \iota_i$ and $\check{\delta}_X = \sum_{i=1}^n p_i$.

Definition 1.3.3 (Additive category) An Ab-category $\mathcal{C}$ is **additive** if:

- ◇ it has a zero object 0;
- ◇ every two objects X, Y have a biproduct $X \oplus Y$.

An Ab-subcategory of an additive category that is itself additive is an **additive subcategory**. Every finite family $X_1, \ldots, X_n$ in an additive category has a biproduct $X_1 \oplus \cdots \oplus X_n$, also called their **direct sum**; the X_i are its direct summands.

An additive functor $F : \mathcal{C}_1 \to \mathcal{C}_2$ between additive categories preserves all finite products and coproducts, because:

- ◇ F preserves biproducts; see [51, Proposition 3.4.13];
- ◇ F preserves zero objects, since $F(\mathrm{id}_X) = \mathrm{id}_{FX}$ and a zero object is characterized by $\mathrm{id} = 0$; see [51, Lemma 3.4.11].

If $\mathcal{C}$ is an Ab-category (respectively additive category), then $\mathcal{C}^{\mathrm{op}}$ is naturally one as well, with $f \mapsto f^{\mathrm{op}}$ an isomorphism $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(Y, X) \to \mathrm{Hom}_{\mathcal{C}}(X, Y)$ of groups.

Henceforth, without ambiguity, 0 denotes both zero objects and zero morphisms in additive categories. A family $f_i : X_i \to X'_i$, $i = 1, \ldots, n$, naturally induces $f_1 \oplus \cdots \oplus f_n : \bigoplus_i X_i \to \bigoplus_i X'_i$, explicitly $\sum_{i=1}^n \iota'_i f_i p_i$.

Example 1.3.4 The category Ab is additive: addition on Hom sets is pointwise, $(f + g)(x) = f(x) + g(x)$; the zero object is the zero group; and biproducts are the usual direct sums. Similarly, R-Mod is additive for any ring R, and the forgetful functor R-$\mathsf{Mod} \to \mathsf{Ab}$ is additive. The category $\mathsf{Ban}_{\mathbb{C}}$ of complex Banach spaces and continuous linear maps is also additive. Addition is again pointwise and the zero object is the zero space. For Banach spaces $(X_i, \|\cdot\|_i)$, $i = 1, 2$, the biproduct may be taken to be the vector-space direct sum $X_1 \oplus X_2$ equipped with the norm $\|(x_1, x_2)\| := \max\{\|x_1\|_1, \|x_2\|_2\}$, where $x_i \in X_i$.

Proposition 1.3.5 For morphisms $f, g : X \to Y$ in an additive category $\mathcal{C}$, $f + g$ is the composite along the top row of the following commutative diagram:

$$\begin{tikzcd}
X \arrow[r, "\delta_X"] \arrow[dr] & X \oplus X \arrow[rr, "f \oplus g"] \arrow[d, "\wr"] & & Y \oplus Y \arrow[r, "\check{\delta}_Y"] & Y \\
& X \times X \arrow[r, "f \times g"] & Y \times Y \arrow[r, "\sim"] & Y \sqcup Y \arrow[u, "\wr"] \arrow[ur] &
\end{tikzcd}$$

where $f \times g$ comes from functoriality of products, and $Y \times Y \xrightarrow{\sim} Y \sqcup Y$ is the inverse of δ in (1.1.1).

Proof Write the biproduct maps as $\iota_i^X, \iota_i^Y, p_i^X, p_i^Y$, $i = 1, 2$. Expressing δ_X and $\check{\delta}_Y$ through them verifies the two triangles. Since $f \oplus g = \iota_1^Y f p_1^X + \iota_2^Y g p_2^X$,

$$\check{\delta}_Y (f \oplus g) \delta_X = \check{\delta}_Y \iota_1^Y f p_1^X \delta_X + \check{\delta}_Y \iota_2^Y g p_2^X \delta_X = f + g. \tag{1.3.2}$$

Identifying the biproducts with coproducts, commutativity of the square becomes

$$\begin{array}{ccc} X \sqcup X & \xrightarrow{f \sqcup g} & Y \sqcup Y \\ \downarrow & & \downarrow \\ X \times X & \xrightarrow[f \times g]{} & Y \times Y \end{array}$$

which is precisely functoriality of δ in (1.1.1). □

Corollary 1.3.6 Let $\mathcal{C}, \mathcal{C}'$ be additive categories and $F : \mathcal{C} \to \mathcal{C}'$ a functor.

(i) F preserves finite products if and only if it preserves finite coproducts.

(ii) If F preserves finite products, equivalently finite coproducts, then it is additive.

(iii) If F preserves zero objects and biproducts, then it is additive.

(iv) If F is an equivalence, then F and its quasi-inverse are additive.

(v) If

$$F : \mathcal{C} \rightleftarrows \mathcal{C}' : G$$

is an adjoint pair, then F, G are additive; Proposition 1.3.1 therefore applies.

Proof For (i), a zero object is both the empty product and empty coproduct, so it remains only to consider the product and coproduct of X, Y. The canonical isomorphism $\delta : FX \sqcup FY \xrightarrow{\sim} FX \times FY$ factors as

$$FX \sqcup FY \to F(X \sqcup Y) \xrightarrow{\sim} F(X \times Y) \to FX \times FY.$$

For (ii), the composite in Proposition 1.3.5 along does not use addition on Hom sets, only finite products and coproducts, the zero object, and their canonical maps. For (iii), all finite products or coproducts are built from zero objects and binary biproducts. Finally, a left adjoint preserves $\varinjlim$, a right adjoint preserves $\varprojlim$ ([51, Theorem 2.8.12]), and equivalences preserve all limits. Thus (iv) and (v) follow from (ii). □

In fact, Proposition 1.3.5 shows that "additivity" is a property of the underlying category $\mathcal{C}$, not additional structure.

Definition 1.3.7 Let $\mathcal{C}$ be an Ab-category with a zero object, and $f : X \to Y$ a morphism.

- $\diamond$ If the equalizer $\ker(f, 0)$ exists, then $\ker(f) := \ker(f, 0)$ together with $\ker(f) \hookrightarrow X$ is the **kernel** of f.
- $\diamond$ If the coequalizer $\operatorname{coker}(f, 0)$ exists, then $\operatorname{coker}(f) := \operatorname{coker}(f, 0)$ together with $Y \twoheadrightarrow \operatorname{coker}(f)$ is the **cokernel** of f.

Thus the composites $\ker(f) \to X \to Y$ and $X \to Y \to \operatorname{coker}(f)$ are both 0.

Kernels and cokernels admit several complementary viewpoints.

- $\diamond$ For a morphism $g : T \to X$ satisfying $fg = 0$,[1] the universal property of $\ker(f) = \ker(f, 0)$ is the commutative diagram

$$\begin{tikzcd} & T \arrow[dl, dashed, "\exists! g'"'] \arrow[d, "g"] \arrow[dr, "0"] & \\ \ker(f) \arrow[r] & X \arrow[r, "f"'] & Y. \end{tikzcd}$$

- $\diamond$ For a morphism $h : Y \to T$ satisfying $hf = 0$, the universal property of $\operatorname{coker}(f) = \operatorname{coker}(f, 0)$ is

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[dr, "0"'] & Y \arrow[r] \arrow[d, "h"] & \operatorname{coker}(f). \arrow[dl, dashed, "\exists! h'"] \\ & T & \end{tikzcd}$$

- $\diamond$ The constructions are dual: $\operatorname{coker}(f) = \ker(f^{\mathrm{op}})$ and $\ker(f) = \operatorname{coker}(f^{\mathrm{op}})$.

Since precisely the zero morphisms factor through a zero object, the universal properties are also characterized by the pullback and pushout diagrams

$$\begin{tikzcd} \ker(f) \simeq & X \times_Y 0 \arrow[r] \arrow[d] \arrow[dr, phantom, "\square"] & X \arrow[d, "f"] & X \arrow[r, "f"] \arrow[d] \arrow[dr, phantom, "\boxplus"] & Y \arrow[d] & \\ & 0 \arrow[r] & Y & 0 \arrow[r] & Y \sqcup_X 0 & \simeq \operatorname{coker}(f). \end{tikzcd} \tag{1.3.3}$$

Remark 1.3.8 An arbitrary equalizer reduces to a kernel. For

$$T \xrightarrow{h} X \overset{f}{\underset{g}{\rightrightarrows}} Y \ ,$$

[1] Translator's note: the source says "for every morphism" without including the condition $fg = 0$, or dually $hf = 0$. These conditions have been restored because they are precisely the equalizer and coequalizer hypotheses; see correction O014-C006.

the condition $fh = gh$ is equivalent to $(f-g)h = 0$, hence $\ker(f,g) = \ker(f-g, 0) =: \ker(f-g)$. Dually, $\operatorname{coker}(f,g) = \operatorname{coker}(f-g,0) =: \operatorname{coker}(f-g)$.

From now on, $\mathcal{C}$ will mainly be additive.

Proposition 1.3.9 For morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$ in an additive category $\mathcal{C}$, there are commutative diagrams

$$\begin{tikzcd} \ker(gf) \arrow[rr, "\sim"] \arrow[dr, hook] & & X \underset{Y}{\times} \ker(g) \arrow[dl, hook'] \\ & X & \end{tikzcd} \qquad \begin{tikzcd} Z \underset{Y}{\sqcup} \operatorname{coker}(f) \arrow[rr, "\sim"] & & \operatorname{coker}(gf) \\ & Z \arrow[ul, two heads] \arrow[ur, two heads] & \end{tikzcd}$$

where $X \underset{Y}{\times} \ker(g) \to X$ (respectively $Z \to Z \underset{Y}{\sqcup} \operatorname{coker}(f)$) is the canonical morphism of the fiber product (respectively fiber coproduct),[2)] provided all indicated kernels, cokernels, and limits exist.

Proof By (1.3.3), iterated fiber products give canonical isomorphisms $\ker(gf) \simeq X \underset{Z}{\times} 0 \simeq X \underset{Y}{\times} (Y \underset{Z}{\times} 0) \simeq X \underset{Y}{\times} \ker(g)$. Duality gives the cokernel statement. $\square$

Functoriality of equalizers and coequalizers induces functoriality of kernels and cokernels, characterized by

$$\begin{tikzcd} \ker(f) \arrow[r] \arrow[d, dashed, "\exists!"'] & X \arrow[r, "f"] \arrow[d, "a"'] & Y \arrow[d, "b"] \\ \ker(g) \arrow[r] & X' \arrow[r, "g"'] & Y' \end{tikzcd} \qquad \begin{tikzcd} X \arrow[r, "f"] \arrow[d, "a"'] & Y \arrow[r] \arrow[d, "b"] & \operatorname{coker}(f) \arrow[d, dashed, "\exists!"'] \\ X' \arrow[r, "g"'] & Y' \arrow[r] & \operatorname{coker}(g). \end{tikzcd} \tag{1.3.4}$$

The next result says that pullback preserves kernels and pushout preserves cokernels.

Proposition 1.3.10 If

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "a"'] & Y \arrow[d, "b"] \\ X' \arrow[r, "g"'] & Y' \end{tikzcd}$$

is a pullback (respectively pushout) square in an additive category, then the morphism $\ker(f) \to \ker(g)$ (respectively $\operatorname{coker}(f) \to \operatorname{coker}(g)$) of (1.3.4) is an isomorphism, provided the indicated objects exist.

---

2) Translator's note: the source says "coproduct," but the displayed object is the fiber coproduct $Z \underset{Y}{\sqcup} \operatorname{coker}(f)$ and the proof uses duality with the fiber product; see correction O014-C005.

Proof By duality it suffices to treat pullbacks. For every T, the universal properties give canonical isomorphisms

$$\begin{aligned}\operatorname{Hom}(T,\ker(f)) &\simeq \ker\left[\operatorname{Hom}(T,X) \to \operatorname{Hom}(T,Y)\right] \\ &\simeq \ker\left[\operatorname{Hom}(T,Y) \underset{\operatorname{Hom}(T,Y')}{\times} \operatorname{Hom}(T,X') \to \operatorname{Hom}(T,Y)\right] \\ &= \ker\left[\operatorname{Hom}(T,X') \to \operatorname{Hom}(T,Y')\right] \simeq \operatorname{Hom}(T,\ker(g)),\end{aligned}$$

and the Yoneda lemma gives $\ker(f) \xrightarrow{\sim} \ker(g)$; see Theorem A.1.1. □

Lemma 1.3.11 Let $\mathcal{C}$ be additive and $f : X \to Y$ a morphism.

(i) f is monic (respectively epic) if and only if $\ker(f) = 0$ (respectively $\operatorname{coker}(f) = 0$).

(ii) f is zero if and only if $\ker(f) \xrightarrow{\sim} X$, equivalently $Y \xrightarrow{\sim} \operatorname{coker}(f)$.

(iii) Given $k : Y \to Z$ (respectively $k' : W \to X$), there is a unique morphism making the corresponding diagram commute:

$$\begin{tikzcd} \ker(kf) \arrow[dr] & & \ker(f) \arrow[ll, hook', "\exists!"'] \arrow[dl] \\ & X & \end{tikzcd} \quad \text{or} \quad \begin{tikzcd} \operatorname{coker}(fk') \arrow[rr, two heads, "\exists!"] & & \operatorname{coker}(f) \\ & Y \arrow[ul] \arrow[ur] & \end{tikzcd}$$

provided the indicated kernels or cokernels exist. If k is monic (respectively k' is epic), then $\ker(f) \xrightarrow{\sim} \ker(kf)$ (respectively $\operatorname{coker}(fk') \xrightarrow{\sim} \operatorname{coker}(f)$).

Proof For (i), by duality consider only monomorphisms. By definition, f is monic precisely when $f(g_1 - g_2) = 0 \iff g_1 - g_2 = 0$ for every $g_1, g_2 : T \to X$. Equivalently, every $g : T \to X$ with $fg = 0$ factors through $T \to 0 \to X$, exactly the universal property of $\ker(f) = 0$. For (ii), $f = 0 \iff \ker(f, 0) \xrightarrow{\sim} X$ follows from the definition of an equalizer. For (iii), use

$$\begin{tikzcd} \{g \in \operatorname{Hom}(T,X) : fg = 0\} & \subset & \{g \in \operatorname{Hom}(T,X) : kfg = 0\} \\ \operatorname{Hom}(T,\ker(f)) \arrow[u, "1:1"] & & \operatorname{Hom}(T,\ker(kf)) \arrow[u, "1:1"'] \end{tikzcd}$$

The top inclusion is equality when k is monic. Apply the Yoneda lemma. □

In an additive category, the image and coimage of Definition 1.2.1 have simple descriptions.

Proposition 1.3.12 For $f : X \to Y$ in an additive category $\mathcal{C}$, there are canonical isomorphisms

$$\begin{aligned}\operatorname{im}(f) &\simeq \ker\left[Y \to \operatorname{coker}(f)\right] \hookrightarrow Y, \\ \operatorname{coim}(f) &\simeq \operatorname{coker}\left[\ker(f) \to X\right] \twoheadleftarrow X;\end{aligned}$$

provided the indicated kernels and cokernels exist.

Proof By duality consider only $\mathrm{im}(f)$. Let $i_1, i_2 : Y \to Y \sqcup_X Y$ be canonical. By Yoneda it suffices to verify, for every T,

$$\begin{aligned}
\mathrm{Hom}\,(T, \mathrm{im}(f)) &= \{h : T \to Y \mid i_1 h = i_2 h\} \\
&= \left\{ h : T \to Y \;\middle|\; \begin{array}{l} \forall S \in \mathrm{Ob}(\mathcal{C}),\ \forall s_1, s_2 : Y \to S, \\ s_1 f = s_2 f \implies s_1 h = s_2 h \end{array} \right\} \\
&\overset{s := s_1 - s_2}{=\!=} \left\{ h : T \to Y \;\middle|\; \begin{array}{l} \forall S \in \mathrm{Ob}(\mathcal{C}),\ \forall s : Y \to S, \\ sf = 0 \implies sh = 0 \end{array} \right\} \\
&= \left\{ h : T \to Y \;\middle|\; \begin{array}{l} T \xrightarrow{h} Y \twoheadrightarrow \mathrm{coker}(f) \\ \text{has composite } 0 \end{array} \right\} \xleftarrow{\sim} \mathrm{Hom}\,(T, \ker[Y \to \mathrm{coker}(f)])\,.
\end{aligned}$$

The first equality is $\mathrm{im}(f) := \ker(i_1, i_2)$. For the second, rewrite the condition as $s' i_1 h = s' i_2 h$ for every $s' : Y \sqcup_X Y \to S$ and use

$$\begin{aligned}
\mathrm{Hom}\left(Y \sqcup_X Y, S\right) &\xrightarrow{1:1} \mathrm{Hom}(Y, S) \underset{\mathrm{Hom}(X,S)}{\times} \mathrm{Hom}(Y, S) \\
s' &\longmapsto (s' i_1, s' i_2) =: (s_1, s_2).
\end{aligned}$$

The fourth equality follows from the cokernel's universal property, reducing to $S = \mathrm{coker}(f)$. $\square$

If $f = 0$, Lemma 1.3.11(ii) gives the simplest example: $\mathrm{im}(f) = 0 = \mathrm{coim}(f)$.

Example 1.3.13 Continuing Example 1.3.4, consider R-Mod. A module homomorphism $f : A \to B$ always has $\ker(f) = f^{-1}(0)$ and $\mathrm{coker}(f) = B/\{f(a) : a \in A\}$. Its $\mathrm{im}(f)$ is the set-theoretic image, $\mathrm{coim}(f) = A/\ker(f)$, and the comparison in (1.2.1) is the canonical module isomorphism $A/\ker(f) \xrightarrow{\sim} \mathrm{im}(f)$. Thus every morphism in R-Mod is strict.

Next consider the additive category $\mathsf{Ban}_{\mathbb{C}}$ from Example 1.3.4. A continuous linear map $f : X \to Y$ has the usual kernel $\ker(f) = f^{-1}(0)$, which is again a Banach space, while $\mathrm{coker}(f) = Y/\overline{\{f(x) : x \in X\}}$ carries the quotient norm. The comparison morphism in (1.2.1) becomes

$$\begin{array}{ccccccc}
\mathrm{coim}(f) & = & X/f^{-1}(0) & \longrightarrow & \overline{\{f(x) : x \in X\}} & = & \mathrm{im}(f) \\
 & & \Cup & & \Cup & & \\
 & & x + f^{-1}(0) & \longmapsto & f(x) & &
\end{array}$$

where the right side has the topology induced from Y and the left the quotient topology from X. If $f : X \to Y$ is a continuous injective linear map of Banach spaces with dense, non-surjective image, then $\mathrm{coim}(f) = X \to Y = \mathrm{im}(f)$ is not an isomorphism. Hence $\mathsf{Ban}_{\mathbb{C}}$ has non-strict morphisms.

1.4 Generalization: Linear Categories over a Commutative Ring

The Hom sets of an additive category have the structure of abelian groups. In many situations they also carry scalar multiplication by some commutative ring $\Bbbk$. For example, every Hom set in $\Bbbk$-Mod is naturally a $\Bbbk$-module, and composition of morphisms is $\Bbbk$-bilinear.

Definition 1.4.1 Let $\Bbbk$ be a commutative ring. Suppose that every Hom set in a category $\mathcal{A}$ is equipped with a $\Bbbk$-module structure such that composition of morphisms $\mathrm{Hom}(Y,Z) \times \mathrm{Hom}(X,Y) \to \mathrm{Hom}(X,Z)$ is a $\Bbbk$-bilinear map for all objects X, Y, Z; in other words, the following diagram commutes:

$$\begin{tikzcd}
\mathrm{Hom}(Y,Z) \times \mathrm{Hom}(X,Y) \arrow[r, "\text{composition}"] \arrow[d] & \mathrm{Hom}(X,Z) \\
\mathrm{Hom}(Y,Z) \underset{\Bbbk}{\otimes} \mathrm{Hom}(X,Y) \arrow[ur, bend right, "\exists!\ \Bbbk\text{-module homomorphism}"'] &
\end{tikzcd}$$

With these data, $\mathcal{A}$ is called a **$\Bbbk$-Mod-category**.

Let $F : \mathcal{A} \to \mathcal{A}'$ be a functor between $\Bbbk$-Mod-categories. If the map $\mathrm{Hom}(X,Y) \to \mathrm{Hom}(FX,FY)$ is a $\Bbbk$-module homomorphism for every $X, Y \in \mathrm{Ob}(\mathcal{A})$, then F is called a **$\Bbbk$-linear functor**.

Observe that $\Bbbk$-Mod is a monoidal category under the tensor product $\otimes := \underset{\Bbbk}{\otimes}$. Thus the term "$\Bbbk$-$\mathsf{Mod}$-category" agrees with the terminology for enriched categories in [51, §3.4]. The Ab-categories introduced earlier are precisely the $\mathbb{Z}$-Mod-categories. Conversely, forgetting scalar multiplication on a $\Bbbk$-Mod-category leaves an Ab-category.

Example 1.4.2 A $\Bbbk$-Mod-category with a single object is nothing but a $\Bbbk$-algebra. The correspondence is given by $\mathcal{A} \mapsto \mathrm{End}_{\mathcal{A}}(\star)$, where $\mathrm{Ob}(\mathcal{A}) = \{\star\}$.

The argument of Proposition 1.3.1 carries over verbatim and gives the following result.

Proposition 1.4.3 Let $\mathcal{A}$ and $\mathcal{A}'$ be $\Bbbk$-Mod-categories.

1. Given a pair of adjoint functors

$$F : \mathcal{A} \rightleftarrows \mathcal{A}' : G\ ,$$

with both F and G $\Bbbk$-linear, the adjunction isomorphism $\mathrm{Hom}_{\mathcal{A}'}(F(\cdot), \cdot) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{A}}(\cdot, G(\cdot))$ is $\Bbbk$-linear.

2. For every $\alpha : I \to \mathcal{A}$, the bijection

$$\mathrm{Hom}_{\mathcal{A}}\left(\varinjlim \alpha, T\right) \simeq \varprojlim_{i \in \mathrm{Ob}(I)} \mathrm{Hom}_{\mathcal{A}}(\alpha(i), T), \quad T \in \mathrm{Ob}(\mathcal{A})$$

is $\Bbbk$-linear whenever $\varinjlim \alpha$ exists. The same statement holds for $\varprojlim$.

Unless otherwise specified, all functors and equivalences between $\Bbbk$-Mod-categories will henceforth be understood to be $\Bbbk$-linear.

Definition 1.4.4 If an additive category $\mathcal{A}$ is equipped with a $\Bbbk$-linear structure compatible with its existing Ab-category structure, then $\mathcal{A}$ together with these data is called a **$\Bbbk$-linear category**.

This amounts to replacing the Ab-category by a $\Bbbk$-Mod-category in the discussion of §1.3. Most results about additive categories in this chapter extend to the $\Bbbk$-linear setting by exactly the same arguments. The sole exception concerns Corollary 1.3.6: in an additive category, addition on the Hom sets can be characterized by properties of the category itself, but this is not true of scalar multiplication by $\Bbbk$. Consequently, functors—and even equivalences—between $\Bbbk$-linear categories are not automatically $\Bbbk$-linear.

We next collect a few results concerning $\Bbbk$-linearity. Recall that the **center** of a category $\mathcal{A}$ is defined by $Z(\mathcal{A}) := \mathrm{End}(\mathrm{id}_{\mathcal{A}})$. Its elements are endomorphisms of the identity functor $\mathrm{id}_{\mathcal{A}}$, written as $(a_X)_{X \in \mathrm{Ob}(\mathcal{A})}$ with $a_X \in \mathrm{End}_{\mathcal{A}}(X)$. Composition makes the center a commutative monoid; see [51, Definition 2.3.8, Proposition 2.3.9]. If $\mathcal{A}$ is an Ab-category, then $Z(\mathcal{A})$ becomes a commutative ring under composition and addition $(a_X)_X + (b_X)_X := (a_X + b_X)_X$.

Proposition 1.4.5 Let $\mathcal{A}$ be an Ab-category. Then there is a bijection

$$\{\Bbbk\text{-}\mathsf{Mod}\text{-category structures on } \mathcal{A}\} \xleftrightarrow{1:1} \mathrm{Hom}_{\mathrm{rings}}(\Bbbk, Z(\mathcal{A})).$$

The map is given as follows. If a $\Bbbk$-Mod-category structure on $\mathcal{A}$ has been fixed, define, for every $a \in \Bbbk$ and $X \in \mathrm{Ob}(\mathcal{A})$, $a_X := a \cdot \mathrm{id}_X \in \mathrm{End}_{\mathcal{A}}(X)$. Then $a \mapsto (a_X)_{X \in \mathrm{Ob}(\mathcal{A})} \in Z(\mathcal{A})$ is a ring homomorphism. Conversely, given a ring homomorphism on the right, $a \mapsto (a_X)_{X \in \mathrm{Ob}(\mathcal{A})}$, define a $\Bbbk$-Mod-category structure on $\mathcal{A}$ by $a \cdot f := f \circ a_X = a_Y \circ f$, for $f \in \mathrm{Hom}(X, Y)$.

Proof The condition for $(a_X)_{X \in \mathrm{Ob}(\mathcal{A})}$ to belong to $Z(\mathcal{A})$ is precisely $f \circ a_X = a_Y \circ f$ for every $f \in \mathrm{Hom}(X, Y)$. If $\mathcal{A}$ is equipped with a $\Bbbk$-Mod-category structure, define

$a_X := a \cdot \mathrm{id}_X$. Bilinearity gives

$$f \circ a_X = f \circ (a \cdot \mathrm{id}_X) = (af) \circ \mathrm{id}_X = (a \cdot \mathrm{id}_Y) \circ f = a_Y \circ f,$$

so $(a_X)_{X \in \mathrm{Ob}(\mathcal{A})} \in Z(\mathcal{A})$. The remaining verifications are immediate. $\square$

Consequently, the Ab-category $\mathcal{A}$ is naturally a $Z(\mathcal{A})$-Mod-category.

1.5 Viewing Limits through Functors

There are two broad kinds of question about the relationship between a functor $F : \mathcal{C} \to \mathcal{D}$ and limits. Taking $\varinjlim$ as an example, let $\alpha : I \to \mathcal{C}$ be a functor. One may ask:

- ◇ Suppose $\varinjlim \alpha$ exists. Does its image $F\left(\varinjlim \alpha\right)$ yield $\varinjlim (F\alpha)$?
- ◇ Suppose $\varinjlim F\alpha$ exists. Can this limit be "lifted" to $\mathcal{C}$? In what sense is such a lift unique?

We focus on the case of $\varinjlim$. These results of course have corresponding versions for $\varprojlim$; by duality, there is no need to repeat the details.

We organize the discussion of $\varinjlim$ in terms of cocones and the category (α/Δ).

Convention 1.5.1 For a fixed functor $\alpha : I \to \mathcal{C}$, a collection of data $(L, (f_i)_{i \in \mathrm{Ob}(I)})$ in $\mathcal{C}$ satisfying the following conditions is called a **cocone** with base α and vertex L:

- ◇ L is an object of $\mathcal{C}$;
- ◇ each $f_i : \alpha(i) \to L$ is a morphism in $\mathcal{C}$ and satisfies the compatibility condition $f_j \circ \alpha(i \to j) = f_i$, as $[i \to j]$ ranges over $\mathrm{Mor}(I)$.

In the terminology of [51, §2.7], these cocones form a category[3] (α/Δ). Concretely, once the base α is fixed, a morphism from a cocone $(L, (f_i)_i)$ to a cocone $(M, (g_i)_i)$ is a commutative diagram

$$\begin{tikzcd} & L \arrow[d] & \\ \alpha(i) \arrow[ur, "f_i"] \arrow[r, "g_i"] \arrow[rr, bend right, "\alpha(i\to j)"'] & M & \alpha(j) \arrow[ul, "f_j"'] \arrow[l, "g_j"'] \end{tikzcd} \tag{1.5.1}$$

[3] More precisely, Δ here denotes the diagonal functor $\mathcal{C} \to \mathcal{C}^I$; see Example 1.6.3. The notation Δ was chosen from the initial letter of the pinyin word **duìjiǎo** ("diagonal"). By a pictorial analogy, however, one may just as well picture Δ as a "cocone"; see diagram (1.5.1).

as $i \to j$ ranges over $\mathrm{Mor}(I)$; composition of morphisms is defined in the usual way. By definition, $\varinjlim \alpha$, together with its family of canonical morphisms $\iota_i : \alpha(i) \to \varinjlim \alpha$, is an initial object of (α/Δ).

In the evident way, the functor F induces a functor $(\alpha/\Delta) \to (F\alpha/\Delta)$ that takes a cocone $(L, (f_i)_i)$ to $(FL, (Ff_i)_i)$. Suppose $\varinjlim \alpha$ exists. If $\varinjlim F\alpha$ also exists, there is a unique morphism in $(F\alpha/\Delta)$

$$\varinjlim F\alpha \to F(\varinjlim \alpha),$$

and this morphism is an isomorphism if and only if $\left(F(\varinjlim \alpha), (F\iota_i)_i\right)$ gives $\varinjlim F\alpha$.

Dually, $\varprojlim \alpha$ is characterized as a terminal object of (Δ/α). Assuming that both projective limits under discussion exist, there is a unique morphism in $(\Delta/F\alpha)$

$$F \varprojlim \alpha \to \varprojlim F\alpha.$$

Definition 1.5.2 Given $F : \mathcal{C} \to \mathcal{D}$ and a functor $\alpha : I \to \mathcal{C}$, consider the induced functor between cocone categories $(\alpha/\Delta) \to (F\alpha/\Delta)$.

- ◇ The functor F is said to **preserve** this $\varinjlim$ if $(\alpha/\Delta) \to (F\alpha/\Delta)$ takes an initial object, when one exists, to an initial object.
- ◇ The functor F is said to **reflect** this $\varinjlim$ if any object of (α/Δ) whose image is an initial object of $(F\alpha/\Delta)$ is itself an initial object.
- ◇ The functor F is said to **create** this $\varinjlim$ if, whenever $\varinjlim F\alpha$ exists, $\varinjlim \alpha$ also exists, and in this situation F preserves this $\varinjlim$ and also reflects this $\varinjlim$.

There are corresponding notions for $\varprojlim$. The notions for $\varinjlim$ and $\varprojlim$ are obtained from one another by replacing $\mathcal{C}$ and $\mathcal{D}$ with $\mathcal{C}^{\mathrm{op}}$ and $\mathcal{D}^{\mathrm{op}}$.

Notice that these notions depend on the particular α under consideration. For example, F may well preserve every finite $\varinjlim$ without preserving $\varinjlim$ in general.

For a fixed α, saying that a functor creates $\varinjlim$ (or $\varprojlim$) means that the cocone (or cone) in $\mathcal{D}$ defining the limit can be lifted uniquely to $\mathcal{C}$, and that the lifted cocone (or cone) gives the corresponding limit in $\mathcal{C}$. This uniqueness comes from the condition in the definition concerning reflection of $\varinjlim$ (or $\varprojlim$). This point of view will be evident in Example 1.5.6 and Proposition 1.5.7 below.

Proposition 1.5.3 Every fully faithful functor reflects every $\varinjlim$ and every $\varprojlim$.

Proof It suffices to consider $\varinjlim$. If F is fully faithful, the universal property of an initial object of (α/Δ) can be checked, after applying F, in $(F\alpha/\Delta)$. □

We shall also use the following notion.

Definition 1.5.4 A functor F is called **conservative** if it satisfies the following condition: a morphism $f \in \mathrm{Mor}(\mathcal{C})$ is an isomorphism if and only if $Ff \in \mathrm{Mor}(\mathcal{D})$ is an isomorphism.

Remark 1.5.5 Suppose F is a conservative functor. If $\varinjlim \alpha$ exists and F preserves this $\varinjlim$, the functor F must also reflect it. Indeed, suppose $L \in \mathrm{Ob}((\alpha/\Delta))$ is mapped to an initial object of $(F\alpha/\Delta)$. Consider the unique morphism $\varinjlim \alpha \to L$ in (α/Δ). Since F preserves this $\varinjlim$, the image of that morphism under F must be an isomorphism. Conservativity then ensures that $\varinjlim \alpha \to L$ is also an isomorphism.

Under these hypotheses, the condition that F reflect $\varinjlim$ may be omitted from the definition of what it means for F to create $\varinjlim$.

Example 1.5.6 We verify that the forgetful functor U from the category of groups Grp to the category of sets Set creates all small $\varprojlim$.

This follows from the concrete constructions of $\varprojlim$ in Grp and Set; see [51, Examples 2.7.5 and 2.8.7]. More explicitly, take a small category I and a functor $\beta : I^{\mathrm{op}} \to \mathsf{Grp}$. As a set,

$$\varprojlim U\beta = \left\{ (x_i)_i \in \prod_{i \in \mathrm{Ob}(I)} U\beta(i) \;\middle|\; \begin{array}{l} \forall [i \to j] \in \mathrm{Mor}(I), \\ \beta(i \to j)(x_j) = x_i \end{array} \right\},$$

and the projection $p_i : \varprojlim U\beta \to U\beta(i)$ maps $(x_j)_j$ to x_i. Our task is now to:

1. equip $\varprojlim U\beta$ with a group structure such that every p_i is a group homomorphism, and show that this group structure is unique;

2. verify the universal property of $\varprojlim \beta$ for this group together with the family of projection homomorphisms $(p_i)_i$.

Clearly, $\varprojlim U\beta$ is a subgroup of the direct product $\prod_i \beta(i)$. Regard this subgroup as $\varprojlim \beta \in \mathrm{Ob}(\mathsf{Grp})$; this is the unique group structure making every p_i a group homomorphism. This completes the first task.

We next verify the universal property of the cone $\left(\varprojlim \beta, (p_i)_i\right)$. Consider any cone $(L, (q_i)_i)$ in Grp, with $q_i : L \to \beta(i)$. The universal property gives a unique map $\varphi : UL \to \varprojlim U\beta$ such that $p_i \varphi = q_i$ for every i. Concretely, φ is in fact given by $y \mapsto (q_i(y))_{i \in \mathrm{Ob}(I)}$, so this map is a group homomorphism $L \to \varprojlim \beta$.[4] The verification is complete.

[4] Translator's note: the source writes $L \to \varinjlim \beta$; the target corrects this to $L \to \varprojlim \beta$, in accordance with the projective limit being proved (O014-C007).

By an entirely similar argument, one can show that the forgetful functor from the category of compact Hausdorff spaces **CHaus** to **Set** also creates all small $\varprojlim$. The corresponding verification is left as an exercise in this chapter.

The next result concerns a functor category of the form $\mathcal{C}^J$ (which may be a "large" category, but this causes no problem). We need two preparatory steps.

◇ Let J be a set. The product of J copies of $\mathcal{C}$, namely $\mathcal{C}^J$, is defined by $\mathrm{Ob}\left(\mathcal{C}^J\right) = \mathrm{Ob}(\mathcal{C})^J$ and $\mathrm{Mor}\left(\mathcal{C}^J\right) = \mathrm{Mor}(\mathcal{C})^J$. For every $j \in J$ there is an evident projection functor $p_j : \mathcal{C}^J \to \mathcal{C}$.

  Given a category I and a functor $\alpha : I \to \mathcal{C}^J$, if $\varinjlim(p_j\alpha)$ exists for every j, then $\varinjlim \alpha$ also exists. It is constructed by taking colimits "pointwise" or "objectwise": $\varinjlim \alpha = \left(\varinjlim(p_j\alpha)\right)_{j\in J}$.

◇ Let J be a category. We may consider the functor category $\mathcal{C}^J$. For each $j \in \mathrm{Ob}(J)$ there is an evaluation functor $\mathrm{ev}_j : \mathcal{C}^J \to \mathcal{C}$, which maps an object F to Fj and a morphism $\varphi = (\varphi_{j'})_{j'\in\mathrm{Ob}(J)}$ to φ_j. We shall later consider functors of the form $\alpha : I \to \mathcal{C}^J$ and their $\varinjlim$. Observe that α can be viewed as a functor $I \times J \to \mathcal{C}$, whose value is denoted $\alpha(i,j)$.

The two systems of notation are compatible. If J is a discrete category and is viewed as a set, then $\mathcal{C}^J$ is simply the product discussed above. For a general J, the forgetful functor $\mathcal{F} : \mathcal{C}^J \to \mathcal{C}^{\mathrm{Ob}(J)}$ maps an object F to $(Fj)_{j\in\mathrm{Ob}(J)}$ and a morphism φ to $(\varphi_j)_{j\in\mathrm{Ob}(J)}$. Thus $\mathrm{ev}_j = p_j\mathcal{F}$.

Proposition 1.5.7 Let J and $\mathcal{C}$ be categories.

(i) The forgetful functor $\mathcal{F} : \mathcal{C}^J \to \mathcal{C}^{\mathrm{Ob}(J)}$ creates $\varinjlim$ and $\varprojlim$.

(ii) Let I be a category and suppose every functor $I \to \mathcal{C}$ has a $\varinjlim$ (or $\varprojlim$). Then all functors of the form $I \to \mathcal{C}^J$ also have a $\varinjlim$ (or $\varprojlim$) in $\mathcal{C}^J$.

(iii) Under the assumption in (ii), for every $j \in \mathrm{Ob}(J)$ the evaluation functor $\mathrm{ev}_j : \mathcal{C}^J \to \mathcal{C}$ preserves $\varinjlim$ (or $\varprojlim$) for diagrams of shape I. In other words, limits in $\mathcal{C}^J$ are likewise defined "pointwise" or "objectwise."

Proof It suffices to consider $\varinjlim$. For (i), let $\alpha : I \to \mathcal{C}^J$ and suppose $\varinjlim \mathcal{F}\alpha$ exists and is given by the cocone

$$\left((X(j))_{j\in\mathrm{Ob}(J)},\ \left(\iota_i = (\iota_{i,j})_{j\in\mathrm{Ob}(J)}\right)_{i\in\mathrm{Ob}(I)}\right)$$

with $\iota_{i,j} : \alpha(i,j) \to X(j)$. We wish to lift it to a cocone in $\mathcal{C}^J$ that gives $\varinjlim \alpha$.

First observe that $X(j) = \varinjlim \alpha(\cdot, j)$ for every j. For each morphism $j \to j'$, functoriality of colimits (see [51, Lemma 2.7.4]) gives a unique morphism $X(j \to j') : X(j) \to X(j')$ making the following diagram commute:

$$\begin{tikzcd} \alpha(i,j) \arrow[r] \arrow[d, "\iota_{i,j}"'] & \alpha(i,j') \arrow[d, "\iota_{i,j'}"] \\ X(j) \arrow[r, "X(j\to j')"'] & X(j') \end{tikzcd} \qquad i \in \mathrm{Ob}(I).$$

In this way, $j \mapsto X(j)$ extends to a functor $X : J \to \mathcal{C}$, while $\iota_i : \alpha(i, \cdot) \to X$ are morphisms in $\mathcal{C}^J$. It is immediate that $(X, (\iota_i)_i) \in \mathrm{Ob}(\alpha/\Delta)$; we next prove that this object is initial. Since the functor $\mathcal{F}$ is conservative, reflection of $\varinjlim$ need not be checked separately.

Given $(L, (f_i)_i) \in \mathrm{Ob}(\alpha/\Delta)$, the universal property of $\varinjlim \mathcal{F}\alpha$ determines a unique morphism $\varphi = (\varphi_j)_j : \mathcal{F}X \to \mathcal{F}L$ such that the triangular part of the following diagram commutes for every (i,j):

$$\begin{tikzcd} & X(j) \arrow[r, "X(j\to j')"] \arrow[d, "\varphi_j"] & X(j') \arrow[d, "\varphi_{j'}"] \\ \alpha(i,j) \arrow[ur, "\iota_{i,j}"] \arrow[r, "f_{i,j}"'] & L(j) \arrow[r, "L(j\to j')"'] & L(j') \end{tikzcd}$$

It remains to show that the square commutes for every $j \to j'$, after which the family of components φ forms a morphism $X \to L$. The outer frame is already known to commute, so

$$\varphi_{j'} X(j \to j') \iota_{i,j} = L(j \to j') f_{i,j} = L(j \to j') \varphi_j \iota_{i,j}$$

for every i. Morphisms with domain $X(j)$ are determined by their composites with all the $\iota_{i,j}$, so the square commutes.

For (ii), consider a functor $\alpha : I \to \mathcal{C}^J$. By the preceding discussion, $\mathcal{F}\alpha : I \to \mathcal{C}^{\mathrm{Ob}(J)}$ has a $\varinjlim$, defined objectwise. By (i), $\mathcal{F}$ creates $\varinjlim \alpha$ in $\mathcal{C}^J$,[5] while the assertion in (iii) that ev_j preserves this $\varinjlim$ follows immediately from the construction. □

[5] Translator's note: the source writes $\mathcal{C}$ here. The target corrects this to $\mathcal{C}^J$, the domain of the forgetful functor $\mathcal{F}$ and the category containing the newly constructed functor $X : J \to \mathcal{C}$; see correction O014-C009.

1.6 Filtered Inductive Limits

A familiar fact from mathematical analysis is that, when a sequence converges, its limit may be computed from any subsequence. In category theory, this idea is embodied by cofinality. We begin with connectedness.

Definition 1.6.1 Let I be a category. If $\mathrm{Ob}(I) \neq \varnothing$ and, for every $i, i' \in \mathrm{Ob}(I)$, there exist $n \geq 1$ and morphisms

$$i = i_0 \leftarrow i_1 \rightarrow i_2 \leftarrow i_3 \rightarrow i_4 \leftarrow \cdots \rightarrow i_{2n} = i',$$

then I is called connected.

We introduce an auxiliary concept that will be used repeatedly in the next several sections.

Definition 1.6.2 (Comma category; see [51, Definition 2.4.7]) Given categories I, J, and a functor $H : J \to I$, let $i \in \mathrm{Ob}(I)$.

- By definition, the objects of the comma category (i/H) (or (H/i)) are data $(j, i \to Hj)$ (or $(j, Hj \to i)$), where $j \in \mathrm{Ob}(J)$ and $i \to Hj$ (or $Hj \to i$) is a morphism in I. A morphism from $(j, i \to Hj)$ to $(j', i \to Hj')$ (or from $(j, Hj \to i)$ to $(j', Hj' \to i)$) is a morphism $f : j \to j'$ in J making the following diagram commute:

$$\begin{tikzcd} & i \arrow[dl] \arrow[dr] & \\ Hj \arrow[rr, "Hf"'] & & Hj' \end{tikzcd} \quad \text{or} \quad \begin{tikzcd} & i & \\ Hj \arrow[ur] \arrow[rr, "Hf"'] & & Hj' \arrow[ul] \end{tikzcd}.$$

  Composition and identity morphisms are defined in the usual way.

- The two categories above come respectively with projection functors $\Pi_{i/} : (i/H) \to J$ and $\Pi_{/i} : (H/i) \to J$, taking an object $(j, i \to Hj)$ (or $(j, Hj \to i)$) to j and a morphism f to f.

- Every morphism $i \to i'$ in I naturally induces functors $(i'/H) \to (i/H)$ and $(H/i) \to (H/i')$, making the following diagrams commute:

$$\begin{tikzcd} (i'/H) \arrow[rr] \arrow[dr, "\Pi_{i'/}"'] & & (i/H) \arrow[dl, "\Pi_{i/}"] \\ & J & \end{tikzcd} \quad \begin{tikzcd} (H/i) \arrow[rr] \arrow[dr, "\Pi_{/i}"'] & & (H/i') \arrow[dl, "\Pi_{/i'}"] \\ & J & \end{tikzcd}.$$

Example 1.6.3 Regard a functor $\alpha : I \to \mathcal{C}$ as an object of $\mathcal{C}^I$, and consider the diagonal functor $\Delta : \mathcal{C} \to \mathcal{C}^I$ taking every object to the corresponding constant functor. By Definition 1.6.2, (α/Δ) is precisely the category of "cocones" introduced in Convention 1.5.1, while the functor $\Pi_{\alpha/}$ maps each cocone to its vertex.

Definition 1.6.4 For a functor $H : J \to I$, if (i/H) is connected for every $i \in \mathrm{Ob}(I)$, then H is called **cofinal**; if H is the inclusion functor of a subcategory, the subcategory J is called cofinal.

It is easy to verify that if $I \to J$ and $J \to K$ are both cofinal, then the composite functor $I \to K$ is also cofinal. The verification is left to the reader as an exercise.

The next result shows that a limit may be computed by restricting the diagram along a cofinal functor. Recall that H induces $H^{\mathrm{op}} : J^{\mathrm{op}} \to I^{\mathrm{op}}$.

Proposition 1.6.5 If $H : J \to I$ is cofinal, then:

- ◇ for a given functor $\alpha : I \to \mathcal{C}$, $\varinjlim \alpha$ exists if and only if $\varinjlim \alpha H$ exists; in that case there is a canonical isomorphism $\varinjlim \alpha H \simeq \varinjlim \alpha$;
- ◇ dually, for a given functor $\beta : I^{\mathrm{op}} \to \mathcal{C}$, $\varprojlim \beta$ exists if and only if $\varprojlim \beta H^{\mathrm{op}}$ exists; in that case there is a canonical isomorphism $\varprojlim \beta H^{\mathrm{op}} \simeq \varprojlim \beta$.

Proof We discuss only $\varinjlim$. In the language of §1.5, it suffices to prove that the functor $(\alpha/\Delta) \to (\alpha H/\Delta)$ is an equivalence. It takes a cocone with base α, namely $(L, (f_i)_i)$, to the cocone with base αH, namely $(L, (f_{Hj})_j)$.

We first prove that this functor is essentially surjective. Consider $L \in \mathrm{Ob}(\mathcal{C})$ and a family of morphisms $(g_j : \alpha H(j) \to L)_{j \in \mathrm{Ob}(J)}$ satisfying the compatibility condition $g_{j'} \circ \alpha H(j \to j') = g_j$. For every $i \in \mathrm{Ob}(I)$, since (i/H) is nonempty, we may choose j and a morphism $i \to H(j)$. Define $f_i := g_j \circ \alpha(i \to H(j))$. This definition is independent of the choice of comma object $(j, i \to H(j))$: indeed, if there are morphisms in (i/H) as in the diagram

$$\begin{tikzcd} & i \arrow[dl] \arrow[d] \arrow[dr] & \\ H(j) & H(j'') \arrow[l, "H(j'' \to j)"'] \arrow[r, "H(j'' \to j')"'] & H(j') \end{tikzcd}$$

then the construction immediately gives

$$g_j \circ \alpha(i \to H(j)) = g_{j''} \circ \alpha(i \to H(j'')) = g_{j'} \circ \alpha(i \to H(j')).$$

Since (i/H) is connected, this suffices to show that f_i is well-defined. The preceding discussion also implies that $(f_i)_i$ satisfies the compatibility condition $f_i = f_{i'} \circ \alpha(i \to i')$: for every $i \to i'$ and $i' \to H(j')$, we may compute f_i using the comma object $(j', i \to i' \to H(j'))$. Hence $(L, (f_i)_i) \in \mathrm{Ob}((\alpha/\Delta))$ maps to $(L, (g_j)_j)$.

We next show that the functor is full and faithful. Giving a morphism in (α/Δ), namely $(L, (f_i)_i) \to (L', (f'_i)_i)$, is equivalent to giving a morphism $\theta : L \to L'$ in $\mathcal{C}$ such that $\theta f_i = f'_i$ for every i. It suffices to check this condition on the objects $H(j)$ in the cofinal image of H: for every i, choose $j \in \mathrm{Ob}(J)$ and $i \to H(j)$; the condition then becomes $\theta f_{H(j)} = f'_{H(j)}$, which says precisely that θ is a morphism in $(\alpha H/\Delta)$. $\square$

Definition 1.6.6 Following [51, Definition 2.7.6], a category I with $\mathrm{Ob}(I) \neq \varnothing$ satisfying the following conditions is called **filtered**[6)].

- ◇ For every $i, j \in \mathrm{Ob}(I)$, there is an object $k \in \mathrm{Ob}(I)$ together with morphisms $i \to k$ and $j \to k$.
- ◇ For every pair of morphisms $f, g : i \to j$ in I, there is an object $k \in \mathrm{Ob}(I)$ and a morphism $h : j \to k$ such that $hf = hg$.

If I is a filtered category and $\alpha : I \to \mathcal{C}$ is a functor, the corresponding $\varinjlim \alpha$ is called a **filtered inductive limit**.

Example 1.6.7 One common class of filtered categories comes from filtered partially ordered sets[7)]. Every partially ordered set $(P, \leq)$ determines a corresponding category $\mathcal{P}$ such that $x \leq y \iff \mathrm{Hom}_{\mathcal{P}}(x, y) \neq \varnothing$. If $\mathcal{P}$ is a filtered category, then $(P, \leq)$ is called a **filtered partially ordered set**. A partially ordered set is filtered if and only if it is nonempty and every pair of elements i, j has a common upper bound k. Every nonempty totally ordered set is plainly filtered.

Proposition 1.6.8 Let I be a filtered category. A full subcategory $J \subset I$ is cofinal if and only if, for every $i \in \mathrm{Ob}(I)$, there are $j \in \mathrm{Ob}(J)$ and a morphism $i \to j$; in that case J is also filtered.

Proof The "only if" direction is clear. For the converse, given $j \leftarrow i \to j'$, with $j, j' \in \mathrm{Ob}(J)$, filteredness of I together with the hypothesis ensures the existence of a commutative diagram

$$\begin{tikzcd}
 & j \arrow[r] & k & j' \arrow[l] & \\
j \arrow[ur, equal] & & & & j' \arrow[ul, equal] \\
 & & i \arrow[ull] \arrow[uu] \arrow[urr] & &
\end{tikzcd}$$

with $k \in \mathrm{Ob}(J)$. This shows that J is cofinal.

To prove that J is filtered, take arbitrary $i, j \in \mathrm{Ob}(J)$. Since I is filtered, there are $k_0 \in \mathrm{Ob}(I)$ and morphisms $i \to k_0$ and $j \to k_0$; also choose $k \in \mathrm{Ob}(J)$ and a morphism

6) Translator's note: the source does not state the condition $\mathrm{Ob}(I) \neq \varnothing$, so the empty category would satisfy both items vacuously; in the second item, the source also calls $k \in \mathrm{Ob}(I)$ a morphism. The target adds the nonemptiness condition and corrects the type of k to an object (O014-C010 and O014-C011).

7) When used to index limits, a filtered category can often be replaced by a filtered partially ordered set; see Remark A.2.3 and the explanations for the exercises in Appendix A.

$k_0 \to k$. Composing these morphisms yields morphisms $i \to k \leftarrow j$ in J.[8)] This proves the first condition for a filtered category. The second is proved by the same idea. $\square$

Example 1.6.9 Define $\mathrm{OFin}_I := \{\text{full subcategories } J \subset I : \mathrm{Ob}(J) \text{ is finite}\}$. Ordered by $\subset$, this is a filtered partially ordered set: for any $J, K \in \mathrm{OFin}_I$, the subcategory obtained by taking the union of their sets of objects is a common upper bound for J and K.

To see the usefulness of this construction, consider an arbitrary functor $\alpha : I \to \mathcal{C}$ (or $\beta : I^{\mathrm{op}} \to \mathcal{C}$). Its restriction to a full subcategory J gives $\alpha|_J$ (or $\beta|_{J^{\mathrm{op}}}$). If $J \subset K$, there is a natural morphism $\varinjlim \alpha|_J \to \varinjlim \alpha|_K$ (or $\varprojlim \beta|_{K^{\mathrm{op}}} \to \varprojlim \beta|_{J^{\mathrm{op}}}$), provided these limits exist. The next result explains how an arbitrary limit can be approximated "in a filtered manner" by limits over subcategories having only finitely many objects.

Proposition 1.6.10 In the situation above, there is a canonical isomorphism

$$\varinjlim_{\substack{J \in \mathrm{OFin}_I \\ \subset}} \varinjlim \alpha|_J \xrightarrow{\sim} \varinjlim \alpha$$

(or $\varprojlim \beta \xrightarrow{\sim} \varprojlim_{J \in \mathrm{OFin}_I^{\mathrm{op}}} \varprojlim \beta|_{J^{\mathrm{op}}}$), provided that the iterated limit on the left (or on the right) exists.[9)]

Proof Consider the case of $\varinjlim$. The universal property gives canonical bijections

$$\begin{aligned} \mathrm{Hom}\left(\varinjlim_{J \in \mathrm{OFin}_I} \varinjlim \alpha|_J, S \right) &\simeq \varprojlim_{J \in \mathrm{OFin}_I^{\mathrm{op}}} \mathrm{Hom}\left(\varinjlim \alpha|_J, S \right) \\ &\simeq \varprojlim_{J \in \mathrm{OFin}_I^{\mathrm{op}}} \varprojlim \mathrm{Hom}(\alpha|_J, S). \end{aligned}$$

for arbitrary $S \in \mathrm{Ob}(\mathcal{C})$, while the right-hand side is an iterated $\varprojlim$ of sets. It remains to check that, for a family of sets $(X_i)_{i \in \mathrm{Ob}(I)}$ together with a compatible family of maps $(f_{i \to j} : X_j \to X_i)_{[i \to j] \in \mathrm{Mor}(I)}$, there is a bijection

$$\begin{aligned} \varprojlim_{i \in \mathrm{Ob}(I)} X_i &\xrightarrow{\sim} \varprojlim_{J \in \mathrm{OFin}_I^{\mathrm{op}}} \varprojlim_{j \in \mathrm{Ob}(J)} X_j \\ (x_i)_i &\longmapsto \big((x_j)_{j \in \mathrm{Ob}(J)} \big)_{J \in \mathrm{OFin}_I}. \end{aligned}$$

8) Translator's note: the source calls $i \to k_0 \leftarrow j$ morphisms in J, although k_0 is only known to lie in I. The target uses the composites into $k \in \mathrm{Ob}(J)$ (O014-C012).

9) Translator's note: in the dual formula, the source writes an outer inductive limit. For $J \subset K$, however, the transition morphism goes from the K-term to the J-term, so the terms form a diagram on $\mathrm{OFin}_I^{\mathrm{op}}$ and the outer construction is a projective limit (O014-C013).

For every J, the concrete construction of the $\varprojlim$ of sets is

$$\varprojlim_{j\in \mathrm{Ob}(J)} X_j = \left\{ (x_j)_j \in \prod_{j\in \mathrm{Ob}(J)} X_j : \forall [i \to j] \in \mathrm{Mor}(J),\ f_{i\to j}(x_j) = x_i \right\},$$

so the bijection is clear: every piece of data defining $\varprojlim_i X_i$, whether a coordinate x_i or a compatibility condition $f_{i\to j}(x_j) = x_i$, is contained in some $J \in \mathrm{OFin}_I$. $\square$

Recall that the category Set has all small $\varinjlim$ and $\varprojlim$. Small filtered inductive limits have a particularly simple description.

Proposition 1.6.11 Let I be a small filtered category and $\alpha : I \to \mathsf{Set}$ a functor. Define the following binary relation on the set $\bigsqcup_{i\in \mathrm{Ob}(I)} \alpha(i)$: for $x \in \alpha(i)$ and $y \in \alpha(j)$, the relation $x \sim y$ means that there is an object $k \in \mathrm{Ob}(I)$ together with morphisms $i \to k$ and $j \to k$ such that $\alpha(i \to k)(x) = \alpha(j \to k)(y)$. Then $\sim$ is an equivalence relation, and

$$\varinjlim \alpha = \left(\bigsqcup_{i\in \mathrm{Ob}(I)} \alpha(i) \right) / \sim .$$

Proof See the discussion following [51, Definition 2.7.6]. $\square$

An element of the preceding $\varinjlim \alpha$ will henceforth be written $[x_{i_0}]$, where $i_0 \in \mathrm{Ob}(I)$ and $[x_{i_0}]$ is the equivalence class containing $x_{i_0} \in \alpha(i_0)$. On the other hand, for a general small category J and functor $\beta : J^{\mathrm{op}} \to \mathsf{Set}$, we write an element of $\varprojlim \beta$ as $(y_j)_{j\in \mathrm{Ob}(J)}$, where $y_j \in \beta(j)$ satisfies the compatibility condition $\beta(j' \to j)(y_j) = y_{j'}$.

Next consider a functor of the form $\alpha : I \times J^{\mathrm{op}} \to \mathsf{Set}$, with I, J small categories; one may take $\varinjlim$ and $\varprojlim$ in the variables I and J, respectively. For every $(i_0, j_0) \in \mathrm{Ob}(I \times J)$, there is a composite of natural maps

$$\varprojlim_j \alpha(i_0, j) \to \alpha(i_0, j_0) \to \varinjlim_i \alpha(i, j_0);$$

this composite is natural in i_0 and j_0. Varying j_0 gives a canonical map $\varprojlim_j \alpha(i_0, j) \to \varprojlim_j \varinjlim_i \alpha(i, j)$; then varying i_0 gives a canonical map

$$e : \varinjlim_i \varprojlim_j \alpha(i, j) \to \varprojlim_j \varinjlim_i \alpha(i, j). \tag{1.6.1}$$

In terms of representatives, e maps $[(x_{i_0,j})_j]$ to $([x_{i_0,j}])_j$.

If $\mathrm{Mor}(J)$ is a finite set, the category J is called finite; equivalently, its object set is finite and every morphism set between two objects is finite. The next result will be used in §1.9.

Proposition 1.6.12 Let I be a small filtered category, J a finite category, and $\alpha : I \times J^{\mathrm{op}} \to \mathsf{Set}$ a functor. Then the map e in (1.6.1) is a bijection.

Proof We first show that e is surjective. Given $(b_j)_{j \in \mathrm{Ob}(J)} \in \varprojlim_j \varinjlim_i \alpha(i,j)$, write each b_j as $[x_{i_j,j}]$. Since I is filtered and J is finite, we may choose a common target object $i \in \mathrm{Ob}(I)$ for all i_j, and replace each representative by its image along a morphism $i_j \to i$. Thus all representatives may be written $x_{i,j}$ with the same index i. For every morphism $j \to j'$ in J, we have

$$[\alpha(i, j \to j')(x_{i,j'})] = [x_{i,j}] .$$

Since $\mathrm{Mor}(J)$ is finite and I is filtered, we may choose one morphism $i \to i_1$ that equalizes all these pairs, replace all representatives by their images at index i_1, and then rename i_1 as i. Thus $\alpha(i, j \to j')(x_{i,j'}) = x_{i,j}$ holds for every $[j \to j'] \in \mathrm{Mor}(J)$. Consequently, $e\left([(x_{i,j})_j]\right) = (b_j)_j$.

We next show that e is injective. Suppose $e(x) = e(y)$. Since I is filtered, we may replace representatives of x and y by their images at one common index $i \in \mathrm{Ob}(I)$, so that they have representatives $(x_{i,j})_j$ and $(y_{i,j})_j$. Thus $[x_{i,j}] = [y_{i,j}]$ for every $j \in \mathrm{Ob}(J)$. Since I is filtered and J is finite, we may choose one common later index, replace all components by their images there, and rename it i, so that $x_{i,j} = y_{i,j}$ for every j. Hence $x = y$. $\square$

1.7 Kan Extensions

This section begins with the problem of extending a functor. Consider categories $\mathcal{C}$, $\mathcal{D}$, $\mathcal{E}$ and functors K and F as in the following diagram:

$$\begin{tikzcd} \mathcal{C} \arrow[d, "K"'] \arrow[dr, bend left, "F"] & \\ \mathcal{D} \arrow[r, dashed, "\exists ?\, L"'] & \mathcal{E} \end{tikzcd}$$

In the basic spirit of category theory, we would like to find a functor L indicated by the dashed arrow, together with a natural isomorphism $F \simeq LK$. This problem generally has no solution: for example, there may be $f \in \mathrm{Mor}(\mathcal{C})$ such that Kf is an isomorphism while Ff is not. Failing that, we may at least ask whether there is a best approximation. The answer is characterized by a universal property, with left and right versions.

Definition 1.7.1 (D. Kan) Consider categories $\mathcal{C}$, $\mathcal{D}$, $\mathcal{E}$ and functors $K : \mathcal{C} \to \mathcal{D}$ and $F : \mathcal{C} \to \mathcal{E}$.

◇ A **left Kan extension** of the functor F along K is the following data $(\operatorname{Lan}_K F, \eta)$:

– $\operatorname{Lan}_K F : \mathcal{D} \to \mathcal{E}$ is a functor;
– $\eta : F \to (\operatorname{Lan}_K F)K$ is a natural transformation.

These data must satisfy the following universal property: for every $L : \mathcal{D} \to \mathcal{E}$ and $\xi : F \to LK$, there is a unique natural transformation $\chi : \operatorname{Lan}_K F \to L$ such that $\xi = (\chi K)\eta$ (using vertical and horizontal composition of natural transformations), or, in 2-cell diagrams:

$$\begin{tikzcd} \mathcal{C} \arrow[d, "K"'] \arrow[dr, bend left, "F"] \arrow[dr, phantom, "\Swarrow \xi"] & \\ \mathcal{D} \arrow[r, "L"'] & \mathcal{E} \end{tikzcd} = \begin{tikzcd} \mathcal{C} \arrow[d, "K"'] \arrow[dr, bend left, "F"] \arrow[dr, phantom, "\Swarrow \eta"] & \\ \mathcal{D} \arrow[r, "\operatorname{Lan}_K F"'] \arrow[r, bend right=80, "L"'] \arrow[r, phantom, bend right=40, "\Downarrow \chi"] & \mathcal{E} \end{tikzcd} .$$

◇ A **right Kan extension** of the functor F along K is the following data $(\operatorname{Ran}_K F, \varepsilon)$:

– $\operatorname{Ran}_K F : \mathcal{D} \to \mathcal{E}$ is a functor;
– $\varepsilon : (\operatorname{Ran}_K F)K \to F$ is a natural transformation.

These data must satisfy the following universal property: for every $R : \mathcal{D} \to \mathcal{E}$ and $\delta : RK \to F$, there is a unique natural transformation $\theta : R \to \operatorname{Ran}_K F$ such that $\delta = \varepsilon(\theta K)$, or, in 2-cell diagrams:

$$\begin{tikzcd} \mathcal{C} \arrow[d, "K"'] \arrow[dr, bend left, "F"] \arrow[dr, phantom, "\Nearrow \delta"] & \\ \mathcal{D} \arrow[r, "R"'] & \mathcal{E} \end{tikzcd} = \begin{tikzcd} \mathcal{C} \arrow[d, "K"'] \arrow[dr, bend left, "F"] \arrow[dr, phantom, "\Nearrow \varepsilon"] & \\ \mathcal{D} \arrow[r, "\operatorname{Ran}_K F"'] \arrow[r, bend right=80, "R"'] \arrow[r, phantom, bend right=40, "\Uparrow \theta"] & \mathcal{E} \end{tikzcd} .$$

If $\mathcal{C}$, $\mathcal{D}$, and $\mathcal{E}$ in the definition are replaced respectively by $\mathcal{C}^{\mathrm{op}}$, $\mathcal{D}^{\mathrm{op}}$, and $\mathcal{E}^{\mathrm{op}}$, the pattern of directions of the functors remains the same, while the natural transformations reverse direction. Thus left and right Kan extensions are dual notions.

Using the functor categories $\mathcal{E}^{\mathcal{C}}$ and $\mathcal{E}^{\mathcal{D}}$, the universal properties of left and right Kan extensions state, respectively, the bijections

$$\begin{aligned} \operatorname{Hom}_{\mathcal{E}^{\mathcal{D}}}(\operatorname{Lan}_K F, L) &\xrightarrow{1:1} \operatorname{Hom}_{\mathcal{E}^{\mathcal{C}}}(F, LK) \\ \chi &\longmapsto (\chi K)\eta, \\ \operatorname{Hom}_{\mathcal{E}^{\mathcal{D}}}(R, \operatorname{Ran}_K F) &\xrightarrow{1:1} \operatorname{Hom}_{\mathcal{E}^{\mathcal{C}}}(RK, F) \\ \theta &\longmapsto \varepsilon(\theta K); \end{aligned} \tag{1.7.1}$$

their inverses are the maps $(L, \xi) \mapsto \chi$ and $(R, \delta) \mapsto \theta$, respectively, from Definition 1.7.1.

Proposition 1.7.2 Given categories $\mathcal{C}$, $\mathcal{D}$, $\mathcal{E}$ and functors $K : \mathcal{C} \to \mathcal{D}$ and $F : \mathcal{C} \to \mathcal{E}$, consider the pullback (that is, precomposition) functor $K^* : \mathcal{E}^{\mathcal{D}} \to \mathcal{E}^{\mathcal{C}}$ between the functor categories.

(i) If it exists, a left Kan extension $(\operatorname{Lan}_K F, \eta)$ is unique up to a unique isomorphism in $\mathcal{E}^{\mathcal{D}}$; the same holds for a right Kan extension.

(ii) If the left Kan extension (or right Kan extension) along K exists for every F, these extensions assemble into a left adjoint to K^*, $\operatorname{Lan}_K : \mathcal{E}^{\mathcal{C}} \to \mathcal{E}^{\mathcal{D}}$ (or a right adjoint $\operatorname{Ran}_K : \mathcal{E}^{\mathcal{C}} \to \mathcal{E}^{\mathcal{D}}$); the family of natural transformations assembled from η (or ε) in Definition 1.7.1 is exactly the unit (or counit) of this adjunction.

Proof For (i), simply apply the universal property in Definition 1.7.1 in the usual way.

For (ii), observe that $LK = K^*(L)$ and $RK = K^*(R)$; the desired adjunction is precisely the content of (1.7.1). The assertion about the unit or counit follows similarly; see [51, Proposition 2.6.5]. □

By convention, the data η or ε in a Kan extension are often omitted from the notation. If the left Kan extension (or right Kan extension) along K exists for every F, the functor Lan_K (or Ran_K) obtained in (ii) also has the corresponding uniqueness property, by uniqueness of adjoints [51, Proposition 2.6.10].

Remark 1.7.3 Since the definition of a Kan extension involves only composition of 2-cells, it extends to general 2-categories; the present situation is the special case of the 2-category Cat. See [51, §3.5] for details.

For every category $\mathcal{C}$, denote the unique functor $\mathcal{C} \to \mathbf{1}$ by $!$; specifying a functor $\mathbf{1} \to \mathcal{C}$ is equivalent to specifying an object of $\mathcal{C}$. For every category $\mathcal{E}$ and object X in it, write $\Delta(X) : \mathcal{C} \to \mathcal{E}$ for the composite $\mathcal{C} \xrightarrow{!} \mathbf{1} \xrightarrow{\text{constant } X} \mathcal{E}$; this is the constant functor that sends every object to X and every morphism to id_X.

Example 1.7.4 (Limits as Kan extensions) Let $F : \mathcal{C} \to \mathcal{E}$ be a functor. Consider the diagram

$$\begin{tikzcd} \mathcal{C} \arrow[d, "!"'] \arrow[dr, bend left, "F"] & \\ \mathbf{1} & \mathcal{E} \end{tikzcd}$$

Then $\operatorname{Lan}_! F$ (or $\operatorname{Ran}_! F$) exists if and only if $\varinjlim F$ (or $\varprojlim F$) exists in $\mathcal{E}$, and in that case the Kan extension is precisely the limit in question.

For example, consider $(\mathrm{Lan}_! F, \eta)$. The functor $\mathrm{Lan}_! F : \mathbf{1} \to \mathcal{E}$ may be viewed as an object of $\mathcal{E}$, while η is a natural transformation $F \to \Delta(\mathrm{Lan}_! F)$ in $\mathcal{E}^{\mathcal{C}}$. Its universal property is equivalent to the assertion that the data $(\mathrm{Lan}_! F, \eta)$ give an initial object of the category (F/Δ); this is precisely the description of $\varinjlim F$ in §1.5.

Example 1.7.5 (Adjunctions as Kan extensions) Consider a pair of functors:

$$F : \mathcal{C} \rightleftarrows \mathcal{D} : G$$

If (F, G) is equipped with the structure of an adjunction with unit $\eta : \mathrm{id}_{\mathcal{C}} \to GF$ and counit $\varepsilon : FG \to \mathrm{id}_{\mathcal{D}}$, then (G, η) gives $\mathrm{Lan}_F(\mathrm{id}_{\mathcal{C}})$ and (F, ε) gives $\mathrm{Ran}_G(\mathrm{id}_{\mathcal{D}})$.

To explain this, first we show that the following pullback functors

$$G^* : \mathcal{C}^{\mathcal{C}} \rightleftarrows \mathcal{C}^{\mathcal{D}} : F^*$$

themselves form a natural adjoint pair. Observe that η induces $\mathrm{id}_{\mathcal{C}}^* = \mathrm{id}_{\mathcal{C}^{\mathcal{C}}} \to F^*G^* = (GF)^*$, and similarly ε induces $G^*F^* \to \mathrm{id}_{\mathcal{C}^{\mathcal{D}}}$; we continue to denote both by η and ε. We must check the triangle identities characterizing the adjunction $(G^*, F^*, \eta, \varepsilon)$. In brief, these identities follow formally by applying pullback $(-)^*$ in functor categories to the triangle identities for $(F, G, \eta, \varepsilon)$; the details are left to the reader.

This adjunction gives a canonical bijection $\mathrm{Hom}_{\mathcal{C}^{\mathcal{D}}}(\underbrace{G^*\mathrm{id}_{\mathcal{C}}}_{=G}, L) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}^{\mathcal{C}}}(\mathrm{id}_{\mathcal{C}}, \underbrace{F^*L}_{=LF})$, for arbitrary $L : \mathcal{D} \to \mathcal{C}$; when $L = G$, id_G maps to η. Writing the universal property of $\mathrm{Lan}_F(\mathrm{id}_{\mathcal{C}})$ as the bijection in (1.7.1), one sees at once that (G, η) gives $\mathrm{Lan}_F(\mathrm{id}_{\mathcal{C}})$. The argument for $\mathrm{Ran}_G(\mathrm{id}_{\mathcal{D}})$ is entirely similar.[10)]

In practice, some situations require a further strengthening of the notion of Kan extension[11)], especially in the later treatment of derived functors.

Definition 1.7.6 Consider categories $\mathcal{C}$, $\mathcal{D}$, $\mathcal{E}$, $\mathcal{F}$ and functors $K : \mathcal{C} \to \mathcal{D}$ and $F : \mathcal{C} \to \mathcal{E}$.[12)]

- ◇ Suppose the right Kan extension $(\mathrm{Ran}_K F, \varepsilon)$ exists. A functor $M : \mathcal{E} \to \mathcal{F}$ is said to **preserve** $\mathrm{Ran}_K F$ if the composite 2-cell

$$\begin{tikzcd} \mathcal{C} \arrow[d, "K"'] \arrow[dr, bend left, "F"] & & \\ \mathcal{D} \arrow[r, "\mathrm{Ran}_K F"'] \arrow[ur, phantom, "\Nearrow\varepsilon"] & \mathcal{E} \arrow[r, "M"'] & \mathcal{F} \end{tikzcd} \quad =: \quad \begin{tikzcd} \mathcal{C} \arrow[rr, "MF"] \arrow[dr, "K"'] & & \mathcal{F} \\ & \mathcal{D} \arrow[ur, "M\,\mathrm{Ran}_K F"'] \arrow[u, phantom, "\Uparrow M\varepsilon"] & \end{tikzcd}$$

10) Translator's note: the source writes $\mathrm{Ran}_F(\mathrm{id}_{\mathcal{C}})$ in the final sentence, contrary to the preceding assertion concerning (F, ε). The target corrects this to $\mathrm{Ran}_G(\mathrm{id}_{\mathcal{D}})$ (O014-C014).

11) One common strengthening is called a pointwise Kan extension; it is not discussed here.

12) Translator's note: the source does not include the category $\mathcal{F}$ in the initial list, even though it is used as the codomain of M. The target adds it (O014-C015).

gives a right Kan extension of MF along K.

- Suppose the left Kan extension $(\mathrm{Lan}_K F, \eta)$ exists. A functor $M : \mathcal{E} \to \mathcal{F}$ is said to **preserve** $\mathrm{Lan}_K F$ if the composite 2-cell

$$\begin{tikzcd} \mathcal{C} \arrow[d, "K"'] \arrow[drr, bend left, "F"] & & \\ \mathcal{D} \arrow[rr, "\mathrm{Lan}_K F"'] \arrow[urr, phantom, "\swarrow\eta"] & & \mathcal{E} \arrow[r, "M"'] & \mathcal{F} \end{tikzcd} \quad =: \quad \begin{tikzcd} \mathcal{C} \arrow[rr, "MF"] \arrow[dr, "K"'] & \arrow[d, phantom, "\Downarrow M\eta"] & \mathcal{F} \\ & \mathcal{D} \arrow[ur, "M\,\mathrm{Lan}_K F"'] & \end{tikzcd}$$

gives a left Kan extension of MF along K.

- If a left Kan extension $(\mathrm{Lan}_K F, \eta)$ (or a right Kan extension $(\mathrm{Ran}_K F, \varepsilon)$) is preserved by every functor whose domain is $\mathcal{E}$, the Kan extension is called **absolute**.

The next step is to explain the relationship between adjunctions and absolute Kan extensions. The result is extremely useful in the later study of derived functors; studying its proof also helps one become accustomed to Kan extensions and the manipulation of 2-cells. Consider a pair of adjoint functors $F : \mathcal{C} \rightleftarrows \mathcal{C}' : G$, specified by a unit $\eta : \mathrm{id}_{\mathcal{C}} \to GF$ and a counit $\varepsilon : FG \to \mathrm{id}_{\mathcal{C}'}$. Also given are functors

$$K : \mathcal{C} \to \mathcal{D}, \quad K' : \mathcal{C}' \to \mathcal{D}'.$$

We shall now use 2-cell diagrams systematically to represent functors, natural transformations between them, and their composites, making the argument more concise. In manipulating 2-cells, we interchange vertical and horizontal composition without further comment; this is valid, as explained in [51, Lemma 2.2.7].

Theorem 1.7.7 (G. Maltsiniotis [34]) Suppose $K'F$ has an absolute right Kan extension along K, denoted $\mathrm{L}F$, while KG has an absolute left Kan extension along K', denoted $\mathrm{R}G$. The associated data are displayed in the following diagrams:

$$\begin{tikzcd} \mathcal{C} \arrow[r, "F"] \arrow[d, "K"'] & \mathcal{C}' \arrow[d, "K'"] \\ \mathcal{D} \arrow[r, "\mathrm{L}F"'] \arrow[ur, phantom, "\nearrow\alpha"] & \mathcal{D}' \end{tikzcd} \qquad \begin{tikzcd} \mathcal{C}' \arrow[r, "G"] \arrow[d, "K'"'] & \mathcal{C} \arrow[d, "K"] \\ \mathcal{D}' \arrow[r, "\mathrm{R}G"'] \arrow[ur, phantom, "\swarrow\beta"] & \mathcal{D}. \end{tikzcd}$$

In this situation, there are unique transformations $\underline{\eta} : \mathrm{id}_{\mathcal{D}} \to \mathrm{R}G \circ \mathrm{L}F$ and $\underline{\varepsilon} : \mathrm{L}F \circ \mathrm{R}G \to$

$\mathrm{id}_{\mathcal{D}'}$ such that the following equations of composite 2-cells hold:

$$
\begin{tikzcd}
\mathcal{C} \arrow[r] \arrow[rr, bend left=60, "\mathrm{id}_{\mathcal{C}}"] \arrow[rr, phantom, bend left=30, "\Downarrow\underline{\eta}"] & \mathcal{C}' \arrow[r] \arrow[d] \arrow[dr, phantom, "\swarrow\beta"] & \mathcal{C} \arrow[d] \\
& \mathcal{D}' \arrow[r, "\mathrm{R}G"'] & \mathcal{D}
\end{tikzcd}
=
\begin{tikzcd}
\mathcal{C} \arrow[r] \arrow[d] \arrow[dr, phantom, "\nearrow\alpha"] & \mathcal{C}' \arrow[d] & \\
\mathcal{D} \arrow[r, "\mathrm{L}F"'] \arrow[rr, bend right=60, "\mathrm{id}_{\mathcal{D}}"'] \arrow[rr, phantom, bend right=30, "\Uparrow\underline{\eta}"] & \mathcal{D}' \arrow[r, "\mathrm{R}G"'] & \mathcal{D}
\end{tikzcd}
$$

$$
\begin{tikzcd}
\mathcal{C}' \arrow[r] \arrow[rr, bend left=60, "\mathrm{id}_{\mathcal{C}'}"] \arrow[rr, phantom, bend left=30, "\Uparrow\underline{\varepsilon}"] & \mathcal{C} \arrow[r] \arrow[d] \arrow[dr, phantom, "\nearrow\alpha"] & \mathcal{C}' \arrow[d] \\
& \mathcal{D} \arrow[r, "\mathrm{L}F"'] & \mathcal{D}'
\end{tikzcd}
=
\begin{tikzcd}
\mathcal{C}' \arrow[r] \arrow[d] \arrow[dr, phantom, "\swarrow\beta"] & \mathcal{C} \arrow[d] & \\
\mathcal{D}' \arrow[r, "\mathrm{R}G"'] \arrow[rr, bend right=60, "\mathrm{id}_{\mathcal{D}'}"'] \arrow[rr, phantom, bend right=30, "\Downarrow\underline{\varepsilon}"] & \mathcal{D} \arrow[r, "\mathrm{L}F"'] & \mathcal{D}'
\end{tikzcd}
$$

Moreover, $\underline{\eta}$ and $\underline{\varepsilon}$ make $(\mathrm{L}F, \mathrm{R}G)$ an adjunction.

Proof The first task is to prove the existence and uniqueness of $\underline{\eta}$ and $\underline{\varepsilon}$. Since $\mathrm{R}G$ is an absolute left Kan extension, the 2-cell

$$
\begin{tikzcd}
\mathcal{C}' \arrow[r, "G"] \arrow[d, "K'"'] \arrow[dr, phantom, "\swarrow\beta"] & \mathcal{C} \arrow[d, "K"] & \\
\mathcal{D}' \arrow[r, "\mathrm{R}G"'] & \mathcal{D} \arrow[r, "\mathrm{L}F"'] & \mathcal{D}'
\end{tikzcd}
= (\mathrm{L}F)\beta : \mathrm{L}F \circ K \circ G \to \mathrm{L}F \circ \mathrm{R}G \circ K'
$$

makes $\mathrm{L}F \circ \mathrm{R}G$ a left Kan extension of $\mathrm{L}F \circ K \circ G$ along K'. On the other hand, we also have

$$
\begin{tikzcd}
\mathcal{C}' \arrow[r, "G"] \arrow[rr, bend left=60, "\mathrm{id}_{\mathcal{C}'}"] \arrow[rr, phantom, bend left=30, "\Uparrow\underline{\varepsilon}"] & \mathcal{C} \arrow[r, "F"] \arrow[d, "K"'] \arrow[dr, phantom, "\nearrow\alpha"] & \mathcal{C}' \arrow[d, "K'"] \\
& \mathcal{D} \arrow[r, "\mathrm{L}F"'] & \mathcal{D}'
\end{tikzcd}
= \underline{\varepsilon}(\alpha G) : \mathrm{L}F \circ K \circ G \to K' \circ \mathrm{id}_{\mathcal{C}'} = K'.
$$

Thus the universal property of the left Kan extension determines a unique $\underline{\varepsilon}$ satisfying the equation of composite 2-cells in the statement of the theorem.

The case of $\underline{\eta}$ is dual. It suffices to observe that $(\mathrm{R}G)\alpha : \mathrm{R}G \circ \mathrm{L}F \circ K \to \mathrm{R}G \circ K' \circ F$ also makes $\mathrm{R}G \circ \mathrm{L}F$ a right Kan extension, because $\mathrm{L}F$ is an absolute right Kan extension.

It remains to verify the triangle identities for the unit $\underline{\eta}$ and counit $\underline{\varepsilon}$, namely the following equations of composite 2-cells:

$$
\begin{tikzcd}[column sep=small]
\mathcal{D}' \arrow[r] \arrow[rr, bend right=50, "\mathrm{id}"'] \arrow[rr, phantom, bend right=25, "\Downarrow"] & \mathcal{D} \arrow[r] \arrow[rr, bend left=50, "\mathrm{id}"] \arrow[rr, phantom, bend left=25, "\Downarrow"] & \mathcal{D}' \arrow[r] & \mathcal{D}
\end{tikzcd}
=
\begin{tikzcd}
\mathcal{D}' \arrow[r, bend left=50, "\mathrm{R}G"] \arrow[r, bend right=50, "\mathrm{R}G"'] \arrow[r, phantom, "\overset{\|}{\mathrm{id}}\,\Downarrow"] & \mathcal{D}
\end{tikzcd}
,\qquad
\begin{tikzcd}[column sep=small]
\mathcal{D} \arrow[r] \arrow[rr, bend left=50, "\mathrm{id}"] \arrow[rr, phantom, bend left=25, "\Downarrow"] & \mathcal{D}' \arrow[r] \arrow[rr, bend right=50, "\mathrm{id}"'] \arrow[rr, phantom, bend right=25, "\Downarrow"] & \mathcal{D} \arrow[r] & \mathcal{D}'
\end{tikzcd}
=
\begin{tikzcd}
\mathcal{D} \arrow[r, bend left=50, "\mathrm{L}F"] \arrow[r, bend right=50, "\mathrm{L}F"'] \arrow[r, phantom, "\overset{\|}{\mathrm{id}}\,\Downarrow"] & \mathcal{D}'
\end{tikzcd}
,
$$

where unambiguous arrow labels have been omitted. We prove the first equation. By the universal property of $(\mathrm{R}G, \beta)$ as a left Kan extension, it suffices to show that the 2-cells obtained by composing both sides with β are equal. For the left side of the first equation, begin with

$$
\begin{tikzcd}
\mathcal{C}' \arrow[r] \arrow[d] & \mathcal{C} \arrow[d] \arrow[dl, Rightarrow] & & \\
\mathcal{D}' \arrow[r] \arrow[rr, bend right=50, "\mathrm{id}"'] \arrow[rr, phantom, bend right=25, "\Downarrow\underline{\varepsilon}"] & \mathcal{D} \arrow[r] \arrow[rr, bend left=50, "\mathrm{id}"] \arrow[rr, phantom, bend left=25, "\Downarrow\underline{\eta}"] & \mathcal{D}' \arrow[r] & \mathcal{D}
\end{tikzcd}
=
\begin{tikzcd}
\mathcal{C}' \arrow[r] \arrow[rr, bend left=50, "\mathrm{id}"] \arrow[rr, phantom, bend left=25, "\varepsilon\Uparrow"] & \mathcal{C} \arrow[r] \arrow[d] & \mathcal{C}' \arrow[d] & \\
& \mathcal{D} \arrow[r] \arrow[ur, Rightarrow] \arrow[rr, bend right=50, "\mathrm{id}"'] \arrow[rr, phantom, bend right=25, "\Uparrow\underline{\eta}"] & \mathcal{D}' \arrow[r] & \mathcal{D}
\end{tikzcd}
$$

where we used the characterization of $\underline{\varepsilon}$. Next, using the characterization of $\underline{\eta}$ and the triangle identities satisfied by η and ε, the right-hand side above becomes

$$
\begin{tikzcd}[column sep=small]
\mathcal{C}' \arrow[r] \arrow[rr, bend right=50, "\mathrm{id}"'] \arrow[rr, phantom, bend right=25, "\Downarrow\varepsilon"] & \mathcal{C} \arrow[r] \arrow[rr, bend left=50, "\mathrm{id}"] \arrow[rr, phantom, bend left=25, "\Downarrow\eta"] & \mathcal{C}' \arrow[r] \arrow[d] & \mathcal{C} \arrow[d] \arrow[dl, Rightarrow, "\beta"'] \\
& & \mathcal{D}' \arrow[r] & \mathcal{D}
\end{tikzcd}
=
\begin{tikzcd}
\mathcal{C}' \arrow[r] \arrow[r, bend left=60] \arrow[r, phantom, bend left=30, "\Downarrow\mathrm{id}"] \arrow[d] & \mathcal{C} \arrow[d] \arrow[dl, Rightarrow, "\beta"'] \\
\mathcal{D}' \arrow[r] & \mathcal{D}
\end{tikzcd}
=
\begin{tikzcd}
\mathcal{C}' \arrow[r] \arrow[d] & \mathcal{C} \arrow[d] \arrow[dl, Rightarrow, "\beta"'] \\
\mathcal{D}' \arrow[r] \arrow[r, bend right=60] \arrow[r, phantom, bend right=30, "\Downarrow\mathrm{id}"] & \mathcal{D}
\end{tikzcd}
$$

This is the first triangle identity. The proof of the second is dual. □

1.8 Constructing Kan Extensions by Limits

As in §1.7, throughout this section we continue to consider functors $K : \mathcal{C} \to \mathcal{D}$ and $F : \mathcal{C} \to \mathcal{E}$, with the aim of studying the existence of $\mathrm{Lan}_K F$ and $\mathrm{Ran}_K F$.

Recall the discussion at the beginning of §1.7. The motivation for Kan extensions is to seek a functor $G : \mathcal{D} \to \mathcal{E}$ such that $F \simeq GK$, or at least to find the best approximation to one. How should Gd be defined for $d \in \mathrm{Ob}(\mathcal{D})$? The only clue is that when $d = Kc$, the object Gd ought to be Fc, up to isomorphism. This suggests approximating Gd by a colimit $\varinjlim$ (or a limit $\varprojlim$) of the objects Fc, indexed by all $c \in \mathrm{Ob}(\mathcal{C})$ and morphisms $Kc \to d$ (or $d \to Kc$). These two constructions have dual universal properties. We now make the construction precise.

Given $d \in \mathrm{Ob}(\mathcal{D})$, define the categories (K/d) and (d/K) as in Definition 1.6.2, together with the projection functors

$$\Pi_{/d} : (K/d) \to \mathcal{C}, \quad \Pi_{d/} : (d/K) \to \mathcal{C}.$$

Our focus will be on the composite functors $F\Pi_{/d} : (K/d) \to \mathcal{E}$ and $F\Pi_{d/} : (d/K) \to \mathcal{E}$, which send the objects $(c, Kc \to d)$ and $(c, d \to Kc)$, respectively, to Fc. Assuming that the limits under discussion exist, we first record a few observations.

- Every morphism $d \to d'$ induces a canonical morphism $\varinjlim F\Pi_{/d} \to \varinjlim F\Pi_{/d'}$; it is characterized by requiring the diagram

$$\begin{tikzcd} Fc \arrow[r, equal] \arrow[d] & Fc \arrow[d] \\ \varinjlim F\Pi_{/d} \arrow[r] & \varinjlim F\Pi_{/d'} \end{tikzcd} \tag{1.8.1}$$

  to commute for every $(c, Kc \to d) \in \mathrm{Ob}((K/d))$ (which induces an object $(c, Kc \to d') \in \mathrm{Ob}((K/d')))$, where the vertical arrows are the structure morphisms of the colimits. This is an instance of the functoriality of colimits; see [51, Lemma 2.7.4][13] and its proof.

- Considering $(c, Kc \xrightarrow{\mathrm{id}} Kc) \in \mathrm{Ob}(K/Kc)$ gives a canonical morphism $\eta_c : Fc \to \varinjlim F\Pi_{/Kc}$.

- Every morphism $c \to c'$ induces a canonical morphism $\varinjlim F\Pi_{/Kc} \to \varinjlim F\Pi_{/Kc'}$ that makes the following diagram commute:

$$\begin{tikzcd} Fc \arrow[r] \arrow[d, "\eta_c"'] & Fc' \arrow[d, "\eta_{c'}"] \\ \varinjlim F\Pi_{/Kc} \arrow[r] & \varinjlim F\Pi_{/Kc'}. \end{tikzcd}$$

By duality, $\varprojlim F\Pi_{d/}$ has the corresponding properties, including a canonical morphism $\varepsilon_c : \varprojlim F\Pi_{Kc/} \to Fc$. The next theorem is based on this functoriality.

Theorem 1.8.1 Consider functors $K : \mathcal{C} \to \mathcal{D}$ and $F : \mathcal{C} \to \mathcal{E}$.

[13] Translator's note: the source gives the locator as Lemma 2.7,4; it has been corrected to Lemma 2.7.4 here (O014-C016).

(i) Suppose that for every $d \in \mathrm{Ob}(\mathcal{D})$, the colimit $\varinjlim(F\Pi_{/d})$ exists in $\mathcal{E}$. Then $(\mathrm{Lan}_K F)(d) := \varinjlim(F\Pi_{/d})$, together with the functoriality given by (1.8.1), determines the left Kan extension $\mathrm{Lan}_K F : \mathcal{D} \to \mathcal{E}$. The associated natural transformation $\eta : F \to (\mathrm{Lan}_K F) \circ K$ comes from the family $(\eta_c)_{c \in \mathrm{Ob}(\mathcal{C})}$ discussed above.

(ii) Suppose that for every $d \in \mathrm{Ob}(\mathcal{D})$, the limit $\varprojlim(F\Pi_{d/})$ exists in $\mathcal{E}$. Then $(\mathrm{Ran}_K F)(d) := \varprojlim(F\Pi_{d/})$, together with the dual of (1.8.1), determines the right Kan extension $\mathrm{Ran}_K F : \mathcal{D} \to \mathcal{E}$. The associated natural transformation $\varepsilon : (\mathrm{Ran}_K F) \circ K \to F$ comes from the family $(\varepsilon_c)_{c \in \mathrm{Ob}(\mathcal{C})}$ discussed above.

(iii) If $K : \mathcal{C} \to \mathcal{D}$ is fully faithful, then the η constructed in (i) (or the ε constructed in (ii)) is an isomorphism.

Proof By duality, we treat only the case of $\mathrm{Lan}_K F$. For (i), we verify the universal property in Definition 1.7.1 step by step.

Let $L : \mathcal{D} \to \mathcal{E}$ be a functor and let $\xi : F \to LK$. For every $d \in \mathrm{Ob}(\mathcal{D})$, every object $(c, Kc \to d)$ of (K/d), and every morphism in (K/d) determined by $f : c \to c'$, namely $(c, Kc \to d) \to (c', Kc' \to d)$, there is a commutative diagram

$$\begin{tikzcd} Fc \arrow[r, "\xi_c"] \arrow[d, "Ff"'] & LKc \arrow[r] \arrow[d, "LKf"'] & Ld. \\ Fc' \arrow[r, "\xi_{c'}"'] & LKc' \arrow[ur] & \end{tikzcd}$$

The universal property of $\varinjlim$ therefore gives $\chi_d : (\mathrm{Lan}_K F)(d) \to Ld$, characterized by requiring the diagram

$$\begin{tikzcd} Fc \arrow[r] \arrow[d, "\xi_c"'] & (\mathrm{Lan}_K F)(d) \arrow[d, "\chi_d"] \\ LKc \arrow[r, "L(Kc\to d)"'] & Ld \end{tikzcd} \tag{1.8.2}$$

to commute for all $(c, Kc \to d) \in \mathrm{Ob}((K/d))$; the arrow in the first row is the structure morphism of the colimit.

We now prove that $(\chi_d)_{d \in \mathrm{Ob}(\mathcal{D})}$ defines a natural transformation $\chi : \mathrm{Lan}_K F \to L$. Given $d \to d'$, take $(c, Kc \to d)$ and its image $(c, Kc \to d')$. Comparing (1.8.1) with (1.8.2), the question reduces to proving that the diagram

$$\begin{tikzcd} & & Ld \arrow[d, "L(d\to d')"] \\ Fc \arrow[r, "\xi_c"] & LKc \arrow[ur, "L(Kc\to d)"] \arrow[r, "L(Kc\to d')"'] & Ld' \end{tikzcd}$$

commutes, which is immediate.

The next step is to verify

$$\forall c \in \mathrm{Ob}(\mathcal{C}), \quad \xi_c = \chi_{Kc}\eta_c : Fc \to LKc. \tag{1.8.3}$$

Since η_c is the canonical morphism from $Fc = F\Pi_{/Kc}(c, Kc \xrightarrow{\mathrm{id}} Kc)$ to $(\mathrm{Lan}_K F)(Kc)$, equation (1.8.3) is simply a special case of (1.8.2).

Finally, we show that a natural transformation $\chi : \mathrm{Lan}_K F \to L$ satisfying (1.8.3) is unique. Fix d and consider any object $(c, Kc \to d)$ of (K/d) and the associated diagram

$$\begin{tikzcd} & Fc \arrow[dl, "\eta_c"'] \arrow[dr] & \\ (\mathrm{Lan}_K F)(Kc) \arrow[rr, "(\mathrm{Lan}_K F)(Kc \to d)"'] \arrow[d, "\chi_{Kc}"'] & & (\mathrm{Lan}_K F)(d) \arrow[d, "\chi_d"] \\ LKc \arrow[rr, "L(Kc \to d)"'] & & Ld \end{tikzcd}$$

where $Fc \to (\mathrm{Lan}_K F)(d)$ is the morphism appearing in (1.8.2). The square commutes by the naturality of χ, while the triangle commutes by (1.8.1). Since $\chi_{Kc}\eta_c = \xi_c$, the composite $Fc \to (\mathrm{Lan}_K F)(d) \xrightarrow{\chi_d} Ld$ equals $L(Kc \to d)\xi_c$. As $(c, Kc \to d)$ varies, this family of equalities uniquely determines χ_d.

Thus $(\mathrm{Lan}_K F, \eta)$ is indeed a left Kan extension.

Now consider (iii), and fix $c \in \mathrm{Ob}(\mathcal{C})$. Since K is fully faithful, it is easy to see that (K/Kc) has the terminal object $(c, Kc \xrightarrow{\mathrm{id}} Kc)$. The morphism $\eta_c : Fc \to \varinjlim F\Pi_{/Kc}$ comes from this terminal object and is therefore an isomorphism. $\square$

Remark 1.8.2 If $\mathcal{C}$ is a small category, then (K/d) and (d/K) are also small categories for every $d \in \mathrm{Ob}(\mathcal{D})$. Consequently, when $\mathcal{E}$ is cocomplete (or complete), Theorem 1.8.1 and Proposition 1.7.2 show that $K^* : \mathcal{E}^{\mathcal{D}} \to \mathcal{E}^{\mathcal{C}}$ has a left adjoint Lan_K (or a right adjoint Ran_K).

Example 1.8.3 (Inverse image of presheaves) Take $\mathcal{E} := \mathsf{Set}$, which is both complete and cocomplete. Let $f : X \to Y$ be a continuous map of topological spaces. Let $\mathcal{C} := \mathsf{Open}_Y$ be the category whose objects are the open subsets of Y and whose morphisms are inclusions of open subsets; similarly, let $\mathcal{D} := \mathsf{Open}_X$. Now define a functor $\mathsf{Open}_Y \to \mathsf{Open}_X$ that sends an open subset $V \subset Y$ to $f^{-1}V$, and regard it below as a functor $K : \mathsf{Open}_Y^{\mathrm{op}} \to \mathsf{Open}_X^{\mathrm{op}}$. Readers familiar with sheaf theory will recognize that the objects of $\mathsf{Open}_Y^{\wedge} := \mathsf{Set}^{\mathsf{Open}_Y^{\mathrm{op}}}$ are, by definition, precisely the **presheaves** on Y, while $K^* : \mathsf{Open}_X^{\wedge} \to \mathsf{Open}_Y^{\wedge}$ is the direct-image functor between the presheaf categories: it sends a presheaf $\mathcal{F}$ on X to the presheaf on Y given by $V \mapsto \mathcal{F}(f^{-1}V)$.

In this situation, the left adjoint of the direct image is Lan_K, given by Theorem 1.8.1; it sends a presheaf $\mathcal{G}$ on Y to

$$U \mapsto \varinjlim_{\substack{V\subset Y:\text{open subset}\\ \text{such that } U\subset f^{-1}V}} \mathcal{G}(V), \quad U \subset X : \text{open subset},$$

$$U \subset f^{-1}V \xleftrightarrow{\text{corresponds to}} \text{a morphism } K(V) \to U \text{ in } \mathsf{Open}_X^{\mathrm{op}}.$$

Unsurprisingly, this is precisely the **inverse image** of $\mathcal{G}$ in sheaf theory.

1.9 Gabriel–Zisman Localization

Gabriel–Zisman localization, henceforth simply called localization, is the most economical way to adjoin inverses to a family of morphisms in a category. The theory originates in [14] and is characterized by a universal property.

Definition 1.9.1 (P. Gabriel, M. Zisman) Let $\mathcal{C}$ be a category and let S be a subset of $\mathrm{Mor}(\mathcal{C})$ that contains all identity morphisms and is closed under composition. A **localization** of $\mathcal{C}$ at S means a category $\mathcal{C}[S^{-1}]$ (which may be a "large" category), together with a functor $Q : \mathcal{C} \to \mathcal{C}[S^{-1}]$, called the localization functor, satisfying the following conditions.

- ◇ For every $s \in S$, its image $Q(s)$ is an isomorphism in $\mathcal{C}[S^{-1}]$.
- ◇ For every category $\mathcal{D}$ (again possibly "large") and functor $F : \mathcal{C} \to \mathcal{D}$, if F sends every morphism in S to an isomorphism, then there is a unique functor $F[S^{-1}] : \mathcal{C}[S^{-1}] \to \mathcal{D}$ such that $F = F[S^{-1}]Q$.

By convention, we often omit Q from the data. Some sources, such as [23, Definition 7.1.1], give a somewhat weaker universal property for localization[14].

We next show that localization is unique up to a unique isomorphism.

Proposition 1.9.2 If $(\mathcal{C}[S^{-1}], Q)$ and $(\mathcal{C}[S^{-1}]', Q')$ are both localizations of $\mathcal{C}$ at S, then there is a unique pair of functors $\mathcal{C}[S^{-1}] \underset{G'}{\overset{G}{\rightleftarrows}} \mathcal{C}[S^{-1}]'$ such that $GQ = Q'$, $G'Q' = Q$, and $G'G = \mathrm{id}_{\mathcal{C}[S^{-1}]}$, $GG' = \mathrm{id}_{\mathcal{C}[S^{-1}]'}$.

Proof Since Q' sends S to isomorphisms, the universal property gives a unique G such that $GQ = Q'$. Similarly, there is a unique G' such that $G'Q' = Q$. Since $G'GQ = Q$,

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14) From the perspective of 2-categories, the more natural universal property is this: for every $\mathcal{D}$, the functor $Q^* : \mathcal{D}^{\mathcal{C}[S^{-1}]} \to \mathcal{D}^{\mathcal{C}}$ is fully faithful, and $G \in \mathrm{Ob}(\mathcal{D}^{\mathcal{C}})$ is isomorphic to some $Q^*(F) = FQ$ if and only if G sends the elements of S to isomorphisms. The functor Q in Definition 1.9.1 automatically has this property—the fully faithful part is Proposition 1.9.4, and the rest is easy.

taking $\mathcal{D} = \mathcal{C}[S^{-1}]$ and $F = Q$ in the universal property gives $G'G = \mathrm{id}_{\mathcal{C}[S^{-1}]}$. Likewise, $GG' = \mathrm{id}_{\mathcal{C}[S^{-1}]'}$[15]. □

Proposition 1.9.3 Suppose that the localization $(\mathcal{C}[S^{-1}], Q)$ exists. Let S^{op} denote the corresponding image of S in $\mathrm{Mor}(\mathcal{C}^{\mathrm{op}})$. Then there is an equivalence of categories, compatible with Q, $\mathcal{C}[S^{-1}]^{\mathrm{op}} \leftrightarrows \mathcal{C}^{\mathrm{op}}[(S^{\mathrm{op}})^{-1}]$.

Proof By the uniqueness of localization, it suffices to apply $(-)^{\mathrm{op}}$ to all the categories and functors in Definition 1.9.1, since this operation does not reverse the direction of a functor. □

The problem is thus reduced to the existence and construction of localization. For general $\mathcal{C}$ and S, one can construct $\mathcal{C}[S^{-1}]$ by formally adjoining inverses to $\mathcal{C}$, much as in the construction of a free group. More concretely, $\mathrm{Ob}(\mathcal{C}[S^{-1}]) = \mathrm{Ob}(\mathcal{C})$, while morphisms in $\mathcal{C}[S^{-1}]$ are represented formally by finite zigzags

$$\cdots bt^{-1}as^{-1}\cdots = \left(\begin{tikzcd}[column sep=small] \cdots \arrow[dr] & & Y \arrow[dl, "s"'] \arrow[dr, "a"] & & W \arrow[dl, "t"'] \arrow[dr, "b"] & & \cdots \arrow[dl] \\ & X & & Z & & U & \end{tikzcd}\right)$$

with $a, b, \ldots \in \mathrm{Mor}(\mathcal{C})$ and $s, t, \ldots \in S$. Composition and Q are defined in the evident way, but one must quotient by the equivalence relation generated by

$$s^{-1}t^{-1} = (ts)^{-1}, \quad ss^{-1} = \mathrm{id}, \quad s^{-1}s = \mathrm{id}$$

and so on; see the sketch in [14, I.1.1]. Localization may therefore be viewed as a kind of calculus of fractions for morphisms.

This construction is completely general, but its drawbacks are equally clear, chiefly because the equivalence relation is hard to work with. For example, it is difficult to characterize those $f, g \in \mathrm{Mor}(\mathcal{C})$ for which $Qf = Qg$. Fortunately, in practice S is often a left (or right) multiplicative system, to be introduced in Definition 1.9.5. In that case the description of $\mathcal{C}[S^{-1}]$ can be simplified: the zigzags above can be **straightened**, so that every morphism can be written in the form as^{-1} (or $s^{-1}a$).

Before introducing multiplicative systems, we record a simple property of the construction above.

Proposition 1.9.4 The localization functor $Q : \mathcal{C} \to \mathcal{C}[S^{-1}]$ is essentially surjective. Moreover, for every category $\mathcal{D}$ and functors $A, B : \mathcal{C}[S^{-1}] \to \mathcal{D}$, the evident map

$$\mathrm{Hom}_{\mathcal{D}^{\mathcal{C}[S^{-1}]}}(A, B) \to \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(AQ, BQ)$$

is a bijection. In other words, $Q^* : \mathcal{D}^{\mathcal{C}[S^{-1}]} \to \mathcal{D}^{\mathcal{C}}$ is a fully faithful functor.

15) Translator's note: the source has $\mathrm{id}_{\mathcal{C}[S^{-1}]}$ for this last identity. Since GG' is an endofunctor of $\mathcal{C}[S^{-1}]'$, the prime has been added to the target localization here (O014-C017).

Proof We may realize the localization concretely as above. Write the map induced by Q on objects as $X \mapsto \underline{X}$. It is a bijection, hence Q is essentially surjective. It follows at once that the map $\mathrm{Hom}_{\mathcal{D}^{\mathcal{C}[S^{-1}]}}(A, B) \to \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(AQ, BQ)$ is injective.

For surjectivity, let $\varphi = (\varphi_X)_{X \in \mathrm{Ob}(\mathcal{C})} \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(AQ, BQ)$. For each $\underline{X} = QX$, define a morphism $\tilde{\varphi}_{\underline{X}} : A\underline{X} \to B\underline{X}$ in $\mathcal{D}$ to be φ_X. It remains to show that $\tilde{\varphi} := (\tilde{\varphi}_{\underline{X}})_{\underline{X}}$ lies in $\mathrm{Hom}_{\mathcal{D}^{\mathcal{C}[S^{-1}]}}(A, B)$. In other words, for every morphism $f : \underline{X} \to \underline{Y}$ in $\mathcal{C}[S^{-1}]$, we must verify that the following diagram commutes in $\mathcal{D}$:

$$\begin{tikzcd} A\underline{X} \arrow[r, "\tilde{\varphi}_{\underline{X}}"] \arrow[d, "Af"'] & B\underline{X} \arrow[d, "Bf"] \\ A\underline{Y} \arrow[r, "\tilde{\varphi}_{\underline{Y}}"'] & B\underline{Y}. \end{tikzcd}$$

By the zigzag construction of morphisms, f decomposes into a string of morphisms of the form Qa or $(Qs)^{-1}$, with $a \in \mathrm{Mor}(\mathcal{C})$ and $s \in S$. The commutativity of the diagram therefore reduces to the corresponding property of φ. $\square$

Definition 1.9.5 A subset $S \subset \mathrm{Mor}(\mathcal{C})$ satisfying the following properties is called a **left multiplicative system** in $\mathcal{C}$.

(S1) For every $X \in \mathrm{Ob}(\mathcal{C})$, one has $\mathrm{id}_X \in S$.

(S2) For all $f, g \in S$, if the composite gf is defined, then $gf \in S$.

(S3) Given morphisms $X \xrightarrow{s \in S} Z \xleftarrow{f} Y$, there are morphisms in $\mathcal{C}$, $X \xleftarrow{f'} W \xrightarrow{s' \in S} Y$, such that $sf' = fs'$, depicted by the commutative diagram

$$\begin{tikzcd} X \arrow[d, "s \in S"'] & W \arrow[l, dashed, "f'"'] \arrow[d, dashed, "s' \in S"] \\ Z & Y \arrow[l, "f"] \end{tikzcd} \quad \text{commutative.}$$

(S4) Let $f, g : X \to Y$ be morphisms in $\mathcal{C}$ and let $s : Y \to W$ belong to S. If $sf = sg$, then there is a morphism $t : Z \to X$ in S such that $ft = gt$. Diagrammatically,

$$\underbrace{Z \overset{t \in S}{\dashrightarrow} X \underset{g}{\overset{f}{\rightrightarrows}} Y}_{\text{commutative}} \xrightarrow{s \in S} W.$$

The dashed parts of these diagrams are the morphisms whose existence is asserted.

If the image S^{op} of S in $\mathrm{Mor}(\mathcal{C}^{\mathrm{op}})$ is a left multiplicative system, then S is called a **right multiplicative system** in $\mathcal{C}$. Equivalently, reverse all arrows in conditions (S3) and (S4). A subset S that is both a left and a right multiplicative system is called a **multiplicative system** in $\mathcal{C}$.

Convention 1.9.6 For a given left or right multiplicative system S, from now on arrows belonging to S will always be drawn as $\rightarrowtail$ in commutative diagrams, to distinguish them from the other arrows.

Fix $X \in \mathrm{Ob}(\mathcal{C})$. In accordance with this convention, define the category $S_{/X}$ (or $S_{X/}$) for a left (or right) multiplicative system S as follows.

$$\mathrm{Ob}\left(S_{/X}\right) := \{\text{morphisms } X \leftarrowtail Z\}, \quad \mathrm{Mor}(S_{/X}) := \left\{ \text{commutative diagrams} \quad \begin{tikzcd}[row sep=small, column sep=small] X & Z \arrow[l, tail] \arrow[d] \\ & W \arrow[ul, tail] \end{tikzcd} \right\},$$

$$\mathrm{Ob}\left(S_{X/}\right) := \{\text{morphisms } X \rightarrowtail Z\}, \quad \mathrm{Mor}(S_{X/}) := \left\{ \text{commutative diagrams} \quad \begin{tikzcd}[row sep=small, column sep=small] X \arrow[r, tail] \arrow[dr, tail] & Z \arrow[d] \\ & W \end{tikzcd} \right\}.$$

Note that if S is a left multiplicative system, then $(S^{\mathrm{op}})_{X/} = (S_{/X})^{\mathrm{op}}$.

Lemma 1.9.7 If S is a left (or right) multiplicative system, then $S^{\mathrm{op}}_{/X}$ (or $S_{X/}$) is a filtered category.

Proof By the duality observed above, we need only discuss the case of a right multiplicative system. For any objects $i : X \to Z$ and $j : X \to Z'$ of $S_{X/}$, use the corresponding version of (S3) to construct a commutative diagram

$$\begin{tikzcd} Z \arrow[r, "j'"] & W \\ X \arrow[u, tail, "i"] \arrow[r, tail, "j"'] & Z' \arrow[u, tail, "i'"'] \end{tikzcd}$$

Then $k := i'j \in S$ gives an object of $S_{X/}$, together with morphisms $i \xrightarrow{j'} k \xleftarrow{i'} j$ in $S_{X/}$.

Let i, j be as above. For any pair of morphisms in $S_{X/}$, $\begin{tikzcd}[column sep=small] Z \arrow[rr, shift left, "f"] \arrow[rr, shift right, "g"'] & & Z' \\ & X \arrow[ul, tail, "i"] \arrow[ur, tail, "j"'] & \end{tikzcd}$

(a commutative diagram), the corresponding version of (S4) gives $h : Z' \rightarrowtail W$ such that $hf = hg$. Put $k := hj : X \rightarrowtail W$ and regard this as an object of $S_{X/}$. We then obtain the commutative diagram in $S_{X/}$ $\; i \overset{f}{\underset{g}{\rightrightarrows}} j \xrightarrow{h} k\;$. These are exactly the defining conditions for a filtered category. $\square$

Let $X, Y \in \mathrm{Ob}(\mathcal{C})$. We now define sets $M^l_{X,Y}$ and $M^r_{X,Y}$ for left and right multiplicative systems, respectively. Note that they need not be small sets.

◇ Let S be a left multiplicative system. Abbreviate a diagram in $\mathcal{C}$ of the form

$$X \xleftarrowtail{\;s\;} Z \xrightarrow{\;a\;} Y$$

by $(Z; s, a)$. These data form the set $M^l_{X,Y}$.

◇ Let S be a right multiplicative system. Abbreviate a diagram in $\mathcal{C}$ of the form

$$X \xrightarrow{a} Z \xleftarrow{s} Y$$

by $(Z; a, s)$. These data form the set $M^r_{X,Y}$.

Define a binary relation $\sim$ as follows: $(Z; s, a) \sim (Z'; s', a')$ (or $(Z; a, s) \sim (Z'; a', s')$) if and only if there is a commutative diagram of the form

$$\begin{tikzcd} & Z \arrow[dl, tail, "s"'] \arrow[dr, "a"] & \\ X & W \arrow[l, tail] \arrow[u] \arrow[d] \arrow[r] & Y \\ & Z' \arrow[ul, tail, "s'"] \arrow[ur, "a'"'] & \end{tikzcd} \qquad \begin{tikzcd} & Z \arrow[d] & \\ X \arrow[ur, "a"] \arrow[r] \arrow[dr, "a'"'] & W & Y \arrow[l, tail] \arrow[ul, tail, "s"'] \arrow[dl, tail, "s'"] \\ & Z' \arrow[u] & \end{tikzcd} \tag{1.9.1}$$

(left multiplicative system) (right multiplicative system)

For every $f : X \to Y$, condition (S1) implies $(X; \mathrm{id}_X, f) \in M^l_{X,Y}$ and $(X; f, \mathrm{id}_Y) \in M^r_{X,Y}$. The two definitions are plainly dual.

Lemma 1.9.8 For a left (or right) multiplicative system S and arbitrary X, Y, the relation $\sim$ defined above is an equivalence relation on $M^l_{X,Y}$ (or $M^r_{X,Y}$). In fact, there are bijections

$$\begin{aligned} M^l_{X,Y}/\sim \ &\xrightarrow{1:1} \varinjlim_{\substack{[X \leftarrowtail Z] \\ \in \mathrm{Ob}(S^{\mathrm{op}}_{/X})}} \mathrm{Hom}(Z, Y), & M^r_{X,Y}/\sim \ &\xrightarrow{1:1} \varinjlim_{\substack{[Y \rightarrowtail Z] \\ \in \mathrm{Ob}(S_{Y/})}} \mathrm{Hom}(X, Z) \\ [Z; s, a] &\longmapsto \left[Z \xrightarrow{a} Y\right] & [Z; a, s] &\longmapsto \left[X \xrightarrow{a} Z\right] \end{aligned}$$

where $[Z; s, a]$ (or $[Z; a, s]$) denotes the equivalence class containing $(Z; s, a)$ (or $(Z; a, s)$).

Proof Consider the functor $\alpha : S^{\mathrm{op}}_{/X} \to \mathsf{Set}$ sending an object $X \leftarrowtail Z$ to $\mathrm{Hom}(Z, Y)$. By definition,

$$M^l_{X,Y} = \bigsqcup_{X \leftarrowtail Z} \alpha(X \leftarrowtail Z),$$

and since $S^{\mathrm{op}}_{/X}$ is filtered (Lemma 1.9.7), it is easy to check that the binary relation $\sim$ on $M^l_{X,Y}$ arising from (1.9.1) translates into the equivalence relation in Proposition 1.6.11. This also proves the bijection with $\varinjlim$. The argument for the $M^r_{X,Y}$ version is identical.

When discussing filtered colimits of sets, §1.6 assumes that the indexing category is small, so that the resulting colimit still lies in **Set**. Here $S_{/X}$ and $S_{Y/}$ need not be small categories. The assertion itself, however, does not depend on set size, and the argument can be formulated so as not to depend on a choice of Grothendieck universe.□

The equivalence class $[Z; s, a]$ (or $[Z; a, s]$) should be understood as the morphism as^{-1} (or $s^{-1}a$) in the category $\mathcal{C}[S^{-1}]$ to be constructed.

Definition–Proposition 1.9.9 Let S be a left multiplicative system, let X, Y be arbitrary, and take $(U; s, a) \in M^l_{X,Y}$ and $(V; t, b) \in M^l_{Y,Z}$. Use (S3) to extend them to the following commutative diagram:

$$\begin{tikzcd}
& & W \arrow[dl, tail, "r"'] \arrow[dr, "c"] & & \\
& U \arrow[dl, tail, "s"'] \arrow[dr, "a"'] & & V \arrow[dl, tail, "t"] \arrow[dr, "b"] & \\
X & & Y & & Z
\end{tikzcd}$$

Then $[V; t, b] \circ [U; s, a] := [W; sr, bc] \in M^l_{X,Z}/\sim$ depends only on $[U; s, a]$ and $[V; t, b]$. The dual statement holds for right multiplicative systems.

Proof First fix $(V; t, b)$ and vary $(U; s, a)$ within its equivalence class. In view of (1.9.1), it suffices, for a commutative diagram

$$\begin{tikzcd}
& U \arrow[dl, tail, "s"'] \arrow[dr, "a"] & W \arrow[l, tail, "r"'] \arrow[dr, "c"] & & \\
X & & Y & V \arrow[l, tail, "t"'] \arrow[r, "b"] & Z. \\
& U' \arrow[ul, tail, "s'"] \arrow[uu, "x"] \arrow[ur, "a'"] & W' \arrow[l, tail, "r'"] \arrow[ur, "c'"'] & &
\end{tikzcd}$$

to prove $(W; sr, bc) \sim (W'; s'r', bc')$. Apply (S3) to obtain an object Q and morphisms q, k such that

$$\begin{tikzcd}
Q \arrow[r, "q"] \arrow[d, tail, "k"'] & W \arrow[d, tail, "r"] \\
W' \arrow[r, "xr'"'] & U
\end{tikzcd} \quad \text{commutes.}$$

Combining this with the preceding diagram gives $tc'k = a'r'k = axr'k = arq =$

tcq. By (S4), there is a $w : P \rightarrowtail Q$ such that $c'kw = cqw$. It is then easy to verify that

$$\begin{tikzcd}
 & W \arrow[dl, tail reversed, "sr"'] \arrow[dr, "bc"] & \\
X & P \arrow[l, "s'r'kw"] \arrow[r, "bc'kw"'] \arrow[u, "qw"] \arrow[d, "kw"'] & Z \\
 & W' \arrow[ul, "s'r'"] \arrow[ur, "bc'"'] &
\end{tikzcd}$$

commutes, that is, $(W; sr, bc) \sim (W'; s'r', bc')$.

The case in which $(U; s, a)$ is fixed and $(V; t, b)$ varies within its equivalence class is similar. $\square$

Definition–Theorem 1.9.10 Let S be a left multiplicative system in a category $\mathcal{C}$.

(i) There is a category $\mathcal{C}[S^{-1}]^l$ (possibly a large category) whose object set is $\mathrm{Ob}(\mathcal{C})$ and whose Hom set between two objects X, Y is $M^l_{X,Y}/\sim$. The identity morphism of X is $[X; \mathrm{id}_X, \mathrm{id}_X]$, and composition is given by the binary operation in Definition–Proposition 1.9.9.

(ii) There is a functor $Q^l : \mathcal{C} \to \mathcal{C}[S^{-1}]^l$ that is the identity on objects and sends a morphism $f : X \to Y$ to $[X; \mathrm{id}_X, f]$.

The dual statement holds for a right multiplicative system S; the corresponding data are denoted $Q^r : \mathcal{C} \to \mathcal{C}[S^{-1}]^r$.

Proof By duality, we need only consider a left multiplicative system.

For (i), it suffices to verify the properties of composition in $\mathcal{C}[S^{-1}]^l$. The operation in Definition–Proposition 1.9.9 plainly makes $[X; \mathrm{id}_X, \mathrm{id}_X]$ an identity morphism, so only associativity remains. Take $(A; s, a) \in M^l_{X,Y}$, $(B; t, b) \in M^l_{Y,Z}$, and $(C; u, c) \in M^l_{Z,W}$. Applying (S3) three times gives the commutative diagram

$$\begin{tikzcd}[column sep=small]
 & & & \bullet \arrow[dl] \arrow[dr] & & & \\
 & & \bullet \arrow[dl] \arrow[dr] & & \bullet \arrow[dl] \arrow[dr] & & \\
 & A \arrow[dl, "s"'] \arrow[dr, "a"] & & B \arrow[dl, "t"'] \arrow[dr, "b"] & & C \arrow[dl, "u"'] \arrow[dr, "c"] & \\
X & & Y & & Z & & W
\end{tikzcd}$$

The whole diagram gives an element of $M^l_{X,W}$; its lower-left part gives a representative of $[B; t, b] \circ [A; s, a]$, and its lower-right part gives a representative of $[C; u, c] \circ [B; t, b]$. This proves associativity:

$$[C; u, c] \circ ([B; t, b] \circ [A; s, a]) = ([C; u, c] \circ [B; t, b]) \circ [A; s, a].$$

For (ii), it remains to check that Q^l respects identities and composition. It preserves identities because $[X; \mathrm{id}_X, \mathrm{id}_X]$ has the required identity property. Now consider morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$ in $\mathcal{C}$. The diagram

$$\begin{tikzcd}[column sep=small]
 & & X \arrow[dr, "f"] & & \\
 & X \arrow[ur, equal] \arrow[dr, "f"] & & Y \arrow[dr, "g"] & \\
X \arrow[ur, equal] & & Y \arrow[ur, equal] & & Z
\end{tikzcd}$$

shows that $Q^l(gf) = Q^l(g)Q^l(f)$. □

Theorem 1.9.11 (P. Gabriel, M. Zisman) Let S be a left (or right) multiplicative system in $\mathcal{C}$. Then $\left(\mathcal{C}[S^{-1}]^l, Q^l\right)$ (or $\left(\mathcal{C}[S^{-1}]^r, Q^r\right)$) is a localization of $\mathcal{C}$ at S.

Proof By duality, we only verify the conditions of Definition 1.9.1 for a left multiplicative system. Write $Q = Q^l$. If $f : X \to Y$ belongs to S, the diagram

$$\begin{tikzcd}
 & X \arrow[dl, tail, "f"'] \arrow[d, dashed, "f"] \arrow[dr, "f"] & \\
Y \arrow[r, equal] & Y \arrow[r, equal] & Y
\end{tikzcd}$$

shows that $[X; f, f] = [Y; \mathrm{id}_Y, \mathrm{id}_Y]$. Hence the following diagrams show that $[X; f, \mathrm{id}_X] = [X; \mathrm{id}_X, f]^{-1}$:

$$\begin{tikzcd}[column sep=small]
 & & X \arrow[dr, equal] & & \\
 & X \arrow[ur, equal] \arrow[dr, "f"] & & X \arrow[dl, tail, "f"'] \arrow[dr, equal] & \\
X \arrow[ur, equal] & & Y & & X
\end{tikzcd}
\qquad
\begin{tikzcd}[column sep=small]
 & & X \arrow[dr, equal] & & \\
 & X \arrow[ur, equal] \arrow[dl, tail, "f"'] \arrow[dr, equal] & & X \arrow[dr, tail, "f"] & \\
Y & & X \arrow[ur, equal] & & Y
\end{tikzcd}$$

Thus Q does send the elements of S to isomorphisms.

Now let $F : \mathcal{C} \to \mathcal{D}$ be a functor sending S to isomorphisms. Define $F[S^{-1}] : \mathcal{C}[S^{-1}]^l \to \mathcal{D}$ as follows: on objects it sends X to FX, and on morphisms it sends $[U; s, a] \in \mathrm{Hom}_{\mathcal{C}[S^{-1}]^l}(X, Y)$ to $(Fa)(Fs)^{-1} \in \mathrm{Hom}_{\mathcal{D}}(FX, FY)$[16]. Clearly, $F[S^{-1}]$ sends identity morphisms in $\mathcal{C}[S^{-1}]^l$ to identity morphisms in $\mathcal{D}$. Apply F to the diagram in Definition–Proposition 1.9.9 and invert the arrows in $F(S)$; it follows at once that $F[S^{-1}]$ preserves composition. The equality $F[S^{-1}]Q = F$ is also clear. Since every morphism in $\mathcal{C}[S^{-1}]^l$ has the form $(Qa)(Qs)^{-1}$, no other definition of $F[S^{-1}]$ is possible. □

Convention 1.9.12 If S is a multiplicative system, Proposition 1.9.2 shows that the two localizations can be identified; we write the resulting functor as $Q : \mathcal{C} \to \mathcal{C}[S^{-1}]$.

[16] Translator's note: the source has $\mathrm{Hom}_{\mathcal{D}}(X, Y)$. Since $(Fa)(Fs)^{-1} : FX \to FY$, the source and target objects have been corrected to FX and FY here (O014-C018).

Corollary 1.9.13 Let S be a left (or right) multiplicative system in $\mathcal{C}$. Morphisms $f, g : X \to Y$ in $\mathcal{C}$ satisfy $Q^l f = Q^l g$ (or $Q^r f = Q^r g$) if and only if there exists $s \in S$ such that $fs = gs$ (or $sf = sg$).

Proof It suffices to check the "only if" direction for a left multiplicative system S. If $Q^l f = Q^l g$, the construction of $\mathcal{C}[S^{-1}]^l$ gives a commutative diagram

$$\begin{tikzcd}
 & X \arrow[dr, "f"] & \\
X \arrow[ur, equal] \arrow[dr, equal] & U \arrow[l, tail, "s"'] \arrow[u] \arrow[d] \arrow[r, "a"] & Y \\
 & X \arrow[ur, "g"'] &
\end{tikzcd}$$

from which $fs = a = gs$. $\square$

Corollary 1.9.14 Let S be a left (or right) multiplicative system in $\mathcal{C}$. Then Q^l (or Q^r) sends monomorphisms to monomorphisms (or epimorphisms to epimorphisms).

Proof We prove only the case of a left multiplicative system. Let $f : X \to Y$ be a monomorphism in $\mathcal{C}$. Take $\alpha, \beta : Q^l W \rightrightarrows Q^l X$ such that $(Q^l f)\alpha = (Q^l f)\beta$. Conditions (S2) and (S3) allow us to represent α and β by elements $(U; s, a)$ and $(U; s, b)$, respectively, of $M^l_{W,X}$; the details are left to the reader as an exercise. The given equality then implies $Q^l(fa) = Q^l(fb)$. By Corollary 1.9.13, there is a $t \in S$ such that $fat = fbt$. Hence $at = bt$, and therefore $\alpha = \beta$. $\square$

Remark 1.9.15 In the basic operations of category theory, we want to avoid large categories whenever possible. Definition–Theorem 1.9.10 says that $\mathcal{C}[S^{-1}]^l$ (or $\mathcal{C}[S^{-1}]^r$) may be a large category. The reason is that the $\varinjlim$ in Lemma 1.9.8 need not be small, so its value need not lie in **Set**. This is a somewhat delicate issue.

If S satisfies the following condition, however, then $\mathcal{C}[S^{-1}]^l$ (or $\mathcal{C}[S^{-1}]^r$) is not large: assume that $S^{\mathrm{op}}_{/X}$ (or $S_{X/}$) has a small cofinal subcategory (Definition 1.6.4) for every $X \in \mathrm{Ob}(\mathcal{C})$. By Proposition 1.6.5, the colimit in question may be restricted to that subcategory, yielding an object of **Set**. If $\mathcal{C}$ itself is small, this condition holds automatically.

Lemma 1.9.16 Let S be a left (or right) multiplicative system in $\mathcal{C}$. Then the localization functor Q^l (or Q^r) preserves finite $\varprojlim$ (or finite $\varinjlim$).

Proof We discuss only the case of a left multiplicative system. Let J be a finite category, and suppose that the limit associated with a functor $\beta : J^{\mathrm{op}} \to \mathcal{C}$ exists. By

Lemma 1.9.8, for every $X \in \mathrm{Ob}(\mathcal{C})$ there are canonical bijections

$$\begin{aligned}\mathrm{Hom}_{\mathcal{C}[S^{-1}]^l}\left(Q^l X, Q^l \varprojlim \beta\right) &\simeq \varinjlim_{X \leftarrow Z} \mathrm{Hom}_{\mathcal{C}}\left(Z, \varprojlim \beta\right) \\ &\simeq \varinjlim_{X \leftarrow Z} \varprojlim_{j \in \mathrm{Ob}(J)} \mathrm{Hom}_{\mathcal{C}}\left(Z, \beta(j)\right).\end{aligned}$$

Applying Lemma 1.9.7 and Proposition 1.6.12 to the last term transforms it further into

$$\varprojlim_{j \in \mathrm{Ob}(J)} \varinjlim_{X \leftarrow Z} \mathrm{Hom}_{\mathcal{C}}\left(Z, \beta(j)\right) \simeq \varprojlim_{j \in \mathrm{Ob}(J)} \mathrm{Hom}_{\mathcal{C}[S^{-1}]^l}\left(Q^l X, Q^l \beta(j)\right).$$

Strictly speaking, §1.6 works in **Set**, whereas the Hom sets here need not lie in **Set**. This is only a minor technical issue; see the related discussion in the proof of Lemma 1.9.8. □

Theorem 1.9.17 Let $\mathcal{C}$ be an **Ab**-category and let S be a left (or right) multiplicative system in it.

(i) The category $\mathcal{C}[S^{-1}]^l$ (or $\mathcal{C}[S^{-1}]^r$) has a canonical **Ab**-category structure making Q^l (or Q^r) an additive, essentially surjective functor.

(ii) If $\mathcal{C}$ is an additive category and S is a multiplicative system, then $\mathcal{C}[S^{-1}]$ is also an additive category.

(iii) The preceding statements remain true if **Ab**-categories are replaced by $\Bbbk$-linear categories, where $\Bbbk$ is a commutative ring. If F in the universal property of Definition 1.9.1 is $\Bbbk$-linear, then so is the associated functor $F[S^{-1}]$.

Proof For (i), it suffices to treat a left multiplicative system. We must define addition on the morphism sets of $\mathcal{C}[S^{-1}]$. Let

$$f, g \in \mathrm{Hom}_{\mathcal{C}[S^{-1}]^l}(X, Y).$$

Since $S^{\mathrm{op}}_{/X}$ is filtered (Lemma 1.9.7), there are a morphism $s : U \rightarrowtail X$ and morphisms $a_1, a_2 : U \to Y$ in $\mathcal{C}$ such that $f = [U; s, a_1]$ and $g = [U; s, a_2]$. Define

$$f + g := [U; s, a_1 + a_2] \in \mathrm{Hom}_{\mathcal{C}[S^{-1}]}(X, Y).$$

In particular, a zero morphism can be written $[U; s, 0]$, and $-[U; s, a] = [U; s, -a]$. One must still prove that $f + g$ is independent of the choice of representatives; this too follows from the filteredness of $S^{\mathrm{op}}_{/X}$, and the routine details are omitted. Once this operation is known to be well defined, the axioms for an **Ab**-category and the additivity of Q are easy to verify.

Statement (ii) follows from (i), the definition of an additive category (Definition 1.3.3), and Lemma 1.9.16.

For (iii), if $\mathcal{C}$ is $\Bbbk$-linear, define scalar multiplication on $\mathrm{Hom}_{\mathcal{C}[S^{-1}]}$ by $t \cdot [U; s, a] = [U; s, ta]$ for $t \in \Bbbk$. The required properties are immediate. □

Example 1.9.18 (Localization of rings) Giving a ring R is equivalent to giving an Ab-category $\mathcal{C}$ with a single object $\star$ such that $R = \mathrm{End}_{\mathcal{C}}(\star)$. If R is commutative, the conditions that $S \subset R = \mathrm{Mor}(\mathcal{C})$ be a left or a right multiplicative system coincide. In this case, the ring corresponding to the Ab-category $\mathcal{C}[S^{-1}]$ is precisely the localization $R[S^{-1}]$ of R; see [51, §5.3]. For a general ring R, the theory in this section provides a construction of noncommutative localizations. The related theory is also a branch of noncommutative ring theory; see [28, §10A] for a detailed account.

Example 1.9.19 (Central localization) Write the center of an additive category $\mathcal{C}$ as $Z(\mathcal{C})$; this is a commutative ring. See the discussion preceding Proposition 1.4.5. For any multiplicative subset S of $Z(\mathcal{C})$, it is easy to check that the set of morphisms

$$\{s_X \in \mathrm{End}_{\mathcal{C}}(X) : s \in S,\ X \in \mathrm{Ob}(\mathcal{C})\}$$

is a multiplicative system. The corresponding category $\mathcal{C}[S^{-1}]$ can of course be constructed using the calculus of fractions above, but a shorter construction is

$$\mathrm{Ob}(\mathcal{C}[S^{-1}]) := \mathrm{Ob}(\mathcal{C}), \quad \mathrm{Hom}_{\mathcal{C}[S^{-1}]}(X, Y) := \mathrm{Hom}_{\mathcal{C}}(X, Y) \underset{Z(\mathcal{C})}{\otimes} Z(\mathcal{C})[S^{-1}],$$

with composition defined in the usual way. The relevant checks are left as an exercise for this chapter.

Remark 1.9.20 For general $\mathcal{C}$ and S, the case in which the localization functor $Q : \mathcal{C} \to \mathcal{C}[S^{-1}]$ has a fully faithful right adjoint is especially interesting and important. A localization with this property is called a **reflective localization**. The Gabriel–Popescu theorem to be introduced in Appendix § A.3 is a typical example.

Remark 1.9.21 (Localization of product categories) Finally, let $\mathcal{C}_i$ be categories equipped with left (or right) multiplicative systems S_i, for $i = 1, 2$, and write their localization functors as $Q_i : \mathcal{C}_i \to \mathcal{C}_i[S_i^{-1}]$. Clearly, $S_1 \times S_2$ is also a left (or right) multiplicative system in $\mathcal{C}_1 \times \mathcal{C}_2$. From the construction in Definition–Theorem 1.9.10, one readily checks that

$$(Q_1, Q_2) : \mathcal{C}_1 \times \mathcal{C}_2 \to \mathcal{C}_1[S_1^{-1}] \times \mathcal{C}_2[S_2^{-1}]$$

is the corresponding localization. This statement generalizes to products of arbitrary families of categories.

1.10 Kan Extensions Along Localization

Suppose that the localization of $\mathcal{C}$ at $S \subset \mathrm{Mor}(\mathcal{C})$, namely $Q : \mathcal{C} \to \mathcal{C}[S^{-1}]$, exists. This section investigates the problem of extending functors: given a category $\mathcal{E}$ and a functor $F : \mathcal{C} \to \mathcal{E}$, can one obtain the following commutative diagram?

$$\begin{tikzcd} \mathcal{C} \arrow[d, "Q"'] \arrow[dr, bend left, "F"] & \\ \mathcal{C}[S^{-1}] \arrow[r, dashed, "\exists?"'] & \mathcal{E} \end{tikzcd}$$

Since F need not send the morphisms in S to isomorphisms, the functor indicated by the dashed arrow need not exist. Nevertheless, we may still seek the best possible approximations to such an extension: the left Kan extension $(\mathrm{Lan}_Q F, \eta)$ and the right Kan extension $(\mathrm{Ran}_Q F, \varepsilon)$. We will omit η and ε from the notation from now on. The aim of this section is to give sufficient conditions, for fixed $\mathcal{C}$, S, and F, under which $\mathrm{Lan}_Q F$ and $\mathrm{Ran}_Q F$ exist; once they do, Proposition 1.7.2 guarantees their uniqueness. These results will be used in the study of derived functors; see § 4.6.

The idea is to choose a suitable subcategory of $\mathcal{C}$. We retain the convention of §1.9: morphisms belonging to S are marked by $\rightarrowtail$.

The arguments in this section make no assumptions about set size.

Proposition 1.10.1 Let $\mathcal{I}$ be a full subcategory of $\mathcal{C}$, let $S \subset \mathrm{Mor}(\mathcal{C})$ be a left (or right) multiplicative system, and set $T := S \cap \mathrm{Mor}(\mathcal{I})$.

(i) If T is assumed to be a left (or right) multiplicative system in $\mathcal{I}$, these data induce a functor $\mathcal{I}[T^{-1}]^l \to \mathcal{C}[S^{-1}]^l$ (or $\mathcal{I}[T^{-1}]^r \to \mathcal{C}[S^{-1}]^r$[17]).

(ii) Assume the following condition: for every morphism $s : W \to Y$ (or $s : Y \to W$) in S with $Y \in \mathrm{Ob}(\mathcal{I})$, there is a morphism $g : V \to W$ such that $sg \in T$ (or $g : W \to V$ such that $gs \in T$). Then T is a left (or right) multiplicative system, and the functor in (i) is fully faithful.

Proof By duality, it suffices to consider a left multiplicative system. The functor in (i) comes from the universal property of localization; on morphisms it sends every equivalence class $[U; s, a]$ relative to T to the same class $[U; s, a]$ relative to S.

For (ii), first verify that T is a left multiplicative system. Conditions (S1) and (S2) of Definition 1.9.5 are immediate. For (S3), given $X \overset{t}{\rightarrowtail} Z \overset{f}{\leftarrow} Y$, construct in $\mathcal{C}$

[17] Translator's note: in the right-hand case, the source gives the target as $\mathcal{C}[T^{-1}]^r$. It has been corrected to $\mathcal{C}[S^{-1}]^r$, since $T \subset \mathrm{Mor}(\mathcal{I})$ whereas $S \subset \mathrm{Mor}(\mathcal{C})$ (O014-C019).

the diagram

$$\begin{tikzcd} X & W \arrow[l, "f'"'] \arrow[d, tail, "s' \in S"] \\ Z \arrow[from=u, tail, "t \in T"'] & Y; \arrow[l, "f"] \end{tikzcd}$$

then choose $g : V \to W$ such that $s'g \in T$. The morphisms $s'g$ and $f'g$ meet the requirements of (S3). The argument for (S4) is the same.

Let $X, Y \in \mathrm{Ob}(\mathcal{I})$. For a morphism in $\mathcal{C}[S^{-1}]^l$ represented by $X \leftarrowtail U \to Y$, the condition allows us to choose $V \to U$ so that the composite $V \to U \to X$ belongs to T. Thus $\mathrm{Hom}_{\mathcal{I}[T^{-1}]^l}(X, Y) \to \mathrm{Hom}_{\mathcal{C}[S^{-1}]^l}(X, Y)$ is surjective. A similar argument shows that every equivalence $\sim$ in $M^l_{X,Y}$ can be realized inside $\mathcal{I}$, proving that the same map is injective. $\square$

The following lemma is stated only for a right multiplicative system S; the case of a left multiplicative system is dual.

Lemma 1.10.2 Let $S \subset \mathrm{Mor}(\mathcal{C})$ be a right multiplicative system. For a given $\underline{X} := QX \in \mathrm{Ob}\left(\mathcal{C}[S^{-1}]\right)$, define the category $(Q/\underline{X})$ as in Definition 1.6.2. We shall represent an object $(Y, QY \to \underline{X})$ of $(Q/\underline{X})$, nonuniquely, by a diagram $Y \xrightarrow{a} U \overset{s}{\leftarrowtail} X$ in $\mathcal{C}$.

Define $S_{X/}$ as in §1.9. The functor $S_{X/} \to (Q/\underline{X})$ that sends an object $[X \overset{t}{\rightarrowtail} V]$ to the diagram $V = V \overset{t}{\leftarrowtail} X$, which determines $(V, QV \to \underline{X})$, is cofinal (Definition 1.6.4).

Proof For $[X \overset{t}{\rightarrowtail} V]$ as in the statement, specifying a morphism from $(Y, QY \to \underline{X})$ to the corresponding $(V, QV \to \underline{X})$ amounts to specifying a morphism $b : Y \to V$ in $\mathcal{C}$ such that the diagram $Y \xrightarrow{b} V \overset{t}{\leftarrowtail} X$ represents the morphism $QY \to \underline{X}$ in $\mathcal{C}[S^{-1}]$.

Given $Y \xrightarrow{a} U \overset{s}{\leftarrowtail} X$, the following diagram maps $(Y, QY \to \underline{X})$ to an object coming from $S_{X/}$:

$$\begin{tikzcd} Y \arrow[r] \arrow[d] & U \arrow[d, equal] & X \arrow[l, tail] \arrow[d, equal] \\ U \arrow[r, equal] & U & X \arrow[l, tail] \end{tikzcd} \qquad \begin{tikzcd} (Y, QY \to \underline{X}) \arrow[d, "a"] \\ (U, QU \to \underline{X}) \end{tikzcd}$$

Next, consider objects $[X \overset{t_i}{\rightarrowtail} V_i]$ of $S_{X/}$ and morphisms from $(Y, QY \to \underline{X})$ to their images, determined by $b_i : Y \to V_i$ for $i = 1, 2$. Using the right-hand diagram in

(1.9.1), choose a commutative diagram

$$\begin{tikzcd}
 & V_1 \arrow[d, "u_1"] & \\
Y \arrow[ur, "b_1"] \arrow[r, "b"'] \arrow[dr, "b_2"'] & W & X \arrow[ul, tail, "t_1"'] \arrow[l, tail, "u"] \arrow[dl, tail, "t_2"] \\
 & V_2 \arrow[u, "u_2"'] &
\end{tikzcd}$$

Thus $u_i : [X \rightarrowtail V_i] \to [X \overset{u}{\rightarrowtail} W]$, while b also determines a morphism from $(Y, QY \to \underline{X})$ to the object $(W, QW \to \underline{X})$ corresponding to u. All the conditions needed for cofinality now follow readily. □

Proposition 1.10.3 Let $\mathcal{I}$ be a full subcategory of $\mathcal{C}$, let $S \subset \operatorname{Mor}(\mathcal{C})$ be a left (or right) multiplicative system, and put $T := S \cap \operatorname{Mor}(\mathcal{I})$. Assume the following "resolution condition": for every $X \in \operatorname{Ob}(\mathcal{C})$, there is a morphism $s : U \to X$ (or $s : X \to U$), where $U \in \operatorname{Ob}(\mathcal{I})$ and $s \in S$.

(i) Then $T \subset \operatorname{Mor}(\mathcal{I})$ is a left (or right) multiplicative system, and $\mathcal{I}[T^{-1}] \to \mathcal{C}[S^{-1}]$ is an equivalence.

(ii) Suppose further that a functor $F : \mathcal{C} \to \mathcal{E}$ sends the elements of T to isomorphisms. Then $\operatorname{Ran}_Q F$ (or $\operatorname{Lan}_Q F$) exists, and the following diagram commutes up to isomorphism:

$$\begin{tikzcd}
\mathcal{C}[S^{-1}] \arrow[r, "\operatorname{Ran}_Q F \text{ or } \operatorname{Lan}_Q F"] & \mathcal{E} \\
\mathcal{I}[T^{-1}] \arrow[u, "\text{equivalence}"] \arrow[ur] &
\end{tikzcd}$$

Here $\mathcal{I}[T^{-1}] \to \mathcal{E}$ is determined by the universal property of localization.

(iii) For $X \in \operatorname{Ob}(\mathcal{C})$, define the categories $S_{/X}$ and $S_{X/}$ as in §1.9. Under the assumptions of (ii), there are canonical isomorphisms[18)]

$$(\operatorname{Ran}_Q F)(QX) \simeq \varprojlim_{[X \leftarrowtail Y] \in \operatorname{Ob}(S_{/X}^{\mathrm{op}})} FY,$$

$$(\operatorname{Lan}_Q F)(QX) \simeq \varinjlim_{[X \rightarrowtail Y] \in \operatorname{Ob}(S_{X/})} FY.$$

The canonical morphism of the Kan extension $\varepsilon : (\operatorname{Ran}_Q F)Q \to F$ (or $\eta : F \to (\operatorname{Lan}_Q F)Q$) is, in the situation of (iii), determined by the term FX corresponding to $\operatorname{id}_X \in \operatorname{Ob}(S_{/X}^{\mathrm{op}})$ (or $\operatorname{id}_X \in \operatorname{Ob}(S_{X/})$).

18) The limit in the first formula is indexed by $\operatorname{Ob}(S_{/X}^{\mathrm{op}})$ because we use $\varprojlim_i \beta(i)$ to denote the limit of a functor $\beta : I^{\mathrm{op}} \to \mathcal{C}$.

Proof It suffices to discuss a right multiplicative system. The condition in Proposition 1.10.1(ii) plainly holds, so T is a right multiplicative system. Denote its localization by $Q' : \mathcal{I} \to \mathcal{I}[T^{-1}]$, and denote the fully faithful functor $\mathcal{I}[T^{-1}] \to \mathcal{C}[S^{-1}]$ of Proposition 1.10.1(ii) by ι_Q. The resolution condition also ensures that ι_Q is essentially surjective, hence an equivalence. This proves (i).

Now consider (ii). Write $\iota : \mathcal{I} \to \mathcal{C}$ for the inclusion functor. By the universal property, there is a functor $F' : \mathcal{I}[T^{-1}] \to \mathcal{E}$ making the following diagram commute:

$$\begin{tikzcd}
 & \mathcal{C} \arrow[d, "Q"] \arrow[dr, "F"] & \\
\mathcal{I} \arrow[ur, "\iota"] \arrow[dr, "Q'"'] & \mathcal{C}[S^{-1}] & \mathcal{E} \\
 & \mathcal{I}[T^{-1}] \arrow[u, "\iota_Q"] \arrow[ur, "F'"'] &
\end{tikzcd}$$

Choose any quasi-inverse ι_Q^{-1} to ι_Q. We will show that $F' \circ \iota_Q^{-1}$ gives the left Kan extension $\mathrm{Lan}_Q F$; equivalently, $\mathrm{Lan}_Q F$ exists and $F' \simeq (\mathrm{Lan}_Q F) \circ \iota_Q$.

To prove the existence of $\mathrm{Lan}_Q F$, define $\Pi_{/\underline{X}} : (Q/\underline{X}) \to \mathcal{C}$ as in Definition 1.6.2, where $\underline{X} := QX \in \mathrm{Ob}(\mathcal{C}[S^{-1}])$. In view of Theorem 1.8.1(i), it is enough to show that $\varinjlim F\Pi_{/\underline{X}}$ exists for every $\underline{X}$.

Apply the cofinal functor $S_{X/} \to (Q/\underline{X})$ of Lemma 1.10.2 and Proposition 1.6.5. This reduces $\varinjlim F\Pi_{/\underline{X}}$ to $\varinjlim_{[X \rightarrowtail U]} FU$, with the colimit taken over $S_{X/}$.

Since $S_{X/}$ is filtered (Lemma 1.9.7), the resolution condition together with Proposition 1.6.8 ensures that the objects $[X \rightarrowtail U]$ with $U \in \mathrm{Ob}(\mathcal{I})$ form a cofinal full subcategory of $S_{X/}$. Another application of Proposition 1.6.5 therefore lets us restrict the colimit to $U \in \mathrm{Ob}(\mathcal{I})$.

We seek to prove that $\varinjlim F\Pi_{/\underline{X}}$, or equivalently $\varinjlim_{[X \rightarrowtail U]} FU$, exists. The original problem depends only on the isomorphism class of $\underline{X}$, so (i) reduces it to the case $X \in \mathrm{Ob}(\mathcal{I})$. But the following diagram gives a morphism in $S_{X/}$:

$$\begin{tikzcd}
U & X \arrow[l, tail] \arrow[d, equal] \\
X \arrow[u, tail] \arrow[r, equal] & X
\end{tikzcd} \quad \text{where} \quad U \in \mathrm{Ob}(\mathcal{I})$$

All arrows marked $\rightarrowtail$ become isomorphisms under F. Thus in this case $\varinjlim FU$ not only exists but is isomorphic to FX. This also proves $F' \simeq (\mathrm{Lan}_Q F) \circ \iota_Q$. Statement (ii) follows.

Finally consider (iii). For $X \in \mathrm{Ob}(\mathcal{C})$, the preceding argument shows that $\varinjlim_{[X \rightarrowtail Y] \in \mathrm{Ob}(S_{X/})} FY$ exists and gives $(\mathrm{Lan}_Q F)(QX)$. Under this correspondence, the canonical morphism η_X of the left Kan extension is naturally determined by the term corresponding to $[X \overset{\mathrm{id}}{\rightarrowtail} X]$; see the construction in §1.8. □

Proposition 1.10.4 Under the hypotheses of Proposition 1.10.3, $\operatorname{Ran}_Q F$ (or $\operatorname{Lan}_Q F$) is the absolute right (or left) Kan extension of F along Q; see Definition 1.7.6.

Proof The first condition in Proposition 1.10.3 concerns only the left (or right) multiplicative system S in $\mathcal{C}$ and the subcategory $\mathcal{I}$, not F; the second only requires F to send $T := S \cap \operatorname{Mor}(\mathcal{I})$ to isomorphisms. In particular, for any functor $M : \mathcal{E} \to \mathcal{F}$, the condition on F is inherited at once by $MF : \mathcal{C} \to \mathcal{F}$.

We discuss only $\operatorname{Lan}_Q F$. With the notation from the proof of Proposition 1.10.3, $\operatorname{Lan}_Q F$ is concretely taken to be the composite

$$\mathcal{C}[S^{-1}] \xrightarrow{\iota_Q^{-1}} \mathcal{I}[T^{-1}] \xrightarrow{F'} \mathcal{E}$$

Here ι_Q and F' both come from the universal property of localization. Hence $M(\operatorname{Lan}_Q F) = MF'\iota_Q^{-1}$. The commutative diagram

$$\begin{tikzcd}
\mathcal{I} \arrow[r, hook] \arrow[dr] & \mathcal{C} \arrow[r, "F"] & \mathcal{E} \arrow[r, "M"] & \mathcal{F} \\
& \mathcal{I}[T^{-1}] \arrow[ur, "F'"] \arrow[urr, "MF'"'] & &
\end{tikzcd}$$

shows that MF' is also induced by the universal property of localization. Apply the construction of Proposition 1.10.3 to $\mathcal{C} \xrightarrow{MF} \mathcal{F}$; it yields $\operatorname{Lan}_Q(MF) = MF'\iota_Q^{-1}$. In other words, $\operatorname{Lan}_Q F$ is preserved by M. □

1.11 Adjoint Functor Theorems

Functors often occur in adjoint pairs. Consider categories $\mathcal{C}$ and $\mathcal{D}$ and a pair of functors

$$F : \mathcal{C} \rightleftarrows \mathcal{D} : G$$

such that (F, G) is an adjoint pair. Then F preserves $\varinjlim$ and G preserves $\varprojlim$, whenever the limits in question exist. Conversely, one naturally asks: if $F : \mathcal{C} \to \mathcal{D}$ preserves colimits, what ensures that it has a right adjoint? If $G : \mathcal{D} \to \mathcal{C}$ preserves limits, what ensures that it has a left adjoint? In many cases the adjoint can be written down explicitly, but sometimes an abstract existence result is needed. Such results are collectively called adjoint functor theorems. Their hypotheses fall roughly into two classes. First, to use preservation of colimits (or limits), one assumes that $\mathcal{C}$ (or $\mathcal{D}$) is cocomplete (or complete). Second, one needs conditions closely related to set size.

Our first task is to reformulate the existence of an adjoint appropriately. By duality, we state only the case of a left adjoint.

For a functor $G : \mathcal{D} \to \mathcal{C}$ and $c \in \mathrm{Ob}(\mathcal{C})$, Definition 1.6.2 gives a category (c/G) whose objects are data $(d, c \xrightarrow{f} Gd)$.

Lemma 1.11.1 A functor $G : \mathcal{D} \to \mathcal{C}$ has a left adjoint if and only if (c/G) has an initial object for every $c \in \mathrm{Ob}(\mathcal{C})$.

Proof Fix c. Specifying a morphism $(d_0, c \xrightarrow{\beta_0} Gd_0) \to (d, c \xrightarrow{\beta} Gd)$ in (c/G) amounts to specifying $\alpha : d_0 \to d$ such that

$$\begin{tikzcd} Gd_0 \arrow[rr, "G\alpha"] & & Gd \\ & c \arrow[ul, "\beta_0"] \arrow[ur, "\beta"'] & \end{tikzcd}$$

commutes. Thus (d_0, β_0) is initial in (c/G) if and only if

$$\begin{array}{ccc} \mathrm{Hom}_{\mathcal{D}}(d_0, d) & \longrightarrow & \mathrm{Hom}_{\mathcal{C}}(c, Gd) \\ \alpha & \longmapsto & \beta := (G\alpha)\beta_0 \end{array} \tag{1.11.1}$$

is a bijection for every $d \in \mathrm{Ob}(\mathcal{D})$. This is the basis of the argument.

Suppose that G has a left adjoint F, and let $c \in \mathrm{Ob}(\mathcal{C})$. We claim that Fc, together with the unit $\eta_c : c \to G(Fc)$, is initial in (c/G). For any $(d, c \xrightarrow{\beta} Gd)$, the map in (1.11.1), $\mathrm{Hom}_{\mathcal{D}}(Fc, d) \to \mathrm{Hom}_{\mathcal{C}}(c, Gd)$, sends α to $(G\alpha)\eta_c$. This is exactly the bijection on Hom sets supplied by the adjunction, so (Fc, η_c) is initial.

Conversely, suppose that (c/G) has an initial object for every c, and write it as $(Fc, \eta_c : c \to G(Fc))$. Since initial objects are unique up to unique isomorphism, these objects form a functor $F : \mathcal{C} \to \mathcal{D}$, and $(\eta_c)_{c \in \mathrm{Ob}(\mathcal{C})}$ gives $\eta : \mathrm{id}_{\mathcal{C}} \to GF$. For every $(c, d) \in \mathrm{Ob}(\mathcal{C}) \times \mathrm{Ob}(\mathcal{D})$, define the bijection from (1.11.1):

$$\begin{aligned} \varphi_{c,d} : \mathrm{Hom}_{\mathcal{D}}(Fc, d) &\to \mathrm{Hom}_{\mathcal{C}}(c, Gd) \\ \alpha &\mapsto (G\alpha)\eta_c. \end{aligned}$$

These bijections are plainly functorial in c and d. Hence (F, G, φ) is an adjoint pair with unit η. $\square$

Definition 1.11.2 Let $\mathcal{E}$ be a category. A nonempty small subset Γ of $\mathrm{Ob}(\mathcal{E})$ is called a **weakly initial family** if, for every $e' \in \mathrm{Ob}(\mathcal{E})$, there is an $e \in \Gamma$ such that $\mathrm{Hom}_{\mathcal{E}}(e, e')$ is nonempty.[19)] If $e \in \mathrm{Ob}(\mathcal{E})$ and $\{e\}$ is weakly initial, then e itself is called a **weakly initial object**.

[19)] Translator's note: the source has the subscript $\mathcal{C}$; it has been corrected to $\mathcal{E}$, the category introduced in this definition (O014-C020).

Every initial object is weakly initial. Weakly terminal families and weakly terminal objects are defined similarly.

Lemma 1.11.3 If Γ is a weakly initial family in $\mathcal{E}$ and $e := \prod_{x\in\Gamma} x$ exists, then e is weakly initial.

Proof For every $e' \in \mathrm{Ob}(\mathcal{E})$, there are $y \in \Gamma$ and a morphism $y \to e'$. Take the composite $e \xrightarrow{\text{projection}} y \to e'$. $\square$

Return to adjoint functor theorems. To apply Lemma 1.11.1, we must study the projection functor $\Pi_{c/} : (c/G) \to \mathcal{D}$ from Definition 1.6.2.

Proposition 1.11.4 Let $\mathcal{D}$ be complete, let $G : \mathcal{D} \to \mathcal{C}$ preserve all small limits, and fix $c \in \mathrm{Ob}(\mathcal{C})$.

(i) The functor $\Pi_{c/}$ is conservative (Definition 1.5.4) and creates all small limits (Definition 1.5.2).

(ii) The category (c/G) is complete.

(iii) The functor $\Pi_{c/}$ sends monomorphisms to monomorphisms.

Proof For (i), conservativity is clear. The key point is that, given $\beta : J \to (c/G)$ sending $j \in \mathrm{Ob}(J)$ to $(d_j, c \xrightarrow{f_j} Gd_j)$, one can lift $\varprojlim_j d_j$ and its projection morphisms to $(\varprojlim_j d_j, c \xrightarrow{f} G(\varprojlim_j d_j))$ and show that it is $\varprojlim \beta$. Explicitly, take f to be the composite in the first row of the commutative diagram

$$\begin{tikzcd} c \arrow[r, "\exists!"] \arrow[dr, "f_{j'}"'] & \varprojlim_j Gd_j \arrow[d] & G(\varprojlim_j d_j) \arrow[l, "\exists!"', "\sim"] \arrow[dl] \\ & Gd_{j'} & \end{tikzcd} \qquad j' \in \mathrm{Ob}(J).$$

The remaining checks are routine and are omitted; compare the examples in §1.5.

Statement (ii) follows from the completeness of $\mathcal{D}$ and (i).

For (iii), first observe that for every morphism $f : a \to b$ in any category, a direct manipulation of the definitions gives

$$f \text{ is monic} \iff \begin{tikzcd} a \arrow[r, "\mathrm{id}_a"] \arrow[d, "\mathrm{id}_a"'] & a \arrow[d, "f"] \\ a \arrow[r, "f"'] & b \end{tikzcd} \text{ is a pullback diagram.}$$

Now consider a monomorphism in (c/G) from $(d', c \to Gd')$ to $(d, c \to Gd)$, determined by $\iota : d' \to d$ in $\mathcal{D}$. By (i),

$$(d', c \to Gd') \underset{(d, c\to Gd)}{\times} (d', c \to Gd') = \left(d' \underset{d}{\times} d', \ c \to G\left(d' \underset{d}{\times} d' \right) \right).$$

The left-hand side identifies with $(d', c \to Gd')$, with both projections corresponding to $\mathrm{id}_{d'}$. Thus $d' \underset{d}{\times} d'$ identifies with d' in the same way. In other words, ι is monic. □

Lemma 1.11.5 If a complete category $\mathcal{E}$ has a weakly initial family Γ, then $\mathcal{E}$ has an initial object.

Proof Completeness and Lemma 1.11.3 ensure that $\mathcal{E}$ has a weakly initial object e. Let I be the full subcategory of $\mathcal{E}$ with $\mathrm{Ob}(I) = \{e\}$; it is small. Completeness gives a limit x of the inclusion $I \to \mathcal{E}$, with canonical morphism $i : x \to e$. Its universal property says

$$\begin{aligned} i_* : \mathrm{Hom}_{\mathcal{E}}(t, x) &\xrightarrow{1:1} \{\phi \in \mathrm{Hom}_{\mathcal{E}}(t, e) : \forall \theta, \theta' \in \mathrm{Hom}_{\mathcal{E}}(e, e),\ \theta\phi = \theta'\phi\} \\ \psi &\longmapsto i\psi \end{aligned} \tag{1.11.2}$$

for arbitrary $t \in \mathrm{Ob}(\mathcal{E})$. This immediately shows that i is monic.

We show that x is initial. Given $y \in \mathrm{Ob}(\mathcal{E})$, choose any morphism $e \to y$ and call the composite $x \xrightarrow{i} e \to y$ by f. For any $g : x \to y$, consider the equalizer $j : \ker(f, g) \hookrightarrow x$. There is a morphism $k : e \to \ker(f, g)$, hence $ijk : e \to e$. Taking $\psi = \mathrm{id}_x$ in (1.11.2) gives $(ijk)i = (\mathrm{id}_e)i = i$. Cancelling the monomorphism i on the left gives $jki = \mathrm{id}_x$, and consequently $f = fjki = gjki = g$. This proves the claim. □

The following result is due to P. Freyd and is also known as the General Adjoint Functor Theorem.

Theorem 1.11.6 (P. Freyd's Adjoint Functor Theorem) Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

(i) Let $\mathcal{D}$ be complete and let $G : \mathcal{D} \to \mathcal{C}$ preserve all small limits. Then G has a left adjoint if and only if the following **solution set condition** holds: for every $c \in \mathrm{Ob}(\mathcal{C})$, there is a family of morphisms $(f_i : c \to Gd_i)_{i \in I}$, indexed by a small set I, such that for every $f : c \to Gd$ there are $i \in I$ and $\alpha : d_i \to d$ for which f factors as $c \xrightarrow{f_i} Gd_i \xrightarrow{G\alpha} Gd$.

(ii) Let $\mathcal{C}$ be cocomplete and let $F : \mathcal{C} \to \mathcal{D}$ preserve all small colimits. Then F has a right adjoint if and only if the following **cosolution set condition** holds: for every $d \in \mathrm{Ob}(\mathcal{D})$, there is a family of morphisms $(g_i : Fc_i \to d)_{i \in I}$, indexed by a small set I, such that for every $g : Fc \to d$ there are $i \in I$ and $\beta : c \to c_i$ for which g factors as $Fc \xrightarrow{F\beta} Fc_i \xrightarrow{g_i} d$.

Proof We prove only (i). Fix $c \in \mathrm{Ob}(\mathcal{C})$. By Lemma 1.11.1, G has a left adjoint if and only if (c/G) has an initial object. The solution set condition says exactly that (c/G) has the weakly initial family $\{(d_i, c \to Gd_i) : i \in I\}$, where I is small. Since every initial object is weakly initial, the "only if" direction is clear. In the other direction, Lemma 1.11.5 reduces the problem to the completeness of (c/G), which is precisely Proposition 1.11.4(ii). $\square$

Example 1.11.7 Consider the category of groups Grp and the forgetful functor $U : \mathsf{Grp} \to \mathsf{Set}$. The category Grp is complete and U preserves all small limits. To deduce from Theorem 1.11.6 that U has the left adjoint $\mathrm{F} : \mathsf{Set} \to \mathsf{Grp}$, which constructs the free group on a set, the key is to verify the solution set condition for every small set c.

Take any $d \in \mathrm{Ob}(\mathsf{Grp})$ and map $f : c \to Ud$. Factor f as $c \to Us \xrightarrow{U\iota} Ud$, where s is the subgroup of d generated by $f(c)$ and $\iota : s \to d$ is the inclusion. Presenting s by the generating set $f(c)$ and relations shows that the isomorphism classes of all such data $(s, c \to Us)$ form a small set I. This establishes the solution set condition.

The same procedure works for other algebraic structures, such as Ab and R-Mod, again giving left adjoints to the forgetful functors; see the latter part of [33, Chapter V, §6]. Likewise, for any commutative ring $\Bbbk$, taking the multiplicative monoid of a $\Bbbk$-algebra gives a forgetful functor $U : \Bbbk\text{-}\mathsf{Alg} \to \mathsf{Mon}$. The same technique shows that U has a left adjoint sending a monoid M to $\Bbbk[M]$; see [51, Definition 5.6.1].

The approach through the Adjoint Functor Theorem 1.11.6 not only applies uniformly to many algebraic structures; in the free-group case it is also shorter than the classical construction in [51, §4.8].

The Special Adjoint Functor Theorem 1.11.13, to be introduced shortly, replaces the solution set condition by conditions on generators and subobjects. We pause to define generators and cogenerators in a category; these notions will also be used in the later section on Grothendieck categories, among other places.

Definition 1.11.8 Let $\mathcal{E}$ be a category and let Σ be a nonempty small subset of $\mathrm{Ob}(\mathcal{E})$.

- ⋄ The set Σ is called a **generating family** if the following holds: given any pair $f, g : x \to y$ in $\mathcal{E}$, if $f\epsilon = g\epsilon$ for every $s \in \Sigma$ and every $\epsilon : s \to x$, then $f = g$. If $\{s\}$ is a generating family for $\mathcal{E}$, then s is called a **generator** of $\mathcal{E}$.
- ⋄ The set Σ is called a **cogenerating family** if the following holds: given any pair $f, g : x \to y$ in $\mathcal{E}$, if $\delta f = \delta g$ for every $s \in \Sigma$ and every $\delta : y \to s$, then $f = g$. If $\{s\}$ is a cogenerating family for $\mathcal{E}$, then s is called a **cogenerator** of $\mathcal{E}$.

The two notions are plainly dual.

Lemma 1.11.9 Let Σ be a small subset of $\mathrm{Ob}(\mathcal{C})$. If every object of $\mathcal{C}$ can be expressed as a colimit (or limit) of elements of Σ, then Σ is a generating (or cogenerating) family.

Proof For colimits, a morphism $\varinjlim \alpha =: x \to y$ is uniquely determined by its composites with all $\alpha(i) \xrightarrow{\iota_i} \varinjlim \alpha$. $\square$

Example 1.11.10 Here are some basic examples of generators and cogenerators.

- In **Set**, the singleton $\{\mathrm{pt}\}$ is a generator: for maps $f, g : X \to Y$, specifying $\epsilon : \{\mathrm{pt}\} \to X$ amounts to specifying $x \in X$, so $f\epsilon = g\epsilon$ for every ϵ is equivalent to pointwise equality of f and g. The set $\{0, 1\}$ is a cogenerator, because if $f(x) \neq g(x)$ for some $x \in X$, one can choose $\delta : Y \to \{0, 1\}$ such that $\delta(f(x)) \neq \delta(g(x))$.

- Let **CHaus** be the category of compact Hausdorff spaces and continuous maps. The singleton $\{\mathrm{pt}\}$ is again a generator, while $[0, 1]$ is a cogenerator: for every compact Hausdorff space Y and distinct $a, b \in Y$, Urysohn's lemma [52, Theorem 6.3.1 and Corollary 7.2.6] gives a continuous map $\delta : Y \to [0, 1]$ with $\delta(a) = 0$ and $\delta(b) = 1$.

- Let R be a ring. In R-**Mod**, the module R is a generator because specifying an R-module homomorphism $R \to M$ amounts to specifying an element of M. Taking coproducts shows that every nonzero free R-module is also a generator.

- Consider the Yoneda embedding $h_{\mathcal{E}} : \mathcal{E} \to \mathcal{E}^\wedge$ of a small category $\mathcal{E}$. Then $h_{\mathcal{E}}(\mathrm{Ob}(\mathcal{E}))$ is a generating family for $\mathcal{E}^\wedge$. This follows immediately from the density theorem for the Yoneda embedding in Appendix § A.1 and Lemma 1.11.9.

The Special Adjoint Functor Theorem involves one more set-theoretic notion. Recall the posets $(\mathrm{Sub}_e, \subset)$ and $(\mathrm{Quot}_e, \twoheadleftarrow)$ introduced in Definition 1.1.3, where e is an object of a category $\mathcal{E}$.

Definition 1.11.11 A category $\mathcal{E}$ is called **well-powered** (or **well-copowered**) if Sub_e (or Quot_e) is a small set for every $e \in \mathrm{Ob}(\mathcal{E})$.

In this section, well-poweredness will be paired with completeness. Choose a representative $s \hookrightarrow e$ of every element of Sub_e. Thus for any subset $S \subset \mathrm{Sub}_e$ it makes sense, independently of the representatives chosen, to ask whether the fiber product $\prod_{s \in S}(s \hookrightarrow e)$ exists. Dually, for a subset $S \subset \mathrm{Quot}_e$, one may ask whether the fiber coproduct $\coprod_{s \in S}(e \twoheadrightarrow s)$ exists.

- If $\mathcal{E}$ is complete, then well-poweredness implies that for every e, all subsets $S \subset \mathrm{Sub}_e$ have fiber products.

- If $\mathcal{E}$ is cocomplete, then well-copoweredness implies that for every e, all subsets $S \subset \mathrm{Quot}_e$ have fiber coproducts.

Lemma 1.11.12 Let $\mathcal{E}$ be complete and well-powered. If there is a cogenerating family $\Sigma \subset \mathrm{Ob}(\mathcal{E})$, then $\mathcal{E}$ has an initial object.

Proof Take $e := \prod_{s \in \Sigma} s$, and let x be the fiber product of all elements of Sub_e; only here is well-poweredness used. We show that x is initial in $\mathcal{E}$.

By construction, $x \hookrightarrow e$ is the infimum of the poset $(\mathrm{Sub}_e, \subset)$; in particular, $\mathrm{Sub}_x = \{x\}$. For any object y and $f, g \in \mathrm{Hom}_{\mathcal{E}}(x, y)$, the equalizer $\ker(f, g)$ must therefore be x itself, so $f = g$. It remains to show that $\mathrm{Hom}_{\mathcal{E}}(x, y)$ is nonempty for every y.

Form in $\mathcal{E}$ the product $\prod_{s \in \Sigma} s^{\mathrm{Hom}_{\mathcal{E}}(y,s)}$. Define $i : y \to \prod_{s \in \Sigma} s^{\mathrm{Hom}_{\mathcal{E}}(y,s)}$ so that its projection onto the component corresponding to $s \in \Sigma$ and $\delta \in \mathrm{Hom}_{\mathcal{E}}(y, s)$ is exactly δ. Since Σ is cogenerating, i is easily seen to be monic. For every $s \in \Sigma$, take the diagonal morphism $d_s : s \to s^{\mathrm{Hom}_{\mathcal{E}}(y,s)}$, and form the pullback

$$
\begin{tikzcd}
z \arrow[r] \arrow[d, "j"'] \arrow[dr, phantom, "\square"] & y \arrow[d, hook, "i"] \\
e \arrow[r, "\prod_{s \in \Sigma} d_s"'] & \prod_{s \in \Sigma} s^{\mathrm{Hom}_{\mathcal{E}}(y,s)}.
\end{tikzcd}
$$

Lemma 1.1.5 ensures that j is also monic, giving a subobject of e. Since x is the infimum of Sub_e, there is a morphism $x \hookrightarrow z$. The composite $x \hookrightarrow z \to y$ is the required morphism. $\square$

Theorem 1.11.13 (Special Adjoint Functor Theorem) Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

(i) A functor $G : \mathcal{D} \to \mathcal{C}$ has a left adjoint if it satisfies the following conditions.

- $\diamond$ $\mathcal{D}$ is complete and well-powered, and G preserves all small limits;
- $\diamond$ there is a cogenerating family $\Sigma \subset \mathrm{Ob}(\mathcal{D})$.

(ii) A functor $F : \mathcal{C} \to \mathcal{D}$ has a right adjoint if it satisfies the following conditions.

- $\diamond$ $\mathcal{C}$ is cocomplete and well-copowered, and F preserves all small colimits;
- $\diamond$ there is a generating family $\Sigma \subset \mathrm{Ob}(\mathcal{C})$.

Proof By duality, it suffices to prove (i). Lemma 1.11.1 again reduces the problem to showing that (c/G) has an initial object for $c \in \mathrm{Ob}(\mathcal{C})$; the strategy is to apply Lemma 1.11.12.

Proposition 1.11.4(ii) shows that (c/G) is complete. Moreover, Definition 1.11.8 and the definition of (c/G) show at once that all data $(d, c \to Gd)$ with $d \in \Sigma$ form a cogenerating family for (c/G). Since Σ is small, this is a small set.

Let $e = (d, c \to Gd) \in \mathrm{Ob}((c/G))$. Subobjects are isomorphism classes of monomorphisms. By Proposition 1.11.4, the projection $\Pi_{c/} : (c/G) \to \mathcal{D}$ is conservative and preserves monomorphisms, so it induces an order-preserving embedding $\mathrm{Sub}_e \hookrightarrow \mathrm{Sub}_d$. Thus (c/G) is well-powered. $\square$

Corollary 1.11.14 If a complete well-powered category $\mathcal{D}$ has a cogenerating family Σ, then $\mathcal{D}$ is cocomplete.

Dually, if a cocomplete well-copowered category $\mathcal{C}$ has a generating family Σ, then $\mathcal{C}$ is complete.

Proof It suffices to prove the assertion for $\mathcal{D}$. Let I be a small category. By Example 1.6.3, it is enough to show that (α/Δ) has an initial object for every $\alpha : I \to \mathcal{D}$. By Lemma 1.11.1, this is equivalent to saying that the diagonal functor Δ has a left adjoint. Clearly Δ preserves all small limits and colimits, so Theorem 1.11.13 applies.□

Example 1.11.15 Let $\iota : \mathsf{CHaus} \to \mathsf{Top}$ be the inclusion of compact Hausdorff spaces into topological spaces. Example 1.11.10 showed that CHaus has the cogenerator $[0, 1]$. By the Tychonoff theorem [52, Theorem 7.7.2], CHaus is readily seen to be complete, and ι preserves all small limits; details are left to the reader. Furthermore, the power set of a small set is small, so CHaus is well-powered. Theorem 1.11.13(i) therefore gives a left adjoint $C : \mathsf{Top} \to \mathsf{CHaus}$ to ι. In point-set topology, $C(X)$ is called the **Stone–Čech compactification** of X; the unit gives a canonical morphism $\eta_X : X \to \iota C(X)$. The constructions in Lemma 1.11.12 and Theorem 1.11.13 agree with the classical construction of the Stone–Čech compactification.

Adjoint functor theorems provide sufficient conditions for representability. For representable functors, see [51, Definition 2.5.2] or the definition of a representable functor in Appendix § A.1. The key observation is this: if $G : \mathcal{D} \to \mathsf{Set}$ has a left adjoint F, then G is representable. Indeed, consider the singleton $\{\mathrm{pt}\}$ and put $r_G := F(\{\mathrm{pt}\})$. The canonical bijections

$$\mathrm{Hom}_{\mathcal{D}}(r_G, d) \simeq \mathrm{Hom}_{\mathsf{Set}}(\{\mathrm{pt}\}, Gd) \simeq Gd, \quad d \in \mathrm{Ob}(\mathcal{D})$$

imply $G \simeq \mathrm{Hom}_{\mathcal{D}}(r_G, \cdot)$.

Corollary 1.11.16 If a complete well-powered category $\mathcal{D}$ has a cogenerating family Σ, then a functor $G : \mathcal{D} \to \mathsf{Set}$ is representable if and only if it preserves all small limits.

Proof The "only if" direction is a general property of representable functors. For the converse, Theorem 1.11.13 ensures that G has a left adjoint $F : \mathsf{Set} \to \mathcal{D}$; now apply the observation above. □

Exercises

1. Consider the category of groups Grp. Prove that a group homomorphism $\varphi : H \to G$ is a monomorphism (or epimorphism) in Grp if and only if its underlying map is injective (or surjective). Some references, including [51], do not adequately explain this fact.

Hint〉 It suffices to prove the "only if" direction. For monomorphisms, consider homomorphisms from $\ker\varphi$ to H. For epimorphisms, one must show that if $\varphi(H) \neq G$, then there are distinct group homomorphisms $f, g : G \rightrightarrows K$ whose restrictions to $\varphi(H)$ agree. One approach is to take K to be the amalgamated free product defined by two copies of the embedding $\varphi(H) \hookrightarrow G$ in [51, Definition 4.8.5], and to take f and g as the embeddings of G into the two copies. Uniqueness of reduced expressions [51, Lemma 4.8.7] implies $f \neq g$.

2. Show that in the category Top, the image of $f : X \to Y$ is $f(X)$ with the subspace topology from Y, while its coimage is $f(X)$ with the quotient topology from X. Give the corresponding topological interpretations of strict monomorphisms and strict epimorphisms.

3. Let $\mathcal{C}$ have finite limits and finite colimits. Prove that the following statements about a morphism $f : X \to Y$ are equivalent.

(i) The morphism f is a strict epimorphism.

(ii) The canonical morphism $\mathrm{coim}(f) \to Y$ is an isomorphism.

(iii) $X \underset{Y}{\times} X \rightrightarrows X \xrightarrow{f} Y$ is a coequalizer diagram.

(iv) The morphism $f : X \to Y$ is the coequalizer of some pair $a, b : W \rightrightarrows X$.

Hint〉 Recall that being epic is equivalent to $\mathrm{im}(f) \xrightarrow{\sim} Y$ (Lemma 1.2.6), and use Proposition 1.1.2. It is easy to prove (i) $\iff$ (ii) $\iff$ (iii) $\implies$ (iv). For (iv) $\implies$ (ii), consider the solid part of the commutative diagram

$$\begin{tikzcd}
W \arrow[r, shift left, "a"] \arrow[r, shift right, "b"'] \arrow[d, "{(a,b)}"'] & X \arrow[r, two heads, "f"] \arrow[d, "\mathrm{id}"] & Y \arrow[d, dashed] \\
X \underset{Y}{\times} X \arrow[r, shift left] \arrow[r, shift right] & X \arrow[r, two heads] & \mathrm{coim}(f)
\end{tikzcd}$$

The universal property of the coequalizer gives the dashed arrow and shows that it and $\mathrm{coim}(f) \to Y$ are inverse to each other.

4. Let $\mathcal{C}$ have finite limits and finite colimits. Prove that if a morphism f has a right inverse (or left inverse), then f is a strict epimorphism (or strict monomorphism). Hint〉 If $fg = \mathrm{id}_Y$, then f is the coequalizer of gf and id_X.

5. Prove that an additive-category structure on any category, if it exists, is unique. Hint〉 Apply Proposition 1.3.5.

6. Compare Example 1.5.6. Let U be the forgetful functor from compact Hausdorff spaces CHaus to sets Set. Prove that U creates all small limits. Show that it does not create all small colimits.

7. Let $\mathcal{I}$ be the category of all finite sets and surjective maps. Prove that $\mathcal{I}^{\mathrm{op}}$ is not filtered in the sense of §1.6.

8. In Definition 1.7.1, prove that every morphism $F \to F'$ induces canonical morphisms $\mathrm{Lan}_K F \to \mathrm{Lan}_K F'$ and $\mathrm{Ran}_K F \to \mathrm{Ran}_K F'$, provided these Kan extensions exist. Characterize the induced morphisms.

9. Let K and F be as in §1.7. Prove that if $M : \mathcal{E} \to \mathcal{F}$ has a left adjoint (or right adjoint), then M preserves $\mathrm{Ran}_K F$ (or $\mathrm{Lan}_K F$).

Hint〉 By duality, suppose that M has a right adjoint N, with unit $\eta' : \mathrm{id} \to NM$ and counit $\varepsilon' : MN \to \mathrm{id}$. For any functor $L : \mathcal{D} \to \mathcal{F}$, there are canonical bijections

$$\begin{aligned}\mathrm{Hom}_{\mathcal{F}^{\mathcal{D}}}\left(M(\mathrm{Lan}_K F), L\right) &\simeq \mathrm{Hom}_{\mathcal{E}^{\mathcal{D}}}\left(\mathrm{Lan}_K F, NL\right) \\ &\overset{(1.7.1)}{\simeq} \mathrm{Hom}_{\mathcal{E}^{\mathcal{C}}}\left(F, NLK\right) \simeq \mathrm{Hom}_{\mathcal{F}^{\mathcal{C}}}\left(MF, LK\right).\end{aligned}$$

Using the general properties of η' and ε', verify that when $L = M(\mathrm{Lan}_K F)$ this bijection sends id to $M\eta$, where η is the morphism belonging to $\mathrm{Lan}_K F$.

For general L, use the familiar technique from the proof of the Yoneda lemma to verify that an arbitrary morphism

$$\varphi : M(\mathrm{Lan}_K F) \to L$$

corresponds to the composite

$$\begin{aligned}(\varphi K)(M\eta) : \quad MF &\xrightarrow{M\eta} M(\mathrm{Lan}_K F)K, \\ M(\mathrm{Lan}_K F)K &\xrightarrow{\varphi K} LK.\end{aligned}$$

10. Consider the diagram of functors

$$\begin{tikzcd} & \mathcal{D} \arrow[dr, "E"] & \\ \mathcal{C} \arrow[ur, "F"] & & \mathcal{D}' \arrow[ll, "G"] \end{tikzcd}$$

where EF is left adjoint to G, with counit $\varepsilon : EFG \to \mathrm{id}_{\mathcal{D}'}$. Prove that if, for every category $\mathcal{X}$, the functor $F^* : \mathcal{X}^{\mathcal{D}} \to \mathcal{X}^{\mathcal{C}}$ is fully faithful, then E can also be realized as a left adjoint to FG, with the same counit ε.

Hint〉 The following argument is from [14, I.1.3.1 Lemma]. Let $\eta : \mathrm{id}_{\mathcal{C}} \to GEF$ be the unit of (EF, G). Taking $\mathcal{X} = \mathcal{D}$ gives $\eta' : \mathrm{id}_{\mathcal{D}} \to FGE$ such that $\eta' F = F\eta$. Check the triangle identities for (E, FG) with η' and ε. First, $(FG\varepsilon)(\eta' FG) = (FG\varepsilon)(F\eta G) = \mathrm{id}_{FG}$. On the other hand, taking $\mathcal{X} = \mathcal{D}'$ shows that $(\varepsilon E)(E\eta') = \mathrm{id}_E$ is equivalent to $(\varepsilon EF)(E\eta' F) = \mathrm{id}_{EF}$; but $(\varepsilon EF)(E\eta' F) = (\varepsilon EF)(EF\eta) = \mathrm{id}_{EF}$.

11. Consider an adjoint pair $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$, and put

$$S := \{s \in \mathrm{Mor}(\mathcal{C}) : Fs \text{ is invertible}\};$$

this factors F uniquely as $\mathcal{C} \xrightarrow{Q} \mathcal{C}[S^{-1}] \xrightarrow{H} \mathcal{D}$. Prove that the following statements are equivalent.

(i) The functor G is fully faithful.

(ii) The counit $\varepsilon : FG \to \mathrm{id}_{\mathcal{D}}$ is an isomorphism.

(iii) The functor H is an equivalence of categories, with QG as a quasi-inverse.

(iv) For every category $\mathcal{X}$, the functor $F^* : \mathcal{X}^{\mathcal{D}} \to \mathcal{X}^{\mathcal{C}}$ is fully faithful.

Also state the dual version of this result.

Hint The following argument is from [14, I.1.3 Proposition]. First, (i) $\iff$ (ii) is fairly easy; see the hint in [51, Chapter 2, Exercise 8].

For (ii) $\implies$ (iii), one must prove $QGH \simeq \mathrm{id}_{\mathcal{C}[S^{-1}]}$. Proposition 1.9.4 reduces this to $QGHQ \simeq Q$. Consider the unit $\eta : \mathrm{id}_{\mathcal{C}} \to GF$. Since $(\varepsilon F)(F\eta) = \mathrm{id}_F$, the morphism $F\eta$ is an isomorphism, hence $Q\eta : Q \xrightarrow{\sim} QGF = QGHQ$.

For (iii) $\implies$ (iv), the fact that H is an equivalence implies that H^* is fully faithful, while Proposition 1.9.4 applies to Q^*.

For (iv) $\implies$ (ii), put $E = \mathrm{id}_{\mathcal{D}}$ in the preceding exercise. Then FG is a right adjoint to $\mathrm{id}_{\mathcal{D}}$, with counit ε. Since right adjoints to $\mathrm{id}_{\mathcal{D}}$ are unique up to isomorphism, ε is an isomorphism.

12. Let S be a left multiplicative system in $\mathcal{C}$. Prove that every commutative diagram in $\mathcal{C}[S^{-1}]^l$ of the form

$$\begin{tikzcd} Q^l X \arrow[r, "Q^l f"] \arrow[d, "\alpha"'] & Q^l X' \arrow[d, "\beta"] \\ Q^l Y \arrow[r, "Q^l g"'] & Q^l Y' \end{tikzcd}$$

arises from a commutative diagram in $\mathcal{C}$ of the form

$$\begin{tikzcd} X \arrow[r, "f"] & X' \\ U \arrow[u, tail] \arrow[r] \arrow[d] & U' \arrow[u, tail] \arrow[d] \\ Y \arrow[r, "g"'] & Y' \end{tikzcd}$$

where $f, g \in \mathrm{Mor}(\mathcal{C})$ are given and, as usual, $\rightarrowtail$ denotes a morphism in S. State the corresponding version for a right multiplicative system.

Hint Represent α and β by $(U^\dagger; s, a)$ and $(U'; t, b)$, respectively. Apply (S3) to obtain W, h, r and a commutative diagram in $\mathcal{C}$

$$\begin{tikzcd} W \arrow[r, "h"] \arrow[d, tail, "r"'] & U' \arrow[d, tail, "t"] \\ U^\dagger \arrow[r, "fs"'] & X'. \end{tikzcd}$$

This shows that the upper square in the following diagram commutes:

$$\begin{tikzcd}
X \arrow[r, "f"] & X' \\
W \arrow[u, tail, "sr"] \arrow[r, "h"] \arrow[d, "ar"'] & U' \arrow[u, tail, "t"'] \arrow[d, "b"] \\
Y \arrow[r, "g"'] & Y'
\end{tikzcd}$$

For the lower square, first show that it commutes after applying Q^l; then use Corollary 1.9.13 to choose $U \rightarrowtail W$ and adjust it to the required commutative diagram.

13. Verify the details of central localization in Example 1.9.19. Prove that the category $\mathcal{C}[S^{-1}]$ defined there and the localization supplied by Theorem 1.9.17 are equivalent as $Z(\mathcal{C})[S^{-1}]$-linear categories.

14. Complete the details in the proof of Proposition 1.11.4(i).

15. Use the Adjoint Functor Theorem 1.11.6 to construct the free product $G \star H$ of any two groups G and H; see [51, Definition 4.8.9].

16. Let $\mathrm{F}(X)$ be the free group on a set X. Prove that the canonical map $\iota : X \to U(\mathrm{F}(X))$ is injective, where U is the forgetful functor from groups to sets. Hint For distinct $x, y \in X$, there is a group H and a map of sets $f : X \to U(H)$ such that $f(x) \neq f(y)$.

17. (P. Freyd) Let $\mathcal{D}$ be complete and let $G : \mathcal{D} \to \mathsf{Set}$ preserve all small limits and satisfy the following solution set condition: there is a small subset $\Gamma \subset \mathrm{Ob}(\mathcal{D})$ such that for every $d \in \mathrm{Ob}(\mathcal{D})$ and $p \in Gd$, there are $x \in \Gamma$, $q \in Gx$, and $f : x \to d$ with $(Gf)(q) = p$. Prove that G is representable.

Hint Write an object of $(\{\mathrm{pt}\}/G)$ as (d, p), where $d \in \mathrm{Ob}(\mathcal{D})$ and $p \in Gd$. The condition says exactly that $\Gamma^\natural := \{(x, q) : x \in \Gamma,\ q \in Gx\}$ is a weakly initial family for $(\{\mathrm{pt}\}/G)$. Show that this category has an initial object (r_G, u_G). Following the argument for (1.11.1), show that for every d there is a bijection

$$\begin{aligned}
\mathrm{Hom}_{\mathcal{D}}(r_G, d) \longrightarrow \mathrm{Hom}_{\mathsf{Set}}(\{\mathrm{pt}\}, Gd) &\xrightarrow{\sim} Gd \\
\alpha \longmapsto\quad &(G\alpha)(u_G).
\end{aligned}$$

Chapter 2 Abelian Categories

An abelian category is an additive category with kernels and cokernels in which every morphism is strict. Abelian categories generalize the category R-Mod of left modules or Mod-R of right modules over a ring R, and support the study of such notions as injective and projective objects, complexes, homology and cohomology, and exact sequences.

Module categories are the prototype of abelian categories, but do not exhaust their substance. There are two reasons for this.

1. In a general abelian category, objects have no elements and morphisms are not maps. Diagram-chasing methods from module categories are therefore no longer available; the most basic exact sequences must instead be derived by operations on objects and functors. Prominent examples are the snake lemma and the five lemma discussed in §2.3, whose proofs use a substitute for diagram chasing (Lemma 2.3.1). Even so, the arguments are considerably more involved than their module-category counterparts.

2. Even when one starts with a module category or one of its abelian subcategories, abstract abelian categories arise when studying functor categories, localizations, or Serre quotients. In geometry and topology, sheaf theory naturally leads to such constructions.

The Freyd–Mitchell embedding theorem B.6.3 states that every small abelian category admits a fully faithful embedding into some $\mathsf{Mod}\text{-}R$, and that the embedding functor preserves exact sequences. Once this result is accepted, many assertions about exact sequences reduce readily to the case of $\mathsf{Mod}\text{-}R$. This reduction keeps diagram chasing available, but its chief benefit may be psychological reassurance: on the one hand it sidesteps the essence of abelian categories, and on the other hand any proof of the embedding theorem must still establish several core facts about abelian categories from scratch. This chapter does not depend on the embedding theorem, nor logically on module theory, but readers are expected to have a basic knowledge of homological algebra in module categories. Diagram-chasing skills remain useful and may even be indispensable in the natural order of study; readers are encouraged to consult [51] or the relevant parts of another textbook. Appendix §B.6 derives the embedding theorem as an application of the Ind construction.

A cocomplete abelian category with a generator in which filtered small colimits are exact is called a Grothendieck category. Such categories are the subject of the final section, §2.10; these conditions are directly related to assumptions on set size. Grothendieck categories occur frequently in geometry and enjoy better properties, such as completeness (Corollary 2.10.9) and the existence of canonical injective resolutions (Theorem 2.10.14). They are a closer analogue of module categories than general abelian categories; the precise meaning of this statement is explained by the Gabriel–Popescu theorem A.3.2 in the appendix.

The basic isomorphism theorems of module theory remain valid for general abelian categories. The same is true of such fundamental notions as subobjects, quotient objects, direct-sum decompositions, indecomposable objects, simple objects, and composition series, all central concerns of algebra. They will be treated in §§2.5— 2.7. For example, the Krull–Remak–Schmidt theorem on direct-sum decompositions has an abstract generalization (Theorem 2.5.9). The language of lattice theory is useful here; see §2.4. It should be stressed that statements involving infinite direct sums often hold only in Grothendieck categories.

Section 2.9 begins with the definition and characterization of abelian subcategories, and then introduces quotients of an abelian category by a certain kind of abelian subcategory, called a Serre subcategory; the result is a Serre quotient. These quotients are characterized by a universal property and can be constructed by the localization introduced in §1.9. The section also introduces the group K_0 of an abelian category and its basic properties, including the Euler–Poincaré principle (Theorem 2.9.11) and the exact sequence for K_0 of a Serre quotient (Proposition 2.9.13). These techniques are common in applications. The construction of K_0 applies to a broader class of structures than abelian categories, called **exact categories**; see [7] for the theory. Moreover, K_0 is only the beginning of a sequence of abelian groups K_n for $n \in \mathbb{Z}_{\geq 0}$. A satisfactory interpretation of the higher K-groups requires homotopy-theoretic ideas, especially the notion of a "spectrum", and lies beyond the scope of this book.

Reading Guide

As noted above, readers should have some familiarity with concrete operations in module categories. For the basic theory of complexes and their cohomology, the material in §§2.1—2.8 is essential. Apart from the treatment of abelian subcategories, the material in §2.9 is used less often later, but it remains standard knowledge and is related to several definitions in the discussion of derived categories. Section 2.10 on Grothendieck categories is supplementary; it requires relatively involved arguments and material from the appendices, and may be skipped on a first reading.

A complex written as data $(X^n, d^n)_n$, where $d^n : X^n \to X^{n+1}$, is also called a cochain complex. If instead one uses subscripts, writing $(X_n, d_n)_n$ with $d_n : X_n \to X_{n-1}$, it is called a chain complex. Except in Chapter 8, this book primarily uses cochain notation.

2.1 Definition of an Abelian Category

Section 1.3 introduced Ab-categories and additive categories. The abelian categories studied in this chapter form a special class of additive categories.

Definition 2.1.1 (Abelian category) Let $\mathcal{A}$ be an additive category. If every morphism in $\mathcal{A}$ has a kernel and a cokernel, and every morphism is strict (Definition 1.2.4), then $\mathcal{A}$ is called an **abelian category**.

If $\mathcal{A}$ is also a $\Bbbk$-linear category (Definition 1.4.4), it is called a **$\Bbbk$-linear abelian category**.

Remark 2.1.2 An abelian category has all finite limits and colimits. Indeed, finite products and coproducts exist (both are biproducts), as do equalizers and coequalizers (Remark 1.3.8); these suffice to construct all finite limits and colimits. See [51, Theorem 2.8.3].

By Example 1.3.13, R-Mod is an abelian category for every ring R. Replacing R by R^{op} shows the same for the category Mod-R of right modules; the special case $R = \mathbb{Z}$ shows that Ab is abelian. By contrast, the category $\mathsf{Ban}_{\mathbb{C}}$ from the same example is not abelian, since it has nonstrict morphisms.

Proposition 2.1.3 If an additive category $\mathcal{A}$ is abelian, then so is $\mathcal{A}^{\mathrm{op}}$.

Proof Kernels and cokernels are dual, while Remark 1.2.2 shows that the morphism $\mathrm{coim}(f) \to \mathrm{im}(f)$ in $\mathcal{A}$ is exactly the morphism $\mathrm{coim}(f^{\mathrm{op}}) \to \mathrm{im}(f^{\mathrm{op}})$ in $\mathcal{A}^{\mathrm{op}}$. $\square$

One abstract way to construct new abelian categories from existing ones is to form functor categories. First note that if $\mathcal{A}$ is an Ab-category, then for every category I[1], the functor category $\mathcal{A}^I$ is naturally an Ab-category: for functors $F, G : I \rightrightarrows \mathcal{A}$, addition in $\mathrm{Hom}_{\mathcal{A}^I}(F, G)$ is defined objectwise by $(\varphi_i)_i + (\psi_i)_i := (\varphi_i + \psi_i)_i$.

Proposition 2.1.4 If $\mathcal{A}$ is an abelian category, then $\mathcal{A}^I$ is an abelian category for every category I.

Proof By Proposition 1.5.7, finite products and coproducts, and kernels and cokernels, are constructed objectwise in $\mathcal{A}^I$. The image and coimage of a morphism, and hence the canonical morphism coim $\to$ im, are likewise constructed objectwise. Thus every abelian category axiom for $\mathcal{A}^I$ reduces to the corresponding axiom for $\mathcal{A}$. $\square$

Every morphism $f : X \to Y$ in an abelian category has a unique epi–mono factorization (Proposition 1.2.10), concretely $X \twoheadrightarrow (\mathrm{coim}(f) \simeq \mathrm{im}(f)) \hookrightarrow Y$. Lemma 1.3.11(iii) therefore gives immediately

$$\ker(f) = \ker[X \to \mathrm{im}(f)], \quad \mathrm{coker}(f) = \mathrm{coker}[\mathrm{im}(f) \to Y].$$

Remark 2.1.5 For morphisms $X \xrightarrow{f} Z \xleftarrow{g} Y$ in an abelian category, the construction of the fiber product (see Remark 2.1.2) shows that $X \underset{Z}{\times} Y$ is the equalizer of $X \oplus Y \underset{(0,g)}{\overset{(f,0)}{\rightrightarrows}} Z$, that is, $\ker[X \oplus Y \xrightarrow{(f,-g)} Z]$. The morphisms $X \leftarrow X \underset{Z}{\times} Y \to Y$ carried by the fiber product come from the projections $X \xleftarrow{p_1} X \oplus Y \xrightarrow{p_2} Y$.

For $X' \xleftarrow{f'} Z' \xrightarrow{g'} Y'$, the fiber coproduct construction realizes $X' \underset{Z'}{\sqcup} Y'$ as $\mathrm{coker}[Z' \xrightarrow{(-f',g')} X' \oplus Y']$, and the morphisms $X' \to X' \underset{Z'}{\sqcup} Y' \leftarrow Y'$ come from $X' \xrightarrow{\iota_1} X' \oplus Y' \xleftarrow{\iota_2} Y'$. These properties are of course evident for modules.

The following result will be used repeatedly.

[1] If one does not want $\mathcal{A}^I$ to be large, one should assume that I is small. This does not affect the substance here.

Proposition 2.1.6 Suppose that a diagram in an abelian category $\mathcal{A}$

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "a"'] & Y \arrow[d, "b"] \\ X' \arrow[r, "g"'] & Y' \end{tikzcd}$$

is a pullback (or pushout), and that g is epic (or f is monic). Then the diagram is also a pushout (or pullback), and f is epic (or g is monic).

Proof It suffices to treat the pullback case. We claim that

- ◇ $X \xrightarrow{(a,f)} X' \oplus Y$ gives $\ker[X' \oplus Y \xrightarrow{(g,-b)} Y']$;
- ◇ $(g,-b) : X' \oplus Y \to Y'$ is epic.

The assertion about the kernel follows from the description of $X \simeq X' \underset{Y'}{\times} Y$ as a kernel in Remark 2.1.5. Since $g = (g,-b)\iota_1$, the fact that g is epic implies that $(g,-b)$ is epic as well.

Consequently, $Y' = \operatorname{im}((g,-b)) \overset{\sim}{\leftarrow} \operatorname{coker}[X \xrightarrow{(a,f)} X' \oplus Y]$. Viewing this from the other direction and applying the description of a fiber coproduct as a cokernel in Remark 2.1.5 shows at once that

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "-a"'] & Y \arrow[d, "-b"] \\ X' \arrow[r, "g"'] & Y' \end{tikzcd}$$

is a pushout. Composing the two columns with the automorphisms $-\mathrm{id}_X$ and $-\mathrm{id}_Y$, respectively, recovers the original diagram, which is therefore also a pushout. In the pushout case we already know that $\operatorname{coker}(f) \overset{\sim}{\to} \operatorname{coker}(g)$ (Proposition 1.3.10); hence f is epic (Lemma 1.3.11). □

Corollary 2.1.7 A subquotient in an abelian category can equivalently be defined as a quotient object of a subobject or as a subobject of a quotient object.

Proof Proposition 2.1.6 shows that Proposition 1.1.8 applies to every abelian category. □

For the category R-**Mod** of left modules over any ring R, all the results above are of course familiar; the same holds for right modules.

2.2 A First Look at Complexes

The starting point of this section is a pair of morphisms in an abelian category satisfying $gf = 0$, namely $X \xrightarrow{f} Y \xrightarrow{g} Z$. The composite $\ker(g) \hookrightarrow Y \twoheadrightarrow \operatorname{coker}(f)$ gives a morphism $\ker(g) \to \operatorname{coker}(f)$; what interests us is its epi–mono factorization. There are two mutually dual points of view.

Lemma 2.2.1 Let $f : X \to Y$ and $g : Y \to Z$ be morphisms in an abelian category $\mathcal{A}$ such that $gf = 0$.

(i) There are unique morphisms φ and ψ that make the following diagrams commute:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, two heads] & Y \\ \operatorname{coim}(f) \arrow[r, "\varphi"'] & \ker(g) \arrow[u, hook] \end{tikzcd} \qquad \begin{tikzcd} Z & Y \arrow[l, "g"'] \arrow[d, two heads] \\ \operatorname{im}(g) \arrow[u, hook] & \operatorname{coker}(f) \arrow[l, "\psi"] \end{tikzcd} \tag{2.2.1}$$

Moreover, φ is a monomorphism and ψ is an epimorphism.

(ii) The morphism $\ker(g) \to \operatorname{coker}(f)$ has the two epi–mono factorizations displayed in the following diagram; they are connected by a unique isomorphism $\operatorname{coker}(\varphi) \simeq \ker(\psi)$:

$$\begin{tikzcd} & \ker(\psi) \arrow[dd, "\wr"] \arrow[dr, hook, "\text{canonical}"] & \\ \ker(g) \arrow[ur, two heads] \arrow[dr, two heads, "\text{canonical}"'] & & \operatorname{coker}(f). \\ & \operatorname{coker}(\varphi) \arrow[ur, hook] & \end{tikzcd}$$

Proof First consider (i). The two diagrams are dual to one another (Remark 1.2.2 and Proposition 2.1.3): the diagram on the right is equivalent to considering in $\mathcal{A}^{\mathrm{op}}$ the sequence $Z \xrightarrow{g^{\mathrm{op}}} Y \xrightarrow{f^{\mathrm{op}}} X$. Thus, for (2.2.1), it suffices to consider φ.

The condition $gf = 0$ says precisely that f factors through $\ker(g) \hookrightarrow Y$. Applying Lemma 1.2.3 on coimages to this factorization gives φ. Next, because $X \to \operatorname{coim}(f)$ is an epimorphism, the factorization of f as $X \twoheadrightarrow \operatorname{coim}(f) \xrightarrow{\overline{f}} Y$ is unique. Here $\overline{f}$ may be taken either as the composite $\operatorname{coim}(f) \xrightarrow{\varphi} \ker(g) \hookrightarrow Y$ or as the composite $\operatorname{coim}(f) \xrightarrow{\sim} \operatorname{im}(f) \hookrightarrow Y$. The latter composite is a monomorphism, and hence so is φ.

To obtain (ii), we show that there is a unique monomorphism $\operatorname{coker}(\varphi) \hookrightarrow \operatorname{coker}(f)$

(or epimorphism $\ker(g) \twoheadrightarrow \ker(\psi)$) that makes the following diagram commute:

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, two heads] & Y \arrow[r, two heads] & \operatorname{coker}(f) \\
\operatorname{coim}(f) \arrow[r, hook, "\varphi"'] & \ker(g) \arrow[u, hook] \arrow[r, two heads, "\text{canonical}"'] & \operatorname{coker}(\varphi) \arrow[u, hook]
\end{tikzcd}
\qquad
\begin{tikzcd}
Z & Y \arrow[l, "g"'] \arrow[d, two heads] & \ker(g) \arrow[l, hook'] \arrow[d, two heads] \\
\operatorname{im}(g) \arrow[u, hook] & \operatorname{coker}(f) \arrow[l, two heads, "\psi"] & \ker(\psi) \arrow[l, hook', "\text{canonical}"]
\end{tikzcd}$$

Once this is proved, the two epi–mono factorizations in (ii) can be read directly from the outer frames of the diagrams. The isomorphism along the middle path is then uniquely determined by Proposition 1.2.10.

It remains to construct the diagrams above. Since the two diagrams are dual, it suffices to discuss the one on the left. Lemma 1.3.11 (iii) shows that $\operatorname{coker}(\varphi)$ can also be identified with the cokernel of the composite $X \to \ker(g)$. The desired morphism therefore comes from the functoriality of cokernels and is characterized by (1.3.4). Why is this morphism a monomorphism? Proposition 1.3.9 expresses the cokernel of a composite morphism as the pushout of the original cokernel; explicitly, the diagram is

$$\begin{tikzcd}
\ker(g) \arrow[r, hook] \arrow[d, two heads] \arrow[dr, phantom, "\boxplus"] & Y \arrow[d, two heads] \\
\operatorname{coker}[X \to \ker(g)] \arrow[r] & \operatorname{coker}(f)
\end{tikzcd} \qquad \text{(pushout diagram)}$$

with all morphisms canonical. Proposition 2.1.6 then shows that the morphism in the second row is a monomorphism. $\square$

In the situation of Lemma 2.2.1, write

$$\begin{aligned}
\mathrm{H}\left[X \xrightarrow{f} Y \xrightarrow{g} Z\right] &:= \operatorname{coker}\left[\operatorname{im}(f) \xrightarrow{\varphi} \ker(g)\right] \\
&\simeq \ker\left[\operatorname{coker}(f) \xrightarrow{\psi} \operatorname{im}(g)\right];
\end{aligned} \tag{2.2.2}$$

this object can also be characterized as the middle term in the epi–mono factorization of $\ker(g) \to \operatorname{coker}(f)$. As expected, (2.2.2) is functorial.

Proposition 2.2.2 Given a commutative diagram in an abelian category

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, "a"'] & Y \arrow[r, "g"] \arrow[d, "b"] & Z \arrow[d, "c"] \\
X' \arrow[r, "f'"'] & Y' \arrow[r, "g'"'] & Z'
\end{tikzcd} \qquad g'f' = 0, \quad gf = 0,$$

there is a unique morphism $\Phi : \mathrm{H}[X \to Y \to Z] \to \mathrm{H}[X' \to Y' \to Z']$ that makes the following diagram commute:

$$\begin{tikzcd} \ker(g) \arrow[r, two heads] \arrow[d, "\text{induced by }(b,c)"'] & \mathrm{H}[X \to Y \to Z] \arrow[r, hook] \arrow[d, "\Phi"] & \operatorname{coker}(f) \arrow[d, "\text{induced by }(a,b)"] \\ \ker(g') \arrow[r, two heads] & \mathrm{H}[X' \to Y' \to Z'] \arrow[r, hook] & \operatorname{coker}(f') \end{tikzcd}$$

The two outer vertical arrows come from the functoriality of kernels and cokernels, while the horizontal arrows are those of Lemma 2.2.1.

Proof As in the argument of Lemma 1.2.8, if Φ exists then it is unique, and it suffices to check that Φ makes the right-hand square commute. Here is the construction. Since $\mathrm{im}(g) = \ker[Z \to \operatorname{coker}(g)]$ and $\mathrm{im}(g') = \ker[Z' \to \operatorname{coker}(g')]$, the functoriality of kernels gives a unique morphism $\mathrm{im}(g) \to \mathrm{im}(g')$ that makes the right half of the following diagram commute:

$$\begin{tikzcd} \operatorname{coker}(f) \arrow[r, two heads, "\psi"] \arrow[d] & \mathrm{im}(g) \arrow[r, hook] \arrow[d] & Z \arrow[d, "c"] \\ \operatorname{coker}(f') \arrow[r, two heads, "\psi'"'] & \mathrm{im}(g') \arrow[r, hook] & Z'; \end{tikzcd}$$

The pattern of Lemma 1.2.8 shows that the left half also commutes. The morphism $\ker(\psi) \to \ker(\psi')$ induced by it is the desired morphism. □

Proposition 2.2.3 Given morphisms satisfying $gf = 0$ and $g'f' = 0$, there is a canonical isomorphism

$$\mathrm{H}\left[X \xrightarrow{f} Y \xrightarrow{g} Z\right] \oplus \mathrm{H}\left[X' \xrightarrow{f'} Y' \xrightarrow{g'} Z'\right] \simeq \mathrm{H}\left[X \oplus X' \xrightarrow{f \oplus f'} Y \oplus Y' \xrightarrow{g \oplus g'} Z \oplus Z'\right].$$

Proof Use the morphisms ι, p, and so forth associated with biproducts, as in (1.3.1). By the functoriality in Proposition 2.2.2, these morphisms also induce

$$\mathrm{H}[X \to Y \to Z] \underset{p}{\overset{\iota}{\rightleftarrows}} \mathrm{H}[X \oplus X' \to Y \oplus Y' \to Z \oplus Z'] \underset{p}{\overset{\iota}{\leftrightarrows}} \mathrm{H}[X' \to Y' \to Z'].$$

The characterization of the induced morphism Φ in Proposition 2.2.2 also tells us that $(a, b, c) \mapsto \Phi$ is compatible with composition, preserves addition, sends 0 to 0, and sends id to id. Hence the morphisms induced at the level of $\mathrm{H}[\cdots]$ also satisfy the biproduct identities (1.3.1). □

Notice that the preceding statement need not hold for infinite coproducts or infinite products in a general abelian category.

Definition 2.2.4 (Complexes) Let $\mathcal{A}$ be an additive category. Consider a sequence of morphisms in $\mathcal{A}$

$$\cdots \to X^n \xrightarrow{d^n} X^{n+1} \xrightarrow{d^{n+1}} X^{n+2} \to \cdots .$$

The sequence may have finitely many terms, extend infinitely in one or both directions, or form a cycle. If $d^{n+1}d^n = 0$ for every n, the data $(X^n, d^n)_n$ are called a **complex**; the customary notation is $(X^\bullet, d^\bullet)$, $X^\bullet$, or X. The term X^n is conventionally called the degree-n term of the complex X.

This chapter is chiefly concerned with abelian categories, where the cohomology and exactness of complexes can be studied.

Definition 2.2.5 (Cohomology and exact sequences) Let $\mathcal{A}$ be an abelian category.

- ◇ For a complex X, in accordance with (2.2.2), its **cohomology** at any non-end term X^n is defined by

$$\mathrm{H}^n(X) = \mathrm{H}^n(X^\bullet, d^\bullet) := \mathrm{H}\left[X^{n-1} \xrightarrow{d^{n-1}} X^n \xrightarrow{d^n} X^{n+1}\right].$$

- ◇ If $\mathrm{H}^n(X) = 0$, the complex X is said to be exact at X^n; this is equivalent to $\mathrm{im}(d^{n-1}) = \ker(d^n)$. A complex that is exact at every term is called an **exact sequence**; an exact complex is also called **acyclic**.

The properties of cohomology are one of the main themes of this book and will be studied further in §3.6.

Remark 2.2.6 (Homology of chain complexes) In Definition 2.2.4, choose decreasing indices and use subscripts, namely

$$\cdots \to X_n \xrightarrow{d_n} X_{n-1} \to \cdots, \quad d_{n-1}d_n = 0,$$

and all the preceding definitions apply unchanged. By conventions originating in topology, the version with increasing indices $(X^\bullet, d^\bullet)$ is usually called a cochain complex, while the version with decreasing indices $(X_\bullet, d_\bullet)$ is called a chain complex. For a chain complex X in an abelian category, the **homology** object at X_n is defined as the following object of $\mathcal{A}$:

$$\mathrm{H}_n(X) = \mathrm{H}_n(X_\bullet, d_\bullet) := \mathrm{H}\left[X_{n+1} \xrightarrow{d_{n+1}} X_n \xrightarrow{d_n} X_{n-1}\right].$$

In algebra, complexes (that is, cochain complexes) and chain complexes differ only in notation, and passing between them is straightforward. For an additive category $\mathcal{A}$, one way to remain entirely within the same category is to reflect the indices by setting

$$X^n := X_{-n}, \quad d^n := d_{-n}.$$

Another way is to reverse the arrows, which involves the opposite category. If a complex in $\mathcal{A}$

$$\cdots \to X^n \xrightarrow{d^n} X^{n+1} \to \cdots$$

is viewed in $\mathcal{A}^{\mathrm{op}}$, it becomes the chain complex

$$\begin{gathered}\cdots \leftarrow X_n \xleftarrow{d_{n+1}} X_{n+1} \leftarrow \cdots\,, \\ X_n := X^n, \quad d_{n+1} := d^{n,\mathrm{op}}.\end{gathered}$$

If $\mathcal{A}$ is an abelian category and the chain complex $(X_\bullet, d_\bullet)$ in $\mathcal{A}^{\mathrm{op}}$ is defined by reversing the arrows, there is a canonical isomorphism $\mathrm{H}^n(X^\bullet) \simeq \mathrm{H}_n(X_\bullet)$. In particular, $X^\bullet$ is exact if and only if $X_\bullet$ is exact. To see this, it suffices to note that $\mathrm{H}[X \xrightarrow{f} Y \xrightarrow{g} Z]$ can be characterized as the middle term in the epi–mono factorization of $\ker(g) \to \operatorname{coker}(f)$. Viewed in $\mathcal{A}^{\mathrm{op}}$, the same object is also the middle term in the epi–mono factorization of $\ker(f^{\mathrm{op}}) \to \operatorname{coker}(g^{\mathrm{op}})$.

Exactness of a complex can be checked term by term. The following standard forms will henceforth be used without further explanation.

- ▷ Monomorphism The complex $0 \to X' \xrightarrow{f} X$ is exact if and only if $\ker(f) = 0$, that is, if and only if f is a monomorphism (Lemma 1.3.11). This is also equivalent to $X' \xrightarrow{\sim} \operatorname{coim}(f)$ (Lemma 1.2.6).

- ▷ Epimorphism Dually, the complex $X \xrightarrow{g} X'' \to 0$ is exact if and only if g is an epimorphism, and this is also equivalent to $\operatorname{im}(g) \xrightarrow{\sim} X''$.

- ▷ Kernel Consider the complex $0 \to X' \xrightarrow{f} X \xrightarrow{g} X''$. In accordance with (2.2.1), factor f through $X' \twoheadrightarrow \operatorname{im}(f) \hookrightarrow \ker(g)$. The claim is that $0 \to X' \to X \to X''$ is exact if and only if $X' \to \ker(g)$ is an isomorphism. In other words, up to isomorphism, an exact sequence of this form is always $0 \to \ker(g) \to X \xrightarrow{g} X''$ and is given by the kernel of a morphism.

  The argument is easy. Consider (2.2.1). The complex is exact at X if and only if $\operatorname{im}(f) \xrightarrow{\sim} \ker(g)$, while exactness at X' is equivalent to $X' \xrightarrow{\sim} \operatorname{im}(f)$. Now apply Proposition 1.1.2 to $X' \twoheadrightarrow \operatorname{im}(f) \hookrightarrow \ker(g)$.

- ▷ Cokernel Dually, the complex $X' \xrightarrow{f} X \xrightarrow{g} X'' \to 0$ induces a morphism $\operatorname{coker}(f) \to X''$ through (2.2.1). This complex is exact if and only if $\operatorname{coker}(f) \xrightarrow{\sim} X''$. Thus, up to isomorphism, an exact sequence of this form arises from a cokernel.

- ▷ Short exact sequence An exact sequence of the form $0 \to X' \to X \to X'' \to 0$ is called a **short exact sequence**. The two preceding items together give two equivalent, mutually dual characterizations of exactness:

◇ $X' \to X$ is a monomorphism and $X'' = \mathrm{coker}[X' \hookrightarrow X]$;
◇ $X \to X''$ is an epimorphism and $X' = \ker[X \twoheadrightarrow X'']$.

As an application, every morphism $f : X \to Y$ gives short exact sequences

$$0 \to \ker(f) \to X \to \mathrm{im}(f) \to 0, \quad 0 \to \mathrm{im}(f) \to Y \to \mathrm{coker}(f) \to 0,$$

which can be spliced into the exact sequence $0 \to \ker(f) \to X \xrightarrow{f} Y \to \mathrm{coker}(f) \to 0$.

Next, suppose that f can be embedded in an exact sequence of the form

$$L' \xrightarrow{\lambda} L \to X \xrightarrow{f} Y \to R \xrightarrow{\rho} R''$$

with both λ and ρ isomorphisms. Exactness then forces both $L \to X$ and $Y \to R$ to be zero, so $\ker(f) = 0$ and $\mathrm{im}(f) = Y$; in other words, f is an isomorphism. This is a commonly used technique for testing whether a morphism is an isomorphism.

The biproducts considered in §1.3 give a very simple class of short exact sequences; Proposition 2.5.3 will characterize them.

Proposition 2.2.7 Let X_1, X_2 be objects of an abelian category $\mathcal{A}$. Then $0 \to X_1 \xrightarrow{\iota_1} X_1 \oplus X_2 \xrightarrow{p_2} X_2 \to 0$ is a short exact sequence.

Proof Take the short exact sequences $0 \to X_1 \xrightarrow{\mathrm{id}} X_1 \to 0 \to 0$ and $0 \to 0 \to X_2 \xrightarrow{\mathrm{id}} X_2 \to 0$. Proposition 2.2.3 shows that their direct sum is still exact. □

2.3 Some Diagram Lemmas

The arguments in this section follow [23, Chapters 8 and 12]. We begin with a criterion for exactness that may be viewed as a substitute for diagram chasing [51, § 6.8] in general abelian categories. Throughout this section, fix an abelian category $\mathcal{A}$.

Lemma 2.3.1 Let $X' \xrightarrow{f} X \xrightarrow{g} X''$ be a complex. This complex is exact if and only if, for every morphism $S \xrightarrow{h} X$ satisfying $gh = 0$, there are a morphism $S' \to X'$ and an epimorphism $S' \twoheadrightarrow S$ that make the following diagram commute:

$$\begin{tikzcd} S' \arrow[r, two heads] \arrow[d] & S \arrow[d, "h"] \\ X' \arrow[r, "f"'] & X. \end{tikzcd}$$

Proof First consider the "only if" direction. The morphisms f and h both factor through $\ker(g)$. On the basis of these factorizations, take $S' := S \underset{\ker(g)}{\times} X'$, and take

$S' \to S$ and $S' \to X'$ to be the natural morphisms from the fiber product. Exactness implies that $X' \twoheadrightarrow \ker(g)$, so Proposition 2.1.6 implies that $S' \to S$ is also an epimorphism.

Now consider the "if" direction. For $S := \ker(g)$ with its natural morphism $h : S \hookrightarrow X$, the hypothesis supplies the solid part of the following diagram:

$$\begin{tikzcd} S' \arrow[r, two heads] \arrow[d] & S \arrow[d, "h"] \\ X' \arrow[r, "f"'] \arrow[ur, dashed, two heads, "\alpha"] & X \end{tikzcd}$$

The dashed part, namely α, comes from the factorization of f given by $gf = 0$. We show that the diagram commutes. The outer frame and the lower-right triangle are already known to commute. Thus the two morphisms in the upper-left triangle become equal after composition with h. Since h is a monomorphism, the upper-left triangle also commutes, proving the claim. Finally, the diagram shows that the composite $S' \to X' \xrightarrow{\alpha} S$ is an epimorphism, and hence α is an epimorphism. Thus f has an epi–mono factorization $X' \twoheadrightarrow \ker(g) \hookrightarrow X$, which gives $\mathrm{im}(f) = \ker(g)$. The complex is therefore exact. ☐

For another approach that replaces diagram chasing in general abelian categories, see [42, Tag 05PL].

We now discuss the Snake Lemma, which is indispensable in applications. Consider the diagram

$$\begin{tikzcd} & \ker' \arrow[r] \arrow[d] & \ker \arrow[r] \arrow[d] & \ker'' \arrow[d] \arrow[ddll, dashed, rounded corners, to path={ -- ([xshift=4ex]\tikztostart.east) |- ([yshift=-2ex]\tikztostart.south |- X''.south) -| ([xshift=-6ex]\tikztotarget.west) |- (\tikztotarget.west) }, "\delta"] & \\ & X' \arrow[r, "f"] \arrow[d, dashed] & X \arrow[r, "g"] \arrow[d, dashed] & |[alias=X'']| X'' \arrow[r] \arrow[d, dashed] & 0 \\ 0 \arrow[r] & Y' \arrow[r, "u"] \arrow[d] & Y \arrow[r, "v"] \arrow[d] & Y'' \arrow[d] & \\ & \mathrm{coker}' \arrow[r] & \mathrm{coker} \arrow[r] & \mathrm{coker}'' & \end{tikzcd} \tag{2.3.1}$$

subject to the following assumptions:

- $X' \xrightarrow{f} X \xrightarrow{g} X'' \to 0$ and $0 \to Y' \xrightarrow{u} Y \xrightarrow{v} Y''$ are given exact sequences;
- $\ker' := \ker[X' \to Y'] \hookrightarrow X'$, with $\ker$ and $\ker''$ defined in the same way;
- $Y' \twoheadrightarrow \mathrm{coker}' := \mathrm{coker}[X' \to Y']$, with coker and coker'' defined in the same way;
- the morphisms $\ker' \to \ker$, $\mathrm{coker}' \to \mathrm{coker}$, and so forth come from the functoriality of kernels and cokernels in (1.3.4).

Thus (2.3.1) is a commutative diagram; every column and both middle rows are exact. We now explain how to construct the dashed **connecting morphism** $\delta : \ker'' \to \operatorname{coker}'$.

The first step is to derive the following commutative diagram from (2.3.1); all its rows are exact:

$$\begin{tikzcd}
0 \arrow[r] & \ker(g) \arrow[r, hook, "s"] \arrow[d, equal] & X \underset{X''}{\times} \ker'' \arrow[r, two heads] \arrow[d] \arrow[dr, phantom, "\square"] & \ker'' \arrow[d] & \\
0 \arrow[r] & \ker(g) \arrow[r, hook] \arrow[d, dashed] & X \arrow[r, "g"'] \arrow[d] & X'' \arrow[r] \arrow[d, dashed] & 0 \\
0 \arrow[r] & Y' \arrow[r, "u"] \arrow[d] \arrow[dr, phantom, "\boxplus"] & Y \arrow[r, two heads] \arrow[d] & \operatorname{coker}(u) \arrow[r] \arrow[d, equal] & 0 \\
 & \operatorname{coker}' \arrow[r, hook] & \operatorname{coker}' \underset{Y'}{\sqcup} Y \arrow[r, two heads, "t"'] & \operatorname{coker}(u) \arrow[r] & 0
\end{tikzcd} \tag{2.3.2}$$

Here the following points apply:

- $\square$ and $\boxplus$ indicate that the corresponding squares are, respectively, a pullback and a pushout diagram. Pullbacks preserve kernels, while pushouts preserve cokernels; this explains the upper-left and lower-right squares in the diagram;
- Proposition 2.1.6 applies to the $\square$ and $\boxplus$ parts of (2.3.2). This explains why $X \underset{X''}{\times} \ker'' \to \ker''$ is an epimorphism and $\operatorname{coker}' \to \operatorname{coker}' \underset{Y'}{\sqcup} Y$ is a monomorphism;
- the dashed arrow $\ker(g) \dashrightarrow Y'$ is none other than the natural morphism $\ker(g) \to \ker(v)$ coming from the functoriality of kernels in (1.3.4);
- similarly, the functoriality of cokernels gives the dashed arrow $X'' \simeq \operatorname{coker}(f) \to \operatorname{coker}(u)$.

Notice that the composite $\ker(g) \dashrightarrow Y' \to \operatorname{coker}'$ in (2.3.2) is zero, because it becomes zero after precomposition with $X' \twoheadrightarrow \operatorname{im}(f) \xrightarrow{\sim} \ker(g)$. Similarly, the composite $\ker'' \to X'' \dashrightarrow \operatorname{coker}(u)$ is zero, because its further composite with $\operatorname{coker}(u) \xrightarrow{\sim} \operatorname{im}(v) \hookrightarrow Y''$ is zero.

Denote the composite along the middle path in (2.3.2) by $\delta_0 : X \underset{X''}{\times} \ker'' \to \operatorname{coker}' \underset{Y'}{\sqcup} Y$. The preceding observation immediately gives

$$\delta_0 s = 0, \quad t\delta_0 = 0.$$

Consequently, δ_0 has a unique factorization

$$X \underset{X''}{\times} \ker'' \twoheadrightarrow \operatorname{coker}(s) \xrightarrow{\exists!} \ker(t) \hookrightarrow \operatorname{coker}' \underset{Y'}{\sqcup} Y.$$

The horizontal morphisms in the upper-right and lower-left corners of (2.3.2) are known to be an epimorphism and a monomorphism, respectively. Hence $\operatorname{coker}(s) \xrightarrow{\sim}$

$\ker''$ and $\mathrm{coker}' \xrightarrow{\sim} \ker(t)$. We have therefore obtained the desired canonical morphism $\delta : \ker'' \to \mathrm{coker}'$.

Remark 2.3.2 The canonical nature of the connecting morphism can be stated more explicitly as follows. Consider the commutative diagram

$$
\begin{tikzcd}
 & & X' \arrow[rr] \arrow[dl] \arrow[dd] & & X \arrow[rr] \arrow[dl] \arrow[dd] & & X'' \arrow[r] \arrow[dl] \arrow[dd] & 0 \\
0 \arrow[r] & Y' \arrow[rr, crossing over] \arrow[dd] & & Y \arrow[rr, crossing over] \arrow[dd] & & Y'' \arrow[dd] & & \\
 & & \underline{X}' \arrow[rr] \arrow[dl] & & \underline{X} \arrow[rr] \arrow[dl] & & \underline{X}'' \arrow[r] \arrow[dl] & 0 \\
0 \arrow[r] & \underline{Y}' \arrow[rr] & & \underline{Y} \arrow[rr] & & \underline{Y}'' & &
\end{tikzcd}
$$

with every row exact. The diagram gives rise to $\ker'$, $\underline{\ker}'$, and so forth. The corresponding connecting morphisms δ and $\underline{\delta}$ fit into the commutative diagram

$$
\begin{tikzcd}
\ker'' \arrow[r, "\delta"] \arrow[d, "\text{natural morphism}"'] & \mathrm{coker}' \arrow[d, "\text{natural morphism}"] \\
\underline{\ker}'' \arrow[r, "\underline{\delta}"'] & \underline{\mathrm{coker}}'.
\end{tikzcd}
$$

By definition, to prove this it suffices to show that δ_0 and $\underline{\delta}_0$ also fit into the corresponding commutative square. Everything follows from the functoriality of kernels, cokernels, products, and coproducts. The details are left to the reader.

The following two observations will be used in the proof below.

- ◇ The construction of the connecting morphism is self-dual: when (2.3.1) is viewed in $\mathcal{A}^{\mathrm{op}}$, all arrows are reversed, while the roles of $\ker'$ and coker', and so forth, are interchanged. The resulting diagram in $\mathcal{A}^{\mathrm{op}}$ still resembles (2.3.1), except that it is rotated through half a turn. When the corresponding connecting morphism is viewed back in $\mathcal{A}$, it still runs from $\ker''$ to coker', and it agrees with the δ constructed above.

- ◇ The morphism $\ker \to \ker''$ factors as $\ker \to X \underset{X''}{\times} \ker'' \to \ker''$. Inspection of (2.3.2) shows that the two composites

$$\ker \to \ker'' \xrightarrow{\delta} \mathrm{coker}' \hookrightarrow \mathrm{coker}' \underset{Y'}{\sqcup} Y, \qquad \ker \to X \underset{X''}{\times} \ker'' \xrightarrow{\delta_0} \mathrm{coker}' \underset{Y'}{\sqcup} Y$$

are equal. But the second composite is also equal to $\ker \to X \to Y \to \mathrm{coker}' \underset{Y'}{\sqcup} Y$, which is zero. We therefore obtain

$$\ker \to \ker'' \xrightarrow{\delta} \mathrm{coker}' \quad \text{has composite } 0. \tag{2.3.3}$$

Theorem 2.3.3 (Snake Lemma) Consider diagram (2.3.1) in an abelian category $\mathcal{A}$. Then $\ker' \to \ker \to \ker'' \xrightarrow{\delta} \operatorname{coker}' \to \operatorname{coker} \to \operatorname{coker}''$ is an exact sequence. If $f : X' \to X$ in the diagram is a monomorphism (or $v : Y \to Y''$ is an epimorphism), then $\ker' \to \ker$ is also a monomorphism (or $\operatorname{coker} \to \operatorname{coker}''$ is also an epimorphism).

Proof The last assertion is easy to prove: if f is a monomorphism, then the composite $\ker' \hookrightarrow X' \xrightarrow{f} X$ is also a monomorphism. Since (2.3.1) commutes, this implies that $\ker' \to \ker$ is a monomorphism. The case in which v is an epimorphism is obtained by reversing all arrows.

The main part of the proof is the exactness of the first sequence. By the duality observed above, it suffices to prove that $\ker' \to \ker \to \ker'' \to \operatorname{coker}'$ is exact.

First, $\ker' \to \ker \to \ker''$ is a complex. Indeed, it is induced by $X' \to X \to X''$, whose composite is zero. We now use Lemma 2.3.1 to prove exactness. Suppose that a morphism $\psi : S \to \ker$ makes the composite $S \xrightarrow{\psi} \ker \to \ker''$ zero. We must construct a commutative diagram

$$\begin{tikzcd}
S' \arrow[r, two heads, "h"] \arrow[d] & S \arrow[d, "\psi"] \arrow[dr, "0"] & \\
\ker' \arrow[r] \arrow[d, hook] & \ker \arrow[r] \arrow[d, hook] & \ker'' \arrow[d, hook] \\
X' \arrow[r, "f"'] & X \arrow[r, "g"'] & X''
\end{tikzcd} \tag{2.3.4}$$

where $\ker' \leftarrow S' \xrightarrow{h} S$ remains to be constructed. To do so, first apply Lemma 2.3.1 to the composite $S \xrightarrow{\psi} \ker \to X$ to obtain the commutative diagram

$$\begin{tikzcd}
S' \arrow[r, two heads, "h"] \arrow[d] & S \arrow[d] \arrow[dr, "0"] & \\
X' \arrow[r, "f"'] & X \arrow[r, "g"'] & X''.
\end{tikzcd}$$

Thus the composite $S' \to X' \to Y' \xrightarrow{u} Y$ is equal to the composite $S' \xrightarrow{\psi h} \ker \to X \to Y$, namely zero. Since u is a monomorphism, the composite $S' \to X' \to Y'$ is also zero. Hence $S' \to X'$ factors through $\ker'$, giving all the arrows in (2.3.4).

Now consider the left part of (2.3.4): the outer frame and the lower square commute. The familiar Lemma 1.2.8 shows that the upper square also commutes. Therefore $\ker' \to \ker \to \ker''$ is exact.

Next consider $\ker \to \ker'' \xrightarrow{\delta} \operatorname{coker}'$. Equation (2.3.3) first shows that this sequence is a complex. To prove its exactness, we again use Lemma 2.3.1. Suppose that a morphism $\psi : S \to \ker''$ satisfies $\delta\psi = 0$. We must construct a commutative diagram of the form

$$\begin{tikzcd} S_0 \arrow[r, two heads] \arrow[d] & S \arrow[d, "\psi"] \\ \ker \arrow[r] & \ker'' \end{tikzcd} \tag{2.3.5}$$

The construction of δ showed that $X \underset{X''}{\times} \ker'' \to \ker'' \to 0$ is exact. Hence there is a commutative diagram

$$\begin{tikzcd} S_1 \arrow[r, two heads] \arrow[d] & S \arrow[d, "\psi"'] \arrow[dr, "0"] & \\ X \underset{X''}{\times} \ker'' \arrow[r] & \ker'' \arrow[r] & 0. \end{tikzcd} \tag{2.3.6}$$

Consider the composite $S_1 \to X \underset{X''}{\times} \ker'' \to X \to Y \xrightarrow{v} Y''$. Comparing the diagram above with (2.3.1) and (2.3.2), we see that this composite is zero. Hence $S_1 \to X \underset{X''}{\times} \ker'' \to X \to Y$ factors as $S_1 \xrightarrow{\xi} Y' \xrightarrow{u} Y$. We now prove that the composite $S_1 \xrightarrow{\xi} Y' \to \operatorname{coker}'$ is zero. Begin with the commutative diagram

$$\begin{tikzcd} S \arrow[r, "\psi"] & \ker'' \arrow[r, "\delta"] & \operatorname{coker}' \arrow[d, hook] \\ S_1 \arrow[u, two heads] \arrow[r] & X \underset{X''}{\times} \ker'' \arrow[u] \arrow[r, "\delta_0"] \arrow[d] & \operatorname{coker}' \underset{Y'}{\sqcup} Y \\ & X \arrow[r] & Y \arrow[u] \end{tikzcd}$$

The composite along the first row is $\delta\psi = 0$, so the composite along the second row is also zero. Consequently, the composite $S_1 \xrightarrow{\xi} Y' \xrightarrow{u} Y \to \operatorname{coker}' \underset{Y'}{\sqcup} Y$ is zero. On the other hand, the composite $Y' \xrightarrow{u} Y \to \operatorname{coker}' \underset{Y'}{\sqcup} Y$ equals $Y' \to \operatorname{coker}' \to \operatorname{coker}' \underset{Y'}{\sqcup} Y$. In the construction of δ we proved that $\operatorname{coker}' \to \operatorname{coker}' \underset{Y'}{\sqcup} Y$ is a monomorphism. Thus the composite $S_1 \xrightarrow{\xi} Y' \to \operatorname{coker}'$ is indeed zero.

The next step is to apply Lemma 2.3.1 to the exact sequence $X' \to Y' \to \operatorname{coker}'$. This gives a commutative diagram

$$\begin{tikzcd} S_0 \arrow[r, two heads] \arrow[d, "k"'] & S_1 \arrow[d, "\xi"] \arrow[dr, "0"] & \\ X' \arrow[r] & Y' \arrow[r] & \operatorname{coker}'. \end{tikzcd}$$

Write λ for the composite $S_0 \twoheadrightarrow S_1 \to X \underset{X''}{\times} \ker'' \to X$. Also write the morphisms $X' \to Y'$ and $X \to Y$ as a and b, respectively. The preceding discussion gives the commutative diagram

$$\begin{tikzcd}
S_0 \arrow[r, two heads] \arrow[d, "k"'] \arrow[drr, "\lambda"] & S_1 \arrow[r] \arrow[d, "\xi"'] & X \underset{X''}{\times} \ker'' \arrow[d] \\
X' \arrow[r, "a"] \arrow[d, "f"'] & Y' \arrow[d, "u"'] & X \arrow[dl, "b"] \\
X \arrow[r, "b"] & Y &
\end{tikzcd}$$

Both squares in this diagram are known to commute; commutativity of the triangle comes from the definition of λ, while commutativity of the trapezoid on the right comes from the preceding discussion of ξ.

It follows at once that $b\lambda = bfk$, so $\lambda - fk : S_0 \to X$ factors through $\ker = \ker(b) \hookrightarrow X$. Finally, consider the diagram

$$\begin{tikzcd}
S_0 \arrow[r, two heads] \arrow[d] \arrow[dd, bend right=60, "\lambda - fk"'] & S_1 \arrow[r, two heads] & S \arrow[d, "\psi"] \\
\ker \arrow[rr] \arrow[d, hook] & & \ker'' \arrow[d, hook] \\
X \arrow[rr, "g"] & & X''.
\end{tikzcd}$$

To obtain the diagram in (2.3.5), it suffices to show that its upper square commutes. The lower square and the curved part are known to commute; the usual argument reduces the problem to commutativity of the outer frame. Since $g(\lambda - fk) = g\lambda$, this in turn reduces to the known commutative diagram (see (2.3.6)):

$$\begin{tikzcd}
S_0 \arrow[r, two heads] \arrow[ddr, bend right, "\lambda"'] & S_1 \arrow[r, two heads] \arrow[d] & S \arrow[d, "\psi"] \\
& X \underset{X''}{\times} \ker'' \arrow[r] \arrow[d] & \ker'' \arrow[d] \\
& X \arrow[r, "g"] & X''.
\end{tikzcd}$$

All the required conditions are therefore satisfied. □

The connecting homomorphism δ and Theorem 2.3.3 are much simpler in the category R-**Mod** of R-modules; see [51, Proposition 6.8.6].

Proposition 2.3.4 (Five Lemma) Consider a commutative diagram with exact rows in an abelian category $\mathcal{A}$:

$$\begin{tikzcd}
X_1 \arrow[r] \arrow[d, "f_1"'] & X_2 \arrow[r] \arrow[d, "f_2"'] & X_3 \arrow[r] \arrow[d, dashed, "f_3"] & X_4 \arrow[r] \arrow[d, "f_4"] & X_5 \arrow[d, "f_5"] \\
Y_1 \arrow[r] & Y_2 \arrow[r] & Y_3 \arrow[r] & Y_4 \arrow[r] & Y_5
\end{tikzcd}$$

(i) If f_1 is an epimorphism and f_2, f_4 are monomorphisms, then f_3 is a monomorphism (only the first four columns are involved);

(ii) if f_5 is a monomorphism and f_2, f_4 are epimorphisms, then f_3 is an epimorphism (only the last four columns are involved);

(iii) if f_1 is an epimorphism, f_5 is a monomorphism, and f_2, f_4 are both isomorphisms, then f_3 is an isomorphism.

Proof Clearly (i) and (ii) are dual to each other, while (iii) follows from (i)–(ii) and Proposition 1.2.7. It therefore suffices to prove (i).

Suppose that a morphism $h : S \to X_3$ satisfies $f_3 h = 0$; we must prove that $h = 0$. The composite $S \xrightarrow{h} X_3 \to X_4 \xrightarrow{f_4} Y_4$ is zero, and therefore so is $S \xrightarrow{h} X_3 \to X_4$. Applying Lemma 2.3.1 to the exact sequence $X_2 \to X_3 \to X_4$ gives a commutative diagram

$$\begin{tikzcd}
S' \arrow[r, two heads] \arrow[d] & S \arrow[d, "h"] \\
X_2 \arrow[r] & X_3.
\end{tikzcd}$$

We now construct a commutative diagram

$$\begin{tikzcd}
S''' \arrow[r, two heads] \arrow[dd] & S'' \arrow[r, two heads] \arrow[dd] & S' \arrow[d] \\
 & & X_2 \arrow[d, "f_2"] \\
X_1 \arrow[r, "f_1"'] & Y_1 \arrow[r] & Y_2.
\end{tikzcd}$$

Here is how. Since $f_3 h = 0$, the composite $S' \to X_2 \xrightarrow{f_2} Y_2 \to Y_3$ is zero. Applying Lemma 2.3.1 to $S' \to Y_2$ and the exact sequence $Y_1 \to Y_2 \to Y_3$ gives the right-hand part of the diagram. Applying Lemma 2.3.1 to $S'' \to Y_1$ and the exact sequence $X_1 \xrightarrow{f_1} Y_1 \to 0$ gives the left-hand part.

Take the composite $S''' \to S'$ and consider the following diagram:

$$\begin{tikzcd}
S''' \arrow[r, two heads] \arrow[d] & S' \arrow[r, two heads] \arrow[d] & S \arrow[d, "h"] & \\
X_1 \arrow[r] \arrow[d, "f_1"'] & X_2 \arrow[r] \arrow[d, "f_2"] & X_3 \arrow[r] & X_4 \\
Y_1 \arrow[r] & Y_2 & &
\end{tikzcd}$$

We show that this diagram commutes. The only issue is the upper-left square. Compose the two paths ↴ and ↳ in that square with f_2 on the right. Since the two squares

$\begin{array}{ccc} X_1 & \longrightarrow & X_2 \\ \downarrow & & \downarrow \\ Y_1 & \longrightarrow & Y_2 \end{array}$ and $\begin{array}{ccc} S''' & \longrightarrow & S' \\ \downarrow & & \downarrow \\ Y_1 & \longrightarrow & Y_2 \end{array}$ commute and f_2 is a monomorphism, the upper-left square also commutes.

Thus the composite $S''' \twoheadrightarrow S' \twoheadrightarrow S \xrightarrow{h} X_3$ is zero, and hence $h = 0$. This proves the result. $\square$

This section has introduced only two of the most commonly used diagram lemmas. Using the language of double complexes, G. Bergman found the comprehensive Salamander Lemma, which subsumes the Snake Lemma and several classical diagram lemmas of homological algebra. Interested readers may consult [3].

2.4 A Glimpse of Lattice Theory

This section gives a brief introduction to a class of partially ordered structures called lattices. The discussion will help clarify the structure of abelian categories.

Throughout this section we consider various partially ordered sets $(P, \leq)$; the corresponding category is denoted by $\mathcal{P}$.

For every subset S of a partially ordered set $(P, \leq)$, there are notions of upper and lower bounds, a supremum $\sup S$, and an infimum $\inf S$. A subset $S \subset P$ has a supremum (or infimum) if and only if the coproduct (or product) of all the objects in S exists in the category $\mathcal{P}$; in that case the supremum (or infimum) is precisely that coproduct (or product).

Definition 2.4.1 If every two elements a, b of a partially ordered set $(P, \leq)$ have a supremum $a \vee b$ and an infimum $a \wedge b$, then $(P, \leq)$ is called a **lattice**. If, in addition, $(P, \leq)$ itself has an upper bound and a lower bound, they are both unique and are denoted by 1 and 0, respectively; in this case P is called a **bounded lattice**.

If a, b are elements of a partially ordered set $(P, \leq)$ and $a \leq b$, then $[a, b] := \{x \in P : a \leq x \leq b\}$ is also a partially ordered set under $\leq$ and is called the **interval** determined by a, b.

If P is a lattice, every interval in P is closed under $\vee$ and $\wedge$ and is a bounded lattice. If P is a bounded lattice, then $P = [0, 1]$, and for every $x \in P$ one has $x \wedge 0 = 0$, $x \vee 0 = x = x \wedge 1$, and $x \vee 1 = 1$.

From the categorical point of view, $a \wedge b$, $a \vee b$, 1, and 0 correspond, respectively, to the product of two objects, the coproduct of two objects, a terminal object (that is, an empty product), and an initial object (that is, an empty coproduct) in the category; this can be checked directly.

As an illustration of the operations $\wedge$ and $\vee$, prove the following fact. If $a^\flat, a, b \in P$ satisfy $a^\flat \leq a$, then

$$a^\flat \vee (a \wedge b) \leq (a^\flat \vee b) \wedge a. \tag{2.4.1}$$

By the definition of a supremum, it first suffices to prove $a^\flat \leq (a^\flat \vee b) \wedge a$ and $(a \wedge b) \leq (a^\flat \vee b) \wedge a$. By the definition of an infimum, it then suffices to prove, respectively, the inequalities

$$a^\flat \leq a^\flat \vee b, \quad a^\flat \leq a, \quad a \wedge b \leq a^\flat \vee b, \quad a \wedge b \leq a;$$

the third follows from $a \wedge b \leq b \leq a^\flat \vee b$, and the others are clear.

Definition 2.4.2 Let P be a lattice.

- P is called a **modular lattice** if the following property holds: whenever $a^\flat, a, b \in P$ satisfy $a^\flat \leq a$, then
$$a^\flat \vee (a \wedge b) = (a^\flat \vee b) \wedge a.$$

- Suppose that P is a bounded lattice. If $x, c \in P$ satisfy $x \vee c = 1$ and $x \wedge c = 0$, then c is called a **complement** of x in P.

We now consider some basic examples of lattices.

- The set of positive integers $\mathbb{Z}_{\geq 1}$ is a lattice under divisibility ($x \mid y \iff x \leq y$): $x \vee y = \mathrm{lcm}(x, y)$ and $x \wedge y = \gcd(x, y)$. The reader can check that this lattice is modular, has lower bound 1, but has no upper bound.

- Let R be a ring and M a left R-module. Its set of submodules Sub_M is a bounded modular lattice under $\subset$: $x \vee y = x + y$ and $x \wedge y = x \cap y$; its upper bound is M and its lower bound is $\{0\}$. This is the origin of the term "modular lattice."

- All closed subspaces of a Hilbert space H form a bounded lattice under $\subset$: $x \vee y := \overline{x + y}$ and $x \wedge y = x \cap y$. One can prove that this lattice is not modular when H is infinite-dimensional (Birkhoff–von Neumann).

Remark 2.4.3 The definition of a lattice also contains a duality. Given the partial order $\leq$ on a set P, we may consider the opposite partial order $\leq^{\mathrm{op}}$: $x \leq^{\mathrm{op}} y \iff x \geq y$; this is equivalent to replacing $\mathcal{P}$ by $\mathcal{P}^{\mathrm{op}}$. If $(P, \leq)$ is a lattice (or a bounded lattice), then so is $(P, \leq^{\mathrm{op}})$. Moreover, $\wedge, \vee, 0, 1$ in $(P, \leq)$ correspond, respectively, to $\vee, \wedge, 1, 0$ in $(P, \leq^{\mathrm{op}})$.

Lemma 2.4.4 Let P be a lattice. Then P is modular if and only if the following property holds for every interval I in P: if $c^\flat, c$ are both complements of $x \in I$ in I and $c^\flat \leq c$, then $c^\flat = c$.

Proof If P is a modular lattice, then every interval I is also a modular lattice. Without loss of generality, assume $I = P$; for the $x, c^\flat, c$ in the statement, we have

$$c = c \wedge 1 = c \wedge (x \vee c^\flat) = c^\flat \vee (x \wedge c) = c^\flat \vee 0 = c^\flat.$$

Conversely, assume that the condition on complements holds for every interval. Let $a^\flat, a, b \in P$ satisfy $a^\flat \leq a$. Set

$$b \wedge a \leq c_1 := a^\flat \vee (b \wedge a) \underset{\because(2.4.1)}{\leq} a \wedge (b \vee a^\flat) =: c_2 \leq b \vee a^\flat.$$

We prove that c_1, c_2 are both complements of b in $\left[b \wedge a,\ b \vee a^\flat\right]$, so that $c_1 = c_2$. First, the definition immediately gives $c_1 \leq a$; therefore

$$b \wedge a \geq b \wedge c_1 = (a^\flat \vee (b \wedge a)) \wedge b \underset{\because(2.4.1)}{\geq} (a^\flat \wedge b) \vee (b \wedge a) = b \wedge a,$$

so $b \wedge c_1 = b \wedge a$. On the other hand, $b \vee c_1 = a^\flat \vee (b \wedge a) \vee b = a^\flat \vee b$. Thus c_1 is indeed a complement of b in $\left[b \wedge a,\ b \vee a^\flat\right]$.

Lattice duality (Remark 2.4.3) reverses the order relation between $a^\flat, a$, interchanges $\wedge$ and $\vee$, and interchanges the roles of c_1 and c_2. Thus the preceding argument, carried out in $(P, \leq^{\mathrm{op}})$, shows that c_2 is a complement of b in $\left[b \wedge a,\ b \vee a^\flat\right]$. $\square$

Proposition 2.4.5 Let a, b be elements of a modular lattice $(P, \leq)$. Then there is an isomorphism of partially ordered sets

$$\begin{aligned}
[a \wedge b, a] &\xleftrightarrow{1:1} [b, a \vee b] \\
x &\longmapsto x \vee b \\
y \wedge a &\longleftarrow\!\shortmid\; y
\end{aligned}$$

called the **standard isomorphism**.

Proof The two maps are clearly well defined and order-preserving. For $x \in [a \wedge b, a]$, the definition of a modular lattice gives $(x \vee b) \wedge a = (a \wedge b) \vee x$, whose right-hand side is x. Dually, $y \in [b, a \vee b]$ implies $(y \wedge a) \vee b = y$ (Remark 2.4.3); thus the two maps are mutually inverse. $\square$

The Zassenhaus lemma of elementary algebra (see [51, Lemma 4.6.4], especially its module-theoretic version) and its consequences can be distilled using lattice theory.

Theorem 2.4.6 (Zassenhaus Lemma for Modular Lattices) Let $(P, \leq)$ be a modular lattice. For given elements $u^\flat \leq u$ and $v^\flat \leq v$, there is a diagram

$$\begin{tikzcd}[arrows=dash]
u^\flat \vee (u \wedge v) \arrow[dd] \arrow[dr] & & (u \wedge v) \vee v^\flat \arrow[dd] \arrow[dl] \\
 & u \wedge v \arrow[dd] & \\
u^\flat \vee (u \wedge v^\flat) \arrow[dr] & & (u^\flat \wedge v) \vee v^\flat \arrow[dl] \\
 & (u^\flat \wedge v) \vee (u \wedge v^\flat) &
\end{tikzcd}$$

meaning that its elements satisfy the following patterns:

$$\begin{tikzcd}[arrows=dash]
x \arrow[d, phantom, "\rotatebox{90}{$\geq$}"] \\
y
\end{tikzcd}
\qquad
\begin{tikzcd}[arrows=dash]
x \arrow[dr] & & y \arrow[dl] \\
 & x \wedge y &
\end{tikzcd}
\qquad
\begin{tikzcd}[arrows=dash]
 & x \vee y \arrow[dl] \arrow[dr] & \\
x & & y
\end{tikzcd}
. \tag{2.4.2}$$

Moreover, there are standard isomorphisms between the intervals

$$\left[u^\flat \vee (u \wedge v^\flat),\ u^\flat \vee (u \wedge v)\right] \xleftarrow{\sim} \left[(u^\flat \wedge v) \vee (u \wedge v^\flat),\ u \wedge v\right] \\ \xrightarrow{\sim} \left[(u^\flat \wedge v) \vee v^\flat,\ (u \wedge v) \vee v^\flat\right],$$

realized by the maps $x \vee u^\flat \vee (u \wedge v^\flat) \leftarrow\!\shortmid x \mapsto x \vee (u^\flat \wedge v) \vee v^\flat$ in Proposition 2.4.5.

Proof The isomorphisms in the last part are obtained by applying Proposition 2.4.5 to the two parallelograms in the diagram; what remains to be checked are the relations in (2.4.2). Notice that the diagram is left–right symmetric under $u \leftrightarrow v$ and $u^\flat \leftrightarrow v^\flat$; it therefore suffices to check its left half. The partial-order relations among its elements are immediate. For the $\wedge$ case, the definition of a modular lattice gives

$$\left(u^\flat \vee (u \wedge v^\flat)\right) \wedge (u \wedge v) = \left(u^\flat \wedge (u \wedge v)\right) \vee (u \wedge v^\flat) = (u^\flat \wedge v) \vee (u \wedge v^\flat).$$

For the $\vee$ case, the inequality $u \wedge v^\flat \leq u \wedge v$ immediately gives

$$u^\flat \vee (u \wedge v^\flat) \vee (u \wedge v) = u^\flat \vee (u \wedge v).$$

This proves (2.4.2). $\square$

Next consider a finite descending chain in a lattice, written $x_0 \geq x_1 \geq \cdots \geq x_r$ ($r \in \mathbb{Z}_{\geq 0}$). A descending chain obtained by inserting finitely many intermediate elements is called a **refinement** of the original descending chain; if the inserted elements include some $y \notin \{x_0, \ldots, x_r\}$, it is called a proper refinement.

Definition 2.4.7 Two descending chains of the same length in a modular lattice, $x_0 \geq \cdots \geq x_r$ and $x'_0 \geq \cdots \geq x'_r$, are called **equivalent** if the following condition holds:

there is a bijection σ from $\{0, \dots, r-1\}$ to itself (that is, a permutation)[2)] such that, for every $0 \le i < r$, there is an isomorphism of partially ordered sets

$$\tau_i : [x_{i+1}, x_i] \simeq \left[x'_{\sigma(i)+1}, x'_{\sigma(i)}\right],$$

and τ_i decomposes into a finite composite of standard isomorphisms (Proposition 2.4.5) or their inverses.

Theorem 2.4.8 (Schreier Refinement Theorem for Modular Lattices)

Consider descending chains

$$x_0 \ge \cdots \ge x_r, \quad y_0 \ge \cdots \ge y_s,$$

in a modular lattice $(P, \le)$ such that $x_0 = y_0$ and $x_r = y_s$, with $r, s \in \mathbb{Z}_{\ge 0}$. Then each chain has a refinement, and the two refinements are equivalent.

Proof The argument is the same as in the case of groups [51, Theorem 4.6.6]; both are based on Theorem 2.4.6. We give the outline. Define

$$x_{i,j} := x_{i+1} \vee (x_i \wedge y_j), \quad y_{j,i} := (x_i \wedge y_j) \vee y_{j+1},$$

with $0 \le i < r$ and $0 \le j \le s$ for $x_{i,j}$, and with $0 \le i \le r$ and $0 \le j < s$ for $y_{j,i}$. Then $x_{i,j+1} \le x_{i,j}$, $x_{i,0} = x_i$, and $x_{i,s} = x_{i+1}$, so $\left(x_{i,j}\right)_{i,j}$, in that order, refines $x_0 \ge \cdots \ge x_r$. Similarly, $\left(y_{j,i}\right)_{j,i}$ refines $y_0 \ge \cdots \ge y_s$. Taking $u^\flat = x_{i+1}$, $u = x_i$, $v^\flat = y_{j+1}$, and $v = y_j$ in Theorem 2.4.6, we obtain

$$[x_{i,j+1}, x_{i,j}] \simeq [y_{j,i+1}, y_{j,i}],$$

and this isomorphism decomposes into standard isomorphisms and their inverses. The assertion follows. □

From now on, we consider only strictly ascending or strictly descending chains.

Definition 2.4.9 Fix a partially ordered set P and elements $a \le b$ in it.[3)] If a descending chain $b = x_0 > \cdots > x_r = a$ in P has no proper refinement, it is called a **composition series** of $[a, b]$.

The length of this composition series is defined to be $r \in \mathbb{Z}_{\ge 0}$; notice that $r = 0$ if and only if $a = b$.

[2)] Translator's note (O014-C021): the source has $\{0, \dots, r\}$, but the successive factors being paired are indexed by $0 \le i < r$; the index $i = r$ would leave x_{i+1} undefined.

[3)] Translator's note (O014-C022): the source has $a < b$, but the next sentence explicitly includes the case $a = b$ and $r = 0$; the target uses the weaker hypothesis consistent with that case.

Theorem 2.4.10 (Jordan–Hölder Theorem for Modular Lattices) Let $a < b$ be elements of a modular lattice $(P, \leq)$. Then any two composition series of $[a, b]$ have the same length and are equivalent in the sense of Definition 2.4.7.

Proof This follows immediately from Theorem 2.4.8. □

The next important question is to determine which intervals in a partially ordered set have composition series. As in the familiar theory of modules, their existence can be guaranteed by ascending- and descending-chain conditions.

Definition 2.4.11 Let $(P, \leq)$ be a partially ordered set. If there is no infinite ascending chain $x_1 < x_2 < x_3 < \cdots$ in P, then P is called **Noetherian**; if there is no infinite descending chain $x_1 > x_2 > x_3 > \cdots$, then P is called **Artinian**. A partially ordered set that is both Artinian and Noetherian is said to have **finite length**.

If $(P, \leq)$ is Noetherian (or Artinian, or of finite length), then every subset of it has the same property. It is easy to prove that the Noetherian (or Artinian) condition is equivalent to the assertion that every nonempty subset of P has a maximal (or minimal) element with respect to $\leq$. In this chapter these notions will be applied chiefly to partially ordered sets of subobjects; see Definition 1.1.3.

Definition 2.4.12 Fix a category $\mathcal{C}$. An object X of $\mathcal{C}$ is called Noetherian (or Artinian, or of finite length) if its partially ordered set of subobjects $(\mathrm{Sub}_X, \subset)$ is Noetherian (or Artinian, or of finite length).

We now return to general partially ordered sets.

Lemma 2.4.13 Let $a \leq b$ be elements of a partially ordered set $(P, \leq)$.

(i) If $[a, b]$ has finite length, then every chain in it can be refined to a composition series; in particular, $[a, b]$ has a composition series.

(ii) If $[a, b]$ is a modular lattice and has a composition series, then $[a, b]$ has finite length.

Proof Without loss of generality, assume $a < b$. For (i), given a chain $y_0 > \cdots > y_r$, it suffices to show that every $[y_{i+1}, y_i]$ has a composition series.[4] Without loss of generality, assume $y_{i+1} = a$ and $y_i = b$. The Artinian condition guarantees a minimal element $x^0 \in [a, b]$ with $x^0 > a$. Similarly, if $x^0 \neq b$, there is a minimal element $x^1 \in [a, b]$ with $x^1 > x^0$, and so on. By the Noetherian condition, this process must

[4] Translator's note (O014-C023): the source has $[y_i, y_{i+1}]$ and then sets $y_i = a$, $y_{i+1} = b$. Since $y_i > y_{i+1}$, both orders are reversed; the target uses the interval $[y_{i+1}, y_i]$ and the well-typed assignment.

stop after finitely many steps, producing a chain $b > \cdots > x^0 > a$ with no proper refinement.

For (ii), choose a composition series $b = x_0 > \cdots > x_r = a$. Theorem 2.4.8 shows that every finite chain $\cdots > y_i > y_{i+1} > \cdots$ has a refinement equivalent to the composition series above; hence the length of that chain cannot exceed the length of the composition series. Thus $[a, b]$ is a partially ordered set of finite length. □

Definition 2.4.14 Let $(P, \leq)$ be a bounded modular lattice of finite length. Choose any composition series $1 = x_0 > \cdots > x_r = 0$ of P. The number $r \in \mathbb{Z}_{\geq 0}$ is called the **length** of $(P, \leq)$.

Theorem 2.4.10 shows that this length is well defined and does not depend on the chosen composition series; the length of $(P, \leq)$ is 0 if and only if P is a one-point set.

2.5 Direct-Sum Decompositions

We begin by discussing how morphisms between direct sums are represented by matrices. The argument is exactly the same as in the module-theoretic case [51, §6.3]; only a brief review is given here.

Fix an additive category $\mathcal{A}$. Consider objects $X_1, \ldots, X_n$ and $X'_1, \ldots, X'_m$ of $\mathcal{A}$. The direct sum $\bigoplus_{i=1}^n X_i$ introduced in Definition 1.3.3 comes with a family of morphisms $X_j \xrightarrow{\iota_j} \bigoplus_{i=1}^n X_i \xrightarrow{p_j} X_j$. Likewise, $\bigoplus_{i=1}^m X'_i$ has morphisms p'_j, ι'_j, and so on. The universal properties of products and coproducts give

$$\mathrm{Hom}\left(\bigoplus_{j=1}^n X_j, \bigoplus_{i=1}^m X'_i\right) \xrightarrow[\sim]{\phi \mapsto (p'_i\phi)_i} \prod_{i=1}^m \mathrm{Hom}\left(\bigoplus_{j=1}^n X_j, X'_i\right) \xrightarrow[\sim]{(p'_i\phi)_i \mapsto (p'_i\phi\iota_j)_{i,j}} \prod_{j=1}^n \prod_{i=1}^m \mathrm{Hom}(X_j, X'_i).$$

Thus every morphism $\phi : \bigoplus_{j=1}^n X_j \to \bigoplus_{i=1}^m X'_i$ corresponds to the matrix

$$\mathcal{M}(\phi) = (\phi_{ij})_{\substack{1\leq i\leq m\\ 1\leq j\leq n}} = \begin{pmatrix} \phi_{11} & \cdots & \phi_{1n} \\ \vdots & \ddots & \vdots \\ \phi_{m1} & \cdots & \phi_{mn} \end{pmatrix}, \quad \phi_{ij} := p'_i\phi\iota_j : X_j \to X'_i.$$

Conversely, every matrix $\mathcal{M} = (\phi_{ij})_{\substack{1\leq i\leq m\\ 1\leq j\leq n}}$ formed from morphisms $\phi_{ij} : X_j \to X'_i$ uniquely determines a morphism ϕ such that $\mathcal{M} = \mathcal{M}(\phi)$. Composition of morphisms agrees with matrix multiplication:

$$\mathcal{M}(\psi\phi) = \mathcal{M}(\psi)\mathcal{M}(\phi)$$

where multiplication of matrix entries is given by composition $\mathrm{Hom}(X'_j, X''_i) \times \mathrm{Hom}(X_k, X'_j) \to \mathrm{Hom}(X_k, X''_i)$.

Lemma 2.5.1 Given a family of morphisms $\phi_i : X_i \to X'_i$ in $\mathcal{A}$, $i = 1, \ldots, n$, set $\phi := (\phi_1, \ldots, \phi_n) : \bigoplus_{i=1}^n X_i \to \bigoplus_{i=1}^n X'_i$. Then ϕ is an isomorphism if and only if each ϕ_i is an isomorphism.

Proof The implication $\Longleftarrow$ is clear. For $\Longrightarrow$, write write the matrix $\mathcal{M}(\phi^{-1})$ as $(\psi_{ij})_{i,j}$. Then

$$\mathcal{M}(\phi^{-1})\mathcal{M}(\phi) = \begin{pmatrix} \psi_{11}\phi_1 & \cdots & \psi_{1n}\phi_n \\ \vdots & \ddots & \vdots \\ \psi_{n1}\phi_1 & \cdots & \psi_{nn}\phi_n \end{pmatrix} = \begin{pmatrix} \mathrm{id}_{X_1} & & \\ & \ddots & \\ & & \mathrm{id}_{X_n} \end{pmatrix}$$

so $\phi_1, \ldots, \phi_n$ all have left inverses. The same argument shows that they all have right inverses. □

Continue to consider a direct-sum decomposition $X \simeq \bigoplus_{i=1}^n X_i$, where $n \geq 1$ and $X_i \neq 0$ for every i. It gives morphisms $\iota_i : X_i \to X$ and $p_i : X \to X_i$. For each $1 \leq i \leq n$, set $e_i := \iota_i p_i \in \mathrm{End}(X)$. These morphisms have the following properties. Recall that an **idempotent** in the ring $\mathrm{End}(X)$ is an element $e \in \mathrm{End}(X)$ satisfying $e^2 = e$; when no confusion can arise, we also call e an idempotent in $\mathcal{A}$.

(E1) Every e_i is idempotent: $e_i^2 = (\iota_i p_i)(\iota_i p_i) = \iota_i(p_i\iota_i)p_i = \iota_i p_i$.

(E2) Orthogonality: $i \neq j \implies e_i e_j = \iota_i p_i \iota_j p_j = 0$.

(E3) The equality $\sum_{i=1}^n e_i = 1$ holds in the ring $\mathrm{End}(X)$.

From now on we regard the direct summands X_i as subobjects of X and write the decomposition as an equality $X = \bigoplus_{i=1}^n X_i$. Conversely, if $\mathcal{A}$ is abelian, a family of idempotents $e_1, \ldots, e_n \in \mathrm{End}(X)$ satisfying (E1)–(E3) determines subobjects of X by

$$X_i := \ker\left(\sum_{j \neq i} e_j\right) = \mathrm{im}(e_i), \quad i = 1, \ldots, n.$$

On the one hand they come with monomorphisms $X_i \xrightarrow{\iota_i} X$; on the other, $e_i : X \to X$ factors uniquely as $X \xrightarrow{p_i} X_i \xrightarrow{\iota_i} X$. This is precisely the data required for a direct sum.

Passing from the idempotents $e_1, \ldots, e_n$ to the direct summands $X_1, \ldots, X_n$ does not require all the properties of an abelian category. It is enough to assume that $\mathcal{A}$ is an Ab-category with a zero object in which every idempotent has a kernel. Such a category is called a **Karoubian category** or a **pseudo-abelian category**. The exercises of this chapter discuss this further.

Proposition 2.5.2 Let $X \neq 0$ be an object of an abelian category (or, more generally, a Karoubian category) $\mathcal{A}$, and let $n \in \mathbb{Z}_{\geq 1}$. The constructions above give a bijection

$$\left\{ \begin{array}{c} \text{direct-sum decompositions} \\ X = \displaystyle\bigoplus_{i=1}^{n} X_i \end{array} \right\} \xleftrightarrow{1:1} \left\{ \begin{array}{l} (e_i)_{i=1}^{n} \in \operatorname{End}(X)^n : \\ \text{satisfying (E1)–(E3)} \end{array} \right\}.$$

Reordering the direct summands $X_1, \dots, X_n$ corresponds to reordering the idempotents $e_1, \dots, e_n$.

Proof See [51, Proposition 6.12.4]. □

In the special case $n = 2$, direct-sum decompositions are related to a special class of short exact sequences, called split short exact sequences.

Proposition 2.5.3 For a short exact sequence $0 \to X' \xrightarrow{f} X \xrightarrow{g} X'' \to 0$ in an abelian category, the following statements are equivalent.

(i) There is an $s : X'' \to X$ such that $gs = \mathrm{id}_{X''}$.

(ii) There is an $r : X \to X'$ such that $rf = \mathrm{id}_{X'}$.

(iii) There is a diagram $X' \underset{f}{\overset{r}{\leftrightarrows}} X \underset{g}{\overset{s}{\leftrightarrows}} X''$ for which $X \simeq X' \oplus X''$; see the review of biproducts in §1.3.

(iv) The map $g_* : \operatorname{Hom}(S, X) \to \operatorname{Hom}(S, X'')$, sending φ to $g\varphi$, is surjective for every object S.

(v) The map $f^* : \operatorname{Hom}(X, S) \to \operatorname{Hom}(X', S)$, sending ψ to ψf, is surjective for every object S.

If any of these conditions holds, the short exact sequence $0 \to X' \to X \to X'' \to 0$ is called **split**. The morphism $s : X'' \to X$ in (i), the morphism $r : X \to X'$ in (ii),[5] and the isomorphism $X \xrightarrow{\sim} X' \oplus X''$ in (iii), denoted by Φ, correspond as follows:

- from Φ to s, take the composite $X'' \xrightarrow{\iota_2} X' \oplus X'' \xrightarrow{\Phi^{-1}} X$;
- from s to r, factor $\mathrm{id}_X - sg : X \to X$ uniquely as $X \xrightarrow{r} X' \xrightarrow{f} X$; this is the desired r;

[5] Editorial correction (O014-C024): the source writes $r : X' \to X$ here, but (ii), the equality $rf = \mathrm{id}_{X'}$, and the subsequent construction all require $r : X \to X'$. The type-correct direction is used here.

◇ from r to Φ, take $(r, g) : X \to X' \oplus X''$, which is an isomorphism.

Moreover, for a given s or its corresponding r, the orthogonal idempotents corresponding to the direct-sum decomposition in (iii) are

$$e' := fr = \mathrm{id}_X - sg, \quad e'' := sg = \mathrm{id}_X - fr.$$

Proof For modules, see [51, Proposition 6.8.5]. Its proof uses only algebraic operations in Hom sets and the characterization of biproducts, all of which also hold in an abelian category, so the argument carries over without change. We omit the details.□

In the setting of Proposition 2.5.3, given $f : X' \to X$ and $g : X \to X''$, an s satisfying $gs = \mathrm{id}_{X''}$ is called a **section** of g, while an r satisfying $rf = \mathrm{id}_{X'}$ is called a **retraction** of f; these terms come from topology. The existence of a section or retraction implies, respectively, that g is an epimorphism or f is a monomorphism, but the converses do not hold.

Notice also that the five statements (i)–(v) of the proposition are collectively self-dual.

Definition 2.5.4 Let S be a nonzero object of an abelian category $\mathcal{A}$. If $S = S' \oplus S''$ implies $S' = 0$ or $S'' = 0$, then S is called **indecomposable**.

Corollary 2.5.5 A nonzero object X of an abelian category is indecomposable if and only if

$$\forall e \in \mathrm{End}(X), \quad e^2 = e \iff (e = 0 \ \vee \ e = 1).$$

Proof Apply Proposition 2.5.2. □

We wish to study decompositions $X = X_1 \oplus \cdots \oplus X_n$, where $X \neq 0$ and every X_i is indecomposable. There are two questions: existence and uniqueness. For modules this is the content of the Krull–Remak–Schmidt theorem; see [51, Corollary 6.12.9]. For a general abelian category, some preparation is needed. We follow the approach of [25].

Definition 2.5.6 Let $\mathcal{A}$ be any category. A **bi-chain** in $\mathcal{A}$ is data $(X_n, \alpha_n, \beta_n)_{n=0}^{\infty}$, where the X_n are objects and $X_n \underset{\beta_n}{\overset{\alpha_n}{\rightleftarrows}} X_{n+1}$ are morphisms between them, with α_n an epimorphism and β_n a monomorphism for $n \in \mathbb{Z}_{\geq 0}$. An object X of $\mathcal{A}$ is said to satisfy the **bi-chain condition** if, for every bi-chain $(X_n, \alpha_n, \beta_n)_{n=0}^{\infty}$ with $X_0 = X$, both α_n and β_n are isomorphisms for $n \gg 0$.

The following result generalizes [51, Lemma 6.11.5].

Lemma 2.5.7 (Fitting's lemma in an abelian category) Let $\mathcal{A}$ be an abelian category, let X be a nonzero object satisfying the bi-chain condition, and let $f \in \mathrm{End}(X)$.

(i) For $n \gg 0$, $\ker(f^n)$ and $\operatorname{im}(f^n)$ are independent of n; denote them by $\ker(f^\infty)$ and $\operatorname{im}(f^\infty)$, respectively. Then $X = \ker(f^\infty) \oplus \operatorname{im}(f^\infty)$.

(ii) If X is indecomposable, then f is either invertible or nilpotent in the ring $\operatorname{End}(X)$.

Proof For $n \in \mathbb{Z}_{\geq 0}$, define $X_n := \operatorname{im}(f^n)$, so $X_0 = X$. Proposition 1.2.5 and coim $\xrightarrow{\sim}$ im give an epimorphism $\alpha_n : X_n \to X_{n+1}$ induced by f and a monomorphism $\beta_n : X_{n+1} \to X_n$. Thus we obtain a bi-chain $(X_n, \alpha_n, \beta_n)_{n=0}^\infty$.

For $n \gg 0$, α_n and β_n are isomorphisms. Hence the subobject $\operatorname{im}(f^\infty) := \operatorname{im}(f^n)$ of X, for $n \gg 0$, is well defined; write its monomorphism as $\iota : \operatorname{im}(f^\infty) \hookrightarrow X$. Likewise, $\ker(f^n) = \ker[X \to \operatorname{im}(f^n)]$ is a subobject $\ker(f^\infty)$ independent of n for $n \gg 0$.

For $n \gg 0$, the morphism $\alpha_{2n-1} \cdots \alpha_n : X_n \to X_{2n}$ is invertible. Denote its inverse by $\psi : \operatorname{im}(f^\infty) \xrightarrow{\sim} \operatorname{im}(f^\infty)$ and set

$$p := \psi f^n : X \to \operatorname{im}(f^\infty), \quad \ker(p) = \ker(f^\infty) \qquad (n \gg 0).$$

Since $p\iota = \mathrm{id}_{\operatorname{im}(f^\infty)}$, the endomorphism $e := \iota p \in \operatorname{End}(X)$ is idempotent, with $\operatorname{im}(e) = \operatorname{im}(f^\infty)$ and $\ker(e) = \ker(f^\infty)$. Proposition 2.5.2 gives the decomposition in (i).

If X is indecomposable, then either $\operatorname{im}(f^\infty) = 0$ and $\ker(f^\infty) = X$, in which case f is nilpotent, or $\operatorname{im}(f^\infty) = X$ and $\ker(f^\infty) = 0$, in which case f is invertible. This proves (ii). $\square$

Recall from [51, Definition 6.12.3] that if S is a ring and $S \smallsetminus S^\times$ is a two-sided ideal, then S is called **local**.

Corollary 2.5.8 Let X be a nonzero object in an abelian category satisfying the bi-chain condition. Then X is indecomposable if and only if $\operatorname{End}(X)$ is a local ring.

Proof Lemma 2.5.7 shows that if X is indecomposable, every element of $\operatorname{End}(X)$ is either nilpotent or invertible. The remainder of the argument concerns only the ring structure of $\operatorname{End}(X)$ and is the same as [51, Lemma 6.12.6]. $\square$

We can now state the version of the Krull–Remak–Schmidt theorem for abelian categories.

Theorem 2.5.9 (M. Atiyah) Let X be a nonzero object of an abelian category.

(i) If X satisfies the bi-chain condition, then there exist $n \in \mathbb{Z}_{\geq 1}$ and indecomposable subobjects $X_1, \ldots, X_n$ such that $X = \bigoplus_{i=1}^n X_i$, and every $\operatorname{End}(X_i)$ is a local ring.

(ii) Suppose that X has decompositions $\bigoplus_{i=1}^n X_i$ and $\bigoplus_{j=1}^m X'_j$, where every X_i and X'_j is indecomposable and every $\operatorname{End}(X_i)$ and $\operatorname{End}(X'_j)$ is local. Then $n = m$, and there is a bijection σ of $\{1, \ldots, n\}$ such that $X_i \simeq X'_{\sigma(i)}$ for every i.

Proof The main point is to prove that an X satisfying the bi-chain condition has a decomposition as in (i). Everything else involves only algebraic operations in the rings End and is the same as [51, Theorem 6.12.8]. We therefore assume the bi-chain condition in what follows.

If $X = Y \oplus Z$ with $Y \neq 0$, then Y also satisfies the bi-chain condition. Indeed, given a bi-chain $(Y_n, \alpha_n, \beta_n)_{n=0}^{\infty}$ with $Y_0 = Y$, take $X_n := Y_n \oplus Z$, $\tilde{\alpha}_n := \alpha_n \oplus \mathrm{id}_Z$, and $\tilde{\beta}_n := \beta_n \oplus \mathrm{id}_Z$. This gives a bi-chain with $X_0 = X$, while Lemma 2.5.1 shows that $\tilde{\alpha}_n$ (or $\tilde{\beta}_n$) is an isomorphism if and only if α_n (or β_n) is.

If X is already indecomposable, Corollary 2.5.8 shows that $\mathrm{End}(X)$ is local, proving (i). If (i) were false for X, there would be nonzero subobjects X_1, Y_1 such that $X = X_1 \oplus Y_1$ and (i) remained false for X_1. Repeating the process on X_1 gives $X_1 = X_2 \oplus Y_2$, and iteration produces a bi-chain $(X_n, \alpha_n, \beta_n)_{n=0}^{\infty}$, where $\alpha_n : X_n \twoheadrightarrow X_{n+1}$ and $\beta_n : X_{n+1} \hookrightarrow X_n$ arise from $X_n = X_{n+1} \oplus Y_{n+1}$. None is an isomorphism, and $X_0 := X$, contradicting the bi-chain condition. This proves the assertion. □

In applications, the key issue is knowing when the bi-chain condition holds. Here is one sufficient condition.

Proposition 2.5.10 Let X be an object of finite length in an abelian category (Definition 2.4.12). Then X satisfies the bi-chain condition.

Proof Let $(X_n, \alpha_n, \beta_n)_{n=0}^{\infty}$ be a bi-chain with $X_0 = X$. Then

$$(\mathrm{im}(\beta_0 \cdots \beta_n))_{n=0}^{\infty}$$

is a descending chain in the partially ordered set Sub_X. Since all β_n are monomorphisms, for $n \gg 0$ the Artinian condition gives a commutative diagram

$$\begin{tikzcd}
X_{n+1} \arrow[r, "\sim"] \arrow[d, "\beta_n"'] & \mathrm{im}(\beta_0 \cdots \beta_n) \arrow[r, hook] \arrow[d, "\wr"] & X_0 \\
X_n \arrow[r, "\sim"] & \mathrm{im}(\beta_0 \cdots \beta_{n-1}) \arrow[ur, hook] &
\end{tikzcd}$$

and β_n is an isomorphism. Similarly, apply the Noetherian condition to the ascending chain $(\ker(\alpha_n \cdots \alpha_0))_{n=0}^{\infty}$ and use the fact that every α_n is an epimorphism. For $n \gg 0$ this gives a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & \ker(\alpha_{n-1} \cdots \alpha_0) \arrow[r] \arrow[d, "\wr"] & X_0 \arrow[r, "\alpha_{n-1} \cdots \alpha_0"] \arrow[d, equal] & X_n \arrow[r] \arrow[d, "\alpha_n"] & 0 \\
0 \arrow[r] & \ker(\alpha_n \cdots \alpha_0) \arrow[r] & X_0 \arrow[r, "\alpha_n \cdots \alpha_0"'] & X_{n+1} \arrow[r] & 0
\end{tikzcd}$$

and α_n is an isomorphism. □

The exercises introduce further ways to deduce the bi-chain condition.

2.6 Subobjects and the Isomorphism Theorems

Throughout this section, let $\mathcal{A}$ be an abelian category. For every object X of $\mathcal{A}$, the partially ordered set of its subobjects is denoted by $(\mathrm{Sub}_X, \subset)$, following the convention of §1.1.

The results of this section are familiar when $\mathcal{A} = R\text{-}\mathsf{Mod}$, where R is any ring, but different arguments are required for an abelian category.

Definition 2.6.1 Let X be an object of $\mathcal{A}$.

- ◇ For a subobject X' of X, the corresponding **quotient** is defined by $X/X' := \operatorname{coker}[X' \to X]$; it is naturally a quotient object of X. Every epimorphism $f : X \twoheadrightarrow X''$ may be viewed as the quotient of X by $\ker(f)$.
- ◇ For subobjects $X_1, \ldots, X_n$ of X, their **intersection** and **sum** are respectively defined by

$$X_1 \cap \cdots \cap X_n := X_1 \underset{X}{\times} \cdots \underset{X}{\times} X_n,$$
$$X_1 + \cdots + X_n := \operatorname{im}\left[X_1 \oplus \cdots \oplus X_n \xrightarrow{\sigma} X\right].$$

  Here σ is induced by the morphisms $X_i \hookrightarrow X$ and may be viewed as summation. Both are naturally subobjects of X: the case of $X_1 \cap \cdots \cap X_n$ follows from Lemma 1.1.6, while the case of $X_1 + \cdots + X_n$ follows directly from the definition.

For example, the definition of cohomology can be rewritten using a quotient:

$$\mathrm{H}^n(X) := \ker(d_X^n)/\operatorname{im}(d_X^{n-1}).$$

More generally, let I be any set and let $(X_i)_{i \in I}$ be a family of subobjects of X. The same method gives a well-defined $\bigcap_{i\in I} X_i$ (respectively, $\sum_{i\in I} X_i$), provided that their fiber product over X (respectively, their coproduct $\bigsqcup_{i \in I} X_i$ in $\mathcal{A}$) exists. The role of this definition is clarified by the following fact.

Proposition 2.6.2 Let $(X_i)_{i\in I}$ be a family of subobjects of X. If their fiber product over X (respectively, their coproduct $\bigsqcup$ in $\mathcal{A}$) exists, it gives the infimum (respectively, supremum) of $(X_i)_{i\in I}$ in the partially ordered set $(\mathrm{Sub}_X, \subset)$.

Proof By construction, $\bigcap_{j\in I} X_j \subset X_i \subset \sum_{j \in I} X_j$ for every i.

Now consider a subobject $Y \hookrightarrow X$. For the intersection, suppose that $\forall i \in I$, $Y \subset X_i$, that is, that there is a family of commutative diagrams $\begin{array}{ccc} Y & \hookrightarrow & X \\ \downarrow & \nearrow & \\ X_i & & \end{array}$. The universal property of the fiber product gives a morphism $Y \to \bigcap_{i\in I} X_i$, and hence $Y \subset \bigcap_{i\in I} X_i$. For the sum, suppose instead that $\forall i \in I$, $X_i \subset Y$. The universal property of the coproduct gives a commutative diagram

$$\begin{tikzcd} \coprod_{i\in I} X_i \arrow[r, "\sigma"] \arrow[dr] & X \\ & Y. \arrow[u, hook] \end{tikzcd}$$

By Lemma 1.2.3, the morphism $\coprod_{i\in I} X_i \to Y$ factors uniquely through $\mathrm{coim}(\sigma) \simeq \mathrm{im}(\sigma)$, compatibly with the morphism to X. Thus $\sum_{i\in I} X_i := \mathrm{im}(\sigma) \subset Y$. $\square$

Convention 2.6.3 In view of Proposition 2.6.2, we may also dispense with fiber products or coproducts and define the intersection $\bigcap_i X_i$ (respectively, the sum $\sum_i X_i$) of a family of subobjects $(X_i)_{i\in I}$ directly as their infimum (respectively, supremum) in Sub_X, whenever it exists. The indexing set I is always assumed to be small. If I has a filtered partial order and $i \leq j \implies X_i \subset X_j$, the supremum $\sum_{i\in I} X_i$ may equally well be denoted by the increasing union $\bigcup_{i\in I} X_i$.

Corollary 2.6.4 For every object X, the partially ordered set $(\mathrm{Sub}_X, \subset)$ is a bounded lattice in the sense of Definition 2.4.1.

Proof The supremum of two subobjects Y, Z is $Y + Z$, and their infimum is $Y \cap Z$; the greatest element of Sub_X is X, and the least is 0. $\square$

This section focuses on $(\mathrm{Sub}_X, \subset)$, but the version in terms of quotient objects $(\mathrm{Quot}_X, \twoheadleftarrow)$ is entirely equivalent. Indeed, there is an evident order-reversing bijection $\mathrm{Sub}_X \simeq \mathrm{Quot}_X$: a subobject X' and a quotient object X'' correspond if and only if they can be placed in a short exact sequence $0 \to X' \to X \to X'' \to 0$. We shall therefore not discuss the quotient version separately.

Definition 2.6.5 Given a morphism $f : X \to Y$ and subobjects $X' \hookrightarrow X$ and $Y' \hookrightarrow Y$, write

$$f^{-1}(Y') := X \underset{Y}{\times} Y', \quad f(X') := \mathrm{im}\left[X' \hookrightarrow X \xrightarrow{f} Y\right];$$

the first is the preimage of Y' and a subobject of X, while the second is the image of X' and a subobject of Y. These constructions give order-preserving maps in both directions: $\mathrm{Sub}_X \underset{f^{-1}(\cdot)}{\overset{f(\cdot)}{\rightleftarrows}} \mathrm{Sub}_Y$.

Lemma 2.6.6 Let $f : X \to Y$ be as above.

(i) For every $X' \in \mathrm{Sub}_X$ and $Y' \in \mathrm{Sub}_Y$,

$$X' \subset f^{-1}(Y') \iff f(X') \subset Y'.$$

(ii) For a family $(X'_i)_{i\in I}$ in Sub_X (respectively, $(Y'_i)_{i\in I}$ in Sub_Y),[6] we have

$$f\left(\sum_{i\in I} X'_i\right) = \sum_{i\in I} f(X'_i), \quad f^{-1}\left(\bigcap_{i\in I} Y'_i\right) = \bigcap_{i\in I} f^{-1}(Y'_i),$$

provided the intersections and sums in question exist.

Proof For (i), by definition $X' \subset f^{-1}(Y')$ is equivalent to the existence of a commutative diagram

$$\begin{tikzcd} X' \arrow[r, dashed, "\exists"] \arrow[d, hook] & Y' \arrow[d, hook] \\ X \arrow[r, "f"'] & Y. \end{tikzcd}$$

Since $f(X')$ is the image of $X' \hookrightarrow X \xrightarrow{f} Y$,[7] Lemma 1.2.3 shows that such diagrams correspond bijectively to

$$\begin{tikzcd} X' \arrow[r] \arrow[d, hook] & f(X') \arrow[r, dashed, "\exists"] \arrow[d, hook] & Y' \arrow[dl, hook] \\ X \arrow[r, "f"'] & Y & \end{tikzcd}$$

which proves (i). If the partially ordered sets are regarded as categories Sub_X, and so on, (i) says that $f(\cdot) : \mathsf{Sub}_X \to \mathsf{Sub}_Y$ is left adjoint to $f^{-1}(\cdot) : \mathsf{Sub}_Y \to \mathsf{Sub}_X$. Since products (respectively, coproducts) correspond to infima (respectively, suprema), (ii) follows from [51, Theorem 2.8.12]. $\square$

Proposition 2.6.7 Consider subobjects $i : A \hookrightarrow X$ and $j : B \hookrightarrow X$. The morphism $(i, j) : A \oplus B \to X$ is an isomorphism if and only if

$$A \cap B = 0, \quad A + B = X.$$

[6] Editorial correction (O014-C025): the source introduces the families (X_i) and (Y_i) without primes, while the two formulas that follow use X'_i and Y'_i. Primes are added to the declaration to match the formulas.

[7] Editorial correction (O014-C026): the source calls $f(X')$ the coimage, but Definition 2.6.5 defines it as $\mathrm{im}[X' \to X \to Y]$, namely the image and a subobject of Y. The term consistent with that object is used here.

Moreover, the commutative diagram

$$\begin{tikzcd}
A \cap B \arrow[r, "\delta_1"] \arrow[d, "\delta_2"'] & A \arrow[d, "i"] \\
B \arrow[r, "j"'] & A + B
\end{tikzcd}$$

is both a pushout and a pullback; here δ_1 and δ_2 are the canonical morphisms.

Proof Define the diagonal morphism $\Delta : A \cap B \to A \oplus B$ by $p_i \Delta = \delta_i$ for $i = 1, 2$. Also define the antidiagonal morphism $\Delta^- : A \cap B \to A \oplus B$ by $p_1 \Delta^- = -\delta_1$ and $p_2 \Delta^- = \delta_2$.

Remark 2.1.5 says that Δ gives an isomorphism $A \cap B := A \underset{X}{\times} B \xrightarrow{\sim} \ker \left[A \oplus B \xrightarrow{(i,-j)} X \right]$. Replacing $(i, -j)$ by (i, j) leaves the image of $A \oplus B \to X$ equal to $A + B$, while the kernel morphism changes from Δ to Δ^-. We therefore obtain a commutative diagram with an exact row

$$\begin{tikzcd}
 & & & X & \\
0 \arrow[r] & A \cap B \arrow[r, "\Delta^-"'] & A \oplus B \arrow[r] \arrow[ur, "{(i,j)}"] & A + B \arrow[r] \arrow[u, hook] & 0
\end{tikzcd}$$

Thus $(i, j) : A \oplus B \to X$ is an isomorphism if and only if $A + B = X$ and $A \cap B = 0$.

Remark 2.1.5 also identifies $A \underset{A \cap B}{\sqcup} B$ with $(A \oplus B)/\operatorname{im}(\Delta^-)$. Hence we have a commutative diagram

$$\begin{tikzcd}
A \cap B \arrow[r, "\delta_1"] \arrow[d, "\delta_2"'] & A \arrow[d] \arrow[ddr, bend left, "i"] & \\
B \arrow[r] \arrow[drr, bend right, "j"'] & (A \oplus B)/\operatorname{im}(\Delta^-) \arrow[dr, "\simeq"] & \\
 & & A + B
\end{tikzcd}$$

whose upper-left square is a pushout; its outer frame is therefore also a pushout. Since δ_1 is a monomorphism, Proposition 2.1.6 shows that the outer frame is also a pullback.$\square$

In other words, B is a complement of A in the lattice Sub_X (Definition 2.4.2) if and only if $(i, j) : A \oplus B \xrightarrow{\sim} X$; the latter is often written as $A \oplus B = X$. Starting from this fact, one can recursively use lattice-theoretic language to determine whether X is a direct sum of finitely many prescribed subobjects $X_1, \ldots, X_n$; the result is analogous to the module case. For an infinite family of subobjects one generally needs to work in a Grothendieck category; see §2.10. The related material is left to the exercises.

Module theory rests on several basic isomorphism theorems; see [51, Propositions 6.1.11, 6.1.12, 6.1.13]. We now extend them to an arbitrary abelian category $\mathcal{A}$.

Theorem 2.6.8 Fix an object X.

(i) For every morphism $f : X \to Y$, there is a canonical isomorphism $X/\ker(f) \xrightarrow{\sim} \mathrm{im}(f)$ making the diagram

$$\begin{tikzcd} X \arrow[r, two heads] \arrow[d, two heads] & \mathrm{im}(f) \\ X/\ker(f) \arrow[ur, "\sim"] & \end{tikzcd} \quad \text{commutative.}$$

(ii) Fix a subobject $Z \hookrightarrow X$, and denote the natural morphism $X \twoheadrightarrow \overline{X} := X/Z$ by π. There is an order-preserving bijection

$$\begin{aligned} \{Y \in \mathrm{Sub}_X : Z \subset Y\} &\xleftrightarrow{1:1} \mathrm{Sub}_{\overline{X}} \\ Y &\longmapsto \pi(Y) = Y/Z =: \overline{Y} \\ \pi^{-1}(\overline{Y}) &\longleftarrow\!\shortmid \overline{Y}, \end{aligned}$$

with notation as in Definition 2.6.5. If $Y \supset Z$, there is a canonical isomorphism $X/Y \xrightarrow{\sim} \overline{X}/\overline{Y}$ making the diagram

$$\begin{tikzcd} X \arrow[r] \arrow[d, two heads] & \overline{X} \arrow[d, two heads] \\ X/Y \arrow[r, "\sim"'] & \overline{X}/\overline{Y} \end{tikzcd}$$

commutative. Moreover, $\overline{Y_1 \cap Y_2} = \overline{Y_1} \cap \overline{Y_2}$ and $\overline{Y_1 + Y_2} = \overline{Y_1} + \overline{Y_2}$.

(iii) For $Y, Z \in \mathrm{Sub}_X$, there is a canonical isomorphism $Y/(Y \cap Z) \xrightarrow{\sim} (Y + Z)/Z$ making the diagram

$$\begin{tikzcd} Y \arrow[r, hook] \arrow[d, two heads] & Y + Z \arrow[d, two heads] \\ Y/(Y \cap Z) \arrow[r, "\sim"'] & (Y + Z)/Z \end{tikzcd} \quad \text{commutative.}$$

Proof In view of the short exact sequence $0 \to \ker(f) \to X \to \mathrm{im}(f) \to 0$, assertion (i) merely restates the definition.

For (ii), given Y, first form the following commutative diagram with exact rows:

$$\begin{tikzcd} 0 \arrow[r] & Z \arrow[r] \arrow[d, equal] & Y \arrow[r] \arrow[d, hook] & Y/Z \arrow[r] \arrow[d, dashed, "\exists!"] & 0 \\ 0 \arrow[r] & Z \arrow[r] & X \arrow[r, "\pi"'] & \overline{X} \arrow[r] & 0 \end{tikzcd}$$

The dashed arrow comes from functoriality of cokernels (1.3.4). Theorem 2.3.3 immediately gives $\ker[Y/Z \to \overline{X}] = 0$ and $X/Y \xrightarrow{\sim} \operatorname{coker}[Y/Z \to \overline{X}]$. In particular,

$$Y/Z = \operatorname{im}[Y \to \overline{X}] = \pi(Y) \ \in \mathrm{Sub}_{\overline{X}}, \quad X/Y \simeq \overline{X}/\pi(Y).$$

We have already observed that both maps in (ii) preserve order. We now show that they are inverse. Given Y, the monomorphism $\overline{Y} \hookrightarrow \overline{X}$ is the kernel of $\overline{X} \to \overline{X}/\overline{Y}$. Hence Proposition 1.3.9 says that $\pi^{-1}(\overline{Y}) := X \underset{\overline{X}}{\times} \overline{Y} \hookrightarrow X$ is the kernel of the composite $X \to \overline{X} \to \overline{X}/\overline{Y}$. By the preceding isomorphism, this is also the kernel of $X \to X/Y$, identifying $\pi^{-1}(\overline{Y})$ with Y.

Conversely, given $\overline{Y}$, set $Y := X \underset{\overline{X}}{\times} \overline{Y} = \pi^{-1}(\overline{Y})$ and consider the pullback diagram

$$\begin{tikzcd} Y \arrow[r] \arrow[d, hook] \arrow[dr, phantom, "\square"] & \overline{Y} \arrow[d, hook] \\ X \arrow[r, "\pi"'] & \overline{X}. \end{tikzcd}$$

Since π in the lower row is an epimorphism, Proposition 2.1.6 says that the morphism in the upper row is also an epimorphism. Thus $\overline{Y} = \pi(Y)$.

Finally, the equalities $\overline{Y_1 \cap Y_2} = \overline{Y_1} \cap \overline{Y_2}$ and $\overline{Y_1 + Y_2} = \overline{Y_1} + \overline{Y_2}$ follow immediately from the order-preserving bijection $Y \leftrightarrow \overline{Y}$.

For (iii), consider the commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & Y \cap Z \arrow[r] \arrow[d, hook] & Y \arrow[r] \arrow[d, hook] & Y/(Y \cap Z) \arrow[r] \arrow[d, dashed, "\exists!\theta"] & 0 \\ 0 \arrow[r] & Z \arrow[r] & Y + Z \arrow[r] & (Y+Z)/Z \arrow[r] & 0 \end{tikzcd}$$

where θ comes from functoriality of cokernels. Proposition 2.6.7 says that the left square is a pushout. Pushouts preserve cokernels (Proposition 1.3.10), so θ is an isomorphism. $\square$

Corollary 2.6.9 Consider a commutative diagram

$$\begin{tikzcd} \ker(g') \arrow[r, hook] \arrow[d, "k"'] & W \arrow[r, "g'"] \arrow[d, "f'"'] & Y \arrow[r, two heads] \arrow[d, "f"] & \operatorname{coker}(g') \arrow[d, "c"] \\ \ker(g) \arrow[r, hook] & X \arrow[r, "g"'] & Z \arrow[r, two heads] & \operatorname{coker}(g) \end{tikzcd}$$

where k, c are the morphisms characterized by functoriality (1.3.4). If the middle square is a pullback (respectively, a pushout), then c is a monomorphism (respectively, k is an epimorphism).

Proof It is enough to handle the pullback case. Since pullback along f can be performed in stages, it suffices to consider two cases: f is an epimorphism or f is a monomorphism. If f is an epimorphism, Proposition 2.1.6 says that the middle square is also a pushout, and then c is an isomorphism. If f is a monomorphism, factor the pullback along g into stages as

$$\begin{tikzcd} W \arrow[r] \arrow[d, "f'"'] \arrow[dr, phantom, "\square"] & \operatorname{im}(g) \underset{Z}{\times} Y \arrow[r, "g''"] \arrow[d] \arrow[dr, phantom, "\square"] & Y \arrow[d, "f"] \\ X \arrow[r, two heads] & \operatorname{im}(g) \arrow[r, hook] & Z; \end{tikzcd}$$

Proposition 2.1.6 says that $W \to \operatorname{im}(g) \underset{Z}{\times} Y$ is an epimorphism, so $\operatorname{coker}(g') = \operatorname{coker}(g'')$. It remains to consider the case in which both f and g are monomorphisms. Then $W = Y \cap X$, and c factors as $Y/(Y \cap X) \xrightarrow{\sim} (Y + X)/X \hookrightarrow Z/X$. $\square$

Theorem 2.6.10 For every object X, the partially ordered set $(\mathrm{Sub}_X, \subset)$ is a bounded modular lattice in the sense of Definition 2.4.2.

Proof We already know that $(\mathrm{Sub}_X, \subset)$ is a bounded lattice. Consider any interval $[Y_1, Y_2]$ in Sub_X. Lemma 2.4.4 reduces the problem to the following assertion: for all $A, B, C \in [Y_1, Y_2]$,

$$(A, B \text{ are both complements of } C,\ A \subset B) \implies A = B. \tag{2.6.1}$$

First, the order-preserving bijection in Theorem 2.6.8(ii) reduces (2.6.1) to the case $Y_1 = 0$ and $Y_2 = X$. By Proposition 2.6.7, the hypothesis about complements becomes $A \oplus C \xrightarrow{\sim} X \xleftarrow{\sim} B \oplus C$, where the two isomorphisms are induced by the monomorphisms $\iota_A, \iota_B, \iota_C$ from A, B, C to X. By hypothesis, there is an $\alpha : A \to B$ such that $\iota_A = \iota_B \alpha$. Hence the diagram

$$\begin{tikzcd} B \oplus C \arrow[r, "(\iota_B, \iota_C)"] & X \\ A \oplus C \arrow[u, "(\alpha, \mathrm{id}_C)"] \arrow[ur, "(\iota_A, \iota_C)"'] & \end{tikzcd}$$

is commutative. Thus (α, id_C) is an isomorphism, and Lemma 2.5.1 then implies that α is an isomorphism. This proves (2.6.1). $\square$

Modularity of the subobject lattice is an important bridge for transporting standard arguments from module theory to abelian categories.

2.7 Simple and Semisimple Objects

Throughout this section, let $\mathcal{A}$ be an abelian category.

Convention 2.7.1 By convention, the coproduct $\coprod_{i\in I} X_i$ of a family of objects $(X_i)_{i\in I}$ in an abelian category, whenever it exists, is written $\bigoplus_{i\in I} X_i$ and called their **direct sum**. If I is finite, all the conventions of Definition 1.3.3 apply.

Definition 2.7.2 A nonzero object X of $\mathcal{A}$ is called **simple** if $\mathrm{Sub}_X = \{0, X\}$.

Simplicity of X means that for every short exact sequence $0 \to X' \to X \to X'' \to 0$, exactly one of $X' = 0$ and $X' \xrightarrow{\sim} X$ holds; equivalently, exactly one of $X \xrightarrow{\sim} X''$ and $X'' = 0$ holds. Consequently, X is simple in $\mathcal{A}$ if and only if it is simple in $\mathcal{A}^{\mathrm{op}}$.

Notice that $X \neq 0$ if and only if $\mathrm{id}_X \neq 0$, and this is also equivalent to $\mathrm{End}(X)$ being a nonzero ring.

Lemma 2.7.3 (Schur's lemma) Let X, Y be objects. If X (respectively, Y) is simple, then every nonzero morphism in $\mathrm{Hom}(X, Y)$ is a monomorphism (respectively, an epimorphism). Consequently, if X is simple, then $\mathrm{End}(X)$ is a division ring.

Proof Suppose that X is simple and that $f : X \to Y$ is nonzero. Then necessarily $\ker(f) = 0$. The case in which Y is simple follows by duality. A morphism that is both a monomorphism and an epimorphism is an isomorphism (Proposition 1.2.7); taking $X = Y$ therefore shows that $\mathrm{End}(X)$ is a division ring. □

We know that Sub_X is a bounded modular lattice (Theorem 2.6.10). Lemma 2.4.13 shows that X has finite length if and only if the partially ordered set $[0, X]$ has a composition series; such a series is also called a composition series of X. For an object X of finite length, its **length** is defined by

$$\ell(X) := \text{the length of } \mathrm{Sub}_X \in \mathbb{Z}_{\geq 0};$$

see Definition 2.4.14. Notice that $\ell(X) = 0 \iff X = 0$. By convention, $Y \subsetneq Z$ means $Y \subset Z$ and $Y \neq Z$.

Definition–Theorem 2.7.4 (Jordan–Hölder theorem) Let X be an object of finite length, and choose a composition series $X = X_0 \supsetneq \cdots \supsetneq X_r = 0$. The objects X_i/X_{i+1}, considered up to isomorphism, are called the **composition factors** of X. They are simple. Their multiset, with multiplicities, is denoted $\mathrm{JH}(X)$ and is independent of the chosen composition series.

Proof The composition factors must be simple, for otherwise $X_0 \supsetneq \cdots \supsetneq X_r$ would admit a proper refinement. Let $X = X'_0 \supsetneq \cdots \supsetneq X'_s$ be another composition series. Theorem 2.4.10 says that it is equivalent to $X_0 \supsetneq \cdots \supsetneq X_r$. In particular, there is a bijection $i \leftrightarrow j$ between their index sets such that the intervals $[X_{i+1}, X_i]$ and $\left[X'_{j+1}, X'_j\right]$ can be linked by standard isomorphisms $[a \wedge b, a] \xrightarrow{\sim} [b, a \vee b]$ in a modular lattice; see Proposition 2.4.5.

So far everything has been phrased in lattice-theoretic language. However, Theorem 2.6.8(iii) shows that such interval isomorphisms correspond to isomorphisms of quotient objects in $\mathcal{A}$,

$$X_i/X_{i+1} \simeq X'_j/X'_{j+1}.$$

As $i \leftrightarrow j$ varies, this proves that the composition factors are independent of the chosen composition series, up to isomorphism and reordering. □

The number of elements of $\mathrm{JH}(X)$, counted with multiplicity, is exactly $\ell(X)$. Multisets also have a union operation, which adds multiplicities; it will again be denoted by $\cup$.

Lemma 2.7.5 Given a short exact sequence $0 \to X' \to X \to X'' \to 0$, the object X has finite length if and only if both X' and X'' do. In that case, $\mathrm{JH}(X) = \mathrm{JH}(X') \cup \mathrm{JH}(X'')$, and hence $\ell(X) = \ell(X') + \ell(X'')$.

Proof Theorem 2.6.8(ii) identifies $\mathrm{Sub}_{X'}$ and $\mathrm{Sub}_{X''}$ with the intervals $[0, X']$ and $[X', X]$ in Sub_X, respectively. Thus finite length of X implies finite length of X' and X''. Conversely, suppose that X' and X'' have finite length, and choose composition series

$$X' = X'_0 \supsetneq \cdots \supsetneq X'_r = 0, \quad X'' = X''_0 \supsetneq \cdots \supsetneq X''_s = 0.$$

By Theorem 2.6.8(ii), lift every X''_i to a subobject Y_i of X satisfying $Y_i \supset X'$. In particular, $Y_0 = X$ and $Y_s = X'$, so

$$X = Y_0 \supsetneq \cdots \supsetneq Y_s \supsetneq X'_1 \supsetneq \cdots \supsetneq X'_r = 0$$

is a composition series whose subquotients form $\mathrm{JH}(X') \cup \mathrm{JH}(X'')$. □

Definition 2.7.6 An object X is called

- **semisimple** if $X = \bigoplus_{i \in I} X_i$ for some family of simple objects $(X_i)_{i \in I}$;
- **split** if every short exact sequence $0 \to X' \to X \to X'' \to 0$ splits.

If every object of $\mathcal{A}$ is semisimple (respectively, split), then $\mathcal{A}$ is called a semisimple (respectively, split) abelian category.[8)]

Many sources require I to be finite in the definition of a semisimple object.

Lemma 2.7.7 If $X = \bigoplus_{i=1}^n X_i$, where $X_1, \dots, X_n$ are simple, then X has finite length, $\mathrm{JH}(X) = \{X_1, \dots, X_n\}$, counted with multiplicity, and $\ell(X) = n$.

Proof Consider the composition series $\bigoplus_{i=1}^n X_i \supsetneq \bigoplus_{i=1}^{n-1} X_i \supsetneq \cdots \supsetneq 0$. □

Remark 2.7.8 For a general semisimple object $X = \bigoplus_{i \in I} X_i$, the Noetherian, Artinian, and finite-length properties of X are all equivalent to finiteness of I. Indeed, by adding (or removing) direct summands indexed by I, one can readily construct a strictly ascending (or strictly descending) chain in Sub_X.

We wish to understand the relation between split and semisimple objects. The argument is similar to the module case [51, Proposition 6.11.4].

Proposition 2.7.9 Let X be a split object. Then every subobject and every quotient object of X is split. If, in addition, X is Artinian, then X is a semisimple object of finite length.

Proof Given a short exact sequence $0 \to X' \to X \to X'' \to 0$, the hypothesis implies $X \simeq X' \oplus X''$, so X'' embeds as a subobject of X. It is therefore enough to show that every subobject X' of X is split. Given $X_0' \subset X'$, there is a $Y \subset X$ such that $X = X_0' \oplus Y$. It remains to prove

$$X' = X_0' \oplus (Y \cap X').$$

This follows from Proposition 2.6.7. On the one hand, $X_0' \cap (Y \cap X') \subset X_0' \cap Y = 0$. On the other hand, Sub_X is modular, and hence $X_0' + (Y \cap X') = X' \cap (Y + X_0') = X' \cap X = X'$. This proves the displayed decomposition.

Now suppose also that X is Artinian. If $X \neq 0$, it has a minimal nonzero subobject X_1, which must be simple, and there is a direct-sum decomposition $X = X_1 \oplus Y_1$. The object Y_1 is still Artinian and split. If $Y_1 \neq 0$, continue by writing $Y_1 = X_2 \oplus Y_2$, and so on. The Artinian condition ensures that the strictly descending chain $X \supsetneq Y_1 \supsetneq Y_2 \cdots$ terminates after finitely many steps, giving the required decomposition $X = X_1 \oplus \cdots \oplus X_n$. □

We may ask conversely whether semisimple objects are split. For a general abelian category, a finiteness assumption is again needed.

---

8) Terminology in the literature is not uniform; some authors call what is termed a split abelian category here a semisimple abelian category.

Proposition 2.7.10 For an object X, consider the following properties.

(i) $X = \sum_{Y \in \mathcal{F}} Y$, where $\mathcal{F}$ is a finite subset of Sub_X and every $Y \in \mathcal{F}$ is simple;

(ii) $X = \bigoplus_{Y \in \mathcal{F}} Y$, where $\mathcal{F}$ is a finite subset of Sub_X and every $Y \in \mathcal{F}$ is simple;

(iii) X is split.

Then (i) $\implies$ (ii) $\implies$ (iii).

Proof Repeat the proof of (i) $\implies$ (ii) $\implies$ (iii) in [51, Proposition 6.11.4]. □

Remark 2.7.11 To extend Proposition 2.7.10 to an infinite subset $\mathcal{F} \subset \mathrm{Sub}_X$, one must require $\mathcal{A}$ to be a Grothendieck category, as introduced in § 2.10. Every semisimple object in a Grothendieck category is then split, and every semisimple Grothendieck category is automatically split. Once the relevant definitions are in place, the argument is similar to the module case and is left as an exercise in this chapter.

2.8 Exact Functors, Injective Objects, and Projective Objects

Given a functor $F : \mathcal{A} \to \mathcal{B}$, one may ask whether it preserves $\varinjlim$ or $\varprojlim$; see [51, §2.8] or the discussion in §1.5. This section focuses on the case in which $\mathcal{A}$ and $\mathcal{B}$ are abelian categories and F is an additive functor. Taking the special instances of $\varprojlim$ (or $\varinjlim$) given by ker (or coker), every morphism $f : X \to Y$ in $\mathcal{A}$ yields a canonical morphism $F\ker(f) \to \ker F(f)$ and its dual $\operatorname{coker} F(f) \to F\operatorname{coker}(f)$; they make the following diagrams commute.

$$
\begin{tikzcd}
F\ker(f) \arrow[rr] \arrow[dr, "{F[\ker(f)\hookrightarrow X]}"'] & & \ker F(f) \arrow[dl] \\
& FX &
\end{tikzcd}
\qquad
\begin{tikzcd}
\operatorname{coker} F(f) \arrow[rr] & & F\operatorname{coker}(f) \\
& FY. \arrow[ul] \arrow[ur, "{F[Y\twoheadrightarrow\operatorname{coker}(f)]}"'] &
\end{tikzcd}
$$

◇ If $F\ker(f) \xrightarrow{\sim} \ker F(f)$ for every f, we say that ***F* preserves kernels**;

◇ if $\operatorname{coker} F(f) \xrightarrow{\sim} F\operatorname{coker}(f)$ for every f, we say that ***F* preserves cokernels**.

If $(X^\bullet, d^\bullet)$ is a complex in an abelian category, then $(FX^\bullet, Fd^\bullet)$ is also a complex; the issue to be examined is exactness.

Proposition 2.8.1 For an additive functor $F : \mathcal{A} \to \mathcal{B}$ between abelian categories, the following statements are equivalent:

(L1) F preserves kernels;

(L2) if $0 \to X' \xrightarrow{f} X \xrightarrow{g} X'' $ is exact in $\mathcal{A}$, then $0 \to F(X') \xrightarrow{Ff} F(X) \xrightarrow{Fg} F(X'')$ is exact in $\mathcal{B}$;

(L3) F preserves all finite $\varprojlim$.

Dually, the following statements are also equivalent:

(R1) F preserves cokernels;

(R2) if $X' \xrightarrow{f} X \xrightarrow{g} X'' \to 0$ is exact in $\mathcal{A}$, then $F(X') \xrightarrow{Ff} F(X) \xrightarrow{Fg} F(X'') \to 0$ is exact in $\mathcal{B}$;

(R3) F preserves all finite $\varinjlim$.

Proof By duality, it suffices to discuss (L1)–(L3).

(L1) $\implies$ (L2). An exact sequence of the form $0 \to \bullet \to \bullet \to \bullet$ is precisely a kernel diagram of a morphism; see the explanation at the end of §2.2.

(L2) $\implies$ (L3). Every finite $\varprojlim$ can be constructed from finite products and equalizers. We already know that additive functors preserve biproducts; since F preserves ker, it preserves all equalizers (Remark 1.3.8). Thus F preserves all finite $\varprojlim$.

(L3) $\implies$ (L1) is immediate. □

For an additive functor F as above, if every exact sequence $(X^\bullet, d^\bullet)$ gives an exact sequence $(FX^\bullet, Fd^\bullet)$, we say that F preserves exact sequences. Likewise, one may ask whether F preserves short exact sequences; these two properties turn out to be equivalent.

Proposition 2.8.2 For an additive functor $F : \mathcal{A} \to \mathcal{B}$ between abelian categories, the following statements are equivalent:

(E1) F preserves short exact sequences;

(E2) F preserves exact sequences;

(E3) F preserves all finite $\varprojlim$ and finite $\varinjlim$.

Proof (E1) $\implies$ (E2). Take an exact sequence $(X^\bullet, d^\bullet)$ in $\mathcal{A}$ and decompose the complex into short exact sequences

$$0 \longrightarrow \ker(d^n) \xrightarrow{\iota^n} X^n \xrightarrow{\mathbf{d}^n} \ker(d^{n+1}) \longrightarrow 0$$

where the morphism $\mathbf{d}^n$ is characterized by $d^n = \iota^{n+1}\mathbf{d}^n$; the index n may be arbitrary, except that X^n cannot be the rightmost term of the exact sequence.

By assumption, the sequence $0 \to F(\ker(d^n)) \xrightarrow{F\iota^n} F(X^n) \xrightarrow{F\mathbf{d}^n} F(\ker(d^{n+1})) \to 0$ remains exact, and $Fd^n = F\iota^{n+1}F\mathbf{d}^n$. The short exact sequences can therefore be reassembled into the exact sequence $(FX^\bullet, Fd^\bullet)$ in $\mathcal{B}$.

(E2) $\implies$ (E3). If F preserves exact sequences, it satisfies conditions (L2) and (R2) of Proposition 2.8.1.

(E3) $\implies$ (E1). Suppose that $0 \to X' \to X \to X'' \to 0$ is exact in $\mathcal{A}$. Applying Proposition 2.8.1 once more, we see that $0 \to F(X') \to F(X) \to F(X'')$ and $F(X') \to F(X) \to F(X'') \to 0$ are both exact; hence $0 \to F(X') \to F(X) \to F(X'') \to 0$ is exact. $\square$

Notice that (E3) $\iff$ (L3) $\wedge$ (R3).

Definition 2.8.3 (Exact functors) For an additive functor $F : \mathcal{A} \to \mathcal{B}$ between abelian categories, consider conditions (L1)–(L3), (R1)–(R3) of Proposition 2.8.1, and conditions (E1)–(E3) of Proposition 2.8.2.

- ◇ If any of (L1)–(L3) holds, then F is called **left exact**.
- ◇ If any of (R1)–(R3) holds, then F is called **right exact**.
- ◇ If any of (E1)–(E3) holds, then F is called **exact**; this is equivalent to saying that F is both left exact and right exact.

For example, every equivalence between abelian categories is of course an exact functor. Another extreme example is the zero functor: it sends every object to a zero object and every morphism to a zero morphism; this functor is also exact.

By Lemma 1.3.11, a left exact functor preserves monomorphisms, while a right exact functor preserves epimorphisms.

Remark 2.8.4 In view of (E1), if F is known to be left exact (or right exact), then F is exact if and only if F preserves epimorphisms (or monomorphisms).

A standard way to check left or right exactness is by means of adjoint functors.

Theorem 2.8.5 Consider a pair of additive functors between abelian categories $\mathcal{A}$ and $\mathcal{B}$:

$$F : \mathcal{A} \rightleftarrows \mathcal{B} : G.$$

Suppose that F and G can be equipped as an adjoint pair (F, G, φ). Then F is right exact and G is left exact; indeed, F preserves $\varinjlim$ and G preserves $\varprojlim$.

Proof Apply [51, Theorem 2.8.12] and conditions (L3), (R3) of Proposition 2.8.1. $\square$

Example 2.8.6 (Left/right exactness of limits) Let $\mathcal{A}$ be an abelian category, let I be any category, and suppose that every functor $\alpha : I \to \mathcal{A}$ has a $\varinjlim$ (or a $\varprojlim$). Endow $\mathcal{A}^I$ with the abelian-category structure of Proposition 2.1.4. We shall show that the functor $\varinjlim : \mathcal{A}^I \to \mathcal{A}$ (or $\varprojlim$) is right exact (or left exact).

It suffices to discuss $\varinjlim$. By Theorem 2.8.5, one strategy is to show that $\varinjlim$ has a right adjoint. Define the diagonal functor $\Delta : \mathcal{A} \to \mathcal{A}^I$, which sends $L \in \mathrm{Ob}(\mathcal{A})$ to the constant functor $\mathrm{Ob}(I) \ni i \mapsto L$. We have an adjoint pair

$$\varinjlim : \mathcal{A}^I \rightleftarrows \mathcal{A} : \Delta.$$

This follows immediately from the universal property of $\varinjlim$: in $\mathcal{A}^I$, giving a morphism from a functor $\alpha : I \to \mathcal{A}$ to $\Delta(L)$ is equivalent to giving a compatible family of morphisms $f_i : \alpha(i) \to L$; in other words, a cone with base α and vertex L. See the related review in §1.5.

Example 2.8.7 Let $f : R \to S$ be a ring homomorphism. Consider the following functors between categories of left modules:

$$\begin{tikzcd} R\text{-}\mathsf{Mod} \arrow[r, bend left, "S \underset{R}{\otimes} (\cdot)"] \arrow[r, bend right, "\mathrm{Hom}_R({}_R S, \cdot)"'] & S\text{-}\mathsf{Mod} \arrow[l, "{}_{R\to S}\mathcal{F}"] \end{tikzcd}$$

Here the forgetful functor ${}_{R\to S}\mathcal{F}$ is simply the operation of regarding an S-module as an R-module via f. The functors $S \underset{R}{\otimes} (\cdot)$ and $\mathrm{Hom}_R({}_R S, \cdot)$ are discussed in [51, §6.6]; here ${}_R S$ means S regarded as a left R-module. According to [51, Corollary 6.6.8],

$$\left(S \underset{R}{\otimes} -,\ {}_{R\to S}\mathcal{F} \right) \quad \text{and} \quad \left({}_{R\to S}\mathcal{F},\ \mathrm{Hom}_R({}_R S, -) \right)$$

are adjoint pairs. Theorem 2.8.5 then implies

$$S \underset{R}{\otimes} (\cdot) \text{ is right exact,} \quad \mathrm{Hom}({}_R S, \cdot) \text{ is left exact,} \quad {}_{R\to S}\mathcal{F} \text{ is exact.}$$

This exactness can also be checked directly by algebraic means. For example, for the forgetful functor ${}_{R\to S}\mathcal{F}$, whether a sequence of module homomorphisms $\cdots \to M^n \xrightarrow{d^n} M^{n+1} \to \cdots$ is a complex (that is, $d^{n+1}d^n = 0$), or is exact (that is, $\mathrm{im}(d^n) = \ker(d^{n+1})$), depends only on the additive-group structure of M^n, not on multiplication in R or S. In other words, ${}_{\mathbb{Z}\to S}\mathcal{F} : S\text{-}\mathsf{Mod} \to \mathsf{Ab}$ is already an exact functor. Alternatively, one can check directly that ${}_{R\to S}\mathcal{F}$ preserves all limits; this follows directly from the construction in [51, Theorem 6.2.2].

An exact functor automatically preserves the im $\simeq$ coim of a morphism (Definition 1.2.1, Proposition 1.3.12); it also preserves cohomology.

Proposition 2.8.8 Let $F : \mathcal{A} \to \mathcal{B}$ be an exact functor between abelian categories. For every complex $(X^\bullet, d^\bullet)$ in $\mathcal{A}$, there is a canonical isomorphism in $\mathcal{B}$

$$F\,\mathrm{H}^n(X^\bullet, d^\bullet) \xrightarrow{\sim} \mathrm{H}^n(FX^\bullet, Fd^\bullet), \qquad n \in \mathbb{Z}.$$

Proof It suffices to consider a three-term complex $X' \xrightarrow{f} X \xrightarrow{g} X''$. Returning to definition (2.2.2),

$$\begin{aligned} F\left(\mathrm{H}\left[X' \xrightarrow{f} X \xrightarrow{g} X''\right]\right) &= F\,\mathrm{coker}\,[\mathrm{im}(f) \to \ker(g)] \\ &\simeq \mathrm{coker}\,[F\,\mathrm{im}(f) \to F\ker(g)] \\ &\simeq \mathrm{coker}\,[\mathrm{im}(Ff) \to \ker(Fg)] = \mathrm{H}\left[FX' \xrightarrow{Ff} FX \xrightarrow{Fg} FX''\right], \end{aligned}$$

and all the isomorphisms that occur are canonical. □

Faithful exact functors have particularly useful properties.

Proposition 2.8.9 For an additive functor $F : \mathcal{A} \to \mathcal{B}$ between abelian categories, the following statements are equivalent:

(i) F is exact and faithful;

(ii) F is exact, and $FX = 0 \iff X = 0$ for every $X \in \mathrm{Ob}(\mathcal{A})$;

(iii) $X' \to X \to X''$ is exact in $\mathcal{A}$ if and only if $FX' \to FX \to FX''$ is exact in $\mathcal{B}$.

Proof (i) $\implies$ (ii): If $FX = 0$, then $F(\mathrm{id}_X) = \mathrm{id}_{FX} = 0$, whence $\mathrm{id}_X = 0$ and $X = 0$.

(ii) $\implies$ (iii): Apply Proposition 2.8.8 to the cohomology.

(iii) $\implies$ (i): This condition immediately implies that F is exact. Suppose that $u : X \to Y$ satisfies $Fu = 0$. Since

$$X \xrightarrow{\mathrm{id}} X \xrightarrow{u} Y \xrightarrow{\mathrm{id}} Y,$$

the image of this sequence under F is exact; hence the sequence itself is exact and $u = 0$. □

Localization (see §1.9) provides an important example of an exact functor.

Proposition 2.8.10 (Exactness of localization) Let $\mathcal{A}$ be an abelian category and let $S \subset \mathrm{Mor}(\mathcal{A})$ be a multiplicative family (Definition 1.9.5). Then the category $\mathcal{A}[S^{-1}]$ supplied by Theorem 1.9.11 has a canonical abelian-category structure, and the localization functor $Q : \mathcal{A} \to \mathcal{A}[S^{-1}]$ is exact.

Proof Theorem 1.9.17 gives $\mathcal{A}[S^{-1}]$ a canonical additive-category structure under which Q is an additive functor. We shall show that every morphism f in $\mathcal{A}[S^{-1}]$ has a kernel and cokernel and is strict. By the construction of $\mathcal{A}[S^{-1}]$, after composing f with an isomorphism from S, we may assume that f is the image under Q of a morphism in $\mathcal{A}$; this does not change the property to be proved. Lemma 1.9.16 says that Q preserves all finite $\varinjlim$ and $\varprojlim$. In particular, Q sends ker, coker, im, and coim in $\mathcal{A}$ to the corresponding constructions in $\mathcal{A}[S^{-1}]$, and therefore also preserves diagram (1.2.1). This shows that $\mathcal{A}[S^{-1}]$ is an abelian category; condition (E3) of Proposition 2.8.2 shows that Q is exact. $\square$

If, in addition, $\mathcal{A}$ is a $\Bbbk$-linear abelian category, where $\Bbbk$ is a commutative ring, then Q is a functor between $\Bbbk$-linear abelian categories. This too is part of Theorem 1.9.17.

Another extremely important example is the Hom functor. If T is an object of an abelian category $\mathcal{A}$, then $\mathrm{Hom}(T, \cdot)$ defines a functor $\mathcal{A} \to \mathsf{Ab}$, while $\mathrm{Hom}(\cdot, T)$ defines a functor $\mathcal{A}^{\mathrm{op}} \to \mathsf{Ab}$; on morphisms, they send $f : X \to Y$ to f_* and f^* on Hom, respectively. Both are plainly additive functors.

If $\mathcal{A}$ is a $\Bbbk$-linear abelian category, the Hom functors may take values in $\Bbbk$-Mod and become $\Bbbk$-linear. This extension has little effect on the discussion below, so we do not treat it separately.

Proposition 2.8.11 (Left exactness of Hom) Let $\mathcal{A}$ be an abelian category and let T be an object of $\mathcal{A}$. Then $\mathrm{Hom}(T, \cdot) : \mathcal{A} \to \mathsf{Ab}$ and $\mathrm{Hom}(\cdot, T) : \mathcal{A}^{\mathrm{op}} \to \mathsf{Ab}$ are both left exact functors.

Proof By duality (replacing $\mathcal{A}$ by $\mathcal{A}^{\mathrm{op}}$), it suffices to treat $\mathrm{Hom}(T, \cdot)$. The point is to show that $\mathrm{Hom}(T, \cdot)$ preserves ker. Since ker in Ab is simply the kernel from group theory, this follows immediately from the universal property of ker. $\square$

Definition 2.8.12 Let X be an object of an abelian category $\mathcal{A}$. If $\mathrm{Hom}(X, \cdot) : \mathcal{A} \to \mathsf{Ab}$ is an exact functor, then X is called a **projective object**; if $\mathrm{Hom}(\cdot, X) : \mathcal{A}^{\mathrm{op}} \to \mathsf{Ab}$ is an exact functor, then X is called an **injective object**.

By Remark 2.8.4 and Proposition 2.8.11, to decide whether an object X is projective (or injective), it suffices to check whether the functor $\mathrm{Hom}(X, \cdot)$ (or $\mathrm{Hom}(\cdot, X)$) preserves epimorphisms. Consequently:

- ◇ An object P is projective if and only if, for every exact sequence in $\mathcal{A}$ of the form $Y \xrightarrow{g} X \to 0$ and every morphism $P \to X$, there exists a $P \to Y$ making the following diagram commute.

$$\begin{tikzcd} & P \arrow[dl, "\exists"'] \arrow[d] & \\ Y \arrow[r, "g"'] & X \arrow[r] & 0 \end{tikzcd}$$

(Equivalently, $g_* : \mathrm{Hom}(P, Y) \to \mathrm{Hom}(P, X)$ is surjective.)

- An object I is injective if and only if, for every exact sequence in $\mathcal{A}$ of the form $0 \to X \xrightarrow{f} Y$ and every morphism $X \to I$, there exists a $Y \to I$ making the following diagram commute.

$$\begin{tikzcd} 0 \arrow[r] & X \arrow[r, "f"] \arrow[d] & Y \arrow[dl, "\exists"] \\ & I & \end{tikzcd}$$

(Equivalently, $f^* : \mathrm{Hom}(Y, I) \to \mathrm{Hom}(X, I)$ is surjective.)

Lemma 2.8.13 Consider a short exact sequence $0 \to X' \xrightarrow{f} X \xrightarrow{g} X'' \to 0$ in an abelian category. If X' is injective or X'' is projective, then this short exact sequence splits.

Proof By duality, we may assume without loss of generality that X' is injective. Apply the characterization above to the commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & X' \arrow[r, "f"] \arrow[d, "\mathrm{id}_{X'}"'] & X \arrow[dl, "\exists r"] \\ & X' & \end{tikzcd}$$

The relation $rf = \mathrm{id}_{X'}$ then satisfies the condition of Proposition 2.5.3. $\square$

Lemma 2.8.14 Consider a family of objects $(X_i)_{i \in I}$ in an abelian category $\mathcal{A}$. If the coproduct $\coprod_{i \in I} X_i$ (or the product $\prod_{i \in I} X_i$) exists in $\mathcal{A}$, then the coproduct is projective (or the product is injective) if and only if every X_i is so.

Proof By duality, it suffices to consider the coproduct $\coprod_{i \in I} X_i$. The universal property gives an isomorphism of functors

$$\mathrm{Hom}_{\mathcal{A}} \left(\coprod_{i \in I} X_i, - \right) \xrightarrow{\sim} \prod_{i \in I} \mathrm{Hom}_{\mathcal{A}} (X_i, -) : \mathcal{A} \to \mathsf{Ab}.$$

It remains only to use the following elementary observation: for a family of maps $(f_i : A_i \to B_i)_{i \in I}$, where A_i, B_i are sets, the induced map $(f_i)_{i \in I} : \prod_{i \in I} A_i \to \prod_{i \in I} B_i$ is surjective if and only if every f_i is surjective. $\square$

For example, let R be a ring. Every free R-module is a projective object of R-Mod. Indeed, it suffices to show that R itself is projective; but $\mathrm{Hom}(R,\cdot) : R\text{-}\mathsf{Mod} \to \mathsf{Ab}$ is isomorphic to the forgetful functor $\mathcal{F} : R\text{-}\mathsf{Mod} \to \mathsf{Ab}$ via the isomorphism that sends a homomorphism $\varphi : R \to X$ to $\varphi(1) \in X$. Thus $\mathrm{Hom}(R,\cdot)$ is exact.

Free modules are also generators for R-Mod; see Definition 1.11.8 and Example 1.11.10. An injective object that is also a cogenerator (or a projective object that is also a generator) is very useful. Here is a simple characterization.

Proposition 2.8.15 (Injective cogenerators and projective generators) In an abelian category $\mathcal{A}$, an injective (or projective) object X is a cogenerator (or generator) if and only if $\mathrm{Hom}(T,X) \neq 0$ (or $\mathrm{Hom}(X,T) \neq 0$) for every $T \in \mathrm{Ob}(\mathcal{A})$ with $T \neq 0$.

Equivalently, $\mathrm{Hom}(\cdot,X)$ (or $\mathrm{Hom}(X,\cdot)$) is a faithful exact functor.

Proof We treat only the case in which X is injective. First suppose that X is a generator. Apply the definition of cogenerator to

$$T \underset{0}{\overset{\mathrm{id}_T}{\rightrightarrows}} T$$

There is then a $\delta \in \mathrm{Hom}(T,X)$ such that $\delta = \delta \circ \mathrm{id}_T \neq \delta \circ 0 = 0$.

Conversely, we wish to show that if $h : S \to T$ is nonzero, then there is a $\delta \in \mathrm{Hom}(T,X)$ such that $\delta h \neq 0$. Observe that there is a $\delta' \in \mathrm{Hom}(\mathrm{im}(h),X)$ with $\delta' \neq 0$. Since $S \twoheadrightarrow \mathrm{im}(h)$ is an epimorphism, the composite $S \twoheadrightarrow \mathrm{im}(h) \xrightarrow{\delta'} X$ is nonzero. Now use the injectivity of X (see the discussion following Definition 2.8.12) to extend δ' to $\delta : T \to X$.

The assertion about faithful exact functors is an immediate application of Proposition 2.8.9. $\square$

To do homological algebra in an abelian category, one often needs enough injective or projective objects.

Definition 2.8.16 Let $\mathcal{A}$ be an abelian category.

- ⋄ If, for every $X \in \mathrm{Ob}(\mathcal{A})$, there are an injective object I and a monomorphism $X \hookrightarrow I$, then $\mathcal{A}$ is said to **have enough injectives**.

- ⋄ If, for every $X \in \mathrm{Ob}(\mathcal{A})$, there are a projective object P and an epimorphism $P \twoheadrightarrow X$, then $\mathcal{A}$ is said to **have enough projectives**.

These two notions are dual to each other. A standard way to construct injective or projective objects is to use adjoints of exact functors; here are the details.

Proposition 2.8.17 Consider a pair of functors between abelian categories

$$\mathcal{A} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{B}$$

and suppose that F is exact.

- ⋄ If G is a left adjoint of F, then G sends projective objects in $\mathcal{B}$ to projective objects in $\mathcal{A}$;
- ⋄ if G is a right adjoint of F, then G sends injective objects in $\mathcal{B}$ to injective objects in $\mathcal{A}$.

Proof By duality, it suffices to consider the case in which G is a left adjoint. In this case G is necessarily additive (Corollary 1.3.6). Let P be a projective object of $\mathcal{B}$. For every exact sequence $(X^\bullet, d^\bullet)$ in $\mathcal{A}$, there is an isomorphism of complexes in Ab

$$\mathrm{Hom}_{\mathcal{A}}(GP, X^\bullet) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{B}}(P, FX^\bullet).$$

Since F is exact, $(FX^\bullet, Fd^\bullet)$ is an exact sequence, and hence the right-hand side is exact in Ab. Therefore $\mathrm{Hom}_{\mathcal{A}}(GP, \cdot) : \mathcal{A} \to \mathsf{Ab}$ is an exact functor. □

For example, consider a ring R and the forgetful functor $\mathcal{F} : R\text{-}\mathsf{Mod} \to \mathsf{Ab}$. By Example 2.8.7, $\mathcal{F}$ is exact and has the right adjoint $G := \mathrm{Hom}_{\mathsf{Ab}}(R, \cdot)$. Proposition 2.8.17 shows that G sends injective objects in Ab to injective objects in R-Mod. Injective objects in Ab are easy to characterize: they are precisely the divisible $\mathbb{Z}$-modules. This is the standard way to construct injective R-modules in module theory and to show that R-Mod has enough injectives; see [51, Theorem 6.9.14].

On the other hand, R-Mod also has enough projectives: for every R-module X, choose a set of generators $A \subset X$; then $R^{\oplus A} \twoheadrightarrow X$.

Example 2.8.18 The following combined exercise involves an abstract abelian category and will be used in §3.12 to study long exact sequences of derived functors. Let $\mathcal{A}$ be an abelian category. Consider the category $\mathbf{2}$ (whose diagram is $0 \to 1$; see the book's introduction on categories). The functor category $\mathcal{A}^{\mathbf{2}}$ is again abelian (Proposition 2.1.4); it is the arrow category: its objects are morphisms $X_0 \to X_1$ in $\mathcal{A}$, and its morphisms are commutative squares in $\mathcal{A}$

$$\begin{array}{ccc} X_0 & \longrightarrow & Y_0 \\ \downarrow & & \downarrow \\ X_1 & \longrightarrow & Y_1 \end{array}$$

For $i \in \{0, 1\}$, the evaluation functor $\mathrm{ev}_i : \mathcal{A}^{\mathbf{2}} \to \mathcal{A}$ sends an object $X_0 \to X_1$ to

X_i. In addition, define additive functors from $\mathcal{A}$ to $\mathcal{A}^{\mathbf{2}}$ by [9)]

$$L_0 : X \mapsto [X \to 0], \quad L_1 : X \mapsto [0 \to X], \quad H : X \mapsto [X \xrightarrow{\mathrm{id}_X} X],$$

for $X \in \mathrm{Ob}(\mathcal{A})$; the definitions on morphisms are clear. We shall prove the following assertions.

(i) The functors ev_0 and ev_1 are both exact and both send injective (or projective) objects of $\mathcal{A}^{\mathbf{2}}$ to injective (or projective) objects of $\mathcal{A}$. Moreover, H, L_0, and L_1 are all exact.

(ii) The functors L_0, H (or H, L_1) send injective (or projective) objects of $\mathcal{A}$ to injective (or projective) objects of $\mathcal{A}^{\mathbf{2}}$.

(iii) If $\mathcal{A}$ has enough injectives (or projectives), then so does $\mathcal{A}^{\mathbf{2}}$.

Since $\varinjlim$ and $\varprojlim$ in $\mathcal{A}^{\mathbf{2}}$ are constructed objectwise, the functors $\mathrm{ev}_0, \mathrm{ev}_1$ and H, L_0, L_1 are indeed exact. The remainder of (i) and (ii) follows from the adjoint relations satisfied by ev_i; the diagrams are as follows, and verification is a simple exercise:

$$\begin{tikzcd} \mathcal{A}^{\mathbf{2}} \arrow[r, "\mathrm{ev}_0" description] & \mathcal{A} \arrow[l, bend right, "H\text{:left adjoint}"'] \arrow[l, bend left, "L_0\text{:right adjoint}"] \end{tikzcd} \qquad \begin{tikzcd} \mathcal{A}^{\mathbf{2}} \arrow[r, "\mathrm{ev}_1" description] & \mathcal{A}. \arrow[l, bend right, "L_1\text{:left adjoint}"'] \arrow[l, bend left, "H\text{:right adjoint}"] \end{tikzcd}$$

For (iii), first consider injective objects. Take an object $[X_0 \xrightarrow{f} X_1]$ of $\mathcal{A}^{\mathbf{2}}$, and choose monomorphisms $\epsilon_i : X_i \hookrightarrow I_i$, where I_i is injective, $i \in \{0, 1\}$. These give two morphisms in $\mathcal{A}^{\mathbf{2}}$, which may be written as the commutative diagrams

$$\begin{tikzcd} X_0 \arrow[r, hook, "\epsilon_0"] \arrow[d, "f"'] & I_0 \arrow[d] \\ X_1 \arrow[r] & 0 \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} X_0 \arrow[r, "\epsilon_1 f"] \arrow[d, "f"'] & I_1 \arrow[d, "\mathrm{id}"] \\ X_1 \arrow[r, hook, "\epsilon_1"'] & I_1 \end{tikzcd}$$

The two framed columns are $L_0(I_0)$ and $H(I_1)$, respectively. By (ii), both are injective objects in $\mathcal{A}^{\mathbf{2}}$, and so is their direct sum (Lemma 2.8.14). Hence the commutative diagram

$$\begin{tikzcd} X_0 \arrow[r, hook, "(\epsilon_0, \epsilon_1 f)"] \arrow[d, "f"'] & I_0 \oplus I_1 \arrow[d, "\text{projection}"] \\ X_1 \arrow[r, hook, "\epsilon_1"'] & I_1 \end{tikzcd}$$

embeds $[X_0 \xrightarrow{f} X_1]$ into an injective object. For projective objects, use the functors H and L_1.

[9)] Editorial correction (O014-C027): the source writes $H : X \to [X \xrightarrow{\mathrm{id}_X} X]$, whereas the context defines the object map of the functor H, in parallel with L_0 and L_1. The assignment arrow $\mapsto$ is used here.

The discussion in Example 2.8.18 extends from $\mathcal{A}^{\mathbf{2}}$ to a general functor category $\mathcal{A}^{\mathcal{C}}$; the argument presents no essentially new difficulty. The exercises in this chapter give the details.

2.9 Serre Subcategories and K_0 Groups

A full subcategory of an abelian category automatically inherits the structure of an Ab-category, so one may ask whether the full subcategory is an additive category or an abelian category.

Definition 2.9.1 If $\mathcal{B}$ is a full subcategory of an abelian category $\mathcal{A}$, $\mathcal{B}$ is itself abelian, and the inclusion functor $\iota : \mathcal{B} \to \mathcal{A}$ is exact (Definition 2.8.3), then $\mathcal{B}$ is called an **abelian subcategory** of $\mathcal{A}$.

Proposition 2.9.2 A full subcategory $\mathcal{B}$ of an abelian category $\mathcal{A}$ is an abelian subcategory if and only if the following conditions hold.

- $0 \in \mathrm{Ob}(\mathcal{B})$;
- if $X, Y \in \mathrm{Ob}(\mathcal{B})$, then $X \oplus Y$ may also be chosen as an object of $\mathcal{B}$;
- for every morphism $f : X \to Y$, if $X, Y \in \mathrm{Ob}(\mathcal{B})$, then $\ker(f)$ and $\mathrm{coker}(f)$ may also be chosen as objects of $\mathcal{B}$.

Proof Notice that these conditions are self-dual. The "only if" direction is clear. Now suppose that the conditions hold. Then $\mathcal{B}$ is an additive category and $\iota : \mathcal{B} \to \mathcal{A}$ is an additive functor. We next show that ι preserves all finite $\varprojlim$ and $\varinjlim$. First, ι preserves 0 and finite direct sums. Moreover, equalizers can be expressed as ker (Remark 1.3.8), while the stated condition says that $\mathcal{B}$ is closed under taking ker. Thus ι preserves all finite $\varprojlim$; the case of $\varinjlim$ is dual.

Recalling the definitions of im and coim in Definition 1.2.1 and using the preceding step, we see that $\mathcal{B}$ is closed under both constructions. For every morphism f in $\mathcal{B}$, the canonical morphism $\mathrm{coim}(f) \xrightarrow{\sim} \mathrm{im}(f)$ in diagram (1.2.1) is an isomorphism in $\mathcal{A}$, hence also in $\mathcal{B}$. Therefore $\mathcal{B}$ is an abelian subcategory. $\square$

Definition 2.9.3 (J.-P. Serre) A full subcategory $\mathcal{T}$ of an abelian category $\mathcal{A}$ is called a **Serre subcategory** of $\mathcal{A}$ if it satisfies the following conditions.

- $0 \in \mathrm{Ob}(\mathcal{T})$;
- for every short exact sequence $0 \to X' \to X \to X'' \to 0$ in $\mathcal{A}$, one has $X \in \mathrm{Ob}(\mathcal{T})$ if and only if $X', X'' \in \mathrm{Ob}(\mathcal{T})$.

If the last condition is weakened as follows: for every exact sequence in $\mathcal{A}$

$$W \to X' \to X \to X'' \to Y,$$

if $W, X', X'', Y \in \mathrm{Ob}(\mathcal{T})$, then $X \in \mathrm{Ob}(\mathcal{T})$, then $\mathcal{T}$ is called a **weak Serre subcategory** of $\mathcal{A}$ [10)].

For example, if R is a commutative ring, all Noetherian (or Artinian) modules form a Serre subcategory of R-Mod. This is a standard fact in module theory [51, Lemma 6.10.2].

If $\mathcal{T}$ is a Serre subcategory (or weak Serre subcategory) of $\mathcal{A}$, then $\mathcal{T}^{\mathrm{op}}$ is a Serre subcategory (or weak Serre subcategory) of $\mathcal{A}^{\mathrm{op}}$.

A weak Serre subcategory $\mathcal{T}$ has the following saturation property: if $X \in \mathrm{Ob}(\mathcal{A})$ is isomorphic to an object of $\mathcal{T}$, then $X \in \mathrm{Ob}(\mathcal{T})$. This follows by applying the last condition to the exact sequence $0 \to 0 \to X \xrightarrow{\sim} Y \to 0$.

Corollary 2.9.4 If $\mathcal{T}$ is a weak Serre subcategory of an abelian category $\mathcal{A}$, then $\mathcal{T}$ is an abelian subcategory of $\mathcal{A}$.

Proof It suffices to verify the conditions of Proposition 2.9.2. First, $0 \in \mathrm{Ob}(\mathcal{T})$. Next, insert the following exact sequences into the definition of a weak Serre subcategory:

$$0 \to X \to X \oplus Y \to Y \to 0 \quad \text{(Proposition 2.2.7)},$$
$$0 \to 0 \to \ker(f) \to X \xrightarrow{f} Y, \quad X \xrightarrow{f} Y \to \operatorname{coker}(f) \to 0 \to 0,$$

which shows that $\mathcal{T}$ is closed under direct sums, ker, and coker. □

Example 2.9.5 If $F : \mathcal{A} \to \mathcal{B}$ is an exact functor between abelian categories, then the objects satisfying $FX = 0$ form a Serre subcategory of $\mathcal{A}$, denoted by $\ker(F)$.

Theorem 2.9.6 (Serre quotient) Let $\mathcal{T}$ be a Serre subcategory of an abelian category $\mathcal{A}$. Then there is an abelian category $\mathcal{A}/\mathcal{T}$ (which may be a large category), together with an exact, essentially surjective functor $Q : \mathcal{A} \to \mathcal{A}/\mathcal{T}$, such that $\ker(Q) = \mathcal{T}$ and with the following universal property: for every abelian category $\mathcal{B}$ and every exact functor $F : \mathcal{A} \to \mathcal{B}$ with $\ker(F) \supset \mathcal{T}$, there is a unique exact functor $G : \mathcal{A}/\mathcal{T} \to \mathcal{B}$ such that $F = GQ$.

In this universal property, G is faithful if and only if $\ker(F) = \mathcal{T}$.

Proof Define $S := \{f \in \mathrm{Mor}(\mathcal{A}) : \ker(f), \operatorname{coker}(f) \in \mathrm{Ob}(\mathcal{T})\}$. We shall show that S is a multiplicative system in the sense of Definition 1.9.5.

10) This somewhat abrupt definition is designed specifically for derived categories; see §4.4.

Clearly S contains all identity morphisms, so (S1) holds. Let $f : X \to Y$ and $g : Y \to Z$ belong to S. The mutually dual exact sequences

$$0 \to \ker(f) \to \ker(gf) \xrightarrow{f} \ker(g),$$
$$\operatorname{coker}(f) \xrightarrow{g} \operatorname{coker}(gf) \to \operatorname{coker}(g) \to 0,$$

immediately give $gf \in S$, so (S2) holds. Next consider morphisms $X \xrightarrow{s \in S} Z \xleftarrow{f} Y$. Define $W := X \underset{Z}{\times} Y$, with morphism $s' : W \to Y$. Observe that $\ker(s') \simeq \ker(s)$ (Proposition 1.3.10), and f induces $\operatorname{coker}(s') \hookrightarrow \operatorname{coker}(s)$ (Corollary 2.6.9); hence $s' \in S$. Thus (S3) holds. Finally, consider morphisms $X \underset{g}{\overset{f}{\rightrightarrows}} Y \xrightarrow{s \in S} W$, with $sf = sg$. Since $\operatorname{im}(f-g) \hookrightarrow \ker(s)$, we have $\operatorname{im}(f-g) \in \operatorname{Ob}(\mathcal{T})$; therefore $Z := \ker(f-g) \hookrightarrow X$ is a morphism in S, proving (S4).

Thus S is a left multiplicative system. All the conditions are self-dual, so S is also a right multiplicative system. Now take the localization functor $Q : \mathcal{A} \to \mathcal{A}/\mathcal{T} := \mathcal{A}[S^{-1}]$. Proposition 1.9.4 says that Q is essentially surjective (indeed, $\operatorname{Ob}(\mathcal{A}) = \operatorname{Ob}(\mathcal{A}/\mathcal{T})$), while Proposition 2.8.10 says that Q is an exact functor between abelian categories. Notice that $\mathrm{id}_{QX} = Q(\mathrm{id}_X)$ equals 0 if and only if there is a $U \in \operatorname{Ob}(\mathcal{A})$ such that $U \xrightarrow{0} X$ lies in S (this can be checked directly using Corollary 1.9.13); this implies $X = \operatorname{coker}[U \xrightarrow{0} X] \in \operatorname{Ob}(\mathcal{T})$. Conversely, if $X \in \operatorname{Ob}(\mathcal{T})$, we may take $U = X$. Hence $\ker(Q) = \mathcal{T}$.

We next verify the universal property. If $\ker(F) \supset \mathcal{T}$, then the exact sequence $\ker(s) \to X \xrightarrow{s \in S} Y \to \operatorname{coker}(s)$ shows that F sends every element of S to an isomorphism, and the universal property of localization determines the required G.

It remains to verify that G is exact. Theorem 1.9.17 first ensures that G is additive. Since every morphism in $\mathcal{A}[S^{-1}]$ can, after composition with an isomorphism arising from S, be represented by a morphism in $\mathcal{A}$, checking that G preserves kernels reduces to the exactness of F and Q. The same applies to cokernels. Since Q is essentially surjective and $\ker(Q) = \mathcal{T}$, the characterization of faithfulness reduces to statement (ii) of Proposition 2.8.9. □

Notice that $\mathcal{A}/\mathcal{T}$ is constructed as a localization and may therefore be a "large category." The exercises give another description of $\mathcal{A}/\mathcal{T}$ that controls the size of the Serre quotient under the assumption that $\mathcal{A}$ is a well-powered category (Definition 1.11.11).

Example 2.9.7 Let S be a multiplicative subset of a commutative ring R. Define a full subcategory $\mathcal{T}$ of R-Mod by declaring that $M \in \operatorname{Ob}(\mathcal{T})$ if and only if, for every $m \in M$, there is an $s \in S$ such that $sm = 0$. It is easy to verify that $\mathcal{T}$ is a Serre subcategory. We now show that R-$\mathsf{Mod}/\mathcal{T}$ is equivalent to $R[S^{-1}]$-Mod.

The assignment $M \mapsto M[S^{-1}] := M \underset{R}{\otimes} R[S^{-1}]$ defines an exact functor $F : R\text{-}\mathsf{Mod} \to R[S^{-1}]\text{-}\mathsf{Mod}$ that sends $\mathcal{T}$ to zero, so the universal property gives an exact functor $G : R\text{-}\mathsf{Mod}/\mathcal{T} \to R[S^{-1}]\text{-}\mathsf{Mod}$. Clearly $\ker(F) = \mathcal{T}$, so G is faithful. Moreover, G is essentially surjective: regard any $R[S^{-1}]$-module N as an R-module; one checks directly that there is an isomorphism of $R[S^{-1}]$-modules $N \underset{R}{\otimes} R[S^{-1}] \simeq N$.

It remains to prove that G is full and faithful. Given an $R[S^{-1}]$-module homomorphism $\varphi : M_1[S^{-1}] \to M_2[S^{-1}]$, form the fiber product [11] $M_0 := M_1 \underset{M_2[S^{-1}]}{\times} M_2$, where the map $M_1 \to M_2[S^{-1}]$ is the composite $M_1 \to M_1[S^{-1}] \xrightarrow{\varphi} M_2[S^{-1}]$. Write its projections as $p_1 : M_0 \to M_1$ and $p_2 : M_0 \to M_2$. The kernel and cokernel of p_1 are S-torsion modules, so p_1 lies in the multiplicative system defining the Serre quotient. Hence the roof $M_1 \xleftarrow{p_1} M_0 \xrightarrow{p_2} M_2$ defines a morphism in $R\text{-}\mathsf{Mod}/\mathcal{T}$ that is sent to φ. This proves the claim.

We now introduce an important construction closely related to Serre subcategories.

Definition 2.9.8 Let $\mathcal{A}$ be an abelian category. Define the commutative group $\mathrm{K}_0(\mathcal{A})$ as follows, writing the group operation additively. Take the free $\mathbb{Z}$-module F with basis $\mathrm{Ob}(\mathcal{A})$, and write the element corresponding to $X \in \mathrm{Ob}(\mathcal{A})$ as $\langle X \rangle \in F$. Let $R \subset F$ be the submodule generated by the elements

$$\langle X \rangle - \langle X' \rangle - \langle X'' \rangle\,, \quad 0 \to X' \to X \to X'' \to 0 : \text{ a short exact sequence in } \mathcal{A},$$

and call $\mathrm{K}_0(\mathcal{A}) := F/R$ the **$\mathbf{K_0}$ group** of $\mathcal{A}$. We henceforth denote the image of $X \in \mathrm{Ob}(\mathcal{A})$ in $\mathrm{K}_0(\mathcal{A})$ by $[X]$.

Lemma 2.9.9 For every abelian category $\mathcal{A}$, the following equalities hold in $\mathrm{K}_0(\mathcal{A})$.

(i) $[0] = 0$;

(ii) if $X, Y \in \mathrm{Ob}(\mathcal{A})$ and $X \simeq Y$, then $[X] = [Y]$;

(iii) $[X \oplus Y] = [X] + [Y]$;

(iv) if $0 \to X^1 \xrightarrow{f^1} \cdots \xrightarrow{f^{n-1}} X^n \to 0$ is an exact sequence in $\mathcal{A}$, then $\sum_{i=1}^{n}(-1)^i[X^i] = 0$;

(v) in the notation of §2.6, for a family of subobjects $X = X_0 \supset \cdots \supset X_r = 0$, one has

$$[X] = \sum_{i=0}^{r-1}[X_i/X_{i+1}].$$

[11] Editorial correction (O014-C029): the source defines M_0 as the set of elements $m \in M_1$ whose images “come from” M_2, and then uses a restriction $\varphi|_{M_0} : M_0 \to M_2$. A preimage in M_2 is not canonical when $M_2 \to M_2[S^{-1}]$ has an S-torsion kernel, so that morphism is not well-defined. We use instead the canonical fiber product and its two projections.

Proof Take the short exact sequence $0 \to 0 \to 0 \to 0 \to 0$; this gives (i). If $X \simeq Y$, then $0 \to X \xrightarrow{\sim} Y \to 0 \to 0$, together with (i), gives (ii). The short exact sequence of Proposition 2.2.7 gives (iii).

For (iv), append zero terms and extend the exact sequence infinitely in both directions. Consider the short exact sequences

$$0 \to \ker(f^i) \to X^i \to \mathrm{im}(f^i) \to 0, \quad i = 1, \ldots, n.$$

Then $\left[X^i\right] = \left[\ker(f^i)\right] + \left[\mathrm{im}(f^i)\right] = \left[\ker(f^i)\right] + \left[\ker(f^{i+1})\right]$; now take the alternating sum.

Finally, (v) follows by induction on r, using the short exact sequence $0 \to X_1 \to X_0 \to X_0/X_1 \to 0$. □

Remark 2.9.10 Since this book requires groups to be realized on small sets, and in view of Lemma 2.9.9 (ii), a fully precise formulation of $\mathrm{K}_0(\mathcal{A})$ should require $\mathcal{A}$ to have a small skeleton [51, Lemma 2.2.12]. This issue has little effect here.

Theorem 2.9.11 (Euler–Poincaré principle) Let $\cdots \to X^i \xrightarrow{d_X^i} X^{i+1} \to \cdots$ be a complex in an abelian category $\mathcal{A}$ with only finitely many nonzero terms. Then the equality

$$\sum_i (-1)^i \left[\mathrm{H}^i(X)\right] = \sum_i (-1)^i \left[X^i\right], \quad \mathrm{H}^i(X) := \mathrm{H}\left[X^{i-1} \xrightarrow{d_X^{i-1}} X^i \xrightarrow{d_X^i} X^{i+1}\right]$$

holds in $\mathrm{K}_0(\mathcal{A})$.

Proof For every i there is an exact sequence

$$0 \to \ker\left(d_X^{i-1}\right) \to X^{i-1} \xrightarrow{d_X^{i-1}} \ker\left(d_X^i\right) \to \mathrm{H}^i(X) \to 0,$$

and all terms are 0 for $|i| \gg 0$. Apply Lemma 2.9.9 (iv) and sum in $\mathrm{K}_0(\mathcal{A})$ to obtain

$$\sum_i (-1)^i \left(\left[X^{i-1}\right] + \left[\mathrm{H}^i(X)\right]\right) = \sum_i (-1)^i \left(\left[\ker\left(d_X^{i-1}\right)\right] + \left[\ker\left(d_X^i\right)\right]\right).$$

The terms on the right-hand side cancel in pairs; rearranging gives $\sum_i (-1)^i [\mathrm{H}^i(X)] = \sum_i (-1)^i [X^i]$. □

Example 2.9.12 Let $\mathcal{A}$ be the abelian category of finite-dimensional vector spaces over a division ring D. Since every vector space has a basis, $\mathrm{K}_0(\mathcal{A}) = \mathbb{Z} \cdot [D]$. On the other hand, the assignment $[X] \mapsto \dim_D X$ defines a group homomorphism $\dim : \mathrm{K}_0(\mathcal{A}) \to \mathbb{Z}$ sending $[D]$ to 1. Thus $\dim : \mathrm{K}_0(\mathcal{A}) \xrightarrow{\sim} \mathbb{Z}$.

If infinite-dimensional vector spaces are allowed, the K_0 group becomes trivial. Indeed, for every D-vector space V, a cardinality argument gives a D-vector space W such that $V \oplus W \simeq W$ and $\dim_D W = \max\{\dim_D V, \aleph_0\}$; consequently, $[V] = 0$.

For an exact functor $F : \mathcal{A} \to \mathcal{B}$ between abelian categories, since F preserves short exact sequences, the assignment $[X] \mapsto [FX]$ defines a group homomorphism [12] $\mathrm{K}_0(F) : \mathrm{K}_0(\mathcal{A}) \to \mathrm{K}_0(\mathcal{B})$. Therefore, if $\mathcal{A}$ is an abelian subcategory of $\mathcal{B}$, there is a natural homomorphism $\mathrm{K}_0(\mathcal{A}) \to \mathrm{K}_0(\mathcal{B})$.

Proposition 2.9.13 Let $\mathcal{T}$ be a Serre subcategory of an abelian category $\mathcal{A}$, and write its Serre quotient as $\mathcal{B} := \mathcal{A}/\mathcal{T}$. Then the exact functors $\mathcal{T} \to \mathcal{A} \xrightarrow{Q} \mathcal{B}$ induce an exact sequence of additive groups

$$\mathrm{K}_0(\mathcal{T}) \to \mathrm{K}_0(\mathcal{A}) \xrightarrow{\mathrm{K}_0(Q)} \mathrm{K}_0(\mathcal{B}) \to 0.$$

Proof First, $Q : \mathcal{A} \to \mathcal{B}$ is essentially surjective, so $\mathrm{K}_0(\mathcal{A}) \to \mathrm{K}_0(\mathcal{B})$ is surjective. The composite $\mathrm{K}_0(\mathcal{T}) \to \mathrm{K}_0(\mathcal{A}) \to \mathrm{K}_0(\mathcal{B})$ is plainly 0. It remains to prove $\ker(\mathrm{K}_0(Q)) \subset \operatorname{im}[\mathrm{K}_0(\mathcal{T}) \to \mathrm{K}_0(\mathcal{A})]$.

Let $\sum_{i=1}^n a_i[X_i] \in \ker(\mathrm{K}_0(Q))$, where $a_1, \ldots, a_n \in \mathbb{Z}$. This is equivalent to the existence of a family of short exact sequences in $\mathcal{B}$, $0 \to Y'_j \xrightarrow{f_j} Y_j \xrightarrow{g_j} Y''_j \to 0$, and integers $b_j \in \mathbb{Z}$ for $j = 1, \ldots, m$, such that

$$\sum_{i=1}^n a_i \langle QX_i \rangle = \sum_{j=1}^m b_j \left(\langle Y_j \rangle - \langle Y'_j \rangle - \langle Y''_j \rangle \right) \tag{2.9.1}$$

in the free $\mathbb{Z}$-module with basis $\mathrm{Ob}(\mathcal{B})$. Since $Q : \mathcal{A} \to \mathcal{B}$ is constructed by localization in Theorem 2.9.6, that theorem together with §1.9 gives:

- ◇ Q identifies $\mathrm{Ob}(\mathcal{A})$ with $\mathrm{Ob}(\mathcal{B})$, so (2.9.1) gives the equality in $\mathrm{K}_0(\mathcal{A})$

$$\sum_{i=1}^n a_i[X_i] = \sum_{j=1}^m b_j \left([Y_j] - [Y'_j] - [Y''_j] \right); \tag{2.9.2}$$

- ◇ moreover, after replacing Y_j, Y'_j, Y''_j, and f_j and g_j by isomorphic objects and morphisms in $\mathcal{B}$ arising from S, we may arrange that there are morphisms u_j and v_j in $\mathcal{A}$ such that $f_j = Q(u_j)$ and $g_j = Q(v_j)$. From $Q(v_j u_j) = g_j f_j = 0$ and Corollary 1.9.13, a further adjustment by members of S lets us arrange that $v_j u_j = 0$.

Thus, for every $1 \le j \le m$, there are exact sequences in $\mathcal{A}$

$$0 \to \ker(v_j) \to Y_j \xrightarrow{v_j} Y''_j \to \operatorname{coker}(v_j) \to 0,$$
$$0 \to \ker(u_j) \to Y'_j \xrightarrow{u_j} \ker(v_j) \to \frac{\ker(v_j)}{\operatorname{im}(u_j)} \to 0.$$

[12] Editorial correction (O014-C028): the source prints $\mathrm{K}_0(f)$, but this homomorphism is induced by the functor F; we use $\mathrm{K}_0(F)$.

Since Q is exact, $\operatorname{coker}(v_j)$, $\ker(u_j)$, and $\frac{\ker(v_j)}{\operatorname{im}(u_j)}$ all belong to $\mathcal{T} = \ker(Q)$. Therefore, in $\mathrm{K}_0(\mathcal{A})$,

$$[Y_j] - [Y'_j] - [Y''_j] \in \operatorname{im}\left[\mathrm{K}_0(\mathcal{T}) \to \mathrm{K}_0(\mathcal{A})\right].$$

Substituting these relations into (2.9.2) gives $\ker(\mathrm{K}_0(Q)) \subset \operatorname{im}\left[\mathrm{K}_0(\mathcal{T}) \to \mathrm{K}_0(\mathcal{A})\right]$. □

For an exact sequence such as the one in Proposition 2.9.13, a strategy that invariably succeeds is to try to extend it to the left: seek a family of higher K-groups $\mathrm{K}_i(\cdot)$ for $i \in \mathbb{Z}_{\geq 0}$ together with a canonical exact sequence

$$\cdots \to \mathrm{K}_{i+1}(\mathcal{B}) \to \mathrm{K}_i(\mathcal{T}) \to \mathrm{K}_i(\mathcal{A}) \to \mathrm{K}_i(\mathcal{B}) \to \cdots$$

One also hopes that the family $(\mathrm{K}_i(\mathcal{A}))_{i\geq 0}$ contains deep information about $\mathcal{A}$ and is, at least to some extent, computable. This is the subject of K-theory, whose applications also encompass exact categories, which are more general than abelian categories. Because the relevant constructions are based on the viewpoint of homotopy theory, this book cannot treat them in detail; interested readers may consult [53], or the summary in [7, §13.6].

2.10 Grothendieck Categories

All the main statements in this section are formulated relative to a chosen Grothendieck universe, rather than in terms of unqualified "large" categories.

Recall the notions of small limits, completeness, and generators (Definition 1.11.8).

Definition 2.10.1 (Grothendieck category [16]) Let $\mathcal{A}$ be an abelian category. It is called a **Grothendieck category** if the following conditions hold.

- ◇ $\mathcal{A}$ is cocomplete; in other words, it has all small $\varinjlim$;
- ◇ $\mathcal{A}$ has a generator;
- ◇ for every small filtered category I (see §1.6), the functor $\varinjlim : \mathcal{A}^I \to \mathcal{A}$ is exact. Equivalently, for every diagram $\alpha \to \beta \to \gamma$ in $\mathcal{A}^I$,

$$\begin{aligned} &\forall i \in \mathrm{Ob}(I), \quad 0 \to \alpha(i) \to \beta(i) \to \gamma(i) \to 0 \quad \text{is exact} \\ &\qquad \implies 0 \to \varinjlim \alpha \to \varinjlim \beta \to \varinjlim \gamma \to 0 \quad \text{is exact.} \end{aligned}$$

Example 2.8.6 has already shown that $\varinjlim$ preserves cokernels. Therefore, by Remark 2.8.4, the last condition is also equivalent to saying that every small filtered $\varinjlim$ preserves monomorphisms. The following argument illustrates the usefulness of this condition.

Proposition 2.10.2 Let $\mathcal{A}$ be a Grothendieck category. For every small set I, taking direct sums (that is, coproducts) gives an exact functor $\bigoplus_I : \mathcal{A}^I \to \mathcal{A}$.

Proof The finite case is clear, while Proposition 1.6.10 expresses $\bigoplus_I$ as a filtered $\varinjlim$ of finite direct sums. $\square$

We next obtain a seemingly simple but very useful property; it too rests on the exactness of filtered $\varinjlim$.

Proposition 2.10.3 Fix a small filtered partially ordered set $(I, \leq)$. Let $\mathcal{A}$ be a Grothendieck category, or more generally suppose that $\varinjlim : \mathcal{A}^{(I,\leq)} \to \mathcal{A}$ exists and is exact. For every

- ◇ $X \in \mathrm{Ob}(\mathcal{A})$ and subobject $Y \subset X$,
- ◇ family of subobjects $(X_i)_{i \in I}$ of X satisfying $i \leq j \implies X_i \subset X_j$,

with the notation of Convention 2.6.3, one has

$$\varinjlim_{i \in I} X_i \xrightarrow{\sim} \bigcup_{i \in I} X_i \subset X,$$

$$Y \cap \left(\bigcup_{i \in I} X_i \right) = \bigcup_{i \in I} (Y \cap X_i) \in \mathrm{Sub}_X.$$

Proof Since $\varinjlim_i$ is exact, the canonical morphism $\iota : \varinjlim_i X_i \to X$ determined by all the maps $X_i \hookrightarrow X$ and the universal property of $\varinjlim$ remains a monomorphism. If all $X_i \hookrightarrow X$ factor through a subobject $Z \subset X$, that universal property gives a factorization of ι as $\varinjlim_i X_i \to Z \subset X$. Thus $\iota : \varinjlim_i X_i \hookrightarrow X$ indeed gives the supremum of $(X_i)_{i \in I}$ in Sub_X. This proves the first equality.

Next write the quotient morphism $X \to X/Y$ as q. Then $Y = \ker(q)$ and $Y \cap X_i = \ker(q|_{X_i})$. Consider the compatible family of morphisms $q|_{X_i} : X_i \to X/Y$. Since taking $\varinjlim_i$ preserves kernels, the preceding paragraph gives $Y \cap \left(\bigcup_{i \in I} X_i \right) = \bigcup_{i \in I} (Y \cap X_i)$. $\square$

Example 2.10.4 (Module categories) Let R be a ring. Then R-Mod is a Grothendieck category. Indeed, its cocompleteness is already known [51, Theorem 6.2.2], while exactness of filtered $\varinjlim$ is given by [51, Lemma 6.8.3]. In the special case $R = \mathbb{Z}$, the category Ab is a Grothendieck category.

To develop the theory further, we need some preparation concerning generators.

Definition 2.10.5 Let s be a generator of a category $\mathcal{C}$. If, for every monomorphism $i : S_1 \hookrightarrow S_2$ in $\mathcal{C}$, the associated map $i_* : \mathrm{Hom}(s, S_1) \hookrightarrow \mathrm{Hom}(s, S_2)$ is bijective if and only if i is an isomorphism, then s is called a **strong generator** of $\mathcal{C}$.

Proposition 2.10.6 Let s be a strong generator of a category $\mathcal{C}$ and suppose that, for every object X and monomorphisms $S_i \hookrightarrow X$ with $i = 1, 2$, the fiber product $S_1 \cap S_2 := S_1 \underset{X}{\times} S_2$ exists. If $\kappa := |\operatorname{Hom}(s, X)|$, then $|\mathrm{Sub}_X| \leq 2^\kappa$.

Proof Let s be a strong generator and X any object. We shall show that

$$\begin{aligned} \mathrm{Sub}_X &\longrightarrow \{\text{subsets of } \operatorname{Hom}(s, X)\} \\ S &\longmapsto \operatorname{Hom}(s, S) \end{aligned}$$

is injective, which gives the desired inequality.

Take $S_1, S_2 \in \mathrm{Sub}_X$ such that $\operatorname{Hom}(s, S_1) = \operatorname{Hom}(s, S_2)$. If $S_1 \subset S_2$, the definition of a strong generator immediately gives $S_1 = S_2$. In general, $S_1 \cap S_2 \subset S_i$ for $i = 1, 2$ (see Definition 2.6.1), and as subsets of $\operatorname{Hom}(s, X)$,

$$\operatorname{Hom}(s, S_1 \cap S_2) = \operatorname{Hom}(s, S_1) \cap \operatorname{Hom}(s, S_2) = \operatorname{Hom}(s, S_i), \qquad i = 1, 2.$$

The preceding step gives $S_1 = S_1 \cap S_2 = S_2$, as claimed. □

Proposition 2.10.7 Every generator in an abelian category is automatically a strong generator.

Proof Let $\mathcal{A}$ be an abelian category and let s be a generator. Given a monomorphism $i : S_1 \hookrightarrow S_2$, consider the pair of morphisms

$$S_2 \underset{0}{\overset{q:\ \text{quotient morphism}}{\rightrightarrows}} S_2/S_1$$

If i is not an isomorphism, then $q \neq 0$. Thus, by the definition of a generator, there is an $\epsilon \in \operatorname{Hom}(s, S_2)$ such that $q\epsilon \neq 0\epsilon = 0$; this morphism ϵ cannot factor through $S_1 = \ker(q)$. □

Corollary 2.10.8 Every Grothendieck category is both well-powered and co-well-powered (Definition 1.11.11).

Proof An abelian category has all finite fiber products, while every Hom set is small by assumption. Thus well-poweredness follows from Proposition 2.10.6 and Proposition 2.10.7. Moreover, in an abelian category Sub_X and Quot_X always have the same cardinality. □

Corollary 2.10.9 Every Grothendieck category $\mathcal{A}$ is complete; in other words, it has all small $\varprojlim$.

Proof The category $\mathcal{A}$ is cocomplete and co-well-powered. Apply Corollary 1.11.14.□

Thus a Grothendieck category has all small direct sums and small direct products. The exercises in this chapter show that the canonical morphism $\delta : \bigoplus_{i\in I} X_i \to \prod_{i\in I} X_i$ in (1.1.1) is a monomorphism; for module categories this is of course already clear.

Corollary 2.10.10 Let $\mathcal{A}$ be a Grothendieck category. A functor $G : \mathcal{A}^{\mathrm{op}} \to \mathsf{Set}$ is representable if and only if G preserves all small $\varprojlim$; more concretely, G turns every small $\varinjlim$ in $\mathcal{A}$ into a small $\varprojlim$ in Set.

Proof Apply Corollary 1.11.16 to $\mathcal{A}^{\mathrm{op}}$. □

Although the definition of a Grothendieck category is not self-dual, the dual version of the preceding result still holds; see Corollary 2.10.16.

We shall now show that Grothendieck categories have enough injectives. In fact, not only does a monomorphism $X \hookrightarrow I$ as in Definition 2.8.16 exist, but I can also be chosen functorially in X.

Lemma 2.10.11 Let $\mathcal{A}$ be a Grothendieck category and let s be a generator. An object I is injective if and only if, for every monomorphism $X \hookrightarrow s$, every morphism $X \to I$ extends to a morphism $s \to I$; in other words, the data can be completed to a commutative diagram
$$\begin{tikzcd} X \arrow[r] \arrow[d, hook] & I \\ s \arrow[ur, dashed, "\exists"'] & \end{tikzcd}.$$

Proof Injectivity is equivalent to the following statement: for every monomorphism $A \hookrightarrow B$, every morphism $f : A \to I$ extends to a morphism $B \to I$. Thus the "only if" direction is clear.

We now prove the converse. Recall that $\mathcal{A}$ is well-powered. Given data $I \xleftarrow{f} A \hookrightarrow B$ as above, consider the small set

$$\mathcal{S} := \left\{ (A', f') \;\middle|\; \begin{array}{ll} A \subset A' \subset B & \text{(as subobjects)}, \\ f' : A' \to I & \text{extends } f \end{array} \right\},$$

partially ordered by extension. By exactness of filtered $\varinjlim$, every totally ordered subset $\mathcal{T}$ of $\mathcal{S}$ has an upper bound

$$\widetilde{A} := \varinjlim_{(A',f')\in\mathcal{T}} A', \qquad \widetilde{f} := \varinjlim_{(A',f')\in\mathcal{T}} f' : \widetilde{A} \to I.$$

Zorn's lemma shows that $\mathcal{S}$ has a maximal element (A', f'). We claim that $A' = B$. Otherwise, because s is a strong generator, there is a $g : s \to B$ that does not factor through $A' \hookrightarrow B$. We shall show that f' extends to $A' + g(s) \to I$, contradicting the maximality of (A', f').

Set $Y := A' \cap g(s)$. Consider the solid part of the following commutative diagram:

$$\begin{tikzcd}
\ker(g) \arrow[r, hook] & g^{-1}(Y) \arrow[r, two heads, "g"] \arrow[d, hook] & Y \arrow[r, hook] \arrow[d, hook] & A' \arrow[r, dashed, "f'"] & I \\
& s \arrow[r, two heads, "g"'] \arrow[urrr, dashed, "\varphi"] & g(s) \arrow[urr, dashed, bend right, "f''"'] & &
\end{tikzcd}$$

By assumption, a φ as shown by the dashed arrow exists and makes the triangular part commute. Since φ vanishes on $\ker(g)$, there is also an f'' as shown by the dashed arrow such that $\varphi = f''g$. Pulling back along $g^{-1}(Y) \twoheadrightarrow Y$ (an epimorphism by Proposition 2.1.6), we find that the restrictions of f' and f'' to Y agree. By the universal property of the fibered coproduct, they can therefore be glued to a morphism $A' + g(s) \to I$. This proves the claim. □

We now introduce a variant of the so-called "small object argument." The reader should briefly review Definition A.2.7 and the related discussion of regular cardinals; what follows uses their basic operations and properties.

Definition 2.10.12 Let $\mathcal{C}$ be a category with all small filtered $\varinjlim$, let $I \subset \mathrm{Mor}(\mathcal{C})$ be a subset, and let α be a small regular cardinal.

(i) An object $X \in \mathrm{Ob}(\mathcal{C})$ is called ***α*-small relative to *I*** if the following condition holds. Let $\widetilde{\alpha}$ be a small regular cardinal with $\widetilde{\alpha} \geq \alpha$. For every functor $\beta \mapsto Y_\beta$ from the ordinal $\widetilde{\alpha}$, regarded as a small filtered category, to $\mathcal{C}$, if $Y_\beta \to Y_{\beta'}$ belongs to I whenever $\beta \leq \beta'$, then the canonical map

$$\varinjlim_{\beta} \mathrm{Hom}\,(X, Y_\beta) \longrightarrow \mathrm{Hom}\left(X, \varinjlim_{\beta} Y_\beta\right)$$

(see Equation A.2.1) is a bijection.

(ii) If X is α-small relative to $\mathrm{Mor}(\mathcal{C})$, then X is called an ***α*-small object**.

Relative to the chosen I, if $\alpha \leq \alpha'$, then α-smallness implies α'-smallness.

Lemma 2.10.13 Let $\mathcal{A}$ be a Grothendieck category, let X be an object of $\mathcal{A}$, and let α be a small regular cardinal. If $\alpha > \kappa := |\mathrm{Sub}_X|$, then X is α-small relative to all monomorphisms (Definition 2.10.12).

Proof Consider data $\beta \mapsto Y_\beta$ from Definition 2.10.12 (i), with the condition that $Y_\beta \to Y_{\beta'}$ is a monomorphism whenever $\beta \leq \beta'$. We may take $\alpha = \widetilde{\alpha}$ in that definition without loss of generality.

Since filtered $\varinjlim$ in $\mathcal{A}$ is exact, every $Y_\beta \to \varinjlim_\gamma Y_\gamma$ remains a monomorphism. Exactness of filtered $\varinjlim$ in Ab also implies that the map

$$\varinjlim_{\gamma<\alpha} \mathrm{Hom}(X, Y_\gamma) \longrightarrow \mathrm{Hom}\left(X, \varinjlim_{\gamma<\alpha} Y_\gamma\right)$$

is injective; it therefore remains to prove surjectivity. We henceforth regard each Y_β as a subobject of $\varinjlim_\gamma Y_\gamma$.

For a morphism $f : X \to \varinjlim_\gamma Y_\gamma$, every $\beta < \alpha$ determines a subobject $f^{-1}(Y_\beta)$ of X. We may assume that $f^{-1}(Y_\beta) \neq X$ for every β, since otherwise there is nothing left to prove. Recall that f^{-1} is given by a fiber product. Using exactness of filtered $\varinjlim$ once more, we obtain a natural isomorphism

$$\varinjlim_{\beta} f^{-1}(Y_\beta) \simeq f^{-1}\left(\varinjlim_{\beta} Y_\beta\right) = X.$$

There are at most κ subobjects $f^{-1}(Y_\beta)$, so there is a subset S of α with $|S| \leq \kappa$ such that the left-hand side may be replaced by $\varinjlim_{\beta\in S}$.

Consider the ordinal $\sigma := \sup S \leq \alpha$. We claim that $\sigma < \alpha$. Otherwise, $|S| \leq \kappa < \alpha$ and $|S| \geq \mathrm{cf}(\alpha)$ (Proposition A.2.8 (ii)) would give $\mathrm{cf}(\alpha) < \alpha$, contradicting the regularity of α. This claim ensures that $f^{-1}(Y_\beta)$ is a subobject of $f^{-1}(Y_\sigma)$ for every $\beta \in S$. Consequently, $f^{-1}(Y_\sigma) = X$, so f factors through Y_σ. □

Theorem 2.10.14 (A. Grothendieck) Let $\mathcal{A}$ be a Grothendieck category. Write $\mathcal{I}$ for the full subcategory of $\mathcal{A}$ consisting of injective objects, and write its inclusion functor as $\iota : \mathcal{I} \to \mathcal{A}$. There are a functor $F : \mathcal{A} \to \mathcal{I}$ and a natural transformation $\varphi : \mathrm{id}_{\mathcal{A}} \to \iota F$ such that every

$$\varphi_X : X \to F(X), \qquad X \in \mathrm{Ob}(\mathcal{A})$$

is a monomorphism in $\mathcal{A}$. In particular, $\mathcal{A}$ has enough injectives.

Proof Choose a generator s. Let F_0 be the identity functor $\mathrm{id}_{\mathcal{A}}$. The first step is to form, for every object X, the pushout diagram

$$\begin{tikzcd}
\bigoplus_{Y\in\mathrm{Sub}_s} \bigoplus_{\varphi:Y\to X} Y \arrow[r] \arrow[d, hook] \arrow[dr, phantom, "\boxplus"] & X \arrow[d, "\varphi(0,1)_X"] \\
\bigoplus_{Y\in\mathrm{Sub}_s} \bigoplus_{\varphi:Y\to X} s \arrow[r] & F_1(X)
\end{tikzcd}$$

Proposition 2.1.6 ensures that $\varphi(0,1)_X : X \to F_1(X)$ in this diagram is a monomorphism. As X varies, the assignment $X \mapsto F_1(X)$ becomes an endofunctor of $\mathcal{A}$, and $\varphi(0,1) : F_0 \to F_1$ becomes a natural transformation.

More generally, for every small ordinal α we shall construct a functor $F_\alpha : \mathcal{A} \to \mathcal{A}$ together with a family of natural transformations $\varphi(\beta, \alpha) : F_\beta \to F_\alpha$ for $\beta \leq \alpha$, having the following properties.

- ◇ $\varphi(\alpha, \alpha) = \mathrm{id}_{F_\alpha}$;
- ◇ $\varphi(\beta, \alpha)_X : F_\beta(X) \to F_\alpha(X)$ is a monomorphism for every X;
- ◇ if $\gamma \leq \beta \leq \alpha$, then $\varphi(\beta, \alpha)\varphi(\gamma, \beta) = \varphi(\gamma, \alpha)$.

The construction proceeds by transfinite recursion [51, §1.3]. Starting with $\alpha = 0$, suppose that F_β and the family of transformations $\varphi(\beta', \beta)$ have been defined for every ordinal $\beta < \alpha$. Set

$$F_\alpha(X) := \begin{cases} F_1(F_\beta(X)), & \alpha = \beta + 1 : \text{ a successor ordinal,} \\ \varinjlim_{\beta<\alpha} F_\beta(X), & \alpha : \text{a limit ordinal,} \end{cases}$$

where the $\varinjlim$ is formed with respect to $(\varphi(\beta', \beta))_{\beta' \leq \beta < \alpha}$. The properties of F_1 and the fact that filtered $\varinjlim$ preserves monomorphisms determine the evident choices of $\varphi(\beta, \alpha)$.

By Lemma A.2.10, choose a small regular cardinal α such that $\alpha > |\mathrm{Sub}_s|$. We show that $F_\alpha X$ is injective for every X. Given $Y \hookrightarrow s$ and $f : Y \to F_\alpha X$, the definition of a regular cardinal A.2.7 implies that α is a limit ordinal. The construction of F_α, the inclusion $\mathrm{Sub}_Y \subset \mathrm{Sub}_s$, and Lemma 2.10.13 show that f factors through some $\varphi : Y \to F_\beta X$ with $\beta + 1 < \alpha$. Consider the commutative diagram

$$\begin{tikzcd}
Y \arrow[r, "\text{through }(Y{,}\varphi)"] \arrow[d, hook] & \bigoplus_{Y'\in\mathrm{Sub}_s}\bigoplus_{\varphi':Y'\to F_\beta X} Y' \arrow[r] \arrow[d, hook] \arrow[dr, phantom, "\boxplus"] & F_\beta X \arrow[d] \arrow[dr] & \\
s \arrow[r, "\text{through }(Y{,}\varphi)"'] & \bigoplus_{Y'\in\mathrm{Sub}_s}\bigoplus_{\varphi':Y'\to F_\beta X} s \arrow[r] & F_{\beta+1}X \arrow[r] & F_\alpha X
\end{tikzcd}$$

This diagram shows directly that f extends to a morphism $s \to F_\alpha X$. Lemma 2.10.11 then implies that $F_\alpha X$ is injective. Finally, take $F := F_\alpha : \mathcal{A} \to \mathcal{I}$ and $\varphi := \varphi(0, \alpha)$. □

Corollary 2.10.15 Every Grothendieck category has an injective cogenerator.

Proof Choose a generator s of the Grothendieck category $\mathcal{A}$. The set Quot_s is known to be small. Choose an injective object I and a monomorphism $\bigoplus_{Q\in\mathrm{Quot}_s} Q \hookrightarrow I$. We use the criterion in Proposition 2.8.15 to show that I is a cogenerator.

Let $T \in \mathrm{Ob}(\mathcal{A})$ be nonzero. There is a nonzero morphism $s \to T$, which factors as $s \twoheadrightarrow Q' \hookrightarrow T$. Consequently there is a

$$Q' \overset{\text{evident}}{\hookrightarrow} \bigoplus_{Q\in\mathrm{Quot}_s} Q \hookrightarrow I.$$

By the definition of an injective object, this composite extends to a morphism $T \to I$, and the extension is plainly nonzero. This proves the claim. □

Corollary 2.10.16 Let $\mathcal{A}$ be a Grothendieck category. A functor $F : \mathcal{A} \to \mathsf{Set}$ is representable if and only if it preserves all small $\varprojlim$.

Proof The category $\mathcal{A}$ has a cogenerator; apply Corollary 1.11.16. □

Another characterization of Grothendieck categories is given by the Gabriel–Popescu–Kuhn theorem A.3.2 in the appendix to this book.

Exercises

1. Show that the full additive subcategory of $\mathbb{Z}$-Mod formed by free $\mathbb{Z}$-modules is not an abelian category.
2. Let X be an object of an abelian category and let $Y, Z \in \mathrm{Sub}_X$. Consider the isomorphisms of partially ordered sets

$$\begin{tikzcd} {[Y \cap Z, Y]} \arrow[r, "\sim"] \arrow[d, "\simeq"] & {[Z, Y + Z]} \arrow[d, "\simeq"] \\ \mathrm{Sub}_{Y/(Y\cap Z)} \arrow[r, "\sim"] & \mathrm{Sub}_{(Y+Z)/Z} \end{tikzcd}$$

where the vertical arrows come from Theorem 2.6.8 (ii), the lower horizontal arrow comes from the isomorphism $Y/(Y \cap Z) \simeq (Y + Z)/Z$ in Theorem 2.6.8 (iii), and the upper horizontal arrow is the map $W \mapsto W + Z$ of Proposition 2.4.5. Show that the diagram commutes.
3. Let $\Bbbk$ be a commutative ring and $\mathcal{A}$ a $\Bbbk$-linear abelian category. Prove that if $\mathrm{Hom}(X, X)$ is an Artinian $\Bbbk$-module for every $X \in \mathrm{Ob}(\mathcal{A})$, then $\mathcal{A}$ satisfies the bi-chain condition of Definition 2.5.6. Hint For a bi-chain $(X_n, \alpha_n, \beta_n)_{n=0}^{\infty}$, the assignment $f \mapsto \beta_n f \alpha_n$ defines a sequence of monomorphisms of $\Bbbk$-modules $\mathrm{Hom}(X_{n+1}, X_{n+1}) \hookrightarrow \mathrm{Hom}(X_n, X_n)$. Show that this morphism is an isomorphism if and only if both α_n and β_n are isomorphisms.
4. Let $\mathcal{K}$ be an Ab-category with a zero object. If $\ker(e)$ exists for every $X \in \mathrm{Ob}(\mathcal{K})$ and every idempotent $e \in \mathrm{End}(X)$, then $\mathcal{K}$ is called a **Karoubian category**.

   (i) Prove that every idempotent e in a Karoubian category has a cokernel, and that there is a canonical isomorphism $\ker(e) \simeq \mathrm{coker}(e)$.

(ii) For every Ab-category $\mathcal{C}$ with a zero object, canonically construct a Karoubian category $\mathrm{kar}(\mathcal{C})$ together with a fully faithful additive functor $\varphi_{\mathcal{C}} : \mathcal{C} \to \mathrm{kar}(\mathcal{C})$ such that, for every Karoubian category $\mathcal{K}$, the functor

$$\varphi_{\mathcal{C}}^* : \mathcal{K}^{\mathrm{kar}(\mathcal{C})} \to \mathcal{K}^{\mathcal{C}}, \quad F \mapsto F\varphi_{\mathcal{C}}$$

is an equivalence. Here $\mathcal{K}^{\mathcal{C}}$ denotes the functor category whose objects are all additive functors $\mathcal{C} \to \mathcal{K}$, and similarly for the other notation.

(iii) If $\Bbbk$ is a commutative ring and $\mathcal{C}$ is $\Bbbk$-linear, then $\mathrm{kar}(\mathcal{C})$ also carries a natural $\Bbbk$-linear structure.

The category $\mathrm{kar}(\mathcal{C})$ is usually called the **Karoubi envelope** of $\mathcal{C}$; it is obtained by formally adjoining all direct summands to $\mathcal{C}$.

Hint〉 Take the objects of $\mathrm{kar}(\mathcal{C})$ to be pairs (X, e), where $X \in \mathrm{Ob}(\mathcal{C})$ and $e \in \mathrm{End}(X)$ is idempotent. A morphism $(X, p) \to (Y, q)$ is a morphism $f : X \to Y$ in $\mathcal{C}$ satisfying $qf = f = fp$, while $\varphi_{\mathcal{C}}(X) = (X, \mathrm{id}_X)$.

5. For a ring R, verify that all projective R-modules form a Karoubian category that is not abelian.

6. Let the abelian category $\mathcal{A}$ have all small direct sums $\bigoplus_{i\in I}$ (respectively, all small products $\prod_{i\in I}$). Prove that a small set $\Sigma \subset \mathrm{Ob}(\mathcal{A})$ is a family of generators (respectively, a family of cogenerators) if and only if, for every $X \in \mathrm{Ob}(\mathcal{A})$, there is a small direct sum (respectively, small product) S of members of Σ, together with an epimorphism $S \twoheadrightarrow X$ (respectively, a monomorphism $X \hookrightarrow S$).

Hint〉 For a family of generators, the "if" direction is easy. For the "only if" direction, take the evident morphism

$$f : \bigoplus_{s\in\Sigma} \bigoplus_{\varphi\in\mathrm{Hom}(s,X)} s \to X;$$

every morphism φ factors through f. Hence every composite $s \xrightarrow{\varphi'} X \to \mathrm{coker}(f)$ is 0; conclude that $\mathrm{coker}(f) = 0$.

7. Let A, B, C be subobjects of an object X in an abelian category, and suppose that $A \cap (B + C) = 0 = B \cap C$. Prove that $A \cap B = 0$ and $(A + B) \cap C = 0$.

Hint〉 Construct a monomorphism $(A + B) \cap C \to A$, then show that it factors through $A \cap (B + C)$. You may first try the case of a module category.

8. Prove that the abelian category of finitely generated $\mathbb{Z}$-modules has enough projective objects but does not have enough injective objects.

9. (Schanuel's lemma) Let P, Q be projective objects in an abelian category and consider the short exact sequences

$$0 \to K \to P \xrightarrow{\phi} M \to 0,$$
$$0 \to L \to Q \xrightarrow{\psi} M \to 0.$$

From these data, form the pullback diagram

$$\begin{tikzcd} X \arrow[r, "\psi'"] \arrow[d, "\phi'"'] \arrow[dr, phantom, "\square"] & P \arrow[d, "\phi"] \\ Q \arrow[r, "\psi"'] & M. \end{tikzcd}$$

Show that there are short exact sequences $0 \to L \to X \xrightarrow{\psi'} P \to 0$ and $0 \to K \to X \xrightarrow{\phi'} Q \to 0$, and then prove that there is an isomorphism $K \oplus Q \simeq L \oplus P$. Hint Apply Propositions 1.3.10 and 2.1.6.

10. Prove that $\mathbb{Q}/\mathbb{Z}$ is an injective cogenerator for Ab.

11. Verify in detail the adjunctions in Example 2.8.18.

12. Let $\mathcal{C}$ be a nonempty category and $\mathcal{A}$ an abelian category. For every $c \in \mathrm{Ob}(\mathcal{C})$, the evaluation functor $\mathrm{ev}_c : \mathcal{A}^{\mathcal{C}} \to \mathcal{A}$ sends F to Fc.

(i) Suppose that $\mathcal{A}$ has all direct sums of the forms $\bigoplus_{c \in \mathrm{Ob}(\mathcal{C})}$ and $\bigoplus_{f \in \mathrm{Hom}_{\mathcal{C}}(c,c')}$. For every $c \in \mathrm{Ob}(\mathcal{C})$, define a functor $\mathcal{L}_c : \mathcal{A} \to \mathcal{A}^{\mathcal{C}}$ on objects by $(\mathcal{L}_c X)(c') = \bigoplus_{\mathrm{Hom}(c,c')} X$. Complete its definition on morphisms so that $\mathcal{L}_c$ is left adjoint to ev_c.

(ii) Under the preceding assumptions, prove that if $\mathcal{A}$ has enough projective objects, then so does $\mathcal{A}^{\mathcal{C}}$.

(iii) Investigate the dual version. Suppose that $\mathcal{A}$ has all products of the forms $\prod_{c \in \mathrm{Ob}(\mathcal{C})}$ and $\prod_{f \in \mathrm{Hom}_{\mathcal{C}}(c',c)}$. Define a functor $\mathcal{R}_c : \mathcal{A} \to \mathcal{A}^{\mathcal{C}}$ by $(\mathcal{R}_c X)(c') = \prod_{\mathrm{Hom}(c',c)} X$ that is right adjoint to ev_c. Then prove that if $\mathcal{A}$ has enough injective objects, so does $\mathcal{A}^{\mathcal{C}}$.

(iv) Investigate the relation of these constructions to Example 2.8.18 when $\mathcal{C} = \mathbf{2}$.

13. With the notation of the preceding exercise, let $\mathcal{C}$ be a small category. Prove that if s is a generator (respectively, cogenerator) for $\mathcal{A}$, then all the $\mathcal{L}_c(s)$ (respectively, $\mathcal{R}_c(s)$), as c ranges over $\mathrm{Ob}(\mathcal{C})$, form a family of generators (respectively, cogenerators) for $\mathcal{A}^{\mathcal{C}}$, provided the required direct sums (respectively, direct products) exist.

Hint For the generator case, for every $c \in \mathrm{Ob}(\mathcal{C})$ and every morphism $f : X \to Y$ in $\mathcal{A}^{\mathcal{C}}$ there is a commutative diagram

$$\begin{tikzcd} \mathrm{Hom}_{\mathcal{A}}(s, Y(c)) \arrow[r, "\sim"] & \mathrm{Hom}_{\mathcal{A}^{\mathcal{C}}}(\mathcal{L}_c(s), Y) \\ \mathrm{Hom}_{\mathcal{A}}(s, X(c)) \arrow[r, "\sim"'] \arrow[u, "f(c)_*"] & \mathrm{Hom}_{\mathcal{A}^{\mathcal{C}}}(\mathcal{L}_c(s), X). \arrow[u, "f_*"'] \end{tikzcd}$$

The condition for a family to be a family of generators is equivalent to saying that, as c ranges over $\mathrm{Ob}(\mathcal{C})$, all the maps f_* in the second column determine f uniquely.

14. (Direct construction of the Serre quotient) For a Serre subcategory $\mathcal{T}$ of an abelian category $\mathcal{A}$, define a new category whose set of objects is $\mathrm{Ob}(\mathcal{A})$, and whose set of morphisms from X to Y is

$$\varinjlim_{X' \subset X, Y' \subset Y} \mathrm{Hom}_{\mathcal{A}}(X', Y/Y'), \quad \text{where } X/X',\ Y' \in \mathrm{Ob}(\mathcal{T}).$$

(i) Complete the definition of composition of morphisms and show that this category is equivalent to the Serre quotient $\mathcal{A}/\mathcal{T}$ of Theorem 2.9.6. Hint〉 Verify that this category has the universal property stated in Theorem 2.9.6.

(ii) Use this construction to show that if $\mathcal{A}$ is well-powered (Definition 1.11.11), then $\mathcal{A}/\mathcal{T}$ is also a $\mathcal{U}$-category, where $\mathcal{U}$ is the chosen Grothendieck universe.

15. Let $\mathcal{B}$ be an abelian subcategory of $\mathcal{A}$, and suppose that for every $X \in \mathrm{Ob}(\mathcal{A})$ there is a sequence of subobjects $0 = X_0 \subset \cdots \subset X_n = X$ such that $X_i/X_{i-1} \in \mathrm{Ob}(\mathcal{B})$. Prove that the inclusion functor $\mathcal{B} \to \mathcal{A}$ induces an isomorphism of groups $\mathrm{K}_0(\mathcal{B}) \xrightarrow{\sim} \mathrm{K}_0(\mathcal{A})$.

Hint〉 Use the Schreier refinement theorem 2.4.8 to show that $[X] \mapsto \sum_{i=1}^{n} [X_i/X_{i-1}]$ gives an inverse map independent of the choice of the sequence of subobjects.

16. Let isom be the set of isomorphism classes of all objects in an additive category $\mathcal{A}$. Write $\langle X \rangle$ for the image of the isomorphism class of an object X in the free $\mathbb{Z}$-module $\mathbb{Z}^{\oplus \mathrm{isom}}$. Define the direct-sum version of the K_0 group by

$$\mathrm{K}_{\oplus}(\mathcal{A}) := \mathbb{Z}^{\oplus \mathrm{isom}} / \text{the submodule generated by all } \langle X \oplus Y \rangle - \langle X \rangle - \langle Y \rangle .$$

Let ind be the set of isomorphism classes of indecomposable objects. Prove that if $\mathcal{A}$ is Karoubian and every nonzero object has a finite direct-sum decomposition $X \simeq X_1 \oplus \cdots \oplus X_k$, with $X_i \neq 0$ and $\mathrm{End}(X_i)$ a local ring (see the discussion before Corollary 2.5.8), then there is a canonical isomorphism

$$\mathbb{Z}^{\oplus \mathrm{ind}} \xrightarrow{\sim} \mathrm{K}_{\oplus}(\mathcal{A}).$$

Hint〉 Recall that if $\mathrm{End}(X_i)$ is a local ring, then X_i is indecomposable; prove this directly or see [51, Lemma 6.12.6]. Then use [51, Theorem 6.12.8] to obtain uniqueness of the decomposition.

17. Let $\mathcal{A}$ be a Grothendieck category and $(X_i)_{i \in I}$ a family of objects indexed by a small set I. Prove that the canonical morphism $\delta : \bigoplus_{i \in I} X_i \to \prod_{i \in I} X_i$ in (1.1.1) is a monomorphism.

Hint〉 For every finite subset $F \subset I$, consider the commutative diagram

$$\begin{tikzcd}
\bigoplus_{i \in I} X_i \arrow[r, "\delta"] & \prod_{i \in I} X_i \\
\bigoplus_{i \in F} X_i \arrow[u, hook] \arrow[r, "\sim", "\delta_F"'] & \prod_{i \in F} X_i \arrow[u, hook, "\iota_F"']
\end{tikzcd}$$

Take the filtered $\varinjlim$ of $\iota_F \delta_F$ over all finite $F \subset I$, and conclude that δ is a monomorphism; see Proposition 1.6.10.

18. Prove that if $\mathcal{A}$ is a Grothendieck category, then the category of complexes $\mathsf{C}(\mathcal{A})$ (see §3.1) and the functor category $\mathcal{A}^{\mathcal{C}}$ are both Grothendieck categories, where $\mathcal{C}$ is any small category.

19. (Characterization of direct sums) Let $\mathcal{A}$ be a Grothendieck category. Consider $X \in \mathrm{Ob}(\mathcal{A})$ and a family of subobjects $(X_i)_{i \in I}$, where I is a small set. The restriction of the canonical morphism $\bigoplus_{i \in I} X_i \to X$ to each X_i is the inclusion $X_i \hookrightarrow X$. Prove that $\bigoplus_{i \in I} X_i \xrightarrow{\sim} X$ if and only if the following properties hold.

(i) $X = \sum_{i \in I} X_i$;

(ii) for every $i \in I$ one has $X_i \cap \sum_{\substack{j \in I \\ j \neq i}} X_j = 0$.

Observe that condition (ii) need only be checked on finite subsets of I. If I itself is finite, the preceding statement holds in any abelian category.

Hint Property (i) is equivalent to the canonical morphism being an epimorphism, while (ii) is equivalent to its being a monomorphism. The first is immediate; for the second, use filtered $\varinjlim$ to reduce to the case of finite I.

20. Consider an object X in a Grothendieck category $\mathcal{A}$.

(i) Supply a detailed proof of Remark 2.7.11. Hint Use the preceding exercise to handle infinite direct sums.

(ii) Under the following hypothesis, prove that if X is split, then X is semisimple: for every nonzero $Y \in \mathrm{Ob}(\mathcal{A})$, there is a nonzero subobject $Y_0 \subset Y$ such that every chain in the partially ordered set $\mathrm{Sub}_{Y_0} \smallsetminus \{Y_0\}$ has an upper bound. Verify this hypothesis for $\mathcal{A} = R\text{-}\mathsf{Mod}$. Hint You may follow the implication (iii) $\Longrightarrow$ (i) in the proof of [51, Proposition 6.11.4], together with the proof of [51, Lemma 6.11.3].

Chapter 3 Complexes

The notion of a complex was introduced briefly in § 2.2. For a complex $X = (X^n, d^n)_n$ in an abelian category $\mathcal{A}$, the cohomology $\mathrm{H}^n(X) := \ker(d^n)/\operatorname{im}(d^{n-1})$[1) extracts important invariants from X. Conversely,[2) a complex whose cohomology vanishes in every degree is an exact sequence, also called an acyclic complex.

Much of §§ 3.2–3.5 does not involve cohomology and applies to an arbitrary additive category $\mathcal{A}$. The category of complexes over $\mathcal{A}$ is denoted by $\mathsf{C}(\mathcal{A})$. Unless stated otherwise, all complexes in this chapter are cochain complexes in the sense of Remark 2.2.6; whenever $\Bbbk$ occurs, it denotes an arbitrary commutative ring.

For complexes X and Y over an additive category $\mathcal{A}$, homotopy is an equivalence relation on $\mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X, Y)$. It can also be understood through the Hom complex $\mathrm{Hom}^\bullet(X, Y)$, studied in § 3.2. If $\mathcal{A}$ is abelian, homotopic morphisms induce the same maps on cohomology. The mapping cone $\mathrm{Cone}(f)$ introduced in § 3.3 will subsequently play a major role in the study of derived categories. Mapping cones are closely related to homotopy, and both have direct topological interpretations.

A double complex $X = (X^{p,q}, {}^{\triangleright}d^{p,q}, {}^{\vartriangle}d^{p,q})_{p,q}$ may be viewed as a complex with two

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[1)] Editorial correction (O014-C032): the source prints $\operatorname{im}(d^{n+1})$. Under the cochain convention $X^{n-1} \xrightarrow{d^{n-1}} X^n \xrightarrow{d^n} X^{n+1}$, the subobject contained in $\ker(d^n)$ is $\operatorname{im}(d^{n-1})$; the corrected index agrees with the book's earlier definition.

[2)] Editorial correction (O014-C030): the source has a full stop followed by a stray character before the clause about the vanishing of all cohomology, leaving the transition ungrammatical. The context gives the converse direction, restored here without changing the mathematical claim.

independent directions, or equivalently as a "complex of complexes." A double complex X can be flattened by direct sums (respectively, direct products) into a complex $\mathrm{tot}_{\oplus} X$ (respectively, $\mathrm{tot}_{\Pi} X$), called a total complex of X. The precise definitions involve several signs; these details are the subject of § 3.5.

One of the basic techniques of homological algebra is to derive a long exact sequence in cohomology from a short exact sequence of complexes $0 \to X' \xrightarrow{f} X \xrightarrow{g} X'' \to 0$:

$$\cdots \to \mathrm{H}^n(X') \xrightarrow{\mathrm{H}^n(f)} \mathrm{H}^n(X) \xrightarrow{\mathrm{H}^n(g)} \mathrm{H}^n(X'') \xrightarrow{\text{connecting morphism}} \mathrm{H}^{n+1}(X') \to \cdots .$$

Long exact sequences can also be understood from the viewpoint of mapping cones, although this requires a nontrivial argument. These topics are treated in §§ 3.6–3.7.

To help the reader become familiar with the basic operations on complexes, § 3.8 offers an integrated exercise based on the Hochschild homology $\mathrm{HH}_n(M)$ and Hochschild cohomology $\mathrm{HH}^n(M)$ of an (R, R)-bimodule M. Hochschild homology and cohomology have many uses throughout mathematics. This chapter gives only a brief pedagogical introduction; both will reappear as examples in later chapters.

Using the truncation functors introduced in § 3.9, we shall relate the vertical or horizontal cohomology of a double complex to the cohomology of its total complex in § 3.10. Certain boundedness conditions on the double complex are needed. Some textbooks treat these properties with spectral sequences; following [23], this book takes a somewhat indirect route and proves them directly.

For an object X of an abelian category $\mathcal{A}$, a resolution of X has left and right versions, concretely given by exact sequences[3)]

$$0 \to X \to I^0 \to I^1 \to I^2 \to \cdots \quad \text{or} \quad \cdots \to P_1 \to P_0 \to X \to 0.$$

Equivalently, one replaces X by a complex with better properties through a quasi-isomorphism $X \to I := (I^n)_n$ or $P := (P_n)_n \to X$, where P is a chain complex. A quasi-isomorphism is a morphism inducing isomorphisms on cohomology or homology. If every I^n (respectively, P_n) is required to be injective (respectively, projective), the construction is called an injective (respectively, projective) resolution of X. Resolutions are studied in depth in § 3.11. They are also functorial and unique up to homotopy in the following sense. For the injective version, suppose the solid part of the following

3) Editorial correction (O014-C031): the source prints $0 \to X \to I^0 \to I^2 \to \cdots$, omitting I^1. The diagram immediately following uses I^0, I^1, I^2 in order; the missing term I^1 is restored here.

diagram lies in $\mathcal{A}$.

$$\begin{tikzcd} 0 \arrow[r] & X' \arrow[r] & (I')^0 \arrow[r] & (I')^1 \arrow[r] & (I')^2 \arrow[r] & \cdots \\ 0 \arrow[r] & X \arrow[r] \arrow[u, "f"] & I^0 \arrow[r] \arrow[u, dashed, "\beta^0"] & I^1 \arrow[r] \arrow[u, dashed, "\beta^1"] & I^2 \arrow[r] \arrow[u, dashed, "\beta^2"] & \cdots \end{tikzcd}$$

Assume that both rows are exact and that every $(I')^n$ is injective. There are dashed arrows $\beta^0, \beta^1, \ldots$ making the entire diagram commutative, and the morphism of complexes $\beta : I \to I'$ is unique up to homotopy. This is a joint application of Theorem 3.11.5 and Lemma 3.11.7, both special cases of broader statements.

In fact, § 3.11 treats not only resolutions of objects X of $\mathcal{A}$, but also resolutions of complexes. Its results include the useful Cartan–Eilenberg resolution (Theorem 3.11.10). Some arguments for complexes inevitably become lengthy, even though their underlying idea remains clear.

Replacing objects of $\mathcal{A}$ by injective (respectively, projective) resolutions, applying a left exact (respectively, right exact) functor $F : \mathcal{A} \to \mathcal{B}$ termwise, and then taking H^n (respectively, $\mathrm{H}_n = \mathrm{H}^{-n}$) leads to the classical definition of the right derived functors $\mathrm{R}^n F$ (respectively, the left derived functors $\mathrm{L}_n F$). Short exact sequences in $\mathcal{A}$ then induce long exact sequences of derived functors; see § 3.12. Resolutions of complexes also allow derived functors to be evaluated on bounded-below or bounded-above complexes; in the classical literature these constructions are called hyperderived functors.

Right (respectively, left) derived functors and their long exact sequences naturally give rise to cohomological (respectively, homological) δ-functors, among which derived functors are "universal." Proposition 3.12.14 characterizes universal δ-functors by the erasability (respectively, co-erasability) condition of Definition 3.12.11. This result is due to Grothendieck. The associated collection of techniques forms the core of classical homological algebra.

Another important construction consists of the right (respectively, left) derived functors of a bifunctor $F : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}$ such that F is left exact (respectively, right exact) in both variables. Standard examples are the Ext and Tor functors of § 3.14; the former necessarily involves complexes over an opposite category. Theorem 3.14.2 studies a class of bifunctors called balanced and shows that deriving in either variable gives the same result. Its proof rests on basic properties of the cohomology of double complexes.

Suppose that $\mathcal{A}$ has exact countable products. The construction $\lim^1$ discussed in § 3.13 is another example of a right derived functor: it is the first right derived functor of $\varprojlim$, while all higher right derived functors vanish (Theorem 3.13.8). To obtain practical criteria ensuring $\lim^1 = 0$, we shall introduce the Mittag–Leffler condition in Definition 3.13.11.

Finally, in § 3.15 we study K-injective and K-projective complexes and define K-injective and K-projective resolutions. These are natural generalizations of injective and projective resolutions to unbounded complexes. Because K-injectivity and K-projectivity are properties of an entire complex, their definitions are more concise and natural than the termwise definitions of injective and projective resolutions. They extend the definition of derived functors to unbounded complexes. The issue is existence: Theorems 3.15.11 and 3.15.12 give partial results. Their proofs are comparatively intricate and require some properties of $\lim^1$.

Reading Guide

Experience with complexes of $\Bbbk$-modules, where $\Bbbk$ is a commutative ring, helps one understand this chapter more quickly but is not strictly necessary. After Chapter 2, working with abelian categories more general than module categories presents no fundamental difficulty.

Readers who want to reach the classical definition of derived functors quickly and are not interested in its topological motivation may first skip the material on mapping cones and mapping cylinders in § 3.3, § 3.4, and § 3.7. In addition, § 3.4 relates $\mathsf{C}(\mathcal{A}^{\mathrm{op}})$ to $\mathsf{C}(\mathcal{A})^{\mathrm{op}}$. Some of the signs there require care, but they do not greatly affect the main ideas. On a first reading, one may study only the statements and skip the proofs.

Since the later material does not depend on Hochschild homology or cohomology, readers may decide whether to skip it according to their time and interests. It is not difficult and will recur as examples and remarks in subsequent chapters. Similarly, the material of § 3.13 appears only as examples later, except in § 3.15. The material of § 3.15 is cited only in § 4.11 and § 4.12. Readers who are not particularly interested in unbounded derived functors may first study the definitions and Example 3.15.3, then omit the remainder.

3.1 Complexes over Additive Categories

Definition 2.2.4 briefly introduced complexes and their usual notation. Let $\mathcal{A}$ be an additive category, and consider unbounded complexes over it, that is, complexes without endpoint terms, $X = (X^n, d^n)_{n\in\mathbb{Z}}$. The morphism d^n is often denoted by d^n_X. For historical reasons, the morphisms d^n_X are also called "differentials."

Definition 3.1.1 A morphism from a complex X to a complex Y is defined to be data $(f^n)_{n\in\mathbb{Z}}$, also abbreviated to f, where $f^n \in \mathrm{Hom}_{\mathcal{A}}(X^n, Y^n)$ satisfies

$$d^n_Y f^n = f^{n+1} d^n_X, \quad n \in \mathbb{Z}.$$

All complexes and their morphisms form a category $\mathsf{C}(\mathcal{A})$. The identity morphism of a complex X is determined by $(\mathrm{id}_X)^n := \mathrm{id}_{X^n}$, and composition is defined by $(gf)^n = g^n f^n$. Addition $f + g := (f^n + g^n)_{n\in\mathbb{Z}}$ makes $\mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X, Y)$ an abelian group.

If $\mathcal{A}$ is a $\Bbbk$-linear category in the sense of Definition 1.4.4, for a fixed commutative ring $\Bbbk$, then $\mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X, Y)$ is also a $\Bbbk$-module under termwise scalar multiplication. In every result of this section, "additive" may be replaced by the more general term "$\Bbbk$-linear." For brevity, only the additive version will be stated.

The key property in the definition of a complex is $d^{n+1}d^n = 0$. It can also be recast in the language of differential graded objects.[4)]

Definition 3.1.2 (Graded objects) Let $\mathcal{A}$ be an arbitrary category. Recall the construction of the product category $\mathcal{A}^{\mathbb{Z}}$:

- ◇ its objects are of the form $X = (X^n)_{n\in\mathbb{Z}}$, where $X^n \in \mathrm{Ob}(\mathcal{A})$;
- ◇ its morphisms are of the form $f = (f^n : X^n \to Y^n)_{n\in\mathbb{Z}}$, with no further condition, and composition is performed termwise (or degreewise), so that $(gf)^n = g^n f^n$;
- ◇ define an automorphism T of $\mathcal{A}^{\mathbb{Z}}$ by the following "shift" functor: $(TX)^n = X^{n+1}$ and $(Tf)^n = f^{n+1}$.

An object $X = (X^n)_n$ is also called a **$\mathbb{Z}$-graded object** over $\mathcal{A}$, or simply a **graded object**.

More generally, objects of $\mathcal{A}^{\mathbb{Z}^m}$ are called $\mathbb{Z}^m$-graded objects and are written $(X^{n_1,\ldots,n_m})_{n_1,\ldots,n_m}$; their morphisms are written $(f^{n_1,\ldots,n_m})_{n_1,\ldots,n_m}$. This category carries a family of mutually commuting shift functors $T_1, \ldots, T_m$. The special case $m = 2$ is also called a **bigraded object**.

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4) The related discussion continues in § 5.2.

If $\mathcal{A}$ is additive, then so is $\mathcal{A}^{\mathbb{Z}^m}$: addition of morphisms, direct sums of objects, and all other operations are performed degreewise.

Definition 3.1.3 (Differential graded objects) Let $\mathcal{A}$ be an additive category. A differential graded object over $\mathcal{A}$ is data (X, d), where $X \in \mathrm{Ob}(\mathcal{A}^{\mathbb{Z}})$ and $d : X \to TX$ satisfies $(Td)d = 0$. A morphism $(X, d_X) \to (Y, d_Y)$ is a morphism $f \in \mathrm{Hom}_{\mathcal{A}^{\mathbb{Z}}}(X, Y)$ satisfying $(Tf)d_X = d_Y f$.

Writing d in a differential graded object (X, d) concretely as a family of morphisms $d_X^n : X^n \to X^{n+1}$, the condition on d is $d_X^{n+1} d_X^n = 0$, while the condition on a morphism is $f^{n+1} d_X^n = d_Y^n f^n$. We have thus recovered Definition 3.1.1. Hence the category of differential graded objects over $\mathcal{A}$ is isomorphic to the category of complexes. We shall pass between these two viewpoints without further comment.

The first step in studying complexes is to understand the various limits in $\mathsf{C}(\mathcal{A})$. The version for the product category $\mathcal{A}^{\mathbb{Z}}$ is straightforward: limits can be formed at each $n \in \mathbb{Z}$, that is, termwise or degreewise in $\mathcal{A}$. We now explain how to lift this construction to $\mathsf{C}(\mathcal{A})$. Write $U : \mathsf{C}(\mathcal{A}) \to \mathcal{A}^{\mathbb{Z}}$ for the forgetful functor sending a differential graded object (X, d) to its underlying graded object X.

Lemma 3.1.4 (Termwise construction of limits) For any functor $\alpha : I \to \mathsf{C}(\mathcal{A})$, write $\overline{\alpha} := U\alpha$. Suppose that $\varinjlim \overline{\alpha}$ exists. It admits a unique structure as an object of $\mathsf{C}(\mathcal{A})$ for which it is $\varinjlim \alpha$. The corresponding statement holds for $\varprojlim$.

In the language of Definition 1.5.2, this says that the forgetful functor U creates $\varinjlim$ and $\varprojlim$.

Proof Suppose that $\alpha : I \to \mathsf{C}(\mathcal{A})$ sends $i \in \mathrm{Ob}(I)$ to the differential graded object $\alpha(i) = (\overline{\alpha}(i), d_i)$. If $\varinjlim \overline{\alpha}$ exists, the morphisms d_i determine $d : \varinjlim \overline{\alpha} \to (T \varinjlim \overline{\alpha} \simeq \varinjlim T\overline{\alpha})$; here the automorphism T automatically preserves $\varinjlim$. This morphism is characterized by the commutative diagram

$$\begin{tikzcd} \varinjlim \overline{\alpha} \arrow[r, "d"] & \varinjlim T\overline{\alpha} \arrow[r, "\sim"] & T \varinjlim \overline{\alpha} \\ \overline{\alpha}(i) \arrow[u] \arrow[r, "d_i"'] & T\overline{\alpha}(i) \arrow[u] \arrow[ur] & \end{tikzcd} \qquad (i \in \mathrm{Ob}(I)).$$

The relations $(Td_i)d_i = 0$ imply $(Td)d = 0$, so $(\varinjlim \overline{\alpha}, d)$ is an object of $\mathsf{C}(\mathcal{A})$. The diagram also shows that this is the unique choice making all maps $(\overline{\alpha}(i), d_i) \to (\varinjlim \overline{\alpha}, d)$ morphisms.

We verify its universal property as $\varinjlim \alpha$ in $\mathsf{C}(\mathcal{A})$. Suppose a compatible family of morphisms $f_i : \alpha(i) \to L$ in $\mathsf{C}(\mathcal{A})$ is given. Forgetting the differentials gives $f_i : \overline{\alpha}(i) \to U(L)$, which determines a unique $f : \varinjlim \overline{\alpha} \to U(L)$. The key point is to show that f is a morphism in $\mathsf{C}(\mathcal{A})$. Although this is to be expected, we write the diagram explicitly:

$$\begin{tikzcd}
\varinjlim \overline{\alpha} \arrow[rr, "d"] \arrow[dd, "f"'] & & \varinjlim T\overline{\alpha} \arrow[r, "\sim"] & T \varinjlim \overline{\alpha} \arrow[dd, "Tf"] \\
& \overline{\alpha}(i) \arrow[ul] \arrow[dl, "f_i"] \arrow[r, "d_i"] & T\overline{\alpha}(i) \arrow[u] \arrow[ur] \arrow[dr, "Tf_i"] & \\
U(L) \arrow[rrr, "d_L"'] & & & TU(L)
\end{tikzcd} \qquad (i \in \mathrm{Ob}(I)).$$

Commutativity of every part follows directly from the definitions or the construction. Thus the composites of $d_L f$ and $(Tf)d$ with every map $\overline{\alpha}(i) \to \varinjlim \overline{\alpha}$ agree. Therefore $d_L f = (Tf)d$, as required. $\square$

Proposition 3.1.5 The category $\mathsf{C}(\mathcal{A})$ is additive. If $\mathcal{A}$ is complete (respectively, cocomplete), then $\mathsf{C}(\mathcal{A})$ is also complete (respectively, cocomplete).

Proof Use Lemma 3.1.4 to reduce every required limit to a degreewise limit in $\mathcal{A}$. $\square$

Definition 3.1.6 (Translation functors) Let X be an object of $\mathsf{C}(\mathcal{A})$ and let $n \in \mathbb{Z}$. Define the complex $X[n]$ as follows:[5)]

$$(X[n])^k := X^{k+n}, \quad d^k_{X[n]} := (-1)^n d_X^{k+n}.$$

This construction gives an additive functor $[n] : \mathsf{C}(\mathcal{A}) \to \mathsf{C}(\mathcal{A})$. If $f : X \to Y$ is a morphism of complexes, then $f[n] : X[n] \to Y[n]$ is given by $f[n]^k := f^{n+k}$.

It is clear that $X[n]$ is again a complex, so the functor is well defined. The following properties are also immediate.

- ◇ $[0] = \mathrm{id}_{\mathsf{C}(\mathcal{A})}$.
- ◇ $[n][m] = [n+m]$. Thus $[n]$ is an automorphism of $\mathsf{C}(\mathcal{A})$, with inverse $[-n]$.
- ◇ The family $(d^n_X)_{n\in\mathbb{Z}}$ gives a morphism of complexes $d_X : X \to X[1]$ (Hint: $d_X^{n+1} d_X^n = 0$).

Convention 3.1.7 Every object S of $\mathcal{A}$ may be regarded as a complex $(S^n, d^n_S)_{n\in\mathbb{Z}}$ with $S^0 := S$, every other term equal to 0, and every d^n_S the zero morphism. This identification gives a fully faithful additive functor $\mathcal{A} \hookrightarrow \mathsf{C}(\mathcal{A})$. More generally, for every $n \in \mathbb{Z}$, $S[-n]$ is the complex concentrated in degree n: it places S in degree n and 0 in every other degree.

[5)] Inserting $(-1)^n$ in the definition of $d_{X[n]}$ is useful. This version is also isomorphic to the signless version $X[n]^\circ := \left(X^{n+k}, d_X^{k+n}\right)_k$. For $n = 1$, define $X[1] \xrightarrow{\sim} X[1]^\circ$ using the sign pattern $\cdots, +, -, +, \cdots$.

The following result is immediate.

Proposition 3.1.8 For an additive functor $F : \mathcal{A} \to \mathcal{A}'$, the assignment $(X^n, d_X^n)_n \mapsto (FX^n, Fd_X^n)_n$ defines a functor $\mathsf{C}F : \mathsf{C}(\mathcal{A}) \to \mathsf{C}(\mathcal{A}')$. It commutes with translation functors: $[m] \circ \mathsf{C}F = \mathsf{C}F \circ [m]$ for every $m \in \mathbb{Z}$.

Remark 3.1.9 Every result in this section also applies to the chain complexes $(X_\bullet, d_\bullet)$ mentioned in Remark 2.2.6. The translation functor on the category of chain complexes is defined by

$$X[n]_k := X_{k+n}, \quad d_k^{X[n]} = (-1)^n d_{n+k}^X$$

If one passes between the two conventions via $X_n = X^{-n}$ and $d_n = d^{-n}$, then the translation $[1]$ for chain complexes corresponds to the translation $[-1]$ for cochain complexes.

3.2 Hom Complexes and Homotopy

We previously defined Hom between two complexes. This section explains how to promote it to a complex and how to promote composition of morphisms to a kind of multiplication on complexes. We continue to let $\mathcal{A}$ be an additive category. If $\mathbb{k}$ is any commutative ring and $\mathcal{A}$ is a $\mathbb{k}$-linear category, every statement in this section has an evident $\mathbb{k}$-linear version.

Recall that taking Hom sets gives a functor $\mathrm{Hom} : \mathcal{A}^{\mathrm{op}} \times \mathcal{A} \to \mathsf{Ab}$. The notation Hom without a superscript refers to the Hom set in $\mathcal{A}$ or in $\mathsf{C}(\mathcal{A})$; when necessary, a subscript distinguishes the category.

Definition 3.2.1 (Hom complex) Let X, Y be objects of $\mathsf{C}(\mathcal{A})$. For every $n \in \mathbb{Z}$, define

$$\mathrm{Hom}^n(X,Y) := \prod_{k\in\mathbb{Z}} \mathrm{Hom}_{\mathcal{A}}\left(X^k, Y^{k+n}\right);$$

termwise composition of morphisms gives

$$\begin{aligned} \mathrm{Hom}^n(Y,Z) \times \mathrm{Hom}^m(X,Y) &\longrightarrow \mathrm{Hom}^{n+m}(X,Z) \\ (g,f) &\longmapsto gf := \left(g^{k+m} f^k\right)_{k\in\mathbb{Z}}. \end{aligned} \tag{3.2.1}$$

Notice that $d_X = (d_X^k)_k \in \mathrm{Hom}^1(X,X)$, and similarly for d_Y. Accordingly, define

$$\begin{aligned} d^n = d^n_{\mathrm{Hom}^\bullet(X,Y)} : \mathrm{Hom}^n(X,Y) &\longrightarrow \mathrm{Hom}^{n+1}(X,Y) \\ f &\longmapsto d_Y f - (-1)^n f d_X. \end{aligned}$$

The operation (3.2.1) is plainly associative and distributive over addition. Given morphisms $u : \underline{X} \to X$ and $v : Y \to \underline{Y}$ in $\mathsf{C}(\mathcal{A})$, regard them respectively as elements of $\mathrm{Hom}^0(\underline{X}, X)$ and $\mathrm{Hom}^0(Y, \underline{Y})$. For every $n \in \mathbb{Z}$ there is then a homomorphism

$$\begin{aligned} \mathrm{Hom}^n(u,v) : \mathrm{Hom}^n(X,Y) &\longrightarrow \mathrm{Hom}^n(\underline{X},\underline{Y}) \\ f &\longmapsto vfu. \end{aligned} \tag{3.2.2}$$

Definition–Proposition 3.2.2 The data $(\mathrm{Hom}^n(X,Y), d^n)_{n\in\mathbb{Z}}$ above form an object of $\mathsf{C}(\mathsf{Ab})$, called the **Hom complex**. As X and Y vary, this construction together with (3.2.2) gives an additive functor

$$\mathrm{Hom}^\bullet : \mathsf{C}(\mathcal{A})^{\mathrm{op}} \times \mathsf{C}(\mathcal{A}) \to \mathsf{C}(\mathsf{Ab}).$$

Proof First verify that $d^{n+1}d^n = 0$. Since $d_Y^2 = 0 = d_X^2$, a direct calculation gives

$$\begin{aligned} d^{n+1}(d^n f) &= d_Y\left(d_Y f - (-1)^n f d_X\right) - (-1)^{n+1}\left(d_Y f - (-1)^n f d_X\right) d_X \\ &= (-1)^{n+1} d_Y f d_X - (-1)^{n+1} d_Y f d_X = 0. \end{aligned}$$

Next consider functoriality. Place the map $\mathrm{Hom}^\bullet(u,v)$ defined in (3.2.2) into the diagram[6)]

$$\begin{tikzcd} \mathrm{Hom}^n(X^\bullet, Y^\bullet) \arrow[r, "\mathrm{Hom}^n(u,v)"] \arrow[d, "d^n"'] & \mathrm{Hom}^n(\underline{X}^\bullet, \underline{Y}^\bullet) \arrow[d, "\underline{d}^n"] \\ \mathrm{Hom}^{n+1}(X^\bullet, Y^\bullet) \arrow[r, "\mathrm{Hom}^{n+1}(u,v)"'] & \mathrm{Hom}^{n+1}(\underline{X}^\bullet, \underline{Y}^\bullet); \end{tikzcd}$$

Because $d_X u = u d_{\underline{X}}$ and $v d_Y = d_{\underline{Y}} v$, this diagram is plainly commutative. □

The differential $d_{\mathrm{Hom}^\bullet(X,Y)}$ is also compatible with shifts. Its proof is much easier than its statement.

Lemma 3.2.3 Let $n, m \in \mathbb{Z}$ and $f \in \mathrm{Hom}^n(X,Y)$. Identify f with the element $\underline{f} = (f^k)_k$ of $\mathrm{Hom}^{n-m}(X, Y[m])$, or with the element $\overline{f} = (f^{k-m})_k$ of $\mathrm{Hom}^{n-m}(X[-m], Y)$. Then

$$\begin{aligned} d^{n-m}_{\mathrm{Hom}^\bullet(X,Y[m])}\underline{f} &\xlongequal{\text{under the identification}} (-1)^m d^n_{\mathrm{Hom}^\bullet(X,Y)} f, \\ d^{n-m}_{\mathrm{Hom}^\bullet(X[-m],Y)}\overline{f} &\xlongequal{\text{under the identification}} d^n_{\mathrm{Hom}^\bullet(X,Y)} f. \end{aligned}$$

Consequently, $\mathrm{Hom}^\bullet(X, Y[m]) = \mathrm{Hom}^\bullet(X,Y)[m] \simeq \mathrm{Hom}^\bullet(X[-m], Y)$.

6) Translator's note (O014-C033): the source labels both horizontal arrows $\mathrm{Hom}^{n+1}(u,v)$. Since the upper arrow connects terms of degree n, this translation restores its label to $\mathrm{Hom}^n(u,v)$; the lower arrow retains the label $\mathrm{Hom}^{n+1}(u,v)$.

Proof For example, for the first equality,

$$(d^{n-m}_{\mathrm{Hom}^\bullet(X,Y[m])}\underline{f})^k = d^{k+n-m}_{Y[m]}\underline{f}^k - (-1)^{n-m}\underline{f}^{k+1}d^k_X : X^k \to Y^{k+n+1},$$

which also equals $(-1)^m\left(d^{k+n}_Y f^k - (-1)^n f^{k+1}d^k_X\right)$. The argument for $\overline{f}$ is similar: although the case of $\overline{f}$ has no factor $(-1)^m$, there is still an isomorphism of Hom complexes; see the footnote to Definition 3.1.6. □

The multiplication defined in (3.2.1) also obeys the Leibniz rule.

Lemma 3.2.4 Let X, Y, Z be complexes and $(g, f) \in \mathrm{Hom}^n(Y, Z) \times \mathrm{Hom}^m(X, Y)$. With respect to the multiplication in (3.2.1), the following equality holds:

$$\begin{aligned} d^{n+m}(gf) = {} & (d^n g)f \\ & + (-1)^n g(d^m f) \end{aligned}$$

in $\mathrm{Hom}^{n+m+1}(X, Z)$.

Proof This is a direct calculation, left to the reader. □

Lemma 3.2.5 Let $n \in \mathbb{Z}$ and $f \in \mathrm{Hom}^n(X, Y)$. Then $d^n f = 0$ if and only if $f \in \mathrm{Hom}(X, Y[n])$.

Proof Via Lemma 3.2.3, identify f with an element of $\mathrm{Hom}^0(X, Y[n])$, reducing the question to the case $n = 0$. Clearly, $d^0 f = 0 \iff d_Y f = f d_X$. □

Definition 3.2.6 Let $n \in \mathbb{Z}$. Regard $\mathrm{Hom}(X, Y[n])$ as a subset of $\mathrm{Hom}^n(X, Y)$.

- ◇ If $f, g \in \mathrm{Hom}(X, Y[n])$ and $h \in \mathrm{Hom}^{n-1}(X, Y)$ (equivalently, $h \in \mathrm{Hom}^{-1}(X, Y[n])$)[7] satisfies $g - f = d^{n-1}h$, then h is called a **homotopy** from f to g.
- ◇ For f, g as above, if there exists a homotopy from f to g, then f and g are called homotopic. A morphism f homotopic to the zero morphism is called **null-homotopic**.

As the special case $n = 0$, a morphism $f \in \mathrm{Hom}(X, Y)$ is null-homotopic if and only if there is a family of morphisms $h^m : X^m \to Y^{m-1}$ such that

$$\forall m \in \mathbb{Z}, \quad f^m = d^{m-1}_Y h^m + h^{m+1} d^m_X.$$

Homotopy is an equivalence relation. Lemma 3.2.5 gives the key equality in Ab (or $\Bbbk$-Mod):

$$\mathrm{H}^n\left(\mathrm{Hom}^\bullet(X, Y), d^\bullet\right) = \mathrm{Hom}\left(X, Y[n]\right) / \{\text{null-homotopic morphisms}\}.$$

The next result shows that null-homotopic morphisms are not merely closed under addition but form an ideal.

[7] Translator's note (O014-C034): the source does not state the degree of h, although $d^{n-1}h$ requires $h \in \mathrm{Hom}^{n-1}(X, Y)$. This translation makes its type explicit.

Lemma 3.2.7 Let $X \xrightarrow{f} Y[m]$ and $Y \xrightarrow{g} Z[n]$ be morphisms of complexes. If f or g is null-homotopic, then $g[m] \circ f : X \to Z[n+m]$ is null-homotopic.

Proof We suppress the superscripts on d. Suppose $f = dh$, where $h \in \mathrm{Hom}^{-1}(X, Y[m])$. Interpreting composition inside $\mathrm{Hom}^\bullet$, we may abbreviate $g[m] \circ f$ as gf. Lemma 3.2.4 gives $d(gh) = (dg)h + (-1)^n g(dh) = (-1)^n gf$, so gf is null-homotopic. The case $g = dh$ is similar. □

Suppressing the subscript on $d_{\mathrm{Hom}^\bullet}$, Lemmas 3.2.4 and 3.2.7 show that the multiplication on $\mathrm{Hom}^\bullet$ in (3.2.1) induces

$$\mathrm{H}^n\left(\mathrm{Hom}^\bullet(Y,Z)\right) \times \mathrm{H}^m\left(\mathrm{Hom}^\bullet(X,Y)\right) \to \mathrm{H}^{n+m}\left(\mathrm{Hom}^\bullet(X,Z)\right), \tag{3.2.3}$$

and the binary operation (3.2.3) inherits associativity and distributivity over addition from (3.2.1).

Definition 3.2.8 Define the category $\mathsf{K}(\mathcal{A})$ from $\mathsf{C}(\mathcal{A})$ by setting

$$\mathrm{Ob}\left(\mathsf{K}(\mathcal{A})\right) := \mathrm{Ob}\left(\mathsf{C}(\mathcal{A})\right),$$
$$\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}\left(X, Y\right) := \mathrm{H}^0\left(\mathrm{Hom}^\bullet\left(X, Y\right), d^\bullet_{\mathrm{Hom}}\right),$$

with composition determined by (3.2.3) for $n = m = 0$.

Observe that $\mathsf{K}(\mathcal{A})$ is an additive category: addition on morphism sets is clear, the zero object comes from the zero object of $\mathsf{C}(\mathcal{A})$, and the existence of products (or coproducts) follows from the more general property that there are natural isomorphisms

$$\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}\left(X, \prod_{i\in I} Y_i\right) \simeq \prod_{i\in I} \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X, Y_i)$$
$$\text{or} \quad \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}\left(\coprod_{i\in I} X_i, Y\right) \simeq \prod_{i\in I} \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X_i, Y),$$

where I is any set, provided $\prod_{i\in I} Y_i$ (or $\coprod_i X_i$) exists at the level of $\mathsf{C}(\mathcal{A})$. Since taking products and taking H^0 commute in $\mathsf{C}(\mathsf{Ab})$, everything reduces to the evident isomorphisms of Hom complexes

$$\mathrm{Hom}^\bullet\left(X, \prod_i Y_i\right) \simeq \prod_i \mathrm{Hom}^\bullet(X, Y_i)$$
$$\text{or} \quad \mathrm{Hom}^\bullet\left(\coprod_i X_i, Y\right) \simeq \prod_i \mathrm{Hom}^\bullet(X_i, Y).$$

The shift functor $[n]$ induces an automorphism of the additive category $\mathsf{K}(\mathcal{A})$, still denoted by $[n]$. The additive functor $\mathsf{C}(\mathcal{A}) \to \mathsf{K}(\mathcal{A})$ is the identity on objects and the quotient map on morphism sets. The following universal property is immediate.

Proposition 3.2.9 If $\mathcal{B}$ is an Ab-category and the additive functor $F : \mathsf{C}(\mathcal{A}) \to \mathcal{B}$ sends every null-homotopic morphism to the zero morphism, then F factors uniquely through $\mathsf{C}(\mathcal{A}) \to \mathsf{K}(\mathcal{A})$.

The exercises in the next chapter will show that $\mathsf{K}(\mathcal{A})$ is generally not an abelian category.

Remark 3.2.10 For chain complexes (Remark 2.2.6) there is a corresponding $\mathrm{Hom}_\bullet$:

$$\begin{gathered}\mathrm{Hom}_n(X, Y) := \prod_{k\in\mathbb{Z}} \mathrm{Hom}\left(X_k, Y_{k+n}\right), \\ d_n f = d_Y f - (-1)^n f d_X, \quad f \in \mathrm{Hom}_n(X, Y).\end{gathered}$$

Indeed, use the convention in Remark 2.2.6, setting $X^n = X_{-n}$, $d^n = d_{-n}$, and so on. Then $(\mathrm{Hom}_n(X, Y), d_n)_n$ is precisely the chain complex corresponding to $(\mathrm{Hom}^n(X, Y), d^n)_n$:

$$\begin{aligned}\mathrm{Hom}_n\left(X, Y\right) &= \prod_{k\in\mathbb{Z}} \mathrm{Hom}\left(X_k, Y_{n+k}\right) \xlongequal{h=-k} \prod_{h\in\mathbb{Z}} \mathrm{Hom}\left(X^h, Y^{h-n}\right) \\ &= \mathrm{Hom}^{-n}\left(X, Y\right),\end{aligned}$$

and the relation between d_n and d^{-n} is similar. Of course, chain complexes have a corresponding version of $\mathsf{K}(\mathcal{A})$ as well.

Given an additive functor $F : \mathcal{A} \to \mathcal{B}$ between additive categories and any $X, Y \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$, applying F termwise gives a morphism in $\mathsf{C}(\mathsf{Ab})$

$$\begin{gathered}\mathrm{Hom}^\bullet\left(X, Y\right) \to \mathrm{Hom}^\bullet\left(\mathsf{C}F(X), \mathsf{C}F(Y)\right) \\ (f^k)_{k\in\mathbb{Z}} \in \mathrm{Hom}^r(X, Y) \mapsto (F(f^k))_{k\in\mathbb{Z}} \in \mathrm{Hom}^r\left(\mathsf{C}F(X), \mathsf{C}F(Y)\right) \qquad (r \in \mathbb{Z}).\end{gathered}$$

Taking H^0 on both sides gives

$$\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X, Y) \longrightarrow \mathrm{Hom}_{\mathsf{K}(\mathcal{B})}\big(\mathsf{C}F(X), \mathsf{C}F(Y)\big).$$

Thus we obtain a functor $\mathsf{K}F : \mathsf{K}(\mathcal{A}) \to \mathsf{K}(\mathcal{B})$ that makes the following diagram commute:

$$\begin{tikzcd} \mathsf{C}(\mathcal{A}) \arrow[r, "\mathsf{C}F"] \arrow[d] & \mathsf{C}(\mathcal{B}) \arrow[d] \\ \mathsf{K}(\mathcal{A}) \arrow[r, "\mathsf{K}F"'] & \mathsf{K}(\mathcal{B}). \end{tikzcd} \tag{3.2.4}$$

The top row is the functor in Proposition 3.1.8. These constructions are compatible with adjoint pairs.

Proposition 3.2.11 Consider an adjoint pair (F, G, φ), where $F : \mathcal{A} \rightleftarrows \mathcal{B} : G$ are additive functors between additive categories. Then

$$\mathsf{K}F : \mathsf{K}(\mathcal{A}) \rightleftarrows \mathsf{K}(\mathcal{B}) : \mathsf{K}G$$

naturally forms an adjoint pair as well.

Proof The adjunction data φ are a family of bijections $\varphi_{A,B} : \mathrm{Hom}_{\mathcal{B}}(FA, B) \simeq \mathrm{Hom}_{\mathcal{A}}(A, GB)$, functorial in A and B. Proposition 1.4.3 ensures that these bijections are linear, and hence they induce a canonical isomorphism of complexes

$$\mathrm{Hom}^\bullet(\mathsf{C}F(X), Y) \simeq \mathrm{Hom}^\bullet(X, \mathsf{C}G(Y)).$$

In degree r, this isomorphism sends

$$(f^k)_{k\in\mathbb{Z}} \in \mathrm{Hom}^r(\mathsf{C}F(X), Y) \longmapsto (\varphi_{X^k, Y^{k+r}}(f^k))_{k\in\mathbb{Z}} \in \mathrm{Hom}^r(X, \mathsf{C}G(Y)).$$

[8] By (3.2.4), taking H^0 on both sides makes $\mathsf{K}F$ left adjoint to $\mathsf{K}G$. □

3.3 Mapping Cones

We continue to let $\mathcal{A}$ be an additive category, with its associated category of complexes $\mathsf{C}(\mathcal{A})$.

Definition 3.3.1 Given a morphism $f : X \to Y$ in $\mathsf{C}(\mathcal{A})$, its **mapping cone** $\mathrm{Cone}(f)$ is defined to be the following complex:

$$\mathrm{Cone}(f)^n := X^{n+1} \oplus Y^n,$$

$$d^n_{\mathrm{Cone}(f)} := \text{ in matrix notation } \begin{array}{c|cc} & X^{n+1} & Y^n \\ \hline X^{n+2} & -d_X^{n+1} & 0 \\ Y^{n+1} & f^{n+1} & d_Y^n \end{array}$$

$$: X^{n+1} \oplus Y^n \to X^{n+2} \oplus Y^{n+1},$$

for $n \in \mathbb{Z}$. Since $d^n_{X[1]} = -d^{n+1}_X$, there is also the more concise notation

$$\mathrm{Cone}(f) := \left(X[1] \oplus Y, \begin{pmatrix} d_{X[1]} & 0 \\ f[1] & d_Y \end{pmatrix} \right).$$

[8] Translator's note (O014-C035): the source writes the component as $\varphi_{A,B}(f^n)$ without tying A and B to the source and target of the relevant graded component. This translation uses r for the complex degree and k for the family index, so the component is typed as $\varphi_{X^k, Y^{k+r}}(f^k)$.

It is easy to prove that Cone(f) is indeed a complex, as the following matrix calculation shows:

$$\begin{pmatrix} -d_X^{n+1} & 0 \\ f^{n+1} & d_Y^n \end{pmatrix} \begin{pmatrix} -d_X^n & 0 \\ f^n & d_Y^{n-1} \end{pmatrix} = \begin{pmatrix} d_X^{n+1} d_X^n & 0 \\ -f^{n+1} d_X^n + d_Y^n f^n & d_Y^n d_Y^{n-1} \end{pmatrix}.$$

Example 3.3.2 Let $f : X \to Y$ be a morphism in $\mathcal{A}$. Regard X, Y as objects of $\mathsf{C}(\mathcal{A})$ concentrated in degree 0. Then Cone(f) is precisely the complex $[X \xrightarrow{f} Y]$, in degrees $-1, 0$, with all other terms equal to 0.

The following two functoriality properties follow immediately from the definition.

Proposition 3.3.3 Given a commutative diagram in $\mathsf{C}(\mathcal{A})$

$$\begin{tikzcd} X \arrow[r, "\varphi"] \arrow[d, "f"'] & X' \arrow[d, "f'"] \\ Y \arrow[r, "\psi"'] & Y' \end{tikzcd}$$

the matrix $\left(\begin{smallmatrix} \varphi[1] & 0 \\ 0 & \psi \end{smallmatrix}\right)$ defines a morphism $\mathrm{Cone}(f) \to \mathrm{Cone}(f')$.

Proposition 3.3.4 Let $F : \mathcal{A} \to \mathcal{A}'$ be an additive functor and let $f : X \to Y$ be a morphism in $\mathsf{C}(\mathcal{A})$. The associated functor $\mathsf{C}F : \mathsf{C}(\mathcal{A}) \to \mathsf{C}(\mathcal{A}')$ satisfies the following canonical isomorphism in $\mathsf{C}(\mathcal{A}')$:

$$(\mathsf{C}F)(\mathrm{Cone}(f)) \simeq \mathrm{Cone}(\mathsf{C}F(f)).$$

The mapping cone can be placed in the following "triangle," which underlies all the theory to come.

Definition 3.3.5 Consider a morphism $f : X \to Y$ in $\mathsf{C}(\mathcal{A})$. It gives the following canonical morphisms in $\mathsf{C}(\mathcal{A})$:

$$Y \xrightarrow{\alpha(f)} \mathrm{Cone}(f) \xrightarrow{\beta(f)} X[1],$$

explicitly expressed in matrix notation as

$$\alpha(f)^n := \text{inclusion} \begin{pmatrix} 0 \\ \mathrm{id}_{Y^n} \end{pmatrix} : Y^n \to X[1]^n \oplus Y^n,$$

$$\beta(f)^n := \text{projection} \begin{pmatrix} \mathrm{id}_{X[1]^n} & 0 \end{pmatrix} : X[1]^n \oplus Y^n \to X[1]^n.$$

It is clear that these are indeed morphisms in $\mathsf{C}(\mathcal{A})$.

We next collect several homotopy properties of mapping cones. First we show that a mapping cone may be regarded as a "homotopy cokernel" or a "homotopy kernel" of a morphism. This is the realization, at the level of complexes, of a basic idea in homotopy theory.

Proposition 3.3.6 Let $f : X \to Y$ be a morphism in $\mathsf{C}(\mathcal{A})$. There are canonical bijections

$$\mathrm{Hom}\,(\mathrm{Cone}(f), T) \xleftrightarrow{1:1} \{(u,h) : u \in \mathrm{Hom}(Y,T),\ h : \text{a homotopy from } uf \text{ to } 0\}\,,$$

$$\mathrm{Hom}\,(T, \mathrm{Cone}(f)[-1]) \xleftrightarrow{1:1} \{(v,k) : v \in \mathrm{Hom}(T,X),\ k : \text{a homotopy from } fv \text{ to } 0\}\,,$$

where T is any object of $\mathsf{C}(\mathcal{A})$ and $\mathrm{Hom} := \mathrm{Hom}_{\mathsf{C}(\mathcal{A})}$. More explicitly,

- ⋄ the image (u,h) of $\tilde{u} : \mathrm{Cone}(f) \to T$ satisfies $u = \tilde{u} \circ \alpha(f)$;
- ⋄ the image (v,k) of $\tilde{v} : T \to \mathrm{Cone}(f)[-1]$ satisfies $v = \beta(f)[-1] \circ \tilde{v}$.

Proof First consider $\mathrm{Hom}(\mathrm{Cone}(f), T)$. A morphism $\tilde{u} : \mathrm{Cone}(f) \to T$ is equivalent to families of morphisms in $\mathcal{A}$, $h^n : X^{n+1} \to T^n$ and $u^n : Y^n \to T^n$, for $n \in \mathbb{Z}$, satisfying the matrix equation

$$\begin{pmatrix} h^{n+1} & u^{n+1} \end{pmatrix} \begin{pmatrix} -d_X^{n+1} & 0 \\ f^{n+1} & d_Y^n \end{pmatrix} = d_T^n \begin{pmatrix} h^n & u^n \end{pmatrix}.$$

Expanding it shows that this equation is equivalent to saying that $u = (u^n)_n : Y \to T$ is a morphism in $\mathsf{C}(\mathcal{A})$ and that $h := (h^{n-1})_n$ satisfies $d^{-1}_{\mathrm{Hom}^\bullet(X,T)} h = uf$. This is the required bijection; the equality $u = \tilde{u} \circ \alpha(f)$ follows immediately from the definition of $\alpha(f)$.

Next consider a morphism $\tilde{v} : T \to \mathrm{Cone}(f)[-1]$. It is equivalent to families of morphisms in $\mathcal{A}$, $v^n : T^n \to X^n$ and $\underline{k}^n : T^n \to Y^{n-1}$, satisfying

$$\begin{pmatrix} d_X^n & 0 \\ -f^n & -d_Y^{n-1} \end{pmatrix} \begin{pmatrix} v^n \\ \underline{k}^n \end{pmatrix} = \begin{pmatrix} v^{n+1} \\ \underline{k}^{n+1} \end{pmatrix} d_T^n \quad (n \in \mathbb{Z}).$$

This equation is equivalent to saying that $v = (v^n)_n : T \to X$ is a morphism in $\mathsf{C}(\mathcal{A})$ and that $k := (-\underline{k}^n)_n$ satisfies $d^{-1}_{\mathrm{Hom}^\bullet(T,Y)} k = fv$. The equality $v = \beta(f)[-1] \circ \tilde{v}$ also follows immediately from the definition of $\beta(f)$. □

Remark 3.3.7 (Homotopy cokernel and homotopy kernel) Proposition 3.3.6 shows that a morphism $u : Y \to T$ (or $v : T \to X$) has uf (or fv) null-homotopic if and only if it factors through $\alpha(f) : Y \to \mathrm{Cone}(f)$ (or through $\beta(f)[-1] : \mathrm{Cone}(f)[-1] \to X$). This naturally recalls the universal property of a cokernel (or kernel), with the following differences:

- ◇ the requirement that uf (or fv) be null-homotopic replaces the strict equality $= 0$;
- ◇ the factorization $\tilde{u}$ (or $\tilde{v}$) supplied by the proposition not only witnesses the fact that uf (or fv) is null-homotopic but also records how it is homotoped to 0; this is data one level higher.

These properties cannot simply be handled in $\mathsf{C}(\mathcal{A})$ or $\mathsf{K}(\mathcal{A})$ using elementary category-theoretic notions; in fact, kernels and cokernels seldom exist in $\mathsf{K}(\mathcal{A})$. For this reason, $Y \to \mathrm{Cone}(f)$ is also called the **homotopy cokernel** of f, while $\beta(f)[-1] : \mathrm{Cone}(f)[-1] \to X$ is called the **homotopy kernel** of f. Remarkably, the mapping cone $\mathrm{Cone}(f)$ and its shifts play both the "cokernel" and "kernel" roles in this sense.

A morphism f in $\mathsf{C}(\mathcal{A})$ is said to be canonically homotopic to g if there is a canonical h such that $g - f = d^{-1}h$; if f is canonically homotopic to 0, it is called canonically null-homotopic.

Proposition 3.3.8 Let $\mathcal{A}$ be an additive category.

(i) If $f : X \to Y$ is an isomorphism in $\mathsf{C}(\mathcal{A})$, then $\mathrm{id}_{\mathrm{Cone}(f)}$ is canonically null-homotopic.

(ii) For any morphism $f : X \to Y$ in $\mathsf{C}(\mathcal{A})$, the morphisms in Definition 3.3.5 satisfy

$$\beta(f) \circ \alpha(f) = 0,$$

while $\alpha(f) \circ f$ and $f[1] \circ \beta(f)$ are both canonically null-homotopic.

Proof First consider (i). By functoriality of mapping cones (Proposition 3.3.3), it suffices to consider $f = \mathrm{id}_X$. Define

$$s^n := \begin{pmatrix} 0 & \mathrm{id}_{X^n} \\ 0 & 0 \end{pmatrix} : X^{n+1} \oplus X^n \to X^n \oplus X^{n-1}, \quad n \in \mathbb{Z}.$$

A direct calculation gives

$$d^{n-1}_{\mathrm{Cone}(\mathrm{id}_X)} s^n + s^{n+1} d^n_{\mathrm{Cone}(\mathrm{id}_X)} = \\ \begin{pmatrix} -d^n_X & 0 \\ \mathrm{id}_{X^n} & d^{n-1}_X \end{pmatrix} \begin{pmatrix} 0 & \mathrm{id}_{X^n} \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & \mathrm{id}_{X^{n+1}} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -d^{n+1}_X & 0 \\ \mathrm{id}_{X^{n+1}} & d^n_X \end{pmatrix} = \mathrm{id}_{X^{n+1} \oplus X^n},$$

so $s = (s^n)_{n \in \mathbb{Z}}$ makes $\mathrm{id}_{\mathrm{Cone}(\mathrm{id}_X)}$ null-homotopic.

Next consider (ii). Clearly $\beta(f) \circ \alpha(f) = 0$. The remaining homotopies come from Proposition 3.3.6: take $T = \mathrm{Cone}(f)$, so that $\tilde{u} := \mathrm{id}_{\mathrm{Cone}(f)} \in \mathrm{Hom}(\mathrm{Cone}(f), T)$ makes $\alpha(f) \circ f$ null-homotopic; take $T = \mathrm{Cone}(f)[-1]$, so that $\tilde{v} := \mathrm{id}_{\mathrm{Cone}(f)[-1]} \in$

$\mathrm{Hom}(T, \mathrm{Cone}(f)[-1])$ makes $f \circ \beta(f)[-1]$ null-homotopic, and hence also makes $f[1] \circ \beta(f)$ null-homotopic. □

Definition 3.3.5 produces two new morphisms $\alpha(f)$ and $\beta(f)$ from f. Up to homotopy, the mapping cones of these two morphisms produce no new objects. We begin with $\mathrm{Cone}(\alpha(f))$. First observe that

$$\mathrm{Cone}(\alpha(f))^n = Y^{n+1} \oplus \mathrm{Cone}(f)^n = Y^{n+1} \oplus X^{n+1} \oplus Y^n.$$

In matrix notation,

$$\alpha(\alpha(f)) = \begin{pmatrix} 0 & 0 \\ \mathrm{id}_{X[1]} & 0 \\ 0 & \mathrm{id}_Y \end{pmatrix} : \mathrm{Cone}(f) \to \mathrm{Cone}(\alpha(f)),$$
$$\beta(\alpha(f)) = \begin{pmatrix} \mathrm{id}_{Y[1]} & 0 & 0 \end{pmatrix} : \mathrm{Cone}(\alpha(f)) \to Y[1].$$

Lemma 3.3.9 For a morphism $f : X \to Y$ in $\mathsf{C}(\mathcal{A})$, matrix notation defines a pair of morphisms

$$\begin{pmatrix} -f[1] \\ \mathrm{id}_{X[1]} \\ 0 \end{pmatrix} : \ X[1] \underset{\psi}{\overset{\phi}{\rightleftarrows}} \mathrm{Cone}(\alpha(f)) \ : \begin{pmatrix} 0 & \mathrm{id}_{X[1]} & 0 \end{pmatrix}.$$

They satisfy $\psi \circ \phi = \mathrm{id}_{X[1]}$ and make the following diagram commute in $\mathsf{K}(\mathcal{A})$:

$$\begin{tikzcd}
\mathrm{Cone}(f) \arrow[r, "\beta(f)"] \arrow[d, "\mathrm{id}_{\mathrm{Cone}(f)}"'] & X[1] \arrow[r, "-f[1]"] \arrow[d, "\phi"] & Y[1] \arrow[d, "\mathrm{id}_{Y[1]}"] \\
\mathrm{Cone}(f) \arrow[r, "\alpha(\alpha(f))"'] & \mathrm{Cone}(\alpha(f)) \arrow[r, "\beta(\alpha(f))"'] & Y[1]
\end{tikzcd}$$

Moreover, ϕ and ψ are inverse to each other in $\mathsf{K}(\mathcal{A})$.

Proof By the preceding description of $\mathrm{Cone}(\alpha(f))$, its differential $d^n_{\mathrm{Cone}(\alpha(f))}$ is, in matrix notation,

$$\begin{pmatrix} -d_Y^{n+1} & 0 & 0 \\ 0 & -d_X^{n+1} & 0 \\ \mathrm{id}_{Y^{n+1}} & f^{n+1} & d_Y^n \end{pmatrix} : Y^{n+1} \oplus X^{n+1} \oplus Y^n \to Y^{n+2} \oplus X^{n+2} \oplus Y^{n+1}.$$

It suffices to verify the following statements in $\mathsf{C}(\mathcal{A})$:

- ◇ $\phi := (\phi^n)_{n \in \mathbb{Z}}$ and $\psi := (\psi^n)_{n \in \mathbb{Z}}$ are both morphisms of complexes;

- ◇ $\psi \circ \phi = \mathrm{id}_{X[1]}$;
- ◇ $\psi \circ \alpha(\alpha(f)) = \beta(f)$;
- ◇ $\beta(\alpha(f)) \circ \phi = -f[1]$;
- ◇ there is an $s = (s^n)_{n\in\mathbb{Z}} \in \mathrm{Hom}^{-1}(\mathrm{Cone}(\alpha(f)), \mathrm{Cone}(\alpha(f)))$ such that $\mathrm{id}_{\mathrm{Cone}(\alpha(f))} - \phi \circ \psi = d^{-1}_{\mathrm{Hom}^\bullet}(s)$.

The first four statements are elementary. For the last, take, in matrix notation,

$$s^n := \begin{pmatrix} 0 & 0 & \mathrm{id}_{Y^n} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} : \mathrm{Cone}(\alpha(f))^n \to \mathrm{Cone}(\alpha(f))^{n-1}$$

and verify it directly. □

For the case of $\beta(f)$, we make a slight modification and consider the mapping cone of $-\beta(f)[-1] : \mathrm{Cone}(f)[-1] \to X$.

Definition 3.3.10 For a morphism $f : X \to Y$ in $\mathsf{C}(\mathcal{A})$, its **mapping cylinder** is defined by

$$\mathrm{Cyl}(f) := \mathrm{Cone}\left(\mathrm{Cone}(f)[-1] \xrightarrow{-\beta(f)[-1]} X\right).$$

It is equipped with canonical morphisms $X \to \mathrm{Cyl}(f) \to \mathrm{Cone}(f)$.

The mapping cylinder has functoriality of the same form as that of the mapping cone (Proposition 3.3.3). In view of Remark 3.3.7, $\mathrm{Cyl}(f)$ can be compared with a coimage in an additive category—roughly speaking, it is a "cokernel of a kernel" (Proposition 1.3.12)—and may therefore be viewed as the homotopy coimage of f.

More explicitly, $\mathrm{Cyl}(f)^n = \mathrm{Cone}(f)^n \oplus X^n = X^{n+1} \oplus Y^n \oplus X^n$. The morphism $X \to \mathrm{Cyl}(f)$ (or $\mathrm{Cyl}(f) \to \mathrm{Cone}(f)$) is the inclusion into the third direct summand (or the projection onto the first two), and

$$d^n_{\mathrm{Cyl}(f)} = \begin{pmatrix} d^n_{\mathrm{Cone}(f)} & 0 \\ -\beta(f)^n & d^n_X \end{pmatrix} = \begin{pmatrix} -d^{n+1}_X & 0 & 0 \\ f^{n+1} & d^n_Y & 0 \\ -\mathrm{id}_{X^{n+1}} & 0 & d^n_X \end{pmatrix}.$$

Lemma 3.3.11 For a morphism $f : X \to Y$ in $\mathsf{C}(\mathcal{A})$, one can define a pair of morphisms

$$\begin{pmatrix} 0 \\ \mathrm{id}_Y \\ 0 \end{pmatrix} : Y \underset{\psi}{\overset{\phi}{\rightleftarrows}} \mathrm{Cyl}(f) : \begin{pmatrix} 0 & \mathrm{id}_Y & f \end{pmatrix}.$$

They satisfy $\psi \circ \phi = \mathrm{id}_Y$ and make the following diagram commute in $\mathsf{K}(\mathcal{A})$:

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, "\mathrm{id}_X"'] & Y \arrow[r, "\alpha(f)"] \arrow[d, "\phi"] & \mathrm{Cone}(f) \arrow[d, "\mathrm{id}_{\mathrm{Cone}(f)}"] \\
X \arrow[r] & \mathrm{Cyl}(f) \arrow[r] & \mathrm{Cone}(f)
\end{tikzcd}$$

Moreover, ϕ and ψ are inverse to each other in $\mathsf{K}(\mathcal{A})$. In addition, f factors as the composite $X \to \mathrm{Cyl}(f) \xrightarrow{\psi} Y$.

Proof The reader may verify directly from the definition that ϕ and ψ are morphisms of complexes. Then $\psi \circ \phi = \mathrm{id}_Y$ is immediate. To prove that ϕ and ψ are inverse in $\mathsf{K}(\mathcal{A})$, as in Lemma 3.3.9, take

$$s^n := \begin{pmatrix} 0 & 0 & -\mathrm{id}_{X^n} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} : X^{n+1} \oplus Y^n \oplus X^n \to X^n \oplus Y^{n-1} \oplus X^{n-1},$$
$$s := (s^n)_{n \in \mathbb{Z}},$$

and verify $\mathrm{id}_{\mathrm{Cyl}(f)} - \phi \circ \psi = d^{-1}_{\mathrm{Hom}^\bullet}(s)$.

The right-hand square in the diagram already commutes in $\mathsf{C}(\mathcal{A})$. For the left-hand square, simply observe that the composite $X \to \mathrm{Cyl}(f) \xrightarrow{\psi} Y$ equals f. □

Lemmas 3.3.9 and 3.3.11 show that, up to isomorphism in $\mathsf{K}(\mathcal{A})$, any morphism $f : X \to Y$ can be replaced by the much simpler projection $\mathrm{Cone}(\alpha(f))[-1] \to Y$ (or by the inclusion $X \to \mathrm{Cyl}(f)$), and this construction is functorial in f. This is an important idea in homological algebra and homotopy theory.

The special case $f = \mathrm{id}_X$ of the mapping cylinder has another use: it can be used to interpret homotopy. For readers familiar with homology theory, all of this has a topological interpretation, but here we discuss only the algebraic version.

Proposition 3.3.12 Let X be an object of $\mathsf{C}(\mathcal{A})$ and set $\mathrm{Cyl}_X := \mathrm{Cyl}(\mathrm{id}_X)$. Notice that $\mathrm{Cyl}^n_X = X^{n+1} \oplus X^n \oplus X^n$.

(i) In $\mathsf{C}(\mathcal{A})$ there are morphisms $X \overset{i_0}{\underset{i_1}{\rightrightarrows}} \mathrm{Cyl}_X \xrightarrow{j} X$, defined in matrix notation, for every $n \in \mathbb{Z}$, by

$$i^n_0 := \begin{pmatrix} 0 \\ 0 \\ \mathrm{id}_{X^n} \end{pmatrix}, \quad i^n_1 := \begin{pmatrix} 0 \\ \mathrm{id}_{X^n} \\ 0 \end{pmatrix}, \quad j^n := \begin{pmatrix} 0 & \mathrm{id}_{X^n} & \mathrm{id}_{X^n} \end{pmatrix}.$$

These morphisms satisfy $ji_0 = \mathrm{id}_X = ji_1$, and the image of j in $\mathsf{K}(\mathcal{A})$ is an isomorphism.

(ii) For every object Y of $\mathsf{C}(\mathcal{A})$, there is a bijection

$$\left\{ (f,g,h) \;\middle|\; \begin{array}{l} f,g \in \mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X,Y) \\ h \in \mathrm{Hom}^{-1}(X,Y) \\ g - f = d^{-1}_{\mathrm{Hom}^\bullet(X,Y)} h \end{array} \right\} \xrightarrow{1:1} \mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(\mathrm{Cyl}_X, Y)$$

$$(f,g,h) \longmapsto \tilde{h} = (\tilde{h}^n)_{n\in\mathbb{Z}},\ \tilde{h}^n := \begin{pmatrix} h^n & g^n & f^n \end{pmatrix}.$$

In particular, $f = \tilde{h} i_0$ and $g = \tilde{h} i_1$.

Proof For (i), note that i_0 is precisely the canonical morphism $X \to \mathrm{Cyl}(\mathrm{id}_X)$ in Definition 3.3.10, while i_1 is the morphism $\phi : X \to \mathrm{Cyl}(\mathrm{id}_X)$ in Lemma 3.3.11. The commutative diagram in Lemma 3.3.11 shows that they give the same isomorphism in $\mathsf{K}(\mathcal{A})$.

For j, it is easy to verify that $j^{n+1} d^n_{\mathrm{Cyl}_X} = \begin{pmatrix} 0 & d^n_X & d^n_X \end{pmatrix} = d^n_X j^n$. Thus j is indeed a morphism, while $ji_0 = \mathrm{id}_X = ji_1$ is immediate. Since i_0 and i_1 are isomorphisms in $\mathsf{K}(\mathcal{A})$, so is j.

For (ii), write any element $\tilde{h}$ of $\mathrm{Hom}^0(\mathrm{Cyl}_X, Y)$ as

$$\tilde{h} = ((h^n, g^n, f^n))_{n\in\mathbb{Z}},$$

where

$$h^n : X^{n+1} \to Y^n, \quad f^n : X^n \to Y^n, \quad g^n : X^n \to Y^n$$

are morphisms in $\mathcal{A}$. Suppressing superscripts and using matrix notation, calculate

$$\tilde{h}\, d_{\mathrm{Cyl}_X} = \begin{pmatrix} h & g & f \end{pmatrix} \begin{pmatrix} -d_X & 0 & 0 \\ \mathrm{id}_X & d_X & 0 \\ -\mathrm{id}_X & 0 & d_X \end{pmatrix} = \begin{pmatrix} -hd_X + g - f & gd_X & fd_X \end{pmatrix},$$

$$d_Y\, \tilde{h} = \begin{pmatrix} d_Y h & d_Y g & d_Y f \end{pmatrix}.$$

Therefore $\tilde{h}$ is a morphism of complexes if and only if $f, g \in \mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X,Y)$ and $g - f = hd_X + d_Y h$. $\square$

Remark 3.3.13 Mapping cones and mapping cylinders of course have versions for chain complexes (see Remark 2.2.6). Given a morphism of chain complexes $f : X \to Y$, define

$$\begin{aligned} \mathrm{Cone}(f)_n &:= (X[-1] \oplus Y)_n, & \mathrm{Cyl}(f)_n &:= (X[-1] \oplus Y \oplus X)_n, \\ d_{\mathrm{Cone}(f)} &:= \begin{pmatrix} d_{X[-1]} & 0 \\ f[-1] & d_Y \end{pmatrix}, & d_{\mathrm{Cyl}(f)} &:= \begin{pmatrix} d_{X[-1]} & 0 & 0 \\ f[-1] & d_Y & 0 \\ -\mathrm{id}_{X[-1]} & 0 & d_X \end{pmatrix}. \end{aligned}$$

Every statement in this section carries over to chain complexes. In particular, there are canonical morphisms

$$X \xrightarrow{f} Y \xrightarrow{\alpha(f)} \mathrm{Cone}(f) \xrightarrow{\beta(f)} X[-1], \quad X \to \mathrm{Cyl}(f) \to \mathrm{Cone}(f).$$

3.4 Complexes on the Opposite Category

Let $\mathcal{A}$ be an additive category. This section relates complexes on $\mathcal{A}$ and on $\mathcal{A}^{\mathrm{op}}$. Since functors involving $\mathcal{A}^{\mathrm{op}}$, such as the Hom functor, occur frequently, this procedure is simple but necessary; some of the signs involved also require a little care. Beginning readers are advised to skip this section or merely skim it on a first reading.

The discussion has three aspects: complexes, homotopy, and mapping cones. The derived-category version considered later (Proposition 4.4.16) is a direct application of these results. Recall that if $f : X \to Y$ is a morphism in $\mathcal{A}$, then f^{op} denotes the corresponding morphism $Y \to X$ in $\mathcal{A}^{\mathrm{op}}$.

Definition–Proposition 3.4.1 The isomorphism of additive categories $\sigma : \mathsf{C}(\mathcal{A}^{\mathrm{op}}) \to \mathsf{C}(\mathcal{A})^{\mathrm{op}}$ is defined as follows. For an object X of $\mathsf{C}(\mathcal{A}^{\mathrm{op}})$, define in $\mathcal{A}$

$$(\sigma X)^n := X^{-n}, \quad d^n_{\sigma X} = (-1)^{n+1} \left[d_X^{-n-1,\mathrm{op}} : (\sigma X)^n \to (\sigma X)^{n+1} \right], \quad n \in \mathbb{Z}.$$

For a morphism $f = (f^n : X^n \to Y^n)_n$ in $\mathsf{C}(\mathcal{A}^{\mathrm{op}})$, take its image σf to be the following morphism in $\mathsf{C}(\mathcal{A})$:

$$(\sigma f)^n := \left(f^{-n}\right)^{\mathrm{op}} : (\sigma Y)^n \to (\sigma X)^n,$$

that is, a morphism $\sigma X \to \sigma Y$ in $\mathsf{C}(\mathcal{A})^{\mathrm{op}}$. For every $m \in \mathbb{Z}$, there is a natural isomorphism

$$s_m : \sigma \circ [m] \xrightarrow{\sim} [-m] \circ \sigma,$$

where $[-m]$ is regarded as a functor from $\mathsf{C}(\mathcal{A})^{\mathrm{op}}$ to itself (strictly speaking, it should be written $[-m]^{\mathrm{op}}$).

Proof The general case of s_m reduces step by step to the case $m = 1$. Define $s_1 = (s_{1,X})_X$ as follows. For every n, take

$$s^n_{1,X} : \sigma(X[1])^n = X^{1-n} \xrightarrow[\sim]{(-1)^{n-1}} X^{1-n} = (\sigma X)[-1]^n. \tag{3.4.1}$$

The entire verification reduces to $s^{n+1}_{1,X} d^n_{\sigma(X[1])} = -d^{-n-1,\mathrm{op}}_{X[1]} = d^{-n,\mathrm{op}}_X = (-1)^n d^{n-1}_{\sigma X} = d^n_{(\sigma X)[-1]} s^n_{1,X}$. □

If the construction σ with coefficient $(-1)^{n+1}$ is applied in both directions, $\sigma^2 X$ is naturally isomorphic to X through the isomorphism whose degree-n component is $(-1)^n \mathrm{id}_{X^n}$. To obtain an inverse, define the reverse construction by the same formulas on objects and morphisms, but use the coefficient $(-1)^n$ on the differential. The composites in both orders are then exactly the identity functor, so σ is indeed an isomorphism of categories.[9)]

Remark 3.4.2 If $\mathcal{A}$ is an abelian category, the signs on $d^\bullet_{\sigma X}$ do not change the cohomology introduced in Definition 2.2.5; thus $\mathrm{H}^{-n}(\sigma X) \in \mathrm{Ob}(\mathcal{A})$ corresponds to $\mathrm{H}^n(X) \in \mathrm{Ob}(\mathcal{A}^{\mathrm{op}})$.

We next examine the relation between σ and homotopy.

Proposition 3.4.3 The functor σ defined above induces an equivalence of additive categories $\mathsf{K}(\mathcal{A}^{\mathrm{op}}) \xrightarrow{\sim} \mathsf{K}(\mathcal{A})^{\mathrm{op}}$.

Proof Let X, Y be objects of $\mathsf{C}(\mathcal{A}^{\mathrm{op}})$. For $f = (f^k)_k \in \mathrm{Hom}^{-1}_{\mathsf{C}(\mathcal{A}^{\mathrm{op}})}(X, Y)$, use the following table to define the corresponding $\sigma f \in \mathrm{Hom}^{-1}_{\mathsf{C}(\mathcal{A})}(\sigma Y, \sigma X)$ in $\mathcal{A}$:

Category	Morphism		
$\mathcal{A}^{\mathrm{op}}$	$X^k \xrightarrow{(-1)^{k+1} f^k} Y^{k-1}$	$d^{k-1}_Y f^k + f^{k+1} d^k_X$	$(d^{-1} f)^k$
$\mathcal{A}$	$(\sigma Y)^{-k+1} \xrightarrow{(\sigma f)^{-k+1}} (\sigma X)^{-k}$	$(\sigma f)^{-k+1} d^{-k}_{\sigma Y} + d^{-k-1}_{\sigma X} (\sigma f)^{-k}$	$d^{-1} (\sigma f)^{-k}$

This suffices to show

$$\begin{aligned} \mathrm{Hom}_{\mathsf{K}(\mathcal{A}^{\mathrm{op}})}(X, Y) &\xrightarrow{\sim} \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(\sigma Y, \sigma X) \\ &= \mathrm{Hom}_{\mathsf{K}(\mathcal{A})^{\mathrm{op}}}(\sigma X, \sigma Y). \end{aligned}$$ □

Finally, using σ and the family of isomorphisms $(s_m)_m$ constructed above, we compare mapping cones in $\mathsf{C}(\mathcal{A})$ and $\mathsf{C}(\mathcal{A}^{\mathrm{op}})$. The details are somewhat routine.

[9)] Translator's note (O014-C036): arguing from $n(n+1) \equiv 0 \pmod 2$, the source states that applying σ twice returns X strictly. Using the same formula in both directions instead gives the differential $-d_X$, not d_X. The signed natural isomorphism and the corrected inverse construction are recorded above.

Proposition 3.4.4 For every morphism $f : X \to Y$ in $\mathsf{C}(\mathcal{A}^{\mathrm{op}})$, there is a commutative diagram in $\mathsf{C}(\mathcal{A})$[10]:

$$\begin{tikzcd}
\sigma(X[1]) \arrow[r, "\sigma(\beta(f))"] \arrow[d, "s_{1,X}"'] & \sigma(\mathrm{Cone}(f)) \arrow[r, "\sigma(\alpha(f))"] \arrow[d, "\theta"] & \sigma Y \arrow[r, "\sigma(f)"] \arrow[d, "\mathrm{id}"] & \sigma X \arrow[d, "\mathrm{id}"] \\
(\sigma X)[-1] \arrow[r, "\alpha(\sigma(f))[-1]"'] & \mathrm{Cone}(\sigma(f))[-1] \arrow[r, "\beta(\sigma(f))[-1]"'] & \sigma Y \arrow[r, "\sigma(f)"'] & \sigma X
\end{tikzcd}$$

where θ is a canonical isomorphism and $s_{1,X}$ is the isomorphism supplied by Definition–Proposition 3.4.1.

Proof If $\mathrm{Cone}(f)$ were replaced by $X[1] \oplus Y$, the statement would be easy and the required isomorphism would simply be $(s_{1,X}, \mathrm{id}_{\sigma Y})$. The complication is that the matrix $d_{\mathrm{Cone}(f)}$ has the off-diagonal entry $f[1]$. Nevertheless, we follow the same method: using (3.4.1), for every $n \in \mathbb{Z}$ define the isomorphism

$$\begin{aligned}
&\theta^n : \sigma(\mathrm{Cone}(f))^n = \sigma(X[1])^n \oplus (\sigma Y)^n \\
&\xrightarrow[\sim]{(s_{1,X},\mathrm{id})^n = ((-1)^{n-1}\mathrm{id},\mathrm{id})} (\sigma X)[-1]^n \oplus (\sigma Y)^n \overset{\text{interchange}}{=\!=\!=} ((\sigma Y) \oplus (\sigma X)[-1])^n .
\end{aligned}$$

Under this identification, the morphism $d^n_{\sigma(\mathrm{Cone}(f))}$ in $\mathcal{A}$ corresponds to

$$\begin{pmatrix} d^n_{\sigma Y} & 0 \\ * & d^n_{(\sigma X)[-1]} \end{pmatrix} : ((\sigma Y) \oplus (\sigma X)[-1])^n \to ((\sigma Y) \oplus (\sigma X)[-1])^{n+1} .$$

As noted at the start of the proof, the diagonal entries pose no problem; we need only determine the lower-left entry. By Definition–Proposition 3.4.1, this entry is $(-1)^{n+1}$ times the following composite in $\mathcal{A}$:

$$\begin{tikzcd}
(\sigma Y)^n \arrow[r] \arrow[d, equal] & (\sigma(X[1]))^{n+1} \arrow[r, "\sim"] \arrow[d, equal] & (\sigma X)[-1]^{n+1} \arrow[d, equal] \\
Y^{-n} \arrow[r, "(f^{-n})^{\mathrm{op}}"'] & X^{-n} \arrow[r, "s^{n+1}_{1,X} = (-1)^n"'] & X^{-n}
\end{tikzcd}$$

Thus the result is $-\sigma(f)^n$. It follows that

$$\theta := (\theta^n)_{n\in\mathbb{Z}} : \sigma(\mathrm{Cone}(f)) \xrightarrow{\sim} \mathrm{Cone}(\sigma(f))[-1]$$

is indeed an isomorphism in $\mathsf{C}(\mathcal{A})$. Once this is established, verifying commutativity presents no essential difficulty. □

Of course, the results of this section also apply to chain complexes and extend to the case in which $\mathcal{A}$ is a $\Bbbk$-linear category.

[10] Here $\alpha(\cdot)$ and $\beta(\cdot)$ are both defined relative to $\mathsf{C}(\mathcal{A})$. If the diagram is viewed in $\mathsf{C}(\mathcal{A})^{\mathrm{op}}$, all its arrows must be reversed.

3.5 Double Complexes

Throughout this section, $\mathcal{A}$ remains an additive category. Roughly speaking, a double complex may be thought of as a two-dimensional version of a complex, with differentials in the horizontal and vertical directions.

Definition 3.5.1 (Double complex) A double complex on an additive category $\mathcal{A}$ consists of a family of objects $(X^{p,q})_{(p,q)\in\mathbb{Z}^2}$ of $\mathcal{A}$, together with morphisms ${}^{\triangleright}d^{p,q} : X^{p,q} \to X^{p+1,q}$ and ${}^{\vartriangle}d^{p,q} : X^{p,q} \to X^{p,q+1}$ satisfying

$$ {}^{\triangleright}d^{p+1,q}\,{}^{\triangleright}d^{p,q} = 0, \quad {}^{\vartriangle}d^{p,q+1}\,{}^{\vartriangle}d^{p,q} = 0, \quad {}^{\triangleright}d^{p,q+1}\,{}^{\vartriangle}d^{p,q} = {}^{\vartriangle}d^{p+1,q}\,{}^{\triangleright}d^{p,q}. $$

These data are usually abbreviated as $(X^{\bullet,\bullet}, {}^{\triangleright}d, {}^{\vartriangle}d)$, $X^{\bullet,\bullet}$, or X.

The conditions on ${}^{\triangleright}d, {}^{\vartriangle}d$ may be abbreviated as

$$ {}^{\triangleright}d^2 = 0, \quad {}^{\vartriangle}d^2 = 0, \quad {}^{\triangleright}d\,{}^{\vartriangle}d = {}^{\vartriangle}d\,{}^{\triangleright}d. $$

Definition 3.5.2 Given an additive category $\mathcal{A}$, a morphism from a double complex X to a double complex Y is a family of morphisms

$$ f = (f^{p,q} : X^{p,q} \to Y^{p,q})_{(p,q)\in\mathbb{Z}^2}\,, $$

such that, for all p, q,

$$ {}^{\triangleright}d_Y^{p,q} f^{p,q} = f^{p+1,q}\,{}^{\triangleright}d_X^{p,q}, \quad {}^{\vartriangle}d_Y^{p,q} f^{p,q} = f^{p,q+1}\,{}^{\vartriangle}d_X^{p,q}. $$

These relations may be abbreviated as ${}^{\triangleright}d_Y f = f\,{}^{\triangleright}d_X$ and ${}^{\vartriangle}d_Y f = f\,{}^{\vartriangle}d_X$.

As in Proposition 3.1.5, these definitions make all double complexes on $\mathcal{A}$ into an additive category $\mathsf{C}^2(\mathcal{A})$.

Visually, a double complex X is displayed as

$$
\begin{tikzcd}
 & \vdots & \vdots & \\
\cdots \arrow[r] & X^{p,q+1} \arrow[u] \arrow[r] & X^{p+1,q+1} \arrow[u] \arrow[r] & \cdots \\
\cdots \arrow[r] & X^{p,q} \arrow[u, "{}^{\vartriangle}d^{p,q}"] \arrow[r, "{}^{\triangleright}d^{p,q}"'] & X^{p+1,q} \arrow[u] \arrow[r] & \cdots \\
 & \vdots \arrow[u] & \vdots \arrow[u] &
\end{tikzcd}
$$

Every row $(X^{\bullet,q}, d^{\bullet,q})$ and every column $(X^{p,\bullet}, d^{p,\bullet})$ is a complex. Thus a double complex can be assembled by columns or by rows.

Definition 3.5.3 Define the additive functor $F_{\mathrm{I}} : \mathsf{C}^2(\mathcal{A}) \to \mathsf{C}(\mathsf{C}(\mathcal{A}))$ as follows: for a double complex X, set $(F_{\mathrm{I}}X)^p = X^{p,\bullet}$ and $d^p_{F_{\mathrm{I}}X} = {}^{\triangleright}d^{p,\bullet} : (F_{\mathrm{I}}X)^p \to (F_{\mathrm{I}}X)^{p+1}$. Similarly, the formulas $(F_{\mathrm{II}}X)^q = X^{\bullet,q}$ and $d^q_{F_{\mathrm{II}}X} = {}^{\triangle}d^{\bullet,q}$ define an additive functor $\mathsf{C}^2(\mathcal{A}) \to \mathsf{C}(\mathsf{C}(\mathcal{A}))$. Here $p, q \in \mathbb{Z}$.

Definition 3.5.2 implies that both F_{I} and F_{II} are isomorphisms of additive categories. This is depicted as follows:

$$
\begin{tikzcd}
\vdots & & & & \\
(F_{\mathrm{II}}X)^q \arrow[u, "d^q_{F_{\mathrm{II}}X}"] & := & \cdots \arrow[r] & X^{p,q} \arrow[u, "{}^{\triangle}d^{p,q}"] \arrow[r, "{}^{\triangleright}d^{p,q}"'] & \cdots \\
\vdots \arrow[u] & & & \arrow[u] & \\
 & & & \| & \\
 & & \cdots \arrow[r] & (F_{\mathrm{I}}X)^p \arrow[r, "d^p_{F_{\mathrm{I}}X}"] & \cdots
\end{tikzcd}
\tag{3.5.1}
$$

Definition 3.5.4 (Total complex) Suppose that $\mathcal{A}$ has countable coproducts, denoted by the direct sum symbol $\bigoplus$, and let X be an object of $\mathsf{C}^2(\mathcal{A})$. Define a complex $\mathrm{tot}_{\oplus}(X)$ on $\mathcal{A}$ as follows:

- ◇ $\mathrm{tot}_{\oplus}(X)^n := \bigoplus_{p+q=n} X^{p,q}$;
- ◇ the restriction of $d^n : \mathrm{tot}_{\oplus}(X)^n \to \mathrm{tot}_{\oplus}(X)^{n+1}$ to the summand $X^{p,q}$ is ${}^{\triangleright}d^{p,q} + (-1)^{p\,\triangle}d^{p,q}$.

Suppressing superscripts, writing out $d^2 : X^{p,q} \to X^{p+2,q} \oplus X^{p+1,q+1} \oplus X^{p,q+2}$ gives

$$d^2 = \left({}^{\triangleright}d^2,\ (-1)^{p\,\triangleright}d^{\triangle}d + (-1)^{p+1\,\triangle}d^{\triangleright}d,\ {}^{\triangle}d^2\right) = (0,0,0).$$

Replacing coproducts in the construction of $\mathrm{tot}_{\oplus}$ by products $\prod$ similarly gives $\mathrm{tot}_{\Pi} X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$, provided the required products exist. The definition of $d^n : \mathrm{tot}_{\Pi}(X)^n \to \mathrm{tot}_{\Pi}(X)^{n+1}$ is analogous to the $\mathrm{tot}_{\oplus}$ case: require that the projection of d^{n-1} to $X^{p,q}$ be the composite

$$\mathrm{tot}^{n-1}_{\Pi}(X) \xrightarrow{\text{projection}} X^{p-1,q} \oplus X^{p,q-1} \xrightarrow{({}^{\triangleright}d^{p-1,q},(-1)^{p\,\triangle}d^{p,q-1})} X^{p,q};$$

the same reasoning gives $d^2 = 0$. When there is no danger of confusion, both $\mathrm{tot}_{\oplus} X$ and $\mathrm{tot}_{\Pi} X$ are called the **total complex** of X.

The following property should be clear.

Proposition 3.5.5 Under the respective hypotheses, $\mathrm{tot}_\oplus$ and tot_Π are both additive functors from $\mathsf{C}^2(\mathcal{A})$ to $\mathsf{C}(\mathcal{A})$.

In the definition of the total complex, the superscripts p and q may at first appear asymmetric. To dispel this impression, define an additive automorphism swap of $\mathsf{C}^2(\mathcal{A})$ such that $\mathrm{swap}(X)^{p,q} = X^{q,p}$ and ${}^{\triangleright}d$ and ${}^{\vartriangle}d$ are interchanged. The accompanying sign $(-1)^{pq}$ is essentially the same mechanism as the Koszul sign rule in [51, Definition–Theorem 7.4.4].

Proposition 3.5.6 For every $(p,q) \in \mathbb{Z}^2$ and every object X of $\mathsf{C}^2(\mathcal{A})$, define a family of isomorphisms

$$r_X^{p,q} := (-1)^{pq}\mathrm{id}_{X^{p,q}} : X^{p,q} \to \mathrm{swap}(X)^{q,p}.$$

- ◇ If $\mathcal{A}$ has countable coproducts, then $(r_X^{p,q})_{(p,q)\in\mathbb{Z}^2}$ induces an isomorphism $r_X : \mathrm{tot}_\oplus(X) \xrightarrow{\sim} \mathrm{tot}_\oplus(\mathrm{swap}(X))$.
- ◇ If $\mathcal{A}$ has countable products, then $(r_X^{p,q})_{(p,q)\in\mathbb{Z}^2}$ induces an isomorphism $r_X : \mathrm{tot}_\Pi(X) \xrightarrow{\sim} \mathrm{tot}_\Pi(\mathrm{swap}(X))$.

Proof For every $(p,q) \in \mathbb{Z}^2$, the diagram

$$\begin{tikzcd}
X^{p,q} \arrow[r, "({}^{\triangleright}d, (-1)^p\, {}^{\vartriangle}d)"] \arrow[d, "(-1)^{pq}\mathrm{id}"'] & X^{p+1,q} \oplus X^{p,q+1} \arrow[d, "((-1)^{(p+1)q}\mathrm{id}, (-1)^{p(q+1)}\mathrm{id})"] \\
X^{p,q} \arrow[r, "((-1)^q\, {}^{\triangleright}d, {}^{\vartriangle}d)"'] & X^{p+1,q} \oplus X^{p,q+1}
\end{tikzcd}$$

commutes, so r_X is indeed a morphism of complexes. □

The requirement that countable coproducts (or countable products) exist in the definition of $\mathrm{tot}_\oplus X$ (or $\mathrm{tot}_\Pi X$) can be weakened. If, for each n, only finitely many pairs (p,q) satisfy $p+q=n$ and $X^{p,q} \neq 0$, then the coproduct (or product) in the definition of the total complex becomes a finite direct sum. In this case, both total complexes may be denoted uniformly by $\mathrm{tot}(X)$. There will be further discussion after Definition 3.10.1.

Remark 3.5.7 By the same idea, for every $k \in \mathbb{Z}_{\geq 1}$ one can define a k-fold complex as data

$$\left(X^{p_1,\ldots,p_k} \in \mathrm{Ob}(\mathcal{A})\right)_{p_1,\ldots,p_k\in\mathbb{Z}}, \quad {}^1d, \ldots, {}^kd,$$

where ${}^id^{p_1,\ldots,p_k} : X^{p_1,\ldots,p_k} \to X^{p_1,\ldots,p_i+1,\ldots,p_k}$ and

$${}^id^2 = 0, \quad {}^id\,{}^jd = {}^jd\,{}^id \quad \text{(superscripts suppressed)}.$$

All k-fold complexes form an additive category $\mathsf{C}^k(\mathcal{A})$. Following Definition 3.5.3, one can also define a family of additive functors $F_i : \mathsf{C}^k(\mathcal{A}) \to \mathsf{C}(\mathsf{C}^{k-1}(\mathcal{A}))$ such that every F_i is an equivalence of categories ($k \geq 2$ and $i = 1, \ldots, k$).

In this setting, if $\mathcal{A}$ has countable coproducts or products, the total-complex functor may be defined in the same way:

$$\mathrm{tot}_{\oplus} \text{ or } \mathrm{tot}_{\Pi} : \mathsf{C}^k(\mathcal{A}) \to \mathsf{C}(\mathcal{A}).$$

The details are the same as for double complexes and need not be repeated.

Define isomorphisms from $\mathsf{C}^2(\mathcal{A})$ to itself by $[m]_{\mathrm{I}} := F_{\mathrm{I}}^{-1} \circ [m] \circ F_{\mathrm{I}}$ and $[m]_{\mathrm{II}} := F_{\mathrm{II}}^{-1} \circ [m] \circ F_{\mathrm{II}}$, corresponding respectively to horizontal and vertical shifts of double complexes. For all $a, b \in \mathbb{Z}$, $[a]_{\mathrm{I}}[b]_{\mathrm{II}} = [b]_{\mathrm{II}}[a]_{\mathrm{I}}$. We now examine the effect of these functors on total complexes and their relation to the automorphism swap.

Proposition 3.5.8 For all $X \in \mathrm{Ob}(\mathsf{C}^2(\mathcal{A}))$ and $(p, q) \in \mathbb{Z}^2$, denote the standard embedding $X^{p,q} \hookrightarrow \mathrm{tot}_{\oplus}(X)^{p+q}$ by $\iota_{p,q}$. For every $n \in \mathbb{Z}$, define

- ◇ $\theta_X^n : \mathrm{tot}_{\oplus}\left(X[1]_{\mathrm{I}}\right)^n \to \mathrm{tot}_{\oplus}(X)[1]^n$ so that its restriction to $X[1]_{\mathrm{I}}^{p,q} = X^{p+1,q}$ is $\iota_{p+1,q}$;
- ◇ $(\theta'_X)^n : \mathrm{tot}_{\oplus}\left(X[1]_{\mathrm{II}}\right)^n \to \mathrm{tot}_{\oplus}(X)[1]^n$ so that its restriction to $X[1]_{\mathrm{II}}^{p,q} = X^{p,q+1}$ is $(-1)^p \iota_{p,q+1}$.

These define canonical isomorphisms in $\mathsf{C}(\mathcal{A})$:

$$\begin{aligned} \theta_X &= (\theta_X^n)_n : \mathrm{tot}_{\oplus}\left(X[1]_{\mathrm{I}}\right) \xrightarrow{\sim} \mathrm{tot}_{\oplus}\left(X\right)[1], \\ \theta'_X &= ((\theta'_X)^n)_n : \mathrm{tot}_{\oplus}\left(X[1]_{\mathrm{II}}\right) \xrightarrow{\sim} \mathrm{tot}_{\oplus}\left(X\right)[1], \end{aligned}$$

together with the anticommutative diagram (that is, its two composites differ by a minus sign)

$$\begin{tikzcd} \mathrm{tot}_{\oplus}\left(X[1]_{\mathrm{I}}[1]_{\mathrm{II}}\right) \arrow[r, "\theta_{X[1]_{\mathrm{II}}}"] \arrow[d, "\theta'_{X[1]_{\mathrm{I}}}"'] & \mathrm{tot}_{\oplus}\left(X[1]_{\mathrm{II}}\right)[1] \arrow[d, "\theta'_X[1]"] \\ \mathrm{tot}_{\oplus}\left(X[1]_{\mathrm{I}}\right)[1] \arrow[r, "\theta_X[1]"'] & \mathrm{tot}_{\oplus}(X)[2] \end{tikzcd}$$

and the commutative diagram

$$\begin{tikzcd} \mathrm{tot}_{\oplus}\left(X[1]_{\mathrm{I}}[1]_{\mathrm{II}}\right) \arrow[r, "\theta \text{ then } \theta'"] \arrow[d, "r_{X[1]_{\mathrm{I}}[1]_{\mathrm{II}}}"'] & \mathrm{tot}_{\oplus}(X)[2] \arrow[d, "r_X[2]"] \\ \mathrm{tot}_{\oplus}\left(\mathrm{swap}(X)[1]_{\mathrm{I}}[1]_{\mathrm{II}}\right) \arrow[r, "\theta' \text{ then } \theta"'] & \mathrm{tot}_{\oplus}\left(\mathrm{swap}(X)\right)[2], \end{tikzcd}$$

where r is as defined in Proposition 3.5.6. If $\mathrm{tot}_{\oplus}$ is replaced by tot_{Π}, one obtains the same θ_X, θ'_X and the corresponding anticommutative diagram.

Proof Direct verification from the definition of the total complex (Definition 3.5.4) and the definition of the shift functor (Definition 3.1.6) shows that θ_X and θ'_X are morphisms of complexes; the details are lengthy but not difficult. Anticommutativity reduces to the observation that the diagram

$$\begin{tikzcd} X^{p+1,q+1} \arrow[r, "\mathrm{id}"] \arrow[d, "(-1)^p \mathrm{id}"'] & X^{p+1,q+1} \arrow[d, "(-1)^{p+1}\mathrm{id}"] \\ X^{p+1,q+1} \arrow[r, "\mathrm{id}"'] & X^{p+1,q+1} \end{tikzcd}$$

is anticommutative.

Now consider the assertion about the commutative diagram. Let $(p, q) \in \mathbb{Z}^2$ and compare the restrictions of the two composites to $(X[1]_{\mathrm{I}}[1]_{\mathrm{II}})^{p,q}$. As explained above, θ followed by θ' (for X) and θ' followed by θ (for $\operatorname{swap}(X)$) give respectively

$$(-1)^{p+1},\ (-1)^q \ \in \operatorname{Aut}(X^{p+1,q+1}) \quad (\text{id suppressed}).$$

On the other hand, the restrictions of $r_{X[1]_{\mathrm{I}}[1]_{\mathrm{II}}}$ and $r_X[2]$ to $X^{p+1,q+1}$ are respectively $(-1)^{pq}$ and $(-1)^{(p+1)(q+1)}$. But $(p+1)(q+1) - pq \equiv (p+1) - q \pmod 2$. This proves the claim. $\square$

As a simple generalization of the anticommutative diagram above, the reader is asked to prove that, for all $a, b \in \mathbb{Z}$, the two morphisms

$$\operatorname{tot}_{\oplus}(X[a]_{\mathrm{I}}[b]_{\mathrm{II}}) \rightrightarrows \operatorname{tot}_{\oplus}(X)[a+b]$$

defined in the two orders (θ then θ', and θ' then θ) differ by the sign $(-1)^{ab}$; on the other hand, the commutative diagram above remains valid for $X[a]_{\mathrm{I}}[b]_{\mathrm{II}}$. The same statements hold with tot_{Π} in place of $\operatorname{tot}_{\oplus}$.

One of the main uses of double complexes is the study of bifunctors. What is a bifunctor?

Convention 3.5.9 A functor of the form $F : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}$ is called a **bifunctor**. Once objects $X_i \in \operatorname{Ob}(\mathcal{A}_i)$ are fixed, we obtain one-variable functors $F(X_1, \cdot) : \mathcal{A}_2 \to \mathcal{B}$ and $F(\cdot, X_2) : \mathcal{A}_1 \to \mathcal{B}$. We can therefore discuss additivity, exactness, and many other notions in each variable of F.

Definition–Proposition 3.5.10 Let $\mathcal{A}_1, \mathcal{A}_2, \mathcal{B}$ be additive categories and let $F : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}$ be a bifunctor additive in each variable. There is an associated functor

$$\begin{aligned} \mathsf{C}^2 F : \mathsf{C}(\mathcal{A}_1) \times \mathsf{C}(\mathcal{A}_2) &\longrightarrow \mathsf{C}^2(\mathcal{B}) \\ (X, Y) &\longmapsto (F(X,Y))^{p,q} := F(X^p, Y^q), \\ &\qquad {}^{\triangleright}d^{p,q} = F\left(d_X^p, \mathrm{id}\right),\ {}^{\vartriangle}d^{p,q} = F\left(\mathrm{id}, d_Y^q\right). \end{aligned}$$

Consequently, if $\mathcal{B}$ has countable coproducts or countable products, one can define respectively

$$\begin{aligned} \mathsf{C}_{\oplus}F &:= \mathrm{tot}_{\oplus} \circ \mathsf{C}^2 F : \mathsf{C}(\mathcal{A}_1) \times \mathsf{C}(\mathcal{A}_2) \to \mathsf{C}(\mathcal{B}), \\ \mathsf{C}_{\Pi}F &:= \mathrm{tot}_{\Pi} \circ \mathsf{C}^2 F : \mathsf{C}(\mathcal{A}_1) \times \mathsf{C}(\mathcal{A}_2) \to \mathsf{C}(\mathcal{B}). \end{aligned}$$

If the totalizations under discussion involve only finite direct sums, then $\mathsf{C}_{\oplus}F = \mathsf{C}_{\Pi}F$.[11)]

Proof The double-complex condition ${}^{\triangleright}d\,{}^{\triangle}d = {}^{\triangle}d\,{}^{\triangleright}d$ follows directly from the definition of a bifunctor; the rest is clear. □

The category of double complexes also has a notion of homotopy. Let $f, g : X \to Y$ be morphisms in $\mathsf{C}^2(\mathcal{A})$. A homotopy from f to g is a pair of families of morphisms satisfying the following conditions.

$$\begin{tikzcd} Y^{p-1,q} & X^{p,q} \arrow[l, "h^{p,q}"'] \arrow[d, "k^{p,q}"] \\ & Y^{p,q-1} \end{tikzcd} \qquad (p,q) \in \mathbb{Z}^2,$$

$${}^{\triangle}d^{p-1,q}h^{p,q} = h^{p,q+1}\,{}^{\triangle}d^{p,q}, \qquad {}^{\triangleright}d^{p,q-1}k^{p,q} = k^{p+1,q}\,{}^{\triangleright}d^{p,q},$$
$$g^{p,q} - f^{p,q} = {}^{\triangleright}d^{p-1,q}h^{p,q} + h^{p+1,q}\,{}^{\triangleright}d^{p,q} + {}^{\triangle}d^{p,q-1}k^{p,q} + k^{p,q+1}\,{}^{\triangle}d^{p,q}.$$

This is reasonable: from ${}^{\triangle}dh = h{}^{\triangle}d$, ${}^{\triangleright}dk = k{}^{\triangleright}d$, and the definition of a double complex, it is easy to check that ${}^{\triangleright}dh + h{}^{\triangleright}d + {}^{\triangle}dk + k{}^{\triangle}d$ always defines a morphism in $\mathsf{C}^2(\mathcal{A})$.

Homotopies of double complexes are reflected by total complexes. More precisely, given $(h^{p,q}, k^{p,q})_{p,q\in\mathbb{Z}}$, one can define $\hat{h} \in \mathrm{Hom}^{-1}(\mathrm{tot}_{\oplus}(X), \mathrm{tot}_{\oplus}(Y))$ so that the restriction of $\hat{h}$ to the direct summand $X^{p,q}$ is

$$(h^{p,q}, (-1)^p k^{p,q}) : X^{p,q} \to Y^{p-1,q} \oplus Y^{p,q-1},$$

which makes $\mathrm{tot}_{\oplus}(g) - \mathrm{tot}_{\oplus}(f) = d^{-1}\hat{h}$; the case of tot_{Π} is handled in the same way.

Thus the homotopy relation on double complexes defines $\mathsf{K}^2(\mathcal{A})$ together with a functor $\mathsf{C}^2(\mathcal{A}) \to \mathsf{K}^2(\mathcal{A})$. If countable coproducts (or products) exist, the functor $\mathrm{tot}_{\oplus}$ (or tot_{Π}) descends to a functor $\mathsf{K}^2(\mathcal{A}) \to \mathsf{K}(\mathcal{A})$.

Return to a bifunctor $F : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}$ between additive categories, assumed additive in each variable. The following is the bifunctor version of (3.2.4).

[11)] Translator's note (O014-C038): the source omits F from this equality and from the rightmost term of each of the two isomorphisms below. Since the functors being defined are $\mathsf{C}_{\oplus}F$ and $\mathsf{C}_{\Pi}F$, this translation restores F in all three places.

Proposition 3.5.11 Let F be as above. Then C^2F factors through $\mathsf{K}^2F : \mathsf{K}(\mathcal{A}_1) \times \mathsf{K}(\mathcal{A}_2) \to \mathsf{K}^2(\mathcal{B})$.

Similarly, $\mathsf{C}_\oplus F$ or $\mathsf{C}_\Pi F$ factors through $\mathsf{K}(\mathcal{A}_1) \times \mathsf{K}(\mathcal{A}_2) \to \mathsf{K}(\mathcal{B})$ and is denoted respectively by $\mathsf{K}_\oplus F$ or $\mathsf{K}_\Pi F$, provided the functor $\mathsf{C}_\oplus F$ or $\mathsf{C}_\Pi F$ is defined.

Proof Take $\mathsf{C}_\oplus F$ as an example. We must show that, for a null-homotopic morphism $f_1 = d^{-1}g : X_1 \to Y_1$ in $\mathsf{C}(\mathcal{A}_1)$ and an arbitrary morphism $f_2 : X_2 \to Y_2$ in $\mathsf{C}(\mathcal{A}_2)$, the corresponding morphism $(\mathsf{C}^2F)(f_1, f_2) : \mathsf{C}^2F(X_1, X_2) \to \mathsf{C}^2F(Y_1, Y_2)$ is null-homotopic. The evident choice is $h^{p,q} := F(g^p, f_2^q)$ and $k^{p,q} := 0$. Since f_2 is a morphism, h indeed commutes with ${}^\triangle d = F(\mathrm{id}, d)$. The second variable is handled similarly. $\square$

Proposition 3.5.8 gives canonical isomorphisms

$$\mathsf{C}_\oplus F(X[1], Y) \simeq \mathsf{C}_\oplus F(X, Y)[1] \simeq \mathsf{C}_\oplus F(X, Y[1]),$$
$$\mathsf{K}_\oplus F(X[1], Y) \simeq \mathsf{K}_\oplus F(X, Y)[1] \simeq \mathsf{K}_\oplus F(X, Y[1]).$$

The same statements hold with Π in place of $\oplus$.

Example 3.5.12 (Hom double complex) Consider the bifunctor $\mathrm{Hom}(\cdot, \cdot) : \mathcal{A}^{\mathrm{op}} \times \mathcal{A} \to \mathsf{Ab}$, which sends a pair of objects (S, T) to $\mathrm{Hom}_{\mathcal{A}}(S, T)$. We will show that the Hom complex $\mathrm{Hom}^\bullet(X, Y)$ is canonically isomorphic to the value at (X, Y) of the composite functor

$$\mathsf{C}(\mathcal{A})^{\mathrm{op}} \times \mathsf{C}(\mathcal{A}) \xrightarrow{(\sigma^{-1}, \mathrm{id})} \mathsf{C}(\mathcal{A}^{\mathrm{op}}) \times \mathsf{C}(\mathcal{A}) \xrightarrow{\mathsf{C}_\Pi \mathrm{Hom}(\cdot,\cdot)} \mathsf{C}(\mathsf{Ab}),$$

where σ is as in Definition–Proposition 3.4.1.

To this end, first consider the bifunctor $F : \mathcal{A} \times \mathcal{A}^{\mathrm{op}} \to \mathsf{Ab}$ defined by $F(T, S) = \mathrm{Hom}_{\mathcal{A}}(S, T)$, and the composite functor

$$\mathsf{C}(\mathcal{A}) \times \mathsf{C}(\mathcal{A})^{\mathrm{op}} \xrightarrow{(\mathrm{id}, \sigma^{-1})} \mathsf{C}(\mathcal{A}) \times \mathsf{C}(\mathcal{A}^{\mathrm{op}}) \xrightarrow{\mathsf{C}^2F} \mathsf{C}^2(\mathsf{Ab}).$$

Let $X, Y \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$. The image of (Y, X) under the composite functor above is called the **Hom double complex** and is denoted by $\mathrm{Hom}^{\bullet,\bullet}(X, Y)$. In view of the evident commutative diagram

$$\begin{tikzcd}
\mathsf{C}(\mathcal{A})^{\mathrm{op}} \times \mathsf{C}(\mathcal{A}) \arrow[r, "(\sigma^{-1}, \mathrm{id})"] \arrow[d, "\simeq"', "\text{interchange}"] & \mathsf{C}(\mathcal{A}^{\mathrm{op}}) \times \mathsf{C}(\mathcal{A}) \arrow[r, "\mathsf{C}^2 \mathrm{Hom}(\cdot,\cdot)"] \arrow[d, "\simeq"', "\text{interchange}"] & \mathsf{C}^2(\mathsf{Ab}) \arrow[d, "\text{swap}"] \\
\mathsf{C}(\mathcal{A}) \times \mathsf{C}(\mathcal{A})^{\mathrm{op}} \arrow[r, "(\mathrm{id}, \sigma^{-1})"'] & \mathsf{C}(\mathcal{A}) \times \mathsf{C}(\mathcal{A}^{\mathrm{op}}) \arrow[r, "\mathsf{C}^2F"'] & \mathsf{C}^2(\mathsf{Ab})
\end{tikzcd}$$

and Proposition 3.5.6, it suffices to prove that there is a canonical isomorphism

$$\operatorname{tot}_\Pi \operatorname{Hom}^{\bullet,\bullet}(X,Y) \xrightarrow{\sim} \operatorname{Hom}^\bullet(X,Y).$$

Inspecting the definitions gives

$$\begin{aligned}
\operatorname{Hom}^{p,q}(X,Y) &= \operatorname{Hom}_{\mathcal{A}}\left(X^{-q}, Y^p\right), \\
{}^{\triangleright}d^{p,q} &= (d_Y^p)_* : \operatorname{Hom}_{\mathcal{A}}\left(X^{-q}, Y^p\right) \to \operatorname{Hom}_{\mathcal{A}}\left(X^{-q}, Y^{p+1}\right), \\
{}^{\vartriangle}d^{p,q} &= (-1)^q (d_X^{-q-1})^* : \operatorname{Hom}_{\mathcal{A}}\left(X^{-q}, Y^p\right) \to \operatorname{Hom}_{\mathcal{A}}\left(X^{-q-1}, Y^p\right).
\end{aligned}$$

[12] Now determine $\operatorname{tot}_\Pi \operatorname{Hom}^{\bullet,\bullet}(X,Y)$. First,

$$\left(\operatorname{tot}_\Pi \operatorname{Hom}^{\bullet,\bullet}(X,Y)\right)^n = \prod_{p+q=n} \operatorname{Hom}_{\mathcal{A}}\left(X^{-q}, Y^p\right) = \prod_{k\in\mathbb{Z}} \operatorname{Hom}\left(X^k, Y^{k+n}\right)$$

under the substitution $k = -q$, which is precisely $\operatorname{Hom}^n(X,Y)$. Next describe $d^n_{\operatorname{tot}_\Pi \operatorname{Hom}^{\bullet,\bullet}(X,Y)}$: its (p,q)-coordinate in $\operatorname{Hom}^{n+1}(X,Y)$, where $p+q = n+1$, comes from

$$\begin{tikzcd}
\operatorname{Hom}^{p-1,q}(X,Y) \arrow[r, "{}^{\triangleright}d^{p-1,q}"] & \operatorname{Hom}^{p,q}(X,Y) \\
& \operatorname{Hom}^{p,q-1}(X,Y). \arrow[u, "(-1)^p\, {}^{\vartriangle}d^{p,q-1}"']
\end{tikzcd}$$

The horizontal arrow is $(d_Y^{p-1})_*$, while the vertical arrow is $(-1)^{p+q-1}(d_X^{-q})^* = (-1)^n (d_X^{-q})^*$. The isomorphism that multiplies the factor $\operatorname{Hom}^{p,q}(X,Y)$ by $(-1)^q$ changes the sign of the vertical term. Comparing with Definition 3.2.1 gives the canonical isomorphism $\operatorname{tot}_\Pi \operatorname{Hom}^{\bullet,\bullet}(X,Y) \xrightarrow{\sim} \operatorname{Hom}^\bullet(X,Y)$, as required.

Finally, if $\mathcal{A}$ is $\Bbbk$-linear, then $\operatorname{Hom}^{\bullet,\bullet}(X,Y)$ naturally becomes a functor with values in $\mathsf{C}^2(\Bbbk\text{-}\mathsf{Mod})$.

3.6 Complexes on an Abelian Category

In this section, $\mathcal{A}$ is assumed to be an abelian category. As usual, all statements about additivity in this section also extend to the case in which $\mathcal{A}$ is a $\Bbbk$-linear category.

For a complex X on an abelian category, we can discuss its cohomology (Definition 2.2.5). Recall that $[n]$ not only shifts the superscripts of a complex but also

[12] Translator's note (O014-C039): the source uses the coefficient $(-1)^{q+1}$, which corresponds to reusing the formula for σ but not to the strict inverse σ^{-1} obtained from the sign correction in the preceding section. This translation uses the inverse coefficient $(-1)^q$; consequently, the totalization is identified with the Hom complex by component signs $(-1)^q$, rather than by literal equality.

multiplies $d_X^\bullet$ by the sign $(-1)^n$. This sign change does not alter the kernel or image of any d_X. Thus

$$\mathrm{H}^k\left(X[n]\right) = \mathrm{H}^{n+k}\left(X\right), \quad k, n \in \mathbb{Z}.$$

Proposition 3.1.5 has already shown that $\mathsf{C}(\mathcal{A})$ is an additive category. We now show that it is also abelian.

Proposition 3.6.1 The category $\mathsf{C}(\mathcal{A})$ is abelian. More precisely, for all $X, Y \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$, $X \oplus Y = (X^n \oplus Y^n, (d_X^n, d_Y^n))_{n\in\mathbb{Z}}$, and for every morphism $f : X \to Y$ one may take

$$\begin{aligned} \ker(f) &= \left(\ker(f^n)\right)_{n\in\mathbb{Z}}, \quad & \operatorname{coker}(f) &= \left(\operatorname{coker}(f^n)\right)_{n\in\mathbb{Z}}, \\ \operatorname{im}(f) &= \left(\operatorname{im}(f^n)\right)_{n\in\mathbb{Z}}, \quad & \operatorname{coim}(f) &= \left(\operatorname{coim}(f^n)\right)_{n\in\mathbb{Z}}. \end{aligned}$$

[13]

Suppose morphisms f and g in $\mathsf{C}(\mathcal{A})$ are composable and $gf = 0$. Then $H := \mathrm{H}\left[X \xrightarrow{f} Y \xrightarrow{g} Z\right]$ is the following complex:

- ◇ its degree-n term is $H^n := \mathrm{H}\left[X^n \xrightarrow{f^n} Y^n \xrightarrow{g^n} Z^n\right]$;
- ◇ the morphism $d_H^n : H^n \to H^{n+1}$ is determined by d_X^n, d_Y^n, and d_Z^n, together with the functoriality of $\mathrm{H}[\cdots]$ (Proposition 2.2.2).[14]

Consequently, $X \xrightarrow{f} Y \xrightarrow{g} Z$ is exact if and only if $X^n \xrightarrow{f^n} Y^n \xrightarrow{g^n} Z^n$ is exact for every $n \in \mathbb{Z}$.

Proof Proposition 3.1.5 explained how to use the forgetful functor from complexes to graded objects, $U : \mathsf{C}(\mathcal{A}) \to \mathcal{A}^{\mathbb{Z}}$, to reduce the required $\varinjlim$ and $\varprojlim$ to degreewise constructions in $\mathcal{A}$. The point is that U creates $\varinjlim$ and $\varprojlim$ (Lemma 3.1.4). This gives degreewise constructions of $X \oplus Y$, $\ker(f)$, $\operatorname{coker}(f)$, and so forth.

In particular, by the characterization (1.2.1), the canonical morphism $\operatorname{coim}(f) \to \operatorname{im}(f)$ is given by the morphisms $\operatorname{coim}(f^n) \to \operatorname{im}(f^n)$ in $\mathcal{A}$. A degreewise isomorphism is an isomorphism of complexes. Thus every morphism in $\mathsf{C}(\mathcal{A})$ is strict. Therefore $\mathsf{C}(\mathcal{A})$ is an abelian category.[15] The description of $\mathrm{H}[X \to Y \to Z]$ follows in the same way. □

[13] Translator's note (O014-C040): the source writes $\ker(f)^n$, $\operatorname{coker}(f)^n$, $\operatorname{im}(f)^n$, and $\operatorname{coim}(f)^n$ on the left but equates each with an entire family indexed by n on the right. This translation removes the superscript n on the left so that all four are identities of complexes; their degree-n components remain exactly those shown on the right.

[14] Translator's note (O014-C041): the source mentions only d_X^n and d_Y^n. This translation adds d_Z^n, which is needed to give a morphism between the two three-term diagrams and hence to induce d_H^n.

[15] Translator's note (O014-C042): the closing sentence of the source names $\mathcal{A}$, although that category was already assumed abelian; the category just proved abelian is $\mathsf{C}(\mathcal{A})$. This translation corrects the category name.

Remark 3.6.2 The same method proves that the category of double complexes $\mathsf{C}^2(\mathcal{A})$, and even the category of k-fold complexes $\mathsf{C}^k(\mathcal{A})$, remains abelian for $k \in \mathbb{Z}_{\geq 1}$; see Remark 3.5.7.

Next consider a morphism $f : X \to Y$ in $\mathsf{C}(\mathcal{A})$. For every $n \in \mathbb{Z}$, Proposition 2.2.2 uniquely determines $\mathrm{H}^n(f) : \mathrm{H}^n(X) \to \mathrm{H}^n(Y)$ so that the following diagram commutes:

$$\begin{tikzcd}
\ker(d_X^n) \arrow[r, two heads] \arrow[d, "\text{induced by } f"'] & \mathrm{H}^n(X) \arrow[r, hook] \arrow[d, "\mathrm{H}^n(f)"] & \operatorname{coker}(d_X^{n-1}) \arrow[d, "\text{induced by } f"] \\
\ker(d_Y^n) \arrow[r, two heads] & \mathrm{H}^n(Y) \arrow[r, hook] & \operatorname{coker}(d_Y^{n-1}).
\end{tikzcd} \tag{3.6.1}$$

For composable morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$, this characterization immediately gives $\mathrm{H}^n(gf) = \mathrm{H}^n(g)\,\mathrm{H}^n(f)$; moreover, $\mathrm{H}^n(\mathrm{id}_X) = \mathrm{id}_{\mathrm{H}^n(X)}$.

Proposition 3.6.3 (Cohomology as a functor) For every $n \in \mathbb{Z}$, the definition above gives an additive functor $\mathrm{H}^n : \mathsf{C}(\mathcal{A}) \to \mathcal{A}$.

Proof The properties $\mathrm{H}^n(gf) = \mathrm{H}^n(g)\,\mathrm{H}^n(f)$ and $\mathrm{H}^n(\mathrm{id}) = \mathrm{id}$ follow directly from the characterization (3.6.1). In the same way one obtains $\mathrm{H}^n(f_1 + f_2) = \mathrm{H}^n(f_1) + \mathrm{H}^n(f_2)$ and $\mathrm{H}^n(tf) = t\,\mathrm{H}^n(f)$ when $\mathcal{A}$ is $\Bbbk$-linear and $t \in \Bbbk$. $\square$

A short exact sequence of complexes automatically induces a long exact sequence in cohomology. This is the most elementary form of a long exact sequence in homology theory.

Proposition 3.6.4 (A short exact sequence induces a long exact sequence) Let $0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$ be a short exact sequence in $\mathsf{C}(\mathcal{A})$. Then Theorem 2.3.3 gives the following canonical exact sequence in $\mathcal{A}$:

$$\begin{tikzcd}
\cdots \arrow[r] & \mathrm{H}^{n-1}(Y) \arrow[r, "\mathrm{H}^{n-1}(g)"] & \mathrm{H}^{n-1}(Z) \arrow[dll, "\delta^{n-1}"', rounded corners] \\
\mathrm{H}^n(X) \arrow[r, "\mathrm{H}^n(f)"] & \mathrm{H}^n(Y) \arrow[r, "\mathrm{H}^n(g)"] & \mathrm{H}^n(Z) \arrow[dll, "\delta^n"', rounded corners] \\
\mathrm{H}^{n+1}(X) \arrow[r, "\mathrm{H}^{n+1}(f)"'] & \mathrm{H}^{n+1}(Y) \arrow[r] & \cdots
\end{tikzcd}$$

This sequence has the following functoriality. If there is a commutative diagram in $\mathsf{C}(\mathcal{A})$ whose rows are exact,

$$\begin{tikzcd} 0 \arrow[r] & X \arrow[r] \arrow[d] & Y \arrow[r] \arrow[d] & Z \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & \underline{X} \arrow[r] & \underline{Y} \arrow[r] & \underline{Z} \arrow[r] & 0 \end{tikzcd}$$

then one obtains a commutative diagram

$$\begin{tikzcd} \cdots \arrow[r] & \mathrm{H}^n(X) \arrow[r] \arrow[d] & \mathrm{H}^n(Y) \arrow[r] \arrow[d] & \mathrm{H}^n(Z) \arrow[r] \arrow[d] & \mathrm{H}^{n+1}(X) \arrow[r] \arrow[d] & \cdots \\ \cdots \arrow[r] & \mathrm{H}^n(\underline{X}) \arrow[r] & \mathrm{H}^n(\underline{Y}) \arrow[r] & \mathrm{H}^n(\underline{Z}) \arrow[r] & \mathrm{H}^{n+1}(\underline{X}) \arrow[r] & \cdots . \end{tikzcd}$$

Proof For every $n \in \mathbb{Z}$ there is an exact sequence $\operatorname{coker}(d_X^{n-1}) \to \operatorname{coker}(d_Y^{n-1}) \to \operatorname{coker}(d_Z^{n-1}) \to 0$. Indeed, take the coker part of the exact sequence in Theorem 2.3.3 applied to the diagram

$$\begin{tikzcd} 0 \arrow[r] & X^{n-1} \arrow[r] \arrow[d] & Y^{n-1} \arrow[r] \arrow[d] & Z^{n-1} \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & X^n \arrow[r] & Y^n \arrow[r] & Z^n \arrow[r] & 0 \end{tikzcd}$$

[16)] Similarly, there is an exact sequence $0 \to \ker(d_X^n) \to \ker(d_Y^n) \to \ker(d_Z^n)$. The two sequences fit into the following commutative diagram with exact rows:

$$\begin{tikzcd} & \operatorname{coker}(d_X^{n-2}) \arrow[r] \arrow[d] & \operatorname{coker}(d_Y^{n-2}) \arrow[r] \arrow[d] & \operatorname{coker}(d_Z^{n-2}) \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & \ker(d_X^n) \arrow[r] & \ker(d_Y^n) \arrow[r] & \ker(d_Z^n) & \end{tikzcd} \tag{3.6.2}$$

The vertical arrows are induced respectively by d_X^{n-1}, d_Y^{n-1}, and d_Z^{n-1} and have epi–mono factorizations of the form $\operatorname{coker}(d^{n-2}) \overset{d^{n-1}}{\twoheadrightarrow} \operatorname{im}(d^{n-1}) \hookrightarrow \ker(d^n)$, with subscripts suppressed.

By Lemma 1.3.11 (iii), the epimorphism $\operatorname{coker}(d^{n-2}) \twoheadrightarrow \operatorname{im}(d^{n-1})$ determines the kernels of the vertical arrows in (3.6.2); then (2.2.2) gives in turn $\mathrm{H}^{n-1}(X)$, $\mathrm{H}^{n-1}(Y)$, and $\mathrm{H}^{n-1}(Z)$. Likewise, the monomorphism $\operatorname{im}(d^{n-1}) \hookrightarrow \ker(d^n)$ determines their cokernels, giving in turn $\mathrm{H}^n(X)$, $\mathrm{H}^n(Y)$, and $\mathrm{H}^n(Z)$. Applying Theorem 2.3.3 now yields the connecting morphism $\delta^{n-1} : \mathrm{H}^{n-1}(Z) \to \mathrm{H}^n(X)$ and the portion of the long exact sequence containing the terms H^n and H^{n-1}. Its functoriality follows from Remark 2.3.2. $\square$

[16)] Translator's note (O014-C043): the source states the sequence $\operatorname{coker}(d_X^n) \to \operatorname{coker}(d_Y^n) \to \operatorname{coker}(d_Z^n) \to 0$, while the vertical arrows in the diagram immediately following it are d_X^{n-1}, d_Y^{n-1}, and d_Z^{n-1}. This translation aligns the cokernel indices with the diagram; renaming the index then gives the row indexed by $n-2$ used below.

Definition 3.6.5 (Quasi-isomorphism) Let $f : X \to Y$ be a morphism in $\mathsf{C}(\mathcal{A})$. If $\mathrm{H}^n(f)$ is an isomorphism for every $n \in \mathbb{Z}$, then f is called a **quasi-isomorphism**.

Proposition 3.6.6 Let $f : X \to Y$ be a morphism in $\mathsf{C}(\mathcal{A})$ and let $n \in \mathbb{Z}$. If f is null-homotopic, then $\mathrm{H}^n(f) = 0$.

Proof Choose $h \in \mathrm{Hom}^{-1}(X, Y)$ such that $f = d_Y h + h d_X$. The composite $\ker(d_X^n) \hookrightarrow X^n \xrightarrow{h^{n+1} d_X^n} Y^n$ is zero. On the other hand, $d_Y^{n-1} h^n$ factors through $\mathrm{im}(d_Y^{n-1})$. Together these facts give $\mathrm{H}^n(f) = 0$. □

Corollary 3.6.7 For every $n \in \mathbb{Z}$, the cohomology functor $\mathrm{H}^n : \mathsf{C}(\mathcal{A}) \to \mathcal{A}$ factors uniquely through $\mathsf{K}(\mathcal{A})$. The notion of quasi-isomorphism therefore extends to morphisms in $\mathsf{K}(\mathcal{A})$. If $f : X \to Y$ is an isomorphism in $\mathsf{K}(\mathcal{A})$, then f is a quasi-isomorphism.

Proof Combine Propositions 3.2.9 and 3.6.6. □

3.7 Mapping Cones and Long Exact Sequences

Long exact sequences are a principal tool of homological algebra. This section compares three ways of constructing a long exact sequence from a short exact sequence. Up to certain signs, all three give the same result. Throughout the section, $\mathcal{A}$ is assumed to be an abelian category.

Proposition 3.7.1 Let $f : X \to Y$ be a morphism in $\mathsf{C}(\mathcal{A})$. Take $\alpha(f)$ and $\beta(f)$ as in Definition 3.3.5. Then

$$0 \to Y \xrightarrow{\alpha(f)} \mathrm{Cone}(f) \xrightarrow{\beta(f)} X[1] \to 0$$

is an exact sequence in $\mathsf{C}(\mathcal{A})$.

Proof Recall the definitions of $\alpha(f)$ and $\beta(f)$. Since exactness can be checked degreewise (Proposition 3.6.1), it suffices to prove the exactness of

$$0 \to Y^n \xrightarrow{(0,\mathrm{id})} X^{n+1} \oplus Y^n \xrightarrow{\text{projection}} X^{n+1} \to 0,$$

for every $n \in \mathbb{Z}$. This is Proposition 2.2.7. □

Given a morphism $f : X \to Y$ in $\mathsf{C}(\mathcal{A})$, the short exact sequence in Proposition 3.7.1, together with Proposition 3.6.4, gives the following long exact sequence in $\mathcal{A}$:

$$\cdots \to \mathrm{H}^n(Y) \xrightarrow{\mathrm{H}^n(\alpha(f))} \mathrm{H}^n(\mathrm{Cone}(f)) \xrightarrow{\mathrm{H}^n(\beta(f))} \underbrace{\mathrm{H}^n(X[1])}_{=\mathrm{H}^{n+1}(X)} \xrightarrow{\xi^n} \mathrm{H}^{n+1}(Y) \to \cdots \quad (3.7.1)$$

The morphism labeled ξ^n is the connecting morphism induced by this short exact sequence. Corollary 3.7.5 will prove that ξ^n is precisely $\mathrm{H}^{n+1}(f) : \mathrm{H}^{n+1}(X) \to \mathrm{H}^{n+1}(Y)$. Thus every part of the mapping-cone long exact sequence (3.7.1) is explicit and comes from f, $\alpha(f)$, and $\beta(f)$.

The next result goes in the opposite direction: every short exact sequence in $\mathsf{C}(\mathcal{A})$ also has a natural relation with a mapping cone. This relation can be constructed in two ways.

Lemma 3.7.2 Given a short exact sequence $0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$ in $\mathsf{C}(\mathcal{A})$, the two morphisms

$$\Phi := (0, g) : \mathrm{Cone}(f) \to Z, \quad \Phi' := (f[1], 0) : X[1] \to \mathrm{Cone}(g)$$

are quasi-isomorphisms of complexes and have the following properties:

(i) $\Phi \circ \alpha(f) = g$ and $\beta(g) \circ \Phi' = f[1]$;

(ii) $\alpha(g)\Phi + \Phi'\beta(f) : \mathrm{Cone}(f) \to \mathrm{Cone}(g)$ is canonically null-homotopic.

Proof Consider the following commutative diagrams in $\mathsf{C}(\mathcal{A})$:

$$\begin{tikzcd}
0 \arrow[r] & X \arrow[r, "\mathrm{id}_X"] \arrow[d, "\mathrm{id}_X"'] & X \arrow[r] \arrow[d, "f"] & 0 \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & X \arrow[r, "f"'] & Y \arrow[r, "g"'] & Z \arrow[r] & 0
\end{tikzcd}
\qquad
\begin{tikzcd}
0 \arrow[r] & X \arrow[r, "f"] \arrow[d] & Y \arrow[r, "g"] \arrow[d, "g"'] & Z \arrow[r] \arrow[d, "\mathrm{id}_Z"] & 0 \\
0 \arrow[r] & 0 \arrow[r] & Z \arrow[r, "\mathrm{id}_Z"'] & Z \arrow[r] & 0
\end{tikzcd}$$

Every row is exact. Hence functoriality of mapping cones (Proposition 3.3.3) gives morphisms in $\mathsf{C}(\mathcal{A})$ arranged as

$$\begin{array}{c}
0 \longrightarrow \mathrm{Cone}\,(\mathrm{id}_X) \longrightarrow \mathrm{Cone}\left(X \xrightarrow{f} Y\right) \longrightarrow \mathrm{Cone}\,(0 \to Z) \longrightarrow 0 \\
0 \longrightarrow \mathrm{Cone}(X \to 0) \longrightarrow \mathrm{Cone}\left(Y \xrightarrow{g} Z\right) \longrightarrow \mathrm{Cone}\,(\mathrm{id}_Z) \longrightarrow 0
\end{array} \tag{3.7.2}$$

Checking degree by degree and using the diagrams above and the definition of a mapping cone shows that both rows of (3.7.2) are exact. It is also easy to see that $\mathrm{Cone}(f) \to \mathrm{Cone}(0 \to Z) \simeq Z$ (or $X[1] \simeq \mathrm{Cone}(X \to 0) \to \mathrm{Cone}(g)$) is exactly Φ (or Φ'). Once they are known to be morphisms of complexes, the equalities in (i) are immediate.

Applying Proposition 3.6.4 to (3.7.2) gives, for every $n \in \mathbb{Z}$, exact sequences

$$\mathrm{H}^n\left(\mathrm{Cone}(\mathrm{id}_X)\right) \to \mathrm{H}^n\left(\mathrm{Cone}(f)\right) \xrightarrow{\mathrm{H}^n(\Phi)} \mathrm{H}^n(Z) \to \mathrm{H}^{n+1}\left(\mathrm{Cone}(\mathrm{id}_X)\right),$$
$$\mathrm{H}^{n-1}\left(\mathrm{Cone}(\mathrm{id}_Z)\right) \to \mathrm{H}^n(X[1]) \xrightarrow{\mathrm{H}^n(\Phi')} \mathrm{H}^n\left(\mathrm{Cone}(g)\right) \to \mathrm{H}^n\left(\mathrm{Cone}(\mathrm{id}_Z)\right).$$

Lemma 3.3.8 (i) and Proposition 3.6.6 show that the terms at both ends are all 0. Thus Φ and Φ' are quasi-isomorphisms.

Finally, verify (ii). Using $\mathrm{Cone}(f)^n = X^{n+1} \oplus Y^n$ and $\mathrm{Cone}(g)^n = Y^{n+1} \oplus Z^n$, write the morphisms as matrices:

$$\begin{tikzcd} & Z \arrow[dr, "\alpha(g)", "\begin{pmatrix} 0 \\ \mathrm{id} \end{pmatrix}"'] & \\ \mathrm{Cone}(f) \arrow[ur, "\Phi"', "\begin{pmatrix} 0 & g \end{pmatrix}"] \arrow[dr, "\beta(f)", "\begin{pmatrix} \mathrm{id} & 0 \end{pmatrix}"'] & & \mathrm{Cone}(g) \\ & X[1] \arrow[ur, "\Phi'"', "\begin{pmatrix} f[1] \\ 0 \end{pmatrix}"] & \end{tikzcd}, \quad \alpha(g)\Phi + \Phi'\beta(f) = \begin{pmatrix} f[1] & 0 \\ 0 & g \end{pmatrix}.$$

On the other hand, take $s = (s^n)_n \in \mathrm{Hom}^{-1}(\mathrm{Cone}(f), \mathrm{Cone}(g))$ with

$$s^n := \begin{pmatrix} 0 & \mathrm{id}_{Y^n} \\ 0 & 0 \end{pmatrix} : X^{n+1} \oplus Y^n \to Y^n \oplus Z^{n-1},$$

which is also the only natural choice. A direct matrix calculation shows that $d^{n-1}_{\mathrm{Cone}(g)} s^n + s^{n+1} d^n_{\mathrm{Cone}(f)}$ equals

$$\begin{pmatrix} -d_Y^n & 0 \\ g^n & d_Z^{n-1} \end{pmatrix} \begin{pmatrix} 0 & \mathrm{id}_{Y^n} \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & \mathrm{id}_{Y^{n+1}} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -d_X^{n+1} & 0 \\ f^{n+1} & d_Y^n \end{pmatrix} = \begin{pmatrix} f^{n+1} & 0 \\ 0 & g^n \end{pmatrix}.$$

Thus $\alpha(g)\Phi + \Phi'\beta(f)$ is indeed null-homotopic. □

In summary, a short exact sequence $0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$ in $\mathsf{C}(\mathcal{A})$ gives three ways to obtain the following long exact sequence in $\mathcal{A}$:

$$\cdots \longrightarrow \mathrm{H}^{n-1}(Y) \longrightarrow \mathrm{H}^{n-1}(Z) \longrightarrow \mathrm{H}^n(X) \longrightarrow \mathrm{H}^n(Y) \longrightarrow \mathrm{H}^n(Z) \longrightarrow \mathrm{H}^{n+1}(X) \longrightarrow \mathrm{H}^{n+1}(Y) \longrightarrow \cdots.$$

(A) Applying Proposition 3.6.4 directly gives the long exact sequence

$$\cdots \longrightarrow \mathrm{H}^n(Y) \xrightarrow{\mathrm{H}^n(g)} \mathrm{H}^n(Z) \xrightarrow{\delta^n} \mathrm{H}^{n+1}(X) \xrightarrow{\mathrm{H}^{n+1}(f)} \mathrm{H}^{n+1}(Y) \longrightarrow \cdots$$

(B) Use the quasi-isomorphism $\Phi : \mathrm{Cone}(f) \to Z$ from Lemma 3.7.2 together with

(3.7.1) to obtain the commutative diagram

$$\begin{tikzcd}
\cdots \arrow[r] & \mathrm{H}^n(Y) \arrow[r, "\mathrm{H}^n(\alpha(f))"] \arrow[d, equal] & \mathrm{H}^n(\mathrm{Cone}(f)) \arrow[r, "\mathrm{H}^n(\beta(f))"] \arrow[d, "\mathrm{H}^n(\Phi)"] & \mathrm{H}^n(X[1]) \arrow[r] \arrow[d, equal] & \mathrm{H}^{n+1}(Y) \arrow[r] \arrow[d, equal] & \cdots \\
\cdots \arrow[r] & \mathrm{H}^n(Y) \arrow[r, "\mathrm{H}^n(g)"'] & \mathrm{H}^n(Z) \arrow[r, "\eta^n"'] & \mathrm{H}^{n+1}(X) \arrow[r, "\xi^n"'] & \mathrm{H}^{n+1}(Y) \arrow[r] & \cdots
\end{tikzcd}$$

where

- $\eta^n := \mathrm{H}^n(\beta(f))\,\mathrm{H}^n(\Phi)^{-1}$;
- ξ^n is the connecting morphism in (3.7.1), of the form

$$\mathrm{H}^{n+1}(X) = \mathrm{H}^n(X[1]) \to \mathrm{H}^{n+1}(Y).$$

Since the first row is known to be exact, the second is exact as well.

(C) Proceed similarly, now using the quasi-isomorphism $\Phi' : X[1] \to \mathrm{Cone}(g)$ from Lemma 3.7.2. This gives the commutative diagram

$$\begin{tikzcd}
\cdots \arrow[r] & \mathrm{H}^n(Z) \arrow[r, "\mathrm{H}^n(\alpha(g))"] \arrow[d, equal] & \mathrm{H}^n(\mathrm{Cone}(g)) \arrow[r, "\mathrm{H}^n(\beta(g))"] & \mathrm{H}^n(Y[1]) \arrow[r] \arrow[d, equal] & \mathrm{H}^{n+1}(Z) \arrow[r] \arrow[d, equal] & \cdots \\
\cdots \arrow[r] & \mathrm{H}^n(Z) \arrow[r, "(\eta')^n"'] & \mathrm{H}^n(X[1]) \arrow[r, "\mathrm{H}^n(f[1])"'] \arrow[u, "\simeq"', "\mathrm{H}^n(\Phi')"] & \mathrm{H}^{n+1}(Y) \arrow[r, "(\xi')^n"'] & \mathrm{H}^{n+1}(Z) \arrow[r] & \cdots
\end{tikzcd}$$

where

- $(\eta')^n := \mathrm{H}^n(\Phi')^{-1}\,\mathrm{H}^n(\alpha(g))$;
- $(\xi')^n$ comes from the connecting morphism $\mathrm{H}^{n+1}(Y) = \mathrm{H}^n(Y[1]) \to \mathrm{H}^{n+1}(Z)$ in (3.7.1), with g in place of f.

Since the first row is known to be exact, the second is exact as well.

How are these three long exact sequences alike, and how do they differ? We first state the conclusions.

- Constructions (A) and (B) differ only by certain minus signs: Proposition 3.7.3 will show that $\eta^n = -\delta^n$, while Corollary 3.7.5 implies $\xi^n = \mathrm{H}^{n+1}(f)$.
- Constructions (A) and (C) give the same result: Corollary 3.7.4, based on the result for (B), will show that $(\eta')^n = \delta^n$, while Corollary 3.7.5 implies $(\xi')^n = \mathrm{H}^{n+1}(g)$.

◇ Thus (B) and (C) differ only by a sign in the connecting morphism $\mathrm{H}^n(Z) \to \mathrm{H}^{n+1}(X)$. This can be explained by the rotation axiom (TR3) for triangulated categories; see Proposition 4.4.7.

We now prove these relations.

Proposition 3.7.3 In the situation above, $\eta^n = -\delta^n$ for every $n \in \mathbb{Z}$.

Proof The following argument is taken from [23, Proposition 12.3.6]. Because it is rather intricate, readers are encouraged first to try the concrete case $\mathcal{A} = R\text{-}\mathsf{Mod}$, with R a ring; the diagram-chasing method of [51, §6.8] gives a direct proof in that case.

For a general abelian category $\mathcal{A}$, fix $n \in \mathbb{Z}$ and take the fiber product

$$S := \operatorname{coker}(d_Y^{n-1}) \underset{\operatorname{coker}(d_Z^{n-1})}{\times} \mathrm{H}^n(Z).$$

How is the connecting morphism δ^n constructed? It is obtained by applying (2.3.2) to the diagram in (3.6.2),

$$\begin{tikzcd}
 & \operatorname{coker}(d_X^{n-1}) \arrow[r] \arrow[d] & \operatorname{coker}(d_Y^{n-1}) \arrow[r] \arrow[d] & \operatorname{coker}(d_Z^{n-1}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \ker(d_X^{n+1}) \arrow[r] & \ker(d_Y^{n+1}) \arrow[r] & \ker(d_Z^{n+1}) &
\end{tikzcd}$$

here rearranged into the form

$$\begin{tikzcd}
 & S \arrow[r, two heads, "w"] \arrow[d, "v"] \arrow[ddl, bend right, "u"'] \arrow[dr, phantom, "\square"] & \mathrm{H}^n(Z) \arrow[d, hook] \\
 & \operatorname{coker}(d_Y^{n-1}) \arrow[r, two heads] \arrow[d, "a"] & \operatorname{coker}(d_Z^{n-1}) \arrow[d] \\
\ker(d_X^{n+1}) \arrow[r, hook, "b"'] \arrow[d, two heads] & \ker(d_Y^{n+1}) \arrow[r, "c"'] & \ker(d_Z^{n+1}) \\
\mathrm{H}^{n+1}(X) & &
\end{tikzcd} \tag{3.7.3}$$

All rows and columns here are exact. The arrow a is induced by d_Y^n, the arrow b is induced by f^{n+1}, and the existence and uniqueness of u follow because commutativity of the right-hand part implies $cav = 0$. Inspecting the construction in (2.3.2) shows that the diagram

$$\begin{tikzcd}
 & \mathrm{H}^n(Z) \arrow[dr, "\delta^n"] & \\
S \arrow[ur, two heads, "w"] \arrow[dr, "u"'] & & \mathrm{H}^{n+1}(X) \\
 & \ker\left(d_X^{n+1}\right) \arrow[ur, two heads] &
\end{tikzcd} \quad \text{commutes;} \tag{3.7.4}$$

since w is known to be an epimorphism, this diagram also uniquely determines δ^n.

To compare it with the mapping cone, verify directly from the definitions that the following diagram commutes:

$$\begin{tikzcd}
\ker(d_X^{n+1}) \oplus \operatorname{coker}(d_Y^{n-1}) \arrow[r, hook] \arrow[d, "{(b,a)}"'] & X^{n+1} \oplus \operatorname{coker}\left(d_Y^{n-1}\right) \arrow[r, two heads] & \operatorname{coker}\left(d_{\mathrm{Cone}(f)}^{n-1}\right) \arrow[d, "d^n_{\mathrm{Cone}(f)}"] \\
\ker(d_Y^{n+1}) \arrow[r, "\sim"] & 0 \oplus \ker\left(d_Y^{n+1}\right) \arrow[r, hook] & \ker\left(d_{\mathrm{Cone}(f)}^{n+1}\right)
\end{tikzcd}$$

The definitions of the horizontal arrows should be clear. Since the fan-shaped part of (3.7.3) commutes, the diagram above implies that the composite

$$S \xrightarrow{(-u,v)} \ker\left(d_X^{n+1}\right) \oplus \operatorname{coker}\left(d_Y^{n-1}\right) \xrightarrow[\text{top row}]{\text{of the diagram above}} \operatorname{coker}\left(d_{\mathrm{Cone}(f)}^{n-1}\right) \xrightarrow{d^n_{\mathrm{Cone}(f)}} \ker\left(d_{\mathrm{Cone}(f)}^{n+1}\right)$$

is 0. Therefore $S \xrightarrow{(-u,v)} \ker\left(d_X^{n+1}\right) \oplus \operatorname{coker}\left(d_Y^{n-1}\right) \to \operatorname{coker}\left(d_{\mathrm{Cone}(f)}^{n-1}\right)$ factors uniquely as $S \to \mathrm{H}^n(\mathrm{Cone}(f)) \hookrightarrow \operatorname{coker}\left(d_{\mathrm{Cone}(f)}^{n-1}\right)$. We claim that the following diagram commutes:

$$\begin{tikzcd}
S \arrow[r, "{(-u,v)}"] \arrow[d] \arrow[dd, bend right, "w"'] & \ker\left(d_X^{n+1}\right) \oplus \operatorname{coker}\left(d_Y^{n-1}\right) \arrow[r, "\text{projection}"] \arrow[d, "\text{evident}"] & \ker\left(d_X^{n+1}\right) \arrow[d, two heads] \\
 & & \mathrm{H}^{n+1}(X) \arrow[d, hook] \\
\mathrm{H}^n\left(\mathrm{Cone}(f)\right) \arrow[r, hook] \arrow[d, "\mathrm{H}^n(\Phi)", "\simeq"'] & \operatorname{coker}\left(d_{\mathrm{Cone}(f)}^{n-1}\right) \arrow[r, "\text{induced by } \beta(f)"'] \arrow[d, "\text{induced by } \Phi"'] & \operatorname{coker}\left(d_X^n\right) \\
\mathrm{H}^n(Z) \arrow[r, hook] & \operatorname{coker}\left(d_Z^{n-1}\right).
\end{tikzcd} \tag{3.7.5}$$

Indeed, commutativity of each square is routine to verify. By the upper-right square in (3.7.3) and $\Phi = (0, g)$, the part

$$\begin{tikzcd}
S \arrow[r, "{(-u,v)}"] \arrow[d, bend right, "w"'] & \ker\left(d_X^{n+1}\right) \oplus \operatorname{coker}\left(d_Y^{n-1}\right) \arrow[d] \\
\mathrm{H}^n(Z) \arrow[r, hook] & \operatorname{coker}\left(d_Z^{n-1}\right)
\end{tikzcd}$$

of diagram (3.7.5) also commutes. Moreover, because $\mathrm{H}^n(Z) \to \operatorname{coker}(d_Z^{n-1})$ is a monomorphism, composing with this arrow shows that the remaining curved left-hand part of (3.7.5) commutes as well.

Notice that the two composites

$$\mathrm{H}^n(\mathrm{Cone}(f)) \hookrightarrow \mathrm{coker}\left(d^{n-1}_{\mathrm{Cone}(f)}\right) \to \mathrm{coker}(d^n_X),$$
$$\mathrm{H}^n(\mathrm{Cone}(f)) \xrightarrow{\mathrm{H}^n(\beta(f))} \mathrm{H}^{n+1}(X) \hookrightarrow \mathrm{coker}(d^n_X)$$

are equal, since both are induced by projection onto X^{n+1}. Hence (3.7.5) gives the commutative diagram

$$\begin{tikzcd}
S \arrow[rr, "-u"] \arrow[d, "w"'] & & \ker\left(d^{n+1}_X\right) \arrow[d] \\
\mathrm{H}^n(Z) \arrow[r, "\sim", "\mathrm{H}^n(\Phi)^{-1}"'] & \mathrm{H}^n\left(\mathrm{Cone}(f)\right) \arrow[r, "\mathrm{H}^n(\beta(f))"'] & \mathrm{H}^{n+1}(X).
\end{tikzcd}$$

By (3.7.4), this implies that the composites $S \xrightarrow{w} \mathrm{H}^n(Z) \xrightarrow{\delta^n} \mathrm{H}^{n+1}(X)$ and $S \xrightarrow{w} \mathrm{H}^n(Z) \xrightarrow{\eta^n} \mathrm{H}^{n+1}(X)$ differ by a minus sign. Since w is an epimorphism, $\eta^n = -\delta^n$, as required. $\square$

Corollary 3.7.4 In the situation above, $(\eta')^n = \delta^n$ for every $n \in \mathbb{Z}$.

Proof Use the notation of Lemma 3.7.2. By Proposition 3.7.3, it suffices to prove that $(\eta')^n + \eta^n = 0$. This equality follows immediately from the fact that $\alpha(g)\Phi + \Phi'\beta(f) : \mathrm{Cone}(f) \to \mathrm{Cone}(g)$ is null-homotopic (Lemma 3.7.2 (ii)). $\square$

Corollary 3.7.5 For any morphism $f : X \to Y$ in $\mathsf{C}(\mathcal{A})$, the connecting morphism $\xi^n : \mathrm{H}^{n+1}(X) \simeq \mathrm{H}^n(X[1]) \to \mathrm{H}^{n+1}(Y)$ in the long exact sequence (3.7.1) equals $\mathrm{H}^{n+1}(f)$.

Proof Apply Lemma 3.7.2 to the short exact sequence $0 \to Y \to \mathrm{Cone}(f) \to X[1] \to 0$. This gives the quasi-isomorphism $(0, \beta(f))$ and the following commutative diagram with exact rows:

$$\begin{tikzcd}
0 \arrow[r] & Y \arrow[r, "\alpha(f)"] \arrow[d, equal] & \mathrm{Cone}(f) \arrow[r, "\alpha(\alpha(f))"] \arrow[d, equal] & \mathrm{Cone}(\alpha(f)) \arrow[r] \arrow[d, "\Phi=(0,\beta(f))"] & 0 \\
0 \arrow[r] & Y \arrow[r, "\alpha(f)"'] & \mathrm{Cone}(f) \arrow[r, "\beta(f)"'] & X[1] \arrow[r] & 0
\end{tikzcd}$$

The morphism $\psi : \mathrm{Cone}(\alpha(f)) \to X[1]$ discussed in Lemma 3.3.9 is precisely $(0, \beta(f))$ here: its degree-n component is the projection from $Y^{n+1} \oplus \mathrm{Cone}(f)^n = Y^{n+1} \oplus X^{n+1} \oplus Y^n$ onto X^{n+1}. The commutative diagram in that lemma therefore gives

$$\text{in } \mathsf{K}(\mathcal{A}) \quad f[1] \circ (0, \beta(f)) + \beta(\alpha(f)) = 0. \tag{3.7.6}$$

Now recall that

$$\xi^n : \mathrm{H}^n(X[1]) \to \mathrm{H}^{n+1}(Y)$$

is the connecting morphism of the short exact sequence

$$0 \to Y \xrightarrow{\alpha(f)} \mathrm{Cone}(f) \xrightarrow{\beta(f)} X[1] \to 0,$$

and applying Proposition 3.7.3 to this sequence shows immediately that the composite

$$\mathrm{H}^n(\mathrm{Cone}(\alpha(f))) \xrightarrow[\sim]{\mathrm{H}^n((0,\beta(f)))} \mathrm{H}^n(X[1]) \xrightarrow{\xi^n} \mathrm{H}^{n+1}(Y)$$

equals $-\mathrm{H}^n(\beta(\alpha(f)))$. But (3.7.6) implies $\mathrm{H}^n(\beta(\alpha(f))) = -\mathrm{H}^n(f[1]) \circ \mathrm{H}^n((0,\beta(f)))$. Thus $\xi^n = \mathrm{H}^n(f[1]) = \mathrm{H}^{n+1}(f)$. $\square$

Corollary 3.7.6 With the notation above, f is a quasi-isomorphism if and only if $\mathrm{H}^n(\mathrm{Cone}(f)) = 0$ for every $n \in \mathbb{Z}$.

Proof Substitute Corollary 3.7.5 into the long exact sequence (3.7.1); the conclusion follows at once. $\square$

3.8 Exercises: Hochschild Homology and Cohomology

This section returns to concrete mathematics. Let $\Bbbk$ be a commutative ring. Tensor products $\underset{\Bbbk}{\otimes}$ of $\Bbbk$-modules will be abbreviated to $\otimes$, the n-th tensor power of a $\Bbbk$-module M will be written $M^{\otimes n}$, and we set $M^{\otimes 0} := \Bbbk$. For a $\Bbbk$-algebra R, as usual, the left and right actions of $\Bbbk$ on every (R,R)-bimodule are assumed to agree. Thus an (R,R)-bimodule is also identified with a left $R \otimes R^{\mathrm{op}}$-module.

Definition 3.8.1 (Bar complex) Let R be a $\Bbbk$-algebra. For every $n \in \mathbb{Z}_{\geq 0}$, define the (R,R)-bimodule

$$\mathsf{B}_n R := R \otimes R^{\otimes n} \otimes R = R^{\otimes(n+2)},$$
$$r(r_0 \otimes \cdots \otimes r_{n+1})r' = rr_0 \otimes \cdots \otimes r_{n+1}r',$$

where $r, r', r_0, \ldots, r_{n+1} \in R$. For every $n \geq 1$, define the bimodule homomorphism

$$b_n : \mathsf{B}_n R \to \mathsf{B}_{n-1} R,$$
$$b_n(r_0 \otimes \cdots \otimes r_{n+1}) := \sum_{k=0}^{n} (-1)^k \cdots \otimes r_k r_{k+1} \otimes \cdots .$$

We call $\mathsf{B}R := (\mathsf{B}_n R, b_n)_{n \geq 0}$ the **bar complex** of R; it is a chain complex in the sense of Remark 2.2.6.

The classical notation for $r_0 \otimes \cdots \otimes r_{n+1} \in \mathsf{B}_n R$ is $(r_0 | \cdots | r_{n+1})$; this is the origin of the name "bar." The homomorphism b acts by deleting one separator in every possible way and then summing the results with alternating signs. From this it is easy to see that $b^2 = 0$. Indeed, $b^2(r_0|\cdots|r_{n+1})$ is a linear combination of elements of the form

$$(\cdots|r_{h-1}r_h|\cdots|r_{k-1}r_k|\cdots) \quad \text{or} \quad (\cdots|r_{h-1}r_hr_{h+1}|\cdots).$$

There are exactly two ways to obtain any such term from $(\cdots|r_{h-1}|\cdots|r_k|\cdots)$ by deleting separators, and their signs cancel. Thus $\mathsf{B}R$ is indeed a chain complex.

Now augment the bar complex as in the following diagram and denote the result by $\mathsf{B}'R$. The morphism b_0 in the diagram is also called the augmentation homomorphism.

$$\begin{array}{ccccccccc} \cdots & \xrightarrow{b_2} & \mathsf{B}_1R & \xrightarrow{b_1} & \mathsf{B}_0R & \xrightarrow{b_0} & (\mathsf{B}_{-1}R := R) & \longrightarrow & 0 \\ & & & & \uplus & & \uplus & & \\ & & & & (r_0|r_1) & \longmapsto & r_0r_1 & & \end{array}$$

Lemma 3.8.2 The augmented chain complex $\mathsf{B}'R$ is exact. More precisely, if $\mathsf{B}'R$ is regarded as a chain complex of $\Bbbk$-modules, then $\mathrm{id}_{\mathsf{B}'R}$ is null-homotopic.

Proof We construct a family of $\Bbbk$-module homomorphisms $h_n : \mathsf{B}'_nR \to \mathsf{B}'_{n+1}R$, for $n \geq -1$, such that $b_{n+1}h_n + h_{n-1}b_n = \mathrm{id}_{\mathsf{B}'_nR}$ (with, of course, the conventions $h_{-2} = 0$ and $b_{-1} = 0$). This will prove that $\mathrm{id}_{\mathsf{B}'R}$ is null-homotopic. Concretely, take

$$h_n(r_0|\cdots|r_{n+1}) = (1_R|r_0|\cdots|r_{n+1}),$$

and extend linearly on B'_nR. For every $n \geq 0$ and $r_0, \ldots, r_{n+1} \in R$, we obtain

$$\begin{aligned} b_{n+1}h_n\,(r_0|\cdots|r_{n+1}) &= (r_0|\cdots|r_{n+1}) + \sum_{k=0}^{n}(-1)^{k+1}(1_R|\cdots|r_kr_{k+1}|\cdots), \\ h_{n-1}b_n\,(r_0|\cdots|r_{n+1}) &= \sum_{k=0}^{n}(-1)^{k}(1_R|\cdots|r_kr_{k+1}|\cdots). \end{aligned}$$

It follows immediately that

$$(b_{n+1}h_n + h_{n-1}b_n)(r_0|\cdots|r_{n+1}) = (r_0|\cdots|r_{n+1}).$$

Since $b_0h_{-1}(r) = b_0(1_R|r) = r$, the equality also holds directly for $n = -1$. $\square$

At first sight, the definitions of $\mathsf{B}R$ and $\mathsf{B}'R$ may seem like unexpected inventions. Example 8.7.7 will later view these chain complexes from the perspective of comonads and give a natural explanation.

Convention 3.8.3 Define $R^e := R \otimes R^{\mathrm{op}}$. For every (R, R)-bimodule M, including the special case $M = R$, henceforth:

- ⋄ regard M as a left R^e-module through $(r \otimes r')m = rmr'$;

◇ regard M as a right R^e-module through $m(r\otimes r') := r'mr$.

With this convention, for every M we can define the chain complex of $\Bbbk$-modules $M \underset{R^e}{\otimes} \mathsf{B}R$ and the complex $\mathrm{Hom}_{R^e}(\mathsf{B}R, M)$.

Definition 3.8.4 (G. Hochschild) For every n, define the $\Bbbk$-linear functors $\mathrm{HH}_n, \mathrm{HH}^n : (R,R)\text{-}\mathsf{Mod} \rightrightarrows \Bbbk\text{-}\mathsf{Mod}$ as follows:

$$\begin{aligned}&\text{Hochschild homology} && \mathrm{HH}_n(M) := \mathrm{H}_n\left(M \underset{R^e}{\otimes} \mathsf{B}R\right),\\ &\text{Hochschild cohomology} && \mathrm{HH}^n(M) := \mathrm{H}^n\left(\mathrm{Hom}_{R^e}(\mathsf{B}R, M)\right).\end{aligned}$$

In the special case $M = R$, these objects are called respectively the **Hochschild homology** $\mathrm{HH}_n(R)$ and **Hochschild cohomology** $\mathrm{HH}^n(R)$ of R.

To simplify the descriptions of $\mathrm{HH}_n(M)$ and $\mathrm{HH}^n(M)$, introduce the following two Hochschild complexes:

$$\begin{aligned}C_\bullet(R,M) &:= \left[\cdots \to M\otimes R^{\otimes n} \xrightarrow{d_n} \cdots \xrightarrow{d_2} M\otimes R \xrightarrow{d_1} M\right],\\ C^\bullet(R,M) &:= \left[M \xrightarrow{d^0} \mathrm{Hom}_\Bbbk(R,M) \xrightarrow{d^1} \cdots \to \mathrm{Hom}_\Bbbk(R^{\otimes n}, M) \xrightarrow{d^n} \cdots\right].\end{aligned} \tag{3.8.1}$$

Their degrees are respectively $\ldots, 2, 1, 0$ and $0, 1, 2, \ldots$; all other terms are set equal to 0. The homomorphisms d_n and d^n are defined as follows.

1. In bar notation, continue to write an element $m\otimes r_1\otimes\cdots\otimes r_n$ of $M\otimes R^{\otimes n}$ as $(m|r_1|\cdots|r_n)$. Define

$$d_n(m|r_1|\cdots|r_n) = \underbrace{(mr_1|r_2|\cdots|r_n) + \sum_{k=1}^{n-1}(-1)^k(m|\cdots|r_kr_{k+1}|\cdots)}_{=:d_n'(m|r_1|\cdots|r_n)} + (-1)^n(r_nm|r_1|\cdots|r_{n-1}). \tag{3.8.2}$$

2. Identify an element of $\mathrm{Hom}_\Bbbk(R^{\otimes n}, M)$ with a $\Bbbk$-multilinear map $R^n \to M$ (see [51, §7.5]). Define

$$(d^nf)(r_1,\ldots,r_{n+1}) = r_1f(r_2,\ldots,r_{n+1}) + \sum_{k=1}^{n}(-1)^kf(\ldots,r_kr_{k+1},\ldots) + (-1)^{n+1}f(r_1,\ldots,r_n)r_{n+1}.$$

For every $n \in \mathbb{Z}_{\geq 0}$, there are isomorphisms

$$\begin{array}{ccc} M \underset{R^e}{\otimes} \mathsf{B}_n R & \xleftarrow{\quad\sim\quad} & M \otimes R^{\otimes n} \\ m \otimes (r_0 \otimes \cdots \otimes r_{n+1}) & \longmapsto & (r_{n+1} m r_0 | r_1 | \cdots | r_n) \\ m \otimes (1_R \otimes r_1 \otimes \cdots \otimes r_n \otimes 1_R) & \longleftarrow\!\shortmid & (m | r_1 | \cdots | r_n) \end{array}$$

[17] and

$$\begin{array}{ccc} \operatorname{Hom}_{R^e}(\mathsf{B}_n R, M) & \xleftarrow{\quad\sim\quad} & \operatorname{Hom}_{\Bbbk}(R^{\otimes n}, M) \\ \varphi & \longmapsto & [(r_1, \ldots, r_n) \mapsto \varphi(1_R, r_1, \ldots, r_n, 1_R)] \\ [r_0 \otimes \cdots \otimes r_{n+1} \mapsto r_0 f(r_1, \ldots, r_n) r_{n+1}] & \longleftarrow\!\shortmid & f. \end{array}$$

A short calculation shows that, through these isomorphisms, $\mathrm{id}_M \otimes b_n$ corresponds to d_n, while b_{n+1}^* corresponds to d^n.[18] Thus the isomorphisms above are isomorphisms of complexes and give the following result. To display the parameter R explicitly once again, write $\mathrm{HH}_n(R, M) := \mathrm{HH}_n(M)$ and $\mathrm{HH}^n(R, M) := \mathrm{HH}^n(M)$. [19] With this notation,

$$\mathrm{HH}^n(R, M) \simeq \mathrm{H}^n(C^\bullet(R, M)), \quad \mathrm{HH}_n(R, M) \simeq \mathrm{H}_n(C_\bullet(R, M)).$$

Readers new to the subject are encouraged to check the calculation in detail.

Remark 3.8.5 Suppose that R is commutative. Every R-module M becomes an (R, R)-bimodule by $rmr' := rr'm$. In this case, the complex $C_\bullet(R, M)$ (or $C^\bullet(R, M)$)[20] is a complex of R-modules if we set

$$r \cdot (m | r_1 | \cdots | r_n) = (rm | r_1 | \cdots | r_n) \quad \text{or} \quad (r \cdot f)(r_1, \ldots, r_n) = r f(r_1, \ldots, r_n).$$

Consequently, $\mathrm{HH}_n(M)$ and $\mathrm{HH}^n(M)$ both acquire R-module structures. This can of course also be proved at the level of the bar complex.

Example 3.8.6 Take $M = R = \Bbbk$. It follows immediately from the construction above that $d_1, d_2, \ldots$ are successively $0, \mathrm{id}, 0, \mathrm{id}, \ldots$. Thus $\mathrm{HH}_0(\Bbbk) = \Bbbk$ and $\mathrm{HH}_{\geq 1}(\Bbbk) = 0$. A similar argument gives $\mathrm{HH}^0(\Bbbk) = \Bbbk$ and $\mathrm{HH}^{\geq 1}(\Bbbk) = 0$.

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[17] Translator's note (O014-C044): the source appends a final $|1_R$ to the image in the middle row, even though the codomain is $M \otimes R^{\otimes n}$; that surplus component is removed here.

[18] Translator's note (O014-C045): the source writes b_n^*, which starts at $\operatorname{Hom}_{R^e}(\mathsf{B}_{n-1} R, M)$; the degree-$n$ differential comes from $b_{n+1} : \mathsf{B}_{n+1} R \to \mathsf{B}_n R$.

[19] Translator's note (O014-C046): in the formula below, the source switches to two-argument notation without defining it; these two abbreviations make that transition explicit.

[20] Translator's note (O014-C047): the source reverses the two arguments as $C_\bullet(M, R)$ and $C^\bullet(M, R)$ in this sentence; the notation here is aligned with the preceding definitions.

In higher degrees n, computing $\mathrm{HH}^n(M)$ and $\mathrm{HH}_n(M)$ from the original definition is generally difficult. Example 3.14.11 and 3.14.6 will later show respectively that

$$\mathrm{HH}_n(M) \simeq \mathrm{Tor}_n^{R^e}(M, R), \quad \mathrm{HH}^n(M) \simeq \mathrm{Ext}^n_{R^e}(R, M),$$

provided that R, as a $\Bbbk$-module, is respectively flat and projective. At that point, some special cases of HH_n and HH^n can be computed with simpler complexes. For general R, in the exercises of 8 we will explain HH_n and HH^n as relative Tor and relative Ext.

Example 3.8.7 (Degree zero: center and cocenter) First consider $\mathrm{HH}^0(M)$. By definition, $d^0 : M = C^0(R, M) \to C^1(R, M) = \mathrm{Hom}_\Bbbk(R, M)$ maps m to $[r \mapsto rm - mr]$. Therefore

$$\mathrm{HH}^0(M) = \{m \in M : \forall r \in R,\ rm = mr\}.$$

The right-hand side may naturally be called the **center** of M; when $M = R$, this is precisely the center in the sense of ring theory.

Next consider $\mathrm{HH}_0(M)$. Write $[M, R]$ for the $\Bbbk$-submodule of M generated by all elements of the form $mr - rm$, with $m \in M$ and $r \in R$. Since $d_1(m|r) = mr - rm$, we obtain

$$\mathrm{im}\,[d_1 : M \otimes R \to M] = [M, R], \quad \mathrm{HH}_0(M) = M/[M, R].$$

In particular, the case $M = R$ gives the submodule $[R, R]$ of R generated by all $[r', r] := r'r - rr'$. The corresponding $\Bbbk$-module $R/[R, R]$ is called the **cocenter** of R. Every $\Bbbk$-module homomorphism $\varphi : R \to N$ satisfying $\varphi(rr') = \varphi(r'r)$, that is, having a "trace-like" property, factors uniquely through $R/[R, R] = \mathrm{HH}_0(R)$.

Example 3.8.8 (Degree one: derivations) Now consider $\mathrm{HH}^1(M)$. Write $[r, m] := rm - mr$. Then

$$\begin{aligned} \ker(d^1) &= \{D \in \mathrm{Hom}_\Bbbk(R, M) : \forall r_1, r_2 \in R,\ r_1 D(r_2) - D(r_1 r_2) + D(r_1) r_2 = 0\}, \\ \mathrm{im}(d^0) &= \{[\cdot, m] \in \mathrm{Hom}_\Bbbk(R, M) : m \in M\}. \end{aligned}$$

The condition defining $\ker(d^1)$ may be rewritten as the Leibniz rule $D(r_1 r_2) = r_1 D(r_2) + D(r_1) r_2$. A $\Bbbk$-module homomorphism D with this property is regarded as a derivation on R with values in M. The $\Bbbk$-module formed by these homomorphisms is denoted by $\mathrm{Der}_\Bbbk(R, M)$, while the elements of the form $[\cdot, m]$ form the submodule $\mathrm{Inn}_\Bbbk(R, M)$. Thus the quotient module

$$\mathrm{HH}^1(M) = \mathrm{Der}_\Bbbk(R, M)/\mathrm{Inn}_\Bbbk(R, M)$$

classifies all derivations on R with values in M, modulo $\mathrm{Inn}_\Bbbk(R, M)$.

Now suppose that R is commutative, in order to interpret $\mathrm{HH}_1(M)$. We need a little preparation. Define the free R-module $\bigoplus_{r\in R} R\widetilde{\mathrm{d}r}$ with basis the symbols $\widetilde{\mathrm{d}r}$, and define the submodule N generated by the following elements:

$$\widetilde{\mathrm{d}(r+r')} - \widetilde{\mathrm{d}r} - \widetilde{\mathrm{d}r'}, \quad \widetilde{\mathrm{d}tr} - t\widetilde{\mathrm{d}r}, \quad \widetilde{\mathrm{d}rr'} - r\widetilde{\mathrm{d}r'} - r'\widetilde{\mathrm{d}r},$$

where $r, r' \in R$ and $t \in \Bbbk$. Thus define the R-module

$$\begin{aligned} \Omega_{R|\Bbbk} &:= \bigoplus_{r\in R} R\widetilde{\mathrm{d}r} \Big/ N \\ &= \sum_{r\in R} R\,\mathrm{d}r, \qquad \mathrm{d}r := \text{the image of } \widetilde{\mathrm{d}r}. \end{aligned}$$

This is called the **module of Kähler differentials** of the $\Bbbk$-algebra R and is characterized by the following universal property:

$$\begin{aligned} \mathrm{Hom}_R(\Omega_{R|\Bbbk}, M) &\xleftrightarrow{\sim} \mathrm{Der}_\Bbbk(R, M) \\ \varphi &\longmapsto [r \mapsto \varphi(\mathrm{d}r)] \qquad M : \text{an } R\text{-module.} \\ [\mathrm{d}r \mapsto D(r)] &\longleftarrow D. \end{aligned}$$

The reader is asked to verify this directly. This isomorphism shows that $R \to \Omega_{R|\Bbbk}$ given by $r \mapsto \mathrm{d}r$ (corresponding to $\varphi = \mathrm{id}$) is a "universal derivation." Topics related to this properly belong to the theory of commutative rings; we merely record the result here.

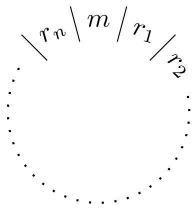
Return to $\mathrm{HH}_1(M)$, still assuming that R is commutative. Let M be an R-module made into an (R,R)-bimodule by $rmr' := (rr')m$. It is immediate that $d_1 : M\otimes R \to M$ is 0, while the image of $d_2 : M \otimes R^{\otimes 2} \to M \otimes R$ is generated by elements of the form $(rm|r') - (m|rr') + (r'm|r)$. We therefore obtain homomorphisms of $\Bbbk$-modules in both directions

$$\begin{aligned} (M\otimes R)/\operatorname{im}(d_2) &\longleftrightarrow M \underset{R}{\otimes} \Omega_{R|\Bbbk} \\ (m|r) + \operatorname{im}(d_2) &\longmapsto m \otimes \mathrm{d}r \\ (r'm|r) + \operatorname{im}(d_2) &\longleftarrow m \otimes r'\,\mathrm{d}r. \end{aligned}$$

From the descriptions of $\operatorname{im}(d_2)$ and $\Omega_{R|\Bbbk}$, a routine check shows that these homomorphisms are well defined and mutually inverse. In fact, both are isomorphisms of R-modules. Thus, when R is commutative, we obtain $\mathrm{HH}_1(M) \simeq M \underset{R}{\otimes} \Omega_{R|\Bbbk}$. In particular, $\mathrm{HH}_1(R) \simeq \Omega_{R|\Bbbk}$.

Hochschild homology and cohomology contain a wealth of information; their connection with derivations and differential forms is no accident. The exercises will give further interpretations of Hochschild cohomology.

Return to the Hochschild chain complex. Keeping the actions of R^e on $\mathsf{B}_n R$ and on M in mind, the intuitive idea is to arrange an element $(m|r_1|\cdots|r_n)$ of $C_n(R,M)$ in a circle:



Thus $d_n(m|r_1|\cdots|r_n)$ is obtained by deleting a separator in $n+1$ ways and summing the results with alternating signs. In the special case $M = R$, the picture has an evident rotational symmetry. Algebraically, for every $n \in \mathbb{Z}_{\geq 0}$ define the $\Bbbk$-module homomorphism

$$\begin{aligned} t_n : R^{\otimes(n+1)} &\longrightarrow R^{\otimes(n+1)} \\ (r_0|\cdots|r_n) &\longmapsto (-1)^n \cdot (r_n|r_0|\cdots|r_{n-1}); \end{aligned}$$

then $t_n^{n+1} = \mathrm{id}$. This homomorphism realizes the symmetry.

Rotational symmetry leads to the theory of **cyclic homology**. Historically, there have been at least two motivations for studying cyclic homology. The first is to seek an analogue of de Rham theory in the noncommutative setting. The second is to study and apply K-theory, including various generalizations of the index theorem. This section uses Hochschild homology and cohomology and cyclic homology only as a light vehicle for familiarizing the reader with operations on complexes; the treatment is merely introductory. Readers wishing to go further may consult monographs such as [29, 47].

All complexes (or double complexes) mentioned below are understood to be chain complexes (or chain double complexes).

For every $n \geq 0$, define $N_n := \mathrm{id} + t_n + \cdots + (t_n)^n \in \mathrm{End}_{\Bbbk}(R^{\otimes(n+1)})$. Define the **cyclic double complex** $\mathrm{CC}(R) = (\mathrm{CC}(R)_{p,q})_{(p,q)\in\mathbb{Z}^2}$ to be the following double complex.

$$\begin{tikzcd}
\vdots & & \vdots \arrow[d, "d_{q+1}"] & \vdots \arrow[d, "d'_{q+1}"] & \\
q & \cdots & R^{\otimes(q+1)} \arrow[l, "N_q"'] \arrow[d, "d_q"] & R^{\otimes(q+1)} \arrow[l, "\mathrm{id}-t_q"'] \arrow[d, "d'_q"] & \cdots \arrow[l] \\
q-1 & \cdots & R^{\otimes q} \arrow[l, "N_{q-1}"'] \arrow[d] & R^{\otimes q} \arrow[l, "\mathrm{id}-t_{q-1}"'] \arrow[d] & \cdots \arrow[l] \\
\vdots & & \vdots & \vdots & \\
 & \cdots & p : \text{even} & p+1 : \text{odd} & \cdots
\end{tikzcd}$$

All its terms are $\Bbbk$-modules, and all terms with $q < 0$ are defined to be 0. The morphisms d_q and d'_q are defined in (3.8.2). Notice that:

- ◇ the even columns are the Hochschild chain complex $C_\bullet(R, R)$ of (3.8.1);
- ◇ it is easy to prove that the odd columns are the augmented bar complex $\mathsf{B}'R$, with the indices shifted so that it begins in degree 0. Hence Lemma 3.8.2 implies that all odd columns are exact; in fact, they are 0 in $\mathsf{K}(\Bbbk\text{-}\mathsf{Mod})$.

To show that $\mathrm{CC}(R)$ is indeed a double complex, we need the following observation. Its proof is elementary and interesting, with no essential difficulty, so it is left as an exercise in this chapter.

Lemma 3.8.9 Let R be a $\Bbbk$-algebra. For every $q \in \mathbb{Z}_{\geq 1}$, the following equalities hold in $\mathrm{Hom}_\Bbbk\left(R^{\otimes(q+1)}, R^{\otimes q}\right)$:

$$d_q(\mathrm{id} - t_q) = (\mathrm{id} - t_{q-1})d'_q, \quad d'_q N_q = N_{q-1} d_q.$$

For every double complex $C = (C_{i,j})_{(i,j)\in\mathbb{Z}^2}$ and $m \in \mathbb{Z}$, define its horizontal shift[21] $C_{\mathrm{I}}[m]$ to be the double complex $(C_{m+i,j})_{i,j}$. For every $p \in \mathbb{Z}$, define the horizontal brutal truncation functor $\sigma_{\mathrm{I},\leq p}C$ (or $\sigma_{\mathrm{I},\geq p}C$) by replacing with 0 every $C_{i,j}$ for which $i > p$ (or $i < p$) and leaving all other terms unchanged. This gives a short exact sequence of double complexes (note the order):

$$0 \to \sigma_{\mathrm{I},\leq p}C \to C \to \sigma_{\mathrm{I},\geq p+1}C \to 0. \tag{3.8.3}$$

Moreover, for every $a \leq b$, define $\sigma_{\mathrm{I},[a,b]} := \sigma_{\mathrm{I},\leq b}\sigma_{\mathrm{I},\geq a} = \sigma_{\mathrm{I},\geq a}\sigma_{\mathrm{I},\leq b}$.

Apply these functors to $\mathrm{CC}(R)$ and then take the total complex tot_Π (Definition 3.5.4); this leads to the following definition.

[21] This book's convention uses the subscript I for horizontal operations and II for vertical operations.

Definition 3.8.10 (B. Feigin, B. Tsygan; A. Connes) For a $\Bbbk$-algebra R and every $n \in \mathbb{Z}$, define

$$\begin{aligned} \mathrm{HP}_n(R) &:= \mathrm{H}_n\left(\mathrm{tot}_\Pi(\mathrm{CC}(R))\right) && \text{(periodic cyclic homology)},\\ \mathrm{HC}_n(R) &:= \mathrm{H}_n\left(\mathrm{tot}\left(\sigma_{\mathrm{I},\geq 0}\mathrm{CC}(R)\right)\right) && \text{(cyclic homology)}. \end{aligned}$$

Both are functors $\Bbbk\text{-}\mathsf{Alg} \to \Bbbk\text{-}\mathsf{Mod}$.

Since $\sigma_{\mathrm{I},\geq 0}\mathrm{CC}(R)$ lies in the first quadrant, its total complex involves only the finite direct sums $\bigoplus_{\substack{p+q=n\\ p,q\geq 0}} \mathrm{CC}_{p,q}(R)$. The subscript on tot is therefore unnecessary.

The definition immediately gives $\mathrm{HC}_{<0}(R) = 0$, while $\mathrm{HC}_0(R)$ is the quotient of R by $\mathrm{im}[d_1 : R^{\otimes 2} \to R]$ and $\mathrm{im}[\mathrm{id} - t_0] = 0$, namely $R/[R,R]$. The exercises will give computations in more special cases.

The cyclic complex has the periodicity $\mathrm{CC}(R) = \mathrm{CC}(R)_{\mathrm{I}}[-2]$, which yields an isomorphism $\mathrm{HP}_n(R) \xrightarrow{\sim} \mathrm{HP}_{n-2}(R)$. For cyclic homology, the corresponding construction is the epimorphism "shift two columns to the left"

$$\begin{aligned} \sigma_{\mathrm{I},\geq 0}\mathrm{CC}(R) &\to \left(\sigma_{\mathrm{I},\geq 0}\mathrm{CC}(R)\right)_{\mathrm{I}}[-2]\\ \mathrm{CC}(R)_{p,q} &\to \begin{cases} \mathrm{CC}(R)_{p-2,q} \text{ (identity)}, & p \geq 2,\\ 0, & 0 \leq p < 2. \end{cases} \end{aligned}$$

Its kernel is given by columns 0 and 1 of $\mathrm{CC}(R)$, namely the sub-double-complex $\sigma_{\mathrm{I},[0,1]}\mathrm{CC}(R)$. This defines Connes's periodicity operator $S : \mathrm{HC}_n(R) \to \mathrm{HC}_{n-2}(R)$ on cyclic homology.

Theorem 3.8.11 (A. Connes) Let R be a $\Bbbk$-algebra. There is a canonical long exact sequence

$$\cdots \xrightarrow{S} \mathrm{HC}_{n-1}(R) \xrightarrow{B} \mathrm{HH}_n(R) \xrightarrow{I} \mathrm{HC}_n(R) \xrightarrow{S} \mathrm{HC}_{n-2}(R) \to \cdots,$$

where S is Connes's periodicity operator and I is induced by the inclusion

$$C_\bullet(R,R) \xrightarrow{\text{column } 0} \sigma_{\mathrm{I},\geq 0}\mathrm{CC}(R).$$

The map B is the corresponding connecting homomorphism.

Proof The first step is to apply (3.8.3) to obtain the short exact sequence of double complexes

$$0 \longrightarrow \text{column } 0 \longrightarrow \sigma_{\mathrm{I},[0,1]}\mathrm{CC}(R) \longrightarrow \text{column } 1 \longrightarrow 0.$$

Taking total complexes involves only finite direct sums. Checking degree by degree shows that the result remains a short exact sequence

$$\begin{tikzcd} 0 \arrow[r] & \text{column } 0 \arrow[r] & \operatorname{tot}\left(\sigma_{\mathrm{I},[0,1]}\mathrm{CC}(R)\right) \arrow[r] & \text{column } 1 \arrow[r] & 0. \\ & C_\bullet(R,R) \arrow[u, "\wr"] & & \mathsf{B}'R \arrow[u, "\wr"] & \end{tikzcd}$$

We already know that $\mathsf{B}'R$ is exact. Applying the long exact sequence of Proposition 3.6.4 shows that $C_\bullet(R,R) \to \operatorname{tot}\left(\sigma_{\mathrm{I},[0,1]}\mathrm{CC}(R)\right)$ is a quasi-isomorphism of complexes.

Next consider the short exact sequence mentioned when defining S:

$$0 \longrightarrow \sigma_{\mathrm{I},[0,1]}\mathrm{CC}(R) \longrightarrow \sigma_{\mathrm{I},\geq 0}\mathrm{CC}(R) \longrightarrow \left(\sigma_{\mathrm{I},\geq 0}\mathrm{CC}(R)\right)_{\mathrm{I}}[-2] \longrightarrow 0.$$

After taking total complexes, this remains a short exact sequence. By the preceding step, the degree-n homology groups of the three total complexes are identified respectively with $\mathrm{HH}_n(R)$, $\mathrm{HC}_n(R)$, and $\mathrm{HC}_{n-2}(R)$.[22] This proves the theorem. □

Remark 3.8.12 Define the cyclic double complex in the cochain sense, $\mathrm{CC}'(R)$, by $\mathrm{CC}'(R)^{p,q} := \operatorname{Hom}_{\Bbbk}\left(\mathrm{CC}(R)_{p,q}, \Bbbk\right)$. In analogy with Definition 3.8.10, define

$$\begin{aligned} \mathrm{HP}^n(R) &:= \mathrm{H}^n\left(\operatorname{tot}_{\oplus}(\mathrm{CC}'(R))\right) && \text{(periodic cyclic cohomology)}, \\ \mathrm{HC}^n(R) &:= \mathrm{H}^n\left(\operatorname{tot}\left(\sigma_{\mathrm{I}}^{\geq 0}\mathrm{CC}'(R)\right)\right) && \text{(cyclic cohomology)} \end{aligned}$$

and so forth, where $\sigma_{\mathrm{I}}^{\geq 0}$ still denotes the horizontal brutal-truncation functor.

Write $R^\vee := \operatorname{Hom}_{\Bbbk}(R, \Bbbk)$ and make it an (R,R)-bimodule by

$$(r\varphi r')(x) := \varphi(r'xr).$$

By currying, the $\Bbbk$-linear dual of the Hochschild column is identified with $C^\bullet(R, R^\vee)$; this identification requires no finiteness hypothesis. Consequently, the long exact sequence in Theorem 3.8.11 has a cohomological version

$$\cdots \xrightarrow{S} \mathrm{HC}^{n+1}(R) \xrightarrow{I} \mathrm{HH}^{n+1}(R, R^\vee) \xrightarrow{B} \mathrm{HC}^n(R) \xrightarrow{S} \mathrm{HC}^{n+2}(R) \to \cdots.$$

[23] The corresponding Connes periodicity operator S comes from the monomorphism "shift two columns to the right," namely

$$\sigma_{\mathrm{I}}^{\geq 0}\mathrm{CC}'(R) \to \left(\sigma_{\mathrm{I}}^{\geq 0}\mathrm{CC}'(R)\right)_{\mathrm{I}}[2].$$

[22] Translator's note (O014-C048): the source says "degree-n cohomology" in this sentence, but all three objects are chain complexes and the formulas use homological subscripts; "homology" is used here.

[23] Translator's note (O014-C049): the source prints $\mathrm{HH}^{n+1}(R) = \mathrm{HH}^{n+1}(R,R)$ as the middle term. However, the $\Bbbk$-linear dual of $C_\bullet(R,R)$ has coefficients in $R^\vee$, not in the regular bimodule R, unless an additional identification $R^\vee \simeq R$ is supplied.

As our toolkit grows, we will return repeatedly to Hochschild homology and cohomology and to cyclic homology and cohomology.

3.9 Truncation Functors

At the beginning of this section, let $\mathcal{A}$ be an additive category.

Definition 3.9.1 For a complex $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$, use the following terms and corresponding full subcategories:

Term	bounded	bounded below	bounded above
Condition	$\lvert n\rvert \gg 0 \implies X^n = 0$	$n \ll 0 \implies X^n = 0$	$n \gg 0 \implies X^n = 0$
Full subcategory	$\mathsf{C}^{\mathrm{b}}(\mathcal{A})$	$\mathsf{C}^{+}(\mathcal{A})$	$\mathsf{C}^{-}(\mathcal{A})$

More generally, for $-\infty \le s \le t \le +\infty$, let $\mathsf{C}^{[s,t]}(\mathcal{A})$ denote the full subcategory consisting of complexes satisfying $n \notin [s,t] \implies X^n = 0$, and define

$$\mathsf{C}^{\ge s}(\mathcal{A}) := \mathsf{C}^{[s,+\infty]}(\mathcal{A}), \quad \mathsf{C}^{\le t}(\mathcal{A}) := \mathsf{C}^{[-\infty,t]}(\mathcal{A}).$$

All these full subcategories are additive categories. If $\mathcal{A}$ is an abelian category, then $\mathsf{C}^{+}(\mathcal{A})$, $\mathsf{C}^{-}(\mathcal{A})$, and $\mathsf{C}^{\mathrm{b}}(\mathcal{A}) = \mathsf{C}^{+}(\mathcal{A}) \cap \mathsf{C}^{-}(\mathcal{A})$ are abelian subcategories.[24] The shift functor $[n]$ preserves $\mathsf{C}^{+}(\mathcal{A})$, $\mathsf{C}^{-}(\mathcal{A})$, and $\mathsf{C}^{\mathrm{b}}(\mathcal{A})$, but maps $\mathsf{C}^{[a,b]}(\mathcal{A})$ to $\mathsf{C}^{[a-n,b-n]}(\mathcal{A})$.

Moreover, $\mathcal{A}$ is naturally identified with $\mathsf{C}^{\ge 0}(\mathcal{A}) \cap \mathsf{C}^{\le 0}(\mathcal{A})$.

For every $\star \in \{+,-,\mathrm{b}\}$, the notion of homotopy between morphisms (Definition 3.2.6) restricts to $\mathsf{C}^{\star}(\mathcal{A})$. Thus Definition 3.2.8 has the following variant.

Definition 3.9.2 For $\star \in \{+,-,\mathrm{b}\}$, $\mathsf{K}(\mathcal{A})$ has a full additive subcategory $\mathsf{K}^{\star}(\mathcal{A})$ that is closed under the shift functors.

From now on, we require $\mathcal{A}$ to be an abelian category. We study how to truncate the part of a complex X in degrees $< n$ (or $> n$). The naive idea is to replace every term in that part by 0. Although simple, this disrupts cohomology, so we introduce a more refined version.

Definition 3.9.3 (Truncation functors) Let $\mathcal{A}$ be an abelian category and let $n \in \mathbb{Z}$.

[24] Translator's note (O014-C050): the source states the second assertion without requiring $\mathcal{A}$ to be abelian, although at this point $\mathcal{A}$ has only been assumed additive. The assertion is made conditional here.

For a complex X, define

$$\begin{tikzcd}
\tau^{\leq n}X \arrow[d, hook] & \cdots \arrow[r] & X^{n-2} \arrow[r] & X^{n-1} \arrow[r] & \ker(d^n) \arrow[r] \arrow[d, hook] & 0 \arrow[r] & 0 \arrow[r] & \cdots \\
\tilde{\tau}^{\leq n}X \arrow[d, hook] & \cdots \arrow[r] & X^{n-2} \arrow[r] & X^{n-1} \arrow[r] & X^n \arrow[r] & \operatorname{coim}(d^n) \arrow[r] \arrow[d, hook] & 0 \arrow[r] & \cdots \\
X \arrow[d, two heads] & \cdots \arrow[r] & X^{n-2} \arrow[r] & X^{n-1} \arrow[r] \arrow[d, two heads] & X^n \arrow[r] & X^{n+1} \arrow[r] & X^{n+2} \arrow[r] & \cdots \\
\tilde{\tau}^{\geq n}X \arrow[d, two heads] & \cdots \arrow[r] & 0 \arrow[r] & \operatorname{im}(d^{n-1}) \arrow[r] & X^n \arrow[r] \arrow[d, two heads] & X^{n+1} \arrow[r] & X^{n+2} \arrow[r] & \cdots \\
\tau^{\geq n}X & \cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \operatorname{coker}(d^{n-1}) \arrow[r] & X^{n+1} \arrow[r] & X^{n+2} \arrow[r] & \cdots
\end{tikzcd}$$

The omitted terms are clear, and in the vertical direction only morphisms other than id and 0 are marked. We thus obtain morphisms between the complexes listed on the left; in this notation, every $\operatorname{coim}(d^i)$ is identified with $\operatorname{im}(d^i)$ through the canonical comparison isomorphism.

It is clear that $\tau^{\leq n}$, $\tilde{\tau}^{\leq n}$, $\tau^{\geq n}$, and $\tilde{\tau}^{\geq n}$ are all additive functors. We have $\tau^{\leq n} = [-n] \circ \tau^{\leq 0} \circ [n]$; the same holds for $\tilde{\tau}^{\leq n}$, $\tau^{\geq n}$, and $\tilde{\tau}^{\geq n}$.

Moreover, if $m \leq n$, there are evident epimorphisms $\tau^{\geq m}X \twoheadrightarrow \tau^{\geq n}X$ and $\tilde{\tau}^{\geq m}X \twoheadrightarrow \tilde{\tau}^{\geq n}X$, and evident monomorphisms $\tau^{\leq m}X \hookrightarrow \tau^{\leq n}X$ and $\tilde{\tau}^{\leq m}X \hookrightarrow \tilde{\tau}^{\leq n}X$.

Lemma 3.9.4 For every $k \in \mathbb{Z}$, the comparison morphisms $\tau^{\leq n}X \to \tilde{\tau}^{\leq n}X$ and $\tilde{\tau}^{\geq n}X \to \tau^{\geq n}X$ in the diagram above induce the following isomorphisms in $\mathcal{A}$[25]

$$\mathrm{H}^k\left(\tau^{\leq n}X\right) \xrightarrow{\sim} \mathrm{H}^k\left(\tilde{\tau}^{\leq n}X\right) \simeq \begin{cases} \mathrm{H}^k(X), & k \leq n \\ 0, & k > n \end{cases},$$

$$\mathrm{H}^k\left(\tilde{\tau}^{\geq n}X\right) \xrightarrow{\sim} \mathrm{H}^k\left(\tau^{\geq n}X\right) \simeq \begin{cases} \mathrm{H}^k(X), & k \geq n \\ 0, & k < n \end{cases},$$

and the following short exact sequences in $\mathsf{C}(\mathcal{A})$:

$$0 \to \tilde{\tau}^{\leq n-1}X \to \tau^{\leq n}X \to \mathrm{H}^n(X)[-n] \to 0,$$
$$0 \to \mathrm{H}^n(X)[-n] \to \tau^{\geq n}X \to \tilde{\tau}^{\geq n+1}X \to 0,$$
$$0 \to \tau^{\leq n}X \to X \to \tilde{\tau}^{\geq n+1}X \to 0,$$
$$0 \to \tilde{\tau}^{\leq n-1}X \to X \to \tau^{\geq n}X \to 0,$$
$$0 \to \tau^{\leq n}X \to \tilde{\tau}^{\leq n}X \to \operatorname{Cone}\left(\mathrm{id}_{\operatorname{im}(d_X^n)[-n-1]}\right) \to 0.$$

[25] Translator's note (O014-C051): the source refers to "the morphisms above," which could be read as the transition morphisms for $m \leq n$ in the immediately preceding paragraph; only the two comparison morphisms displayed here induce these cohomology isomorphisms.

Here $\mathrm{H}^n(X)[-n]$ is understood according to Convention 3.1.7. All these morphisms are functorial in X.

Proof The degree-n component of the morphism $\tilde{\tau}^{\leq n-1}X \to \tau^{\leq n}X$ is $\mathrm{coim}(d^{n-1}) \xrightarrow{\sim} \mathrm{im}(d^{n-1}) \hookrightarrow \ker(d^n)$,[26] and every other component is id. The degree-n component of the morphism $\tau^{\leq n}X \to \mathrm{H}^n(X)[-n]$ is

$$\ker(d^n) \twoheadrightarrow \ker(d^n)/\operatorname{im}(d^{n-1}) = \mathrm{H}^n(X).$$

The degree-n component of the morphism $\tau^{\geq n}X \to \tilde{\tau}^{\geq n+1}X$ is the epimorphism induced by d^n, $\mathrm{coker}(d^{n-1}) \twoheadrightarrow \mathrm{im}(d^n)$, and every other component is id. The degree-n component of the morphism $\mathrm{H}^n(X)[-n] \to \tau^{\geq n}X$ is

$$\ker(d^n)/\operatorname{im}(d^{n-1}) \hookrightarrow X^n/\operatorname{im}(d^{n-1}) = \mathrm{coker}(d^{n-1}).$$

All remaining verifications are routine and are left to the reader as exercises. $\square$

Thus $\tau^{\leq n}$ and $\tilde{\tau}^{\leq n}$ (or $\tau^{\geq n}$ and $\tilde{\tau}^{\geq n}$) do indeed truncate cohomology in degrees $> n$ (or $< n$).

Remark 3.9.5 For a complex $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$, applying the functor σ of Definition–Proposition 3.4.1 gives

$$\sigma X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}^{\mathrm{op}})) = \mathrm{Ob}(\mathsf{C}(\mathcal{A}^{\mathrm{op}})^{\mathrm{op}}).$$

Ignoring some $\pm$ signs that have no effect in this context, this operation amounts to reversing the arrows in the complex and then reversing the degree indices. By the duality between ker and coker, under this operation $\tau^{\geq n}(X)$ corresponds to $\tau^{\leq -n}(\sigma X) \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}^{\mathrm{op}}))$, and so forth. Similarly, because im and coim are dual to each other, $\tilde{\tau}^{\geq n}(X) \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$ corresponds to $\tilde{\tau}^{\leq -n}(\sigma X) \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}^{\mathrm{op}}))$.

Compared with $\tilde{\tau}^{\leq n}$ and $\tilde{\tau}^{\geq n}$, the functors $\tau^{\leq n}$ and $\tau^{\geq n}$ have some properties that are easier to use. The first is adjunction.

Proposition 3.9.6 For every $n \in \mathbb{Z}$, there are the following adjunctions:

$$\begin{aligned} \mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X, Y) &\simeq \mathrm{Hom}_{\mathsf{C}^{\leq n}(\mathcal{A})}\left(X, \tau^{\leq n}Y\right), \quad X \in \mathrm{Ob}(\mathsf{C}^{\leq n}(\mathcal{A})) \\ \mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X, Y) &\simeq \mathrm{Hom}_{\mathsf{C}^{\geq n}(\mathcal{A})}\left(\tau^{\geq n}X, Y\right), \quad Y \in \mathrm{Ob}(\mathsf{C}^{\geq n}(\mathcal{A})). \end{aligned}$$

Proof This is clear from the definition of truncation. The reader may write down the corresponding unit and counit morphisms: they are the morphisms shown in Definition 3.9.3, or id. $\square$

[26] Translator's note (O014-C052): the source writes $\mathrm{im}(d^{n-1})$ directly as the degree-n term of $\tilde{\tau}^{\leq n-1}X$; that term is actually $\mathrm{coim}(d^{n-1})$, canonically identified with its image.

Next, $\tau^{\leq n}$ and $\tau^{\geq n}$ descend to $\mathsf{K}(\mathcal{A})$.

Definition–Proposition 3.9.7 For every $n \in \mathbb{Z}$, the functor $\tau^{\leq n}$ (or $\tau^{\geq n}$) naturally induces a functor from $\mathsf{K}(\mathcal{A})$ to itself, still denoted by $\tau^{\leq n}$ (or $\tau^{\geq n}$).

Proof Let $h \in \mathrm{Hom}^{-1}(X, Y)$. For the case of $\tau^{\leq n}$, define $\overline{h}^n := h^n \circ \left(\ker(d_X^n) \hookrightarrow X^n\right)$. Define $\overline{h} \in \mathrm{Hom}^{-1}\left(\tau^{\leq n}X, \tau^{\leq n}Y\right)$ by the following diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & X^{n-2} \arrow[r] \arrow[dl, "h^{n-2}"'] & X^{n-1} \arrow[r] \arrow[dl, "h^{n-1}"'] & \ker(d_X^n) \arrow[r] \arrow[dl, "\overline{h}^n"'] & 0 \arrow[r] \arrow[dl] & 0 \arrow[r] \arrow[dl] & \cdots \\
\cdots \arrow[r] & Y^{n-2} \arrow[r] & Y^{n-1} \arrow[r] & \ker(d_Y^n) \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots ;
\end{tikzcd}$$

It is easy to see that $\tau^{\leq n}\left(d^{-1}h\right) = d^{-1}\overline{h}$.

For the case of $\tau^{\geq n}$, instead define $\underline{h}^{n+1} := \left(Y^n \twoheadrightarrow \operatorname{coker}(d_Y^{n-1})\right) \circ h^{n+1}$ and use this to define $\underline{h} \in \mathrm{Hom}^{-1}\left(\tau^{\geq n}X, \tau^{\geq n}Y\right)$; the remainder is similar. The two cases are of course dual. $\square$

The following simple but important property follows immediately from Definition 3.9.3.

Proposition 3.9.8 For all integers $a < b$ and n, and $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$, we have[27)]

$$\tau^{\leq a}\tau^{\geq b}(X) = 0 = \tau^{\geq b}\tau^{\leq a}(X),$$
$$\tau^{\leq n}\tau^{\geq n}(X) \simeq \mathrm{H}^n(X)[-n] \simeq \tau^{\geq n}\tau^{\leq n}(X) \quad \text{(canonical isomorphisms)}.$$

3.10 Cohomology of double complexes

This section examines the cohomology of a double complex X in the horizontal and vertical directions, and its relationship with the cohomology of the total complex. The results will be used to study derived functors of bifunctors; the argument follows [23, §12.5].

Let $\mathcal{A}$ be an abelian category. As always, when $\mathcal{A}$ is a $\Bbbk$-linear abelian category, all the results have corresponding generalizations.

Recall that Definition 3.5.3 (or the more explicit diagram (3.5.1)) introduced a pair of invertible additive functors

$$\mathsf{C}^2(\mathcal{A}) \underset{F_{\mathrm{II}}}{\overset{F_{\mathrm{I}}}{\rightrightarrows}} \mathsf{C}(\mathsf{C}(\mathcal{A})).$$

27) Translator's note (O014-C053): the source writes $X \in \mathrm{Ob}(\mathcal{A})$, but the truncation functors and H^n in this proposition act on complexes.

Using them, define for every double complex X (with $p, q, n \in \mathbb{Z}$ below)

$$\mathrm{H}^p_{\mathrm{I}}(X) := (\mathrm{H}^p \circ F_{\mathrm{I}})(X) = \text{the complex} \left[\begin{array}{c} \vdots \\ {}^{\vartriangle}d \uparrow \\ \mathrm{H}^p(X^{\bullet,q+1}, {}^{\triangleright}d) \\ {}^{\vartriangle}d \uparrow \\ \mathrm{H}^p(X^{\bullet,q}, {}^{\triangleright}d) \\ {}^{\vartriangle}d \uparrow \\ \vdots \end{array} \right] \quad \text{(see Proposition 3.6.1)},$$

$$\begin{aligned} \mathrm{H}^q_{\mathrm{II}}(X) &:= (\mathrm{H}^q \circ F_{\mathrm{II}})(X) \\ &= \text{the complex} \left[\cdots \xrightarrow{{}^{\triangleright}d} \mathrm{H}^q(X^{p,\bullet}, {}^{\vartriangle}d) \xrightarrow{{}^{\triangleright}d} \mathrm{H}^q(X^{p+1,\bullet}, {}^{\vartriangle}d) \xrightarrow{{}^{\triangleright}d} \cdots \right], \\ \tau^{\leq n}_{\mathrm{I}} &:= (F_{\mathrm{I}})^{-1} \circ \tau^{\leq n} \circ F_{\mathrm{I}}, \\ \tau^{\leq n}_{\mathrm{II}} &:= (F_{\mathrm{II}})^{-1} \circ \tau^{\leq n} \circ F_{\mathrm{II}}, \end{aligned}$$

[28] There are likewise functors $\tilde{\tau}^{\leq n}_{\mathrm{I}}$, $\tau^{\geq n}_{\mathrm{I}}$, $\tilde{\tau}^{\geq n}_{\mathrm{I}}$ and $\tilde{\tau}^{\leq n}_{\mathrm{II}}$, $\tau^{\geq n}_{\mathrm{II}}$, $\tilde{\tau}^{\geq n}_{\mathrm{II}}$. Thus there are still morphisms of functors

$$\begin{aligned} \tau^{\leq n}_{\mathrm{I}} \to \tilde{\tau}^{\leq n}_{\mathrm{I}} \to \mathrm{id}_{\mathsf{C}^2(\mathcal{A})} \to \tilde{\tau}^{\geq n}_{\mathrm{I}} \to \tau^{\geq n}_{\mathrm{I}}, \\ \tau^{\leq n}_{\mathrm{II}} \to \tilde{\tau}^{\leq n}_{\mathrm{II}} \to \mathrm{id}_{\mathsf{C}^2(\mathcal{A})} \to \tilde{\tau}^{\geq n}_{\mathrm{II}} \to \tau^{\geq n}_{\mathrm{II}}. \end{aligned}$$

Definition 3.10.1 For a double complex X, write $\mathrm{Supp}(X) := \{(p,q) \in \mathbb{Z}^2 : X^{p,q} \neq 0\}$. Let $\mathsf{C}^2_f(\mathcal{A})$ be the full subcategory consisting of those objects X of $\mathsf{C}^2(\mathcal{A})$ for which the set $\{(p,q) \in \mathrm{Supp}(X) : p+q = n\}$ is finite for every $n \in \mathbb{Z}$.

Example 3.10.2 If $\mathrm{Supp}(X) \subset \mathbb{Z}^2_{\geq 0}$ (the first quadrant), if $\mathrm{Supp}(X) \subset \mathbb{Z}^2_{\leq 0}$ (the third quadrant), or if $\mathrm{Supp}(X)$ is the union of finitely many rows or columns, then X is an object of $\mathsf{C}^2_f(\mathcal{A})$.

The total complexes $\mathrm{tot}_{\oplus}(X)$ and $\mathrm{tot}_{\Pi}(X)$ are defined for every object X of $\mathsf{C}^2_f(\mathcal{A})$, with no additional assumption on $\mathcal{A}$. Moreover, $\mathrm{tot}_{\oplus}(X) = \mathrm{tot}_{\Pi}(X)$ in this case. We therefore obtain an additive functor $\mathrm{tot} : \mathsf{C}^2_f(\mathcal{A}) \to \mathsf{C}(\mathcal{A})$.

We now make a simple but necessary observation. Given an object X of $\mathsf{C}^2_f(\mathcal{A})$ and $n \in \mathbb{Z}$, there is a finite subset $S \subset \mathbb{Z}^2$ such that $\mathrm{H}^n(\mathrm{tot}(X))$ is completely determined by the data $(X^{p,q}, {}^{\triangleright}d^{p,q}, {}^{\vartriangle}d^{p,q})_{(p,q)\in S}$.

Lemma 3.10.3 The total-complex functor $\mathrm{tot} : \mathsf{C}^2_f(\mathcal{A}) \to \mathsf{C}(\mathcal{A})$ is exact (Definition 2.8.3).

[28] Translator's note (O014-C054): in the first two definitions the source prints only the composite functors $\mathrm{H}^p \circ F_{\mathrm{I}}$ and $\mathrm{H}^q \circ F_{\mathrm{II}}$ on the right, although the left-hand sides are their values on the double complex X. The argument X has been restored.

Proof Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be an exact sequence in $\mathsf{C}^2_f(\mathcal{A})$. Recalling the relevant definitions, such as Proposition 3.6.1, it suffices to show that for every $n \in \mathbb{Z}$ the sequence

$$\bigoplus_{p+q=n} X^{p,q} \xrightarrow{\bigoplus_{p+q=n} f^{p,q}} \bigoplus_{p+q=n} Y^{p,q} \xrightarrow{\bigoplus_{p+q=n} g^{p,q}} \bigoplus_{p+q=n} Z^{p,q}$$

is exact in $\mathcal{A}$. Since these direct sums are finite, this reduces to the exactness of $X^{p,q} \xrightarrow{f^{p,q}} Y^{p,q} \xrightarrow{g^{p,q}} Z^{p,q}$. □

Although the next lemma may look complicated, it is merely an exercise in the definitions and applies to every additive category $\mathcal{A}$.

Lemma 3.10.4 Let $h \in \mathbb{Z}$ and let Y be an object of $\mathsf{C}(\mathcal{A})$. Form

- ◇ the object $Y[-h]$ of $\mathsf{C}(\mathcal{A})$;
- ◇ the object $Y_{\mathrm{I}}[-h]$ of $\mathsf{C}(\mathsf{C}(\mathcal{A}))$, whose term in degree h is Y and whose other terms are 0.

There are then canonical isomorphisms in $\mathsf{C}(\mathcal{A})$

$$(\operatorname{tot} \circ F_{\mathrm{I}}^{-1}) \operatorname{Cone}\left(\mathrm{id}_{Y_{\mathrm{I}}[-h]}\right) \simeq \operatorname{Cone}\left(\mathrm{id}_{Y[-h]}\right),$$
$$(\operatorname{tot} \circ F_{\mathrm{I}}^{-1})\,(Y_{\mathrm{I}}[-h]) \simeq Y[-h].$$

Proof By definition, $\operatorname{Cone}\left(\mathrm{id}_{Y_{\mathrm{I}}[-h]}\right)$ is the object $Y \xrightarrow{\mathrm{id}_Y} Y$ of $\mathsf{C}^{[h-1,h]}(\mathsf{C}(\mathcal{A}))$, concentrated in degrees $h-1$ and h. Hence $Z := F_{\mathrm{I}}^{-1} \operatorname{Cone}\left(\mathrm{id}_{Y_{\mathrm{I}}[-h]}\right)$ is the double complex shown below:

$$\begin{array}{ccc}
n \in \mathbb{Z} & \vdots & \vdots \\
 & \uparrow{\scriptstyle d_Y} & \uparrow{\scriptstyle d_Y} \\
(q = n-h+1) & Y^{n-h+1} \xrightarrow{\ \mathrm{id}\ } & Y^{n-h+1} \\
 & \uparrow{\scriptstyle d_Y} & \uparrow{\scriptstyle d_Y} \\
(q = n-h) & Y^{n-h} \xrightarrow{\ \mathrm{id}\ } & Y^{n-h} \\
 & \vdots & \vdots \\
 & (p = h-1) & (p = h)
\end{array}$$

All columns corresponding to $p \notin \{h-1, h\}$ are zero. The definition of the total complex (Definition 3.5.4) now gives at once

$$\begin{aligned}
\operatorname{tot}(Z)^n &= Y^{n-h+1} \oplus Y^{n-h} = Y[-h]^{n+1} \oplus Y[-h]^n \\
&= \operatorname{Cone}\left(\mathrm{id}_{Y[-h]}\right)^n, \\
d^n_{\operatorname{tot}(Z)} &= \begin{pmatrix} (-1)^{h-1} d_Y^{n-h+1} & 0 \\ \mathrm{id}_{Y^{n-h+1}} & (-1)^h d_Y^{n-h} \end{pmatrix} = \begin{pmatrix} -d^{n+1}_{Y[-h]} & 0 \\ \mathrm{id}_{Y[-h]^{n+1}} & d^n_{Y[-h]} \end{pmatrix} \\
&= d^n_{\operatorname{Cone}\left(\mathrm{id}_{Y[-h]}\right)}.
\end{aligned}$$

The second isomorphism can be checked similarly, but more easily, and is left to the reader. □

Lemma 3.10.5 Let X be an object of $\mathsf{C}^2_f(\mathcal{A})$ and let $q \in \mathbb{Z}$.

(i) The natural morphism $\operatorname{tot}\left(\tau_{\mathrm{I}}^{\leq q} X\right) \to \operatorname{tot}\left(\tilde{\tau}_{\mathrm{I}}^{\leq q} X\right)$ is a quasi-isomorphism.

(ii) There is a canonical short exact sequence

$$0 \to \operatorname{tot}\left(\tilde{\tau}_{\mathrm{I}}^{\leq q-1}(X)\right) \to \operatorname{tot}\left(\tau_{\mathrm{I}}^{\leq q}(X)\right) \to \mathrm{H}^q_{\mathrm{I}}(X)[-q] \to 0.$$

Proof Apply Lemma 3.9.4 in $\mathsf{C}(\mathsf{C}(\mathcal{A}))$ to obtain the short exact sequences

$$\begin{gathered}
0 \to \tau^{\leq q} F_{\mathrm{I}}(X) \to \tilde{\tau}^{\leq q} F_{\mathrm{I}}(X) \to \operatorname{Cone}\left(\mathrm{id}_{\operatorname{im}\left(d^q_{F_{\mathrm{I}}X}\right)_{\mathrm{I}}[-q-1]}\right) \to 0, \\
0 \to \tilde{\tau}^{\leq q-1} F_{\mathrm{I}}(X) \to \tau^{\leq q} F_{\mathrm{I}}(X) \to \mathrm{H}^q_{\mathrm{I}}(X)_{\mathrm{I}}[-q] \to 0.
\end{gathered}$$

The notation $(\cdots)_{\mathrm{I}}[\cdots]$ has the meaning specified in Lemma 3.10.4. Apply $\operatorname{tot} \circ F_{\mathrm{I}}^{-1}$ to these short exact sequences. Lemma 3.10.3 says that tot is exact, and F_{I}^{-1} is of course exact as well. Substitution from Lemma 3.10.4 gives short exact sequences in $\mathsf{C}(\mathcal{A})$

$$\begin{gathered}
0 \to \operatorname{tot}\left(\tau_{\mathrm{I}}^{\leq q} X\right) \to \operatorname{tot}\left(\tilde{\tau}_{\mathrm{I}}^{\leq q} X\right) \to \operatorname{Cone}\left(\mathrm{id}_{\operatorname{im}(d^q_{F_{\mathrm{I}}X})[-q-1]}\right) \to 0, \\
0 \to \operatorname{tot}\left(\tilde{\tau}_{\mathrm{I}}^{\leq q-1} X\right) \to \operatorname{tot}\left(\tau_{\mathrm{I}}^{\leq q} X\right) \to \mathrm{H}^q_{\mathrm{I}}(X)[-q] \to 0.
\end{gathered}$$

The second is precisely (ii). Part (i) follows from the first together with Propositions 3.6.4 and 3.3.8 (i). □

For any object X of $\mathsf{C}^2(\mathcal{A})$, let $\mathrm{H}_{\mathrm{I}}(X)$ and $\mathrm{H}_{\mathrm{II}}(X)$ denote the following double complexes ($p, q \in \mathbb{Z}$):

$$\begin{aligned}
\left(\mathrm{H}_{\mathrm{I}}(X)^{p,\bullet}, {}^{\vartriangle} d^{p,\bullet}_{\mathrm{H}_{\mathrm{I}}(X)}\right) &:= \mathrm{H}^p_{\mathrm{I}}(X), \quad {}^{\vartriangleright} d^{\bullet,\bullet}_{\mathrm{H}_{\mathrm{I}}(X)} := 0, \\
\left(\mathrm{H}_{\mathrm{II}}(X)^{\bullet,q}, {}^{\vartriangleright} d^{\bullet,q}_{\mathrm{H}_{\mathrm{II}}(X)}\right) &:= \mathrm{H}^q_{\mathrm{II}}(X), \quad {}^{\vartriangle} d^{\bullet,\bullet}_{\mathrm{H}_{\mathrm{II}}(X)} := 0,
\end{aligned} \tag{3.10.1}$$

In other words, $\mathrm{H}_{\mathrm{I}}(X)$ (respectively $\mathrm{H}_{\mathrm{II}}(X)$) is obtained by taking the cohomology of X in the ${}^{\triangleright}d$-direction (respectively the ${}^{\vartriangle}d$-direction). Both are endofunctors of $\mathsf{C}^2(\mathcal{A})$. We may then define $\mathrm{H}_{\mathrm{II}}\,\mathrm{H}_{\mathrm{I}}(X)$ and $\mathrm{H}_{\mathrm{I}}\,\mathrm{H}_{\mathrm{II}}(X)$; both satisfy ${}^{\triangleright}d = 0 = {}^{\vartriangle}d$.

Theorem 3.10.6 Let $f : X \to Y$ be a morphism in $\mathsf{C}^2_f(\mathcal{A})$. If the induced morphism $\mathrm{H}_{\mathrm{II}}\,\mathrm{H}_{\mathrm{I}}(X) \to \mathrm{H}_{\mathrm{II}}\,\mathrm{H}_{\mathrm{I}}(Y)$ is an isomorphism, then $\mathrm{tot}(f) : \mathrm{tot}(X) \to \mathrm{tot}(Y)$ is a quasi-isomorphism.

The same conclusion holds if instead the induced morphism $\mathrm{H}_{\mathrm{I}}\,\mathrm{H}_{\mathrm{II}}(X) \to \mathrm{H}_{\mathrm{I}}\,\mathrm{H}_{\mathrm{II}}(Y)$ is assumed to be an isomorphism.

Proof The condition is equivalent to saying that the morphism $\mathrm{H}^n_{\mathrm{I}}(f) : \mathrm{H}^n_{\mathrm{I}}(X) \to \mathrm{H}^n_{\mathrm{I}}(Y)$ in $\mathsf{C}(\mathcal{A})$ is a quasi-isomorphism for every $n \in \mathbb{Z}$. The first step is to reduce to the case in which

$$q \ll 0 \implies \tilde{\tau}^{\leq q}_{\mathrm{I}}(X) = 0 = \tilde{\tau}^{\leq q}_{\mathrm{I}}(Y). \tag{3.10.2}$$

Indeed, let $r \in \mathbb{Z}$. Functorial truncation as in Lemma 3.9.4 shows that the original condition implies that $\mathrm{H}^n_{\mathrm{I}}(\tau^{\geq r}_{\mathrm{I}} f) : \mathrm{H}^n_{\mathrm{I}}(\tau^{\geq r}_{\mathrm{I}} X) \to \mathrm{H}^n_{\mathrm{I}}(\tau^{\geq r}_{\mathrm{I}} Y)$ is also a quasi-isomorphism for every n. For a fixed n, however, the definition of $\mathsf{C}^2_f(\mathcal{A})$ shows that when $r \ll 0$ there is a commutative diagram

$$\begin{tikzcd}
\mathrm{H}^n(\mathrm{tot}(X)) \arrow[r, "\mathrm{H}^n(\mathrm{tot}(f))"] \arrow[d, "\wr"'] & \mathrm{H}^n(\mathrm{tot}(Y)) \arrow[d, "\wr"] \\
\mathrm{H}^n\left(\mathrm{tot}(\tau^{\geq r}_{\mathrm{I}} X)\right) \arrow[r, "\mathrm{H}^n\left(\mathrm{tot}(\tau^{\geq r}_{\mathrm{I}} f)\right)"'] & \mathrm{H}^n\left(\mathrm{tot}(\tau^{\geq r}_{\mathrm{I}} Y)\right)
\end{tikzcd} \tag{3.10.3}$$

The double complexes in the second row satisfy (3.10.2). We may therefore replace f by $\tau^{\geq r}_{\mathrm{I}} f$ from now on and assume (3.10.2). Our goal is to prove that $\mathrm{H}^n(\mathrm{tot}(f))$ is an isomorphism for the chosen n.

For every $q \in \mathbb{Z}$, Lemma 3.10.5 (ii) gives a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{tot}\left(\tilde{\tau}^{\leq q-1}_{\mathrm{I}} X\right) \arrow[r] \arrow[d, "\alpha_{q-1}"] & \mathrm{tot}\left(\tau^{\leq q}_{\mathrm{I}} X\right) \arrow[r] \arrow[d, "\beta_q"] & \mathrm{H}^q_{\mathrm{I}}(X)[-q] \arrow[r] \arrow[d, "\mathrm{H}^q_{\mathrm{I}}(f)[-q]"] & 0 \\
0 \arrow[r] & \mathrm{tot}\left(\tilde{\tau}^{\leq q-1}_{\mathrm{I}} Y\right) \arrow[r] & \mathrm{tot}\left(\tau^{\leq q}_{\mathrm{I}} Y\right) \arrow[r] & \mathrm{H}^q_{\mathrm{I}}(Y)[-q] \arrow[r] & 0
\end{tikzcd}$$

[29] The right-hand vertical arrow is a quasi-isomorphism, and all the vertical arrows come from f. We shall prove that both α_q and β_q are quasi-isomorphisms. If α_{q-1} is a

[29] Translator's note (O014-C055): the source labels the right-hand vertical arrow only by $\simeq$. The theorem's hypothesis says that this morphism is the quasi-isomorphism $\mathrm{H}^q_{\mathrm{I}}(f)[-q]$, not necessarily an isomorphism of complexes; the morphism has therefore been named and its precise property stated.

quasi-isomorphism, then functoriality of the associated long exact sequences (Proposition 3.6.4), together with Proposition 2.3.4, implies that β_q is also a quasi-isomorphism. On the other hand, Lemma 3.10.5 (i) gives the commutative diagram

$$\begin{tikzcd}
\operatorname{tot}\left(\tau_{\mathrm{I}}^{\le q} X\right) \arrow[r, "\text{quasi-isomorphism}"] \arrow[d, "\beta_q"'] & \operatorname{tot}\left(\tilde{\tau}_{\mathrm{I}}^{\le q} X\right) \arrow[d, "\alpha_q"] \\
\operatorname{tot}\left(\tau_{\mathrm{I}}^{\le q} Y\right) \arrow[r, "\text{quasi-isomorphism}"'] & \operatorname{tot}\left(\tilde{\tau}_{\mathrm{I}}^{\le q} Y\right)
\end{tikzcd}$$

Thus, if β_q is a quasi-isomorphism, then so is α_q. Since α_q is simply $0 \xrightarrow{\sim} 0$ when $q \ll 0$, recursion shows that α_q and β_q are quasi-isomorphisms for all q.

Recall that $n \in \mathbb{Z}$ has been fixed. By an argument similar to that for (3.10.3), when $q \gg 0$ there is a commutative diagram

$$\begin{tikzcd}
\mathrm{H}^n(\operatorname{tot}(X)) \arrow[r, "\mathrm{H}^n(\operatorname{tot}(f))"] & \mathrm{H}^n(\operatorname{tot}(Y)) \\
\mathrm{H}^n\left(\operatorname{tot}(\tau_{\mathrm{I}}^{\le q} X)\right) \arrow[u, "\wr"] \arrow[r, "\sim", "\mathrm{H}^n(\beta_q)"'] & \mathrm{H}^n\left(\operatorname{tot}(\tau_{\mathrm{I}}^{\le q} Y)\right) \arrow[u, "\wr"']
\end{tikzcd}$$

Hence $\mathrm{H}^n(\operatorname{tot}(f))$ is indeed an isomorphism.

Finally, the same argument applies when $\mathrm{H_I}\,\mathrm{H_{II}}(X) \xrightarrow{\sim} \mathrm{H_I}\,\mathrm{H_{II}}(Y)$. Alternatively, one may use Proposition 3.5.6 to interchange $\mathrm{H_I}$ and $\mathrm{H_{II}}$. □

The proof above is rather involved. Example 5.6.3 will give another proof based on spectral sequences.

Corollary 3.10.7 Let X be an object of $\mathsf{C}^2_f(\mathcal{A})$. If X is row-exact, that is, if $(X^{\bullet,q}, {}^{\triangleright}d)$ is exact for every $q \in \mathbb{Z}$, then $\operatorname{tot}(X)$ is exact. Likewise, if X is column-exact, then $\operatorname{tot}(X)$ is exact.

Proof If X is row-exact, then $\mathrm{H_{II}}\,\mathrm{H_I}(X) = 0$. Theorem 3.10.6 therefore implies that $\operatorname{tot}(X) \to \operatorname{tot}(0) = 0$ is a quasi-isomorphism; in other words, $\operatorname{tot}(X)$ is exact.

The argument in the column-exact case is identical. Alternatively, the two cases can be deduced from one another by Proposition 3.5.6. □

3.11 Resolutions

Let $\mathcal{A}$ be an abelian category, and consider a complex $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$. If X^n is an injective object (respectively a projective object) of $\mathcal{A}$ for every $n \in \mathbb{Z}$, we say that X consists of injective objects (respectively projective objects). Recall the trivial fact that 0 is both injective and projective.

Definition 3.11.1 Let $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$.

- ◇ If $X \to I$ is a quasi-isomorphism in $\mathsf{C}(\mathcal{A})$, where $I \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$ consists of injective objects, then $X \to I$ is called an **injective resolution** of X.
- ◇ If $P \to X$ is a quasi-isomorphism in $\mathsf{C}(\mathcal{A})$, where $P \in \mathrm{Ob}(\mathsf{C}^-(\mathcal{A}))$ consists of projective objects, then $P \to X$ is called a **projective resolution** of X.

Replacing $\mathcal{A}$ by $\mathcal{A}^{\mathrm{op}}$, that is, reversing all arrows, shows that injective and projective resolutions are dual notions.

Example 3.11.2 As the most elementary and classical special case, consider an injective resolution $X \to I$ of $X \in \mathrm{Ob}(\mathcal{A})$. The definition requires only $I \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$, but we would like to take $I \in \mathrm{Ob}(\mathsf{C}^{\geq 0}(\mathcal{A}))$. This is possible: use the truncation

$$X \to I \to \tau^{\geq 0} I.$$

To see that the composite is an injective resolution, two properties must be checked. First, Lemma 3.9.4 shows that it remains a quasi-isomorphism. Second, $\tau^{\geq 0} I$ still consists of injective objects. Indeed, consider the short exact sequence $0 \to \mathrm{im}(d_I^{n-1}) \to I^n \to \mathrm{coker}(d_I^{n-1}) \to 0$. Since $d_I^{n-1} = 0$ when $n \ll 0$, while $\mathrm{coker}(d_I^{n-1}) \simeq \mathrm{im}(d_I^n)$ when $n < 0$, Lemmas 2.8.13 and 2.8.14 show recursively that for $n \leq 0$

$$I^n \simeq \mathrm{im}\left(d_I^{n-1}\right) \oplus \mathrm{coker}\left(d_I^{n-1}\right), \quad \text{and hence } \mathrm{coker}\left(d_I^{n-1}\right) \text{ is injective;}$$

the recursion starts at $n \ll 0$, where this formula becomes $0 = 0 \oplus 0$. Taking $n = 0$ shows that $\mathrm{coker}(d_I^{-1})$ is injective, so $\tau^{\geq 0} I$ consists of injective objects.

In practice one is interested in resolutions that extend sufficiently far. The observation above shows that we may replace $X \to I$ by $X \to \tau^{\geq 0} I$, so that the injective resolution has the form

$$\begin{array}{ccccccccccc}
\cdots 0 & \longrightarrow & 0 & \longrightarrow & X & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
\cdots 0 & \longrightarrow & 0 & \longrightarrow & I^0 & \longrightarrow & I^1 & \longrightarrow & I^2 & \longrightarrow & \cdots
\end{array}$$

This can also be flattened and viewed as the exact sequence in $\mathcal{A}$

$$0 \to X \to I^0 \to I^1 \to \cdots, \quad \text{every } I^n \text{ is injective, } n \geq 0.$$

Similarly, for a projective resolution $P \to X$, replace it by the composite $\tau^{\leq 0}P \to P \to X$. We may thus assume that it comes from the exact sequence in $\mathcal{A}$

$$\cdots \to P^{-1} \to P^0 \to X \to 0, \quad \text{every } P^n \text{ is projective, } n \leq 0.$$

This is usually written in chain-complex notation as $\cdots \to P_1 \to P_0 \to X \to 0$, where $P_n := P^{-n}$.

The question is whether injective or projective resolutions exist. By duality, it is enough to discuss the construction of injective resolutions first. The simplest case is again $X \in \mathrm{Ob}(\mathcal{A})$. Assume that $\mathcal{A}$ has enough injectives; see Definition 2.8.16. Choose a monomorphism $X \to I^0$ with I^0 injective. Suppose recursively that an exact sequence

$$0 \to X \to I^0 \to \cdots \to I^n, \quad n \geq 0,$$

has been constructed with $I^0, \dots, I^n$ injective. At the initial step $n = 0$, use the cokernel $\mathrm{coker}[X \to I^0]$; for $n \geq 1$, use $\mathrm{coker}[I^{n-1} \to I^n]$. In either case choose an injective object I^{n+1} and a monomorphism from that cokernel to I^{n+1}. Composition gives $I^n \to I^{n+1}$ such that $0 \to X \to \cdots \to I^{n+1}$ is still exact. Repeating the construction yields an injective resolution of X. [30)]

If $\mathcal{A}$ has enough projectives, the construction of a projective resolution $\cdots \to P^0 \to X \to 0$ for $X \in \mathrm{Ob}(\mathcal{A})$ is entirely dual. These constructions extend to complexes, subject to boundedness conditions, as follows.

Theorem 3.11.3 Let $\mathcal{A}^\flat$ be an additive full subcategory of $\mathcal{A}$, and suppose that for every object A of $\mathcal{A}$ there is a monomorphism $A \hookrightarrow B$ (respectively an epimorphism $B \twoheadrightarrow A$) with $B \in \mathrm{Ob}(\mathcal{A}^\flat)$. Then:

(i) For every object X of $\mathsf{C}^+(\mathcal{A})$ (respectively $\mathsf{C}^-(\mathcal{A})$), there is a quasi-isomorphism

$$f : X \to I \quad (\text{respectively } f : P \to X),$$

where f is a monomorphism (respectively an epimorphism), and $I \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}^\flat))$ (respectively $P \in \mathrm{Ob}(\mathsf{C}^-(\mathcal{A}^\flat))$).

(ii) More precisely, if $m \in \mathbb{Z}$ and $X \in \mathrm{Ob}(\mathsf{C}^{\geq m}(\mathcal{A}))$ (respectively $X \in \mathrm{Ob}(\mathsf{C}^{\leq m}(\mathcal{A}))$), then one may take $I \in \mathrm{Ob}(\mathsf{C}^{\geq m}(\mathcal{A}))$ (respectively $P \in \mathrm{Ob}(\mathsf{C}^{\leq m}(\mathcal{A}))$).

[30)] Translator's note (O014-C056): the source applies the formula $\mathrm{coker}[I^{n-1} \to I^n]$ also when $n = 0$, introducing an undefined I^{-1}. The initial step has been written with $\mathrm{coker}[X \to I^0]$, and the printed formula restricted to $n \geq 1$.

Consequently, if $\mathcal{A}$ has enough injectives (respectively enough projectives), then every such X has an injective resolution $X \hookrightarrow I$ (respectively a projective resolution $P \twoheadrightarrow X$).

Proof By duality, we treat only the version in which X belongs to $\mathsf{C}^+(\mathcal{A})$. For (i), suppose recursively that we have a commutative diagram

$$\begin{tikzcd}
\cdots \arrow[r] & X^{n-1} \arrow[r, "d_X^{n-1}"] \arrow[d, "f^{n-1}"'] & X^n \arrow[r, "d_X^n"] \arrow[d, "f^n"] & X^{n+1} \arrow[r] & \cdots \\
\cdots \arrow[r] & I^{n-1} \arrow[r, "d_I^{n-1}"'] & I^n & &
\end{tikzcd}$$

where the second row is a complex of objects of $\mathcal{A}^\flat$, $m \ll 0 \implies I^m = 0$, every f^m is a monomorphism for $m \leq n$, and for $m \leq n-1$ the morphism f^m induces an isomorphism $\ker(d_X^m)/\operatorname{im}(d_X^{m-1}) \xrightarrow{\sim} \ker(d_I^m)/\operatorname{im}(d_I^{m-1})$.

We wish to extend the second row to the right. First suppose that

$$f^n\left(\operatorname{im}\left(d_X^{n-1}\right)\right) = \operatorname{im}\left(d_I^{n-1}\right) \cap f^n\left(\ker\left(d_X^n\right)\right) = \operatorname{im}\left(d_I^{n-1}\right) \cap f^n(X^n); \tag{3.11.1}$$

note that the inclusions $\cdots \subset \cdots \subset \cdots$ always hold. The pushout construction gives the diagram

$$\begin{tikzcd}
X^n/\ker\left(d_X^n\right) \arrow[r, hook, "\delta\text{:induced by } d_X^n"] \arrow[d, "\alpha\text{:induced by } f^n"'] \arrow[dr, phantom, "\boxplus"] & X^{n+1} \arrow[d, "\beta"] \\
I^n/\left(f^n\left(\ker\left(d_X^n\right)\right) + \operatorname{im}\left(d_I^{n-1}\right)\right) \arrow[r, "\eta"'] & T
\end{tikzcd}$$

Since f^n is a monomorphism and (3.11.1) implies

$$\begin{aligned}
&\left(f^n\left(\ker\left(d_X^n\right)\right) + \operatorname{im}\left(d_I^{n-1}\right)\right) \cap f^n(X^n) \\
&\quad \overset{\text{Theorem 2.6.10}}{=\!=\!=} f^n\left(\ker\left(d_X^n\right)\right) + \left(\operatorname{im}\left(d_I^{n-1}\right) \cap f^n(X^n)\right) \\
&\qquad = f^n\left(\ker\left(d_X^n\right)\right) + f^n\left(\operatorname{im}\left(d_X^{n-1}\right)\right) = f^n\left(\ker\left(d_X^n\right)\right),
\end{aligned}$$

the basic operations discussed in §2.6 show that α is also a monomorphism.

Applying Proposition 2.1.6 twice shows that β and η in the pushout diagram are monomorphisms as well. Choose a monomorphism $T \hookrightarrow I^{n+1}$ with $I^{n+1} \in \operatorname{Ob}(\mathcal{A}^\flat)$. This gives the composite morphisms

$$\begin{gathered}
d_I^n : I^n \twoheadrightarrow I^n/\left(f^n\left(\ker\left(d_X^n\right)\right) + \operatorname{im}\left(d_I^{n-1}\right)\right) \xrightarrow{\eta} T \hookrightarrow I^{n+1}, \\
f^{n+1} : X^{n+1} \xrightarrow{\beta} T \hookrightarrow I^{n+1}.
\end{gathered}$$

Clearly $d_I^n d_I^{n-1} = 0$, f^{n+1} is a monomorphism, and $d_I^n f^n = f^{n+1} d_X^n$. On cohomology, the monomorphism f^n induces

$$\begin{aligned} \frac{\ker\left(d_X^n\right)}{\operatorname{im}\left(d_X^{n-1}\right)} \longrightarrow \frac{\ker\left(d_I^n\right)}{\operatorname{im}\left(d_I^{n-1}\right)} &= \frac{f^n\left(\ker d_X^n\right) + \operatorname{im}\left(d_I^{n-1}\right)}{\operatorname{im}\left(d_I^{n-1}\right)} \\ (\because \text{ Theorem 2.6.8 (iii)}) \quad &\simeq \frac{f^n\left(\ker d_X^n\right)}{f^n\left(\ker d_X^n\right) \cap \operatorname{im}\left(d_I^{n-1}\right)}; \end{aligned}$$

by (3.11.1) and the monicity of f^n, this is plainly an isomorphism. The extension to the right is complete.

We now explain how, in general, to modify I^n so as to reduce to the case where (3.11.1) holds. Consider the canonical morphisms

$$I^n \underset{p_1}{\overset{\iota_1}{\rightleftarrows}} I^n \oplus \left(X^n / \operatorname{im}\left(d_X^{n-1}\right)\right) \underset{p_2}{\overset{\iota_2}{\leftrightarrows}} X^n / \operatorname{im}\left(d_X^{n-1}\right) \overset{q}{\twoheadleftarrow} X^n.$$

Choose $J \in \mathrm{Ob}(\mathcal{A}^\flat)$ and a monomorphism $j : I^n \oplus (X^n / \operatorname{im}(d_X^{n-1})) \hookrightarrow J$. Define the composite morphisms

$$\begin{aligned} f' : X^n &\xrightarrow{(f^n, q)} I^n \oplus \left(X^n / \operatorname{im}\left(d_X^{n-1}\right)\right) \xrightarrow{j} J, \\ d' : I^{n-1} &\xrightarrow{d_I^{n-1}} I^n \xrightarrow{\iota_1} I^n \oplus \left(X^n / \operatorname{im}\left(d_X^{n-1}\right)\right) \xrightarrow{j} J. \end{aligned}$$

It is easy to see that f' is a monomorphism, $\ker(d') = \ker(d_I^{n-1})$, and $f' d_X^{n-1} = d' f^{n-1}$.

Moreover, working in $I^n \oplus (X^n / \operatorname{im}(d_X^{n-1}))$ shows that

$$f'(X^n) \cap \operatorname{im}(d') = f'\left(\operatorname{im}\left(d_X^{n-1}\right)\right).$$

Thus replacing $d_I^{n-1} : I^{n-1} \to I^n$ (respectively $f^n : X^n \to I^n$) by $d' : I^{n-1} \to J$ (respectively $f' : X^n \to J$) ensures (3.11.1).

Finally consider (ii). In the construction above set $\cdots = I^{m-2} = I^{m-1} = 0$. Equation (3.11.1) plainly holds for $n = m - 1$ (all its terms are zero), so extending the diagram to the right produces no nonzero terms in degrees below m. This proves the claim. □

Corollary 3.11.4 Suppose that $\mathcal{A}$ has enough injectives (respectively enough projectives). Then $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$ has an injective resolution $X \to I$ (respectively a projective resolution $P \to X$) if and only if $\mathrm{H}^n(X) = 0$ for $n \ll 0$ (respectively $n \gg 0$).

Proof Consider the injective case. The quasi-isomorphism condition in the definition of a resolution gives the "only if" direction. Conversely, for $m \ll 0$ the truncation functor gives a quasi-isomorphism $X \to \tau^{\geq m} X \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$, reducing the "if" direction to Proposition 3.11.3. □

The theoretical value of injective and projective resolutions is illustrated by the following fundamental result, which also implies their uniqueness up to homotopy.

Theorem 3.11.5 Fix a quasi-isomorphism $\alpha : X \to Y$ in $\mathsf{C}(\mathcal{A})$.

- ⋄ Given a morphism $\gamma : X \to I$, where I is an object of $\mathsf{C}^+(\mathcal{A})$ consisting of injective objects, there is a unique morphism β in $\mathsf{K}(\mathcal{A})$ making the diagram commute:

$$\begin{tikzcd} & Y \arrow[d, "\beta"] \\ X \arrow[ur, "\alpha"] \arrow[r, "\gamma"'] & I \end{tikzcd}$$

- ⋄ Given a morphism $\gamma : P \to Y$, where P is an object of $\mathsf{C}^-(\mathcal{A})$ consisting of projective objects, there is a unique morphism β in $\mathsf{K}(\mathcal{A})$ making the diagram commute:

$$\begin{tikzcd} & X \arrow[dl, "\alpha"'] \\ Y & P \arrow[l, "\gamma"] \arrow[u, "\beta"'] \end{tikzcd}$$

The two assertions are of course dual. A concise treatment will be given after Theorem 4.4.1. When α is a monomorphism (respectively an epimorphism), one can moreover choose β so that the first (respectively second) diagram commutes already in $\mathsf{C}(\mathcal{A})$; this is the content of Lemma 3.11.7 below and its dual. In fact, the exercises for this chapter give a direct proof of Theorem 3.11.5 based on that lemma, with a hint, and the reader is encouraged to try it.

Whichever approach one takes, the following property is indispensable.

Lemma 3.11.6 Let X be an acyclic object of $\mathsf{C}(\mathcal{A})$.

- ⋄ If an object I of $\mathsf{C}^+(\mathcal{A})$ consists of injective objects, then $\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X, I) = 0$.
- ⋄ If an object P of $\mathsf{C}^-(\mathcal{A})$ consists of projective objects, then $\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(P, X) = 0$.

Proof It is enough to treat the case of I. Fix $\alpha \in \mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X, I)$. We seek morphisms $h^n : X^n \to I^{n-1}$ such that

$$\alpha^n = d_I^{n-1} h^n + h^{n+1} d_X^n, \quad n \in \mathbb{Z}. \tag{3.11.2}$$

Suppose recursively that $\dots, h^{n-1}, h^n$ have been found so that (3.11.2) holds for $\dots, \alpha^{n-2}, \alpha^{n-1}$. This is legitimate because all these h's may be taken to be zero when

$n \ll 0$. We seek the dashed morphisms

$$\begin{tikzcd}
 & I^n & \\
X^n \arrow[ur, "\alpha^n - d_I^{n-1}h^n"] \arrow[r, two heads, "d_X^n"'] & \operatorname{im}(d_X^n) \arrow[u, dashed, "\beta"] \arrow[r, hook] & X^{n+1} \arrow[ul, dashed, "h^{n+1}"']
\end{tikzcd}$$

[31] that make the whole diagram commute. First,

$$\left(\alpha^n - d_I^{n-1}h^n\right) d_X^{n-1} = d_I^{n-1}\alpha^{n-1} - d_I^{n-1}\left(\alpha^{n-1} - d_I^{n-2}h^{n-1}\right) = 0.$$

Since X is acyclic, this induces the displayed morphism β. Since I^n is injective, β extends to a morphism $h^{n+1} : X^{n+1} \to I^n$, as required. $\square$

From the perspective of derived categories, Theorems 3.11.3 and 3.11.5 already provide enough information to begin studying derived functors. The kinds of resolutions introduced next are better suited for use with spectral sequences; see §5.6. We state the injective versions and first make some preparations.

Lemma 3.11.7 Let $\alpha : X \to Y$ be a quasi-isomorphism and a monomorphism in $\mathsf{C}(\mathcal{A})$. Given a morphism $\gamma : X \to I$ in $\mathsf{C}(\mathcal{A})$, where $I \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$ consists of injective objects, there is a morphism $\beta : Y \to I$ in $\mathsf{C}(\mathcal{A})$ such that $\gamma = \beta\alpha$.

Proof Using α^n, regard each X^n as a subobject of Y^n, and omit α^n from the notation. We construct β^n step by step, taking $\beta^n = 0$ when $n \ll 0$. Suppose suitable morphisms $\ldots, \beta^{n-2}, \beta^{n-1}$ have been found. The desired morphism $\beta^n : Y^n \to I^n$ can and must be defined on the subobjects X^n and $\operatorname{im}(d_Y^{n-1})$ as follows.

- ⋄ On X^n, the morphism β^n equals γ^n.
- ⋄ On $\operatorname{im}(d_Y^{n-1})$, it is induced by the composite $Y^{n-1} \xrightarrow{\beta^{n-1}} I^{n-1} \xrightarrow{d_I^{n-1}} I^n$. For this to make sense, we claim that $d_I^{n-1}\beta^{n-1}$ restricts to zero on $\ker(d_Y^{n-1})$. Consider the commutative diagram

$$\begin{tikzcd}
\operatorname{im}(d_X^{n-2}) \arrow[r] \arrow[d, hook] & \ker(d_X^{n-1}) \arrow[r] \arrow[d, hook] & X^{n-1} \arrow[d, hook] \arrow[dr, "\gamma^{n-1}"] & & \\
\operatorname{im}(d_Y^{n-2}) \arrow[r] & \ker(d_Y^{n-1}) \arrow[r] & Y^{n-1} \arrow[r, "\beta^{n-1}"'] & I^{n-1} \arrow[r, "d_I^{n-1}"'] & I^n.
\end{tikzcd}$$

The inductive hypothesis on $\ldots, \beta^{n-2}, \beta^{n-1}$ and the equality $d_I^{n-1}d_I^{n-2} = 0$ show that the composite along the second row is zero. Since γ is a morphism of

[31] Translator's note (O014-C057): the lower-left node is printed as X in the source, although d_X^n, α^n, and h^n all have domain X^n. The node has been restored to X^n.

complexes, the composite $\ker(d_X^{n-1}) \to \cdots \to I^n$ is also zero. We obtain a commutative diagram

$$\begin{tikzcd} \mathrm{H}^{n-1}(X) \arrow[r, "0"] \arrow[d] & I^n \\ \mathrm{H}^{n-1}(Y) \arrow[ur, "\text{induced by } d_I^{n-1}\beta^{n-1}"'] & \end{tikzcd}$$

But the quasi-isomorphism condition says $\mathrm{H}^{n-1}(X) \xrightarrow{\sim} \mathrm{H}^{n-1}(Y)$, proving the claim.

If we can prove the equality of subobjects

$$X^n \cap \mathrm{im}\left(d_Y^{n-1}\right) = \mathrm{im}\left(d_X^{n-1}\right),$$

then the pushout diagram of Proposition 2.6.7 and the universal property of the pushout extend the morphism to $X^n + \mathrm{im}(d_Y^{n-1})$; injectivity of I^n then extends it to $\beta^n : Y^n \to I^n$. The inclusion $\supset$ is clear.

The point is to show that any morphism $\phi : T \to Y^n$ in $\mathcal{A}$ which factors through $X^n \cap \mathrm{im}(d_Y^{n-1})$ automatically factors through $\mathrm{im}(d_X^{n-1})$. Such a ϕ induces a morphism $T \to \ker(d_X^n) = X^n \cap \ker(d_Y^n)$. Composing along the second row of the commutative diagram

$$\begin{tikzcd} T \arrow[r, "\phi"] \arrow[d, "\phi"'] & \ker(d_X^n) \arrow[r, two heads] \arrow[d, hook] & \mathrm{H}^n(X) \arrow[d] \\ \mathrm{im}(d_Y^{n-1}) \arrow[r, hook] & \ker(d_Y^n) \arrow[r, two heads] & \mathrm{H}^n(Y) \end{tikzcd}$$

gives zero. Since $\mathrm{H}^n(X) \xrightarrow{\sim} \mathrm{H}^n(Y)$, the composite along the first row is zero as well. Thus ϕ factors through $\mathrm{im}(d_X^{n-1})$.

By construction, $\beta = (\beta^n)_n$ is a morphism $Y \to I$, and $\gamma = \beta\alpha$ follows from the construction too. $\square$

Lemma 3.11.8 Let $A, C \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$. For every $n \in \mathbb{Z}$, define $B^n := A^n \oplus C^n$, with its evident morphisms $A^n \xrightarrow{i^n} B^n \xrightarrow{p^n} C^n$. There is a bijection

$$\mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(C, A[1]) \xrightarrow{1:1} \left\{ \text{families of morphisms } (d_B^n)_{n\in\mathbb{Z}} \;\middle|\; \begin{array}{l} \text{such that } (B^n, d_B^n)_n \text{ is a complex and} \\ 0 \to A \xrightarrow{(i^n)_n} B \xrightarrow{(p^n)_n} C \to 0 \text{ is exact} \end{array} \right\},$$

explicitly sending $\delta : C \to A[1]$ to the differential given in matrix form by

$$d_B^n = \begin{pmatrix} d_A^n & \delta^n \\ 0 & d_C^n \end{pmatrix} : B^n \to B^{n+1}.$$

Proof Once $(B^n, d_B^n)_n$ is known to be a complex and $(i^n)_n$ and $(p^n)_n$ are morphisms of complexes, exactness of $0 \to A \to B \to C \to 0$ follows by checking it termwise.

The equalities $d_B^n i^n = i^{n+1} d_A^n$ and $d_C^n p^n = p^{n+1} d_B^n$ hold for all n if and only if there are morphisms $\delta^n : C^n \to A^{n+1}$ for which d_B^n has the upper-triangular matrix in the statement. It remains to characterize the families $(\delta^n)_n$ for which $d_B^{n+1} d_B^n = 0$. A routine calculation gives the necessary and sufficient condition $d_A^{n+1}\delta^n + \delta^{n+1} d_C^n = 0$, equivalently $d_{A[1]}^n \delta^n = \delta^{n+1} d_C^n$. □

Proposition 3.11.9 (Horseshoe Lemma) Let $0 \to A \to B \to C \to 0$ be a short exact sequence in $\mathsf{C}(\mathcal{A})$, with $A, B, C \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$. Choose injective resolutions $\epsilon : A \to I$ and $\eta : C \to K$ with the properties stated in Theorem 3.11.3. These data extend to a commutative diagram with exact rows in $\mathsf{C}^+(\mathcal{A})$

$$\begin{tikzcd} 0 \arrow[r] & I \arrow[r] & J \arrow[r] & K \arrow[r] & 0 \\ 0 \arrow[r] & A \arrow[r] \arrow[u, "\epsilon"] & B \arrow[r] \arrow[u, "\kappa"] & C \arrow[r] \arrow[u, "\eta"'] & 0 \end{tikzcd}$$

such that $\kappa : B \to J$ is also an injective resolution and is a monomorphism.

The reader may wish to give a relatively simple proof of the special case $A, B, C \in \mathrm{Ob}(\mathcal{A})$.

Proof Construct the pushout diagram in $\mathsf{C}(\mathcal{A})$

$$\begin{tikzcd} 0 \arrow[r] & I \arrow[r] & L \arrow[r] & C \arrow[r] & 0 \\ 0 \arrow[r] & A \arrow[r] \arrow[u, "\epsilon"] \arrow[ur, phantom, "\boxplus"] & B \arrow[r] \arrow[u] & C \arrow[r] \arrow[u, "\mathrm{id}"'] & 0 \end{tikzcd}$$

Here we use two facts:

- ◇ a pushout preserves the cokernel C;
- ◇ Proposition 2.1.6 and monicity of $A \to B$ imply that $I \to L$ is also a monomorphism.

Thus the diagram still has exact rows, and $L \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$. Another application of Proposition 2.1.6, now using the monicity of ϵ, shows that $B \to L$ is a monomorphism too.

Because I^n is injective, Lemma 2.8.13 implies that $0 \to I^n \to L^n \to C^n \to 0$ splits for every n, giving an isomorphism $\Phi^n : L^n \xrightarrow{\sim} I^n \oplus C^n$. Lemma 3.11.8 then determines $\delta : C \to I[1]$ so that

$$\begin{tikzcd} L^n \arrow[r, "\Phi^n", "\sim"'] \arrow[d, "d_L^n"'] & I^n \oplus C^n \arrow[d, "e^n"] \\ L^{n+1} \arrow[r, "\sim", "\Phi^{n+1}"'] & I^{n+1} \oplus C^{n+1} \end{tikzcd} \quad \text{always commutes, where } e^n := \begin{pmatrix} d_I^n & \delta^n \\ 0 & d_C^n \end{pmatrix}. \tag{3.11.3}$$

Likewise, Lemma 3.11.8 constructs from any $\theta : K \to I[1]$ a complex J such that

$$J^n := I^n \oplus K^n, \quad d_J^n := \begin{pmatrix} d_I^n & \theta^n \\ 0 & d_K^n \end{pmatrix}, \quad 0 \to I \to J \to K \to 0 \text{ is exact.}$$

We want to choose θ so that the composite $L^n \xrightarrow[\sim]{\Phi^n} I^n \oplus C^n \xrightarrow{(\mathrm{id}, \eta^n)} J^n$ defines a morphism $L \to J$. If it does, then $L \to J$ is a monomorphism. Let κ be the composite $B \to L \to J$; it remains a monomorphism, and the diagram in the statement is readily checked to commute. Since ϵ and η are quasi-isomorphisms, functoriality of the long exact sequences in Proposition 3.6.4, together with the Five Lemma (Proposition 2.3.4), shows that κ is a quasi-isomorphism as well, and hence is the desired injective resolution.

How should θ be chosen? By (3.11.3), the displayed maps $L^n \to J^n$ form a morphism of complexes if and only if

$$\begin{pmatrix} \mathrm{id}_{I^{n+1}} & 0 \\ 0 & \eta^{n+1} \end{pmatrix} \begin{pmatrix} d_I^n & \delta^n \\ 0 & d_C^n \end{pmatrix} = \begin{pmatrix} d_I^n & \theta^n \\ 0 & d_K^n \end{pmatrix} \begin{pmatrix} \mathrm{id}_{I^n} & 0 \\ 0 & \eta^n \end{pmatrix}, \quad n \in \mathbb{Z};$$

[32] in other words, $\delta = \theta\eta$. Since the quasi-isomorphism η is a monomorphism in $\mathsf{C}(\mathcal{A})$ and $I[1]$ consists of injective objects, Lemma 3.11.7 gives such a $\theta : K \to I[1]$. □

To state the next result, for a given complex X and every $p \in \mathbb{Z}$ define

$$B^p := \mathrm{im}\left(d_X^{p-1}\right), \quad Z^p := \ker\left(d_X^p\right), \quad H^p := Z^p / B^p = \mathrm{H}^p(X),$$

thereby decomposing X into the short exact sequences

$$0 \to Z^p \to X^p \xrightarrow{d_X^p} B^{p+1} \to 0, \quad 0 \to B^p \to Z^p \to H^p \to 0.$$

The Cartan–Eilenberg resolution introduced next may be viewed as a simultaneous resolution of these short exact sequences.

Theorem 3.11.10 (H. Cartan, S. Eilenberg) Suppose that $\mathcal{A}$ has enough injectives. For every object X of $\mathsf{C}^+(\mathcal{A})$, there is a double complex I satisfying the following conditions, together with a morphism $\epsilon : X \to (I^{\bullet,0}, {}^{\triangleright}d^{\bullet,0})$ in $\mathsf{C}(\mathcal{A})$, where ${}^{\triangleright}d$ and ${}^{\triangle}d$ come from the double-complex structure on I (Definition 3.5.1):

(i) For all $(p, q) \in \mathbb{Z}^2$, one has $q < 0 \implies I^{p,q} = 0$.

[32] Translator's note (O014-C058): the lower-right entry of the left-hand inner matrix is printed as d_K^n in the source. For the composition to be well typed on $I^n \oplus C^n$ and yield the displayed equality, it has been restored to d_C^n.

(ii) Choose $N \in \mathbb{Z}$ such that $n < N \implies X^n = 0$. Then, for all $(p, q) \in \mathbb{Z}^2$, one has $p < N \implies I^{p,q} = 0$.

(iii) For every $p \in \mathbb{Z}$, there is an injective resolution of $X^p \in \mathrm{Ob}(\mathcal{A})$

$$0 \to X^p \xrightarrow{\epsilon^p} I^{p,0} \xrightarrow{{}^{\vartriangle}d^{p,0}} I^{p,1} \xrightarrow{{}^{\vartriangle}d^{p,1}} \cdots .$$

(iv) The preceding morphisms induce an injective resolution of $Z^p := \ker(d_X^p)$

$$0 \to \ker\left(d_X^p\right) \to \ker\left({}^{\triangleright}d^{p,0}\right) \to \ker\left({}^{\triangleright}d^{p,1}\right) \to \cdots .$$

(v) Likewise, $B^{p+1} := \mathrm{im}(d_X^p)$ has an injective resolution

$$0 \to \mathrm{im}\left(d_X^p\right) \to \mathrm{im}\left({}^{\triangleright}d^{p,0}\right) \to \mathrm{im}\left({}^{\triangleright}d^{p,1}\right) \to \cdots .$$

(vi) Likewise, $H^p := \mathrm{H}^p(X)$ has an injective resolution

$$0 \to \mathrm{H}^p\left(X\right) \to \underbrace{\mathrm{H}^p\left(I^{\bullet,0}, {}^{\triangleright}d\right) \to \mathrm{H}^p\left(I^{\bullet,1}, {}^{\triangleright}d\right) \to \cdots}_{=\mathrm{H}^p_{\mathrm{I}}(I),\text{ see §3.10}} .$$

Data (I, ϵ) with these properties are called a **Cartan–Eilenberg resolution** of X.

Proof Choose $N \in \mathbb{Z}$ such that $n < N \implies X^n = 0$. We have already defined the short exact sequences

$$\begin{array}{ll} 0 \to Z^N \to X^N \to B^{N+1} \to 0, & 0 \to B^{N+1} \to Z^{N+1} \to H^{N+1} \to 0, \\ 0 \to Z^{N+1} \to X^{N+1} \to B^{N+2} \to 0, & 0 \to B^{N+2} \to Z^{N+2} \to H^{N+2} \to 0, \\ & \vdots \end{array}$$

For every H^n (respectively B^n), choose an injective resolution as in Example 3.11.2; when $n < N$ (respectively $n \leq N$), take it to be $0 \to 0$.

- ◇ Note that $Z^N = H^N$. Proposition 3.11.9 extends the injective resolutions of Z^N and B^{N+1} simultaneously to an injective resolution $I^{N,\bullet}$ of X^N, compatible with the first short exact sequence.
- ◇ Next use Proposition 3.11.9 to extend the injective resolutions of B^{N+1} and H^{N+1} simultaneously to an injective resolution of Z^{N+1}, compatible with the second short exact sequence.
- ◇ Repeat the same operation to obtain an injective resolution $I^{N+1,\bullet}$ of X^{N+1} together with an injective resolution of Z^{N+2}, compatible with the third and fourth short exact sequences, respectively.

Continuing in this way gives, for every n, an injective resolution $X^n \to I^{n,0} \to I^{n,1} \to \cdots$; put $I^{n,\bullet} := 0$ when $n < N$. The construction also specifies maps $I^{n,m} \to I^{n+1,m}$ so as to form a double complex $(I, {}^{\triangleright}d, {}^{\vartriangle}d)$. One must check ${}^{\triangleright}d^2 = 0$, but there is no substantive difficulty. [33)] □

One may also regard ϵ as a morphism of double complexes $X \to I$, provided X is identified with the double complex concentrated in row 0 (the horizontal axis $q = 0$).

Remark 3.11.11 For all $p, q \in \mathbb{Z}$, consider the short exact sequences occurring in the Cartan–Eilenberg resolution

$$0 \to \ker\left({}^{\triangleright}d^{p,q}\right) \to I^{p,q} \xrightarrow{{}^{\triangleright}d^{p,q}} \operatorname{im}\left({}^{\triangleright}d^{p,q}\right) \to 0,$$
$$0 \to \operatorname{im}\left({}^{\triangleright}d^{p-1,q}\right) \to \ker\left({}^{\triangleright}d^{p,q}\right) \to \underbrace{\mathrm{H}^p\left(I^{\bullet,q}, {}^{\triangleright}d\right)}_{=:\mathrm{H}_{\mathrm{I}}(I)^{p,q}} \to 0,$$
$$0 \to \operatorname{im}\left({}^{\triangleright}d^{p-1,q}\right) \to I^{p,q} \to \operatorname{coker}\left({}^{\triangleright}d^{p-1,q}\right) \to 0.$$

Their leftmost terms are all injective, so Lemma 2.8.13 implies that they split. Recall that every additive functor $F : \mathcal{A} \to \mathcal{B}$ preserves split short exact sequences. For fixed q, the images of the displayed sequences under F give the corresponding decompositions of the complex $(F(I^{\bullet,q}), F({}^{\triangleright}d))$. In particular,

$$F\,\mathrm{H}^p\left(I^{\bullet,q}, {}^{\triangleright}d\right) \simeq \mathrm{H}^p\left(F(I^{\bullet,q}), F({}^{\triangleright}d)\right).$$

In other words, F preserves horizontal cohomology in a Cartan–Eilenberg resolution.

Remark 3.11.12 A Cartan–Eilenberg resolution gives another, indirect way to construct an injective resolution of $X \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$. First regard X as a double complex concentrated in row 0. It is easy to see that $\epsilon : X \to I$ is a morphism in $\mathsf{C}^2_f(\mathcal{A})$. In the notation of §3.10, first take vertical cohomology H_{II} of X and I, and then horizontal cohomology H_{I}. The induced morphism $\mathrm{H}_{\mathrm{I}}\,\mathrm{H}_{\mathrm{II}}(X) \to \mathrm{H}_{\mathrm{I}}\,\mathrm{H}_{\mathrm{II}}(I)$ is an isomorphism. Hence Theorem 3.10.6 shows that

$$\operatorname{tot}(\epsilon) : X = \operatorname{tot}(X) \to \operatorname{tot}(I)$$

is a quasi-isomorphism in $\mathsf{C}^+(\mathcal{A})$. For every $n \in \mathbb{Z}$, $\operatorname{tot}^n(I)$ is a finite direct sum of injective objects. We have therefore obtained an injective resolution $X \to \operatorname{tot}(I)$.

[33)] Translator's note (O014-C059): where the horizontal differential is being constructed, the source repeats the vertical map $I^{n,m} \to I^{n,m+1}$. The following check of ${}^{\triangleright}d^2 = 0$ determines the intended direction, $I^{n,m} \to I^{n+1,m}$, which has been restored.

3.12 Classical derived functors

The purpose of this section is to explain, without using the language of derived categories, how to study right derived functors (respectively left derived functors) for abelian categories with enough injectives (respectively projectives). This already encompasses many cohomology theories commonly encountered in applications. The definitions use the injective resolutions (respectively projective resolutions) introduced in §3.11.

Definition 3.12.1 Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor between abelian categories.

- Suppose that $\mathcal{A}$ has enough injectives. For every $X \in \mathsf{C}^+(\mathcal{A})$, choose an injective resolution $X \to I$. For each $n \in \mathbb{Z}$, define the value at X of the ***n*-th right derived functor** $\mathrm{R}^n F$ by

$$\mathrm{R}^n F(X) := \mathrm{H}^n(\mathsf{C}F(I)).$$

- Suppose that $\mathcal{A}$ has enough projectives. For every $X \in \mathsf{C}^-(\mathcal{A})$, choose a projective resolution $P \to X$. For each $n \in \mathbb{Z}$, define the value at X of the ***n*-th left derived functor** $\mathrm{L}_n F$ by

$$\mathrm{L}_n F(X) := \mathrm{H}^{-n}(\mathsf{C}F(P)).$$

These functors all take values in $\mathcal{B}$. By construction, they come with canonical families of morphisms $\mathrm{H}^n(\mathsf{C}F(X)) \to \mathrm{R}^n F(X)$ and $\mathrm{L}_n F(X) \to \mathrm{H}^{-n}(\mathsf{C}F(X))$.

This formulation leaves several questions. In what sense do these objects depend on the chosen resolutions? And how does one promote a definition on objects to a functor? Right and left derived functors are plainly dual, so we discuss only right derived functors below. We begin with the second question, encapsulating the basic tool in a lemma.

Lemma 3.12.2 Consider the solid part of the following diagram in $\mathsf{C}^+(\mathcal{A})$:

$$\begin{tikzcd} X \arrow[r] \arrow[d, "f"'] & I \arrow[d, dashed, "\beta"] \\ Y \arrow[r] & J \end{tikzcd} \qquad \begin{aligned} X \to I &: \text{ a quasi-isomorphism,} \\ Y \to J &: \text{ an injective resolution,} \\ f &: \text{ an arbitrary morphism.} \end{aligned}$$

There is a unique morphism β in $\mathsf{K}(\mathcal{A})$ for which the diagram commutes in $\mathsf{K}(\mathcal{A})$.

Proof Apply Theorem 3.11.5 to $X \to I$ and to the composite $X \xrightarrow{f} Y \to J$. □

Reversing arrows gives the projective-resolution version in $\mathsf{C}^-(\mathcal{A})$. In fact, the exercises for this chapter show that when $X \hookrightarrow I$, one may choose β so that the diagram commutes in $\mathsf{C}(\mathcal{A})$. Since this section uses only $\mathrm{H}^n(\beta)$, the version in $\mathsf{K}(\mathcal{A})$ suffices.

Return to derived functors. Let $f : X \to Y$ be a morphism in $\mathsf{C}^+(\mathcal{A})$, and choose injective resolutions $X \to I$ and $Y \to J$. Lemma 3.12.2 shows that, for every $n \in \mathbb{Z}$, the morphism

$$\mathrm{R}^n F(f) := \mathrm{H}^n(\mathsf{K}F\beta) : \ \mathrm{H}^n(\mathsf{K}F(I)) \to \mathrm{H}^n(\mathsf{K}F(J))$$

depends only on f and the resolutions $X \to I$, $Y \to J$. Once the resolutions are fixed, one has

$$\mathrm{R}^n F(f_1 + f_2) = \mathrm{R}^n F(f_1) + \mathrm{R}^n F(f_2),$$
$$\mathrm{R}^n F(gf) = \mathrm{R}^n F(g)\, \mathrm{R}^n F(f), \quad \mathrm{R}^n F(\mathrm{id}) = \mathrm{id}.$$

Taking $X = Y$ and $f = \mathrm{id}_X$ also shows that different injective resolutions give the same $(\mathrm{R}^n F)(X)$ up to a unique isomorphism. Thus, like objects characterized by universal properties throughout category theory, $\mathrm{R}^n F(X)$ is independent of its auxiliary data (the injective resolution) up to canonical isomorphism, and $\mathrm{R}^n F : \mathsf{C}^+(\mathcal{A}) \to \mathcal{B}$ is an additive functor.

By construction, if $F, G : \mathcal{A} \to \mathcal{B}$ are additive functors, every morphism $F \to G$ canonically induces $\mathrm{R}^n F \to \mathrm{R}^n G$ for all $n \in \mathbb{Z}$. Once an injective resolution $X \to I$ is fixed, the morphism $\mathrm{R}^n F(X) \to \mathrm{R}^n G(X)$ is concretely determined by $\mathsf{C}F(I) \to \mathsf{C}G(I)$. Similarly, it induces $\mathrm{L}_n F \to \mathrm{L}_n G$.

Convention 3.12.3 Whenever we speak below of right derived functors (respectively left derived functors) of an additive functor $F : \mathcal{A} \to \mathcal{B}$, we implicitly assume that $\mathcal{A}$ has enough injectives (respectively enough projectives).

Theorem 3.12.4 (Long exact sequence of derived functors) Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor between abelian categories, and consider a short exact sequence $0 \to X \to Y \to Z \to 0$ in $\mathsf{C}(\mathcal{A})$.

- ⋄ If $X, Y, Z \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$, there are canonical morphisms $\delta^n = \delta^n_{X,Y,Z} : \mathrm{R}^n F(Z) \to \mathrm{R}^{n+1} F(X)$ for $n \in \mathbb{Z}$, and an exact sequence

$$\cdots \to \mathrm{R}^{n-1} F(Z) \xrightarrow{\delta^{n-1}} \mathrm{R}^n F(X) \to \mathrm{R}^n F(Y) \to \mathrm{R}^n F(Z) \xrightarrow{\delta^n} \mathrm{R}^{n+1} F(X) \to \cdots .$$

- ⋄ If $X, Y, Z \in \mathrm{Ob}(\mathsf{C}^-(\mathcal{A}))$, there are canonical morphisms $\partial_n = \partial_n^{X,Y,Z} : \mathrm{L}_n F(Z) \to \mathrm{L}_{n-1} F(X)$ for $n \in \mathbb{Z}$, and an exact sequence

$$\cdots \to \mathrm{L}_{n+1} F(Z) \xrightarrow{\partial_{n+1}} \mathrm{L}_n F(X) \to \mathrm{L}_n F(Y) \to \mathrm{L}_n F(Z) \xrightarrow{\partial_n} \mathrm{L}_{n-1} F(X) \to \cdots .$$

The connecting morphisms δ^n and ∂_n are functorial as follows. If

$$\begin{tikzcd} 0 \arrow[r] & X \arrow[r] \arrow[d, "\alpha"] & Y \arrow[r] \arrow[d, "\beta"] & Z \arrow[r] \arrow[d, "\gamma"] & 0 \\ 0 \arrow[r] & X' \arrow[r] & Y' \arrow[r] & Z' \arrow[r] & 0 \end{tikzcd}$$

is a commutative diagram with exact rows, then, in the respective cases, the two diagrams

$$\begin{tikzcd} \mathrm{R}^n F(Z) \arrow[r, "\delta^n"] \arrow[d, "\mathrm{R}^n F(\gamma)"'] & \mathrm{R}^{n+1} F(X) \arrow[d, "\mathrm{R}^{n+1} F(\alpha)"] \\ \mathrm{R}^n F(Z') \arrow[r, "\delta^n"'] & \mathrm{R}^{n+1} F(X') \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} \mathrm{L}_n F(Z) \arrow[r, "\partial_n"] \arrow[d, "\mathrm{L}_n F(\gamma)"'] & \mathrm{L}_{n-1} F(X) \arrow[d, "\mathrm{L}_{n-1} F(\alpha)"] \\ \mathrm{L}_n F(Z') \arrow[r, "\partial_n"'] & \mathrm{L}_{n-1} F(X') \end{tikzcd}$$

commute for every $n \in \mathbb{Z}$.

Proof We treat only $\mathrm{R}^n F$. Apply Proposition 3.11.9 to $0 \to X \to Y \to Z \to 0$ to obtain a commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & I \arrow[r] & J \arrow[r] & K \arrow[r] & 0 \\ 0 \arrow[r] & X \arrow[r] \arrow[u] & Y \arrow[r] \arrow[u] & Z \arrow[r] \arrow[u] & 0 \end{tikzcd}$$

in which every column is an injective resolution. Lemma 2.8.13 ensures that $0 \to I^n \to J^n \to K^n \to 0$ splits. Its image under F is therefore still a split short exact sequence, so $0 \to \mathsf{C}F(I) \to \mathsf{C}F(J) \to \mathsf{C}F(K) \to 0$ remains exact. Proposition 3.6.4 now gives the morphisms δ^n and the long exact sequence

$$\cdots \to \mathrm{R}^{n-1} F(Z) \xrightarrow{\delta^{n-1}} \mathrm{R}^n F(X) \to \mathrm{R}^n F(Y) \to \mathrm{R}^n F(Z) \xrightarrow{\delta^n} \cdots$$

Functoriality of δ^n is more involved. We use the abelian category $\mathcal{A}^{\mathbf{2}}$ of Example 2.8.18. First express the given commutative diagram with exact rows as the short exact sequence in $\mathcal{A}^{\mathbf{2}}$

$$0 \to [X \xrightarrow{\alpha} X'] \to [Y \xrightarrow{\beta} Y'] \to [Z \xrightarrow{\gamma} Z'] \to 0.$$

Example 2.8.18 says that $\mathcal{A}^{\mathbf{2}}$ has enough injectives. Choose injective resolutions $[X \to X'] \hookrightarrow \mathcal{I}$ and $[Z \to Z'] \hookrightarrow \mathcal{K}$, and use Proposition 3.11.9 to extend them to a commutative diagram with exact rows in $\mathsf{C}(\mathcal{A}^{\mathbf{2}})$:

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{I} \arrow[r] & \mathcal{J} \arrow[r] & \mathcal{K} \arrow[r] & 0 \\ 0 \arrow[r] & {[X \to X']} \arrow[r] \arrow[u, hook] & {[Y \to Y']} \arrow[r] \arrow[u, hook] & {[Z \to Z']} \arrow[r] \arrow[u, hook] & 0 \end{tikzcd} \tag{3.12.1}$$

such that $[Y \to Y'] \to \mathcal{J}$ is also an injective resolution. Write $\mathcal{J}^n = [J^n \to (J')^n]$, and similarly for the other terms. The evaluation functors ev_0 and ev_1 from Example 2.8.18 show that J^n, $(J')^n$, and the corresponding terms are injective objects of $\mathcal{A}$, while $Y \to J$, $Y' \to J'$, and the corresponding maps are injective resolutions in $\mathsf{C}(\mathcal{A})$. Thus compatible injective resolutions have been erected above the original diagram.

Now expand the first row of (3.12.1) as the following commutative diagram with exact rows in $\mathsf{C}(\mathcal{A})$:

$$\begin{tikzcd}
0 \arrow[r] & I \arrow[r] \arrow[d] & J \arrow[r] \arrow[d] & K \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & I' \arrow[r] & J' \arrow[r] & K' \arrow[r] & 0
\end{tikzcd}$$

For every n, both rows split because I^n and $(I')^n$ are injective. Applying F degreewise therefore preserves their exactness. Functoriality of δ^n now reduces to the corresponding assertion of Proposition 3.6.4. $\square$

In the classical setting, one is mainly interested in the restrictions of $\mathrm{R}^n F$ and $\mathrm{L}_n F$ to $\mathcal{A}$, still written $\mathrm{R}^n F(X)$ and $\mathrm{L}_n F(X)$ for $X \in \mathrm{Ob}(\mathcal{A})$. They are computed by flattening an injective resolution to an exact sequence $0 \to X \to I^0 \to I^1 \to \cdots$ (or a projective resolution to $\cdots \to P^1 \to P^0 \to X \to 0$); see Example 3.11.2. Derived functors defined on complexes were called **hyper-derived functors** in the early literature and are more naturally treated using derived categories or spectral sequences. We therefore focus below on the classical case $X \in \mathrm{Ob}(\mathcal{A})$.

The first step is to distill the long exact sequence of Theorem 3.12.4 into the notion of a δ-functor.

Definition 3.12.5 Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. A **cohomological δ-functor** from $\mathcal{A}$ to $\mathcal{B}$ consists of the following data:

- ◇ additive functors $F^n : \mathcal{A} \to \mathcal{B}$ for $n \in \mathbb{Z}_{\geq 0}$;
- ◇ for each short exact sequence $0 \to X \to Y \to Z \to 0$ in $\mathcal{A}$, specified morphisms $\delta^n : F^n Z \to F^{n+1} X$, called connecting morphisms, for $n \in \mathbb{Z}_{\geq 0}$. They are functorial in morphisms of short exact sequences and yield an exact sequence

$$\begin{tikzcd}
0 \arrow[r] & F^0(X) \arrow[r] & F^0(Y) \arrow[r] & F^0(Z) \arrow[dll, rounded corners, "\delta^0"] \\
 & F^1(X) \arrow[r] & F^1(Y) \arrow[r] & F^1(Z) \arrow[dll, rounded corners, "\delta^1"] \\
 & F^2(X) \arrow[r] & F^2(Y) \arrow[r] & \cdots .
\end{tikzcd}$$

A morphism from a cohomological δ-functor $(F^n, \delta^n)_{n\geq 0}$ to $(G^n, \eta^n)_{n\geq 0}$ is a family of morphisms $\varphi^n : F^n \to G^n$ compatible with the connecting morphisms of short exact sequences. In other words, the diagram

$$\begin{tikzcd} G^{n-1}(Z) \arrow[r, "\eta^{n-1}"] & G^n(X) \\ F^{n-1}(Z) \arrow[u, "\varphi^{n-1}"] \arrow[r, "\delta^{n-1}"'] & F^n(X) \arrow[u, "\varphi^n"'] \end{tikzcd}$$

[34] must commute for all $n \geq 1$ and every short exact sequence $0 \to X \to Y \to Z \to 0$.

Dually, a **homological δ-functor** from $\mathcal{A}$ to $\mathcal{B}$ consists of the following data:

◇ additive functors $F_n : \mathcal{A} \to \mathcal{B}$ for $n \in \mathbb{Z}_{\geq 0}$;

◇ for each short exact sequence $0 \to X \to Y \to Z \to 0$ in $\mathcal{A}$, specified connecting morphisms $\partial_n : F_n Z \to F_{n-1}X$ for $n \geq 1$, functorial in morphisms of short exact sequences and yielding an exact sequence

$$\begin{tikzcd} \cdots \arrow[r] & F_2(Y) \arrow[r] & F_2(Z) \arrow[dll, rounded corners, "\partial_2"] \\ F_1(X) \arrow[r] & F_1(Y) \arrow[r] & F_1(Z) \arrow[dll, rounded corners, "\partial_1"] \\ F_0(X) \arrow[r] & F_0(Y) \arrow[r] & F_0(Z) \arrow[r] & 0. \end{tikzcd}$$

Morphisms of homological δ-functors are defined analogously, with compatibility with connecting morphisms.

The long exact sequences in the definition automatically imply that F^0 is left exact and F_0 is right exact. We now return to derived functors.

Proposition 3.12.6 Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor between abelian categories. If $\mathcal{A}$ has enough injectives, then:

◇ $n < 0 \implies \mathrm{R}^n F = 0$;

◇ if I is injective in $\mathcal{A}$, then $n > 0 \implies \mathrm{R}^n F(I) = 0$;

◇ $(\mathrm{R}^n F, \delta^n)_{n\geq 0}$ is a cohomological δ-functor;

◇ if F is left exact, there is a canonical isomorphism $F \xrightarrow{\sim} \mathrm{R}^0 F$.

Dually, if $\mathcal{A}$ has enough projectives, then:

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[34] Translator's note (O014-C060): both objects in the left column are printed as X in the source. For a short exact sequence $0 \to X \to Y \to Z \to 0$, the connecting morphisms δ^{n-1} and η^{n-1} have domains $F^{n-1}(Z)$ and $G^{n-1}(Z)$, respectively, and codomains $F^n(X)$ and $G^n(X)$. Both left-hand objects have therefore been restored to Z.

- ◇ $n < 0 \implies \mathrm{L}_n F = 0$;
- ◇ if P is projective in $\mathcal{A}$, then $n > 0 \implies \mathrm{L}_n F(P) = 0$;
- ◇ $(\mathrm{L}_n F, \partial_n)_{n \geq 0}$ is a homological δ-functor;
- ◇ if F is right exact, there is a canonical isomorphism $\mathrm{L}_0 F \xrightarrow{\sim} F$.

Proof By duality, it suffices to treat $\mathrm{R}^n F$. Choose an injective resolution $0 \to X \to I^0 \to I^1 \to \cdots$ of $X \in \mathrm{Ob}(\mathcal{A})$. Then

$$\mathrm{R}^n F(X) = \mathrm{H}^n(\cdots \to 0 \to 0 \to \underbrace{FI^0}_{\text{term in degree 0}} \to \underbrace{FI^1}_{\text{term in degree 1}} \to \cdots).$$

This immediately gives $\mathrm{R}^n F(X) = 0$ when $n < 0$. If $X = I$ is injective, take the resolution $0 \to I \xrightarrow{\mathrm{id}} I \to 0 \to \cdots$, which gives $\mathrm{R}^n F(I) = 0$ for $n > 0$.

In view of this, the assertion about cohomological δ-functors is just a restatement of Theorem 3.12.4.

Finally suppose that F is left exact. Since $X \xrightarrow{\sim} \ker[I^0 \to I^1]$ and a left exact functor preserves kernels, $\mathrm{R}^0 F(X) = \ker[FI^0 \to FI^1]$ is canonically isomorphic to FX. □

In general, one considers right derived functors only for left exact functors, and left derived functors only for right exact functors. Why? An elementary explanation comes from the problem of extending exact sequences. For example, if F is left exact and $0 \to X \to Y \to Z \to 0$ is a short exact sequence in $\mathcal{A}$, we would like to extend the exact sequence $0 \to FX \to FY \to FZ$ as far as possible to the right. Right derived functors do exactly this.

For $n \geq 1$, the functors $\mathrm{R}^n F$ and $\mathrm{L}_n F$ are customarily called the **higher derived functors** of F, with domain understood to be the objects of $\mathcal{A}$ rather than complexes. As an application of extending an exact sequence, the following corollary says that higher derived functors are precisely the obstructions to exactness.

Corollary 3.12.7 For a left exact (respectively right exact) additive functor $F : \mathcal{A} \to \mathcal{B}$, the following are equivalent.

(i) $F : \mathcal{A} \to \mathcal{B}$ is exact.

(ii) For every $X \in \mathrm{Ob}(\mathcal{A})$ and $n > 0$, $\mathrm{R}^n F(X) = 0$ (respectively $\mathrm{L}_n F(X) = 0$).

(iii) For every $X \in \mathrm{Ob}(\mathcal{A})$, $\mathrm{R}^1 F(X) = 0$ (respectively $\mathrm{L}_1 F(X) = 0$).

Proof Consider the left exact case. For (i) $\implies$ (ii), the complex $I^0 \to I^1 \to \cdots$ used to compute the derived functors is exact away from its zeroth term, and therefore so is its image under F. The implication (ii) $\implies$ (iii) is trivial. For (iii) $\implies$ (i), use

$\mathrm{R}^0 F \simeq F$ and apply the exact sequence

$$0 \to FX \to FY \to FZ \to \mathrm{R}^1 F(X)$$

to each short exact sequence $0 \to X \to Y \to Z \to 0$ in $\mathcal{A}$. □

Convention 3.12.8 Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact (respectively right exact) additive functor. An object $A \in \mathrm{Ob}(\mathcal{A})$ is called ***F*-acyclic** if all its higher right derived functors (respectively left derived functors) under F vanish. For example, when F is left exact (respectively right exact), Proposition 3.12.6 implies that every injective (respectively projective) object is F-acyclic.

The following classical technique, called dimension shifting, is often used in recursive arguments with derived functors.

Proposition 3.12.9 (Dimension shifting) Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact (respectively right exact) additive functor, and suppose that $\mathcal{A}$ has enough injectives (respectively enough projectives). Let

$$0 \to X \to A \to B \to 0, \quad (\text{respectively } 0 \to B \to A \to X \to 0),$$

be a short exact sequence in $\mathcal{A}$, with A F-acyclic. Then for $n \geq 1$ there are natural isomorphisms

$$\mathrm{R}^n F(X) \simeq \begin{cases} \mathrm{R}^{n-1} F(B), & n \geq 2, \\ \operatorname{coker}\left[FA \to FB\right], & n = 1, \end{cases}$$

respectively

$$\mathrm{L}_n F(X) \simeq \begin{cases} \mathrm{L}_{n-1} F(B), & n \geq 2, \\ \ker\left[FB \to FA\right], & n = 1. \end{cases}$$

Proof By duality, prove only the assertion about $\mathrm{R}^n F(X)$. Inspect the corresponding long exact sequence

$$\mathrm{R}^{n-1} F(A) \to \mathrm{R}^{n-1} F(B) \to \mathrm{R}^n F(X) \to \underbrace{\mathrm{R}^n F(A)}_{=0}, \quad n \in \mathbb{Z}_{\geq 1},$$

and use $\mathrm{R}^0 F(A) \simeq FA$ and $\mathrm{R}^0 F(B) = FB$. □

The next key result shows that F-acyclic objects may also be used to compute derived functors.

Corollary 3.12.10 Let $F : \mathcal{A} \to \mathcal{B}$ be left exact (respectively right exact), and suppose there is an exact sequence in $\mathcal{A}$

$$0 \to X \to A^0 \to A^1 \to \cdots \quad (\text{respectively } \cdots \to A_1 \to A_0 \to X \to 0),$$

where every A^n (respectively A_n) is F-acyclic for $n \geq 0$. In addition, set $A^{-n} = 0$ (respectively $A_{-n} = 0$) for every $n \geq 1$.[35] Then, for every $n \geq 0$, there are natural isomorphisms

$$\mathrm{R}^n F(X) \simeq \mathrm{H}^n\left(FA^\bullet\right) \quad \text{or} \quad \mathrm{L}_n F(X) \simeq \mathrm{H}_n\left(FA_\bullet\right).$$

Proof We prove only the assertion about $\mathrm{R}^n F(X)$. For $m \geq 1$, put

$$B^m := \operatorname{im}\left[A^{m-1} \to A^m\right] = \ker\left[A^m \to A^{m+1}\right];$$

also put $B^0 := X$. The original exact sequence then decomposes into

$$0 \to B^m \to A^m \to B^{m+1} \to 0, \quad m \in \mathbb{Z}_{\geq 0}.$$

When $n = 0$, the claim reduces to preservation of kernels: $FX \simeq \ker[FA^0 \to FA^1]$. For $n \geq 1$, repeated dimension shifting using Proposition 3.12.9 gives natural isomorphisms

$$\mathrm{R}^n F(X) = \mathrm{R}^n F(B^0) \simeq \cdots \simeq \mathrm{R}^1 F(B^{n-1}) \simeq \operatorname{coker}\left[FA^{n-1} \to FB^n\right];$$

and because F preserves kernels, the definition of B^n identifies FB^n with $\ker[FA^n \to FA^{n+1}]$. Thus, for $n \geq 1$,

$$\mathrm{R}^n F(X) \simeq \frac{\ker\left[FA^n \to FA^{n+1}\right]}{\operatorname{im}\left[FA^{n-1} \to FA^n\right]}.$$

□

The preceding argument relies mainly on the long exact sequence and hardly at all on the definition of derived functors. This suggests characterizing the derived functors of F among general cohomological (respectively homological) δ-functors. The key requirement is that their positive-degree terms be effaceable (respectively co-effaceable).

Definition 3.12.11 (A. Grothendieck) Let $E : \mathcal{A} \to \mathcal{B}$ be an additive functor between abelian categories. It is called **effaceable** (respectively **co-effaceable**) if for every $X \in \mathrm{Ob}(\mathcal{A})$ there is a monomorphism $a : X \hookrightarrow Y$ (respectively an epimorphism $a : Y \twoheadrightarrow X$) such that $Ea = 0$.

We shall derive the following universal property from effaceability.

Definition 3.12.12 Let $(R^n, \delta^n)_{n\geq 0}$ be a cohomological δ-functor from $\mathcal{A}$ to $\mathcal{B}$. It is called a **universal cohomological δ-functor** if, for every cohomological δ-functor $(F^n, \delta^n_F)_{n\geq 0}$ from $\mathcal{A}$ to $\mathcal{B}$, every morphism $\varphi^0 : R^0 \to F^0$ extends uniquely to a morphism $(\varphi^n)_{n\geq 0}$ of δ-functors.

[35] Translator's note (O014-C062): the source gives no range for the convention $A^{-n} = 0$ (respectively $A_{-n} = 0$). If it were applied also at $n = 0$, it would force $A^0 = 0$ (respectively $A_0 = 0$), contradicting the resolution term just introduced. The intended range $n \geq 1$ has been made explicit.

Dually, let $(L_n, \partial_n)_{n\geq 0}$ be a homological δ-functor with the following property: for every homological δ-functor $(F_n, \partial_n^F)_{n\geq 0}$, every morphism $\varphi_0 : F_0 \to L_0$ extends uniquely to $(\varphi_n)_{n\geq 0}$. Then $(L_n, \partial_n)_{n\geq 0}$ is called a **universal homological δ-functor**.

Our goal is to show that, for a left exact (respectively right exact) functor $F : \mathcal{A} \to \mathcal{B}$, a universal cohomological δ-functor with $R^0 = F$ (respectively a universal homological δ-functor with $L_0 = F$), if it exists, is unique up to a unique isomorphism.

Lemma 3.12.13 Let $E : \mathcal{A} \to \mathcal{B}$ be effaceable, let $0 \to X \xrightarrow{u} I \xrightarrow{v} B \to 0$ be a short exact sequence in $\mathcal{A}$, and let $f : X \to Y$ be any morphism. There is a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & X \arrow[r, "u"] \arrow[d, "f"'] & I \arrow[r, "v"] \arrow[d] & B \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & Y \arrow[r, "\iota"'] & J \arrow[r] & C \arrow[r] & 0,
\end{tikzcd}$$

such that $E\iota = 0$.

Proof Choose a monomorphism $h : Y \hookrightarrow J_0$ such that $Eh = 0$, and form the pushout diagram

$$\begin{tikzcd}
X \arrow[r, hook, "u"] \arrow[d, "hf"'] \arrow[dr, phantom, "\boxplus"] & I \arrow[d] \\
J_0 \arrow[r, "k"'] & J,
\end{tikzcd}$$

where k is a monomorphism by Proposition 2.1.6. Set $\iota = kh : Y \to J$ and $C := \operatorname{coker}(\iota)$. This gives the left-hand square of the required diagram; functoriality of cokernels gives the right-hand square. $\square$

Proposition 3.12.14 Let $(R^n, \delta^n)_{n\geq 0}$ be a cohomological δ-functor from $\mathcal{A}$ to $\mathcal{B}$. If R^n is effaceable for every $n > 0$, then $(R^n, \delta^n)_{n\geq 0}$ is universal.

Dually, a homological δ-functor whose positive-degree parts are co-effaceable is universal.

Proof We prove only the cohomological version. Let $(F^n, \delta_F^n)_{n\geq 0}$ be another cohomological δ-functor from $\mathcal{A}$ to $\mathcal{B}$, and fix $\varphi^0 : R^0 \to F^0$. We construct φ^n recursively for $n \geq 1$. Given $X \in \operatorname{Ob}(\mathcal{A})$, the hypothesis supplies a short exact sequence

$$0 \to X \to I \to B \to 0$$

such that $R^n(X \to I) = 0$. The corresponding long exact sequences give the solid part of a commutative diagram with exact rows

$$\begin{tikzcd}
F^{n-1}I \arrow[r] & F^{n-1}B \arrow[r, "\delta_F^{n-1}"] & F^nX \arrow[r] & F^nI \\
R^{n-1}I \arrow[r] \arrow[u, "\varphi_I^{n-1}"] & R^{n-1}B \arrow[r, "\delta^{n-1}"'] \arrow[u, "\varphi_B^{n-1}"] & R^nX \arrow[r] \arrow[u, dashed, "\varphi_X^n"'] & 0.
\end{tikzcd}$$

Exactness of the rows and the universal property of the cokernel give a unique dashed arrow φ_X^n making the whole diagram commute. This characterizes φ_X^n, although it still appears to depend on the choice of $X \hookrightarrow I$.

Given any morphism $f : X \to Y$, we claim there is a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & X \arrow[r] \arrow[d, "f"'] & I \arrow[r] \arrow[d] & B \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & Y \arrow[r] & J \arrow[r] & C \arrow[r] & 0,
\end{tikzcd} \tag{3.12.2}$$

such that both $R^n(X \to I)$ and $R^n(Y \to J)$ are zero. Indeed, first choose $X \hookrightarrow I$, then apply Lemma 3.12.13. Substituting this into the constructions of φ_X^n and φ_Y^n yields

$$\begin{tikzcd}
& F^{n-1}B \arrow[rr] \arrow[dd] & & F^nX \arrow[dd] \\
R^{n-1}B \arrow[ur] \arrow[rr] \arrow[dd] & & R^nX \arrow[ur, "\varphi_X^n"'] \arrow[dd] & \\
& F^{n-1}C \arrow[rr] & & F^nY \\
R^{n-1}C \arrow[ur] \arrow[rr] & & R^nY \arrow[ur, "\varphi_Y^n"'] &
\end{tikzcd}$$

Every face except the right one commutes by construction, definition, or the induction hypothesis. Since $R^{n-1}B \to R^nX$ is an epimorphism, the usual argument shows that the right face commutes too.

This simultaneously proves that φ_X^n is independent of the choice of $X \hookrightarrow I$ (take $f = \mathrm{id}_X$ and observe from the proof of Lemma 3.12.13 that any two maps $X \hookrightarrow I_i$ can be sent by pushouts into a common $X \hookrightarrow J$) and that it is functorial in X (take arbitrary f). This recursively completes the construction of φ^n.

It remains to show that $(\varphi^n)_{n\geq 0}$ is compatible with the connecting morphisms. Given a short exact sequence $0 \to X \to Y \to Z \to 0$, use Lemma 3.12.13 to choose a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & X \arrow[r] \arrow[d, "\mathrm{id}_X"'] & Y \arrow[r] \arrow[d] & Z \arrow[r] \arrow[d, "\alpha"] & 0 \\
0 \arrow[r] & X \arrow[r] & I \arrow[r] & B \arrow[r] & 0
\end{tikzcd}$$

such that $R^n(X \to I) = 0$. For each $n \geq 1$, form the diagram

$$\begin{tikzcd} F^{n-1}Z \arrow[r, "F^{n-1}\alpha"] & F^{n-1}B \arrow[r, "\delta_F^{n-1}"] & F^n X \\ R^{n-1}Z \arrow[u, "\varphi_Z^{n-1}"] \arrow[r, "R^{n-1}\alpha"'] & R^{n-1}B \arrow[u, "\varphi_B^{n-1}"] \arrow[r, "\delta^{n-1}"'] & R^n X \arrow[u, "\varphi_X^n"'] \end{tikzcd}$$

[36] By the preceding commutative diagram and the axioms of a cohomological δ-functor, the composites along the two rows are the connecting morphisms $F^{n-1}Z \to F^n X$ and $R^{n-1}Z \to R^n X$ whose compatibility is at issue. The left square commutes by functoriality of φ^{n-1}, and the right square by the construction of φ_X^n. □

Corollary 3.12.15 (Characterization of derived functors) Let $F : \mathcal{A} \to \mathcal{B}$ be left exact and suppose that $\mathcal{A}$ has enough injectives. Then $(\mathrm{R}^n F, \delta^n)_{n\geq 0}$ is the universal cohomological δ-functor satisfying $\mathrm{R}^0 F \simeq F$.

Dually, if F is right exact and $\mathcal{A}$ has enough projectives, then $(\mathrm{L}_n F, \partial_n)_{n\geq 0}$ is the universal homological δ-functor satisfying $\mathrm{L}_0 F \simeq F$.

Proof It is enough to discuss the left exact case. For $n > 0$, the functor $\mathrm{R}^n F$ is effaceable: every $X \in \mathrm{Ob}(\mathcal{A})$ embeds into an injective object I, and Proposition 3.12.6 gives $\mathrm{R}^n F(I) = 0$. The same proposition gives $\mathrm{R}^0 F \simeq F$. Apply Proposition 3.12.14. □

3.13 Example: $\lim^1$

Our first example of derived functors comes from a common class of inverse limits. Regard $\mathbb{Z}_{\geq 0}$ as a category under its usual total order; then $\mathbb{Z}_{\geq 0}^{\mathrm{op}}$ may be displayed as $\cdots \to 2 \to 1 \to 0$. For any category $\mathcal{A}$, consider the functor category

$$\mathsf{InvSys}(\mathcal{A}) := \mathcal{A}^{\mathbb{Z}_{\geq 0}^{\mathrm{op}}}. \tag{3.13.1}$$

Its objects may be identified with data $(A_n, f_n)_{n\geq 0}$, where $A_n \in \mathrm{Ob}(\mathcal{A})$ and $f_n \in \mathrm{Hom}(A_{n+1}, A_n)$. Such data are also called **inverse systems** in $\mathcal{A}$. Using derived functors, this section studies their inverse limits when $\mathcal{A}$ is abelian.

Convention 3.13.1 An abelian category $\mathcal{A}$ is said to have **exact countable products** (respectively **exact countable coproducts**) if it has countable products (respectively countable coproducts, or direct sums) and the product functor $\prod : \mathcal{A}^{\mathbb{Z}_{\geq 0}} \to \mathcal{A}$ (respectively the direct-sum functor $\oplus : \mathcal{A}^{\mathbb{Z}_{\geq 0}} \to \mathcal{A}$) is exact.

[36] Translator's note (O014-C061): the source labels the two connecting arrows in this diagram δ_F^n and δ^n. Since their domains are $F^{n-1}B$ and $R^{n-1}B$, and their codomains are $F^n X$ and $R^n X$, the well-typed indices are $n-1$; the labels have been restored to δ_F^{n-1} and δ^{n-1}.

The product functor is always left exact. Thus, assuming countable products exist, its exactness is equivalent to preservation of epimorphisms: for every family of epimorphisms $(g_n : A_n \to B_n)_{n\geq 0}$, the corresponding map $\prod_{n\geq 0} g_n : \prod_{n\geq 0} A_n \to \prod_{n\geq 0} B_n$ must also be an epimorphism. The assertion for countable coproducts is dual.

Hypothesis 3.13.2 Throughout this section, the abelian category $\mathcal{A}$ is assumed to have exact countable products.

The category R-Mod of modules over any ring R has this property.

Definition 3.13.3 Given an object $A = (A_n, f_n)_n$ of $\mathsf{InvSys}(\mathcal{A})$, define the shift morphism $T_A : \prod_{n\geq 0} A_n \to \prod_{n\geq 0} A_n$ as follows. When $\mathcal{A} = \mathsf{Ab}$, $T_A((a_n)_{n\geq 0}) := (f_n(a_{n+1}))_{n\geq 0}$; the general definition is analogous. Define further the canonical morphism

$$\Delta_A := T_A - \mathrm{id} : \prod_{n\geq 0} A_n \to \prod_{n\geq 0} A_n.$$

As A varies, these maps give endomorphisms T and $\Delta = T - \mathrm{id}$ of the product functor $\prod_{n\geq 0}$.

Notice that this definition requires only that $\mathcal{A}$ be an Ab-category with countable products.

Recall that inverse limits are generally constructed from equalizers and products. Here $\mathsf{InvSys}(\mathcal{A})$ is naturally an abelian category (Proposition 2.1.4), so the construction reduces to $\varprojlim A = \ker[\Delta_A : \prod_n A_n \to \prod_n A_n]$.

Moreover, $\mathsf{InvSys}(\mathcal{A})$ also has exact countable products, since limits in a functor category are constructed pointwise; see §1.5.

Lemma 3.13.4 The construction above gives a left exact additive functor $\varprojlim$: $\mathsf{InvSys}(\mathcal{A}) \to \mathcal{A}$. On the other hand, the product functor $\prod_{n\geq 0} : \mathsf{InvSys}(\mathcal{A}) \to \mathcal{A}$ is exact.

Proof Both functors are plainly additive, and left exactness follows from Example 2.8.6. By hypothesis, the functor $\prod_{n\geq 0} : (A_n, f_n)_n \mapsto \prod_n A_n$ also preserves epimorphisms, since it does so pointwise. Remark 2.8.4 therefore shows that it is exact. $\square$

Definition 3.13.5 Define additive functors $\lim^n : \mathsf{InvSys}(\mathcal{A}) \to \mathcal{A}$ by

$$\begin{aligned}
\lim{}^0 &:= \varprojlim = \ker\Delta : \ A \mapsto \ker\Delta_A, \\
\lim{}^1 &:= \operatorname{coker}\Delta : \ A \mapsto \operatorname{coker}\Delta_A, \\
\lim{}^n &:= 0, \quad n \in \mathbb{Z}_{\geq 2}.
\end{aligned}$$

Let $0 \to X \to Y \to Z \to 0$ be a short exact sequence in $\mathsf{InvSys}(\mathcal{A})$; equivalently, it is exact pointwise. Applying the exact functor $\prod_{n\geq 0}$ and its endomorphism Δ gives the commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & \prod_n X_n \arrow[r] \arrow[d, "\Delta_X"'] & \prod_n Y_n \arrow[r] \arrow[d, "\Delta_Y"] & \prod_n Z_n \arrow[r] \arrow[d, "\Delta_Z"] & 0 \\ 0 \arrow[r] & \prod_n X_n \arrow[r] & \prod_n Y_n \arrow[r] & \prod_n Z_n \arrow[r] & 0. \end{tikzcd}$$

The Snake Lemma (Theorem 2.3.3) gives the long exact sequence

$$0 \to \lim^0 X \to \lim^0 Y \to \lim^0 Z \xrightarrow[\text{connecting morphism}]{\delta^0} \lim^1 X \to \lim^1 Y \to \lim^1 Z \to 0;$$

it is functorial in short exact sequences. For $n > 0$, set $\delta^n := 0$. We summarize this as follows.

Lemma 3.13.6 The data $(\lim^n, \delta^n)_{n\geq 0}$ form a cohomological δ-functor from $\mathsf{InvSys}(\mathcal{A})$ to $\mathcal{A}$ (Definition 3.12.5), with $\lim^0 = \varprojlim$.

We now turn to the right derived functors of $\varprojlim$. If $\mathcal{A}$ has enough injectives, then so does $\mathsf{InvSys}(\mathcal{A})$. Indeed, the exercises in Chapter 2 already discuss the general functor category $\mathcal{A}^{\mathcal{C}}$; here $\mathcal{C} = \mathbb{Z}_{\geq 0}^{\mathrm{op}}$. We give a direct proof of the following slightly more precise result.

[Translator's clarification: the following results are under the standing assumption that $\mathcal{A}$ has enough injective objects.]

Lemma 3.13.7 Under Hypothesis 3.13.2, define the additive full subcategory $\mathcal{R}$ of $\mathsf{InvSys}(\mathcal{A})$ by

$$\mathrm{Ob}(\mathcal{R}) = \left\{ \begin{array}{l|l} R = (R_k, r_k)_{k\geq 0} & R \text{ is injective, } \Delta_R \text{ is an epimorphism,} \\ \in \mathrm{Ob}\left(\mathsf{InvSys}(\mathcal{A})\right) & \forall k,\ R_k \text{ is injective in } \mathcal{A} \end{array} \right\}.$$

For every $A \in \mathrm{Ob}(\mathsf{InvSys}(\mathcal{A}))$, there are an object $R \in \mathrm{Ob}(\mathcal{R})$ and a monomorphism $A \hookrightarrow R$.

Proof For every $m \in \mathbb{Z}_{\geq 0}$, define a functor $\mathcal{R}_m : \mathcal{A} \to \mathsf{InvSys}(\mathcal{A})$ by sending an object X to

$$\begin{array}{ccccccccc} \cdots & \xrightarrow{\mathrm{id}} & X & \xrightarrow{\mathrm{id}} & X & \longrightarrow & 0 \longrightarrow \cdots \longrightarrow 0. \\ \cdots & & \text{term } m+1 & & \text{term } m & & \end{array} \tag{3.13.2}$$

This is plainly right adjoint to $\mathrm{ev}_m : (A_n, f_n)_{n\geq 0} \mapsto A_m$, and hence preserves injective objects (Proposition 2.8.17). It is also straightforward to check that $\Delta_{\mathcal{R}_m X}$ is an epimorphism; one can write down a right inverse explicitly.

Given $A = (A_n, f_n)_{n\geq 0}$, choose for each m an injective object I_m of $\mathcal{A}$ and a monomorphism $A_m \hookrightarrow I_m$. Then $A \hookrightarrow \prod_{m\geq 0} \mathcal{R}_m(I_m) =: R$. By Lemma 2.8.14, the object on the right is still injective; in fact, by construction every R_k is injective in $\mathcal{A}$. Since $\prod_m$ is exact, $\Delta_R = \prod_m \Delta_{\mathcal{R}_m(I_m)}$ remains an epimorphism. Thus $R \in \mathrm{Ob}(\mathcal{R})$. □

We may therefore discuss $\mathrm{R}^n \varprojlim$ for $n \geq 0$. Recall that the right derived functors of $\varprojlim$ are characterized as a universal cohomological δ-functor (Corollary 3.12.15).

Theorem 3.13.8 (S. Eilenberg) Under Hypothesis 3.13.2, $\lim^n$ is effaceable in the sense of Definition 3.12.11 for every $n > 0$. Consequently there is a canonical isomorphism $\mathrm{R}^1 \varprojlim \simeq \lim^1$, while $\mathrm{R}^n \varprojlim = 0$ for $n \notin \{0, 1\}$.

Proof Lemma 3.13.7 shows that $\lim^1$ is effaceable. Hence $(\lim^n)_{n\geq 0}$ is a universal δ-functor by Proposition 3.12.14. □

Having identified $\lim^1$ as the obstruction to exactness of $\varprojlim$, we now study conditions that ensure $\lim^1 = 0$.

Example 3.13.9 If an object A of $\mathsf{InvSys}(\mathcal{A})$ satisfies $\lim^1 A = 0$, then $\lim^1 Q = 0$ for every quotient object Q of A. Indeed, the long exact sequence induced by $0 \to K \to A \to Q \to 0$ ends with $\lim^1 A \to \lim^1 Q \to 0$.

Example 3.13.10 If every $f_n : A_{n+1} \to A_n$ has a section (a right inverse) s_n, then Δ_A also has a section $\Sigma_A : \prod_n A_n \to \prod_n A_n$, and hence Δ_A is an epimorphism. In element notation, one may recursively define $\Sigma_A((b_n)_n) = (a_n)_n$ by $a_0 = 0$ and $a_{n+1} := s_n(a_n + b_n)$. The general case is analogous.

Definition 3.13.11 (Mittag-Leffler condition) Suppose that every morphism in a category $\mathcal{C}$ has an image (Definition 1.2.1), and let $X = (X_k, f_k)_{k\geq 0}$ be an object of $\mathsf{InvSys}(\mathcal{C})$. For $b \geq a$, let $f_a^b : X_b \to X_a$ be the composite $f_a \cdots f_{b-1}$, with $f_a^a = \mathrm{id}$. The inverse system X is called **Mittag-Leffler** if

$$\forall k \geq 0,\ \exists N \geq k, \quad n \geq N \implies \mathrm{im}\, f_k^n = \mathrm{im}\, f_k^N.$$

For example, if every f_k is an epimorphism, then $(X_k, f_k)_{k\geq 0}$ is automatically Mittag-Leffler.

For a Mittag-Leffler system $(X_k, f_k)_{k\geq 0}$, each X_k has a well-defined subobject $X_k^\flat := \mathrm{im}\, f_k^N$ for $N \gg k$. It is easy to see that every morphism

$$\cdots \to X_2^\flat \xrightarrow{f_1^\flat} X_1^\flat \xrightarrow{f_0^\flat} X_0^\flat$$

in the resulting inverse system, where $f_k^\flat := f_k|_{X_{k+1}^\flat}$, is an epimorphism.

Lemma 3.13.12 Let $\mathcal{C}$ be Set or R-Mod for an arbitrary ring R, and suppose that an object $(X_n, f_n)_{n\geq 0}$ of $\mathsf{InvSys}(\mathcal{C})$ is Mittag-Leffler.

(i) The inclusions $X_n^\flat \hookrightarrow X_n$ induce a canonical isomorphism $\varprojlim_n X_n^\flat \xrightarrow{\sim} \varprojlim_n X_n$.

(ii) If $\mathcal{C} = \mathsf{Set}$ and every X_n is nonempty, then $\varprojlim_n X_n \neq \varnothing$.

Proof For (i), write down the explicit construction of the inverse limit in these categories:

$$\varprojlim_n X_n = \left\{ (x_n)_{n\geq 0} \in \prod_{n\geq 0} X_n : \forall n \geq 0,\ f_n(x_{n+1}) = x_n \right\};$$

every x_n in this expression automatically belongs to $\bigcap_{N\geq n} \mathrm{im}(f_n^N)$, which immediately gives $\varprojlim_n X_n = \varprojlim_n X_n^\flat$.

For (ii), the definition shows that every $X_n^\flat$ is nonempty. By (i), the problem reduces to the case where every f_n is surjective, when the claim is clear. $\square$

Proposition 3.13.13 Let R be a ring and $\mathcal{A} := R$-Mod. If an object $A = (A_n, f_n)_{n\geq 0}$ of $\mathsf{InvSys}(\mathcal{A})$ is Mittag-Leffler, then $\lim^1 A = 0$. Equivalently, for every short exact sequence $0 \to A \to B \to C \to 0$ in $\mathsf{InvSys}(\mathcal{A})$, the sequence $0 \to \varprojlim A \to \varprojlim B \to \varprojlim C \to 0$ is exact.

Proof First prove the equivalence of the two assertions. Let $A \in \mathsf{InvSys}(\mathcal{A})$. If $\lim^1 A = 0$, exactness of $0 \to \varprojlim A \to \varprojlim B \to \varprojlim C \to 0$ follows immediately from the long exact sequence. Conversely, suppose this sequence is always exact. Choose a monomorphism $A \hookrightarrow B$ with B injective in $\mathsf{InvSys}(\mathcal{A})$. Dimension shifting (Proposition 3.12.9) gives $\mathrm{R}^1 \varprojlim A = 0$, and Theorem 3.13.8 then gives $\lim^1 A = 0$.

Now prove the main assertion. Suppose that A is Mittag-Leffler in the short exact sequence $0 \to \underbrace{(A_n, f_n)_n}_{=A} \xrightarrow{\varphi} \underbrace{(B_n, g_n)_n}_{=B} \xrightarrow{\psi} \underbrace{(C_n, h_n)_n}_{=C} \to 0.$ [37] We must show that $\varprojlim \psi$ is an epimorphism. By Lemma 3.13.12 (ii), it is enough to show, for every $c = (c_n)_{n\geq 0} \in \varprojlim C$, that the inverse system of nonempty sets

$$\cdots \to \psi_2^{-1}(c_2) \xrightarrow{g_1} \psi_1^{-1}(c_1) \xrightarrow{g_0} \psi_0^{-1}(c_0)$$

is Mittag-Leffler, since this makes $(\varprojlim \psi)^{-1}(c)$ nonempty.

Given $k \geq 0$, choose $N \geq k$ large enough that $\mathrm{im}\, f_k^n = \mathrm{im}\, f_k^N$ for all $n \geq N$. Fix $b_N \in \psi_N^{-1}(c_N)$. To verify the Mittag-Leffler condition, we show that for every $n \geq N$ there is a $b_n \in \psi_n^{-1}(c_n)$ such that $g_k^n(b_n) = g_k^N(b_N)$.

[37] Translator's note (O014-C063): under the brace for C, the source writes the inverse system as (C_n, h_n) without its indexing subscript. Since C is an object of $\mathsf{InvSys}(\mathcal{A})$, the notation has been restored to $(C_n, h_n)_n$.

Choose any $b'_n \in \psi_n^{-1}(c_n)$. Then $\psi_N(g_N^n(b'_n)) = c_N = \psi_N(b_N)$, so there is an a_N such that $g_N^n(b'_n) - b_N = \varphi_N(a_N)$. Choose a_n with $f_k^n(a_n) = f_k^N(a_N)$. Then $b_n := b'_n - \varphi_n(a_n)$ has the required properties. $\square$

For a general abelian category $\mathcal{A}$ satisfying Hypothesis 3.13.2, there are counterexamples in which the Mittag-Leffler condition does not imply $\lim^1 = 0$; see [38] for a detailed discussion.

The following elementary exercise will be used in §3.15.

Corollary 3.13.14 Suppose that $\mathcal{A}$ satisfies Hypothesis 3.13.2, and let $A \to B \to C \to D$ be an exact sequence in $\mathsf{InvSys}(\mathcal{A})$. If $\lim^1 A = 0$, then

$$\varprojlim B \to \varprojlim C \to \varprojlim D$$

is exact in $\mathcal{A}$.

Proof Put $H := \ker[C \to D]$ and $X := \operatorname{im}[A \to B]$. Since $\varprojlim$ is left exact, there is a canonical isomorphism

$$\ker\left[\varprojlim C \to \varprojlim D\right] \simeq \varprojlim H.$$

The morphism $B \to C$ factors as $B \twoheadrightarrow H \hookrightarrow C$, so it remains to prove that $\varprojlim B \to \varprojlim H$ is an epimorphism. Example 3.13.9 gives $\lim^1 X = 0$. Applying $\varprojlim$ to $0 \to X \to B \to H \to 0$ then shows that $\varprojlim B \to \varprojlim H$ is an epimorphism. $\square$

3.14 Example: Ext and Tor

We shall study the derived functors of Hom and $\otimes$. It is useful to introduce a few auxiliary concepts first.

Definition 3.14.1 Let $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{B}$ be abelian categories, and let the bifunctor $F : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}$ (Convention 3.5.9) be left exact in both variables. If, for every injective object $I_i \in \operatorname{Ob}(\mathcal{A}_i)$ $(i = 1, 2)$, both functors

$$F(I_1, \cdot) : \mathcal{A}_2 \to \mathcal{B} \quad \text{and} \quad F(\cdot, I_2) : \mathcal{A}_1 \to \mathcal{B}$$

are exact, then F is called **balanced**.

Dually, if F is right exact in both variables, and if for every projective object $P_i \in \operatorname{Ob}(\mathcal{A}_i)$ the functors $F(P_1, \cdot)$ and $F(\cdot, P_2)$ are both exact, then F is again called balanced.

By duality, the following theorem is stated only for bifunctors that are left exact in both variables.

Theorem 3.14.2 Let $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{B}$ be abelian categories, and suppose that $\mathcal{A}_1$ and $\mathcal{A}_2$ have enough injective objects. Let the bifunctor $F : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}$ be left exact in both variables. We may therefore take right derived functors in either variable. For fixed $X_i \in \mathrm{Ob}(\mathcal{A}_i)$, $i = 1, 2$, write

$$\mathrm{R}^n_{\mathrm{I}} F(\cdot, X_2) : \mathcal{A}_1 \to \mathcal{B}, \qquad \mathrm{R}^n_{\mathrm{II}} F(X_1, \cdot) : \mathcal{A}_2 \to \mathcal{B}.$$

If F is balanced, then there is a canonical isomorphism of bifunctors

$$\mathrm{R}^n_{\mathrm{I}} F(X_1, X_2) \simeq \mathrm{R}^n_{\mathrm{II}} F(X_1, X_2).$$

Proof The following argument relies on the theory of double complexes; see §3.5 and §3.10. For $i = 1, 2$, choose an injective resolution $0 \to X_i \to I_i^0 \to I_i^1 \to \cdots$; define $I_i^n := 0$ for $n < 0$. This defines the double complex $F(I_1^\bullet, I_2^\bullet)$.

On the other hand, form the complexes $F(X_1, I_2^\bullet)$ and $F(I_1^\bullet, X_2)$. Regard them as double complexes concentrated, respectively, on the vertical axis $(0, \bullet)$ and the horizontal axis $(\bullet, 0)$. Since $X_i \xrightarrow{\sim} \ker[I_i^0 \to I_i^1] \hookrightarrow I_i^0$, there are morphisms in $\mathsf{C}^2_f(\mathcal{B})$

$$F(X_1, I_2^\bullet) \xrightarrow{\lambda} F(I_1^\bullet, I_2^\bullet) \xleftarrow{\rho} F(I_1^\bullet, X_2).$$

We shall show that $\mathrm{tot}(\lambda)$ and $\mathrm{tot}(\rho)$ are quasi-isomorphisms.

First consider λ. Take horizontal cohomology H_{I} on both sides; this leaves $F(X_1, I_2^\bullet)$, which lies on the vertical axis, unchanged. Since $F(\cdot, I_2^q)$ is exact for every q, we obtain

$$(\mathrm{H}_{\mathrm{I}}\, F(I_1^\bullet, I_2^\bullet))^{p,q} = \begin{cases} 0, & p \neq 0, \\ \ker\left[F(I_1^0, I_2^q) \to F(I_1^1, I_2^q)\right] = F(X_1, I_2^q), & p = 0. \end{cases}$$

Thus $\mathrm{H}_{\mathrm{I}}(\lambda)$ is an isomorphism from $\mathrm{H}_{\mathrm{I}}\, F(X_1, I_2^\bullet)$ to $\mathrm{H}_{\mathrm{I}}\, F(I_1^\bullet, I_2^\bullet)$, so $\mathrm{H}_{\mathrm{II}}\, \mathrm{H}_{\mathrm{I}}(\lambda)$ is also an isomorphism. Theorem 3.10.6 now shows that $\mathrm{tot}(\lambda)$ is a quasi-isomorphism.

For ρ, the same argument shows that $\mathrm{H}_{\mathrm{I}}\, \mathrm{H}_{\mathrm{II}}(\rho)$ is an isomorphism; Theorem 3.10.6 then implies that $\mathrm{tot}(\rho)$ is also a quasi-isomorphism.

The total complex of $F(X_1, I_2^\bullet)$ is still that complex itself, and taking H^n gives $\mathrm{R}^n_{\mathrm{II}} F(X_1, X_2)$. Likewise, taking H^n of $F(I_1^\bullet, X_2)$, or of its total complex, gives $\mathrm{R}^n_{\mathrm{I}} F(X_1, X_2)$. Using $\mathrm{tot}(F(I_1^\bullet, I_2^\bullet))$ as an intermediary yields

$$\mathrm{R}^n_{\mathrm{I}} F(X_1, X_2) \simeq \mathrm{R}^n_{\mathrm{II}} F(X_1, X_2).$$

Because every morphism $X_i \to Y_i$ lifts to a morphism between injective resolutions, uniquely up to homotopy, what we have obtained is in fact an isomorphism of bifunctors $\mathrm{R}^n_{\mathrm{I}} F \simeq \mathrm{R}^n_{\mathrm{II}} F$. □

Remark 3.14.3 The argument actually gives a method for resolving both variables simultaneously and then deriving F in a balanced way by means of the total complex of the double complex $F(I_1^\bullet, I_2^\bullet)$.

Now let $\mathcal{A}$ be an abelian category and consider the bifunctor $\mathrm{Hom} = \mathrm{Hom}_{\mathcal{A}} : \mathcal{A}^{\mathrm{op}} \times \mathcal{A} \to \mathsf{Ab}$. Proposition 2.8.11 says that Hom is left exact in both variables. For every $n \in \mathbb{Z}$, make the following definitions.

- If $\mathcal{A}$ has enough projective objects, define $\mathrm{Ext}^n_{\mathcal{A},\mathrm{I}}(X,Y) := \mathrm{R}^n_{\mathrm{I}} \mathrm{Hom}(X,Y)$, computed using a projective resolution of X in $\mathcal{A}$.
- If $\mathcal{A}$ has enough injective objects, define $\mathrm{Ext}^n_{\mathcal{A},\mathrm{II}}(X,Y) := \mathrm{R}^n_{\mathrm{II}} \mathrm{Hom}(X,Y)$, computed using an injective resolution of Y in $\mathcal{A}$.

Definition–Proposition 3.14.4 (The Ext functor) The bifunctor $\mathrm{Hom}_{\mathcal{A}}$ is balanced. Consequently, whenever the abelian category $\mathcal{A}$ has enough injective and projective objects, there is, for every $n \in \mathbb{Z}$, a bifunctor

$$\mathrm{Ext}^n = \mathrm{Ext}^n_{\mathcal{A}} : \mathcal{A}^{\mathrm{op}} \times \mathcal{A} \to \mathsf{Ab}$$

such that $\mathrm{Ext}^n_{\mathcal{A}}(X,Y) := \mathrm{Ext}^n_{\mathrm{I}}(X,Y) \simeq \mathrm{Ext}^n_{\mathrm{II}}(X,Y)$ (canonically).

Proof Balancedness is an immediate consequence of the definitions of injective and projective objects, so Theorem 3.14.2 applies. □

If R is a ring and $\mathcal{A} = R\text{-}\mathsf{Mod}$ or $\mathsf{Mod}\text{-}R$, we write $\mathrm{Ext}^n_{\mathcal{A}}$ as Ext^n_R.

Observe that $\mathrm{Ext}^0 = \mathrm{Hom}$, while $(\mathrm{Ext}^n)_{n\geq 0}$ has, in the variables X and Y respectively, long exact sequences of the form

$$\cdots \to \mathrm{Ext}^{n-1}(X',Y) \xrightarrow{\delta^{n-1}} \mathrm{Ext}^n(X'',Y) \to \mathrm{Ext}^n(X,Y) \to \mathrm{Ext}^n(X',Y) \to \cdots,$$
$$\cdots \to \mathrm{Ext}^{n-1}(X,Y'') \xrightarrow{\delta^{n-1}} \mathrm{Ext}^n(X,Y') \to \mathrm{Ext}^n(X,Y) \to \mathrm{Ext}^n(X,Y'') \to \cdots,$$

where $0 \to X' \to X \to X'' \to 0$ and $0 \to Y' \to Y \to Y'' \to 0$ are given short exact sequences.

If $\mathcal{A}$ is a $\Bbbk$-linear abelian category, the values of Ext^n may naturally be regarded as objects of $\Bbbk\text{-}\mathsf{Mod}$.

Remark 3.14.5 The symbol Ext stands for "extension," because the elements of $\mathrm{Ext}^1(X,Y)$ correspond bijectively to the isomorphism classes of extensions $0 \to Y \to E \to X \to 0$ (that is, short exact sequences); the higher Ext^n have a similar interpretation. This fact is best understood through the derived category; see §4.5.

The characterization of exact functors in Corollary 3.12.7 immediately gives the following.

- ◇ An object I is injective $\iff \operatorname{Ext}^1(\cdot, I) = 0 \iff \operatorname{Ext}^{\geq 1}(\cdot, I) = 0$.
- ◇ An object P is projective $\iff \operatorname{Ext}^1(P, \cdot) = 0 \iff \operatorname{Ext}^{\geq 1}(P, \cdot) = 0$.

Example 3.14.6 (HH^n as Ext^n) Return to the basic setting for Hochschild homology and cohomology from §3.8: $\Bbbk$ is a commutative ring and R is a $\Bbbk$-algebra; now assume in addition that R is projective as a $\Bbbk$-module. Write $\otimes := \otimes_{\Bbbk}$. Since projective modules are direct summands of free modules, $R^{\otimes n}$ is still projective as a $\Bbbk$-module. [38] Regard $R \otimes R^{\otimes n} \otimes R$ as a left R^e-module, where $R^e := R \otimes R^{\mathrm{op}}$. Since $R \otimes (\cdot) \otimes R : \Bbbk\text{-}\mathsf{Mod} \to R^e\text{-}\mathsf{Mod}$ is left adjoint to the forgetful functor [51, Corollary 6.6.8], it carries projective objects to projective objects (Proposition 2.8.17). Thus Lemma 3.8.2 shows that the bar complex $\mathsf{B}R$ gives a projective resolution of R as a left R^e-module,

$$\cdots \to \mathsf{B}_1 R \to \mathsf{B}_0 R \to R \to 0. \tag{3.14.1}$$

Now let M be an (R, R)-bimodule. Substituting the original definition of Hochschild cohomology from Definition 3.8.4 gives

$$\mathrm{HH}^n(M) = \mathrm{H}^n\left(\operatorname{Hom}_{R^e}(\mathsf{B}R, M)\right) = \operatorname{Ext}^n_{R^e}(R, M).$$

We next turn to the Tor functor. Let R be a ring. The module categories R-Mod (and Mod-R) have enough projective objects[39]. The tensor product gives a bifunctor

$$\otimes_R : \mathsf{Mod}\text{-}R \times R\text{-}\mathsf{Mod} \to \mathsf{Ab},$$

which is right exact in either variable (see [51, Proposition 6.9.2]). Apply the right-exact version of the preceding theory to define the left derived functors in both variables, $\mathrm{L}_{\mathrm{I},n}\otimes_R$ and $\mathrm{L}_{\mathrm{II},n}\otimes_R$; these can be nonzero only for $n \geq 0$. Recall also the definition of a flat module [51, Definition 6.9.4].

Definition–Proposition 3.14.7 (The Tor functor) For every ring R, the bifunctor $\otimes_R$ is balanced. Hence, for every $n \in \mathbb{Z}$, one may define a bifunctor

$$\operatorname{Tor}_n = \operatorname{Tor}^R_n : \mathsf{Mod}\text{-}R \times R\text{-}\mathsf{Mod} \to \mathsf{Ab}$$

by $\operatorname{Tor}^R_n(X, Y) := (\mathrm{L}_{\mathrm{I},n}\otimes_R)(X, Y) \simeq (\mathrm{L}_{\mathrm{II},n}\otimes_R)(X, Y)$ (canonically).

Proof Balancedness follows from the fact that every projective module is flat [51, Corollary 6.9.9]. Apply the left-derived-functor version of Theorem 3.14.2. □

[38] Alternatively, use the tensor adjunction [51, Theorem 6.6.5].

[39] Indeed, free modules are always projective; see [51, Example 6.9.6].

We have $\operatorname{Tor}_0^R(X,Y) = X \underset{R}{\otimes} Y$, while $(\operatorname{Tor}_n)_{n\geq 0}$ has the following long exact sequences:

$$\cdots \to \operatorname{Tor}_{n+1}^R(X'',Y) \xrightarrow{\partial_{n+1}} \operatorname{Tor}_n^R(X',Y) \to \operatorname{Tor}_n^R(X,Y) \to \operatorname{Tor}_n^R(X'',Y) \to \cdots,$$
$$\cdots \to \operatorname{Tor}_{n+1}^R(X,Y'') \xrightarrow{\partial_{n+1}} \operatorname{Tor}_n^R(X,Y') \to \operatorname{Tor}_n^R(X,Y) \to \operatorname{Tor}_n^R(X,Y'') \to \cdots,$$

where $0 \to X' \to X \to X'' \to 0$ and $0 \to Y' \to Y \to Y'' \to 0$ are given short exact sequences.

As in the case of Ext, we obtain the following characterizations.

- ⋄ A right R-module M is flat $\iff \operatorname{Tor}_1^R(M,\cdot) = 0 \iff \operatorname{Tor}_{\geq 1}^R(M,\cdot) = 0$.
- ⋄ A left R-module N is flat $\iff \operatorname{Tor}_1^R(\cdot,N) = 0 \iff \operatorname{Tor}_{\geq 1}^R(\cdot,N) = 0$.

Let M be an R-module. In analogy with the definition of a projective resolution, a **flat resolution** of M is an exact sequence of the form

$$\cdots \to F^2 \to F^1 \to F^0 \to M \to 0, \qquad \text{each } F^n \text{ is a flat } R\text{-module.}$$

By Corollary 3.12.10 and the preceding observation, $\operatorname{Tor}_n^R(X,Y)$ can also be computed using a flat resolution of X or Y. This is the practical technique for computing Tor, whereas projective resolutions play a chiefly theoretical role.

Such computations can naturally be balanced as well. Choose arbitrary flat resolutions $\cdots \to P_1 \to P_0 \to X \to 0$ and $\cdots \to Q_1 \to Q_0 \to Y \to 0$. Form the double complex, in the sense of chain complexes, $P_\bullet \underset{R}{\otimes} Q_\bullet$, and denote its total complex by $P \otimes Q$. The homological version of Remark 3.14.3 gives $\operatorname{Tor}_n^R(X,Y) \simeq \mathrm{H}_n(P \otimes Q)$; the only properties needed are the exactness of $P_p \underset{R}{\otimes} (\cdot)$ and $(\cdot) \underset{R}{\otimes} Q_q$, precisely the properties guaranteed by flatness.

Remark 3.14.8 (Bimodule structure) Let A and B be rings, let X be an (A,R)-bimodule, and let Y be an (R,B)-bimodule. Since Tor_n^R is a bifunctor, $\operatorname{Tor}_n^R(X,Y)$ carries a canonical (A,B)-bimodule structure. For example, consider left multiplication by A: every $a \in A$ gives an endomorphism of X as a right R-module and hence induces an endomorphism L_a of $\operatorname{Tor}_n^R(X,Y)$; one has $L_a L_{a'} = L_{aa'}$ and $L_1 = \mathrm{id}$. Right multiplication by B is handled similarly in the second variable. For $n = 0$, this recovers the natural bimodule structure on $X \underset{R}{\otimes} Y$.

If Y, as a left R-module, is supplied with a flat resolution $\cdots \to P_0 \to Y \to 0$, the left A-action can be read directly from the complex $X \underset{R}{\otimes} P_\bullet$: it comes from left multiplication by A on X. Right multiplication is treated in the same way.

As a special case, if a ring homomorphism $R \to S$ is fixed, then $\operatorname{Tor}_n^R(\cdot,S)$ (or $\operatorname{Tor}_n^R(S,\cdot)$) naturally becomes a functor $\mathsf{Mod}\text{-}R \to \mathsf{Mod}\text{-}S$ (or $R\text{-}\mathsf{Mod} \to S\text{-}\mathsf{Mod}$).

A similar argument shows that, when X is an (R, A)-bimodule and Y is an (R, B)-bimodule, every $\operatorname{Ext}^n_R(X, Y)$ naturally carries an (A, B)-bimodule structure.

For a commutative ring R, there is no need to distinguish left from right R-modules; the corresponding category is denoted by $R\text{-}\mathsf{Mod}$. Every R-module is then naturally an (R, R)-bimodule, so Tor^R_n becomes a bifunctor $(R\text{-}\mathsf{Mod})^2 \to R\text{-}\mathsf{Mod}$.

Proposition 3.14.9 For a commutative ring R, there is a canonical isomorphism of bifunctors

$$\operatorname{Tor}^R_n(X, Y) \simeq \operatorname{Tor}^R_n(Y, X)$$

for every $n \in \mathbb{Z}$.

Proof Choose flat resolutions $P_\bullet$ and $Q_\bullet$ of X and Y, respectively. Write $P \otimes Q$ for the total complex of the double complex $P_\bullet \underset{R}{\otimes} Q_\bullet$, and define $Q \otimes P$ similarly. The symmetry constraint for the tensor product [51, Proposition 6.5.14] gives canonical isomorphisms

$$P_p \underset{R}{\otimes} Q_q \simeq Q_q \underset{R}{\otimes} P_p, \qquad p, q \in \mathbb{Z}.$$

In other words, $P_\bullet \underset{R}{\otimes} Q_\bullet \simeq \operatorname{swap}(Q_\bullet \underset{R}{\otimes} P_\bullet)$. Apply Proposition 3.5.6 to obtain a canonical isomorphism of total complexes $P \otimes Q \simeq Q \otimes P$. This proves the assertion. $\square$

Remark 3.14.10 The symbol Tor stands for "torsion," as suggested by the following example. Let $\mathfrak{a}$ be a left ideal of a ring R, and let X be any right R-module. Since the free left module R is flat, apply Proposition 3.12.9 to $0 \to \mathfrak{a} \to R \to R/\mathfrak{a} \to 0$ to shift dimensions. This gives

$$\begin{aligned}
\operatorname{Tor}^R_1(X, R/\mathfrak{a}) &\simeq \ker\left[X \underset{R}{\otimes} \mathfrak{a} \to X \underset{R}{\otimes} R \simeq X\right] \\
&\simeq \left\{\sum_i x_i \otimes a_i \in X \underset{R}{\otimes} \mathfrak{a} : \sum_i x_i a_i = 0\right\}, \\
\operatorname{Tor}^R_n(X, R/\mathfrak{a}) &\simeq \operatorname{Tor}^R_{n-1}(X, \mathfrak{a}), \qquad (n > 1).
\end{aligned}$$

If, moreover, $t \in R$ is not a right zero divisor and we take $\mathfrak{a} := Rt \xleftarrow{\sim} R$ (mapping r to rt), then $\operatorname{Tor}^R_1(X, R/Rt) \simeq \{x \in X : xt = 0\}$, the t-torsion submodule of X; for $n > 1$, $\operatorname{Tor}^R_n(X, R/Rt) = 0$.

Example 3.14.11 (HH_n as Tor_n) Continue from the opening setup of Example 3.14.6, but weaken the condition on R by requiring only that R be flat as a $\Bbbk$-module. Then $R^{\otimes n}$ is flat as well. The associativity constraint $(\cdot) \underset{R^e}{\otimes} (R \otimes R^{\otimes n} \otimes R) \simeq (\cdot) \otimes R^{\otimes n}$ shows that $R \otimes R^{\otimes n} \otimes R$ is a flat left R^e-module. Thus $\mathsf{B}R$ is a flat resolution of R as a left R^e-module, as in (3.14.1).

For every (R, R)-bimodule M, the original definition of Hochschild homology from Definition 3.8.4 gives

$$\mathrm{HH}_n(M) = \mathrm{H}_n\left(M \underset{R^e}{\otimes} \mathsf{B}R\right) = \mathrm{Tor}_n^{R^e}(M, R).$$

We can now combine Examples 3.14.6 and 3.14.11 to compute further examples of Hochschild homology and cohomology. First take the polynomial algebra $R = \Bbbk[t]$ and identify $R^e = R \otimes R^{\mathrm{op}}$ with $\Bbbk[x, y]$. The scalar action on R as an R^e-module is $h(x, y) \cdot f(t) = h(t, t)f(t)$, and R has the free resolution

$$0 \to R^e \xrightarrow{\text{multiplication by } x-y} R^e \xrightarrow{f \mapsto f(t,t)} R \to 0.$$

Computing $\mathrm{Ext}^n_{R^e}(R, \cdot)$ and $\mathrm{Tor}_n^{R^e}(\cdot, R)$ from this resolution immediately gives $\mathrm{HH}_n(R) \simeq R \simeq \mathrm{HH}^n(R)$ for $n \in \{0, 1\}$, while $\mathrm{HH}_n(R) = 0 = \mathrm{HH}^n(R)$ for $n > 1$.

Next take $R = \Bbbk[t]/(t^2)$, so that $R^e = \Bbbk[x, y]/(x^2, y^2)$. As an R^e-module, R has a free resolution of period 2,

$$\cdots \xrightarrow{x+y} R^e \xrightarrow{x-y} R^e \xrightarrow{x+y} R^e \xrightarrow{\text{multiplication by } x-y} R^e \xrightarrow{f \mapsto f(t,t)} R \to 0,$$

whose exactness is left to the reader to verify directly. For every R-module M, applying $M \underset{R^e}{\otimes} (\cdot)$ and $\mathrm{Hom}_{R^e}(\cdot, M)$ gives, respectively,

$$\cdots \xrightarrow{2t} M \xrightarrow{0} M \xrightarrow{2t} M \xrightarrow{0} M \to M \underset{R^e}{\otimes} R \to 0,$$

$$0 \to \mathrm{Hom}_{R^e}(R, M) \to M \xrightarrow{0} M \xrightarrow{2t} M \xrightarrow{0} M \xrightarrow{2t} \cdots.$$

Thus $\mathrm{HH}_k(M)$ and $\mathrm{HH}^k(M)$ depend only on k mod 2; this periodicity might be difficult to see from the definitions alone.

If $\Bbbk$ is a field, $\mathrm{HH}_n(M)$ and $\mathrm{HH}^n(M)$ can be read off from these complexes, but the answer depends on $\mathrm{char}(\Bbbk)$. The exercises will treat the general case $R = \Bbbk[t]/(t^n)$.

The following result is often used in algebraic topology and also provides a small check on the preceding material; it concerns chain complexes and their homology (Remark 2.2.6). Let $C = (C_\bullet, d_\bullet^C)$ (respectively, $D = (D_\bullet, d_\bullet^D)$) be a chain complex of right R-modules (respectively, left R-modules). Form the double chain complex $C_\bullet \otimes D_\bullet$ and denote its total complex by

$$C \otimes D := \mathrm{tot}_\oplus (C_\bullet \otimes D_\bullet);$$

the subscript R on the tensor product is omitted. Concretely, the degree-n term of the total complex $C \otimes D$ is $\bigoplus_{p+q=n} C_p \otimes D_q$, while the restriction of d_n to $C_p \otimes D_q$ is $d_p^C \otimes \mathrm{id} + (-1)^p \mathrm{id} \otimes d_q^D$. This gives a canonical homomorphism

$$\kappa : \bigoplus_{p+q=n} \mathrm{H}_p(C) \otimes \mathrm{H}_q(D) \to \mathrm{H}_n(C \otimes D).$$

Theorem 3.14.12 (Künneth theorem for homology) Let C, D be chain complexes as above. If every C_p and every $\operatorname{im}(d_p^C)$ is a flat right R-module, then there is a canonical short exact sequence

$$0 \longrightarrow \bigoplus_{p+q=n} \mathrm{H}_p(C) \otimes \mathrm{H}_q(D) \xrightarrow{\kappa} \mathrm{H}_n(C \otimes D) \longrightarrow \bigoplus_{p+q=n-1} \operatorname{Tor}_1^R(\mathrm{H}_p(C), \mathrm{H}_q(D)) \longrightarrow 0.$$

Proof For every $p \in \mathbb{Z}$, define submodules of C_p by $B_p := \operatorname{im}(d_{p+1}^C)$ and $Z_p := \ker(d_p^C)$. Regard these as chain complexes $Z = Z_\bullet$ and $B = B_\bullet$ by setting $d^Z = d^B := 0$. There is then a short exact sequence of chain complexes

$$0 \longrightarrow Z \longrightarrow C \xrightarrow{d^C = (d_p^C)_p} B[-1] \longrightarrow 0.$$

The short exact sequence $0 \to Z_p \to C_p \xrightarrow{d_p^C} B_{p-1} \to 0$ and the long exact sequence for Tor imply that Z_p is also flat. Since B_{p-1} is flat, the sequence

$$0 \to Z_p \otimes D_q \to C_p \otimes D_q \to B_{p-1} \otimes D_q \to 0$$

remains exact. Since direct sums of modules, including infinite direct sums, preserve exactness, we obtain a short exact sequence of chain complexes $0 \to Z \otimes D \to C \otimes D \xrightarrow{d^C \otimes \mathrm{id}_D} B[-1] \otimes D \to 0$.

Because $d^Z = d^{B[-1]} = 0$, the associated long exact sequence in homology simplifies, by flatness, to

$$\overbrace{\bigoplus_{p+q=n+1} B_{p-1} \otimes \mathrm{H}_q(D)}^{=\bigoplus_{p+q=n} B_p \otimes \mathrm{H}_q(D)} \xrightarrow{\partial_{n+1}} \bigoplus_{p+q=n} Z_p \otimes \mathrm{H}_q(D) \to \mathrm{H}_n(C \otimes D)$$
$$\to \underbrace{\bigoplus_{p+q=n} B_{p-1} \otimes \mathrm{H}_q(D)}_{=\bigoplus_{p+q=n-1} B_p \otimes \mathrm{H}_q(D)} \xrightarrow{\partial_n} \bigoplus_{p+q=n-1} Z_p \otimes \mathrm{H}_q(D).$$

It remains only to explain the connecting morphisms ∂_{n+1} and ∂_n. Up to possible signs, they arise from the inclusions $B_p \hookrightarrow Z_p$. To see this, return to the explicit construction of the long exact sequence, namely the homological version of Proposition 3.6.4; here one can use a diagram chase, as in the discussion preceding [51, Proposition 6.8.6]. The details are left to the reader.

It follows that $\operatorname{coker}(\partial_{n+1})$ may be identified with $\bigoplus_{p+q=n} \mathrm{H}_p(C) \otimes \mathrm{H}_q(D)$, and the induced monomorphism to $\mathrm{H}_n(C \otimes D)$ is κ.

Finally, $0 \to B_p \to Z_p \to \mathrm{H}_p(C) \to 0$ is a flat resolution of $\mathrm{H}_p(C)$. Hence $\ker(\partial_n) \simeq \bigoplus_{p+q=n-1} \operatorname{Tor}_1^R(\mathrm{H}_p(C), \mathrm{H}_q(D))$. This gives the asserted short exact sequence. $\square$

Notice that if R is a division ring, then κ is necessarily an isomorphism.

Because D plays the role of homology coefficients in topology, the special case of Theorem 3.14.12 in which D is concentrated in degree zero is also called the **universal coefficient theorem**. Results of this kind have many variants. Moreover, the ring structure supplies operations on Tor that are absent for other derived bifunctors. These structures are better formulated in the language of derived categories or spectral sequences.

3.15 K-injective and K-projective complexes

Injective and projective resolutions of complexes supplied the initial definition of derived functors in §3.12. That construction requires bounded-below or bounded-above complexes, whereas applications involving unbounded complexes and their derived categories require a broader class of resolutions. The theory of K-injective and K-projective resolutions originates with [41]. This section provides some preparation for that theory; the arguments follow [42], and the reader may also consult [23, Chapter 14] or [5].

Definition 3.15.1 (N. Spaltenstein) Let $\mathcal{A}$ be an abelian category.

◇ A complex $I \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$ is called a **K-injective complex**[40)] if, for every acyclic complex $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$, the complex $\mathrm{Hom}^\bullet(X, I)$ is also acyclic.

  If $f : A \to I$ is a quasi-isomorphism in $\mathsf{C}(\mathcal{A})$ and I is K-injective, then f is called a **K-injective resolution** of A.

◇ A complex $P \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$ is called a **K-projective complex** if, for every acyclic complex $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$, the complex $\mathrm{Hom}^\bullet(P, X)$ is also acyclic.

  If $f : P \to A$ is a quasi-isomorphism in $\mathsf{C}(\mathcal{A})$ and P is K-projective, then f is called a **K-projective resolution** of A.

If every object of $\mathsf{C}(\mathcal{A})$ has a K-injective resolution (respectively, a K-projective resolution), then $\mathcal{A}$ is said to have **enough K-injective complexes** (respectively, **enough K-projective complexes**).

The two sets of definitions are plainly dual. In practice, the following criterion is often easier to use.

40) A perhaps more natural term would be homotopically injective complex, and similarly homotopically projective complex.

Lemma 3.15.2 A complex I (respectively, P) is K-injective (respectively, K-projective) if and only if, for every acyclic complex X, $\operatorname{Hom}_{\mathsf{K}(\mathcal{A})}(X, I) = 0$ (respectively, $\operatorname{Hom}_{\mathsf{K}(\mathcal{A})}(P, X) = 0$).

Proof Let $n \in \mathbb{Z}$. Lemma 3.2.3 gives

$$\mathrm{H}^n \operatorname{Hom}^\bullet(X, I) \simeq \operatorname{Hom}_{\mathsf{K}(\mathcal{A})}(X[-n], I), \quad \mathrm{H}^n \operatorname{Hom}^\bullet(P, X) \simeq \operatorname{Hom}_{\mathsf{K}(\mathcal{A})}(P, X[n]);$$

but X is acyclic if and only if $X[-n]$ is acyclic, and this is also equivalent to $X[n]$ being acyclic. □

Example 3.15.3 Lemma 3.11.6 can immediately be restated as follows.

- ⋄ If $I \in \operatorname{Ob}(\mathsf{C}^+(\mathcal{A}))$ consists of injective objects, then I is K-injective.
- ⋄ If $P \in \operatorname{Ob}(\mathsf{C}^-(\mathcal{A}))$ consists of projective objects, then P is K-projective.

Thus an injective (respectively, projective) resolution is a special case of a K-injective (respectively, K-projective) resolution.

Example 3.15.4 (A. Dold) The one-sided boundedness condition in Example 3.15.3 is essential. For example, take $\mathcal{A} = \mathbb{Z}/4\mathbb{Z}\text{-}\mathsf{Mod}$ and consider the acyclic complex Q in it:

$$\cdots \xrightarrow{2} \mathbb{Z}/4\mathbb{Z} \xrightarrow{2} \mathbb{Z}/4\mathbb{Z} \xrightarrow{2} \mathbb{Z}/4\mathbb{Z} \xrightarrow{2} \cdots .$$

Every term is a free module. On the other hand, the termwise tensor product $Q \underset{\mathbb{Z}/4\mathbb{Z}}{\otimes} \mathbb{Z}/2\mathbb{Z}$ is the complex

$$\cdots \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \cdots .$$

Therefore Q is not K-projective. Indeed, if Q were K-projective, then id_Q would be homotopic to 0, and hence so would $\mathrm{id}_{Q \otimes \mathbb{Z}/2\mathbb{Z}}$, contradicting $\mathrm{H}^n(Q \underset{\mathbb{Z}/4\mathbb{Z}}{\otimes} \mathbb{Z}/2\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$.

From another point of view, this example also shows that an unbounded complex consisting of projective objects need not be an appropriate resolution: $0 \to Q$ and $0 \to 0$ are both quasi-isomorphisms, but the preceding paragraph shows that id_Q is not homotopic to 0. Thus Q and 0 are not isomorphic in $\mathsf{K}(\mathcal{A})$, and Theorem 3.11.5 cannot simply be extended to the unbounded case.

The following result is the natural generalization of Theorem 3.11.5; its proof is likewise postponed until the discussion following Theorem 4.4.1.

Theorem 3.15.5 Let $\alpha : X \to Y$ be a quasi-isomorphism in $\mathsf{C}(\mathcal{A})$.

- ⋄ Given a morphism $\gamma : X \to I$, where I is a K-injective complex, there is a commutative diagram in $\mathsf{K}(\mathcal{A})$

$$\begin{tikzcd} & Y \arrow[dd, "\beta"] \\ X \arrow[ur, "\alpha"] \arrow[dr, "\gamma"'] & \\ & I \end{tikzcd}$$

- ◇ Given a morphism $\gamma : P \to Y$, where P is a K-projective complex, there is a commutative diagram in $\mathsf{K}(\mathcal{A})$

$$\begin{tikzcd} & X \arrow[dl, "\alpha"'] \\ Y & \\ & P \arrow[uu, "\beta"'] \arrow[ul, "\gamma"] \end{tikzcd}$$

In both cases, β is unique as a morphism in $\mathsf{K}(\mathcal{A})$.

Proposition 3.15.6 The equivalence σ of Definition–Proposition 3.4.1 has the following property: $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}^{\mathrm{op}}))$ is K-injective (respectively, K-projective) if and only if $\sigma X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A})^{\mathrm{op}}) = \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$ is K-projective (respectively, K-injective).

Proof The functor σ preserves acyclic complexes and induces $\mathsf{K}(\mathcal{A}^{\mathrm{op}}) \xrightarrow{\sim} \mathsf{K}(\mathcal{A})^{\mathrm{op}}$ (Proposition 3.4.3); hence σ preserves the criterion of Lemma 3.15.2. □

Next consider adjoint functors. Proposition 2.8.17 has a corresponding version in this setting.

Proposition 3.15.7 Consider a pair of functors between abelian categories $\mathcal{A} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{B}$, and suppose that F is exact.

- ◇ If G is left adjoint to F, then $\mathsf{K}G : \mathsf{K}(\mathcal{B}) \to \mathsf{K}(\mathcal{A})$ carries K-projective complexes to K-projective complexes.
- ◇ If G is right adjoint to F, then $\mathsf{K}G : \mathsf{K}(\mathcal{B}) \to \mathsf{K}(\mathcal{A})$ carries K-injective complexes to K-injective complexes.

Proof By duality it suffices to consider the case in which G is a left adjoint. In this case G is automatically additive (Corollary 1.3.6). Let $P \in \mathrm{Ob}(\mathsf{C}(\mathcal{B}))$ be K-projective. For every acyclic complex $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$, Proposition 3.2.11 gives an isomorphism

$$\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(\mathsf{K}G(P), X) \simeq \mathrm{Hom}_{\mathsf{K}(\mathcal{B})}(P, \mathsf{K}F(X)).$$

Since F is exact, $\mathsf{K}F(X)$ is acyclic, so the right-hand side is 0. Apply Lemma 3.15.2.□

The next task in this section is to investigate the existence of K-injective or K-projective resolutions. The argument is somewhat intricate; we begin with the K-injective case.

Lemma 3.15.8 Consider a sequence of complexes over $\mathcal{A}$, $\cdots \to I_2 \to I_1 \to I_0$. Suppose that:

- every I_k is K-injective;
- for every k and n, the morphism $I^n_{k+1} \to I^n_k$ is an epimorphism admitting a section (in other words, it has a right inverse; see the discussion after Proposition 2.5.3);
- for every n, the object $I^n := \varprojlim_k I^n_k$ exists in $\mathcal{A}$.

Then the induced differentials $d^n_I := \varprojlim_k d^n_{I_k} : I^n \to I^{n+1}$ make $I := (I^n, d^n_I)_n$ a K-injective complex.

Proof Let $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$ be acyclic. For each $k \in \mathbb{Z}_{\geq 0}$, take the following portion of the complex $\mathrm{Hom}^\bullet(X, I_k)$:

$$\begin{array}{ccccccc}
\prod_n \mathrm{Hom}\left(X^n, I_k^{n-2}\right) & \to & \prod_n \mathrm{Hom}\left(X^n, I_k^{n-1}\right) & \to & \prod_n \mathrm{Hom}\left(X^n, I_k^{n}\right) & \to & \prod_n \mathrm{Hom}\left(X^n, I_k^{n+1}\right) \\
\| & & \| & & \| & & \| \\
\mathrm{Hom}^{-2}(X, I_k) & & \mathrm{Hom}^{-1}(X, I_k) & & \mathrm{Hom}^{0}(X, I_k) & & \mathrm{Hom}^{1}(X, I_k)
\end{array} \tag{3.15.1}$$

This sequence is exact because I_k is K-injective. Now consider the homomorphism induced by $I_{k+1} \to I_k$,

$$\mathrm{Hom}\left(X^n, I^{n-2}_{k+1}\right) \to \mathrm{Hom}\left(X^n, I^{n-2}_{k}\right).$$

Since there is a section $I^{n-2}_k \to I^{n-2}_{k+1}$, this homomorphism is an epimorphism. The same remains true after taking $\prod_n$. Thus, as k varies, the leftmost term in (3.15.1) satisfies the Mittag–Leffler condition (Definition 3.13.11).

Now let $k \in \mathbb{Z}_{\geq 0}$ vary in (3.15.1). Corollary 3.13.14 gives an exact sequence

$$\varprojlim_k \prod_n \mathrm{Hom}\left(X^n, I_k^{n-1}\right) \longrightarrow \varprojlim_k \prod_n \mathrm{Hom}\left(X^n, I_k^{n}\right) \longrightarrow \varprojlim_k \prod_n \mathrm{Hom}\left(X^n, I_k^{n+1}\right)$$

Interchanging $\varprojlim_k$ and $\prod_n$ now gives an exact sequence[41)]

$$\begin{array}{ccccc}
\prod_n \varprojlim_k \mathrm{Hom}\left(X^n, I_k^{n-1}\right) & \longrightarrow & \prod_n \varprojlim_k \mathrm{Hom}\left(X^n, I_k^{n}\right) & \longrightarrow & \prod_n \varprojlim_k \mathrm{Hom}\left(X^n, I_k^{n+1}\right) \\
\wr\uparrow & & \wr\uparrow & & \wr\uparrow \\
\prod_n \mathrm{Hom}\left(X^n, I^{n-1}\right) & & \prod_n \mathrm{Hom}\left(X^n, I^{n}\right) & & \prod_n \mathrm{Hom}\left(X^n, I^{n+1}\right)
\end{array}$$

41) Translator's note (O014-C064): in the third term the source omits the subscript k on $\varprojlim_k$. This sentence explicitly interchanges $\varprojlim_k$ with $\prod_n$, and the vertical isomorphism below identifies the limit with I^{n+1}, so the target restores that subscript.

This too is a portion of $\operatorname{Hom}^\bullet(X, I)$, and its cohomology at the middle term is precisely $\operatorname{Hom}_{\mathsf{K}(\mathcal{A})}(X, I)$. Hence $\operatorname{Hom}_{\mathsf{K}(\mathcal{A})}(X, I) = 0$. Apply Lemma 3.15.2 to finish the proof.□

Lemma 3.15.9 Suppose that $\mathcal{A}$ has enough injective objects. For every $A \in \operatorname{Ob}(\mathsf{C}(\mathcal{A}))$, there is a commutative diagram in $\mathsf{C}(\mathcal{A})$

$$\begin{tikzcd} \cdots \arrow[r] & I_2 \arrow[r] & I_1 \arrow[r] & I_0 \\ \cdots \arrow[r] & \tau^{\geq -2}A \arrow[r] \arrow[u, "f_2"'] & \tau^{\geq -1}A \arrow[r] \arrow[u, "f_1"'] & \tau^{\geq 0}A \arrow[u, "f_0"'] \end{tikzcd}$$

with $\tau^{\geq k}A$ and the arrows between them as in Definition 3.9.3, such that:

- ◇ every I_k is an object of $\mathsf{C}^+(\mathcal{A})$ consisting of injective objects;
- ◇ every vertical arrow f_k is both a monomorphism and a quasi-isomorphism;
- ◇ for every k and n, the morphism $I^n_{k+1} \to I^n_k$ is an epimorphism admitting a section.

Proof We construct the diagram from right to left. All injective resolutions discussed below are implicitly monomorphisms.

First apply Theorem 3.11.3 to $\tau^{\geq 0}A \in \operatorname{Ob}(\mathsf{C}^+(\mathcal{A}))$ to obtain an injective resolution $f_0 : \tau^{\geq 0}A \to I_0$.

Next, the definitions directly show that $\tau^{\geq -k-1}A \to \tau^{\geq -k}A$ is an epimorphism in every degree, and hence an epimorphism in $\mathsf{C}^+(\mathcal{A})$. Suppose that the required quasi-isomorphisms

$$\tau^{\geq 0}A \xrightarrow{f_0} I_0, \quad \ldots, \quad \tau^{\geq -k}A \xrightarrow{f_k} I_k$$

and the corresponding commutative diagram have been constructed. Choose an injective resolution $H \xrightarrow{g} J$ of $H := \ker[\tau^{\geq -k-1}A \to \tau^{\geq -k}A]$ by Theorem 3.11.3. The Horseshoe Lemma, Proposition 3.11.9, then gives a commutative diagram with exact rows in $\mathsf{C}(\mathcal{A})$:

$$\begin{tikzcd} 0 \arrow[r] & J \arrow[r] & I_{k+1} \arrow[r] & I_k \arrow[r] & 0 \\ 0 \arrow[r] & H \arrow[r] \arrow[u, "g"] & \tau^{\geq -k-1}A \arrow[r] \arrow[u, "f_{k+1}"] & \tau^{\geq -k}A \arrow[r] \arrow[u, "f_k"'] & 0 \end{tikzcd}$$

such that f_{k+1} is an injective resolution. Exactness of the first row is equivalent to exactness of $0 \to J^n \to I^n_{k+1} \to I^n_k \to 0$ for every n. Since all its terms are injective objects of $\mathcal{A}$, Lemma 2.8.13 shows that $I^n_{k+1} \to I^n_k$ admits a section. This proves the result. □

The following discussion uses notation from §3.13.

Take A as above. As $k \geq 0$ varies, the complexes $\tau^{\geq -k}A$ form an object of $\mathsf{InvSys}(\mathsf{C}(\mathcal{A}))$, which we denote by τA. The definition of the truncation functors gives a family of canonical morphisms $A \to \tau^{\geq -k}A$. Considering the complexes $\tau^{\geq -k}A$ degree by degree shows that their $\varprojlim_k$ exists and that these canonical morphisms induce an isomorphism

$$A \xrightarrow{\sim} \varprojlim_{k\geq 0} \left(\tau^{\geq -k}A\right). \tag{3.15.2}$$

Now consider the I_k. Suppose that $\mathcal{A}$ has countable products; then $\mathsf{C}(\mathcal{A})$ also has countable products, constructed degree by degree. For $A \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$, consider the diagram in Lemma 3.15.9. It follows that

$$I := \varprojlim_k I_k \quad \text{exists.}$$

More concretely, Definition 3.13.3 and the discussion following it place I and A in exact sequences

$$0 \to I \to \prod_k I_k \xrightarrow{\Delta_I} \prod_k I_k,$$
$$0 \to A \to \prod_k \tau^{\geq -k}A \xrightarrow{\Delta_{\tau A}} \prod_k \tau^{\geq -k}A,$$

where Δ_I and $\Delta_{\tau A}$ are as in that definition. We obtain commutative diagrams

$$\begin{tikzcd} I \arrow[r, hook] \arrow[d] & \prod_k I_k \\ \mathrm{Cone}(\Delta_I)[-1] \arrow[ur, "\beta(\Delta_I)[-1]"'] & \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r, hook] \arrow[d] & \prod_k \tau^{\geq -k}A \\ \mathrm{Cone}(\Delta_{\tau A})[-1] \arrow[ur, "\beta(\Delta_{\tau A})[-1]"'] & \end{tikzcd} \tag{3.15.3}$$

The two vertical arrows are defined by inclusion into the first coordinate of the mapping cone; the reader may check this directly. Another explanation, though more elaborate than necessary, can be given using the universal property of the homotopy kernel (Proposition 3.3.6).

On the other hand, $(f_k)_{k\geq 0}$ and the universal property of $\varprojlim$ give a morphism

$$f : A \simeq \varprojlim_k (\tau^{\geq -k}A) \to \varprojlim_k I_k = I,$$

which makes the following diagram commute:

$$\begin{tikzcd} I \arrow[r, "\text{as in (3.15.3)}"] & \mathrm{Cone}\,(\Delta_I)\,[-1] \\ A \arrow[u, "f"] \arrow[r] & \mathrm{Cone}\,(\Delta_{\tau A})\,[-1]. \arrow[u, "\text{induced by all the } f_k"'] \end{tikzcd} \tag{3.15.4}$$

Lemmas 3.15.8 and 3.15.9 show that I is a K-injective complex. If f is a quasi-isomorphism, then f is a K-injective resolution of A. This last step requires an additional hypothesis.

Lemma 3.15.10 In the situation above, suppose in addition that $\mathcal{A}$ has exact countable products (Convention 3.13.1). Then f is a quasi-isomorphism if and only if $A \to \operatorname{Cone}(\Delta_{\tau A})[-1]$ is a quasi-isomorphism.

Proof Consider the natural commutative diagram[42]

$$\begin{tikzcd}
\prod_k I_k \arrow[r, "\Delta_I"] & \prod_k I_k \arrow[r, "\alpha(\Delta_I)"] & \operatorname{Cone}(\Delta_I) \\
\prod_k \tau^{\geq -k} A \arrow[u, "\prod_k f_k"] \arrow[r, "\Delta_{\tau A}"'] & \prod_k \tau^{\geq -k} A \arrow[u, "\prod_k f_k"] \arrow[r, "\alpha(\Delta_{\tau A})"'] & \operatorname{Cone}(\Delta_{\tau A}) \arrow[u]
\end{tikzcd}$$

Taking the long exact sequences of the mapping cones (3.7.1) gives a commutative diagram with exact rows:

$$\begin{tikzcd}
\cdots \mathrm{H}^n\left(\prod_k I_k\right) \arrow[r] & \mathrm{H}^n(\operatorname{Cone}(\Delta_I)) \arrow[r] & \mathrm{H}^{n+1}\left(\prod_k I_k\right) \arrow[r, "\mathrm{H}^{n+1}(\Delta_I)"] & \mathrm{H}^{n+1}\left(\prod_k I_k\right) \\
\cdots \mathrm{H}^n\left(\prod_k \tau^{\geq -k} A\right) \arrow[u, "\mathrm{H}^n(\prod_k f_k)"] \arrow[r] & \mathrm{H}^n(\operatorname{Cone}(\Delta_{\tau A})) \arrow[u] \arrow[r] & \mathrm{H}^{n+1}\left(\prod_k \tau^{\geq -k} A\right) \arrow[u, "\mathrm{H}^{n+1}(\prod_k f_k)"] \arrow[r, "\mathrm{H}^{n+1}(\Delta_{\tau A})"'] & \mathrm{H}^{n+1}\left(\prod_k \tau^{\geq -k} A\right). \arrow[u, "\mathrm{H}^{n+1}(\prod_k f_k)"']
\end{tikzcd}$$

Every f_k is a quasi-isomorphism, and exactness of countable products ensures that $\prod_k f_k$ remains a quasi-isomorphism. Applying Proposition 2.3.4 shows that $\operatorname{Cone}(\Delta_{\tau A}) \to \operatorname{Cone}(\Delta_I)$[43] is also a quasi-isomorphism.

Next we prove that $I \to \operatorname{Cone}(\Delta_I)[-1]$ is a quasi-isomorphism. First, Δ_I is an epimorphism; this can be checked degree by degree. Indeed, $\Delta_{I^n} : \prod_k I_k^n \to \prod_k I_k^n$ is an epimorphism because $I_{k+1}^n \to I_k^n$ admits a section; see Example 3.13.10. Compare the short exact sequence $0 \to I \to \prod_k I_k \xrightarrow{\Delta_I} \prod_k I_k \to 0$ with Lemma 3.7.2. It follows that the previously defined morphism $I \to \operatorname{Cone}(\Delta_I)[-1]$ is indeed a quasi-isomorphism.

[42] Translator's note (O014-C065): in the source, the first lower horizontal arrow is labelled Δ_A. The lower row belongs to the inverse system τA, and the right side of that row already uses $\alpha(\Delta_{\tau A})$ and $\operatorname{Cone}(\Delta_{\tau A})$; the target therefore restores the label $\Delta_{\tau A}$.

[43] Translator's note (O014-C066): the source writes Δ_{τ_A} at this occurrence, although the inverse-system object that was defined is τA and all surrounding notation uses $\Delta_{\tau A}$. The target makes this consistent with the defined notation.

Insert these results into the commutative diagram (3.15.4) and then take cohomology. This gives the desired necessary and sufficient condition. □

Theorem 3.15.11 An abelian category $\mathcal{A}$ satisfying the following two conditions has enough K-injective complexes:

- ◇ $\mathcal{A}$ has exact countable products;
- ◇ $\mathcal{A}$ has enough injective objects.

Proof Lemma 3.15.10 shows that it suffices to prove that $A \to \operatorname{Cone}(\Delta_{\tau A})[-1]$ is a quasi-isomorphism for every $A \in \operatorname{Ob}(\mathsf{C}(\mathcal{A}))$. Let $n \in \mathbb{Z}$. Since countable products are exact, Lemma 3.9.4 gives canonical isomorphisms

$$\mathrm{H}^n\left(\prod_k \tau^{\geq -k} A\right) \simeq \prod_k \mathrm{H}^n\left(\tau^{\geq -k} A\right) \simeq \prod_{k \geq -n} \mathrm{H}^n(A).$$

If elementwise reasoning is available in $\mathcal{A}$, then after taking H^n, the morphism $\Delta_{\tau A}$ has the form

$$\begin{aligned} \textstyle\prod_{k\geq -n} \mathrm{H}^n(A) &\xrightarrow{\nabla^n} \textstyle\prod_{k\geq -n} \mathrm{H}^n(A) \\ (a_k)_{k\geq -n} &\longmapsto (a_{k+1} - a_k)_{k\geq -n}. \end{aligned}$$

Clearly ∇^n is an epimorphism (a section can be written down explicitly), while $\ker(\nabla^n) = \{(a)_{k\geq -n} : a \in \mathrm{H}^n(A)\}$ is the diagonal subobject. This argument readily translates to a general category $\mathcal{A}$. Now consider the long exact sequence of the mapping cone

$$\cdots \to \mathrm{H}^n \operatorname{Cone}(\Delta_{\tau A})[-1] \to \prod_{k\geq -n} \mathrm{H}^n(A) \xrightarrow[\text{epi}]{\nabla^n} \prod_{k\geq -n} \mathrm{H}^n(A) \to \mathrm{H}^n \operatorname{Cone}(\Delta_{\tau A}) \cdots$$

This sequence identifies the morphism $\mathrm{H}^n \operatorname{Cone}(\Delta_{\tau A})[-1] \to \prod_{k\geq -n} \mathrm{H}^n(A)$ with the inclusion of $\ker(\nabla^n)$. Moreover, (3.15.3) gives a commutative diagram

$$\begin{tikzcd} \mathrm{H}^n(A) \arrow[r, hook, "\text{diagonal morphism}"] \arrow[d] & \prod_{k\geq -n} \mathrm{H}^n(A) \\ \mathrm{H}^n \operatorname{Cone}(\Delta_{\tau A})[-1] \arrow[ur, hook] & \end{tikzcd}$$

Thus, for every n, the morphism $\mathrm{H}^n(A) \to \mathrm{H}^n(\operatorname{Cone}(\Delta_{\tau A})[-1])$ is an isomorphism. □

Reversing the arrows by means of Proposition 3.15.6 gives the dual version of Theorem 3.15.11.

Theorem 3.15.12 An abelian category $\mathcal{A}$ satisfying the following two conditions has enough K-projective complexes:

◇ $\mathcal{A}$ has exact countable coproducts;

◇ $\mathcal{A}$ has enough projective objects.

Corollary 3.15.13 If $\mathcal{A}$ is a Grothendieck category with enough projective objects (Definition 2.10.1), then $\mathcal{A}$ also has enough K-projective complexes.

Proof By definition, a Grothendieck category $\mathcal{A}$ has countable coproducts. Proposition 2.10.2 says that those countable coproducts are exact. Apply Theorem 3.15.12. □

Example 3.15.14 Let R be a ring. Theorem 3.15.11 and Corollary 3.15.13 show that R-Mod has enough K-injective and K-projective complexes.

In fact, every complex in a Grothendieck category has a K-injective resolution, and these K-injective resolutions can be made functorial, as in Theorem 2.10.14. This is the main theorem of [40]; see also [42, Tag 079P]. That result covers many situations arising in geometry that are not covered by Theorem 3.15.11.

Exercises

1. Let $\Bbbk$ be a commutative ring. Thus $\mathsf{C}(\Bbbk\text{-}\mathsf{Mod})$ and $\mathsf{K}(\Bbbk\text{-}\mathsf{Mod})$ are naturally $\Bbbk$-Mod-categories in the sense of Definition 1.4.1. Prove that, for every $n \in \mathbb{Z}$ and every object X of $\mathsf{C}(\Bbbk\text{-}\mathsf{Mod})$, there are canonical isomorphisms of $\Bbbk$-modules
$$\mathrm{H}^n(X) \simeq \mathrm{Hom}_{\mathsf{K}(\Bbbk\text{-}\mathsf{Mod})}(\Bbbk, X[n]) \simeq \mathrm{Hom}_{\mathsf{K}(\Bbbk\text{-}\mathsf{Mod})}(\Bbbk[-n], X).$$
On the right, $\Bbbk$ is regarded as a complex concentrated in degree 0.

2. This exercise relates homotopies between morphisms to isomorphisms between their mapping cones. Let $\mathcal{A}$ be an additive category and take any pair of morphisms $f, g \in \mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X, Y)$. Construct a bijection
$$\left\{ \begin{array}{c} s \in \mathrm{Hom}^{-1}(X,Y) : \\ d^{-1}_{\mathrm{Hom}^\bullet(X,Y)}(s) = f - g \end{array} \right\} \xrightarrow{1:1} \left\{ \begin{array}{c} h : \mathrm{Cone}(f) \to \mathrm{Cone}(g), \\ \text{a morphism of complexes making} \\ \text{the following diagram commute} \end{array} \right\},$$
$$\begin{tikzcd} & \mathrm{Cone}(f) \arrow[dd, "h"] \arrow[dr, "\beta(f)"] & \\ Y \arrow[ur, "\alpha(f)"] \arrow[dr, "\alpha(g)"'] & & X[1] \\ & \mathrm{Cone}(g) \arrow[ur, "\beta(g)"'] & \end{tikzcd}$$
and prove that every such morphism h is an isomorphism of complexes. In particular, f and g are homotopic if and only if such an h exists.

$\boxed{\text{Hint}}$ Commutativity of the diagram in degree n is equivalent to the matrix equation

$$h^n = \begin{pmatrix} \mathrm{id}_{X^{n+1}} & 0 \\ s^{n+1} & \mathrm{id}_{Y^n} \end{pmatrix}, \quad \text{where} \quad s^{n+1} : X^{n+1} \to Y^n.$$

3. (Split exactness) Let $\mathcal{A}$ be an Abelian category and X a complex over it. Prove that the following statements are equivalent.

(i) X is equal to 0 in $\mathsf{K}(\mathcal{A})$;

(ii) id_X is null-homotopic;

(iii) X is acyclic and there exists $(s^n)_n \in \mathrm{Hom}^{-1}(X, X)$ such that $d^n s^{n+1} d^n = d^n$ for every n;

(iv) there is a family of objects $(Y^n)_n$ in $\mathcal{A}$ and an isomorphism of complexes $\Phi : X \xrightarrow{\sim} (Y^n \oplus Y^{n+1}, \overline{d}^n)_n$, where $\overline{d}$ is represented by the matrix $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$.

A complex satisfying any of these conditions is also called **split exact**.[44] For $X = [\cdots 0 \to A \to B \to C \to 0 \cdots]$, prove that this condition is equivalent to X being a split short exact sequence.

$\boxed{\text{Hint}}$ It suffices to prove (iii) $\implies$ (iv). The condition implies that $d^{n-1}s^n \in \mathrm{End}_{\mathcal{A}}(X^n)$ is idempotent. This gives a decomposition $X^n = Y^n \oplus Z^n$ with $Y^n = \mathrm{im}(d^{n-1}s^n)$. Prove that $Y^n = \mathrm{im}(d^{n-1})$ and that the restriction of d^n gives an isomorphism $Z^n \xrightarrow{\sim} \mathrm{im}(d^n)$; then take Φ corresponding to $\left(\begin{smallmatrix} ds \\ d \end{smallmatrix}\right)$.

4. Let $\mathcal{A}$ be an Abelian category. Prove that X is an injective object of $\mathsf{C}(\mathcal{A})$ if and only if every term of X is injective in $\mathcal{A}$ and X is equal to 0 in $\mathsf{K}(\mathcal{A})$. State the version for projective objects.

$\boxed{\text{Hint}}$ For the only-if direction, observe that the short exact sequence in $\mathsf{C}(\mathcal{A})$ $0 \to X \to \mathrm{Cone}(\mathrm{id}_X) \to X[1] \to 0$ splits. Proposition 3.3.8 then shows that X is acyclic. Choose a section $X[1] \to \mathrm{Cone}(\mathrm{id}_X)$ and write it as $\left(\begin{smallmatrix} 1 \\ -\theta \end{smallmatrix}\right)$, where $\theta^n : X^{n+1} \to X^n$. Verify that $d_X^{n-1}\theta^{n-1} + \theta^n d_X^n = \mathrm{id}_{X^n}$ to prove that id_X is null-homotopic.

Fix n. By the preceding exercise, decompose $X^n = Y^n \oplus Y^{n+1}$; it remains to prove that Y^n is injective. For any morphism $f : A \to Y^n$ and monomorphism $g : A \hookrightarrow B$ in $\mathcal{A}$, consider the commutative diagram with exact row in $\mathsf{C}(\mathcal{A})$

$$\begin{tikzcd} 0 \arrow[r] & A[-n] \arrow[r, "{g[-n]}"] \arrow[d, "\Psi"'] & B[-n] \arrow[dl, dashed, "\exists"] \\ & X & \end{tikzcd}$$

where $\Psi^n = \left(\begin{smallmatrix} f \\ 0 \end{smallmatrix}\right)$. Use this diagram to verify the injectivity criterion for Y^n.

For the if direction, again write $X^n = Y^n \oplus Y^{n+1}$ and describe explicitly all possible morphisms $f : Z \to X$.

[44] More generally, let X be a complex over an additive category $\mathcal{A}$. If there are two families of objects $(Y^n)_n$ and $(H^n)_n$ such that $X^n \simeq Y^n \oplus H^n \oplus Y^{n+1}$ and d is identified with $\left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, then X is called **split**.

5. Let $\Bbbk$ be a commutative ring and let R be a unital $\Bbbk$-algebra, in accordance with the convention of this book. Write down the canonical isomorphisms $\mathrm{HH}_0(M_n(R)) \simeq \mathrm{HH}_0(R)$ and $\mathrm{HH}^0(M_n(R)) \simeq \mathrm{HH}^0(R)$, where $M_n(R)$ is the $\Bbbk$-algebra of $n \times n$ matrices over R. In fact, analogous isomorphisms hold in every degree m, for both HH_m and HH^m; see [29, Theorem 1.2.4].

6. In the setting of §3.8, define the sub-bimodule $\mathrm{Triv}_n R$ of $\mathsf{B}_n R$, for every $n \in \mathbb{Z}_{\geq 0}$, by

$$\mathrm{Triv}_n R := \sum_{h=1}^{n} R \otimes (\cdots \otimes \underbrace{\Bbbk}_{h\text{-th factor}} \otimes \cdots) \otimes R, \qquad \mathrm{Triv}_0 R = 0.$$

(i) Verify that this defines a subcomplex $\mathrm{Triv} R$ of $\mathsf{B}R$, and that $\mathrm{Triv} R$ is exact. Hint Restrict $(h^n)_n$ from the proof of Lemma 3.8.2 to $\mathrm{Triv} R$.

(ii) Write $\overline{R} := R/\Bbbk$ as a $\Bbbk$-module and define the **reduced bar complex** of R to be the quotient

$$\overline{\mathsf{B}}R := \mathsf{B}R/\mathrm{Triv}R = \left(R \otimes \overline{R}^{\otimes n} \otimes R, b_n\right)_{n \geq 0}.$$

Following the pattern of (3.8.1), simplify $M \underset{R^e}{\otimes} \overline{\mathsf{B}}R$ and $\mathrm{Hom}_{R^e}(\overline{\mathsf{B}}R, M)$.

(iii) Give canonical isomorphisms $\mathrm{H}_n(M \underset{R^e}{\otimes} \overline{\mathsf{B}}R) \simeq \mathrm{HH}_n(M)$ and $\mathrm{H}^n(\mathrm{Hom}_{R^e}(\overline{\mathsf{B}}R, M)) \simeq \mathrm{HH}^n(M)$. Hint In general, this follows from the theorem on normalized complexes in the Dold–Kan correspondence, 8.5.12, provided that $\mathsf{B}R$ is understood from the viewpoint of 8.7.7.

7. Let $\Bbbk$ be a commutative ring, and let $\widetilde{R}$ and R be unital $\Bbbk$-algebras, in accordance with the convention of this book. Suppose there is a split short exact sequence of $\Bbbk$-modules

$$0 \longrightarrow M \longrightarrow \widetilde{R} \underset{s}{\overset{p}{\rightleftarrows}} R \longrightarrow 0,$$

where M is a two-sided ideal of $\widetilde{R}$, $ps = \mathrm{id}_R$, and $M^2 = \{0\}$. Use s to identify $\widetilde{R}$ with $R \oplus M$.

(i) For $m \in M$ and $r \in R$, choose any $\widetilde{r} \in p^{-1}(r)$. Prove that $\widetilde{r}m$ and $m\widetilde{r}$ depend only on m and r, and that these operations make M an (R,R)-bimodule.

(ii) Prove that there is a bilinear map $f : R^2 \to M$ such that multiplication in $\widetilde{R}$ is given by

$$(r_1, m_1)(r_2, m_2) = (r_1 r_2, r_1 m_2 + m_1 r_2 + f(r_1, r_2)), \quad r_1, r_2 \in R, \quad m_1, m_2 \in M.$$

(iii) Such data $(R, \widetilde{R}, M)$ are called a **square-zero extension** of R by M. Prove that, up to the appropriate notion of isomorphism, square-zero extensions are classified by $\mathrm{HH}^2(M)$, with the zero element corresponding to the trivial extension $R \times M$. Hint Compute $\mathrm{HH}^2(M)$ using the reduced bar complex from the preceding exercise.

8. Complete the proof of Lemma 3.8.9.

9. In the setting of Definition 3.8.10, verify for the special case $R = \Bbbk$ that:

(i) if n is odd, then $\mathrm{HP}_n(\Bbbk) = \mathrm{HC}_n(\Bbbk) = 0$;

(ii) if n is even (or nonnegative even), then $\mathrm{HP}_n(\Bbbk)$ (or $\mathrm{HC}_n(\Bbbk)$) is isomorphic to $\Bbbk$.

10. In the setting of Definition 3.8.10 and Theorem 3.8.11, take the polynomial ring $R = \Bbbk[t]$.

(i) Prove that $\Omega_{R|\Bbbk}$ in Example 3.8.8 is a free R-module of rank 1, with basis $\mathrm{d}t$.

(ii) Describe the connecting morphism $B : \mathrm{HC}_0(R) \to \mathrm{HH}_1(R)$; note that both sides are isomorphic to R.

(iii) Describe all the $\mathrm{HC}_n(R)$ as explicitly as possible. For $\Bbbk = \mathbb{Z}$, prove that $\mathrm{HC}_1(\mathbb{Z}[t]) = \bigoplus_{n\geq 2} \mathbb{Z}/n\mathbb{Z}$.

Hint〉 Use the long exact sequence of Theorem 3.8.11 and the calculation of $\mathrm{HH}_n(\Bbbk[t])$ in high degrees in Example 3.14.11.

11. Let $\mathcal{A}$ be an Abelian category and let $0 \to X \to I^0 \to I^1 \to \cdots$ be an exact sequence in $\mathsf{C}(\mathcal{A})$. Regard $I = [I^0 \to I^1 \to \cdots]$ as a double complex, with $I^{p,q} := (I^p)^q$. Assuming that $I \in \mathrm{Ob}(\mathsf{C}^2_f(\mathcal{A}))$, prove that the evident morphism $X \to \mathrm{tot}(I)$ is a quasi-isomorphism.

12. Let $X \in \mathrm{Ob}(\mathsf{C}^2_f(\mathcal{A}))$, where $\mathcal{A}$ is an Abelian category. Prove that, if the i-th column $\left(X^{i,\bullet}, {}^{\vartriangle}d\right)$ and the j-th row $\left(X^{\bullet,j}, {}^{\triangleright}d\right)$ are exact for every $i, j \in \mathbb{Z} \smallsetminus \{0\}$, then for every $n \in \mathbb{Z}$ there is a canonical isomorphism $\mathrm{H}^n(X^{0,\bullet}, {}^{\vartriangle}d) \simeq \mathrm{H}^n(X^{\bullet,0}, {}^{\triangleright}d)$.

Hint〉 Define the brutal truncation functor $\sigma_{\mathrm{I}}^{\leq n}$ (respectively $\sigma_{\mathrm{I}}^{\geq n}$) from $\mathsf{C}^2(\mathcal{A})$ to itself by replacing the terms $(X^{i,j})_{(i,j)\in\mathbb{Z}^2}$ for $i > n$ (respectively $i < n$) by 0 and leaving all other terms unchanged. Use Theorem 3.10.6 to prove that both morphisms in

$$0\text{-th column} = \sigma_{\mathrm{I}}^{\leq 0}\sigma_{\mathrm{I}}^{\geq 0}(X) \twoheadleftarrow \sigma_{\mathrm{I}}^{\geq 0} X \hookrightarrow X$$

induce quasi-isomorphisms on total complexes. Do the same after interchanging the roles of rows and columns.

13. Use Lemma 3.11.7 to prove the existence part for β in Theorem 3.11.5.

Hint〉 It suffices to treat the first case. The mapping-cylinder lemma, 3.3.11, factors $X \xrightarrow{\alpha} Y$ as $X \xrightarrow{\alpha'} \mathrm{Cyl}(\alpha) \xrightarrow{\psi} Y$, where α' is a monomorphism and ψ is invertible in $\mathsf{K}(\mathcal{A})$.

14. With the notation of the preceding exercise, prove the uniqueness part for β in Theorem 3.11.5.

Hint〉 It suffices to treat the first case. Use the mapping cylinder again to reduce to the case in which every $\alpha^n : X^n \hookrightarrow Y^n$ is the inclusion of a direct summand. Suppose the morphisms $\beta_i : Y \to I$ in $\mathsf{C}(\mathcal{A})$, for $i = 1, 2$, satisfy $\gamma - \beta_i\alpha = d^{-1}_{\mathrm{Hom}^\bullet(X,I)} h_i$, where $h_i \in \mathrm{Hom}^{-1}(X, I)$. Let $C := \mathrm{coker}(\alpha)$ be the acyclic complex and define $h^n : Y^n \simeq X^n \oplus C^n \to I^{n-1}$ so that its restriction to X^n (respectively C^n) is $h_1^n - h_2^n$ (respectively 0). Verify that

$$\beta_1 - \beta_2 - d^{-1}_{\mathrm{Hom}^\bullet(Y,I)} h : Y \to I$$

becomes 0 after composition with α. Thus, up to homotopy, $\beta_1 - \beta_2$ factors as $Y \twoheadrightarrow C \to I$; apply Lemma 3.11.6 to the second morphism.

15. Let $\mathcal{A}$ be an Abelian category. Prove that $(\mathrm{H}^n)_{n\geq 0}$ is a universal cohomological δ-functor from the Abelian category $\mathsf{C}^{\geq 0}(\mathcal{A})$ (see Definition 3.9.1) to $\mathcal{A}$. Hint To show that H^n is effaceable when $n > 0$, take $\alpha(\mathrm{id}_X) : X \hookrightarrow \mathrm{Cone}(\mathrm{id}_X)$. This morphism factors through the brutal truncation $\sigma^{\geq 0}\,\mathrm{Cone}(\mathrm{id}_X)$. Use Proposition 3.3.8 (i).

16. (Complex version of the Milnor exact sequence) Let R be any ring and let $(X_k, f_k)_{k\geq 0}$ be an object of $\mathsf{InvSys}(\mathsf{C}(R\text{-}\mathsf{Mod}))$. Suppose $(X_k^n, f_k^n)_{k\geq 0}$ satisfies the Mittag–Leffler condition for every $n \in \mathbb{Z}$. The notation f_k will henceforth be suppressed. Follow these steps to construct a short exact sequence

$$0 \to \lim\nolimits^1_k \mathrm{H}^{n-1}(X_k) \to \mathrm{H}^n(\varprojlim_k X_k) \to \varprojlim_k \mathrm{H}^n(X_k) \to 0.$$

(i) As in Theorem 3.11.10, define $B_k^n \subset Z_k^n \subset X_k^n$ and H_k^n, and make $B_k \subset Z_k \subset X_k$ and H_k complexes with $d_{B_k} = d_{Z_k} = d_{H_k} = 0$. Prove that $(B_k^n)_{k\geq 0}$ satisfies the Mittag–Leffler condition for every n.

(ii) Prove that $0 \to \varprojlim_k Z_k \to \varprojlim_k X_k \xrightarrow{d} (\varprojlim_k X_k)[1]$ is exact.

(iii) Define $B^n := \mathrm{im}(d^{n-1} : \varprojlim_k X_k^{n-1} \to \varprojlim_k X_k^n)$[45], and make B a complex with $d_B = 0$. Prove that there is a short exact sequence $0 \to B[1] \to \varprojlim_k B_k[1] \to \lim^1_k Z_k \to 0$. Hint Use the short exact sequence $0 \to Z_k \to X_k \xrightarrow{d} B_k[1] \to 0$ and the Mittag–Leffler condition on $(X_k^n)_{k\geq 0}$.

(iv) Deduce the isomorphism $\lim^1_k Z_k \xrightarrow{\sim} \lim^1_k H_k$ and the short exact sequence $0 \to \varprojlim_k B_k \to \varprojlim_k Z_k \to \varprojlim_k H_k \to 0$. Hint Use the short exact sequence $0 \to B_k \to Z_k \to H_k \to 0$ and the Mittag–Leffler condition on $(B_k^n)_{k\geq 0}$.

(v) Take the appropriate inclusions $B \subset \varprojlim_k B_k \subset \varprojlim_k Z_k$ and consider the corresponding quotients to obtain the required short exact sequence.

17. Let $\Bbbk$ be a field. Define objects A and B of $\mathsf{InvSys}(\Bbbk\text{-}\mathsf{Mod})$ by $A_k := X^k\Bbbk[\![X]\!]$ and $B_k := X^k\Bbbk[X]$, taking the transition morphisms to be the inclusions. Neither satisfies the Mittag–Leffler condition. Prove that $\lim^1 A = 0$, whereas $\lim^1 B \simeq \Bbbk[\![X]\!]/\Bbbk[X]$. Hint Embed these systems of ideals into systems of rings, and use the dimension-shifting technique of Proposition 3.12.9 to compute $\lim^1$.

18. Let D be a division ring. Prove that Ext^n_D and Tor^D_n vanish for $n > 0$.

19. For every ring R, formulate precisely and prove the canonical isomorphism $\mathrm{Tor}^R_n(X, Y) \simeq \mathrm{Tor}^{R^{\mathrm{op}}}_n(Y, X)$.

20. Let the integral domain R be a principal ideal domain. For all finitely generated R-modules M, N and every $n \in \mathbb{Z}_{\geq 0}$, describe $\mathrm{Ext}^n_R(M, N)$ and $\mathrm{Tor}^R_n(M, N)$ explicitly.

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[45] Translator's note (O014-C067): the source omits the subscript k from the limit in the codomain, even though the domain, the system under discussion, and the subsequent exact sequence all use $\varprojlim_k$. The target restores that subscript.

21. Let M be a $\mathbb{Z}$-module. Prove that $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, M)$ is the subgroup of M consisting of all its torsion elements.

22. (Universal coefficient theorem for cohomology) Let R be a ring and let $C = (C_\bullet, d_\bullet^C)$ be a chain complex of left R-modules. For any left R-module M, applying $\mathrm{Hom}_R(\cdot, M)$ termwise gives a cochain complex $C^\bullet := \mathrm{Hom}(C_\bullet, M)$; denote its degree-$n$ cohomology by $\mathrm{H}^n(C, M)$. Suppose that $B_n := \mathrm{im}(d_{n+1})$ and $Z_n := \ker(d_n)$ are projective modules for every $n \in \mathbb{Z}$, and define $H_n := Z_n/B_n = \mathrm{H}_n(C)$. Prove that, for every $n \in \mathbb{Z}$, there is a canonical short exact sequence

$$0 \to \mathrm{Ext}_R^1(\mathrm{H}_{n-1}(C), M) \to \mathrm{H}^n(C, M) \to \mathrm{Hom}_R(\mathrm{H}_n(C), M) \to 0.$$

Prove also that, if R is a principal ideal domain and every C_n is free, then these hypotheses always hold.

Hint〉 Define the functor $D(\cdot) := \mathrm{Hom}_R(\cdot, M)$ from $(R\text{-}\mathsf{Mod})^{\mathrm{op}}$ to Ab. Use the short exact sequence $0 \to H_n \to C_n/B_n \to B_{n-1} \to 0$ and the hypotheses to obtain a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & D(Z_{n-1}) \arrow[r, "\mathrm{id}"] \arrow[d] & D(Z_{n-1}) \arrow[r] \arrow[d, "D(d_n^C)"] & 0 \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & D(B_{n-1}) \arrow[r, "D(d_n^C)"'] & D(C_n/B_n) \arrow[r] & D(H_n) \arrow[r] & 0.
\end{tikzcd}$$

Prove that $d_{D(C)}^{n-1}$ factors as $D(C_{n-1}) \twoheadrightarrow D(Z_{n-1}) \xrightarrow{D(d_n^C)} D(C_n/B_n) \hookrightarrow D(C_n)$ and that $D(C_n/B_n) \simeq \ker(d_C^n)$. Thus the cokernel of the middle morphism in the diagram is $\mathrm{H}^n(C, M)$. On the other hand, the sequence $0 \to B_{n-1} \to Z_{n-1} \to H_{n-1} \to 0$ shows that the cokernel of the left morphism is $\mathrm{Ext}_R^1(\mathrm{H}_{n-1}(C), M)$. Apply the snake lemma, Theorem 2.3.3.

23. In the setting of §3.8, take $R = \Bbbk[t]/(t^n)$, where t is a polynomial variable and $n \in \mathbb{Z}_{\geq 1}$, so that $R^e = \Bbbk[x, y]/(x^n, y^n)$.[46)]

(a) Prove that the following chain complex is exact:

$$\cdots \xrightarrow{v} R^e \xrightarrow{u} R^e \xrightarrow{v} R^e \xrightarrow{u} R^e \to R \to 0,$$

where $R^e \to R$ maps $f(x, y)$ to $f(t, t)$ and

$$u := x - y, \qquad v := \sum_{\substack{a+b=n-1 \\ a,b \geq 0}} x^a y^b.$$

Deduce that, for every R-module M, $\mathrm{HH}_{k+2}(M) \simeq \mathrm{HH}_k(M)$ and $\mathrm{HH}^{k+2}(M) \simeq \mathrm{HH}^k(M)$.

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46) Translator's note (O014-C068): the source gives the coefficient ring as $R[x, y]$. However, for $R = \Bbbk[t]/(t^n)$, the ring $R^e = R \otimes_{\Bbbk} R^{\mathrm{op}}$ is canonically $\Bbbk[x, y]/(x^n, y^n)$. The target restores the required coefficient ring $\Bbbk$.

(b) Let $\Bbbk$ be a field. Describe $\mathrm{HH}_k(M)$ and $\mathrm{HH}^k(M)$, distinguishing the cases $k \geq 1$ and whether $\mathrm{char}(\Bbbk)$ is relatively prime to n.

24. For any ring R and $n \in \mathbb{Z}_{\geq 0}$, prove that Tor_n^R commutes with small filtered inductive limits; that is,

$$\mathrm{Tor}_n^R(\varinjlim_i X_i, Y) \simeq \varinjlim_i \mathrm{Tor}_n^R(X_i, Y), \qquad \mathrm{Tor}_n^R(X, \varinjlim_i Y_i) \simeq \varinjlim_i \mathrm{Tor}_n^R(X, Y_i).$$

Discuss whether there is an analogous statement for the Ext functors.

25. Prove that an object I of an Abelian category $\mathcal{A}$ is injective if and only if, regarded as a complex, I is K-injective. State the dual version. Hint〉 One direction follows from Example 3.15.3. For the converse, regard a short exact sequence $0 \to A \to B \to C \to 0$ in $\mathcal{A}$ as an acyclic complex, and construct a morphism from that complex to I from the given morphism $A \to I$.

26. (N. Nitsure) Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor between Abelian categories. Suppose $\mathcal{A}$ has enough injective objects and $X \in \mathrm{Ob}(\mathcal{A})$. Take an injective resolution $0 \to X \to I^0 \to I^1 \to \cdots$ and an F-acyclic resolution $0 \to X \to A^0 \to A^1 \to \cdots$. Write

$$I = [\cdots \to 0 \to I^0 \to I^1 \to \cdots], \qquad A = [\cdots \to 0 \to A^0 \to A^1 \to \cdots].$$

- ◇ Lemma 3.12.2 determines a unique morphism $\beta : A \to I$ in $\mathsf{K}(\mathcal{A})$ compatible with $A \leftarrow X \to I$. After applying F, this morphism gives[47)]

$$c^n := \mathrm{H}^n(F(\beta)) : \mathrm{H}^n(F(A)) \to \mathrm{H}^n(F(I)) = \mathrm{R}^n F(X).$$

- ◇ On the other hand, Corollary 3.12.10 gives, by dimension shifting, an isomorphism

$$d^n : \mathrm{H}^n(F(A)) \xrightarrow{\sim} \mathrm{R}^n F(X).$$

Prove that $d^n = (-1)^{n(n+1)/2} c^n$ for every $n \geq 0$. Hint〉 This is the content of [36].

47) Translator's note (O014-C069): the source writes $\mathrm{H}^n(\beta) : \mathrm{H}^n(A) \to \mathrm{H}^n(I)$ and, in the following item, $\mathrm{H}^n(A)$. Since F-acyclicity is being used to compute the derived functors of F, the morphism and complexes whose cohomology must be taken are $F(\beta)$, $F(A)$, and $F(I)$. The target restores the application of F.

Chapter 4 Triangulated Categories and Derived Categories

Derived categories were pioneered by A. Grothendieck and J.-L. Verdier in the 1960s to study duality theorems in algebraic geometry. Subsequent developments have shown that they provide a convenient and powerful language in geometry, topology, and representation theory alike. The information recorded by the derived category $\mathsf{D}(\mathcal{A})$ of an Abelian category $\mathcal{A}$ lies between a complex and its cohomology. It is expressed precisely in the language of triangulated categories introduced in §§ 4.1–4.2: $\mathsf{D}(\mathcal{A})$ is equipped with the translation autoequivalence $X \mapsto X[1]$ and a class of sequences of morphisms called distinguished triangles (also called exact triangles),

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1], \quad \text{or, for short,} \quad X \to Y \to Z \xrightarrow{+1} .$$

These triangles satisfy axioms (TR0)–(TR5). For example, a distinguished triangle can be rotated into

$$Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-f[1]} Y[1] \quad \text{or} \quad Z[-1] \xrightarrow{-h[-1]} X \xrightarrow{f} Y \xrightarrow{g} Z.$$

Distinguished triangles replace short exact sequences in the category of complexes $\mathsf{C}(\mathcal{A})$, though the more accurate analogy is with cofiber sequences in homotopy theory.

Cohomological functors from a triangulated category to an Abelian category provide a mechanism for producing long exact sequences from distinguished triangles.

The passage from $\mathsf{C}(\mathcal{A})$ to $\mathsf{D}(\mathcal{A})$ is the subject of § 4.4 and proceeds in two steps.

1. The first step passes from $\mathsf{C}(\mathcal{A})$ to $\mathsf{K}(\mathcal{A})$. The latter is equipped with the translation functor $X \mapsto X[1]$ and the structure of a triangulated category. For every morphism $f : X \to Y$, its mapping cone together with the canonical morphisms
$$X \xrightarrow{f} Y \xrightarrow{\alpha(f)} \mathrm{Cone}(f) \xrightarrow{\beta(f)} X[1]$$
forms the corresponding distinguished triangle, up to isomorphism. The tools needed for this step were prepared in the first half of Chapter 3.

2. The second step formally adjoins inverses to all quasi-isomorphisms, or equivalently, formally makes every acyclic complex zero, by localization. The result is $\mathsf{D}(\mathcal{A})$, which has the same collection of objects as $\mathsf{C}(\mathcal{A})$ and inherits its triangulated structure from $\mathsf{K}(\mathcal{A})$. This step rests on the general theory of Verdier localization in § 4.3. It is here that the octahedral axiom (TR5), the most intricate axiom in the definition of a triangulated category, plays its role.

Taking cohomology of complexes gives the cohomological functor $\mathrm{H}^0 : \mathsf{D}(\mathcal{A}) \to \mathcal{A}$ at the level of the derived category, with $\mathrm{H}^n(X) := \mathrm{H}^0(X[n])$. Starting instead from the category of bounded-below (or bounded-above, or bounded) complexes, the same procedure gives the triangulated category $\mathsf{D}^+(\mathcal{A})$ (or $\mathsf{D}^-(\mathcal{A})$, or $\mathsf{D}^{\mathrm{b}}(\mathcal{A})$). These may also be regarded as full triangulated subcategories of $\mathsf{D}(\mathcal{A})$, characterized respectively by $n \ll 0$ (or $n \gg 0$, or $|n| \gg 0$) $\implies \mathrm{H}^n(X) = 0$.

More generally, cohomology-vanishing conditions can be used to define the full subcategories $\mathsf{D}^{\geq 0}(\mathcal{A})$ and $\mathsf{D}^{\leq 0}(\mathcal{A})$ of $\mathsf{D}(\mathcal{A})$, and the truncation functors on complexes can still be defined on the derived category. The canonical functor $\mathcal{A} \to \mathsf{D}(\mathcal{A})$ is fully faithful and identifies $\mathcal{A}$ with $\mathsf{D}^{\leq 0}(\mathcal{A}) \cap \mathsf{D}^{\geq 0}(\mathcal{A})$. If $\mathcal{A}$ has enough injective or projective objects, then $\mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y[n]) \simeq \mathrm{Ext}^n_{\mathcal{A}}(X, Y)$, where X and Y are objects of $\mathcal{A}$ viewed as complexes concentrated in degree zero.[1)] The definition of $\mathrm{Ext}^n_{\mathcal{A}}$ is given in § 3.14, and all these matters are treated in § 4.5.

For a left exact functor $F : \mathcal{A} \to \mathcal{A}'$, its classical right derived functor is realized at the level of bounded-below derived categories as $\mathrm{R}F : \mathsf{D}^+(\mathcal{A}) \to \mathsf{D}^+(\mathcal{A}')$. It can be characterized as a left Kan extension; the relevant data are expressed by the following

[1)] Translator's note (O014-C071): the source describes X and Y as complexes over $\mathcal{A}$. The fully faithful embedding $\mathcal{A} \to \mathsf{D}(\mathcal{A})$ in the preceding sentence and the definition of $\mathrm{Ext}^n_{\mathcal{A}}$ in § 3.14, however, give this formula for objects of $\mathcal{A}$ viewed as degree-zero complexes; the intended scope has been restored here.

2-cell diagram.

$$\begin{tikzcd} \mathsf{K}^+(\mathcal{A}) \arrow[r, "\mathsf{K}F"] \arrow[d, "Q"'] & \mathsf{K}^+(\mathcal{A}') \arrow[d, "Q'"] \arrow[dl, Rightarrow] \\ \mathsf{D}^+(\mathcal{A}) \arrow[r, "\mathrm{R}F"'] & \mathsf{D}^+(\mathcal{A}') \end{tikzcd}$$

Here Q and Q' denote the localization functors, while $\mathrm{R}F$ must also be equipped with a triangulated-functor structure. The classical $\mathrm{R}^n F$ is obtained by restricting $\mathrm{R}F$ to $\mathcal{A}$ and then taking H^n. If instead F is right exact, the lift of the classical left derived functor $\mathrm{L}_n F$ is represented as follows.[2)]

$$\begin{tikzcd} \mathsf{K}^-(\mathcal{A}) \arrow[r, "\mathsf{K}F"] \arrow[d, "Q"'] & \mathsf{K}^-(\mathcal{A}') \arrow[d, "Q'"] \\ \mathsf{D}^-(\mathcal{A}) \arrow[r, "\mathrm{L}F"'] \arrow[ur, Rightarrow] & \mathsf{D}^-(\mathcal{A}'). \end{tikzcd}$$

As in the classical theory, the existence of derived functors and the detailed structure of derived categories depend on resolutions of complexes. Resolutions provide the scaffolding needed to work concretely with derived categories. Details appear in §§ 4.6–4.8. The unbounded version $\mathsf{D}(\mathcal{A}) \to \mathsf{D}(\mathcal{A}')$ has a similar formulation and will be discussed in § 4.11.

By considering the amplitude of a derived functor, that is, the "shift" it produces at the level of cohomology, one can discuss the dimension of a derived functor (Definition–Proposition 4.8.9). Under suitable conditions, taking derived functors is also compatible with composition (Theorem 4.6.4). This provides a uniform explanation for several spectral sequences in the classical theory, called Grothendieck spectral sequences and described in Theorem 5.6.6. The general theory of spectral sequences is deferred to Chapter 5.

All these constructions can be formulated for general triangulated categories and their Verdier localizations. It is worth noting that applications often involve triangulated categories that do not arise directly from Abelian categories, or cohomological functors other than H^n. In this general setting, the roles of truncation functors and of $\mathsf{D}^{\leq 0}$ and $\mathsf{D}^{\geq 0}$ are taken by a t-structure on the triangulated category, but this book does not treat that topic.

The derived functors discussed concretely in this chapter are limited to examples arising in algebra. The first is the functor RHom in § 4.9. This is the derived-category version of the Ext functor and involves the theory of derived bifunctors. Several adjunctions will also be lifted to derived categories, following the principle of replacing Hom

[2)] Translator's note (O014-C070): in the following diagram, the source labels the right vertical arrow Q, even though its source and target are $\mathsf{K}^-(\mathcal{A}')$ and $\mathsf{D}^-(\mathcal{A}')$, respectively. In accord with the preceding diagram and the names used in the prose, the label Q' has been restored here.

by RHom. Next, § 4.10 treats products and coproducts in $\mathsf{D}(\mathcal{A})$ and then introduces the lift of $\lim^n$, denoted $\mathrm{R}\lim$. Like $\lim^n$, this functor has a concise description in terms of the homotopy limit (Definition 4.10.3), though at the level of derived categories the description can only be given in a coarse form.

Then § 4.12 introduces the derived-category version of the Tor functor, namely the derived tensor product $\overset{\mathrm{L}}{\otimes}$, a triangulated bifunctor. That section also proves many facts, including the associativity constraint (Proposition 4.12.13) and the RHom version of the adjunction (Theorem 4.12.17). As in the classical theory, RHom and $\overset{\mathrm{L}}{\otimes}$ are the two fundamental examples of derived functors. A complete and clean description requires unbounded derived categories and the derived functors between them. This in turn requires the theory of § 4.11, especially the notions of K-projective and K-flat resolutions. In the module-theoretic setting considered here, the existence of these resolutions poses no problem.

Finally, several shortcomings of the derived category $\mathsf{D}(\mathcal{A})$ should be noted.

1. Among the axioms of a triangulated category, (TR2) says that every morphism can be placed in a distinguished triangle. Such a triangle is unique up to isomorphism, but not "up to unique isomorphism" in the sense familiar from algebra.

2. Triangulated categories usually have neither kernels nor cokernels, making the operations $\varinjlim$ and $\varprojlim$ extremely difficult to carry out. Mapping cones provide substitutes in the homotopical sense, but, as already noted, their uniqueness in the derived category is problematic.

3. Although derived functors at the level of derived categories have a clear definition, unlike universal cohomological δ-functors they are difficult to characterize simply by effaceability; see Proposition 3.12.14.

The root of the problem is that the homotopy relation is erased too crudely when $\mathsf{D}(\mathcal{A})$ is constructed. These shortcomings must be remedied by a construction that is higher than the derived category and perhaps also more natural; the derived category should be only the shadow cast by that construction.

阅读提示

Most of this chapter consists of core concepts in the theory of triangulated and derived categories. Side topics include the treatment of n-extensions in § 4.5, centered mainly on the theorem of Nobuo Yoneda (Theorem 4.5.13), and the treatment of unbounded derived functors in § 4.11, which rests on the theory of § 3.15. Both topics are common knowledge among specialists, but readers should judge for themselves whether they need them. If one considers only derived functors between bounded-below or bounded-above derived categories, every statement in § 4.12 concerning K-projective (or K-flat) resolutions may be replaced by a projective (or flat) resolution by bounded-above complexes without disrupting the exposition. The validity of this replacement is guaranteed by Proposition 4.12.4.

4.1 Definition of a Triangulated Category

A triangulated category is an additive category with translation, equipped with a class of distinguished triangles. We begin by explaining what a category with translation is.

Definition 4.1.1 A **category with translation** is data $(\mathcal{D}, T)$ consisting of a category $\mathcal{D}$ and an equivalence $T : \mathcal{D} \to \mathcal{D}$.[3)] The equivalence T is called the **translation functor** of $\mathcal{D}$. In addition, choose a quasi-inverse T^{-1} of T and isomorphisms $T^{-1}T \simeq \mathrm{id}_{\mathcal{D}} \simeq TT^{-1}$ so that these data form an adjoint equivalence in the sense of [51, Theorem 2.6.12].

- A functor from $(\mathcal{D}, T)$ to $(\mathcal{D}', T')$ is a functor $F : \mathcal{D} \to \mathcal{D}'$ together with a chosen isomorphism $FT \simeq T'F$. By convention, the isomorphism is suppressed and these data are abbreviated as $F : (\mathcal{D}, T) \to (\mathcal{D}', T')$.

- A morphism between two functors $(\mathcal{D}, T) \overset{F}{\underset{F'}{\rightrightarrows}} (\mathcal{D}', T')$ is a morphism $\varphi : F \to F'$ that makes the following diagram commute:

$$\begin{tikzcd} FT \arrow[r, "\varphi T"] \arrow[d, "\wr"'] & F'T \arrow[d, "\wr"] \\ T'F \arrow[r, "T'\varphi"'] & T'F'. \end{tikzcd}$$

- A subcategory of a category with translation $(\mathcal{D}, T)$ is a subcategory $\mathcal{D}'$ of $\mathcal{D}$ that is closed under T and for which $T' := T|_{\mathcal{D}'}$ makes $(\mathcal{D}', T')$ a category with translation.

3) Many references require T to be an automorphism. In that case one may choose an inverse functor T^{-1} satisfying $T^{-1}T = \mathrm{id}_{\mathcal{D}} = TT^{-1}$. This condition covers every situation treated in this book and avoids possible subtleties from 2-category theory, so the reader may harmlessly make the same assumption. The study of stable model categories or stable ∞-categories, however, cannot avoid the case in which the equivalence is not an isomorphism, especially in topology.

If $\mathcal{D}$ is additive and T is an additive functor (as it is automatically; see Corollary 1.3.6), then $(\mathcal{D}, T)$ is called an **additive category with translation**; functors between such categories are also required to be additive. After fixing a commutative ring $\Bbbk$, the same definition applies in the $\Bbbk$-linear setting.

If $(\mathcal{D}, T)$ is a category with translation, then so is $(\mathcal{D}^{\mathrm{op}}, T^{-1})$. By convention, the data $(\mathcal{D}, T)$ are often abbreviated to $\mathcal{D}$. The following notion is particularly useful.

Definition 4.1.2 (Morphisms with degree) A morphism of degree m in a category with translation $(\mathcal{D}, T)$ is a morphism of the form $f : X \to T^m Y$, also written $f : X \xrightarrow{+m} Y$. Given $X \xrightarrow[f]{+m} Y \xrightarrow[g]{+n} Z$, their composite gf is defined to be the morphism of degree $m+n$ given by $T^m(g)f$. Commutative diagrams can likewise be considered for morphisms of this kind.

In the multivariable setting, suppose that $\mathcal{D}$ is equipped with strictly commuting automorphisms $T_1, \ldots, T_n$, so that $T_i T_j = T_j T_i$. A morphism of degree $(m_1, \ldots, m_n)$ is defined to be a morphism of the form $X \to T_1^{m_1} \cdots T_n^{m_n} Y$. It is written $X \xrightarrow{+(m_1,\ldots,m_n)} Y$.

For example, given a category $\mathcal{A}$, the category $(\mathcal{A}^{\mathbb{Z}}, T)$ of graded objects from Definition 3.1.2 is a category with translation. More generally, one can consider the category $\mathcal{A}^{\mathbb{Z}^n}$ of $\mathbb{Z}^n$-graded objects, together with a translation functor $T_1, \ldots, T_n$ in each variable, and then discuss morphisms with degree between $\mathbb{Z}^n$-graded objects. The multivariable case, however, is not a focus of this chapter.

Example 4.1.3 The category of complexes $\mathsf{C}(\mathcal{A})$ over an additive category $\mathcal{A}$, together with the translation functor $T : X \mapsto X[1]$ from Definition 3.1.6, is an additive category with translation. The category $\mathsf{K}(\mathcal{A})$ from Definition 3.2.8 inherits the translation functor from $\mathsf{C}(\mathcal{A})$, which is still denoted by T. The canonical functor $F : \mathsf{C}(\mathcal{A}) \to \mathsf{K}(\mathcal{A})$ plainly satisfies $FT = TF$. The same applies, for every $\star \in \{+, -, \mathrm{b}\}$, to the categories $\mathsf{C}^\star(\mathcal{A})$ (or $\mathsf{K}^\star(\mathcal{A})$) introduced in Definition 3.9.1 (or Definition 3.9.2).

In this chapter we mainly consider additive categories with translation.

Definition 4.1.4 Let $(\mathcal{D}, T)$ be an additive category with translation. A **triangle** in it is a sequence of morphisms in $\mathcal{D}$ of the form

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} TX.$$

A morphism between two triangles is a commutative diagram of the form

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\alpha"'] & Y \arrow[r, "g"] \arrow[d, "\beta"] & Z \arrow[r, "h"] \arrow[d, "\gamma"] & TX \arrow[d, "T\alpha"] \\ X' \arrow[r, "f'"'] & Y' \arrow[r, "g'"'] & Z' \arrow[r, "h'"'] & TX'. \end{tikzcd}$$

The two rows are the given triangles. Such a morphism is also abbreviated to $(X, Y, Z) \to (X', Y', Z')$. Isomorphisms between triangles are defined in the same way.

The same notion is available if the additive category is replaced by a more general $\Bbbk$-linear category, where $\Bbbk$ is any commutative ring.

Clearly, triangles in $(\mathcal{D}, T)$ are the same as triangles in $(\mathcal{D}^{\mathrm{op}}, T^{-1})$.

Remark 4.1.5 Consider a triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} TX$ in a $\Bbbk$-linear category with translation $(\mathcal{D}, T)$, and let $\epsilon, \zeta, \eta \in \Bbbk^\times$. If $\epsilon\zeta\eta = 1$, then the original triangle is isomorphic to $X \xrightarrow{\epsilon f} Y \xrightarrow{\zeta g} Z \xrightarrow{\eta h} TX$, as shown by the commutative diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "1"'] & Y \arrow[r, "g"] \arrow[d, "\epsilon"] & Z \arrow[r, "h"] \arrow[d, "\epsilon\zeta"] & TX \arrow[d, "\epsilon\zeta\eta"] \\ X \arrow[r, "\epsilon f"'] & Y \arrow[r, "\zeta g"'] & Z \arrow[r, "\eta h"'] & TX. \end{tikzcd}$$

In the notation of Definition 4.1.2, a triangle can also be written $X \to Y \to Z \xrightarrow{+1}$, or more pictorially as $\begin{tikzcd} & Y \arrow[dr] & \\ X \arrow[ur] & & Z \arrow[ll, "+1"] \end{tikzcd}$. Thus a triangle may be pictured as an upward-spiraling chain, while a morphism between triangles resembles the structure of deoxyribonucleic acid: the arrows α, β, γ, and so forth can be compared to the hydrogen bonds between bases.

Definition 4.1.6 (Triangulated categories and triangulated functors) A **triangulated category** is an additive category with translation $(\mathcal{D}, T)$ equipped with a class $\mathcal{H}$ of triangles, called **distinguished triangles**, satisfying the following axioms.

(TR0) Every triangle isomorphic to a distinguished triangle is distinguished.

(TR1) For every $X \in \mathrm{Ob}(\mathcal{D})$, the triangle $X \xrightarrow{\mathrm{id}_X} X \to 0 \to TX$ is distinguished.

(TR2) Every morphism $f : X \to Y$ can be extended to a distinguished triangle of the form $X \xrightarrow{f} Y \to Z \to TX$, where $Z \in \mathrm{Ob}(\mathcal{D})$.

(TR3) A triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} TX$ is distinguished if and only if its "counterclockwise rotation"

$$Y \xrightarrow{g} Z \xrightarrow{h} TX \xrightarrow{-Tf} TY$$

is distinguished. The sign pattern may equivalently be chosen as $(-++)$, $(+-+)$, or $(---)$; see Remark 4.1.5.

(TR4) Suppose the solid part of the following diagram is commutative:

$$\begin{tikzcd}
X \arrow[r] \arrow[d, "\alpha"'] & Y \arrow[r] \arrow[d, "\beta"] & Z \arrow[r] \arrow[d, dashed, "\gamma"] & TX \arrow[d, dashed, "T\alpha"] \\
X' \arrow[r] & Y' \arrow[r] & Z' \arrow[r] & TX'.
\end{tikzcd}$$

If both rows are distinguished triangles, there exists a morphism $\gamma : Z \to Z'$ as indicated by the dashed arrow such that the whole diagram containing α, β, γ, and $T\alpha$ commutes. Thus one obtains a morphism of triangles.

(TR5) Suppose that distinguished triangles

$$X \xrightarrow{f} Y \xrightarrow{h} Z' \to TX,$$

$$Y \xrightarrow{g} Z \xrightarrow{k} X' \to TY,$$

$$X \xrightarrow{gf} Z \xrightarrow{m} Y' \to TX$$

are given. There exists a distinguished triangle

$$Z' \xrightarrow{u} Y' \xrightarrow{v} X' \xrightarrow{w} TZ'$$

such that the following diagram commutes:

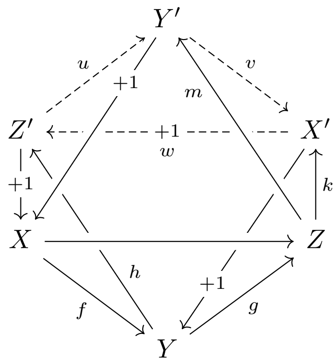
$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, equal] & Y \arrow[r, "h"] \arrow[d, "g"] & Z' \arrow[r] \arrow[d, "u"] & TX \arrow[d, equal] \\
X \arrow[r, "gf"] \arrow[d, "f"'] & Z \arrow[r, "m"] \arrow[d, equal] & Y' \arrow[r] \arrow[d, "v"] & TX \arrow[d, "Tf"] \\
Y \arrow[r, "g"] \arrow[d, "h"'] & Z \arrow[r, "k"] \arrow[d, "m"] & X' \arrow[r] \arrow[d, equal] & TY \arrow[d, "Th"] \\
Z' \arrow[r, "u"'] & Y' \arrow[r, "v"'] & X' \arrow[r, "w"'] & TZ'.
\end{tikzcd}$$

Data $(\mathcal{D}, T, \mathcal{H})$ satisfying only (TR0)–(TR4) are called a **pretriangulated category**. The data T and $\mathcal{H}$ are often omitted from the notation.

A **triangulated functor** from $\mathcal{D}$ to $\mathcal{D}'$ is a functor $F : \mathcal{D} \to \mathcal{D}'$ between additive categories with translation that maps distinguished triangles to distinguished triangles. Morphisms between triangulated functors are the same as the morphisms in Definition 4.1.1; this also defines a quasi-inverse of a triangulated functor. If two triangulated functors $\mathcal{D} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}'$ are quasi-inverse to one another, they are said to form an **equivalence** between $\mathcal{D}$ and $\mathcal{D}'$.

If $(\mathcal{D}, T)$ is $\Bbbk$-linear, all the definitions above have corresponding extensions.

Remark 4.1.7 In the notation of Definition 4.1.2, axiom (TR5) can be rewritten as the diagram



The dashed part represents the morphisms whose existence is asserted by the axiom. Four faces of this octahedron are distinguished triangles, and the other four faces are commutative; the details are left to the reader. For this reason, (TR5) is also called the **octahedral axiom**.

Remark 4.1.8 (Duality) Let $(\mathcal{D}, T, \mathcal{H})$ be a pretriangulated (or triangulated) category. Regard T^{-1} as an endofunctor of $\mathcal{D}^{\mathrm{op}}$ and denote it by S. Define a class of triangles $\mathcal{K}$ on $(\mathcal{D}^{\mathrm{op}}, S)$ as follows. If

$$[X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} TX] \in \mathcal{H},$$

then its clockwise rotation $[T^{-1}Z \xrightarrow{-T^{-1}h} X \xrightarrow{f} Y \xrightarrow{g} Z] \in \mathcal{H}$ may be regarded as the following triangle in $\mathcal{D}^{\mathrm{op}}$:

$$Z \xrightarrow{g^{\mathrm{op}}} Y \xrightarrow{f^{\mathrm{op}}} X \xrightarrow{-(T^{-1}h)^{\mathrm{op}}} SZ. \tag{4.1.1}$$

Define $\mathcal{K}$ to be the class of all triangles of the form (4.1.1). It is straightforward to verify that $(\mathcal{D}^{\mathrm{op}}, S, \mathcal{K})$ is a pretriangulated (or triangulated) category. It is also clear that applying this procedure twice recovers the original $(\mathcal{D}, T, \mathcal{H})$.

The notions of pretriangulated subcategory and triangulated subcategory will be introduced in Definition–Proposition 4.2.10; we postpone their discussion until then.

Lemma 4.1.9 If $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{+1}$ is a distinguished triangle in a pretriangulated category, then $gf = 0$.

Proof Axioms (TR1) and (TR4) give the commutative diagram

$$\begin{tikzcd} X \arrow[r, equal] \arrow[d, equal] & X \arrow[r] \arrow[d, "f"] & 0 \arrow[r] \arrow[d] & TX \arrow[d, equal] \\ X \arrow[r, "f"'] & Y \arrow[r, "g"'] & Z \arrow[r] & TX \end{tikzcd}$$

from which $gf = 0$ is immediate. □

Definition 4.1.10 (Cohomological functor) Let $\mathcal{D}$ be a pretriangulated category and $\mathcal{A}$ an Abelian category. An additive functor $H : \mathcal{D} \to \mathcal{A}$ is called a **cohomological functor** if it has the following property: for every distinguished triangle $X \to Y \to Z \xrightarrow{+1}$ in $\mathcal{D}$, the corresponding sequence $HX \to HY \to HZ$ is exact in $\mathcal{A}$.

Remark 4.1.11 (Long exact sequence of a cohomological functor) For a cohomological functor H and any distinguished triangle $X \to Y \to Z \xrightarrow{+1}$ in $\mathcal{D}$, repeated rotation by (TR3) extends the exact sequence in Definition 4.1.10 to the long exact sequence

$$\cdots \to HT^{-1}Z \to HX \to HY \to HZ \to HTX \to HTY \to \cdots .$$

Rotating the triangle introduces some minus signs, but these do not affect the exactness of the sequence above.

The basic examples of cohomological functors are the Hom functors.

Proposition 4.1.12 Let $\mathcal{D}$ be a pretriangulated category and let S be an object of $\mathcal{D}$. Then both $\mathrm{Hom}(S, \cdot) : \mathcal{D} \to \mathsf{Ab}$ and $\mathrm{Hom}(\cdot, S) : \mathcal{D}^{\mathrm{op}} \to \mathsf{Ab}$ are cohomological functors.

If $(\mathcal{D}, T)$ is $\Bbbk$-linear, the corresponding assertion holds with Ab replaced by $\Bbbk$-Mod.

Proof By duality it suffices to prove the assertion for $\mathrm{Hom}(S, \cdot)$. Consider a distinguished triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{+1}$ and the corresponding sequence

$$\mathrm{Hom}(S, X) \xrightarrow{f_*} \mathrm{Hom}(S, Y) \xrightarrow{g_*} \mathrm{Hom}(S, Z).$$

Lemma 4.1.9 implies $g_* f_* = (gf)_* = 0$. Conversely, suppose that $h \in \mathrm{Hom}(S, Y)$ satisfies $g_*(h) = gh = 0$, and consider the following diagram, whose solid part is commutative.[4)]

$$\begin{tikzcd} S \arrow[r, equal] \arrow[d, dashed, "k"'] & S \arrow[r] \arrow[d, "h"] & 0 \arrow[r] \arrow[d] & TS \arrow[d, dashed, "Tk"] \\ X \arrow[r, "f"'] & Y \arrow[r, "g"'] & Z \arrow[r] & TX \end{tikzcd}$$

[4)] Translator's note (O014-C072): at the vertex TS, the source draws two coincident dashed arrows to TX, one unlabeled and one labeled Tk. A morphism of triangles has only the single component Tk in this position; the redundant unlabeled arrow has been removed here.

If a morphism $k : S \to X$ can be found as indicated by the dashed arrow so that the entire diagram commutes, then $f_*(k) = h$. Axiom (TR1) ensures that both rows are distinguished triangles. After applying suitable rotations (TR3), the existence of k follows from (TR4). $\square$

Corollary 4.1.13 In a pretriangulated category, if a triangle has the form $X \xrightarrow{f} Y \to 0 \xrightarrow{+1}$, then f must be an isomorphism.

Proof For every object S, Proposition 4.1.12 gives an exact sequence in Ab

$$\underbrace{\mathrm{Hom}(S, T^{-1}0)}_{=0} \to \mathrm{Hom}(S, X) \xrightarrow{f_*} \mathrm{Hom}(S, Y) \to \underbrace{\mathrm{Hom}(S, 0)}_{=0},$$

so f_* is an isomorphism. Since S is arbitrary, f is an isomorphism. $\square$

4.2 Basic Properties

The preceding section gave only the definitions and initial properties of triangulated categories and cohomological functors. This section develops their structure further.

Proposition 4.2.1 Consider the following commutative diagram in a pretriangulated category $\mathcal{D}$:

$$\begin{tikzcd}
X \arrow[r] \arrow[d, "\alpha"'] & Y \arrow[r] \arrow[d, "\beta"] & Z \arrow[r] \arrow[d, "\gamma"] & TX \arrow[d, "T\alpha"] \\
X' \arrow[r] & Y' \arrow[r] & Z' \arrow[r] & TX'
\end{tikzcd}$$

If each row is a distinguished triangle and any two of α, β, and γ are isomorphisms, then the remaining morphism is also an isomorphism.

Proof After suitable rotations (TR3), we may assume without loss of generality that α and γ are isomorphisms. To show that β is an isomorphism, it suffices to prove, for every object S, that $\mathrm{Hom}(S, Y) \xrightarrow{\beta_*} \mathrm{Hom}(S, Y')$ is an isomorphism. Proposition 4.1.12 gives the following commutative diagram with exact rows:

$$\begin{tikzcd}
\mathrm{Hom}(S, T^{-1}Z) \arrow[r] \arrow[d, "\wr"] & \mathrm{Hom}(S, X) \arrow[r] \arrow[d, "\wr"] & \mathrm{Hom}(S, Y) \arrow[r] \arrow[d, "\beta_*"] & \mathrm{Hom}(S, Z) \arrow[r] \arrow[d, "\wr"] & \mathrm{Hom}(S, TX) \arrow[d, "\wr"] \\
\mathrm{Hom}(S, T^{-1}Z') \arrow[r] & \mathrm{Hom}(S, X') \arrow[r] & \mathrm{Hom}(S, Y') \arrow[r] & \mathrm{Hom}(S, Z') \arrow[r] & \mathrm{Hom}(S, TX').
\end{tikzcd}$$

Now apply Proposition 2.3.4. $\square$

In the argument above, $\mathrm{Hom}(S, \cdot)$ may also be replaced by $\mathrm{Hom}(\cdot, S)$.

Remark 4.2.2 For the moment, call a triangle $X \to Y \to Z \xrightarrow{+1}$ a **special triangle** (or a **cospecial triangle**) if, for every $S \in \mathrm{Ob}(\mathcal{D})$, the sequence

$$\cdots \to \mathrm{Hom}(S, X) \to \mathrm{Hom}(S, Y) \to \mathrm{Hom}(S, Z) \to \mathrm{Hom}(S, TX) \to \cdots$$

(or the corresponding version with $\mathrm{Hom}(\cdot, S)$) is exact. If $X_i \to Y_i \to Z_i \xrightarrow{+1}$ is a family of distinguished triangles ($i \in I$), then $\prod_i X_i \to \prod_i Y_i \to \prod_i Z_i \xrightarrow{+1}$ (or $\coprod_i X_i \to \coprod_i Y_i \to \coprod_i Z_i \xrightarrow{+1}$) is a special triangle (or a cospecial triangle), provided the products (or coproducts) in question exist. This follows simply from the universal properties of products and coproducts together with [51, Lemma 6.8.3]. Notice that because $T : \mathcal{D} \to \mathcal{D}$ is an equivalence, it necessarily preserves products and coproducts. Proposition 4.2.1 extends to the case in which both rows are special (or cospecial) triangles.

Corollary 4.2.3 For every morphism $f : X \to Y$ in a pretriangulated category $\mathcal{D}$, the distinguished triangle $X \xrightarrow{f} Y \to Z \xrightarrow{+1}$ in axiom (TR2) is unique up to isomorphism.

Proof Given distinguished triangles $X \xrightarrow{f} Y \to Z \xrightarrow{+1}$ and $X \xrightarrow{f} Y \to Z' \xrightarrow{+1}$, axiom (TR4) gives a morphism γ that makes the diagram

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, "\mathrm{id}_X"'] & Y \arrow[r] \arrow[d, "\mathrm{id}_Y"] & Z \arrow[r] \arrow[d, "\gamma"] & TX \arrow[d, "\mathrm{id}_{TX}"] \\
X \arrow[r, "f"'] & Y \arrow[r] & Z' \arrow[r] & TX
\end{tikzcd}$$

commute. Proposition 4.2.1 then implies that γ is an isomorphism. □

It must be emphasized that the isomorphism in Corollary 4.2.3 is **not canonical**.

Lemma 4.2.4 Let $\mathcal{D}$ be a pretriangulated category and let $X_i \xrightarrow{f_i} Y_i \xrightarrow{g_i} Z_i \xrightarrow{h_i} TX_i$ be a family of distinguished triangles. Suppose that $\prod_i X_i$, $\prod_i Y_i$, and $\prod_i Z_i$ all exist. Then

$$\prod_{i\in I} X_i \xrightarrow{\prod_i f_i} \prod_{i\in I} Y_i \xrightarrow{\prod_i g_i} \prod_{i\in I} Z_i \xrightarrow{\prod_i h_i} \underbrace{\prod_{i\in I} TX_i}_{\simeq T(\prod_{i\in I} X_i)}$$

is also a distinguished triangle. The corresponding assertion holds with $\prod_{i\in I}$ replaced by $\coprod_{i\in I}$.

Proof By duality, we need only treat the case of $\prod_{i\in I}$. From now on we always identify $\prod_i TX_i$ with $T(\prod_i X_i)$. By (TR2), choose a distinguished triangle $\prod_i X_i \xrightarrow{\prod_i f_i}$

$\prod_i Y_i \to Q \xrightarrow{h} \prod_i TX_i$. For every $i \in I$, apply (TR4) to obtain a commutative diagram

$$\begin{tikzcd}
\prod_{j\in I} X_j \arrow[r, "\prod_j f_j"] \arrow[d] & \prod_j Y_j \arrow[r] \arrow[d] & Q \arrow[r, "h"] \arrow[d, "\exists\gamma_i"] & \prod_{j\in I} TX_j \arrow[d] \\
X_i \arrow[r, "f_i"'] & Y_i \arrow[r, "g_i"'] & Z_i \arrow[r, "h_i"'] & TX_i
\end{tikzcd}$$

in which all vertical morphisms other than γ_i are the canonical projections. The universal property of the product then gives a commutative diagram

$$\begin{tikzcd}
\prod_{i\in I} X_i \arrow[r, "\prod_i f_i"] \arrow[d, "\mathrm{id}"'] & \prod_{i\in I} Y_i \arrow[r] \arrow[d, "\mathrm{id}"] & Q \arrow[r, "h"] \arrow[d, "\prod_i \gamma_i"] & \prod_{i\in I} TX_i \arrow[d, "\mathrm{id}"] \\
\prod_{i\in I} X_i \arrow[r, "\prod_i f_i"'] & \prod_{i\in I} Y_i \arrow[r, "\prod_i g_i"'] & \prod_{i\in I} Z_i \arrow[r, "\prod_i h_i"'] & \prod_{i\in I} TX_i.
\end{tikzcd}$$

The second row is a special triangle in the sense of Remark 4.2.2. The extended version of Proposition 4.2.1 now implies that $\prod_{i\in I} \gamma_i$ is an isomorphism. □

Corollary 4.2.5 In every pretriangulated category, a triangle of the form $X \xrightarrow{\iota_1} X \oplus Y \xrightarrow{p_2} Y \xrightarrow{0} TX$ is always distinguished; here ι_i and p_j are defined as in §1.3.

Proof Apply Lemma 4.2.4 to the distinguished triangles $X \xrightarrow{\mathrm{id}_X} X \to 0 \to TX$ and $0 \to Y \xrightarrow{\mathrm{id}_Y} Y \to 0$. □

Conversely, if one morphism in a distinguished triangle is 0, then the triangle comes from a direct sum. This fact can be verified by rotating the triangle into the following form.

Proposition 4.2.6 In every pretriangulated category, a distinguished triangle of the form $X \to M \to Y \xrightarrow{0} TX$ must be isomorphic to the triangle $X \to X \oplus Y \to Y \xrightarrow{0} TX$ from Corollary 4.2.5.

Proof A suitably rotated version of (TR4) shows that there is a morphism $\alpha : X \oplus Y \to M$ such that the following diagram is a morphism of distinguished triangles:

$$\begin{tikzcd}
X \arrow[r, "\iota_1"] \arrow[d, "\mathrm{id}_X"'] & X \oplus Y \arrow[r, "p_2"] \arrow[d, "\alpha"] & Y \arrow[r, "0"] \arrow[d, "\mathrm{id}_Y"] & TX \arrow[d, "\mathrm{id}_{TX}"] \\
X \arrow[r] & M \arrow[r] & Y \arrow[r, "0"'] & TX
\end{tikzcd}$$

Proposition 4.2.1 implies that α is an isomorphism. □

The next result shows that additivity in the definition of a triangulated functor is actually redundant.

Corollary 4.2.7 Let $\mathcal{D}$ and $\mathcal{D}'$ be pretriangulated categories, with both translation functors denoted by T. Let $F : (\mathcal{D}, T) \to (\mathcal{D}', T)$ be a functor between categories with translation. If F maps distinguished triangles to distinguished triangles, then F is additive and hence is automatically a triangulated functor.

Proof The distinguished triangle $0 \xrightarrow{\mathrm{id}} 0 \xrightarrow{\mathrm{id}} 0 \xrightarrow{\mathrm{id}} 0$ in $\mathcal{D}$ gives the distinguished triangle $F(0) \xrightarrow{\mathrm{id}} F(0) \xrightarrow{\mathrm{id}} F(0) \xrightarrow{\mathrm{id}} F(0)$ in $\mathcal{D}'$. Lemma 4.1.9 then gives $\mathrm{id}_{F(0)} = \mathrm{id}_{F(0)} \circ \mathrm{id}_{F(0)} = 0$, so $F(0) \simeq 0$. This also implies that F maps zero morphisms to zero morphisms.

Next consider the distinguished triangle in $\mathcal{D}$ supplied by Corollary 4.2.5, namely $X \xrightarrow{\iota_1} X \oplus Y \xrightarrow{p_2} Y \xrightarrow{0} TX$. The corresponding triangle

$$FX \xrightarrow{F\iota_1} F(X \oplus Y) \xrightarrow{Fp_2} FY \xrightarrow{0} TFX$$

is distinguished in $\mathcal{D}'$. Take $M := F(X \oplus Y)$ in Proposition 4.2.6. The morphism α in the proof of that proposition can clearly be taken to be the canonical morphism $FX \oplus FY \to F(X \oplus Y)$ determined by the universal property of the direct sum. Thus α is an isomorphism. By Corollary 1.3.6 (iii), F is additive. □

The next two results show that, in an equivalence of pretriangulated categories (Definition 4.1.6), it is enough to assume that the functor in one direction is triangulated. This is another application of Proposition 4.1.12.

Proposition 4.2.8 Let $\mathcal{D}$ and $\mathcal{D}'$ be pretriangulated categories, and consider functors $\mathcal{D} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}'$ such that (F, G) is an adjunction. Then F is a triangulated functor if and only if G is a triangulated functor. In this case the unit $\eta : \mathrm{id}_{\mathcal{D}} \to GF$ and counit $\varepsilon : FG \to \mathrm{id}_{\mathcal{D}'}$ of the adjunction are compatible with the translation functors; see Definition 4.1.1.

Proof By duality, it suffices to consider the case in which F is a triangulated functor. Corollary 1.3.6 shows that G is additive. Denote both translation functors on $\mathcal{D}$ and $\mathcal{D}'$ by T. Transporting the structure gives a new adjunction

$$\hat{F} := TFT^{-1}, \quad \hat{G} := TGT^{-1}, \quad \hat{\eta} := T\eta T^{-1}, \quad \hat{\varepsilon} := T\varepsilon T^{-1}.$$

Since $\hat{F} \simeq F$, uniqueness of right adjoints [51, Proposition 2.6.10] gives a corresponding isomorphism $\hat{G} \simeq G$ and commutative diagrams

$$\begin{tikzcd} \hat{F}\hat{G} \arrow[r, "\hat{\varepsilon}"] \arrow[d, "\wr"] & \mathrm{id}_{\mathcal{D}'} \\ FG \arrow[ur, "\varepsilon"'] & \end{tikzcd} \qquad \begin{tikzcd} \hat{G}\hat{F} \arrow[d, "\wr"] & \mathrm{id}_{\mathcal{D}} \arrow[l, "\hat{\eta}"'] \arrow[dl, "\eta"] \\ GF & \end{tikzcd}$$

In particular, $GT \simeq TG$ and $FGT \simeq FTG \simeq TFG$. After composing the commutative diagram for $\hat{\varepsilon}$ and ε on the right with T, it can be rewritten as

$$\begin{tikzcd} TFG \arrow[r, "T\varepsilon"] \arrow[d, "\wr"] & T \\ FGT \arrow[ur, "\varepsilon T"'] & \end{tikzcd} \tag{4.2.1}$$

This equation and its dual version for η show that η and ε are compatible with the translation functors.

We now show that G preserves distinguished triangles. Consider a distinguished triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{+1}$ in $\mathcal{D}'$. Choose a distinguished triangle $GX \xrightarrow{Gf} GY \xrightarrow{h} A \xrightarrow{+1}$ in $\mathcal{D}$, apply F to it, and then use (TR3) to obtain the commutative diagram

$$\begin{tikzcd} FGX \arrow[r, "FGf"] \arrow[d, "\varepsilon_X"'] & FGY \arrow[r, "Fh"] \arrow[d, "\varepsilon_Y"] & FA \arrow[r] \arrow[d] & TFGX \arrow[d, "T\varepsilon_X"] \\ X \arrow[r, "f"'] & Y \arrow[r, "g"'] & Z \arrow[r] & TX. \end{tikzcd}$$

Apply the adjunction to this diagram, and adjust the rightmost square using (4.2.1) and $GT \simeq TG$. One readily obtains a commutative diagram

$$\begin{tikzcd} GX \arrow[r, "Gf"] \arrow[d, "\mathrm{id}_X"'] & GY \arrow[r, "h"] \arrow[d, "\mathrm{id}_Y"] & A \arrow[r] \arrow[d] & TGX \arrow[d, "\mathrm{id}_{TGX}"] \\ GX \arrow[r, "Gf"'] & GY \arrow[r, "Gg"'] & GZ \arrow[r] & TGX \end{tikzcd}$$

whose first row is a distinguished triangle. For every $B \in \mathrm{Ob}(\mathcal{D})$, apply $\mathrm{Hom}_{\mathcal{D}}(B, \cdot)$ to obtain a commutative diagram

$$\begin{tikzcd} \mathrm{Hom}(B, GX) \arrow[r, "(Gf)_*"] \arrow[d, "\simeq"'] & \mathrm{Hom}(B, GY) \arrow[r, "h_*"] \arrow[d, "\simeq"] & \mathrm{Hom}(B, A) \arrow[r] \arrow[d] & \mathrm{Hom}(B, TGX) \arrow[r] \arrow[d, "\simeq"] & \mathrm{Hom}(B, TGY) \arrow[d, "\simeq"] \\ \mathrm{Hom}(B, GX) \arrow[r, "(Gf)_*"'] & \mathrm{Hom}(B, GY) \arrow[r, "(Gg)_*"'] & \mathrm{Hom}(B, GZ) \arrow[r] & \mathrm{Hom}(B, TGX) \arrow[r] & \mathrm{Hom}(B, TGY) \end{tikzcd}$$

Proposition 4.1.12 shows that the first row is exact. On the other hand, by $TG \simeq GT$ and the adjunction, the second row is isomorphic to the sequence induced by the distinguished triangle $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{+1}$, namely

$$\mathrm{Hom}(FB, X) \xrightarrow{f_*} \mathrm{Hom}(FB, Y) \xrightarrow{g_*} \mathrm{Hom}(FB, Z) \longrightarrow \mathrm{Hom}(FB, TX) \longrightarrow \mathrm{Hom}(FB, TY),$$

and hence is also exact. Since B is arbitrary, Proposition 2.3.4 implies that $A \to GZ$ is an isomorphism. Thus $GX \xrightarrow{Gf} GY \xrightarrow{Gg} GZ \to TGX$ is indeed a distinguished triangle in $\mathcal{D}$. □

Corollary 4.2.9 Let $\mathcal{D}$ and $\mathcal{D}'$ be pretriangulated categories and let the functors $\mathcal{D} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}'$ be quasi-inverse to one another. Then:

(i) F is a triangulated functor if and only if G is a triangulated functor;

(ii) if F is a triangulated functor, then F and G give an equivalence of the pretriangulated categories.

Proof Use [51, Theorem 2.6.12] to realize (F, G) and (G, F) as adjunctions, and then apply Proposition 4.2.8. □

Definition–Proposition 4.2.10 (Pretriangulated subcategories) Let $\mathcal{D}'$ be a full additive subcategory of a pretriangulated category $\mathcal{D}$ that is closed under the translation functor T. Denote the inclusion functor by $\iota : \mathcal{D}' \to \mathcal{D}$.

(i) If a pretriangulated structure on $\mathcal{D}'$ making ι a triangulated functor exists, it is unique. In this case, a triangle $X \to Y \to Z \xrightarrow{+1}$ in $\mathcal{D}'$ is distinguished if and only if it is distinguished in $\mathcal{D}$.

(ii) This pretriangulated structure exists if and only if the following condition holds: whenever $X \to Y \to Z \xrightarrow{+1}$ is a distinguished triangle in $\mathcal{D}$ and any two of its terms belong to $\mathrm{Ob}(\mathcal{D}')$, the remaining term is isomorphic to an object of $\mathcal{D}'$.

(iii) If, in addition, $\mathcal{D}$ is assumed to be triangulated, then $\mathcal{D}'$ is also triangulated.

Such a pretriangulated (or triangulated) category $\mathcal{D}'$ is called a **pretriangulated subcategory** (or **triangulated subcategory**) of $\mathcal{D}$.

Proof First consider (i). Suppose that ι is triangulated. Every distinguished triangle in $\mathcal{D}'$ is of course distinguished in $\mathcal{D}$. Conversely, suppose that a distinguished triangle $X \xrightarrow{f} Y \to Z \xrightarrow{+1}$ in $\mathcal{D}$ satisfies $X, Y, Z \in \mathrm{Ob}(\mathcal{D}')$. Choose any distinguished triangle $X \xrightarrow{f} Y \to Z' \xrightarrow{+1}$ in $\mathcal{D}'$. Applying Corollary 4.2.3 in $\mathcal{D}$ shows that the two triangles are isomorphic. Since $\mathcal{D}'$ is full, (TR0) implies that $X \xrightarrow{f} Y \to Z \xrightarrow{+1}$ is also distinguished in $\mathcal{D}'$. This is precisely the pretriangulated structure asserted in (i).

If this pretriangulated structure exists, then Corollary 4.2.3 shows that the condition in (ii) is necessary, since after rotation we may always assume that $X, Y \in \mathrm{Ob}(\mathcal{D}')$. Conversely, assume the condition in (ii) and define the distinguished triangles in $\mathcal{D}'$ as stated in (i). Axioms (TR0), (TR1), (TR3), and (TR4) for $\mathcal{D}'$ are inherited directly from $\mathcal{D}$. For (TR2), given a morphism $f : X \to Y$ in $\mathcal{D}'$, choose a distinguished triangle $X \xrightarrow{f} Y \to Z \xrightarrow{+1}$ in $\mathcal{D}$. The condition says that Z is isomorphic to an object of $\mathcal{D}'$; we may therefore assume that $Z \in \mathrm{Ob}(\mathcal{D}')$ and thus obtain a distinguished triangle in $\mathcal{D}'$. This proves that $\mathcal{D}'$ is pretriangulated and establishes sufficiency.

For (iii), the point is to verify (TR5) for $\mathcal{D}'$. Indeed, the triangle $Z' \to Y' \to X' \xrightarrow{+1}$ obtained by applying (TR5) in $\mathcal{D}$ is automatically distinguished in $\mathcal{D}'$ by (i). □

Convention 4.2.11 Fix a category $\mathcal{C}$. A subcategory $\mathcal{C}'$ is called a **saturated subcategory** if it is closed under isomorphisms, that is,

$$\forall X \in \mathrm{Ob}(\mathcal{C}'),\ \forall Y \in \mathrm{Ob}(\mathcal{C}), \quad X \simeq Y \implies Y \in \mathrm{Ob}(\mathcal{C}').$$

For every nonempty family of full subcategories $(\mathcal{C}_i)_{i\in I}$, their intersection $\bigcap_{i\in I} \mathcal{C}_i$ is by definition the full subcategory of $\mathcal{C}$ satisfying $\mathrm{Ob}\left(\bigcap_{i\in I} \mathcal{C}_i\right) = \bigcap_{i\in I} \mathrm{Ob}(\mathcal{C}_i)$.

Proposition 4.2.12 Let $\mathcal{D}$ be a pretriangulated (or triangulated) category, and let $(\mathcal{D}_i)_{i\in I}$ be a family of saturated pretriangulated (or saturated triangulated) subcategories, with $I \neq \varnothing$. Then $\bigcap_{i\in I} \mathcal{D}_i$ is also a saturated pretriangulated (or saturated triangulated) subcategory.

Proof An intersection of saturated subcategories plainly preserves the condition in Definition–Proposition 4.2.10 (ii). □

A common technique is to cut out a saturated pretriangulated (or triangulated) subcategory using a cohomological functor.

Proposition 4.2.13 Let $\mathcal{D}$ be a pretriangulated (or triangulated) category, $H : \mathcal{D} \to \mathcal{A}$ a cohomological functor, and $\mathcal{T}$ a weak Serre subcategory of $\mathcal{A}$ (Definition 2.9.3). Define a full subcategory $\mathcal{D}_{H,\mathcal{T}}$ of $\mathcal{D}$ by

$$X \in \mathrm{Ob}(\mathcal{D}_{H,\mathcal{T}}) \iff \forall n \in \mathbb{Z},\ H(T^n X) \in \mathrm{Ob}(\mathcal{T}).$$

Then $\mathcal{D}_{H,\mathcal{T}}$ is a saturated pretriangulated (or saturated triangulated) subcategory of $\mathcal{D}$.

Proof It suffices to verify the condition in Definition–Proposition 4.2.10 (ii). Recall that a weak Serre subcategory is saturated. Hence saturation and T-invariance of $\mathcal{D}_{H,\mathcal{T}}$ are clear. Now consider a distinguished triangle $X \to Y \to Z \xrightarrow{+1}$ in $\mathcal{D}$, two of whose terms lie in $\mathcal{D}_{H,\mathcal{T}}$. After rotation, we may assume without loss of generality that $X, Z \in \mathrm{Ob}(\mathcal{D}_{H,\mathcal{T}})$. For every $n \in \mathbb{Z}$, there is an exact sequence

$$H(T^{n-1}Z) \to H(T^n X) \to H(T^n Y) \to H(T^n Z) \to H(T^{n+1}X).$$

The definition of a weak Serre subcategory immediately gives $H(T^n Y) \in \mathrm{Ob}(\mathcal{T})$. □

Example 4.2.14 Let $\mathcal{D}$ be a pretriangulated (or triangulated) category and let $H : \mathcal{D} \to \mathcal{A}$ be a cohomological functor. Define $\mathcal{N}_H$ to be the full subcategory whose objects satisfy $H(T^n X) = 0$ for every $n \in \mathbb{Z}$. This is the case of Proposition 4.2.13 in which $\mathcal{T}$ is the subcategory consisting of zero objects. Thus $\mathcal{N}_H$ is a pretriangulated (or triangulated) subcategory.

The following technical result will be used later in § 4.3; it depends on the octahedral axiom (TR5).

Lemma 4.2.15 (J.-L. Verdier) Every commutative diagram in a triangulated category $\mathcal{D}$

$$\begin{tikzcd}
X \arrow[r, "f"] & Y \\
X' \arrow[u, "u"] \arrow[r, "f'"'] & Y' \arrow[u, "v"']
\end{tikzcd}$$

can be extended to a diagram

$$\begin{tikzcd}
TX' \arrow[r, "Tf'"] & TY' \arrow[r, "Tg'"] & TZ' \arrow[r, "-Th'"] & T^2X' \\
X'' \arrow[u, "o"] \arrow[r, "f''"] & Y'' \arrow[u, "p"] \arrow[r, "g''"] & Z'' \arrow[u, "q"] \arrow[r, "h''"] \arrow[ur, phantom, "\star"] & TX'' \arrow[u, "-To"'] \\
X \arrow[u, "r"] \arrow[r, "f"] & Y \arrow[u, "s"] \arrow[r, "g"] & Z \arrow[u, "t"] \arrow[r, "h"] & TX \arrow[u, "Tr"'] \\
X' \arrow[u, "u"] \arrow[r, "f'"'] & Y' \arrow[u, "v"] \arrow[r, "g'"'] & Z' \arrow[u, "w"] \arrow[r, "h'"'] & TX' \arrow[u, "Tu"']
\end{tikzcd}$$

such that every row and column is a distinguished triangle, the square marked $\star$ is anticommutative (that is, $(To)h'' = -(Th')q$), and every other square is commutative.

Proof We construct the required distinguished triangles step by step and then prove their commutativity and anticommutativity properties. First apply (TR2) separately to u, v, f, and f' to obtain the portion

$$\begin{tikzcd}
\bullet & \bullet & & \\
\bullet \arrow[u] & \bullet \arrow[u] & & \\
\bullet \arrow[u] \arrow[r] & \bullet \arrow[u] \arrow[r] & \bullet \arrow[r] & \bullet \\
\bullet \arrow[u] \arrow[r] & \bullet \arrow[u] \arrow[r] & \bullet \arrow[r] & \bullet
\end{tikzcd}$$

of the diagram, with every row and column distinguished. Next apply (TR2) to $fu = vf' : X' \to Y$ to obtain a distinguished triangle $X' \to Y \xrightarrow{m} A \xrightarrow{n} TX'$. The following construction is based on (TR5).

1. Apply (TR5) to the three distinguished triangles containing f', v, and vf' to

obtain the following morphism of distinguished triangles:

$$\begin{tikzcd}
X' \arrow[r, "f'"] \arrow[d, equal] & Y' \arrow[r, "g'"] \arrow[d, "v"] & Z' \arrow[r, "h'"] \arrow[d, "i"] & TX' \arrow[d, equal] \\
X' \arrow[r, "vf'"] \arrow[d, "f'"'] & Y \arrow[r, "m"] \arrow[d, equal] & A \arrow[r, "n"] \arrow[d, "j"] & TX' \arrow[d, "Tf'"] \\
Y' \arrow[r, "v"] \arrow[d, "g'"'] & Y \arrow[r, "s"] \arrow[d, "m"] & Y'' \arrow[r, "p"] \arrow[d, equal] & TY' \arrow[d, "Tg'"] \\
Z' \arrow[r, "i"'] & A \arrow[r, "j"'] & Y'' \arrow[r, "k"'] & TZ'
\end{tikzcd} \tag{4.2.2}$$

The last row is a newly constructed distinguished triangle.

2. Apply (TR5) to the three distinguished triangles containing u, f, and fu to obtain the following morphism of distinguished triangles:

$$\begin{tikzcd}
X' \arrow[r, "u"] \arrow[d, equal] & X \arrow[r, "r"] \arrow[d, "f"] & X'' \arrow[r, "o"] \arrow[d, "a"] & TX' \arrow[d, equal] \\
X' \arrow[r, "fu"] \arrow[d, "u"'] & Y \arrow[r, "m"] \arrow[d, equal] & A \arrow[r, "n"] \arrow[d, "b"] & TX' \arrow[d, "Tu"] \\
X \arrow[r, "f"] \arrow[d, "r"'] & Y \arrow[r, "g"] \arrow[d, "m"] & Z \arrow[r, "h"] \arrow[d, equal] & TX \arrow[d, "Tr"] \\
X'' \arrow[r, "a"'] & A \arrow[r, "b"'] & Z \arrow[r, "c"'] & TX''
\end{tikzcd} \tag{4.2.3}$$

The last row is a newly constructed distinguished triangle.

3. Apply (TR2) to $f'' := ja : X'' \to Y''$ to obtain a distinguished triangle $X'' \xrightarrow{f''} Y'' \xrightarrow{g''} Z'' \xrightarrow{h''} TX''$.

4. Consider the following distinguished triangles obtained from the preceding steps and rotations (TR3):

$$X'' \xrightarrow{a} A \xrightarrow{b} Z \xrightarrow{c} TX'', \quad A \xrightarrow{j} Y'' \xrightarrow{k} TZ' \xrightarrow{-Ti} TA,$$
$$X'' \xrightarrow{f''=ja} Y'' \xrightarrow{g''} Z'' \xrightarrow{h''} TX''.$$

Applying (TR5) then gives a morphism of distinguished triangles

$$
\begin{tikzcd}
X'' \arrow[r, "a"] \arrow[d, equal] & A \arrow[r, "b"] \arrow[d, "j"] & Z \arrow[r, "c"] \arrow[d, "t"] & TX'' \arrow[d, equal] \\
X'' \arrow[r, "f''"] \arrow[d, "a"'] & Y'' \arrow[r, "g''"] \arrow[d, equal] & Z'' \arrow[r, "h''"] \arrow[d, "q"] & TX'' \arrow[d, "Ta"] \\
A \arrow[r, "j"] \arrow[d, "b"'] & Y'' \arrow[r, "k"] \arrow[d, "m"] & TZ' \arrow[r, "-Ti"] \arrow[d, equal] & TA \arrow[d, "Tb"] \\
Z \arrow[r, "t"'] & Z'' \arrow[r, "q"'] & TZ' \arrow[r, "l"'] & TZ
\end{tikzcd}
\tag{4.2.4}
$$

The last row is a newly constructed distinguished triangle. Set $w := bi$. Commutativity of (4.2.4) implies $l = -Tw$; rotation therefore gives the distinguished triangle

$$Z' \xrightarrow{w} Z \xrightarrow{t} Z'' \xrightarrow{q} TZ'.$$

The operations above give the second row and third column of the required diagram. The remaining first row (or fourth column) is obtained by rotating the fourth row (or first column) according to (TR3), and is therefore also a distinguished triangle (see Remark 4.1.5).

We now verify the commutativity and anticommutativity properties. From $w = bi$ and $s = jm$, as implied by (4.2.2), it follows that (u, v, w) and (r, s, t) are respectively the composites

$$
\begin{gathered}
(X', Y', Z') \xrightarrow[(4.2.2)]{(\mathrm{id}, v, i)} (X', Y, A) \xrightarrow[(4.2.3)]{(u, \mathrm{id}, b)} (X, Y, Z), \\
(X, Y, Z) \xrightarrow[(4.2.3)]{(r, m, \mathrm{id})} (X'', A, Z) \xrightarrow[(4.2.4)]{(\mathrm{id}, j, t)} (X'', Y'', Z''),
\end{gathered}
$$

so both are morphisms of triangles. Finally, consider the morphism $(o, p, q, -To)$ in the diagram. It remains to prove that the following diagram is commutative:

$$
\begin{tikzcd}
TX' \arrow[r, "Tf'"] & TY' \arrow[r, "Tg'"] & TZ' \arrow[r, "-Th'"] & T^2X' \\
A \arrow[r, "j"] \arrow[u, "n"] & Y'' \arrow[r, "k"] \arrow[u, "p"] & TZ' \arrow[r, "-Ti"] \arrow[u, equal] & TA \arrow[u, "Tn"'] \\
X'' \arrow[r, "f''"'] \arrow[u, "a"] \arrow[uu, bend left=60, "o"] & Y'' \arrow[r, "g''"'] \arrow[u, equal] & Z'' \arrow[r, "h''"'] \arrow[u, "q"] & TX''. \arrow[u, "Ta"'] \arrow[uu, bend right=60, "To"']
\end{tikzcd}
$$

Indeed, (4.2.3) implies $o = na$, so both wings commute. Since the three squares on the right of (4.2.2) commute, the upper part also commutes; commutativity of the lower part follows from (4.2.4). This proves the assertion. $\square$

Except for Proposition 4.2.8, every assertion in this section also holds for $\Bbbk$-linear categories, where $\Bbbk$ is a commutative ring.

Remark 4.2.16 Generators and compact objects in triangulated categories are important topics not treated in this book. Although the ideas resemble the versions in Definition 1.11.8 and A.2.2, these notions require different definitions in a triangulated category. This leads to an important form of the Brown Representability Theorem for triangulated categories: if $\mathcal{D}$ is a "compactly generated" triangulated category with small $\coprod$, and a cohomological functor $H : \mathcal{D}^{\mathrm{op}} \to \mathsf{Ab}$ maps small $\coprod$ in $\mathcal{D}$ to $\prod$ in Ab, then H is representable. Readers who need this result are advised to consult [42, Tags 09SI, 09SM, 0A8E].

Notice the family resemblance between the Brown Representability Theorem and the various representability and adjoint-functor theorems in §1.11, §2.10, and § A.2.

4.3 Localization of Triangulated Categories

This section examines the relation between triangulated categories and Gabriel–Zisman localization. More precisely,

- ◇ the construction of categorical localization in §1.9 formally adjoins inverses to the morphisms in a multiplicative system, and our first task is to show that this construction preserves the (pre)triangulated structure;
- ◇ for a triangulated category, localization can also be viewed as formally making a triangulated subcategory zero; this idea is analogous to the construction of Serre quotients in §2.9.

Throughout what follows, $(\mathcal{D}, T)$ is assumed to be a pretriangulated category. If one passes from additive categories to the $\Bbbk$-linear setting, where $\Bbbk$ is any commutative ring, every result in this section has a corresponding generalization; we omit the details.

Definition 4.3.1 Let $S \subset \mathrm{Mor}(\mathcal{D})$ be a multiplicative system as in Definition 1.9.5. We say that S is **compatible with triangulation** if the following conditions hold.

(ST1) For $s \in \mathrm{Mor}(\mathcal{D})$, one has $s \in S$ if and only if $Ts \in S$.

(ST2) Consider the following diagram, in which the solid part is given:

$$\begin{tikzcd} X \arrow[r] \arrow[d, "\alpha"'] & Y \arrow[r] \arrow[d, "\beta"] & Z \arrow[r] \arrow[d, dashed, "\gamma"] & TX \arrow[d, dashed, "T\alpha"] \\ X' \arrow[r] & Y' \arrow[r] & Z' \arrow[r] & TX' \end{tikzcd}$$

If $\alpha, \beta \in S$, then there is a morphism γ, as indicated by the dashed arrow, such that the entire diagram commutes and $\gamma \in S$.

Notice that (ST2) is a strengthening of (TR4) for S.

Proposition 4.3.2 Let $S \subset \mathrm{Mor}(\mathcal{D})$ be a multiplicative system compatible with triangulation. Then the localization $\mathcal{D}[S^{-1}]$ has a unique pretriangulated-category structure for which $Q : \mathcal{D} \to \mathcal{D}[S^{-1}]$ is a triangulated functor. Up to isomorphism, the distinguished triangles in $\mathcal{D}[S^{-1}]$ are precisely the images of distinguished triangles in $\mathcal{D}$. If $\mathcal{D}$ is triangulated, then so is $\mathcal{D}[S^{-1}]$.

Proof Theorem 1.9.17 shows that $\mathcal{D}[S^{-1}]$ is an additive category. Moreover, (ST1) and the universal property of localization give a unique pair of endofunctors of $\mathcal{D}[S^{-1}]$ that make the following diagram commute:

$$\begin{tikzcd} \mathcal{D} \arrow[r, shift left, "T"] \arrow[d, "Q"'] & \mathcal{D} \arrow[l, shift left, "T^{-1}"] \arrow[d, "Q"] \\ \mathcal{D}[S^{-1}] \arrow[r, shift left] & \mathcal{D}[S^{-1}] \arrow[l, shift left] \end{tikzcd}$$

The universal property also shows that the two functors in the second row are mutually inverse. To simplify notation, we continue to write them as $\mathcal{D}[S^{-1}] \underset{T^{-1}}{\overset{T}{\rightleftarrows}} \mathcal{D}[S^{-1}]$. Thus $(\mathcal{D}[S^{-1}], T)$ is an additive category with translation.

We first prove uniqueness of the pretriangulated structure. By definition, the triangulated functor Q must preserve distinguished triangles. Conversely, let $X \xrightarrow{f} Y \to Z \xrightarrow{+1}$ be a distinguished triangle in $\mathcal{D}[S^{-1}]$. After replacing it by an isomorphic triangle, we may assume that $f : X \to Y$ comes from a morphism in $\mathcal{D}$, which we denote by $f_0 : X_0 \to Y_0$ to avoid ambiguity. By (TR2), extend f_0 to a distinguished triangle $X_0 \xrightarrow{f_0} Y_0 \to Z_0 \xrightarrow{+1}$. Corollary 4.2.3 then guarantees that its image under Q is isomorphic to the original triangle.

We now prove existence of the pretriangulated structure. Define the distinguished triangles in $\mathcal{D}[S^{-1}]$ as stated in the proposition. Axiom (TR0) is immediate, while (TR1) and (TR3) are inherited directly from $\mathcal{D}$. For (TR2), consider a morphism $f : X \to Y$ in $\mathcal{D}[S^{-1}]$. As usual, after replacing its objects by isomorphic ones using morphisms from S, we may assume that f comes from $\mathcal{D}$; thus (TR2) reduces to the assertion in $\mathcal{D}$.

For (TR4), consider the following commutative diagram in $\mathcal{D}[S^{-1}]$; the dashed part will be dealt with below.

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\alpha"'] & Y \arrow[r, "g"] \arrow[d, "\beta"] & Z \arrow[r, "h"] \arrow[d, dashed, "\gamma"] & TX \arrow[d, dashed, "T\alpha"] \\ X' \arrow[r, "f'"'] & Y' \arrow[r, "g'"'] & Z' \arrow[r, "h'"'] & TX' \end{tikzcd} \tag{4.3.1}$$

Here each row is a distinguished triangle and all horizontal arrows come from $\mathcal{D}$. From the construction in §1.9, one can show that there are objects $A, B \in \mathrm{Ob}(\mathcal{D})$, a morphism $[A \to B] \in \mathrm{Mor}(\mathcal{D})$, and $s, t \in S$ such that $\alpha = as^{-1}$ and $\beta = bt^{-1}$ in $\mathcal{D}[S^{-1}]$, and such that the solid part of the following diagram commutes in $\mathcal{D}$. The dashed part and the object C will be discussed presently.

$$\begin{tikzcd}
X \arrow[r, "f"] & Y \arrow[r, "g"] & Z \arrow[r, "h"] & TX \\
A \arrow[r] \arrow[u, "s"] \arrow[d, "a"'] & B \arrow[r, dashed] \arrow[u, "t"] \arrow[d, "b"'] & C \arrow[r, dashed] \arrow[u, dashed, "u"] \arrow[d, dashed, "c"'] & TA \arrow[u, dashed, "Ts"'] \arrow[d, dashed, "Ta"] \\
X' \arrow[r, "f'"'] & Y' \arrow[r, "g'"'] & Z' \arrow[r, "h'"'] & TX'.
\end{tikzcd}$$

The explicit construction is somewhat cumbersome and appears as a guided exercise in Chapter 1.

Apply (TR2) to $A \to B$ in $\mathcal{D}$ to obtain a distinguished triangle $A \to B \to C \xrightarrow{+1}$ as the second row. Then apply (ST2) to obtain $C \xrightarrow{u \in S} Z$, and apply (TR4) to obtain $C \xrightarrow{c} Z'$, so that the entire diagram commutes in $\mathcal{D}$. In $\mathcal{D}[S^{-1}]$, set $\gamma := cu^{-1}$. With this choice the whole diagram (4.3.1) commutes. This verifies (TR4).

If $\mathcal{D}$ is triangulated, verification of (TR5) for $\mathcal{D}[S^{-1}]$ likewise reduces to verification in $\mathcal{D}$. □

Proposition 4.3.3 Let $\mathcal{N}$ be a saturated triangulated subcategory of a triangulated category $\mathcal{D}$ (Definition–Proposition 4.2.10, Convention 4.2.11). Then

$$S\mathcal{N} := \left\{ s \in \mathrm{Mor}(\mathcal{D}) \;\middle|\; \begin{array}{l} \exists \text{ a distinguished triangle } X \xrightarrow{s} Y \to Z \xrightarrow{+1} \\ \text{such that } Z \in \mathrm{Ob}(\mathcal{N}) \end{array} \right\}$$

is a multiplicative system compatible with triangulation.

Proof Since $\mathcal{N}$ is closed under translation, rotating triangles shows that $S\mathcal{N}$ satisfies (ST1). Next consider the commutative diagram in (ST2), and suppose that $\alpha, \beta \in S\mathcal{N}$. Lemma 4.2.15 gives a diagram

$$\begin{tikzcd}
{} & {} & {} & \\
X'' \arrow[r] \arrow[u, "+1"] & Y'' \arrow[r] \arrow[u, "+1"] & Z'' \arrow[r, "+1"] \arrow[u, "+1"] & {} \\
X \arrow[r] \arrow[u] & Y \arrow[r] \arrow[u] & Z \arrow[r, "+1"] \arrow[u] & {} \\
X' \arrow[r] \arrow[u, "\alpha"] & Y' \arrow[r] \arrow[u, "\beta"] & Z' \arrow[r, "+1"] \arrow[u, "\gamma"] & {}
\end{tikzcd}$$

in which every row and column is a distinguished triangle, all squares commute, and (α, β, γ) is a morphism of triangles. Using the saturation of $\mathcal{N}$, the fact that $\alpha, \beta \in S$, and Corollary 4.2.3, we see that $X'', Y'' \in \mathrm{Ob}(\mathcal{N})$. The definition of a saturated triangulated subcategory then implies $Z'' \in \mathrm{Ob}(\mathcal{N})$, hence $\gamma \in S$. This verifies (ST2).

We next verify the axioms for a multiplicative system (Definition 1.9.5). Since $0 \in \mathrm{Ob}(\mathcal{N})$, axiom (S1) follows from (TR1).

To verify (S2), let $f, g \in S\mathcal{N}$. Choose distinguished triangles $X \xrightarrow{f} Y \to Z' \xrightarrow{+1}$ and $Y \xrightarrow{g} Z \to X' \xrightarrow{+1}$ with $Z', X' \in \mathrm{Ob}(\mathcal{N})$, and use (TR2) to choose a distinguished triangle $X \xrightarrow{gf} Z \to Y' \xrightarrow{+1}$. Applying (TR5) gives a distinguished triangle $Z' \to Y' \to X' \xrightarrow{+1}$. The saturated triangulated-subcategory property therefore implies $Y' \in \mathrm{Ob}(\mathcal{N})$, that is, $gf \in S\mathcal{N}$.

To verify (S3), consider morphisms $X \xrightarrow{s \in S\mathcal{N}} Z \xleftarrow{f} Y$. Choose a distinguished triangle $X \xrightarrow{s} Z \xrightarrow{h} N \xrightarrow{+1}$ with $N \in \mathrm{Ob}(\mathcal{N})$, and use (TR2) to choose a triangle $Y \xrightarrow{hf} N \to U \xrightarrow{+1}$. Apply (TR4) to the solid part of the diagram

$$\begin{tikzcd}
Y \arrow[r, "hf"] \arrow[d, "f"'] & N \arrow[r] \arrow[d, equal] & U \arrow[r, "+1"] \arrow[d, dashed] & TY \arrow[d, dashed, "Tf"] \\
Z \arrow[r, "h"'] & N \arrow[r] & TX \arrow[r, "-Ts"'] & TZ
\end{tikzcd}$$

to obtain the morphisms indicated by the dashed arrows, and hence a morphism of distinguished triangles. After rotation, this yields an element $s' : W := T^{-1}U \to Y$ of $S\mathcal{N}$, a corresponding morphism $f' : W \to X$, and the commutative diagram required in (S3):

$$\begin{tikzcd}
X \arrow[d, "s \in S\mathcal{N}"'] & W \arrow[l, "f'"'] \arrow[d, "s' \in S\mathcal{N}"] \\
Z & Y. \arrow[l, "f"]
\end{tikzcd}$$

To verify (S4), replace (f, g) in that axiom by $(f - g, 0)$. It suffices to prove the following assertion: for a morphism $f : X \to Y$, if there is an $s : Y \to W$ with $s \in S\mathcal{N}$ and $sf = 0$, then there is a $t : Z \to X$ with $t \in S\mathcal{N}$ and $ft = 0$. To prove this, choose a distinguished triangle $N \to Y \xrightarrow{s} W \xrightarrow{+1}$ with $N \in \mathrm{Ob}(\mathcal{N})$. Since $\mathrm{Hom}(X, \cdot)$ is cohomological and $sf = 0$, there is an $h \in \mathrm{Hom}(X, N)$ such that f factors as $X \xrightarrow{h} N \to Y$. Choose a distinguished triangle $Z \xrightarrow{t} X \xrightarrow{h} N \xrightarrow{+1}$; then $t \in S\mathcal{N}$. On the other hand, $ht = 0$ (Lemma 4.1.9) implies $ft = 0$. Thus (S4) holds.

The arguments for the dual versions of (S3) and (S4) are identical. □

Theorem 4.3.4 (Verdier localization) Let $\mathcal{N}$ be a saturated triangulated subcategory of $\mathcal{D}$. Define $\mathcal{D}/\mathcal{N} := \mathcal{D}\left[(S\mathcal{N})^{-1}\right]$ (which is allowed to be a "large" category), equipped with the triangulated structure from Proposition 4.3.2. Then the localization functor $Q : \mathcal{D} \to \mathcal{D}/\mathcal{N}$ has the following properties.

(i) For every morphism $f : X \to Y$ in $\mathcal{D}$, one has $Qf = 0$ if and only if f factors as $X \to N \to Y$ with $N \in \mathrm{Ob}(\mathcal{N})$; in particular, Q maps $\mathcal{N}$ to 0.

(ii) For every pretriangulated category $\mathcal{E}$ and every triangulated functor $F : \mathcal{D} \to \mathcal{E}$, if F maps $\mathcal{N}$ to 0, then F factors uniquely as $\mathcal{D} \xrightarrow{Q} \mathcal{D}/\mathcal{N} \xrightarrow{\overline{F}} \mathcal{E}$, where $\overline{F}$ is a triangulated functor.

(iii) For every Abelian category $\mathcal{A}$ and every cohomological functor $H : \mathcal{D} \to \mathcal{A}$, if H maps $\mathcal{N}$ to 0, then H factors uniquely as $\mathcal{D} \xrightarrow{Q} \mathcal{D}/\mathcal{N} \xrightarrow{\overline{H}} \mathcal{A}$, where $\overline{H}$ is a cohomological functor.

Proof For (i), if f factors as $X \to N \to Y$, then $Qf = 0$. Indeed, the rotation $N \to 0 \to TN \xrightarrow{+1}$ of the distinguished triangle $N \xrightarrow{\mathrm{id}_N} N \to 0 \xrightarrow{+1}$ shows that $N \to 0$ is mapped to an isomorphism in $\mathcal{D}/\mathcal{N}$. Conversely, if $Qf = 0$, there is a morphism $s : M \to X$ with $s \in S\mathcal{N}$ and $fs = 0$ (Corollary 1.9.13). By the definition of $S\mathcal{N}$, there is a commutative diagram whose rows are distinguished triangles (the solid part):

$$\begin{tikzcd}
M \arrow[r, "s"] \arrow[d] & X \arrow[r] \arrow[d, "f"] & N \arrow[r] \arrow[d, dashed] & TM \arrow[d, dashed] \\
0 \arrow[r] & Y \arrow[r, "\mathrm{id}_Y"'] & Y \arrow[r] & 0
\end{tikzcd} \qquad (N \in \mathrm{Ob}(\mathcal{N})),$$

By (TR4), the morphisms indicated by the dashed arrows can be supplied so that the entire diagram commutes. This diagram factors f as $X \to N \to Y$.

Consider the situation in (ii). Put $s \in S\mathcal{N}$ into a distinguished triangle $X \xrightarrow{s} Y \to N \xrightarrow{+1}$ with $N \in \mathrm{Ob}(\mathcal{N})$. Then $FX \xrightarrow{Fs} FY \to 0 \xrightarrow{+1}$ is a distinguished triangle in $\mathcal{E}$. Corollary 4.1.13 shows that Fs is an isomorphism. The universal property then gives a unique factorization $H = \overline{F} \circ Q$, where $\overline{F}$ is an additive functor. In view of the description of the triangulated structure on $\mathcal{D}/\mathcal{N}$ (Proposition 4.3.2), it is straightforward to verify that $\overline{F}$ preserves translations and triangles; the details are left to the reader.

Consider the situation in (iii). Again put $s \in S\mathcal{N}$ into a distinguished triangle $X \xrightarrow{s} Y \to N \xrightarrow{+1}$. Applying H gives the exact sequence $\underbrace{HT^{-1}N}_{=0} \to HX \xrightarrow{Hs} HY \to \underbrace{HN}_{=0}$, so Hs is an isomorphism. The universal property then gives a unique factorization $H = \overline{H} \circ Q$, where $\overline{H}$ is additive. To verify that $\overline{H}$ is cohomological, recall that the distinguished triangles in $\mathcal{D}/\mathcal{N}$ are isomorphic to images of distinguished triangles in $\mathcal{D}$. $\square$

Example 4.3.5 Cohomological functors give an important class of multiplicative systems compatible with triangulation. Let $H : \mathcal{D} \to \mathcal{A}$ be a cohomological functor. A morphism $f : X \to Y$ in $\mathcal{D}$ is called an **H-quasi-isomorphism** if $H(T^n f)$ is an isomorphism in $\mathcal{A}$ for every $n \in \mathbb{Z}$. All H-quasi-isomorphisms form a subset $S_H \subset \mathrm{Mor}(\mathcal{D})$. It plainly satisfies (ST1), while (ST2) follows from the long exact sequence of a cohomological functor (Remark 4.1.11) together with Proposition 2.3.4.

On the other hand, define $\mathcal{N}_H$ as in Example 4.2.14. It is a saturated triangulated subcategory, so $S\mathcal{N}_H$ is defined. The two are related as follows.

Proposition 4.3.6 Let $H : \mathcal{D} \to \mathcal{A}$ be a cohomological functor. Then $S\mathcal{N}_H = S_H$.

Proof Let $f : X \to Y$ be a morphism in $\mathcal{D}$. Extend f to a distinguished triangle $X \xrightarrow{f} Y \to N \xrightarrow{+1}$. Recall that N is unique up to isomorphism. The long exact sequence of a cohomological functor therefore implies

$$f \text{ is an } H\text{-quasi-isomorphism} \iff \forall n,\ H(T^n N) = 0,$$

and the right-hand side is equivalent to $N \in \mathrm{Ob}(\mathcal{N}_H)$. This proves the assertion. $\square$

Thus there are at least two ways to localize at S_H: adjoin inverses to the elements of S_H, or make $\mathcal{N}_H$ zero. These give two universal properties of the Verdier localization $Q : \mathcal{D} \to \mathcal{D}/\mathcal{N}_H = \mathcal{D}[S_H^{-1}]$.

Finally, we discuss the relation between localization and subcategories. Let $\mathcal{N}$ and $\mathcal{I}$ be triangulated subcategories of a triangulated category $\mathcal{D}$, with $\mathcal{N}$ saturated. Then $\mathcal{N} \cap \mathcal{I}$ is also a saturated triangulated subcategory of $\mathcal{I}$. The universal property of Verdier localization determines a triangulated functor $i : \mathcal{I}/(\mathcal{N} \cap \mathcal{I}) \to \mathcal{D}/\mathcal{N}$.

Note the following simple fact. The system $S(\mathcal{N}\cap\mathcal{I})$, defined relative to $\mathcal{N}\cap\mathcal{I} \subset \mathcal{I}$, satisfies

$$S(\mathcal{N} \cap \mathcal{I}) = S\mathcal{N} \cap \mathrm{Mor}(\mathcal{I}). \tag{4.3.2}$$

The inclusion $\subset$ is clear. For $\supset$, suppose that a morphism $f : X \to Y$ in $\mathcal{I}$ can be placed in a distinguished triangle $X \xrightarrow{f} Y \to N \xrightarrow{+1}$ in $\mathcal{D}$ with $N \in \mathrm{Ob}(\mathcal{N})$. Corollary 4.2.3 shows that N is isomorphic to an object in $\mathcal{I}$. Since $\mathcal{N}$ is saturated, we may take $N \in \mathrm{Ob}(\mathcal{N} \cap \mathcal{I})$, and hence $f \in S(\mathcal{N} \cap \mathcal{I})$.

Proposition 4.3.7 Let $\mathcal{N}$, $\mathcal{D}$, and $\mathcal{I}$ be as above, and assume the following "resolution condition" : for every $X \in \mathrm{Ob}(\mathcal{D})$, there exist $U \in \mathrm{Ob}(\mathcal{I})$ and a morphism $U \to X$ belonging to $S\mathcal{N}$. Then $i : \mathcal{I}/(\mathcal{N} \cap \mathcal{I}) \to \mathcal{D}/\mathcal{N}$ is an equivalence of triangulated categories.

The same conclusion holds if the resolution condition is replaced by: for every $X \in \mathrm{Ob}(\mathcal{D})$, there exist $U \in \mathrm{Ob}(\mathcal{I})$ and a morphism $X \to U$ belonging to $S\mathcal{N}$.

Proof Proposition 1.10.3 implies that i is an equivalence, and Corollary 4.2.9 shows that the equivalence is triangulated. $\square$

The result above will be used in the study of derived functors.

4.4 Derived Categories

The first half of this section considers an additive category $\mathcal{A}$; when derived categories are defined later, $\mathcal{A}$ will be required to be Abelian. Example 4.1.3 has already shown that $\mathsf{C}(\mathcal{A})$ and $\mathsf{K}(\mathcal{A})$ are additive categories with translation; the translation functor in both is denoted by $X \mapsto X[1]$.

Consider a morphism $f : X \to Y$ in $\mathsf{C}(\mathcal{A})$ and its mapping cone $\mathrm{Cone}(f)$. Definition 3.3.5 gives the following triangle in $\mathsf{C}(\mathcal{A})$:

$$X \xrightarrow{f} Y \xrightarrow{\alpha(f)} \mathrm{Cone}(f) \xrightarrow{\beta(f)} X[1] \tag{4.4.1}$$

Passing to the quotient gives a triangle in $\mathsf{K}(\mathcal{A})$.

Theorem 4.4.1 Regard $\mathsf{K}(\mathcal{A})$ as an additive category with translation $T : X \mapsto X[1]$. Call a triangle $X \to Y \to Z \xrightarrow{+1}$ in $\mathsf{K}(\mathcal{A})$ distinguished if it is isomorphic to a triangle of the form (4.4.1). These distinguished triangles form a set $\mathcal{H}$ for which $(\mathsf{K}(\mathcal{A}), T, \mathcal{H})$ is a triangulated category.

Proof We verify the axioms in Definition 4.1.6 one by one. Axioms (TR0) and (TR2) are immediate from the definition, while (TR3) follows from the commutative diagram in $\mathsf{K}(\mathcal{A})$ supplied by Lemma 3.3.9:

$$\begin{tikzcd}
Y \arrow[r, "\alpha(f)"] \arrow[d, "\mathrm{id}"'] & \mathrm{Cone}(f) \arrow[r, "\beta(f)"] \arrow[d, "\mathrm{id}"] & X[1] \arrow[r, "-f[1]"] \arrow[d, "\wr"] & Y[1] \arrow[d, "\mathrm{id}"] \\
Y \arrow[r, "\alpha(f)"'] & \mathrm{Cone}(f) \arrow[r, "\alpha(\alpha(f))"'] & \mathrm{Cone}(\alpha(f)) \arrow[r, "\beta(\alpha(f))"'] & Y[1].
\end{tikzcd}$$

For (TR1), given $X \in \mathrm{Ob}(\mathsf{K}(\mathcal{A}))$, consider the mapping-cone triangle of the zero morphism $0 \to X \to X \xrightarrow{+1}$. Rotating it by (TR3) gives the distinguished triangle $X \xrightarrow{\mathrm{id}_X} X \to 0 \xrightarrow{+1}$.

For (TR4), consider the following commutative diagram in $\mathsf{K}(\mathcal{A})$ (the solid part):

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, "\alpha"'] & Y \arrow[r, "\alpha(f)"] \arrow[d, "\beta"] & \mathrm{Cone}(f) \arrow[r, "\beta(f)"] \arrow[d, dashed, "\gamma"] & TX \arrow[d, dashed, "T\alpha"] \\
X' \arrow[r, "f'"'] & Y' \arrow[r, "\alpha(f')"'] & \mathrm{Cone}(f') \arrow[r, "\beta(f')"'] & TX'.
\end{tikzcd} \tag{4.4.2}$$

Commutativity of the left square is equivalent to the existence of a family of morphisms in $\mathcal{A}$, $h^n : X^n \to (Y')^{n-1}$, such that

$$\beta^n f^n - (f')^n \alpha^n = h^{n+1} d_X^n + d_{Y'}^{n-1} h^n, \quad n \in \mathbb{Z}.$$

In matrix notation, define for every $n \in \mathbb{Z}$ the morphism

$$\gamma^n = \begin{pmatrix} \alpha^{n+1} & 0 \\ h^{n+1} & \beta^n \end{pmatrix} : X^{n+1} \oplus Y^n \to (X')^{n+1} \oplus (Y')^n.$$

Check that

$$\begin{pmatrix} \alpha^{n+1} & 0 \\ h^{n+1} & \beta^n \end{pmatrix} \begin{pmatrix} -d_X^n & 0 \\ f^n & d_Y^{n-1} \end{pmatrix} = \begin{pmatrix} -d_{X'}^n & 0 \\ (f')^n & d_{Y'}^{n-1} \end{pmatrix} \begin{pmatrix} \alpha^n & 0 \\ h^n & \beta^{n-1} \end{pmatrix},$$

so these matrices determine a morphism $\gamma : \mathrm{Cone}(f) \to \mathrm{Cone}(f')$. It is clear that the two right-hand squares in (4.4.2) commute in $\mathsf{C}(\mathcal{A})$. Notice that the homotopy $(h^n)_{n\in\mathbb{Z}}$ can be chosen in several ways, so γ is not unique.

It remains to verify (TR5). In the statement of that axiom, without loss of generality take $Z' = \mathrm{Cone}(f)$, $X' = \mathrm{Cone}(g)$, and $Y' = \mathrm{Cone}(gf)$, while h, k, and m are respectively $\alpha(f)$, $\alpha(g)$, and $\alpha(gf)$; the corresponding morphisms of degree $+1$ have the form $\beta(\cdot)$. Functoriality of the mapping cone (Proposition 3.3.3), or a direct check, gives morphisms

$$\begin{tikzcd} \mathrm{Cone}(f) \arrow[r, "u:=(\mathrm{id}_{X[1]},g)"] \arrow[d, equal] & \mathrm{Cone}(gf) \arrow[r, "v:=(f[1],\mathrm{id}_Z)"] \arrow[d, equal] & \mathrm{Cone}(g) \arrow[d, equal] \\ Z' & Y' & X'. \end{tikzcd}$$

By functoriality of the mapping cone, every square in diagram (TR5) not involving w commutes. For the remaining part, namely the lower-right square in (TR5), to commute as well, the morphism $w : X' \to TZ'$ must, and can only, be the composite

$$X' \xrightarrow{\beta(g)} TY \xrightarrow{T(\alpha(f))} TZ'.$$

It therefore remains to prove that $Z' \xrightarrow{u} Y' \xrightarrow{v} X' \xrightarrow{w} TZ'$ is a distinguished triangle. Notice that for every $n \in \mathbb{Z}$,

$$\mathrm{Cone}(u)^n = \mathrm{Cone}(f)^{n+1} \oplus \mathrm{Cone}(gf)^n = X^{n+2} \oplus Y^{n+1} \oplus X^{n+1} \oplus Z^n,$$
$$(X')^n = \mathrm{Cone}(g)^n = Y^{n+1} \oplus Z^n.$$

In matrix notation, define

$$\varphi^n := \begin{pmatrix} 0 & \mathrm{id}_{Y^{n+1}} & f^{n+1} & 0 \\ 0 & 0 & 0 & \mathrm{id}_{Z^n} \end{pmatrix} : \mathrm{Cone}(u)^n \to (X')^n,$$

$$\psi^n := \begin{pmatrix} 0 & 0 \\ \mathrm{id}_{Y^{n+1}} & 0 \\ 0 & 0 \\ 0 & \mathrm{id}_{Z^n} \end{pmatrix} : (X')^n \to \mathrm{Cone}(u)^n.$$

A direct check shows that these define morphisms $\varphi : \mathrm{Cone}(u) \leftrightarrows X' : \psi$ in $\mathsf{C}(\mathcal{A})$ satisfying $\varphi \circ \psi = \mathrm{id}_{X'}$, as well as $\varphi \circ \alpha(u) = v$ and $\beta(u) \circ \psi = w$. Next consider the diagram

$$\begin{tikzcd} Y' \arrow[r, "v"] \arrow[d, "\mathrm{id}_{Y'}"'] & X' \arrow[r, "w"] \arrow[d, shift left, "\psi"] & TZ' \arrow[d, "\mathrm{id}_{TZ'}"] \\ Y' \arrow[r, "\alpha(u)"'] & \mathrm{Cone}(u) \arrow[r, "\beta(u)"'] \arrow[u, shift left, "\varphi"] & TZ' \end{tikzcd}$$

It suffices to show that this diagram commutes in $\mathsf{K}(\mathcal{A})$ and that ψ and φ are inverse in $\mathsf{K}(\mathcal{A})$. The only point still to be proved is that $\psi \circ \varphi$ is homotopic to $\mathrm{id}_{\mathrm{Cone}(u)}$. Let $h^n : \mathrm{Cone}(u)^n \to \mathrm{Cone}(u)^{n-1}$ be the evident composite

$$X^{n+2} \oplus Y^{n+1} \oplus X^{n+1} \oplus Z^n \twoheadrightarrow X^{n+1} \hookrightarrow X^{n+1} \oplus Y^n \oplus X^n \oplus Z^{n-1}.$$

A routine verification gives $\mathrm{id}_{\mathrm{Cone}(u)^n} - \psi^n \varphi^n = h^{n+1} d^n_{\mathrm{Cone}(u)} + d^{n-1}_{\mathrm{Cone}(u)} h^n$. This proves the assertion. $\square$

Homotopy equivalence is indispensable in the proof. If one works at the level of $\mathsf{C}(\mathcal{A})$, mapping cones cannot produce a triangulated category.

Proof (Proof of Theorems 3.11.5 and 3.15.5) We can now complete the previously deferred part of the proof of Theorem 3.11.5. By duality, it suffices to consider the situation $Y \xleftarrow{\alpha} X \xrightarrow{\gamma} I$, where α is a quasi-isomorphism and $I \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$ consists of injective objects. The cohomological functor $\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(\cdot, I)$ (Proposition 4.1.12) gives a long exact sequence

$$\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(\mathrm{Cone}(\alpha), I) \to \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(Y, I) \xrightarrow{\alpha^*} \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X, I) \to \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(\mathrm{Cone}(\alpha)[-1], I);$$

but $\mathrm{Cone}(\alpha)$ is acyclic (Corollary 3.7.6), so Lemma 3.11.6 implies that α^* is an isomorphism.

The argument for Theorem 3.15.5, concerning K-injective or K-projective complexes, is entirely similar. $\square$

Corollary 4.4.2 For $\star \in \{+,-,\mathrm{b}\}$, the category $\mathsf{K}^\star(\mathcal{A})$ introduced in Definition 3.9.2 is a triangulated subcategory of $\mathsf{K}(\mathcal{A})$ (Definition–Proposition 4.2.10).

Proof Clearly $\mathsf{K}^\star(\mathcal{A})$ is closed under the translation functor. Moreover, for a mapping-cone triangle $X \xrightarrow{f} Y \to \mathrm{Cone}(f) \xrightarrow{+1}$, it is clear that X, Y belong to $\mathsf{C}^\star(\mathcal{A})$ if and only if $\mathrm{Cone}(f) \in \mathsf{C}^\star(\mathcal{A})$. This proves that condition (ii) in Definition–Proposition 4.2.10 holds. □

Proposition 4.4.3 Let $F : \mathcal{A}_1 \to \mathcal{A}_2$ be an additive functor between additive categories. Then $\mathsf{K}F : \mathsf{K}(\mathcal{A}_1) \to \mathsf{K}(\mathcal{A}_2)$ is a triangulated functor. The same assertion holds with $\mathsf{K}^\star(\cdot)$ in place of $\mathsf{K}(\cdot)$, for $\star \in \{+,-,\mathrm{b}\}$.

Proof This follows from Proposition 3.1.8 (for translation) and Proposition 3.3.4 (for mapping cones). □

Corollary 4.4.4 Let $\mathcal{A}'$ be a full additive subcategory of $\mathcal{A}$. Then $\mathsf{K}(\mathcal{A}')$ embeds as a triangulated subcategory of $\mathsf{K}(\mathcal{A})$. The same assertion holds with $\mathsf{K}^\star(\cdot)$ in place of $\mathsf{K}(\cdot)$, for $\star \in \{+,-,\mathrm{b}\}$.

Proposition 4.4.5 Let $\mathcal{A}$ be an Abelian category. For every $n \in \mathbb{Z}$, the functor $\mathrm{H}^n : \mathsf{K}(\mathcal{A}) \to \mathcal{A}$ is cohomological in the sense of Definition 4.1.10. The same assertion holds with $\mathsf{K}^\star(\mathcal{A})$ in place of $\mathsf{K}(\mathcal{A})$, for $\star \in \{+,-,\mathrm{b}\}$.

Proof Consider the distinguished mapping-cone triangle in $\mathsf{K}(\mathcal{A})$, $X \xrightarrow{f} Y \xrightarrow{\alpha(f)} \mathrm{Cone}(f) \xrightarrow{+1}$. By the long exact sequence (3.7.1) and Corollary 3.7.5 (the heart of §3.7), the sequence

$$\mathrm{H}^n(X) \xrightarrow{\mathrm{H}^n(f)} \mathrm{H}^n(Y) \xrightarrow{\mathrm{H}^n(\alpha(f))} \mathrm{H}^n(\mathrm{Cone}(f))$$

is indeed exact. □

The definition of the derived category will involve Verdier localization. In the notation of Example 4.3.5, the cohomological functor $\mathrm{H}^0 : \mathsf{K}(\mathcal{A}) \to \mathcal{A}$ determines a multiplicative system compatible with triangulation,

$$\mathrm{qis} := S_{\mathrm{H}^0} \subset \mathrm{Mor}(\mathsf{K}(\mathcal{A})),$$

whose elements are precisely the quasi-isomorphisms. Another way to understand qis is to consider the saturated triangulated subcategory $\mathcal{N} := \mathcal{N}_{\mathrm{H}^0}$ of $\mathsf{K}(\mathcal{A})$, consisting of the acyclic complexes. Proposition 4.3.6 implies $S\mathcal{N} = \mathrm{qis}$.

For $\star \in \{+,-,\mathrm{b}\}$, we similarly define the multiplicative system compatible with triangulation $\mathrm{qis}^\star := \mathrm{qis} \cap \mathrm{Mor}(\mathsf{K}^\star(\mathcal{A}))$ and the subcategory $\mathcal{N}^\star := \mathcal{N} \cap \mathsf{K}^\star(\mathcal{A})$; they still satisfy $S\mathcal{N}^\star = \mathrm{qis}^\star$.

Definition 4.4.6 The **derived category** of an Abelian category $\mathcal{A}$ is defined to be the triangulated category

$$\begin{aligned}\mathsf{D}(\mathcal{A}) &:= \mathsf{K}(\mathcal{A})\left[\mathrm{qis}^{-1}\right]\\ &= \mathsf{K}(\mathcal{A})/\mathcal{N}.\end{aligned}$$

Likewise, for $\star \in \{+, -, \mathrm{b}\}$, define the triangulated category

$$\begin{aligned}\mathsf{D}^\star(\mathcal{A}) &:= \mathsf{K}^\star(\mathcal{A})\left[(\mathrm{qis}^\star)^{-1}\right]\\ &= \mathsf{K}^\star(\mathcal{A})/\mathcal{N}^\star.\end{aligned}$$

We call $\mathsf{D}^+(\mathcal{A})$ (respectively, $\mathsf{D}^-(\mathcal{A})$ and $\mathsf{D}^{\mathrm{b}}(\mathcal{A})$) the bounded-below (respectively, bounded-above and bounded) derived category of $\mathcal{A}$.

An issue that is difficult to avoid is controlling the size of a derived category in the set-theoretic sense, since we do not want $\mathsf{D}(\mathcal{A})$ to be a "large" category; see the related discussion in Remark 1.9.15. We defer this issue until Corollary 4.5.2.

Proposition 4.4.7 Every short exact sequence $0 \to X \xrightarrow{f} Y \xrightarrow{g} Z \to 0$ in $\mathsf{C}(\mathcal{A})$ extends canonically to a distinguished triangle $X \to Y \to Z \xrightarrow{+1}$ in $\mathsf{D}(\mathcal{A})$. More precisely, there is a canonical commutative diagram in $\mathsf{K}(\mathcal{A})$

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, equal] & Y \arrow[r, "\alpha(f)"] \arrow[d, equal] & \mathrm{Cone}(f) \arrow[r, "\beta(f)"] \arrow[d, "\Phi"] & X[1] \arrow[dd, "\Phi'"] \\
X \arrow[r, "f"] \arrow[d, "\Phi'[-1]"'] & Y \arrow[r, "g"] \arrow[d, equal] & Z \arrow[d, equal] & \\
\mathrm{Cone}(g)[-1] \arrow[r, "\beta(g)[-1]"'] & Y \arrow[r, "g"'] & Z \arrow[r, "-\alpha(g)"'] & \mathrm{Cone}(g)
\end{tikzcd}$$

in which the first and third rows are distinguished triangles, and both Φ and Φ' are quasi-isomorphisms.

Proof The commutative diagram is simply a restatement of Lemma 3.7.2. The first row is the distinguished mapping-cone triangle of f, while the third row is a rotation of the mapping-cone triangle of g. $\square$

Thus short exact sequences in the category of complexes are reflected in the triangulated structure on the derived category. Conversely, it is very rare for an Abelian category $\mathcal{A}$ to have the property that $\mathsf{D}(\mathcal{A})$ is Abelian; the exercises in this chapter will characterize such categories.

The exercises in this chapter will also introduce the group $\mathrm{K}_0(\mathsf{D}^{\mathrm{b}}(\mathcal{A}))$ and relate it to $\mathrm{K}_0(\mathcal{A})$ from Definition 2.9.8; the reader is encouraged to work this out.

By the characterization of $\mathcal{N}$ and Theorem 4.3.4, the cohomological functor H^n : $\mathsf{K}(\mathcal{A}) \to \mathcal{A}$ factors through the derived category as

$$\mathrm{H}^n : \mathsf{D}(\mathcal{A}) \to \mathcal{A}, \quad \mathrm{H}^n(X) = \mathrm{H}^0(X[n]),$$

where $n \in \mathbb{Z}$. The same applies to $\star \in \{+, -, \mathrm{b}\}$ and $\mathsf{D}^\star(\mathcal{A})$.

The universal property determines triangulated functors $\mathsf{D}^{\mathrm{b}}(\mathcal{A}) \to \mathsf{D}^{\pm}(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$ that commute with the cohomology functor H^0. We want to regard these functors as embeddings of triangulated subcategories; for this we need the following abstract result.

Lemma 4.4.8 Let $\mathcal{N}$ and $\mathcal{I}$ be triangulated subcategories of a triangulated category $\mathcal{D}$, with $\mathcal{N}$ saturated. If either of the following conditions holds in $\mathcal{D}$, then the functor $i : \mathcal{I}/(\mathcal{N} \cap \mathcal{I}) \to \mathcal{D}/\mathcal{N}$ is fully faithful:

(i) every morphism $Y \to N$ factors as $Y \to Y' \to N$;

(ii) every morphism $N \to Y$ factors as $N \to Y' \to Y$;

here $Y \in \mathrm{Ob}(\mathcal{I})$ and $N \in \mathrm{Ob}(\mathcal{N})$ are given, and $Y' \in \mathrm{Ob}(\mathcal{N} \cap \mathcal{I})$.

Proof By duality, it suffices to consider the case in which (i) holds. Recall that (4.3.2) guarantees $S(\mathcal{N} \cap \mathcal{I}) = S\mathcal{N} \cap \mathrm{Mor}(\mathcal{I})$. We wish to apply the criterion in Proposition 1.10.1 (ii). Thus it suffices to prove the following: suppose $s : W \to Y$ belongs to the multiplicative system $S\mathcal{N}$ and $Y \in \mathrm{Ob}(\mathcal{I})$. Then there is a $g : V \to W$ such that $V \in \mathrm{Ob}(\mathcal{I})$ and $sg \in S\mathcal{N}$.

There is clearly a distinguished triangle $W \xrightarrow{-s} Y \to N \xrightarrow{+1}$ with $N \in \mathrm{Ob}(\mathcal{N})$. Factor $Y \to N$ as $Y \xrightarrow{\alpha} Y' \xrightarrow{\beta} N$, with $Y' \in \mathrm{Ob}(\mathcal{N} \cap \mathcal{I})$. Extend α to a distinguished triangle $V \to Y \xrightarrow{\alpha} Y' \xrightarrow{+1}$ in $\mathcal{I}$ to obtain the following diagram, whose solid part is determined:

$$\begin{tikzcd}
Y \arrow[r, "\alpha"] \arrow[d, "\mathrm{id}"] & Y' \arrow[r] \arrow[d, "\beta"] & TV \arrow[r] \arrow[d, dashed, "Tg"] & TY \arrow[d, dashed, "\mathrm{id}"] \\
Y \arrow[r] & N \arrow[r] & TW \arrow[r, "Ts"'] & TY
\end{tikzcd}$$

each row being a distinguished triangle. Use (TR4) to choose $Tg : TV \to TW$ so that the entire diagram commutes. Notice that $V \in \mathrm{Ob}(\mathcal{I})$ and $TV \to TY$ belongs to $S\mathcal{N}$; therefore $sg : V \to Y$ also belongs to $S\mathcal{N}$. This proves the assertion. $\square$

Proposition 4.4.9 For every $\star \in \{+, -, \mathrm{b}\}$, the functor $\mathsf{D}^\star(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$ is fully faithful. For every $X \in \mathrm{Ob}(\mathsf{D}(\mathcal{A}))$, the object X is isomorphic to an object of $\mathsf{D}^+(\mathcal{A})$ (respectively, $\mathsf{D}^-(\mathcal{A})$ and $\mathsf{D}^{\mathrm{b}}(\mathcal{A})$) if and only if $n \ll 0$ (respectively, $n \gg 0$ and $|n| \gg 0$) implies $\mathrm{H}^n(X) = 0$.

Proof The first assertion is an application of Lemma 4.4.8: take $\mathcal{D} = \mathsf{K}(\mathcal{A})$, $\mathcal{I} = \mathsf{K}^\star(\mathcal{A})$, and $\mathcal{N} = \mathcal{N}_{\mathrm{H}^0}$. For example, when $\star = +$, we must show that for every $Y \in \mathrm{Ob}(\mathsf{K}^+(\mathcal{A}))$, $N \in \mathrm{Ob}(\mathcal{N})$, and morphism $N \to Y$, there is a factorization

$$N \to Y' \to Y, \quad Y' \in \mathrm{Ob}(\mathcal{N}^+).$$

This holds because, for $n \ll 0$, the morphism $N \to Y$ naturally factors as $N \to \tau^{\geq n}N \to \tau^{\geq n}Y = Y$, where $\tau^{\geq n}$ is the truncation functor from Definition 3.9.3 and $\tau^{\geq n}N \in \mathrm{Ob}(\mathcal{N}^+)$.

Next, if $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$ satisfies $n \ll 0 \implies \mathrm{H}^n(X) = 0$, then for $n \ll 0$ the morphism $X \to \tau^{\geq n}X$ is a quasi-isomorphism and hence an isomorphism in $\mathsf{D}(\mathcal{A})$. This characterizes, up to isomorphism, the image of $\mathsf{D}^+(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$.

The argument for $\star = -$ is dual. In the same way, $\mathsf{K}^{\mathrm{b}}(\mathcal{A}) \to \mathsf{K}^\pm(\mathcal{A})$ induces a fully faithful functor $\mathsf{D}^{\mathrm{b}}(\mathcal{A}) \to \mathsf{D}^\pm(\mathcal{A})$, whose image is also characterized by H^n. This gives the case of $\mathsf{D}^{\mathrm{b}}(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$. □

Corollary 4.4.10 Every functor in the sequence $\mathsf{D}^{\mathrm{b}}(\mathcal{A}) \to \mathsf{D}^\pm(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$ is an embedding of a triangulated subcategory.

Proof Use the long exact sequence of H^n to verify, for $\star \in \{+, -, \mathrm{b}\}$, that $\mathsf{D}^\star(\mathcal{A})$, as a full additive subcategory of $\mathsf{D}(\mathcal{A})$, satisfies condition (ii) in Definition–Proposition 4.2.10. It is therefore a triangulated subcategory; the case $\mathsf{D}^{\mathrm{b}}(\mathcal{A}) \to \mathsf{D}^\pm(\mathcal{A})$ is also clear. □

Definition 4.4.11 For $-\infty \leq s \leq t \leq +\infty$, let $\mathsf{D}^{[s,t]}(\mathcal{A})$ (or $\mathsf{K}^{[s,t]}(\mathcal{A})$) denote the full subcategory of $\mathsf{D}(\mathcal{A})$ (or $\mathsf{K}(\mathcal{A})$) defined by the condition

$$n \notin [s,t] \implies \mathrm{H}^n(X) = 0.$$

In addition, set

$$\begin{aligned} \mathsf{D}^{\geq s}(\mathcal{A}) &:= \mathsf{D}^{[s,+\infty]}(\mathcal{A}), & \mathsf{D}^{\leq t}(\mathcal{A}) &:= \mathsf{D}^{[-\infty,t]}(\mathcal{A}), \\ \mathsf{K}^{\geq s}(\mathcal{A}) &:= \mathsf{K}^{[s,+\infty]}(\mathcal{A}), & \mathsf{K}^{\leq t}(\mathcal{A}) &:= \mathsf{K}^{[-\infty,t]}(\mathcal{A}). \end{aligned}$$

There are natural additive functors $\mathcal{A} \to \mathsf{C}(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$ that regard each object as a complex concentrated in degree zero. Theorem 4.5.4 will show that under this identification $\mathcal{A}$ is equivalent to $\mathsf{D}^{\leq 0}(\mathcal{A}) \cap \mathsf{D}^{\geq 0}(\mathcal{A})$.

Remark 4.4.12 If $\mathcal{A}$ is $\Bbbk$-linear, where $\Bbbk$ is a commutative ring, then $\mathsf{D}(\mathcal{A})$, $\mathsf{D}^{\mathrm{b}}(\mathcal{A})$, and so on, as well as the embeddings between them, are also $\Bbbk$-linear; see Theorem 1.9.17 (iii).

Since the truncation functors $\tau^{\leq n}$ and $\tau^{\geq n}$ from Definition–Proposition 3.9.7 preserve acyclic complexes, they induce functors $\tau^{\leq n} : \mathsf{D}(\mathcal{A}) \to \mathsf{D}^{\leq n}(\mathcal{A})$ and $\tau^{\geq n} : \mathsf{D}(\mathcal{A}) \to \mathsf{D}^{\geq n}(\mathcal{A})$.

Proposition 4.4.13 For every $n \in \mathbb{Z}$, there are canonical adjoint pairs and a distinguished triangle

$$\text{inclusion} : \mathsf{D}^{\leq n}(\mathcal{A}) \rightleftarrows \mathsf{D}(\mathcal{A}) : \tau^{\leq n},$$

$$\tau^{\geq n} : \mathsf{D}(\mathcal{A}) \rightleftarrows \mathsf{D}^{\geq n}(\mathcal{A}) : \text{inclusion},$$

$$\tau^{\leq n} X \to X \to \tau^{\geq n+1} X \xrightarrow{+1}, \quad X \in \mathrm{Ob}(\mathsf{D}(\mathcal{A})).$$

Proof It suffices to discuss the first adjoint pair. Let $X \in \mathrm{Ob}(\mathsf{D}^{\leq n}(\mathcal{A}))$ and $Y \in \mathrm{Ob}(\mathsf{D}(\mathcal{A}))$. The unit and counit of the proposed adjunction are respectively $\eta_X : X \to \tau^{\leq n} X$ and $\varepsilon_Y : \tau^{\leq n} Y \to Y$. The morphism ε_Y is the canonical morphism already defined at the level of complexes, while η_X is the inverse of the quasi-isomorphism $\tau^{\leq n} X \to X$. It remains to verify the triangle identities. After truncating at the level of complexes, it suffices to check them when X comes from $\mathsf{C}^{\leq n}(\mathcal{A})$; in that case $\eta_X = \mathrm{id}_X$. The required triangle identities then follow from the complex-level version in Proposition 3.9.6.

For the distinguished triangle, recall that at the level of $\mathsf{C}(\mathcal{A})$ there is a short exact sequence $0 \to \tau^{\leq n} X \to X \to \tilde{\tau}^{\geq n+1} X \to 0$ and a quasi-isomorphism $\tilde{\tau}^{\geq n+1} X \to \tau^{\geq n+1} X$ (Lemma 3.9.4). □

Remark 4.4.14 (Direct construction of the derived category) Let $\mathrm{Qis} \subset \mathrm{Mor}(\mathsf{C}(\mathcal{A}))$ denote the set of all quasi-isomorphisms. Suppose that, in defining $\mathsf{D}(\mathcal{A})$, instead of first passing to $\mathsf{K}(\mathcal{A})$, we directly adjoin inverses to quasi-isomorphisms to obtain $\mathsf{C}(\mathcal{A})[\mathrm{Qis}^{-1}]$. Is the result equivalent to $\mathsf{D}(\mathcal{A})$? The answer is yes. A little thought about the universal property shows that the key is to verify the following property for every category $\mathcal{D}$ and functor $G : \mathsf{C}(\mathcal{A}) \to \mathcal{D}$: suppose G maps Qis to isomorphisms and $f, g \in \mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X, Y)$ are homotopic. Then $Gf = Gg$.

The key is to use the mapping cylinder. Let h be a homotopy from f to g. As in Proposition 3.3.12 (ii), this homotopy determines a morphism $\tilde{h} : \mathrm{Cyl}_X \to Y$ such that $f = \tilde{h} i_0$ and $g = \tilde{h} i_1$. Part (i) of that proposition, however, states that $j i_0 = j i_1$, where $j : \mathrm{Cyl}_X \to X$ is a quasi-isomorphism. Hence Gj is an isomorphism, which in turn implies $G i_0 = G i_1$. This proves the property.

Although $\mathsf{D}(\mathcal{A})$ can be constructed directly as $\mathsf{C}(\mathcal{A})[\mathrm{Qis}^{-1}]$, the advantage of passing through $\mathsf{K}(\mathcal{A})$ is that the latter has a triangulated structure; another reason is that Qis is not a multiplicative system. The same assertion holds for $\mathsf{D}^{\star}(\mathcal{A})$, with $\star \in \{+, -, \mathrm{b}\}$.

Another important kind of triangulated subcategory is one defined by cohomology.

Definition 4.4.15 (Triangulated subcategories defined by cohomology) Let $\mathcal{T}$ be a weak Serre subcategory of $\mathcal{A}$ (Definition 2.9.3). Define the full subcategory $\mathsf{D}_{\mathcal{T}}(\mathcal{A})$ of $\mathsf{D}(\mathcal{A})$ by

$$X \in \mathrm{Ob}(\mathsf{D}_{\mathcal{T}}(\mathcal{A})) \iff \forall n \in \mathbb{Z},\ \mathrm{H}^n(X) \in \mathrm{Ob}(\mathcal{T}).$$

By Proposition 4.2.13, this is a saturated triangulated subcategory of $\mathsf{D}(\mathcal{A})$.

For $\star \in \{+, -, \mathrm{b}\}$, define the saturated triangulated subcategory $\mathsf{D}^\star_{\mathcal{T}}(\mathcal{A}) := \mathsf{D}_{\mathcal{T}}(\mathcal{A}) \cap \mathsf{D}^\star(\mathcal{A})$.

Finally, there is a simple relation between the derived categories of $\mathcal{A}$ and $\mathcal{A}^{\mathrm{op}}$. Equip the opposite of a triangulated category with the triangulated structure described in Remark 4.1.8.

Proposition 4.4.16 The equivalence $\sigma : \mathsf{C}(\mathcal{A}^{\mathrm{op}}) \xrightarrow{\sim} \mathsf{C}(\mathcal{A})^{\mathrm{op}}$ from Definition–Proposition 3.4.1 induces equivalences of triangulated categories

$$\mathsf{D}(\mathcal{A}^{\mathrm{op}}) \simeq \mathsf{D}(\mathcal{A})^{\mathrm{op}}, \quad \mathsf{D}^{\pm}(\mathcal{A}^{\mathrm{op}}) \simeq \mathsf{D}^{\mp}(\mathcal{A})^{\mathrm{op}}, \quad \mathsf{D}^{\mathrm{b}}(\mathcal{A}^{\mathrm{op}}) \simeq \mathsf{D}^{\mathrm{b}}(\mathcal{A})^{\mathrm{op}}.$$

Moreover, for every $n \in \mathbb{Z}$, $\tau^{\leq n} \circ \sigma = \sigma \circ \tau^{\geq -n}$ and $\tau^{\geq n} \circ \sigma = \sigma \circ \tau^{\leq -n}$.

Proof Proposition 3.4.3 shows that σ induces $\mathsf{K}(\mathcal{A}^{\mathrm{op}}) \simeq \mathsf{K}(\mathcal{A})^{\mathrm{op}}$. This functor preserves distinguished triangles (compare carefully the diagram in Proposition 3.4.4 with Remark 4.1.8: after reversing the first row of the former diagram, it is the image of a distinguished triangle; the second row is distinguished in $\mathsf{K}(\mathcal{A})$ and, after reversal, becomes distinguished in $\mathsf{K}(\mathcal{A})^{\mathrm{op}}$). It also preserves cohomology (Remark 3.4.2), and therefore induces $\mathsf{D}(\mathcal{A}^{\mathrm{op}}) \xrightarrow{\sim} \mathsf{D}(\mathcal{A})^{\mathrm{op}}$. The assertions about the truncation functors follow from Remark 3.9.5. □

4.5 Morphisms and Extensions

Throughout this section, $\mathcal{A}$ is an Abelian category. Without further comment, we identify $\mathrm{Ob}(\mathsf{C}(\mathcal{A}))$, $\mathrm{Ob}(\mathsf{K}(\mathcal{A}))$, and $\mathrm{Ob}(\mathsf{D}(\mathcal{A}))$; when necessary, we distinguish the morphism sets in these categories by writing $\mathrm{Hom}_{\mathsf{D}(\mathcal{A})}$, $\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}$, $\mathrm{Hom}_{\mathsf{C}(\mathcal{A})}$, and $\mathrm{Hom}_{\mathcal{A}}$. Readers not yet familiar with K-injective and K-projective complexes may skip the statements concerning them.

Proposition 4.5.1 Suppose $X \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$ consists of injective objects, or, more generally, suppose $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$ is a K-injective complex as in Definition 3.15.1. Then $\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(\cdot, X) \simeq \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(\cdot, X)$.

Dually, suppose $X \in \mathrm{Ob}(\mathsf{C}^-(\mathcal{A}))$ consists of projective objects, or suppose $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$ is a K-projective complex. Then $\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X, \cdot) \simeq \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, \cdot)$.

Proof By duality, it suffices to treat $\mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(\cdot, X)$. Lemma 1.9.8 gives

$$\mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(T, X) \simeq \varinjlim_{\substack{\alpha: S \to T \\ \text{quasi-isomorphism in } \mathsf{K}(\mathcal{A})}} \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(S, X), \quad T \in \mathrm{Ob}(\mathsf{K}(\mathcal{A})). \tag{4.5.1}$$

Suppose $X \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$ consists of injective objects. For a quasi-isomorphism $\alpha \in \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(S, T)$, Theorem 3.11.5 shows that $\alpha^* : \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(T, X) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(S, X)$. Hence the $\varinjlim$ in (4.5.1) is equal to $\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(T, X)$.

If $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$ is K-injective, use Theorem 3.15.5 instead. □

One immediate application of the preceding proposition is to control the size of a derived category. Here we again use the chosen Grothendieck universe $\mathcal{U}$, together with the terminology of $\mathcal{U}$-categories (that is, "categories" in the standard convention of this book) and small $\mathcal{U}$-categories.

Corollary 4.5.2 Use the notation above. Let $\mathcal{A}$ be a $\mathcal{U}$-category.

(i) If $\mathcal{A}$ has enough injective objects (respectively projective objects), then $\mathsf{D}^+(\mathcal{A})$ (respectively $\mathsf{D}^-(\mathcal{A})$) is also a $\mathcal{U}$-category.

(ii) If $\mathcal{A}$ has enough K-injective or K-projective complexes (Definition 3.15.1), then $\mathsf{D}(\mathcal{A})$ is also a $\mathcal{U}$-category.

(iii) If $\mathcal{A}$ is a small $\mathcal{U}$-category, then so is $\mathsf{D}(\mathcal{A})$.

Proof It suffices to verify that $\mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y)$ is a small $\mathcal{U}$-set. Notice that the $\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}$ version is easy to handle, since it is straightforward to prove that $\mathsf{C}(\mathcal{A})$ is a $\mathcal{U}$-category.

Suppose $\mathcal{A}$ has enough injective objects and $X \in \mathrm{Ob}(\mathsf{D}^+(\mathcal{A}))$. Without loss of generality, we may assume that $X \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$ consists of injective objects; (i) then follows from Proposition 4.5.1. If X has a K-injective resolution (respectively Y has a K-projective resolution), without loss of generality we may assume that X is K-injective (respectively Y is K-projective); the same argument gives (ii).

For (iii), observe that $\mathsf{K}(\mathcal{A})$ is then a small $\mathcal{U}$-category. Apply this observation to Remark 1.9.15. □

Proposition 4.5.3 (Orthogonality) Let $a, b, n \in \mathbb{Z}$, with $a < b$.

(i) If $X \in \mathrm{Ob}(\mathsf{D}^{\leq a}(\mathcal{A}))$ and $Y \in \mathrm{Ob}(\mathsf{D}^{\geq b}(\mathcal{A}))$, then $\mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y) = 0$.

(ii) If $X \in \mathrm{Ob}(\mathsf{D}^{\leq n}(\mathcal{A}))$ and $Y \in \mathrm{Ob}(\mathsf{D}^{\geq n}(\mathcal{A}))$, then H^n induces a canonical isomorphism

$$\mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X,Y) \xrightarrow{\simeq} \mathrm{Hom}_{\mathcal{A}}\left(\mathrm{H}^n(X), \mathrm{H}^n(Y)\right).$$

Proof Let $a \leq b$. Represent a morphism $f \in \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X,Y)$ by a diagram $X \xleftarrow{s} Z \xrightarrow{g} Y$ in $\mathsf{K}(\mathcal{A})$, where s is a quasi-isomorphism. Replacing f by g, we may reduce to the case in which f comes from $\mathsf{C}(\mathcal{A})$. Next replace f by the composite $\tau^{\leq a}X \to X \xrightarrow{f} Y \to \tau^{\geq b}Y$ in $\mathsf{C}(\mathcal{A})$. The two morphisms at the ends of this composite are quasi-isomorphisms. We may therefore reduce further to the case $X \in \mathrm{Ob}(\mathsf{C}^{\leq a}(\mathcal{A}))$ and $Y \in \mathrm{Ob}(\mathsf{C}^{\geq b}(\mathcal{A}))$.

If $a < b$, every morphism $X \to Y$ in $\mathsf{C}(\mathcal{A})$ must be zero. This proves (i).

Now take X and Y as above, but set $a = n = b$ to treat (ii). Consider the canonical morphisms

$$\mathrm{Hom}_{\mathcal{A}}\left(\mathrm{H}^n(X), \mathrm{H}^n(Y)\right) \xleftarrow{\mathrm{H}^n} \mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X,Y) \to \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X,Y) \to \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X,Y). \tag{4.5.2}$$

- ◇ The first morphism H^n is an isomorphism: because $X \in \mathrm{Ob}(\mathsf{C}^{\leq n}(\mathcal{A}))$ and $Y \in \mathrm{Ob}(\mathsf{C}^{\geq n}(\mathcal{A}))$, any $f \in \mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X,Y)$ can be nonzero only in degree n, and the only conditions on f^n are

$$f^n\left(\mathrm{im}(d_X^{n-1})\right) = 0, \quad f\left(X^n\right) \subset \ker\left(d_Y^n\right).$$

  But $X^n / \mathrm{im}(d_X^{n-1}) = \mathrm{H}^n(X)$ and $\ker(d_Y^n) = \mathrm{H}^n(Y)$.

- ◇ The second morphism is also an isomorphism, since the conditions on X, Y imply $\mathrm{Hom}^{-1}(X,Y) = \{0\}$.

Thus, to prove (ii), it remains to show that the last morphism in (4.5.2) is an isomorphism. Applying Lemma 1.9.8 gives

$$\mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X,Y) \simeq \varinjlim_{[Y \rightarrowtail Z] \in \mathrm{Ob}(\mathrm{qis}_{Y/})} \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X,Z).$$

If $Y \to Z$ is a quasi-isomorphism, then both morphisms in $Y \to Z \to \tau^{\geq n}Z$ are quasi-isomorphisms in $\mathsf{K}(\mathcal{A})$. This shows that the quasi-isomorphisms $Y \to Z$ with Z coming from $\mathsf{C}^{\geq n}(\mathcal{A})$ form a cofinal full subcategory of the filtered category $\mathrm{qis}_{Y/}$ (Definition 1.6.4 and Proposition 1.6.8). By Proposition 1.6.5, we may henceforth restrict the $\varinjlim$ to this subcategory.

We already know that the first two morphisms in (4.5.2) are canonical isomorphisms. Take $Y \to Z$ as above. This gives the commutative diagram

$$\begin{tikzcd} \mathrm{Hom}_{\mathcal{A}}(\mathrm{H}^n(X), \mathrm{H}^n(Y)) \arrow[d, "\wr"] & \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X, Y) \arrow[l, "\mathrm{H}^n"', "\sim"] \arrow[d] \\ \mathrm{Hom}_{\mathcal{A}}(\mathrm{H}^n(X), \mathrm{H}^n(Z)) & \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X, Z) \arrow[l, "\sim"', "\mathrm{H}^n"] \end{tikzcd}$$

which shows that $\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X, Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X, Z)$. Hence the preceding $\varinjlim$ is constant with value $\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X, Y)$. The last morphism in (4.5.2) is therefore an isomorphism. □

Theorem 4.5.4 Regarding the objects of $\mathcal{A}$ as complexes concentrated in degree zero gives an equivalence of additive categories $\mathcal{A} \to \mathsf{D}^{\leq 0}(\mathcal{A}) \cap \mathsf{D}^{\geq 0}(\mathcal{A})$, with quasi-inverse given by H^0.

Proof It is clear that the objects of $\mathcal{A}$ give objects of $\mathsf{D}^{\leq 0}(\mathcal{A}) \cap \mathsf{D}^{\geq 0}(\mathcal{A})$. Thus we obtain an additive functor $\Phi : \mathcal{A} \to \mathsf{D}^{\leq 0}(\mathcal{A}) \cap \mathsf{D}^{\geq 0}(\mathcal{A})$. Proposition 4.5.3 (ii) implies that Φ is fully faithful.

We next show that Φ is essentially surjective. Let X be an object of $\mathsf{D}^{\leq 0}(\mathcal{A}) \cap \mathsf{D}^{\geq 0}(\mathcal{A})$. Applying the truncation functors to the complex X in $\mathsf{C}(\mathcal{A})$, we see that every morphism in $X \to \tau^{\geq 0} X \leftarrow \tau^{\leq 0}\tau^{\geq 0} X$ is a quasi-isomorphism, while Proposition 3.9.8 gives $\tau^{\leq 0}\tau^{\geq 0} X \simeq \mathrm{H}^0(X)$. Hence Φ is an equivalence and H^0 is its quasi-inverse. □

More generally, for every $n \in \mathbb{Z}$, the functor $S \mapsto S[-n]$ gives an additive equivalence from $\mathcal{A}$ to $\mathsf{D}^{\geq n}(\mathcal{A}) \cap \mathsf{D}^{\leq n}(\mathcal{A})$, with H^n as quasi-inverse. This is simply the shifted version of the result above.

Henceforth, without further comment, we identify $\mathcal{A}$ with the full additive subcategory $\mathsf{D}^{\geq 0}(\mathcal{A}) \cap \mathsf{D}^{\leq 0}(\mathcal{A})$ of $\mathsf{D}^{\mathrm{b}}(\mathcal{A})$.

Definition 4.5.5 Let $\star \in \{+, -, \mathrm{b}, \ \}$ (where $\star = \ $ means that there is no superscript), and let $\mathcal{A}$ and $\mathcal{A}'$ be Abelian categories. If a triangulated functor $R : \mathsf{D}^{\star}(\mathcal{A}) \to \mathsf{D}^{\star}(\mathcal{A}')$ restricts to $\mathcal{A} \to \mathsf{D}^{[a,b]}(\mathcal{A}')$, where $-\infty \leq a \leq b \leq +\infty$, then R is said to have **amplitude** contained in $[a, b]$.

Here is a simple observation: if $X' \to X \to X'' \xrightarrow{+1}$ is a distinguished triangle in $\mathsf{D}(\mathcal{A})$ and $X', X'' \in \mathrm{Ob}(\mathsf{D}^{[a,b]}(\mathcal{A}))$, then $X \in \mathrm{Ob}(\mathsf{D}^{[a,b]}(\mathcal{A}))$. This follows immediately from the long exact sequence for H^n.

Lemma 4.5.6 If the amplitude of R is contained in $[a, b]$, then for every $-\infty < c \leq d < +\infty$ one has

$$R \text{ restricts to } \mathsf{D}^{[c,d]}(\mathcal{A}) \to \mathsf{D}^{[c+a,d+b]}(\mathcal{A}').$$

Proof The case $c = d$ is immediate. If $c < d$, then for every $X \in \mathrm{Ob}(\mathsf{D}^{[c,d]}(\mathcal{A}))$, take the distinguished triangle from Proposition 4.4.13

$$\tau^{\leq d-1}X \to X \to \tau^{\geq d}X \xrightarrow{+1}, \quad \tau^{\geq d}X \simeq \mathrm{H}^d(X)[-d]$$

and apply the preceding observation to its image under R to carry out the induction. □

Consequently, if R has finite amplitude, then it restricts to $\mathsf{D}^{\mathrm{b}}(\mathcal{A}) \to \mathsf{D}^{\mathrm{b}}(\mathcal{A}')$.

We next give a related result explaining how a triangulated functor is determined by its values on $\mathcal{A}$.

Proposition 4.5.7 (Way-out Lemma [18, Chapter I, §7]) Let $\mathcal{T}$ be a weak Serre subcategory of $\mathcal{A}$ and let $\star \in \{\ \ , +, \mathrm{b}\}$. Consider triangulated functors $F, G : \mathsf{D}^{\star}_{\mathcal{T}}(\mathcal{A}) \to \mathsf{D}(\mathcal{A}')$ and a morphism $\eta : F \to G$ between them such that $\eta_X : FX \to GX$ is an isomorphism for every $X \in \mathrm{Ob}(\mathcal{T})$. Then η is also an isomorphism if one of the following conditions holds:

(i) $\star = \mathrm{b}$;

(ii) $\star = +$, and there exists $k \in \mathbb{Z}$ such that F, G restrict to $\mathsf{D}^{\geq 0}_{\mathcal{T}}(\mathcal{A}) \to \mathsf{D}^{\geq k}(\mathcal{A}')$;

(iii) $\star = \ \ $, and there exist $k, \ell \in \mathbb{Z}$ such that F, G respectively restrict to $\mathsf{D}^{\geq 0}_{\mathcal{T}}(\mathcal{A}) \to \mathsf{D}^{\geq k}(\mathcal{A}')$ and $\mathsf{D}^{\leq 0}_{\mathcal{T}}(\mathcal{A}) \to \mathsf{D}^{\leq \ell}(\mathcal{A}')$.

In addition, let $\mathfrak{I}$ be a subset of $\mathrm{Ob}(\mathcal{T})$ such that for every $X \in \mathrm{Ob}(\mathcal{T})$ there exist $I \in \mathfrak{I}$ and a monomorphism $X \hookrightarrow I$. If the condition on η_X is replaced by the condition that η_I be an isomorphism for every $I \in \mathfrak{I}$, then η is still an isomorphism under assumption (i) or (ii).

Of course, (ii) also has a dual version for $\star = -$, which we do not write out.

Proof For (i), simply repeat the argument of Lemma 4.5.6; use the distinguished triangle $\tau^{\leq d}X \to X \to \tau^{\geq d+1}X \xrightarrow{+1}$ and Proposition 4.2.1 inductively to reduce to the case $X \simeq \mathrm{H}^{d+1}(X)[-d-1]$.

For (ii), the goal is to prove that $\mathrm{H}^j(FX) \to \mathrm{H}^j(GX)$ is an isomorphism for every $X \in \mathrm{Ob}(\mathsf{D}^{+}_{\mathcal{T}}(\mathcal{A}))$ and $j \in \mathbb{Z}$. Consider the distinguished triangle above once more, but take $d \gg 0$ so that $F\tau^{\geq d+1}X$ and $G\tau^{\geq d+1}X$ both lie in $\mathsf{D}^{\geq j+1}(\mathcal{A}')$. We obtain a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] \arrow[d] & \mathrm{H}^j\left(F\tau^{\leq d}X\right) \arrow[r] \arrow[d] & \mathrm{H}^j(FX) \arrow[r] \arrow[d] & 0 \arrow[d] \\
0 \arrow[r] & \mathrm{H}^j\left(G\tau^{\leq d}X\right) \arrow[r] & \mathrm{H}^j(GX) \arrow[r] & 0
\end{tikzcd}$$

But $\tau^{\leq d}X \in \mathrm{Ob}(\mathsf{D}^{\mathrm{b}}_{\mathcal{T}}(\mathcal{A}))$, so this case follows from (i).

For (iii), choose any $d \in \mathbb{Z}$. Using $\tau^{\leq d}X \to X \to \tau^{\geq d+1}X \xrightarrow{+1}$ and Proposition 4.2.1, this case follows from (ii) and its dual.

Now consider the final assertion. Since (ii) has been proved, it suffices to show that η_X is an isomorphism for every $X \in \mathrm{Ob}(\mathcal{T})$. Take an exact sequence $0 \to X \to I^0 \to I^1 \to \cdots$ with every $I^n \in \mathfrak{I}$. Write this sequence as a quasi-isomorphism of complexes $X \to I$; it remains to prove that η_I is an isomorphism. To do this, define the brutal truncation $\sigma^{\leq d}I$ of the complex, which agrees with I in degrees $\leq d$ and has zero terms in all other degrees; define $\sigma^{\geq d+1}I$ similarly. We can now follow the arguments for (i) and (ii), using instead the canonical short exact sequence in $\mathsf{C}(\mathcal{A})$

$$0 \to \sigma^{\geq d+1}I \to I \to \sigma^{\leq d}I \to 0$$

and its corresponding distinguished triangle. The details are left to the reader. $\square$

Recall Theorem 4.5.4. Its key ingredient is the full faithfulness $\mathrm{Hom}_{\mathcal{A}}(X, Y) \simeq \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y)$. What information do we obtain when the degrees of X and Y are shifted relative to one another? This is our next topic.

Definition 4.5.8 (Ext functors: General case) For $X, Y \in \mathrm{Ob}(\mathcal{A})$ and $n \in \mathbb{Z}$, define $\mathrm{Ext}^n_{\mathcal{A}}(X, Y) := \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y[n])$. The functor $\mathrm{Ext}^n_{\mathcal{A}} : \mathcal{A}^{\mathrm{op}} \times \mathcal{A} \to \mathsf{Ab}$ is additive in each variable.

The notation $\mathrm{Ext}^n_{\mathcal{A}}(X, Y)$ already appeared in Definition–Proposition 3.14.4. The discussion of RHom will reconcile the two definitions; see Corollary 4.9.4. Here we do not assume that $\mathcal{A}$ has enough injective or projective objects.

As for Hom, functoriality of Ext^n in its two variables is still called pullback and pushout, respectively. When no confusion can arise, $\mathrm{Ext}^n_{\mathcal{A}}$ is also abbreviated to Ext^n. We now show that its properties resemble those of the Ext^n defined in §3.14.

Proposition 4.5.9 Below, X, Y are arbitrary objects of $\mathcal{A}$. The family of bifunctors $(\mathrm{Ext}^n)_{n\in\mathbb{Z}}$ has the following properties.

(i) If $n < 0$, then $\mathrm{Ext}^n(X, Y) = 0$.

(ii) There is a canonical isomorphism $\mathrm{Ext}^0(X, Y) \simeq \mathrm{Hom}_{\mathcal{A}}(X, Y)$.

(iii) Short exact sequences $0 \to X' \to X \to X'' \to 0$ and $0 \to Y' \to Y \to Y'' \to 0$ in $\mathcal{A}$ induce the respective long exact sequences

$$\cdots \to \mathrm{Ext}^{n-1}(X', Y) \xrightarrow{\delta^{n-1}} \mathrm{Ext}^n(X'', Y) \to \mathrm{Ext}^n(X, Y) \to \mathrm{Ext}^n(X', Y) \to \cdots,$$

$$\cdots \to \mathrm{Ext}^{n-1}(X, Y'') \xrightarrow{\delta^{n-1}} \mathrm{Ext}^n(X, Y') \to \mathrm{Ext}^n(X, Y) \to \mathrm{Ext}^n(X, Y'') \to \cdots,$$

with every connecting morphism δ^n functorial in the short exact sequence.

(iv) If $\mathrm{Ext}^1(X,\cdot) = 0$ (respectively $\mathrm{Ext}^1(\cdot,Y) = 0$), then X (respectively Y) is a projective (respectively injective) object of $\mathcal{A}$.

Proof Assertions (i) and (ii) follow immediately from Proposition 4.5.3.

For (iii), first use Proposition 4.4.7 to turn the short exact sequences canonically into distinguished triangles $X' \to X \to X'' \xrightarrow{+1}$ and $Y' \to Y \to Y'' \xrightarrow{+1}$ in $\mathsf{D}(\mathcal{A})$. Then use the fact that Hom is a cohomological functor in each variable (Proposition 4.1.12).

Finally, (iv) follows by applying (i) and (ii) to the long exact sequences in (iii). $\square$

Since Ext^n is given by Hom in $\mathsf{D}(\mathcal{A})$, for $f \in \mathrm{Ext}^n(X,Y)$ and $g \in \mathrm{Ext}^n(X',Y')$ there is an evident direct sum $f \oplus g \in \mathrm{Ext}^n(X \oplus X', Y \oplus Y')$. On the other hand, Ext has a composition operation,

$$\begin{array}{ccc} \mathrm{Ext}^m(Y,Z) \times \mathrm{Ext}^n(X,Y) & & \\ \| & & \\ \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(Y,Z[m]) \times \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X,Y[n]) & \longrightarrow & \mathrm{Ext}^{m+n}(X,Z) \\ (f,g) & \longmapsto & fg := f[n] \circ g, \end{array} \tag{4.5.3}$$

where $m,n \in \mathbb{Z}$.

It is easy to see that this operation is associative and bilinear. Consequently, the Abelian group

$$\mathrm{Ext}(X) := \bigoplus_{n \in \mathbb{Z}} \mathrm{Ext}^n(X,X)$$

has a natural ring structure, with $\mathrm{id}_A \in \mathrm{End}_{\mathcal{A}}(A)$ as multiplicative identity, while $\mathrm{Ext}(X,Y) := \bigoplus_{n\in\mathbb{Z}} \mathrm{Ext}^n(X,Y)$ becomes an $(\mathrm{Ext}(Y), \mathrm{Ext}(X))$-bimodule.

More generally, if $\mathcal{A}$ is $\Bbbk$-linear, where $\Bbbk$ is a commutative ring, then the functors Ext^n take values in $\Bbbk$-Mod, while $\mathrm{Ext}(X)$ becomes a $\Bbbk$-algebra. The definition of multiplication in (4.5.3) shows that $\mathrm{Ext}(X)$ is in fact a graded $\Bbbk$-algebra; see [51, Definition 7.4.1]. Such constructions are extremely useful in representation theory.

Definition 4.5.10 Let $X \in \mathrm{Ob}(\mathcal{A})$. The graded algebra $\mathrm{Ext}(X)$ defined above is called the **Ext-algebra** of the object X.

The notation Ext abbreviates "extension." To explain the concrete connection, we first define what an extension means.

Definition 4.5.11 Let X and Y be objects of $\mathcal{A}$ and let $n \in \mathbb{Z}_{\geq 1}$. An exact sequence in $\mathcal{A}$ of the form

$$\mathcal{E}: \quad 0 \to Y \to E^1 \to \cdots \to E^n \to X \to 0$$

is called an ***n*-extension** of X by Y. For fixed X, Y, consider the following binary relation on the set of all n-extensions:

$\mathcal{E} \sim_0 \mathcal{E}' \iff$ there is a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & Y \arrow[r] \arrow[d, equal] & E^1 \arrow[r] \arrow[d] & \cdots \arrow[r] & E^n \arrow[r] \arrow[d] & X \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & Y \arrow[r] & (E')^1 \arrow[r] & \cdots \arrow[r] & (E')^n \arrow[r] & X \arrow[r] & 0
\end{tikzcd}$$

This binary relation generates an equivalence relation $\sim$.[5)] The equivalence classes of n-extensions form the set $\mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y)$.

Proposition 2.3.4 implies that equivalence of 1-extensions is the same as isomorphism of short exact sequences

$$\begin{tikzcd}
0 \arrow[r] & Y \arrow[r] \arrow[d, equal] & E \arrow[r] \arrow[d, "\wr"] & X \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & Y \arrow[r] & E' \arrow[r] & X \arrow[r] & 0.
\end{tikzcd}$$

Convention 4.5.12 Henceforth a 1-extension is simply called an extension, and equivalence between such extensions is called isomorphism. A split short exact sequence (Proposition 2.5.3) is also called a split extension; all such extensions are isomorphic.

The following operations are available on n-extensions.

- ▷ Composition For $[\mathcal{E}_1] \in \mathrm{Ext}^{n,\mathrm{Yoneda}}(Y, Z)$ and $[\mathcal{E}_2] \in \mathrm{Ext}^{m,\mathrm{Yoneda}}(X, Y)$, define $[\mathcal{E}_1] \circ [\mathcal{E}_2] \in \mathrm{Ext}^{n+m,\mathrm{Yoneda}}(X, Z)$ by splicing the two exact sequences end-to-end at Y. This operation is also called the **Yoneda product**.

- ▷ Pullback For a morphism $f : X' \to X$, the corresponding map $f^* : \mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y) \to \mathrm{Ext}^{n,\mathrm{Yoneda}}(X', Y)$ is obtained as follows. Given $[\mathcal{E}] \in \mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y)$, consider the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & Y \arrow[r] \arrow[d, equal] & E^1 \arrow[r] \arrow[d, equal] & \cdots \arrow[r] & E^{n-1} \arrow[r] \arrow[d, equal] & E^n \underset{X}{\times} X' \arrow[r] \arrow[d] \arrow[dr, phantom, "\square"] & X' \arrow[r] \arrow[d, "f"] & 0 \\
0 \arrow[r] & Y \arrow[r] & E^1 \arrow[r] & \cdots \arrow[r] & E^{n-1} \arrow[r] & E^n \arrow[r] & X \arrow[r] & 0
\end{tikzcd}$$

The square marked $\square$ is a pullback square, while $E^{n-1} \to E^n \underset{X}{\times} X'$ is determined by $E^{n-1} \to E^n$ and $E^{n-1} \xrightarrow{0} X'$. Pullback preserves kernels, and $E^n \underset{X}{\times} X' \to X'$ is still an epimorphism (Proposition 2.1.6); from this one checks that the first row remains exact. Its equivalence class is $f^*[\mathcal{E}]$.

5) In other words, $\mathcal{E}_1 \sim \mathcal{E}_2$ if and only if they can be connected by a chain of commutative diagrams of this kind.

▷ Pushout For a morphism $g : Y \to Y'$, the corresponding map $g_* : \mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y) \to \mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y')$ is obtained as follows. Given $[\mathcal{E}] \in \mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y)$, pushout gives the commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & Y \arrow[r] \arrow[d, "g"'] \arrow[dr, phantom, "\boxplus"] & E^1 \arrow[r] \arrow[d] & E^2 \arrow[r] \arrow[d, equal] & \cdots \arrow[r] & E^n \arrow[r] \arrow[d, equal] & X \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & Y' \arrow[r] & E^1 \coprod_Y Y' \arrow[r] & E^2 \arrow[r] & \cdots \arrow[r] & E^n \arrow[r] & X \arrow[r] & 0 \end{tikzcd}$$

Here $E^1 \coprod_Y Y' \to E^2$ is determined by $E^1 \to E^2$ and $Y' \xrightarrow{0} E^2$. Dually, one checks that the second row remains exact. Its equivalence class is $g_*[\mathcal{E}]$.

▷ Direct sum Take the termwise direct sum of two extensions, $\mathcal{E} \oplus \mathcal{E}'$, and then take its equivalence class. This gives a map

$$\oplus : \mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y) \times \mathrm{Ext}^{n,\mathrm{Yoneda}}(X', Y') \longrightarrow \mathrm{Ext}^{n,\mathrm{Yoneda}}(X \oplus X', Y \oplus Y').$$

▷ Baer sum This is the following map:

$$\dot{+} : \mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y) \times \mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y) \longrightarrow \mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y).$$

It is defined by pulling $[\mathcal{E}_1] \oplus [\mathcal{E}_2]$ back along $X \hookrightarrow X \times X$, then pushing it out along $Y \oplus Y \twoheadrightarrow Y$ (these morphisms come from Convention 1.3.2). The result is $[\mathcal{E}_1] \dot{+} [\mathcal{E}_2] \in \mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y)$.

One checks that pullback and pushout commute: $g_* f^* = f^* g_*$, while the Baer sum $\dot{+}$ is commutative. These properties also follow from those of Ext^n by the following theorem.

Theorem 4.5.13 (Nobuo Yoneda) For all $X, Y \in \mathrm{Ob}(\mathcal{A})$ and $n \in \mathbb{Z}_{\geq 1}$, there is a canonical bijection

$$\mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y) \xrightarrow{1:1} \mathrm{Ext}^n(X, Y),$$

which sends the equivalence class of an n-extension $0 \to Y \to E^1 \to \cdots \to E^n \to X \to 0$ to $as^{-1} \in \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y[n])$, where

$$\begin{tikzcd} X & & & & X \\ Z \arrow[u, "s:\text{quasi-isomorphism}"] \arrow[d, "a"'] & Y \arrow[r] \arrow[d, "\mathrm{id}"'] & E^1 \arrow[r] & \cdots \arrow[r] & E^n \arrow[u] \\ Y[n] & Y & & & \end{tikzcd}$$

and all omitted terms are zero. Under this bijection, composition, direct sum, pullback, and pushout of n-extensions correspond to the respective operations on Ext^n, while the Baer sum $\dot{+}$ corresponds to the additive group law on $\mathrm{Ext}^n(X, Y)$.

Proof An equivalence of extensions $\mathcal{E} \sim \mathcal{E}'$ is reflected by a quasi-isomorphism between the corresponding complexes Z and Z'. Thus the displayed map is well defined. It remains to prove that it is surjective and injective. We begin with surjectivity.

Every $f \in \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y[n])$ is represented by a diagram $X \xleftarrow{s} Z \xrightarrow{a} Y[n]$ in $\mathsf{C}(\mathcal{A})$, where s is a quasi-isomorphism. After precomposing with the quasi-isomorphism $\tau^{\leq 0}Z \to Z$, we may assume that $Z \in \mathrm{Ob}(\mathsf{C}^{\leq 0}(\mathcal{A}))$. Next, the adjunction in Proposition 3.9.6 factors s and a through the quasi-isomorphism $Z \twoheadrightarrow \tau^{\geq -n}Z$. Hence we may assume that $Z \in \mathrm{Ob}(\mathsf{C}^{[-n,0]}(\mathcal{A}))$.

Since Z is exact outside degree 0, imitate the definition of the pushout of an extension above and define a complex Z' to be the second row of the following diagram:

$$\begin{tikzcd}
\cdots 0 \arrow[r] & Z^{-n} \arrow[r] \arrow[d, "a^{-n}"'] \arrow[dr, phantom, "\boxplus"] & Z^{-n+1} \arrow[r] \arrow[d] & Z^{-n+2} \arrow[r] \arrow[d, equal] & \cdots \\
\cdots 0 \arrow[r] & Y \arrow[r] & Z^{-n+1} \underset{Z^{-n}}{\sqcup} Y \arrow[r] & Z^{-n+2} \arrow[r] & \cdots
\end{tikzcd}$$

The vertical morphisms give a quasi-isomorphism $Z \to Z'$, and a factors as $Z \to Z' \xrightarrow{a'} Y[n]$.

Moreover, $Z \xrightarrow{s} X$ also factors canonically as $Z \to Z' \xrightarrow{s'} X$. This is immediate when $n > 1$; when $n = 1$, the morphism $(s')^0 : Z^0 \underset{Z^{-1}}{\sqcup} Y \to X$ is determined by $s^0 : Z^0 \to X$ and $Y \xrightarrow{0} X$.

In short, we have reduced to the case in which $X \xleftarrow{s} Z \xrightarrow{a} Y[n]$ has the form

$$\begin{tikzcd}
\cdots 0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots \arrow[r] & X \arrow[r] & 0 \cdots \\
\cdots 0 \arrow[r] & Z^{-n} \arrow[r] \arrow[u] \arrow[d, equal] & Z^{-n+1} \arrow[r] \arrow[u] \arrow[d] & \cdots \arrow[r] & Z^0 \arrow[r] \arrow[u] \arrow[d] & 0 \cdots \\
\cdots 0 \arrow[r] & Y \arrow[r] & 0 \arrow[r] & \cdots \arrow[r] & 0 \arrow[r] & 0 \cdots
\end{tikzcd}$$

Flattening this diagram, however, gives the n-extension $0 \to Y \to Z^{-n+1} \to \cdots \to Z^0 \to X \to 0$. This proves surjectivity.

We now prove injectivity. In view of (1.9.1), the starting point is a commutative diagram in $\mathsf{K}(\mathcal{A})$

$$\begin{tikzcd}
 & Z_2 \arrow[dl, "s_2"'] \arrow[dr, "a_2"] & \\
X & W \arrow[l, "t"'] \arrow[r, "b"] \arrow[u] \arrow[d] & Y[n] \\
 & Z_1 \arrow[ul, "s_1"] \arrow[ur, "a_1"'] &
\end{tikzcd}$$

$W \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$, $s_i, t :$ quasi-isomorphisms,
$(Z_i; s_i, a_i)$ is a reduced n-extension,
$(i = 1, 2)$.

We wish to show that this diagram gives an equivalence of n-extensions. As before, first precompose with $\tau^{\leq 0}W \to W$ to reduce to $W \in \mathrm{Ob}(\mathsf{C}^{\leq 0}(\mathcal{A}))$. Next, use the truncation adjunction to factor all the morphisms in $\mathsf{C}(\mathcal{A})$ through $W \twoheadrightarrow \tau^{\geq -n}W$. Let K be the kernel of $W \twoheadrightarrow \tau^{\geq -n}W$. Since $\mathrm{Hom}^{-1}(K, V) = 0$ for every $V \in \mathrm{Ob}(\mathsf{C}^{[-n,0]}(\mathcal{A}))$, the homotopies between these morphisms factor in the same way. This operation therefore does not change the commutativity of the diagram in $\mathsf{K}(\mathcal{A})$. We have reduced to the case $W \in \mathrm{Ob}(\mathsf{C}^{[-n,0]}(\mathcal{A}))$.

Similarly, $\mathrm{Hom}^{-1}(W, X) = 0$, so the left half of the diagram already commutes in $\mathsf{C}(\mathcal{A})$. For the right half, since $a_i^{-n} = \mathrm{id}_Y$, we may modify the morphisms $W \to Z_i$ by suitable homotopies so that the right half also commutes in $\mathsf{C}(\mathcal{A})$. Now repeat the pushout operation from the surjectivity proof to reduce to $W^{-n} = Y$ and $b^{-n} = \mathrm{id}_Y$, without changing the commutativity of the left half, as one verifies directly. Thus $(W; t, b)$ also corresponds to an n-extension, and the commutative diagram witnesses the equivalence relation $\sim$ in Definition 4.5.11. This proves injectivity.

Compatibility of composition, direct sum, pullback, and pushout with this bijection is a routine verification. For the Baer sum $\dot{+}$, Proposition 1.3.5 realizes the sum $f + g$ in $\mathrm{Ext}^n(X, Y) := \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y[n])$ as the composite $X \to X \oplus X \xrightarrow{f \oplus g} (Y \oplus Y)[n] \to Y[n]$, which agrees exactly with the definition of $\dot{+}$. ☐

Dually, an element of $\mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y[n])$ can also be represented by a diagram of the form $X \xrightarrow{b} Z \xleftarrow{t} Y[n]$ in $\mathsf{C}(\mathcal{A})$, where t is a quasi-isomorphism. The exercises for this chapter compare these two constructions.

Proposition 4.5.14 Consider an extension $0 \to Y \xrightarrow{f} E \xrightarrow{g} X \to 0$. In the exact sequence from Proposition 4.5.9 (iii),

$$\mathrm{Hom}_{\mathcal{A}}(X, E) \xrightarrow{g_*} \mathrm{Hom}_{\mathcal{A}}(X, X) \to \mathrm{Ext}^1(X, Y),$$

the element of $\mathrm{Ext}^1(X, Y)$ determined by this extension is the image of id_X.

Proof The connecting homomorphism $\mathrm{Hom}_{\mathcal{A}}(X, X) \to \mathrm{Ext}^1(X, Y)$ is precisely e_*, where $e \in \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y[1])$ is determined by the diagram in $\mathsf{C}(\mathcal{A})$

$$X \underset{\text{quasi-isomorphism}}{\xleftarrow{\Phi}} \mathrm{Cone}(f) \xrightarrow{\beta(f)} Y[1]$$

as in Proposition 4.4.7. But $\mathrm{Cone}(f)$ is the complex $Y \xrightarrow{f} E$ concentrated in degrees -1 and 0. Comparison with Theorem 4.5.13 immediately shows that e comes from the extension $0 \to Y \xrightarrow{f} E \xrightarrow{g} X \to 0$. ☐

Proposition 4.5.15 The zero element of the additive group $\left(\mathrm{Ext}^{n,\mathrm{Yoneda}}(X, Y), \dot{+}\right)$ is represented by the following extensions:

$n=1$	The split extension $0 \to Y \to X \oplus Y \to X \to 0$
$n=2$	$0 \to Y \xrightarrow{\mathrm{id}_Y} Y \xrightarrow{0} X \xrightarrow{\mathrm{id}_X} X \to 0$
$n \geq 3$	$0 \to Y \xrightarrow{\mathrm{id}_Y} Y \to 0 \to \cdots \to 0 \to X \xrightarrow{\mathrm{id}_X} X \to 0$.

Proof For $n = 1$, suppose that the short exact sequence $0 \to Y \xrightarrow{f} E \xrightarrow{g} X \to 0$ corresponds to the zero element. The exact sequence in Proposition 4.5.14 shows that there is an $s \in \mathrm{Hom}_{\mathcal{A}}(X, E)$ such that $gs = \mathrm{id}_X$; hence the short exact sequence splits.

Next consider $n = 2$. Both $\mathrm{Ext}^1(0, Y)$ and $\mathrm{Ext}^1(X, 0)$ are plainly zero groups; corresponding extensions may be taken to be $0 \to Y \xrightarrow{\mathrm{id}} Y \to 0 \to 0$ and $0 \to 0 \to X \xrightarrow{\mathrm{id}} X \to 0$. Since $\mathrm{Ext}^1 \times \mathrm{Ext}^1 \to \mathrm{Ext}^2$ is bilinear, splicing these two extensions gives the zero element of $\mathrm{Ext}^2(X, Y)$. For $n \geq 3$, consider the $(n-1)$-extension $0 \to \cdots \to 0 \to X \xrightarrow{\mathrm{id}} X \to 0$; the same argument gives the zero element of $\mathrm{Ext}^n(X, Y)$. □

4.6 Triangulated Functors and Localization

Fix a triangulated category $\mathcal{D}$ and a saturated triangulated subcategory $\mathcal{N}$, and denote the corresponding Verdier localization by $Q : \mathcal{D} \to \mathcal{D}/\mathcal{N}$. Likewise, consider a triangulated category $\mathcal{D}'$ and $Q' : \mathcal{D}' \to \mathcal{D}'/\mathcal{N}'$.

Let $F : \mathcal{D} \to \mathcal{D}'$ be a triangulated functor. Consider the diagram whose solid part is given:

$$\begin{tikzcd} \mathcal{D} \arrow[r, "F"] \arrow[d, "Q"'] & \mathcal{D}' \arrow[d, "Q'"] \\ \mathcal{D}/\mathcal{N} \arrow[r, dashed] & \mathcal{D}'/\mathcal{N}' \end{tikzcd} \tag{4.6.1}$$

The naive hope is to complete the dashed part so that the entire diagram commutes. If $F(\mathrm{Ob}(\mathcal{N})) \subset \mathrm{Ob}(\mathcal{N}')$, then $Q'F$ annihilates $\mathcal{N}$, and the universal property in Theorem 4.3.4 induces the dashed functor. In practice, however, F rarely has this property. We therefore settle for less and turn to the Kan extensions introduced in §1.7.

Definition 4.6.1 (Derived functors of a triangulated functor) Let $F : \mathcal{D} \to \mathcal{D}'$ be as above.

- ◇ If $\mathrm{Lan}_Q(Q'F)$ exists and is a triangulated functor, denote it by $\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} F : \mathcal{D}/\mathcal{N} \to \mathcal{D}'/\mathcal{N}'$;
- ◇ if $\mathrm{Ran}_Q(Q'F)$ exists and is a triangulated functor, denote it by $\mathrm{L}_{\mathcal{N}}^{\mathcal{N}'} F : \mathcal{D}/\mathcal{N} \to \mathcal{D}'/\mathcal{N}'$.

We call $\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} F$ (respectively, $\mathrm{L}_{\mathcal{N}}^{\mathcal{N}'} F$) the **right derived functor** (respectively, **left derived functor**) of the triangulated functor F. Its uniqueness up to unique isomorphism follows from the uniqueness of Kan extensions.

By Definition 1.7.1 of Kan extensions, these data fit into the 2-cell diagrams

$$
\begin{tikzcd}
\mathcal{D} \arrow[r, "F"] \arrow[d, "Q"'] & \mathcal{D}' \arrow[d, "Q'"] \arrow[dl, Rightarrow] \\
\mathcal{D}/\mathcal{N} \arrow[r, "\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} F"'] & \mathcal{D}'/\mathcal{N}'
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{D} \arrow[r, "F"] \arrow[d, "Q"'] & \mathcal{D}' \arrow[d, "Q'"] \\
\mathcal{D}/\mathcal{N} \arrow[r, "\mathrm{L}_{\mathcal{N}}^{\mathcal{N}'} F"'] \arrow[ur, Rightarrow, shorten <=2em] & \mathcal{D}'/\mathcal{N}'.
\end{tikzcd}
\tag{4.6.2}
$$

The universal property says that every way of filling (4.6.1) with a 2-cell is obtained uniquely by "pushing out" the diagram for $\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} F$, or uniquely by "pulling into" the diagram for $\mathrm{L}_{\mathcal{N}}^{\mathcal{N}'} F$, according to the direction of the filling.

As an illustration, we explain how the universal property of the 2-cell derives a canonical morphism $\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} F \to \mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} G$ from a morphism of functors $F \to G$, assuming that these derived functors exist. Examine the diagram on the left:

$$
\begin{tikzcd}
\mathcal{D} \arrow[r, bend left=50, "F"] \arrow[r, "G"'] \arrow[r, phantom, bend left=25, "\Downarrow"] \arrow[d] & \mathcal{D}' \arrow[d] \arrow[dl, Rightarrow] \\
\mathcal{D}/\mathcal{N} \arrow[r, "\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} G"'] & \mathcal{D}'/\mathcal{N}'
\end{tikzcd}
\quad
\xlongequal{\text{vertical composition}}
\quad
\begin{tikzcd}
\mathcal{D} \arrow[r, "F"] \arrow[d] & \mathcal{D}' \arrow[d] \arrow[dl, Rightarrow] \\
\mathcal{D}/\mathcal{N} \arrow[r, "\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} F"] \arrow[r, bend right=50, "\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} G"'] \arrow[r, phantom, bend right=25, "\exists!\Downarrow"] & \mathcal{D}'/\mathcal{N}'
\end{tikzcd}
$$

By the universal property of (4.6.2), the vertical composite in the left-hand diagram is uniquely the pushout of the diagram corresponding to $\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} F$, as shown on the right. This "push" is the desired canonical morphism. The case of left derived functors is of course dual. The next remark shows that these canonical morphisms are compatible with translation.

Remark 4.6.2 Suppose that $\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} F$ (respectively, $\mathrm{L}_{\mathcal{N}}^{\mathcal{N}'} F$) exists. If (L, ξ) (respectively, (R, δ)) in the universal property of the Kan extension is taken to consist of a triangulated functor and a morphism compatible with translation (see Definition 4.1.1), then the determined morphism $\chi : \mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} F \to L$ (respectively, $\theta : R \to \mathrm{L}_{\mathcal{N}}^{\mathcal{N}'} F$) is automatically compatible with translation as well. This follows immediately from uniqueness.

Definition 4.6.1 raises two questions. First, how can the existence of $\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} F$ (or $\mathrm{L}_{\mathcal{N}}^{\mathcal{N}'} F$) be guaranteed? Second, how do these functors relate to composition of functors? We address the two questions in order.

Definition 4.6.3 Let $\mathcal{N}$, $\mathcal{N}'$, and $F : \mathcal{D} \to \mathcal{D}'$ be as above, and let $\mathcal{I}$ be a triangulated subcategory of $\mathcal{D}$. If the following conditions hold,

- ▷ Resolution condition for every $X \in \mathrm{Ob}(\mathcal{D})$ there is a morphism $X \to Y$ (respectively, $Y \to X$) in $S\mathcal{N}$ with $Y \in \mathrm{Ob}(\mathcal{I})$;
- ▷ $\mathcal{N}$-preservation condition $F\,(\mathrm{Ob}(\mathcal{N} \cap \mathcal{I})) \subset \mathrm{Ob}(\mathcal{N}')$,

then $\mathcal{I}$ is called ***F*-injective** (respectively, ***F*-projective**).

For example, if $F\,(\mathrm{Ob}(\mathcal{N})) \subset \mathrm{Ob}(\mathcal{N}')$, then $\mathcal{D}$ itself is both F-injective and F-projective.

The condition $F\,(\mathrm{Ob}(\mathcal{N} \cap \mathcal{I})) \subset \mathrm{Ob}(\mathcal{N}')$, together with the universal property of Verdier localization, gives a triangulated functor $F^\flat : \mathcal{I}/(\mathcal{I} \cap \mathcal{N}) \to \mathcal{D}'/\mathcal{N}'$.

Proposition 4.6.4 If $\mathcal{I}$ is an F-injective (respectively, F-projective) triangulated subcategory, then $\mathrm{R}^{\mathcal{N}'}_{\mathcal{N}} F$ (respectively, $\mathrm{L}^{\mathcal{N}'}_{\mathcal{N}} F$) exists, and the following diagram commutes up to isomorphism:

$$\begin{tikzcd}
\mathcal{D}/\mathcal{N} \arrow[r, "\text{desired functor}"] & \mathcal{D}'/\mathcal{N}'. \\
\mathcal{I}/(\mathcal{I} \cap \mathcal{N}) \arrow[u, "i\text{:equivalence}"] \arrow[ur, "F^\flat"'] &
\end{tikzcd}$$

Proof Consider the case of $\mathrm{R}^{\mathcal{N}'}_{\mathcal{N}} F$. The existence of the left Kan extension $\mathrm{Lan}_Q(Q'F)$ follows from Proposition 1.10.3 (ii). The only point requiring verification is that $Q'F$ sends $S\mathcal{N} \cap \mathrm{Mor}(\mathcal{I}) \overset{(4.3.2)}{=\!=} S(\mathcal{N} \cap \mathcal{I})$ to isomorphisms. Take $f : A \to B$ in this set and complete it to a distinguished triangle $A \xrightarrow{f} B \to C \xrightarrow{+1}$, where $C \in \mathrm{Ob}(\mathcal{N} \cap \mathcal{I})$. Then the distinguished triangle $FA \xrightarrow{Ff} FB \to FC \xrightarrow{+1}$ has $FC \in \mathrm{Ob}(\mathcal{N}')$, ensuring that $(Q'F)(f)$ is an isomorphism.

Proposition 4.3.7 implies that i has a triangulated quasi-inverse i^{-1}. Consequently, $\mathrm{R}^{\mathcal{N}'}_{\mathcal{N}} F := F^\flat \circ i^{-1}$ gives $\mathrm{Lan}_Q(Q'F)$ and is at the same time a triangulated functor. □

Theorem 4.6.5 Consider triangulated categories $\mathcal{D}$, $\mathcal{D}'$, and $\mathcal{D}''$, with saturated triangulated subcategories $\mathcal{N}$, $\mathcal{N}'$, and $\mathcal{N}''$. Let triangulated functors $\mathcal{D} \xrightarrow{F} \mathcal{D}' \xrightarrow{F'} \mathcal{D}''$ be given.

(i) Suppose that the right derived functors $\mathrm{R}^{\mathcal{N}'}_{\mathcal{N}} F$, $\mathrm{R}^{\mathcal{N}''}_{\mathcal{N}'} F'$, and $\mathrm{R}^{\mathcal{N}''}_{\mathcal{N}} (F'F)$ exist. There is then a corresponding canonical morphism

$$\mathrm{R}^{\mathcal{N}''}_{\mathcal{N}} (F'F) \to \left(\mathrm{R}^{\mathcal{N}''}_{\mathcal{N}'} F'\right) \left(\mathrm{R}^{\mathcal{N}'}_{\mathcal{N}} F\right).$$

(ii) The analogous statement holds for left derived functors, with the corresponding canonical morphism

$$\left(\mathrm{L}_{\mathcal{N}'}^{\mathcal{N}''} F'\right)\left(\mathrm{L}_{\mathcal{N}}^{\mathcal{N}'} F\right) \to \mathrm{L}_{\mathcal{N}}^{\mathcal{N}''}(F'F).$$

(iii) Fix a triangulated subcategory $\mathcal{I}$ of $\mathcal{D}$ (and a triangulated subcategory $\mathcal{I}'$ of $\mathcal{D}'$). If $\mathcal{I}$ is F-injective, $\mathcal{I}'$ is F'-injective, and $F(\mathrm{Ob}(\mathcal{I})) \subset \mathrm{Ob}(\mathcal{I}')$, then $\mathcal{I}$ is $F'F$-injective and the canonical morphism in (i) is an isomorphism.

(iv) The analogous statement holds with "injective" replaced by "projective." In this case, $\mathcal{I}$ is $F'F$-projective and the canonical morphism in (ii) is an isomorphism.

Proof Clearly, (i), (iii) are respectively dual to (ii), (iv), so it suffices to consider the former pair.

For (i), write $R := \mathrm{R}_{\mathcal{N}}^{\mathcal{N}'} F$ and $R' := \mathrm{R}_{\mathcal{N}'}^{\mathcal{N}''} F'$. Consider the 2-cell diagram

$$\begin{tikzcd}
\mathcal{D} \arrow[r, "F"] \arrow[d, "Q"'] & \mathcal{D}' \arrow[r, "F'"] \arrow[d, "Q'"] \arrow[dl, Rightarrow] & \mathcal{D}'' \arrow[d, "Q''"] \arrow[dl, Rightarrow] \\
\mathcal{D}/\mathcal{N} \arrow[r, "R"'] & \mathcal{D}'/\mathcal{N}' \arrow[r, "R'"'] & \mathcal{D}''/\mathcal{N}''
\end{tikzcd}$$

The universal property of the left Kan extension gives $\mathrm{R}_{\mathcal{N}}^{\mathcal{N}''}(F'F) \to R'R$, characterized by

$$\begin{tikzcd}
\mathcal{D} \arrow[r, "F'F"] \arrow[d, "Q"'] & \mathcal{D}'' \arrow[d, "Q''"] \arrow[dl, Rightarrow] \\
\mathcal{D}/\mathcal{N} \arrow[r, "\mathrm{R}_{\mathcal{N}}^{\mathcal{N}''}(F'F)"'{name=U}] \arrow[r, bend right=60, "R'R"'{name=D}] & \mathcal{D}''/\mathcal{N}''
\arrow[Rightarrow, from=U, to=D]
\end{tikzcd}
\text{ composed with } =
\begin{tikzcd}
\mathcal{D} \arrow[r, "F'F"] \arrow[d, "Q"'] & \mathcal{D}'' \arrow[d, "Q''"] \arrow[dl, Rightarrow] \\
\mathcal{D}/\mathcal{N} \arrow[r, "R'R"'] & \mathcal{D}''/\mathcal{N}''.
\end{tikzcd}
\tag{4.6.3}$$

For (iii), the fact that $\mathcal{I}$ is $F'F$-injective follows directly from Definition 4.6.3 and the condition $F(\mathrm{Ob}(\mathcal{I})) \subset \mathrm{Ob}(\mathcal{I}')$. Let $Y \in \mathrm{Ob}(\mathcal{I})$. The construction in Proposition 4.6.4 gives

$$\begin{gathered}
RQY = Q'FY \in \mathrm{Ob}\left(\mathcal{I}'/(\mathcal{N}' \cap \mathcal{I}')\right), \\
(R'R)QY = R'(Q'FY) = Q''F'FY = \mathrm{R}_{\mathcal{N}}^{\mathcal{N}''}(F'F)(QY).
\end{gathered}$$

For a general object $QX \in \mathrm{Ob}(\mathcal{D}/\mathcal{N})$, with $X \in \mathrm{Ob}(\mathcal{D})$, there is a morphism $X \to Y$ in $S\mathcal{N}$ with $Y \in \mathrm{Ob}(\mathcal{I})$, so that $QX \xrightarrow{\sim} QY$. Applying the preceding step yields an isomorphism $\mathrm{R}_{\mathcal{N}}^{\mathcal{N}''}(F'F)(QX) \simeq R'R(QX)$. The interested reader may check that these isomorphisms are indeed compatible with the morphism $\mathrm{R}_{\mathcal{N}}^{\mathcal{N}''}(F'F) \to R'R$ characterized by (4.6.3). □

Remark 4.6.6 (P. Deligne's definition) By Proposition 1.10.3 (iii), under the hypotheses of Proposition 4.6.4, the derived functors can be written as

$$(\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'}F)(QX) \simeq \varinjlim_{(X\to Y)\in \mathrm{Ob}(S\mathcal{N}_{X/})} (Q'F)Y,$$

$$(\mathrm{L}_{\mathcal{N}}^{\mathcal{N}'}F)(QX) \simeq \varprojlim_{(Y\to X)\in \mathrm{Ob}(S\mathcal{N}_{/X}^{\mathrm{op}})} (Q'F)Y.$$

These isomorphisms are canonical and, by means of resolutions, each limit is attained by some Y. In [45, Exp XVII, Déf 1.2.1], Deligne defines $(\mathrm{R}F_{\mathcal{N}}^{\mathcal{N}'}F)(QX)$ and $(\mathrm{L}_{\mathcal{N}}^{\mathcal{N}'}F)(QX)$ respectively as the above limits, provided that the limit in question is actually attained by some Y. One advantage of this definition is that it can be applied to a single object X, and hence can be made "partial."

The formulas above also describe the morphisms that come with the derived functors as Kan extensions, $Q'F \to (\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'}F)Q$ and $(\mathrm{L}_{\mathcal{N}}^{\mathcal{N}'}F)Q \to Q'F$. They correspond, respectively, to the terms $Y = X$ in $\varinjlim$ and $\varprojlim$.

We now turn to bifunctors. Let $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}'$ be triangulated categories, with translation functors T_1, T_2, and T', respectively.

Definition 4.6.7 A **triangulated bifunctor** from $\mathcal{D}_1 \times \mathcal{D}_2$ to $\mathcal{D}'$ consists of the following data.

- ◇ A bifunctor $F : \mathcal{D}_1 \times \mathcal{D}_2 \to \mathcal{D}'$ endowed with the structure of a triangulated functor in each variable.
- ◇ For all $X \in \mathrm{Ob}(\mathcal{D}_1)$ and $Y \in \mathrm{Ob}(\mathcal{D}_2)$, the following diagram is anticommutative (that is, its two composites differ by a minus sign)[6)]

$$\begin{array}{ccc} F(T_1X, T_2Y) & \longrightarrow & T'F(X, T_2Y) \\ \downarrow & & \downarrow \\ T'F(T_1X, Y) & \longrightarrow & (T')^2(X, Y) \end{array}$$

  The arrows come from the triangulated-functor structures of F in the two variables.

In the situation above, take $\mathcal{N}_1$, $\mathcal{N}_2$, and $\mathcal{N}'$ to be saturated triangulated subcategories of $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}'$, respectively. Denote the corresponding localization functors by $Q_i : \mathcal{D}_i \to \mathcal{D}_i/\mathcal{N}_i$ $(i = 1, 2)$ and $Q' : \mathcal{D}' \to \mathcal{D}'/\mathcal{N}'$. Now fix a triangulated bifunctor $F : \mathcal{D}_1 \times \mathcal{D}_2 \to \mathcal{D}'$. For this bifunctor we may study the functor-extension problem in

[6)] Compare Proposition 3.5.8.

the diagram

$$\begin{tikzcd} \mathcal{D}_1 \times \mathcal{D}_2 \arrow[r, "F"] \arrow[d, "{(Q_1,Q_2)}"'] & \mathcal{D}' \arrow[d, "Q'"] \\ (\mathcal{D}_1/\mathcal{N}_1) \times (\mathcal{D}_2/\mathcal{N}_2) \arrow[r, dashed] & \mathcal{D}'/\mathcal{N}'. \end{tikzcd}$$

Since (Q_1, Q_2) is the localization at the multiplicative system $S\mathcal{N}_1 \times S\mathcal{N}_2$ (Remark 1.9.21), we can still speak of the two Kan extensions of $Q'F$ along (Q_1, Q_2).

Definition 4.6.8 (Derived functors of a triangulated bifunctor) For the data above, if

$$\mathrm{Lan}_{(Q_1,Q_2)}(Q'F), \quad (\text{respectively, } \mathrm{Ran}_{(Q_1,Q_2)}(Q'F))$$

exists and is a triangulated bifunctor, denote it by $\mathrm{R}^{\mathcal{N}'}_{\mathcal{N}_1\times\mathcal{N}_2}F$ (respectively, $\mathrm{L}^{\mathcal{N}'}_{\mathcal{N}_1\times\mathcal{N}_2}F$) and call it the **right derived bifunctor** (respectively, **left derived bifunctor**) of F.

Definition 4.6.9 Consider a triangulated bifunctor $F : \mathcal{D}_1 \times \mathcal{D}_2 \to \mathcal{D}'$, together with $\mathcal{N}_1$, $\mathcal{N}_2$, and $\mathcal{N}'$ as above. Let $\mathcal{I}_i$ be a triangulated subcategory of $\mathcal{D}_i$. We call $(\mathcal{I}_1, \mathcal{I}_2)$ ***F*-injective** (respectively, ***F*-projective**) if the following properties hold simultaneously:

- ◇ for every $X_1 \in \mathrm{Ob}(\mathcal{I}_1)$, the subcategory $\mathcal{I}_2$ is $F(X_1, \cdot)$-injective (respectively, projective);
- ◇ for every $X_2 \in \mathrm{Ob}(\mathcal{I}_2)$, the subcategory $\mathcal{I}_1$ is $F(\cdot, X_2)$-injective (respectively, projective).

For $\mathcal{I}_j$ as above ($j = 1, 2$), the universal property of Verdier localization gives $i_j : \mathcal{I}_j/(\mathcal{I}_j \cap \mathcal{N}_j) \to \mathcal{D}_j/\mathcal{N}_j$. The following is the counterpart of Proposition 4.6.4.

Proposition 4.6.10 Suppose that $(\mathcal{I}_1, \mathcal{I}_2)$ is F-injective.

(i) The right derived bifunctor $\mathrm{R}F := \mathrm{R}^{\mathcal{N}'}_{\mathcal{N}_1\times\mathcal{N}_2}F$ exists, and the following diagram commutes up to isomorphism:

$$\begin{tikzcd} \mathcal{D}_1/\mathcal{N}_1 \times \mathcal{D}_2/\mathcal{N}_2 \arrow[r, "\mathrm{R}F"] & \mathcal{D}'/\mathcal{N}' \\ \mathcal{I}_1/(\mathcal{I}_1 \cap \mathcal{N}_1) \times \mathcal{I}_2/(\mathcal{I}_2 \cap \mathcal{N}_2) \arrow[u, "{i=(i_1,i_2):\text{equivalence}}"] \arrow[ur, "F^\flat"'] & \end{tikzcd}$$

where $F^\flat$ is determined by the universal property of localization.

(ii) For $X_1 \in \mathrm{Ob}(\mathcal{I}_1)$ and $X_2 \in \mathrm{Ob}(\mathcal{I}_2)$, there are canonical isomorphisms

$$(\mathrm{R}F)(Q_1X_1, \cdot) \simeq \mathrm{R}^{\mathcal{N}'}_{\mathcal{N}_2}F(X_1, \cdot), \quad (\mathrm{R}F)(\cdot, Q_2X_2) \simeq \mathrm{R}^{\mathcal{N}'}_{\mathcal{N}_1}F(\cdot, X_2).$$

The corresponding statements remain valid if F-injective is replaced by F-projective and $\mathrm{R}F$ by $\mathrm{L}F$.

Proof It suffices to treat the F-injective case. First, Proposition 1.10.3 (i) implies that i is an equivalence. The hypotheses show that $Q'F$ sends $S\mathcal{N}_1 \cap \mathrm{Mor}(\mathcal{I}_1) \times S\mathcal{N}_2 \cap \mathrm{Mor}(\mathcal{I}_2)$ to isomorphisms; this need only be checked in each variable separately. Proposition 1.10.3 (ii) implies that $Q'F$ has a left Kan extension along (Q_1, Q_2), namely $\mathrm{R}F$, which makes the diagram above commute up to isomorphism.

We next show that $\mathrm{R}F$ is a triangulated bifunctor. By Proposition 4.3.2, the relevant properties can be transported along the equivalence i and checked for F on $\mathcal{I}_1 \times \mathcal{I}_2$. This proves (i).

For (ii), by symmetry it suffices to fix $X_1 \in \mathrm{Ob}(\mathcal{I}_1)$. In this case $F^\flat(Q_1X_1, \cdot) : \mathcal{I}_2/(\mathcal{I}_2 \cap \mathcal{N}_2) \to \mathcal{D}'/\mathcal{N}'$ is determined from $Q'F(X_1, \cdot)$ by the universal property of localization, and the following diagram of functors commutes up to isomorphism:

$$\begin{tikzcd}
\mathcal{D}_2/\mathcal{N}_2 \arrow[r, "\mathrm{R}F(Q_1X_1,\cdot)"] & \mathcal{D}'/\mathcal{N}' \\
\mathcal{I}_2/\mathcal{I}_2 \cap \mathcal{N}_2 \arrow[u, "i_2:\text{equivalence}"] \arrow[ur, "F^\flat(Q_1X_1,\cdot)"] & \mathcal{I}_2 \arrow[l] \arrow[u, "Q'F(X_1,\cdot)"']
\end{tikzcd}$$

Since $\mathcal{I}_2$ is $F(X_1, \cdot)$-injective, comparison with the construction in Proposition 4.6.4 shows that $(\mathrm{R}F)(Q_1X_1, \cdot)$ is precisely $\mathrm{R}^{\mathcal{N}'}_{\mathcal{N}_2}F(X_1, \cdot)$. □

Once F-injective (respectively, F-projective) data $(\mathcal{I}_1, \mathcal{I}_2)$ are available, the limit formulas in Remark 4.6.6 also apply to the left (respectively, right) derived bifunctor of F.

4.7 General Theory of Derived Functors

Let $\mathcal{A}$ and $\mathcal{A}'$ be abelian categories, and fix an additive functor $F : \mathcal{A} \to \mathcal{A}'$. We now apply the theory of §4.6, and in particular the setting of (4.6.1), to

$$\begin{tikzcd}
\mathsf{K}^\star(\mathcal{A}) \arrow[r, "\mathsf{K}^\star F"] \arrow[d, "Q"'] & \mathsf{K}^\star(\mathcal{A}') \arrow[d, "Q'"] \\
\mathsf{D}^\star(\mathcal{A}) & \mathsf{D}^\star(\mathcal{A}')
\end{tikzcd} \qquad (\star \in \{+, -, \quad \})$$

Here Q and Q' are localization functors of triangulated categories, and $\mathsf{K}^\star F$ is a triangulated functor (Proposition 4.4.3). The blank superscript $\star = \quad$ corresponds to the case of $\mathsf{K}(\mathcal{A})$ and $\mathsf{D}(\mathcal{A})$. Recall that $\mathsf{D}^\star(\mathcal{A}) = \mathsf{K}^\star(\mathcal{A})/\mathcal{N}^\star$, where $\mathcal{N}^\star$ is the saturated triangulated subcategory of acyclic complexes; likewise for $\mathcal{A}'$.

Unless F is exact, in general there is no triangulated functor $\mathsf{D}^\star(\mathcal{A}) \to \mathsf{D}^\star(\mathcal{A}')$ that makes the diagram above commute. The natural approach is to study the right and left derived functors in the sense of Definition 4.6.1, using the two kinds of Kan extension respectively.

Definition 4.7.1 (Derived functors between derived categories) For $\star \in \{+, -, \ \}$, if

$$\begin{aligned} {}^\star\mathrm{R}F : \mathsf{D}^\star(\mathcal{A}) \to \mathsf{D}^\star(\mathcal{A}') &: \quad \text{the right derived functor of } \mathsf{K}^\star F, \quad \text{or} \\ {}^\star\mathrm{L}F : \mathsf{D}^\star(\mathcal{A}) \to \mathsf{D}^\star(\mathcal{A}') &: \quad \text{the left derived functor of } \mathsf{K}^\star F \end{aligned}$$

exists, it is called the **right derived functor** (respectively, **left derived functor**) of $F : \mathcal{A} \to \mathcal{A}'$. When the derived functor exists, define, for every $n \in \mathbb{Z}$,

$$\begin{aligned} {}^\star\mathrm{R}^n F &:= \mathrm{H}^n \circ {}^\star\mathrm{R}F, \\ {}^\star\mathrm{L}_n F &:= \mathrm{H}^{-n} \circ {}^\star\mathrm{L}F. \end{aligned}$$

Both are functors with values in $\mathcal{A}'$. The subscript n is used to agree with the customary notation for chain complexes when studying left derived functors. The left superscript $\star$ is often omitted when no confusion can arise.

Example 4.7.2 (Deriving an exact functor) The simplest case is that in which F is exact. For every $\star \in \{+, -, \ \}$, the derived functors ${}^\star\mathrm{R}F$ and ${}^\star\mathrm{L}F$ always exist. Indeed, $\mathsf{K}^\star F$ preserves acyclic complexes, so its derived functor is determined directly by the universal property of localization.

Theorem 4.7.3 (Long exact sequence of derived functors) Fix $\star \in \{+, -, \ \}$. If ${}^\star\mathrm{R}F$ or ${}^\star\mathrm{L}F$ exists, then every distinguished triangle $X \to Y \to Z \xrightarrow{+1}$ in $\mathsf{D}^\star(\mathcal{A})$ gives rise to a canonical long exact sequence

$$\cdots \to {}^\star\mathrm{R}^{n-1}F(Z) \to {}^\star\mathrm{R}^n F(X) \to {}^\star\mathrm{R}^n F(Y) \to {}^\star\mathrm{R}^n F(Z) \to {}^\star\mathrm{R}^{n+1}F(X) \to \cdots,$$
$$\cdots \to {}^\star\mathrm{L}_{n+1}F(Z) \to {}^\star\mathrm{L}_n F(X) \to {}^\star\mathrm{L}_n F(Y) \to {}^\star\mathrm{L}_n F(Z) \to {}^\star\mathrm{L}_{n-1}F(X) \to \cdots.$$

Proof By definition, both ${}^\star\mathrm{R}F$ and ${}^\star\mathrm{L}F$ are triangulated functors. Now apply the cohomology functor H^0. □

How do we ensure that derived functors exist, and how do we compute them? Proposition 4.6.4 already contains the general idea. For $\star \in \{+, -, \ \}$, take $\mathcal{N}^\star$ as in Definition 4.4.6. We seek a triangulated subcategory $\mathcal{I}$ of $\mathsf{K}^\star(\mathcal{A})$ such that ${}^\star\mathrm{R}F$ or ${}^\star\mathrm{L}F$ can be defined as the composite in the diagram of functors

$$\mathsf{D}^\star(\mathcal{A}) = \mathsf{K}^\star(\mathcal{A})/\mathcal{N}^\star \xleftarrow{\text{equivalence}} \mathcal{I}/\mathcal{I}\cap\mathcal{N}^\star \xrightarrow{\text{universal property of localization}} \mathsf{K}^\star(\mathcal{A}')/\mathcal{N}^\star = \mathsf{D}^\star(\mathcal{A}').$$

How should $\mathcal{I}$ be chosen? It should be F-injective (respectively, F-projective) in the following sense.

Convention 4.7.4 Fix $\star$. If a triangulated subcategory $\mathcal{I}$ of $\mathsf{K}^\star(\mathcal{A})$ is a $\mathsf{K}^\star F$-injective (respectively, $\mathsf{K}^\star F$-projective) triangulated subcategory in the sense of Definition 4.6.3, we call $\mathcal{I}$ an F-injective (respectively, F-projective) subcategory for short.

If $\mathsf{K}^\star(\mathcal{A})$ has an F-injective (respectively, F-projective) subcategory $\mathcal{I}$, Proposition 4.6.4 guarantees the existence of ${}^\star\mathrm{R}F$ (respectively, ${}^\star\mathrm{L}F$). Its restriction to $\mathcal{I}$ is isomorphic to $Q'\mathsf{K}^\star F$, while the limit formulas in Remark 4.6.6 make explicit the canonical morphism that accompanies the derived functor as a Kan extension:

$$Q'\mathsf{K}^\star F \to ({}^\star\mathrm{R}F)Q \quad \text{or} \quad ({}^\star\mathrm{L}F)Q \to Q'(\mathsf{K}^\star F). \tag{4.7.1}$$

The entire general theory developed in §4.6 applies directly. For example, consider additive functors between abelian categories

$$\mathcal{A} \xrightarrow{F} \mathcal{A}' \xrightarrow{F'} \mathcal{A}''.$$

Provided the derived functors in question exist, there are canonical morphisms

$${}^\star\mathrm{R}(F'F) \to ({}^\star\mathrm{R}F')({}^\star\mathrm{R}F), \quad ({}^\star\mathrm{L}F')\,({}^\star\mathrm{L}F) \to {}^\star\mathrm{L}(F'F),$$

and Theorem 4.6.5 gives sufficient conditions for these morphisms to be isomorphisms.

There is also the following compatibility.

Lemma 4.7.5 If $\mathsf{K}(\mathcal{A})$ and $\mathsf{K}^+(\mathcal{A})$ each have an F-injective subcategory, then the restriction of $\mathrm{R}F$ to $\mathsf{D}^+(\mathcal{A})$ is ${}^+\mathrm{R}F$.

The corresponding statement also holds for F-projective subcategories, $\mathrm{L}F$, and ${}^-\mathrm{L}F$.

Proof We first consider $\mathrm{R}F$. Let $X \in \mathrm{Ob}(\mathsf{K}^+(\mathcal{A}))$. Write $\mathrm{qis}_{X/}$ for the category of all quasi-isomorphisms $X \to Y$ in $\mathsf{K}(\mathcal{A})$, and $\mathrm{qis}^+_{X/}$ for its version in $\mathsf{K}^+(\mathcal{A})$. Remark 4.6.6 gives a canonical isomorphism

$$\mathrm{R}F(QX) \simeq \varinjlim_{[X\to Y]\in\mathrm{Ob}(\mathrm{qis}_{X/})} (Q'\mathsf{K}F)(Y). \tag{4.7.2}$$

For every object $X \to Y$ of $\mathrm{qis}_{X/}$, if $m \ll 0$ (depending only on X), then $X \to Y \to \tau^{\geq m}Y$ is an object of $\mathrm{qis}^+_{X/}$. Lemma 1.9.7 ensures that $\mathrm{qis}_{X/}$ is filtered, so Proposition 1.6.8 ensures that the full subcategory $\mathrm{qis}^+_{X/}$ is cofinal in it. Thus the $\varinjlim$ in (4.7.2) may be taken over $\mathrm{qis}^+_{X/}$ (Proposition 1.6.5), and its value is precisely ${}^+\mathrm{R}F(QX)$.

For $\mathrm{L}F$, use the left-derived-functor version of Remark 4.6.6 and truncate in the opposite direction. $\square$

We next discuss the bifunctors introduced in Convention 3.5.9. The corresponding theory has also been prepared in §4.6. Let $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{B}$ be abelian categories, and let $F : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}$ be additive in each variable. As in Definition–Proposition 3.5.10, define

$$\begin{aligned} \mathsf{C}_{\oplus} F &:= \mathrm{tot}_{\oplus} \circ \mathsf{C}^2 F \quad (\text{if } \mathcal{B} \text{ has countable coproducts}), \\ \mathsf{C}_{\Pi} F &:= \mathrm{tot}_{\Pi} \circ \mathsf{C}^2 F \quad (\text{if } \mathcal{B} \text{ has countable products}). \end{aligned} \tag{4.7.3}$$

By Proposition 3.5.11, both factor through functors at the level of $\mathsf{K}(\cdot)$, denoted by $\mathsf{K}_{\oplus} F$ and $\mathsf{K}_{\Pi} F$.

Proposition 4.7.6 The functors $\mathsf{K}_{\Pi} F$, $\mathsf{K}_{\oplus} F : \mathsf{K}^{\star}(\mathcal{A}_1) \times \mathsf{K}^{\star}(\mathcal{A}_2) \to \mathsf{K}^{\star}(\mathcal{B})$ above are triangulated bifunctors in the sense of Definition 4.6.7.

Proof Take $\mathsf{K}_{\Pi} F$ as an example (assuming that $\mathcal{B}$ has countable products). The discussion following Proposition 3.5.11 shows that this functor is compatible with translation in each variable, while Proposition 3.5.8 gives the anticommutative diagram required in the definition of a triangulated bifunctor. It remains to prove that $\mathsf{C}_{\Pi} F$ preserves mapping cones in each variable. For the first variable, take a morphism $f : X_1 \to X_1'$ in $\mathsf{C}(\mathcal{A}_1)$ and an object X_2 of $\mathsf{C}(\mathcal{A}_2)$. There are canonical isomorphisms in $\mathcal{B}$

$$\begin{aligned} \mathsf{C}_{\Pi} F(\mathrm{Cone}(f), X_2)^n &\simeq \prod_{p+q=n} F(X_1^{p+1}, X_2^q) \times F((X_1')^p, X_2^q) \\ &\simeq \prod_{p+q=n+1} F(X_1^p, X_2^q) \times \prod_{p+q=n} F((X_1')^p, X_2^q) \\ &\simeq \mathrm{Cone}\left(\mathsf{C}_{\Pi}(f, \mathrm{id}_{X_2})\right)^n, \quad n \in \mathbb{Z}. \end{aligned}$$

It remains to verify that these isomorphisms form an isomorphism of complexes and are compatible with the morphisms $\alpha(\cdot)$ and $\beta(\cdot)$ accompanying the mapping cones. There is no essential difficulty, and the details are left to the reader. □

Next, let $Q_i : \mathsf{K}(\mathcal{A}_i) \to \mathsf{D}(\mathcal{A}_i)$ and $Q : \mathsf{K}(\mathcal{B}) \to \mathsf{D}(\mathcal{B})$ be the localization functors $(i = 1, 2)$. Place these data in the framework of Definition 4.6.8, taking $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}'$ to be, respectively, $\mathsf{K}^{\star}(\mathcal{A}_1)$, $\mathsf{K}^{\star}(\mathcal{A}_2)$, and $\mathsf{K}^{\star}(\mathcal{B})$, while the acyclic complexes form $\mathcal{N}_1^{\star}$, $\mathcal{N}_2^{\star}$, and $(\mathcal{N}')^{\star}$.

Definition 4.7.7 (Derived bifunctors) Fix $\star \in \{+, -, \ \}$ and consider an additive bifunctor $F : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}$.

- ◇ Suppose that $\mathcal{B}$ has countable products. If the right derived bifunctor of $\mathsf{K}_{\Pi} F$ in the sense of Definition 4.6.8,

$$\mathsf{D}^{\star}(\mathcal{A}_1) \times \mathsf{D}^{\star}(\mathcal{A}_2) \to \mathsf{D}^{\star}(\mathcal{B}),$$

exists, it is called the **right derived bifunctor** of F and is denoted by ${}^{\star}\mathrm{R}F$.

◇ Suppose that $\mathcal{B}$ has countable coproducts. If the left derived bifunctor of $\mathsf{K}_{\oplus}F$,

$$\mathsf{D}^{\star}(\mathcal{A}_1) \times \mathsf{D}^{\star}(\mathcal{A}_2) \to \mathsf{D}^{\star}(\mathcal{B}),$$

exists, it is called the **left derived bifunctor** of F and is denoted by ${}^{\star}\mathrm{L}F$.

We continue to write ${}^{\star}\mathrm{R}^nF := \mathrm{H}^n \circ {}^{\star}\mathrm{R}F$ and ${}^{\star}\mathrm{L}_nF := \mathrm{H}^{-n} \circ {}^{\star}\mathrm{L}F$.

Fix $\star \in \{+, -, \ \}$, and let $\mathcal{I}_i$ be a triangulated subcategory of $\mathsf{K}^{\star}(\mathcal{A}_i)$, $i = 1, 2$. As usual, if $(\mathcal{I}_1, \mathcal{I}_2)$ is a pair of $\mathsf{K}^{\star}F$-injective (respectively, projective) subcategories in the sense of Definition 4.6.9, we call the pair F-injective (respectively, F-projective). Proposition 4.6.10 immediately gives the following conclusions, and all the isomorphisms are canonical.

◇ If $(\mathcal{I}_1, \mathcal{I}_2)$ is F-injective, then ${}^{\star}\mathrm{R}F$ exists and

$$(X_1, X_2) \in \mathrm{Ob}(\mathcal{I}_1) \times \mathrm{Ob}(\mathcal{I}_2) \implies {}^{\star}\mathrm{R}F(Q_1X_1, Q_2X_2) \simeq Q\mathsf{K}_{\Pi}F(X_1, X_2). \quad (4.7.4)$$

◇ If $(\mathcal{P}_1, \mathcal{P}_2)$ is F-projective, then ${}^{*}\mathrm{L}F$ exists and

$$(X_1, X_2) \in \mathrm{Ob}(\mathcal{P}_1) \times \mathrm{Ob}(\mathcal{P}_2) \implies {}^{\star}\mathrm{L}F(Q_1X_1, Q_2X_2) \simeq Q\mathsf{K}_{\oplus}F(X_1, X_2). \quad (4.7.5)$$

We next discuss a product on derived bifunctors. We begin with its version at the level of complexes. Keep the notation above, but now assume that F is right exact in each variable. For every complex X, write $Z^n(X) := \ker(d_X^n)$ and $B^n(X) := \mathrm{im}(d_X^{n-1})$. For every $(p, q) \in \mathbb{Z}^2$, there is a natural morphism

$$F\left(Z^p(X_1), Z^q(X_2)\right) \to Z^{p+q}\left(\mathsf{C}_{\oplus}F(X_1, X_2)\right).$$

This morphism sends the images of $F(B^p(X_1), Z^q(X_2))$ and $F(Z^p(X_1), B^q(X_2))$ into $B^{p+q}(\mathsf{C}_{\oplus}F(X_1, X_2))$. Taking quotients and using the right exactness of F gives a canonical morphism

$$\kappa : F\left(\mathrm{H}^p(X_1), \mathrm{H}^q(X_2)\right) \to \mathrm{H}^{p+q}\left(\mathsf{C}_{\oplus}F(X_1, X_2)\right).$$

The discussion preceding the Künneth theorem for homology, Theorem 3.14.12, is the special case of this construction with $F = \otimes_R$.

Now change the hypothesis: let F be left exact in each variable. Dualizing the construction above gives a canonical morphism

$$\lambda : \mathrm{H}^{p+q}\left(\mathsf{C}_{\Pi}F(X_1, X_2)\right) \to F\left(\mathrm{H}^p(X_1), \mathrm{H}^q(X_2)\right).$$

The existence of the countable $\oplus$ or $\prod$ under discussion is still assumed. Since only cohomology matters here, C may also be replaced by K.

Proposition 4.7.8 Suppose that F is right exact (respectively, left exact) in each variable, that $\star \in \{+, -, \quad\}$, and that there is an F-projective pair $(\mathcal{P}_1, \mathcal{P}_2)$ (respectively, an F-injective pair $(\mathcal{I}_1, \mathcal{I}_2)$). Then there is a canonical morphism

$$F\left(\mathrm{H}^p(\cdot), \mathrm{H}^q(\cdot)\right) \to \mathrm{H}^{p+q}\, {}^\star\mathrm{L}F \quad \text{or} \quad \mathrm{H}^{p+q}\, {}^\star\mathrm{R}F \to F\left(\mathrm{H}^p(\cdot), \mathrm{H}^q(\cdot)\right),$$

where both sides are functors from $\mathsf{D}^\star(\mathcal{A}_1) \times \mathsf{D}^\star(\mathcal{A}_2)$ to $\mathcal{B}$. These morphisms are characterized by the commutative diagram

$$\begin{tikzcd}
 & \mathrm{H}^{p+q}\, \mathsf{K}_\oplus F(X_1, X_2) \\
F\left(\mathrm{H}^p(X_1), \mathrm{H}^q(X_2)\right) \arrow[r] \arrow[ur, "\kappa"] & \mathrm{H}^{p+q}\, {}^\star\mathrm{L}F(Q_1X_1, Q_2X_2) \arrow[u, "\mathrm{can}"']
\end{tikzcd}$$

or by

$$\begin{tikzcd}
 & \mathrm{H}^{p+q}\, \mathsf{K}_\Pi F(X_1, X_2) \arrow[dl, "\lambda"'] \arrow[d, "\mathrm{can}"] \\
F\left(\mathrm{H}^p(X_1), \mathrm{H}^q(X_2)\right) & \mathrm{H}^{p+q}\, {}^\star\mathrm{R}F(Q_1X_1, Q_2X_2), \arrow[l]
\end{tikzcd}$$

where $X_i \in \mathrm{Ob}(\mathsf{K}^\star(\mathcal{A}_i))$. The arrows labeled can are the canonical morphisms accompanying the derived functors as Kan extensions (after applying H^{p+q}); see (4.7.1) for the one-variable case.

Proof By duality, it suffices to discuss the case involving κ. Observe that, apart from $\mathsf{K}_\oplus F(X_1, X_2)$, the other two vertices in the diagram depend only on Q_1X_1 and Q_2X_2.

Fix the pair $(\mathcal{P}_1, \mathcal{P}_2)$. Since all morphisms in the diagram are functorial and $\mathcal{P}_i/(\mathcal{P}_i \cap \mathcal{N}_i^\star)$ is equivalent to $\mathsf{D}^\star(\mathcal{A}_i)$, it suffices to determine the desired morphism for $X_i \in \mathrm{Ob}(\mathcal{P}_i)$; this can be checked by drawing a diagram. In this case, however, ${}^\star\mathrm{L}F(Q_1X_1, Q_2X_2) = \mathsf{K}_\oplus F(X_1, X_2)$, so the claim is immediate. □

In short, everything can be computed by taking resolutions, and the final result is independent of the choice of resolutions.

4.8 Bounded Derived Functors

This section explains how to study ${}^+\mathrm{R}F$ and ${}^-\mathrm{L}F$ by means of injective and projective resolutions. The case of unbounded derived categories is deferred to §4.11.

Convention 4.8.1 This section focuses on $\star \in \{+, -\}$ and adopts the following conventions.

- ◇ For right derived functors we always take $\star = +$ and use the abbreviation $\mathrm{R}F := {}^{+}\mathrm{R}F : \mathsf{D}^{+}(\mathcal{A}) \to \mathsf{D}^{+}(\mathcal{A}')$.
- ◇ For left derived functors we always take $\star = -$ and use the abbreviation $\mathrm{L}F := {}^{-}\mathrm{L}F : \mathsf{D}^{-}(\mathcal{A}) \to \mathsf{D}^{-}(\mathcal{A}')$.

In view of Lemma 4.7.5, these conventions cause no confusion.

Definition–Proposition 4.8.2 Let $\mathcal{A}^\flat$ be a full additive subcategory of $\mathcal{A}$. We call $\mathcal{A}^\flat$ **of type I** relative to F if the following conditions hold; in this case $\mathsf{K}^{+}(\mathcal{A}^\flat)$ is an F-injective subcategory of $\mathsf{K}^{+}(\mathcal{A})$:

(I1) for every $X \in \mathrm{Ob}(\mathcal{A})$, there is a monomorphism $X \hookrightarrow I$ with $I \in \mathrm{Ob}(\mathcal{A}^\flat)$;

(I2) if $0 \to X' \to X \to X'' \to 0$ is a short exact sequence in $\mathcal{A}$ and $X', X \in \mathrm{Ob}(\mathcal{A}^\flat)$, then $X'' \in \mathrm{Ob}(\mathcal{A}^\flat)$ and $0 \to FX' \to FX \to FX'' \to 0$ is exact.

Dually, we call $\mathcal{A}^\flat$ **of type P** relative to F if the following conditions hold; in this case $\mathsf{K}^{-}(\mathcal{A}^\flat)$ is an F-projective subcategory of $\mathsf{K}^{-}(\mathcal{A})$:

(P1) for every $X \in \mathrm{Ob}(\mathcal{A})$, there is an epimorphism $P \twoheadrightarrow X$ with $P \in \mathrm{Ob}(\mathcal{A}^\flat)$;

(P2) if $0 \to X' \to X \to X'' \to 0$ is a short exact sequence in $\mathcal{A}$ and $X, X'' \in \mathrm{Ob}(\mathcal{A}^\flat)$, then $X' \in \mathrm{Ob}(\mathcal{A}^\flat)$ and $0 \to FX' \to FX \to FX'' \to 0$ is exact.

Proof It suffices to treat the type-I version. First, $\mathsf{K}^{+}(\mathcal{A}^\flat)$ is a triangulated subcategory of $\mathsf{K}^{+}(\mathcal{A})$ (Corollary 4.4.4). Theorem 3.11.3 shows that for every $X \in \mathrm{Ob}(\mathsf{K}^{+}(\mathcal{A}))$ there is a quasi-isomorphism $X \to I$ with $I \in \mathsf{K}^{+}(\mathcal{A}^\flat)$. This is the resolution condition for an F-injective subcategory. It remains to prove that if $X \in \mathsf{K}^{+}(\mathcal{A}^\flat)$ is acyclic, then FX is also acyclic.

For every $n \in \mathbb{Z}$, there is a short exact sequence $0 \to \ker(d^n) \to X^n \xrightarrow{d^n} \ker(d^{n+1}) \to 0$. If $n \ll 0$, then $\ker(d^n) = X^n = 0$. Applying (I2) recursively for all n gives $\ker(d^n) \in \mathrm{Ob}(\mathcal{A}^\flat)$ together with short exact sequences in $\mathcal{A}'$

$$0 \to F\ker(d^n) \to FX^n \xrightarrow{Fd^n} F\ker(d^{n+1}) \to 0.$$

Splicing these short exact sequences shows that FX is acyclic. □

The terms type I and type P are merely convenient labels. There is a complete characterization of the subcategories $\mathcal{A}^\flat$ that give F-injective or F-projective subcategories; see [23, Proposition 13.3.5].

The next result shows that derived functors can be computed by taking resolutions in a full subcategory of type I or P.

Theorem 4.8.3 Suppose that $\mathcal{A}$ has a full additive subcategory $\mathcal{A}^\flat$ of type I (respectively, type P) relative to the additive functor $F : \mathcal{A} \to \mathcal{A}'$. Then the derived functor $\mathrm{R}F$ (respectively, $\mathrm{L}F$) exists. More precisely:

(i) for every $X \in \mathrm{Ob}(\mathsf{K}^+(\mathcal{A}))$ (respectively, $X \in \mathrm{Ob}(\mathsf{K}^-(\mathcal{A}))$), there is a quasi-isomorphism $X \to I$ (respectively, $P \to X$) with $I \in \mathrm{Ob}(\mathsf{K}^+(\mathcal{A}^\flat))$ (respectively, $P \in \mathrm{Ob}(\mathsf{K}^-(\mathcal{A}^\flat))$);

(ii) for the quasi-isomorphism in (i), the morphism in (4.6.2) gives

$$\mathrm{R}F(QX) \xrightarrow{\sim} \mathrm{R}F(QI) \xleftarrow{\sim} Q'\mathsf{K}^+F(I), \quad \mathrm{L}F(QX) \xleftarrow{\sim} \mathrm{L}F(QP) \xrightarrow{\sim} Q'\mathsf{K}^-F(P);$$

(iii) $\mathrm{R}F$ restricts to $\mathsf{D}^{\geq 0}(\mathcal{A}) \to \mathsf{D}^{\geq 0}(\mathcal{A}')$ (respectively, $\mathrm{L}F$ restricts to $\mathsf{D}^{\leq 0}(\mathcal{A}) \to \mathsf{D}^{\leq 0}(\mathcal{A}')$);

(iv) suppose that F is left exact (respectively, right exact). If $X \in \mathrm{Ob}(\mathcal{A})$, the morphism in (4.6.2) gives an isomorphism $FX \xrightarrow{\sim} \mathrm{R}^0F(QX)$ (respectively, $FX \xleftarrow{\sim} \mathrm{L}_0F(QX)$);

(v) if $X \in \mathrm{Ob}(\mathcal{A}^\flat)$, then, for $n > 0$, $\mathrm{R}^nF(X) = 0$ (respectively, $\mathrm{L}_nF(X) = 0$).

Proof It suffices to consider $\mathrm{R}F$. Since $\mathsf{K}^+(\mathcal{A}^\flat)$ is F-injective, a quasi-isomorphism $X \to I$ always exists. The existence of $\mathrm{R}F$ and the isomorphism $\mathrm{R}F(QX) \simeq Q'\mathsf{K}^+F(I)$ merely restate Proposition 4.6.4. This proves (i) and (ii).

For (iii), if $X \in \mathrm{Ob}(\mathsf{D}^{\geq 0}(\mathcal{A}))$, we may assume that X comes from an object of $\mathsf{C}^{\geq 0}(\mathcal{A})$. Theorem 3.11.3 allows us to choose a quasi-isomorphism $X \to I$ with $I \in \mathrm{Ob}(\mathsf{C}^{\geq 0}(\mathcal{A}^\flat))$. Hence

$$\mathrm{R}F(QX) \simeq Q'\mathsf{K}^+F(I) \in \mathrm{Ob}\left(\mathsf{D}^{\geq 0}(\mathcal{A}')\right).$$

If, in addition, $X \in \mathrm{Ob}(\mathcal{A}^\flat)$, the quasi-isomorphism $X \to I$ may be taken to be id_X. In that case $\mathrm{R}F(QX) = Q'\mathsf{K}^+F(X) = Q'FX$ is concentrated in degree zero. This also proves (v).

Finally, consider (iv). Suppose that F is left exact. For $X \in \mathrm{Ob}(\mathcal{A})$, choose an exact sequence $0 \to X \to I^0 \to I^1 \to \cdots$, with every $I^n \in \mathrm{Ob}(\mathcal{A}^\flat)$. Put $I = [I^0 \to I^1 \to \cdots]$. By (ii) and left exactness, $FX \xrightarrow{\sim} \ker[Fd_I^0 : F(I^0) \to F(I^1)] \simeq \mathrm{R}^0F(X)$, as desired. $\square$

Example 4.8.4 (Injective and projective objects) Suppose that $\mathcal{A}$ has enough injective objects, and let $\mathcal{A}^\flat$ be the full additive subcategory of injective objects. It is of type I relative to every F: condition (I1) holds by assumption, while for (I2), Lemma 2.8.13

shows that the short exact sequence in that condition splits. Thus $X' \oplus X''$ is injective, and Lemma 2.8.14 implies that X'' is injective as well.

Dually, if $\mathcal{A}$ has enough projective objects, the full additive subcategory consisting of those objects is of type P relative to every F.

Example 4.8.4 connects this discussion with the classical definition of derived functors in §3.12. Suppose that $\mathcal{A}$ has enough injective objects. Consider the restriction $\mathrm{R}^n F|_{\mathcal{A}}$ of the right derived functor. Since a short exact sequence in $\mathcal{A}$ extends canonically to a distinguished triangle (Proposition 4.4.7), the long exact sequence immediately makes $(\mathrm{R}^n F|_{\mathcal{A}})_{n\geq 0}$ a cohomological δ-functor in the sense of Definition 3.12.5. Dually, the left derived functors form the homological δ-functor $(\mathrm{L}_n F|_{\mathcal{A}})_{n\geq 0}$.

Proposition 4.8.5 Suppose that F is left exact (respectively, right exact) and that $\mathcal{A}$ has enough injective (respectively, projective) objects. Then $\mathrm{R}^n F : \mathsf{C}^+(\mathcal{A}) \to \mathcal{A}'$ (respectively, $\mathrm{L}_n F : \mathsf{C}^-(\mathcal{A}) \to \mathcal{A}'$) agrees with the version previously defined in Definition 3.12.1.

More generally, if there is a subcategory $\mathcal{A}^\flat$ of type I (respectively, type P), as in Definition–Proposition 4.8.2, then $(\mathrm{R}^n F|_{\mathcal{A}})_{n\geq 0}$ (respectively, $(\mathrm{L}_n F|_{\mathcal{A}})_{n\geq 0}$) is a universal δ-functor in the sense of Definition 3.12.12.

Proof It suffices to treat the left-exact case. For $X \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$, choose an injective resolution $X \to I$. Theorem 4.8.3 (ii) gives canonical isomorphisms $\mathrm{R}^n F(QX) \simeq \mathrm{H}^n \,\mathsf{K}^+ F(I)$ for every $n \in \mathbb{Z}$. The right-hand side is precisely the version of the derived functor from Definition 3.12.1.

More generally, suppose that $\mathcal{A}^\flat$ is of type I relative to F. To prove the universality of $(\mathrm{R}^n F|_{\mathcal{A}})_{n\geq 0}$, by Proposition 3.12.14 it suffices to show that $\mathrm{R}^n F|_{\mathcal{A}}$ is effaceable for $n > 0$. This follows directly from condition (I1) in Definition–Proposition 4.8.2 and Theorem 4.8.3 (v). □

To derive a composite of functors, one often needs a larger class of objects for resolutions. We need the following concept.

Definition 4.8.6 Suppose that $\mathrm{R}F$ (respectively, $\mathrm{L}F$) exists. An object $X \in \mathrm{Ob}(\mathcal{A})$ is called ***F*-acyclic** if $\mathrm{R}F(X)$ (respectively, $\mathrm{L}F(X)$) is an object of $\mathsf{D}^{\geq 0}(\mathcal{A}') \cap \mathsf{D}^{\leq 0}(\mathcal{A}') \simeq \mathcal{A}'$.

The next result shows that derived functors can be computed using F-acyclic resolutions; compare Corollary 3.12.10.

Corollary 4.8.7 Let $\mathcal{A}^\flat$ be a full additive subcategory of $\mathcal{A}$ of type I (respectively, type P) relative to F. Define $\mathcal{A}^\natural$ to be the full additive subcategory consisting of all

F-acyclic objects of $\mathcal{A}$. Then:

(i) $\mathcal{A}^\natural \supset \mathcal{A}^\flat$;

(ii) $\mathcal{A}^\natural$ is also of type I (respectively, type P);

(iii) for every quasi-isomorphism $X \to I$ (respectively, $P \to X$), with $I \in \mathrm{Ob}(\mathsf{K}^+(\mathcal{A}^\natural))$ (respectively, $P \in \mathrm{Ob}(\mathsf{K}^-(\mathcal{A}^\natural))$), the morphism in (4.6.2) gives

$$\mathrm{R}F(QX) \xrightarrow{\sim} \mathrm{R}F(QI) \xleftarrow{\sim} Q'\mathsf{K}^+F(I), \quad \mathrm{L}F(QX) \xleftarrow{\sim} \mathrm{L}F(QP) \xrightarrow{\sim} Q'\mathsf{K}^-F(P).$$

Proof Theorem 4.8.3 (v) implies (i), so condition (I1) or (P1) also holds for $\mathcal{A}^\natural$. For (I2), suppose that $0 \to X' \to X \to X'' \to 0$ is a short exact sequence in $\mathcal{A}$ and that X', X are F-acyclic. The long exact sequence from Theorem 4.7.3 gives exact sequences

$$\underbrace{\mathrm{R}^nF(X)}_{=0} \to \mathrm{R}^nF(X'') \to \underbrace{\mathrm{R}^{n+1}F(X')}_{=0}, \quad n \in \mathbb{Z}_{\geq 1},$$

so X'' is F-acyclic. On the other hand, the $n = 0$ portion of the long exact sequence, together with $\mathrm{R}^1F(X') = 0$, shows that $0 \to FX' \to FX \to FX'' \to 0$ is exact. The proof of (P2) is dual. This proves (ii).

Finally, (iii) follows by applying Theorem 4.8.3 (ii) to $\mathcal{A}^\natural$. $\square$

We now discuss deriving a composite of functors. The basic tool is Theorem 4.6.5.

Theorem 4.8.8 (Deriving a composite of functors) Consider additive functors between abelian categories

$$\mathcal{A} \xrightarrow{F} \mathcal{A}' \xrightarrow{F'} \mathcal{A}''.$$

Suppose that there are full subcategories $\mathcal{A}^\flat$ of $\mathcal{A}$ and $(\mathcal{A}')^\flat$ of $\mathcal{A}'$, of type I relative to F and F', respectively, in the sense of Definition–Proposition 4.8.2. If F sends the objects of $\mathcal{A}^\flat$ to F'-acyclic objects of $\mathcal{A}'$, then the canonical morphism $\mathrm{R}(F'F) \to (\mathrm{R}F')(\mathrm{R}F)$ is an isomorphism.

There is a corresponding version for the left derived functors $\mathrm{L}F$, $\mathrm{L}F'$, and the canonical morphism $(\mathrm{L}F')(\mathrm{L}F) \to \mathrm{L}(F'F)$, using subcategories of type P.

Proof It suffices to treat right derived functors. Let $(\mathcal{A}')^\natural$ be the full additive subcategory of F'-acyclic objects in $\mathcal{A}'$. Corollary 4.8.7, together with Definition–Proposition 4.8.2, shows that $\mathsf{K}^+(\mathcal{A}^\flat)$ and $\mathsf{K}^+((\mathcal{A}')^\natural)$ are, respectively, F-injective and F'-injective subcategories of $\mathsf{K}^+(\mathcal{A})$ and $\mathsf{K}^+(\mathcal{A}')$. Apply Theorem 4.6.5. $\square$

Using the amplitude from Definition 4.5.5, we can define the dimensions of the right and left derived functors of F.

Definition–Proposition 4.8.9 Suppose that $F : \mathcal{A} \to \mathcal{A}'$ is left exact (respectively, right exact), and that the derived functor $\mathrm{R}F$ (respectively, $\mathrm{L}F$) exists and is determined as in Theorem 4.8.3. There is a $d \in \mathbb{Z}_{\geq 0} \sqcup \{+\infty\}$ such that the amplitude of the derived functor is contained in $[0, d]$ (respectively, $[-d, 0]$). The infimum of all such d is called the **dimension** of $\mathrm{R}F$ (respectively, $\mathrm{L}F$).

Proof It suffices to treat the left-exact case. Theorem 4.8.3 implies that $\mathrm{R}F$ restricts to $\mathcal{A} \to \mathsf{D}^{\geq 0}(\mathcal{A}')$, so its amplitude is contained in an interval $[0, d]$. $\square$

Consequently, if $\mathrm{R}F$ (respectively, $\mathrm{L}F$) exists and has finite dimension, it restricts to $\mathsf{D}^{\mathrm{b}}(\mathcal{A}) \to \mathsf{D}^{\mathrm{b}}(\mathcal{A}')$.

Example 4.8.10 (Tor-dimension) Let R be a ring and X a right R-module. Consider the right-exact functor from R-Mod to Ab that sends $Y \mapsto X \otimes Y := X \underset{R}{\otimes} Y$. Since R-Mod has enough projective objects, $\mathrm{L}(X \otimes \cdot)$ exists, and $\mathrm{L}_n(X \otimes \cdot)$ is precisely $\mathrm{Tor}_n^R(X, \cdot)$ from Definition–Proposition 3.14.7. The dimension of $\mathrm{L}(X \otimes \cdot)$ is called the **Tor-dimension** of X. A similar definition applies to left R-modules, and Proposition 3.14.9 implies that the two agree when R is commutative. The derived tensor-product functor is discussed further in §4.12.

Now fix abelian categories $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{B}$, and an additive bifunctor

$$F : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}.$$

We do not require $\mathcal{B}$ to have countable coproducts or products, because C^2F restricts to

$$\mathsf{C}^2F : \mathsf{C}^{\pm}(\mathcal{A}_1) \times \mathsf{C}^{\pm}(\mathcal{A}_2) \to \mathsf{C}^2_f(\mathcal{B}).$$

On $\mathsf{C}^2_f(\mathcal{B})$, the total-complex functor $\mathrm{tot}_{\oplus} = \mathrm{tot}_{\Pi}$ is always defined; denote this functor by tot. Likewise, denote $\mathsf{C}_{\oplus}F = \mathsf{C}_{\Pi}F$ by $\mathsf{C}F$, and $\mathsf{K}_{\oplus}F = \mathsf{K}_{\Pi}F$ by $\mathsf{K}F$.

To study the derived bifunctors of Definition 4.7.7, we consider triangulated subcategories $\mathcal{I}_i$ (respectively, $\mathcal{P}_i$) of $\mathsf{K}^+(\mathcal{A}_i)$ (respectively, $\mathsf{K}^-(\mathcal{A}_i)$). As usual, the sign is determined by whether we are considering the right or left derived bifunctor, and $i = 1, 2$. How should we choose the pair $(\mathcal{I}_1, \mathcal{I}_2)$ or $(\mathcal{P}_1, \mathcal{P}_2)$? Definition–Proposition 4.8.2 and the techniques of §3.10 give the following answer.

Lemma 4.8.11 Let $\mathcal{A}_i^\flat \subset \mathcal{A}_i$ be full additive subcategories, $i = 1, 2$. If the following conditions hold simultaneously, then the pair $\left(\mathsf{K}^+(\mathcal{A}_1^\flat), \mathsf{K}^+(\mathcal{A}_2^\flat)\right)$ (respectively, $\left(\mathsf{K}^-(\mathcal{A}_1^\flat), \mathsf{K}^-(\mathcal{A}_2^\flat)\right)$) is F-injective (respectively, F-projective) in the sense of Definition 4.6.9:

- ⋄ for fixed $X_1 \in \mathrm{Ob}(\mathcal{A}_1^\flat)$, $\mathcal{A}_2^\flat$ is of type I (respectively, type P) relative to $F(X_1, \cdot)$;

◇ for fixed $X_2 \in \mathrm{Ob}(\mathcal{A}_2^\flat)$, $\mathcal{A}_1^\flat$ is of type I (respectively, type P) relative to $F(\cdot, X_2)$.

In particular, if $\mathcal{A}_1$ and $\mathcal{A}_2$ each have enough injective (respectively, projective) objects, then $\mathrm{R}F$ (respectively, $\mathrm{L}F$) exists.

Proof Set $\mathcal{I}_i := \mathsf{K}^+(\mathcal{A}_i^\flat)$, $i = 1, 2$. The only point to prove is that, for every $X_1 \in \mathrm{Ob}(\mathsf{K}^+(\mathcal{A}_1^\flat))$, the functor $(\mathsf{K}F)(X_1, \cdot) : \mathsf{K}^+(\mathcal{A}_2^\flat) \to \mathsf{K}^+(\mathcal{B})$ sends acyclic complexes to acyclic complexes; the same statement holds after interchanging the indices 1 and 2. The existence of $\mathrm{R}F$ (respectively, $\mathrm{L}F$) then follows directly from Proposition 4.6.10.

Take $Y \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}_2^\flat))$. Then $(\mathsf{K}^2 F)(X_1, Y)$ belongs to $\mathrm{Ob}(\mathsf{C}_f^2(\mathcal{B}))$. By condition (I2), each column $F(X_1^p, Y^\bullet)$ is acyclic. Therefore Corollary 3.10.7 implies that $\mathsf{K}F(X_1, Y) := \mathrm{tot}(\mathsf{K}^2 F(X_1, Y))$ is exact, as desired. □

If F-injective (respectively, F-projective) subcategories are available, the functor $\mathrm{R}F$ (respectively, $\mathrm{L}F$) can be determined by (4.7.4) (respectively, (4.7.5)).

4.9 Example: RHom

Throughout this section, $\mathcal{A}$ remains an abelian category. If $\mathcal{A}$ is moreover $\Bbbk$-linear, where $\Bbbk$ is a commutative ring, every statement in this section can be strengthened by replacing Ab with $\Bbbk\text{-}\mathsf{Mod}$.

The bifunctor $\mathrm{Hom} := \mathrm{Hom}_{\mathcal{A}} : \mathcal{A}^{\mathrm{op}} \times \mathcal{A} \to \mathsf{Ab}$ is left exact in each variable. To study its derived bifunctor in the sense of Definition 4.7.7, we need some preparation. First, the general theory of §4.8 gives

$$\mathsf{C}\,\mathrm{Hom} : \mathsf{C}^+(\mathcal{A}^{\mathrm{op}}) \times \mathsf{C}^+(\mathcal{A}) \to \mathsf{C}^+(\mathsf{Ab}),$$
$$\mathsf{K}\,\mathrm{Hom} : \mathsf{K}^+(\mathcal{A}^{\mathrm{op}}) \times \mathsf{K}^+(\mathcal{A}) \to \mathsf{K}^+(\mathsf{Ab}).$$

Denote all localization functors by Q. Recall the isomorphism $\sigma : \mathsf{C}^\pm(\mathcal{A}^{\mathrm{op}}) \xrightarrow{\sim} \mathsf{C}^\mp(\mathcal{A})^{\mathrm{op}}$ of Definition–Proposition 3.4.1. The isomorphisms it induces at the level of the triangulated categories $\mathsf{K}^\pm$ and $\mathsf{D}^\pm$ are also denoted by σ (Propositions 3.4.3 and 4.4.16).

Definition 4.9.1 If the right derived bifunctor of Hom in the sense of Definition 4.7.7 exists, denote it by $\mathrm{RHom} = \mathrm{RHom}_{\mathcal{A}}$. It is a functor from $\mathsf{D}^+(\mathcal{A}^{\mathrm{op}}) \times \mathsf{D}^+(\mathcal{A})$ to $\mathsf{D}^+(\mathsf{Ab})$ and, by means of $\sigma : \mathsf{D}^\pm(\mathcal{A}^{\mathrm{op}}) \xrightarrow{\sim} \mathsf{D}^\mp(\mathcal{A})^{\mathrm{op}}$, can also be regarded as a functor

$$\mathrm{RHom} : \mathsf{D}^-(\mathcal{A})^{\mathrm{op}} \times \mathsf{D}^+(\mathcal{A}) \to \mathsf{D}^+(\mathsf{Ab}).$$

Lemma 4.9.2 Write $\mathcal{I}_{\mathcal{A}}$ (respectively, $\mathcal{P}_{\mathcal{A}}$) for the full subcategory of injective (respectively, projective) objects of $\mathcal{A}$.

- If $\mathcal{A}$ has enough injective objects, then $(\mathsf{K}^+(\mathcal{A}^{\mathrm{op}}), \mathsf{K}^+(\mathcal{I}_{\mathcal{A}}))$ is Hom-injective.
- If $\mathcal{A}$ has enough projective objects, then $(\mathsf{K}^+(\mathcal{P}_{\mathcal{A}}^{\mathrm{op}}), \mathsf{K}^+(\mathcal{A}))$ is Hom-injective.

Proof It suffices to treat the case of enough injectives; the projective case is entirely similar. We verify the conditions of Lemma 4.8.11.

First, fix $X_1 \in \mathrm{Ob}(\mathcal{A})$. Recall that $\mathcal{I}_{\mathcal{A}}$ is of type I for every additive functor $F : \mathcal{A} \to \mathsf{Ab}$ (Example 4.8.4); take the special case $F = \mathrm{Hom}(X_1, \cdot)$.

Second, fix $X_2 \in \mathrm{Ob}(\mathcal{I}_{\mathcal{A}})$. To verify that $\mathcal{A}^{\mathrm{op}}$ is of type I relative to $\mathrm{Hom}(\cdot, X_2)$, observe that the resolution condition (I1) holds trivially. For (I2), it suffices to show that $\mathrm{Hom}(\cdot, X_2) : \mathcal{A}^{\mathrm{op}} \to \mathsf{Ab}$ is exact; this is precisely the characterization of X_2 as an injective object. $\square$

The following result involves the Hom complex of Definition 3.2.1, denoted by $\mathrm{Hom}^\bullet$.

Theorem 4.9.3 Suppose that $\mathcal{A}$ has enough injective objects or enough projective objects.

(i) The right derived bifunctor RHom exists.

(ii) If $\mathcal{A}$ has enough injective objects and $X_2 \to I_2$ is an injective resolution, then it induces an isomorphism

$$\mathrm{RHom}(QX_1, QX_2) \xrightarrow{\sim} Q\,\mathrm{Hom}^\bullet(X_1, I_2).$$

In particular, if $X_1 \in \mathrm{Ob}(\mathcal{A})$, then $\mathrm{RHom}(QX_1, \cdot) \simeq \mathrm{R}(\mathrm{Hom}(X_1, \cdot)) : \mathsf{D}^+(\mathcal{A}) \to \mathsf{D}^+(\mathsf{Ab})$.

(iii) If $\mathcal{A}$ has enough projective objects and $P_1 \to X_1$ is a projective resolution, then it induces an isomorphism

$$\mathrm{RHom}(QX_1, QX_2) \xrightarrow{\sim} Q\,\mathrm{Hom}^\bullet(P_1, X_2).$$

In particular, if $X_2 \in \mathrm{Ob}(\mathcal{A})$, then $\mathrm{RHom}(\cdot, QX_2) \simeq \mathrm{R}(\mathrm{Hom}(\cdot, X_2)) : \mathsf{D}^-(\mathcal{A})^{\mathrm{op}} \to \mathsf{D}^+(\mathsf{Ab})$.

(iv) There is a canonical isomorphism $\mathrm{H}^n\,\mathrm{RHom}(X, Y) \simeq \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y[n])$, where $X \in \mathrm{Ob}(\mathsf{D}^-(\mathcal{A}))$ and $Y \in \mathrm{Ob}(\mathsf{D}^+(\mathcal{A}))$. If $X, Y \in \mathrm{Ob}(\mathcal{A})$, the right-hand side is precisely $\mathrm{Ext}^n(X, Y)$ from Definition 4.5.8.

Proof By Lemma 4.9.2 and the following observations, statements (i)–(iii) all follow by applying Lemma 4.8.11:

- at the level of complexes, Example 3.5.12 gives a canonical isomorphism $\mathsf{C}\,\mathrm{Hom}(\sigma^{-1}(X_1), X_2) \simeq \mathrm{Hom}^\bullet(X_1, X_2)$;

- ◇ $P_1 \to X_1$ is a projective resolution in $\mathsf{K}^-(\mathcal{A})$ if and only if the corresponding morphism $\sigma^{-1}(X_1) \to \sigma^{-1}(P_1)$ is an injective resolution in $\mathsf{K}^+(\mathcal{A}^{\mathrm{op}})$.

For (iv), first suppose that $\mathcal{A}$ has enough injective objects. We may assume that $Y \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$ consists of injective objects. Identifying the objects of $\mathsf{K}(\mathcal{A})$ and $\mathsf{D}(\mathcal{A})$, Proposition 4.5.1 gives

$$\mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y[n]) \simeq \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X, Y[n]) \simeq \mathrm{H}^n \,\mathrm{Hom}^\bullet(X, Y) \xrightarrow[\sim]{\text{(ii)}} \mathrm{H}^n \,\mathrm{RHom}(X, Y).$$

If $\mathcal{A}$ has enough projective objects, the argument is entirely similar. □

Corollary 4.9.4 Let $X, Y \in \mathrm{Ob}(\mathcal{A})$. If $\mathcal{A}$ has enough injective (respectively, projective) objects, then $\mathrm{Ext}^n(X, Y)$ from Definition 4.5.8 is canonically isomorphic to $\mathrm{Ext}^n_{\mathcal{A},\mathrm{II}}(X, Y)$ (respectively, $\mathrm{Ext}^n_{\mathcal{A},\mathrm{I}}(X, Y)$) as defined in §3.14.

Proof Apply Theorem 4.9.3 (ii) and (iii) directly. □

Definition 4.9.5 Let $X \in \mathrm{Ob}(\mathcal{A})$. The following dimensions are dimensions of right derived functors in the sense of Definition–Proposition 4.8.9.

- ◇ If $\mathcal{A}$ has enough injective objects, the dimension of $\mathrm{R}(\mathrm{Hom}(\cdot, X)) : \mathsf{D}^-(\mathcal{A})^{\mathrm{op}} \to \mathsf{D}^+(\mathsf{Ab})$ is called the **injective dimension** of X and is denoted by $\mathrm{inj.dim}(X)$.

- ◇ If $\mathcal{A}$ has enough projective objects, the dimension of $\mathrm{R}(\mathrm{Hom}(X, \cdot)) : \mathsf{D}^+(\mathcal{A}) \to \mathsf{D}^+(\mathsf{Ab})$ is called the **projective dimension** of X and is denoted by $\mathrm{proj.dim}(X)$.

For $X \in \mathrm{Ob}(\mathcal{A})$ and $n \in \mathbb{Z}_{\geq 0} \sqcup \{\infty\}$, an injective resolution $0 \to X \to I^0 \to I^1 \to \cdots$ is said to have length $\leq n$ if $k > n \Rightarrow I^k = 0$; projective resolutions are treated in the same way. The notation $\mathrm{Ext}^{\geq m}(Y, X) = 0$ means that $k \geq m \Rightarrow \mathrm{Ext}^k(Y, X) = 0$, and similarly in the other cases.

Proposition 4.9.6 Let $n \in \mathbb{Z}_{\geq 0}$. If $\mathcal{A}$ has enough injective objects, then

$$\begin{aligned} \mathrm{inj.dim}(X) \leq n &\iff \mathrm{Ext}^{\geq n+1}(\cdot, X) = 0 \\ &\iff \mathrm{Ext}^{n+1}(\cdot, X) = 0 \iff X \text{ has an injective resolution of length} \leq n. \end{aligned}$$

If $\mathcal{A}$ has enough projective objects, then

$$\begin{aligned} \mathrm{proj.dim}(X) \leq n &\iff \mathrm{Ext}^{\geq n+1}(X, \cdot) = 0 \\ &\iff \mathrm{Ext}^{n+1}(X, \cdot) = 0 \iff X \text{ has a projective resolution of length} \leq n. \end{aligned}$$

Proof It suffices to discuss injective resolutions. Suppose that $\mathrm{inj.dim}(X) \leq n$. The definition of dimension implies that, for every $Y \in \mathrm{Ob}(\mathcal{A})$, $\big(\mathrm{R}\,\mathrm{Hom}(\cdot, X)\big)(Y)$ lies in $\mathrm{Ob}(\mathsf{D}^{[0,n]}(\mathsf{Ab}))$, and hence $\mathrm{Ext}^{\geq n+1}(Y, X) = 0$.

Now suppose that $\mathrm{Ext}^{n+1}(\cdot, X) = 0$. Take an injective resolution $0 \to X \to I^0 \xrightarrow{d^0} \cdots$. Repeatedly applying dimension shifting from Proposition 3.12.9 to the short exact sequences

$$0 \to X \to I^0 \to \mathrm{im}\left(d_I^0\right) \to 0, \quad 0 \to \mathrm{im}\left(d_I^{k-2}\right) \to I^{k-1} \to \mathrm{im}\left(d_I^{k-1}\right) \to 0$$

for $1 < k \leq n$ gives

$$\mathrm{Ext}^1\left(\cdot, \mathrm{im}\left(d_I^{n-1}\right)\right) \simeq \cdots \simeq \mathrm{Ext}^{n+1}\left(\cdot, X\right) = 0.$$

Thus $\mathrm{im}(d_I^{n-1})$ is injective (Corollary 3.12.7). Replacing I^n by this object gives an injective resolution of length $\leq n$.

If X has an injective resolution of length $\leq n$, computing the derived functor with this resolution shows that, for every $Y \in \mathrm{Ob}(\mathcal{A})$, $\big(\mathrm{R}\,\mathrm{Hom}(\cdot, X)\big)(Y)$ lies in $\mathrm{Ob}(\mathsf{D}^{[0,n]}(\mathsf{Ab}))$. $\square$

In keeping with Example 4.8.10, injective and projective dimension may also be viewed as two kinds of "Ext-dimension," although this terminology is not standard.

Definition–Proposition 4.9.7 (Global dimension) Suppose that $\mathcal{A}$ has enough injective and projective objects. Then

$$\begin{array}{ccccc}
\sup\limits_{X \in \mathrm{Ob}(\mathcal{A})} \mathrm{inj.dim}(X) & = & \inf\left\{n \in \mathbb{Z}_{\geq 0} \;\middle|\; \mathrm{Ext}^{\geq n+1}(\cdot,\cdot) = 0\right\} & = & \sup\limits_{X \in \mathrm{Ob}(\mathcal{A})} \mathrm{proj.dim}(X) \\
& & \| & & \\
& & \sup\left\{ n \in \mathbb{Z}_{\geq 0} \;\middle|\; \begin{array}{l} \exists X, Y \in \mathrm{Ob}(\mathcal{A}) \\ \mathrm{Ext}^n(X,Y) \neq 0 \end{array} \right\} & & (\text{with the convention } \inf \varnothing := \infty)
\end{array}$$

This value is denoted by $\mathrm{gl.dim}(\mathcal{A}) \in \mathbb{Z}_{\geq 0} \sqcup \{\infty\}$ and is called the **global dimension**, or **global homological dimension**, of $\mathcal{A}$.

Proof For every $n \in \mathbb{Z}_{\geq 0}$, Proposition 4.9.6 shows that $n \geq \sup_X \mathrm{inj.dim}(X)$ is equivalent to $\mathrm{Ext}^{\geq n+1}(\cdot,\cdot) = 0$. Take the infimum over all n satisfying this condition. The case of proj.dim is exactly the same, and the vertical equality is immediate. $\square$

For example, if R is a principal ideal domain, every submodule of a free R-module is again free [51, Lemma 6.7.4]. Thus every R-module, say M, has a free resolution of the form $0 \to F_1 \to F_0 \to M \to 0$, which shows that $\mathrm{gl.dim}(R\text{-}\mathsf{Mod}) \leq 1$.

Global dimension is an upper bound for the dimensions of all derived functors.

Corollary 4.9.8 Let $F : \mathcal{A} \to \mathcal{A}'$ be a left-exact (respectively, right-exact) functor between abelian categories, and suppose that $\mathcal{A}$ has enough injective and projective objects. Then the dimension of $\mathrm{R}F$ (respectively, $\mathrm{L}F$) is at most $\mathrm{gl.dim}(\mathcal{A})$.

Proof The definition of dimension involves the amplitude of $\mathrm{R}F$ (respectively, $\mathrm{L}F$); see Definition 4.5.5. To control the amplitude, for every $X \in \mathrm{Ob}(\mathcal{A})$ use Proposition 4.9.6 to choose an injective (respectively, projective) resolution of length at most $\mathrm{gl.dim}(\mathcal{A})$, and compute $\mathrm{R}F(X)$ (respectively, $\mathrm{L}F(X)$) using that resolution. $\square$

We conclude this section with a result of adjunction type.

Theorem 4.9.9 Consider an adjoint pair of additive functors between abelian categories $F : \mathcal{A} \rightleftarrows \mathcal{A}' : G$. Suppose that $\mathcal{A}$ has enough projective objects and $\mathcal{A}'$ has enough injective objects. Then there is a canonical isomorphism in $\mathsf{D}^+(\mathsf{Ab})$

$$\mathrm{RHom}_{\mathcal{A}'}\left(\mathrm{L}F(X), Y\right) \simeq \mathrm{RHom}_{\mathcal{A}}\left(X, \mathrm{R}G(Y)\right),$$

where $X \in \mathrm{Ob}(\mathsf{D}^-(\mathcal{A}))$ and $Y \in \mathrm{Ob}(\mathsf{D}^+(\mathcal{A}'))$. Consequently, there is a canonical isomorphism in Ab

$$\mathrm{Hom}_{\mathsf{D}(\mathcal{A}')}\left(\mathrm{L}F(X), Y\right) \simeq \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}\left(X, \mathrm{R}G(Y)\right).$$

Proof Choose a projective resolution $P \to X$ and an injective resolution $Y \to I$. Applying Theorem 4.9.3 gives

$$\mathrm{RHom}_{\mathcal{A}}(\mathrm{L}F(QX), QY) \simeq Q\,\mathrm{Hom}^{\bullet}\left(\mathsf{K}F(P), I\right).$$

But $\mathsf{K}F(P)$ is obtained by applying F degree by degree to the complex. From the definition of the Hom complex and the adjunction, the right-hand side is isomorphic to $Q\,\mathrm{Hom}^{\bullet}(P, \mathsf{K}G(I))$, which is precisely $\mathrm{RHom}_{\mathcal{A}}(QX, \mathrm{R}G(QY))$.

Theorem 3.11.5 shows that a projective resolution of X is unique up to unique isomorphism in $\mathsf{K}^-(\mathcal{A})$, and every morphism in $\mathsf{D}^-(\mathcal{A})$ lifts uniquely to projective resolutions (Proposition 4.5.1). The same holds for injective resolutions of Y. This suffices to show that the isomorphism constructed above is functorial in (X, Y).

The final assertion follows by applying H^0 to both sides. $\square$

Here $\mathrm{L}F$ and $\mathrm{R}G$ are not adjoint functors in the strict sense: the former is a functor between D^- categories, whereas the latter is a functor between D^+ categories. The version for unbounded derived categories in §4.11 will remove this discrepancy.

4.10 Example: $\mathrm{R}\lim$ as a Homotopy Limit

This section continues the discussion of §3.13. Let $\mathcal{A}$ be an abelian category.

Lemma 4.10.1 Suppose that $\mathcal{A}$ has exact countable products (respectively, exact countable coproducts); see Convention 3.13.1. Then every functor in the sequence $\mathsf{C}(\mathcal{A}) \to \mathsf{K}(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$ preserves countable products (respectively, countable coproducts). In this case, there are canonical isomorphisms $\mathrm{H}^n(\prod_k X_k) \xrightarrow{\sim} \prod_k \mathrm{H}^n(X_k)$ (respectively, $\mathrm{H}^n(\bigoplus_k X_k) \xrightarrow{\sim} \bigoplus_k \mathrm{H}^n(X_k)$).

Proof It suffices to prove the product case. Exactness of countable products directly gives $\mathrm{H}^n(\prod_k X_k) \xrightarrow{\sim} \prod_k \mathrm{H}^n(X_k)$.

For every complex Y, the definition shows that $\mathrm{Hom}^\bullet(Y, \prod_k X_k)$ is isomorphic to the product of the $\mathrm{Hom}^\bullet(Y, X_k)$ formed degree by degree in $\mathsf{C}(\mathsf{Ab})$. Apply exactness of countable products when taking H^0 to obtain a canonical isomorphism $\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(Y, \prod_k X_k) \simeq \prod_k \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(Y, X_k)$. Thus $\mathsf{C}(\mathcal{A}) \to \mathsf{K}(\mathcal{A})$ preserves $\prod_k$.

We now prove that $\mathsf{K}(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$ preserves $\prod_k$; the localization functor Q is suppressed. Suppose we are given a family of morphisms $f_k : Y \to X_k$ in $\mathsf{D}(\mathcal{A})$. We wish to prove that there is a unique $f \in \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(Y, \prod_k X_k)$ satisfying $p_h f = f_h$ for every h, where $p_h : \prod_k X_k \to X_h$ is the canonical projection in $\mathsf{K}(\mathcal{A})$. We may assume that each f_k is represented by a diagram in $\mathsf{K}(\mathcal{A})$

$$Y \xrightarrow{b_k} Z_k \xleftarrow{s_k} X_k, \quad s_k: \text{ a quasi-isomorphism.} \tag{4.10.1}$$

The morphism $s := \prod_k s_k : \prod_k X_k \to \prod_k Z_k$ remains a quasi-isomorphism. The diagram

$$Y \xrightarrow{b:=(b_k)_k} \prod_k Z_k \xleftarrow{s} \prod_k X_k$$

determines $f \in \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(Y, \prod_k X_k)$. Notice that f depends only on $(f_k)_k$, not on the choices made in (4.10.1). The commutativity of the following diagram implies that $p_h f = f_h$ for every h:

$$\begin{tikzcd} & Z_h & X_h \arrow[l, "s_h"'] \\ Y \arrow[ur, "b_h"] \arrow[r, "b"'] & \prod_k Z_k \arrow[u] & \prod_k X_k. \arrow[l, "s"] \arrow[u, "p_h"'] \end{tikzcd}$$

If $g \in \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(Y, \prod_k X_k)$ also satisfies $p_h g = f_h$ for every h, there is a commutative diagram in $\mathsf{K}(\mathcal{A})$

$$
\begin{tikzcd}
 & M_h & X_h \arrow[l, "t_h"'] \\
Y \arrow[ur, "c_h"] \arrow[r, "c"] \arrow[dr, "(c_k)_k"'] & M \arrow[u, "r_h"] \arrow[d, "(r_k)_k"] & \prod_k X_k \arrow[l, "t"'] \arrow[u, "p_h"'] \arrow[dl, "u"] \\
 & \prod_k M_k &
\end{tikzcd}
\qquad
\begin{aligned}
& h \in \mathbb{Z}, \\
& t, t_h : \text{ quasi-isomorphisms}, \\
& u := (r_k)_k t = \textstyle\prod_k t_k,
\end{aligned}
$$

such that the second row represents g, while the $\nearrow$ part represents f_h. The square is constructed using the version of condition (S3) in Definition 1.9.5 for a right multiplicative system, while commutativity of the lower part is immediate.

Exactness of $\prod_k$ shows that u is a quasi-isomorphism. Consequently, g is also represented by the $\searrow\!\nearrow$ part of the diagram. But $Y \xrightarrow{c_k} M_k \xleftarrow{t_k} X_k$ represents f_k. Comparing this with the construction of f above gives $f = g$. $\square$

Remark 4.10.2 If the indexing set for the product (respectively, coproduct) is replaced by an arbitrary small set I, the corresponding statement remains valid.

Let $\mathcal{D}$ be an Ab-category. Consider the category $\mathsf{InvSys}(\mathcal{D}) := \mathcal{D}^{\mathbb{Z}_{\geq 0}^{\mathrm{op}}}$ from (3.13.1). Its objects are data $X = (X_k, f_k)_{k \geq 0}$, where $X_k \in \mathrm{Ob}(\mathcal{D})$ and $f_k \in \mathrm{Hom}_{\mathcal{D}}(X_{k+1}, X_k)$. If $\mathcal{D}$ has countable products, define the canonical morphism as in Definition 3.13.3:

$$\Delta_X := T_X - \mathrm{id} : \prod_k X_k \to \prod_k X_k.$$

Dually, an object of the category $\mathsf{DirSys}(\mathcal{D}) := \mathcal{D}^{\mathbb{Z}_{\geq 0}}$ consists of data $X = (X_k, g_k)_{k \geq 0}$[7], where $X_k \in \mathrm{Ob}(\mathcal{D})$ and $g_k \in \mathrm{Hom}_{\mathcal{D}}(X_k, X_{k+1})$. Similarly, define $\nabla_X := T_X - \mathrm{id} : \bigoplus_k X_k \to \bigoplus_k X_k$, assuming that $\mathcal{D}$ has countable coproducts; T_X is again the analogous shift morphism.

If $\ker(\Delta_X)$ or $\mathrm{coker}(\nabla_X)$ exists, there are canonical isomorphisms

$$
\begin{aligned}
X \in \mathrm{Ob}\,(\mathsf{InvSys}(\mathcal{D})) &\implies \varprojlim_k X_k \simeq \ker(\Delta_X), \\
X \in \mathrm{Ob}\,(\mathsf{DirSys}(\mathcal{D})) &\implies \varinjlim_k X_k \simeq \mathrm{coker}(\nabla_X).
\end{aligned}
$$

The naive hope is to apply this to $\mathcal{D} = \mathsf{D}(\mathcal{A})$. Derived categories are rarely abelian categories. Nevertheless, Remark 3.3.7 shows that mapping cones can serve as homotopical versions of kernels and cokernels. Roughly speaking, the analogous construction in a triangulated category is a distinguished triangle.

[7] Such data are also called a direct system in $\mathcal{D}$.

Definition 4.10.3 (M. Bökstedt, A. Neeman [5]) Let $\mathcal{D}$ be a triangulated category.

- ◇ If $\mathcal{D}$ has countable products, a **homotopy limit** (or homotopy $\varprojlim$) of an object $X \in \mathrm{Ob}(\mathsf{InvSys}(\mathcal{D}))$ means a distinguished triangle

$$\mathrm{holim}(X) \to \prod_k X_k \xrightarrow{\Delta_X} \prod_k X_k \xrightarrow{+1} .$$

- ◇ If $\mathcal{D}$ has countable coproducts, a **homotopy colimit** (or homotopy $\varinjlim$) of an object $X \in \mathrm{Ob}(\mathsf{DirSys}(\mathcal{D}))$ means a distinguished triangle

$$\bigoplus_k X_k \xrightarrow{\nabla_X} \bigoplus_k X_k \to \mathrm{hocolim}(X) \xrightarrow{+1} .$$

The distinguished triangles occurring in this definition are isomorphic to one another (Corollary 4.2.3), but the isomorphisms are not canonical.

Return to complexes. Observe that $\mathsf{InvSys}(\mathsf{C}(\mathcal{A})) = \mathsf{C}(\mathsf{InvSys}(\mathcal{A}))$; the objects of both categories are the same commutative diagrams

$$\begin{tikzcd}
 & \vdots \arrow[d, "f_1^n"'] & \vdots \arrow[d, "f_1^{n+1}"] & \\
\cdots \arrow[r] & X_1^n \arrow[r, "d_{X_1}^n"] \arrow[d, "f_0^n"'] & X_1^{n+1} \arrow[r] \arrow[d, "f_0^{n+1}"] & \cdots \\
\cdots \arrow[r] & X_0^n \arrow[r, "d_{X_0}^n"] & X_0^{n+1} \arrow[r] & \cdots
\end{tikzcd}$$

with each row a complex.

Morphisms in the two categories are likewise the same two-level commutative diagrams. Thus, in $\mathcal{D} := \mathsf{D}(\mathcal{A})$, we can compare $\mathrm{holim}(X)$ with the right derived functor $\mathrm{R}\lim(X)$ of $\varprojlim$, provided that both exist.

Also observe that the cohomology H^m of an object $X \in \mathsf{C}(\mathsf{InvSys}(\mathcal{A}))$ is $(\mathrm{H}^m(X_k), \mathrm{H}^m(f_k))_{k\geq 0}$. Thus $X \to Y$ is a quasi-isomorphism if and only if every $X_k \to Y_k$ is a quasi-isomorphism in $\mathsf{C}(\mathcal{A})$.

Theorem 4.10.4 Suppose that $\mathcal{A}$ is an abelian category with exact countable products and enough injective objects.

(i) The right derived functor of the left-exact functor $\varprojlim : \mathsf{InvSys}(\mathcal{A}) \to \mathcal{A}$ exists and is written

$$\mathrm{R}\lim : \mathsf{D}^+(\mathsf{InvSys}(\mathcal{A})) \to \mathsf{D}^+(\mathcal{A}).$$

(ii) For every object $X = (X_k, f_k)_{k\geq 0}$ of $\mathsf{C}^+(\mathsf{InvSys}(\mathcal{A}))$, there is a distinguished triangle in $\mathsf{D}(\mathcal{A})$

$$\mathrm{R}\lim(X) \to \prod_k X_k \xrightarrow{\Delta_X} \prod_k X_k \xrightarrow{+1} .$$

In particular, $\mathrm{R}\lim(X) \simeq \mathrm{holim}(X)$.

(iii) The dimension of $\mathrm{R}\lim$ in the sense of Definition–Proposition 4.8.9 is at most 1.

Proof Since $\mathsf{InvSys}(\mathcal{A})$ has enough injective objects, (i) holds. More precisely, take the category $\mathcal{R}$ from Lemma 3.13.7 and apply Theorem 3.11.3 to $\mathsf{InvSys}(\mathcal{A})$ and $\mathcal{R}$. For every $X = (X_k, f_k)_{k\geq 0}$ in $\mathsf{C}^+(\mathsf{InvSys}(\mathcal{A}))$, there is a quasi-isomorphism $X \to Y$ with $Y \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{R}))$. Then $\mathrm{R}\lim(X) = \varprojlim Y$. The definition of $\mathcal{R}$ ensures that Δ_Y is surjective degree by degree, so there is a short exact sequence in $\mathsf{C}^+(\mathcal{A})$

$$0 \to \varprojlim Y \to \prod_{k\geq 0} Y_k \xrightarrow{\Delta_Y} \prod_{k\geq 0} Y_k \to 0.$$

Furthermore, every $X_k \to Y_k$ is a quasi-isomorphism, so $\prod_{k\geq 0} X_k \to \prod_{k\geq 0} Y_k$ is also a quasi-isomorphism. This gives the distinguished triangle in (ii):

$$\mathrm{R}\lim(X) \to \prod_{k\geq 0} X_k \xrightarrow{\Delta_X} \prod_{k\geq 0} X_k \xrightarrow{+1}.$$

For (iii), if $X \in \mathrm{Ob}(\mathsf{InvSys}(\mathcal{A}))$, then Δ_X is a morphism in $\mathsf{InvSys}(\mathcal{A})$, and $\mathrm{holim}(X) \simeq \mathrm{Cone}(\Delta_X)[-1]$ lies in $\mathrm{Ob}(\mathsf{D}^{[0,1]}(\mathcal{A}))$. This is also a restatement of Theorem 3.13.8. □

There is a dual version of Theorem 4.10.4 for $\varinjlim : \mathsf{DirSys}(\mathcal{A}) \to \mathcal{A}$; we do not repeat the details.

Example 4.11.11 will show how to obtain $\mathrm{R}\lim$ on the unbounded derived category so that Theorem 4.10.4 (ii) remains valid.

4.11 Unbounded Derived Functors

Consider an additive functor $F : \mathcal{A} \to \mathcal{A}'$ between abelian categories. This section returns to Definition 4.7.1 and focuses on the versions of derived functors for unbounded derived categories,

$$\mathrm{R}F,\ \mathrm{L}F : \mathsf{D}(\mathcal{A}) \to \mathsf{D}(\mathcal{A}').$$

To guarantee the existence of $\mathrm{R}F$ or $\mathrm{L}F$ and to compute with it, we need the notions of K-injective and K-projective complexes from Definition 3.15.1, together with the corresponding resolutions.

Lemma 4.11.1 Let $\mathcal{A}$ be an abelian category. If $Z \in \mathrm{Ob}(\mathsf{K}(\mathcal{A}))$ is both K-injective (respectively, K-projective) and acyclic, then $Z = 0$.

Proof Lemma 3.15.2 implies that $\mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(Z, Z) = 0$. □

We first state a general result. Fix triangulated categories $\mathcal{D}_1$, $\mathcal{D}_2$, and $\mathcal{D}'$, together with their saturated triangulated subcategories $\mathcal{N}_1$, $\mathcal{N}_2$, and $\mathcal{N}'$.

Lemma 4.11.2 Given a triangulated bifunctor $G : \mathcal{D}_1 \times \mathcal{D}_2 \to \mathcal{D}'$, define a full subcategory $\mathcal{K}_2^G$ of $\mathcal{D}_2$ by

$$\mathrm{Ob}(\mathcal{K}_2^G) = \{X_2 \in \mathrm{Ob}(\mathcal{D}_2) : X_1 \in \mathrm{Ob}(\mathcal{N}_1) \implies G(X_1, X_2) \in \mathrm{Ob}(\mathcal{N}')\}.$$

Then $\mathcal{K}_2^G$ is a saturated triangulated subcategory. After interchanging the two variables, the corresponding statement holds for $\mathcal{K}_1^G \subset \mathcal{D}_1$.

Proof The isomorphism $G(X_1, X_2[1]) \simeq G(X_1, X_2)[1]$ shows that $\mathcal{K}_2^G$ is closed under translation. Similarly, suppose that $I \to J \to K \xrightarrow{+1}$ is a distinguished triangle with $I, K \in \mathrm{Ob}(\mathcal{K}_2^G)$. From the distinguished triangle

$$G(X_1, I) \to G(X_1, J) \to G(X_1, K) \xrightarrow{+1}$$

it follows immediately that $X_1 \in \mathrm{Ob}(\mathcal{N}_1) \implies G(X_1, J) \in \mathrm{Ob}(\mathcal{N}')$. Saturation is clear. ☐

This result can be applied to Hom : $\mathcal{A}^{\mathrm{op}} \times \mathcal{A} \to \mathsf{Ab}$ and the corresponding triangulated bifunctor $\mathsf{K}_\Pi \,\mathrm{Hom}$. Example 3.5.12 has shown that the latter bifunctor is essentially the Hom complex:

$$\begin{aligned} (\mathsf{C}_\Pi \,\mathrm{Hom})(\sigma^{-1}X_1, X_2) &\simeq \mathrm{Hom}^\bullet(X_1, X_2), \\ (\mathsf{K}_\Pi \,\mathrm{Hom})(\sigma^{-1}X_1, X_2) &\simeq \mathrm{Hom}^\bullet(X_1, X_2) \text{ viewed in } \mathsf{K}(\mathsf{Ab}), \end{aligned} \tag{4.11.1}$$

where $\sigma : \mathsf{C}(\mathcal{A}^{\mathrm{op}}) \xrightarrow{\sim} \mathsf{C}(\mathcal{A})^{\mathrm{op}}$ and $\mathsf{K}(\mathcal{A}^{\mathrm{op}}) \xrightarrow{\sim} \mathsf{K}(\mathcal{A})^{\mathrm{op}}$ are the isomorphisms from Definition–Proposition 3.4.1 and Proposition 3.4.3. We shall later need the corresponding version for derived categories.

Proposition 4.11.3 All K-injective (respectively, K-projective) complexes form a saturated triangulated subcategory $\mathcal{I}$ (respectively, $\mathcal{P}$) of $\mathsf{K}(\mathcal{A})$. If $\mathcal{A}$ has enough K-injective (respectively, K-projective) complexes, then for every additive functor $F : \mathcal{A} \to \mathcal{A}'$, the subcategory $\mathcal{I}$ (respectively, $\mathcal{P}$) is F-injective (respectively, F-projective).

Proof It suffices to treat the K-injective case. The first assertion follows by applying Lemma 4.11.2 to $G = \mathsf{K}_\Pi \,\mathrm{Hom}$, using the definition of K-injective complexes in Definition 3.15.1 and (4.11.1): the subcategory $\mathcal{K}_2^G$ there is precisely $\mathcal{I}$ here.

Suppose that $\mathcal{A}$ has enough K-injective complexes. If I is an acyclic K-injective complex, Lemma 4.11.1 implies that $I = 0$, so $(\mathsf{K}F)I$ is certainly acyclic. Thus all the conditions in Definition 4.6.3 for an F-injective triangulated subcategory are satisfied. ☐

Write $Q : \mathsf{K}(\mathcal{A}) \to \mathsf{D}(\mathcal{A})$ and $Q' : \mathsf{K}(\mathcal{A}') \to \mathsf{D}(\mathcal{A}')$ for the localization functors.

Corollary 4.11.4 If $\mathcal{A}$ has enough K-injective complexes, then F has a right derived functor $\mathrm{R}F : \mathsf{D}(\mathcal{A}) \to \mathsf{D}(\mathcal{A}')$, such that $\mathrm{R}F(QX) \simeq Q'\mathsf{K}F(X)$ whenever X is K-injective.

Dually, if $\mathcal{A}$ has enough K-projective complexes, then F has a left derived functor $\mathrm{L}F : \mathsf{D}(\mathcal{A}) \to \mathsf{D}(\mathcal{A}')$, such that $\mathrm{L}F(QX) \simeq Q'\mathsf{K}F(X)$ whenever X is K-projective.

Proof Apply Proposition 4.11.3 in the general framework of Proposition 4.6.4. $\square$

Now fix abelian categories $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{B}$. For an additive bifunctor $F : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}$, consider the unbounded version of the derived bifunctor $\mathrm{R}F$ (respectively, $\mathrm{L}F$) from Definition 4.7.7. From now on, we always assume that $\mathcal{B}$ has countable products (respectively, countable coproducts).

Concretely, we wish to determine $\mathrm{R}F$ (respectively, $\mathrm{L}F$) using the subcategories $\mathcal{I}_1$, $\mathcal{I}_2$ (respectively, $\mathcal{P}_1$, $\mathcal{P}_2$) above.

Lemma 4.11.5 For $i = 1, 2$, take the triangulated subcategory $\mathcal{I}_i$ (respectively, $\mathcal{P}_i$) of $\mathsf{K}(\mathcal{A}_i)$ from Proposition 4.11.3.

- ◇ If $\mathcal{A}_1$ and $\mathcal{A}_2$ have enough K-injective complexes, then $\mathcal{I}_1 \times \mathcal{I}_2$ is F-injective.
- ◇ If $\mathcal{A}_1$ and $\mathcal{A}_2$ have enough K-projective complexes, then $\mathcal{P}_1 \times \mathcal{P}_2$ is F-projective.

Proof It suffices to treat the K-injective case. The key is to show that if $X_1 \in \mathrm{Ob}(\mathcal{I}_1)$, then $(\mathsf{K}_\Pi F)(X_1, \cdot)$ sends every acyclic object Z of $\mathcal{I}_2$ to an acyclic object of $\mathsf{K}(\mathcal{B})$. But Lemma 4.11.1 gives $Z = 0$, and $(\mathsf{K}_\Pi F)(X_1, \cdot)$ is additive, so the assertion follows immediately. The corresponding result follows after interchanging the indices 1 and 2. $\square$

Write $Q : \mathsf{K}(\mathcal{B}) \to \mathsf{D}(\mathcal{B})$ and $Q_i : \mathsf{K}(\mathcal{A}_i) \to \mathsf{D}(\mathcal{A}_i)$ for the localization functors ($i = 1, 2$).

Proposition 4.11.6 For $F : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}$, suppose that $\mathcal{A}_1$ and $\mathcal{A}_2$ both have enough K-injective (respectively, K-projective) complexes. Then the derived bifunctor $\mathrm{R}F$ (respectively, $\mathrm{L}F$) exists. If $X_i \in \mathrm{Ob}(\mathsf{K}(\mathcal{A}))$ is K-injective (respectively, K-projective) for $i = 1, 2$, then there is a canonical isomorphism

$$\mathrm{R}F(Q_1X_1, Q_2X_2) \simeq Q(\mathsf{K}_\Pi F)(X_1, X_2) \quad \text{(respectively, } \mathrm{L}F(Q_1X_1, Q_2X_2) \simeq Q(\mathsf{K}_\oplus F)(X_1, X_2)\text{)}\ .$$

Proof Apply Lemma 4.11.5 directly to Proposition 4.6.10. $\square$

For a particular bifunctor F, one can often choose F-injective (respectively, F-projective) subcategories larger than the subcategories of K-injective (respectively, K-projective) complexes. The RHom functor to be discussed next is one example.

Theorem 4.11.7 (Unbounded RHom) Suppose that the abelian category $\mathcal{A}$ has enough K-injective or enough K-projective complexes. For simplicity, denote every localization functor by Q.

(i) The bifunctor $\mathrm{Hom} = \mathrm{Hom}_{\mathcal{A}} : \mathcal{A}^{\mathrm{op}} \times \mathcal{A} \to \mathsf{Ab}$ has a right derived bifunctor $\mathsf{D}(\mathcal{A}^{\mathrm{op}}) \times \mathsf{D}(\mathcal{A}) \to \mathsf{D}(\mathsf{Ab})$. Via the equivalence σ of Proposition 4.4.16, it can also be written as

$$\mathrm{RHom} : \mathsf{D}(\mathcal{A})^{\mathrm{op}} \times \mathsf{D}(\mathcal{A}) \to \mathsf{D}(\mathsf{Ab}).$$

Moreover, if $\mathcal{A}$ has enough K-injective (respectively, K-projective) complexes, then $\mathsf{K}(\mathcal{A})^{\mathrm{op}} \times \mathcal{I}$ (respectively, $\mathcal{P}^{\mathrm{op}} \times \mathsf{K}(\mathcal{A})$) is Hom-injective.

(ii) Suppose that $\mathcal{A}$ has enough K-injective complexes and that $X_2 \to I_2$ is a K-injective resolution. This resolution induces

$$\mathrm{RHom}(QX_1, QX_2) \xrightarrow{\sim} Q\,\mathrm{Hom}^\bullet(X_1, I_2).$$

In particular, $\mathrm{RHom}(QX_1, \cdot) \simeq \mathrm{R}\,(\mathrm{Hom}(X_1, \cdot))$.

(iii) Suppose that $\mathcal{A}$ has enough K-projective complexes and that $P_1 \to X_1$ is a K-projective resolution. This resolution induces

$$\mathrm{RHom}(QX_1, QX_2) \xrightarrow{\sim} Q\,\mathrm{Hom}^\bullet(P_1, X_2).$$

In particular, $\mathrm{RHom}(\cdot, QX_2) \simeq \mathrm{R}\,(\mathrm{Hom}(\cdot, X_2))$.

(iv) There is a canonical isomorphism $\mathrm{H}^n\,\mathrm{RHom}(X,Y) \simeq \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y[n]) =: \mathrm{Ext}^n(X,Y)$ (Definition 4.5.8).

Proof The argument is nearly identical to that for Theorem 4.9.3. For example, suppose that $\mathcal{A}$ has enough K-injective complexes. The key is to show that $\mathsf{K}(\mathcal{A})^{\mathrm{op}} \times \mathcal{I}$ is Hom-injective.

◇ Fix a complex X_1. The functor $(\mathsf{K}_\Pi\,\mathrm{Hom})(X_1, \cdot)$ sends every acyclic K-injective complex X_2 to an acyclic complex, because X_2 is 0 in $\mathsf{K}(\mathcal{A})$ (Lemma 4.11.1).

◇ Fix a K-injective complex X_2. If X_1 is acyclic, the definition of a K-injective complex implies that $\mathrm{Hom}^\bullet\,(X_1, X_2)$ is acyclic. By (4.11.1), $(\mathsf{K}_\Pi\,\mathrm{Hom})(X_1, X_2)$ is therefore acyclic as well.

Thus the acyclicity conditions required for the Hom-injective property have been verified, while the resolution conditions hold directly. □

Theorem 4.11.8 Consider an adjoint pair of additive functors between abelian categories $F : \mathcal{A} \rightleftarrows \mathcal{A}' : G$. Suppose that $\mathcal{A}$ has enough K-projective complexes and $\mathcal{A}'$ has enough K-injective complexes. Then there is a canonical isomorphism in $\mathsf{D}(\mathsf{Ab})$

$$\mathrm{RHom}_{\mathcal{A}'}\,(\mathrm{L}F(X), Y) \simeq \mathrm{RHom}_{\mathcal{A}}\,(X, \mathrm{R}G(Y)),$$

where $X \in \mathrm{Ob}\,(\mathsf{D}(\mathcal{A}))$ and $Y \in \mathrm{Ob}\,(\mathsf{D}(\mathcal{A}'))$.

Proof The argument is the same as for Theorem 4.9.9. The only difference is that injective (respectively, projective) resolutions are replaced by K-injective (respectively, K-projective) resolutions, and Theorem 3.15.5 replaces Theorem 3.11.5 in the discussion of uniqueness of resolutions. $\square$

Remark 4.11.9 Applying H^0 to both sides of Theorem 4.11.8 gives the adjunction

$$\mathrm{Hom}_{\mathsf{D}(\mathcal{A}')}\left(\mathrm{L}F(X), Y\right) \simeq \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}\left(X, \mathrm{R}G(Y)\right).$$

Since derived functors obtained through resolutions are always absolute Kan extensions in the sense of Definition 1.7.6 (Proposition 1.10.4), Theorem 1.7.7 also says that $(\mathrm{L}F, \mathrm{R}G)$ is an adjoint pair. The two approaches give the same adjunction isomorphism. Why? It suffices to consider the case in which X is K-projective and Y is K-injective. In this case, the adjunction isomorphism of Theorem 4.11.8 is realized concretely by the unit η and counit ε at the level of $\mathsf{K}(\cdot)$ as

$$\mathrm{Hom}_{\mathsf{K}(\mathcal{A}')}\left(\mathsf{K}F(X), Y\right) \simeq \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}\left(X, \mathsf{K}G(Y)\right).$$

On the other hand, Theorem 1.7.7 determines $\underline{\eta}$ and $\underline{\varepsilon}$ at the level of derived categories. The characterization given there suffices to show that they are compatible with η and ε. The details are left to the interested reader.

Finally, another set of sufficient conditions for the existence of unbounded derived functors is based on finite dimension (Definition–Proposition 4.8.9). Although the proof technique does not go beyond the scope of this book, we state the result without proof to avoid a digression; see [42, Tags 07K6, 07K7]. The starting point is the construction of bounded derived functors in Theorem 4.8.3.

Proposition 4.11.10 Let $F : \mathcal{A} \to \mathcal{A}'$ be a left-exact (respectively, right-exact) functor between abelian categories. Assume that

- ◇ relative to F, there is a full additive subcategory $\mathcal{A}^\flat$ of $\mathcal{A}$ of type I (respectively, type P), as in Definition–Proposition 4.8.2;
- ◇ the corresponding bounded derived functor ${}^+\mathrm{R}F$ (respectively, ${}^-\mathrm{L}F$) has finite dimension.

Then $\mathrm{R}F$ (respectively, $\mathsf{L}F$): $\mathsf{D}(\mathcal{A}) \to \mathsf{D}(\mathcal{A}')$ exists. Moreover, $\mathsf{K}(\mathcal{A}^\flat)$ is an F-injective (respectively, F-projective) subcategory of $\mathsf{K}(\mathcal{A})$.

Example 4.11.11 Suppose that $\mathcal{A}$ has exact countable products and enough injective objects. Take the left-exact functor F to be $\varprojlim : \mathsf{InvSys}(\mathcal{A}) \to \mathcal{A}$. Theorem 4.10.4 (iii) shows that the bounded version of $\mathrm{R}\lim$ has dimension at most 1, while the subcategory $\mathcal{R} \subset \mathsf{InvSys}(\mathcal{A})$ is of type I relative to $\varprojlim$. The preceding result therefore gives the unbounded derived functor

$$\mathrm{R}\lim : \mathsf{D}(\mathsf{InvSys}(\mathcal{A})) \to \mathsf{D}(\mathcal{A}).$$

As in Theorem 4.10.4 (ii), for every object X of $\mathsf{C}(\mathsf{InvSys}(\mathcal{A}))$ there is a distinguished triangle in $\mathsf{D}(\mathcal{A})$

$$\mathrm{R}\lim(X) \to \prod_k X_k \xrightarrow{\Delta_X} \prod_k X_k \xrightarrow{+1} .$$

In particular, $\mathrm{R}\lim(X) \simeq \mathrm{holim}(X)$. By Proposition 4.11.10, the original proof applies with no other changes: simply replace every $\mathsf{C}^+(\cdots)$ by $\mathsf{C}(\cdots)$.

4.12 Example: K-Flat Complexes and $\overset{\mathbf{L}}{\otimes}$

Throughout this section, $\Bbbk$ is a commutative ring.

Convention 4.12.1 Let R and S be $\Bbbk$-algebras. By the convention of this book, every (R,S)-bimodule M is tacitly assumed to satisfy $tm = mt$ for $t \in \Bbbk$ and $m \in M$. In other words, $(R,S)\text{-}\mathsf{Mod} \simeq (R \underset{\Bbbk}{\otimes} S^{\mathrm{op}})\text{-}\mathsf{Mod}$. When $\Bbbk = \mathbb{Z}$, this condition is redundant.

From now on, the forgetful functor $(R,S)\text{-}\mathsf{Mod} \to R\text{-}\mathsf{Mod}$ (respectively, $(R,S)\text{-}\mathsf{Mod} \to \mathsf{Mod}\text{-}S$) is written $M \mapsto {}_RM$ (respectively, $M \mapsto M_S$); the same notation applies to complexes. These forgetful functors are exact, and the triangulated functors they induce between derived categories are denoted in the same way.

Notice that, for every complex M of (R,S)-bimodules, one has

$$M_S \text{ acyclic} \iff M \text{ acyclic} \iff {}_RM \text{ acyclic}.$$

From now on, let R, A, and B be $\Bbbk$-algebras. Our starting point is the bifunctor given by the tensor product,

$$\otimes_R : (A,R)\text{-}\mathsf{Mod} \times (R,B)\text{-}\mathsf{Mod} \to (A,B)\text{-}\mathsf{Mod},$$

which is right exact in each variable. The simplest case is $A = \Bbbk = B$; the corresponding bifunctor becomes $\otimes_R : \mathsf{Mod}\text{-}R \times R\text{-}\mathsf{Mod} \to \Bbbk$.

For convenience, if X is a complex of (A,R)-bimodules and Y is a complex of (R,B)-bimodules, we henceforth write the bicomplex obtained by taking tensor products term by term as $X^\bullet \underset{R}{\otimes} Y^\bullet$, and write its total complex as

$$X \underset{R}{\otimes} Y := \mathrm{tot}_\oplus\left(X^\bullet \underset{R}{\otimes} Y^\bullet\right).$$

Categories of bimodules have enough K-projective complexes (apply Example 3.15.14); hence Proposition 4.11.6 ensures the existence of the unbounded left derived bifunctor.

Definition 4.12.2 Denote the left derived bifunctor $\mathrm{L}\left(\cdot \underset{R}{\otimes} \cdot\right)$ of $\otimes_R$ by

$$\underset{R}{\overset{\mathrm{L}}{\otimes}} : \mathsf{D}((A,R)\text{-}\mathsf{Mod}) \times \mathsf{D}((R,B)\text{-}\mathsf{Mod}) \to \mathsf{D}((A,B)\text{-}\mathsf{Mod}).$$

When no confusion can arise, $X \underset{R}{\overset{\mathrm{L}}{\otimes}} Y$ is also abbreviated to $X \overset{\mathrm{L}}{\otimes} Y$.

The abstract definition above is straightforward, but to study $\otimes_R$ further we still need to introduce a class of quasi-isomorphisms called K-flat resolutions. The reasons include the following:

- ◇ the definition of a K-flat resolution is directly related to the tensor product and is more useful for establishing important properties and making concrete computations, as already seen in §3.14;
- ◇ in geometric settings such as sheaf theory, K-injective resolutions may be available while K-projective resolutions are not; in that situation, K-flat resolutions are the only means of studying the derived tensor product.

Definition 4.12.3 (N. Spaltenstein [41, Definition 5.1]) Let X be a complex of right R-modules. If, for every complex Y of left R-modules,

$$Y \text{ acyclic} \implies X \underset{R}{\otimes} Y \text{ acyclic},$$

then X is called **K-flat**[8]; this property depends only on the isomorphism class of X in $\mathsf{K}(\mathsf{Mod}\text{-}R)$. K-flatness is defined similarly for complexes of left R-modules.

For bounded-above complexes, K-flatness follows from the familiar notion of flatness in module theory.

Proposition 4.12.4 Let X be an object of $\mathsf{C}^-(\mathsf{Mod}\text{-}R)$, and suppose that every X^n is flat. Then X is K-flat. The same statement holds for objects of $\mathsf{C}^-(R\text{-}\mathsf{Mod})$.

Proof Let Y be an acyclic complex of left R-modules; we shall prove that $X \underset{R}{\otimes} Y$ is acyclic. Using truncation functors, write Y as $\varinjlim_m \tau^{\leq m} Y$. Degree by degree, $\varinjlim_m$ commutes both with tensor products [51, Proposition 6.9.2] and with $\bigoplus$; it therefore suffices to consider the case in which Y is bounded above. Then $X^\bullet \underset{R}{\otimes} Y^\bullet$ belongs to

[8] A more reasonable name might be a homotopy-flat complex.

the category $\mathsf{C}^2_f(\Bbbk\text{-}\mathsf{Mod})$ of Definition 3.10.1. Since $X^p \underset{R}{\otimes} Y^\bullet$ is acyclic for every p, Corollary 3.10.7 ensures that $X \underset{R}{\otimes} Y$ is acyclic. □

In view of this result, if the reader is interested only in bounded-above derived categories, every K-flat complex appearing in this section may be replaced by a bounded-above complex of flat modules.

Definition 4.12.5 If $P \to X$ is a quasi-isomorphism in $\mathsf{C}((A,R)\text{-}\mathsf{Mod})$ and P_R is K-flat, then this morphism is called a **K-flat resolution** of X relative to R. If every $X \in \mathsf{C}((A,R)\text{-}\mathsf{Mod})$ has a K-flat resolution relative to R, then $(A,R)\text{-}\mathsf{Mod}$ is said to have enough K-flat complexes relative to R.

A K-flat resolution relative to R is defined similarly for $X \in \mathrm{Ob}(\mathsf{C}((R,B)\text{-}\mathsf{Mod}))$.

A natural question is how to guarantee that a category of bimodules has enough K-flat complexes relative to R. What is the relation between K-flat and K-projective complexes? As expected, the bridge comes from the classical adjunction between $\otimes$ and Hom [51, Theorem 6.6.5].

The first step is to lift that adjunction to the level of complexes. If Y is an object of $\mathsf{C}((R,B)\text{-}\mathsf{Mod})$ and Z is an object of $\mathsf{C}((A,B)\text{-}\mathsf{Mod})$, the Hom complex $\mathrm{Hom}^\bullet(Y_B, Z_B)$ can be endowed degree by degree with an (A,R)-bimodule structure, making it an object of $\mathsf{C}((A,R)\text{-}\mathsf{Mod})$.

Proposition 4.12.6 (Hom–⊗ adjunction: complex version) Let X be an object of $\mathsf{C}((A,R)\text{-}\mathsf{Mod})$, let Y be an object of $\mathsf{C}((R,B)\text{-}\mathsf{Mod})$, and let Z be an object of $\mathsf{C}((A,B)\text{-}\mathsf{Mod})$. There are canonical isomorphisms in $\mathsf{C}(\Bbbk\text{-}\mathsf{Mod})$

$$\begin{aligned}\mathrm{Hom}^\bullet\left(X \underset{R}{\otimes} Y,\ Z\right) &\simeq \mathrm{Hom}^\bullet\left(X, \mathrm{Hom}^\bullet(Y_B, Z_B)\right)\\ &\simeq \mathrm{Hom}^\bullet\left(Y, \mathrm{Hom}^\bullet({}_AX, {}_AZ)\right).\end{aligned}$$

Consequently, taking $\ker(d^0)$ gives isomorphisms of $\Bbbk$-modules

$$\begin{aligned}\mathrm{Hom}_{\mathsf{C}((A,B)\text{-}\mathsf{Mod})}\left(X \underset{R}{\otimes} Y,\ Z\right) &\simeq \mathrm{Hom}_{\mathsf{C}((A,R)\text{-}\mathsf{Mod})}\left(X, \mathrm{Hom}^\bullet(Y_B, Z_B)\right)\\ &\simeq \mathrm{Hom}_{\mathsf{C}((R,B)\text{-}\mathsf{Mod})}\left(Y, \mathrm{Hom}^\bullet({}_AX, {}_AZ)\right).\end{aligned}$$

Proof It suffices to treat the first isomorphism. To simplify notation, write Y and Z for Y_B and Z_B below, and omit the subscript on Hom. Fix $n \in \mathbb{Z}$. The degree-n term on the left-hand side is

$$\prod_k \operatorname{Hom}\left((X \underset{R}{\otimes} Y)^k, Z^{k+n}\right) \simeq \prod_k \prod_{p+q=k} \operatorname{Hom}\left(X^p \underset{R}{\otimes} Y^q, Z^{k+n}\right)$$
$$= \prod_{p,q} \operatorname{Hom}\left(X^p \underset{R}{\otimes} Y^q, Z^{p+q+n}\right) \simeq \prod_p \prod_q \operatorname{Hom}\left(X^p, \operatorname{Hom}\left(Y^q, Z^{p+q+n}\right)\right)$$
$$\simeq \prod_p \operatorname{Hom}\left(X^p, \operatorname{Hom}^{p+n}(Y, Z)\right), \tag{4.12.1}$$

and the last term is precisely the degree-n term on the right-hand side. Concretely, under the second isomorphism $\simeq$, the homomorphism of (A, B)-bimodules $\varphi^{p,q} : X^p \underset{R}{\otimes} Y^q \to Z^{p+q+n}$ corresponds to

$$[x \mapsto [y \mapsto \varphi^{p,q}(x \otimes y)]] \ \in \operatorname{Hom}\left(X^p, \operatorname{Hom}\left(Y^q, Z^{p+q+n}\right)\right).$$

It remains only to verify that (4.12.1) determines an isomorphism of complexes.

Take an element of the degree-n term on the left and write it as $(\varphi^{p,q})_{p,q\in\mathbb{Z}}$. The (p, q)-component of its image under $d_{\operatorname{Hom}^\bullet(X\otimes Y,Z)}$ sends $x \otimes y \in X^p \underset{R}{\otimes} Y^q$ to

$$d_Z\varphi^{p,q}(x \otimes y) - (-1)^n \left(\varphi^{p+1,q}(d_X x \otimes y) + (-1)^p\varphi^{p,q+1}(x \otimes d_Y y)\right).$$

For the corresponding element of degree n on the right, the p-component of its image under $d_{\operatorname{Hom}^\bullet(X,\operatorname{Hom}^\bullet(Y,Z))}$ sends $x \in X^p$ to

$$d_{\operatorname{Hom}^\bullet(Y,Z)} \left[y \mapsto \varphi^{p,q}(x \otimes y)\right]_{q\in\mathbb{Z}} - (-1)^n \left[y \mapsto \varphi^{p+1,q}(d_X x \otimes y)\right]_{q\in\mathbb{Z}};$$

expanding the definition of $d^{p+n}_{\operatorname{Hom}^\bullet(Y,Z)}$ further, for the fixed pair (p, q) this becomes

$$y \mapsto d_Z\varphi^{p,q}(x \otimes y) - (-1)^{p+n}\varphi^{p,q+1}(x \otimes d_Y y) - (-1)^n\varphi^{p+1,q}(d_X x \otimes y).$$

This proves the assertion. □

We now return to the relation between K-projective and K-flat complexes.

Lemma 4.12.7 Let X be a K-projective complex of (A, R)-bimodules. If A is flat as a $\Bbbk$-module, then X_R is K-flat.

Let Y be a K-projective complex of (R, B)-bimodules. If B is flat as a $\Bbbk$-module, then ${}_RY$ is K-flat.

Proof It suffices to prove the first assertion. Take an acyclic complex Y of left R-modules. Regard every left A-module I (that is, every $(A, \Bbbk)$-bimodule) as a complex concentrated in degree zero. Proposition 4.12.6, with $B = \Bbbk$, gives

$$\operatorname{Hom}^\bullet\left(X \underset{R}{\otimes} Y,\ I\right) \simeq \operatorname{Hom}^\bullet\left(X, \operatorname{Hom}^\bullet(Y_\Bbbk, I_\Bbbk)\right).$$

Suppose that I is an injective left A-module. Then $I_{\Bbbk}$ is an injective $\Bbbk$-module, because the forgetful functor $A\text{-}\mathsf{Mod} \to \Bbbk\text{-}\mathsf{Mod}$ has the exact left adjoint $A \underset{\Bbbk}{\otimes} (\cdot)$; see Proposition 2.8.17 and [51, Corollary 6.6.8]. Hence the complex of $\Bbbk$-modules $\mathrm{Hom}^\bullet(Y_{\Bbbk}, I_{\Bbbk})$ is acyclic. It remains acyclic as a complex of (A,R)-bimodules, so the K-projectivity of X implies that $\mathrm{Hom}^\bullet(X \underset{R}{\otimes} Y, I)$ is acyclic.

Because I is injective, it is easy to see that, for every $n \in \mathbb{Z}$,

$$\mathrm{Hom}\left(\mathrm{H}^n\left(X \underset{R}{\otimes} Y\right), I\right) \simeq \mathrm{H}^{-n}\,\mathrm{Hom}^\bullet\left(X \underset{R}{\otimes} Y,\ I\right) = 0.$$

Since every left A-module embeds into an injective module, $X \underset{R}{\otimes} Y$ must be acyclic. This proves the assertion. □

If the reader is interested only in tensor products over commutative rings, that is, in the special case $A = R = B = \Bbbk$, or only in the case where $\Bbbk$ is a field, all the flatness assumptions on A or B above are automatically satisfied.

Corollary 4.12.8 If A (respectively, B) is flat as a $\Bbbk$-module, then $(A,R)\text{-}\mathsf{Mod}$ (respectively, $(R,B)\text{-}\mathsf{Mod}$) has enough K-flat complexes relative to R.

Taking $A = \Bbbk$ (respectively, $B = \Bbbk$) further shows that K-projective complexes of right (respectively, left) R-modules are always K-flat relative to R, and that $\mathsf{Mod}\text{-}R$ (respectively, $R\text{-}\mathsf{Mod}$) has enough K-flat complexes.

We now explain how to determine the derived functor of $\otimes_R$ by means of K-flat resolutions. To simplify notation, all localization functors Q will henceforth be suppressed.

Proposition 4.12.9 Define the full subcategory $\mathcal{F}_{A,R}$ of $\mathsf{K}((A,R)\text{-}\mathsf{Mod})$ by

$$\mathrm{Ob}(\mathcal{F}_{A,R}) = \{X : X_R \text{ K-flat}\};$$

and define $\mathcal{F}_{R,B}$ similarly. Both are saturated triangulated subcategories. We now determine subcategories of

$$\mathsf{K}((A,R)\text{-}\mathsf{Mod}) \times \mathsf{K}((R,B)\text{-}\mathsf{Mod}) \quad \text{that are } \otimes_R\text{-projective.}$$

(i) Suppose that $(A,R)\text{-}\mathsf{Mod}$ has enough K-flat complexes relative to R. Then $(\mathcal{F}_{A,R}, \mathsf{K}((R,B)\text{-}\mathsf{Mod}))$ is an $\otimes_R$-projective subcategory. Consequently, if Y is a complex of (R,B)-bimodules, then $\cdot \overset{\mathrm{L}}{\underset{R}{\otimes}} Y$ is the left derived functor of $\cdot \underset{R}{\otimes} Y$ and can be computed using K-flat resolutions.

(ii) Suppose that $(R,B)\text{-}\mathsf{Mod}$ has enough K-flat complexes relative to R. Then $(\mathsf{K}((A,R)\text{-}\mathsf{Mod}), \mathcal{F}_{R,B})$ is an $\otimes_R$-projective subcategory. Consequently, if X is a complex of (A, R)-bimodules, then $X \overset{\mathrm{L}}{\underset{R}{\otimes}} \cdot$ is the left derived functor of $X \underset{R}{\otimes} \cdot$ and can be computed using K-flat resolutions.

Proof Lemma 4.11.2 shows that $\mathcal{F}_{A,R}$ and $\mathcal{F}_{R,B}$ are saturated triangulated subcategories. For the remaining assertions, it suffices to treat (i). Recalling the relevant definitions in §4.7, the key is to verify the following properties:

- ◇ for a complex X in $\mathcal{F}_{A,R}$, if Y is acyclic, then $X \underset{R}{\otimes} Y$ is acyclic;
- ◇ for a fixed Y, if X is an acyclic complex in $\mathcal{F}_{A,R}$, then $X \underset{R}{\otimes} Y$ is acyclic.

The first property follows directly from the definition of K-flatness. Now consider the second. Since $X \underset{R}{\otimes} Y$ uses only the left R-action on Y, we may assume that $B = \Bbbk$. In this case, Corollary 4.12.8 ensures the existence of a K-flat resolution $Q \to Y$ in $\mathsf{C}(R\text{-}\mathsf{Mod})$. We thus obtain

$$\text{a distinguished triangle in } \mathsf{K}(R\text{-}\mathsf{Mod}) \quad Q \to Y \to N \xrightarrow{+1}, \quad N : \text{an acyclic complex},$$

$$\text{a distinguished triangle in } \mathsf{K}(A\text{-}\mathsf{Mod}) \quad X \underset{R}{\otimes} Q \to X \underset{R}{\otimes} Y \to X \underset{R}{\otimes} N \xrightarrow{+1} .$$

Since X is acyclic and Q is K-flat, $X \underset{R}{\otimes} Q$ is acyclic. On the other hand, since N is acyclic and X is K-flat relative to R, $X \underset{R}{\otimes} N$ is acyclic. Thus $X \underset{R}{\otimes} Y$ is acyclic as well.

Finally, the assertion about $\overset{\mathrm{L}}{\underset{R}{\otimes}}$ is simply a direct application of Proposition 4.6.10 (ii). □

Example 4.12.10 (Change of rings and $\overset{\mathrm{L}}{\otimes}$) For a homomorphism $R \to S$ of $\Bbbk$-algebras, regard S as an (S, R)-bimodule. Following [51, Corollary 6.6.8], consider the right-exact additive functor

$$_{R\to S}P := S \underset{R}{\otimes} (\cdot) : R\text{-}\mathsf{Mod} \to S\text{-}\mathsf{Mod}.$$

This functor has a left derived functor $\mathrm{L}_{R\to S}P$. Taking $A = S$ and $B = \Bbbk$ in Corollary 4.12.8 and Proposition 4.12.9 (ii) shows that flatness causes no difficulty here and that

$$\mathrm{L}_{R\to S}P \simeq S \overset{\mathrm{L}}{\underset{R}{\otimes}} (\cdot).$$

Change of rings is also transitive: given homomorphisms $R \to S \to T$, there is a canonical isomorphism

$$\mathrm{L}_{S\to T}P \circ \mathrm{L}_{R\to S}P \xrightarrow{\sim} \mathrm{L}_{R\to T}P.$$

Indeed, the displayed morphism comes from Theorem 4.6.5 on deriving a composite of functors. Moreover, since

$$X \underset{S}{\otimes} (S \underset{R}{\otimes} Y) \simeq X_R \underset{R}{\otimes} Y, \quad X \in \mathrm{Ob}(\mathsf{C}(\mathsf{Mod}\text{-}S)),\ Y \in \mathrm{Ob}(\mathsf{C}(R\text{-}\mathsf{Mod})),$$

the functor ${}_{R\to S}P$ preserves K-flat complexes. This ensures that the canonical morphism is an isomorphism.

In exactly the same way, one can define the left derived functor $\mathrm{L}P_{R\to S}$ of $P_{R\to S} = (\cdot) \underset{R}{\otimes} S : \mathsf{Mod}\text{-}R \to \mathsf{Mod}\text{-}S$, which is given by $(\cdot) \overset{\mathrm{L}}{\underset{R}{\otimes}} S$.

Proposition 4.12.11 If A or B is flat as a $\Bbbk$-module, there is a canonical 2-cell diagram

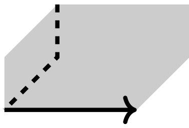
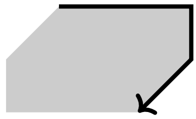
$$\begin{tikzcd} \mathsf{D}((A,R)\text{-}\mathsf{Mod}) \times \mathsf{D}((R,B)\text{-}\mathsf{Mod}) \arrow[r, "\overset{\mathrm{L}}{\underset{R}{\otimes}}"] \arrow[d, "\text{forgetful}"'] & \mathsf{D}((A,B)\text{-}\mathsf{Mod}) \arrow[d, "\text{forgetful}"] \\ \mathsf{D}(\mathsf{Mod}\text{-}R) \times \mathsf{D}(R\text{-}\mathsf{Mod}) \arrow[r, "\overset{\mathrm{L}}{\underset{R}{\otimes}}"'] \arrow[ur, Rightarrow, "\sim"] & \mathsf{D}(\mathsf{Ab}). \end{tikzcd}$$

Proof Properties of this kind are usually proved with the aid of resolutions. In brief, let X be a complex of (A,R)-bimodules and Y a complex of (R,B)-bimodules. By Proposition 4.12.9, we may assume that X_R (respectively, ${}_RY$) is K-flat. Then $X \underset{R}{\otimes} Y$ computes $\overset{\mathrm{L}}{\underset{R}{\otimes}}$ in both the first and second rows, giving the isomorphism $\Rightarrow$, which we provisionally denote by α.

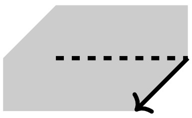
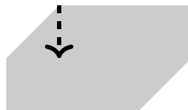
To characterize α abstractly, one may take the following viewpoint. Add another layer above the diagram under consideration, written schematically as

$$\begin{tikzcd} & \mathsf{K}(\cdots) \times \mathsf{K}(\cdots) \arrow[rr, "\underset{R}{\otimes}"] \arrow[dl, "\text{forgetful}"'] \arrow[dd] & & \mathsf{K}(\cdots) \arrow[dl, "\text{forgetful}"'] \arrow[dd] \\ \mathsf{K}(\cdots) \times \mathsf{K}(\cdots) \arrow[rr, "\underset{R}{\otimes}"] \arrow[dd] & & \mathsf{K}(\mathsf{Ab}) \arrow[dd] & \\ & \mathsf{D}(\cdots) \times \mathsf{D}(\cdots) \arrow[rr] \arrow[dl] & & \mathsf{D}(\cdots) \arrow[dl] \\ \mathsf{D}(\cdots) \times \mathsf{D}(\cdots) \arrow[rr] & & \mathsf{D}(\mathsf{Ab}) & \end{tikzcd}$$

All vertical arrows here are localization functors. We wish to fill the bottom face with $\begin{tikzcd}[sep=small] \bullet \arrow[r] \arrow[d] & \bullet \arrow[d] \\ \bullet \arrow[r] \arrow[ur, Rightarrow] & \bullet \end{tikzcd}$. The other faces of the cube are easy to handle: by the definition of a left derived bifunctor, the front and back faces have such fillings; by the universal property of localization, the left and right faces commute up to isomorphism; and the top face commutes by the definition of the tensor product. Tracing these faces gives a morphism between composite functors

 $\Rightarrow \cdots \Rightarrow$  $=: \sqsupset$.

Recall that $\mathsf{D}(\cdots) \times \mathsf{D}(\cdots) \to \mathsf{D}(\cdots)$ is absolute as a right Kan extension (Proposition 1.10.4). Therefore the composite

 is the right Kan extension of $\sqsupset$ along .

Applying the morphism into $\sqsupset$ and the universal property of the right Kan extension (Definition 1.7.1) gives the desired α.

Once the existence of α has been established, take K-flat complexes as in the first paragraph of the proof and compute all the functors in the upper layer. This proves that α is an isomorphism. □

From now on, arguments of this kind based on universal properties will often be omitted; instead, we shall verify statements directly using resolutions.

Corollary 4.12.12 The functor $\overset{\mathrm{L}}{\underset{R}{\otimes}}$ of Definition 4.12.2, under the assumption that A or B is flat as a $\Bbbk$-module, satisfies the canonical isomorphism

$$\mathrm{H}^{-n}\left(X \overset{\mathrm{L}}{\underset{R}{\otimes}} Y\right) \simeq \mathrm{Tor}_n^R(X, Y), \quad n \in \mathbb{Z},$$

where X is an (A, R)-bimodule, Y is an (R, B)-bimodule, and Tor_n^R is as in Definition–Proposition 3.14.7.

Proof The left-hand side is obtained by composing the diagram of Proposition 4.12.11 in the direction $\urcorner\!\downarrow$ whereas the right-hand side is obtained by composing it in the direction $\llcorner\!\rightarrow$. □

Proposition 4.12.13 (Associativity constraint) Let A, B, R, and S be $\Bbbk$-algebras. Consider the following two functors:

$$\mathsf{D}((A, R)\text{-}\mathsf{Mod}) \times \mathsf{D}((R, S)\text{-}\mathsf{Mod}) \times \mathsf{D}((S, B)\text{-}\mathsf{Mod}) \to \mathsf{D}((A, B)\text{-}\mathsf{Mod}),$$
$$(X, Y, Z) \mapsto (X \overset{\mathrm{L}}{\underset{R}{\otimes}} Y) \overset{\mathrm{L}}{\underset{S}{\otimes}} Z,$$
$$(X, Y, Z) \mapsto X \overset{\mathrm{L}}{\underset{R}{\otimes}} (Y \overset{\mathrm{L}}{\underset{S}{\otimes}} Z).$$

If A and B are both flat as $\Bbbk$-modules, there is a canonical isomorphism

$$a(X, Y, Z) : (X \overset{\mathrm{L}}{\underset{R}{\otimes}} Y) \overset{\mathrm{L}}{\underset{S}{\otimes}} Z \xrightarrow{\sim} X \overset{\mathrm{L}}{\underset{R}{\otimes}} (Y \overset{\mathrm{L}}{\underset{S}{\otimes}} Z).$$

Proof Fix Y. By Proposition 4.12.9, we may assume that X_R and ${}_RZ$ are K-flat. Then every $\overset{\mathrm{L}}{\underset{R}{\otimes}}$ can be computed at the level of complexes, and the assertion reduces to the associativity constraint for tensor products [51, Proposition 6.5.12]. □

Under these flatness assumptions, taking $\overset{\mathrm{L}}{\otimes}$ of three or more objects in different orders yields results that are mutually isomorphic through the associativity constraint. These isomorphisms also satisfy coherence, which can be formulated by imitating the pentagon axiom for monoidal categories; see [51, Definition 3.1.1]. Again, one takes K-flat resolutions and checks the assertion at the level of complexes. To keep the discussion brief, this book presents only the special case of a commutative ring.

Example 4.12.14 ($\overset{\mathrm{L}}{\otimes}$ over a commutative ring) Let R be a commutative ring. In the definition $\overset{\mathrm{L}}{\otimes} := \overset{\mathrm{L}}{\underset{R}{\otimes}}$, take every ring to be R, in particular $\Bbbk = R$. Then $(R, R)\text{-}\mathsf{Mod} = R\text{-}\mathsf{Mod}$, and all the preceding flatness assumptions are automatically satisfied. Consequently, $\mathrm{H}^{-n}(X \overset{\mathrm{L}}{\otimes} Y) \simeq \mathrm{Tor}_n^R(X, Y)$ for all R-modules X and Y.

This makes $\mathsf{D}(R\text{-}\mathsf{Mod})$ a monoidal category with respect to $\overset{\mathrm{L}}{\otimes}$ in the sense of [51, §3.1]. Its unit object is R itself, and its associativity constraint comes from $a = (a(X, Y, Z))_{X,Y,Z}$ in Proposition 4.12.13. Since R is a flat R-module, it is K-flat (Proposition 4.12.4); hence both $R \overset{\mathrm{L}}{\otimes} (\cdot)$ and $(\cdot) \overset{\mathrm{L}}{\otimes} R$ can be computed at the level of complexes, making all the monoidal-category axioms easy to prove. Furthermore, $\mathsf{D}(R\text{-}\mathsf{Mod})$ is a symmetric monoidal category in the sense of [51, Definition 3.3.1]. Its braiding (or commutativity constraint) $c(X, Y) : X \overset{\mathrm{L}}{\otimes} Y \xrightarrow{\sim} Y \overset{\mathrm{L}}{\otimes} X$ comes from the commutativity constraint at the level of complexes, and its axioms are also checked there. In view of Proposition 3.5.6, swapping the superscripts in the total complex causes no difficulty here.

Example 4.12.15 (Tor-algebra) Continue with the setting of Example 4.12.14, with R commutative. Identify the complexes of R-modules X, Y, Z, W with their images in the derived category. For every $(p, q) \in \mathbb{Z}^2$, there is a canonical morphism

$$\begin{aligned}\mathrm{H}^{-p}\left(X \overset{\mathrm{L}}{\otimes} Y\right) \otimes \mathrm{H}^{-q}\left(Z \overset{\mathrm{L}}{\otimes} W\right) &\to \mathrm{H}^{-p-q}\left((X \overset{\mathrm{L}}{\otimes} Y) \overset{\mathrm{L}}{\otimes} (Z \overset{\mathrm{L}}{\otimes} W)\right) \\ &\simeq \mathrm{H}^{-p-q}\left((X \overset{\mathrm{L}}{\otimes} Z) \overset{\mathrm{L}}{\otimes} (Y \overset{\mathrm{L}}{\otimes} W)\right) \to \mathrm{H}^{-p-q}\left((X \otimes Z) \overset{\mathrm{L}}{\otimes} (Y \otimes W)\right),\end{aligned}$$

where the first part applies Proposition 4.7.8 (take $F = \otimes$), the second uses the associativity and commutativity constraints on $(\mathsf{D}(R\text{-}\mathsf{Mod}), \overset{\mathrm{L}}{\otimes})$, and the third comes from the canonical morphism accompanying $\overset{\mathrm{L}}{\otimes}$ as a left derived bifunctor:

$$QX_1 \overset{\mathrm{L}}{\otimes} QX_2 \to Q(X_1 \otimes X_2), \quad X_1, X_2 \in \mathrm{Ob}(\mathsf{K}(R\text{-}\mathsf{Mod})),$$

For precision, the localization functor $Q : \mathsf{K}(R\text{-}\mathsf{Mod}) \to \mathsf{D}(R\text{-}\mathsf{Mod})$ has been written explicitly again here.

- ◇ The composite of the three canonical morphisms above gives an "external product" for the Tor functor:

$$\mathrm{Tor}_p^R(X, Y) \otimes \mathrm{Tor}_q^R(Z, W) \to \mathrm{Tor}_{p+q}^R(X \otimes Z, Y \otimes W).$$

- ◇ Take $Z = X$ and $W = Y$ to be R-algebras (concentrated in degree 0). Their multiplications give homomorphisms of R-modules $X \otimes X \to X$ and $Y \otimes Y \to Y$. Composing the external product with these homomorphisms gives

$$\mathrm{Tor}_p^R(X, Y) \otimes \mathrm{Tor}_q^R(X, Y) \to \mathrm{Tor}_{p+q}^R(X, Y),$$

called the "internal product" for the Tor functor. This makes $\bigoplus_{n\in\mathbb{Z}} \mathrm{Tor}_n^R(X, Y)$ a $\mathbb{Z}$-graded R-algebra in the sense of [51, Definition 7.4.1], called the **Tor-algebra**. It has nonzero components only in nonnegative degrees, and its degree-zero component is the subalgebra $\mathrm{Tor}_0^R(X, Y) = X \otimes Y$, whose definition may be found in [51, Definition 7.3.1]. The associativity required for the product can be verified by reducing it to the associativity constraint for $\overset{\mathrm{L}}{\otimes}$ and to the associativity of multiplication on X and Y separately.

- ◇ Suppose further that X and Y are commutative R-algebras. In this case, the Tor-algebra is also an "anticommutative" $\mathbb{Z}$-graded algebra in the sense of [51, Definition 7.4.3][9]: for all $p, q \in \mathbb{Z}$,

$$a \in \mathrm{Tor}_p^R(X, Y),\ b \in \mathrm{Tor}_q^R(X, Y) \implies ab = (-1)^{pq} ba.$$

Since both $X \otimes X \to X$ and $Y \otimes Y \to Y$ are commutative, where does the sign come from? Its source is the definition of the external product. Let C_1 and C_2 be two copies of $X \overset{\mathrm{L}}{\otimes} Y$. Then ab and ba differ through the isomorphism $C_1 \overset{\mathrm{L}}{\otimes} C_2 \xrightarrow{\sim} C_2 \overset{\mathrm{L}}{\otimes} C_1$, namely the commutativity constraint for $\overset{\mathrm{L}}{\otimes}$. If C_1 and C_2 are represented by K-flat complexes, this isomorphism of total complexes is exactly the isomorphism $r_{C_1^\bullet \otimes C_2^\bullet}$ of Proposition 3.5.6; on the component of bidegree (p, q), it produces the sign $(-1)^{pq}$.

[9] The literal term is somewhat misleading, because this property is a natural manifestation of commutativity. Graded-commutative is more appropriate, as in Example 7.1.11.

Return to general $\Bbbk$-algebras. To discuss the adjunction between $\overset{\mathrm{L}}{\otimes}$ and RHom, one more lemma is needed. Below, $\mathrm{RHom}_{(A,B)}$ denotes RHom in (A,B)-Mod; analogous notation is used for other categories of bimodules (or of left or right modules).

Lemma 4.12.16 Fix $\Bbbk$-algebras A, B, and R. Suppose that B is flat as a $\Bbbk$-module. Then the forgetful functor from (A,B)-Mod to A-Mod preserves K-injective complexes, and RHom_A has a canonical lift to a functor

$$\mathsf{D}\left((A,R)\text{-}\mathsf{Mod}\right)^{\mathrm{op}} \times \mathsf{D}\left((A,B)\text{-}\mathsf{Mod}\right) \to \mathsf{D}\left((R,B)\text{-}\mathsf{Mod}\right),$$

in other words, there is a canonical 2-cell diagram

$$\begin{tikzcd}
\mathsf{D}\left((A,R)\text{-}\mathsf{Mod}\right)^{\mathrm{op}} \times \mathsf{D}\left((A,B)\text{-}\mathsf{Mod}\right) \arrow[r] \arrow[d, "\text{forgetful}"'] & \mathsf{D}\left((R,B)\text{-}\mathsf{Mod}\right) \arrow[d, "\text{forgetful}"] \arrow[dl, Rightarrow, "\sim"'] \\
\mathsf{D}(A\text{-}\mathsf{Mod})^{\mathrm{op}} \times \mathsf{D}(A\text{-}\mathsf{Mod}) \arrow[r, "\mathrm{RHom}_A"'] & \mathsf{D}(\mathsf{Ab}).
\end{tikzcd}$$

Likewise, if A is flat as a $\Bbbk$-module, then RHom_B has an analogous lift.

Proof Because B is a flat $\Bbbk$-module, the functor $Y \mapsto {}_AY$ has the exact left adjoint $(\cdot) \underset{\Bbbk}{\otimes} B$, and therefore preserves K-injective complexes.

Next take $X \in \mathrm{Ob}(\mathsf{K}((A,R)\text{-}\mathsf{Mod}))$ and $Y \in \mathrm{Ob}(\mathsf{K}(A,B)\text{-}\mathsf{Mod})$, with Y K-injective. By the preceding step, $\mathrm{Hom}^\bullet\left({}_AX, {}_AY\right)$ computes $\mathrm{RHom}_A({}_AX, {}_AY)$, while at the same time it is a complex of (R,B)-bimodules. This completes the lift. $\square$

Theorem 4.12.17 (Adjunction between RHom and $\overset{\mathbf{L}}{\otimes}$) Let A, B, and R be $\Bbbk$-algebras. Below, X, Y, and Z denote objects of $\mathsf{D}((A,R)\text{-}\mathsf{Mod})$, $\mathsf{D}((R,B)\text{-}\mathsf{Mod})$, and $\mathsf{D}((A,B)\text{-}\mathsf{Mod})$, respectively. Suppose that B is flat as a $\Bbbk$-module. Then there is a canonical isomorphism in $\mathsf{D}(\Bbbk\text{-}\mathsf{Mod})$

$$\mathrm{RHom}_{(A,B)}\left(X \overset{\mathrm{L}}{\underset{R}{\otimes}} Y, Z\right) \xrightarrow{\sim} \mathrm{RHom}_{(R,B)}\left(Y, \mathrm{RHom}_A\left({}_AX, {}_AZ\right)\right).$$

If instead A is assumed flat as a $\Bbbk$-module, there is a canonical isomorphism in $\mathsf{D}(\Bbbk\text{-}\mathsf{Mod})$

$$\mathrm{RHom}_{(A,B)}\left(X \overset{\mathrm{L}}{\underset{R}{\otimes}} Y, Z\right) \xrightarrow{\sim} \mathrm{RHom}_{(A,R)}\left(X, \mathrm{RHom}_B\left(Y_B, Z_B\right)\right).$$

Proof Consider the first equality. Suppose that Y is K-projective and Z is K-injective. Lemma 4.12.7 ensures that ${}_RY$ is K-flat, while Lemma 4.12.16 ensures that ${}_AZ$ remains K-injective. Every displayed RHom and $\overset{\mathrm{L}}{\underset{R}{\otimes}}$ can therefore be computed at the level of complexes. The desired isomorphism then follows from Proposition 4.12.6. $\square$

Example 4.12.18 (Adjunction for change of rings) In the setting of Example 4.12.10 (take a homomorphism $R \to S$ of $\Bbbk$-algebras, $A = S$, $B = \Bbbk$, and $X = S$), the first equality in Theorem 4.12.17 becomes the canonical isomorphism

$$\begin{gathered} \mathrm{RHom}_S\left(\mathrm{L}_{R\to S}P(Y), Z\right) \simeq \mathrm{RHom}_R\left(Y, {}_RZ\right), \\ Y \in \mathrm{Ob}(\mathsf{D}(R\text{-}\mathsf{Mod})), \quad Z \in \mathrm{Ob}(\mathsf{D}(S\text{-}\mathsf{Mod})). \end{gathered}$$

The only point still requiring explanation is $\mathrm{RHom}_S({}_SS, {}_SZ) \simeq {}_RZ$, which follows immediately from the isomorphism at the level of complexes

$$\mathrm{Hom}^\bullet\left({}_SS, \cdot\right) \simeq {}_R(\cdot) : \mathsf{C}(S\text{-}\mathsf{Mod}) \to \mathsf{C}(R\text{-}\mathsf{Mod}).$$

If, furthermore, both R and S are commutative rings and $\Bbbk := R$, then RHom_S (respectively, RHom_R) takes values in $\mathsf{D}(S\text{-}\mathsf{Mod})$ (respectively, $\mathsf{D}(R\text{-}\mathsf{Mod})$). The adjunction isomorphism can then be written as the canonical isomorphism in $\mathsf{D}(R\text{-}\mathsf{Mod})$:

$${}_R\,\mathrm{RHom}_S\left(\mathrm{L}_{R\to S}P(Y), Z\right) \simeq \mathrm{RHom}_R\left(Y, {}_RZ\right).$$

Everything can be verified at the level of complexes by taking K-injective and K-projective resolutions. The case of right modules is similar.

Exercises

1. Let $(\mathcal{D}, T, \mathcal{H})$ be a pretriangulated category (respectively, a triangulated category). Define a new family of triangles $\mathcal{H}^-$ by

$$[X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} TX] \in \mathcal{H} \iff [X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{-h} TX] \in \mathcal{H}^-.$$

(i) Prove that $(\mathcal{D}, T, \mathcal{H}^-)$ is also a pretriangulated category (respectively, a triangulated category).

(ii) Prove that $(\mathcal{D}, T, \mathcal{H})$ and $(\mathcal{D}, T, \mathcal{H}^-)$ are equivalent as pretriangulated categories.

2. Let $(\mathcal{D}, T, \mathcal{H})$ be a pretriangulated category. Prove that every monomorphism (respectively, epimorphism) $u : X \to Y$ in $\mathcal{D}$ has a left inverse (respectively, a right inverse). Use this to show that a pretriangulated category that is also abelian must be split (Definition 2.7.6).

Hint Place the monomorphism u in a distinguished triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} TX$. From $u \circ T^{-1}w = 0$, deduce that $w = 0$, and then apply Proposition 4.2.6.

3. Apply the preceding exercise to give an example of an abelian category $\mathcal{A}$ for which neither $\mathsf{K}(\mathcal{A})$ nor $\mathsf{D}(\mathcal{A})$ is an abelian category.

Hint〉 Consider the morphism $f : \mathbb{Z}/p^2\mathbb{Z} \twoheadrightarrow \mathbb{Z}/p\mathbb{Z}$ in $\mathcal{A} = \mathsf{Ab}$, where p is prime. If $\mathsf{K}(\mathcal{A})$ were abelian, consider the epi–mono factorizations of $\mathsf{K}f$ and $\mathrm{H}^0(\mathsf{K}f)$, then express both as projections onto direct summands and inclusion morphisms to obtain a contradiction.

4. Let $\mathcal{A}$ be an abelian category. Prove that $\mathsf{D}(\mathcal{A})$ is an abelian category if and only if $\mathcal{A}$ is split.

Hint〉 For the "only if" direction, embed $\mathcal{A}$ in $\mathsf{D}(\mathcal{A})$ and apply the preceding exercises. For the "if" direction, show that when $\mathcal{A}$ is split, taking cohomology gives an equivalence from $\mathsf{D}(\mathcal{A})$ to $\mathcal{A}^{\mathbb{Z}}$.

5. It is often said that the octahedral axiom (TR5) for a triangulated category replaces the isomorphism theorem $\dfrac{X/Z}{Y/Z} \simeq X/Y$ in an abelian category, where $X \supset Y \supset Z$ (Theorem 2.6.8 (ii)). Explain explicitly what this statement means.

6. (K_0 of a pretriangulated category) For every pretriangulated category $(\mathcal{D}, T, \mathcal{H})$, define $\mathrm{K}_0(\mathcal{D})$ to be the quotient of the free $\mathbb{Z}$-module generated by $\mathrm{Ob}(\mathcal{D})$ by the relations

$$\text{there is a distinguished triangle } X \to Y \to Z \xrightarrow{+1} \implies [X] - [Y] + [Z] = 0,$$

where $[X] \in \mathrm{K}_0(\mathcal{D})$ denotes the equivalence class containing $X \in \mathrm{Ob}(\mathcal{D})$.

(i) Prove that $[0] = 0$, $[TX] = -[X]$, and $[X \oplus Y] = [X] + [Y]$.

(ii) Show that every triangulated functor $F : \mathcal{D} \to \mathcal{D}'$ induces a canonical homomorphism $\mathrm{K}_0(\mathcal{D}) \to \mathrm{K}_0(\mathcal{D}')$.

(iii) Consider a cohomological functor $H : \mathcal{D} \to \mathcal{A}$, where $\mathcal{A}$ is an abelian category. Suppose that, for every X, $|n| \gg_X 0 \implies H(T^n X) = 0$. Define a natural homomorphism $\mathrm{K}_0(\mathcal{D}) \to \mathrm{K}_0(\mathcal{A})$.

Hint〉 Send $[X]$ to $\sum_n (-1)^n [H(T^n X)]$. Apply Lemma 2.9.9 (iv).

(iv) Consider the special case $\mathcal{D} := \mathsf{D}^{\mathrm{b}}(\mathcal{A})$. Give mutually inverse homomorphisms

$$\begin{tikzcd} \mathrm{K}_0(\mathcal{A}) \arrow[r, shift left] & \mathrm{K}_0(\mathsf{D}^{\mathrm{b}}(\mathcal{A})) \arrow[l, shift left] \\ {[A]} \arrow[r, mapsto] & {[A]} \\ \sum_n (-1)^n [\mathrm{H}^n(X)] & {[X]} \arrow[l, mapsto] \end{tikzcd}$$

where $A \in \mathrm{Ob}(\mathcal{A})$, also regarded as a complex concentrated in degree 0. Hint〉 One composite is plainly the identity; the other requires truncating the complex.

7. Let $X \in \mathrm{Ob}(\mathsf{D}(\mathcal{A}))$. Prove that

$$X \in \mathsf{D}^{\leq 0}(\mathcal{A}) \iff \forall Z \in \mathrm{Ob}(\mathsf{D}^{\geq 1}(\mathcal{A})),\ \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Z) = 0,$$
$$X \in \mathsf{D}^{\geq 0}(\mathcal{A}) \iff \forall Z \in \mathrm{Ob}(\mathsf{D}^{\leq -1}(\mathcal{A})),\ \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(Z, X) = 0.$$

Hint〉 Apply Proposition 4.5.3 (i), together with the adjoint pairs and distinguished triangle in Proposition 4.4.13.

8. In the setting of Theorem 4.6.5, suppose that F' satisfies $F'(\mathrm{Ob}(\mathcal{N}')) \subset \mathrm{Ob}(\mathcal{N}'')$ (regard this as a kind of "exactness"). If $\mathcal{D}$ has an F-injective or an F-projective subcategory, prove that the canonical morphism

$$\mathrm{R}_{\mathcal{N}}^{\mathcal{N}''}(F'F) \to \left(\mathrm{R}_{\mathcal{N}'}^{\mathcal{N}''}F'\right)\left(\mathrm{R}_{\mathcal{N}}^{\mathcal{N}'}F\right)$$

or

$$\left(\mathrm{L}_{\mathcal{N}'}^{\mathcal{N}''}F'\right)\left(\mathrm{L}_{\mathcal{N}}^{\mathcal{N}'}F\right) \to \mathrm{L}_{\mathcal{N}}^{\mathcal{N}''}(F'F)$$

is an isomorphism.

9. In the setting of Theorem 4.6.5, show that the following diagrams of functors and morphisms between them commute (the notation $\mathcal{N}$ and so forth is suppressed):

$$\begin{tikzcd} \mathrm{R}(F''F'F) \arrow[r] \arrow[d] & \mathrm{R}(F''F')\mathrm{R}F \arrow[d] \\ \mathrm{R}F''\mathrm{R}(F'F) \arrow[r] & \mathrm{R}F''\mathrm{R}F'\mathrm{R}F \end{tikzcd} \qquad \begin{tikzcd} \mathrm{L}(F''F'F) & \mathrm{L}(F''F')\mathrm{L}F \arrow[l] \\ \mathrm{L}F''\mathrm{L}(F'F) \arrow[u] & \mathrm{L}F''\mathrm{L}F'\mathrm{L}F. \arrow[l] \arrow[u] \end{tikzcd}$$

10. For $F = \mathrm{Hom}$, express explicitly the canonical morphism of Proposition 4.7.8 in the form $\mathrm{Ext}^{q-p}(X,Y) \to \mathrm{Hom}\left(\mathrm{H}^p(X), \mathrm{H}^q(Y)\right)$.

11. Reformulate the bijection in Theorem 4.5.13 as follows. From an n-extension $0 \to Y \to E^1 \to \cdots \to E^n \to X \to 0$, construct $t^{-1}b \in \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y[n])$ as follows:

$$\begin{tikzcd} X \arrow[d, "b"'] \\ W \\ Y[n] \arrow[u, "t\text{:quasi-isomorphism}"] \end{tikzcd} \qquad \Bigg| \qquad \begin{tikzcd} & & & X \arrow[d, "\mathrm{id}"] \\ E^1 \arrow[r] & E^2 \arrow[r] & \cdots \arrow[r] & X \\ Y \arrow[u] \end{tikzcd}$$

This gives $t^{-1}b \in \mathrm{Hom}_{\mathsf{D}(\mathcal{A})}(X, Y[n])$. Prove that this bijection differs by the factor $(-1)^n$ from the original bijection based on as^{-1}. In the special case $n = 1$, use the $t^{-1}b$ version to derive the corresponding version of Proposition 4.5.14 (the one involving the image of id_Y).

Hint〉 Prove that $ta + (-1)^{n-1}bs : Z \to W$ is canonically null-homotopic. For the second part, note the minus sign in Proposition 4.4.7.

12. Suppose that all objects S, T in an abelian category $\mathcal{A}$ satisfy $\mathrm{Ext}^2_{\mathcal{A}}(S,T) = 0$. Prove that every $X \in \mathrm{Ob}(\mathsf{D}^{\mathrm{b}}(\mathcal{A}))$ is isomorphic to $\bigoplus_n \mathrm{H}^n(X)[-n]$. Show that $\mathcal{A} := \mathbb{Z}\text{-}\mathsf{Mod}$ satisfies this condition.

Hint〉 We may assume that X is a bounded complex. Begin with $n \gg 0$. Apply Theorem 4.5.13 and Proposition 4.5.15 to the 2-extension $0 \to \ker(d^n) \to X^n \xrightarrow{d^n} X^{n+1} \to \mathrm{coker}(d^n) \to 0$, then modify X step by step until $d^\bullet = 0$, throughout up to isomorphism.

13. Prove that if $\mathcal{A}$ has enough injective objects, $F : \mathcal{A} \to \mathcal{A}'$ is a left-exact additive functor, and $\mathrm{R}F$ has finite dimension, then the homomorphism $\mathrm{K}_0(\mathcal{A}) \simeq \mathrm{K}_0(\mathsf{D}^{\mathrm{b}}(\mathcal{A})) \to \mathrm{K}_0(\mathsf{D}^{\mathrm{b}}(\mathcal{A}')) \simeq \mathrm{K}_0(\mathcal{A}')$ induced by the triangulated functor $\mathrm{R}F$ is determined by

$$[A] \mapsto \sum_n (-1)^n \left[\mathrm{R}^n F(A)\right], \quad A \in \mathrm{Ob}(\mathcal{A}).$$

14. Let R be a ring. Prove that a filtered direct limit $\varinjlim$ of K-flat complexes in $\mathsf{C}(R\text{-}\mathsf{Mod})$ is again K-flat.

15. Let R be a ring. If a complex P of R-modules is K-flat and every P^n is a flat R-module, then P is called **strongly K-flat**.

(i) Prove that for every $X \in \mathrm{Ob}(\mathsf{C}(R\text{-}\mathsf{Mod}))$ there is a strongly K-flat complex P together with a quasi-isomorphism $P \to X$. Hint〉 Use the dual of the argument for the existence of K-injective resolutions; see §3.15.

(ii) Write $\mathcal{SF}$ for the full subcategory of $\mathsf{K}(R\text{-}\mathsf{Mod})$ formed by the strongly K-flat complexes. Express $\mathsf{D}(R\text{-}\mathsf{Mod})$ as the localization of $\mathcal{SF}$ at the quasi-isomorphisms.

16. (P. Berthelot, A. Ogus) Let f be a non-zero-divisor in a commutative ring R. For every R-module M, write $M[f] := \{m \in M : fm = 0\}$. A module M with $M[f] = 0$ is called ***f*-torsion-free**. First show that an f-torsion-free M can be embedded as a submodule of $M[\frac{1}{f}] = M \underset{R}{\otimes} R[\frac{1}{f}]$. Thus $f^m M$ makes sense for every $m \in \mathbb{Z}$.

Let X be a complex of f-torsion-free R-modules. Using the observation above, define a subcomplex $\eta_f X$ of $X^{\bullet} \underset{R}{\otimes} R[\frac{1}{f}]$ by

$$(\eta_f X)^n := \left\{x \in f^n X^n : d_X^n(x) \in f^{n+1} X^{n+1}\right\}, \quad n \in \mathbb{Z}.$$

(i) Prove that multiplication degree by degree by f gives an isomorphism $\eta_f(X[1]) \xrightarrow{\sim} (\eta_f X)[1]$.

(ii) Show that if $f, g \in R$ are both non-zero-divisors and every X^n is fg-torsion-free, then $\eta_f \eta_g X = \eta_{fg} X$.

(iii) Give a canonical isomorphism $\mathrm{H}^n(X)/\mathrm{H}^n(X)[f] \xrightarrow{\sim} \mathrm{H}^n(\eta_f X)$; thus the effect of η_f is to correct f-torsion. Deduce that if X and Y are complexes of f-torsion-free R-modules and $\alpha : X \to Y$ is a quasi-isomorphism, then α induces a quasi-isomorphism $\eta_f(\alpha) : \eta_f X \to \eta_f Y$.

(iv) Prove that in this way η_f induces an additive functor $\mathrm{L}\eta_f : \mathsf{D}(R\text{-}\mathsf{Mod}) \to \mathsf{D}(R\text{-}\mathsf{Mod})$ whose restriction to $\mathcal{SF}$ from the preceding exercise equals η_f.

(v) Show that $\mathrm{L}\eta_f$ is not a triangulated functor. Thus it is not a left derived functor in the sense of Definition 4.6.1, although it remains a right Kan extension.

Hint〉 Take $R = \mathbb{Z}$ and $f = p$ prime. Verify that $\mathrm{L}\eta_p(\mathbb{Z}/p\mathbb{Z}) = 0$, whereas $\mathrm{L}\eta_p(\mathbb{Z}/p^2\mathbb{Z}) \simeq \mathbb{Z}/p\mathbb{Z}$.

17. Continuing the preceding exercise, give a canonical morphism $\mathrm{L}\eta_f X \overset{\mathrm{L}}{\underset{R}{\otimes}} \mathrm{L}\eta_f Y \to \mathrm{L}\eta_f \left(X \overset{\mathrm{L}}{\underset{R}{\otimes}} Y \right)$ and show that this morphism is symmetric in X and Y in the appropriate sense. In the language of [51, Remark 3.1.8], $\mathrm{L}\eta_f$ is in fact a right-lax monoidal functor with respect to $\overset{\mathrm{L}}{\underset{R}{\otimes}}$.

Chapter 5 Spectral Sequences

Spectral sequences are classical algebraic tools that continually find new uses, and their roots in topology run especially deep. Some textbooks and lecture notes even use them as a basic tool of homological algebra to establish some of the fundamental properties in Chapter 2 and Chapter 3.

Spectral sequences were introduced by J. Leray in the 1940s and gradually assumed their modern form through the work of J.-L. Koszul, H. Cartan, and others. Their chief motivation was the study of the homology groups of fiber bundles. In algebraic terms, the original problem amounts to understanding the homology (or cohomology) of a chain complex (or complex) X. Let us take the case of a complex as our example. Although X itself may be difficult to grasp, it can be studied through a suitable filtration $\cdots \supset \mathrm{F}^p X \supset \mathrm{F}^{p+1} X \supset \cdots$. The filtration $\mathrm{F}^\bullet X$ induces a filtration on every $\mathrm{H}^n(X)$. Information about $\mathrm{F}^\bullet X$ and its subquotients can then be used to approximate, step by step, each subquotient $\mathrm{gr}^p \mathrm{H}^n(X)$ of the filtered cohomology. Spectral sequences are the art of organizing such successive approximations.

A complex may also be generalized to a differential object (X, d) in an abelian category (Definition 5.2.1), leading to the abstract definition of a spectral sequence in Definition 5.2.6. A spectral sequence is a sequence of differential objects $\mathscr{E} = (E_r, d_r)_r$ such that the cohomology of each "page," $\mathrm{H}(E_r, d_r)$, is the next page E_{r+1}. Adding a grading and allowing the differential d to have a degree gives the notions of graded spectral sequences E_r^p and bigraded spectral sequences $E_r^{p,q}$ (Definition 5.2.8). Concretely, the differentials in a bigraded spectral sequence have the form

$$d_r^{p,q} : E_r^{p,q} \to E_r^{p+r,q-r+1},$$
$$E_r = (E_r^{p,q})_{(p,q)\in\mathbb{Z}^2}, \quad d_r = (d_r^{p,q})_{(p,q)\in\mathbb{Z}^2}.$$

A filtered complex gives a filtered differential object and hence naturally produces a bigraded spectral sequence. This is the form of spectral sequence that occurs most often in applications. Every notion has a corresponding version for chain complexes and their homology.

This chapter begins with W. Massey's theory of exact couples. Exact couples are another device for producing spectral sequences, and their scope is wider than that of filtered differential objects. After introducing filtrations and graded structures in §5.1, we give the general definition of a spectral sequence in §5.2, together with its graded and bigraded versions and such basic terminology as degeneration, boundedness, and the limiting page. At the end of that section, Remark 5.2.13 explains how to read exact sequences from the edges of a spectral sequence, an important technique.

Exact couples are defined in §5.3. Next, §5.4 explains how an exact couple produces a graded spectral sequence from a filtered differential object, and §5.5 goes on to explain how a filtered complex produces a bigraded spectral sequence. That section also treats the central notion of convergence and the Classical Convergence Theorem 5.5.5. Some of the verifications are slightly technical, but there is no fundamental difficulty. Readers may first work in a concrete abelian category such as Ab to simplify the arguments.

A double complex gives rise to a filtered complex in two ways, according as the total complex is filtered by the horizontal or the vertical coordinate. The corresponding spectral sequences are denoted by $\mathscr{E}_{\mathrm{I}}$ and $\mathscr{E}_{\mathrm{II}}$, respectively. Their details are the subject of §5.6. We also give standard applications, including the derivation of balanced bifunctors (Example 5.6.4), the spectral sequence of hyperderived functors (Example 5.6.5), and the Grothendieck spectral sequence (Theorem 5.6.6). The Cartan–Eilenberg resolutions introduced earlier (Theorem 3.11.10) play a role here.

The Grothendieck spectral sequence concerns the derivation of a composite of functors and is particularly common in geometry. As an example involving right derived functors, consider left-exact additive functors between abelian categories $\mathcal{A} \xrightarrow{F} \mathcal{A}' \xrightarrow{F'} \mathcal{A}''$, where $\mathcal{A}$ and $\mathcal{A}'$ have enough injective objects. If F sends injective objects to F'-acyclic objects (Convention 3.12.8), there is a canonical strongly convergent bigraded spectral sequence

$$E_2^{p,q} = (\mathrm{R}^p F')(\mathrm{R}^q F)(X) \Rightarrow \mathrm{R}^{p+q}(F'F)(X), \quad X \in \mathrm{Ob}(\mathcal{A}).$$

The derivation of composite functors can also be handled at the level of derived categories; see Theorem 4.6.5. Besides being formally elementary and easy to compute, the spectral-sequence version can provide additional information in concrete situations, such as exact sequences in low degrees.

The final section of the chapter, §5.7, briefly introduces multiplicative structures on spectral sequences. This aspect has received attention since the earliest history of spectral sequences. Proposition 5.7.6 shows that a filtered differential graded algebra gives rise to a spectral sequence with a canonical multiplicative structure; a deeper treatment is left to specialized monographs.

As an exercise in applications, the Lyndon–Hochschild–Serre spectral sequence, which is fundamental in group cohomology, can be obtained both as an application of the Grothendieck spectral sequence and concretely from a filtered complex. The latter approach also equips it with a canonical multiplicative structure. This topic will be discussed in §6.9 in the future.

阅读提示

This chapter requires only a knowledge of abelian categories and complexes. Apart from a few simple definitions, it does not depend on the material in Chapter 4; the two chapters can be read independently. Since the spectral sequence of a filtered differential object can be written down directly, the theory of exact couples is not essential for that construction (see Proposition 5.4.2 and the discussion that follows). Readers may therefore consider temporarily skipping §5.3. Moreover, among the later sections of this book, only §6.9 requires a knowledge of spectral sequences.

5.1 Filtrations and Graded Structures

We encountered graded objects in Definition 3.1.2. A $\mathbb{Z}$-graded object in an arbitrary category $\mathcal{A}$ is simply called a **graded object**; by definition, these are the objects $(X^p)_{p\in\mathbb{Z}}$ of $\mathcal{A}^{\mathbb{Z}}$. A $\mathbb{Z}^2$-graded object is simply called a **bigraded object**; by definition, these are the objects $(X^{p,q})_{(p,q)\in\mathbb{Z}^2}$ of $\mathcal{A}^{\mathbb{Z}\times\mathbb{Z}}$. Morphisms between graded (or bigraded) objects are written as $(f^p)_p$ (or $(f^{p,q})_{p,q}$), as usual, and are composed degree by degree.

If $\mathcal{A}$ is an abelian category, then $\mathcal{A}^{\mathbb{Z}}$ and $\mathcal{A}^{\mathbb{Z}\times\mathbb{Z}}$ are also abelian categories; kernels, cokernels, and all other operations are computed degree by degree.

The category $\mathcal{A}^{\mathbb{Z}}$ has a shift autoequivalence T determined by $(TX)^p = X^{p+1}$ and $(Tf)^p = f^{p+1}$. More generally, $\mathcal{A}^{\mathbb{Z}^n}$ has a family of shift autoequivalences $T_1, \ldots, T_n$

corresponding to shifts along the n coordinates. These shifts commute strictly: $T_iT_j = T_jT_i$.

Within the framework of this chapter, graded structures arise primarily from filtrations.

Definition 5.1.1 Let X be an object of an additive category $\mathcal{A}$.

- ◇ A **decreasing filtration** $\mathrm{F}^\bullet X$ is a sequence of subobjects

$$\cdots \supset \mathrm{F}^p X \supset \mathrm{F}^{p+1} X \supset \cdots \quad (p \in \mathbb{Z}).$$

- ◇ If there exist $a \leq b$ such that $\mathrm{F}^a X = X$ and $\mathrm{F}^b X = 0$, the filtration is called **finite**.

- ◇ Increasing filtrations $\mathrm{F}_\bullet X$ and their finiteness are defined similarly.

Decreasing and increasing filtrations can be distinguished by the position of the index p. When no confusion is likely, both are simply called filtrations. Decreasing filtrations are customary in the study of complexes and cohomology, whereas increasing filtrations are customary for chain complexes and homology. This chapter mainly uses decreasing filtrations.

All filtered objects $(X, \mathrm{F}^\bullet X)$ form a category $\mathrm{Fil}^\bullet(\mathcal{A})$. A morphism $f : (X, \mathrm{F}^\bullet X) \to (Y, \mathrm{F}^\bullet Y)$ is a morphism $f : X \to Y$ that preserves the filtrations; that is, for every p, it restricts to a morphism $\mathrm{F}^p X \to \mathrm{F}^p Y$. The category $\mathrm{Fil}_\bullet(\mathcal{A})$ is defined similarly.

For every X, a decreasing filtration $\mathrm{F}^\bullet X$ (or increasing filtration $\mathrm{F}_\bullet X$) gives a graded object $(\mathrm{F}^p X)_{p\in\mathbb{Z}}$ (or $(\mathrm{F}_p X)_{p\in\mathbb{Z}}$). This defines a functor $\mathrm{Fil}^\bullet(\mathcal{A}) \to \mathcal{A}^{\mathbb{Z}}$ (or $\mathrm{Fil}_\bullet(\mathcal{A}) \to \mathcal{A}^{\mathbb{Z}}$), called the Rees construction.

If $\mathcal{A}$ is an abelian category, there is another way to construct a graded object from a filtration.

Definition 5.1.2 Let $\mathcal{A}$ be an abelian category, and consider a filtration $\mathrm{F}^\bullet X$ (or $\mathrm{F}_\bullet X$) on an object X. Define the graded object $\mathrm{gr}\, X$ by

$$\mathrm{gr}^p X := \mathrm{F}^p X / \mathrm{F}^{p+1} X \quad (\text{or} \quad \mathrm{gr}_p X := \mathrm{F}_p X / \mathrm{F}_{p-1} X).$$

This defines a functor $\mathrm{gr} : \mathrm{Fil}^\bullet(\mathcal{A}) \to \mathcal{A}^{\mathbb{Z}}$ (or $\mathrm{gr} : \mathrm{Fil}_\bullet(\mathcal{A}) \to \mathcal{A}^{\mathbb{Z}}$).

To extract information about X from $(\mathrm{gr}^p X)_p$, we need at least $\bigcup_p \mathrm{F}^p X = X$ and $\bigcap_p \mathrm{F}^p X = 0$, using the notation of Convention 2.6.3. To make the first condition meaningful and useful, we also want $\varinjlim_{p\to-\infty}$ to be exact in $\mathcal{A}$, or else we may simply require that there be an M such that $\mathrm{F}^M X = X$. We begin with some related notions.

Definition 5.1.3 Let $\mathcal{A}$ be an abelian category and let X be an object with a filtration $\mathrm{F}^\bullet X$.

- If $\bigcap_p \mathrm{F}^p X = 0$, the filtration is called **separated**.
- If $\bigcup_p \mathrm{F}^p X = X$ and either of the following conditions holds, the filtration is called **exhaustive**:
  (a) $\mathcal{A}$ has exact countable filtered direct limits $\varinjlim$; more precisely, $\varinjlim : \mathcal{A}^{(\mathbb{Z}_{\geq 0}, \leq)} \to \mathcal{A}$ exists and is exact[1)];
  (b) there is an $M \in \mathbb{Z}$ such that $\mathrm{F}^M X = X$.
- If the family of canonical morphisms $X \twoheadrightarrow X/\mathrm{F}^p X$ induces an isomorphism $X \xrightarrow{\sim} \varprojlim_p X/\mathrm{F}^p X$, the filtration is called **complete**.

Increasing filtrations $\mathrm{F}_\bullet X$ are treated in the same way.

The notion of completeness is inspired by the case of topological groups; see [51, §4.10].

A finite filtration is automatically separated, exhaustive, and complete; this is the principal case needed in this chapter. Observe that if there is an N such that $\mathrm{F}^N X = 0$, then $\mathrm{F}^\bullet X$ is separated and complete. Moreover, since the kernel of $X \to \varprojlim_p X/\mathrm{F}^p X$ is $\bigcap_p \mathrm{F}^p X$, completeness implies separatedness.

The image of an exhaustive filtration is still exhaustive; this follows by interpreting $\bigcup_p$ as $\sum_p$ and then applying Lemma 2.6.6 (ii). Moreover, an exhaustive filtration satisfies

$$\begin{gathered} \varinjlim_{p \to -\infty} \mathrm{F}^p X \xrightarrow{\sim} \bigcup_p \mathrm{F}^p X = X, \\ K = K \cap \left(\bigcup_p \mathrm{F}^p X \right) = \bigcup_p \left(K \cap \mathrm{F}^p X \right), \end{gathered} \tag{5.1.1}$$

where K is any subobject of X. Under condition (a) for an exhaustive filtration, these assertions follow readily from Proposition 2.10.3; under condition (b), they are immediate.

Proposition 5.1.4 Let $\mathcal{A}$ be an abelian category, and let $f : (X, \mathrm{F}^\bullet X) \to (Y, \mathrm{F}^\bullet Y)$ be a morphism in $\mathrm{Fil}^\bullet(\mathcal{A})$.

(i) Suppose that $\mathrm{F}^\bullet X$ is separated and exhaustive. If $\mathrm{gr}(f)$ is a monomorphism, then f is a monomorphism.

1) If $\mathcal{A}$ is a Grothendieck category, then (a) holds automatically.

(ii) Suppose that $\mathrm{F}^\bullet X$ is complete and exhaustive, while $\mathrm{F}^\bullet Y$ is separated and exhaustive. If $\mathrm{gr}(f)$ is an isomorphism, then f is also an isomorphism, and in this case $\mathrm{F}^\bullet Y$ is complete as well.

The corresponding statements hold for increasing filtrations.

Proof For each $p \in \mathbb{Z}$, consider the following commutative diagram with exact rows:

$$\begin{tikzcd} 0 \arrow[r] & \mathrm{F}^{p+1}X \arrow[r] \arrow[d, "f"'] & \mathrm{F}^p X \arrow[r] \arrow[d, "f"] & \mathrm{gr}^p X \arrow[r] \arrow[d, "\mathrm{gr}^p(f)"] & 0 \\ 0 \arrow[r] & \mathrm{F}^{p+1}Y \arrow[r] & \mathrm{F}^p Y \arrow[r] & \mathrm{gr}^p Y \arrow[r] & 0 \end{tikzcd}$$

Write K^p and C^p for the kernel and cokernel of $\mathrm{F}^p X \xrightarrow{f} \mathrm{F}^p Y$.

For (i), apply Theorem 2.3.3 to the diagram above. This gives $K^{p+1} = K^p$. In particular, $K^p \subset \bigcap_{\ell \geq p} \mathrm{F}^\ell X = 0$, so the restriction of f to every $\mathrm{F}^p X$ is a monomorphism. To prove that f itself is a monomorphism, it remains only to use

$$\ker(f) \overset{\because (5.1.1)}{=\!=\!=} \bigcup_p (\ker(f) \cap \mathrm{F}^p X) = 0.$$

For (ii), the diagram gives both $K^{p+1} = K^p$ and $C^{p+1} \xrightarrow{\sim} C^p$. For each $\ell > p$, consider the commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & \mathrm{F}^\ell X \arrow[r] \arrow[d, "f"] & \mathrm{F}^p X \arrow[r] \arrow[d, "f"] & \mathrm{F}^p X/\mathrm{F}^\ell X \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & \mathrm{F}^\ell Y \arrow[r] & \mathrm{F}^p Y \arrow[r] & \mathrm{F}^p Y/\mathrm{F}^\ell Y \arrow[r] & 0 \end{tikzcd}$$

Another application of Theorem 2.3.3 gives the isomorphism induced by f

$$\mathrm{F}^p X/\mathrm{F}^\ell X \xrightarrow{\sim} \mathrm{F}^p Y/\mathrm{F}^\ell Y.$$

Taking $\varinjlim_{p: p<\ell}$ on both sides and using (5.1.1) gives $X/\mathrm{F}^\ell X \xrightarrow{\sim} Y/\mathrm{F}^\ell Y$. Taking $\varprojlim_\ell$ now gives the commutative diagram

$$\begin{tikzcd} X \arrow[r, "\sim"] \arrow[d, "f"'] & \varprojlim_\ell X/\mathrm{F}^\ell X \arrow[d, "\wr"] \\ Y \arrow[r, hook] & \varprojlim_\ell Y/\mathrm{F}^\ell Y \end{tikzcd}$$

It follows that both f and $Y \to \varprojlim_\ell Y/\mathrm{F}^\ell Y$ are isomorphisms. $\square$

5.2 General definition of spectral sequences

We begin with the general definition of a differential object. This involves an additive category with translation $(\mathcal{A}, T)$ as in Definition 4.1.1, as well as morphisms with degree as in Definition 4.1.2.

Definition 5.2.1 A **differential object** over an additive category with translation $(\mathcal{A}, T)$ is data (X, d), where $X \in \mathrm{Ob}(\mathcal{A})$ and $d : X \xrightarrow{+1} X$ satisfies $d^2 = 0$.

- ◇ A morphism from (X, d) to (X', d') is a morphism $\alpha : X \to X'$ satisfying $d'\alpha = \alpha d$.
- ◇ All differential objects form a category $(\mathcal{A}, T)_d$; we also write $\mathcal{A}_d := (\mathcal{A}, \mathrm{id}_{\mathcal{A}})_d$.

If $\mathcal{A}$ is an Ab-category (or a $\Bbbk$-Mod-category, where $\Bbbk$ is a commutative ring), then so is $(\mathcal{A}, T)_d$. The forgetful functor $(\mathcal{A}, T)_d \to \mathcal{A}$ sends (X, d) to X.

Proposition 5.2.2 The forgetful functor $(\mathcal{A}, T)_d \to \mathcal{A}$ creates all $\varinjlim$ and $\varprojlim$ (Definition 1.5.2).

Proof Replay the proof of Lemma 3.1.4. □

It follows that if $\mathcal{A}$ is an Abelian category, then $(\mathcal{A}, T)_d$ is also Abelian and $(\mathcal{A}, T)_d \to \mathcal{A}$ is exact.

In the case of an Abelian category, one distinguishing feature of a differential object is that it has homology or cohomology.

Definition 5.2.3 Let $\mathcal{A}$ be an Abelian category. We define the additive functor

$$\mathrm{H} : (\mathcal{A}, T)_d \to \mathcal{A}, \qquad \mathrm{H}(X, d) := \ker(d)/\operatorname{im}(T^{-1}d).$$

We call a triangular diagram in an Abelian category $\mathcal{A}$ $\begin{tikzcd} A \arrow[rr] & & B \arrow[dl] \\ & C \arrow[ul, "+m"] & \end{tikzcd}$ **exact** if

$$\cdots T^{k-m}C \to T^kA \to T^kB \to T^kC \to T^{k+m}A \cdots$$

is an exact sequence extending infinitely in both directions. More general triangular diagrams with degrees are treated in the same way.

Lemma 5.2.4 Let $(\mathcal{A}, T)$ be an Abelian category with translation. If $0 \to (X', d') \to (X, d) \to (X'', d'') \to 0$ is a short exact sequence in $(\mathcal{A}, T)_d$, then the following diagram is exact at every vertex:

$$\begin{tikzcd} \mathrm{H}(X', d') \arrow[rr] & & \mathrm{H}(X, d). \arrow[dl] \\ & \mathrm{H}(X'', d'') \arrow[ul, "+1"] & \end{tikzcd}$$

Proof In $\mathcal{A}$, consider the sequence of complexes

$$0 \to (T^n X', T^n d')_n \to (T^n X, T^n d)_n \to (T^n X'', T^n d'')_n \to 0.$$

It is exact degree by degree, hence is a short exact sequence. Apply Proposition 3.6.4 to it. □

Example 5.2.5 (Complexes as differential objects) For an additive category $\mathcal{A}$, consider $\mathcal{A}^{\mathbb{Z}}$ together with the translation functor $T : (X^n)_n \mapsto (X^{n+1})_n$. By the discussion in §3.1, objects of $(\mathcal{A}^{\mathbb{Z}}, T)_d$ are also called differential graded objects, and there is an isomorphism of categories $\mathsf{C}(\mathcal{A}) \simeq (\mathcal{A}^{\mathbb{Z}}, T)_d$. If $\mathcal{A}$ is Abelian, then plainly

$$\mathrm{H}(X, d) = (\mathrm{H}^p(X))_{p \in \mathbb{Z}}\,.$$

We can now give the definition of a spectral sequence. To simplify the discussion, we first consider the case without translation, or equivalently the case of morphisms without degree; in other words, we take $T = \mathrm{id}$.

Definition 5.2.6 (J. Leray, J.-L. Koszul) Let $\mathcal{A}$ be an Abelian category and $a \in \mathbb{Z}$. A **spectral sequence** starting at a is data $\mathscr{E} = (E_r, d_r)_{r \in \mathbb{Z}_{\geq a}}$ together with $(t_r)_{r \geq a+1}$, where

- ◇ $(E_r, d_r) \in \mathrm{Ob}(\mathcal{A}_d)$;
- ◇ $t_r : \mathrm{H}(E_{r-1}, d_{r-1}) \xrightarrow{\sim} E_r$ for $r \geq a + 1$.

The data a and $(t_r)_r$ are often omitted from the notation.

A morphism of spectral sequences $\mathscr{E} \to \mathscr{E}'$ is a family of morphisms in $\mathcal{A}_d$ $\varphi_r : (E_r, d_r) \to (E'_r, d'_r)$ satisfying $t_r \,\mathrm{H}(\varphi_{r-1}) = \varphi_r t_r$ for $r \geq a + 1$.

The pair (E_r, d_r) is customarily called the ***r*-th page** of the spectral sequence $\mathscr{E}$; computing E_{r+1} from E_r amounts to turning the page. Abstractly, the precise page at which a spectral sequence starts is immaterial, but in applications it depends on the particular context.

If d_r in a spectral sequence is viewed as a morphism $\mathrm{H}(E_{r-1}, d_{r-1}) \to \mathrm{H}(E_{r-1}, d_{r-1})$, then both $\ker(d_r)$ and $\mathrm{im}(d_r)$ correspond to subobjects of $\ker(d_{r-1})$ that contain $\mathrm{im}(d_{r-1})$. For simplicity, suppose the spectral sequence starts at $a = 0$. Define Z_1 (respectively B_1) to be $\ker(d_0)$ (respectively $\mathrm{im}(d_0)$), then define Z_2 (respectively B_2)

as the inverse image of $\ker(d_1)$ (respectively $\mathrm{im}(d_1)$) under $E_0 \supset Z_1 \twoheadrightarrow E_1$, and so on. Iteration yields

$$\begin{gathered} 0 =: B_0 \subset B_1 \subset B_2 \subset \cdots \subset Z_2 \subset Z_1 \subset Z_0 =: E_0, \\ Z_r \leftrightarrow \ker(d_{r-1}), \quad B_r \leftrightarrow \mathrm{im}(d_{r-1}), \quad Z_r/B_r \simeq E_r. \end{gathered} \tag{5.2.1}$$

- ◇ If $Z_\infty := \bigcap_r Z_r$ and $B_\infty := \bigcup_r B_r$ exist, the **limit** of the spectral sequence $\mathscr{E}$ is defined to be $E_\infty := Z_\infty / B_\infty$.
- ◇ If there is an r such that $r' \geq r \implies d_{r'} = 0$, then $\mathscr{E}$ is said to **degenerate at the E_r page**. In this case clearly $Z_r = Z_{r+1} = \cdots$ and $B_r = B_{r+1} = \cdots$, so $Z_r = Z_\infty$ and $B_r = B_\infty$; consequently, $E_r = E_\infty$.

When discussing homologically graded spectral sequences, we shall instead use the notation $(E^r, d^r)_r$ and B^r, Z^r.

Proposition 5.2.7 Let $\varphi_r : \mathscr{E} \to \mathscr{E}'$ be a morphism of spectral sequences. If φ_r is an isomorphism, then φ_s is also an isomorphism for $s \geq r$; if the limits exist, then $\varphi_\infty : E_\infty \to E'_\infty$ is also an isomorphism.

Proof The isomorphism φ_r induces an isomorphism $\mathrm{H}(E_r, d_r) \to \mathrm{H}(E'_r, d'_r)$, which is precisely φ_{r+1}. Without loss of generality, suppose that the spectral sequences start at r. Then $(\varphi_s)_{s \geq r}$ is an isomorphism of spectral sequences, and the assertion concerning φ_∞ is immediate. $\square$

More generally, for an Abelian category with translation $(\mathcal{A}, T)$, the definition of a spectral sequence extends to the case $(E_r, d_r) \in \mathrm{Ob}((\mathcal{A}, T^{a_r})_d)$, where $a_1, a_2, \ldots$ is a sequence of integers. The definitions of $\mathrm{H}(E_r, d_r)$ and B_r, Z_r above remain unchanged.

In the generalization we shall need later, we may go even further and equip $\mathcal{A}$ with a family of mutually commuting automorphisms $T_1, \ldots, T_n$, with

$$d_r : E_r \xrightarrow{+\vec{a}_r} E_r, \qquad \vec{a}_r = (a_{r,1}, \ldots, a_{r,n}) \in \mathbb{Z}^n.$$

Following this idea, we introduce the two most commonly used versions, both of which involve graded structures.

Definition 5.2.8 (Spectral sequences: graded and bigraded versions) Let $\mathcal{A}$ be an Abelian category. Cohomologically graded and cohomologically bigraded spectral sequences, both written $\mathscr{E} = (E_r, d_r)_r$, are defined as follows.

Version	E_r	d_r	Degree of morphism
Graded	$(E_r^p)_{p \in \mathbb{Z}} \in \mathrm{Ob}(\mathcal{A}^{\mathbb{Z}})$	$(d_r^p)_p$	$d_r^p : E_r^p \xrightarrow{+r} E_r^{p+r}$
Bigraded	$(E_r^{p,q})_{(p,q) \in \mathbb{Z}^2} \in \mathrm{Ob}(\mathcal{A}^{\mathbb{Z} \times \mathbb{Z}})$	$(d_r^{p,q})_{(p,q)}$	$d_r^{p,q} : E_r^{p,q} \xrightarrow{+(r,-r+1)} E_r^{p+r,q-r+1}$

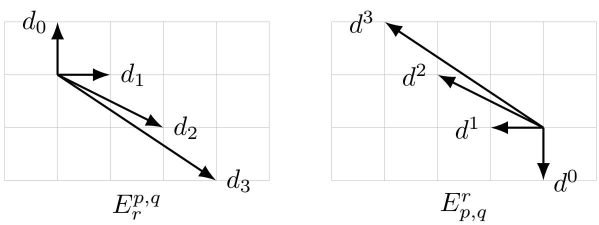
In both cases $(d_r)^2 = 0$ is required, with composition understood as the composition of morphisms with degrees, and the data include a specified isomorphism $\mathrm{H}(E_{r-1}, d_{r-1}) \xrightarrow{\sim} E_r$.

Dually, a homologically graded (or bigraded) spectral sequence is defined as the following data $(E^r, d^r)_{r\geq 1}$, again with $(d^r)^2 = 0$ and a specified isomorphism $\mathrm{H}(E^r, d^r) \simeq E^{r+1}$:

Version	E^r	d^r	Degree of morphism
Graded	$(E^r_p)_{p\in\mathbb{Z}} \in \mathrm{Ob}(\mathcal{A}^{\mathbb{Z}})$	$(d^r_p)_p$	$d^r_p : E^r_p \xrightarrow{-r} E^r_{p-r}$
Bigraded	$(E^r_{p,q})_{(p,q)\in\mathbb{Z}^2} \in \mathrm{Ob}(\mathcal{A}^{\mathbb{Z}\times\mathbb{Z}})$	$(d^r_{p,q})_{(p,q)}$	$d^r_{p,q} : E^r_{p,q} \xrightarrow{+(-r,r-1)} E^r_{p-r,q+r-1}$

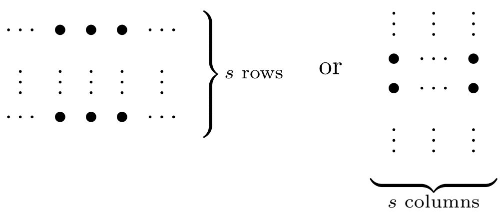
Morphisms between these spectral sequences are defined in the usual way and must be compatible with the specified isomorphisms.

In (p, q) coordinates, the directions of d_r and d^r in a bigraded spectral sequence are as follows.



In either version, following the preceding pattern, we may define the objects $B^p_r \subset Z^p_r$, $B^{p,q}_r \subset Z^{p,q}_r$, E^p_∞, $E^{p,q}_\infty$, as well as $B^r_p \subset Z^r_p$, $B^r_{p,q} \subset Z^r_{p,q}$, E^∞_p, $E^\infty_{p,q}$, and so forth.

Example 5.2.9 (The finite-width case) In many common situations, the nonzero terms of a bigraded spectral sequence are concentrated in a horizontal (or vertical) strip of width s, where $s \in \mathbb{Z}_{\geq 0}$, as in the following figure:



Inspection of the arrow directions shows that $r > s$ (or $r \geq s$) implies $d^{p,q}_r = 0$ for all p, q; in this case the spectral sequence degenerates at the E_r page. The homological case is similar.

Definition 5.2.10 Let $\mathscr{E}$ be a cohomological (or homological) bigraded spectral sequence, and let $r \in \mathbb{Z}$. If for every $n \in \mathbb{Z}$ there are at most finitely many $(p,q) \in \mathbb{Z}^2$ satisfying $p+q=n$ and $E_r^{p,q} \neq 0$ (or $E^r_{p,q} \neq 0$), then E_r (or E^r) is called **bounded**.

It follows from $E_{r+1} \simeq \mathrm{H}(E_r, d_r)$ that $E_r^{p,q} = 0 \implies E_{r+1}^{p,q} = 0$. In particular, if E_r is bounded, then E_{r+1} is bounded. The homological case is similar.

Proposition 5.2.11 Let $\mathscr{E}$ be a cohomological (or homological) bigraded spectral sequence for which E_r (or E^r) is bounded. For every $(p,q) \in \mathbb{Z}^2$, there is an $r(p,q)$ such that, whenever $r \geq r(p,q)$, one has $E_r^{p,q} = E_{r+1}^{p,q}$ (or $E^r_{p,q} = E^{r+1}_{p,q}$). Consequently, the limit of $\mathscr{E}$ exists and satisfies $E_r^{p,q} = E_\infty^{p,q}$ (or $E^r_{p,q} = E^\infty_{p,q}$).

Proof It suffices to discuss the case of E_r. Fix n. By boundedness, for all (p,q) with $p+q=n$ and all sufficiently large r, we have $E_r^{p-r,q+r-1} = 0$, hence $B_r^{p,q} = 0$. Similarly, for sufficiently large r, we have $E_r^{p+r,q-r+1} = 0$, hence $Z_r^{p,q} = E_r^{p,q}$. This proves the assertion. $\square$

Definition 5.2.12 Let $\mathscr{E}$ be a cohomological (or homological) bigraded spectral sequence, and let $r \in \mathbb{Z}$. If $E_r^{p,q} \neq 0$ (or $E^r_{p,q} \neq 0$) implies $p, q \geq 0$, then E_r (or E^r) is said to lie in the first quadrant. The other quadrants are defined similarly.

An E_r lying in the first or third quadrant is clearly bounded. It follows from $E_{r+1} \simeq \mathrm{H}(E_r, d_r)$ that if E_r lies in any given quadrant, then E_{r+1} lies in the same quadrant. The homological case is similar.

Remark 5.2.13 (Edge calculations) Spectral sequences lying in the first quadrant occur especially often. For the cohomological version, the edge terms $E_r^{\bullet,0}$ and $E_r^{0,\bullet}$ have special properties. Fix $p, q \geq 1$. By carefully examining the direction of d_r and recalling that $E_{r+1} \simeq \mathrm{H}(E_r, d_r)$, one verifies that

$$\begin{aligned} E_2^{p,0} \twoheadrightarrow E_3^{p,0} \twoheadrightarrow \cdots \twoheadrightarrow E_{p+1}^{p,0} &= E_\infty^{p,0} && (\because \text{ all outgoing } d \text{ are } 0), \\ E_\infty^{0,q} = E_{q+2}^{0,q} \hookrightarrow E_{q+1}^{0,q} \hookrightarrow \cdots &\hookrightarrow E_2^{0,q} \hookrightarrow E_1^{0,q} && (\because \text{ all incoming } d \text{ are } 0). \end{aligned} \tag{5.2.2}$$

A first-quadrant homological bigraded spectral sequence has the corresponding properties:

$$\begin{aligned} E^\infty_{p,0} = E^{p+1}_{p,0} \hookrightarrow E^p_{p,0} \hookrightarrow \cdots &\hookrightarrow E^3_{p,0} \hookrightarrow E^2_{p,0} && (\because \text{ all incoming } d \text{ are } 0), \\ E^1_{0,q} \twoheadrightarrow E^2_{0,q} \twoheadrightarrow \cdots \twoheadrightarrow E^{q+2}_{0,q} &= E^\infty_{0,q} && (\because \text{ all outgoing } d \text{ are } 0). \end{aligned} \tag{5.2.3}$$

The morphisms arising from (5.2.2) or (5.2.3) are collectively called **edge morphisms**. The preceding discussion gives the following exact sequence ($p \geq 2$):

$$0 \to E_\infty^{0,p-1} \to E_p^{0,p-1} \xrightarrow{d} E_p^{p,0} \to E_\infty^{p,0} \to 0, \tag{5.2.4}$$

in which every morphism except d is an edge morphism. There is also a homological version of this exact sequence ($p \geq 2$):

$$0 \to E^{\infty}_{p,0} \to E^{p}_{p,0} \xrightarrow{d} E^{p}_{0,p-1} \to E^{\infty}_{0,p-1} \to 0. \tag{5.2.5}$$

Beginning readers should be sure to draw the directions of the morphisms d involving these terms, determine when they are 0, and thereby verify all the assertions above.

5.3 Exact couples

Several commonly used kinds of spectral sequence arise from exact couples, a discovery due to W. Massey. To grasp the essential idea, we first return to the case without degrees.

Definition 5.3.1 (Exact couple) An exact couple over an Abelian category $\mathcal{A}$ is an exact diagram in $\mathcal{A}$

$$\begin{tikzcd} D \arrow[rr, "i"] & & D \arrow[dl, "j"] \\ & E \arrow[ul, "k"] & \end{tikzcd}$$

Morphisms between data $\mathscr{C} := (D, E, i, j, k)$ are defined in the evident way.

Given an exact couple $\mathscr{C} = (D, E, i, j, k)$, set $d := jk : E \to E$. Then $dj = jkj = 0$ and $kd = kjk = 0$, whence $d^2 = 0$. The exactness conditions readily give the following factorizations:

$$\begin{tikzcd} D \arrow[r, "j"] \arrow[d, two heads, "i"'] & \ker(d) \arrow[d, two heads] \\ i(D) \arrow[r, "\exists!\, j'"'] & \mathrm{H}(E, d) \end{tikzcd} \qquad \begin{tikzcd} \ker(d) \arrow[r, "k"] \arrow[d, two heads] & i(D) \\ \mathrm{H}(E, d) \arrow[ur, "\exists!\, k'"'] & \end{tikzcd} \tag{5.3.1}$$

Also define $E' := \mathrm{H}(E, d)$, $D' := i(D)$, and $i' := i|_{i(D)} : D' \to D'$.

Lemma 5.3.2 Given an exact couple $\mathscr{C} = (D, E, i, j, k)$, the data constructed above,

$$\mathscr{C}' = (D', E', i', j', k') : \quad \begin{tikzcd} D' \arrow[rr, "i'"] & & D' \arrow[dl, "j'"] \\ & E' \arrow[ul, "k'"] & \end{tikzcd}$$

also form an exact couple.

Proof Examining (5.3.1) and using the exactness conditions, one readily verifies that

$$\begin{aligned}
\ker(i') &= i(D) \cap \ker(i) = \ker(j) \cap k(E) \\
&= \operatorname{im}\left[\ker(d) \xrightarrow{k} i(D)\right] = \operatorname{im}(k'), \\
\ker(j') &= i\left(j^{-1}(jk(E))\right) \\
&= i\left(k(E) + \ker(j)\right) = i\left(i(D)\right) = \operatorname{im}(i'), \\
\ker(k') &= (\ker(k) \cap \ker(d))/\operatorname{im}(d) = (j(D) \cap \ker(jk))/\operatorname{im}(d) \\
&= j(D)/\operatorname{im}(d) = \operatorname{im}(j').
\end{aligned}$$

Thus the new triangular diagram is still exact. □

Iterating this procedure gives a sequence of exact couples $(\mathscr{C}_{(r)})_{r\geq 1}$, with $\mathscr{C}_{(1)} = \mathscr{C}$ and $\mathscr{C}_{(r+1)} = \mathscr{C}'_{(r)}$ for $r \geq 1$. Writing $\mathscr{C}_{(r)} = (D_r, E_r, i_r, j_r, k_r)$, the sequence $(E_r, d_r := j_r k_r)_{r\geq 1}$ is a spectral sequence.

Lemma 5.3.3 For the spectral sequence $(E_r, d_r)_{r\geq 1}$ obtained from an exact couple $\mathscr{C} = (D, E, i, j, k)$, define the family of subobjects $\overline{B}_r \subset \overline{Z}_r$ of $E = E_1$ according to (5.2.1). Bars are used because the indexing starts at $r = 1$; see Proposition 5.3.4 below. For $r \geq 0$,

$$\overline{B}_{r+1} = j\left(\ker(i^r)\right) \subset k^{-1}\left(\operatorname{im}(i^r)\right) = \overline{Z}_{r+1};$$

$$\overline{B}_\infty = j\left(\bigcup_{r\geq 2} \ker(i^r)\right) \subset k^{-1}\left(\bigcap_{r\geq 2} \operatorname{im}(i^r)\right) = \overline{Z}_\infty,$$

provided that the relevant $\bigcup_r$ and $\bigcap_r$ exist. More precisely, the exact couple $\mathscr{C}_{(r+1)}$ is canonically isomorphic to

$$\begin{tikzcd}
i^r D \arrow[rr, "i"] & & i^r D \arrow[dl, "\overline{j}_{r+1}"] \\
& \dfrac{k^{-1}(i^r D)}{j(\ker(i^r))} \arrow[ul, "\overline{k}_{r+1}"] &
\end{tikzcd}$$

where $\overline{k}_{r+1}$ is induced by $k : E \to D$, while $\overline{j}_{r+1}$ is characterized by the commutative diagram

$$\begin{tikzcd}
D \arrow[r, "i^r"] \arrow[d, "j"'] & i^r D \arrow[d, "\overline{j}_{r+1}"] \\
k^{-1}(i^r D) \arrow[r, two heads] & \dfrac{k^{-1}(i^r D)}{j(\ker(i^r))}.
\end{tikzcd}$$

Proof The description of the exact couple $\mathscr{C}_{(r+1)}$ can be proved recursively. The case $r = 0$ is trivial ($i^0 = \mathrm{id}$), and the descriptions of $\overline{B}_{r+1}$ and $\overline{Z}_{r+1}$ are immediate consequences. Since the details are somewhat routine, they are omitted. The assertion $\overline{B}_\infty \subset \overline{Z}_\infty$ follows from Lemma 2.6.6 (ii). □

We next explain how to construct an exact couple from a differential object.

Proposition 5.3.4 Let $\alpha : (D, d) \to (D, d)$ be a monomorphism in $\mathcal{A}_d$, and write d_α for the morphism induced by d on $\operatorname{coker}(\alpha)$. Then there is an exact couple

$$\begin{tikzcd} \mathrm{H}(D,d) \arrow[rr, "i=\mathrm{H}(\alpha)"] & & \mathrm{H}(D,d) \arrow[dl, "j"] \\ & \mathrm{H}(\operatorname{coker}(\alpha), d_\alpha) \arrow[ul, "k"] & \end{tikzcd}$$

Take the inverse images of $\overline{B}_{r+1} \subset \overline{Z}_{r+1}$ in $E_0 := \operatorname{coker}(\alpha)$, and denote them by

$$0 =: B_0 \subset B_1 \subset B_2 \subset \cdots \subset Z_2 \subset Z_1 \subset Z_0 := \operatorname{coker}(\alpha).$$

For every $r \geq 0$:

- ◇ B_{r+1} is the image of $(\alpha^r)^{-1}(dD) \subset D$ in $\operatorname{coker}(\alpha)$;
- ◇ Z_{r+1} is the image of $d^{-1}(\alpha^{r+1}D) \subset D$ in $\operatorname{coker}(\alpha)$;
- ◇ $d_{r+1} \in \operatorname{End}(Z_{r+1}/B_{r+1})$ is induced by $d^{-1}(\alpha^{r+1}D) \xrightarrow{d} \alpha^{r+1}D \xleftarrow[\sim]{\alpha^{r+1}} D$.

In particular, there are canonical isomorphisms

$$E_{r+1} \simeq \frac{Z_{r+1}}{B_{r+1}} \simeq \frac{d^{-1}(\alpha^{r+1}D) + \alpha D}{(\alpha^r)^{-1}(dD) + \alpha D}, \qquad r \in \mathbb{Z}_{\geq 1}.$$

Proof The exact couple is obtained by applying Lemma 5.2.4 (with $T = \mathrm{id}_{\mathcal{A}}$) to the short exact sequence in $\mathcal{A}_d$

$$0 \to (D, d) \xrightarrow{\alpha} (D, d) \to (\operatorname{coker}(\alpha), d_\alpha) \to 0.$$

In particular, j is induced by $D \twoheadrightarrow \operatorname{coker}(\alpha)$, while k is essentially the connecting morphism in the long exact sequence. The description of B_{r+1} is simply the version of Lemma 5.3.3 lifted to E_0. For Z_{r+1}, besides that lemma, the key point is the description of the connecting morphism $k : \mathrm{H}(\operatorname{coker}(\alpha), d_\alpha) \to \mathrm{H}(D, d)$. A careful review of the constructions in (2.3.2) and (3.6.2) shows that this morphism is induced by the middle vertical morphism $\overline{d}$ in the following commutative diagram with exact rows:

$$\begin{tikzcd} & \operatorname{coker}(d) \arrow[r] \arrow[d, "\overline{d}"'] & \operatorname{coker}(d) \arrow[r] \arrow[d, "\overline{d}"'] & \operatorname{coker}(d_\alpha) \arrow[r] \arrow[d, "\overline{d_\alpha}"] & 0 \\ 0 \arrow[r] & \ker(d) \arrow[r, "\alpha"'] & \ker(d) \arrow[r] & \ker(d_\alpha) & \end{tikzcd}$$

and $\overline{d}$ itself comes from $d : D \to D$. The remaining verifications are lengthy but straightforward; the reader may begin with the case where $\mathcal{A}$ is a category of modules. □

Thus the spectral sequence provided by a differential object can be made to start at 0 by setting $E_0 := \operatorname{coker}(\alpha)$ and $d_0 := d_\alpha$.

For an Abelian category with translation $(\mathcal{A}, T)$, morphisms with degrees can still be composed and notions such as exactness remain available, so the theory of exact couples readily extends to the case in which i, j, k have degrees. For example, in the situation to be discussed in §5.4, we can consider the exact couple

$$\begin{tikzcd}[column sep=small] D \arrow[rr, "-1"] & & D \arrow[dl, "+1"] \\ & E \arrow[ul] & \end{tikzcd} \overset{\text{unrolled}}{=\!=\!=} \left[\begin{tikzcd}[column sep=small] \cdots & TD \arrow[dl] \arrow[rr] & & D \arrow[dl] \arrow[rr] & & T^{-1}D \arrow[dl] \arrow[rr] & & T^{-2}D \arrow[dl] & \cdots \\ \cdots & & TE \arrow[ul] & & E \arrow[ul] & & T^{-1}E \arrow[ul] & & \cdots \arrow[ul] \end{tikzcd} \right].$$

Denote this exact couple by $\mathscr{C}$. The natural extension of Lemma 5.3.3 to the case with degrees shows that $\mathscr{C}_{(r)}$ has the form $\begin{tikzcd}[column sep=small] \bullet \arrow[rr, "-1"] & & \bullet \arrow[dl, "+r"] \\ & \bullet \arrow[ul] & \end{tikzcd}$. Thus $d_r = j_r k_r$ is a morphism of degree r. The resulting spectral sequence is therefore graded.

5.4 The spectral sequence of a filtered differential object

Throughout this section, we work with an Abelian category with translation $(\mathcal{A}, T)$.

Definition 5.4.1 An object $(X, d) \in \operatorname{Ob}((\mathcal{A}, T)_d)$ equipped with a filtration $(\mathrm{F}^\bullet X, d_{\mathrm{F}^\bullet X})$ is called a **filtered differential object** over $(\mathcal{A}, T)$. In this case, $\operatorname{gr}^p X$ is also naturally equipped with $\operatorname{gr}^p d : \operatorname{gr}^p X \to T \operatorname{gr}^p X$, so that $(\operatorname{gr}^p X, \operatorname{gr}^p d) \in \operatorname{Ob}((\mathcal{A}, T)_d)$.

Since $d_{\mathrm{F}^p X}$ on the subobject $\mathrm{F}^p X$ is best understood as the restriction of $d : X \to TX$, we continue to denote it by d. A similar definition applies to increasing filtrations, with an entirely dual formulation.

We return to spectral sequences. The preceding section explained how to construct an exact couple from a differential object. Now consider a filtered differential object $(X, d, \mathrm{F}^\bullet X)$ over $\mathcal{A}$. Write $S : (Y^p)_p \mapsto (Y^{p+1})_p$ for the standard translation functor on $\mathcal{A}^{\mathbb{Z}}$; it plainly commutes with T, acting on each Y^p. Consequently, the morphisms considered below have two kinds of degree: the "filtration degree" corresponding to S and the "internal degree" corresponding to T. For the moment we focus on the former.

Since the filtration is decreasing, there is a monomorphism

$$\alpha : \left(\mathrm{F}^{p+1} X, d\right)_{p \in \mathbb{Z}} \hookrightarrow S^{-1} \left(\mathrm{F}^{p+1} X, d\right)_{p \in \mathbb{Z}} = \left(\mathrm{F}^p X, d\right)_{p \in \mathbb{Z}}.$$

Thus $E_0 := \operatorname{coker}(\alpha) = \left(\operatorname{gr}^p X, \operatorname{gr}^p d\right)_{p \in \mathbb{Z}}$ and $E_1^p = \mathrm{H}(\operatorname{gr}^p X, \operatorname{gr}^p d)$.

Regard α as a morphism of degree -1 in $\mathcal{A}^{\mathbb{Z}}$ (with respect to S). Applying Proposition 5.3.4 gives the following exact couple with degrees over $\mathcal{A}^{\mathbb{Z}}$:

$$\mathscr{C} = \left[\begin{tikzcd} \mathrm{H}\left(\mathrm{F}^{p+1}X, d\right)_p \arrow[rr, "\mathrm{H}(\alpha)", "-1"'] & & \mathrm{H}\left(\mathrm{F}^{p+1}X, d\right)_p \arrow[dl, "+1"] \\ & \mathrm{H}\left(\mathrm{gr}^p X, \mathrm{gr}^p d\right)_p \arrow[ul] & \end{tikzcd}\right],$$

where the $\nwarrow$ arrow comes from the connecting morphism induced by the short exact sequence $0 \to (\mathrm{F}^{p+1}X, d) \to (\mathrm{F}^p X, d) \to (\mathrm{gr}^p X, \mathrm{gr}^p d) \to 0$.

The resulting sequence $\mathscr{E} = (E_r^p, d_r^p)_{\substack{r\geq 0\\ p\in\mathbb{Z}}}$ is called the spectral sequence determined by $\mathrm{F}^\bullet X$ in $\mathcal{A}^{\mathbb{Z}}$. At the end of §5.3 we explained that d_r is a morphism of degree r with respect to S, namely

$$d_r = \left(d_r^p : E_r^p \to (S^r E_r)^p = E_r^{p+r}\right)_{p\in\mathbb{Z}};$$

of course, the degree of d_r can also be read from the formula in Proposition 5.3.4.

In other words, a filtered differential object $(X, d, \mathrm{F}^\bullet X)$ gives a cohomologically graded spectral sequence as in Definition 5.2.8. Notice that d_r^p here may also carry an internal degree (unless $T = \mathrm{id}_{\mathcal{A}}$); this is suppressed from the notation and will be set out in detail in §5.5.

Following the pattern of (5.2.1), define subobjects of $Z_0 = (\mathrm{gr}^p X, \mathrm{gr}^p d)_p$ by

$$B_r = (B_r^p)_p \subset (Z_r^p)_p = Z_r.$$

To simplify notation, the statements below suppress the internal degree of d; this amounts to considering the special case $T = \mathrm{id}_{\mathcal{A}}$. The extension to the general case is routine: translation functors need only be inserted appropriately wherever $d^{-1}(\cdots)$ or $d(\cdots)$ occurs, so that the expressions make strict sense.

Proposition 5.4.2 For a filtered differential object $(X, d, \mathrm{F}^\bullet X)$, the associated spectral sequence satisfies

$$Z_r^p = \frac{\left(\mathrm{F}^p X \cap d^{-1}\mathrm{F}^{p+r}X\right) + \mathrm{F}^{p+1}X}{\mathrm{F}^{p+1}X},$$
$$B_r^p = \frac{\left(\mathrm{F}^p X \cap d\mathrm{F}^{p-r+1}X\right) + \mathrm{F}^{p+1}X}{\mathrm{F}^{p+1}X},$$

and $d_r^p : E_r^p \to E_r^{p+r}$ is induced by the morphism $d^{-1}\mathrm{F}^{p+r}X \xrightarrow{d} \mathrm{F}^{p+r}X \cap \ker(d)$.

Proof Observe that the degree-p component of α^r may be identified with the inclusion $\mathrm{F}^p X \hookrightarrow \mathrm{F}^{p-r}X$ or $\mathrm{F}^{p+r}X \hookrightarrow \mathrm{F}^p X$. The assertion reduces to the graded version of Proposition 5.3.4. $\square$

Since $\varinjlim$ and $\varprojlim$ in $\mathcal{A}^{\mathbb{Z}}$ are taken degree by degree, $E_\infty = (E_\infty^p)_{p\in\mathbb{Z}}$ can be written as

$$E_\infty^p = \frac{Z_\infty^p}{B_\infty^p} = \frac{\bigcap_r \left((\mathrm{F}^p X \cap d^{-1}\mathrm{F}^{p+r} X) + \mathrm{F}^{p+1} X\right)}{\bigcup_r \left((\mathrm{F}^p X \cap d\mathrm{F}^{p-r+1} X) + \mathrm{F}^{p+1} X\right)}, \tag{5.4.1}$$

provided the displayed $\bigcap$ and $\bigcup$ exist for every p.

Remark 5.4.3 Proposition 5.4.2 on the spectral sequence of a filtered differential object is the foundation for the later discussions in this chapter. It is an application of exact couples, but we could equally regard the formulas in Proposition 5.4.2 as direct definitions and verify from them all the properties required of a spectral sequence.

From the preceding description, the spectral sequence E_r^p of $(X, d, \mathrm{F}^\bullet X)$ can be viewed as a process of successively approximating $\mathrm{H}(X, d)$. To make precise what "approximating" means, we need the following concept.

Definition 5.4.4 (Induced filtration) For a filtered differential object $(X, d, \mathrm{F}^\bullet X)$ over $(\mathcal{A}, T)$, the object $\mathrm{H}(X, d)$ carries the induced filtration

$$\mathrm{F}^p \mathrm{H}(X, d) := \mathrm{im}\left[\mathrm{F}^p X \cap \ker(d) \to \mathrm{H}(X, d)\right], \qquad p \in \mathbb{Z}.$$

Lemma 5.4.5 If there is an N with $\mathrm{F}^N X = 0$, then $\mathrm{F}^N \mathrm{H}(X, d) = 0$. If $\mathrm{F}^\bullet X$ is an exhaustive filtration in the sense of Definition 5.1.3, then $\mathrm{F}^\bullet \mathrm{H}(X, d)$ is exhaustive as well.

Proof The first part is immediate; we prove the second. By Lemma 2.6.6 (ii), $\bigcup_p \mathrm{F}^p \mathrm{H}(X, d)$ is the image of $\bigcup_p (\mathrm{F}^p X \cap \ker(d))$, while (5.1.1) gives

$$\bigcup_p (\mathrm{F}^p X \cap \ker(d)) = \left(\bigcup_p \mathrm{F}^p X\right) \cap \ker(d) = \ker(d).$$

Moreover, if there is an M with $\mathrm{F}^M X = X$, then of course $\mathrm{F}^M \mathrm{H}(X, d) = \mathrm{H}(X, d)$. ☐

We shall apply the graded version of these observations in §5.5.

Lemma 5.4.6 For a filtered differential object $(X, d, \mathrm{F}^\bullet X)$ and every $p \in \mathbb{Z}$, there is a canonical isomorphism

$$\mathrm{gr}^p \mathrm{H}(X, d) \simeq \frac{\mathrm{F}^p X \cap \ker(d)}{(\mathrm{F}^{p+1} X \cap \ker(d)) + (\mathrm{F}^p X \cap \mathrm{im}(T^{-1} d))}.$$

Proof Expand the definition of $\mathrm{F}^p \mathrm{H}(X, d)/\mathrm{F}^{p+1} \mathrm{H}(X, d)$ and use the standard isomorphism theorem for Abelian categories, Proposition 2.6.8, to obtain

$$\begin{aligned}\operatorname{gr}^p \mathrm{H}(X,d) &= \frac{(\mathrm{F}^p X \cap \ker(d)) + \operatorname{im}(T^{-1}d)}{(\mathrm{F}^{p+1} X \cap \ker(d)) + \operatorname{im}(T^{-1}d)} \\ &= \frac{(\mathrm{F}^p X \cap \ker(d)) + (\mathrm{F}^{p+1} X \cap \ker(d)) + \operatorname{im}(T^{-1}d)}{(\mathrm{F}^{p+1} X \cap \ker(d)) + \operatorname{im}(T^{-1}d)} \\ &\simeq \frac{\mathrm{F}^p X \cap \ker(d)}{((\mathrm{F}^{p+1} X \cap \ker(d)) + \operatorname{im}(T^{-1}d)) \cap (\mathrm{F}^p X \cap \ker(d))} \\ &= \frac{\mathrm{F}^p X \cap \ker(d)}{(\mathrm{F}^{p+1} X \cap \ker(d)) + (\mathrm{F}^p X \cap \operatorname{im}(T^{-1}d))}.\end{aligned}$$

The last step uses the fact that Sub_X is a modular lattice[2)] (Theorem 2.6.10), together with $\mathrm{F}^{p+1} X \cap \ker(d) \subset \mathrm{F}^p X \cap \ker(d)$ and $\operatorname{im}(T^{-1}d) \subset \ker(d)$. The remaining steps are standard. □

Definition–Proposition 5.4.7 For a filtered differential object $(X, d, \mathrm{F}^\bullet X)$, construct the associated cohomologically graded spectral sequence $(E_r, d_r)_r$. If the $\bigcap_r$ and $\bigcup_r$ in (5.4.1) exist, then the limit E_∞ exists, and the $\operatorname{gr} \mathrm{H}(X,d)$ determined by the induced filtration can be canonically realized as a subquotient of E_∞.

- ◇ If $\operatorname{gr} \mathrm{H}(X,d) = E_\infty$, the spectral sequence $(E_r, d_r)_r$ is called **weakly convergent**.
- ◇ If $(E_r, d_r)_r$ is weakly convergent and $\mathrm{F}^\bullet \mathrm{H}(X,d)$ is exhaustive and complete, the spectral sequence $(E_r, d_r)_r$ is called **strongly convergent**.

Proof Fix $p \in \mathbb{Z}$. In the expression for E_∞^p in (5.4.1), the numerator contains $(\mathrm{F}^p X \cap \ker(d)) + \mathrm{F}^{p+1} X$, while the denominator is contained in $(\mathrm{F}^p X \cap \operatorname{im}(T^{-1}d)) + \mathrm{F}^{p+1} X$ (recall that d is a morphism $X \to TX$). This gives the following subquotient of E_∞^p:

$$\begin{aligned}&\frac{(\mathrm{F}^p X \cap \ker(d)) + \mathrm{F}^{p+1} X}{(\mathrm{F}^p X \cap \operatorname{im}(T^{-1}d)) + \mathrm{F}^{p+1} X} \\ &\qquad \simeq \frac{\mathrm{F}^p X \cap \ker(d)}{((\mathrm{F}^p X \cap \operatorname{im}(T^{-1}d)) + \mathrm{F}^{p+1} X) \cap (\mathrm{F}^p X \cap \ker(d))} \\ &\qquad\qquad = \frac{\mathrm{F}^p X \cap \ker(d)}{(\mathrm{F}^p X \cap \operatorname{im}(T^{-1}d)) + (\mathrm{F}^{p+1} X \cap \ker(d))}.\end{aligned}$$

The first isomorphism is standard. The following equality uses the fact that Sub_X is a modular lattice, together with $\mathrm{F}^p X \cap \operatorname{im}(T^{-1}d) \subset \mathrm{F}^p X \cap \ker(d)$ and $\mathrm{F}^{p+1} X \subset \mathrm{F}^p X$. Now apply Lemma 5.4.6. □

Definitions of convergence vary slightly among references. All the statements above have corresponding versions for differential objects with increasing filtrations $(X, d, \mathrm{F}_\bullet X)$ and the associated homologically graded spectral sequences.

[2)] That is, for all subobjects $A, A^\flat, B \subset X$ with $A^\flat \subset A$, one has $A^\flat + (A \cap B) = (A^\flat + B) \cap A$.

5.5 The spectral sequence of a filtered complex

We continue the line of thought from §5.4, still with a fixed Abelian category $\mathcal{A}$. The result of Proposition 5.4.2 for filtered differential objects can be extended to allow the morphism d to have a degree. In particular, this theory applies to filtered complexes.

Concretely, replace the previous $\mathcal{A}$ by $\mathcal{A}^{\mathbb{Z}}$ and denote its translation functor by T. By Example 5.2.5, an object (X, d) of $(\mathcal{A}^{\mathbb{Z}}, T)_d$ can be identified with a complex $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$. If we further take a decreasing filtration $(X, d, \mathrm{F}^\bullet X)$, then

$$\begin{aligned} \mathrm{H}(X, d) &= (\mathrm{H}^n(X))_{n\in\mathbb{Z}}, \\ \mathrm{H}(\mathrm{F}^p X, d) &= (\mathrm{H}^n(\mathrm{F}^p X))_{n\in\mathbb{Z}}, \\ \mathrm{F}^p \mathrm{H}^n(X) &:= \mathrm{im}\left[\mathrm{F}^p X^n \cap \ker(d_X^n) \to \mathrm{H}^n(X)\right] \quad \text{(Definition 5.4.4)}. \end{aligned}$$

Thus the E_r in the spectral sequence $\mathscr{E}$ constructed above in fact takes values in $\mathcal{A}^{\mathbb{Z}\times\mathbb{Z}}$ and is a bigraded object. Besides the filtration degree p, there is an "internal degree" n coming from the complex structure; the corresponding translation functors S and T commute strictly. We already know that d_r has degree r with respect to S; we now determine its degree with respect to T.

- In the exact couple $\mathscr{C}_{(1)}$ used to construct $\mathscr{E}$, the morphism $\nwarrow$ has degree 1 with respect to T, while the other morphisms have degree zero. Indeed, $\nwarrow$ comes from the cohomological connecting morphism and hence has degree 1 with respect to T, while the other arrows plainly have degree zero.
- Using the description in Lemma 5.3.3, the same statement follows recursively for $\mathscr{C}_{(r)}$ for every r.
- Consequently, both d_r and the d in (X, d) are morphisms of degree 1 with respect to T. This is also immediate from the description in Proposition 5.3.4.

For applications, the customary convention is to use the indices p and $q := n - p$. Thus

$$\begin{gathered} E_r = (E_r^{p,q})_{(p,q)\in\mathbb{Z}^2} = (Z_r^{p,q}/B_r^{p,q})_{(p,q)\in\mathbb{Z}^2}, \\ E_0^{p,q} = (\mathrm{gr}^p X)^{p+q}, \\ E_1^{p,q} = \mathrm{H}^{p+q}(\mathrm{gr}^p X, \mathrm{gr}^p d), \\ d_r = \left(d_r^{p,q} : E_r^{p,q} \to E_r^{p+r,q-r+1}\right)_{(p,q)\in\mathbb{Z}^2} : E_r \xrightarrow{(r,-r+1)} E_r. \end{gathered}$$

In other words, a filtered complex gives a cohomological bigraded spectral sequence as in Definition 5.2.8. Dually, a chain complex with an increasing filtration gives a homological bigraded spectral sequence.

Example 5.5.1 Consider a filtered complex $(X, d, \mathrm{F}^\bullet X)$. If for every $n \in \mathbb{Z}$ one has $\mathrm{F}^0 X^n = X^n$ and $\mathrm{F}^{n+1} X^n = 0$, then E_0 lies in the first quadrant (Definition 5.2.12). This follows immediately from $E_0^{p,q} = (\mathrm{gr}^p X)^{p+q}$.

Proposition 5.5.2 For a complex $X \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$ with decreasing filtration $\mathrm{F}^\bullet X$, the associated spectral sequence satisfies

$$Z_r^{p,q} = \frac{\left(\mathrm{F}^p X^{p+q} \cap d^{-1}\mathrm{F}^{p+r} X^{p+q+1}\right) + \mathrm{F}^{p+1} X^{p+q}}{\mathrm{F}^{p+1} X^{p+q}},$$
$$B_r^{p,q} = \frac{\left(\mathrm{F}^p X^{p+q} \cap d\mathrm{F}^{p-r+1} X^{p+q-1}\right) + \mathrm{F}^{p+1} X^{p+q}}{\mathrm{F}^{p+1} X^{p+q}},$$

and $d_r^{p,q} : E_r^{p,q} \to E_r^{p+r,q-r+1}$ is induced by

$$d^{-1}\mathrm{F}^{p+r} X^{p+q+1} \xrightarrow{d} \mathrm{F}^{p+r} X^{p+q+1} \cap \ker(d).$$

Proof In Proposition 5.4.2, include the complex degree $n = p + q$ and observe that d has degree 1 with respect to the complex translation functor T. $\square$

Similarly, (5.4.1) has the bigraded version

$$E_\infty^{p,q} = \frac{Z_\infty^{p,q}}{B_\infty^{p,q}} = \frac{\bigcap_r \left(\left(\mathrm{F}^p X^{p+q} \cap d^{-1}\mathrm{F}^{p+r} X^{p+q+1}\right) + \mathrm{F}^{p+1} X^{p+q}\right)}{\bigcup_r \left(\left(\mathrm{F}^p X^{p+q} \cap d\mathrm{F}^{p-r+1} X^{p+q-1}\right) + \mathrm{F}^{p+1} X^{p,q}\right)}, \tag{5.5.1}$$

provided the displayed $\bigcup$ and $\bigcap$ exist. In this case $E_\infty = (E_\infty^{p,q})_{(p,q)\in\mathbb{Z}^2}$.

Definition–Proposition 5.4.7 now becomes a bigraded statement: for all p, q, the object $\mathrm{gr}^p \mathrm{H}^{p+q}(X)$ can be canonically realized as a subquotient of $E_\infty^{p,q}$. The convergence properties of the spectral sequence are correspondingly refined by the grading.

Definition 5.5.3 For a filtered complex $(X, d, \mathrm{F}^\bullet X)$ over $\mathcal{A}$, construct the associated cohomologically graded spectral sequence $(E_r, d_r)_r$. If the $\bigcap_r$ and $\bigcup_r$ in (5.5.1) exist, then $\mathrm{gr}^p \mathrm{H}^{p+q}(X)$ can be canonically realized as a subquotient of $E_\infty^{p,q}$.

- ◇ If $\mathrm{gr}^p \mathrm{H}^{p+q}(X) = E_\infty^{p,q}$ for every $(p,q) \in \mathbb{Z}^2$, then $(E_r, d_r)_r$ is called **weakly convergent**.
- ◇ If the sequence is weakly convergent and, for every $n \in \mathbb{Z}$, the filtration $\mathrm{F}^\bullet \mathrm{H}^n(X)$ is exhaustive and complete, then $(E_r, d_r)_r$ is called **strongly convergent**.

Convention 5.5.4 Strong convergence of a spectral sequence is also denoted by $E_r^{p,q} \Rightarrow \mathrm{H}^{p+q}(X)$. The subscript r is usually written concretely as 1, 2, and so on, depending on which page of the spectral sequence we wish to describe.

More generally, suppose we are given a cohomological bigraded spectral sequence $\mathscr{E}$, a filtered graded object $(H, \mathrm{F}^\bullet H)$, and isomorphisms $E_\infty^{p,q} \simeq \mathrm{gr}^p H^{p+q}$, with $\mathrm{F}^\bullet H$ exhaustive and complete. We also denote this situation by $E_r^{p,q} \Rightarrow H^{p+q}$. Homological bigraded spectral sequences are treated similarly; in particular, $E_{p,q}^r \Rightarrow H_{p+q}$ entails $E_{p,q}^\infty \simeq \mathrm{gr}_p H_{p+q}$.

The following classical convergence theorem is sufficient for initial applications. For more general convergence conditions, see [4].

Theorem 5.5.5 (Classical convergence theorem) Consider a filtered complex $(X, d, \mathrm{F}^\bullet X)$. Suppose that for every $n \in \mathbb{Z}$,

- ◇ the filtration $\mathrm{F}^\bullet X^n$ is exhaustive (Definition 5.1.3),
- ◇ there is an $N = N(n)$ such that $\mathrm{F}^N X^n = 0$.

Then the associated spectral sequence is strongly convergent (Definition–Proposition 5.4.7).

If the hypotheses are strengthened so that the filtration on every X^n is finite (Definition 5.1.1), then the induced filtration on $\mathrm{H}^n(X)$ is also finite. In this case the associated spectral sequence is bounded (Definition 5.2.10).

Proof By the graded version of Lemma 5.4.5, the induced filtration $\mathrm{F}^\bullet \mathrm{H}^n(X)$ is exhaustive and complete. It therefore remains to prove weak convergence.

We need to recall the proof of Definition–Proposition 5.4.7, which explains how to realize $\mathrm{gr}^p \mathrm{H}^{p+q}(X)$ as a subquotient of $E_\infty^{p,q}$ for each $(p, q) \in \mathbb{Z}^2$. The key point is

$$\begin{aligned}
&\bigcap_r \left(\left(\mathrm{F}^p X^{p+q} \cap d^{-1}\mathrm{F}^{p+r} X^{p+q+1} \right) + \mathrm{F}^{p+1} X^{p+q} \right) \\
&\qquad \supset \left(\mathrm{F}^p X^{p+q} \cap \ker(d) \right) + \mathrm{F}^{p+1} X^{p+q}, \\
&\bigcup_r \left(\left(\mathrm{F}^p X^{p+q} \cap d\mathrm{F}^{p-r+1} X^{p+q-1} \right) + \mathrm{F}^{p+1} X^{p+q} \right) \\
&\qquad \subset \left(\mathrm{F}^p X^{p+q} \cap \mathrm{im}(d) \right) + \mathrm{F}^{p+1} X^{p+q}.
\end{aligned}$$

Proving weak convergence is equivalent to strengthening both inclusions to equalities. But when $r \gg 0$ relative to p, q, the expression inside $\bigcap_r$ in the first line is simply $(\mathrm{F}^p X^{p+q} \cap \ker(d)) + \mathrm{F}^{p+1} X^{p+q}$, so equality holds.

For the second line, the term $+\mathrm{F}^{p+1} X^{p+q}$ can first be moved outside $\bigcup_r$. Recall that $d(\mathrm{F}^\bullet X^n)$ is an exhaustive filtration of $d(X^n)$. Hence

$$\begin{aligned}
&\bigcup_r \left(\mathrm{F}^p X^{p+q} \cap d\mathrm{F}^{p-r+1} X^{p+q-1} \right) \\
&\qquad \xlongequal{\because (5.1.1)} \mathrm{F}^p X^{p+q} \cap \bigcup_r d\mathrm{F}^{p-r+1} X^{p+q-1} \\
&\qquad = \mathrm{F}^p X^{p+q} \cap d \bigcup_r \mathrm{F}^{p-r+1} X^{p+q-1} = \mathrm{F}^p X^{p+q} \cap \mathrm{im}(d).
\end{aligned}$$

Finally, suppose the filtration on every X^n is finite. Then $\mathrm{F}^\bullet \mathrm{H}^n(X)$ is naturally bounded. To show that the spectral sequence is bounded, fix r, n and take $p+q=n$. By the description in Proposition 5.5.2, when $p \gg 0$ the numerator of $Z_r^{p,q}$ is 0, while when $p \ll 0$ its denominator is X^n. Thus only finitely many (p,q) satisfy $E_{r+1}^{p,q} \neq 0$.☐

The dual version is clear and will not be repeated.

Corollary 5.5.6 (Exact sequence of low-degree terms) Let the filtered complex $(X, d, \mathrm{F}^\bullet X)$ satisfy

$$\mathrm{F}^0 X^n = X^n, \qquad \mathrm{F}^{n+1} X^n = 0, \qquad n \in \mathbb{Z},$$

as in Example 5.5.1. The associated spectral sequence gives the canonical exact sequence

$$0 \to E_2^{1,0} \to \mathrm{H}^1(X) \to E_2^{0,1} \xrightarrow{d} E_2^{2,0} \to \mathrm{H}^2(X).$$

Dually, for a chain complex X with increasing filtration, if $\mathrm{F}_{-1}X = 0$ and $\mathrm{F}_n X = X$, there is a canonical exact sequence

$$\mathrm{H}_2(X) \to E_{2,0}^2 \xrightarrow{d} E_{0,1}^2 \to \mathrm{H}_1(X) \to E_{1,0}^2 \to 0.$$

Proof Substitute $p=2$ into (5.2.4) to obtain the exact sequence

$$0 \to E_\infty^{0,1} \to E_2^{0,1} \xrightarrow{d} E_2^{2,0} \to E_\infty^{2,0} \to 0.$$

By the classical convergence theorem 5.5.5, $E_\infty^{0,1} \simeq \mathrm{gr}^0 \mathrm{H}^1(X)$, $E_\infty^{1,0} \simeq \mathrm{gr}^1 \mathrm{H}^1(X)$, and $E_\infty^{2,0} \simeq \mathrm{gr}^2 \mathrm{H}^2(X)$. The hypotheses give

$$\mathrm{gr}^2 \mathrm{H}^2(X) = \mathrm{F}^2 \mathrm{H}^2(X), \qquad \mathrm{gr}^1 \mathrm{H}^1(X) = \mathrm{F}^1 \mathrm{H}^1(X),$$
$$\mathrm{gr}^0 \mathrm{H}^1(X) = \mathrm{H}^1(X)/\mathrm{F}^1 \mathrm{H}^1(X) = \mathrm{H}^1(X)/E_\infty^{1,0}.$$

Moreover, substituting $p=1$ into (5.2.2) gives $E_2^{1,0} = E_\infty^{1,0}$. Splicing these equalities together yields the asserted exact sequence. The homological version is analogous. ☐

If $E_r^{p,q}$ is strongly convergent, then once sufficiently many pages (E_r, d_r) are known, in principle one can read $(\mathrm{gr}^p \mathrm{H}^{p+q}(X))_{p,q\in\mathbb{Z}}$ from E_∞. Passing from $(\mathrm{gr}^p \mathrm{H}^n(X))_p$ to $\mathrm{H}^n(X)$, however, amounts to determining a series of extensions in $\mathcal{A}$, which is generally difficult unless $\mathrm{H}^n(X)$ splits. If we consider only coarser properties, a spectral sequence can sometimes provide a concise answer. The following example is essentially an application of the Euler–Poincaré principle.

Lemma 5.5.7 Let $\mathscr{E} = (E_r, d_r)_{r\geq 1}$ be a cohomological bigraded spectral sequence, and suppose there is an r such that

$$\left\{(p,q) \in \mathbb{Z}^2 : E_r^{p,q} \neq 0\right\} \quad \text{is a finite set.}$$

Then for $r \gg 0$, $\mathscr{E}$ degenerates at the E_r page.

Proof By the hypothesis, for sufficiently large r all nonzero terms of E_r are concentrated in a region of finite length and width. Apply Example 5.2.9. □

Proposition 5.5.8 Let the cohomological bigraded spectral sequence $(E_r, d_r)_r$ satisfy the following conditions:

- ◇ there is an r such that $\{(p,q) \in \mathbb{Z}^2 : E_r^{p,q} \neq 0\}$ is a finite set;
- ◇ there is strong convergence $E_r^{p,q} \Rightarrow H^{p+q}$ in the sense of Convention 5.5.4.

Then in the group $\mathrm{K}_0(\mathcal{A})$ of Definition 2.9.8, for $r \gg 0$ one has

$$\sum_{n\in\mathbb{Z}} (-1)^n [H^n] = \sum_{p,q\in\mathbb{Z}} (-1)^{p+q} [E_r^{p,q}],$$

and both sides are finite sums.

Proof For $r \gg 0$, only finitely many terms on the right are nonzero. Moreover,

$$E_{r+1}^{p,q} \simeq \mathrm{H}\left[E_r^{p-r,q+r-1} \xrightarrow{d} E_r^{p,q} \xrightarrow{d} E_r^{p+r,q-r+1}\right].$$

If degree is counted by $n := p+q$, then d has degree 1. Take the alternating sum over n in $\mathrm{K}_0(\mathcal{A})$ and apply Theorem 2.9.11 to obtain

$$\sum_{p,q\in\mathbb{Z}} (-1)^{p+q} [E_r^{p,q}] = \sum_{p,q\in\mathbb{Z}} (-1)^{p+q} [E_{r+1}^{p,q}].$$

Lemma 5.5.7 shows that the spectral sequence eventually degenerates. Thus, for $r \gg 0$, $E_r^{p,q} = E_\infty^{p,q}$ for every (p,q). Consequently,

$$\sum_{p,q} (-1)^{p+q} [E_r^{p,q}] = \sum_{p,q} (-1)^{p+q} [\mathrm{gr}^p \mathrm{H}^{p+q}(X)] = \sum_n (-1)^n \sum_p [\mathrm{gr}^p H^n],$$

and the hypotheses ensure that all these sums are finite. By Lemma 2.9.9 (v) and the properties of $\mathrm{F}^\bullet H$, we also have $\sum_p [\mathrm{gr}^p H^n(X)] = [\mathrm{H}^n(X)]$. □

If $(E_r, d_r)_r$ comes from a filtered complex $(X, d, \mathrm{F}^\bullet)$ and the filtration $\mathrm{F}^\bullet X^n$ on every X^n is finite, then Theorem 5.5.5 ensures that the strong-convergence hypothesis $E_r^{p,q} \Rightarrow \mathrm{H}^{p+q}(X)$ in Proposition 5.5.8 holds automatically. The condition on $\{(p,q) : E_r^{p,q} \neq 0\}$, however, is not automatic.

Filtered complexes already suffice to produce some simple but useful spectral sequences; see the exercises for this chapter. Since algebra is the subject of this book, several of the best-known spectral sequences arise from bicomplexes. We therefore turn next to the bicomplex case.

5.6 Spectral sequences of bicomplexes and their applications

Throughout this section, we assume that the Abelian category $\mathcal{A}$ has the countable direct sums or countable products under consideration. Let X be a bicomplex over $\mathcal{A}$, written $X \in \mathrm{Ob}(\mathsf{C}^2(\mathcal{A}))$. The total complex $\mathrm{tot}_{\oplus} X$ has two decreasing filtrations,

$$\mathrm{F}^p_{\mathrm{I}}(\mathrm{tot}_{\oplus} X)^n = \bigoplus_{\substack{i+j=n \\ i \geq p}} X^{i,j},$$
$$\mathrm{F}^q_{\mathrm{II}}(\mathrm{tot}_{\oplus} X)^n = \bigoplus_{\substack{i+j=n \\ j \geq q}} X^{i,j}.$$

Examining each (i, j) component separately shows that

$$\bigcap_p \mathrm{F}^p_{\mathrm{I}} = 0 = \bigcap_q \mathrm{F}^q_{\mathrm{II}}, \quad \bigcup_p \mathrm{F}^p_{\mathrm{I}} = \mathrm{tot}_{\oplus} X = \bigcup_q \mathrm{F}^q_{\mathrm{II}}.$$

We thus obtain two filtered complexes. The corresponding spectral sequences are denoted respectively by $\mathscr{E}_{\mathrm{I}} = \mathscr{E}_{\mathrm{I}}(X)$ and $\mathscr{E}_{\mathrm{II}} = \mathscr{E}_{\mathrm{II}}(X)$, or more concretely by $E^{p,q}_{\mathrm{I},r}$ and $E^{p,q}_{\mathrm{II},r}$, where $r \in \mathbb{Z}_{\geq 0}$. If $X^{p,q} \neq 0 \implies p, q \geq 0$, we say that X lies in the first quadrant. In this case, for $\star \in \{\mathrm{I}, \mathrm{II}\}$,

$$\mathrm{F}^0_{\star}(\mathrm{tot}_{\oplus} X)^n = (\mathrm{tot}_{\oplus} X)^n, \quad \mathrm{F}^{n+1}_{\star}(\mathrm{tot}_{\oplus} X)^n = 0,$$

so the corresponding spectral sequences also lie in the first quadrant.

For a chain bicomplex X, one similarly defines two increasing filtrations,

$$\mathrm{F}_{\mathrm{I},p}(\mathrm{tot}_{\oplus} X)_n = \bigoplus_{\substack{i+j=n \\ i \leq p}} X_{i,j},$$
$$\mathrm{F}_{\mathrm{II},q}(\mathrm{tot}_{\oplus} X)_n = \bigoplus_{\substack{i+j=n \\ j \leq q}} X_{i,j},$$

together with the corresponding homological bigraded spectral sequences.

Proposition 5.6.1 For $X \in \mathrm{Ob}(\mathsf{C}^2(\mathcal{A}))$, the first few pages of its two spectral sequences and their corresponding morphisms d are described as follows:

	$E^{p,q}_0$	$d^{p,q}_0$	$E^{p,q}_1$	$d^{p,q}_1$	$E^{p,q}_2$
$\mathscr{E}_{\mathrm{I}}$	$X^{p,q}$	$(-1)^p\, {}^{\vartriangle}d^{p,q}$	$\mathrm{H}^q(X^{p,\bullet}, {}^{\vartriangle}d)$	$\mathrm{H}^q({}^{\triangleright}d^{p,\bullet})$	$\mathrm{H}_{\mathrm{I}}\,\mathrm{H}_{\mathrm{II}}(X)^{p,q}$
$\mathscr{E}_{\mathrm{II}}$	$X^{q,p}$	${}^{\triangleright}d^{q,p}$	$\mathrm{H}^q(X^{\bullet,p}, {}^{\triangleright}d)$	$(-1)^q\,\mathrm{H}^q({}^{\vartriangle}d^{\bullet,p})$	$\mathrm{H}_{\mathrm{II}}\,\mathrm{H}_{\mathrm{I}}(X)^{q,p}$

The functors $\mathrm{H}_{\mathrm{I}}, \mathrm{H}_{\mathrm{II}} : \mathsf{C}^2(\mathcal{A}) \to \mathsf{C}^2(\mathcal{A})$ are defined in §3.10, especially in (3.10.1).

For chain complexes and their homology, the same method gives the following table.

	$E^0_{p,q}$	$d^0_{p,q}$	$E^1_{p,q}$	$d^1_{p,q}$	$E^2_{p,q}$
$\mathscr{E}_{\mathrm{I}}$	$X_{p,q}$	$(-1)^p {}^{\vartriangle}d_{p,q}$	$\mathrm{H}_q(X_{p,\bullet}, {}^{\vartriangle}d)$	$\mathrm{H}_q({}^{\triangleright}d_{p,\bullet})$	$\mathrm{H}_{\mathrm{I}}\,\mathrm{H}_{\mathrm{II}}(X)_{p,q}$
$\mathscr{E}_{\mathrm{II}}$	$X_{q,p}$	${}^{\triangleright}d_{q,p}$	$\mathrm{H}_q(X_{\bullet,p}, {}^{\triangleright}d)$	$(-1)^q\,\mathrm{H}_q({}^{\vartriangle}d_{\bullet,p})$	$\mathrm{H}_{\mathrm{II}}\,\mathrm{H}_{\mathrm{I}}(X)_{q,p}$

Proof This follows by a direct verification from the description in Proposition 5.5.2; the sign on ${}^{\vartriangle}d$ comes from the definition of the total complex. We omit the details. □

The same definitions and results apply to $\mathrm{tot}_{\Pi} X$, with entirely similar properties.

For $X \in \mathrm{Ob}(\mathsf{C}^2_f(\mathcal{A}))$ (Definition 3.10.1), its total complex involves only finite direct sums. Thus there is no need to distinguish the $\oplus$ and Π versions; both are denoted uniformly by $\mathrm{tot}\, X$.

Theorem 5.6.2 Let $X \in \mathrm{Ob}(\mathsf{C}^2_f(\mathcal{A}))$. The two corresponding spectral sequences $\mathscr{E}_{\mathrm{I}}$ and $\mathscr{E}_{\mathrm{II}}$ are bounded and strongly convergent, and the two corresponding induced filtrations on $\mathrm{H}^n(\mathrm{tot}\, X)$ are finite for every $n \in \mathbb{Z}$.

Proof The definition of $\mathsf{C}^2_f(\mathcal{A})$ shows that $\mathrm{F}^{\bullet}_{\mathrm{I}}(\mathrm{tot}\, X)^n$ and $\mathrm{F}^{\bullet}_{\mathrm{II}}(\mathrm{tot}\, X)^n$ are finite filtrations for every n. Apply Theorem 5.5.5. □

We now present several representative applications. Since all the filtrations and spectral sequences involved are finite, these results apply to every Abelian category[3)].

Example 5.6.3 We now reprove Theorem 3.10.6 using spectral sequences: if a morphism $f : X \to Y$ in $\mathsf{C}^2_f(\mathcal{A})$ induces $\mathrm{H}_{\mathrm{II}}\,\mathrm{H}_{\mathrm{I}}(X) \xrightarrow{\sim} \mathrm{H}_{\mathrm{II}}\,\mathrm{H}_{\mathrm{I}}(Y)$ (or $\mathrm{H}_{\mathrm{I}}\,\mathrm{H}_{\mathrm{II}}(X) \xrightarrow{\sim} \mathrm{H}_{\mathrm{I}}\,\mathrm{H}_{\mathrm{II}}(Y)$), then $\mathrm{tot}(f) : \mathrm{tot}(X) \to \mathrm{tot}(Y)$ is a quasi-isomorphism.

First suppose that $\mathrm{H}_{\mathrm{II}}\,\mathrm{H}_{\mathrm{I}}(X) \xrightarrow{\sim} \mathrm{H}_{\mathrm{II}}\,\mathrm{H}_{\mathrm{I}}(Y)$. The morphism f induces a morphism of spectral sequences $\mathscr{E}_{\mathrm{II}}(X) \to \mathscr{E}_{\mathrm{II}}(Y)$. This is already an isomorphism on the E_2 page and hence also gives an isomorphism on the limit page E_∞ (Proposition 5.2.7).

By the convergence guaranteed by Theorem 5.6.2, the morphism $\mathrm{gr}^p\,(\mathrm{H}^n\,\mathrm{tot}(X)) \to \mathrm{gr}^p\,(\mathrm{H}^n\,\mathrm{tot}(Y))$ induced by $\mathrm{tot}(f)$ is an isomorphism for all $p, n \in \mathbb{Z}$. The induced filtrations on H^n are known to be finite. Therefore $\mathrm{H}^n\,\mathrm{tot}(f) : \mathrm{H}^n\,\mathrm{tot}(X) \to \mathrm{H}^n\,\mathrm{tot}(Y)$ is also an isomorphism (Proposition 5.1.4).

If instead we consider the spectral sequence $\mathscr{E}_{\mathrm{I}}$, the analogous argument deduces that $\mathrm{tot}(f)$ is a quasi-isomorphism from $\mathrm{H}_{\mathrm{I}}\,\mathrm{H}_{\mathrm{II}}(X) \xrightarrow{\sim} \mathrm{H}_{\mathrm{I}}\,\mathrm{H}_{\mathrm{II}}(Y)$.

Example 5.6.4 (Deriving a bifunctor) Let the Abelian categories $\mathcal{A}_1$ and $\mathcal{A}_2$ have enough injective objects (or projective objects), and let the bifunctor $F : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{B}$

3) More precisely, neither the existence of the limit terms $Z^{p,q}_\infty$, $B^{p,q}_\infty$, $E^{p,q}_\infty$ nor the condition on exhaustive filtrations in Definition 5.1.3 causes any difficulty.

be left exact (or right exact) in each variable. We discuss the left-exact case. Take $X_i \in \mathrm{Ob}(\mathcal{A}_i)$ and choose injective resolutions $0 \to X_i \to I_i^0 \to \cdots$; define $I_i^n := 0$ for $n < 0$ $(i = 1, 2)$. These data form a first-quadrant bicomplex

$$Y^{p,q} := F\left(I_1^p, I_2^q\right).$$

The corresponding first-quadrant spectral sequence $\mathscr{E}_{\mathrm{I}}$ therefore satisfies

$$E_1^{p,q} = \mathrm{H}^q\left(F(I_1^p, I_2^\bullet)\right) \Rightarrow \mathrm{H}^{p+q}(\mathrm{tot}(Y)), \quad p, q \in \mathbb{Z}.$$

The right-hand side $\mathrm{H}^{p+q}(\mathrm{tot}\, Y)$ is the value of the right-derived bifunctor $\mathrm{R}^{p+q}F(X_1, X_2)$; see Definition 4.7.7.

Now suppose in addition that F is balanced (Definition 3.14.1); then $F(I_1^p, \cdot)$ is exact. This shows that $q \neq 0 \implies E_1^{p,q} = 0$, while $E_1^{p,0} = F(I_1^p, X_2)$. Hence $q \neq 0 \implies E_2^{p,q} = 0$, and

$$\begin{aligned} E_2^{p,0} &= \mathrm{H}^p\left[\cdots \to F(I_1^p, X_2) \to F(I_1^{p+1}, X_2) \to \cdots\right] \\ &= (\mathrm{R}_{\mathrm{I}}^p F)(X_1, X_2), \quad \text{with the notation of Theorem 3.14.2.} \end{aligned}$$

In particular, the spectral sequence degenerates at the E_2 page, so

$$\begin{aligned} &E_2^{p,q} = E_\infty^{p,q}, \\ &E_\infty^{p,q} = \mathrm{gr}^p\, \mathrm{H}^{p+q}\left(\mathrm{tot}(Y)\right) = 0, \quad \text{if } q \neq 0, \\ &E_\infty^{p,0} = \mathrm{H}^p\left(\mathrm{tot}(Y)\right) = \mathrm{R}^p F(X_1, X_2). \end{aligned}$$

Thus $\mathrm{R}^n F(X_1, X_2) \simeq (\mathrm{R}_{\mathrm{I}}^n F)(X_1, X_2)$. If instead we use $\mathscr{E}_{\mathrm{II}}$, the same argument gives $\mathrm{R}^n F(X_1, X_2) \simeq (\mathrm{R}_{\mathrm{II}}^n F)(X_1, X_2)$. Hence $\mathrm{R}_{\mathrm{I}} F \simeq \mathrm{R}_{\mathrm{II}} F$, which is precisely the assertion of Theorem 3.14.2.

Example 5.6.5 (The spectral sequence of hyperderived functors) Let $\mathcal{A}$ and $\mathcal{B}$ be Abelian categories, let $\mathcal{A}$ have enough injective objects (or projective objects), and let $F : \mathcal{A} \to \mathcal{B}$ be a left-exact (or right-exact) additive functor. In the first half of §3.12, for a complex $X \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$ (or $X \in \mathrm{Ob}(\mathsf{C}^-(\mathcal{A}))$), we defined the right-derived functors $\mathrm{R}^n F(X)$ (or the left-derived functors $\mathrm{L}_n F(X)$), where $n \in \mathbb{Z}$. When $X \in \mathrm{Ob}(\mathcal{A})$, these are derived functors in the classical sense. By contrast, the case of a general complex is customarily called that of hyperderived functors. Spectral sequences link the two cases. We describe the case of right-derived functors; interchanging superscripts and subscripts gives the version for left-derived functors.

Take $X \in \mathrm{Ob}(\mathsf{C}^+(\mathcal{A}))$. Regard X as a bicomplex concentrated in row 0, and take the Cartan–Eilenberg resolution $\epsilon : X \to I$ supplied by Theorem 3.11.10; this is a morphism in $\mathsf{C}_f^2(\mathcal{A})$. By Remark 3.11.12 or the result of Example 5.6.3, $\mathrm{tot}(\epsilon) : X =$

$\operatorname{tot}(X) \to \operatorname{tot}(I)$ is a quasi-isomorphism in $\mathsf{C}^+(\mathcal{A})$ and hence is an injective resolution of X.

From the bicomplex $(\mathsf{C}^2 F)(I) \in \operatorname{Ob}\left(\mathsf{C}^2_f(\mathcal{B})\right)$, construct the convergent spectral sequences $\mathscr{E}_{\mathrm{I}}$ and $\mathscr{E}_{\mathrm{II}}$. They converge to the same target, $\mathrm{H}^{p+q} \operatorname{tot}\left((\mathsf{C}^2 F) I\right) = \mathrm{H}^{p+q}\, \mathsf{C}F(\operatorname{tot} I)$, namely the value of the hyperderived functor $\mathrm{R}^{p+q}F(X)$.

First consider $\mathscr{E}_{\mathrm{I}}$. Since $I^{p,\bullet}$ is an injective resolution of X^p, we have $E^{p,q}_{\mathrm{I},1} = \mathrm{H}^q\left(F I^{p,\bullet}\right) \simeq \mathrm{R}^q F(X^p)$; the right-hand side is the value of the classical derived functor $\mathrm{R}^q F$ at X^p.[4)] Similarly,

$$E^{p,q}_{\mathrm{I},2} = \mathrm{H}^p\left[\cdots \to \mathrm{R}^q F(X^p) \to \mathrm{R}^q F(X^{p+1}) \to \cdots\right].$$

Perhaps more interesting is the E_2 page of $\mathscr{E}_{\mathrm{II}}$. By the properties of a Cartan–Eilenberg resolution, taking horizontal cohomology gives $\mathrm{H}_{\mathrm{I}}(I)^{q,\bullet}$ as an injective resolution of $\mathrm{H}^q(X)$, while Remark 3.11.11 shows that F preserves horizontal cohomology: $\mathrm{H}_{\mathrm{I}}(\mathsf{C}^2 F(I))^{q,\bullet} \simeq \mathsf{C}F\left(\mathrm{H}_{\mathrm{I}}(I)^{q,\bullet}\right)$. Thus

$$\begin{aligned} E^{p,q}_{\mathrm{II},2} &= \boxed{\begin{array}{lcr} \text{take H second} & \uparrow p & \\ & \longrightarrow q & \text{take H first} \end{array}} = (\mathrm{R}^p F)\left(\mathrm{H}^q(X)\right) \\ &\Rightarrow \mathrm{R}^{p+q}F(X), \quad p, q \in \mathbb{Z}. \end{aligned}$$

The Grothendieck spectral sequence introduced below concerns deriving a composition of functors. It encompasses a large class of spectral sequences in geometry and algebra, and its argument, like that in Example 5.6.5, is also based on a Cartan–Eilenberg resolution.

Theorem 5.6.6 (Grothendieck spectral sequence) Consider additive functors between Abelian categories

$$\mathcal{A} \xrightarrow{F} \mathcal{A}' \xrightarrow{F'} \mathcal{A}''.$$

Suppose both are left-exact (or right-exact) functors, $\mathcal{A}$ and $\mathcal{A}'$ have enough injective objects (or projective objects), and F sends injective objects (or projective objects) to F'-acyclic objects in the sense of Convention 3.12.8. Then for every $X \in \operatorname{Ob}(\mathcal{A})$ there is a first-quadrant cohomological (or homological) bigraded spectral sequence

$$\begin{aligned} E^{p,q}_2 &= (\mathrm{R}^p F')(\mathrm{R}^q F)(X) \Rightarrow \mathrm{R}^{p+q}(F'F)(X), \\ \text{or} \quad E^2_{p,q} &= (\mathrm{L}_p F')(\mathrm{L}_q F)(X) \Rightarrow \mathrm{L}_{p+q}(F'F)(X). \end{aligned}$$

[4)] Translator's correction: the source prints $\mathrm{R}^p F$, whereas the preceding formula and the vertical cohomological index require $\mathrm{R}^q F$.

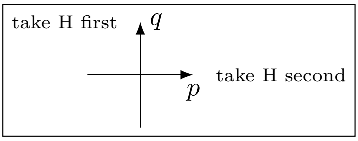
Its convergence properties are as in Theorem 5.6.2. The corresponding exact sequences of low-degree terms (Corollary 5.5.6) can respectively be written as

$$\begin{aligned} 0 \to (\mathrm{R}^1 F')(FX) &\to \mathrm{R}^1(F'F)(X) \\ &\to F'\big((\mathrm{R}^1F)X\big) \to (\mathrm{R}^2F')(FX) \to \mathrm{R}^2(F'F)(X), \end{aligned}$$

or

$$\begin{aligned} \mathrm{L}_2(F'F)(X) \to (\mathrm{L}_2F')(FX) &\to F'\left((\mathrm{L}_1F)X\right) \\ &\to \mathrm{L}_1(F'F)(X) \to (\mathrm{L}_1F')(FX) \to 0. \end{aligned}$$

Proof By duality, it suffices to discuss the cohomological case. Take an injective resolution $0 \to X \to I^0 \to I^1 \to \cdots$ of X and define $I^n := 0$ for $n < 0$. In $\mathcal{A}'$, take a Cartan–Eilenberg resolution $\mathsf{C}F(I) \to J$ of $\mathsf{C}F(I) := (FI^p)_p$ (Theorem 3.11.10), and then consider the first-quadrant bicomplex $\mathsf{C}^2F'(J) = (F'(J^{p,q}))_{p,q}$ and its corresponding convergent spectral sequences $\mathscr{E}_{\mathrm{I}}$ and $\mathscr{E}_{\mathrm{II}}$. First,

$E^{p,q}_{\mathrm{I},2} =$ 

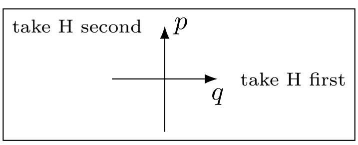
$$\begin{aligned} &= \mathrm{H}^p\left[\cdots \to (\mathrm{R}^qF')(FI^p) \to (\mathrm{R}^qF')(FI^{p+1}) \to \cdots\right] \\ &\Rightarrow \mathrm{H}^{p+q}\,\mathrm{tot}\left(\mathsf{C}^2F'(J)\right). \end{aligned}$$

Since $F(I^p)$ is F'-acyclic by hypothesis, $E^{p,q}_{\mathrm{I},2} = 0$ when $q \neq 0$. The same argument as in Example 5.6.4 then shows that $\mathscr{E}_{\mathrm{I}}$ degenerates at the E_2 page, and

$$E^{p,q}_{\mathrm{I},\infty} = E^{p,q}_{\mathrm{I},2} = \begin{cases} \mathrm{H}^p\left(F'F(I^\bullet)\right) = \mathrm{R}^p(F'F)(X), & q = 0, \\ 0, & q \neq 0, \end{cases}$$

while $\mathrm{H}^p\,\mathrm{tot}\left(\mathsf{C}^2F'(J)\right) = E^{p,0}_{\mathrm{I},\infty} = \mathrm{R}^p(F'F)(X)$.

Now consider $\mathscr{E}_{\mathrm{II}}$. The technique is similar to that in Example 5.6.5: the q-th column of horizontal cohomology, $\mathrm{H}_{\mathrm{I}}(J)^{q,\bullet}$, gives an injective resolution of $\mathrm{H}^q(\mathsf{C}F(I)) = (\mathrm{R}^qF)(X)$. Recalling also that F' preserves horizontal cohomology, we obtain

$E^{p,q}_{\mathrm{II},2} =$ 

$= (\mathrm{R}^pF')(\mathrm{R}^qF)\,(X)$

$$\Rightarrow \mathrm{H}^{p+q}\,\mathrm{tot}\left(\mathsf{C}^2F'(J)\right) = \mathrm{R}^{p+q}(F'F)(X).$$

Thus $\mathscr{E}_{\mathrm{II}}$ is the spectral sequence required in the first part. Substituting the description of $E^{p,q}_{\mathrm{II},2}$ into the low-degree exact sequence of $\mathscr{E}_{\mathrm{II}}$ (Corollary 5.5.6) gives the assertion in the final part immediately. □

The Grothendieck spectral sequence can be regarded as a concrete version of Theorem 4.8.8, but the information it contains is more explicit than an isomorphism in the derived category. We shall give an application in §6.9.

Example 5.6.7 (Change of rings) Let $R \to S$ be a ring homomorphism. Let X be a right S-module and Y a left R-module. Write S_R (or X_R) for S (or X) regarded as a right R-module. We claim that there is a first-quadrant homological bigraded spectral sequence

$$E^2_{p,q} = \mathrm{Tor}^S_p\left(X, \mathrm{Tor}^R_q(S_R, Y)\right) \Rightarrow \mathrm{Tor}^R_{p+q}(X_R, Y);$$

Here $\mathrm{Tor}^R_q(S_R, Y)$ is given its left S-module structure as in Remark 3.14.8. This is a direct application of Theorem 5.6.6, arising from the isomorphism of functors

$$X \underset{S}{\otimes} \left(S \underset{R}{\otimes} (\cdot) \right) \simeq X \underset{R}{\otimes} (\cdot) : R\text{-}\mathsf{Mod} \to \mathsf{Ab},$$

where the inner functor $S \underset{R}{\otimes} (\cdot) : R\text{-}\mathsf{Mod} \to S\text{-}\mathsf{Mod}$ has left-derived functors $\mathrm{Tor}^R_\bullet(S_R, \cdot)$ taking values in $S\text{-}\mathsf{Mod}$. By the isomorphism above, the inner functor sends flat modules to flat modules, so the homological version of the Grothendieck spectral sequence indeed applies.

With entirely analogous notation and techniques, for a right R-module X and a left S-module Y, one also has

$$E^2_{p,q} = \mathrm{Tor}^S_p\left(\mathrm{Tor}^R_q(X, {}_R S), Y\right) \Rightarrow \mathrm{Tor}^R_{p+q}(X, {}_R Y).$$

Next consider the Ext functor. Let X be a left S-module and Y a left R-module, and write ${}_R X$ for X regarded as a left R-module. Give $\mathrm{Ext}^q_R({}_R S, Y)$ its left S-module structure as in Remark 3.14.8. Then there are first-quadrant cohomological bigraded spectral sequences

$$\begin{aligned} E_2^{p,q} &= \mathrm{Ext}^p_S\left(X, \mathrm{Ext}^q_R({}_R S, Y)\right) \Rightarrow \mathrm{Ext}^{p+q}_R({}_R X, Y), \\ E_2^{p,q} &= \mathrm{Ext}^p_S\left(\mathrm{Tor}^R_q(S_R, Y), X\right) \Rightarrow \mathrm{Ext}^{p+q}_R(Y, {}_R X). \end{aligned}$$

They correspond respectively to the following isomorphisms of composite functors:

$$\begin{aligned} \mathrm{Hom}_S\left(X, \mathrm{Hom}_R({}_R S, \cdot)\right) &\simeq \mathrm{Hom}_R({}_R X, \cdot), \\ \mathrm{Hom}_S\left(S \underset{R}{\otimes} (\cdot), X \right) &\simeq \mathrm{Hom}_R(\cdot, {}_R X), \end{aligned}$$

the adjunctions introduced in [51, Corollary 6.6.8]. Mixing the right-derived functors Ext^p_S and the left-derived functors Tor^R_q in the second line causes no problem, because the first variable of Hom_S lies in the opposite category $S\text{-}\mathsf{Mod}^{\mathrm{op}}$.

The spectral sequences above should be compared with the derived-category versions in §4.12.

Example 5.6.8 For the spectral sequence $E_2^{p,q} = \mathrm{Ext}_S^p\left(\mathrm{Tor}_q^R(S_R, Y), X\right) \Rightarrow \mathrm{Ext}_R^{p+q}(Y, {}_RX)$ from Example 5.6.7, we record a useful special case, also as an exercise. Suppose the global dimension of S-Mod is at most 1 (Definition–Proposition 4.9.7). Consequently, the nonzero terms $E_2^{p,q}$ are concentrated in the region $p = 0, 1$. Thus, for $n \geq 0$, the decreasing filtration $\mathrm{F}^\bullet$ on $\mathrm{Ext}_R^n(Y, {}_RX)$ must have the form

$$\mathrm{Ext}_R^n(Y, {}_RX) = \mathrm{F}^0 \underset{E_\infty^{0,n}}{\supsetneq} \mathrm{F}^1 \underset{E_\infty^{1,n-1}}{\supsetneq} \mathrm{F}^2 = \{0\}.$$

Moreover, Example 5.2.9 shows that the spectral sequence degenerates at the E_2 page, so $E_2^{p,q} \simeq E_\infty^{p,q}$. Combining these facts gives the short exact sequence

$$0 \to \mathrm{Ext}_S^1\left(\mathrm{Tor}_{n-1}^R(S_R, Y), X\right) \to \mathrm{Ext}_R^n(Y, {}_RX) \to \mathrm{Hom}_S\left(\mathrm{Tor}_n^R(S_R, Y), X\right) \to 0;$$

by convention $\mathrm{Tor}_{-1}^R = 0$. This may be regarded as a variant of the universal coefficient theorem.

Similarly, suppose the right S-module X has Tor-dimension at most 1 (Example 4.8.10). The spectral sequence $E_{p,q}^2 = \mathrm{Tor}_p^S(X, \mathrm{Tor}_q^R(S_R, Y)) \Rightarrow \mathrm{Tor}_{p+q}^R(X_R, Y)$ degenerates at the E^2 page. Examining the corresponding increasing filtration on $\mathrm{Tor}_n^R(X_R, Y)$ gives the short exact sequence

$$0 \to X \underset{S}{\otimes} \mathrm{Tor}_n^R(S_R, Y) \to \mathrm{Tor}_n^R(X_R, Y) \to \mathrm{Tor}_1^S(X, \mathrm{Tor}_{n-1}^R(S_R, Y)) \to 0.$$

If S is a principal ideal domain, both the global-dimension and Tor-dimension hypotheses hold automatically.

5.7 A glimpse of multiplicative structures

For simplicity and concreteness, in this section we take a commutative ring $\Bbbk$ and $\mathcal{A} = \Bbbk$-Mod; this suffices for the classical applications. We refer to $\Bbbk$-modules simply as modules and to $\Bbbk$-algebras simply as algebras, and write $\otimes := \otimes_\Bbbk$.

Let I be a commutative monoid, with its binary operation written additively. The definitions from [51, §7.4] may be summarized as follows:

- ◇ an I-graded module is a module with a direct-sum decomposition $M = \bigoplus_{i \in I} M^i$; elements of M^i are called homogeneous elements of degree i;

◇ an I-graded algebra is a graded module equipped with a homomorphism $\mu : A \otimes A \to A$, also written as a multiplication $xy := \mu(x \otimes y)$, such that A becomes a ring and

$$1_A \in A^0, \qquad A^i \cdot A^j \subset A^{i+j}, \qquad i, j \in I;$$

thus the multiplication μ is completely determined by the data $\mu^{i,j} : A^i \otimes A^j \to A^{i+j}$;

◇ homomorphisms and isomorphisms between I-graded modules or algebras are defined in the usual way; I-graded subalgebras and ideals are defined similarly.

For $k \in I$, also define a homomorphism of degree k between I-graded modules: this is a module homomorphism $\varphi : M \to N$ satisfying $\varphi(M^i) \subset M^{i+k}$ for every i, and it is determined by the data $(\varphi^i : M^i \to M^{i+k})_{i \in I}$. Taking $k = 0$ recovers homomorphisms in the usual sense. Degrees of homomorphisms add under composition.

Definition 5.7.1 Let I be a commutative monoid equipped with a homomorphism $\epsilon : I \to \mathbb{Z}/2\mathbb{Z}$. A **differential I-graded algebra** whose differential has degree k is an I-graded algebra $A = \bigoplus_{p \in I} A^p$ together with an endomorphism of degree $k \in I$, $d = (d^p)_p : A \to A$, satisfying $d^2 = 0$ and the Leibniz rule

$$d(xy) = (dx) \cdot y + (-1)^{\epsilon(p)} x \cdot dy, \qquad x \in A^p, \quad y \in A^{p'}, \quad p, p' \in I.$$

Observe that $d(1_A) = d(1_A \cdot 1_A) = d(1_A) + d(1_A)$, hence $d(1_A) = 0$. Clearly $\ker(d)$ is an I-graded subalgebra of A, while $\mathrm{im}(d)$ is a two-sided I-graded ideal in $\ker(d)$. Therefore

$$\mathrm{H}(A, d) := \ker(d) / \mathrm{im}(d)$$

is again an I-graded algebra.

Example 5.7.2 Take $I = \mathbb{Z}$ and $\epsilon(p) = p \mod 2$. For any complex X over a $\Bbbk$-linear category $\mathcal{B}$, the Hom complex $\mathrm{Hom}^\bullet(X, X)$ of Definition 3.2.1, together with $d = d_{\mathrm{Hom}^\bullet(X,X)}$, becomes a differential $\mathbb{Z}$-graded algebra whose differential has degree 1. These assertions are essentially the content of Lemma 3.2.4. We have

$$\mathrm{H}(\mathrm{Hom}^\bullet(X, X), d) = \bigoplus_{n \in \mathbb{Z}} \mathrm{Hom}_{\mathsf{K}(\mathcal{B})}(X, X[n]),$$

with multiplication induced by composition of morphisms in $\mathsf{K}(\mathcal{B})$.

Example 5.7.3 Take $\Bbbk = \mathbb{C}$, $I = \mathbb{Z}$, and $\epsilon(p) = p \mod 2$. The $\mathbb{C}$-valued differential forms of degree p on a smooth manifold $\mathfrak{X}$ form a vector space $A^p(\mathfrak{X})$. Set $A(\mathfrak{X}) := \bigoplus_{p=0}^{\dim X} A^p(\mathfrak{X})$. With multiplication $\mu(\omega \otimes \eta) := \omega \wedge \eta$ and exterior differentiation d, this becomes a differential $\mathbb{Z}$-graded algebra whose differential has degree 1. The associated algebra $\mathrm{H}(A(\mathfrak{X}), \mathrm{d}) = \bigoplus_p \mathrm{H}^p_{\mathrm{dR}}(\mathfrak{X})$ is precisely the de Rham cohomology of $\mathfrak{X}$; the multiplication induced by $\wedge$ gives it a graded-algebra structure. This is a fundamental object in topology and is isomorphic as a graded algebra to the singular cohomology ring of $\mathfrak{X}$.

The multiplication on $\mathrm{H}(A(\mathfrak{X}), \mathrm{d})$ also satisfies $xy = (-1)^{pq}yx$, where x and y are homogeneous elements of degrees p and q, respectively, because the multiplication on $A(\mathfrak{X})$ already has this property. As a simple and beautiful example, complex projective space $\mathfrak{X} := \mathbb{P}^n(\mathbb{C})$ gives the graded algebra $\mathrm{H}(A(\mathfrak{X}), \mathrm{d}) \simeq \mathbb{C}[t]/(t^{n+1})$, where the variable t corresponds to a homogeneous element of degree 2.

Returning to our main subject, in this section we consider the following cases:

Abbreviation	I	$\epsilon : I \to \mathbb{Z}/2\mathbb{Z}$
Singly graded	$\mathbb{Z}$	$\epsilon(p) = p \mod 2$
Bigraded	$\mathbb{Z}^2$	$\epsilon(p, q) = p + q \mod 2$

This theory overlaps with Definition 5.2.1, but the focus of this section is the multiplication, which was not mentioned there.

To lighten the notation, below we collectively call these various cases "graded", distinguishing them mainly by the superscript p or (p, q). Likewise, we refer collectively to differential I-graded algebras as differential graded algebras.

Convention 5.7.4 For a cohomological bigraded spectral sequence $\mathscr{E}$ in the case $\mathcal{A} = \Bbbk\text{-}\mathsf{Mod}$, we identify every page E_r with the graded module $\bigoplus_{p,q} E_r^{p,q}$ and regard $d_r = (d_r^{p,q})_{p,q}$ as an endomorphism of degree $(r, -r+1)$ of the graded module E_r. The homological and singly graded cases are treated similarly.

In short, a spectral sequence with a multiplicative structure is a spectral sequence consisting of differential graded algebras.

Definition 5.7.5 A **multiplicative structure** on a cohomological bigraded spectral sequence $\mathscr{E}$ is a family of homomorphisms $\mu_r : E_r \otimes E_r \to E_r$ (that is, "multiplications") such that

- ◇ every (E_r, μ_r, d_r) is a differential graded algebra whose differential has degree $(r, -r+1)$;
- ◇ the map $t_{r+1} : \mathrm{H}(E_r, d_r) \xrightarrow{\sim} E_{r+1}$ in the spectral sequence data is an isomorphism of graded algebras.

Multiplicative structures on singly graded or homological spectral sequences are defined similarly.

The preceding discussion shows that $Z_r = \ker(d_r)$ is a graded subalgebra of E_r, while $B_r = \mathrm{im}(d_r)$ is a two-sided graded ideal of Z_r. Hence $\mathrm{H}(E_r, d_r)$ becomes a graded algebra; this is the precise meaning of Definition 5.7.5.

Furthermore, if the limit exists, then $Z_\infty = \bigoplus_{p,q} Z_\infty^{p,q}$ is also a graded subalgebra, with $B_\infty = \bigoplus_{p,q} B_\infty^{p,q}$ as its graded ideal. Thus E_∞ is also a graded algebra.

As an example, consider differential graded algebras whose differential has degree 1, abbreviated as dg-algebras. Expanding the definition, this is equivalent to a complex of $\Bbbk$-modules $X = (X^n, d_X^n)_{n\in\mathbb{Z}}$ such that $X \xlongequal{\text{identified}} \bigoplus_n X^n$ is equipped with a multiplication $\mu : X \otimes X \to X$ satisfying $X^p \cdot X^{p'} \subset X^{p+p'}$ and the Leibniz rule

$$d(xy) = dx \cdot y + (-1)^p x \cdot dy, \qquad x \in X^p, \quad y \in X^{p'}.$$

Now suppose the data (X, d) are extended to a filtered complex $(X, d, \mathrm{F}^\bullet X)$. Recall that this already entails $d(\mathrm{F}^p X) \subset \mathrm{F}^p X$; we further require the multiplication on X to be compatible with the filtration:

$$\mathrm{F}^p X^n \cdot \mathrm{F}^{p'} X^{n'} \subset \mathrm{F}^{p+p'} X^{n+n'}.$$

In this case $(X, d, \mu, \mathrm{F}^\bullet X)$ is called a filtered differential graded algebra. For the induced filtration on $\mathrm{H}(X, d)$, the condition above ensures that

$$\mathrm{gr}\,\mathrm{H}(X, d) \xlongequal{\text{identified}} \bigoplus_{(p,q)\in\mathbb{Z}^2} \mathrm{gr}^p\,\mathrm{H}^{p+q}(X, d)$$

naturally becomes a graded algebra.

Proposition 5.7.6 Let $(X, d, \mu, \mathrm{F}^\bullet X)$ be a filtered differential graded algebra whose differential has degree 1.

(i) The spectral sequence $\mathscr{E}$ of the filtered complex $(X, d, \mathrm{F}^\bullet X)$ has a canonical multiplicative structure.

(ii) The isomorphisms $E_\infty^{p,q} \simeq \mathrm{gr}^p\,\mathrm{H}^{p+q}(X, d)$ in the classical convergence theorem 5.5.5 in fact give an isomorphism of graded algebras $E_\infty \simeq \mathrm{gr}\,\mathrm{H}(X, d)$.

Proof Take $x \in E_r^{p,q}$ and $y \in E_r^{p',q'}$. By the description in Proposition 5.5.2, choose lifts

$$\tilde{x} \in \mathrm{F}^p X^{p+q} \cap d^{-1}\left(\mathrm{F}^{p+r} X^{p+q+1}\right), \qquad \tilde{y} \in \mathrm{F}^{p'} X^{p'+q'} \cap d^{-1}\left(\mathrm{F}^{p'+r} X^{p'+q'+1}\right).$$

By definition,

$$\begin{gathered} \tilde{x}\tilde{y} \in \mathrm{F}^{p+p'} X^{p+q+p'+q'}, \\ d(\tilde{x}\tilde{y}) = d\tilde{x} \cdot \tilde{y} + (-1)^{p+q} \tilde{x} \cdot d\tilde{y} \in \mathrm{F}^{p+p'+r} X^{p+q+p'+q'+1}. \end{gathered}$$

Thus $\widetilde{x}\widetilde{y}$ determines an element $xy \in E_r^{p+p',q+q'}$. A routine calculation (please verify it) shows that xy depends only on x and y.

Next, $d_r : E_r \xrightarrow{(r,-r+1)} E_r$ is induced by the differential d on X. This gives E_r the structure of a differential graded algebra; associativity, the Leibniz rule, and all other required properties reduce to checks on (X, d). The same argument shows that $\mathrm{H}(E_r, d_r) \simeq E_{r+1}$ is an isomorphism of graded algebras.

The assertion that the canonical isomorphisms $E_\infty^{p,q} \simeq \mathrm{gr}^p \mathrm{H}^{p+q}(X, d)$ in the classical convergence theorem preserve multiplication likewise reduces to a check on X, so we omit the details. □

Remark 5.7.7 Multiplicative structures are an indispensable tool in topology. Algebraic topology without multiplication is almost unimaginable, or at least insipid. Historically, when Leray first encountered spectral sequences, he already considered multiplicative structures, calling them "spectral rings". They often greatly simplify spectral-sequence calculations in topology. This is why the present section gives a brief introduction to multiplicative structures, though it only scratches the surface.

There is an evident difficulty in applying Definition 5.7.5. The differential graded algebra structures on all the E_r must be given at once, since the definition alone provides no reason for the multiplication on one page to induce a multiplication on the next. This can indeed be done in some situations, as in Proposition 5.7.6.

For general cases, or for those that are difficult to compute by hand, spectral sequences are often constructed from simple data, such as the exact couples of §5.3. It is therefore natural to ask whether there are simple conditions on an exact couple ensuring that the corresponding spectral sequence has a multiplicative structure.

The answer appears to be negative: information beyond the exact couple is required. One useful construction is the equally classical **Cartan–Eilenberg system**, together with its spectral product; see [11, II.A]. As an application, the Serre–Atiyah–Hirzebruch spectral sequence, which is of central importance in topology, has a multiplicative structure for every multiplicative generalized cohomology theory. Since this material departs from our main line, we stop here.

Exercises

1. For an Abelian category $\mathcal{A}$, verify the following properties of the category of filtered objects $\mathrm{Fil}^\bullet(\mathcal{A})$.

   (i) It is an additive category, and every morphism has a kernel, cokernel, image, and coimage. Describe them as concretely as possible.

(ii) For a morphism $f : X \to Y$ in $\mathrm{Fil}^\bullet(\mathcal{A})$, if $f(\mathrm{F}^n X) \to f(X) \cap \mathrm{F}^n Y$ is an isomorphism for every n, then f is called a strict morphism. Prove that this notion is equivalent to the version in Definition 1.2.4.

Hint This is equivalent to saying that the two natural filtrations on $f(X)$ coincide: one is the image of $\mathrm{F}^\bullet X$, and the other is the restriction of $\mathrm{F}^\bullet Y$. The former corresponds to $\mathrm{coim}(f)$ in $\mathrm{Fil}^\bullet(\mathcal{A})$, and the latter to $\mathrm{im}(f)$.

(iii) Give an example showing that $\mathrm{Fil}^\bullet(\mathcal{A})$ is not an Abelian category in general.

Although filtered objects do not form an Abelian category, geometry still requires the filtered derived category $\mathsf{DF}(\mathcal{A})$ in the case of finite filtrations. Its definition requires rather deeper techniques; see [42, Tag 05RX].

2. For a filtered differential object $(X, d, \mathrm{F}^\bullet X)$ over an Abelian category $\mathcal{A}$, prove that $d_1^p : E_1^p \to E_1^{p+1}$ in the spectral sequence is the connecting morphism induced by the short exact sequence of differential objects

$$0 \to \mathrm{gr}^{p+1} X \to \mathrm{F}^p X/\mathrm{F}^{p+2} X \to \mathrm{gr}^p X \to 0.$$

3. (Bockstein spectral sequence) Let f be a non-zero-divisor in a commutative ring R. For every R-module M, define $M[f] := \{m \in M : fm = 0\}$; call M ***f*-torsion-free** if $M[f] = \{0\}$. Let C be a chain complex and suppose every C_n is f-torsion-free. Multiplication by f gives a monomorphism of chain complexes $\alpha : C \to C$. For the data (C, α), Proposition 5.3.4 gives the exact couple

$$\begin{tikzcd}
\mathrm{H}_\bullet(C) \arrow[rr, "\mathrm{H}_\bullet(\alpha)"] & & \mathrm{H}_\bullet(C) \arrow[dl] \\
& \mathrm{H}_\bullet\left(C \underset{R}{\otimes} R/(f)\right) \arrow[ul, "-1"] &
\end{tikzcd}$$

and the associated homologically graded spectral sequence $(E_q^r)_{\substack{r\geq 0\\ q\in\mathbb{Z}}}$. Describe every page (E^r, d^r) and E^∞ as explicitly as possible.

4. In the preceding problem, take $R = \mathbb{Z}$ and $f = p$, with p prime, and suppose every C_n is a free $\mathbb{Z}$-module of finite rank. For every $\mathbb{Z}$-module M, write M_{tor} for the submodule of all its torsion elements and define its torsion-free quotient $M_{\mathrm{tf}} := M/M_{\mathrm{tor}}$. Prove that

$$E_q^\infty \simeq \mathrm{H}_q(C)_{\mathrm{tf}} \underset{\mathbb{Z}}{\otimes} \mathbb{F}_p, \qquad q \in \mathbb{Z}.$$

Deduce that

$$\dim_{\mathbb{F}_p} \mathrm{H}_q\left(C \underset{\mathbb{Z}}{\otimes} \mathbb{F}_p\right) \geq \dim_{\mathbb{F}_p}\left(\mathrm{H}_q(C)_{\mathrm{tf}} \underset{\mathbb{Z}}{\otimes} \mathbb{F}_p\right).$$

Give an example in which the inequality is strict.

5. (Two-column and two-row spectral sequences) Let a strongly convergent spectral sequence $E_2^{p,q} \Rightarrow H^{p+q}$ be given, in the sense of Convention 5.5.4.

(i) Prove that if $E_2^{p,q}$ is nonzero only for $p \in \{0,1\}$, then for every $q \in \mathbb{Z}$ there is a canonical short exact sequence

$$0 \to E_2^{1,q-1} \to H^q \to E_2^{0,q} \to 0.$$

(ii) Prove that if $E_2^{p,q}$ is nonzero only for $q \in \{0,1\}$, then for every $p \in \mathbb{Z}$ there is a canonical exact sequence

$$\cdots \to H^{p-1} \to E_2^{p-2,1} \xrightarrow{d_2} E_2^{p,0} \to H^p \to E_2^{p-1,1} \xrightarrow{d_2} E_2^{p+1,0} \to H^{p+1} \to \cdots.$$

(iii) Treat the version $E^2_{p,q} \Rightarrow H_{p+q}$. Hint〉 Interchange superscripts and subscripts, reverse the arrows, and leave everything else unchanged.

6. Consider left-exact functors F' and F satisfying the hypotheses of Theorem 5.6.6, and suppose in addition that $\mathrm{R}F'$ and $\mathrm{R}F$ are both of finite dimension.

(i) Prove that $F'F$ is also of finite dimension, and give an upper bound for its dimension.

(ii) Consider the homomorphism $\chi_F : \mathrm{K}_0(\mathcal{A}) \to \mathrm{K}_0(\mathcal{A}')$ determined by the triangulated functor $\mathrm{R}F$, and the analogous homomorphisms $\chi_{F'}$, $\chi_{F'F}$ determined by $\mathrm{R}F'$, $\mathrm{R}(F'F)$; see the exercises in Chapter 3. Prove that $\chi_{F'F} = \chi_{F'} \circ \chi_F$.

These properties can also be proved using derived categories. The case of right-exact functors and $\mathrm{L}F', \mathrm{L}F$ is of course dual.

7. Let $\mathcal{A}$ be an Abelian category with enough injective objects, and let $X \in \mathrm{Ob}(\mathcal{A})$ carry a finite filtration $X = \mathrm{F}^0 X \supset \cdots \supset \mathrm{F}^{N+1} X = 0$. Let $G : \mathcal{A} \to \mathcal{A}'$ be a left-exact additive functor. Prove that there is a strongly convergent spectral sequence

$$E_1^{p,q} = \mathrm{R}^{p+q} G(\mathrm{gr}^p X) \Rightarrow \mathrm{R}^{p+q} G(X).$$

Hint〉 One method is to imitate the construction of a Cartan–Eilenberg resolution (Theorem 3.11.10, which uses Proposition 3.11.9). Starting from $\mathrm{F}^N X$, choose a suitable series of injective resolutions

$$\begin{array}{ccccc}
\vdots & & & & \vdots \\
\uparrow & & & & \uparrow \\
\mathrm{F}^0 I^1 & \supset & \cdots & \supset & \mathrm{F}^N I^1 \\
\uparrow & & & & \uparrow \\
\mathrm{F}^0 I^0 & \supset & \cdots & \supset & \mathrm{F}^N I^0 \\
\uparrow & & & & \uparrow \\
\mathrm{F}^0 X & \supset & \cdots & \supset & \mathrm{F}^N X \\
\uparrow & & & & \uparrow \\
0 & & & & 0
\end{array}$$

such that applying $\mathsf{C}G$ to $\mathrm{F}^\bullet(I)$ still gives a filtered complex and the associated spectral sequence has the required form.

8. Let R be a ring, let $C = (C_n, d_n^C)_n$ be a chain complex of right R-modules, and let D be a left R-module. Suppose every C_n is flat and $n \ll 0 \implies C_n = 0$. Take a projective resolution $\cdots \to P_1 \to P_0 \to D \to 0$ and construct the bicomplex $C_\bullet \underset{R}{\otimes} P_\bullet$.

(i) Show that the associated homological bigraded spectral sequence $\mathscr{E}_{\mathrm{I}}$ degenerates at the E_{I}^2 page.

(ii) Show that $(E_{\mathrm{II}}^2)_{p,q} = \mathrm{Tor}_p^R(\mathrm{H}_q(C), D) \Rightarrow \mathrm{H}_{p+q}(C_\bullet \underset{R}{\otimes} D)$.

(iii) Prove that if every $\mathrm{im}(d_n^C)$ is flat, this gives the short exact sequence in the homological Künneth theorem 3.14.12. Hint The flat resolution $0 \to \mathrm{im}(d_C^{n+1}) \to \ker(d_C^n) \to \mathrm{H}_n(C) \to 0$ causes the nonzero terms of E_{II}^2 to be concentrated at $p \in \{0, 1\}$.

9. Let R, S be rings, let X be a right R-module, Y an (R, S)-bimodule, and Z a left S-module. Construct a bicomplex such that the corresponding spectral sequences satisfy

$$(E_{\mathrm{I}}^2)_{p,q} = \mathrm{Tor}_p^R\left(X, \mathrm{Tor}_q^S(Y_S, Z)\right), \qquad (E_{\mathrm{II}}^2)_{p,q} = \mathrm{Tor}_p^S\left(\mathrm{Tor}_q^R(X, {}_RY), Z\right).$$

Hint Take resolutions of X and Z. Readers familiar with derived categories may compare this with Proposition 4.12.13 on the associativity constraint (take $A = \Bbbk = B$).

10. Describe explicitly the canonical homomorphism $\mathrm{Ext}_R^n(Y, {}_RX) \to \mathrm{Hom}_S(\mathrm{Tor}_n^R(S_R, Y), X)$ in the short exact sequence of Example 5.6.8.

Advanced Part

Chapter 6 Group Homology and Cohomology

Let G be a group and let $\Bbbk$ be a commutative ring. A G-module means a $\Bbbk$-module equipped with a linear left action of G, written as left multiplication; in many situations $\Bbbk$ is usually taken to be $\mathbb{Z}$. All G-modules form an Abelian category G-Mod, isomorphic to $\Bbbk[G]$-Mod, where $\Bbbk[G]$ is the group algebra of G. For a G-module M, define the $\Bbbk$-modules

$$\begin{aligned} \text{invariants} \quad & M^G := \{x \in M : \forall g \in G,\ gx = x\}, \\ \text{coinvariants} \quad & M_G := M / \langle gx - x : g \in G, x \in M \rangle . \end{aligned}$$

From the algebraic point of view, the cohomology $\mathrm{H}^n(G, M)$ and homology $\mathrm{H}_n(G, M)$ of a G-module M are, respectively, the values at M of the right-derived functors of $(\cdot)^G$ and the left-derived functors of $(\cdot)_G$, where $n \in \mathbb{Z}_{\geq 0}$.

Group homology and cohomology have a long history. Around 1935, W. Hurewicz studied path-connected topological spaces E whose higher homotopy groups satisfy $n > 1 \implies \pi_n(E) = 0$. He proved that their homology and cohomology groups are completely determined by the fundamental group $\pi_1(E)$; this is perhaps the earliest documented formulation of group cohomology. As homological algebra advanced rapidly during the twentieth century, group cohomology and homology soon acquired purely algebraic formulations and were used to address problems within algebra itself. Today they are basic tools in number theory, topology, and geometry.

On the other hand, $\mathrm{H}^n(G, M)$ and $\mathrm{H}_n(G, M)$ can also be identified with $\mathrm{Ext}^n(\Bbbk, M)$ and $\mathrm{Tor}_n(\Bbbk, M)$, respectively, in $\Bbbk[G]$-Mod, with $\Bbbk$ regarded as a trivial G-module. Computing Ext and Tor from a projective resolution of $\Bbbk$ gives another description of $\mathrm{H}^n(G, M)$ and $\mathrm{H}_n(G, M)$. In fact, we shall give a canonical resolution $\mathsf{L} \to \Bbbk$ of $\Bbbk$ as a left (or right) $\Bbbk[G]$-module. It yields the standard cochain complex $(C^n(G, M))_n$ (or chain complex $(C_n(G, M))_n$) representing $\mathrm{H}^n(G, M)$ (or $\mathrm{H}_n(G, M)$); see Propositions 6.2.8 and 6.2.9. Later, §§8.6— 8.7 will interpret these canonical resolutions from the viewpoint of simplicial theory.

The preceding material constitutes the main content of §§6.1—6.2. This chapter adopts two viewpoints at once: the classical theory of derived functors, especially universal δ-functors and their characterization, and derived categories. The classical theory has the advantage of being operational, whereas derived categories provide finer information than the individual cohomology or homology groups and often offer a clearer line of thought. The standard cochain or chain complexes provide the most detailed and canonical information, since they operate at the level of complexes.

In §6.3, we shall give concrete interpretations of H^1 and H^2: they correspond, respectively, to crossed homomorphisms and equivalence classes of extensions of G by an Abelian group. The precise meaning of a group extension is given in Definition 6.3.5. In fact, if all extensions of G by M are organized into a 2-category $\mathsf{Ext}(G, M)$, then the truncation $\tau^{\leq 2}\overline{C}(G, M)$ of the normalized standard complex is exactly a linear incarnation of $\mathsf{Ext}(G, M)$; this is the content of Remark 6.3.10.

For any group homomorphism $\varphi : H \to G$ and any G-module M, its pullback $\varphi^* M$ can be defined as an H-module. General constructions from module theory give simple descriptions of both the right and left adjoints of φ^*. When φ is the inclusion of a subgroup, φ^* is called the restriction functor on G-modules and is also written Res^G_H, while its right (respectively left) adjoint is written Ind^G_H (respectively ind^G_H); both are called induction functors. Their properties are the subject of §6.4. In particular, we shall prove the so-called Shapiro lemma (Theorem 6.4.3):

$$\mathrm{H}^n\left(G, \mathrm{Ind}^G_H(N)\right) \simeq \mathrm{H}^n(H, N), \quad \mathrm{H}_n\left(G, \mathrm{ind}^G_H(N)\right) \simeq \mathrm{H}_n(H, N).$$

Continuing the general discussion of φ^*, §6.5 will define families of canonical homomorphisms

$$\mathrm{H}^n(G, M) \to \mathrm{H}^n(H, \varphi^* M), \quad \mathrm{H}_n(H, \varphi^* M) \to \mathrm{H}_n(G, M).$$

These homomorphisms can be combined with the functoriality of cohomology or homology. Thus every equivariant homomorphism $f : M \to N$ (or $f : N \to M$) induces the corresponding canonical homomorphism $\mathrm{H}^n(f)$ (or $\mathrm{H}_n(f)$), where M is a G-module and N an H-module; see Definition 6.5.7. In the special case $n = 2$, §6.6 will interpret these canonical homomorphisms in terms of group extensions.

For concrete illustrations, §6.7 discusses finite cyclic groups and free groups. The arguments harmoniously combine techniques from group theory and cohomology. The cohomological and homological dimensions of groups (Definition 6.7.8) will also enter at the appropriate point.

The concrete flavor of group theory continues in §6.8. When $(G : H)$ is finite, we shall state further properties of the restriction and induction functors; indeed, in this case $\mathrm{Ind}_H^G = \mathrm{ind}_H^G$. Definition–Proposition 6.8.4 will give families of canonical homomorphisms

$$\mathrm{cor}^n : \mathrm{H}^n(H, M) \to \mathrm{H}^n(G, M), \quad \mathrm{cor}_n : \mathrm{H}_n(G, M) \to \mathrm{H}_n(H, M),$$

where M is a G-module and $\mathrm{Res}_H^G M$ is abbreviated to M to save notation. These are called corestriction homomorphisms, and their direction is opposite to that of the canonical homomorphisms induced by $H \hookrightarrow G$ (namely restriction, written res^n and res_n). They satisfy the fundamental identities $\mathrm{cor}^n \mathrm{res}^n = (G : H)$ and $\mathrm{res}_n \mathrm{cor}_n = (G : H)$ (Proposition 6.8.5), while the special case $\mathrm{cor}_1 : \mathrm{H}_1(G, \mathbb{Z}) \to \mathrm{H}_1(H, \mathbb{Z})$ becomes the classical transfer homomorphism in group theory, $\mathrm{Ver}_{G|H} : G_{\mathrm{ab}} \to H_{\mathrm{ab}}$.

Let H be a normal subgroup of G and let M be a G-module. Then $\mathrm{H}^q(H, M)$ and $\mathrm{H}_q(H, M)$ become (G/H)-modules. The Lyndon–Hochschild–Serre spectral sequences discussed in §6.9,

$$\begin{aligned} E_2^{p,q} &= \mathrm{H}^p(G/H, \mathrm{H}^q(H, M)) \Rightarrow \mathrm{H}^{p+q}(G, M), \\ E_{p,q}^2 &= \mathrm{H}_p(G/H, \mathrm{H}_q(H, M)) \Rightarrow \mathrm{H}_{p+q}(G, M), \end{aligned}$$

are indispensable advanced tools in the theory of group cohomology and homology; see Theorem 6.9.3. The information in the exact sequence of their low-degree terms is extremely useful. The Lyndon–Hochschild–Serre spectral sequence is at once an application of the Grothendieck spectral sequence and the result of a canonical filtration on the standard complex. The latter description gives finer information, such as the multiplicative structure mentioned in Remark 6.9.5.

More precisely, the multiplicative operation on group cohomology is the cup product, which originates in topology. In §6.10, we shall approach it from two viewpoints. First, the cup product is understood as a natural reflection of the tensor-product operation on G-modules; the language of derived categories is useful here. Second, its formula is written directly at the level of the standard complex (Definition–Proposition 6.10.7). The second description is, of course, an explicit realization of the first. The key to deriving it is to lift $\Bbbk \xrightarrow{\sim} \Bbbk \otimes \Bbbk$ appropriately to a "diagonal embedding" $\Delta : \mathsf{L} \to \mathsf{L} \otimes \mathsf{L}$; several signs in the argument require careful handling.

If G is finite, the Tate cohomology $\hat{\mathrm{H}}^n(G, M)$ introduced in §6.11 combines cohomology and homology. Its construction depends on the canonical homomorphism peculiar to finite groups,

$$\nu : M_G \to M^G, \quad (\text{the image of } x \in M) \mapsto \sum_{g \in G} gx.$$

For $n \geq 1$ (respectively $n \leq -2$), Tate cohomology equals $\mathrm{H}^n(G, M)$ (respectively $\mathrm{H}_{-n-1}(G, M)$), while for $n = 0$ (respectively $n = -1$) it equals $\operatorname{coker}(\nu)$ (respectively $\ker(\nu)$). Tate cohomology continues to satisfy such properties as long exact sequences and Shapiro's lemma. In many respects it is even easier to work with than cohomology or homology, and it has especially prominent applications in number theory.

The preceding theory concerns only discrete groups; groups equipped with a topology present a much more intricate case. Profinite groups G and smooth G-modules, discussed in §6.12, form a special case that is both simple and useful; the necessary background is given in §B.1. The theory remains essentially algebraic. The corresponding cohomology can be understood as a right-derived functor or, equivalently, defined as a $\varinjlim$ of the cohomology of finite groups; the continuous version of the standard complex provides another method of calculation. In number theory, the main cases of interest are those in which G is a Galois group or an arithmetic fundamental group.

Finally, §6.13 introduces non-Abelian group cohomology. For a non-Abelian group, usually only H^0 and H^1 can be defined, so its long exact sequence is correspondingly limited (Theorem 6.13.7). This theory is particularly well suited to classifying geometric or algebraic objects, because the symmetry groups of such objects are often non-Abelian. Several examples will be discussed later in §7.11. These constructions also apply to profinite groups.

Reading Guide

The role of the coefficient ring $\Bbbk$ in this chapter is secondary, and it is often taken to be $\mathbb{Z}$; see the explanation in Remark 6.2.12. Readers may adapt the material to their needs and skip arguments involving derived categories; doing so will not interfere with most classical applications. In particular, the cup product on cohomology may be treated entirely through the explicit operations in Definition–Proposition 6.10.7. The notion of a G-module (see §6.1) occasionally appears as an example or exercise in later chapters and usually involves only the basic definitions. The exceptions are §7.10 and §7.11, which use smooth

G-modules for profinite groups and their cohomology. Readers interested in number theory should in addition master §§6.11—6.13.

6.1 G-Modules and Their Resolutions

Unless otherwise stated, throughout this section $\Bbbk$ is a commutative ring and G a group. Write the identity element of G as 1_G.

Definition 6.1.1 A ***G*-module** with coefficients in $\Bbbk$ is a $\Bbbk$-module M equipped with a left action of G. The action $G \times M \to M$ is written as multiplication and satisfies

$$\begin{aligned} g(m+m') &= gm + gm', \\ g(tm) &= t(gm), \end{aligned}$$

for all $g \in G$, $t \in \Bbbk$, and $m, m' \in M$. In other words, M is equipped with a linear action of G.

A homomorphism $f : M_1 \to M_2$ between two G-modules is defined to be a $\Bbbk$-module homomorphism satisfying the equivariance property $f(gm_1) = gf(m_1)$ for all $g \in G$ and $m_1 \in M_1$.

Strictly speaking, the structure above should be called a left G-module. A right G-module is defined in an entirely similar way and is equivalent to a left G^{op}-module. When there is no danger of confusion, the coefficient ring $\Bbbk$ is suppressed from the notation, and the category of all G-modules is written $G\text{-}\mathsf{Mod}$. This category is $\Bbbk$-linear (Definition 1.4.4).

Example 6.1.2 Equip $\Bbbk$ with the trivial action of G, namely $gm = m$. The resulting object is called the **trivial *G*-module** and is denoted simply by $\Bbbk$.

Definition 6.1.3 Let M be a G-module. If a $\Bbbk$-submodule $N \subset M$ is closed under the action of G, then N is called a G-submodule. Equip the quotient $\Bbbk$-module M/N with the action $g(m+N) = gm + N$; this is called the corresponding quotient G-module.

The group algebra $\Bbbk[G]$ can be constructed from G; see [51, Definition 5.6.3]. Each of its elements has a unique expression of the form $\sum_{g \in G} a_g g$, with only finitely many nonzero coefficients $a_g \in \Bbbk$. From now on, we regard G as a subset of $\Bbbk[G]$.

Proposition 6.1.4 There is an isomorphism of categories $G\text{-}\mathsf{Mod} \xrightarrow{\sim} \Bbbk[G]\text{-}\mathsf{Mod}$ as follows. It preserves the underlying $\Bbbk$-module. If M is a G-module, its scalar multiplication as a left $\Bbbk[G]$-module comes from the homomorphism

$$\Bbbk[G] \underset{\Bbbk}{\otimes} M \to M, \quad (\sum_g a_g g) \otimes m \mapsto \sum_g a_g(gm),$$

while G-module homomorphisms correspond to $\Bbbk[G]$-module homomorphisms.

Proof The inverse functor is determined as follows. For a left $\Bbbk[G]$-module M, use the embedding of multiplicative monoids $G \hookrightarrow \Bbbk[G]$ to let G act on M from the left, thereby making M a G-module. It is straightforward to verify that both directions are well defined and inverse to each other. □

Consequently, G-Mod is a $\Bbbk$-linear Abelian category, since $\Bbbk[G]$-Mod is one. Properties of G-submodules and quotient G-modules translate directly into properties of $\Bbbk[G]$-modules.

Definition 6.1.5 Write Hom in G-Mod as Hom_G. Under the correspondence of Proposition 6.1.4, this is also $\mathrm{Hom}_{\Bbbk[G]}$ in $\Bbbk[G]$-Mod.

Several basic operations on $\Bbbk$-modules extend readily to G-modules:

- ▷ Direct sums and direct products For a family of G-modules $(M_i)_{i\in I}$, equip the $\Bbbk$-modules $\prod_{i\in I} M_i$ and $\bigoplus_{i\in I} M_i$ with the left G-action

  $$g(m_i)_{i\in I} := (gm_i)_{i\in I},$$

  called the diagonal action. With this action, $\prod_{i\in I} M_i$ and $\bigoplus_{i\in I} M_i$ are G-modules.

- ▷ Tensor products For G-modules M and N, equip $M \otimes N := M \underset{\Bbbk}{\otimes} N$ with the diagonal left G-action

  $$g(x \otimes y) = gx \otimes gy, \quad x \in M, \quad y \in N;$$

  since $(x, y) \mapsto gx \otimes gy$ is bilinear, this formula is well defined. Plainly $x \otimes y \mapsto y \otimes x$ defines an isomorphism of G-modules $M \otimes N \simeq N \otimes M$.

- ▷ Hom modules For G-modules M and N, the $\Bbbk$-module homomorphisms between them form the $\Bbbk$-module $\mathrm{Hom}(M, N) := \mathrm{Hom}_{\Bbbk}(M, N)$. Equip $\mathrm{Hom}(M, N)$ with the left G-action

  $$({}^g f)(x) = g(f(g^{-1}x)), \quad f \in \mathrm{Hom}(M, N), \quad x \in M,$$

  where g^{-1} and g on the right arise from the left G-actions on M and N, respectively. As a composite of three $\Bbbk$-module homomorphisms, ${}^g f$ belongs to $\mathrm{Hom}(M, N)$. Thus ${}^g f$ can be viewed as the "conjugate" gfg^{-1}, making the properties required for a G-module structure evident.

▷ Contragredient modules In the construction of the Hom module, take $N = \Bbbk$ with its trivial G-action. Then $\mathrm{Hom}(M, \Bbbk)$ becomes a G-module, called the contragredient module of M. Its G-action is given by ${}^g f = f \circ g^{-1}$.

All these constructions are functorial in M and N. The definitions immediately give

$$\begin{aligned}\mathrm{Hom}_G(M, N) &= \mathrm{Hom}(M, N)^G \\ &:= \{f \in \mathrm{Hom}(M, N) : \forall g \in G,\ {}^g f = f\}.\end{aligned}$$

Under the isomorphism of categories in Proposition 6.1.4, direct sums and direct products of G-modules are, respectively, the direct sums and direct products of $\Bbbk[G]$-modules. The G-module structures on tensor products and Hom modules are slightly subtler; from the viewpoint of module theory, they in fact correspond to the Hopf-algebra structure on $\Bbbk[G]$ (Example 7.5.13). Interested readers may compare the related discussions in §7.5 and §9.3; we shall not elaborate here.

Proposition 6.1.6 With respect to the tensor-product operation, G-Mod is a symmetric monoidal category whose unit object is the trivial G-module $\Bbbk$.

Proof This follows directly from the definitions above. □

Moreover, the usual canonical isomorphism in $\Bbbk$-Mod,

$$\begin{aligned}\mathrm{Hom}(L \otimes M, N) &\xleftrightarrow{\sim} \mathrm{Hom}(L, \mathrm{Hom}(M, N)) \\ \varphi &\longmapsto [w \mapsto \varphi(w \otimes (\cdot))] \\ [w \otimes x \mapsto \psi(w)(x)] &\longleftarrow\!\shortmid\ \psi\end{aligned}$$

also lifts to the level of G-modules. If L, M, and N are G-modules and the tensor product and Hom above carry the G-module structures just defined, then both directions are isomorphisms of G-modules; readers are invited to verify this carefully.

Consequently, ${}^g\varphi = \varphi$ is equivalent to ${}^g\psi = \psi$, giving the adjunction

$$\mathrm{Hom}_G(L \otimes M, N) \xleftrightarrow{\sim} \mathrm{Hom}_G(L, \mathrm{Hom}(M, N)), \tag{6.1.1}$$

where L, M, and N are all G-modules. Here is a simple application.

Proposition 6.1.7 Let L and M be G-modules, with L projective as a G-module and M projective as a $\Bbbk$-module. Then $L \otimes M$ is projective as a G-module.

Proof The adjunction (6.1.1) shows that the functor $(\cdot) \otimes M : G\text{-}\mathsf{Mod} \to G\text{-}\mathsf{Mod}$ has the exact right adjoint $\mathrm{Hom}(M, \cdot)$. Proposition 2.8.17 therefore implies that $(\cdot) \otimes M$ preserves projective objects. □

Definition 6.1.8 (Restriction and inflation) For every group homomorphism $\varphi : H \to G$, there is a corresponding functor

$$\varphi^* : G\text{-}\mathsf{Mod} \to H\text{-}\mathsf{Mod};$$

for a G-module M, its image under φ^* is the same $\Bbbk$-module M, with the H-action given by $hm := \varphi(h)m$. Two important cases of the functor φ^* are:

- ▷ Restriction For a subgroup $H \subset G$, take $\varphi : H \hookrightarrow G$. The corresponding functor $\mathrm{Res}^G_H : G\text{-}\mathsf{Mod} \to H\text{-}\mathsf{Mod}$ restricts the group action to H.
- ▷ Inflation For a normal subgroup $H \lhd G$, take $\varphi : G \twoheadrightarrow G/H$. The corresponding functor $\mathrm{Infl}^G_{G/H} : (G/H)\text{-}\mathsf{Mod} \to G\text{-}\mathsf{Mod}$ pulls the group action back to G.

Notice that if there is a sequence of group homomorphisms $H \xrightarrow{\varphi} G \xrightarrow{\psi} F$, then the corresponding functors satisfy $\varphi^*\psi^* = (\psi\varphi)^*$.

From the group-algebra viewpoint, the homomorphism $\varphi : H \to G$ extends to a $\Bbbk$-algebra homomorphism $\Bbbk[H] \to \Bbbk[G]$, and the corresponding functor $\Bbbk[G]\text{-}\mathsf{Mod} \to \Bbbk[H]\text{-}\mathsf{Mod}$ is precisely φ^*. This functor can also be expressed by a tensor product as

$$\varphi^* M \simeq \Bbbk[G] \underset{\Bbbk[G]}{\otimes} M, \quad \Bbbk[G] \text{ regarded via } \varphi \text{ as a bimodule } (\Bbbk[H], \Bbbk[G]). \tag{6.1.2}$$

The left and right adjoints of φ^*, together with the corresponding adjunction isomorphisms, can be written explicitly.

Proposition 6.1.9 Use the notation above. For a group homomorphism $\varphi : H \to G$, any G-module M, and any H-module N, there are canonical isomorphisms of $\Bbbk$-modules

$$\begin{aligned}
\mathrm{Hom}_H(\varphi^* M, N) &\xleftrightarrow{\sim} \mathrm{Hom}_G\left(M, \mathrm{Hom}_H(\Bbbk[G], N)\right) \\
\theta &\longmapsto [x \mapsto [G \ni g \mapsto \theta(gx)]] \\
[x \mapsto \xi(x)(1_G)] &\longleftarrow\!\shortmid\ \xi, \\
\mathrm{Hom}_H(N, \varphi^* M) &\xleftrightarrow{\sim} \mathrm{Hom}_G\left(\Bbbk[G] \underset{\Bbbk[H]}{\otimes} N, M\right) \\
\psi &\longmapsto [g \otimes y \mapsto g\psi(y)] \\
[y \mapsto \eta(1_G \otimes y)] &\longleftarrow\!\shortmid\ \eta.
\end{aligned}$$

Here $\Bbbk[G]$ is regarded both as a $(\Bbbk[H], \Bbbk[G])$-bimodule and as a $(\Bbbk[G], \Bbbk[H])$-bimodule. In this way, $\mathrm{Hom}_H(\Bbbk[G], N)$ and $\Bbbk[G] \underset{\Bbbk[H]}{\otimes} N$ acquire G-module structures.

Proof Expand the definitions directly to verify that both pairs of maps are well-defined $\Bbbk$-module homomorphisms; the calculation is not difficult. Once this has been checked, it is clear that the maps are mutually inverse. Alternatively, apply the general theory in [51, Corollary 6.6.8]. □

Since G-Mod is an Abelian category, projective and injective resolutions of G-modules can be considered in the usual way. For the subject of this chapter, the most important are projective resolutions of the trivial G-module $\Bbbk$. Such resolutions not only exist; they also admit a canonical, explicit choice.

Definition 6.1.10 Let G be a group. For each $n \in \mathbb{Z}_{\geq 0}$, define

$$\mathsf{L}_n := \text{the free } \Bbbk\text{-module with basis } G^{n+1},$$

whose elements are written as $\Bbbk$-linear combinations of $(g_0, \ldots, g_n) \in G^{n+1}$. For every $n \geq 1$, define homomorphisms

$$\begin{aligned} &\partial_n, \partial'_n : \mathsf{L}_n \to \mathsf{L}_{n-1}, \\ &\partial_n\,(g_0, \ldots, g_n) = \sum_{k=0}^{n} (-1)^{n-k} (\ldots, \widehat{g_k}, \ldots), \\ &\partial'_n\,(g_0, \ldots, g_n) = \sum_{k=0}^{n} (-1)^{k} (\ldots, \widehat{g_k}, \ldots), \end{aligned}$$

where $\widehat{g_k}$ indicates that the term is omitted; thus $\partial'_n = (-1)^n \partial_n$. Also define the homomorphism

$$\partial_0 = \partial'_0 : \mathsf{L}_0 \to \Bbbk, \quad (g_0) \mapsto 1.$$

Define the action $g(g_0, \ldots, g_n) = (gg_0, \ldots, gg_n)$ (respectively $(g_0, \ldots, g_n)g = (g_0 g, \ldots, g_n g)$) so that L_n becomes a left (respectively right) G-module, and equip $\Bbbk$ with the trivial G-action. Then every ∂_n and ∂'_n is a homomorphism of left (respectively right) G-modules.

Plainly $(\mathsf{L}_n, \partial_n)_n$ and $(\mathsf{L}_n, \partial'_n)_n$ differ only by the reversal $(g_0, \ldots, g_n) \mapsto (g_n, \ldots, g_0)$. This reversal commutes with both the left and the right G-actions, so for all the properties below it suffices to treat either ∂_n or ∂'_n.

Remark 6.1.11 If L_0 is identified with $\Bbbk[G]$, then $\partial_0 = \partial'_0 : \mathsf{L}_0 \to \Bbbk$ is identified with the map

$$\Bbbk[G] \to \Bbbk, \quad \sum_g a_g g \mapsto \sum_g a_g;$$

this is plainly also a surjective $\Bbbk$-algebra homomorphism. It is called the **augmenta-**

tion homomorphism of $\Bbbk[G]$, and its kernel

$$\mathfrak{I} := \left\{ \sum_g a_g g : \sum_g a_g = 0 \right\}$$

is called the **augmentation ideal** of $\Bbbk[G]$. As a $\Bbbk$-module, $\mathfrak{I}$ plainly decomposes with respect to the following basis:

$$\mathfrak{I} = \bigoplus_{\substack{g \in G \\ g \neq 1_G}} \Bbbk(g - 1_G).$$

Lemma 6.1.12 The complex $\cdots \to \mathsf{L}_n \xrightarrow{\partial_n} \cdots \xrightarrow{\partial_0} \Bbbk \to 0$ defined above is exact. Regarded as a complex of $\Bbbk$-modules, its identity morphism is null-homotopic. The same assertions hold when ∂_n is replaced by ∂'_n.

Proof It suffices to discuss the ∂'_n version. For every $n \geq 1$, observe that $\partial'_{n-1}\partial'_n$ sends $(g_0, \ldots, g_n)$ to a sum of elements of the form

$$\pm(\ldots, \widehat{g_p}, \ldots, \widehat{g_q}, \ldots), \quad 0 \leq p < q \leq n.$$

The signs cancel in pairs, giving a complex; the elementary verification is left to the reader.

Next, for every $n \geq 0$, define a $\Bbbk$-module homomorphism

$$s_n : \mathsf{L}_n \to \mathsf{L}_{n+1}, \quad (g_0, \ldots, g_n) \mapsto (1_G, g_0, \ldots, g_n),$$

and define $s_{-1} : \Bbbk \to \mathsf{L}_0$ by sending $1 \in \Bbbk$ to $1_G \in \mathsf{L}_0$. It is easy to see that

$$\partial'_{n+1} s_n + s_{n-1} \partial'_n = \mathrm{id}_{\mathsf{L}_n}$$

for every $n \geq -1$, with the conventions $\mathsf{L}_{-1} := \Bbbk$ and $\partial'_{-1} := 0$. This shows that

$$\cdots \to \mathsf{L}_n \xrightarrow{\partial'_n} \cdots \xrightarrow{\partial'_0} \Bbbk \to 0$$

has a null-homotopic identity morphism as a complex of $\Bbbk$-modules and is therefore exact. $\square$

Write the complex $\cdots \to \mathsf{L}_n \xrightarrow{\partial_n} \cdots \to \mathsf{L}_0 \to 0$ as L. It may be viewed either as a complex in degrees $\ldots, -n, \ldots, 0$ or as a chain complex in degrees $\ldots, n, \ldots, 0$. In either interpretation, $\partial_0 = \partial'_0$ gives a morphism $\mathsf{L} \to \Bbbk$.

This complex and morphism can be understood either as a complex of left G-modules or as a complex of right G-modules. By symmetry, we shall mainly state the left-handed version.

Proposition 6.1.13 The morphism $\mathsf{L} \to \mathbb{k}$ defined above gives a free resolution of the trivial G-module $\mathbb{k}$.

Proof As $(h_1, \ldots, h_n) \in G^n$ varies, $(1_G, h_1, \ldots, h_n)$ spans a basis of L^n as a left $\mathbb{k}[G]$-module. □

There is also a canonical free resolution of $\mathbb{k}$ that is more concise and sometimes more practical than L.

Definition–Proposition 6.1.14 (Normalized complex) For every $n > 0$, define $\mathrm{Triv}_n \subset \mathsf{L}_n$ to be the G-submodule generated by the elements

$$(g_0, \ldots, g_n) \in G^{n+1}, \quad \exists\, 0 \leq k < n,\ g_k = g_{k+1}.$$

Also define $\mathrm{Triv}_0 = 0$. With respect to either ∂_n or ∂'_n, these objects form a subcomplex Triv of L. The normalization of L is then defined to be the quotient complex $\overline{\mathsf{L}} := \mathsf{L}/\mathrm{Triv}$. Regarded as a complex of $\mathbb{k}$-modules, the identity morphism on Triv is null-homotopic.

Proof Plainly Triv_n is invariant under $(g_0, \ldots, g_n) \mapsto (g_n, \ldots, g_0)$, so the ∂_n and ∂'_n versions do not differ; it suffices to treat the latter. If $(g_0, \ldots, g_n)$ satisfies $g_k = g_{k+1}$, then in $\partial'_n(g_0, \ldots, g_n) = \sum_i (-1)^i (\ldots, \widehat{g_i}, \ldots)$, the terms corresponding to $i = k, k+1$ cancel, while the remaining terms lie in Triv_{n-1}. This proves that the objects form a subcomplex.

To show that the identity morphism of Triv is null-homotopic as a complex of $\mathbb{k}$-modules, simply restrict the homotopy $s_n : \mathsf{L}_n \to \mathsf{L}_{n+1}$ in the proof of Lemma 6.1.12 to Triv_n, for $n \geq 0$. □

By construction, $\mathsf{L} \to \mathbb{k}$ vanishes on Triv and hence factors through $\overline{\mathsf{L}} \to \mathbb{k}$.

Proposition 6.1.15 The morphism $\overline{\mathsf{L}} \to \mathbb{k}$ defined above gives a free resolution of the trivial G-module $\mathbb{k}$.

Proof First, every $\overline{\mathsf{L}}_n = \mathsf{L}_n/\mathrm{Triv}_n$ is a free G-module: a basis is given by the images of $(1_G, h_1, \ldots, h_n) \in G^{n+1}$ with $h_1 \neq 1_G$ and $h_k \neq h_{k+1}$ for $1 \leq k < n$.

Second, the acyclicity of Triv implies that $\mathsf{L} \to \overline{\mathsf{L}}$ is a quasi-isomorphism. Hence Proposition 6.1.13 implies that $\overline{\mathsf{L}} \to \mathbb{k}$ is a quasi-isomorphism. □

For later applications, it is more convenient to express the basis in the following "bar" notation. First, for integers $a \leq b$ and a sequence of elements $(g_a, \ldots, g_b) \in G^{b-a+1}$, introduce the abbreviation

$$g_{[a,b]} := g_a g_{a+1} \cdots g_b.$$

As a left G-module, a basis of L_n may be taken to be

$$(g_1|\cdots|g_n) := \left(1_G, g_{[1,1]}, g_{[1,2]}, \ldots, g_{[1,n]}\right), \quad (g_1, \ldots, g_n) \in G^n. \tag{6.1.3}$$

Observe that

$$\begin{aligned}
\partial_n(g_1|\ldots|g_n) &= (-1)^n \left(g_{[1,1]}, \ldots, g_{[1,n]}\right) + (-1)^{n-1} \left(1_G, g_{[1,2]}, \ldots, g_{[1,n]}\right) \\
&\quad + \cdots + \left(1_G, g_{[1,1]}, \ldots, g_{[1,n-1]}\right) \\
&= (-1)^n g_1(g_2|\cdots|g_n) + \sum_{k=1}^{n-1} (-1)^{n-k}(\cdots|g_k g_{k+1}|\cdots) \\
&\quad + (g_1|\cdots|g_{n-1}), \\
\partial'_n(g_1|\cdots|g_n) &= g_1(g_2|\cdots|g_n) + \sum_{k=1}^{n-1} (-1)^k(\cdots|g_k g_{k+1}|\cdots) \\
&\quad + (-1)^n (g_1|\cdots|g_{n-1}).
\end{aligned} \tag{6.1.4}$$

Now consider the mirror image of (6.1.3). Take as a basis of L_n as a right G-module

$$(g_1|\cdots|g_n)^\dagger := \left(g_{[1,n]}, \ldots, g_{[n-1,n]}, g_{[n,n]}, 1_G\right), \quad (g_1, \ldots, g_n) \in G^n. \tag{6.1.5}$$

Then $\partial'_n(g_1|\cdots|g_n)^\dagger = (-1)^n \partial_n(g_1|\cdots|g_n)^\dagger$ equals

$$\begin{aligned}
&\left(g_{[2,n]}, \ldots, g_{[n,n]}, 1_G\right) - \left(g_{[1,n]}, g_{[3,n]}, \ldots, g_{[n,n]}, 1_G\right) + \cdots \\
&\qquad + (-1)^n \left(g_{[1,n-1]}, \ldots, g_{[n-1,n-1]}\right) g_n \\
&\quad = (g_2|\cdots|g_n)^\dagger + \sum_{k=1}^{n-1} (-1)^k (\cdots|g_k g_{k+1}|\cdots)^\dagger \\
&\qquad + (-1)^n (g_1|\cdots|g_{n-1})^\dagger g_n.
\end{aligned} \tag{6.1.6}$$

The normalized complex is handled in the same way: bases of $\overline{\mathsf{L}}_n$ as a left or right G-module may be taken, respectively, to be

$$(g_1|\cdots|g_n) \quad \text{or} \quad (g_1|\cdots|g_n)^\dagger \quad \bmod \mathrm{Triv}_n, \tag{6.1.7}$$

subject to $g_k \neq 1_G$ for every $1 \leq k \leq n$.

Remark 6.1.16 The resolutions above have a profound topological interpretation: L is the chain complex supplied by the universal torsor $\mathrm{E}G$ over the classifying space $\mathrm{B}G$; see Examples 8.5.2 and 8.6.9.

6.2 Group Homology and Cohomology

Throughout this section, the commutative ring $\Bbbk$ and the group G remain fixed.

Definition 6.2.1 For every G-module M, define the $\Bbbk$-modules

$$M^G := \{m \in M : \forall g \in G,\ gm = m\},$$
$$M_G := M / \langle gm - m : g \in G,\ m \in M \rangle.$$

We call M^G the submodule of **invariants** of M, and M_G the quotient module of **coinvariants** of M.

Taking invariants and coinvariants gives $\Bbbk$-linear functors $G\text{-}\mathsf{Mod} \to \Bbbk\text{-}\mathsf{Mod}$. Both can be characterized as adjoints to the inflation functor Infl of Definition 6.1.8 in the case $H = G$.

Proposition 6.2.2 Consider the functor $\mathrm{Infl} := \mathrm{Infl}^G_{\{1\}} : \Bbbk\text{-}\mathsf{Mod} \to G\text{-}\mathsf{Mod}$, which sends each $\Bbbk$-module N to the G-module with trivial action $gy = y$. There are two adjoint pairs

$$\mathrm{Infl} : \Bbbk\text{-}\mathsf{Mod} \rightleftarrows G\text{-}\mathsf{Mod} : (\cdot)^G,$$

$$(\cdot)_G : G\text{-}\mathsf{Mod} \rightleftarrows \Bbbk\text{-}\mathsf{Mod} : \mathrm{Infl}.$$

Proof Let M be a G-module and N a $\Bbbk$-module. To specify a G-module homomorphism $\varphi : \mathrm{Infl}(N) \to M$ is the same as to specify a $\Bbbk$-module homomorphism $\varphi : N \to M$ such that $\varphi(y) = g\varphi(y)$ always holds, and this in turn is the same as to specify a $\Bbbk$-module homomorphism $\varphi : N \to M^G$.

On the other hand, to specify a $\Bbbk$-module homomorphism $\psi : M_G \to N$ is the same as to specify a $\Bbbk$-module homomorphism $\varphi : M \to N$ such that $\varphi(gx - x) = 0$ always holds. The latter condition is equivalent to $\varphi(gx) = g\varphi(x)$, where the G-action on the right is defined by the G-module structure $\mathrm{Infl}(N)$. □

The functors $(\cdot)^G$ and $(\cdot)_G$ can also be understood from the viewpoint of $\Bbbk[G]$-modules. In what follows, through the augmentation homomorphism of Remark 6.1.11, we regard $\Bbbk$ as a $(\Bbbk[G], \Bbbk[G])$-bimodule; in other words, the action of G on either side of $\Bbbk$ is trivial.

Proposition 6.2.3 Upon identifying G-modules with left $\Bbbk[G]$-modules, there are natural isomorphisms of functors

$$(\cdot)^G \simeq \mathrm{Hom}_G(\Bbbk, \cdot), \quad (\cdot)_G \simeq \Bbbk \underset{\Bbbk[G]}{\otimes} (\cdot).$$

Proof Consider a G-module M. By Proposition 6.2.2, to specify a G-module homomorphism $\varphi : \Bbbk \to M$ is the same as to specify a $\Bbbk$-module homomorphism $\varphi : \Bbbk \to M^G$. The latter is equivalent to specifying the element $\varphi(1)$ of M^G. This gives the first isomorphism.

For the second isomorphism, first identify $\Bbbk$ with $\Bbbk[G]/\mathfrak{I}$, where the augmentation ideal is $\mathfrak{I} = \bigoplus_{g \neq 1_G} \Bbbk(g - 1_G)$. Then $\Bbbk \underset{\Bbbk[G]}{\otimes} M \simeq M/\mathfrak{I}M = M_G$, by the map sending $t \otimes x$ to the image of $tx \in M$ in M_G. This proves the claim. □

Thus the adjunctions of Proposition 6.2.2 can be understood as the special case of Proposition 6.1.9 obtained by taking the trivial homomorphism $\varphi : G \twoheadrightarrow \{1\}$.

Corollary 6.2.4 For every group homomorphism $\varphi : H \to G$, there are isomorphisms of functors

$$\begin{aligned} \mathrm{Hom}_H(\Bbbk[G], \cdot)^G &\xrightarrow{\sim} (\cdot)^H, & (\cdot)_H &\xrightarrow{\sim} (\Bbbk[G] \underset{\Bbbk[H]}{\otimes} \cdot)_G, \\ \varphi &\mapsto \varphi(1_G), & (x \text{ in the quotient}) &\mapsto (1_G \otimes x \text{ in the quotient}). \end{aligned}$$

Both sides are functors from H-Mod to $\Bbbk$-Mod. The left-hand side involves the functor $\mathrm{Hom}_H(\Bbbk[G], \cdot) : H\text{-}\mathsf{Mod} \to G\text{-}\mathsf{Mod}$ introduced in Proposition 6.1.9.

Proof Let N be an H-module. The first isomorphism follows from the adjunction in Proposition 6.1.9:

$$\mathrm{Hom}_G(\Bbbk, \mathrm{Hom}_H(\Bbbk[G], N)) \simeq \mathrm{Hom}_H(\varphi^*\Bbbk, N) = \mathrm{Hom}_H(\Bbbk, N).$$

The second isomorphism follows from the associativity constraint for tensor products, $\Bbbk \underset{\Bbbk[G]}{\otimes} (\Bbbk[G] \underset{\Bbbk[H]}{\otimes} N) \simeq \Bbbk \underset{\Bbbk[H]}{\otimes} N$. Verification of the explicit map is left to the reader. Alternatively, the assertion can be proved directly by showing that the displayed map is well defined and invertible. □

The adjoint pairs in Proposition 6.2.2 also ensure that $(\cdot)_G$ is right exact and $(\cdot)^G$ is left exact; these facts are also easy to verify directly.

Definition 6.2.5 (Group homology and cohomology) Let G be a group. **Group homology** with coefficients in $\Bbbk$ is defined to be the left-derived functors of $(\cdot)_G$:

$$\mathrm{H}_n(G, \cdot) := \mathrm{L}_n(\cdot)_G : G\text{-}\mathsf{Mod} \to \Bbbk\text{-}\mathsf{Mod},$$

while **group cohomology** is defined to be the right-derived functors of $(\cdot)^G$:

$$\mathrm{H}^n(G, \cdot) := \mathrm{R}^n(\cdot)^G : G\text{-}\mathsf{Mod} \to \Bbbk\text{-}\mathsf{Mod},$$

where $n \in \mathbb{Z}$. Both vanish unless $n \geq 0$. Similar definitions apply to right G-modules.

Following the convention of much of the literature, when the coefficients are not specified we assume $\Bbbk = \mathbb{Z}$. In this case, a G-module is simply an additive group equipped with a left G-action that preserves addition.

Example 6.2.6 Consider the augmentation ideal $\mathfrak{I} = \bigoplus_{g \neq 1_G} \Bbbk(g - 1_G)$ of $\Bbbk[G]$. For every G-module M, there is an isomorphism $\mathrm{H}_0(G, M) \simeq M_G \simeq M/\mathfrak{I}M$; in particular, $\mathrm{H}_0(G, \mathfrak{I}) \simeq \mathfrak{I}/\mathfrak{I}^2$.

Now consider the short exact sequence $0 \to \mathfrak{I} \to \Bbbk[G] \to \Bbbk \to 0$, whose final morphism is the augmentation homomorphism. Since $\Bbbk[G]$ is a projective module, the dimension-shifting technique of Proposition 3.12.9 gives

$$\mathrm{H}_n(G, \Bbbk) \simeq \begin{cases} \mathrm{H}_{n-1}(G, \mathfrak{I}), & n \geq 2 \\ \ker\left[\mathfrak{I}/\mathfrak{I}^2 \to \Bbbk[G]/\mathfrak{I}\right] = \mathfrak{I}/\mathfrak{I}^2, & n = 1. \end{cases}$$

By the isomorphism between $\Bbbk[G]$-Mod and G-Mod, together with Proposition 6.2.3, group homology and cohomology can also be interpreted through the functors Tor and Ext introduced in §3.14:

$$\begin{aligned} \mathrm{H}_n(G, M) &\simeq \mathrm{Tor}_n^G(\Bbbk, M) := \mathrm{Tor}_n^{\Bbbk[G]}(\Bbbk, M), \\ \mathrm{H}^n(G, M) &\simeq \mathrm{Ext}_G^n(\Bbbk, M) := \mathrm{Ext}_{\Bbbk[G]}^n(\Bbbk, M). \end{aligned} \tag{6.2.1}$$

This interpretation sometimes reduces the questions at hand to known results in module theory. Here is a fundamental, nontrivial example involving a spectral sequence relating Ext and Tor.

Proposition 6.2.7 (Universal coefficient theorem for group homology and cohomology) Take $\Bbbk = \mathbb{Z}$. Equip the $\mathbb{Z}$-module M with the trivial G-action, making it a G-module. For every $n \in \mathbb{Z}_{\geq 0}$, there are canonical short exact sequences

$$0 \longrightarrow \mathrm{Ext}^1_{\mathbb{Z}}\left(\mathrm{H}_{n-1}(G, \mathbb{Z}), M\right) \longrightarrow \mathrm{H}^n(G, M) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}\left(\mathrm{H}_n(G, \mathbb{Z}), M\right) \longrightarrow 0,$$

$$0 \longrightarrow M \underset{\mathbb{Z}}{\otimes} \mathrm{H}_n(G, \mathbb{Z}) \longrightarrow \mathrm{H}_n(G, M) \longrightarrow \mathrm{Tor}_1^{\mathbb{Z}}(M, \mathrm{H}_{n-1}(G, \mathbb{Z})) \longrightarrow 0;$$

for $n = 0$, use the convention $\mathrm{H}_{-1}(G, \mathbb{Z}) = 0$.

Proof We may assume $n \geq 1$. First consider cohomology. Once the reader is familiar with the standard cochain and chain complexes to be introduced below, the short exact sequence follows easily by applying the universal coefficient theorem for cohomology from the exercises in Chapter 3 to the chain complex $C := \mathsf{L} \underset{\Bbbk[G]}{\otimes} \mathbb{Z}$ and the $\mathbb{Z}$-module M. The details are left to interested readers.

We give another argument here. In Example 5.6.8, take the ring homomorphism $R := \mathbb{Z}[G] \to S := \mathbb{Z}$ to be the augmentation; also take the left R-module $Y = \mathbb{Z}$ (the trivial G-module) and the left S-module $X = M$. The short exact sequence in that example then becomes the required one.

For homology, one may similarly apply the standard chain complex to the homological Künneth theorem 3.14.12, leaving the details to the reader, or understand the result through Example 5.6.8. □

Following the constructions introduced in §3.12, we may also take the relevant derived functors of a bounded-below chain complex (or a bounded-below cochain complex) M. These are called group hyperhomology (or group hypercohomology). In the derived category one may further consider

$$\Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} M, \quad \mathrm{RHom}_G(\Bbbk, M) := \mathrm{RHom}_{\Bbbk[G]}(\Bbbk, M).$$

By (6.2.1) and the fact that Ext (respectively Tor) can be computed using a projective resolution in the first variable, the free resolution $\mathsf{L} \to \Bbbk$ of Proposition 6.1.13 immediately supplies a concrete complex (respectively chain complex) for computing group cohomology (respectively homology). Definition 6.1.10 gives two ways to make L a complex of left or right G-modules: use $\partial_n : \mathsf{L}_n \to \mathsf{L}_{n-1}$ or use $\partial'_n : \mathsf{L}_n \to \mathsf{L}_{n-1}$. We shall choose between them according to the context so as to obtain the standard form of the complex; the choice does not affect its cohomology.

Proposition 6.2.8 (Computing group cohomology by the standard cochain complex) For every G-module M, define the **standard cochain complex** $C(G, M) = (C^n(G, M))_n$ of $\Bbbk$-modules as follows:

$$C^n(G, M) := \{\text{maps } f : G^n \to M\}, \quad n \in \mathbb{Z}_{\geq 0},$$

Each term is made into a $\Bbbk$-module by pointwise addition and scalar multiplication, the terms in negative degrees are defined to be 0, and $d^n : C^n(G, M) \to C^{n+1}(G, M)$ is defined by

$$\begin{aligned}(d^n f)(g_1, \ldots, g_{n+1}) &= g_1 f(g_2, \ldots, g_{n+1}) \\ &\quad + \sum_{k=1}^{n} (-1)^k f(\ldots, g_k g_{k+1}, \ldots) + (-1)^{n+1} f(g_1, \ldots, g_n).\end{aligned}$$

For every n, there is a canonical isomorphism $\mathrm{H}^n(C(G, M)) \simeq \mathrm{H}^n(G, M)$. More precisely, $C(G, M)$ represents $\mathrm{RHom}_G(\Bbbk, M)$ in the derived category.

Proof Consider the complex $\mathsf{L} = (\mathsf{L}_n, \partial_n)_n$. Since $\mathsf{L} \to \Bbbk$ is a free resolution by left G-modules, $\mathrm{RHom}_G(\Bbbk, M)$ is represented by the following Hom complex (Definition 3.2.1):

$$\mathrm{Hom}^\bullet(\mathsf{L}, M) = \left[\cdots \to \mathrm{Hom}_{\Bbbk[G]}(\mathsf{L}_n, M) \xrightarrow{(-1)^{n+1} \partial^*_{n+1}} \mathrm{Hom}_{\Bbbk[G]}(\mathsf{L}_{n+1}, M) \to \cdots \right],$$

Its degrees are $\ldots, -n, -n-1, \ldots$; the sign $(-1)^{n+1}$ comes from the general definition of the Hom complex.

For every $n \geq 0$, use the basis $(g_1|\cdots|g_n)$ of the free left $\Bbbk[G]$-module L_n from (6.1.3). Thus

$$\begin{aligned} \mathrm{Hom}_G(\mathsf{L}_n, M) &\xrightarrow{\quad\sim\quad} C^n(G, M) \\ \phi &\longmapsto [f : (g_1, \ldots, g_n) \mapsto \phi((g_1|\cdots|g_n))], \end{aligned}$$

which is plainly an isomorphism of $\Bbbk$-modules. The key point is to identify $(-1)^{n+1}\partial^*_{n+1}$ on $C^n(G, M)$. Consider $(-1)^{n+1}\partial^*_{n+1}\phi = (-1)^{n+1}\phi \circ \partial_{n+1}$. By (6.1.4), this map sends $(g_1|\cdots|g_{n+1}) \in \mathsf{L}_{n+1}$ to

$$\begin{aligned} &\phi(g_1(g_2|\cdots|g_{n+1})) + \sum_{k=1}^{n}(-1)^k \phi((\cdots|g_k g_{k+1}|\cdots)) + \cdots + (-1)^{n+1}\phi((g_1|\cdots|g_n)) \\ &\quad = g_1 f(g_2, \ldots, g_{n+1}) + \sum_{k=1}^{n}(-1)^k f(\ldots, g_k g_{k+1}, \ldots) + (-1)^{n+1} f(g_1, \ldots, g_n). \end{aligned}$$

This is precisely $(d^n f)(g_1, \ldots, g_{n+1})$ in the statement.[1)] □

The cohomology of a right G-module M has a similar description, which we do not repeat. We next state the corresponding result for the group homology of a left G-module; its proof is immediate.

Proposition 6.2.9 (Computing group homology by the standard chain complex) For every G-module M and $n \in \mathbb{Z}_{\geq 0}$, there is a canonical isomorphism

$$\mathrm{H}_n(G, M) \simeq \mathrm{H}_n\left(\mathsf{L} \underset{\Bbbk[G]}{\otimes} M\right), \quad n \in \mathbb{Z}_{\geq 0},$$

where $(\mathsf{L}_n, \partial'_n)_n$ is used as the chain complex of right G-modules. Take the basis $(g_1|\cdots|g_n)^\dagger$ of the free right $\Bbbk[G]$-module L_n from (6.1.5). Then $\mathsf{L}_n \underset{\Bbbk[G]}{\otimes} M$ is identified with the direct sum[2)] $M^{\oplus G^n}$, and an element in the summand corresponding to $(g_1, \ldots, g_n) \in G^n$ has a unique expression

$$(g_1|\cdots|g_n)^\dagger \otimes x, \quad x \in M,$$

[1)] Translator's correction: the source ends the argument list at g_n, but $d^n f \in C^{n+1}(G, M)$ and the calculation above uses $(g_1, \ldots, g_{n+1})$.

[2)] Translator's correction: the source prints L^n, whereas the chain complex and the basis in the same sentence use L_n.

while, by (6.1.6), $\partial'_n \otimes \mathrm{id}_M : \mathsf{L}_n \underset{\Bbbk[G]}{\otimes} M \to \mathsf{L}_{n-1} \underset{\Bbbk[G]}{\otimes} M$ is given by

$$\begin{aligned}(g_1|\cdots|g_n)^\dagger \otimes x \mapsto \\ &(g_2|\cdots|g_n)^\dagger \otimes x + \sum_{k=1}^{n-1}(-1)^k(\cdots|g_kg_{k+1}|\cdots)^\dagger \otimes x \\ &\qquad + (-1)^n(g_1|\cdots|g_{n-1})^\dagger \otimes g_n x.\end{aligned}$$

For comparison with Proposition 6.2.8, denote the chain complex above by $(C_n(G,M))_n$ and call it the **standard chain complex**. In the derived category it represents $\Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} M$.

Example 8.7.8 below will explain the relation between the standard chain complex and the bar construction.

There is of course a corresponding description for the homology of a right G-module M. Consider $M \underset{\Bbbk[G]}{\otimes} \mathsf{L}$. We have $M \underset{\Bbbk[G]}{\otimes} \mathsf{L}_n \simeq M^{\oplus G^n}$, and an element in the summand corresponding to $(g_1, \ldots, g_n)$ has a unique expression

$$x \otimes (g_1|\cdots|g_n), \quad x \in M,$$

while the map $\mathrm{id}_M \otimes \partial'_n$ is

$$\begin{aligned}x \otimes (g_1|\cdots|g_n) \mapsto \\ &xg_1(g_2|\cdots|g_n) + \sum_{k=1}^{n-1}(-1)^k x \otimes (\cdots|g_kg_{k+1}|\cdots) \\ &\qquad + (-1)^n x \otimes (g_1|\cdots|g_{n-1}).\end{aligned}$$

The corresponding standard chain complex is again denoted by $(C_n(G,M))_n$.

Readers familiar with §3.8 will readily recognize these complexes as special cases of the Hochschild cochain and chain complexes. Here is the precise statement.

Proposition 6.2.10 (Relation with Hochschild theory) Let $R := \Bbbk[G]$. For a left (respectively right) G-module M, write M_+ (respectively ${}_+M$) for the (R,R)-bimodule determined by

$$g_1 x g_2 := g_1 x \text{ (respectively } g_1 x g_2 = x g_2), \quad g_1, g_2 \in G, \ x \in M.$$

There are canonical isomorphisms compatible with long exact sequences,

$$\mathrm{H}^n(G,M) \simeq \mathrm{HH}^n(R, M_+), \quad \text{(respectively } \mathrm{H}_n(G,M) \simeq \mathrm{HH}_n(R, {}_+M)),$$

and in fact the complex $(C^n(G,M))_n$ (respectively the chain complex $(C_n(G,M))_n$) is isomorphic to $(C^n(R,M_+))_n$ (respectively $(C_n(R,{}_+M))_n$) from (3.8.1).

Proof Observe that $R^{\otimes n}$ is a free $\Bbbk$-module with basis G^n: $(g_1, \ldots, g_n)$ corresponds to $g_1 \otimes \cdots \otimes g_n \in R^{\otimes n}$. It remains only to compare the standard cochain complex with the complex described in (3.8.1). Concretely, in the cohomological case, send $f : G^n \to M$ to the corresponding n-multilinear $\Bbbk$-map $R^n \to M$. In the homological case, send $x \otimes (g_1|\cdots|g_n)^\dagger$ to $(x|g_1|\cdots|g_n) \in C_n(R, {}_+M)$. The remaining comparison is left to the reader. $\square$

Returning to our main subject, from now on all G-modules under consideration are left modules unless otherwise stated.

Example 6.2.11 In light of the discussion above, consider

$$\mathrm{H}_1(G, \Bbbk) \simeq \frac{\ker [C_1(G, \Bbbk) \to C_0(G, \Bbbk)]}{\mathrm{im}\, [C_2(G, \Bbbk) \to C_1(G, \Bbbk)]}.$$

Every element of $C_1(G, \Bbbk)$ has a unique expression as a $\Bbbk$-linear combination of the elements $[g] := (g)^\dagger \otimes 1$ for $g \in G$. Observe that $C_0(G, \Bbbk) \simeq \Bbbk$ and

$$(\partial_1' \otimes \mathrm{id}_\Bbbk)\,[g] = 1 - 1 = 0, \quad (\partial_2' \otimes \mathrm{id}_\Bbbk)\,((g_1|g_2)^\dagger \otimes 1) = [g_2] - [g_1 g_2] + [g_1].$$

Therefore $\mathrm{H}_1(G, \Bbbk)$ is isomorphic to the quotient $\Bbbk$-module

$$Q := \bigoplus_{g \in G} \Bbbk \cdot [g] \Big/ \langle [g_1 g_2] - [g_1] + [g_2] : g_1, g_2 \in G \rangle\,;$$

The map $g \mapsto [g]$ induces a group homomorphism $a : G \to (Q, +)$. It has the following universal property: for every $\Bbbk$-module N and group homomorphism $b : G \to (N, +)$, there is a unique $\Bbbk$-module homomorphism $\varphi : Q \to N$ such that $b = \varphi a$. The reader may quickly verify this universal property and deduce from it the canonical isomorphism of $\Bbbk$-modules

$$\begin{aligned} G_{\mathrm{ab}} \underset{\mathbb{Z}}{\otimes} \Bbbk &\xrightarrow{\quad\sim\quad} \mathrm{H}_1(G, \Bbbk) \\ (g \bmod G_{\mathrm{der}}) \otimes 1 &\longmapsto [g] \text{ modulo the relations,} \end{aligned}$$

where $G_{\mathrm{ab}} := G/G_{\mathrm{der}}$ is the abelianization of G and G_{der} is its derived subgroup, as in [51, Definition 4.7.2]. Readers who know the topological background of group homology will recognize the connection with the Hurewicz theorem in homology theory.

It is natural also to ask for the composite of the isomorphism $\mathrm{H}_1(G, \Bbbk) \xrightarrow{\sim} \mathfrak{I}/\mathfrak{I}^2$ from Example 6.2.6 with the isomorphism above. We claim that this composite is given explicitly by

$$\begin{aligned} G_{\mathrm{ab}} \otimes \Bbbk &\xrightarrow{\quad\sim\quad} \mathfrak{I}/\mathfrak{I}^2 \\ (g \bmod G_{\mathrm{der}}) \otimes 1 &\longmapsto 1_G - g \mod \mathfrak{I}^2. \end{aligned}$$

Why is this so? The isomorphism in Example 6.2.6 comes from the map $\mathrm{H}_1(G,\Bbbk) \to \mathrm{H}_0(G,\mathfrak{I}) \simeq \mathfrak{I}/\mathfrak{I}^2$ induced by the short exact sequence $0 \to \mathfrak{I} \to \Bbbk[G] \to \Bbbk \to 0$. Write down the following commutative diagram with exact rows:

$$\begin{tikzcd}
 & & (g)^\dagger \otimes 1_G \arrow[r, mapsto] \arrow[d, phantom, "\rotatebox{90}{$\in$}"] & {[g] = (g)^\dagger \otimes 1} \arrow[d, phantom, "\rotatebox{90}{$\in$}"] & \\
0 \arrow[r] & C_1(G,\mathfrak{I}) \arrow[r] \arrow[d] & C_1(G,\Bbbk[G]) \arrow[r] \arrow[d, "\partial_1'\otimes\mathrm{id}_{\Bbbk[G]}"] & C_1(G,\Bbbk) \arrow[r] \arrow[d, "\partial_0'\otimes\mathrm{id}_{\Bbbk[G]}=0"] & 0 \\
0 \arrow[r] & C_0(G,\mathfrak{I}) \arrow[r] & C_0(G,\Bbbk[G]) \arrow[r] & C_0(G,\Bbbk) \arrow[r] & 0,
\end{tikzcd}$$

The map $\partial_1' \otimes \mathrm{id}_{\Bbbk[G]}$ sends $(g)^\dagger \otimes 1_{\Bbbk[G]}$ to $1_G - g \in C_0(G,\mathfrak{I}) = \mathfrak{I}$, thereby determining an element of $\mathfrak{I}/\mathfrak{I}^2 \simeq \mathrm{H}_0(G,\mathfrak{I})$.[3)] This is precisely the construction in the snake lemma.

Remark 6.2.12 The form of the standard cochain complex (respectively standard chain complex) shows that, as an additive group, $\mathrm{H}^n(G,M)$ (respectively $\mathrm{H}_n(G,M)$) is completely determined by the additive structure of M; multiplication by elements of $\Bbbk$ only supplies its $\Bbbk$-module structure. Thus the case $\Bbbk = \mathbb{Z}$ already captures the essence of group cohomology (respectively homology), although allowing general coefficients causes no additional difficulty.

Remark 6.2.13 (Normalized version) To compute $\mathrm{H}^n(G,M)$ and $\mathrm{H}_n(G,M)$, one may instead use the normalized free resolution $\overline{\mathsf{L}} \to \Bbbk$ supplied by Proposition 6.1.15. Denote the corresponding cochain and chain complexes by $(\overline{C}^n(G,M))_n$ and $(\overline{C}_n(G,M))_n$, respectively, and call them the normalized standard cochain complex and normalized standard chain complex. They are, respectively, a subcomplex and a quotient complex of the original versions. For example, in the cohomological case, the description of L_n in (6.1.7) gives

$$\overline{C}^n(G,M) = \left\{ f \in C^n(G,M) \;\middle|\; \begin{array}{l} \exists 1 \leq k \leq n,\ g_k = 1_G \\ \quad\Longrightarrow f(g_1,\ldots,g_n) = 0 \end{array} \right\}.$$

[3)] Translator's correction: the source uses C^0 and H^0, but the diagram and the connecting morphism being computed are homological, so the correct indices are C_0 and H_0.

6.3 Low-degree cohomology: crossed homomorphisms and group extensions

This section aims to deepen our understanding of H^1 and H^2. Fix a group G and a commutative ring $\Bbbk$. Given a G-module M, define the following submodules of $C^n(G,M)$ for the standard complex $C(G,M)$ of Proposition 6.2.8:

$$Z^n(G,M) := \ker(d^n) \supset \operatorname{im}(d^{n-1}) =: B^n(G,M),$$

and set $B^0(G,M) = 0$. We identify

$$\mathrm{H}^n(G,M) \quad \text{with} \quad Z^n(G,M)/B^n(G,M).$$

In view of their topological origins, the elements of $Z^n(G,M)$ are also called n-**cocycles**, while the elements of $B^n(G,M)$ are called n-**coboundaries**.

Recall that $C^n(G,M)$ consists of all maps $f: G^n \to M$. The low-degree part of the standard complex is

$$\begin{array}{ccccc} C^0(G,M) & \xrightarrow{d^0} & C^1(G,M) & \xrightarrow{\qquad\quad d^1 \qquad\quad} & C^2(G,M) \\ m & \longmapsto & [g \mapsto gm - m] & & \\ & & f & \longmapsto & [(g_1,g_2) \mapsto g_1 f(g_2) - f(g_1 g_2) + f(g_1)], \end{array}$$

In particular, $Z^0(G,M) = M^G$. We next consider the case $n = 1$.

Definition 6.3.1 (Crossed homomorphism) Let M be a G-module. The elements of $Z^1(G,M)$ are also called **crossed homomorphisms** from G to M; in other words, a crossed homomorphism is a map $f: G \to M$ satisfying

$$f(g_1 g_2) = g_1 f(g_2) + f(g_1), \qquad g_1, g_2 \in G.$$

For fixed $\Bbbk$, they form a $\Bbbk$-module under pointwise operations.

Notice that the elements of $B^1(G,M)$ are the crossed homomorphisms of the form $f_m : g \mapsto gm - m$, where $m \in M$.

In general, every crossed homomorphism f satisfies $f(1_G) = 0$ (take $g_1 = g_2 = 1_G$) and $g^{-1} f(g) = -f(g^{-1})$ (take $g_1 = g^{-1}$ and $g_2 = g$).

Example 6.3.2 If G acts trivially on M, then the crossed homomorphisms are precisely the homomorphisms from G to the additive group $(M,+)$, and $B^1(G,M) = \{0\}$. Thus in this case

$$\operatorname{Hom}_{\mathsf{Grp}}(G,M) \simeq \mathrm{H}^1(G,M).$$

Another way to understand crossed homomorphisms is through semidirect products. For a G-module M, use the G-action to form the semidirect product $M \rtimes G$. Its elements have the form (m, g) with $m \in M$ and $g \in G$, and it has a projection homomorphism $\pi : M \rtimes G \to G$ sending (m, g) to g.

Proposition 6.3.3 Let M be a G-module. There is a canonical bijection

$$Z^1(G, M) \xrightarrow{1:1} \{\text{group homomorphisms } \sigma : G \to M \rtimes G \mid \pi\sigma = \mathrm{id}_G\}.$$

A map σ on the right-hand side is also called a section of π; the crossed homomorphism f corresponds to the section $\sigma(g) = (f(g), g)$.

Proof A map σ satisfying $\pi\sigma = \mathrm{id}_G$ must have the form $\sigma(g) = (f(g), g)$ for some map $f : G \to M$. By the definition of the semidirect product, σ is a group homomorphism if and only if the following equality holds in M:

$$f(g_1) + g_1 f(g_2) = f(g_1 g_2), \qquad g_1, g_2 \in G.$$

This is exactly the definition of a crossed homomorphism. $\square$

Crossed homomorphisms are also related to the augmentation ideal $\mathfrak{I} \subset \Bbbk[G]$ introduced in Remark 6.1.11.

Proposition 6.3.4 For every G-module M, there is a canonical isomorphism $\mathrm{Hom}_G(\mathfrak{I}, M) \xrightarrow{\sim} Z^1(G, M)$ sending a homomorphism φ to the crossed homomorphism $f(g) = \varphi(g - 1_G)$.

Proof As a $\Bbbk$-module, $\mathfrak{I}$ has basis $\{g - 1_G : g \in G, \ g \neq 1_G\}$. Thus specifying a $\Bbbk$-module homomorphism $\varphi : \mathfrak{I} \to M$ is equivalent to specifying a map $f : G \to M$ satisfying $f(1_G) = 0$, with $f(g) = \varphi(g - 1_G)$. The condition that φ be a G-module homomorphism is $\varphi(g_1(g_2 - 1_G)) = g_1\varphi(g_2 - 1_G)$. But

$$\text{left-hand side} = \varphi(g_1g_2 - 1_G - g_1 + 1_G) = f(g_1g_2) - f(g_1), \qquad \text{right-hand side} = g_1 f(g_2).$$

This is exactly the crossed-homomorphism condition. $\square$

For the next step, the case $n = 2$, we first recall the notion of a group extension. Take $\Bbbk = \mathbb{Z}$ and introduce some terminology.

Definition 6.3.5 Let A be a group. An **extension** of G by A is a short exact sequence of groups [51, Definition 4.3.7]

$$0 \to A \to E \xrightarrow{\pi} G \to 1.$$

Henceforth we abbreviate the data of the group extension to E. A group homomorphism $s : G \to E$ satisfying $\pi s = \mathrm{id}_G$ is called a **splitting** of the extension. In this case s embeds G as a subgroup of E, thereby identifying E with the semidirect product $A \rtimes G$. An extension possessing a splitting is called **split**. See [51, §4.3] for details.

Broadly speaking, the theory of group extensions studies how to build a larger group from a normal subgroup and a quotient group. To relate this to the cohomology of G-modules, from now on write the group A above as M, require it to be commutative, and write its group operation additively. Such a group extension determines a group homomorphism

$$\alpha : G \to \mathrm{Aut}_{\mathsf{Grp}}(M), \qquad \alpha(g)(x) = exe^{-1}, \qquad e \in \pi^{-1}(g) \text{ chosen arbitrarily.}$$

Thus M becomes a G-module through α; this is the most basic invariant of a group extension.

Given a G-module M, we wish to classify all extensions of G by M whose associated α is precisely the given G-module structure on M, up to equivalence (or isomorphism) of extensions. An equivalence from an extension E to an extension E' is defined in the evident way as a commutative diagram of groups

$$\begin{tikzcd} 0 \arrow[r] & M \arrow[r] \arrow[d, equal] & E \arrow[r] \arrow[d, "\varphi"] & G \arrow[r] \arrow[d, equal] & 1 \\ 0 \arrow[r] & M \arrow[r] & E' \arrow[r] & G \arrow[r] & 1. \end{tikzcd} \tag{6.3.1}$$

The reader is invited to verify that φ in this diagram is automatically a group isomorphism, and that an extension is split if and only if it is equivalent to the semidirect product $M \rtimes G$.

Definition 6.3.6 Define the category $\mathsf{Ext}(G, M)$ of all extensions of G by the given G-module M; its morphisms are the commutative diagrams (6.3.1).

This category is a groupoid: all its morphisms are invertible. Classifying extensions amounts to describing $\mathsf{Ext}(G, M)$ up to equivalence of categories. This divides into two subproblems:

- ◇ describe $\mathrm{Ob}(\mathsf{Ext}(G, M))/ \simeq$, that is, the equivalence classes of extensions, and identify the split extensions among them;
- ◇ describe the morphisms in $\mathsf{Ext}(G, M)$, that is, the isomorphisms between extensions.

We begin with the simpler second subproblem. First, an extension E has a simple class of automorphisms of the form $\mathrm{Ad}_m : e \mapsto mem^{-1}$, where $m \in M$ and the group operation in E is written multiplicatively; these are called inner automorphisms. For abstract reasons, the inner automorphisms form a normal subgroup of $\mathrm{Aut}(E)$, and the corresponding quotient group is called the outer automorphism group of E.

To classify group extensions, we need to compute cohomology using the normalized standard complex $\overline{C}(G, M)$ described in Remark 6.2.13. Accordingly, define

$$\overline{C}^n(G, M) \supset \overline{Z}^n(G, M) \supset \overline{B}^n(G, M),$$
$$\mathrm{H}^n(G, M) \simeq \overline{Z}^n(G, M)/\overline{B}^n(G, M).$$

We have $\overline{C}^0(G, M) = C^0(G, M) = M$. The preceding discussion of crossed homomorphisms has also shown that $\overline{Z}^1(G, M) = Z^1(G, M)$.

Lemma 6.3.7 Let $0 \to M \to E \xrightarrow{\pi} G \to 1$ be an object of $\mathsf{Ext}(G, M)$. Its automorphism group is canonically isomorphic to $\overline{Z}^1(G, M)$, while its outer automorphism group is canonically isomorphic to $\mathrm{H}^1(G, M)$. More explicitly, an element $z \in \overline{Z}^1(G, M)$ corresponds to the automorphism sending $e \in E$ to $z(\pi(e))e$.

Proof For the moment, write the group operation on M multiplicatively. Take $E' = E$ in the commutative diagram (6.3.1). For every $e \in E$, there is a unique $z(e) \in M$ such that $\varphi(e) = z(e)e$. If $m \in M$, then $\varphi(me) = m\varphi(e) = mz(e)e = z(e)me$, so $z(e)$ depends only on the image of e in G; henceforth we write it as a map $z : G \to M$. Conversely, every map $z : G \to M$ determines a map $\varphi : E \to E$ by $\varphi(e) = z(\pi(e))e$, and (6.3.1) commutes at the level of sets.

Consider a map $z : G \to M$ and the corresponding φ. Put $g_i = \pi(e_i)$ for $i = 1, 2$. Then

$$\begin{aligned} \varphi(e_1e_2) &= z(g_1g_2)e_1e_2, \\ \varphi(e_1)\varphi(e_2) &= z(g_1)e_1z(g_2)e_2 \\ &= z(g_1) \underbrace{g_1 \cdot z(g_2)}_{\text{the } G\text{-module structure on } M} e_1e_2. \end{aligned}$$

Thus φ is a group homomorphism if and only if z is a crossed homomorphism. It is clear that composition of group homomorphisms corresponds to addition of crossed homomorphisms. Now take $m \in M$. The corresponding inner automorphism is $e \mapsto mem^{-1} = mem^{-1}e^{-1}e$. Clearly $mem^{-1}e^{-1} \in M$, and in additive notation this element is $m - \pi(e)m$. This proves the assertion. □

Definition 6.3.8 The groupoid $\tau^{\leq 2}\overline{\mathsf{C}}(G, M)$ is defined as follows: its set of objects is $\overline{Z}^2(G, M)$, while its morphism sets are

$$\mathrm{Hom}(f, f') := \{w \in \overline{C}^1(G, M) : f' = d^1w + f\}.$$

Composition of morphisms is defined by addition in $\overline{C}^1(G, M)$. Thus the isomorphism classes of objects correspond bijectively to the elements of $\mathrm{H}^2(G, M)$.

The meaning of the notation is clear: $\tau^{\leq 2}\overline{\mathsf{C}}(G,M)$ is the category determined by the truncated complex $\tau^{\leq 2}\overline{C}(G,M)$.

Theorem 6.3.9 Let M be a G-module. The category $\mathsf{Ext}(G,M)$ is equivalent to $\tau^{\leq 2}\overline{\mathsf{C}}(G,M)$. In particular, equivalence classes of extensions of G by M correspond bijectively to elements of $\mathrm{H}^2(G,M)$, and the equivalence class of split extensions corresponds to $0 \in \mathrm{H}^2(G,M)$.

Proof We first construct a functor $\tau^{\leq 2}\overline{\mathsf{C}}(G,M) \to \mathsf{Ext}(G,M)$. Given a map $f : G^2 \to M$, define a binary operation on the set $M \times G$ by

$$(m_1, g_1) \cdot (m_2, g_2) := (m_1 + g_1 \cdot m_2 + f(g_1, g_2), g_1 g_2)\,.$$

Associativity is equivalent to the identity

$$f(g_1, g_2) + f(g_1 g_2, g_3) = g_1 \cdot f(g_2, g_3) + f(g_1, g_2 g_3), \tag{6.3.2}$$

that is, to $f \in Z^2(G,M)$. Furthermore, $(0, 1_G)$ is the identity if and only if

$$f(g, 1_G) = 0 = f(1_G, g),$$

that is, if and only if $f \in \overline{Z}^2(G,M)$. Taking $(g_1, g_2, g_3) = (g, g^{-1}, g)$ in (6.3.2) and using the last equality gives

$$f(g, g^{-1}) = g \cdot f(g^{-1}, g). \tag{6.3.3}$$

Thus $M \times G$ acquires a monoid structure from $f \in \overline{Z}^2(G,M)$; denote this monoid by E. It is automatically a group, since (6.3.3) gives

$$(0, g)^{-1} = \left(-f(g^{-1}, g), g^{-1}\right), \qquad (m, 1_G)^{-1} = (-m, 1_G).$$

It is easy to prove that the maps $m \mapsto (m, 1_G)$ and $(m, g) \mapsto g$ give a group extension $0 \to M \to E \to G \to 1$ that induces precisely the original G-action on M. Verify that $f = 0$ corresponds to the semidirect product $E = M \rtimes G$.

If, moreover, $w \in \overline{C}^1(G,M)$ and $f' := d^1 w + f$, denote the corresponding extension by E'. The map

$$(m, g) \mapsto (m - w(g), g)$$

gives an isomorphism from E to E'. Indeed, the condition that this map preserve multiplication is equivalent to the identity

$$-w(g_1) - g_1 \cdot w(g_2) + (d^1 w)(g_1, g_2) = -w(g_1 g_2),$$

which is exactly the definition of $d^1 w$.

We have therefore obtained a functor $\tau^{\leq 2}\overline{\mathsf{C}}(G, M) \to \mathsf{Ext}(G, M)$, and both categories are groupoids. Comparing automorphism groups of objects by Lemma 6.3.7 shows that this functor is fully faithful. It remains to show that it is essentially surjective.

Let E be an object of $\mathsf{Ext}(G, M)$. Choose an arbitrary section $s : G \to E$ of the map $\pi : E \to G$ in the extension data; that is, a map satisfying $\pi s = \mathrm{id}_G$. This is equivalent to choosing a representative of every M-coset in E. A section s is called normalized if $s(1_G) = 1_E$. For all $g_1, g_2 \in G$, the elements $s(g_1 g_2)$ and $s(g_1)s(g_2)$ have the same image under π. Hence there is a unique $f(g_1, g_2) \in M$ such that

$$s(g_1)s(g_2) = f(g_1, g_2)s(g_1 g_2).$$

It is easy to prove that associativity, $(s(g_1)s(g_2))s(g_3) = s(g_1)(s(g_2)s(g_3))$, translates into (6.3.2), namely $f \in Z^2(G, M)$. If s is normalized, then $f(1_G, g) = 0 = f(g, 1_G)$ for every g, which is equivalent to $f \in \overline{Z}^2(G, M)$.

For the chosen normalized section s, give $M \times G$ the group structure corresponding to f. Consider the bijection

$$M \times G \xrightarrow{1:1} E, \qquad (m, g) \mapsto ms(g).$$

The preceding discussion shows immediately that this bijection preserves group multiplication and gives an isomorphism in $\mathsf{Ext}(G, M)$. This completes the proof of the equivalence. $\square$

In the final part of the essential-surjectivity argument, changing the section s is equivalent to choosing an arbitrary map $h : G \to M$ and taking $s'(g) := h(g)s(g)$. The corresponding map $f' : G^2 \to M$ becomes

$$\begin{aligned} f'(g_1, g_2) &= (h(g_1) + g_1 \cdot h(g_2) - h(g_1 g_2)) + f(g_1, g_2) \\ &= (d^1 h)(g_1, g_2) + f(g_1, g_2). \end{aligned}$$

If one also requires $h(1_G) = 0$, then $d^1 h \in \overline{B}^2(G, M)$. Historically, this observation was the starting point for the connection between extensions and $\mathrm{H}^2(G, M)$; it also supplies a group-extension interpretation for the definitions of $\overline{Z}^2(G, M)$ and $\overline{B}^2(G, M)$, although this was left implicit in the proof.

Remark 6.3.10 Theorem 6.3.9 says that the complex $\tau^{\leq 2}\overline{C}(G, M)$ is a linear incarnation of the category of group extensions $\mathsf{Ext}(G, M)$: the term in degree 2 describes objects, the term in degree 1 describes morphisms, and the term in degree 0 remains to be interpreted. For $m \in M = \overline{C}^0(G, M)$, there is an inner automorphism of the group extension $\mathrm{Ad}_m : e \mapsto mem^{-1}$, which is trivial if and only if $m \in M^G = \mathrm{H}^0(G, M)$.

From the viewpoint of inner automorphisms, we may regard the elements of M as 2-morphisms in $\mathsf{Ext}(G, M)$: for $m \in M$ and $\varphi_1, \varphi_2 \in \mathrm{Hom}_{\mathsf{Ext}(G,M)}(E, E')$,

$$\text{interpret the 2-cell diagram } \begin{tikzcd} E \arrow[r, bend left, "\varphi_1"] \arrow[r, bend right, "\varphi_2"'] \arrow[r, phantom, "\Downarrow m"] & E' \end{tikzcd} \text{ as } \varphi_2 = \mathrm{Ad}_m \varphi_1.$$

Notice that $\mathrm{Ad}_m \varphi_1 = \varphi_1 \mathrm{Ad}_m$ and $\mathrm{Ad}_{m_1} \mathrm{Ad}_{m_2} = \mathrm{Ad}_{m_1+m_2}$. Thus $\mathrm{H}^0(G, M)$ becomes the "2-automorphism group" of every morphism φ. In this sense, the complex $\tau^{\leq 2}\overline{C}(G, M)$ may be called a linear incarnation of the 2-category $\mathsf{Ext}(G, M)$. See [51, §3.5] for the definition of a 2-category.

Other aspects of group extensions will be discussed in §6.6. Before that, we need to introduce more operations on group homology and cohomology.

6.4 Induced modules

For a group homomorphism $\varphi : H \to G$, Proposition 6.1.9 gives the right adjoint $\mathrm{Hom}_H(\Bbbk[G], \cdot)$ and the left adjoint $\Bbbk[G] \underset{\Bbbk[H]}{\otimes} (\cdot)$ of φ^*. This section focuses on the common case in which $H \subset G$ is a subgroup and φ is the inclusion homomorphism. We begin by introducing notation.

Definition 6.4.1 (Induced module) Let H be a subgroup of G. For an H-module N, define the G-modules

$$\mathrm{Ind}_H^G(N) := \mathrm{Hom}_H(\Bbbk[G], N), \qquad \mathrm{ind}_H^G(N) := \Bbbk[G] \underset{\Bbbk[H]}{\otimes} N.$$

As N varies, both constructions give functors $H\text{-}\mathsf{Mod} \to G\text{-}\mathsf{Mod}$.

Both operations may be called **induction** from H to G in a broad sense. The functor ind_H^G is also sometimes called induction with finite support; this terminology will be explained by Lemma 6.8.1 below.

Proposition 6.4.2 The functors $\mathrm{Res}_H^G : G\text{-}\mathsf{Mod} \to H\text{-}\mathsf{Mod}$ and $\mathrm{Ind}_H^G, \mathrm{ind}_H^G : H\text{-}\mathsf{Mod} \to G\text{-}\mathsf{Mod}$ are all exact. The functor Ind_H^G preserves injective objects, ind_H^G preserves projective objects, and Res_H^G preserves both.

Proof We already know that Res_H^G is exact. Choosing coset representatives shows that $\Bbbk[G]$ is a free left $\Bbbk[H]$-module, so $\mathrm{Hom}_{\Bbbk[H]}(\Bbbk[G], \cdot)$ is exact. Similarly, $\Bbbk[G]$ is a free right $\Bbbk[H]$-module, so $\Bbbk[G] \underset{\Bbbk[H]}{\otimes} (\cdot)$ is exact. By the adjunctions, the assertions about preserving injective and projective objects follow directly from Proposition 2.8.17. $\square$

Proposition 6.4.2 provides a way to construct injective (or projective) resolutions in G-Mod: take an injective (or projective) resolution of the given G-module M in $\Bbbk$-Mod, and then induce that resolution to G using $\mathrm{Ind}_{\{1\}}^G$ (or $\mathrm{ind}_{\{1\}}^G$).

The following result is commonly called **Shapiro's lemma**.

Theorem 6.4.3 (B. Eckmann, D. K. Faddeev, A. Shapiro) Let H be a subgroup of G. For every H-module N and $n \in \mathbb{Z}$, there are canonical isomorphisms

$$\mathrm{H}^n\left(G, \mathrm{Ind}_H^G(N)\right) \simeq \mathrm{H}^n(H, N), \qquad \mathrm{H}_n\left(G, \mathrm{ind}_H^G(N)\right) \simeq \mathrm{H}_n(H, N).$$

The two sides of each isomorphism, viewed as functors in N, have long exact sequences, and the isomorphisms are compatible with those sequences.

Proof We give two methods. For cohomology, the first considers the functors

$$A^n := \mathrm{H}^n\left(G, \mathrm{Ind}_H^G(\cdot)\right).$$

Since Ind_H^G is exact, this family of functors has long exact sequences inherited from $\mathrm{H}^n(G, \cdot)$ and hence is a cohomological δ-functor in the sense of Definition 3.12.5. For $n = 0$, Corollary 6.2.4 gives a canonical isomorphism

$$A^0(N) = (\mathrm{Ind}_H^G(N))^G \xrightarrow{\sim} N^H, \qquad \varphi \mapsto \varphi(1_G).$$

To extend this canonically to an isomorphism of cohomological δ-functors $A^n \xrightarrow{\sim} \mathrm{H}^n(H, \cdot)$, it suffices to prove that A^n is effaceable for $n > 0$ (Definition 3.12.11, Proposition 3.12.14). By Proposition 6.4.2, if N is embedded into an injective H-module I, then $\mathrm{Ind}_H^G(N) \hookrightarrow \mathrm{Ind}_H^G(I)$ and $\mathrm{Ind}_H^G(I)$ is an injective G-module. Hence $n > 0$ implies $\mathrm{H}^n(G, \mathrm{Ind}_H^G(I)) = 0$, proving effaceability.

The idea for homology is similar and uses the coeffaceability of homological δ-functors. The required degree-zero isomorphism $N_H \xrightarrow{\sim} (\mathrm{ind}_H^G(N))_G$ also comes from Corollary 6.2.4.

The second method works in the derived category. For cohomology, we again start from the canonical isomorphism

$$\underbrace{\mathrm{Hom}_G(\Bbbk, \mathrm{Ind}_H^G(\cdot))}_{\simeq \mathrm{Ind}_H^G(\cdot)^G} \simeq \underbrace{\mathrm{Hom}_H(\Bbbk, \cdot)}_{\simeq (\cdot)^H} : H\text{-Mod} \to \Bbbk\text{-Mod}.$$

Both sides are left-exact functors. Since Ind_H^G preserves injective objects, taking right-derived functors in Theorem 4.6.5 (iii) gives

$$\mathrm{RHom}_G(\Bbbk, \mathrm{R}\,\mathrm{Ind}_H^G(\cdot)) \simeq \mathrm{RHom}_H(\Bbbk, \cdot).$$

But Ind_H^G is exact, so $\mathrm{R}\,\mathrm{Ind}_H^G(\cdot)$ is determined by the universal property of localization and may still be denoted by Ind_H^G. Thus at the level of derived categories there is an isomorphism of triangulated functors

$$\mathrm{RHom}_G(\Bbbk, \mathrm{Ind}_H^G(\cdot)) \simeq \mathrm{RHom}_H(\Bbbk, \cdot).$$

Taking H^n on both sides gives the desired result.

The idea for homology is similar; the key is the isomorphism

$$\Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} \left(\Bbbk[G] \overset{\mathrm{L}}{\underset{\Bbbk[H]}{\otimes}} (\cdot) \right) \simeq \Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[H]}{\otimes}} (\cdot).$$

This is also a special case of Example 4.12.10 or of Proposition 4.12.13 on the associativity constraint. Since ind_H^G is exact, $\Bbbk[G] \overset{\mathrm{L}}{\underset{\Bbbk[H]}{\otimes}} (\cdot)$ may also be replaced by $\Bbbk[G] \underset{\Bbbk[H]}{\otimes} (\cdot)$; the remaining steps are the same. □

Notice that if Res_H^G is interpreted as $\Bbbk[G] \underset{\Bbbk[G]}{\otimes} (\cdot)$ according to (6.1.2), then, since $\mathrm{Res}_H^G \Bbbk = \Bbbk$, the isomorphism $\mathrm{RHom}_G(\Bbbk, \mathrm{R}\,\mathrm{Ind}_H^G(\cdot)) \simeq \mathrm{RHom}_H(\Bbbk, \cdot)$ reduces to the adjunction between RHom and $\overset{\mathrm{L}}{\otimes}$ (Theorem 4.12.17).

Group cohomology and homology can be described concretely through the standard complex and the standard chain complex of Propositions 6.2.8 and 6.2.9. We want to make the isomorphisms in Theorem 6.4.3 explicit in terms of these complexes. There is no essential difficulty. To distinguish the group, for every group H write the complex of H-modules L from Definition 6.1.10 as L^H; it gives a free resolution $\mathsf{L}^H \to \Bbbk$.

Lemma 6.4.4 Let $\varphi : H \to G$ be a group homomorphism. The corresponding homomorphism $H^{n+1} \to G^{n+1}$ induces a quasi-isomorphism of complexes $\mathsf{L}^H \to \varphi^*(\mathsf{L}^G)$ whose composite with $\varphi^*(\mathsf{L}^G) \to \varphi^*\Bbbk = \Bbbk$ equals $\mathsf{L}^H \to \Bbbk$.

In particular, if H is a subgroup of G, there is a quasi-isomorphism $\mathsf{L}^H \to \mathrm{Res}_H^G \mathsf{L}^G$ with the above composite property.

Proof It follows directly from the definitions that $\mathsf{L}^H \to \varphi^*(\mathsf{L}^G)$ is a morphism of complexes; the assertion about its composite is equally clear. Since φ^* is exact, while $\mathsf{L}^G \to \Bbbk$ and $\mathsf{L}^H \to \Bbbk$ are both quasi-isomorphisms, $\mathsf{L}^H \to \varphi^*(\mathsf{L}^G)$ is a quasi-isomorphism as well. □

Proposition 6.4.5 Let H be a subgroup of G and N an H-module. Identify $\mathrm{H}^n(G, \mathrm{Ind}_H^G(N))$ (or $\mathrm{H}^n(H,N)$) with the H^n of the standard complex $C(G, \mathrm{Ind}_H^G(N))$ (or $C(H,N)$), as in Proposition 6.2.8. Then the isomorphism

$$\mathrm{H}^n\left(G, \mathrm{Ind}_H^G(N)\right) \simeq \mathrm{H}^n(H,N), \qquad n \in \mathbb{Z}_{\geq 0},$$

is induced by the morphism of complexes

$$\begin{aligned} C^m(G, \mathrm{Ind}_H^G(N)) &\to C^m(H, N), \\ [f : G^m \to \mathrm{Ind}_H^G(N)] &\mapsto [(h_1, \ldots, h_m) \mapsto f(h_1, \ldots, h_m)(1_G)] \\ &\qquad = (f|_{H^m})(\cdot)(1_G), \end{aligned}$$

where $h_1, \ldots, h_m \in H$.

Proof Again there are two methods. First, verify that the displayed maps $C^m(G, \mathrm{Ind}_H^G(N)) \to C^m(H, N)$ do indeed form a morphism of complexes. Because Ind_H^G is exact, as N varies these maps induce a morphism between cohomological δ-functors. In degree zero, this map is clearly the canonical isomorphism $\mathrm{Ind}_H^G(N)^G \xrightarrow{\sim} N^H$, $\varphi \mapsto \varphi(1_G)$. We already know that $\mathrm{H}^n(G, \mathrm{Ind}_H^G(\cdot))$ is a universal cohomological δ-functor (Theorem 6.4.3). Thus the above morphism is unique and must agree with the isomorphism $\mathrm{H}^n(G, \mathrm{Ind}_H^G(N)) \simeq \mathrm{H}^n(H, N)$ in that theorem.

For the second method, let M (respectively, N) be a bounded-below complex of G-modules (respectively, H-modules). Extend the functors Ind_H^G and Res_H^G degreewise to complexes, keeping the same notation. The adjunction of Proposition 6.1.9 lifts to Hom complexes:

$$\mathrm{Hom}_H^\bullet(\mathrm{Res}_H^G M, N) \simeq \mathrm{Hom}_G^\bullet(M, \mathrm{Ind}_H^G(N)).$$

A direct verification is not difficult; this can also be understood as the adjunction between $\mathrm{Hom}^\bullet$ and $\otimes$ described earlier (Proposition 4.12.6).

Now let N be an H-module and take an injective resolution $\iota : N \to I$. Then $\mathrm{Ind}_H^G(N) \to \mathrm{Ind}_H^G(I)$ is still an injective resolution, and naturality of the Hom complexes gives the following commutative diagram of complexes (the solid part):

$$\begin{tikzcd}
\mathrm{Hom}_H^\bullet(\Bbbk, I) \arrow[d, "\alpha"'] & \mathrm{Hom}_G^\bullet(\Bbbk, \mathrm{Ind}_H^G(I)) \arrow[l, "\sim"'] \arrow[d] \\
\mathrm{Hom}_H^\bullet(\mathrm{Res}_H^G \mathsf{L}^G, I) \arrow[d, "\beta"'] & \mathrm{Hom}_G^\bullet(\mathsf{L}^G, \mathrm{Ind}_H^G(I)) \arrow[l, "\sim"] \\
\mathrm{Hom}_H^\bullet(\mathsf{L}^H, I) & \\
\mathrm{Hom}_H^\bullet(\mathsf{L}^H, N) \arrow[u] & \mathrm{Hom}_G^\bullet(\mathsf{L}^G, \mathrm{Ind}_H^G(N)) \arrow[l, dashed, "\gamma"] \arrow[uu]
\end{tikzcd} \tag{6.4.1}$$

where α and β come from the morphism in Lemma 6.4.4. Thus $\beta\alpha$ is the familiar map $\mathrm{Hom}_H^\bullet(\Bbbk, I) \to \mathrm{Hom}_H^\bullet(\mathsf{L}^H, I)$. The solid horizontal isomorphisms come from the adjunction, and all vertical arrows are quasi-isomorphisms.

Recall the second proof of Theorem 6.4.3. The first row of (6.4.1) gives the derived-category form of

$$\mathrm{H}^n(H, N) \simeq \mathrm{H}^n(G, \mathrm{Ind}_H^G(N)), \qquad n \in \mathbb{Z}.$$

On the other hand, the substance of Proposition 6.2.8 is the isomorphisms of complexes

$$\mathrm{Hom}^\bullet_H(\mathsf{L}^H, N) \simeq C(H, N), \qquad \mathrm{Hom}^\bullet_G(\mathsf{L}^G, \mathrm{Ind}^G_H(N)) \simeq C(G, \mathrm{Ind}^G_H(N)).$$

It therefore remains only to supply the arrow γ in (6.4.1) so that the square commutes, and to describe it. Since ι is monic, such a γ, if it exists, is unique.

Take $m \in \mathbb{Z}_{\geq 0}$ and $\phi \in \mathrm{Hom}^m_G(\mathsf{L}^G, \mathrm{Ind}^G_H(N))$. Inspecting the construction shows that the image of ϕ in $\mathrm{Hom}^m_H(\mathsf{L}^H, I)$ is

$$(h_0, \ldots, h_m) \mapsto \iota\left(\phi(h_0, \ldots, h_m)(1_G)\right).$$

Comparison with the isomorphism in the proof of Proposition 6.2.8 immediately shows that γ can be defined as asserted. $\square$

We now state the homological version, using the notation for the standard chain complex from Proposition 6.2.9.

Proposition 6.4.6 Let H be a subgroup of G and N an H-module. Identify $\mathrm{H}_n(G, \mathrm{ind}^G_H(N))$ (or $\mathrm{H}_n(H, N)$) with the H_n of the standard chain complex $C(G, \mathrm{ind}^G_H(N))$ (or $C(H, N)$), as in Proposition 6.2.9. Then the isomorphism

$$\mathrm{H}_n\left(G, \mathrm{ind}^G_H(N)\right) \simeq \mathrm{H}_n(H, N), \qquad n \in \mathbb{Z}_{\geq 0},$$

is induced by the morphism of complexes

$$\begin{gathered} C_m(H, N) \to C_m(G, \mathrm{ind}^G_H(N)), \\ (h_1|\cdots|h_m)^\dagger \otimes y \mapsto (h_1|\cdots|h_m)^\dagger \otimes (1_G \otimes y), \end{gathered}$$

where $h_1, \ldots, h_m \in H$ and $y \in N$.

Proof We explain only the derived-category approach. Take a projective resolution $\pi : P \to N$ in H-modules. The commutative diagram (6.4.1) from the proof of Proposition 6.4.5 must be replaced by

$$\begin{tikzcd}
\Bbbk \underset{\Bbbk[H]}{\otimes} P \arrow[r, "\sim"] & \Bbbk \underset{\Bbbk[G]}{\otimes} \mathrm{ind}^G_H(P) \\
\mathrm{Res}^G_H \mathsf{L}^G \underset{\Bbbk[H]}{\otimes} P \arrow[u, "\alpha"] \arrow[r, "\sim"] & \mathsf{L}^G \underset{\Bbbk[G]}{\otimes} \mathrm{ind}^G_H(P) \arrow[u] \arrow[dd] \\
\mathsf{L}^H \underset{\Bbbk[H]}{\otimes} P \arrow[u, "\beta"] \arrow[d] & \\
\mathsf{L}^H \underset{\Bbbk[H]}{\otimes} N \arrow[r, dashed, "\gamma"'] & \mathsf{L}^G \underset{\Bbbk[G]}{\otimes} \mathrm{ind}^G_H(N),
\end{tikzcd}$$

where the solid horizontal arrows come from the associativity constraint for tensor products at the level of complexes, while all vertical arrows are quasi-isomorphisms. Once again, it remains to describe γ. The rest of the argument is similar to the cohomological version. □

6.5 Change of groups

Consider a group homomorphism $\varphi : H \to G$. In what follows, M denotes an arbitrary G-module and N an arbitrary H-module. Our first task in this section is to define two families of canonical homomorphisms

$$\begin{aligned} \mathrm{H}^n(G, M) &\to \mathrm{H}^n(H, \varphi^* M), \\ \mathrm{H}_n(H, \varphi^* M) &\to \mathrm{H}_n(G, M), \end{aligned} \tag{6.5.1}$$

where $n \in \mathbb{Z}_{\geq 0}$. Since $\varphi^* : G\text{-}\mathsf{Mod} \to H\text{-}\mathsf{Mod}$ is exact, the functors on both sides of these homomorphisms have long exact sequences; in other words, they are cohomological or homological δ-functors. The homomorphisms above will be compatible with the long exact sequences. We present two constructions.

For cohomology, the first method considers the family of functors $B^n := \mathrm{H}^n(H, \varphi^*(\cdot))$. There is a canonical homomorphism

$$M^G \hookrightarrow M^{\varphi(H)} = (\varphi^* M)^H = B^0(M).$$

Because $(\mathrm{H}^n(G, \cdot))_n$ is a universal cohomological δ-functor, this homomorphism extends uniquely to a morphism of cohomological δ-functors $\mathrm{H}^n(G, \cdot) \to B^n$.

The homology case is similar; the key is to use the canonical homomorphism $(\varphi^* M)_H = M_{\varphi(H)} \twoheadrightarrow M_G$.

The second method works in the derived category. We introduce some related notation.

Convention 6.5.1 Write $\mathsf{C}(G) := \mathsf{C}(G\text{-}\mathsf{Mod})$, $\mathsf{K}(G) := \mathsf{K}(G\text{-}\mathsf{Mod})$, and $\mathsf{D}(G) := \mathsf{D}(G\text{-}\mathsf{Mod})$; use similar notation when one-sided or two-sided boundedness conditions are imposed. Likewise, write $\mathsf{C}(\Bbbk) := \mathsf{C}(\Bbbk\text{-}\mathsf{Mod})$, together with $\mathsf{K}(\Bbbk)$, $\mathsf{D}(\Bbbk)$, and so forth.

The functor on derived categories induced by the exact functor φ^* is still denoted $\varphi^* : \mathsf{D}(G) \to \mathsf{D}(H)$. Use the customary notation R and L for right- and left-derived functors, respectively. We seek morphisms between triangulated functors

$$\begin{aligned} \mathrm{R}\left((\cdot)^G\right) &\to \mathrm{R}\left((\cdot)^H\right) \circ \varphi^*, \\ \mathrm{L}\left((\cdot)_H\right) \circ \varphi^* &\to \mathrm{L}\left((\cdot)_G\right). \end{aligned}$$

The key is the universal property. In the cohomology case, lift the functors $(\cdot)^G$, $(\cdot)^H$, and so on to the level of complexes, with the same notation. Consider the following 2-cell diagram involving categories, functors, and morphisms of functors, all understood to be triangulated.

$$\begin{tikzcd}
\mathsf{K}^+(G) \arrow[r, "\varphi^*"] \arrow[d] \arrow[rr, bend left, "(\cdot)^G"] \arrow[rr, phantom, bend left=20, "\Downarrow"] & \mathsf{K}^+(H) \arrow[r, "(\cdot)^H"] \arrow[d] \arrow[dl, Rightarrow, "\sim"'] & \mathsf{K}^+(\Bbbk) \arrow[d] \arrow[dl, Rightarrow] \\
\mathsf{D}^+(G) \arrow[r, "\varphi^*"'] & \mathsf{D}^+(H) \arrow[r, "\mathrm{R}((\cdot)^H)"'] & \mathsf{D}^+(\Bbbk)
\end{tikzcd} \tag{6.5.2}$$

The $\Rightarrow$ on the curved part comes from the previously used morphism $(\cdot)^G \to (\cdot)^H \circ \varphi^*$. The $\overset{\sim}{\Rightarrow}$ on the left square comes from the exactness of φ^*, while the $\Rightarrow$ on the right square is part of the data of the right-derived functor $\mathrm{R}((\cdot)^H)$. The superscript $+$ is used here only for simplicity; the unbounded version requires the theory of §4.11.

Recall that 2-cell diagrams can be composed vertically, meaning that the morphisms $\Rightarrow$ between functors are composed. By the universal property of right-derived functors (see Remark 4.6.2 and the preceding discussion), there is a unique $\Downarrow$ in the left-hand diagram below such that

$$\begin{tikzcd}
\mathsf{K}^+(G) \arrow[r, "(\cdot)^G"] \arrow[d] & \mathsf{K}^+(\Bbbk) \arrow[d] \arrow[dl, Rightarrow] \\
\mathsf{D}^+(G) \arrow[r, "\mathrm{R}((\cdot)^G)"] \arrow[r, bend right=60, "\mathrm{R}((\cdot)^H)\circ\varphi^*"'] \arrow[r, phantom, bend right=30, "\exists!\Downarrow"] & \mathsf{D}^+(\Bbbk)
\end{tikzcd}
\quad \underset{\text{vertical composite}}{=\!=\!=} \quad \text{vertical composite of (6.5.2)}
\quad
\begin{tikzcd}
\mathsf{K}^+(G) \arrow[r, "(\cdot)^G"] \arrow[d] & \mathsf{K}^+(\Bbbk) \arrow[d] \arrow[dl, Rightarrow] \\
\mathsf{D}^+(G) \arrow[r, "\mathrm{R}((\cdot)^H)\circ\varphi^*"'] & \mathsf{D}^+(\Bbbk)
\end{tikzcd} \tag{6.5.3}$$

The symbol $\swarrow\!\!\swarrow$ in the left-hand diagram is part of the data of the right-derived functor $\mathrm{R}\left((\cdot)^G\right)$. This uniquely determines the morphism

$$\mathrm{R}\left((\cdot)^G\right) \to \mathrm{R}\left((\cdot)^H\right) \circ \varphi^*.$$

The method for homology is dual. It uses the universal property of left-derived functors and the previously used morphism $(\cdot)_H \circ \varphi^* \to (\cdot)_G$. We next give a more direct description.

Lemma 6.5.2 For every group homomorphism $\varphi : H \to G$, the morphisms characterized above can be defined as follows.

- In the cohomology case, take the composite

$$\mathrm{R}\left((\cdot)^G\right) \to \mathrm{R}\left((\cdot)^H \circ \varphi^*\right) \to \mathrm{R}\left((\cdot)^H\right) \circ \varphi^*. \tag{6.5.4}$$

The first part comes from $(\cdot)^G \to (\cdot)^H \circ \varphi^*$, since a morphism of functors induces a morphism between their right-derived functors; the second part comes from Theorem 4.6.5 (i).

- ◇ The homology case is similar: take the composite

$$\mathrm{L}\left((\cdot)_H\right) \circ \varphi^* \to \mathrm{L}\left((\cdot)_H \circ \varphi^*\right) \to \mathrm{L}\left((\cdot)_G\right). \tag{6.5.5}$$

The first part comes from Theorem 4.6.5 (ii), and the second from $(\cdot)_H \circ \varphi^* \to (\cdot)_G$.

Proof It suffices to discuss the cohomology case. First, for the morphisms in (6.5.4), check the following equalities of vertical composites:

$$\begin{tikzcd}
\mathsf{K}^+(G) \arrow[r, "(\cdot)^G"] \arrow[d] & \mathsf{K}^+(\Bbbk) \arrow[d] \arrow[dl, Rightarrow] \\
\mathsf{D}^+(G) \arrow[r, "\mathrm{R}((\cdot)^G)"] \arrow[r, phantom, shift right=3ex, "\Downarrow"] \arrow[r, bend right=60, "\mathrm{R}((\cdot)^H\circ\varphi^*)"'] & \mathsf{D}^+(\Bbbk)
\end{tikzcd}
=
\begin{tikzcd}
\mathsf{K}^+(G) \arrow[r, "(\cdot)^H\circ\varphi^*"] \arrow[r, bend left=60, "(\cdot)^G"] \arrow[r, phantom, shift left=3ex, "\Downarrow"] \arrow[d] & \mathsf{K}^+(\Bbbk) \arrow[d] \arrow[dl, Rightarrow] \\
\mathsf{D}^+(G) \arrow[r, "\mathrm{R}((\cdot)^H\circ\varphi^*)"'] & \mathsf{D}^+(\Bbbk)
\end{tikzcd}$$

and

$$\begin{tikzcd}
\mathsf{K}^+(G) \arrow[r, "(\cdot)^H\circ\varphi^*"] \arrow[d] & \mathsf{K}^+(\Bbbk) \arrow[d] \arrow[dl, Rightarrow] \\
\mathsf{D}^+(G) \arrow[r, "\mathrm{R}((\cdot)^H\circ\varphi^*)"'] \arrow[r, phantom, shift right=3ex, "\Downarrow"] \arrow[r, bend right=60, "\mathrm{R}((\cdot)^H)\circ\varphi^*"'] & \mathsf{D}^+(\Bbbk)
\end{tikzcd}
=
\begin{tikzcd}
\mathsf{K}^+(G) \arrow[r, "\varphi^*"] \arrow[d] & \mathsf{K}^+(H) \arrow[r, "(\cdot)^H"] \arrow[d] \arrow[dl, Rightarrow, "\sim"'] & \mathsf{K}^+(\Bbbk) \arrow[d] \arrow[dl, Rightarrow] \\
\mathsf{D}^+(G) \arrow[r, "\varphi^*"'] & \mathsf{D}^+(H) \arrow[r, "\mathrm{R}((\cdot)^H)"'] & \mathsf{D}^+(\Bbbk).
\end{tikzcd}$$

The first equality characterizes the induced morphism $\mathrm{R}((\cdot)^G) \to \mathrm{R}((\cdot)^H \circ \varphi^*)$, while the second characterizes the canonical morphism in Theorem 4.6.5. Both arise from the universal property of right-derived functors; see the explanation at the beginning of §4.6. Splicing these equalities together shows that the composite (6.5.4) satisfies (6.5.3). □

Remark 6.5.3 The first method, based on the universal cohomological δ-functor property, can be combined with derived-category theory to give another concise construction for the cohomology case of (6.5.1). Recall that for every G-module M and every n, there are canonical isomorphisms

$$\mathrm{H}^n(G, M) \simeq \mathrm{Ext}^n_G(\Bbbk, M) \simeq \mathrm{Hom}_{\mathsf{D}(G)}(\Bbbk, M[n]), \quad \mathrm{Ext}^\bullet_G := \mathrm{Ext}^\bullet_{\Bbbk[G]};$$

and likewise for H-modules. Since $\varphi^*\Bbbk \simeq \Bbbk$ and φ^* is a triangulated functor, we assert that the desired canonical homomorphism is the composite

$$\mathrm{Hom}_{\mathsf{D}(G)}(\Bbbk, M[n]) \xrightarrow{\text{functor } \varphi^*} \mathrm{Hom}_{\mathsf{D}(H)}(\varphi^*\Bbbk, \varphi^*(M[n])) \simeq \mathrm{Hom}_{\mathsf{D}(H)}(\Bbbk, (\varphi^*M)[n]).$$

To prove this, it suffices to note that when $n = 0$ the formula becomes the evident embedding $M^G \hookrightarrow (\varphi^*M)^H$ (because $\varphi^*\Bbbk \simeq \Bbbk$ sends 1 to 1), and that every morphism in the composite above is compatible with the long exact sequence in the second variable of the Hom functor in a triangulated category (Proposition 4.1.12).

Remark 6.5.4 If φ is the inclusion of a subgroup, then $\varphi^* = \mathrm{Res}^G_H$ preserves injective and projective objects. By Theorem 4.6.5 (iii) and (iv), the two morphisms in Lemma 6.5.2,

$$\mathrm{R}\left((\cdot)^H \circ \varphi^*\right) \to \mathrm{R}\left((\cdot)^H\right) \circ \varphi^*, \quad \mathrm{L}\left((\cdot)_H\right) \circ \varphi^* \to \mathrm{L}\left((\cdot)_H \circ \varphi^*\right)$$

are isomorphisms. The description of the canonical morphisms therefore simplifies further: take an injective resolution $M \to I := [I^0 \to I^1 \to \cdots]$. Then $\mathrm{Res}^G_H(M) \to \mathrm{Res}^G_H(I)$ is also an injective resolution, and the action on M of the morphism $\mathrm{R}\left((\cdot)^G\right) \to \mathrm{R}\left((\cdot)^H\right) \circ \mathrm{Res}^G_H$ is

$$I^G \to \left(\mathrm{Res}^G_H(I)\right)^H, \quad I^G := \left[I^{0,G} \to I^{1,G} \to \cdots\right], \text{ and similarly.}$$

In the homology case, take a projective resolution $P \to M$. The action on M of the morphism $\mathrm{L}\left((\cdot)_H\right) \circ \mathrm{Res}^G_H \to \mathrm{L}\left((\cdot)_G\right)$ is $\left(\mathrm{Res}^G_H(P)\right)_H \to P_G$.

Lemma 6.5.5 Let $F \xleftarrow{\psi} G \xleftarrow{\varphi} H$ be group homomorphisms, L an F-module, and $n \in \mathbb{Z}_{\geq 0}$. Then the composite $\mathrm{H}^n(F, L) \to \mathrm{H}^n(G, \psi^*L) \to \mathrm{H}^n(H, \varphi^*\psi^*L)$ equals $\mathrm{H}^n(F, L) \to \mathrm{H}^n(H, (\psi\varphi)^*L)$.

Likewise, the composite $\mathrm{H}_n(H, \varphi^*\psi^*L) \to \mathrm{H}_n(G, \psi^*L) \to \mathrm{H}_n(F, L)$ equals $\mathrm{H}_n(H, (\psi\varphi)^*L) \to \mathrm{H}_n(F, L)$.

Analogous equalities also hold at the level of derived categories.

Proof The assertion about H^n (or H_n) is an immediate application of the universal cohomological (or homological) δ-functor property and need only be checked for $n = 0$.

For the derived-category version, the assertion about H^n is replaced by

$$\left[\mathrm{R}((\cdot)^F) \to \mathrm{R}((\cdot)^G) \circ \psi^* \to \mathrm{R}((\cdot)^H) \circ \varphi^* \circ \psi^*\right] \overset{\text{composite}}{=\!=\!=} \left[\mathrm{R}((\cdot)^F) \to \mathrm{R}((\cdot)^H) \circ (\psi\varphi)^*\right].$$

For this, it suffices to check the property (6.5.3) for the left-hand side. This reduces to an interesting and straightforward exercise in the composition of 2-cells, which the reader is invited to try. □

We now generalize the canonical homomorphisms (6.5.1) further.

Definition 6.5.6 Let M be a G-module and N an H-module. Below, f is always a $\Bbbk$-module homomorphism and h is an arbitrary element of H.

- ◇ If $f : M \to N$ satisfies $f(\varphi(h)x) = hf(x)$ for $x \in M$, then f is called **φ-equivariant**.
- ◇ If $f : N \to M$ satisfies $f(hy) = \varphi(h)f(y)$ for $y \in N$, then f is called **φ-equivariant**.

Equivariance is also equivalent to saying that f gives an H-module homomorphism $\varphi^*M \to N$ or $N \to \varphi^*M$. Moreover, $\mathrm{id}_M : M \to \varphi^*M$ and $\mathrm{id}_M : \varphi^*M \to M$ are both φ-equivariant.

Definition 6.5.7 If $f : M \to N$ is φ-equivariant, define a family of canonical homomorphisms

$$\mathrm{H}^n(f) := [\mathrm{H}^n(G, M) \to \mathrm{H}^n(H, \varphi^*M) \to \mathrm{H}^n(H, N)] \text{ as the composite;}$$

if $f : N \to M$ is φ-equivariant, define a family of canonical homomorphisms

$$\mathrm{H}_n(f) := [\mathrm{H}_n(H, N) \to \mathrm{H}_n(H, \varphi^*M) \to \mathrm{H}_n(G, M)] \text{ as the composite.}$$

Notice that f and φ point in opposite directions in the cohomology case and in the same direction in the homology case. The homomorphisms in Definition 6.5.7 are compatible with long exact sequences. They include the functoriality of H^n and H_n (or the natural homomorphisms (6.5.1)) as the special cases $\varphi = \mathrm{id}_G$ (or $f = \mathrm{id}_M$). These homomorphisms also lift to the level of derived categories.

Equivariant module homomorphisms can also be composed in tandem with group homomorphisms; the definition is evident.

Proposition 6.5.8 (Functoriality with respect to equivariant homomorphisms) Consider group homomorphisms $F \xleftarrow{\psi} G \xleftarrow{\varphi} H$ and equivariant module homomorphisms $L \xrightarrow{g} M \xrightarrow{f} N$, where L is an F-module, M a G-module, and N an H-module. For every $n \in \mathbb{Z}$, we have $\mathrm{H}^n(fg) = \mathrm{H}^n(f)\,\mathrm{H}^n(g)$.

If instead the equivariant module homomorphisms point in the reverse direction, $L \xleftarrow{g} M \xleftarrow{f} N$, then $\mathrm{H}_n(gf) = \mathrm{H}_n(g)\,\mathrm{H}_n(f)$. Analogous composition equalities also hold at the level of derived categories.

Proof It suffices to discuss the cohomology case. Fix n. By Lemma 6.5.5, it is enough to prove that the diagram

$$\begin{tikzcd}
\mathrm{H}^n(F,L) \arrow[d] & & \\
\mathrm{H}^n(G,\psi^*L) \arrow[r] \arrow[d] & \mathrm{H}^n(G,M) \arrow[d] & \\
\mathrm{H}^n(H,\varphi^*\psi^*L) \arrow[r] & \mathrm{H}^n(H,\varphi^*M) \arrow[r] & \mathrm{H}^n(H,N)
\end{tikzcd}$$

commutes. But each square commutes by the functoriality of (6.5.1). The derived-category version is entirely similar. □

Example 6.5.9 (Conjugation) Let $H \lhd G$. Every $t \in G$ induces an automorphism of H, $a_t : h \mapsto t^{-1}ht$. Now let M be a G-module and, to simplify notation, continue to write M for $\mathrm{Res}_H^G M$. The homomorphism $c_t : M \xrightarrow{x \mapsto tx} M$ is equivariant with respect to the oppositely directed group homomorphism $H \xleftarrow{a_t} H$. The corresponding homomorphism

$$\mathrm{H}^n(c_t) : \mathrm{H}^n(H,M) \to \mathrm{H}^n(H,M),$$

decomposes by definition as $\mathrm{H}^n(H,M) \to \mathrm{H}^n(H,a_t^*M) \xrightarrow{c_t} \mathrm{H}^n(H,M)$; each part is a morphism between cohomological δ-functors.

- ◇ When $n = 0$, the first part is simply $\mathrm{id}_M : M^H \to a_t^*(M)^H = M^H$.
- ◇ When $n = 0$, the second part is $c_t : M^H \to M^H$, which depends only on the coset tH.

If an injective resolution $M \to I$ of the G-module is chosen, applying $c_t : I^{n,H} \to I^{n,H}$ termwise gives $\mathrm{H}^n(c_t)$. When $G = H$, the coset $tG = G$, so $\mathrm{H}^n(c_t) = \mathrm{id}$.

The homology case is similar, but a_t must be replaced by the group automorphism $b_t : h \mapsto tht^{-1}$, which points in the same direction as the homomorphism c_t; the reader is invited to verify this.

Finally, we explain how to define the canonical homomorphisms of Definition 6.5.6 explicitly using the standard complex (or standard chain complex) of Proposition 6.2.8 (or Proposition 6.2.9). This makes all the operations concrete.

Proposition 6.5.10 Consider a group homomorphism $\varphi : H \to G$. Let M be a G-module, N an H-module, and $n \in \mathbb{Z}_{\geq 0}$. Identify $\mathrm{H}^n(G,M)$ (or $\mathrm{H}_n(G,M)$) with the H^n (or H_n) of the standard complex $C(G,M)$ (or chain complex $C(G,M)$); do the same after replacing (G,M) by (H,N).

◇ Let the module homomorphism $f : M \to N$ be φ-equivariant. Then $\mathrm{H}^n(f)$ is induced by the morphism of complexes

$$\begin{aligned} C^m(G,M) &\longrightarrow C^m(H,N) \\ c &\longmapsto [(h_1,\dots,h_m) \mapsto f(c(\varphi(h_1),\dots,\varphi(h_m)))]. \end{aligned}$$

◇ Let the module homomorphism $f : N \to M$ be φ-equivariant. Then $\mathrm{H}_n(f)$ is induced by the morphism of chain complexes

$$\begin{aligned} C_m(H,N) &\longrightarrow C_m(G,M) \\ (h_1|\cdots|h_m)^\dagger \otimes y &\longmapsto (\varphi(h_1)|\cdots|\varphi(h_m))^\dagger \otimes f(y). \end{aligned}$$

In fact, these formulas give the morphism induced by f at the level of derived categories.

The result above also applies, with the same formulas, to the normalized standard complex $(\overline{C}^m(G,M))_m$ and normalized chain complex $(\overline{C}_m(G,M))_m$ introduced in Remark 6.2.13.

Proof The problem divides into two parts: first the case $f = \mathrm{id}_M$, which is an explicit description of the canonical homomorphisms (6.5.1), and second the case $\varphi = \mathrm{id}$. The latter is straightforward, so we discuss only the former.

Consider the cohomology case. Proposition 6.1.13 gives a projective resolution of G-modules $\mathsf{L}^G \to \Bbbk$ (or a projective resolution of H-modules $\mathsf{L}^H \to \Bbbk$). Now consider

$$\mathrm{Hom}^\bullet_G\left(\mathsf{L}^G, \cdot\right) \xrightarrow{\varphi^*} \mathrm{Hom}^\bullet_H\left(\varphi^*(\mathsf{L}^G), \varphi^*(\cdot)\right) \to \mathrm{Hom}^\bullet_H\left(\mathsf{L}^H, \varphi^*(\cdot)\right),$$

where the second part comes from the morphism $\mathsf{L}^H \to \varphi^*(\mathsf{L}^G)$ (see Lemma 6.4.4). Denote its composite by Θ. The asserted morphism of complexes $C(G,M) \to C(H,\varphi^*(M))$ corresponds to the action of Θ on the G-module M.

We now prove that this formula indeed gives the canonical homomorphism defined above. The first method returns to the definition (6.5.1), which comes from the universal cohomological δ-functor property. It suffices to show that $C(G,M) \to C(H,\varphi^* M)$ induces the embedding $M^G \to (\varphi^* M)^H$ on H^0 and is compatible with long exact sequences. Details are left to the reader.

The second method works in the derived category. The functor $\mathrm{Hom}^\bullet_H(\mathsf{L}^H, \cdot)$ preserves acyclic complexes, so it induces a functor $\mathsf{D}^+(H) \to \mathsf{D}^+(\Bbbk)$ at the level of derived categories, with the same notation. This functor can be identified with $\mathrm{RHom}_H(\Bbbk, \cdot)$. Thus verifying (6.5.3) is equivalent to verifying the following equality of vertical composites:

$$
\begin{tikzcd}
\mathsf{K}^+(G) \arrow[r, "\mathrm{Hom}^\bullet_G(\Bbbk,\cdot)"] \arrow[d] & \mathsf{K}^+(\Bbbk) \arrow[d] \arrow[dl, Rightarrow] \\
\mathsf{D}^+(G) \arrow[r, "\mathrm{Hom}^\bullet_G(\mathsf{L}^G,\cdot)"] \arrow[r, bend right=60, "\mathrm{Hom}^\bullet_H(\mathsf{L}^H,\varphi^*(\cdot))"'] \arrow[r, phantom, shift right=6, "\Theta\Downarrow"] & \mathsf{D}^+(\Bbbk)
\end{tikzcd}
=
\begin{tikzcd}
\mathsf{K}^+(G) \arrow[r, "\varphi^*"] \arrow[d] \arrow[rr, bend left=30, "\mathrm{Hom}^\bullet_G(\Bbbk,\cdot)"] \arrow[rr, phantom, shift left=5, "\Downarrow"] & \mathsf{K}^+(H) \arrow[r, "\mathrm{Hom}^\bullet_H(\Bbbk,\cdot)"] \arrow[d] \arrow[dl, Rightarrow, "\sim"'] & \mathsf{K}^+(\Bbbk) \arrow[d] \arrow[dl, Rightarrow] \\
\mathsf{D}^+(G) \arrow[r, "\varphi^*"'] & \mathsf{D}^+(H) \arrow[r, "\mathrm{Hom}^\bullet_H(\mathsf{L}^H,\cdot)"'] & \mathsf{D}^+(\Bbbk),
\end{tikzcd}
$$

The symbols $\swarrow\!\!\!\swarrow$ involving $\mathrm{Hom}^\bullet$ come from $\mathsf{L}^G \to \Bbbk$ and $\mathsf{L}^H \to \Bbbk$. The complex-level version of the morphism $(\cdot)^G \to (\cdot)^H \circ \varphi^*$ is readily identified with $\varphi^* : \mathrm{Hom}^\bullet_G(\Bbbk, \cdot) \to \mathrm{Hom}^\bullet_H(\varphi^*\Bbbk, \varphi^*(\cdot)) = \mathrm{Hom}^\bullet_H(\Bbbk, \varphi^*(\cdot))$. The desired equality therefore reduces to commutativity of the outer frame of the following diagram of complexes:

$$
\begin{tikzcd}
\mathrm{Hom}^\bullet_G(\Bbbk, \cdot) \arrow[r] \arrow[d, "\varphi^*"'] & \mathrm{Hom}^\bullet_G(\mathsf{L}^G, \cdot) \arrow[d, "\varphi^*"] \arrow[dr, "\Theta"] & \\
\mathrm{Hom}^\bullet_H(\Bbbk, \varphi^*(\cdot)) \arrow[r] \arrow[rr, bend right=25, "\text{induced by } \mathsf{L}^H \to \Bbbk"'] & \mathrm{Hom}^\bullet_H(\varphi^*\mathsf{L}^G, \varphi^*(\cdot)) \arrow[r] & \mathrm{Hom}^\bullet_H(\mathsf{L}^H, \varphi^*(\cdot)).
\end{tikzcd}
$$

It remains only to show that the curved lower part commutes, but this is exactly the content of Lemma 6.4.4. This proves the cohomology case. The homology case is similar.

Finally, the version for $\overline{C}(G, M)$ is obtained by replacing L^G and L^H in the proof with $\overline{\mathsf{L}}^G$ and $\overline{\mathsf{L}}^H$, respectively; everything else is unchanged. $\square$

Once the explicit formulas of Proposition 6.5.10 are available, the equalities $\mathrm{H}^n(fg) = \mathrm{H}^n(f)\,\mathrm{H}^n(g)$ and $\mathrm{H}_n(gf) = \mathrm{H}_n(g)\,\mathrm{H}_n(f)$ in Proposition 6.5.8, as well as their derived-category versions, are immediate at the level of complexes.

6.6 Group extensions revisited

To gain a more concrete understanding of the operations defined in §6.5, and to continue the preliminary discussion in §6.3 of group extensions (Definition 6.3.5), we return to the relationship between group extensions and H^2. We begin from the viewpoint of group theory. In what follows, consider a G-module M and an extension

$0 \to M \to E \xrightarrow{\pi} G \to 1$. Such extensions have natural pullback, pushout, and addition operations, following ideas similar to those of §4.5.

▷ Pullback Let $\varphi : H \to G$ be a group homomorphism. Form the fiber product in the category of groups

$$\varphi^* E := E \underset{G}{\times} H = \{(e, h) \in E \times H : \pi(e) = \varphi(h)\}.$$

The monomorphism $M \to \varphi^* E$, $x \mapsto (x, 1_H)$, determines a group extension $0 \to M \to \varphi^* E \xrightarrow{\text{projection}} H \to 1$. This gives a functor $\mathsf{Ext}(G, M) \to \mathsf{Ext}(H, M)$. If φ is the inclusion of a subgroup H, pullback along φ simply amounts to taking $\pi^{-1}(H)$.

▷ Pushout Let $\theta : M \to N$ be a homomorphism of G-modules. Form the fibered coproduct

$$\theta_* E := N \underset{M}{\sqcup} E = \frac{N \times E}{(y + \theta(x), e) \sim (y, xe)}, \qquad x \in M,\ y \in N,\ e \in E.$$

Write $[y, e]$ for the image of $(y, e) \in N \times E$ in it. Its multiplication is

$$[y_1, e_1][y_2, e_2] = [y_1 + \pi(e_1) y_2, e_1 e_2].$$

There are natural homomorphisms

$$N \xrightarrow{y \mapsto [y, 1_E]} \theta_* E \xrightarrow{[y,e] \mapsto \pi(e)} G,$$

which determine a group extension $\theta_* E$. This gives a functor $\mathsf{Ext}(G, M) \to \mathsf{Ext}(G, N)$.

▷ Taking the inverse As a special case, E can be pushed out along the G-module homomorphism $M \xrightarrow{x \mapsto -x} M$; denote the resulting extension by $-E$.

▷ Direct sum Given objects E_i of $\mathsf{Ext}(G_i, M_i)$ for $i = 1, 2$, there is an evident group extension

$$0 \to M_1 \oplus M_2 \to E_1 \times E_2 \to G_1 \times G_2 \to 1,$$

where $G_1 \times G_2$ acts on $M_1 \oplus M_2$ by $(g_1, g_2)(x_1, x_2) = (g_1 x_1, g_2 x_2)$. Denote this extension by $E_1 \oplus E_2$.

▷ Baer sum Now take objects E_1, E_2 of $\mathsf{Ext}(G, M)$ and form $E_1 \oplus E_2$. The addition map $M \oplus M \to M$ is a G-module homomorphism for the action of the diagonal subgroup G of $G \times G$. Thus first pull $E_1 \oplus E_2$ back along the diagonal inclusion $G \hookrightarrow G \times G$, and then push out along the addition homomorphism $M \oplus M \to M$. Denote the resulting object of $\mathsf{Ext}(G, M)$ by $E_1 \dot{+} E_2$.

Pullback and pushout can be combined into a single operation. Let $\varphi : H \to G$ be a group homomorphism and let the additive-group homomorphism $\theta : M \to N$ be φ-equivariant (Definition 6.5.6). There is a corresponding functor $\mathsf{Ext}(G,M) \to \mathsf{Ext}(H,N)$, defined by first pulling back along φ and then pushing out along $\theta : \varphi^* M \to N$.

The normalized standard complex $\overline{C}(G,M)$ has parallel operations. Given φ and θ as above, Proposition 6.5.10 explicitly gives a morphism of complexes $\overline{C}(G,M) \to \overline{C}(H,N)$; this defines the corresponding pullback, pushout, and inverse operations. The direct sum is defined by

$$\begin{aligned} \oplus : \overline{C}^n(G_1,M_1) \oplus \overline{C}^n(G_2,M_2) &\to \overline{C}^n(G_1 \times G_2, M_1 \oplus M_2), \\ f_1 \oplus f_2 : ((g_{1,i}, g_{2,i}))_{i=1}^n &\mapsto (f_1(g_{1,1},\dots,g_{1,n}), f_2(g_{2,1},\dots,g_{2,n}))\,. \end{aligned}$$

As n varies, this gives a morphism of complexes $\overline{C}(G_1,M_1) \oplus \overline{C}(G_2,M_2) \to \overline{C}(G_1 \times G_2, M_1 \oplus M_2)$. Following the same pattern, define the Baer sum on $\overline{C}(G,M)$ using direct sum, pullback, and pushout. The result is simply addition in $\overline{C}^n(G,M)$.

Proposition 6.6.1 Up to canonical isomorphism, pullback, pushout, taking inverses, direct sum, and Baer sum of group extensions correspond, under the equivalence of Theorem 6.3.9, to the respective operations on the complex $\tau^{\le 2}\overline{C}(G,M)$ (or the category $\tau^{\le 2}\overline{\mathsf{C}}(G,M)$).

Proof Given a group extension $0 \to M \to E \to G \to 1$, choose a normalized section $s : G \to E$ and the corresponding $f \in \overline{Z}^2(G,M)$ as in the proof of Theorem 6.3.9. Let $\varphi : H \to G$ be a group homomorphism. A normalized section of $\varphi^* E$ can be chosen as

$$s' : h \mapsto (s(\varphi(h)), h) \in E \underset{G}{\times} H.$$

Then $s'(h_1)s'(h_2) = f'(h_1,h_2)s'(h_1h_2)$, where

$$f'(h_1,h_2) := f(\varphi(h_1),\varphi(h_2)), \qquad h_1,h_2 \in H.$$

This proves the pullback case. For pushout along a G-module homomorphism $\theta : M \to N$, use the notation above and choose the normalized section of $\theta_* E$ to be

$$s'(g) := [0, s(g)] \in \theta_* E = N \underset{M}{\sqcup} E.$$

Computing in $\theta_* E$ gives

$$\begin{aligned} s'(g_1)s'(g_2) &= [0,s(g_1)][0,s(g_2)] = [0, s(g_1)s(g_2)] \\ &= [0, f(g_1,g_2)s(g_1g_2)] = [\theta f(g_1,g_2), s(g_1g_2)] \\ &= [\theta f(g_1,g_2), 1_E]s'(g_1g_2). \end{aligned}$$

Thus the f' corresponding to s' is precisely θf. This proves the pushout case.

The direct-sum case is checked in the same way, and the details are left to the reader. Taking inverses and forming Baer sums decompose into the operations above. $\square$

As a special case, up to canonical isomorphism, taking inverses and forming Baer sums of group extensions correspond respectively to inversion and addition in the group $\mathrm{H}^2(G, M)$.

Definition 6.6.2 If A is a central subgroup of E in a group extension $0 \to A \to E \to G \to 1$, the extension is called a **central extension**.

Given a central extension as above, suppose that for every central extension $0 \to B \to F \to G \to 1$, there is a unique group homomorphism $\varphi : E \to F$ making the following diagram commute:

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r] \arrow[d, "\varphi|_A"'] & E \arrow[r] \arrow[d, "\varphi"] & G \arrow[r] \arrow[d, equal] & 1 \\ 0 \arrow[r] & B \arrow[r] & F \arrow[r] & G \arrow[r] & 1, \end{tikzcd}$$

Then $0 \to A \to E \to G \to 1$ is called a **universal central extension** of G. If it exists, it is unique up to a unique isomorphism.

By definition, every inner automorphism of a central extension is trivial. For every additive group A, define the corresponding category of central extensions by

$$\mathsf{CExt}(G, A) := \mathsf{Ext}(G, A), \qquad \text{with } A \text{ regarded as a } G\text{-module with trivial action.}$$

Lemma 6.6.3 If a group G has a universal central extension $0 \to A \to E \xrightarrow{\pi} G \to 1$, then $G = G_{\mathrm{der}}$ and $E = E_{\mathrm{der}}$.

Proof Consider the split central extension $0 \to E_{\mathrm{ab}} \to E_{\mathrm{ab}} \times G \to G \to 1$ of G. There are homomorphisms $(0, \pi)$ and (q, π) from the extension E to $E_{\mathrm{ab}} \times G$, where $q : E \to E_{\mathrm{ab}}$ is the quotient homomorphism. The uniqueness part of the universal property implies $q = 0$, that is, $E = E_{\mathrm{der}}$. It follows at once that $G = G_{\mathrm{der}}$. $\square$

Using the pushout construction, it is not difficult to prove that φ in Definition 6.6.2 factors uniquely through $E \to (\varphi|_A)_* E$. Thus the universal property can be rewritten as follows: for every central extension $0 \to B \to F \to G \to 1$, there is a unique group homomorphism $\theta : A \to B$ and a unique isomorphism $\theta_* E \xrightarrow{\sim} F$ in $\mathsf{CExt}(G, B)$. Under the assumption $G = G_{\mathrm{der}}$, the category $\mathsf{CExt}(G, B)$ has no nontrivial automorphisms. The definition of a universal central extension can therefore be reformulated once more:

for every abelian group B, there is a bijection

$$\begin{aligned} \mathrm{Hom}_{\mathsf{Ab}}(A, B) &\xrightarrow{\ 1:1\ } \mathrm{Ob}(\mathsf{CExt}(G, B))/\simeq \\ \theta &\longmapsto \text{the isomorphism class of } \theta_* E. \end{aligned}$$

The left-hand side is a functor of B. The right-hand side likewise gives a functor $\kappa : \mathsf{Ab} \to \mathsf{Set}$, whose functoriality in B is given by pushout of extensions. Thus the existence of a universal central extension becomes the question whether κ is representable.

Theorem 6.6.4 A group G has a universal central extension if and only if $G = G_{\mathrm{der}}$. When this condition holds, the central subgroup A in a universal central extension is isomorphic to $\mathrm{H}_2(G, \mathbb{Z})$.

Proof By Lemma 6.6.3, it suffices to prove the "if" direction. Consider the functor κ above. Theorem 6.3.9 identifies $\kappa(B)$ with $\mathrm{H}^2(G, B)$, while Proposition 6.6.1 shows that pushout corresponds to the respective operation on $\mathrm{H}^2(G, \cdot)$. We know that $\mathrm{H}_1(G, \mathbb{Z}) \simeq G_{\mathrm{ab}} = 0$ (Example 6.2.11). Taking $n = 2$ in Proposition 6.2.7 therefore gives $\mathrm{H}^2(G, B) \simeq \mathrm{Hom}_{\mathsf{Ab}}(\mathrm{H}_2(G, \mathbb{Z}), B)$, naturally in B. Hence κ is representable. □

The group $\mathrm{H}_2(G, \mathbb{Z})$ is also called the **Schur multiplier** of G. Corollary 6.9.4 below is useful for computing it. In practice, the crucial matter is often an explicit description of the universal central extension; the existence criterion above is only the first step.

As another application of H^2, we derive a result in group theory.

Theorem 6.6.5 (Schur–Zassenhaus) Let $1 \to A \to E \xrightarrow{\pi} G \to 1$ be a group extension in which A and G are finite groups of relatively prime orders. Then the extension splits.

Proof The first step is to handle the case in which A is abelian. Give A the G-module structure arising from the group extension. It is enough to prove that $\mathrm{H}^2(G, A)$ is trivial. Since $|A|$ and $|G|$ are relatively prime, it suffices to prove the following equalities for $k = 2$:

$$k \geq 1 \implies |A| \cdot \mathrm{H}^k(G, A) = 0 = |G| \cdot \mathrm{H}^k(G, A).$$

The first equality is easy. For every $t \in \mathbb{Z}$, the endomorphism $A \to A$ given by multiplication by t induces on every $\mathrm{H}^k(G, A)$ the endomorphism that is likewise multiplication by t; this is a simple exercise in this chapter. The second equality is another exercise in this chapter; Corollary 6.8.6 below will give a complete, though somewhat circuitous, proof.

For general A, our next aim is to show that E has a subgroup H of order $|G|$. This will imply $H \cap A = \{1\}$ and make $\pi|_H : H \xrightarrow{\sim} G$ an isomorphism, whose inverse gives a splitting of the group extension. We argue by induction on $|E|$. We may assume without loss of generality that $|A| > 1$.

Choose a prime p dividing $|A|$, so that $p \nmid |G|$, and take a Sylow p-subgroup P of A. Then P is also a Sylow p-subgroup of E. Since $A \lhd E$ and all Sylow p-subgroups are conjugate, all of them are contained in A. Counting the Sylow p-subgroups in A and E, respectively, gives $(E : N_E(P)) = (A : N_A(P))$, where N_E and N_A denote normalizers; rearranging,

$$(N_E(P) : N_A(P)) = (E : A) = |G|.$$

We have $N_A(P) \lhd N_E(P)$. The corresponding group extension

$$1 \to N_A(P) \to N_E(P) \to N_E(P)/N_A(P) \to 1$$

still satisfies the hypotheses of the theorem. If $N_E(P) \neq E$, the induction hypothesis gives a subgroup H of $N_E(P)$ with $|H| = (N_E(P) : N_A(P)) = |G|$, proving the assertion.

We may therefore assume that $P \lhd E$, hence $P \lhd A$. Consider the group extension

$$1 \to A/P \to E/P \to G \to 1.$$

By induction, there is a subgroup H of E such that $H \supset P$ and $|H/P| = |G|$. A basic fact of group theory says that the center $Z := Z_P$ of P is nontrivial. We have $Z \lhd H$ and $(H/Z : P/Z) = (H : P) = |G|$. Applying induction again, there is a subgroup K of H such that $K \supset Z$ and $|K/Z| = |G|$.

We now have a group extension $1 \to Z \to K \to K/Z \to 1$. Since Z is a p-group, if $K \neq E$, induction gives a subgroup of K of order $|K/Z| = |G|$, proving the assertion. Henceforth assume $K = E$. Then necessarily $H = E$, so $(H : P) = |G|$ gives $P = A$; in this case $Z = Z_A \lhd E$.

Finally, consider the group extension $1 \to A/Z \to E/Z \to G \to 1$. We obtain a subgroup Q of E such that $Q \supset Z$ and $|Q/Z| = |G|$. If A is nonabelian, then $Z \neq A$ and $Q \neq E$; considering the group extension $1 \to Z \to Q \to Q/Z \to 1$ yields a subgroup of Q of order $|G|$. If A is abelian, the problem was already solved at the beginning of the proof. □

If A is abelian, all splittings of the extension E in Theorem 6.6.5 are conjugate by A. Indeed, specifying a splitting is equivalent to specifying an isomorphism of group extensions $A \rtimes G \simeq E$, so any two splittings $s, s' : G \to E$ differ by an automorphism of E. But Lemma 6.3.7 or Theorem 6.3.9 shows that the outer automorphism group of the extension is $\mathrm{H}^1(G, A)$, and the first step in the proof of Theorem 6.6.5 showed that this group is trivial.

The conjugacy property of splittings extends to general A, but its proof is more involved.

6.7 Examples: cyclic groups and free groups

For $m \in \mathbb{Z}_{\geq 1}$, in this section C_m denotes the cyclic group of order m with a chosen generator σ, and its group operation is written multiplicatively. We begin with a general construction.

Definition–Proposition 6.7.1 Let $\Bbbk$ be a commutative ring. If G is finite, define

$$\nu = \nu^G := \sum_{g \in G} g \in \Bbbk[G].$$

If G acts on $\Bbbk[G]$ by left multiplication, then $\Bbbk[G]^G = \Bbbk\nu$. If G is infinite, then $\Bbbk[G]^G = \{0\}$.

Proof For $e = \sum_{g \in G} a_g g \in \Bbbk[G]$, the condition $e \in \Bbbk[G]^G$ is equivalent to all the coefficients a_g being equal. $\square$

Similarly, if G is finite, then ν is also invariant under right multiplication by G. It lies in the center of $\Bbbk[G]$.

Convention 6.7.2 Let A be a G-module. For any $e, e' \in \Bbbk[G]$, write

$$A^{e=e'} := \{a \in A : ea = e'a\}.$$

We now consider group cohomology with coefficients in $\mathbb{Z}$. The group algebra $\mathbb{Z}[C_m]$ is commutative, and writing $1 := 1_{C_m} \in \mathbb{Z}[C_m]$ causes no confusion. Clearly $A^{C_m} = A^{\sigma=1}$ for every C_m-module A. In this case,

$$\nu = 1 + \sigma + \cdots + \sigma^{m-1} \in \mathbb{Z}[C_m],$$

and the equality $(\sigma - 1)\nu = \sigma^m - 1 = 0$ in $\mathbb{Z}[C_m]$ implies

$$\nu A \subset A^{\sigma=1}, \qquad (\sigma - 1)A \subset A^{\nu=0}.$$

We now show that, in the special case $A = \mathbb{Z}[C_m]$, both inclusions above are equalities.

Lemma 6.7.3 Write $\mathfrak{I} \subset \mathbb{Z}[C_m]$ for the augmentation ideal of Remark 6.1.11. We have

$$\begin{aligned}\nu\mathbb{Z}[C_m] &= \mathbb{Z}[C_m]^{\sigma=1} = \mathbb{Z}\nu,\\ (\sigma-1)\mathbb{Z}[C_m] &= \mathbb{Z}[C_m]^{\nu=0} = \mathfrak{I}.\end{aligned}$$

Proof For the first part, we have already shown that $\mathbb{Z}[C_m]^{\sigma=1} = \mathbb{Z}[C_m]^{C_m} = \mathbb{Z}\nu$. Since $\nu\sigma^k = \nu = \sigma^k\nu$ for every k, we have $\nu\mathbb{Z}[C_m] = \mathbb{Z}\nu$.

For the second part, take $e = \sum_{k\in\mathbb{Z}/m\mathbb{Z}} e_k\sigma^k$. Then $\nu e = (\sum_{k\in\mathbb{Z}/m\mathbb{Z}} e_k)\nu$ is zero if and only if $\sum_{k\in\mathbb{Z}/m\mathbb{Z}} e_k = 0$, which is equivalent to $e \in \mathfrak{I}$. This gives the second equality. Moreover, $\sigma^k - 1 = (\sigma-1)(1+\cdots+\sigma^{k-1})$ implies $\mathfrak{I} \subset (\sigma-1)\mathbb{Z}[C_m]$, so the first equality holds as well. □

Lemma 6.7.4 The trivial C_m-module $\mathbb{Z}$ has a free resolution

$$\cdots \xrightarrow{\nu} \mathbb{Z}[C_m] \xrightarrow{\sigma-1} \mathbb{Z}[C_m] \xrightarrow{\nu} \mathbb{Z}[C_m] \xrightarrow{\sigma-1} \mathbb{Z}[C_m] \to \mathbb{Z} \to 0,$$

where $\mathbb{Z}[C_m] \twoheadrightarrow \mathbb{Z}$ is the augmentation homomorphism and the remaining arrows repeat with period 2.

Proof Lemma 6.7.3 gives short exact sequences

$$\begin{gathered}0 \to \mathfrak{I} \to \mathbb{Z}[C_m] \xrightarrow{\nu} \mathbb{Z}\nu \to 0,\\ 0 \to \mathbb{Z}\nu \to \mathbb{Z}[C_m] \xrightarrow{\sigma-1} \mathfrak{I} \to 0.\end{gathered}$$

In addition, there is an isomorphism $\mathbb{Z}\nu \xrightarrow[\sim]{\nu\mapsto 1} \mathbb{Z}$. Splicing the two sequences alternately gives the asserted resolution. □

Proposition 6.7.5 Let A be a C_m-module and $n \in \mathbb{Z}_{\geq 0}$. There are canonical isomorphisms

$$\mathrm{H}^n(C_m, A) \simeq \begin{cases} A^{\sigma=1}, & n = 0,\\ A^{\nu=0}/(\sigma-1)A, & n > 0 \text{ odd},\\ A^{\sigma=1}/\nu A, & n > 0 \text{ even},\end{cases}$$

$$\mathrm{H}_n(C_m, A) \simeq \begin{cases} A/(\sigma-1)A, & n = 0,\\ A^{\sigma=1}/\nu A, & n > 0 \text{ odd},\\ A^{\nu=0}/(\sigma-1)A, & n > 0 \text{ even}.\end{cases}$$

Proof By Lemma 6.7.4, the cohomology and homology of A are computed, respectively, by the one-sided, 2-periodic cochain and chain complexes

$$0 \to \underbrace{A}_{\text{degree } 0} \xrightarrow{\sigma-1} \underbrace{A}_{\text{degree } 1} \xrightarrow{\nu} A \xrightarrow{\sigma-1} \cdots,$$

$$\cdots \xrightarrow{\sigma-1} A \xrightarrow{\nu} \underbrace{A}_{\text{degree } 1} \xrightarrow{\sigma-1} \underbrace{A}_{\text{degree } 0} \to 0.$$

□

We next study the homology and cohomology of free groups [51, Definition 4.8.2], with the infinite cyclic group $\mathbb{Z}$ as the simplest special case.

Write the free group on a set X as $G := \mathbf{F}(X)$. We first describe the structure of the augmentation ideal $\mathfrak{I} \subset \mathbb{Z}[G]$.

Lemma 6.7.6 Let X be a set and $G := \mathbf{F}(X)$. Then $\mathfrak{I}$ is the free left $\mathbb{Z}[G]$-module with basis $\{x - 1_G : x \in X\}$.

Proof Write $X - 1_G := \{x - 1_G : x \in X\} \subset \mathfrak{I}$. We first show that this set generates the left $\mathbb{Z}[G]$-module $\mathfrak{I}$. For every $x \in X$, write $G(x)$ (respectively, $G(x^{-1})$) for the set of elements of G whose reduced expression ends in x (respectively, x^{-1}). Then $G \smallsetminus \{1_G\}$ is the disjoint union of all the $G(x)$ and $G(x^{-1})$ as x ranges over X. We know that $\{h - 1_G : h \in G,\ h \neq 1_G\}$ is a $\mathbb{Z}$-basis of $\mathfrak{I}$, and

$$\begin{aligned} gx \in G(x) \implies\ & gx - 1_G = g(x - 1_G) + (g - 1_G), \\ & \ell(gx) = \ell(g) + 1, \\ gx^{-1} \in G(x^{-1}) \implies\ & gx^{-1} - 1_G = -(gx^{-1})(x - 1_G) + (g - 1_G), \\ & \ell(gx^{-1}) = \ell(g) + 1. \end{aligned}$$

Induction on length shows that $X - 1_G$ generates $\mathfrak{I}$.

To show that $X - 1_G$ is a basis, take an arbitrary G-module A and a map $\varphi : X - 1_G \to A$. We claim that there is a G-module homomorphism $\widetilde{\varphi} : \mathfrak{I} \to A$ making the following diagram commute:

$$\begin{tikzcd} X - 1_G \arrow[r, "\text{inclusion}"] \arrow[dr, "\varphi"'] & \mathfrak{I} \arrow[d, "\widetilde{\varphi}"] \\ & A. \end{tikzcd}$$

Since $X - 1_G$ generates $\mathfrak{I}$, the homomorphism $\widetilde{\varphi}$ in this diagram is automatically unique. It therefore suffices to prove the universal property of the free module.

Use the G-module structure on A to form the semidirect product $E := A \rtimes G$. Consider the map of sets $X \to E$ sending x to $(\varphi(x - 1_G), x)$. The universal property of the free group extends this map uniquely to a group homomorphism $\Phi : G \to E$. Looking at the second coordinate shows that the composite $G \xrightarrow{\Phi} E \xrightarrow{\text{projection}} G$ is id_G. Hence Proposition 6.3.3 gives a crossed homomorphism $\psi : G \to A$ such that $\Phi(g) = (\psi(g), g)$. By Proposition 6.3.4, the crossed homomorphism ψ corresponds to a G-module homomorphism $\widetilde{\varphi} : \mathfrak{I} \to A$ satisfying $\widetilde{\varphi}(g - 1_G) = \psi(g)$. Therefore

$$\widetilde{\varphi}(x - 1_G) = \psi(x) = \varphi(x - 1_G), \qquad x \in X.$$

This proves the assertion. ∎

Proposition 6.7.7 Let X be a set and $G := \mathbf{F}(X)$. The short exact sequence from the augmentation homomorphism, $0 \to \mathfrak{I} \to \mathbb{Z}[G] \to \mathbb{Z} \to 0$, is a free resolution of the trivial G-module $\mathbb{Z}$. Consequently:

(i) for every G-module A,

$$n \geq 2 \implies \mathrm{H}^n(G, A) = 0 = \mathrm{H}_n(G, A);$$

(ii) if A is an abelian group with trivial G-action, then $\mathrm{H}^1(G, A) \simeq A^X$ and $\mathrm{H}_1(G, A) \simeq A^{\oplus X}$.

Proof The first part and assertion (i) follow immediately from Lemma 6.7.6. Assertion (ii) can be read off from the free resolution and the description of $\mathfrak{I}$ in Lemma 6.7.6; the reader is invited to verify it. $\square$

Definition 6.7.8 For a group G, define the following elements of $\mathbb{Z}_{\geq 0} \sqcup \{\infty\}$:

$$\begin{aligned}
\mathrm{cd}(G) &:= \sup\{n \in \mathbb{Z}_{\geq 0} : \exists M \in \mathrm{Ob}(G\text{-}\mathsf{Mod}),\ \mathrm{H}^n(G, M) \neq 0\},\\
\mathrm{hd}(G) &:= \sup\{n \in \mathbb{Z}_{\geq 0} : \exists M \in \mathrm{Ob}(G\text{-}\mathsf{Mod}),\ \mathrm{H}_n(G, M) \neq 0\}.
\end{aligned}$$

The quantity $\mathrm{cd}(G)$ (respectively, $\mathrm{hd}(G)$) is called the **cohomological dimension** (respectively, **homological dimension**) of G.

By Proposition 6.2.3, the above $\mathrm{cd}(G)$ (respectively, $\mathrm{hd}(G)$) also equals the projective dimension (respectively, Tor-dimension) of the trivial G-module $\mathbb{Z}$. See Example 4.8.10 and Proposition 4.9.6.

Corollary 6.7.9 If G is a nontrivial free group, then $\mathrm{cd}(G) = \mathrm{hd}(G) = 1$.

For cohomology, the converse of the result above also holds: $\mathrm{cd}(G) = 1$ implies that G is a free group. This is known as the Stallings–Swan theorem [43].

6.8 Finite-index subgroups

In this section, with a commutative ring $\Bbbk$ fixed, we consider a group G and its subgroup H. Our aim is to discuss some special phenomena that occur when the index $(G : H)$ is finite.

Definition 6.4.1 defines the induced G-modules $\mathrm{Ind}_H^G(N)$ and $\mathrm{ind}_H^G(N)$ for every H-module N. Our first step in this section is to realize them as certain spaces of maps; at this stage, $(G : H)$ need not be finite.

For every $\Bbbk$-module V, give the set of maps $\{f : G \to V\}$ the following G-module structure. Its $\Bbbk$-module structure comes from pointwise operations on maps, while the G-action is

$$(gf)(x) = f(xg), \quad f : G \to V,\ g, x \in G.$$

For every map $f : G \to V$, write $\operatorname{Supp}(f) := \{x \in G : f(x) \neq 0\}$. To make the statement convenient, choose representatives $(g_i)_{i \in I}$ for the coset decompositions such that [4)]

$$G = \bigsqcup_{i \in I} g_i H = \bigsqcup_{i \in I} H g_i^{-1}.$$

Lemma 6.8.1 (Induced modules as spaces of maps) There are canonical isomorphisms of G-modules as follows.

$$\begin{aligned}
\operatorname{Ind}_H^G(N) &\xleftrightarrow{\ \sim\ } \left\{ f : G \to N \;\middle|\; \begin{array}{l} \forall h \in H, \forall x \in G, \\ f(hx) = h \cdot f(x) \end{array} \right\} \\
\varphi &\longmapsto \varphi|_{G \subset \Bbbk[G]} \\
\left[\textstyle\sum_g a_g g \mapsto \sum_g a_g f(g)\right] &\longleftarrow f, \\
\operatorname{ind}_H^G(N) &\xleftrightarrow{\ \sim\ } \left\{ f : G \to N \;\middle|\; \text{as above, but } H \backslash \operatorname{Supp}(f) \text{ is finite} \right\} \\
g_i \otimes y &\longmapsto \left[h g_i^{-1} \mapsto hy, \text{ equal to 0 outside } H g_i^{-1}\right] \\
\textstyle\sum_i g_i \otimes f(g_i^{-1}) &\longleftarrow f,
\end{aligned}$$

where $g \in G$, $y \in N$, and $i \in I$. The right-hand sides of the isomorphisms are given the G-module structure discussed above. These maps do not depend on the choice of representatives g_i.

Proof Verify directly that all the maps are well-defined, G-equivariant, mutually inverse, and independent of the choice of representatives. $\square$

The isomorphisms of Proposition 6.1.9 are easily rewritten using the spaces of

[4)] Editorial correction: the Chinese source indexes the representatives by $i \in H$, whereas the decompositions themselves are indexed by the set I.

maps above. For example,

$$\begin{aligned}
\mathrm{Hom}_H\left(\mathrm{Res}^G_H M, N\right) &\xrightarrow{\sim} \mathrm{Hom}_G\left(M, \mathrm{Ind}^G_H N\right) \\
\theta &\longmapsto [x \mapsto [g \mapsto \theta(gx)]], \\
\mathrm{Hom}_H\left(N, \mathrm{Res}^G_H M\right) &\xrightarrow{\sim} \mathrm{Hom}_G\left(\mathrm{ind}^G_H N, M\right) \\
\psi &\longmapsto \left[f \mapsto \textstyle\sum_{\text{cosets } gH} g\psi(f(g^{-1}))\right].
\end{aligned} \tag{6.8.1}$$

Corollary 6.8.2 For every H-module N, $\mathrm{ind}^G_H(N)$ embeds canonically as a G-submodule of $\mathrm{Ind}^G_H(N)$. If $(G:H)$ is finite, the two are equal.

Proof The G-modules on the right-hand sides of Lemma 6.8.1 have an evident inclusion. If $(G:H)$ is finite, every map $f: G \to N$ has $H \backslash \mathrm{Supp}(f)$ finite. □

We next define corestriction maps under the assumption that $(G:H)$ is finite. To simplify notation, for a G-module M we write the H-module $\mathrm{Res}^G_H(M)$ simply as M below.

- ◇ The family of canonical homomorphisms in (6.5.1), with φ taken to be the inclusion $H \hookrightarrow G$, is called **restriction** and is denoted by

$$\mathrm{res}^n : \mathrm{H}^n(G, M) \to \mathrm{H}^n(H, M), \quad n \in \mathbb{Z}_{\geq 0}.$$

- ◇ Likewise, in the homology case there is a canonical homomorphism in the opposite direction,

$$\mathrm{res}_n : \mathrm{H}_n(H, M) \to \mathrm{H}_n(G, M), \quad n \in \mathbb{Z}_{\geq 0}.$$

When res^n is described through the standard cochain complex by Proposition 6.5.10, its effect is simply to restrict a map $G^n \to M$ to H^n, whence the name. Section 6.5 has already lifted these operations to the derived category.

Definition 6.8.3 Assume $(G:H)$ is finite. For every G-module M, define the following $\Bbbk$-module homomorphisms:

$$\begin{aligned}
\nu^{G|H} : M^H &\longrightarrow M^G \\
x &\longmapsto \textstyle\sum_{\bar{g} \in G/H} gx, \\
\nu_{G|H} : M_G &\longrightarrow M_H \\
(x \text{ viewed in the quotient}) &\longmapsto \textstyle\sum_{\bar{g} \in H\backslash G} (gx \text{ viewed in the quotient}),
\end{aligned}$$

where $g \in G$ is any representative of the coset $\bar{g}$. Both homomorphisms are functorial in M.

In the first row of the definition, gx clearly depends only on the coset $\bar{g} = gH$. The second row requires a little explanation. For a given $x \in M$, the image of gx in M_H depends only on the coset Hg. Therefore, for every coset $t \in H\backslash G$, we can choose an arbitrary representative $\theta(t) \in G$ and sum the images of $\theta(t)x$ as t ranges over $H\backslash G$. If x is replaced by $g'x$, with $g' \in G$, then $(\theta(t)g')_{t\in H\backslash G}$ is still a family of representatives ranging over $H\backslash G$, and hence

$$\sum_t (\theta(t)g'x \text{ viewed in the quotient}) = \sum_t (\theta(t)x \text{ viewed in the quotient}).$$

Thus $\nu_{G|H}$ is well-defined at the level of M_G.

Definition–Proposition 6.8.4 Assume $(G : H)$ is finite and let M be a G-module.

(i) There is a family of canonical homomorphisms

$$\mathrm{cor}^n : \mathrm{H}^n(H, M) \to \mathrm{H}^n(G, M), \quad n \in \mathbb{Z}_{\geq 0},$$

compatible with long exact sequences and giving $\nu^{G|H} : M^H \to M^G$ in degree 0.

(ii) Likewise, there is a family of canonical homomorphisms compatible with long exact sequences,

$$\mathrm{cor}_n : \mathrm{H}_n(G, M) \to \mathrm{H}_n(H, M),$$

giving $\nu_{G|H} : M_G \to M_H$ in degree 0.

Both families of homomorphisms are called **corestriction**, since they point in the direction opposite to restriction[5].

Proof First consider cohomology. We give two explanations. For the first, Res^G_H : G-Mod $\to$ H-Mod is an exact functor preserving injective objects. Consequently, $M \mapsto \mathrm{H}^n(H, M)$ is the nth right-derived functor of $M \mapsto M^H$, for $n \in \mathbb{Z}_{\geq 0}$. Functoriality of $\nu^{G|H}$ therefore makes it induce a morphism of cohomological δ-functors $\mathrm{H}^n(H, \cdot) \to \mathrm{H}^n(G, \cdot)$.

The second explanation involves the derived category. The argument above is easily upgraded to the derived-category level, but here we give another viewpoint based on adjunction. First, we assert that the adjunction between ind^G_H and Res^G_H lifts to the RHom level, giving

$$\mathrm{RHom}_G\left(\mathrm{ind}^G_H(\Bbbk), M\right) \simeq \mathrm{RHom}_H(\Bbbk, M).$$

[5] With good reason, many sources call the corestriction homomorphism the "transfer." This book uses the term transfer homomorphism in a narrower sense; see the discussion below.

A similar idea appeared in the proof of Theorem 6.4.3. The key is to interpret ind_H^G as the change-of-rings functor $\Bbbk[H]\text{-}\mathsf{Mod} \to \Bbbk[G]\text{-}\mathsf{Mod}$ induced by $\Bbbk[H] \to \Bbbk[G]$, note that it is exact (Proposition 6.4.2), and then apply Example 4.12.18.

Furthermore, the assumption that $(G : H)$ is finite, together with Corollary 6.8.2, gives

$$\mathrm{RHom}_G\left(\mathrm{ind}_H^G(\Bbbk), M\right) \simeq \mathrm{RHom}_G\left(\mathrm{Ind}_H^G(\Bbbk), M\right) \to \mathrm{RHom}_G(\Bbbk, M),$$

where the last part comes from $\Bbbk \to \mathrm{Ind}_H^G(\Bbbk)$. This is the unit of the adjoint pair $(\mathrm{Res}_H^G, \mathrm{Ind}_H^G)$ evaluated at $\Bbbk$, and it can also be described more concretely using Lemma 6.8.1: an element $t \in \Bbbk$ is sent to the corresponding constant map $G \to \Bbbk$.

This yields a canonical morphism $\mathrm{RHom}_H(\Bbbk, M) \to \mathrm{RHom}_G(\Bbbk, M)$. Taking H^n on both sides gives the desired result. Determining its behavior for $n = 0$ amounts to replacing RHom by Hom in the operations above. More explicitly, if $\mathrm{ind}_H^G \subset \mathrm{Ind}_H^G$ is realized as spaces of maps and the adjunction isomorphisms are described using (6.8.1), then

$$\begin{array}{ccccccc}
\mathrm{Hom}_H(\Bbbk, M) & \xrightarrow{\sim} & \mathrm{Hom}_G\left(\mathrm{ind}_H^G \Bbbk, M\right) & = & \mathrm{Hom}_G\left(\mathrm{Ind}_H^G \Bbbk, M\right) & \longrightarrow & \mathrm{Hom}_G(\Bbbk, M) \\
[1 \mapsto x] & & \longmapsto & & [f \mapsto \sum_{gH} f(g^{-1})gx] & \longmapsto & [1 \mapsto \sum_{gH} gx].
\end{array}$$

[6] The composite above is plainly $\nu^{G|H} : M^H \to M^G$. This proves the assertion. □

Proposition 6.8.5 Assume $(G : H)$ is finite. For every G-module M and $n \in \mathbb{Z}_{\geq 0}$, the composites of homomorphisms

$$\mathrm{H}^n(G, M) \xrightarrow{\mathrm{res}^n} \mathrm{H}^n(H, M) \xrightarrow{\mathrm{cor}^n} \mathrm{H}^n(G, M),$$
$$\mathrm{H}_n(G, M) \xrightarrow{\mathrm{cor}_n} \mathrm{H}_n(H, M) \xrightarrow{\mathrm{res}_n} \mathrm{H}_n(G, M)$$

are both the endomorphism given by multiplication by $(G : H)$.

Proof First consider the case $n = 0$. In cohomology the map is $M^G \hookrightarrow M^H \xrightarrow{\nu^{G|H}} M^G$, which plainly sends $x \in M^G$ to $(G : H)x$. In homology the map is $M_G \xrightarrow{\nu_{G|H}} M_H \twoheadrightarrow M_G$; a direct check gives the same conclusion.

Now suppose $n \geq 1$. In the cohomology case, choose an embedding $M \hookrightarrow I$ with I an injective G-module; then I is also an injective H-module. Write $L := I/M$. Functoriality of restriction and corestriction, together with the long exact sequences,

[6] Editorial correction: the Chinese source writes Res_M^G and $\mathrm{Hom}_H(\mathrm{Ind}_H^G \Bbbk, M)$; the adjunction and module types require Res_H^G and $\mathrm{Hom}_G(\mathrm{Ind}_H^G \Bbbk, M)$, as displayed here.

gives a commutative diagram with exact columns:

$$\begin{tikzcd} \mathrm{H}^{n-1}(G,L) \arrow[r, "\mathrm{res}^{n-1}"] \arrow[d, two heads] & \mathrm{H}^{n-1}(H,L) \arrow[r, "\mathrm{cor}^{n-1}"] \arrow[d, two heads] & \mathrm{H}^{n-1}(G,L) \arrow[d, two heads] \\ \mathrm{H}^{n}(G,M) \arrow[r, "\mathrm{res}^{n}"] & \mathrm{H}^{n}(H,M) \arrow[r, "\mathrm{cor}^{n}"] & \mathrm{H}^{n}(G,M), \end{tikzcd}$$

By induction, the composite in the first row is multiplication by $(G:H)$. Since all the vertical arrows are surjective, the same is true of the composite in the second row. The homology argument is similar: take an epimorphism $P \twoheadrightarrow M$ with P a projective G-module.

Notice that the case $n = 0$ also lets us conclude at the derived-category level that the composite $\mathrm{RHom}_G(\Bbbk, M) \to \mathrm{RHom}_H(\Bbbk, M) \to \mathrm{RHom}_G(\Bbbk, M)$ is $(G:H)\mathrm{id}$. The homology case is analogous. □

Corollary 6.8.6 Let $|G| = m \in \mathbb{Z}_{\geq 1}$. For every G-module M, we have

$$n \geq 1 \implies m\,\mathrm{H}^n(G,M) = m\,\mathrm{H}_n(G,M) = \{0\}.$$

Proof In Proposition 6.8.5, take $H = \{1\}$ and use the elementary fact that if $n \geq 1$, then $\mathrm{H}^n(\{1\}, M)$ and $\mathrm{H}_n(\{1\}, M)$ both vanish. □

For the rest of the section, take $\Bbbk = \mathbb{Z}$ and regard $\mathbb{Z}$ as a trivial G-module. Example 6.2.11 gives canonical isomorphisms $\mathrm{H}_1(G,\mathbb{Z}) \simeq G_{\mathrm{ab}}$ and $\mathrm{H}_1(H,\mathbb{Z}) \simeq H_{\mathrm{ab}}$. If $(G:H)$ is finite, then $\mathrm{cor}_1 : \mathrm{H}_1(G,\mathbb{Z}) \to \mathrm{H}_1(H,\mathbb{Z})$ corresponds to the group homomorphism

$$\mathrm{Ver}_{G|H} : G_{\mathrm{ab}} \to H_{\mathrm{ab}},$$

called the **transfer homomorphism** from G to H (German: *die Verlagerung*).

The key is to give a group-theoretic description of $\mathrm{Ver}_{G|H}$. Recall that G acts on the right of $H\backslash G$ by right multiplication.

Proposition 6.8.7 Assume $(G:H)$ is finite. Choose a representative in G for every coset in $H\backslash G$, expressed as a map $\theta : H\backslash G \to G$. For every $s \in G$ and $t \in H\backslash G$, there is a unique $h_{t,s} \in H$ such that

$$\theta(t)s = h_{t,s}\theta(ts).$$

The transfer homomorphism defined above can be described by

$$\mathrm{Ver}_{G|H}(\underbrace{s \text{ as a class}}_{\in G_{\mathrm{ab}}}) = \prod_{t \in H\backslash G} \underbrace{h_{t,s} \text{ as a class}}_{\in H_{\mathrm{ab}}}.$$

Proof Take the short exact sequence of G-modules $0 \to \mathfrak{I}_G \to \mathbb{Z}[G] \to \mathbb{Z} \to 0$, where $\mathfrak{I}_G$ is the augmentation ideal of $\mathbb{Z}[G]$ (Example 6.2.6). The inclusion $\mathbb{Z}[H] \subset \mathbb{Z}[G]$ gives an H-submodule $\mathfrak{I}_H \subset \mathfrak{I}_G$. Since cor_n is compatible with long exact sequences, and $\mathbb{Z}[G]$ is projective both as a G-module and as an H-module, we obtain the commutative diagram

$$\begin{tikzcd}
G_{\mathrm{ab}} \arrow[r, "\sim"'] \arrow[dd, "\mathrm{Ver}_{G|H}"'] & \mathrm{H}_1(G,\mathbb{Z}) \arrow[r, "\sim"] \arrow[d, "\mathrm{cor}_1"'] & \mathrm{H}_0(G,\mathfrak{I}_G) \arrow[r, equal] \arrow[d, "\mathrm{cor}_0"] & \mathfrak{I}_G/\mathfrak{I}_G^2 \arrow[d, "\nu_{G|H}"] \\
 & \mathrm{H}_1(H,\mathbb{Z}) \arrow[r, hook] \arrow[d, equal] & \mathrm{H}_0(H,\mathfrak{I}_G) \arrow[r, equal] & \mathfrak{I}_G/\mathfrak{I}_H\mathfrak{I}_G \\
H_{\mathrm{ab}} \arrow[r, "\sim"] & \mathrm{H}_1(H,\mathbb{Z}) \arrow[r, "\sim"] & \mathrm{H}_0(H,\mathfrak{I}_H) \arrow[r, equal] \arrow[u] & \mathfrak{I}_H/\mathfrak{I}_H^2 \arrow[u]
\end{tikzcd}$$

We check the required equality in $\mathfrak{I}_G/\mathfrak{I}_H\mathfrak{I}_G$. The end of Example 6.2.11 shows that the image of $s \in G$ in G_{ab} corresponds through the first row to the image of $1_G - s$ in $\mathfrak{I}_G/\mathfrak{I}_G^2$. Applying $\nu_{G|H}$ to it gives

$$\begin{aligned}
\sum_{t \in H\backslash G} \theta(t)(1_G - s) &= \sum_t (\theta(t) - \theta(t)s) = \sum_t (\theta(t) - h_{t,s}\theta(ts)) \\
&= \sum_t \theta(t) - \sum_t h_{t,s}\theta(ts) \\
&= \sum_t \theta(ts) - \sum_t h_{t,s}\theta(ts) \ \in \mathfrak{I}_G
\end{aligned}$$

in $\mathfrak{I}_G/\mathfrak{I}_H\mathfrak{I}_G$. The last equality holds because, as t varies, both t and ts range over $H\backslash G$. The final expression is also equal to $\sum_t (1_G - h_{t,s})\theta(ts)$.

Finally, observe that $(1_G - h_{t,s})\theta(ts) \equiv 1_G - h_{t,s} \pmod{\mathfrak{I}_H\mathfrak{I}_G}$, while the right-hand side corresponds to the image of $h_{t,s} \bmod \mathfrak{I}_H$ in $\mathfrak{I}_H/\mathfrak{I}_H^2$ through the third row. This proves the assertion. $\square$

The map described above, $s \bmod G_{\mathrm{der}} \mapsto \prod_t h_{t,s} \bmod H_{\mathrm{der}}$, is a classical construction in group theory. Having identified it with cor_1, we have also proved that it is independent of the choice of θ.

6.9 The Lyndon–Hochschild–Serre spectral sequence

Throughout this section, fix a commutative ring $\Bbbk$, a group G, and a normal subgroup $H \lhd G$. Recall the restriction functor $\mathrm{Res}^G_H : G\text{-}\mathsf{Mod} \to H\text{-}\mathsf{Mod}$ and the inflation functor $\mathrm{Infl}^G_{G/H} : G/H\text{-}\mathsf{Mod} \to G\text{-}\mathsf{Mod}$ from Definition 6.1.8. For every G-module M, introduce the notation

$$M^{H \lhd G} := (\mathrm{Res}^G_H M)^H, \qquad M_{H \lhd G} := (\mathrm{Res}^G_H M)_H.$$

Notice that $M^{H\lhd G}$ has an evident left G/H-action, $(gH)\cdot x := gx$. Similarly, $M_{H\lhd G}$ has an evident left G/H-action. This gives two functors

$$(\cdot)^{H\lhd G},\ (\cdot)_{H\lhd G} : G\text{-}\mathsf{Mod} \to G/H\text{-}\mathsf{Mod}.$$

Lemma 6.9.1 The functor $(\cdot)^{H\lhd G}$ is right adjoint to $\mathrm{Infl}^G_{G/H}$, while $(\cdot)_{H\lhd G}$ is left adjoint to it.

Proof This is a straightforward extension of Proposition 6.2.2; verification is left to the reader. □

There are evident commutative diagrams of functors

$$\begin{tikzcd} G\text{-}\mathsf{Mod} \arrow[rr, "(\cdot)^{H\lhd G}"] \arrow[dr, "(\cdot)^G"'] & & G/H\text{-}\mathsf{Mod} \arrow[dl, "(\cdot)^{G/H}"] \\ & \Bbbk\text{-}\mathsf{Mod} & \end{tikzcd} \qquad \begin{tikzcd} G\text{-}\mathsf{Mod} \arrow[rr, "(\cdot)_{H\lhd G}"] \arrow[dr, "(\cdot)_G"'] & & G/H\text{-}\mathsf{Mod}. \arrow[dl, "(\cdot)_{G/H}"] \\ & \Bbbk\text{-}\mathsf{Mod} & \end{tikzcd}$$

The functor $(\cdot)^{H\lhd G}$ is left exact and preserves injective objects, since it has the exact left adjoint $\mathrm{Infl}^G_{G/H}$. Similarly, $(\cdot)_{H\lhd G}$ is a right-exact functor that preserves projective objects. Theorem 4.8.8 on deriving composite functors therefore gives canonical isomorphisms in the derived category $\mathsf{D}(\Bbbk\text{-}\mathsf{Mod})$ (with appropriate parentheses omitted):

$$\begin{aligned} \mathrm{R}(\cdot)^G &\xrightarrow{\sim} \mathrm{R}(\cdot)^{G/H} \circ \mathrm{R}(\cdot)^{H\lhd G}, \\ \mathrm{L}(\cdot)_G &\xleftarrow{\sim} \mathrm{L}(\cdot)_{G/H} \circ \mathrm{L}(\cdot)_{H\lhd G}. \end{aligned} \tag{6.9.1}$$

The derived functors $\mathrm{R}(\cdot)^{H\lhd G}$ and $\mathrm{L}(\cdot)_{H\lhd G}$ in these formulas are not new objects. For each of the commutative diagrams

$$\begin{tikzcd} G\text{-}\mathsf{Mod} \arrow[r, "(\cdot)^{H\lhd G}"] \arrow[d, "\mathrm{Res}^G_H"'] & G/H\text{-}\mathsf{Mod} \arrow[d, "\text{forgetful}=\mathrm{Res}^{G/H}_{\{1\}}"] \\ H\text{-}\mathsf{Mod} \arrow[r, "(\cdot)^H"'] & \Bbbk\text{-}\mathsf{Mod} \end{tikzcd} \qquad \begin{tikzcd} G\text{-}\mathsf{Mod} \arrow[r, "(\cdot)_{H\lhd G}"] \arrow[d, "\mathrm{Res}^G_H"'] & G/H\text{-}\mathsf{Mod} \arrow[d, "\text{forgetful}"] \\ H\text{-}\mathsf{Mod} \arrow[r, "(\cdot)_H"'] & \Bbbk\text{-}\mathsf{Mod}, \end{tikzcd}$$

derive both composites according to Theorem 4.8.8. Since Res^G_H is exact and preserves both injective and projective objects (Proposition 6.4.2), while the forgetful functor is exact, that theorem gives

$$\begin{aligned} \text{forgetful} \circ \mathrm{R}(\cdot)^{H\lhd G} &\xleftarrow{\sim} \mathrm{R}(\text{composite functor}) \xrightarrow{\sim} \mathrm{R}(\cdot)^H \circ \mathrm{Res}^G_H, \\ \text{forgetful} \circ \mathrm{L}(\cdot)_{H\lhd G} &\xrightarrow{\sim} \mathrm{L}(\text{composite functor}) \xleftarrow{\sim} \mathrm{L}(\cdot)_H \circ \mathrm{Res}^G_H. \end{aligned}$$

In other words, after applying the forgetful functor, the right-derived functor $\mathrm{R}(\cdot)^{H\lhd G}$ is the same as $\mathrm{R}(\cdot)^H$, and the left-derived functor $\mathrm{L}(\cdot)_{H\lhd G}$ is the same as $\mathrm{L}(\cdot)_H$. It is therefore natural to write their cohomology and homology as

$$\begin{aligned} \mathrm{R}^n(\cdot)^{H \lhd G} &= \mathrm{H}^n(H, \cdot) \quad \text{with its } G/H\text{-action},\\ \mathrm{L}_n(\cdot)_{H \lhd G} &= \mathrm{H}_n(H, \cdot) \quad \text{with its } G/H\text{-action}. \end{aligned}$$

Notice that we have begun to omit the restriction functor Res from the notation.

These actions are easy to describe. For cohomology, let M be a G-module and $n \in \mathbb{Z}_{\geq 0}$. There are at least three viewpoints on the G/H-action:

1. Take an injective resolution $0 \to M \to I^0 \to \cdots$. It is also an injective resolution of H-modules. The group G/H acts naturally on the complex $\cdots \to I^{n,H} \to I^{n+1,H} \to \cdots$, and hence on $\mathrm{H}^n(H, M)$. This description comes directly from the abstract construction above.

2. The group G/H acts on the functor $(\cdot)^H : G\text{-}\mathsf{Mod} \to \Bbbk\text{-}\mathsf{Mod}$. The universal property of universal cohomological δ-functors then makes it act on $\mathrm{H}^n(H, M)$, compatibly with long exact sequences.

3. In Example 6.5.9, every $t \in G/H$ acts on $\mathrm{H}^n(H, M)$ through $\mathrm{H}^n(c_t)$.

The reader is invited to verify that these three descriptions agree. The key is to reduce their comparison to the case $n = 0$.

The next important step is to extract more concrete information from (6.9.1). This involves the cohomological and homological bigraded spectral sequences of Definition 5.2.8, together with the following preparation.

Lemma 6.9.2 For every G-module M and every n, the canonical homomorphism $\mathrm{H}^n(G, M) \to \mathrm{H}^n(H, M)$ (see (6.5.1)) factors through $\mathrm{H}^n(H, M)^{G/H}$. The canonical homomorphism $\mathrm{H}_n(H, M) \to \mathrm{H}_n(G, M)$ factors through $\mathrm{H}_n(H, M)_{G/H}$.

Proof We discuss cohomology. Take an injective resolution $M \to I$, with $I = [I^0 \to I^1 \to \cdots]$, and write $I^H := [I^{0,H} \to \cdots]$. Since I is also an injective resolution of M as an H-module, the inclusion $I^G \hookrightarrow I^H$ induces $\mathrm{H}^n(G, M) \to \mathrm{H}^n(H, M)$.

As expected, Remark 6.5.4 ensures that this map is precisely the canonical homomorphism in question. Now compare it with the discussion of the G/H-action above. □

Theorem 6.9.3 (R. Lyndon, G. Hochschild–J.-P. Serre) Let M be a G-module and $H \lhd G$.

(i) There are first-quadrant cohomological and homological spectral sequences

$$\begin{aligned} E_2^{p,q} &\simeq \mathrm{H}^p\left(G/H, \mathrm{H}^q(H, M)\right) \Rightarrow \mathrm{H}^{p+q}(G, M),\\ E^2_{p,q} &\simeq \mathrm{H}_p\left(G/H, \mathrm{H}_q(H, M)\right) \Rightarrow \mathrm{H}_{p+q}(G, M). \end{aligned}$$

(ii) Let $n \in \mathbb{Z}_{\geq 1}$ and suppose that $\mathrm{H}^i(H, M) = 0$ for all $1 \leq i < n$ in the cohomological case, or $\mathrm{H}_i(H, M) = 0$ for all $1 \leq i < n$ in the homological case. Then there are, respectively, exact sequences

$$\begin{aligned} 0 \to \mathrm{H}^n(G/H, M^H) \to \mathrm{H}^n(G, M) \to \mathrm{H}^n(H, M)^{G/H} \\ \to \mathrm{H}^{n+1}(G/H, M^H) \to \mathrm{H}^{n+1}(G, M), \end{aligned}$$

and

$$\begin{aligned} \mathrm{H}_{n+1}(G, M) \to \mathrm{H}_{n+1}(G/H, M_H) \\ \to \mathrm{H}_n(H, M)_{G/H} \to \mathrm{H}_n(G, M) \to \mathrm{H}_n(G/H, M_H) \to 0. \end{aligned}$$

Notice that for $n = 1$ the hypothesis is vacuous.

(iii) In the cohomological exact sequence above, every morphism other than $\mathrm{H}^n(H, M)^{G/H} \to \mathrm{H}^{n+1}(G/H, M^H)$ comes from the equivariant homomorphism $M^H \hookrightarrow M$ (relative to $G/H \twoheadleftarrow G$; see Definition 6.5.7), the homomorphism $M \xrightarrow{\mathrm{id}} M$ (relative to $G \hookleftarrow H$), and the factorization in Lemma 6.9.2. The homological exact sequence is similar.

Proof For (i), it suffices to express (6.9.1) concretely through the Grothendieck spectral sequence of Theorem 5.6.6.

We now discuss (ii). If $n = 1$, the hypothesis is vacuous and the corresponding exact sequence is precisely the low-degree exact sequence in Theorem 5.6.6; its underlying principle is the edge computation in Remark 5.2.13. We next review and extend that argument to general n. The idea is to exploit the zero terms in the spectral sequence.

It suffices to discuss the cohomological case. The hypothesis shows that the spectral sequence in (i) satisfies

$$1 \leq q < n \implies E_2^{p,q} = E_3^{p,q} = \cdots = E_\infty^{p,q} = 0.$$

Recall that the direction of d_r is $d_r^{p,q} : E_r^{p,q} \to E_r^{p+r, q-r+1}$ and that $E_{r+1}^{p,q}$ is the cohomology of $E_r^{p,q}$ with respect to d_r. Every d leaving $E_{n+1}^{n+1,0}$ (or entering $E_{n+1}^{0,n}$) is zero. We therefore obtain the horizontal exact sequence in the diagram

$$\begin{tikzcd}
 & E_\infty^{0,n} & & & E_\infty^{n+1,0} & \\
 & \wr & & & \wr & \\
 & \vdots & & & \vdots & \\
 & \wr & & & \wr & \\
0 \arrow[r] & E_{n+2}^{0,n} \arrow[r] & E_{n+1}^{0,n} \arrow[r, "d_{n+1}^{0,n}"] & E_{n+1}^{n+1,0} \arrow[r] & E_{n+2}^{n+1,0} \arrow[r] & 0. \\
 & & \wr & \wr & & \\
 & & \vdots & \vdots & & \\
 & & \wr & \wr & & \\
 & \mathrm{H}^n(H, M)^{G/H} \arrow[r, "\sim"] & E_2^{0,n} & E_2^{n+1,0} & \mathrm{H}^{n+1}(G/H, M^H) \arrow[l, "\sim"'] &
\end{tikzcd}$$

The reason for the vertical isomorphisms is similar: once every d entering or leaving $E_r^{p,q}$ is zero, there is an isomorphism $E_r^{p,q} \simeq E_{r+1}^{p,q}$. In the same way one obtains

$$\mathrm{H}^n(G/H, M^H) \simeq E_2^{n,0} \simeq \cdots \simeq E_\infty^{n,0}.$$

Next, recall that the target $\mathrm{H}^{p+q}(G, M)$ of the spectral sequence has a finite decreasing filtration $\cdots \supset \mathrm{F}^k \supset \mathrm{F}^{k+1} \supset \cdots$, with $E_\infty^{p,q} \simeq \mathrm{gr}^p \mathrm{H}^{p+q}(G, M)$. At the edge of the first quadrant, this gives

$$\begin{aligned} E_\infty^{n+1,0} &\simeq \mathrm{F}^{n+1} \subset \mathrm{H}^{n+1}(G, M), \\ E_\infty^{0,n} &\simeq \mathrm{H}^n(G, M)/\mathrm{F}^1. \end{aligned}$$

How can one determine $\mathrm{F}^1 \subset \mathrm{H}^n(G, M)$? On the line $p + q = n$, only $(n, 0)$ and $(0, n)$ can possibly satisfy $E_\infty^{p,q} \neq 0$. Thus the filtration on $\mathrm{H}^n(G, M)$ has the form

$$0 = \mathrm{F}^{n+1} \underset{E_\infty^{n,0}}{\underbrace{\subseteq}} \mathrm{F}^n = \cdots = \mathrm{F}^1 \underset{E_\infty^{0,n}}{\underbrace{\subseteq}} \mathrm{F}^0 = \mathrm{H}^n(G, M).$$

Combining all the isomorphisms and exact sequences above gives the desired result.

Assertion (iii) is as expected, but the following argument is somewhat long and may be skipped. For brevity, we discuss only $\mathrm{H}^n(G, M) \to \mathrm{H}^n(H, M)^{G/H}$; the other cases are similar or easy. Recall the construction in §5.6. Take an injective resolution $M \to I$, with I a complex of G-modules, and then take a Cartan–Eilenberg resolution J of the complex of G/H-modules I^H (Theorem 3.11.10). Regarding the complexes I^G and so forth as double complexes concentrated on the horizontal axis gives a commutative diagram of double complexes of $\Bbbk$-modules

$$\begin{tikzcd} & J^{G/H} \arrow[r, hook] & J \\ I^G \arrow[r, equal] & (I^H)^{G/H} \arrow[u] \arrow[r, hook] & I^H. \arrow[u] \end{tikzcd} \tag{6.9.2}$$

Following the convention of Proposition 5.6.1, write $\mathrm{H}_{\mathrm{II}}\,\mathrm{H}_{\mathrm{I}}(\cdot)$ for the operation of first taking horizontal and then vertical cohomology of a double complex. Note the following points.

- ◇ Since the horizontal cohomology $\mathrm{H}_{\mathrm{I}}(J)$ of the Cartan–Eilenberg resolution gives an injective resolution of $\mathrm{H}^p(I^H)$ in column p ($p \in \mathbb{Z}$), the morphism $I^H \to J$ induces the isomorphism

$$\mathrm{H}^n(H, M) = \mathrm{H}^n(I^H) = \mathrm{H}_{\mathrm{II}}\,\mathrm{H}_{\mathrm{I}}(I^H)^{n,0} \xrightarrow{\sim} \mathrm{H}_{\mathrm{II}}\,\mathrm{H}_{\mathrm{I}}(J)^{n,0}.$$

◇ Remark 3.11.11 gives $\mathrm{H_I}(J^{G/H}) \simeq \mathrm{H_I}(J)^{G/H}$. Since $\mathrm{H_I}(J)$ vertically resolves each $\mathrm{H}^p(H, M)$, while $(\cdot)^{G/H}$ is left exact, applying $\mathrm{H_{II}}\,\mathrm{H_I}(\cdot)^{n,0}$ to $J^{G/H} \hookrightarrow J$ corresponds to

$$\text{the inclusion homomorphism} \qquad \mathrm{H}^n(H, M)^{G/H} \hookrightarrow \mathrm{H}^n(H, M).$$

◇ We have shown that $I^G \hookrightarrow I^H$ induces on H^n (or, equivalently, on $\mathrm{H_{II}}\,\mathrm{H_I}(\cdot)^{n,0}$) the canonical homomorphism $\mathrm{H}^n(G, M) \to \mathrm{H}^n(H, M)$.

Filter the first-quadrant double complex $J^{G/H}$ decreasingly by the vertical coordinate (the pth step is the part with vertical coordinate $\geq p$). The corresponding spectral sequence is $(E_r^{p,q})_{p,q}$, where $r = 0, 1, \ldots$. Write $\ker(d^n)_{n,0}$ for the projection of $\ker(d^n)$ in the total complex $\mathrm{tot}(J^{G/H})$ onto the component $(n, 0)$. The description of $Z_r^{p,q}$ in Proposition 5.5.2, together with the observations above, gives

$$\ker(d^n)_{n,0} \simeq Z_\infty^{0,n} \subset \cdots \subset Z_2^{0,n} \twoheadrightarrow E_2^{0,n} = \mathrm{H_{II}}\,\mathrm{H_I}(J^{G/H})^{n,0} \simeq \mathrm{H}^n(H, M)^{G/H}. \tag{6.9.3}$$

We now describe $\mathrm{H}^n(G, M) \to \mathrm{H}^n(H, M)^{G/H}$. Choose a representative of an element of $\mathrm{H}^n(G, M)$ in $\ker[I^{n,G} \to I^{n+1,G}]$, map it through $I^G \to J^{G/H}$ into $\ker(d^n)_{n,0}$, and then follow (6.9.3) to $\mathrm{H_{II}}\,\mathrm{H_I}(J^{G/H})^{n,0}$. To take its image further in $\mathrm{H}^n(H, M)$, apply $\mathrm{H_{II}}\,\mathrm{H_I}(\cdot)^{n,0}$ to the whole diagram (6.9.2) and compose along . If instead we follow and use the three observations above, the result is the canonical homomorphism $\mathrm{H}^n(G, M) \to \mathrm{H}^n(H, M)$. This proves (iii). □

Remark 6.9.5 will give a canonical and perhaps more natural construction of the Lyndon–Hochschild–Serre spectral sequence.

The cohomological exact sequence above is also called the **inflation–restriction exact sequence**, because the canonical homomorphism $\mathrm{H}^n(G/H, M^H) \to \mathrm{H}^n(G, M)$ is commonly called the inflation homomorphism, while $\mathrm{H}^n(G, M) \to \mathrm{H}^n(H, M)^{G/H}$ is called the restriction homomorphism. The more complicated morphisms are

$$\mathrm{H}^n(H, M)^{G/H} \to \mathrm{H}^{n+1}(G/H, M^H), \qquad \mathrm{H}_{n+1}(G/H, M_H) \to \mathrm{H}_n(H, M)_{G/H}.$$

They are both called **transgression** maps, in keeping with the transgression maps in the theory of characteristic classes.

As an application, we briefly derive a formula of Hopf for computing $\mathrm{H}_2(G, \mathbb{Z})$ from a presentation of the group G. Here we take $\Bbbk = \mathbb{Z}$ and give $\mathbb{Z}$ the trivial G-module structure.

Corollary 6.9.4 (H. Hopf) Let G be a group and $G \simeq F/R$, where F is a free group and $R \lhd F$. Write $[F, R]$ for the subgroup of R generated by the elements $[f, r] :=$

$frf^{-1}r^{-1}$, with $f \in F$ and $r \in R$. This subgroup is normal in R and $[F,R] \subset [F,F] := F_{\mathrm{der}}$. There is an isomorphism

$$\mathrm{H}_2(G,\mathbb{Z}) \simeq (R \cap [F,F])/[F,R].$$

Proof Proposition 6.7.7 gives $\mathrm{H}_2(F,\mathbb{Z}) = 0$. Thus the exact sequence in Theorem 6.9.3 for $n=1$ becomes

$$0 \to \mathrm{H}_2(G,\mathbb{Z}) \to \mathrm{H}_1(R,\mathbb{Z})_G \to \mathrm{H}_1(F,\mathbb{Z}) \to \mathrm{H}_1(G,\mathbb{Z}) \to 0.$$

Interpreting H_1 through Example 6.2.11, we obtain

$$0 \to \mathrm{H}_2(G,\mathbb{Z}) \to \left(\frac{R}{[R,R]}\right)_G \to \frac{F}{[F,F]} \to \frac{G}{[G,G]} \to 0.$$

The action of G on $R/[R,R]$ is as follows. Choose a preimage $f \in F$ of $g \in G$; one can verify that the action of g comes from the conjugation action of f on R (exercise). For an arbitrary $\bar{r} \in R/[R,R]$, choose a preimage $r \in R$. Then $g\bar{r} - \bar{r}$ (with the group operation written additively) is the image of

$$frf^{-1} \cdot r^{-1} \in R \quad \text{(with the group operation written multiplicatively).}$$

As f and r vary, these elements generate $[F,R]$. Hence $(R/[R,R])_G \simeq R/[F,R]$; the rest is immediate. □

Remark 6.9.5 (Multiplicative structure of the Lyndon–Hochschild–Serre spectral sequence) In the cohomological case, write the spectral sequence of a G-module M as $E_r^{p,q}(M)$. The cup-product operation to be introduced in §6.10, applied to the cohomology of G/H and H, equips the E_2 term with a multiplicative structure

$$\cup : E_2^{p_1,q_1}(M_1) \otimes E_2^{p_2,q_2}(M_2) \to E_2^{p_1+p_2,q_1+q_2}(M_1 \otimes M_2),$$

while the convergence target of the spectral sequence also has a cup product, applied to the group G:

$$\cup : \mathrm{H}^{n_1}(G,M_1) \otimes \mathrm{H}^{n_2}(G,M_2) \to \mathrm{H}^{n_1+n_2}(G,M_1 \otimes M_2).$$

The question is whether these structures can be defined on every page and whether d_r has good properties with respect to them, such as the Leibniz rule

$$d_r(\alpha_1 \cup \alpha_2) = d_r(\alpha_1) \cup \alpha_2 + (-1)^{p_1+q_1}\alpha_1 \cup (d_r\alpha_2), \qquad \alpha_i \in E_r^{p_i,q_i}(M_i).$$

In their original work [20, pp. 118–119], Hochschild and Serre wrote down directly a canonical filtration on the normalized standard complex $\overline{C}(G,M)$ that is compatible with the cup product defined at the level of $\overline{C}(G, M)$. Consequently, every page inherits a multiplication with the properties above. Concretely, they use the decreasing filtration $\overline{C}(G, M) = \mathrm{F}^0 \supset \mathrm{F}^1 \supset \cdots$. If $p > n$, the degree-n term of the complex F^p is 0; otherwise, the degree-n term of F^p is

$$\left\{ f \in \overline{C}^n(G, M) \;\middle|\; \begin{array}{l} (g_1, \ldots, g_n) \text{ has at least } n - p + 1 \text{ entries in } H \\ \implies f(g_1, \ldots, g_n) = 0 \end{array} \right\}.$$

The details are left as an exercise for this chapter; interested readers, or those who need them, may consult the original work.

In particular, $\overline{C}(G, \Bbbk)$ becomes a filtered differential graded algebra, and the spectral sequence $E_r^{p,q}(\Bbbk)$ has the corresponding multiplicative structure (Definition 5.7.5 and Proposition 5.7.6). From the topological viewpoint, this is entirely unsurprising.

6.10 The cup-product operation

Let G be an arbitrary group. As usual, write $\otimes = \otimes_{\Bbbk}$; this notation denotes both the tensor product of $\Bbbk$-modules and the tensor product of G-modules.

Convention 6.10.1 Write $\mathrm{Ext}_G^{\bullet} := \mathrm{Ext}_{\Bbbk[G]}^{\bullet}$ and $\mathsf{D}(G) := \mathsf{D}(G\text{-}\mathsf{Mod})$.

The aim of this section is to construct a cup-product operation on cohomology for arbitrary G-modules M_1 and M_2,

$$\cup : \mathrm{H}^p(G, M_1) \otimes \mathrm{H}^q(G, M_2) \to \mathrm{H}^{p+q}(G, M_1 \otimes M_2), \quad p, q \in \mathbb{Z}_{\geq 0},$$

which for $p = q = 0$ gives the evident morphism $M_1^G \otimes M_2^G \to (M_1 \otimes M_2)^G$.

Reformulated in terms of $\mathrm{Ext}_G^{\bullet}$, the aim is equivalently to define

$$\cup : \mathrm{Ext}_G^p(\Bbbk, M_1) \otimes \mathrm{Ext}_G^q(\Bbbk, M_2) \to \mathrm{Ext}_G^{p+q}(\Bbbk, M_1 \otimes M_2).$$

Recall that $\mathrm{Ext}_G^p(\Bbbk, M_1) \simeq \mathrm{Hom}_{\mathsf{D}(G)}(\Bbbk, M_1[p])$. Our first viewpoint on the cup product is based on morphisms in the derived category. Readers interested only in concrete computations with complexes may proceed directly to Definition–Proposition 6.10.7.

Definition 6.10.2 Write the left-derived bifunctor of $\otimes : G\text{-}\mathsf{Mod} \times G\text{-}\mathsf{Mod} \to G\text{-}\mathsf{Mod}$ as $\overset{\mathrm{L}}{\otimes}$. This bifunctor gives $\mathsf{D}^-(G)$ the structure of a symmetric monoidal category, with the trivial G-module $\Bbbk$ as its unit object.

The tensor product $\otimes$ of complexes is defined, as usual, by taking the total complex $\mathrm{tot}_{\oplus}$. The commutativity constraint in the symmetric monoidal structure involves several signs; see Proposition 3.5.6 for details.

A projective G-module remains projective as a $\Bbbk$-module (Proposition 6.4.2, with H trivial), and is therefore also flat. If X and Y are bounded-above complexes of G-modules, and every term of X is flat as a $\Bbbk$-module, then

$$X \otimes Y = X \overset{\mathrm{L}}{\otimes} Y, \quad Y \otimes X = Y \overset{\mathrm{L}}{\otimes} X;$$

see Proposition 4.12.4. In particular,

$$\Bbbk \otimes Y = \Bbbk \overset{\mathrm{L}}{\otimes} Y \simeq Y \simeq Y \overset{\mathrm{L}}{\otimes} \Bbbk = Y \otimes \Bbbk. \tag{6.10.1}$$

Lemma 6.10.3 Let $\varphi : H \to G$ be a group homomorphism. Continue to write the functor on derived categories induced by the exact functor φ^* as $\varphi^* : \mathsf{D}^-(G) \to \mathsf{D}^-(H)$. Then φ^* is a monoidal functor with respect to $\overset{\mathrm{L}}{\otimes}$; see [51, Definition 3.1.7] and the remark following it. In other words, there are canonical isomorphisms satisfying all compatibility conditions,

$$\varphi^*(\Bbbk) \simeq \Bbbk, \quad \varphi^* X_1 \overset{\mathrm{L}}{\otimes} \varphi^* X_2 \simeq \varphi^*(X_1 \overset{\mathrm{L}}{\otimes} X_2).$$

Proof The isomorphism $\varphi^*\Bbbk \simeq \Bbbk$ (indeed, an equality) and all its compatibilities are immediate. Furthermore, if a G-module M is flat as a $\Bbbk$-module, then so is $\varphi^* M$. Since $\overset{\mathrm{L}}{\otimes}$ can be computed using resolutions that are flat over $\Bbbk$, the canonical isomorphism $\varphi^* X_1 \overset{\mathrm{L}}{\otimes} \varphi^* X_2 \simeq \varphi^*(X_1 \overset{\mathrm{L}}{\otimes} X_2)$ is obtained by reducing to the case in which X_1 and X_2 are termwise flat over $\Bbbk$. □

We can now formulate a preliminary version of the cup product in the derived category; the operation simply reflects the tensor-product operation. For the moment, write the isomorphism $\Bbbk \xrightarrow{\sim} \Bbbk \otimes \Bbbk = \Bbbk \overset{\mathrm{L}}{\otimes} \Bbbk$ given by (6.10.1) as δ.

Definition 6.10.4 Let X_1 and X_2 be objects of $\mathsf{D}^-(G)$. Define the $\Bbbk$-module homomorphism

$$\overset{\mathrm{L}}{\cup} : \mathrm{Hom}_{\mathsf{D}(G)}(\Bbbk, X_1) \otimes \mathrm{Hom}_{\mathsf{D}(G)}(\Bbbk, X_2) \to \mathrm{Hom}_{\mathsf{D}(G)}\left(\Bbbk, X_1 \overset{\mathrm{L}}{\otimes} X_2\right)$$

as follows: $\alpha \overset{\mathrm{L}}{\cup} \beta$ is the composite morphism

$$\Bbbk \xrightarrow{\delta} \Bbbk \otimes \Bbbk = \Bbbk \overset{\mathrm{L}}{\otimes} \Bbbk \xrightarrow{\alpha \overset{\mathrm{L}}{\otimes} \beta} X_1 \overset{\mathrm{L}}{\otimes} X_2.$$

Clearly, the map $(\alpha, \beta) \mapsto \alpha \overset{\mathrm{L}}{\cup} \beta$ is bilinear, so the definition is valid. The operation can also be understood as the composition of morphisms. Indeed, from the general properties of monoidal categories, it is easy to see that $\alpha \overset{\mathrm{L}}{\cup} \beta$ is given by the following composite:

$$\begin{array}{c} \mathrm{Hom}_{\mathsf{D}(G)}(\Bbbk, X_1) \otimes \mathrm{Hom}_{\mathsf{D}(G)}(\Bbbk, X_2) \\ \downarrow \\ \mathrm{Hom}_{\mathsf{D}(G)}\left(\Bbbk \otimes X_2, X_1 \overset{\mathrm{L}}{\otimes} X_2\right) \otimes \mathrm{Hom}_{\mathsf{D}(G)}(\Bbbk \otimes \Bbbk, \Bbbk \otimes X_2) \\ \downarrow \\ \mathrm{Hom}_{\mathsf{D}(G)}\left(\Bbbk, X_1 \overset{\mathrm{L}}{\otimes} X_2\right) \end{array}$$

The first part is based on (6.10.1), while the second is composition of morphisms in $\mathsf{D}(G)$.

The following properties are formal consequences of Definition 6.10.4, or of the reformulation above; throughout, X_i and X are objects of $\mathsf{D}^-(G)$.

▷ Functoriality The operation $\overset{\mathrm{L}}{\cup}$ is functorial in X_1 and X_2.

▷ Unit Under the isomorphisms in (6.10.1), both the left and the right action of $\mathrm{id} \in \mathrm{End}_{\mathsf{D}(G)}(\Bbbk)$ through $\overset{\mathrm{L}}{\cup}$ on $\mathrm{Hom}_{\mathsf{D}(G)}(\Bbbk, X)$ are the identity.

▷ Associativity Let $\alpha_i \in \mathrm{Hom}_{\mathsf{D}(G)}(\Bbbk, X_i)$ for $i = 1, 2, 3$. Up to the associativity constraint for $\overset{\mathrm{L}}{\otimes}$,

$$\alpha_1 \overset{\mathrm{L}}{\cup} (\alpha_2 \overset{\mathrm{L}}{\cup} \alpha_3) = (\alpha_1 \overset{\mathrm{L}}{\cup} \alpha_2) \overset{\mathrm{L}}{\cup} \alpha_3.$$

▷ Commutativity Under the commutativity constraint $X_1 \overset{\mathrm{L}}{\otimes} X_2 \xrightarrow{\sim} X_2 \overset{\mathrm{L}}{\otimes} X_1$, the element $\alpha \overset{\mathrm{L}}{\cup} \beta$ is mapped to $\beta \overset{\mathrm{L}}{\cup} \alpha$. To see this, it suffices to apply Definition 6.10.4 and the following commutative diagram:

$$\begin{tikzcd} \Bbbk \arrow[r, "\delta"] \arrow[dr, "\delta"'] & \Bbbk \overset{\mathrm{L}}{\otimes} \Bbbk \arrow[r, "\alpha \overset{\mathrm{L}}{\otimes} \beta"] \arrow[d] & X_1 \overset{\mathrm{L}}{\otimes} X_2 \arrow[d] \\ & \Bbbk \overset{\mathrm{L}}{\otimes} \Bbbk \arrow[r, "\beta \overset{\mathrm{L}}{\otimes} \alpha"] & X_2 \overset{\mathrm{L}}{\otimes} X_1 \end{tikzcd}$$

All vertical arrows are commutativity constraints for $\overset{\mathrm{L}}{\otimes}$. This is a general fact about symmetric monoidal categories.

We next define the cup product at the level of cohomology of G-modules; several signs will be involved.

Definition 6.10.5 (Cup product) Let M_1 and M_2 be G-modules and let $p_1, p_2 \in \mathbb{Z}$. In Definition 6.10.4, take $X_i = M_i[p_i]$, and then compose $\overset{\mathrm{L}}{\cup}$ with the canonical morphisms

$$M_1[p_1] \overset{\mathrm{L}}{\otimes} M_2[p_2] \xrightarrow{\text{can}} M_1[p_1] \otimes M_2[p_2]$$
$$\xrightarrow[\sim]{\theta'} (M_1[p_1] \otimes M_2)[p_2] \xrightarrow[\sim]{\theta} (M_1 \otimes M_2)[p_1 + p_2]$$

in $\mathsf{D}(G)$, where

- ◇ can is the canonical morphism associated with the left-derived bifunctor;
- ◇ the canonical isomorphisms θ and θ' are defined as in Proposition 3.5.8.

This yields the cup product on cohomology,

$$\cup : \mathrm{Ext}^{p_1}_G(\Bbbk, M_1) \otimes \mathrm{Ext}^{p_2}_G(\Bbbk, M_2) \to \mathrm{Ext}^{p_1+p_2}_G(\Bbbk, M_1 \otimes M_2),$$

or, in other notation, $\cup : \mathrm{H}^{p_1}(G, M_1) \otimes \mathrm{H}^{p_2}(G, M_2) \to \mathrm{H}^{p_1+p_2}(G, M_1 \otimes M_2)$.

If $p_1 = p_2 = 0$, the cup product is obtained by taking the tensor product of two morphisms $\Bbbk \to M_1$ and $\Bbbk \to M_2$ in G-Mod, namely $\Bbbk \simeq \Bbbk \otimes \Bbbk \to M_1 \otimes M_2$. The result is precisely the evident map $(M_1)^G \otimes (M_2)^G \to (M_1 \otimes M_2)^G$.

Inspecting the definition, the isomorphism $M_1[p_1] \otimes M_2[p_2] \xrightarrow{\sim} (M_1 \otimes M_2)[p_1 + p_2]$ involved has the following concrete description: on its only nonzero term $M_1[p_1]^{-p_1} \otimes M_2[p_2]^{-p_2} = M_1 \otimes M_2$, it acts as $(-1)^{p_1 p_2}$; the sign comes from θ'.

In addition to the functoriality and unit law already formulated in the derived category, the cup product has the following properties.

▷ Associativity For $\alpha_i \in \mathrm{H}^{p_i}(G, M_i)$ with $i = 1, 2, 3$, one has

$$(\alpha_1 \cup \alpha_2) \cup \alpha_3 = \alpha_1 \cup (\alpha_2 \cup \alpha_3).$$

This property follows easily from the associativity of $\overset{\mathrm{L}}{\cup}$, but the sign mentioned above still needs to be taken into account. In fact, the sign on both sides is $(-1)^{p_1 p_2 + p_2 p_3 + p_1 p_3}$.

▷ Graded commutativity Let $c : M_1 \otimes M_2 \xrightarrow{\sim} M_2 \otimes M_1$ be the isomorphism determined by $x \otimes y \mapsto y \otimes x$. For all $\alpha \in \mathrm{H}^p(G, M_1)$ and $\beta \in \mathrm{H}^q(G, M_2)$, one has

$$\mathrm{H}^{p+q}(c)(\alpha \cup \beta) = (-1)^{pq} \beta \cup \alpha.$$

Indeed, by the commutativity of $\overset{\mathrm{L}}{\cup}$, it suffices to check that the following diagram commutes:

$$\begin{tikzcd} M_1[p] \overset{\mathrm{L}}{\otimes} M_2[q] \arrow[r, "\mathrm{can}"] \arrow[d] & M_1[p] \otimes M_2[q] \arrow[r, "\text{extract } [q] \text{ first}"] \arrow[d] & (M_1 \otimes M_2)[p+q] \arrow[d, "(-1)^{pq} c[p+q]"] \\ M_2[q] \overset{\mathrm{L}}{\otimes} M_1[p] \arrow[r, "\mathrm{can}"'] & M_2[q] \otimes M_1[p] \arrow[r, "\text{extract } [p] \text{ first}"'] & (M_2 \otimes M_1)[p+q] \end{tikzcd}$$

The vertical arrows on the left and in the middle are the commutativity constraints for tensor products in the derived category or the category of complexes. The left square plainly commutes. For the right square, the last part of Proposition 3.5.8 says that the following diagram commutes:

$$\begin{tikzcd} M_1[p] \otimes M_2[q] \arrow[r, "\text{extract } [q] \text{ first}"] \arrow[d] & (M_1 \otimes M_2)[p+q] \arrow[d, "c[p+q]"] \\ M_2[q] \otimes M_1[p] \arrow[r, "\text{extract } [q] \text{ first}"'] & (M_2 \otimes M_1)[p+q]. \end{tikzcd}$$

However, by the anticommutative diagram in Proposition 3.5.8 and its straightforward generalization, the isomorphism $M_2[q] \otimes M_1[p] \xrightarrow{\sim} (M_2 \otimes M_1)[p+q]$ obtained by extracting $[q]$ first differs by $(-1)^{pq}$ from the one obtained by extracting $[p]$ first. This proves commutativity of the right square in the original diagram.

▷ Cohomology algebra In particular, $\mathrm{H}^\bullet(G, \Bbbk) := \bigoplus_p \mathrm{H}^p(G, \Bbbk)$ becomes a graded $\Bbbk$-algebra under $\cup$, and its multiplication satisfies the graded commutativity law above, $\beta \cup \alpha = (-1)^{pq} \alpha \cup \beta$. In fact, one can prove that $\mathrm{H}^\bullet(G, \Bbbk)$ is the same as the graded $\Bbbk$-algebra in Definition 4.5.10,

$$\mathrm{Ext}_G(\Bbbk) := \bigoplus_p \mathrm{Ext}^p_G(\Bbbk, \Bbbk) = \bigoplus_p \mathrm{H}^p(G, \Bbbk).$$

▷ Bimodule structure Under the operation $\cup$, $\bigoplus_p \mathrm{H}^p(G, M)$ becomes a graded bimodule with left and right actions by $\mathrm{H}^\bullet(G, \Bbbk)$.

A group homomorphism $\varphi : H \to G$ induces a homomorphism of cohomology algebras $\varphi^* : \mathrm{H}^\bullet(G, \Bbbk) \to \mathrm{H}^\bullet(H, \Bbbk)$. This is an immediate consequence of the following result.

Proposition 6.10.6 (Change of groups) Use the notation of Definition 6.10.5. Let $\varphi : H \to G$ be a group homomorphism. Write $\varphi^* : \mathrm{H}^{p_i}(G, M_i) \to \mathrm{H}^{p_i}(H, \varphi^* M_i)$ for the canonical homomorphism in (6.5.1), for $i = 1, 2$. Then

$$\varphi^*(\alpha \cup \beta) = \varphi^* \alpha \cup \varphi^* \beta.$$

As a special case, if $H \subset G$ is a subgroup, the restriction map $\mathrm{res}^n : \mathrm{H}^n(G, M) \to \mathrm{H}^n(H, M)$ discussed in §6.8 preserves cup products.

Proof Apply φ^* to every morphism in Definition 6.10.4, and then use the fact that φ^* is a monoidal functor (Lemma 6.10.3). In $\mathrm{Ext}_H^{p_1+p_2}(\Bbbk, \varphi^* M_1 \otimes \varphi^* M_2)$, this gives

$$\varphi^*(\alpha \cup \beta) \xlongequal{\text{identified with}} \text{the composite morphism}$$
$$\left[\Bbbk \xrightarrow{\delta} \Bbbk \overset{\mathrm{L}}{\otimes} \Bbbk \xrightarrow{\varphi^*\alpha \overset{\mathrm{L}}{\otimes} \varphi^*\beta} \varphi^* M_1[p_1] \overset{\mathrm{L}}{\otimes} \varphi^* M_2[p_2] \to (\varphi^* M_1 \otimes \varphi^* M_2)[p_1 + p_2] \right].$$

Here, $\varphi^*\alpha \in \mathrm{Ext}_H^{p_1}(\Bbbk, \varphi^* M_1) \simeq \mathrm{Hom}_H(\varphi^*\Bbbk, \varphi^* M_1[p_1])$ is obtained by applying the functor φ^* to the morphism α. By Remark 6.5.3, however, this element is also the image of α under the canonical homomorphism (6.5.1); the same applies to $\varphi^*\beta$. The result is $\varphi^*\alpha \cup \varphi^*\beta$. $\square$

Our next aim is to describe the cup product through the standard complex $C(G, M)$ of Proposition 6.2.8. This description is not only more concrete; it also lifts the cup product canonically to the level of complexes. We begin with the definition. Later, Proposition 6.10.10 will show that it agrees with the previous version.

Definition–Proposition 6.10.7 (Cup product on the standard complex) Let M_1 and M_2 be G-modules. For all $p_1, p_2 \in \mathbb{Z}$, define the homomorphisms

$$\begin{aligned} \cup : C^{p_1}(G, M_1) \otimes C^{p_2}(G, M_2) &\longrightarrow C^{p_1+p_2}(G, M_1 \otimes M_2) \\ f_1 \otimes f_2 &\longmapsto f_1(g_1, \ldots, g_{p_1}) \otimes (g_1 \cdots g_{p_1} f_2(g_{p_1+1}, \ldots, g_{p_1+p_2})) \end{aligned}$$

where $g_1, \ldots, g_{p_1+p_2} \in G$.

(i) These homomorphisms satisfy associativity (taking M_1, M_2, and M_3 into account) and the Leibniz rule

$$d(f_1 \cup f_2) = df_1 \cup f_2 + (-1)^{p_1} f_1 \cup df_2;$$

in particular, $\cup$ gives a morphism of complexes $\cup : C(G, M_1) \otimes C(G, M_2) \to C(G, M_1 \otimes M_2)$.

(ii) The normalized standard complex introduced in Remark 6.2.13 is closed under $\cup$, and hence there is a morphism

$$\cup : \overline{C}(G, M_1) \otimes \overline{C}(G, M_2) \to \overline{C}(G, M_1 \otimes M_2).$$

Proof Associativity follows immediately from the formula for $\cup$, while the Leibniz rule is also easy to verify. Let

$$\varphi \in C^p(G, M_1), \quad \psi \in C^q(G, M_2).$$

In

$$(d\varphi \cup \psi)(g_1, \ldots, g_{p+q+1}) = \left(g_1\varphi(g_2, \ldots, g_{p+1}) + \sum_{k=1}^{p}(-1)^k\varphi(\ldots, g_k g_{k+1}, \ldots) + (-1)^{p+1}\varphi(g_1, \ldots, g_p)\right) \otimes (g_1 \cdots g_{p+1})\psi(g_{p+2}, \ldots, g_{p+q+1})$$

and

$$(-1)^p(\varphi \cup d\psi)(g_1, \ldots, g_{p+q+1}) = \varphi(g_1, \ldots, g_p) \otimes (g_1 \cdots g_p)\left((-1)^p g_{p+1}\psi(g_{p+2}, \ldots, g_{p+q+1}) + \sum_{k=p+1}^{p+q}(-1)^k\psi(\ldots, g_k g_{k+1}, \ldots) + (-1)^{p+q+1}\psi(g_{p+1}, \ldots, g_{p+q})\right),$$

the term $\varphi(g_1, \ldots, g_p) \otimes (g_1 \cdots g_{p+1})\psi(g_{p+2}, \ldots)$ occurs respectively as the last and first term, and the two cancel. The sum of the two expressions is therefore $d(\varphi \cup \psi)(g_1, \ldots, g_{p+q+1})$.

Notice that the Leibniz rule is equivalent to saying that, as p_1, p_2 vary, $\cup$ gives a morphism of complexes. This proves (i).

If both f_1 and f_2 belong to the normalized standard complex, the definition implies that $(f_1 \cup f_2)(g_1, \ldots, g_{p_1+p_2}) = 0$ whenever one of $g_1, \ldots, g_{p_1+p_2}$ equals 1_G. This also proves (ii). ☐

Recall that the standard complex $C(G, M)$ essentially computes $\mathrm{RHom}_G(\Bbbk, M) = \mathrm{Hom}^\bullet_G(\mathsf{L}, M)$ through the free resolution $\mathsf{L} \to \Bbbk$, where M is a G-module. More precisely, for L we use the chain complex $(\mathsf{L}_n, \partial_n)_n$ in Definition 6.1.10. By setting $\mathsf{L}^n := \mathsf{L}_{-n}$ while retaining the morphisms ∂_n, L may be regarded as a complex.

Lemma 6.10.8 Write the free resolution of Definition 6.1.10 as $\epsilon : \mathsf{L} \to \Bbbk$. Then:

(i) $\epsilon \otimes \epsilon : \mathsf{L} \otimes \mathsf{L} \to \Bbbk \otimes \Bbbk \simeq \Bbbk$ is a projective resolution of the G-module $\Bbbk$;

(ii) one can define a morphism of complexes $\Delta : \mathsf{L} \to \mathsf{L} \otimes \mathsf{L}$ whose degree-$-n$ term is given by

$$\begin{aligned} \Delta_n : \mathsf{L}_n &\longrightarrow \bigoplus\nolimits_{p=0}^{n} \mathsf{L}_p \otimes \mathsf{L}_{n-p} \\ (g_0, \ldots, g_n) &\longmapsto \left((-1)^{p(n-p)}(g_0, \ldots, g_p) \otimes (g_p, \ldots, g_n)\right)_{p=0}^{n} \end{aligned}$$

and which makes the following diagram commute:

$$\begin{tikzcd} \mathsf{L} \arrow[r, "\Delta"] \arrow[d, "\epsilon"'] & \mathsf{L} \otimes \mathsf{L} \arrow[d, "\epsilon\otimes\epsilon"] \\ \Bbbk \arrow[r, "\sim", "\delta"'] & \Bbbk \otimes \Bbbk. \end{tikzcd}$$

Proof For (i), notice that every term of $\mathsf{L}\otimes\mathsf{L}$ remains a projective G-module; one may use Proposition 6.1.7. To prove that $\epsilon\otimes\epsilon$ is a quasi-isomorphism, factor the morphism as

$$\mathsf{L}\otimes\mathsf{L} \to \mathsf{L}\otimes\Bbbk \to \Bbbk\otimes\Bbbk.$$

Since L consists of flat $\Bbbk$-modules, both parts are quasi-isomorphisms.

For (ii), it is clear that every Δ_n is a homomorphism of G-modules. A direct calculation, split into the various cases, verifies that Δ is indeed a morphism of complexes. Commutativity of the diagram is immediate. □

In comparison with the cup product in topology, Δ here is analogous to the "diagonal embedding" of a space. This morphism can also be interpreted through the Alexander–Whitney map to be introduced in §8.8; see the exercises in Chapter 8.

Notice that, because $\epsilon\otimes\epsilon$ is a quasi-isomorphism, the general theory of complexes guarantees the existence of a morphism $\mathsf{L}\to\mathsf{L}\otimes\mathsf{L}$ that makes the diagram in (ii) commute, and this morphism is unique up to homotopy (Theorem 3.11.5). The value of Lemma 6.10.8 (ii) lies in its explicit formula.

Return to the description of $\overset{\mathrm{L}}{\cup}$ in Definition 6.10.4. Represent the morphisms α and β in $\mathsf{D}(G)$ by morphisms of complexes $\mathsf{L}\to X_1$ and $\mathsf{L}\to X_2$, respectively, where the X_i are bounded-above complexes, and lift δ to Δ through Lemma 6.10.8. The composite of $\alpha \overset{\mathrm{L}}{\cup} \beta \in \mathrm{Hom}_{\mathsf{D}(G)}(\Bbbk, X_1 \overset{\mathrm{L}}{\otimes} X_2)$ with the canonical morphism $X_1 \overset{\mathrm{L}}{\otimes} X_2 \xrightarrow{\text{can}} X_1\otimes X_2$ is given concretely by the composite of morphisms of complexes

$$\begin{tikzcd} \mathsf{L} \arrow[r, "\Delta"] & \mathsf{L}\otimes\mathsf{L} \arrow[r, "\alpha\otimes\beta"] \arrow[d, equal] & X_1\otimes X_2. \\ & \mathsf{L}\overset{\mathrm{L}}{\otimes}\mathsf{L} & \end{tikzcd}$$

For G-modules M_i and $p_i\in\mathbb{Z}$, one has $\mathrm{H}^{p_i}\,\mathrm{Hom}^\bullet(\mathsf{L}, M_i) \simeq \mathrm{H}^{p_i}(G, M_i)$, for $i=1,2$.

We next interpret the cup product through Hom complexes. For arbitrary complexes of G-modules X, Y, X', Y' and $m,n\in\mathbb{Z}$, define the $\Bbbk$-module homomorphism

$$\mu^{m,n} : \mathrm{Hom}^m_G(X,Y)\otimes\mathrm{Hom}^n_G(X',Y') \to \mathrm{Hom}^{m+n}_G(X\otimes X', Y\otimes Y'), \qquad (6.10.2)$$

as follows. For an element $f \otimes g$ on the left, define

$$\mu^{m,n}(f \otimes g)(x \otimes x') := (-1)^{pn} f(x) \otimes g(x') \ \in Y^{p+m} \otimes (Y')^{q+n},$$
$$x \otimes x' \in X^p \otimes (X')^q.$$

A direct verification shows that all the $\mu^{m,n}$ give a morphism of complexes

$$\mu : \mathrm{Hom}^\bullet_G(X, Y) \otimes \mathrm{Hom}^\bullet_G(X', Y') \to \mathrm{Hom}^\bullet_G(X \otimes X', Y \otimes Y'),$$

or, in other words, that (6.10.2) satisfies the Leibniz rule. The sign $(-1)^{pn}$ in the definition can be explained by the Koszul sign rule to be discussed in §7.1 and the exercises in Chapter 7, since the formula interchanges the positions of g and x.

Lemma 6.10.9 Apply the operation above with $X = X' = \mathsf{L}$, $Y = M_1$, and $Y' = M_2$. Then the morphism of complexes

$$\mathrm{Hom}^\bullet_G(\mathsf{L}, M_1) \otimes \mathrm{Hom}^\bullet_G(\mathsf{L}, M_2) \xrightarrow{\mu} \mathrm{Hom}^\bullet_G(\mathsf{L} \otimes \mathsf{L}, M_1 \otimes M_2)$$
$$\xrightarrow{\Delta^*} \mathrm{Hom}^\bullet(\mathsf{L}, M_1 \otimes M_2) \quad (6.10.3)$$

induces on cohomology the cup product of Definition 6.10.5.

Proof For $i = 1, 2$, use Lemma 3.2.3 to identify $\mathrm{Hom}^{p_i}_G(\mathsf{L}, M_i)$ with $\mathrm{Hom}^0_G(\mathsf{L}, M_i[p_i])$. Choose f_i in it satisfying $d_{\mathrm{Hom}^\bullet} f_i = 0$, and consider the image of $f_1 \otimes f_2$ under (6.10.2). Since f_i can be nonzero only on L_{p_i}, the sign in this image is $(-1)^{p_1 p_2}$, exactly the sign in Definition 6.10.5. The remaining verification is immediate. □

Recall that $\mathrm{Hom}^\bullet(\mathsf{L}, M_i) \simeq C(G, M_i)$, so that $\mathrm{H}^{p_i}(G, M_i) \simeq \mathrm{H}^{p_i}(C(G, M_i))$. The concrete map is given in the proof of Proposition 6.2.8, for $i = 1, 2$.

Proposition 6.10.10 The cup product in Definition 6.10.5 is the same as the composite homomorphism

$$\mathrm{H}^{p_1}(G, M_1) \otimes \mathrm{H}^{p_2}(G, M_2) \simeq \mathrm{H}^{p_1}(C(G, M_1)) \otimes \mathrm{H}^{p_2}(C(G, M_2))$$
$$\xrightarrow{\kappa} \mathrm{H}^{p_1+p_2}(C(G, M_1) \otimes C(G, M_2))$$
$$\xrightarrow{\mathrm{H}^{p_1+p_2}(\cup)} \mathrm{H}^{p_1+p_2}(C(G, M_1 \otimes M_2)) \simeq \mathrm{H}^{p_1+p_2}(G, M_1 \otimes M_2),$$

where the canonical morphism κ is as described immediately before Theorem 3.14.12. The conclusion is the same when the normalized standard complex is used.

Proof Interpret the cup product as in Lemma 6.10.9. Take $f_i \in C^{p_i}(G, M_i)$. For the bar notation in the display below, put $p := p_1$. For $g_1, \ldots, g_{p_1+p_2} \in G$, the element

$$(g_1 | \cdots | g_{p_1+p_2}) := \left(1_G, g_{[1,1]}, \ldots, g_{[1,p_1+p_2]}\right) \in \mathsf{L}_{p_1+p_2}$$

has, after applying Δ, the following component in $\mathsf{L}_{p_1} \otimes \mathsf{L}_{p_2}$:

$$(-1)^{p_1 p_2} \left(1_G, \ldots, g_{[1,p_1]}\right) \otimes \left(g_{[1,p_1]}, \ldots, g_{[1,p_1+p_2]}\right) \\ = (-1)^{p_1 p_2} (g_1|\cdots|g_p) \otimes g_1 \cdots g_{p_1} \left(g_{p_1+1}|\cdots|g_{p_1+p_2}\right).$$

Under $\mu^{p_1,p_2}(f_1 \otimes f_2)$ in (6.10.2), this element is mapped to

$$f_1(g_1, \ldots, g_{p_1}) \otimes g_1 \cdots g_{p_1} f_2(g_{p_1+1}, \cdots, g_{p_1+p_2}).$$

Thus the operation in (6.10.3) is indeed compatible with $\cup$ on $C(G, M)$. The version for the normalized standard complex is the same. $\square$

Notice that although the description of the cup product on the standard complex is direct and transparent, graded commutativity is difficult to derive directly from it; in the framework of the standard complex, a rather roundabout proof is required. This book does not take that route.

Remark 6.10.11 The structure exhibited by Definition–Proposition 6.10.7 is genuinely richer than its cohomological version in Definition 6.10.5, because it equips the entire standard complex with a differential graded structure. Thus $C(G, \Bbbk)$ under $\cup$ becomes a differential graded algebra in the sense of Definition 5.7.1 or 7.2.1; the same holds for $\overline{C}(G, \Bbbk)$. The cohomological version of the cup product is obtained only after taking $\mathrm{H}^\bullet$: it is graded, but not differential.

The cup product on the standard complex is functorial. Let $\varphi : H \to G$ be a group homomorphism, let M_i (respectively, N_i) be G-modules (respectively, H-modules), and let $f_i : M_i \to N_i$ be a homomorphism equivariant with respect to φ, for $i = 1, 2$; see Definition 6.5.6. Then the following diagram commutes:

$$\begin{tikzcd} C(G, M_1) \otimes C(G, M_2) \arrow[r, "\cup"] \arrow[d] & C(G, M_1 \otimes M_2) \arrow[d] \\ C(H, N_1) \otimes C(H, N_2) \arrow[r, "\cup"'] & C(H, N_1 \otimes N_2); \end{tikzcd}$$

The vertical arrows come from the equivariant homomorphisms f_1, f_2, and $f_1 \otimes f_2$; see Proposition 6.5.10. The normalized standard complex inherits this property.

Corollary 6.10.12 Suppose there is a short exact sequence of G-modules $0 \to M_1' \xrightarrow{f} M_1 \xrightarrow{g} M_1'' \to 0$ such that the corresponding sequence

$$0 \to M_1' \otimes M_2 \to M_1 \otimes M_2 \to M_1'' \otimes M_2 \to 0$$

is also short exact. Write the connecting homomorphism on cohomology as $\delta^p : \mathrm{H}^p(G, M_1'') \to \mathrm{H}^{p+1}(G, M_1')$, and so forth. Then

$$(\delta^p \alpha) \cup \beta = \delta^{p+q}(\alpha \cup \beta), \quad \alpha \in \mathrm{H}^p(G, M_1''),\ \beta \in \mathrm{H}^q(G, M_2).$$

If instead one takes a short exact sequence $0 \to M_2' \to M_2 \to M_2'' \to 0$ such that

$$0 \to M_1 \otimes M_2' \to M_1 \otimes M_2 \to M_1 \otimes M_2'' \to 0$$

remains exact, then

$$\alpha \cup (\delta^q \beta) = (-1)^p \delta^{p+q}(\alpha \cup \beta).$$

Proof Handle the first case using the cup product on the standard complex. Choose a representative $a'' \in C^p(G, M_1'')$ of α. Recall the concrete construction of $\delta^p\alpha$ through the snake lemma: there is an $a \in C^p(G, M_1)$ mapping to a'', and $da'' = 0$ implies that da comes from some $a' \in C^{p+1}(G, M_1')$; the cohomology class of the latter element is $\delta^p\alpha$. Diagrammatically:

$$\begin{tikzcd} & a \arrow[r, mapsto] \arrow[d, mapsto, "d"'] & a'' \\ a' \arrow[r, mapsto] & da & \end{tikzcd}$$

Next, choose a representative $b \in C^q(G, M_2)$ of β. From $db = 0$ and the Leibniz rule, one obtains $d(a'' \cup b) = 0$ and

$$\begin{tikzcd} & a \cup b \arrow[r, mapsto] \arrow[d, mapsto, "d"'] & a'' \cup b \\ a' \cup b \arrow[r, mapsto] & da \cup b & \end{tikzcd}$$

This implies that the cohomology class of $a' \cup b$ is $\delta^{p+q}(\alpha \cup \beta)$.

The argument for the second case is similar, or it can be reduced to the first case by graded commutativity. □

As an application, we now explain the relationship between the corestriction cor^n of Definition–Proposition 6.8.4 and the cup product.

Proposition 6.10.13 (Projection formula) Let M_1 and M_2 be G-modules, let H be a subgroup of G, and suppose $(G : H)$ is finite. For all $\alpha \in \mathrm{H}^p(G, M_1)$ and $\beta \in \mathrm{H}^q(H, M_2)$, the following equality holds in $\mathrm{H}^{p+q}(G, M_1 \otimes M_2)$:

$$\mathrm{cor}^{p+q}(\mathrm{res}^p(\alpha) \cup \beta) = \alpha \cup \mathrm{cor}^q(\beta);$$

and likewise with the left and right positions interchanged.

Proof First handle the case $p = q = 0$. Recall that cor^0 becomes $\nu^{G|H} : M_i^H \to M_i^G$, which sends $x \in M_i^H$ to $\sum_{gH} gx \in M_i^G$. It therefore suffices to prove that the following diagram commutes:

$$\begin{tikzcd} M_1^G \otimes M_2^H \arrow[rr, "\mathrm{id}\otimes\nu^{G|H}"] \arrow[d] & & M_1^G \otimes M_2^G \arrow[d, "\text{evident}"] \\ M_1^H \otimes M_2^H \arrow[r, "\text{evident}"'] & (M_1 \otimes M_2)^H \arrow[r, "\nu^{G|H}"'] & (M_1 \otimes M_2)^G. \end{tikzcd}$$

This is immediate. If $x \in M_1^G$ and $y \in M_2^H$, then $\nu^{G|H}(x \otimes y) = \sum_{gH} gx \otimes gy = \sum_{gH} x \otimes gy = x \otimes \nu^{G|H}(y)$.

For the case $p \geq 1$ or $q \geq 1$, we use dimension shifting and a recursive argument. For example, consider the case $p \geq 1$ and the short exact sequence of G-modules

$$0 \to M_1 \to \mathrm{Ind}_{\{1\}}^G M_1 \to M_1' \to 0,$$

where the first morphism sends x to $[g \mapsto gx]$, for $g \in G$. As a homomorphism of $\Bbbk$-modules, this morphism has the left inverse $\varphi \mapsto \varphi(1_G)$; hence tensoring the short exact sequence with M_2 preserves exactness. On the other hand, for $n \geq 1$, $\mathrm{H}^n(G, \mathrm{Ind}_{\{1\}}^G(M_1)) \simeq \mathrm{H}^n(\{1\}, M_1) = 0$ (Theorem 6.4.3). Thus the connecting homomorphism gives a surjection

$$\delta_G^{p-1} : \mathrm{H}^{p-1}(G, M_1') \twoheadrightarrow \mathrm{H}^p(G, M_1).$$

If this short exact sequence is restricted to H, one still obtains a connecting homomorphism $\delta_H^{p-1} : \mathrm{H}^{p-1}(H, M_1') \to \mathrm{H}^p(H, M_1)$, and so forth. Notice that Res_H^G commutes with the operation $\otimes M_2$.

We may therefore suppose that $\alpha = \delta_G^{p-1}(\alpha')$. Compatibility of res and cor with connecting homomorphisms gives

$$\begin{aligned} \mathrm{cor}^{p+q}\left(\mathrm{res}^p(\delta_G^{p-1}\alpha') \cup \beta\right) &= \mathrm{cor}^{p+q}\left(\delta_H^{p-1}(\mathrm{res}^{p-1}\alpha') \cup \beta\right) \\ &\overset{\text{Corollary 6.10.12}}{=\!=\!=} \mathrm{cor}^{p+q}\delta_H^{p+q-1}\left((\mathrm{res}^{p-1}\alpha') \cup \beta\right) \\ = \delta_G^{p+q-1}\mathrm{cor}^{p+q-1}\left((\mathrm{res}^{p-1}\alpha') \cup \beta\right) &\overset{\text{induction}}{=\!=\!=} \delta_G^{p+q-1}\left(\alpha' \cup \mathrm{cor}^q\beta\right) \\ &\overset{\text{Corollary 6.10.12}}{=\!=\!=} (\delta_G^{p-1}\alpha') \cup \mathrm{cor}^q\beta. \end{aligned}$$

The induction on q is entirely similar. The version with the left and right positions interchanged follows from graded commutativity. □

Example 6.10.14 Let C_m be a cyclic group of order m, where $m \in \mathbb{Z}_{\geq 1}$. Take the coefficient ring $\Bbbk = \mathbb{Z}$. We explain how to understand, through the cup product, the periodic isomorphism $\mathrm{H}^n(C_m, A) \simeq \mathrm{H}^{n+2}(C_m, A)$ implied by Proposition 6.7.5, where A is an arbitrary C_m-module and $n \geq 1$.

Choose any generator t of $\mathrm{H}^2(C_m, \mathbb{Z}) \simeq \mathbb{Z}/m\mathbb{Z}$. We claim that for $n \geq 1$, the cup product induces an isomorphism

$$t \cup (\cdot) : \mathrm{H}^n(C_m, A) \xrightarrow{\sim} \mathrm{H}^{n+2}(C_m, \Bbbk \otimes A) = \mathrm{H}^{n+2}(C_m, A).$$

Choose a generator σ of C_m. There are short exact sequences

$$\begin{aligned} &0 \to \mathfrak{I} \to \mathbb{Z}[C_m] \to \mathbb{Z} \to 0, && 0 \to \mathfrak{I} \otimes A \to \mathbb{Z}[C_m] \otimes A \to A \to 0,\\ &0 \to \mathbb{Z} \xrightarrow{\nu} \mathbb{Z}[C_m] \xrightarrow{\sigma - 1} \mathfrak{I} \to 0, && 0 \to A \xrightarrow{\nu\otimes\mathrm{id}} \mathbb{Z}[C_m] \otimes A \xrightarrow{(\sigma-1)\otimes\mathrm{id}} \mathfrak{I} \otimes A \to 0. \end{aligned}$$

The sequences on the left come from Lemma 6.7.3. Each of their terms is a free $\mathbb{Z}$-module, so the sequences on the right obtained by applying $\otimes A$ are also exact. First, this gives

$$\mathbb{Z} = \mathrm{H}^0(C_m, \mathbb{Z}) \twoheadrightarrow \mathrm{H}^1(C_m, \mathfrak{I}) \xrightarrow{\sim} \mathrm{H}^2(C_m, \mathbb{Z}) = \mathbb{Z}/m\mathbb{Z},$$

where both maps are connecting homomorphisms from the long exact sequences. Choose a preimage $\tilde{t}$ of t in $\mathbb{Z}$; this integer must be relatively prime to m. Corollary 6.10.12 gives the commutative diagram

$$\begin{tikzcd} \mathrm{H}^n(C_m, A) \arrow[r, "\text{connecting homomorphism}"] & \mathrm{H}^{n+1}(C_m, \mathfrak{I} \otimes A) \arrow[r, "\text{connecting homomorphism}"] & \mathrm{H}^{n+2}(C_m, A). \\ & \mathrm{H}^n(C_m, A) \arrow[ul, "\tilde{t}\cup(\cdot)"] \arrow[u] \arrow[ur, "t\cup(\cdot)"'] & \end{tikzcd}$$

The exercises in this chapter will show that the composite in the first row is the periodic isomorphism. It therefore suffices to prove that $\tilde{t} \cup (\cdot)$ is an isomorphism. This is clear: the operation is precisely multiplication by $\tilde{t} \in \mathbb{Z}$, while $m\,\mathrm{H}^n(C_m, A) = \{0\}$ (Corollary 6.8.6). In fact, if t corresponds to 1 mod m, take $\tilde{t} = 1$; it is then immediately clear that the periodic isomorphism is exactly $t \cup (\cdot)$.

Remark 6.10.15 (Cap product) As in topology, there is also an operation between group cohomology and group homology called the cap product,

$$\cap : \mathrm{H}^p(G, M_1) \otimes \mathrm{H}_q(G, M_2) \to \mathrm{H}_{q-p}(G, M_1 \otimes M_2), \quad M_1, M_2 : G\text{-modules}.$$

Since this book does not use the cap product, we mention it only briefly. Regard an element $\alpha \in \mathrm{H}^p(G, M_1)$ as a morphism $\Bbbk \to M_1[p]$ in $\mathsf{D}(G)$. This element induces the following morphism in $\mathsf{D}^-(G)$:

$$\begin{aligned} &\Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} M_2 \xrightarrow{\sim} \Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} (\Bbbk \overset{\mathrm{L}}{\otimes} M_2) \xrightarrow{\mathrm{id}\otimes\alpha\otimes\mathrm{id}} \Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} (M_1[p] \overset{\mathrm{L}}{\otimes} M_2) \\ &\xrightarrow{\mathrm{id}\overset{\mathrm{L}}{\otimes}\mathrm{can}} \Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} (M_1[p] \otimes M_2) \xrightarrow{\mathrm{id}\overset{\mathrm{L}}{\otimes}\theta} \Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} ((M_1 \otimes M_2)[p]) \xrightarrow{\theta'} \left(\Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} (M_1 \otimes M_2)\right)[p]. \end{aligned}$$

Finally, take H^{-q} of this composite to obtain

$$\alpha \cap (\cdot) : \mathrm{H}_q(G, M_2) \to \mathrm{H}_{q-p}(G, M_1 \otimes M_2).$$

In the special case $p = q$, one obtains a pairing

$$\langle \cdot, \cdot \rangle : \mathrm{H}^p(G, M_1) \otimes \mathrm{H}_p(G, M_2) \to (M_1 \otimes M_2)_G .$$

6.11 Tate Cohomology

For a finite group G and a commutative ring $\Bbbk$, Definition–Proposition 6.7.1 introduced the element $\nu = \sum_{g \in G} g$ of $\Bbbk[G]^G$. It lies in the center of $\Bbbk[G]$. Thus, for every G-module M, multiplication $x \mapsto \nu x$ gives a G-module homomorphism $M \to M$, again denoted by ν.

The identities $\nu g = \nu = g\nu$ imply $\nu M \subset M^G$ and $\mathfrak{I}M \subset M^{\nu=0}$, where $\mathfrak{I}$ is the augmentation ideal of $\Bbbk[G]$ and $M^{\nu=0}$ is as in Convention 6.7.2. Consequently, ν induces a $\Bbbk$-module homomorphism

$$\nu : M_G \to M^G. \tag{6.11.1}$$

Since group homology (respectively, cohomology) can be regarded as a derived version of M_G (respectively, M^G), the canonical homomorphism $M_G \to M^G$ suggests a bridge between homology and cohomology. Why not, for example, consider its cokernel in some derived sense[7)]? We set this idea aside for the moment and define the corresponding cohomology in the classical way.

Definition 6.11.1 (Tate cohomology) Let G be a finite group and M a G-module. For every $n \in \mathbb{Z}$, define

$$\begin{aligned}
\hat{\mathrm{H}}^n(G, M) &:= \mathrm{H}^n(G, M) \quad (n \geq 1), \\
\hat{\mathrm{H}}^0(G, M) &:= M^G / \nu M \\
&= \operatorname{coker}\left[\nu : M_G \to M^G\right] \twoheadleftarrow \mathrm{H}^0(G, M), \\
\hat{\mathrm{H}}^{-1}(G, M) &:= M^{\nu=0} / \mathfrak{I}M \\
&= \ker\left[\nu : M_G \to M^G\right] \hookrightarrow \mathrm{H}_0(G, M), \\
\hat{\mathrm{H}}^{-n}(G, M) &:= \mathrm{H}_{n-1}(G, M) \quad (n \geq 2).
\end{aligned}$$

All of these modules are functorial in M.

7) From the topological point of view: in the homotopical sense.

Tate cohomology is often used in Galois theory and algebraic number theory; one example is the $\hat{\mathrm{H}}^{-1}$ appearing in [51, Definition 9.6.3]. It is due to J. Tate.

A useful cohomology theory generally has a long exact sequence, and Tate cohomology is no exception.

Theorem 6.11.2 Let $0 \to M' \to M \to M'' \to 0$ be a short exact sequence of G-modules. There is then a long exact sequence

$$\cdots \to \hat{\mathrm{H}}^n(G, M) \to \hat{\mathrm{H}}^n(G, M'') \xrightarrow{\delta^n} \hat{\mathrm{H}}^{n+1}(G, M') \to \hat{\mathrm{H}}^{n+1}(G, M) \to \cdots$$

extending infinitely in both directions. The connecting homomorphisms δ^n are functorial with respect to morphisms of short exact sequences, that is, with respect to commutative diagrams with exact rows

$$\begin{tikzcd}
0 \arrow[r] & M' \arrow[r] \arrow[d] & M \arrow[r] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & L' \arrow[r] & L \arrow[r] & L'' \arrow[r] & 0
\end{tikzcd}$$

Proof Splice the long exact sequences in homology and cohomology as follows:

$$\begin{tikzcd}
\cdots \arrow[r] & \mathrm{H}_1(G, M'') \arrow[r] \arrow[d] & \mathrm{H}_0(G, M') \arrow[r] \arrow[d, "N_1"] & \mathrm{H}_0(G, M) \arrow[r] \arrow[d, "N_2"] & \mathrm{H}_0(G, M'') \arrow[r] \arrow[d, "N_3"] & 0 \arrow[d] \\
 & 0 \arrow[r] & \mathrm{H}^0(G, M') \arrow[r] & \mathrm{H}^0(G, M) \arrow[r] & \mathrm{H}^0(G, M'') \arrow[r] & \mathrm{H}^1(G, M') \arrow[r] & \cdots
\end{tikzcd}$$

Here N_1, N_2, N_3 are all the homomorphisms previously denoted by ν. The diagram above is commutative and its rows are exact. The only point requiring explanation is the commutativity of the two squares that touch 0. This can be checked explicitly using the standard complex and the chain complex, while recalling the connecting morphisms $\mathrm{H}_1 \to \mathrm{H}_0$ and $\mathrm{H}^0 \to \mathrm{H}^1$; the verification is left as an exercise.

The Snake Lemma now gives the exact sequence

$$\begin{tikzcd}
\ker(N_1) \arrow[r] \arrow[d, equal] & \ker(N_2) \arrow[r] \arrow[d, equal] & \ker(N_3) \arrow[r, "\delta^{-1}"] \arrow[d, equal] & \operatorname{coker}(N_1) \arrow[r] \arrow[d, equal] & \operatorname{coker}(N_2) \arrow[r] \arrow[d, equal] & \operatorname{coker}(N_3) \arrow[d, equal] \\
\hat{\mathrm{H}}^{-1}(G, M') & \hat{\mathrm{H}}^{-1}(G, M) & \hat{\mathrm{H}}^{-1}(G, M'') & \hat{\mathrm{H}}^0(G, M') & \hat{\mathrm{H}}^0(G, M) & \hat{\mathrm{H}}^0(G, M'')
\end{tikzcd}$$

This is one part of the desired exact sequence, and the functoriality of the connecting homomorphism δ^{-1} follows from the Snake Lemma. The remaining parts reduce to the long exact sequences in group homology and group cohomology, together with the commutative diagram with exact rows proved in the first step. □

We next state the Tate-cohomology version of Shapiro's Lemma (Theorem 6.4.3). Recall that if G is finite, then for every subgroup H the two induction functors $\mathrm{Ind}_H^G, \mathrm{ind}_H^G : H\text{-}\mathsf{Mod} \to G\text{-}\mathsf{Mod}$ coincide; see Corollary 6.8.2. From now on both will be denoted by Ind_H^G.

Proposition 6.11.3 For a finite group G, a subgroup H, and an H-module N, there is a canonical isomorphism

$$\hat{\mathrm{H}}^n(H, N) \simeq \hat{\mathrm{H}}^n\left(G, \mathrm{Ind}_H^G(N)\right), \quad n \in \mathbb{Z}.$$

Proof For $n \geq 1$ or $n \leq -2$, the assertion reduces to Theorem 6.4.3. We next handle the cases $n = -1, 0$. Realize $\mathrm{Ind}_H^G(N)$ as a mapping space by Lemma 6.8.1. Write ν^G (respectively, ν^H) for the ν corresponding to G (respectively, H). It remains to prove that the following diagram commutes:

$$\begin{tikzcd} \mathrm{Ind}_H^G(N)_G \arrow[r, "\nu^G"] & \mathrm{Ind}_H^G(N)^G \arrow[d, "\wr"] \\ N_H \arrow[u, "\wr"] \arrow[r, "\nu^H"'] & N^H \end{tikzcd}$$

The vertical isomorphisms come from Corollary 6.2.4. Let $y \in N$, and write $[y]$ for its image in N_H. The image of $[y]$ in $\mathrm{Ind}_H^G(N)_G$ is $[f]$, where $f \in \mathrm{Ind}_H^G(N)$ satisfies $f(1_G) = y$ and $\mathrm{Supp}(f) \subset H$. Thus the image in N^H of $\nu^G f : x \mapsto \sum_{g \in G} f(xg)$ is $(\nu^G f)(1_G) = \sum_g f(g)$. But this also equals $\sum_{h \in H} f(h) = \sum_h hf(1_G) = \nu^H y$. This proves the assertion. $\square$

The argument can be strengthened to show that the canonical isomorphism is compatible with the long exact sequences on both sides.

Corollary 6.11.4 Let G be a finite group. For every $\Bbbk$-module N,

$$\hat{\mathrm{H}}^n\left(G, \mathrm{Ind}_{\{1\}}^G(N)\right) = 0, \quad n \in \mathbb{Z}.$$

Proof Clearly $\hat{\mathrm{H}}^n(\{1\}, N) = 0$ for every $n \in \mathbb{Z}$. $\square$

Example 6.11.5 Let $E|F$ be a finite Galois extension of fields, with Galois group $\mathrm{Gal}(E|F)$ acting on E. Take the coefficient ring to be $\Bbbk = F$ or $\mathbb{Z}$. The Normal Basis Theorem [51, Theorem 9.5.6] is equivalent to the statement $E \simeq \mathrm{Ind}_{\{1\}}^{\mathrm{Gal}(E|F)}(F)$, so Corollary 6.11.4 implies

$$\hat{\mathrm{H}}^n(\mathrm{Gal}(E|F), E) = 0, \quad n \in \mathbb{Z};$$

therefore $n \geq 1 \implies \mathrm{H}^n(\mathrm{Gal}(E|F), E) = 0$. If $E|F$ is cyclic and $n = -1$, this gives another proof of the additive version of Hilbert's Theorem 90 mentioned in [51, Theorem 9.6.10]. For the multiplicative version, see Example 7.11.10.

For a general $H \subset G$ and a G-module M, the left and right adjoints of restriction give canonical morphisms

$$\epsilon_M : \mathrm{ind}_H^G(\mathrm{Res}_H^G(M)) \to M, \quad \eta_M : M \to \mathrm{Ind}_H^G(\mathrm{Res}_H^G(M)).$$

From the concrete description of the adjoint pairs (Proposition 6.1.9), $\epsilon_M(g \otimes x) = gx$, while $\eta_M(x) = [g \mapsto gx]$. Thus ϵ_M is surjective and η_M is injective. Returning to the case of finite groups, we have $\operatorname{ind}_{\{1\}}^G = \operatorname{Ind}_{\{1\}}^G$. Using ϵ_M and η_M, the vanishing in Corollary 6.11.4 shows that each $\hat{\mathrm{H}}^n(G, \cdot)$ is both erasable and coerasable (Definition 3.12.11). Consequently, the dimension-shifting technique of Proposition 3.12.9 has the following simplified form.

Corollary 6.11.6 (Dimension shifting) For every G-module M, as usual write M as an abbreviation for $\operatorname{Res}_{\{1\}}^G M$, and consider the short exact sequences

$$0 \to M' \to \operatorname{Ind}_{\{1\}}^G(M) \xrightarrow{\epsilon_M} M \to 0,$$
$$0 \to M \xrightarrow{\eta_M} \operatorname{Ind}_{\{1\}}^G(M) \to M'' \to 0.$$

The corresponding connecting homomorphisms in the long exact sequences give canonical isomorphisms

$$\hat{\mathrm{H}}^{n-1}(G, M'') \xrightarrow{\sim} \hat{\mathrm{H}}^n(G, M) \xrightarrow{\sim} \hat{\mathrm{H}}^{n+1}(G, M'), \quad n \in \mathbb{Z}.$$

Proof Combine the long exact sequence (Theorem 6.11.2) with the vanishing in Corollary 6.11.4. □

Many properties of Tate cohomology can be reduced by dimension shifting to the case $n = 0$.

Return to the definition. Choose an arbitrary projective resolution $P \to M$ and injective resolution $M \to I$ of the G-module M, regard both as complexes, and then consider the morphism of complexes $P_G \to I^G$, shown in the diagram

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & I^{0,G} \arrow[r] & I^{1,G} \arrow[r] & I^{2,G} \arrow[r] & \cdots \\
\cdots \arrow[r] & P_{2,G} \arrow[r] \arrow[u] & P_{1,G} \arrow[r] \arrow[u] & P_{0,G} \arrow[r] \arrow[u, "\Omega"] & 0 \arrow[r] \arrow[u] & 0 \arrow[r] \arrow[u] & \cdots
\end{tikzcd}$$

Here Ω is the composite $P_{0,G} \twoheadrightarrow M_G \xrightarrow{\nu} M^G \hookrightarrow I^{0,G}$; commutativity is clear. Take the mapping cone $\operatorname{Cone}\left[P_G \to I^G\right]$, or equivalently the total complex of the diagram above (placing P_G in vertical coordinate -1). The reader is asked to verify that

$$\mathrm{H}^n\left(\operatorname{Cone}\left[P_G \to I^G\right]\right) = \hat{\mathrm{H}}^n(G, M). \tag{6.11.2}$$

The equation above in fact shows how to define a canonical morphism $\Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} M \to \operatorname{RHom}_G(\Bbbk, M)$ in the derived category $\mathsf{D}(G)$ of G-modules. Complete this morphism to a distinguished triangle in $\mathsf{D}(G)$,

$$\Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} M \to \operatorname{RHom}_G(\Bbbk, M) \to \operatorname{Tate}(G, M) \xrightarrow{+1},$$

so that $\mathrm{H}^n(\mathrm{Tate}(G, M)) \simeq \hat{\mathrm{H}}^n(G, M)$. The difficulty with this construction is that different choices for $\mathrm{Tate}(G, M)$ in the distinguished triangle are isomorphic, but the isomorphisms are not canonical. This is a conspicuous defect of the theory of derived categories; see Corollary 4.2.3 and the explanation following it.

Nevertheless, suppose that $\Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} M$ (respectively, $\mathrm{RHom}_G(\Bbbk, M)$) is represented by a chosen complex concentrated in nonpositive (respectively, nonnegative) degrees, including but not limited to P_G (respectively, I^G) in (6.11.2). We can still define the morphism Ω, take its mapping cone as $\mathrm{Tate}(G, M)$, and interpret it as the homotopy cokernel in the sense of Remark 3.3.7.[8] The exercises will explain how to use the standard resolution of $\Bbbk$ instead, providing further tools for computing Tate cohomology.

We now discuss the special case of finite cyclic groups in greater detail.

Example 6.11.7 Let $m \in \mathbb{Z}_{\geq 1}$ and let C_m be the cyclic group of order m; choose any generator σ of C_m. For every C_m-module A, we claim that $\hat{\mathrm{H}}^n(C_m, A)$ is the H^n of the following complex ($n \in \mathbb{Z}$):

$$\cdots \xrightarrow{\sigma-1} A \xrightarrow{\nu} \underbrace{A}_{\text{degree-zero term}} \xrightarrow{\sigma-1} A \xrightarrow{\nu} A \xrightarrow{\sigma-1} \cdots$$

This complex repeats with period 2. Indeed, the proof of Proposition 6.7.5 explicitly wrote down the complexes computing homology and cohomology; splicing them by ν yields the mapping cone $\mathrm{Tate}(G, M)$ mentioned above, in precisely the form displayed. We immediately obtain canonical isomorphisms

$$\hat{\mathrm{H}}^n(C_m, A) \simeq \begin{cases} A^{\nu=0}/(\sigma-1)A, & n \text{ odd} \\ A^{\sigma=1}/\nu A, & n \text{ even.} \end{cases}$$

Periodicity shows that if $0 \to A' \to A \to A'' \to 0$ is a short exact sequence of C_m-modules, then the long exact sequence of Theorem 6.11.2 fits into the exact hexagon

$$\begin{tikzcd}
 & \hat{\mathrm{H}}^{\mathrm{even}}(C_m, A') \arrow[r, "f_1"] & \hat{\mathrm{H}}^{\mathrm{even}}(C_m, A) \arrow[dr, "f_2"] & \\
\mathrm{H}^{\mathrm{odd}}(C_m, A'') \arrow[ur, "f_6"] & & & \hat{\mathrm{H}}^{\mathrm{even}}(C_m, A''). \arrow[dl, "f_3"] \\
 & \hat{\mathrm{H}}^{\mathrm{odd}}(C_m, A) \arrow[ul, "f_5"] & \hat{\mathrm{H}}^{\mathrm{odd}}(C_m, A') \arrow[l, "f_4"] &
\end{tikzcd} \tag{6.11.3}$$

[8] Taking the homotopy kernel of $\Bbbk \overset{\mathrm{L}}{\underset{\Bbbk[G]}{\otimes}} M \to \mathrm{RHom}_G(\Bbbk, M)$ produces no new structure; it differs from the homotopy cokernel only by a shift.

Definition 6.11.8 Let A be a C_m-module ($m \in \mathbb{Z}_{\geq 1}$). Assuming that $\hat{\mathrm{H}}^n(C_m, A)$ is finite for every n, define the positive rational number

$$h(C_m, A) := \frac{\left|\hat{\mathrm{H}}^{\mathrm{even}}(C_m, A)\right|}{\left|\hat{\mathrm{H}}^{\mathrm{odd}}(C_m, A)\right|},$$

called the **Herbrand quotient** of A.

If A is finite, then $\hat{\mathrm{H}}^n(C_m, A)$ is clearly finite. Note that the number of its elements is independent of the choice of coefficient ring $\Bbbk$; we may take $\Bbbk = \mathbb{Z}$. In algebraic number theory, the following theorem of Herbrand is a fundamental and important result.

Theorem 6.11.9 (J. Herbrand) Let $m \in \mathbb{Z}_{\geq 1}$ and let C_m be the cyclic group of order m.

(i) Let $0 \to A' \to A \to A'' \to 0$ be a short exact sequence of C_m-modules. If two of $h(C_m, A')$, $h(C_m, A)$, and $h(C_m, A'')$ are defined, then the third is automatically defined; in that case,

$$h(C_m, A) = h(C_m, A')h(C_m, A'').$$

(ii) If A is a finite C_m-module, then $h(C_m, A) = 1$.

Proof For (i), consider the exact hexagon (6.11.3). Write $n_i := |\operatorname{im}(f_i)|$. In the sense of multiplication of cardinalities, the cardinalities of the six terms are, respectively,

$$\begin{array}{cccc}
 & \boxed{n_6 n_1} & n_1 n_2 & \\
n_5 n_6 & & & \boxed{n_2 n_3} \\
 & \boxed{n_4 n_5} & n_3 n_4 &
\end{array}$$

The three Herbrand quotients correspond to the quotients of opposite terms in the diagram above. Thus assertion (i) is equivalent to the following observations:

- ◇ if the cardinalities at both ends of two diagonals are finite, then the same holds at both ends of the remaining diagonal;
- ◇ the product of the terms in square-cornered boxes equals the product of the terms in round-cornered boxes.

For (ii), there are short exact sequences of finite $\mathbb{Z}$-modules

$$0 \to A^{\sigma=1} \to A \xrightarrow{\sigma-1} \mathfrak{I}A \to 0, \quad 0 \to A^{\nu=0} \to A \xrightarrow{\nu} \nu A \to 0;$$

here $(\sigma - 1)A = \mathfrak{I}A$ follows from Lemma 6.7.3. Therefore,

$$|A^{\sigma=1}| \cdot |\mathfrak{I}A| = |A^{\nu=0}| \cdot |\nu A|,$$

which rearranges to $|\hat{\mathrm{H}}^0(C_m, A)| = |\hat{\mathrm{H}}^{-1}(C_m, A)|$. This proves the assertion. □

6.12 Cohomology of Profinite Groups

This section presupposes the basic definitions and results on profinite groups from §B.1.

Let G be a profinite group and M a G-module. It is not difficult to verify that the following conditions are equivalent:

(i) $M = \bigcup_K M^K$, where K ranges over the open normal subgroups of G;

(ii) for every $x \in M$, the stabilizer subgroup $\mathrm{Stab}_G(x)$ is open;

(iii) the action map $G \times M \to M$ is continuous when M is given the discrete topology.

Definition 6.12.1 Let G be a profinite group. If any of the conditions above holds, the G-module M is called **smooth**[9]. All smooth G-modules form a full subcategory $G\text{-}\mathsf{Mod}^\infty$ of $G\text{-}\mathsf{Mod}$.

Although the condition on G itself is topological, characterization (i) or (ii) of a smooth G-module is entirely algebraic: smoothness is a property of a G-module, not additional structure. It follows readily that $G\text{-}\mathsf{Mod}^\infty$ is an abelian subcategory of $G\text{-}\mathsf{Mod}$. The operations of Definition 6.1.8 have natural analogues here.

Definition–Proposition 6.12.2 Let $\varphi : H \to G$ be a continuous homomorphism of profinite groups and let M be a smooth G-module. Then the H-module $\varphi^* M$ is still smooth. This gives a functor

$$\varphi^* : G\text{-}\mathsf{Mod}^\infty \to H\text{-}\mathsf{Mod}^\infty.$$

Proof For every open normal subgroup $K \lhd G$, its inverse image $\varphi^{-1}(K) \lhd H$ is again open and normal. If $x \in M^K$, then $x \in (\varphi^* M)^{\varphi^{-1}K}$. Thus

$$\varphi^* M = \bigcup_{L \lhd H:\,\text{open}} (\varphi^* M)^L,$$

which proves that $\varphi^* M$ is smooth. □

[9] In view of (iii), many sources call it discrete.

Note that the functor φ^* is always exact. We single out two special cases:

▷ Restriction If H is a closed subgroup of G, take φ to be the inclusion homomorphism $H \hookrightarrow G$. The corresponding functor is denoted by

$$\mathrm{Res}^G_H : G\text{-}\mathsf{Mod}^\infty \to H\text{-}\mathsf{Mod}^\infty.$$

▷ Inflation For a closed normal subgroup $H \lhd G$, take φ to be the quotient homomorphism $G \twoheadrightarrow G/H$. The corresponding functor is denoted by

$$\mathrm{Infl}^G_{G/H} : G/H\text{-}\mathsf{Mod}^\infty \to G\text{-}\mathsf{Mod}^\infty.$$

The groups H and G/H mentioned above are automatically profinite, so these definitions make sense.

Definition–Proposition 6.12.3 For every G-module M, define $M^\infty := \bigcup_K M^K$, where K ranges over the open normal subgroups of G. This is a smooth G-module, and M is smooth if and only if $M = M^\infty$.

For every smooth G-module M', we have

$$\mathrm{Hom}_G(M', M) = \mathrm{Hom}_G(M', M^\infty).$$

In other words, inclusion : $G\text{-}\mathsf{Mod}^\infty \leftrightarrows G\text{-}\mathsf{Mod} : (\cdot)^\infty$ is an adjoint pair.

Proof Clear. □

The elements of M^∞ are also called the **smooth elements** of M. Since $(\cdot)^\infty$ is the right adjoint of an exact functor, it preserves injective objects (Proposition 2.8.17).

Corollary 6.12.4 The abelian category $G\text{-}\mathsf{Mod}^\infty$ has enough injective objects.

Proof Let M be a smooth G-module and embed it in an injective G-module I. Then $M = M^\infty \hookrightarrow I^\infty$, while I^∞ is an injective smooth G-module. □

Remark 6.12.5 In general, $G\text{-}\mathsf{Mod}^\infty$ does not have enough projective objects. This is the principal reason why this section studies cohomology rather than homology.

Since $G\text{-}\mathsf{Mod}^\infty$ has enough injective objects, we can discuss the right derived functors of the invariants functor $(\cdot)^G$. Our approach is first to give a direct definition that reduces to the case of finite groups, and then to identify it with the right derived functors in Proposition 6.12.11. For now we use only classical language and do not invoke derived categories.

First take a profinite group G and an open normal subgroup K. For every smooth G-module M, the submodule of K-invariants $M^K = (\mathrm{Res}^G_K M)^K$ naturally becomes a G/K-module. This gives a functor $(\cdot)^K : G\text{-}\mathsf{Mod}^\infty \to G/K\text{-}\mathsf{Mod}$ and the evident adjoint pair

$$\begin{gathered} \mathrm{Infl}^G_{G/K} : G/K\text{-}\mathsf{Mod} \rightleftarrows G\text{-}\mathsf{Mod}^\infty : (\cdot)^K \\ \mathrm{Hom}_G(\mathrm{Infl}^G_{G/K}(M'), M) = \mathrm{Hom}_{G/K}(M', M^K). \end{gathered} \tag{6.12.1}$$

If open normal subgroups K, K' of G satisfy $K' \supset K$, then the inclusion $M^{K'} \hookrightarrow M^K$ is equivariant with respect to the oppositely directed quotient homomorphism $G/K' \twoheadleftarrow G/K$ (Definition 6.5.6). Hence for every $n \in \mathbb{Z}_{\geq 0}$ there is a canonical homomorphism

$$\rho^{K'}_K : \mathrm{H}^n(G/K', M^{K'}) \to \mathrm{H}^n(G/K, M^K).$$

If $K_1 \subset K_2 \subset K_3$ are open normal subgroups of G, then $\rho^{K_2}_{K_1}\rho^{K_3}_{K_2} = \rho^{K_3}_{K_1}$. Thus the following definition makes sense.

Definition 6.12.6 Let G be a profinite group and M a smooth G-module. For every $n \in \mathbb{Z}_{\geq 0}$, define

$$\mathrm{H}^n(G, M) := \varinjlim_K \mathrm{H}^n(G/K, M^K),$$

where K ranges over the open normal subgroups of G, and $\varinjlim$ is taken with respect to the partial order $K' \preceq K \iff K \subset K'$ and the morphisms $\rho^{K'}_K$.

All open normal subgroups form a filtered partially ordered set: they are closed under finite intersections. If G is finite, then $\{1\}$ is its unique maximal element, and $\mathrm{H}^n(G, M)$ reduces to the original version in §6.2. In general,

$$\mathrm{H}^0(G, M) = \varinjlim_K (M^K)^{G/K} = M^G.$$

Let $f : M_1 \to M_2$ be a homomorphism of smooth G-modules. For every open normal subgroup K, it restricts to $f_K : M_1^K \to M_2^K$. Clearly, $K' \supset K$ implies $\mathrm{H}^n(f_K)\rho^{K'}_K = \rho^{K'}_K \mathrm{H}^n(f_{K'})$, so for every $n \in \mathbb{Z}_{\geq 0}$ there is a homomorphism

$$\mathrm{H}^n(f) : \mathrm{H}^n(G, M_1) \to \mathrm{H}^n(G, M_2).$$

Example 6.12.7 Let $E|F$ be a Galois extension of fields. Since $E = \bigcup_{L|F} E^{\mathrm{Gal}(E|L)} = \bigcup_{L|F} L$, where $L|F$ ranges over the finite Galois subextensions, the action of $\mathrm{Gal}(E|F)$ on E is smooth. Applying Example 6.11.5 to every $L|F$ gives

$$n \geq 1 \implies \mathrm{H}^n(\mathrm{Gal}(E|F), E) = \varinjlim_{L|F} \mathrm{H}^n(\mathrm{Gal}(L|F), L) = 0.$$

Just as in Proposition 6.2.8 and Remark 6.2.13 for groups without topology, the cohomology of a profinite group can be computed by the standard complex or its normalized version; the difference lies in the continuity condition.

Definition 6.12.8 For a smooth G-module M, give M the discrete topology and define its **standard complex** $C(G, M) = (C^n(G, M))_n$ by

$$C^n(G, M) := \{\text{continuous maps } f : G^n \to M\}, \qquad n \in \mathbb{Z}_{\geq 0}.$$

Terms of negative degree are defined to be 0, while $d^n : C^n(G, M) \to C^{n+1}(G, M)$ is given by

$$(d^n f)(g_1, \ldots, g_{n+1}) = g_1 f(g_2, \ldots, g_{n+1}) \\ + \sum_{k=1}^{n} (-1)^k f(\ldots, g_k g_{k+1}, \ldots) + (-1)^{n+1} f(g_1, \ldots, g_n).$$

Define the **normalized standard complex** to be the following subcomplex $\overline{C}(G, M)$ of $C(G, M)$ (see Remark 6.2.13):

$$\overline{C}^n(G, M) = \left\{ f \in C^n(G, M) \;\middle|\; \begin{array}{l} \exists 1 \leq k \leq n,\ g_k = 1_G \\ \implies f(g_1, \ldots, g_n) = 0 \end{array} \right\}.$$

For every open normal subgroup $K \lhd G$, there are evident inclusions of complexes $C(G/K, M^K) \hookrightarrow C(G, M)$ and $\overline{C}(G/K, M^K) \hookrightarrow \overline{C}(G, M)$.

Since M is discrete, a map $f : G^n \to M$ is continuous if and only if it is locally constant. Compactness of G then gives an open normal subgroup $K_1 \lhd G$ such that f factors through $(G/K_1)^n \to M$. In particular, f assumes only finitely many values; smoothness gives an open normal subgroup $K_2 \lhd G$ such that the values of f lie in M^{K_2}. Set $K := K_1 \cap K_2$. It follows that

$$C^n(G, M) \simeq \varinjlim_{K \lhd G \text{ open}} C^n(G/K, M^K), \qquad \overline{C}^n(G, M) \simeq \varinjlim_{K \lhd G \text{ open}} \overline{C}^n(G/K, M^K).$$

As n varies, these give isomorphisms of complexes.

As before, define the submodules of $C^n(G, M)$

$$Z^n(G, M) := \ker(d^n), \qquad B^n(G, M) := \operatorname{im}(d^{n-1}),$$

with $B^0(G, M) := \{0\}$. Then $\mathrm{H}^n(C(G, M)) = Z^n(G, M)/B^n(G, M)$. Define the normalized versions $\overline{Z}^n(G, M)$ and $\overline{B}^n(G, M)$ similarly.

Theorem 6.12.9 Let G be a profinite group. For all $n \in \mathbb{Z}_{\geq 0}$ and every smooth G-module M, there are canonical isomorphisms

$$\mathrm{H}^n(C(G, M)) \simeq \mathrm{H}^n(G, M) \simeq \mathrm{H}^n(\overline{C}(G, M)).$$

Proof Cohomology takes values in **Ab**, or in $\Bbbk$-**Mod** if a coefficient ring $\Bbbk$ has been specified. Filtered colimits are exact in these categories [51, Theorem 6.2.2]. Hence

$$Z^n(G,M) = \varinjlim_K Z^n(G/K, M^K), \qquad B^n(G,M) = \varinjlim_K B^n(G/K, M^K),$$

and therefore

$$\mathrm{H}^n(C(G,M)) = \frac{\varinjlim_K Z^n(G/K,M^K)}{\varinjlim_K B^n(G/K,M^K)} \simeq \varinjlim_K \frac{Z^n(G/K,M^K)}{B^n(G/K,M^K)} = \varinjlim_K \mathrm{H}^n(C(G/K,M^K)).$$

Apply Proposition 6.2.8 to the last term to obtain $\varinjlim_K \mathrm{H}^n(G/K,M^K)$, namely $\mathrm{H}^n(G,M)$.

The case of $\overline{C}(G,M)$ is similar. □

Theorem 6.12.10 Let G be a profinite group and let $0 \to M' \xrightarrow{f} M \xrightarrow{f'} M'' \to 0$ be a short exact sequence of smooth G-modules. There is an associated long exact sequence

$$\cdots \to \mathrm{H}^{n-1}(G,M'') \xrightarrow{\delta^{n-1}} \mathrm{H}^n(G,M') \xrightarrow{\mathrm{H}^n(f)} \mathrm{H}^n(G,M) \xrightarrow{\mathrm{H}^n(f')} \mathrm{H}^n(G,M'') \xrightarrow{\delta^n} \mathrm{H}^{n+1}(G,M') \to \cdots,$$

with the convention that $\mathrm{H}^n(G,\cdot) = 0$ for $n < 0$. The connecting homomorphisms δ^n are natural with respect to morphisms of short exact sequences; in other words, they form a cohomological δ-functor.

Proof We work with the standard complex; the argument for the normalized version is similar. By Proposition 3.6.4, it suffices to prove that

$$0 \to C^n(G,M') \to C^n(G,M) \to C^n(G,M'') \to 0$$

is a short exact sequence. The only point that is not immediate is the surjectivity of $C^n(G,M) \to C^n(G,M'')$. Let $u : G^n \to M''$ be continuous. As above, there is an open normal subgroup $K \lhd G$ such that u is constant on each coset of $K^n \lhd G^n$. On every coset, choose an arbitrary inverse image in M of this constant value. □

Let M be a smooth G-module with the discrete topology. The low-degree terms of the complex $C(G,M)$ or $\overline{C}(G,M)$ still have direct interpretations.

▷ $(n = 0)$ Clearly $C^0(G,M) = M$ and $\mathrm{H}^0(G,M) = Z^0(G,M) = M^G$. The normalized standard complex gives the same result.

▷ $(n = 1)$ The elements of $Z^1(G, M) = \overline{Z}^1(G, M)$ are continuous crossed homomorphisms $G \to M$. Equivalently, they are continuous homomorphisms $\sigma : G \to M \rtimes G$ satisfying $\pi\sigma = \mathrm{id}_G$, as in Proposition 6.3.3. Here $\pi : M \rtimes G \to G$ is the projection and $M \rtimes G$ carries the product topology. On the other hand, the elements of $B^1(G, M)$ are generated by the crossed homomorphisms $f_m : g \mapsto gm - m$; these are automatically continuous.

▷ $(n = 2)$ Suppose that M is finite. The elements of

$$\mathrm{H}^2(G, M) \simeq \frac{Z^2(G, M)}{B^2(G, M)} \simeq \frac{\overline{Z}^2(G, M)}{\overline{B}^2(G, M)}$$

correspond bijectively to equivalence classes of extensions of profinite groups

$$1 \to M \to E \xrightarrow{\pi} G \to 1 \qquad \text{(short exact sequence of groups)},$$

where the conjugation action of G on M comes from the G-module structure on M. Compared with the version in Definition 6.3.5 and (6.3.1), a "profinite group extension" and its classification must additionally satisfy:

- ◇ E is a profinite group, π is continuous, and M is embedded as a closed subgroup of E;
- ◇ equivalences, or isomorphisms, of extensions are defined by continuous group homomorphisms $E \to E'$, and the splitting of an extension is likewise defined topologically.

Moreover, the automorphism group of an extension is isomorphic to $\overline{Z}^1(G, M)$. The proofs are not difficult, but require some topological language and are therefore left as exercises for this chapter; the finiteness of M is essential.

We now return to the right derived functors $(\mathrm{R}^n(\cdot)^G)_{n\geq 0}$. One could study their values on arbitrary bounded-below complexes[10)], or even their operation in the derived category. This section, however, discusses only smooth G-modules; the rest is left to the exercises.

Proposition 6.12.11 There is an isomorphism of cohomological δ-functors

$$(\mathrm{H}^n(G, \cdot))_{n\geq 0} \simeq (\mathrm{R}^n(\cdot)^G)_{n\geq 0}.$$

Proof Theorem 6.12.10 shows that both sides are cohomological δ-functors, and at $n = 0$ both are the invariants functor $(\cdot)^G$. By Proposition 3.12.14, it is enough to show that $\mathrm{H}^n(G, \cdot)$ is erasable for $n > 0$ in the sense of Definition 3.12.11. Let M be

[10)] In other words, study the hypercohomology of profinite groups.

a smooth G-module and embed it in an injective smooth G-module I. For every open normal subgroup $K \lhd G$, the adjunction (6.12.1) shows that $(\cdot)^K$ preserves injective objects. Hence I^K is an injective G/K-module, and

$$n > 0 \implies \mathrm{H}^n(G, I) = \varinjlim_{K} \mathrm{H}^n(G/K, I^K) = 0.$$

This proves erasability. □

By Proposition 6.12.11, the method of §6.5 applies directly to a continuous homomorphism $\varphi : H \to G$ of profinite groups and defines a cohomological "pullback" operation

$$\mathrm{H}^n(G, M) \to \mathrm{H}^n(H, \varphi^* M), \qquad M : \text{ a smooth } G\text{-module}, \quad n \in \mathbb{Z}_{\geq 0}.$$

These operations are compatible with long exact sequences. As usual, exactness of φ^* and the universal property of cohomological δ-functors reduce the matter to the evident inclusion at $n = 0$, $M^G \hookrightarrow (\varphi^* M)^H$.

As a special case, for all n and all closed subgroups H there are restriction maps $\mathrm{res}^n : \mathrm{H}^n(G, M) \to \mathrm{H}^n(H, M)$; here $\mathrm{Res}^G_H M$ is abbreviated to M.

The homomorphisms above may also be constructed from the homomorphisms

$$\begin{aligned} \mathrm{H}^n(G/K, M^K) &\xrightarrow{(6.5.1)} \mathrm{H}^n(H/\varphi^{-1}(K), \varphi_K^*(M^K)) \\ &\xrightarrow{M^K \subset (\varphi^* M)^{\varphi^{-1} K}} \mathrm{H}^n(H/\varphi^{-1}(K), (\varphi^* M)^{\varphi^{-1} K}), \end{aligned}$$

where K ranges over the open normal subgroups of G. Reducing to the case of finite groups shows that the homomorphism on the standard complex is computed by the formula of Proposition 6.5.10. To prove that this map is indeed the same as the one constructed earlier by the universal property, it is enough to compare them at $n = 0$ and show compatibility with long exact sequences. The standard complex suffices for both tasks.

Similarly, the corestriction homomorphism of Definition–Proposition 6.8.4 extends to profinite groups. Let H be an open subgroup of G. It is then automatically closed and $(G : H)$ is finite (see §B.1). Define homomorphisms

$$\mathrm{cor}^n : \mathrm{H}^n(H, M) \to \mathrm{H}^n(G, M), \qquad M : \text{ a smooth } G\text{-module}, \quad n \in \mathbb{Z}_{\geq 0},$$

which are compatible with long exact sequences and at $n = 0$ give $\nu^{G|H} : M^H \to M^G$.

The construction again uses the universal property of cohomological δ-functors, but here we need to know that $\mathrm{Res}^G_H : G\text{-}\mathsf{Mod}^\infty \to H\text{-}\mathsf{Mod}^\infty$ preserves injective objects;

this is the content of Corollary 6.12.15. Once this fact is granted, Proposition 6.8.5 still has an analogue: for every n,

$$\operatorname{cor}^n \circ \operatorname{res}^n = (G : H) \cdot \mathrm{id}_{\mathrm{H}^n(G,M)}.$$

As before, the proof reduces to the case $n = 0$.

We now discuss induced modules in the profinite case. Henceforth G is always a profinite group. To distinguish the two settings, denote the induction functor without topology (Definition 6.4.1) by $\operatorname{Ind}_H^{G,\mathrm{alg}}$; following Lemma 6.8.1, we realize it as a mapping space.

Definition 6.12.12 For a closed subgroup H of G and a smooth H-module N, equip the mapping space $\operatorname{Maps}(G, N) := \{f : G \to N\}$ with pointwise addition and scalar multiplication, together with the left G-action

$$(gf)(x) = f(xg), \qquad g, x \in G.$$

Define the smooth G-module

$$\begin{aligned}\operatorname{Ind}_H^G(N) &:= \{f \in \operatorname{Maps}(G, N)^\infty : \forall h \in H,\ \forall x \in G,\ f(hx) = hf(x)\} \\ &= (\operatorname{Ind}_H^{G,\mathrm{alg}}(N))^\infty.\end{aligned}$$

The notation $(\cdot)^\infty$ is from Definition–Proposition 6.12.3. This is called the G-module induced from N.

Since G is compact, smoothness of $f \in \operatorname{Ind}_H^{G,\mathrm{alg}}(N)$ is equivalent to its continuity as a map $G \to N$, where N carries the discrete topology.

Proposition 6.12.13 For every closed subgroup H of G, there is an adjoint pair

$$\operatorname{Res}_H^G : G\text{-}\mathsf{Mod}^\infty \leftrightarrows H\text{-}\mathsf{Mod}^\infty : \operatorname{Ind}_H^G.$$

Consequently, Ind_H^G preserves injective objects.

If H is an open subgroup of G, there is also an adjoint pair

$$\operatorname{Ind}_H^G : H\text{-}\mathsf{Mod}^\infty \leftrightarrows G\text{-}\mathsf{Mod}^\infty : \operatorname{Res}_H^G.$$

In this case Ind_H^G also preserves projective objects.

Proof By the adjunction in the setting without topology and Definition–Proposition 6.12.3, for every smooth G-module M and smooth H-module N there are canonical isomorphisms

$$\begin{aligned}\operatorname{Hom}_H(\operatorname{Res}_H^G(M), N) &\simeq \operatorname{Hom}_G(M, \operatorname{Ind}_H^{G,\mathrm{alg}}(N)) \\ &= \operatorname{Hom}_G(M, \operatorname{Ind}_H^{G,\mathrm{alg}}(N)^\infty) = \operatorname{Hom}_G(M, \operatorname{Ind}_H^G(N)).\end{aligned}$$

Now suppose that H is open, so that $(G : H)$ is finite. From Corollary 6.8.2 we obtain the adjunction without topology

$$\mathrm{Hom}_H(N, \mathrm{Res}^G_H(M)) \simeq \mathrm{Hom}_G(\mathrm{Ind}^{G,\mathrm{alg}}_H(N), M).$$

We claim that $\mathrm{Ind}^{G,\mathrm{alg}}_H(N) = \mathrm{Ind}^G_H(N)$. Take $f \in \mathrm{Ind}^{G,\mathrm{alg}}_H(N)$ and $x \in G$, and choose an open subgroup $K_0 \subset H$ such that $f(x) \in N^{K_0}$. Set $K := x^{-1}K_0 x$, which is an open subgroup of G. Then

$$g \in K \implies xg = xgx^{-1}x \in K_0 x \implies f(xg) = f(x).$$

Since x was arbitrary, the map $f : G \to N$ is locally constant, so $f \in \mathrm{Ind}^G_H(N)$. This proves the second adjunction.

Finally, the claims about preserving injective or projective objects follow from exactness of Res^G_H and Proposition 2.8.17. □

Lemma 6.12.14 If H is a closed subgroup of G, then $\mathrm{Ind}^G_H : H\text{-}\mathsf{Mod}^\infty \to G\text{-}\mathsf{Mod}^\infty$ is exact.

Proof We already know that Ind^G_H has a left adjoint, so it is left exact. It remains to prove that $N_1 \twoheadrightarrow N_2$ implies $\mathrm{Ind}^G_H(N_1) \twoheadrightarrow \mathrm{Ind}^G_H(N_2)$. By Lemma B.1.8, the quotient homomorphism $\pi : G \to H\backslash G$ has a continuous section s. For every smooth H-module N, there is a $\Bbbk$-module isomorphism

$$\begin{aligned} \mathrm{Ind}^G_H(N) &\xleftrightarrow{\sim} \{\text{continuous maps } H\backslash G \to N\} \\ f &\longmapsto [x \mapsto f(s(x))] \\ [hs(x) \mapsto hf'(x)] &\longleftarrow\!\shortmid f' \end{aligned}$$

where $x \in H\backslash G$ and $h \in H$. Although this isomorphism does not preserve the G-action, it is natural in N.

Since continuous maps $H\backslash G \to N$ are always locally constant, working on the right-hand side immediately shows that $\mathrm{Ind}^G_H(N_1) \to \mathrm{Ind}^G_H(N_2)$ is surjective. □

Corollary 6.12.15 If H is an open subgroup of G, then $\mathrm{Res}^G_H : G\text{-}\mathsf{Mod}^\infty \to H\text{-}\mathsf{Mod}^\infty$ preserves injective objects.

Proof By the preceding two results, it has the exact left adjoint Ind^G_H. □

We now obtain the profinite version of Theorem 6.4.3.

Proposition 6.12.16 Let H be a closed subgroup of a profinite group G and let N be a smooth H-module. There are isomorphisms

$$\mathrm{H}^n(G, \mathrm{Ind}^G_H(N)) \simeq \mathrm{H}^n(H, N), \qquad n \in \mathbb{Z}_{\geq 0}.$$

Proof Because Ind_H^G is exact and preserves injective objects (Lemma 6.12.14 and Proposition 6.12.13), the argument in Theorem 6.4.3 using erasable cohomological δ-functors applies unchanged. Recall that the heart of that argument is the case $n = 0$, namely

$$\mathrm{Ind}_H^G(N)^G = (\mathrm{Ind}_H^{G,\mathrm{alg}}(N)^{\infty})^G = (\mathrm{Ind}_H^{G,\mathrm{alg}}(N))^G \xrightarrow{\sim} N^H.$$

The first equality is the definition, the second is immediate, and the last isomorphism comes from Corollary 6.2.4. □

In addition:

- the cohomological Lyndon–Hochschild–Serre spectral sequence (see §6.9) extends to the case where G is a profinite group and H is a closed normal subgroup;
- the cup product (see §6.10) also extends to profinite groups by using the direct definition on the standard complex; see Definition–Proposition 6.10.7.

The statements are unchanged, so we do not repeat them.

6.13 Nonabelian Cohomology

Let G be a group. The cohomology $\mathrm{H}^n(G, M)$ discussed previously always requires its "coefficients" M to have a module structure, or at least to be a $\mathbb{Z}$-module, that is, an abelian group. This section studies the case in which the coefficients form a nonabelian group. From now on we use the symbol A in place of M.

Definition 6.13.1 Let A be a group, with its binary operation written multiplicatively and its identity element denoted by 1. Suppose that A is equipped with a left action of the group G, written

$$G \times A \to A, \quad (g, a) \mapsto {}^g a.$$

If the group action is compatible with the multiplicative structure of A,

$${}^g(a_1 a_2) = {}^g a_1 {}^g a_2, \quad {}^g 1 = 1,$$

then A is called a group with a G-action, or simply a ***G*-group**. A homomorphism of G-groups is defined to be a group homomorphism that preserves the G-action. The resulting category is denoted by G-Grp.

We will soon see that, for a G-group A, without additional structure or commutativity only H^0 and H^1 can be defined in a reasonable and useful way. In this setting H^1 need not be a group; it is instead a pointed set.

Definition 6.13.2 A **pointed set** is data (X, x_0), where X is a set and $x_0 \in X$ is its basepoint. A morphism from (X, x_0) to (Y, y_0) is a map $f : X \to Y$ satisfying $f(x_0) = y_0$. These data form the category $\mathsf{Set}_\bullet$.

Let (X, x_0) be a pointed set. If a group G acts on X from the left, written $(g, x) \mapsto {}^g x$, and ${}^g x_0 = x_0$ always holds, we say that G acts on (X, x_0). In that case we can define the pointed subset of invariants

$$(X, x_0)^G := \{x \in X : \forall g \in G,\ gx = x\}, \quad \text{basepoint} = x_0.$$

All pointed sets with a G-action, together with morphisms compatible with that action, form the category denoted by $G\text{-}\mathsf{Set}_\bullet$.

When no confusion can arise, we often write an object (X, x_0) of $\mathsf{Set}_\bullet$ simply as X, and in $G\text{-}\mathsf{Set}_\bullet$ write $(X, x_0)^G$ as X^G.

Every group becomes a pointed set by choosing its identity element as the basepoint. This gives the evident functor $\mathsf{Grp} \to \mathsf{Set}_\bullet$; similarly, there is a functor $G\text{-}\mathsf{Grp} \to G\text{-}\mathsf{Set}_\bullet$. Note that the invariant subset A^G of a G-group A is a subgroup.

Definition 6.13.3 Let $(X, x_0) \xrightarrow{f} (Y, y_0) \xrightarrow{g} (Z, z_0)$ be morphisms in $\mathsf{Set}_\bullet$ (or $G\text{-}\mathsf{Set}_\bullet$). This sequence of morphisms is called exact if $g^{-1}(z_0) = \operatorname{im}(f)$. In this way, exact sequences in $\mathsf{Set}_\bullet$ (or $G\text{-}\mathsf{Set}_\bullet$) are defined.

For abelian groups this recovers the familiar notion of exactness. We can now define H^0 and H^1 for every G-group A.

Definition 6.13.4 Let A be an object of $G\text{-}\mathsf{Set}_\bullet$. Define the pointed set

$$\mathrm{H}^0(G, A) = Z^0(G, A) := A^G.$$

If A is a G-group, then $\mathrm{H}^0(G, A)$ is a group. In this case, further define

$$\begin{aligned} Z^1(G, A) &:= \left\{ c : G \to A \;\middle|\; \begin{array}{l} \forall g_1, g_2 \in G, \\ c(g_1 g_2) = c(g_1) \cdot {}^{g_1} c(g_2) \end{array} \right\}, \\ \mathrm{H}^1(G, A) &:= Z^1(G, A)/\sim, \end{aligned}$$

where the equivalence relation $c \sim c'$ means that there is an $a \in A$ such that, for every $g \in G$,

$$c'(g) = a^{-1} \cdot c(g) \cdot {}^g a.$$

It is easy to see that this is indeed an equivalence relation—it comes from a right A-action—and $\mathrm{H}^1(G, A)$ naturally becomes a pointed set. Its basepoint is the equivalence class determined by the constant map $c(g) = 1$.

Note that substituting $g_1 = g_2 = 1_G$ into the condition defining $Z^1(G,A)$ gives $c(1_G) = 1$. Henceforth $[c]$ denotes the equivalence class of $c \in Z^1(G,A)$.

Remark 6.13.5 (H^1 as conjugacy classes of sections) Form the semidirect product $A \rtimes G$ from the G-group A, and let π denote the projection homomorphism $A \rtimes G \to G$. The elements of $Z^1(G,A)$ correspond bijectively to group homomorphisms $\sigma : G \to A \rtimes G$ satisfying $\pi\sigma = \mathrm{id}_G$ (that is, "sections" of π), via the map sending $c \in Z^1(G,A)$ to $\sigma(g) = (c(g), g)$. Moreover, $[c'] = [c]$ if and only if there is an $a \in A$ such that $\sigma' = a^{-1}\sigma a$, where A is embedded as a subgroup of $A \rtimes G$ and conjugation is performed pointwise.

If A is an abelian group, these constructions reduce to the descriptions of H^0 and H^1 using the standard cochain complex (Proposition 6.2.8). In the nonabelian case, the constructions remain functorial, as follows.

- Every morphism $f : A_1 \to A_2$ induces
$$\mathrm{H}^0(f) : \mathrm{H}^0(G, A_1) \to \mathrm{H}^0(G, A_2), \quad \mathrm{H}^1(f) : \mathrm{H}^1(G, A_1) \to \mathrm{H}^1(G, A_2);$$
the first map is evident, while the second is given by $\mathrm{H}^1(f)([c]) = [f \circ c]$.
- Let $\varphi : H \to G$ be a group homomorphism. Pull the G-action on the G-group A back to H, giving the H-group $\varphi^* A$.
  - Define the canonical morphism $\mathrm{H}^0(G,A) \to \mathrm{H}^0(H, \varphi^* A)$ to be the evident inclusion $A^G \hookrightarrow A^{\varphi(H)} = (\varphi^* A)^H$.
  - Define the canonical morphism of pointed sets
  $$\mathrm{H}^1(G,A) \to \mathrm{H}^1(H, \varphi^* A).$$
  It is given by $[c] \mapsto [c \circ \varphi]$.

These canonical maps have the various properties seen in §6.5. They can also be summarized as follows. Given $\varphi : H \to G$, let A_1 be an object of G-$\mathsf{Set}_\bullet$ (or G-Grp), let A_2 be an object of H-$\mathsf{Set}_\bullet$ (or H-Grp), and let the morphism $f : A_1 \to A_2$ be equivariant with respect to φ, as in Definition 6.5.6. There are corresponding maps $\mathrm{H}^i(f) : \mathrm{H}^i(G, A_1) \to \mathrm{H}^i(H, A_2)$, and the analogue of Proposition 6.5.10 takes the form:

- $\mathrm{H}^0(f)$ is the restriction of f to $A_1^G \to A_2^H$;
- $\mathrm{H}^1(f)$ maps $[c]$ to $[f \circ c \circ \varphi]$.

Convention 6.13.6 Write 1 for the zero object in $\mathsf{Set}_\bullet$ or G-$\mathsf{Set}_\bullet$ determined by the singleton set.

We now come to the main subject. Henceforth consider a short exact sequence in G-$\mathsf{Set}_\bullet$ (Definition 6.13.3)

$$1 \to A \xrightarrow{u} B \xrightarrow{v} C \to 1,$$

where A and B are assumed to be G-groups and u is a group homomorphism. Exactness therefore implies that u is injective and v is surjective. We further require:

$$\begin{gathered} \forall b, b' \in B,\ [v(b) = v(b') \iff \exists a \in A,\ b' = bu(a)], \\ \text{in other words: } C \text{ is identified with } B/u(A), \text{ basepoint} = 1 \cdot u(A). \end{gathered} \tag{6.13.1}$$

If C is a group and v is a group homomorphism, considering $b^{-1}b' \in v^{-1}(1) = \mathrm{im}(u)$ shows that (6.13.1) holds automatically, and in this case $u(A) \lhd B$. Many applications, however, do not fall into this case.

Our present goal is to construct from these data an exact sequence that is "as long as possible," together with the corresponding connecting morphisms. Without further comment, we identify A with its image as a subgroup of B and omit the symbol u.

The degree-zero case. First define the connecting morphism $\delta^0 : \mathrm{H}^0(G, C) \to \mathrm{H}^1(G, A)$. Let $c \in C^G$. Choose any $b \in v^{-1}(c)$. Since v preserves the G-action, (6.13.1) shows that, for every $g \in G$, there is a unique $a(g) \in A$ such that ${}^g b = ba(g)$. Hence $g \mapsto a(g)$ gives an element a of $Z^1(G, A)$, because

$$a(g_1 g_2) = b^{-1} \cdot {}^{g_1 g_2} b = b^{-1} \cdot {}^{g_1} b \cdot {}^{g_1}(b^{-1} \cdot {}^{g_2} b) = a(g_1) \cdot {}^{g_1} a(g_2).$$

If $b', b \in v^{-1}(c)$, there is an $\alpha \in A$ such that $b' = b\alpha$, and hence

$$(b')^{-1} \cdot {}^g b' = \alpha^{-1} \cdot b^{-1} \cdot {}^g b \cdot {}^g \alpha.$$

This shows that the class $[a] \in \mathrm{H}^1(G, A)$ is determined solely by c. If c is the basepoint of C, we may take $b = 1 \in B$. We have therefore obtained a morphism in $\mathsf{Set}_\bullet$, $\delta^0 : \mathrm{H}^0(G, C) \to \mathrm{H}^1(G, A)$.

The degree-zero case: a central subgroup. Suppose that C is a group, v is a group homomorphism, and A is contained in the center Z_B of the group B; in particular, A is abelian. In this case $\mathrm{H}^0(G, C) = C^G$ and $\mathrm{H}^1(G, A)$ are both groups. From the definition of δ^0 and the condition $A \subset Z_B$, it is easy to check that δ^0 is a group homomorphism.

The degree-one case. Continue to suppose that C is a group, v is a group homomorphism, and $A \subset Z_B$; then $\mathrm{H}^2(G, A)$ can be described using the standard cochain complex. We define $\delta^1 : \mathrm{H}^1(G, C) \to \mathrm{H}^2(G, A)$. Let $c \in Z^1(G, C)$. For every $g \in G$, choose $b(g) \in v^{-1}(c(g))$. Since (6.13.1) always holds and v preserves the G-action, there is a unique map $a : G^2 \to A$ satisfying

$$b(g_1) \cdot {}^{g_1} b(g_2) = a(g_1, g_2) b(g_1 g_2).$$

We now check that $a \in Z^2(G, A)$. This is equivalent to

$$a(g_1, g_2)^{-1} \cdot a(g_1, g_2g_3) \cdot a(g_1g_2, g_3)^{-1} \cdot {}^{g_1}a(g_2, g_3) = 1.$$

Expand the left-hand side as an expression in B:

$$\underbrace{b(g_1g_2) \cdot {}^{g_1}b(g_2)^{-1} \cdot b(g_1)^{-1}}_{a(g_1,g_2)^{-1}} \cdot \underbrace{b(g_1) \cdot {}^{g_1}b(g_2g_3)b(g_1g_2g_3)^{-1}}_{a(g_1,g_2g_3)} \\ \underbrace{b(g_1g_2g_3) \cdot {}^{g_1g_2}b(g_3)^{-1} \cdot b(g_1g_2)^{-1}}_{a(g_1g_2,g_3)^{-1}} \cdot {}^{g_1}a(g_2, g_3),$$

and then use $A \subset Z_B$ to simplify it to

$$\begin{aligned} &b(g_1g_2) \cdot {}^{g_1}b(g_2)^{-1} \cdot {}^{g_1}b(g_2g_3) \cdot {}^{g_1g_2}b(g_3)^{-1}b(g_1g_2)^{-1} \cdot {}^{g_1}a(g_2, g_3) \\ &= b(g_1g_2) \cdot {}^{g_1}b(g_2)^{-1} \cdot \underbrace{{}^{g_1}b(g_2) \cdot {}^{g_1g_2}b(g_3) \cdot {}^{g_1}b(g_2g_3)^{-1}}_{{}^{g_1}a(g_2,g_3)} \cdot {}^{g_1}b(g_2g_3) \cdot {}^{g_1g_2}b(g_3)^{-1}b(g_1g_2)^{-1}. \end{aligned}$$

The last expression is visibly equal to 1, so $a \in Z^2(G, A)$.

If the choice of every $b(g)$ is changed and written $b'(g) = \alpha(g)b(g)$, where $\alpha : G \to A$ is any map, then the corresponding $a'(g_1, g_2)$ is

$$\begin{aligned} a'(g_1, g_2) &= \alpha(g_1) \cdot b(g_1) \cdot {}^{g_1}\alpha(g_2) \cdot {}^{g_1}b(g_2)b(g_1g_2)^{-1}\alpha(g_1g_2)^{-1} \\ &= \alpha(g_1) \cdot {}^{g_1}\alpha(g_2) \cdot \alpha(g_1g_2)^{-1} \cdot a(g_1, g_2) \quad (\because\ A \subset Z_B). \end{aligned}$$

Thus $a, a' \in Z^2(G, A)$ differ only by a factor belonging to $B^2(G, A)$.

Finally, consider the case in which c is replaced by $c'(g) = \gamma^{-1} \cdot c(g) \cdot {}^g\gamma$, with $\gamma \in C$. Choose any $\beta \in v^{-1}(\gamma)$. For the b' corresponding to c', we may take $b'(g) = \beta^{-1} \cdot b(g) \cdot {}^g\beta$. The map $a' : G^2 \to A$ constructed from b' is

$$\begin{aligned} a'(g_1, g_2) &= \left(\beta^{-1} \cdot b(g_1) \cdot {}^{g_1}\beta\right) \cdot {}^{g_1}\left(\beta^{-1} \cdot b(g_2) \cdot {}^{g_2}\beta\right) \cdot \left({}^{g_1g_2}\beta^{-1} \cdot b(g_1g_2)^{-1} \cdot \beta\right) \\ &= \beta^{-1} \cdot \underbrace{b(g_1) \cdot {}^{g_1}b(g_2) \cdot b(g_1g_2)^{-1}}_{=a(g_1,g_2)} \cdot \beta \\ &= a(g_1, g_2). \end{aligned}$$

Thus the class of a in $\mathrm{H}^2(G, A)$ is determined by $[c] \in \mathrm{H}^1(G, C)$. If c is the constant map with value 1, then b may also be chosen to be constantly 1, and likewise for a. We have thus obtained a morphism in $\mathsf{Set}_\bullet$, $\delta^1 : \mathrm{H}^1(G, C) \to \mathrm{H}^2(G, A)$.

Theorem 6.13.7 Let $1 \to A \xrightarrow{u} B \xrightarrow{v} C \to 1$ be a short exact sequence in G-$\mathsf{Set}_\bullet$, where A and B are G-groups and u is a group homomorphism.

(i) If condition (6.13.1) holds, there is an exact sequence in $\mathsf{Set}_\bullet$

$$\begin{tikzcd}
1 \arrow[r] & \mathrm{H}^0(G,A) \arrow[r, "\mathrm{H}^0(u)"] & \mathrm{H}^0(G,B) \arrow[r, "\mathrm{H}^0(v)"] & \mathrm{H}^0(G,C) \arrow[dll, rounded corners, "\delta^0"'] \\
& \mathrm{H}^1(G,A) \arrow[r, "\mathrm{H}^1(u)"] & \mathrm{H}^1(G,B). &
\end{tikzcd}$$

(ii) In addition to the assumptions above, suppose that C is also a group and v is a group homomorphism; this automatically implies (6.13.1). The exact sequence then extends to

$$\begin{tikzcd}
1 \arrow[r] & \mathrm{H}^0(G,A) \arrow[r, "\mathrm{H}^0(u)"] & \mathrm{H}^0(G,B) \arrow[r, "\mathrm{H}^0(v)"] & \mathrm{H}^0(G,C) \arrow[dll, rounded corners, "\delta^0"'] \\
& \mathrm{H}^1(G,A) \arrow[r, "\mathrm{H}^1(u)"] & \mathrm{H}^1(G,B) \arrow[r, "\mathrm{H}^1(v)"] & \mathrm{H}^1(G,C).
\end{tikzcd}$$

(iii) If we further require that $u(A)$ be contained in the center Z_B of B, then δ^0 is a group homomorphism and the exact sequence extends to

$$\begin{tikzcd}
1 \arrow[r] & \mathrm{H}^0(G,A) \arrow[r, "\mathrm{H}^0(u)"] & \mathrm{H}^0(G,B) \arrow[r, "\mathrm{H}^0(v)"] & \mathrm{H}^0(G,C) \arrow[dll, rounded corners, "\delta^0"'] \\
& \mathrm{H}^1(G,A) \arrow[r, "\mathrm{H}^1(u)"] & \mathrm{H}^1(G,B) \arrow[r, "\mathrm{H}^1(v)"] & \mathrm{H}^1(G,C) \arrow[dll, rounded corners, "\delta^1"'] \\
& \mathrm{H}^2(G,A). & &
\end{tikzcd}$$

The connecting morphisms δ^0 and δ^1 are defined as above. Both are canonical, meaning that they are compatible with morphisms of short exact sequences.

Proof The detailed constructions of δ^0 and δ^1 above already show that both are canonical; in case (iii), they also showed that δ^0 is a group homomorphism. We now check exactness. Henceforth regard A as a subgroup of B and omit u.

In case (i), exactness at $\mathrm{H}^0(G,A)$ and $\mathrm{H}^0(G,B)$ is clear. At $\mathrm{H}^0(G,C)$, let $c \in C^G$. By definition, c maps to the basepoint of $\mathrm{H}^1(G,A)$ if and only if there are $\alpha \in A$ and $b \in B$ such that

$$v(b) = c, \quad \forall g \in G,\ b^{-1} \cdot {}^g b = \alpha^{-1} \cdot {}^g \alpha.$$

But this is equivalent to the existence of $\alpha \in A$ and $b \in B$ such that $v(b) = c$ and $b\alpha^{-1} \in B^G$, which is in turn equivalent to $c \in \mathrm{im}(\mathrm{H}^0(u))$.

At $\mathrm{H}^1(G, A)$, let $a \in Z^1(G, A)$. The class $[a] \in \mathrm{H}^1(G, A)$ maps to the basepoint if and only if there is a $b \in B$ such that $b^{-1} \cdot a(g) \cdot {}^g b = 1$ for every g. If this holds, set $c := v(b^{-1})$. Applying v to ${}^g b^{-1} = b^{-1} \cdot a(g)$ shows that $c \in C^G$ and $[a] = \delta^0([c])$. The converse argument is similar.

In case (ii), what remains is exactness at $\mathrm{H}^1(G, B)$. Let $b \in Z^1(G, B)$. Its class $[b]$ maps to the basepoint if and only if there is a $\beta \in B$ such that $\beta^{-1} \cdot b(g) \cdot {}^g \beta \in A$ for every g. Clearly, this is the same as saying that, up to $\sim$, the class comes from $Z^1(G, A)$.

Consider case (iii). Let $c \in Z^1(G, C)$. Its class $[c]$ maps to the basepoint if and only if there are maps $b : G \to B$ and $\alpha : G \to A$ such that $v \circ b = c$ and

$$b(g_1) \cdot {}^{g_1} b(g_2) = \alpha(g_1) \cdot {}^{g_1} \alpha(g_2) \cdot \alpha(g_1 g_2)^{-1} \cdot b(g_1 g_2).$$

Since $A \subset Z_B$, if $\alpha^{-1} b : G \to B$ is defined by pointwise multiplication, the equation above is also equivalent to

$$(\alpha^{-1} b)(g_1) \cdot {}^{g_1} (\alpha^{-1} b)(g_2) = (\alpha^{-1} b)(g_1 g_2),$$

that is, $\alpha^{-1} b \in Z^1(G, B)$. Clearly this is equivalent to saying that $[c]$ comes from $\mathrm{H}^1(G, B)$. This proves exactness. $\square$

If A and B are further required to be abelian, then $C \simeq B/u(A)$ is an abelian group, and we recover the familiar version with coefficients in an abelian group.

Note that the exact sequence in Theorem 6.13.7 (i) describes the inverse image of the basepoint under $\mathrm{H}^1(u)$, but in general it cannot determine whether two elements of $\mathrm{H}^1(G, A)$ have the same image under $\mathrm{H}^1(u)$. The "twisting" construction introduced in the exercises for this chapter helps address this problem.

Example 6.13.8 Let B be a G-group and A a subgroup stable under the G-action. Equip the set B/A with the basepoint $1 \cdot A$ and a left G-action. This gives a short exact sequence in G-$\mathsf{Set}_\bullet$

$$1 \to A \xrightarrow{\text{inclusion}} B \xrightarrow{\text{quotient}} B/A \to 1.$$

Condition (6.13.1) clearly holds. Hence Theorem 6.13.7 (i) gives an exact sequence in $\mathsf{Set}_\bullet$

$$1 \to A^G \to B^G \to (B/A)^G \xrightarrow{\delta^0} \mathrm{H}^1(G, A) \to \mathrm{H}^1(G, B).$$

The evident map $B^G/A^G \hookrightarrow (B/A)^G$ is injective, but it is often not bijective: a coset stable under the G-action need not contain a G-fixed point; the exercises will give an example. Nevertheless, the exact sequence identifies the cohomological obstruction.

For a morphism f in $\mathsf{Set}_\bullet$, define $\ker(f)$ to be the inverse image of the basepoint. Then

$$\begin{aligned}\text{the coset } bA \in (B/A)^G \text{ comes from } B^G &\iff bA \in \ker(\delta^0),\\ (B/A)^G = B^G/A^G &\iff \ker[\mathrm{H}^1(G,A) \to \mathrm{H}^1(G,B)] = 1.\end{aligned}$$

We close this section with the nonabelian version of Shapiro's Lemma (Theorem 6.4.3). As preparation, first define a nonabelian version of induced modules using a mapping space; compare Lemma 6.8.1. The construction applies to objects of H-Grp or H-$\mathsf{Set}_\bullet$, where H is a subgroup of G; the group action is written $(h,a) \mapsto {}^h a$.

Definition 6.13.9 Let H be a subgroup of G and let A be an object of H-Grp (or H-$\mathsf{Set}_\bullet$). Define the object $\mathrm{Ind}_H^G(A)$ of G-Grp (or G-$\mathsf{Set}_\bullet$) as follows. As an object of G-Set, it is

$$\mathrm{Ind}_H^G(A) := \left\{ f : G \to A \;\middle|\; \begin{array}{l} \forall h \in H,\ \forall x \in G \\ f(hx) = {}^h f(x). \end{array} \right\},$$
$$({}^g f)(x) := f(xg), \quad f \in \mathrm{Ind}_H^G(A),\ x, g \in G,$$

with the group operation (respectively, basepoint) defined pointwise on maps (respectively, by the constant map to the basepoint). We call $A \mapsto \mathrm{Ind}_H^G(A)$ induction.

The induction construction gives compatible functors H-$\mathsf{Grp} \to G$-Grp and H-$\mathsf{Set}_\bullet \to G$-$\mathsf{Set}_\bullet$.

Theorem 6.13.10 Let H be a subgroup of G and let A be an object of H-$\mathsf{Set}_\bullet$ or H-Grp. There are canonical isomorphisms in $\mathsf{Set}_\bullet$

$$\mathrm{H}^i\left(G, \mathrm{Ind}_H^G(A)\right) \xrightarrow{\sim} \mathrm{H}^i(H, A),$$

with $i = 0$ if A comes from H-$\mathsf{Set}_\bullet$, or $i = 0, 1$ if A comes from H-Grp. The maps are as follows.

- ▷ $(i = 0)$ Take the map $f \mapsto f(1_G)$, which defines $Z^0\left(G, \mathrm{Ind}_H^G(A)\right) = \mathrm{Ind}_H^G(A)^G \to A^H = Z^0(H, A)$.
- ▷ $(i = 1)$ Take the map $Z^1\left(G, \mathrm{Ind}_H^G(A)\right) \to Z^1(H, A)$ sending $c : G \to \mathrm{Ind}_H^G(A)$ to

$$\begin{aligned}(c|_H)(\cdot)(1_G) : H &\to A\\ h &\mapsto c(h)(1_G).\end{aligned}$$

Both induce morphisms on H^i. If A comes from H-Grp, the isomorphism for $i = 0$ is also an isomorphism of groups.

Proof Routine arguments show that the maps in both cases are well-defined and preserve the basepoint or group structure; details are left to the reader.

For $i = 0$, it is easy to see that $\mathrm{Ind}_H^G(A)^G \to A^H$ is indeed a bijection and gives an isomorphism of groups if A is an H-group. We now focus on the case where A is an object of H-Grp and $i = 1$.

Suppose that $r, s \in Z^1(G, \mathrm{Ind}_H^G(A))$ have the same image in $\mathrm{H}^1(H, A)$. We prove that $r \sim s$. This assumption is equivalent to the existence of an $\alpha \in A$ such that

$$r(h)(1_G) = \alpha^{-1} \cdot s(h)(1_G) \cdot {}^h\alpha$$

for every $h \in H$. There is a $\zeta \in \mathrm{Ind}_H^G(A)$ with $\zeta(1_G) = \alpha$. Adjusting s by ζ within its equivalence class, we may ensure that $r(h)(1_G) = s(h)(1_G)$ for every h.

The definition of $Z^1(G, \mathrm{Ind}_H^G(A))$ implies the identities in A

$$r(g_1g_2)(x) = r(g_1)(x) \cdot r(g_2)(xg_1), \quad s(g_1g_2)(x) = s(g_1)(x) \cdot s(g_2)(xg_1)$$

for all $x, g_1, g_2 \in G$. On the one hand, substituting $g_1 = g \in G$ and $g_2 = g^{-1}x^{-1}$ and rearranging gives

$$\begin{aligned} r(g)(x) &= r(x^{-1})(x) \cdot r(g^{-1}x^{-1})(xg)^{-1}, \\ s(g)(x) &= s(x^{-1})(x) \cdot s(g^{-1}x^{-1})(xg)^{-1}. \end{aligned} \tag{6.13.2}$$

On the other hand, substituting $g_1 = x^{-1}$ and $g_2 = h \in H$ gives

$$s(x^{-1}h)(x) \cdot r(x^{-1}h)(x)^{-1} = s(x^{-1})(x) \cdot r(x^{-1})(x)^{-1}. \tag{6.13.3}$$

For every $x \in G$, define $t(x) := s(x^{-1})(x) \cdot r(x^{-1})(x)^{-1}$. We claim that $t : G \to A$ belongs to $\mathrm{Ind}_H^G(A)$. Indeed, if $h \in H$ and $x \in G$, then

$$\begin{aligned} t(hx) &= s(x^{-1}h^{-1})(hx) \cdot r(x^{-1}h^{-1})(hx)^{-1} \\ &= {}^h\left(s(x^{-1}h^{-1})(x) \cdot r(x^{-1}h^{-1})(x)^{-1}\right) \\ &\overset{(6.13.3)}{=\!=} {}^h\left(s(x^{-1})(x) \cdot r(x^{-1})(x)^{-1}\right) = {}^h t(x). \end{aligned}$$

To prove $r \sim s$, it is enough to check the identity in $\mathrm{Ind}_H^G(A)$ given by $t^{-1} \cdot s(g) \cdot {}^g t = r(g)$. The value of the left-hand side at $x \in G$ is

$$r(x^{-1})(x) \cdot s(x^{-1})(x)^{-1} \cdot s(g)(x) \cdot s(g^{-1}x^{-1})(xg) \cdot r(g^{-1}x^{-1})(xg)^{-1};$$

expanding $s(g)(x)$ and $r(g)(x)$ using (6.13.2), this expression is visibly equal to $r(g)(x)$. This proves injectivity.

Finally, prove surjectivity. Let $s^\flat \in Z^1(H, A)$. Write π for the quotient map $G \twoheadrightarrow H\backslash G$, and choose any map $\sigma : H\backslash G \to G$ satisfying $\pi\sigma = \mathrm{id}$. Without loss of generality, assume $\sigma(H \cdot 1_G) = 1_G$. Every $g \in G$ has a unique representation

$$g = \tau(g)\sigma(Hg), \quad \tau(g) \in H.$$

Note that $\tau(1_G) = 1_G$. For every $g, x \in G$, define

$$s(g)(x) := {}^{\tau(x)}s^\flat\left(\tau(\sigma(Hx)g)\right) \in A.$$

If $h \in H$, then $\tau(hx) = h\tau(x)$ immediately gives $s(g)(hx) = {}^h s(g)(x)$, so $s(g) \in \mathrm{Ind}_H^G(A)$. The next step is to show that $s \in Z^1(G, \mathrm{Ind}_H^G(A))$. The reader is first asked to verify

$$\tau(uv) = \tau(u)\tau(\sigma(Hu)v), \quad u, v \in G. \tag{6.13.4}$$

We need to prove that $s(g_1g_2)(x) = s(g_1)(x) \cdot s(g_2)(xg_1)$ for all $g_1, g_2, x \in G$. By the definition of $Z^1(H, A)$, the left-hand side equals

$$\begin{aligned}
{}^{\tau(x)}s^\flat\left(\tau(\sigma(Hx)g_1g_2)\right) &\xlongequal{(6.13.4)} {}^{\tau(x)}s^\flat\left(\tau(\sigma(Hx)g_1)\tau(\sigma(Hxg_1)g_2)\right) \\
&= {}^{\tau(x)}s^\flat\left(\tau(\sigma(Hx)g_1)\right) \cdot {}^{\tau(x)\tau(\sigma(Hx)g_1)}s^\flat\left(\tau(\sigma(Hxg_1)g_2)\right) \\
&\xlongequal{(6.13.4)} {}^{\tau(x)}s^\flat\left(\tau(\sigma(Hx)g_1)\right) \cdot {}^{\tau(xg_1)}s^\flat\left(\tau(\sigma(Hxg_1)g_2)\right),
\end{aligned}$$

which is the right-hand side.

Moreover, if $h \in H$, then $s(h)(1_G) = s^\flat(h)$, so s maps to $s^\flat$. This proves surjectivity. □

Clearly the isomorphism in Theorem 6.13.10, when A is an abelian H-group, reduces to the special case of Proposition 6.4.5. Moreover, a direct computation shows that these isomorphisms are compatible with the connecting morphisms in Theorem 6.13.7; the details are left as an exercise.

Remark 6.13.11 (Nonabelian cohomology of profinite groups) Let G be a profinite group. An object A of G-$\mathsf{Set}_\bullet$ or G-Grp is called **smooth** if $A = \bigcup_K A^K$, where K ranges over the open normal subgroups of G; in this case A is given the discrete topology. Since all arguments in this section are based on analogies with the standard cochain complex, all the preceding results apply to a profinite group G and smooth A. The only difference is that $c \in Z^i(G, A)$, as a map from G to A, must be continuous, that is, locally constant. Thus $\mathrm{H}^i(G, A)$ and the various canonical morphisms can be defined for $i = 0, 1$, and

$$\begin{aligned}
Z^i(G, A) &= \varinjlim_K Z^i(G/K, A^K), \\
\mathrm{H}^i(G, A) &= \varinjlim_K \mathrm{H}^i(G/K, A^K), \quad i = 0, 1.
\end{aligned}$$

In addition, the definition of $\mathrm{Ind}_H^G(A)$ must require $f : G \to A$ to be locally constant. The proof of Theorem 6.13.10 also requires Lemma B.1.8 to ensure that the $\zeta : G \to A$ in the injectivity part can be chosen locally constant and that the $\sigma : H\backslash G \to G$ in the surjectivity part is continuous.

Finally, Remark 6.13.5 remains valid for profinite groups G: $Z^1(G, A)$ corresponds to the set of continuous sections of the projection homomorphism $\pi : A \rtimes G \to G$, while $\mathrm{H}^1(G, A)$ is the quotient of the set of continuous sections by the conjugation action of A.

Exercises

1. Let $\Bbbk$ be a commutative ring and $t \in \Bbbk$. For every $\Bbbk$-module, write m_t for the endomorphism given by multiplication by t. Let M be a G-module. Show that the endomorphisms induced by $m_t \in \mathrm{End}_G(M)$ on $\mathrm{H}^n(G, M)$ and $\mathrm{H}_n(G, M)$ are again precisely m_t.

Hint For cohomology, either write down the induced endomorphism on the standard complex, or take an injective resolution $0 \to M \to I^0 \to \cdots$ and show that the induced endomorphism comes from the diagram

$$\begin{tikzcd} 0 \arrow[r] & M \arrow[r] \arrow[d, "m_t"] & I^0 \arrow[r] \arrow[d, "m_t"] & I^1 \arrow[r] \arrow[d, "m_t"] & \cdots \\ 0 \arrow[r] & M \arrow[r] & I^0 \arrow[r] & I^1 \arrow[r] & \cdots . \end{tikzcd}$$

2. Let the G-module M be projective as a $\Bbbk$-module. Prove that $\mathsf{L} \otimes M \to \Bbbk \otimes M \simeq M$ gives a projective resolution of the G-module M; the same statement holds with $\overline{\mathsf{L}}$ in place of L.

Hint Apply Proposition 6.1.7.

3. Prove Proposition 6.2.7 using the universal coefficient theorem for the cohomology of chain complexes.

4. Let p be a prime, G a p-group, and M a G-module satisfying $pM = 0$. Prove that the following properties are equivalent: (i) $M = 0$, (ii) $M^G = 0$, (iii) $M_G = 0$.

Hint To prove (ii) $\implies$ (i), take $x \in M \smallsetminus \{0\}$ and consider the $\mathbb{F}_p$-vector subspace E of M generated by $\{gx : g \in G\}$. Use $|E^G| \equiv |E| \pmod p$ to show that $M^G \neq 0$.

To prove (iii) $\implies$ (i), take $M^\vee := \mathrm{Hom}_{\mathbb{F}_p}(M, \mathbb{F}_p)$ with its natural G-module structure. Observe that $(M^\vee)^G \simeq \mathrm{Hom}_{\mathbb{F}_p}(M_G, \mathbb{F}_p)$, and deduce $M_G = 0 \implies M^\vee = 0 \implies M = 0$.

5. Let F be a field. A **representation** of a group G is data (ρ, V), where V is an F-vector space, $\mathrm{GL}(V)$ is its group of linear automorphisms, and $\rho : G \to \mathrm{GL}(V)$ is a group homomorphism. Clearly, this is the same as a G-module with coefficients in F. A **projective representation** is data (π, V) with a homomorphism $\pi : G \to \mathrm{PGL}(V) :=$

$\mathrm{GL}(V)/F^\times\mathrm{id}$. Isomorphisms of representations or projective representations are defined in the evident way. Henceforth identify $F^\times$ with $F^\times\mathrm{id}$.

(i) For a projective representation (π, V) of G, suppose that for every $g \in G$ an inverse image $\widetilde{\pi}(g) \in \mathrm{GL}(V)$ of $\pi(g)$ has been chosen. Verify that

$$\widetilde{\pi}(g_1)\widetilde{\pi}(g_2) = c(g_1, g_2)\widetilde{\pi}(g_1 g_2)$$

defines $c \in Z^2(G, F^\times)$, where G acts trivially on $F^\times$. If $\widetilde{\pi}(1_G) = \mathrm{id}_V$, then $c \in \overline{Z}^2(G, F^\times)$.

(ii) Prove that the image of c in $\mathrm{H}^2(G, F^\times)$ depends only on the isomorphism class of (π, V) and not on any of the choices.

(iii) Consider the central extension of groups $1 \to F^\times \to \widetilde{G} \to G \to 1$ (Definition 6.6.2), where

$$\widetilde{G} := \{(g, \widetilde{\pi}) \in G \times \mathrm{GL}(V) : \pi(g) = \widetilde{\pi} \bmod F^\times\}.$$

The first homomorphism maps $t \in F^\times$ to $(1_G, t)$ and the second is projection. Show that this extension is isomorphic to the central extension determined by the cohomology class in (ii).

(iv) Give a bijection

$$\{\text{representations } (\sigma, V) \text{ lifting } (\pi, V)\} \xleftrightarrow{1:1} \{\text{splittings of the central extension } \widetilde{G}\}.$$

Show also that (π, V) lifts to a representation if and only if the cohomology class in (ii) is trivial.

(v) Explicitly give a group G and a projective representation (π, V) that cannot be lifted to a representation. Try to find an example with nontrivial $\pi : G \to \mathrm{PGL}(V)$.

6. For subgroups $G_1 \subset G_2 \subset G_3$, prove the canonical isomorphisms of functors $\mathrm{Ind}_{G_2}^{G_3} \mathrm{Ind}_{G_1}^{G_2} \simeq \mathrm{Ind}_{G_1}^{G_3}$ and $\mathrm{ind}_{G_2}^{G_3} \mathrm{ind}_{G_1}^{G_2} \simeq \mathrm{ind}_{G_1}^{G_3}$. Describe these isomorphisms as explicitly as possible.

7. Prove that if H is a subgroup of G, then $\mathrm{cd}(H) \leq \mathrm{cd}(G)$ and $\mathrm{hd}(H) \leq \mathrm{hd}(G)$; see Definition 6.7.8. Hint Apply Theorem 6.4.3.

8. Show that for $m \in \mathbb{Z}_{\geq 1}$, $\mathrm{cd}(C_m) = \mathrm{hd}(C_m) = \infty$. Prove that $\mathrm{cd}(G) < \infty$ implies that G is torsion-free, and that $\mathrm{cd}(G) = 0$ if and only if G is the trivial group. Hint For the first part, apply Proposition 6.7.5 with $A = \mathbb{Z}/m\mathbb{Z}$. For the other parts, consider the cyclic subgroups of G.

9. Try to prove Corollary 6.8.6 directly by operations on the standard complex.

10. Let U, V be subgroups of G and suppose that $(G : V)$ is finite. Write the cohomological restriction map as res_U^G and the corestriction map as cor_G^V, and so on; the cohomological degree n is omitted from the notation. Choose a double-coset decomposition

$$G = \bigsqcup_{\sigma \in S} U\sigma V,$$

where the finite set $S \subset G$ consists of representatives of the double cosets. Let M be a G-module.

(i) For every $\sigma \in G$ and $n \in \mathbb{Z}_{\geq 0}$, define in a natural way

$$\mathcal{T}_\sigma : \mathrm{H}^n(V \cap \sigma^{-1} U \sigma, M) \to \mathrm{H}^n(U \cap \sigma V \sigma^{-1}, M)$$

so that for $n = 0$ it becomes

$$M^{V \cap \sigma^{-1} U \sigma} \xrightarrow{\sim} M^{U \cap \sigma V \sigma^{-1}}, \qquad x \mapsto \sigma x.$$

(ii) Prove that

$$\mathrm{res}^G_U \circ \mathrm{cor}^V_G = \sum_{\sigma \in S} \mathrm{cor}^{U \cap \sigma V \sigma^{-1}}_U \circ \mathcal{T}_\sigma \circ \mathrm{res}^V_{V \cap \sigma^{-1} U \sigma}.$$

(iii) Describe the action of $\mathrm{res}^G_U \circ \mathrm{cor}^U_G$ when $U = V \lhd G$.

11. Let G be an abelian group, regarded as a $\mathbb{Z}$-module. Define $G \wedge G$ as the quotient of $G \underset{\mathbb{Z}}{\otimes} G$ by the submodule generated by $\{g \otimes g : g \in G\}$, and write $x \wedge y$ for the image of $x \otimes y$. Prove that $\mathrm{H}_2(G, \mathbb{Z}) \simeq G \wedge G$.

Hint〉 Once a presentation $G = F/R$ of the group has been fixed, the Hopf formula (Corollary 6.9.4) gives $\mathrm{H}_2(G, \mathbb{Z}) \simeq [F, F]/[F, R]$. Since G is abelian, $[F, F] \subset R$. From the identity $[xy, z] = x[y, z]x^{-1} \cdot [x, z]$, the map

$$\psi_0 : G \times G \to [F, F]/[F, R], \qquad (f_1 R, f_2 R) \mapsto [f_1, f_2] \bmod [F, R]$$

is $\mathbb{Z}$-bilinear and induces a surjective homomorphism $\psi : G \wedge G \to [F, F]/[F, R]$. Choose a basis $\{x_i\}_{i \in I}$ of F and give I a total order. We need the following group-theoretic fact: $H := [F, F]/[[F, F], F]$ is a free abelian group with basis the cosets of $[x_i, x_j]$ for $i < j$. We may therefore define $\phi_0 : H \to G \wedge G$ by mapping the coset of $[x_i, x_j]$ to $x_i R \wedge x_j R$. Show that

$$f_1, f_2 \in F \implies \phi_0(\text{the coset of } [f_1, f_2]) = f_1 R \wedge f_2 R.$$

From this, descend ϕ_0 to a homomorphism $\phi : [F, F]/[F, R] \to G \wedge G$, and then show that ϕ and ψ are inverse to one another.

The group-theoretic fact above for finite I appears in [17, Theorem 11.2.4]; the general case follows from it.

12. Let $H \lhd G$ and let M be a G-module. Give the standard complex $C(H, M^H)$ its natural G/H-action, inducing a canonical G/H-action on every $\mathrm{H}^n(H, M^H)$ (see the discussion preceding Lemma 6.9.2).

13. For the filtration in Remark 6.9.5, describe the E_2 page of its spectral sequence and show that the properties in Theorem 6.9.3 continue to hold. Explain the compatibility of the spectral sequence with the cup product as fully as possible.

14. Consider the finite cyclic group C_m of order m. For every C_m-module A and $n \in \mathbb{Z}_{\geq 1}$, prove that the periodicity isomorphism $\mathrm{H}^n(C_m, A) \to \mathrm{H}^{n+2}(C_m, A)$ in Proposition 6.7.5 equals the composite of the two connecting homomorphisms from the canonical short exact sequences in Example 6.10.14:

$$0 \to \mathfrak{I} \otimes A \to \mathbb{Z}[C_m] \otimes A \to A \to 0,$$
$$0 \to A \xrightarrow{\nu \otimes \mathrm{id}} \mathbb{Z}[C_m] \otimes A \xrightarrow{(\sigma-1)\otimes \mathrm{id}} \mathfrak{I} \otimes A \to 0.$$

Hint〉 Compute cohomology with the periodic complex $A \xrightarrow{\sigma-1} A \xrightarrow{\nu} A \to \cdots$ from the proof of Proposition 6.7.5, starting in degree zero, and use it to describe the connecting homomorphisms. Treat the odd and even cases separately; some computation is required.

15. Let $\Bbbk$ be any commutative ring and $m \in \mathbb{Z}_{\geq 1}$. Determine the structure of the cohomology algebra $\mathrm{H}^\bullet(C_m, \Bbbk)$ as a graded $\Bbbk$-algebra.

16. For the cap product introduced in Remark 6.10.15, prove the following adjunction relation with the cup product:

$$\langle \alpha \cup \beta, x \rangle = \langle \alpha, \beta \cap x \rangle \in (M_1 \otimes M_2 \otimes M_3)_G,$$

where M_1, M_2, M_3 are G-modules, $\alpha \in \mathrm{H}^p(G, M_1)$, $\beta \in \mathrm{H}^q(G, M_2)$, and $x \in \mathrm{H}_{p+q}(G, M_3)$.

17. In this problem G is a finite group, the coefficient ring in cohomology and homology is $\mathbb{Z}$, and a free G-module means a free $\mathbb{Z}[G]$-module. Write the contragredient module of a G-module M as $M^* := \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Z})$.

(i) Take a free resolution $\epsilon : P \to \mathbb{Z}$ of the trivial G-module, where $P := [\cdots \to P_n \to P_{n-1} \to \cdots]$ and every P_n is finitely generated. Define $P^* := [\cdots \to P_n^* \to P_{n+1}^* \to \cdots]$. Show that $\epsilon^* : \mathbb{Z} \to P^*$ is a quasi-isomorphism. Hint〉 Every P_n is a free $\mathbb{Z}$-module. If the chain complex of free $\mathbb{Z}$-modules $(C_n, \partial_n)_{n\in\mathbb{Z}}$ is exact and $Z_n := \ker(\partial_n)$, then the short exact sequence $0 \to Z_{n+1} \to C_{n+1} \xrightarrow{\partial_n} Z_n \to 0$ splits for every n.

(ii) Prove that every P_n^* is a finitely generated projective G-module. Hint〉 It is enough to show that $\mathbb{Z}[G]^* \simeq \mathbb{Z}[G]$.

(iii) Let Q be a finitely generated projective G-module. For every G-module M, prove that the canonical morphism $\nu : (Q \otimes M)_G \to (Q \otimes M)^G$ from (6.11.1) is an isomorphism. Hint〉 Use a direct-sum decomposition to reduce to the case $Q = \mathbb{Z}[G]$, and then verify it directly. For greater simplicity, the automorphism

$$\left(\sum_{g\in G} a_g g\right) \otimes x \mapsto \sum_{g\in G} a_g (g \otimes g^{-1}x)$$

turns the diagonal G-action on $\mathbb{Z}[G] \otimes M$ into left multiplication.

(iv) Define $P_{-n} := P_{n-1}^*$ and splice P and P^* into the chain complex

$$\mathcal{P} := [\cdots \to P_1 \to P_0 \xrightarrow{\epsilon^*\epsilon} P_{-1} \to P_{-2} \to \cdots].$$

Prove that for every G-module M there is a canonical isomorphism

$$\mathrm{H}^n \operatorname{Hom}_G(\mathcal{P}, M) \simeq \text{Tate cohomology } \hat{\mathrm{H}}^n(G, M), \qquad n \in \mathbb{Z}.$$

Hint For $n \geq 1$ the statement is clear. For $n \leq -2$, use the following property: if Q is a finitely generated projective G-module, then $Q \otimes M \xrightarrow{\sim} \operatorname{Hom}(Q^*, M)$. Deduce that

$$(Q \otimes M)_G \underset{\nu}{\xrightarrow{\sim}} (Q \otimes M)^G \xrightarrow{\sim} \operatorname{Hom}(Q^*, M)^G = \operatorname{Hom}_G(Q^*, M).$$

For $n = 0, -1$, identify $\operatorname{Hom}_G(P_{-1}, M) \to \operatorname{Hom}_G(P_0, M)$ by means of the isomorphism above with

$$(P_0 \otimes M)_G \xrightarrow{(\epsilon \otimes \mathrm{id})_G} M_G \xrightarrow{\nu} M^G \xrightarrow{\epsilon^*} \operatorname{Hom}_G(P_0, M).$$

(v) Use this result to prove the long exact sequence in Tate cohomology directly.

(vi) Take the free resolution $\epsilon : P \to \mathbb{Z}$ from Proposition 6.1.15, choose its natural basis for every P_n, and then explicitly describe the chain complex $\mathcal{P}$ from (iv).

Based on (vi), a cup product on Tate cohomology can be defined as in Lemma 6.10.8, by appropriately defining $\Delta : \mathcal{P} \to \mathcal{P} \otimes \mathcal{P}$. For explicit formulas, see [9, Chapter IV, §7].

18. Let G be a finite group and H a subgroup. For a G-module M, use dimension shifting (Corollary 6.11.6) to extend the canonical morphisms from §6.8, $\mathrm{res}^n : \hat{\mathrm{H}}^n(G, M) \to \hat{\mathrm{H}}^n(H, M)$ and $\mathrm{res}_n : \hat{\mathrm{H}}^{-n-1}(H, M) \to \hat{\mathrm{H}}^{-n-1}(G, M)$, from the case $n \geq 1$ to all $n \in \mathbb{Z}$. For the moment, denote the former by res_+ and the latter by res_-.

(i) Describe the action of res_+ on $\hat{\mathrm{H}}^0$ and the action of res_- on $\hat{\mathrm{H}}^{-1}$.

(ii) Prove that res_+ on $\hat{\mathrm{H}}^{-1}$ is induced by $\nu_{G|H} : M_G \to M_H$ (Definition 6.8.3), and on $\hat{\mathrm{H}}^{<-1}$ equals cor_n (Definition–Proposition 6.8.4).

(iii) Prove that res_- on $\hat{\mathrm{H}}^0$ is induced by $\nu^{G|H} : M^H \to M^G$, and on $\hat{\mathrm{H}}^{>0}$ equals cor^n.

19. For a group G and a bounded-below complex of G-modules $M = [\cdots \to M^p \to M^{p+1} \to \cdots]$, let p vary in the standard complexes $C(G, M^p)$ to form a double complex. Denote its total complex by $C(G, M) := \mathrm{tot}((C^q(G, M^p))_{p,q})$. Prove the canonical isomorphism

$$\mathrm{H}^n(C(G, M)) \simeq \mathrm{H}^n(G, [\cdots \to M^p \to M^{p+1} \to \cdots]), \qquad n \in \mathbb{Z},$$

where the right-hand side is defined as in §3.12 and is also called group hypercohomology.

Hint By definition, the right-hand side is obtained by taking an injective resolution $M \to I$ (recall that I is a bounded-below complex, every I^q is an injective G-module, and $M \to I$ is a quasi-isomorphism) and then taking the cohomology determined by the Hom complex $\operatorname{Hom}_G^\bullet(\Bbbk, I)$. Show that the same cohomology can be computed by $\operatorname{Hom}_G^\bullet(\mathsf{L}, M)$.

20. Prove the corresponding result when G is a profinite group, requiring every M^p to be a smooth G-module and using the continuous standard complexes $C(G, M^p)$.

Hint〉 Define a functor $C(G, \cdot)$ from $\mathsf{C}^+(G\text{-}\mathsf{Mod}^\infty)$ to $\mathsf{C}^+(\Bbbk\text{-}\mathsf{Mod})$. Show that it induces a triangulated functor at the level of $\mathsf{K}^+(\cdots)$ and preserves acyclic objects, and hence induces a triangulated functor at the level of $\mathsf{D}^+(\cdots)$, still denoted by $C(G, \cdot)$. The natural transformation $(\cdot)^G \to C(G, \cdot)$ and the universal property of right derived functors give a morphism of triangulated functors $\mathrm{R}(\cdot)^G \to C(G, \cdot)$. To prove that this is an isomorphism, use the way-out lemma (Proposition 4.5.7) to reduce to the case handled by Theorem 6.12.9.

21. For a profinite group G and a finite G-module M, give a complete proof of the bijective correspondence mentioned in §6.12 between equivalence classes of profinite group extensions and $\mathrm{H}^2(G, M)$.

Hint〉 Follow the argument in §6.3. In the profinite case the required observations are: (i) the group structure defined by every $f \in \overline{Z}^2(G, M)$ on the product space $M \times G$ makes it a profinite group; (ii) for a profinite group extension E, there is an isomorphism of topological groups $E/M \xrightarrow{\sim} G$; (iii) in this situation, Lemma B.1.8 gives a continuous section $s : G \to E$ that can be adjusted to be normalized.

22. For any field F, define $\mathrm{GL}(n, F)$ to be the group of invertible $n \times n$ matrices over F, and $\mathrm{SL}(n, F)$ to be the subgroup cut out by $\det = 1$. Observe that $G := \mathrm{Gal}(\mathbb{C}|\mathbb{R})$ acts on $\mathrm{GL}(n, \mathbb{C})$ and $\mathrm{SL}(n, \mathbb{C})$, with fixed points $\mathrm{GL}(n, \mathbb{R})$ and $\mathrm{SL}(n, \mathbb{R})$, respectively.

(i) Take $n = 2$. Let $B := \mathrm{SL}(2, \mathbb{C})$ act by conjugation on the space of all 2×2 matrices, and define $A := \mathrm{Stab}\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Show that this subgroup is invariant under the G-action and describe A^G.

(ii) Show that B/A (respectively, B^G/A^G) is naturally identified with the orbit of $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ under conjugation by $\mathrm{SL}(2, \mathbb{C})$ (respectively, $\mathrm{SL}(2, \mathbb{R})$). For B/A, explain how the G-action is reflected on the orbit.

(iii) Show that $(B/A)^G \neq B^G/A^G$. In other words, there is a real 2×2 matrix that is conjugate to $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ by $\mathrm{SL}(2, \mathbb{C})$ but not by $\mathrm{SL}(2, \mathbb{R})$.

(iv) Show that the phenomenon above does not occur if SL is replaced by GL.

23. State and prove the compatibility between the isomorphisms in Theorem 6.13.10 and the connecting homomorphisms in Theorem 6.13.7.

24. ("Twisting" group cohomology) Use the notation of §6.13. Let A be a G-group.

(i) For $c \in Z^1(G, A)$, define a new group action $G \times A \to A$ by

$$ {}^{c,g}a := c(g) \cdot {}^{g}a \cdot c(g)^{-1}, \qquad g \in G,\ a \in A. $$

Show that this indeed gives a G-group, denoted by ${}^{c}A$. Prove also that if $c \sim c'$, there is an isomorphism of G-groups ${}^{c'}A \simeq {}^{c}A$; describe it explicitly.

(ii) Prove that every $c \in Z^1(G, A)$ induces a bijection

$$\begin{aligned} \nu_c : Z^1(G, {}^cA) &\xrightarrow{1:1} Z^1(G, A), \\ c' &\longmapsto c'c \quad \text{(pointwise multiplication)}, \end{aligned}$$

which induces a bijection $\boldsymbol{\nu}_c : \mathrm{H}^1(G, {}^cA) \xrightarrow{1:1} \mathrm{H}^1(G, A)$ that maps the basepoint to $[c]$. Give a simplified description when A is abelian.

(iii) Suppose there is a short exact sequence $1 \to A \to B \to B/A \to 1$ as in Theorem 6.13.7 (i). For $c \in Z^1(G, A)$, prove that the following diagram commutes:

$$\begin{tikzcd} \mathrm{H}^1(G, {}^cA) \arrow[r] \arrow[d, "\boldsymbol{\nu}_c"'] & \mathrm{H}^1(G, {}^cB) \arrow[d, "\boldsymbol{\nu}_c"] \\ \mathrm{H}^1(G, A) \arrow[r] & \mathrm{H}^1(G, B). \end{tikzcd}$$

Use this to describe all fibers of the map $\mathrm{H}^1(G, A) \to \mathrm{H}^1(G, B)$.

(iv) State the modifications required when G is a profinite group.

Chapter 7 Monad Theory

The starting point of this chapter, §7.1, is the notion of an algebra over a monoidal category $(\mathcal{V}, \otimes)$. This means an object $A \in \mathrm{Ob}(\mathcal{V})$ equipped with a multiplication $\mu : A \otimes A \to A$ and a unit morphism $\eta : \mathbf{1} \to A$, satisfying the associative and unit laws. The concept is modeled on that of a monoid, while algebras over a commutative ring $\Bbbk$ are the typical examples. These algebras form a category $\mathsf{Alg}(\mathcal{V})$. When $(\mathcal{V}, \otimes)$ has stronger commutativity properties, $\mathsf{Alg}(\mathcal{V})$ acquires additional structure and properties. The best situation is when $\mathcal{V}$ is a symmetric monoidal category; then $\mathsf{Alg}(\mathcal{V})$ is itself a symmetric monoidal category.

For an algebra A, a left A-module is an object M of $\mathcal{V}$ together with a scalar-multiplication morphism $\mu_M : A \otimes M \to M$ satisfying the associative and unit laws; right modules and bimodules are defined similarly. Since complexes are the principal objects of this book, a natural step is to take an abelian category $\mathcal{A}$ with countable coproducts and a right-exact additive bifunctor $\otimes : \mathcal{A} \times \mathcal{A} \to \mathcal{A}$, extend $\otimes$ to complexes by taking total complexes, and then consider the monoidal category $(\mathsf{C}(\mathcal{A}), \otimes)$, usually endowed with the Koszul braiding (7.1.3). Differential graded algebras and modules thus arise naturally as a continuation of the theory; "dg" abbreviates "differential graded." These are the subject of §7.2.

We also know that the Hom sets in $\mathsf{C}(\mathcal{A})$ can be upgraded to Hom complexes, and that composition can likewise be lifted to the level of Hom complexes. In particular, the endomorphism complex $\mathrm{End}^\bullet(X)$ of an object X becomes a dg-algebra. Generalizing these observations leads to the notion of a dg-category (Definition 7.2.4), and one can speak of dg-functors between dg-categories. After a brief introduction to closed monoidal categories in §7.3, we shall explain in §7.4 that the category dgCat of all small dg-categories is closed. Equivalently, all dg-functors between two small dg-categories $\mathcal{C}$ and $\mathcal{D}$ themselves form a dg-category $\mathcal{H}\mathrm{om}(\mathcal{C}, \mathcal{D})$.

The various dg-structures arise naturally and are important tools in linear algebra. This chapter gives only a brief introduction, however, and §7.5 turns instead to coalgebras and comodules over a monoidal category $\mathcal{V}$. These are simply algebras and modules over the opposite monoidal category $\mathcal{V}^{\mathrm{op}}$. More concretely, a coalgebra C is equipped with a comultiplication $\Delta : C \to C \otimes C$ and a counit morphism $\epsilon : C \to \mathbf{1}$, while a left comodule M is equipped with a scalar-comultiplication morphism $\rho : M \to C \otimes M$; their associative and unit laws are obtained by reversing all the arrows.

If $\mathcal{V}$ is a braided monoidal category, mutually compatible algebra and coalgebra structures $(A, \mu, \eta, \Delta, \epsilon)$ on an object A are called a bialgebra, and a bialgebra equipped with an antipode is called a Hopf algebra (Definition 7.5.8). In the concrete case $\mathcal{V} = \Bbbk\text{-}\mathsf{Mod}$, these concepts are inspired by geometry and topology. For example, the cohomology of a Lie group, or more generally of an H-group, naturally becomes a Hopf algebra; quantum groups form another important class of examples. Moreover, the group algebra $\Bbbk[G]$ of any group G is also a Hopf algebra.

We now introduce the protagonist of §7.6, which also gives this chapter its title: the monad. For any category $\mathcal{C}$, the endofunctor category $\mathcal{C}^{\mathcal{C}}$ is a strict monoidal category under composition, and a monad is simply an algebra in $\mathcal{C}^{\mathcal{C}}$; a comonad is its dual version. Simple though it is, this concept plays a major role in algebra, geometry, and computer science. One notable use of monads is to explain the various "standard resolutions" of homological algebra, such as $\mathsf{B}R$ from §3.8 or L from §6.1. We shall investigate this aspect in §8.7.

Monads and comonads arise chiefly from adjoint pairs

$$F : \mathcal{C} \rightleftarrows \mathcal{D} : G.$$

Such a pair determines a monad T on $\mathcal{C}$ and a comonad L on $\mathcal{D}$, expressed in terms of the unit η and counit ε of the adjunction as

monad	comonad
$T := GF$ $\mu := \left[T^2 \xrightarrow{G\varepsilon F} T\right]$ $\eta : \mathrm{id}_{\mathcal{C}} \to T$	$L := FG$ $\delta := \left[L \xrightarrow{F\eta G} L^2\right]$ $\varepsilon : L \to \mathrm{id}_{\mathcal{D}}$

The adjoint pair naturally induces a functor $\mathbb{K} : \mathcal{D} \to \mathcal{C}^T$ (or $\mathcal{C} \to \mathcal{D}^L$), where $\mathcal{C}^T$ (or $\mathcal{D}^L$) denotes the category of modules under the action of the monad (or its comonadic counterpart). If $\mathbb{K}$ is an equivalence, the adjoint pair is called monadic (or comonadic). This means that the information in $\mathcal{D}$ (or $\mathcal{C}$) can be reconstructed from the action of the monad (or comonad) on the other side. Beck's Theorem 7.6.14 gives a practical characterization of monadicity.

The next step is Morita theory in §7.7, a classical theme in ring theory. The first part considers functors between categories of left or right modules that preserve small $\varinjlim$ (or $\varprojlim$). Theorem 7.7.2 states that they all arise by taking the tensor product (or Hom) with a bimodule P. This also confirms the special position of bimodules in ring and module theory: they provide enough means to "change rings." One thereby obtains a module-theoretic characterization of equivalences between the categories of left modules A-Mod and B-Mod, called Morita equivalences (Definition–Proposition 7.7.4); the characterizations for left and right modules are entirely parallel.

For the adjoint pair determined by an (A, B)-bimodule P, the second part examines the associated monad and comonad and their modules. Assuming that P is a finitely generated projective B-module, Proposition 7.7.12 explicitly presents them as an algebra in the monoidal category (A, A)-Mod and a coalgebra in (B, B)-Mod, respectively. From this foundation one readily obtains a refinement of Morita equivalence (Theorem 7.7.16, Corollary 7.7.17).

Morita theory concerns given module categories. As a complement, in §7.8 we study when an abelian category $\mathcal{A}$ is equivalent to the category of right modules over a ring R. The answer involves a compact projective generator of $\mathcal{A}$ (Theorem 7.8.4). We shall also give a sufficient condition for the finitely generated version (Theorem 7.8.5).

Returning to Beck's monadicity theorem, the first half of §7.9 gives a simple application in module theory, called faithfully flat descent for modules. For a homomorphism of commutative rings $K \to L$, it explains how to reconstruct a K-module from an L-module carrying an action of the change-of-rings comonad $\mathbf{L}$. Descent techniques are widely used in geometry, and the second half of §7.9 reformulates the condition of carrying an $\mathbf{L}$-action in the form customarily used in geometry. All of this rests on the preparation in §7.7.

As a special case of faithfully flat descent, §7.10 takes $L|K$ to be a Galois extension of fields and reformulates the preceding result in terms of the action of the Galois group $\mathrm{Gal}(L|K)$. This technique is called Galois descent. Its main result, Theorem 7.10.7, says that the category of K-vector spaces is equivalent to the category of L-vector spaces equipped with a smooth semilinear action of $\mathrm{Gal}(L|K)$. The exercises give another direct proof.

On the basis of faithfully flat descent, or its special case of Galois descent, one can further descend various structures such as algebras and modules. In §7.11 we consider a kind of data expressed in terms of vector spaces and their tensor powers. Galois descent and group cohomology then reduce the classification of such data to nonabelian group cohomology H^1. Although the purpose is only to illustrate the technique, it already helps clarify several basic questions in algebra, such as the classification of nondegenerate quadratic forms.

Reading Guide

This chapter falls roughly into three parts. The first (§§7.1—7.5) introduces algebras over monoidal categories, dg-structures, and Hopf algebras; these topics occupy a natural and important place in algebra. The second (§§7.6—7.8) centers on monads and the related Morita theory. The third (§§7.9—7.11) discusses descent. The three parts are arranged more or less sequentially, although the dg-structures and dg-categories are not needed for what follows. Chapter 8 and Chapter 9 of this book use several results from the second part. Further material appears in the exercises for this chapter.

7.1 Algebras over Monoidal Categories

For a fixed commutative ring $\Bbbk$, we know that a $\Bbbk$-algebra is a multiplicative structure superimposed on a $\Bbbk$-module. Its definition can be expressed using the tensor-product bifunctor $\otimes = \otimes_{\Bbbk}$ on $\Bbbk\text{-}\mathsf{Mod}$ and commutative diagrams. The essential point is that $(\Bbbk\text{-}\mathsf{Mod}, \otimes)$ is a monoidal category with unit $\mathbf{1} := \Bbbk$. When discussing commutative $\Bbbk$-algebras and their variants, the central role is played by the braiding $c(X,Y) : X \otimes Y \xrightarrow{\sim} Y \otimes X$ on this monoidal category, which has the symmetry property $c(Y,X)c(X,Y) = \mathrm{id}$. See [51, §7.4].

Once formulated in the appropriate language, this theory encompasses general **monoidal categories** and **symmetric monoidal categories**. Following [51, Definitions 3.1.1 and 3.1.2], the convention in this section[1)] defines a monoidal category to be data $(\mathcal{V}, \otimes, a, \mathbf{1}, \iota)$, often abbreviated to $\mathcal{V}$ or $(\mathcal{V}, \otimes)$, where

- ⋄ $\otimes : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$ is a bifunctor;
- ⋄ $a(X,Y,Z) : (X \otimes Y) \otimes Z \xrightarrow{\sim} X \otimes (Y \otimes Z)$ is a canonical morphism called the associativity constraint $(X,Y,Z \in \mathrm{Ob}(\mathcal{V}))$, and it satisfies the pentagon axiom;

1) Definitions in other sources may differ slightly, but the resulting concepts of a monoidal category are equivalent.

◇ $\mathbf{1} = \mathbf{1}_{\mathcal{V}} \in \mathrm{Ob}(\mathcal{V})$ is the unit;

◇ $\iota : \mathbf{1} \otimes \mathbf{1} \xrightarrow{\sim} \mathbf{1}$.

This gives a family of canonical isomorphisms $\mathbf{1} \otimes X \underset{\sim}{\xrightarrow{\lambda_X}} X \underset{\sim}{\xleftarrow{\rho_X}} X \otimes \mathbf{1}$; by [51, Lemma 3.1.5], these isomorphisms satisfy $\lambda_{\mathbf{1}} = \iota = \rho_{\mathbf{1}}$. They are all called unit constraints and will no longer be displayed explicitly.

A **braiding** on a monoidal category is an isomorphism of bifunctors, written

$$c = \left(c(X,Y) : X \otimes Y \xrightarrow{\sim} Y \otimes X \right)_{X,Y \in \mathrm{Ob}(\mathcal{V})},$$

required to be compatible with the associativity and unit constraints; see [51, Definition 3.3.1]. A monoidal category equipped with a specified braiding is called a **braided monoidal category**. If $c(Y,X)c(X,Y) = \mathrm{id}_{X \otimes Y}$ always holds, the data are called a **symmetric monoidal category**.

Definition 7.1.1 (Algebra) An algebra over a monoidal category $\mathcal{V}$ means data (A, μ, η), where

$$A \in \mathrm{Ob}(\mathcal{V}), \quad \mu = \mu_A : A \otimes A \to A, \quad \eta = \eta_A : \mathbf{1} \to A,$$

such that the following diagrams commute:

$$\begin{tikzcd} \mathbf{1} \otimes A \arrow[r, "\eta\otimes\mathrm{id}"] \arrow[dr, "\sim"'] & A \otimes A \arrow[d, "\mu"] & A \otimes \mathbf{1} \arrow[l, "\mathrm{id}\otimes\eta"'] \arrow[dl, "\sim"] \\ & A & \end{tikzcd} \qquad \begin{tikzcd} (A \otimes A) \otimes A \arrow[rr, "a(A,A,A)", "\sim"'] \arrow[d, "\mu\otimes\mathrm{id}"'] & & A \otimes (A \otimes A) \arrow[d, "\mathrm{id}\otimes\mu"] \\ A \otimes A \arrow[r, "\mu"'] & A & A \otimes A. \arrow[l, "\mu"] \end{tikzcd}$$

Thus μ may be viewed as a multiplication on A, and η as its unit morphism.

A morphism from an algebra (A, μ, η) to (A', μ', η') is defined to be a morphism $\phi : A \to A'$ in $\mathcal{V}$ that makes the following diagrams commute:

$$\begin{tikzcd} \mathbf{1} \arrow[r, "\eta"] \arrow[dr, "\eta'"'] & A \arrow[d, "\phi"] \\ & A' \end{tikzcd} \qquad \begin{tikzcd} A \otimes A \arrow[r, "\phi\otimes\phi"] \arrow[d, "\mu"'] & A' \otimes A' \arrow[d, "\mu'"] \\ A \arrow[r, "\phi"'] & A'. \end{tikzcd}$$

We often abbreviate (A, μ, η) to A. In some sources an algebra is also called a ring or a monoid in $\mathcal{V}$, and it is sometimes called an associative algebra to emphasize the associative law displayed by the second commutative diagram.

The concept of a module admits a similar generalization.

Definition 7.1.2 (Module) Let (A, μ, η) be an algebra over $\mathcal{V}$. A left A-module is defined to be data (M, μ_M), where

$$M \in \mathrm{Ob}(\mathcal{V}), \quad \mu_M : A \otimes M \to M,$$

that make the following diagrams commute:

$$
\begin{tikzcd}
\mathbf{1} \otimes M \arrow[r, "\eta\otimes\mathrm{id}"] \arrow[dr, "\sim"'] & A \otimes M \arrow[d, "\mu_M"] \\
 & M
\end{tikzcd}
\qquad
\begin{tikzcd}
(A \otimes A) \otimes M \arrow[rr, "a(A,A,M)", "\sim"'] \arrow[d, "\mu_A\otimes\mathrm{id}"'] & & A \otimes (A \otimes M) \arrow[d, "\mathrm{id}\otimes\mu_M"] \\
A \otimes M \arrow[r, "\mu_M"'] & M & A \otimes M. \arrow[l, "\mu_M"]
\end{tikzcd}
$$

Thus μ_M may be viewed as scalar multiplication on the module.

As usual, the module (M, μ_M) is abbreviated to M. A morphism from M to M' is defined to be a morphism $\psi : M \to M'$ in $\mathcal{V}$ that makes the following diagram commute:

$$
\begin{tikzcd}
A \otimes M \arrow[r, "\mathrm{id}\otimes\psi"] \arrow[d, "\mu_M"'] & A \otimes M' \arrow[d, "\mu_{M'}"] \\
M \arrow[r, "\psi"'] & M'.
\end{tikzcd}
$$

Right A-modules are defined similarly. If M carries both a left A-module structure and a right B-module structure that make the following diagram commute,

$$
\begin{tikzcd}
(A \otimes M) \otimes B \arrow[rr, "a(A,M,B)", "\sim"'] \arrow[d, "\mu^A_M\otimes\mathrm{id}"'] & & A \otimes (M \otimes B) \arrow[d, "\mathrm{id}\otimes\mu^B_M"] \\
M \otimes B \arrow[r, "\mu^B_M"'] & M & A \otimes M, \arrow[l, "\mu^A_M"]
\end{tikzcd}
$$

then M is called an (A, B)-bimodule. A morphism between two such bimodules is a morphism $\psi : M \to M'$ that is simultaneously a morphism of left A-modules and of right B-modules.

If M is a left A-module and N a right B-module, then $M \otimes N$ naturally becomes an (A, B)-bimodule.

Remark 7.1.3 (The unit as an algebra) For example, $\mathbf{1}$ becomes an algebra with $\mu_{\mathbf{1}} := \iota$ and $\eta_{\mathbf{1}} := \mathrm{id}_{\mathbf{1}}$. The required commutative diagrams follow from the standard properties of $\mathbf{1}$ and $\iota : \mathbf{1} \otimes \mathbf{1} \xrightarrow{\sim} \mathbf{1}$; see [51, Lemma 3.1.5]. For every algebra A there is exactly one morphism $\mathbf{1} \to A$, and it must be η_A.

Similarly, it is easy to see that every $M \in \mathrm{Ob}(\mathcal{V})$ has exactly one left (or right) $\mathbf{1}$-module structure, obtained by taking $\mu_M : \mathbf{1} \otimes M \to M$ to be the unit constraint of $\mathcal{V}$.

Our next goal is to discuss (a) commutativity of algebras, (b) their tensor products. The two are closely related, and both require a braiding on $\mathcal{V}$. We begin with commutativity.

Definition 7.1.4 The natural properties of the braiding show that, if A is an algebra over a braided monoidal category $\mathcal{V}$, then replacing μ by $\mu \circ c(A, A)$ while leaving all other data unchanged again gives an algebra over $\mathcal{V}$. This algebra is defined to be the **opposite algebra** A^{op} of A. The verification is lengthy but not difficult, and is left as an exercise for this chapter.

By functoriality of the braiding, if $\phi : A_1 \to A_2$ is a morphism, then it is also a morphism from A_1^{op} to A_2^{op}.

Definition 7.1.5 Let A be an algebra over a braided monoidal category $\mathcal{V}$. It is called a **commutative algebra** if the following diagram commutes in $\mathcal{V}$:

$$\begin{tikzcd}
A \otimes A \arrow[rr, "c(A,A)", "\sim"'] \arrow[dr, "\mu_A"'] & & A \otimes A. \arrow[dl, "\mu_A"] \\
& A &
\end{tikzcd}$$

In other words, we require the multiplication μ_A to satisfy the commutative law. Equivalently, $A = A^{\mathrm{op}}$.

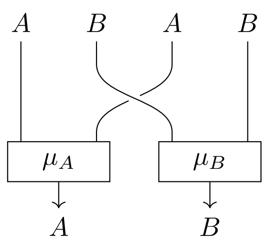
The algebra $\mathbf{1}$ of Remark 7.1.3 is a simple example of a commutative algebra. The required commutative diagram follows from the compatibility of the braiding c with the unit constraints.

Continue to suppose that $\mathcal{V}$ is a braided monoidal category. Just as over a commutative ring, for any two algebras A and B the braiding c gives a canonical algebra structure on $A \otimes B$:

- The multiplication $\mu_{A\otimes B}$ is the composite

$$\begin{aligned}
(A \otimes B) \otimes (A \otimes B) &\xrightarrow{\sim} (A \otimes (B \otimes A)) \otimes B \\
&\xrightarrow{(\mathrm{id}_A \otimes c(B,A)) \otimes \mathrm{id}_B} (A \otimes (A \otimes B)) \otimes B \\
&\xrightarrow{\sim} (A \otimes A) \otimes (B \otimes B) \xrightarrow{\mu_A \otimes \mu_B} A \otimes B,
\end{aligned}$$

  where all unlabeled arrows come from associativity constraints. In braid diagram notation this composite is depicted as follows:



- The unit $\eta_{A\otimes B}$ is the composite

$$\mathbf{1} \xrightarrow[\sim]{\iota^{-1}} \mathbf{1}\otimes\mathbf{1} \xrightarrow{\eta_A\otimes\eta_B} A\otimes B.$$

To verify that this is indeed an algebra, the conditions of Definition 7.1.1 can be reduced to the corresponding properties of (A, μ_A, η_A) and (B, μ_B, η_B) using only c (to interchange the order of $\otimes$), ι (to duplicate the unit), and the associativity constraints (to change the placement of parentheses).

If M is a left A-module and N a left B-module, then $M\otimes N$ naturally becomes a left $(A\otimes B)$-module. Its construction and verification are the same as in the preceding paragraph, using c to interchange positions. The case of right modules is similar.

For the algebra $\mathbf{1}$ defined in Remark 7.1.3, it is clear that $\mathbf{1}\otimes A \simeq A \simeq A\otimes\mathbf{1}$, with the isomorphisms given by the unit constraints of $\mathcal{V}$.

Remark 7.1.6 The construction above is functorial in A and B: if $A\to A'$ and $B\to B'$ are morphisms of algebras, then so is the corresponding morphism $A\otimes B\to A'\otimes B'$. Another lengthy but straightforward fact is that, when A, B, C are all algebras, the associativity constraint $(A\otimes B)\otimes C \xrightarrow{\sim} A\otimes(B\otimes C)$ is a morphism of algebras.

Lemma 7.1.7 Let A be an algebra over a braided monoidal category $\mathcal{V}$. Then A is commutative if and only if its multiplication $\mu_A : A\otimes A\to A$ is a morphism of algebras.

Proof For μ_A to be a morphism of algebras, two conditions are required. The condition concerning the unit is equivalent to commutativity of

$$\begin{tikzcd}
\mathbf{1}\otimes\mathbf{1} \arrow[r, "\eta_A\otimes\eta_A"] \arrow[d, "\iota"'] & A\otimes A \arrow[d, "\mu_A"] \\
\mathbf{1} \arrow[r, "\eta_A"'] & A
\end{tikzcd}$$

and this holds automatically because A is an algebra. The remaining condition is commutativity of

$$\begin{tikzcd}
(A\otimes A)\otimes(A\otimes A) \arrow[r, "\mu_A\otimes\mu_A"] \arrow[d, "\mu_{A\otimes A}"'] & A\otimes A \arrow[d, "\mu_A"] \\
A\otimes A \arrow[r, "\mu_A"'] & A
\end{tikzcd}$$

Identify the upper-left corner with $A\otimes(A\otimes A)\otimes A$ by the associativity constraints. By associativity of μ_A, composition in the diagram along $\urcorner\!\!\downarrow$ equals

$$A\otimes(A\otimes A)\otimes A \xrightarrow{\mathrm{id}\otimes\mu_A\otimes\mathrm{id}} A\otimes A\otimes A \xrightarrow{\text{multiplication}} A.$$

On the other hand, the definition of $\mu_{A\otimes A}$ and the associativity of μ_A show that composition in the diagram along $\lfloor\!\!\longrightarrow$ equals

$$A \otimes (A \otimes A) \otimes A \xrightarrow{\mathrm{id}\otimes(\mu_A c(A,A))\otimes\mathrm{id}} A \otimes A \otimes A \xrightarrow{\text{multiplication}} A.$$

Thus $\mu_A c(A,A) = \mu_A$ implies that the original diagram commutes, or equivalently that μ_A is a morphism of algebras.

Conversely, let $f \in \{\mu_A, \mu_A c(A,A)\}$. We have the elementary commutative diagram

$$\begin{tikzcd}
A \otimes (A \otimes A) \otimes A \arrow[r, "\mathrm{id}\otimes f\otimes\mathrm{id}"] & A \otimes A \otimes A \arrow[r, "\text{multiplication}"] & A. \\
A \otimes A \arrow[u, "\eta_A\otimes\mathrm{id}\otimes\eta_A"] \arrow[r, "f"'] & A \arrow[u, "\eta_A\otimes\mathrm{id}\otimes\eta_A"] \arrow[ur, "\mathrm{id}"'] &
\end{tikzcd}$$

If μ_A is a morphism of algebras, the preceding discussion shows that the first row gives the same composite for the two choices of f. Considering the second row then yields $\mu_A = \mu_A c(A,A)$. □

Lemma 7.1.8 Let A and B be algebras over a symmetric monoidal category $\mathcal{V}$. Then $c(A,B) : A \otimes B \to B \otimes A$ is an isomorphism of algebras.

Proof Two commutative diagrams must be verified for $c(A,B)$. The first is

$$\begin{tikzcd}
 & \mathbf{1}\otimes\mathbf{1} \arrow[dl, "\sim"'] \arrow[r, "\eta_A\otimes\eta_B"] \arrow[dd, "c(\mathbf{1},\mathbf{1})"] & A \otimes B \arrow[dd, "c(A,B)"] \\
\mathbf{1} & & \\
 & \mathbf{1}\otimes\mathbf{1} \arrow[ul, "\sim"] \arrow[r, "\eta_B\otimes\eta_A"'] & B \otimes A
\end{tikzcd}$$

Commutativity on the right follows from functoriality of c, while that on the left follows from compatibility of c with the unit constraints [51, (3.12)]. For the second diagram, which says that $c(A,B)$ preserves multiplication, the argument is based on symmetry of the braiding and is exactly the same as at the end of [51, §7.4]. □

Proposition 7.1.9 Let A and B be commutative algebras over a symmetric monoidal category $\mathcal{V}$. Then $A \otimes B$ is also commutative.

Proof It is enough to show that the multiplication $\mu_{A\otimes B}$ is a morphism of algebras. Recall the definition of $\mu_{A\otimes B}$ and apply Lemma 7.1.8 to $c(B,A)$, the commutativity assumptions on A, B together with Lemma 7.1.7 to μ_A, μ_B, and Remark 7.1.6 on the functoriality of $\otimes$. Every part of the composite is then a morphism of algebras. □

We have defined algebras, modules, and their morphisms. All these structures form categories, with notation

structure	category	condition
algebra	$\mathsf{Alg}(\mathcal{V})$	$\mathcal{V}$ is a monoidal category
commutative algebra	$\mathsf{CAlg}(\mathcal{V})$	$\mathcal{V}$ is a braided monoidal category
left A-module	$A\text{-}\mathsf{Mod}$	A is an algebra
right B-module	$\mathsf{Mod}\text{-}B$	B is an algebra
(A, B)-bimodule	$(A, B)\text{-}\mathsf{Mod}$	A and B are algebras

The preceding discussion also shows that $\mathsf{Alg}(\mathcal{V})$ and $\mathsf{CAlg}(\mathcal{V})$ generally have less structure than $\mathcal{V}$. Labeling the levels of structure by numbers, the situation can be represented schematically as

$$\begin{tikzcd}
0 & 1 & 2 & \infty \\
\text{category} & \text{monoidal category} \arrow[l, "\mathsf{Alg}"'] & \text{braided monoidal category} \arrow[l, "\mathsf{Alg}"'] \arrow[ll, bend left, "\mathsf{CAlg}"] & \text{symmetric monoidal category} \arrow[loop right, "{\mathsf{Alg},\mathsf{CAlg}}"]
\end{tikzcd}$$

Moreover, whenever $\mathsf{Alg}(\mathcal{V})$ or $\mathsf{CAlg}(\mathcal{V})$ becomes a monoidal category under $\otimes$, its unit is always $\mathbf{1}$.

Recall that Lemma 7.1.7 says that a commutative algebra over a braided monoidal category $\mathcal{V}$ automatically gives an algebra over the monoidal category $\mathsf{Alg}(\mathcal{V})$, by taking the multiplication $A \otimes A \to A$ to be that of A. In fact, this is the only possible choice. This leads to a simple, interesting, and important observation: there is an equivalence of categories

$$\mathsf{Alg}\left(\mathsf{Alg}(\mathcal{V})\right) \simeq \mathsf{CAlg}(\mathcal{V}), \quad \mathcal{V}: \text{ braided monoidal category.} \tag{7.1.1}$$

These facts rest on the so-called **Eckmann–Hilton argument**; their proof is deliberately left as an exercise for this chapter, supplied with a hint.

Example 7.1.10 Let the category $\mathcal{C}$ have finite products. In particular, it has an empty product, namely a terminal object, denoted by $\mathbf{1}$. For every two objects X, Y there is a canonical isomorphism $c(X, Y) : X \times Y \xrightarrow{\sim} Y \times X$. These data make $(\mathcal{C}, \times)$ a symmetric monoidal category with unit $\mathbf{1}$. We may discuss algebras over $\mathcal{C}$, also called monoid objects in $\mathcal{C}$[2)], together with their commutative versions.

- ◇ If $\mathcal{C} = \mathsf{Set}$, the one-point set is the unit $\mathbf{1}$. The corresponding algebras are monoids, and the corresponding commutative algebras are commutative monoids.

2) For categories of this kind, [51, §4.11] also discusses group objects, but the inverse property cannot be formulated in an arbitrary monoidal category.

◇ If $\mathcal{C} = \mathsf{Top}$, the corresponding algebras are topological monoids, and the corresponding commutative algebras are commutative topological monoids.

On the other hand, let $\Bbbk$ be a commutative ring and give $\Bbbk\text{-}\mathsf{Mod}$ the structure of a symmetric monoidal category through $\otimes = \otimes_{\Bbbk}$, with $\Bbbk$ as the unit $\mathbf{1}$ and the braiding determined by $x \otimes y \mapsto y \otimes x$. The corresponding algebras (or commutative algebras) are precisely algebras (or commutative algebras) in the usual algebraic sense; see [51, Definition 7.1.1].

In these concrete situations, the multiplication of an algebra A is often written directly as multiplication of elements $xy = x \cdot y$, and the unit is directly identified with the element 1_A, so that the notation μ_A, η_A can be omitted.

Example 7.1.11 (Graded modules and graded algebras) Fix an abelian category $\mathcal{A}$ and a nonempty set I. An object of $\mathcal{A}^I$ is called an I-graded object and is written $M = (M^i)_{i \in I}$, while a morphism is written $(f^i : M^i \to N^i)_{i \in I}$. The object $M^i \in \mathrm{Ob}(\mathcal{A})$ (or the morphism $f^i \in \mathrm{Mor}(\mathcal{A})$) is customarily called the degree-i component of M (or f). The special cases $I = \mathbb{Z}$ and $\mathbb{Z}^n$ have already been discussed repeatedly in §3.1, §4.1, §5.1, and elsewhere.

For concreteness, take for the moment $\mathcal{A} := \Bbbk\text{-}\mathsf{Mod}$, where $\Bbbk$ is a commutative ring. An I-graded object in $\mathcal{A}$ is also called an I-graded module. Suppose further that I is a monoid. Define a bifunctor $\otimes$ on the category $\mathcal{A}^I$ by

$$(M \otimes N)^i := \bigoplus_{jk=i} M^j \otimes N^k, \quad (f \otimes g)^i := \bigoplus_{jk=i} f^j \otimes g^k. \tag{7.1.2}$$

Also define $\mathbf{1}$ to be $\Bbbk$ concentrated in the component $i = 1_I$ (the unit of I). These data make $\mathcal{A}^I$ a monoidal category. The unit condition $\eta_A : \mathbf{1} = \Bbbk \to A$ automatically ensures that $1_A := \eta_A(1) \in A^{1_I}$.

In the special case where I is commutative, this recovers the I-graded modules and I-graded algebras introduced in [51, Definition 7.4.1].

Continue to take $\mathcal{A} = \Bbbk\text{-}\mathsf{Mod}$ and require $(I, +)$ to be a commutative monoid. Every additive homomorphism $\epsilon : I \to \mathbb{Z}/2\mathbb{Z}$ makes $\Bbbk\text{-}\mathsf{Mod}^I$ a symmetric monoidal category with the **Koszul braiding** characterized by

$$\begin{aligned} M^a \otimes N^b &\to N^b \otimes M^a \\ x \otimes y &\mapsto (-1)^{\epsilon(a)\epsilon(b)} y \otimes x, \end{aligned} \tag{7.1.3}$$

where $a, b \in I$. The corresponding commutative algebras are precisely the ϵ-commutative algebras of [51, Definition 7.4.3]. The various equations and definitions that result are collectively called the **Koszul sign rule**. Whenever tensor positions are interchanged, the sign must be changed according to this rule.

For example, the condition for A to be a commutative algebra is written explicitly as

$$xy = (-1)^{\epsilon(a)\epsilon(b)}yx, \quad x \in A^a,\ y \in A^b.$$

The typical situation is

$$I \in \{\mathbb{Z}, \mathbb{Z}_{\geq 0}\}, \quad \epsilon(a) := a \mod 2,$$

where the corresponding ϵ-commutative algebras are called anticommutative graded algebras in the source cited above. Since this braiding appears everywhere in the study of $\mathbb{Z}$-graded objects, the term **graded commutative** is more apt.

Now return to a general abelian category $\mathcal{A}$ and $(I, +, \epsilon)$ as above. To generalize the preceding construction, we need to extend $\mathcal{A}$ to a symmetric monoidal category $(\mathcal{A}, \otimes)$ such that

- ◇ $\mathcal{A}$ is an abelian category, and coproducts (that is, direct sums) of the form in (7.1.2) exist;
- ◇ $\otimes : \mathcal{A} \times \mathcal{A} \to \mathcal{A}$ is additive in each variable and preserves these coproducts.

Under these conditions, $\otimes$ and the Koszul braiding on $\mathcal{A}^I$ can still be defined in the same way, making $(\mathcal{A}^I, \otimes)$ a symmetric monoidal category. Its unit is still the unit $\mathbf{1}$ of $\mathcal{A}$, viewed as an I-graded object concentrated in degree zero.

For general $(\mathcal{A}, \otimes)$, all verifications are the same as in the case of $\Bbbk$-Mod. The only difference is that "elements" of an object no longer have meaning in the general case.

Convention 7.1.12 In the special case $\mathcal{A} = \Bbbk\text{-}\mathsf{Mod}$, the usual convention is to regard an I-graded module $(M^i)_{i \in I}$ as a module equipped with a direct-sum decomposition $M = \bigoplus_{i \in I} M^i$, and to require morphisms to preserve the direct-sum components.

Example 7.1.13 In the construction above ($\mathcal{A} = \Bbbk\text{-}\mathsf{Mod}$), let $\Bbbk$ be a field with $\mathrm{char}(\Bbbk) \neq 2$, take $I = \mathbb{Z}/2\mathbb{Z} = \{0, 1\}$, and set $\epsilon = \mathrm{id}_{\mathbb{Z}/2\mathbb{Z}}$. The corresponding symmetric monoidal category is denoted by $\mathsf{Vect}^{-}_{\mathbb{Z}/2\mathbb{Z}}(\Bbbk)$. Its objects can be viewed as $\Bbbk$-vector spaces equipped with a direct-sum decomposition $V = V_0 \oplus V_1$, and are also called **super vector spaces**. This is the source of terminology common in linear algebra and mathematical physics, such as superalgebra and commutative superalgebra.

Finally, we discuss the effect of lax monoidal functors on algebras.

Definition 7.1.14 Let $\mathcal{V}$ and $\mathcal{V}'$ be monoidal categories. According to [51, Remark 3.1.8], a right lax monoidal functor from $\mathcal{V}$ to $\mathcal{V}'$, abbreviated here to a **lax monoidal functor**, means data (F, ξ_F, φ_F), where $F : \mathcal{V} \to \mathcal{V}'$ is a functor and

$$\xi_F : F(\cdot) \otimes F(\cdot) \to F(\cdot \otimes \cdot), \quad \varphi_F : \mathbf{1}_{\mathcal{V}'} \to F\left(\mathbf{1}_{\mathcal{V}}\right),$$

are morphisms subject to a collection of compatibility conditions with the monoidal structures. The data (F, ξ_F, φ_F) are often abbreviated to F.

If $\mathcal{V}$ and $\mathcal{V}'$ are both braided (or symmetric) monoidal categories, and F also makes the following diagram commute, then F is called a braided functor, or is said to be compatible with the braidings:

$$\begin{tikzcd} F(X) \otimes F(Y) \arrow[r, "\xi_F(X,Y)"] \arrow[d, "c'(FX,FY)"'] & F(X \otimes Y) \arrow[d, "Fc(X,Y)"] \\ F(Y) \otimes F(X) \arrow[r, "\xi_F(Y,X)"'] & F(Y \otimes X). \end{tikzcd}$$

If ξ_F and φ_F in the data of a lax monoidal functor are both isomorphisms, the functor is called a **monoidal functor**.

We may of course also discuss morphisms between lax monoidal functors, also called natural transformations:

$$\theta = (\theta_X)_{X \in \mathrm{Ob}(\mathcal{V})} : F \to G, \quad F, G : \mathcal{V} \to \mathcal{V}'.$$

In addition to the conditions for a natural transformation, we require θ to make the following diagrams commute:

$$\begin{tikzcd} F(X) \otimes F(Y) \arrow[r, "\xi_F(X,Y)"] \arrow[d, "\theta_X \otimes \theta_Y"'] & F(X \otimes Y) \arrow[d, "\theta_{X \otimes Y}"] \\ G(X) \otimes G(Y) \arrow[r, "\xi_G(X,Y)"'] & G(X \otimes Y) \end{tikzcd} \qquad \begin{tikzcd} \mathbf{1}_{\mathcal{V}'} \arrow[r, "\varphi_F"] \arrow[dr, "\varphi_G"'] & F(\mathbf{1}_{\mathcal{V}}) \arrow[d, "\theta_{\mathbf{1}_{\mathcal{V}}}"] \\ & G(\mathbf{1}_{\mathcal{V}}). \end{tikzcd}$$

If F and G are monoidal functors, this definition is in fact equivalent to [51, Definition 3.1.10], where φ_F and φ_G are called "standard" isomorphisms. We may therefore discuss equivalences between monoidal categories or braided monoidal categories.

Proposition 7.1.15 A lax monoidal functor F induces a functor $\mathsf{Alg}(\mathcal{V}) \to \mathsf{Alg}(\mathcal{V}')$. If A is an algebra over $\mathcal{V}$, then F induces A-$\mathsf{Mod} \to F(A)$-Mod, and similarly for the other module categories.

If, in addition, $\mathcal{V}$ and $\mathcal{V}'$ are braided monoidal categories and F is a braided functor, then F also induces $\mathsf{CAlg}(\mathcal{V}) \to \mathsf{CAlg}(\mathcal{V}')$, while $\mathsf{Alg}(\mathcal{V}) \to \mathsf{Alg}(\mathcal{V}')$ is a lax monoidal functor. If, moreover, $\mathcal{V}$ and $\mathcal{V}'$ are both symmetric, then the same holds for $\mathsf{CAlg}(\mathcal{V}) \to \mathsf{CAlg}(\mathcal{V}')$.

Proof Let A be an algebra over $\mathcal{V}$. Define the multiplication μ_{FA} as $FA \otimes FA \to F(A \otimes A) \xrightarrow{F\mu_A} FA$, and the unit η_{FA} as $\mathbf{1}_{\mathcal{V}'} \to F(\mathbf{1}_{\mathcal{V}}) \xrightarrow{F\eta_A} FA$. The remaining verifications are routine. $\square$

Example 7.1.16 Consider the symmetric monoidal abelian category $\mathcal{A}$ and the category $\mathcal{A}^I$ from Example 7.1.11. Define the functor

$$\mathrm{ev}_0 : \mathcal{A}^I \to \mathcal{A}, \quad \left(M^i\right)_{i\in I} \mapsto M^0.$$

This functor is exact and symmetric lax monoidal: simply take

$$\xi_{\mathrm{ev}_0}(M,N) : M^0 \otimes N^0 \to (M\otimes N)^0 := \bigoplus_{i+j=0} M^i \otimes N^j$$

to be the evident morphism, and $\varphi_{\mathrm{ev}_0} : \mathbf{1} \to \mathbf{1}$ to be the identity. Thus $A \mapsto A^0$ induces lax monoidal functors

$$\mathsf{Alg}(\mathcal{A}^I) \to \mathsf{Alg}(\mathcal{A}), \quad \mathsf{CAlg}(\mathcal{A}^I) \to \mathsf{CAlg}(\mathcal{A}).$$

The following remark concerns algebras over a general monoidal category $\mathcal{V}$. Recall that finite ordinals correspond bijectively to $\mathbb{Z}_{\geq 0}$; the finite ordinal $\mathbf{n}$ corresponding to n is identified with the totally ordered n-element set $\{0,\ldots,n-1\}$, with $0 \leq 1 \leq 2 \leq \cdots$.

Remark 7.1.17 (The walking algebra) A map $f : A \to B$ between partially ordered sets is called order-preserving if $a \leq a' \implies f(a) \leq f(a')$. Write FinOrd for the category of finite ordinals whose morphisms are order-preserving maps between ordinals. This is a strict monoidal category under $\mathbf{n} \otimes \mathbf{m} := \mathbf{n} + \mathbf{m}$. A morphism $f \otimes g : \mathbf{n} \otimes \mathbf{m} \to \mathbf{n}' \otimes \mathbf{m}'$ maps the first n elements (or the last m elements) by f (or g), and the object $\mathbf{0}$ is the unit[3)]. Make $\mathbf{1}$ into an algebra over FinOrd: take its unit to be $\varnothing = \mathbf{0} \hookrightarrow \mathbf{1}$ and its multiplication to be $\mathbf{1} \otimes \mathbf{1} = \mathbf{2} \twoheadrightarrow \mathbf{1}$. The associated commutative diagrams are entirely straightforward to verify.

The ordinal $\mathbf{1}$ (or $\mathbf{2}$) may be viewed as the "walking" object (or morphism). Indeed, for any category $\mathcal{C}$, specifying a functor $\mathbf{1} \to \mathcal{C}$ (or $\mathbf{2} \to \mathcal{C}$) is equivalent to specifying an object (or a morphism) in $\mathcal{C}$. Now consider a monoidal category $\mathcal{V}$. We claim that there is an isomorphism of categories

$$\{\mathcal{V}\text{-algebras } A\} \simeq \{\text{monoidal functors } F : \mathsf{FinOrd} \to \mathcal{V}\},$$

which also means that FinOrd is the "walking algebra." First, for a monoidal functor $F : \mathsf{FinOrd} \to \mathcal{V}$, define $A := F(\mathbf{1})$. As a simple special case of Proposition 7.1.15, A naturally becomes an algebra over $\mathcal{V}$: the multiplication $\mu : A \otimes A \to A$ is $F(\mathbf{2} \twoheadrightarrow \mathbf{1})$, while the unit $\eta : \mathbf{1}_{\mathcal{V}} \to A$ is $F(\mathbf{0} \hookrightarrow \mathbf{1})$.

Conversely, from an algebra (A,μ,η) over $\mathcal{V}$ define a monoidal functor $F : \mathsf{FinOrd} \to \mathcal{V}$ by

$$F(\mathbf{n}) = A^{\otimes n} := \underbrace{A \otimes (A \otimes (\cdots))}_{n \text{ copies of } A}, \quad A^{\otimes 0} := \mathbf{1}_{\mathcal{V}},$$

3) Here $\mathbf{1}$ always denotes an ordinal; the unit of the monoidal category $\mathcal{V}$ is denoted separately by $\mathbf{1}_{\mathcal{V}}$.

For $f : \mathbf{n} \to \mathbf{m}$, if f is injective, then $F(f) : A^{\otimes n} \to A^{\otimes m}$ is obtained by inserting $\eta : \mathbf{1}_{\mathcal{V}} \to A$ outside the image of f; if f is surjective, $F(f)$ is obtained by using $\mu : A \otimes A \to A$ to successively combine the factors in each fiber. The associativity constraints and other properties of a monoidal category ensure that these operations are well defined. A general map f has a unique surjective–injective factorization, and this defines F on morphisms. Functoriality is clear. It is not difficult to verify that the two constructions are mutually inverse.

7.2 Example: Differential Graded Structures

Continuing the discussion in Example 7.1.11, consider a symmetric monoidal abelian category $(\mathcal{A}, \otimes)$ that admits countable coproducts. Throughout this section we also assume that $\otimes$ is right exact in each variable. The standard example is, of course, $\mathcal{A} = \Bbbk\text{-}\mathsf{Mod}$.

We previously discussed $\mathbb{Z}$-graded algebras over $\mathcal{A}$, or, equivalently, algebras over the monoidal category $(\mathcal{A}^{\mathbb{Z}}, \otimes)$. More precisely, in the setting of Example 7.1.11, this means taking $I = \mathbb{Z}$ and the Koszul braiding (7.1.3) corresponding to $\epsilon(a) = a \bmod 2$. Henceforth, "graded" always means "$\mathbb{Z}$-graded." The aim of this section is to add a "differential" structure; part of the discussion overlaps with §5.7.

Define the shift functor $T : \mathcal{A}^{\mathbb{Z}} \to \mathcal{A}^{\mathbb{Z}}$ by $(TM)^n = M^{n+1}$ and $(Tf)^n = f^{n+1}$. Thus $\left(\mathcal{A}^{\mathbb{Z}}, T\right)$ is an abelian category with translation, and a differential object over it (Definition 5.2.1) is called a **differential graded object** over $\mathcal{A}$. In fact, the abelian category $\left(\mathcal{A}^{\mathbb{Z}}, T\right)_d$ of differential graded objects is canonically isomorphic to the category of complexes $\mathsf{C}(\mathcal{A})$. This has already been observed in §3.1.

Accordingly, from now on we identify differential graded objects over $\mathcal{A}$ with complexes, without distinguishing between the two.

We next equip $\mathsf{C}(\mathcal{A})$ with a symmetric monoidal structure. For any complexes M and N, the bifunctor $\otimes : \mathcal{A} \times \mathcal{A} \to \mathcal{A}$ gives the double complex $M^\bullet \otimes N^\bullet$; totalization by direct sums gives

$$M \otimes N := \operatorname{tot}_{\oplus}\left(M^\bullet \otimes N^\bullet\right), \quad (M \otimes N)^n = \bigoplus_{a+b=n} M^a \otimes N^b.$$

In the example $\mathcal{A} = \Bbbk\text{-}\mathsf{Mod}$, the differential $d = d_{M \otimes N}$ on the complex (or, equivalently, on the differential object) can be written elementwise as

$$d(x \otimes y) = (d_M x) \otimes y + (-1)^a x \otimes (d_N y), \quad x \in M^a,\ y \in N^b.$$

With these definitions, the Koszul braiding (7.1.3) is a morphism of complexes. The reader may verify this directly or deduce it from Proposition 3.5.6.

Moreover, the unit object $\mathbf{1}$ of $(\mathcal{A}, \otimes)$, viewed as a complex concentrated in degree zero, is clearly the unit object of the monoidal category $(\mathsf{C}(\mathcal{A}), \otimes)$. The following fact is immediate: for every $X = (X^n, d_X^n)_n \in \mathrm{Ob}(\mathsf{C}(\mathcal{A}))$,

$$\mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(\mathbf{1}, X) \simeq \mathrm{Hom}_{\mathcal{A}}\left(\mathbf{1}, \ker\left(d_X^0\right)\right). \tag{7.2.1}$$

Definition 7.2.1 With respect to the symmetric monoidal structure above, the objects of $\mathsf{Alg}\,(\mathsf{C}(\mathcal{A}))$ are called **differential graded algebras** over $\mathcal{A}$, abbreviated **dg algebras**; the objects of $\mathsf{CAlg}\,(\mathsf{C}(\mathcal{A}))$ are called commutative differential graded algebras over $\mathcal{A}$, abbreviated commutative dg algebras.

For a differential graded algebra A, one can therefore also speak of left (or right) A-modules M, and even of bimodules; each forms an abelian category. In this setting M is also called a **differential graded module** over A, abbreviated a **dg module**. This notion combines the theory of complexes with module theory and is extremely useful. The exercises in this chapter give a further introduction; the reader may also consult monographs such as [48]. To emphasize the differential graded structure, we write the category of dg modules as $A\text{-}\mathsf{dgMod}$, and use analogous notation for its variants.

The forgetful functor $\mathsf{C}(\mathcal{A}) \to \mathcal{A}^{\mathbb{Z}}$ preserves $\otimes$, so Proposition 7.1.15 shows that it preserves algebra (or commutative algebra) structures. When $\mathcal{A} = \Bbbk\text{-}\mathsf{Mod}$, to say that A is a dg algebra is precisely to say that A is a graded algebra equipped with a differential d whose multiplication satisfies the Leibniz rule

$$d(xy) = (dx)y + (-1)^a x(dy), \quad x \in A^a,\ y \in A^b.$$

Substituting $x = 1_A = y$ gives $d(1_A) = 0$. The opposite algebra A^{op} is obtained by replacing the multiplication with $x \cdot^{\mathrm{op}} y := (-1)^{ab} yx$.

Similarly, when $\mathcal{A} = \Bbbk\text{-}\mathsf{Mod}$, a left A-module M is equivalently a graded $\Bbbk$-module M with a scalar-multiplication morphism $A \otimes M \to M$ satisfying associativity and the other module axioms, and equipped with a differential d satisfying the Leibniz rule

$$d(tm) = (dt)m + (-1)^a t(dm), \quad t \in A^a,\ m \in M^b;$$

for a right A-module M, the Leibniz rule becomes

$$d(mt) = (dm)t + (-1)^b m(dt), \quad t \in A^a,\ m \in M^b.$$

For a general $(\mathcal{A}, \otimes)$, all these properties can be formulated without elements. For example, the unit of an algebra A corresponds to a morphism $\mathbf{1} \to \ker\left(d_A^0\right)$ in $\mathcal{A}$; see (7.2.1).

Taking cohomology gives an additive functor $\mathrm{H} = (\mathrm{H}^n)_{n \in \mathbb{Z}} : \mathsf{C}(\mathcal{A}) \to \mathcal{A}^{\mathbb{Z}}$. The next result says that multiplication of complexes induces multiplication on cohomology.

Proposition 7.2.2 The functor H induces

$$\mathsf{Alg}\left(\mathsf{C}(\mathcal{A})\right) \to \mathsf{Alg}\left(\mathcal{A}^{\mathbb{Z}}\right), \quad \mathsf{CAlg}\left(\mathsf{C}(\mathcal{A})\right) \to \mathsf{CAlg}\left(\mathcal{A}^{\mathbb{Z}}\right).$$

If A is a dg algebra and M is a left A-module (or a right module, or a bimodule), then $\mathrm{H}(M)$ is respectively a left $\mathrm{H}(A)$-module (or a right module, or a bimodule).

Proof By Proposition 7.1.15, it suffices to prove that H is a lax monoidal functor. The required canonical morphism

$$\xi_{\mathrm{H}}(M,N) : \mathrm{H}(M) \otimes \mathrm{H}(N) \to \mathrm{H}(M \otimes N),$$

is nearly tautological: it is the morphism κ discussed in Theorem 3.14.12 or Proposition 4.7.8. On the other hand, take

$$\varphi_{\mathrm{H}} : \mathbf{1} \to \mathrm{H}(\mathbf{1}) = \mathbf{1} \quad \text{(concentrated in degree zero)}$$

to be id. □

We may also take the lax monoidal functor $\mathrm{ev}_0 : \mathcal{A}^{\mathbb{Z}} \to \mathcal{A}$ of Example 7.1.16. Again, Proposition 7.1.15 shows that it sends graded algebras (or graded commutative algebras) to algebras (or commutative algebras), and modules to modules. Observe that $\mathrm{ev}_0 \mathrm{H} = \mathrm{H}^0$. In applications, each stage of the composite $\mathsf{C}(\mathcal{A}) \xrightarrow{\mathrm{H}} \mathcal{A}^{\mathbb{Z}} \xrightarrow{\mathrm{ev}_0} \mathcal{A}$ discards information; the richest structure remains on the dg algebra itself.

Example 7.2.3 Let $\Bbbk$ be a commutative ring. For any $\Bbbk$-linear category $\mathcal{A}$, consider the multiplication on the Hom complexes from (3.2.1):

$$\mathrm{Hom}^a(Y,Z) \times \mathrm{Hom}^b(X,Y) \to \mathrm{Hom}^{a+b}(X,Z), \quad X,Y,Z \in \mathrm{Ob}(\mathsf{C}(\mathcal{A})).$$

Because this multiplication satisfies the Leibniz rule (Lemma 3.2.4), it determines a morphism in $\mathsf{C}(\Bbbk\text{-}\mathsf{Mod})$

$$\mathrm{Hom}^\bullet(Y,Z) \otimes \mathrm{Hom}^\bullet(X,Y) \to \mathrm{Hom}^\bullet(X,Z).$$

Moreover, the multiplication is associative, so $\mathrm{End}^\bullet(X) := \mathrm{Hom}^\bullet(X,X)$ is an algebra over $(\mathsf{C}(\Bbbk\text{-}\mathsf{Mod}), \otimes)$ with unit $\mathrm{id}_X \in \mathrm{End}^0(X)$. Furthermore, $\mathrm{Hom}^\bullet(X,Y)$ is an $(\mathrm{End}^\bullet(Y), \mathrm{End}^\bullet(X))$-bimodule.

Taking the cohomology $\mathrm{H} = (\mathrm{H}^n)_n$ of $\mathrm{Hom}^\bullet(X,Y)$ gives only

$$\bigoplus_{n \in \mathbb{Z}} \mathrm{Hom}_{\mathsf{K}(\mathcal{A})}(X, Y[n]);$$

whereas applying (7.2.1) to $\mathrm{Hom}^\bullet(X,Y)$ shows that $\mathrm{Hom}_{\mathsf{C}(\mathcal{A})}$ can be reconstructed as

$$\mathrm{Hom}_{\mathsf{C}(\Bbbk\text{-}\mathsf{Mod})}\left(\Bbbk, \mathrm{Hom}^\bullet(X,Y)\right) \simeq \mathrm{Hom}_{\mathsf{C}(\mathcal{A})}(X,Y)$$

with $\Bbbk$ serving as the unit object of $(\mathsf{C}(\Bbbk\text{-}\mathsf{Mod}), \otimes)$.

Thus the sets $\mathrm{Hom}_{\mathsf{C}(\mathcal{A})}$ and composition of morphisms in $\mathsf{C}(\mathcal{A})$ are merely shadows of the Hom complexes and their multiplication, obtained by applying $\mathrm{Hom}_{\mathsf{C}(\Bbbk\text{-}\mathsf{Mod})}(\Bbbk, \cdot)$.

Starting from the example $\mathsf{C}(\mathcal{A})$, we are naturally led to the concept of a differential graded category. This concept involves the **$\mathcal{V}$-enriched categories** introduced in [51, Definition 3.4.1]. Briefly, for any monoidal category $\mathcal{V}$, a $\mathcal{V}$-enriched category $\mathcal{C}$ still has objects and morphisms, but the Hom sets are replaced by Hom objects in $\mathcal{V}$, written $\mathcal{H}\mathrm{om} = \mathcal{H}\mathrm{om}_{\mathcal{C}}$ to distinguish them. Composition of morphisms is replaced by a morphism in $\mathcal{V}$

$$\mathcal{H}\mathrm{om}(Y, Z) \otimes \mathcal{H}\mathrm{om}(X, Y) \to \mathcal{H}\mathrm{om}(X, Z),$$

while an identity morphism $\mathrm{id}_X \in \mathrm{Hom}(X, X)$ is replaced by $\mathrm{id}_X : \mathbf{1} \to \mathcal{H}\mathrm{om}(X, X)$. These are all morphisms in $\mathcal{V}$, and their properties are expressed by commutative diagrams in $\mathcal{V}$.

For enriched versions of functors and morphisms between them, see [51, Definitions 3.4.1, 3.4.2, 3.4.4]. They are defined in the expected way; in brief, an enriched functor lifts maps between Hom sets to morphisms between $\mathcal{H}\mathrm{om}$ objects.

For example, a $\Bbbk$-Mod-category is precisely a category enriched over $(\Bbbk\text{-}\mathsf{Mod}, \otimes)$.

Conversely, let us temporarily call a category without enrichment an ordinary category. If set-theoretic size issues are ignored, an ordinary category is the same thing as a category enriched over $(\mathsf{Set}, \times)$.

Definition 7.2.4 Let $\Bbbk$ be a commutative ring, and consider the corresponding symmetric monoidal category $(\mathsf{C}(\Bbbk\text{-}\mathsf{Mod}), \otimes)$. A category enriched over $(\mathsf{C}(\Bbbk\text{-}\mathsf{Mod}), \otimes)$ is called a **differential graded category** over $\Bbbk$, abbreviated a **dg category** over $\Bbbk$. Functors between such categories and morphisms between those functors are defined as for enriched categories.[4)]

By definition, an object X of a dg category gives a dg algebra $\mathrm{End}^\bullet(X)$.

Example 7.2.5 Example 7.2.3 says that $\mathsf{C}(\mathcal{A})$ naturally becomes a dg category over $\Bbbk$ for any $\Bbbk$-linear category $\mathcal{A}$. For example, for every $\Bbbk$-algebra R, the category of complexes of left (or right) R-modules is automatically a dg category over $\Bbbk$.

Example 7.2.6 Let $\Bbbk$ be a commutative ring and let A be a dg algebra over $\mathcal{A} :=$

[4)] Much of the literature on infinity categories uses the term DG-category, as well as the notation DGCat for the category they form, and this notation may overlap with that of this book. Although those DG-categories are closely related to the dg categories discussed here, the two are not the same notion.

$\Bbbk\text{-}\mathsf{Mod}$. For left A-modules M and N, define the following subcomplex of $\operatorname{Hom}^\bullet_\Bbbk(M,N)$:

$$\operatorname{Hom}^p_A(M,N) := \left\{ f \in \operatorname{Hom}^p_\Bbbk(M,N) \;\middle|\; \begin{array}{l} \forall m \in M,\ \forall k \in \mathbb{Z},\ \forall t \in A^k, \\ f(tm) = (-1)^{pk} t f(m) \end{array} \right\}.$$

For left A-modules L, M, and N, composition on the Hom complexes $\operatorname{Hom}^\bullet_\Bbbk$ induces a morphism of complexes

$$\operatorname{Hom}^p_A(M,N) \otimes \operatorname{Hom}^q_A(L,M) \to \operatorname{Hom}^{p+q}_A(L,N).$$

Thus the category $A\text{-}\mathsf{dgMod}$ of left A-modules is enriched to a dg category. For right A-modules, the condition in the definition of $\operatorname{Hom}^p_A(M,N)$ must be replaced by $f(mt) = f(m)t$ (think about why). Bimodules are handled similarly.

The definition above can also be formulated without reference to elements, and therefore extends to a general $\mathcal{A}$.

More precisely, a dg category enriches every Hom set of the underlying ordinary category to a Hom complex $\operatorname{Hom}^\bullet$, while dg versions of functors and their morphisms must be formulated inside $\mathsf{C}(\Bbbk\text{-}\mathsf{Mod})$. Explicitly:

- A **dg functor** $F : \mathcal{C} \to \mathcal{D}$ between dg categories consists of a map of object sets $F : \operatorname{Ob}(\mathcal{C}) \to \operatorname{Ob}(\mathcal{D})$ and morphisms of Hom complexes $\operatorname{Hom}^\bullet_\mathcal{C}(X,Y) \to \operatorname{Hom}^\bullet_\mathcal{D}(FX,FY)$. The latter are morphisms in $\mathsf{C}(\Bbbk\text{-}\mathsf{Mod})$ and are required to make both diagrams below commute:

$$\begin{tikzcd}
\operatorname{Hom}^\bullet_\mathcal{C}(Y,Z) \otimes \operatorname{Hom}^\bullet_\mathcal{C}(X,Y) \arrow[r] \arrow[d] & \operatorname{Hom}^\bullet_\mathcal{C}(X,Z) \arrow[d] \\
\operatorname{Hom}^\bullet_\mathcal{D}(FY,FZ) \otimes \operatorname{Hom}^\bullet_\mathcal{D}(FX,FY) \arrow[r] & \operatorname{Hom}^\bullet_\mathcal{D}(FX,FZ)
\end{tikzcd}$$

$$\begin{tikzcd}
\Bbbk \arrow[r] \arrow[dr] & \operatorname{Hom}^\bullet_\mathcal{C}(X,X) \arrow[d] \\
& \operatorname{Hom}^\bullet_\mathcal{D}(FX,FX);
\end{tikzcd}$$

- In the classical sense, a morphism, or natural transformation, ϕ between dg functors $F, G : \mathcal{C} \to \mathcal{D}$ is a family of elements

$$\phi_X \in \ker\left[\operatorname{Hom}^0_\mathcal{D}(FX,GX) \to \operatorname{Hom}^1_\mathcal{D}(FX,GX)\right]$$
$$\xlongequal{(7.2.1)} \operatorname{Hom}_{\mathsf{C}(\Bbbk\text{-}\mathsf{Mod})}\left(\Bbbk, \operatorname{Hom}^\bullet_\mathcal{D}(FX,GX)\right),$$

indexed by $X \in \mathrm{Ob}(\mathcal{C})$, such that for every $m \in \mathbb{Z}$ and $f \in \mathrm{Hom}^m_{\mathcal{C}}(X,Y)$,

$$(Gf)\phi_X = \phi_Y(Ff) \ \in \mathrm{Hom}^m_{\mathcal{D}}(FX, GY),$$

where multiplication means multiplication in the Hom complex.

Thus one may speak of isomorphisms, equivalences, adjunctions, and other notions for dg categories. An element of $\mathrm{Hom}^n_{\mathcal{C}}(X,Y)$ is sometimes called a **morphism of degree $\boldsymbol{n}$** from X to Y.

For a general monoidal category $\mathcal{V}$, one may view $\mathrm{Hom}_{\mathcal{V}}(\mathbf{1}, M)$ as the set of "elements" of $M \in \mathrm{Ob}(\mathcal{V})$; this is at least natural when $\mathcal{V} = \mathsf{Set}$ or $\Bbbk\text{-}\mathsf{Mod}$. Following this idea, every $\mathcal{V}$-enriched category $\mathcal{C}$ has an underlying ordinary category defined by

$$\mathrm{Hom}_{\mathcal{C}}(X,Y) := \mathrm{Hom}_{\mathcal{V}}\left(\mathbf{1}, \mathcal{H}\mathrm{om}_{\mathcal{C}}(X,Y)\right),$$

and we continue to denote that ordinary category by $\mathcal{C}$; see [51, Remark 3.4.3] for details. In the example of the dg category $\mathsf{C}(\mathcal{A})$, the resulting ordinary category is the ordinary category of complexes $\mathsf{C}(\mathcal{A})$. Thus our notation is consistent.

Another operation comes from lax monoidal functors. Here is the general statement.

Proposition 7.2.7 Let $F : \mathcal{V} \to \mathcal{V}'$ be a lax monoidal functor between monoidal categories. For every $\mathcal{V}$-enriched category $\mathcal{C}$, define a $\mathcal{V}'$-enriched category $F(\mathcal{C})$ as follows: set $\mathrm{Ob}(F(\mathcal{C})) = \mathrm{Ob}(\mathcal{C})$, and for every pair of objects X, Y, set

$$\begin{aligned} \mathcal{H}\mathrm{om}_{F(\mathcal{C})}(X,Y) &:= F\mathcal{H}\mathrm{om}_{\mathcal{C}}(X,Y), \\ \mathrm{id}_X : \mathbf{1}_{\mathcal{V}'} &\xrightarrow{\varphi_F} F(\mathbf{1}_{\mathcal{V}}) \to F\mathcal{H}\mathrm{om}_{\mathcal{C}}(X,X). \end{aligned}$$

Composition is defined by

$$\begin{aligned} &F\mathcal{H}\mathrm{om}_{\mathcal{C}}(Y,Z) \otimes F\mathcal{H}\mathrm{om}_{\mathcal{C}}(X,Y) \\ &\qquad\qquad \xrightarrow{\xi_F} F\left(\mathcal{H}\mathrm{om}_{\mathcal{C}}(Y,Z) \otimes \mathcal{H}\mathrm{om}_{\mathcal{C}}(X,Y)\right) \to F\mathcal{H}\mathrm{om}_{\mathcal{C}}(X,Z), \end{aligned}$$

and on the corresponding ordinary categories this construction gives an ordinary functor $\mathcal{C} \to F(\mathcal{C})$.

Proof The lax monoidal structure (F, ξ_F, φ_F) supplies all properties required of composition. On the level of ordinary categories, the functor is the identity on objects and sends a morphism $f : \mathbf{1}_{\mathcal{V}} \to \mathcal{H}\mathrm{om}_{\mathcal{C}}(X,Y)$ to the composite $\mathbf{1}_{\mathcal{V}'} \xrightarrow{\varphi_F} F(\mathbf{1}_{\mathcal{V}}) \xrightarrow{Ff} F\mathcal{H}\mathrm{om}_{\mathcal{C}}(X,Y)$. ☐

Comparing this construction with that of Proposition 7.1.15, we see that $\mathcal{H}\mathrm{om}_{F(\mathcal{C})}(X,X)$, as an object of $\mathsf{Alg}(\mathcal{V}')$, is induced from $\mathcal{H}\mathrm{om}_{\mathcal{C}}(X,X)$; the same applies to the bimodule structures on $\mathcal{H}\mathrm{om}$.

Remark 7.2.8 Proposition 7.2.7 gives another way to pass from a dg category to an ordinary category. Apply the lax monoidal functor H of Proposition 7.2.2 to any dg category $\mathcal{C}$ over $\Bbbk$. The result is a category $\mathrm{H}(\mathcal{C})$ enriched over $(\Bbbk\text{-}\mathsf{Mod}^{\mathbb{Z}}, \otimes)$, satisfying

$$\mathrm{Ob}(\mathrm{H}(\mathcal{C})) = \mathrm{Ob}(\mathcal{C}), \quad \mathcal{H}\mathrm{om}_{\mathrm{H}(\mathcal{C})}(X, Y) = (\mathrm{H}^n \operatorname{Hom}^\bullet(X, Y))_{n \in \mathbb{Z}}.$$

Taking the degree-0 part, or equivalently applying the lax monoidal functor H^0 to $\mathcal{C}$, produces an ordinary $\Bbbk$-Mod-category called the **homotopy category** $\mathrm{h}(\mathcal{C})$ of $\mathcal{C}$. It satisfies

$$\mathrm{Ob}(\mathrm{h}\mathcal{C}) = \mathrm{Ob}(\mathcal{C}), \quad \operatorname{Hom}_{\mathrm{h}\mathcal{C}}(X, Y) = \mathrm{H}^0 \operatorname{Hom}^\bullet(X, Y).$$

In the special case $\mathcal{C} = \mathsf{C}(\mathcal{A})$, substituting Definition 3.2.8 immediately gives

$$\mathrm{h}(\mathsf{C}(\mathcal{A})) = \mathsf{K}(\mathcal{A}).$$

As seen above, the world of dg categories supports many homotopy-flavored operations. Their meaning ultimately has to be explained through applications and also involves the abstract language of model categories. We give only a brief outline of some elementary notions, without developing them further.

Definition 7.2.9 Let $F : \mathcal{C} \to \mathcal{D}$ be a dg functor between dg categories. If

$$\operatorname{Hom}^\bullet_{\mathcal{C}}(X, Y) \to \operatorname{Hom}^\bullet_{\mathcal{D}}(FX, FY)$$

is a quasi-isomorphism for all $X, Y \in \mathrm{Ob}(\mathcal{C})$, then F is called **quasi-fully faithful**. If the induced functor $\mathrm{h}F : \mathrm{h}\mathcal{C} \to \mathrm{h}\mathcal{D}$ is essentially surjective, then F is called **quasi-essentially surjective**. A dg functor that is both quasi-fully faithful and quasi-essentially surjective is called a **quasi-equivalence**.

Thus quasi-equivalent dg categories $\mathcal{C}$ and $\mathcal{D}$ have equivalent homotopy categories. We continue this brief discussion of dg categories in §7.4.

7.3 Closed Monoidal Categories

Let $\Bbbk$ be a commutative ring. Then $\Bbbk$-Mod is a symmetric monoidal category under $\otimes := \otimes_{\Bbbk}$. It has the following closedness properties.

- It is enriched over itself: the set of homomorphisms $\operatorname{Hom}(X, Y)$ naturally has the structure of a $\Bbbk$-module;
- the basic properties of the tensor product give a canonical isomorphism in $\Bbbk$-Mod

$$\operatorname{Hom}(X \otimes Y, Z) \simeq \operatorname{Hom}(X, \operatorname{Hom}(Y, Z)).$$

Extending these properties to general monoidal categories leads to the following concept.

Definition 7.3.1 Let $\mathcal{C}$ be a monoidal category. The category $\mathcal{C}$ is called **right closed** (or **left closed**) if, for every object Y, the functor $(\cdot) \otimes Y : \mathcal{C} \to \mathcal{C}$ (or $Y \otimes (\cdot) : \mathcal{C} \to \mathcal{C}$) is equipped with a specified right adjoint. A monoidal category that is both left closed and right closed is called a **closed monoidal category**.

All monoidal categories considered below are braided monoidal categories, so we shall no longer distinguish left closed from right closed. In this situation, the right adjoint of $(\cdot) \otimes Y$ is denoted by $[Y, \cdot] : \mathcal{C} \to \mathcal{C}$.

Recall that if a category $\mathcal{C}$ has finite products, then $\mathcal{C}$ is a symmetric monoidal category under $\times$, with the terminal object as its unit $\mathbf{1}$ (Example 7.1.10).

Definition 7.3.2 Let $\mathcal{C}$ be a category with finite products. If $(\mathcal{C}, \times)$ is a closed monoidal category, then $\mathcal{C}$ is called a **Cartesian closed category**.

More generally, for any two pairs of adjoint functors (F, G) and (F', G') between categories $\mathcal{C}_1$ and $\mathcal{C}_2$, every $\varphi : F \to F'$ naturally induces $\psi : G' \to G$; conversely, every $\psi : G' \to G$ naturally induces $\varphi : F \to F'$. In either case, the induced morphism is characterized by the commutative diagram

$$\begin{tikzcd} \mathrm{Hom}(F'X, Y) \arrow[r, "\sim"] \arrow[d, "(\varphi_X)^*"'] & \mathrm{Hom}(X, G'Y) \arrow[d, "(\psi_Y)_*"] \\ \mathrm{Hom}(FX, Y) \arrow[r, "\sim"'] & \mathrm{Hom}(X, GY) \end{tikzcd}$$

This is simply an application of the Yoneda lemma. The reader may also try to write down these induced morphisms using the unit and counit of the adjoint pairs.

As an application, every morphism $Y \to Y'$ in a closed monoidal category induces $[Y', \cdot] \to [Y, \cdot]$. Thus, being a closed monoidal category is equivalent to the existence of a bifunctor $[\cdot, \cdot] : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{C}$ and a family of canonical bijections

$$\mathrm{Hom}(X \otimes Y, Z) \simeq \mathrm{Hom}(X, [Y, Z]), \tag{7.3.1}$$

functorial in all three variables.

The bifunctor $[\cdot, \cdot]$ is also called the **internal Hom** of the closed monoidal category $\mathcal{C}$. The definition also yields the following conclusions.

- ◇ From $\mathrm{Hom}(X, Z) \simeq \mathrm{Hom}(X \otimes \mathbf{1}, Z) \simeq \mathrm{Hom}(X, [\mathbf{1}, Z])$ and the Yoneda lemma, one obtains $Z \simeq [\mathbf{1}, Z]$;
- ◇ the unit morphism of the adjoint pair gives $\mathrm{coev}_{X,Y} : X \to [Y, X \otimes Y]$, while the counit morphism gives $\mathrm{ev}_{Y,X} : [Y, X] \otimes Y \to X$;

◇ from the composite

$$[Y,Z] \otimes ([X,Y] \otimes X) \xrightarrow{\mathrm{id} \otimes \mathrm{ev}_{X,Y}} [Y,Z] \otimes Y \xrightarrow{\mathrm{ev}_{Y,Z}} Z$$

and the adjunction together with the associativity constraint, one obtains

$$[Y,Z] \otimes [X,Y] \to [X,Z];$$

◇ taking $\mathrm{coev}_{\mathbf{1},X}$ gives a morphism $\mathbf{1} \to [X,X]$.

The term internal Hom and the preceding properties make clear that a closed monoidal category is enriched over itself. We first show how to recover the ordinary Hom from the internal Hom.

Proposition 7.3.3 Let $\mathcal{C}$ be a closed monoidal category. There is a family of canonical bijections

$$\mathrm{Hom}(X,Y) \simeq \mathrm{Hom}(\mathbf{1},[X,Y]), \quad X,Y \in \mathrm{Ob}(\mathcal{C}).$$

Proof By (7.3.1), $\mathrm{Hom}(\mathbf{1},[X,Y]) \simeq \mathrm{Hom}(\mathbf{1} \otimes X, Y) \simeq \mathrm{Hom}(X,Y)$. □

To show that $\mathcal{C}$ is enriched over itself, one must still prove that $[Y,Z] \otimes [X,Z] \to [Y,Z]$ and $\mathbf{1} \to [X,X]$ satisfy axioms such as associativity. This can be checked using the standard properties of the unit and counit. Since the details are rather tedious, the proof is left as an exercise for this chapter.

We next show that the adjunction (7.3.1) is itself automatically internalized in $\mathcal{C}$.

Proposition 7.3.4 Let $\mathcal{C}$ be a closed monoidal category. There is a family of canonical isomorphisms

$$[X \otimes Y, Z] \simeq [X,[Y,Z]];$$

more precisely, this is an isomorphism between functors from $\mathcal{C}^{\mathrm{op}} \times \mathcal{C}^{\mathrm{op}} \times \mathcal{C}$ to $\mathcal{C}$.

Proof Fix X, Y, and Z. For every object T, (7.3.1) and the associativity constraint of $\mathcal{C}$ give natural bijections

$$\begin{aligned}\mathrm{Hom}(T,[X \otimes Y, Z]) &\simeq \mathrm{Hom}(T \otimes (X \otimes Y), Z) \simeq \mathrm{Hom}((T \otimes X) \otimes Y, Z) \\ &\simeq \mathrm{Hom}(T \otimes X, [Y,Z]) \simeq \mathrm{Hom}(T,[X,[Y,Z]]).\end{aligned}$$

Since T is arbitrary, the Yoneda lemma gives the desired isomorphism. □

The following basic examples involve the symmetric monoidal categories introduced in §7.1.

◇ The category of sets Set is Cartesian closed: take $[X,Y]$ to be the set of maps $Y^X = \{\text{maps } X \to Y\}$. Then (7.3.1), or its internalized version in Proposition 7.3.4, gives the natural bijection $Z^{X\times Y} \xrightarrow{1:1} (Z^X)^Y$. In computer science, such a bijection, or its generalization to an arbitrary closed monoidal category, is often called **currying**.

◇ All small categories and the functors between them form the category Cat. Its products are the products of categories $\mathcal{C}_1 \times \cdots \times \mathcal{C}_2$, and its empty product is the category $\mathbf{1}$. The category Cat is Cartesian closed; this is equivalent to the following simple statement: specifying a bifunctor $\mathcal{A} \times \mathcal{B} \to \mathcal{C}$ is equivalent to specifying a functor $\mathcal{A} \to \mathcal{B}^{\mathcal{C}}$, and is also equivalent to specifying $\mathcal{B} \to \mathcal{C}^{\mathcal{A}}$.

◇ Although the category of topological spaces Top has many good properties and is a symmetric monoidal category under the product $\times$, it is not Cartesian closed; see the discussion at the end of [24, §1.5]. Since mapping spaces X^Y occur everywhere in topology, the cost of giving up closedness is too high. A convenient replacement is the category of compactly generated Hausdorff spaces[5)] CGHaus. It is a full subcategory of Top and is itself Cartesian closed. One can prove that there is an adjoint pair $\mathsf{CGHaus} \underset{k}{\overset{\text{inclusion functor}}{\rightleftarrows}} \mathsf{Top}$, where the inclusion functor preserves $\varinjlim$ but not products.

◇ Let G be a group and $\Bbbk$ a commutative ring. The symmetric monoidal category $G\text{-}\mathsf{Mod}$ introduced in Proposition 6.1.6 is closed: its internal Hom is precisely the Hom module introduced in §6.1, and the required adjunction is exactly (6.1.1).

◇ Again let $\Bbbk$ be a commutative ring, and set $\mathcal{A} := \Bbbk\text{-}\mathsf{Mod}$. As seen in §7.2, the category of complexes $\mathsf{C}(\mathcal{A})$ is a symmetric monoidal category. It is closed, with internal Hom given by the Hom complex. The adjunction required in (7.3.1) is the adjunction between Hom and $\otimes$ at the level of complexes. Concretely, it is a corollary of Proposition 4.12.6 in the case $A = R = B = \Bbbk$. The corresponding unit and counit morphisms are

$$\begin{aligned} \mathrm{ev}_{X,Y} : \mathrm{Hom}^n(X,Y) \underset{\Bbbk}{\otimes} X^m &\longrightarrow Y^{n+m} \\ f \otimes x &\longmapsto f(x), \\ \mathrm{coev}_{X,Y} : X^n &\longrightarrow \mathrm{Hom}^n(Y, X \otimes Y) \\ x &\longmapsto [y \mapsto x \otimes y]. \end{aligned}$$

5) A compactly generated Hausdorff space is a Hausdorff space X with the following property: if the intersection of a subset $A \subset X$ with every compact subset $K \subset X$ is closed, then A is closed.

The last example shows that Definition 3.2.1 of the Hom complex is entirely natural, provided that $\mathsf{C}(\mathcal{A})$ is given the symmetric monoidal structure described above.

7.4 Case Study: The Closed Structure on dg-Categories

We previously discussed the closed monoidal structure on Cat, where $\otimes$ comes from the ordinary product of categories $\mathcal{C}_1 \times \mathcal{C}_2$. On the other hand, relative to a fixed commutative ring $\Bbbk$, Definition 7.2.4 introduced dg-categories, namely categories enriched over $(\mathsf{C}(\Bbbk\text{-}\mathsf{Mod}), \otimes)$. The aim of this section is to show that all small dg-categories themselves form a closed monoidal category $\mathsf{dgCat}_\Bbbk$. This category can be regarded as both a linearized and a complex-enriched version of Cat, and it arises naturally and usefully in many situations. All that is needed is a sequence of lengthy but straightforward constructions.

The first step is to upgrade the ordinary product of categories to a tensor product of dg-categories. This involves the symmetric braiding on $(\mathsf{C}(\Bbbk\text{-}\mathsf{Mod}), \otimes)$.

Definition 7.4.1 Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be dg-categories over $\Bbbk$. Define a new dg-category $\mathcal{C}_1 \otimes \mathcal{C}_2$ by

$$\begin{gathered}\mathrm{Ob}(\mathcal{C}_1 \otimes \mathcal{C}_2) := \mathrm{Ob}(\mathcal{C}_1) \times \mathrm{Ob}(\mathcal{C}_2),\\ \mathrm{Hom}^\bullet_{\mathcal{C}_1\otimes\mathcal{C}_2}((X,X'),(Y,Y')) := \mathrm{Hom}^\bullet_{\mathcal{C}_1}(X,Y) \otimes \mathrm{Hom}^\bullet_{\mathcal{C}_2}(X',Y'),\end{gathered}$$

and take $\mathrm{id}_X \otimes \mathrm{id}_{X'}$ as the unit of $\mathrm{Hom}^\bullet_{\mathcal{C}_1\otimes\mathcal{C}_2}((X,X'),(X,X'))$. Composition of morphisms is defined by

$$\begin{gathered}\big(\mathrm{Hom}^\bullet_{\mathcal{C}_1}(Y,Z) \otimes \mathrm{Hom}^\bullet_{\mathcal{C}_2}(Y',Z')\big) \otimes \big(\mathrm{Hom}^\bullet_{\mathcal{C}_1}(X,Y) \otimes \mathrm{Hom}^\bullet_{\mathcal{C}_2}(X',Y')\big)\\ \downarrow\wr\\ \big(\mathrm{Hom}^\bullet_{\mathcal{C}_1}(Y,Z) \otimes \mathrm{Hom}^\bullet_{\mathcal{C}_1}(X,Y)\big) \otimes \big(\mathrm{Hom}^\bullet_{\mathcal{C}_2}(Y',Z') \otimes \mathrm{Hom}^\bullet_{\mathcal{C}_2}(X',Y')\big)\\ \downarrow\\ \mathrm{Hom}^\bullet_{\mathcal{C}_1}(X,Z) \otimes \mathrm{Hom}^\bullet_{\mathcal{C}_2}(X',Z')\end{gathered}$$

where the isomorphism comes from the Koszul braiding and the associativity constraint.

The symmetry of the braiding shows that $\mathcal{C}_1 \otimes \mathcal{C}_2$ is naturally equivalent to $\mathcal{C}_2 \otimes \mathcal{C}_1$. Write $\Bbbk$ for the dg-category with one object and endomorphism complex $\Bbbk$. It is easy to prove that $\Bbbk \otimes \mathcal{C}$ is equivalent to $\mathcal{C} \otimes \Bbbk$.

Definition 7.4.2 Let $\Bbbk$ be a commutative ring. Write $\mathsf{dgCat}_\Bbbk$ for the category of all small dg-categories over $\Bbbk$. This is a symmetric monoidal category with unit $\Bbbk$.

The restriction to small categories is, of course, meant to avoid set-theoretic difficulties. Its chief benefit is that all functors between two given small categories $\mathcal{C}$ and $\mathcal{D}$ form a small set.

The second step is to enrich the Hom set between dg-functors into a complex.

Definition 7.4.3 Let $F, G : \mathcal{C} \to \mathcal{D}$ be dg-functors between dg-categories. For every $n \in \mathbb{Z}$, a morphism of degree n from F to G is defined to be data

$$\phi = (\phi_X)_{X \in \mathrm{Ob}(\mathcal{C})}, \quad \phi_X \in \mathrm{Hom}^n_{\mathcal{D}}(FX, GX),$$

such that, for all $m \in \mathbb{Z}$, $X, Y \in \mathrm{Ob}(\mathcal{C})$, and $f \in \mathrm{Hom}^m_{\mathcal{C}}(X, Y)$, the following equality holds in $\mathrm{Hom}^{n+m}_{\mathcal{D}}(FX, GY)$:

$$(Gf)\phi_X = (-1)^{nm}\phi_Y(Ff);$$

here composition in this formula is understood as the morphisms in $\Bbbk$-**Mod**

$$\begin{gathered}\mathrm{Hom}^m_{\mathcal{D}}(GX, GY) \otimes \mathrm{Hom}^n_{\mathcal{D}}(FX, GX) \to \mathrm{Hom}^{m+n}_{\mathcal{D}}(FX, GY),\\ \mathrm{Hom}^n_{\mathcal{D}}(FY, GY) \otimes \mathrm{Hom}^m_{\mathcal{D}}(FX, FY) \to \mathrm{Hom}^{m+n}_{\mathcal{D}}(FX, GY).\end{gathered}$$

The $\Bbbk$-module of all degree-n morphisms $\phi : F \to G$ is denoted by $\mathcal{H}\mathrm{om}^n(F, G)$.

The condition above may be viewed as saying that the diagram of graded morphisms

$$\begin{tikzcd} FX \arrow[r, "\phi_X"] \arrow[d, "Ff"'] & GX \arrow[d, "Gf"] \\ FY \arrow[r, "\phi_Y"'] & GY \end{tikzcd} \qquad \begin{aligned} &\phi : \text{degree } n \\ &f : \text{degree } m \end{aligned}$$

commutes up to the sign $(-1)^{nm}$ required by the Koszul sign rule. Strictly speaking, these "graded morphisms" are not morphisms, but must be understood as elements of a Hom complex.

Definition–Proposition 7.4.4 For dg-functors $F, G : \mathcal{C} \to \mathcal{D}$ and every $n \in \mathbb{Z}$, one can define a homomorphism

$$d^n = d^n_{\mathcal{H}\mathrm{om}^\bullet(F,G)} : \mathcal{H}\mathrm{om}^n(F, G) \to \mathcal{H}\mathrm{om}^{n+1}(F, G).$$

For every $\phi = (\phi_X)_{X \in \mathrm{Ob}(\mathcal{C})}$, set

$$(d^n\phi)_X := d^n_{\mathrm{Hom}^\bullet(FX,GX)}(\phi_X).$$

This makes $(\mathcal{H}\mathrm{om}^n(F, G), d^n)_{n \in \mathbb{Z}}$ into a complex $\mathcal{H}\mathrm{om}^\bullet(F, G)$.

Proof First check that $d^n\phi$ does indeed belong to $\mathcal{H}\mathrm{om}^{n+1}(F,G)$. Let $f \in \mathrm{Hom}^m_{\mathcal{C}}(X,Y)$. Apply d^{m+n} to both sides of $(Gf)\phi_X = (-1)^{nm}\phi_Y(Ff)$ (suppressing the subscript $\mathrm{Hom}^\bullet(FX,GY)$). This gives

$$\begin{aligned}
d^{m+n}((Gf)\phi_X) &= (d^m Gf)\phi_X + (-1)^m (Gf)(d^n\phi_X) \\
&= G(d^m f)\phi_X + (-1)^m (Gf)(d^n\phi)_X \\
&= (-1)^{n(m+1)}\phi_Y F(d^m f) + (-1)^m (Gf)(d^n\phi)_X, \\
(-1)^{nm} d^{m+n}(\phi_Y(Ff)) &= (-1)^{nm}(d^n\phi_Y)Ff + (-1)^{nm+n}\phi_Y d^n(Ff) \\
&= (-1)^{nm}(d^n\phi)_Y Ff + (-1)^{nm+n}\phi_Y F(d^m f).
\end{aligned}$$

Equality of these two expressions immediately yields $(Gf)(d^n\phi)_X = (-1)^{(n+1)m}(d^n\phi)_Y Ff$.

Second, since $\mathrm{Hom}^\bullet(FX,GX)$ is a complex,

$$\begin{aligned}
(d^{n+1}d^n\phi)_X &= d^{n+1}_{\mathrm{Hom}^\bullet(FX,GX)}((d^n\phi)_X) \\
&= d^{n+1}_{\mathrm{Hom}^\bullet(FX,GX)} d^n_{\mathrm{Hom}^\bullet(FX,GX)}\phi_X = 0.
\end{aligned}$$

Thus we obtain the complex $\mathcal{H}\mathrm{om}^\bullet(F,G)$. $\square$

For three dg-functors $F,G,H : \mathcal{C} \to \mathcal{D}$, there is a composition operation

$$\begin{aligned}
\mathcal{H}\mathrm{om}^a(G,H) \otimes \mathcal{H}\mathrm{om}^b(F,G) &\to \mathcal{H}\mathrm{om}^{a+b}(F,H) \\
\psi \otimes \phi &\mapsto \psi\phi := (\psi_X\phi_X)_{X\in\mathrm{Ob}(\mathcal{C})},
\end{aligned} \tag{7.4.1}$$

where $a,b \in \mathbb{Z}$. This operation is associative and satisfies $d^{a+b}(\psi\phi) = (d^a\psi)\phi + (-1)^a\psi(d^b\phi)$. By definition, everything reduces to a verification in $\mathrm{Hom}^\bullet_{\mathcal{D}}$. This composition should be understood as vertical composition of graded morphisms, depicted as

$$\begin{tikzcd}
\mathcal{C} \arrow[r, bend left=60, "F"{name=F}] \arrow[r, "G"{name=G}] \arrow[r, bend right=60, "H"'{name=H}] & \mathcal{D}
\arrow[from=F, to=G, phantom, "\Downarrow\phi"]
\arrow[from=G, to=H, phantom, "\Downarrow\psi"]
\end{tikzcd} \quad \text{composed to give} \quad
\begin{tikzcd}
\mathcal{C} \arrow[r, bend left=50, "F"] \arrow[r, bend right=50, "H"'] \arrow[r, phantom, "\Downarrow\psi\phi"] & \mathcal{D}.
\end{tikzcd}$$

Consider the special case $n = 0$. The condition on $\phi = (\phi_X)_X \in \mathcal{H}\mathrm{om}^0(F,G)$ says precisely that $(Gf)\phi_X = \phi_Y(Ff)$ for every $f \in \mathrm{Hom}^m(X,Y)$, while the condition $d^0\phi = 0$ says precisely that $\phi_X \in \ker\left(d^0_{\mathrm{Hom}^\bullet(FX,GX)}\right)$ for every $X \in \mathrm{Ob}(\mathcal{C})$. Hence the elements of the $\Bbbk$-module

$$\ker\left[\mathcal{H}\mathrm{om}^0(F,G) \xrightarrow{d^0} \mathcal{H}\mathrm{om}^1(F,G)\right]$$

are exactly the morphisms $\phi : F \to G$ between the dg-functors F and G in the ordinary sense.

The next step is to enrich the functor category between dg-categories into a dg-category.

Definition 7.4.5 For arbitrary dg-categories $\mathcal{C}$ and $\mathcal{D}$, define the dg-category $\mathcal{H}\mathrm{om}(\mathcal{C}, \mathcal{D})$ as follows.

- Its objects are all dg-functors $F : \mathcal{C} \to \mathcal{D}$.
- For any dg-functors $F, G : \mathcal{C} \to \mathcal{D}$, their Hom complex is the complex $\mathcal{H}\mathrm{om}^\bullet(F, G)$ defined above.
- The composition operation $\mathcal{H}\mathrm{om}^\bullet(G, H) \otimes \mathcal{H}\mathrm{om}^\bullet(F, G) \to \mathcal{H}\mathrm{om}^\bullet(F, H)$ is defined as in (7.4.1); it is called vertical composition.

If $\mathcal{C}$ and $\mathcal{D}$ are both small dg-categories, then $\mathcal{H}\mathrm{om}(\mathcal{C}, \mathcal{D})$ is also small: its set of objects is a small set.

In Definition 7.4.1 we defined the tensor product of two dg-categories, making the category $\mathsf{dgCat}_{\Bbbk}$ of Definition 7.4.2 a symmetric monoidal category under $\otimes$. The relation between $\otimes$ and $\mathcal{H}\mathrm{om}$ can now be stated clearly as follows.

Proposition 7.4.6 The symmetric monoidal category $\mathsf{dgCat}_{\Bbbk}$ is closed. Its internal Hom is given by $\mathcal{H}\mathrm{om}(\mathcal{C}, \mathcal{D})$ as defined above, where $\mathcal{C}$ and $\mathcal{D}$ range over all small dg-categories.

Proof The key is to give a canonical isomorphism

$$\mathrm{Hom}_{\mathsf{dgCat}_{\Bbbk}}(\mathcal{C} \otimes \mathcal{D}, \mathcal{E}) \simeq \mathrm{Hom}_{\mathsf{dgCat}_{\Bbbk}}(\mathcal{C}, \mathcal{H}\mathrm{om}(\mathcal{D}, \mathcal{E})).$$

If the dg-structure is ignored, the matter is simple: specifying $F : \mathcal{C} \times \mathcal{D} \to \mathcal{E}$ is equivalent to specifying a functor $\mathcal{C} \to \mathcal{E}^{\mathcal{D}}$ that maps an object X to the functor $F(X, \cdot) : \mathcal{D} \to \mathcal{E}$. The essential issue is the condition at the level of Hom that makes a functor a dg-functor.

By definition, making F into a dg-functor is equivalent to upgrading the maps on Hom sets to morphisms of complexes

$$F_{X,Y,Z,W} : \mathrm{Hom}^\bullet_{\mathcal{C}}(X, Y) \otimes \mathrm{Hom}^\bullet_{\mathcal{D}}(Z, W) \to \mathrm{Hom}^\bullet_{\mathcal{E}}(F(X, Z), F(Y, W)),$$

functorial in all four variables. Treating the two variables of F separately, these data can be split into two parts:

- for every $X \in \mathrm{Ob}(\mathcal{C})$, the functor $F(X, \cdot)$ is upgraded to a dg-functor;
- there is a family of morphisms $F_{X,Y,Z} : \mathrm{Hom}^\bullet_{\mathcal{C}}(X, Y) \to \mathrm{Hom}^\bullet_{\mathcal{E}}(F(X, Z), F(Y, Z))$, functorial in all three variables.

For illustration, $F_{X,Y,Z,W}(f \otimes g)$ is obtained from $F_{X,Y,Z}(f)$ by using $g \in \mathrm{Hom}^b_{\mathcal{D}}(Z, W)$ through $F(Y, \cdot)$. The data in the second item are equivalent to a family of morphisms

$$\mathrm{Hom}^\bullet_{\mathcal{C}}(X, Y) \to \mathcal{H}\mathrm{om}^\bullet(F(X, \cdot), F(Y, \cdot)),$$

functorial in X and Y. In conclusion, specifying the dg-functor F is equivalent to specifying a dg-functor from $\mathcal{C}$ to $\mathcal{H}\mathrm{om}(\mathcal{D}, \mathcal{E})$. □

Together with the theory in §7.3, the preceding discussion shows that $\mathsf{dgCat}_{\Bbbk}$ is enriched over itself: for any small dg-categories $\mathcal{C}$ and $\mathcal{D}$, their $\mathcal{H}\mathrm{om}$ object is again a dg-category.

Since $\otimes$ and $\mathcal{H}\mathrm{om}$ determine each other through the adjunction (7.3.1), once either the definition of $\otimes$ or that of $\mathcal{H}\mathrm{om}$ is accepted, the other definition is at least equally natural.

Remark 7.4.7 With Proposition 7.4.6 in hand, the general properties of closed monoidal categories show that for any three small dg-categories $\mathcal{C}$, $\mathcal{D}$, and $\mathcal{E}$, composition of functors can be upgraded to a dg-functor

$$\mathcal{H}\mathrm{om}(\mathcal{D}, \mathcal{E}) \otimes \mathcal{H}\mathrm{om}(\mathcal{C}, \mathcal{D}) \to \mathcal{H}\mathrm{om}(\mathcal{C}, \mathcal{E}).$$

On objects, this functor maps (G, F) to GF; on Hom complexes, it is written

$$\begin{array}{c} \mathcal{H}\mathrm{om}^\bullet(G_1, G_2) \otimes \mathcal{H}\mathrm{om}^\bullet(F_1, F_2) \longrightarrow \mathcal{H}\mathrm{om}^\bullet(G_1F_1, G_2F_2) \\ \| \\ \mathrm{Hom}^\bullet_{\mathcal{H}\mathrm{om}(\mathcal{D},\mathcal{E})\otimes\mathcal{H}\mathrm{om}(\mathcal{C},\mathcal{D})}((G_1, F_1), (G_2, F_2)) \\ F_i : \mathcal{C} \to \mathcal{D}, \quad G_i : \mathcal{D} \to \mathcal{E}, \quad i = 1, 2. \end{array} \tag{7.4.2}$$

The operation above may be pictured as horizontal composition of morphisms between dg-functors, with diagram

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F_1"] \arrow[r, bend right, "F_2"'] \arrow[r, phantom, "\Downarrow\phi"] & \mathcal{D} \arrow[r, bend left, "G_1"] \arrow[r, bend right, "G_2"'] \arrow[r, phantom, "\Downarrow\psi"] & \mathcal{E} \end{tikzcd} \quad \text{composed to give} \quad \begin{tikzcd} \mathcal{C} \arrow[r, bend left, "G_1F_1"] \arrow[r, bend right, "G_2F_2"'] \arrow[r, phantom, "\Downarrow\psi\phi"] & \mathcal{E}. \end{tikzcd}$$

Because this gives a functor, (7.4.2) must (a) be associative, (b) commute with pullback (or pushforward) along graded morphisms in the dg-category $\mathcal{H}\mathrm{om}(\mathcal{D}, \mathcal{E})$ (or $\mathcal{H}\mathrm{om}(\mathcal{C}, \mathcal{D})$). But the operation in (b) is precisely the vertical composition introduced in (7.4.1). Thus, we see that the order of vertical and horizontal composition can be interchanged.

For functors between ordinary categories, morphisms between the functors likewise have commuting vertical and horizontal compositions. This can be understood as a property of the Cartesian closed category Cat (or by regarding it as a 2-category; see [51, Example 3.5.3]). The property of $\mathsf{dgCat}_{\Bbbk}$ above is its dg version, or its linear algebra version.

7.5 From Coalgebras to Hopf Algebras

The first aim of this section is to dualize Definition 7.1.1 of an algebra. To this end, first note that the definition of a monoidal category is self-dual. Let $\mathcal{V}$ be a monoidal category. On $\mathcal{V}^{\mathrm{op}}$, one can retain the original bifunctor $\otimes$ and unit object $\mathbf{1}$, but replace the associativity constraint a and $\iota : \mathbf{1} \otimes \mathbf{1} \xrightarrow{\sim} \mathbf{1}$ in [51, Definition 3.1.1] by their inverses. In this way, $\mathcal{V}^{\mathrm{op}}$ becomes a monoidal category.

Similarly, if $\mathcal{V}$ has a braiding c as in [51, Definition 3.3.1], taking inverses gives the corresponding braiding on $\mathcal{V}^{\mathrm{op}}$. Under this correspondence, $\mathcal{V}$ is a symmetric monoidal category if and only if $\mathcal{V}^{\mathrm{op}}$ is one.

Definition 7.5.1 (Coalgebra) Let $\mathcal{V}$ be a monoidal category. An algebra over $\mathcal{V}^{\mathrm{op}}$ is called a **coalgebra** over $\mathcal{V}$. In other words, a coalgebra is data (C, Δ, ϵ), where

$$C \in \mathrm{Ob}(\mathcal{V}), \quad \Delta = \Delta_C : C \to C \otimes C, \quad \epsilon = \epsilon_C : C \to \mathbf{1},$$

satisfying the commutative diagrams dual to those in Definition 7.1.1:

$$\begin{tikzcd}
\mathbf{1} \otimes C & C \otimes C \arrow[l, "\epsilon\otimes\mathrm{id}"'] \arrow[r, "\mathrm{id}\otimes\epsilon"] & C \otimes \mathbf{1} \\
& C \arrow[ul, "\sim"] \arrow[u, "\Delta"'] \arrow[ur, "\sim"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
(C \otimes C) \otimes C & & C \otimes (C \otimes C) \arrow[ll, "\sim"] \\
C \otimes C \arrow[u, "\Delta\otimes\mathrm{id}"] & C \arrow[l, "\Delta"] \arrow[r, "\Delta"'] & C \otimes C. \arrow[u, "\mathrm{id}\otimes\Delta"']
\end{tikzcd}$$

The map Δ is customarily called the **comultiplication** on C, and ϵ the **counit** of C.

A morphism from a coalgebra (C, Δ, ϵ) to (C', Δ', ϵ') is defined dually to Definition 7.1.1: it is a morphism $\phi : C \to C'$ for which the following diagrams commute:

$$\begin{tikzcd}
\mathbf{1} & C \arrow[l, "\epsilon"'] \arrow[d, "\phi"] \\
& C' \arrow[ul, "\epsilon'"]
\end{tikzcd}
\qquad
\begin{tikzcd}
C \otimes C \arrow[r, "\phi\otimes\phi"] & C' \otimes C' \\
C \arrow[u, "\Delta"] \arrow[r, "\phi"'] & C'. \arrow[u, "\Delta'"']
\end{tikzcd}$$

If $\mathcal{V}$ is equipped with a braiding c and $c\Delta = \Delta$, then (C, Δ, ϵ) is called **cocommutative**.

The following definition is simply the dual of Definition 7.1.2.

Definition 7.5.2 (Comodule) Let (C, Δ, ϵ) be a coalgebra over $\mathcal{V}$. A left C-comodule is defined to be data (M, ρ), where

$$M \in \mathrm{Ob}(\mathcal{V}), \quad \rho = \rho_M : M \to C \otimes M,$$

and ρ may be viewed as scalar comultiplication on the left comodule. The condition is that the following diagrams commute:

$$\begin{tikzcd} \mathbf{1} \otimes M & C \otimes M \arrow[l, "\epsilon\otimes\mathrm{id}"'] \\ & M \arrow[ul, "\sim"] \arrow[u, "\rho"'] \end{tikzcd} \qquad \begin{tikzcd} (C \otimes C) \otimes M & & C \otimes (C \otimes M) \arrow[ll, "\sim"'] \\ C \otimes M \arrow[u, "\Delta\otimes\mathrm{id}"] & M \arrow[l, "\rho"] \arrow[r, "\rho"'] & C \otimes M. \arrow[u, "\mathrm{id}\otimes\rho"'] \end{tikzcd}$$

A homomorphism from a left comodule (M, ρ) to (M', ρ') is a morphism $f : M \to M'$ satisfying $(\mathrm{id} \otimes f)\rho = \rho' f$. Right C-comodules, bicomodules, and homomorphisms between them are defined similarly.

An elementary but important example is C itself: with $\rho_C := \Delta : C \to C \otimes C$, it becomes both a left and a right comodule.

Although the definition of a comodule may seem contrary to one's usual intuition, working with comodules need not be more complicated than working with modules. For example, take $\mathcal{V} = \Bbbk\text{-}\mathsf{Mod}$, where $\Bbbk$ is a commutative ring. Let M be a free $\Bbbk$-module with basis $(v_i)_{i \in I}$. Specifying $\rho : M \to M \underset{\Bbbk}{\otimes} C$ is equivalent to specifying a family of elements $(t_{ij})_{(i,j) \in I^2}$ of C such that, for each $i \in I$,

$$\rho(v_i) = \sum_{j \in I} v_j \otimes t_{ji} \quad \text{(finite sum)},$$

and the comodule conditions are expressed, for every i, by

$$v_i = \sum_j \epsilon(t_{ji}) v_j, \quad \sum_j v_j \otimes \Delta(t_{ji}) = \sum_{j,k} v_k \otimes t_{kj} \otimes t_{ji}.$$

Renaming the indices appropriately, these conditions simplify further, for all i, j, to

$$\begin{aligned} \epsilon(t_{ji}) &= \delta_{j,i}, \\ \Delta(t_{ji}) &= \sum_k t_{jk} \otimes t_{ki}; \end{aligned} \tag{7.5.1}$$

here $\delta_{j,i}$ is the Kronecker delta, defined to be 1 when $j = i$ and 0 otherwise.

Continuing the preceding discussion, express a $\Bbbk$-linear map $f : M \to M'$ between right comodules relative to the chosen bases as $f(v_i) = \sum_j r_{ji} v'_j$, that is, as a matrix with $r_{ji} \in \Bbbk$. A similar argument proves that f is a comodule homomorphism if and only if, for all i, k,

$$\sum_j r_{kj} t_{ji} = \sum_j r_{ji} t'_{kj},$$

where $\Bbbk$ acts on C from the left by scalar multiplication. The version for left comodules is, of course, similar.

We shall continue the discussion of several aspects of comodule theory in §7.7.

Remark 7.5.3 If (C, Δ, ϵ) is a coalgebra over $\mathcal{V}$ and (A, μ, η) is an algebra over $\mathcal{V}$, then $\mathrm{Hom}(C, A)$ is equipped with a binary operation $\star$, called **convolution**, as follows:

$$f \star g := \mu(f \otimes g)\Delta, \quad f, g \in \mathrm{Hom}(C, A).$$

The properties of Δ and μ readily show that $\star$ is associative, and the following commutative diagram suffices to show that $(\mathrm{Hom}(C, A), \star)$ is a monoid with unit $\eta\epsilon \in \mathrm{Hom}(C, A)$:

$$\begin{tikzcd}
A \otimes \mathbf{1} \arrow[r, "\mu(\mathrm{id}\otimes\eta)"] & A & \mathbf{1} \otimes A \arrow[l, "\mu(\eta\otimes\mathrm{id})"'] \\
C \otimes \mathbf{1} \arrow[u, "f\otimes\mathrm{id}"] \arrow[r] & C \arrow[u, "f"] & \mathbf{1} \otimes C \arrow[l] \arrow[u, "\mathrm{id}\otimes f"'] \\
C \otimes C \arrow[u, "\mathrm{id}\otimes\epsilon"] & C \arrow[l, "\Delta"] \arrow[r, "\Delta"'] \arrow[u, equal] & C \otimes C. \arrow[u, "\epsilon\otimes\mathrm{id}"']
\end{tikzcd}$$

A standard argument shows that if $\varphi : C' \to C$ is a coalgebra morphism and $\psi : A \to A'$ an algebra morphism, then

$$\mathrm{Hom}(C, A) \to \mathrm{Hom}(C', A'), \quad f \mapsto \psi f \varphi \tag{7.5.2}$$

is a homomorphism between the convolution monoids.

From now on, let $\mathcal{V}$ be a braided monoidal category, and write its braiding in the usual form

$$c(X, Y) : X \otimes Y \to Y \otimes X.$$

By §7.1, the category of algebras $\mathsf{Alg}(\mathcal{V})$ (or the category of coalgebras $\mathsf{Alg}(\mathcal{V}^{\mathrm{op}})^{\mathrm{op}}$) has a monoidal structure with unit $\mathbf{1}$. We may therefore consider coalgebras in the category of algebras (or algebras in the category of coalgebras). A simple but interesting observation is that both lead to the same data $(A, \mu, \eta, \Delta, \epsilon)$, with $A \in \mathrm{Ob}(\mathcal{V})$. The conditions required of the morphisms $A \xrightarrow{\Delta} A \otimes A \xrightarrow{\mu} A$ and $\mathbf{1} \xrightarrow{\eta} A \xrightarrow{\epsilon} \mathbf{1}$ in $\mathcal{V}$ are:

(i) (A, μ, η) is an algebra;

(ii) (A, Δ, ϵ) is a coalgebra;

(iii) the comultiplication $\Delta : A \to A \otimes A$ and counit $\epsilon : A \to \mathbf{1}$ are algebra morphisms;

(iv) the multiplication $\mu : A \otimes A \to A$ and unit $\eta : \mathbf{1} \to A$ are coalgebra morphisms.

Once (i) and (ii) are assumed, (iii) and (iv) are equivalent, so it is enough to check either one. Both state the compatibility of (Δ, ϵ) and (μ, η) through four commutative diagrams, one for each possible pair; in formulas,

$$\Delta\mu = (\mu \otimes \mu)(\mathrm{id}_A \otimes c(A, A) \otimes \mathrm{id}_A)(\Delta \otimes \Delta),$$
$$\Delta\eta = \eta \otimes \eta, \quad \epsilon\mu = \epsilon \otimes \epsilon, \quad \epsilon\eta = \mathrm{id}_{\mathbf{1}}.$$

Definition 7.5.4 (Bialgebra) Let $\mathcal{V}$ be a braided monoidal category. If A is an algebra in the category of coalgebras over $\mathcal{V}$, or equivalently a coalgebra in the category of algebras over $\mathcal{V}$, then A is called a **bialgebra** over $\mathcal{V}$. Concretely, a bialgebra consists of the data $(A, \mu, \eta, \Delta, \epsilon)$.

If A and A' are both bialgebras and $\phi : A \to A'$ is simultaneously an algebra morphism and a coalgebra morphism, then ϕ is called a morphism from the bialgebra A to the bialgebra A'.

The full data of a bialgebra are often abbreviated simply as A.

If $\Bbbk$ is a commutative ring and $\mathcal{V} = \Bbbk\text{-}\mathsf{Mod}$, then algebras, coalgebras, and bialgebras over $\mathcal{V}$ are also called $\Bbbk$-algebras, $\Bbbk$-coalgebras, and $\Bbbk$-bialgebras, respectively, and so forth.

Example 7.5.5 Let M be a monoid and $\Bbbk$ a commutative ring. Form the monoid algebra $\Bbbk[M]$ over $\Bbbk$; see [51, Definition 5.6.1]. Define the $\Bbbk$-module homomorphism

$$\Delta : \Bbbk[M] \to \Bbbk[M] \underset{\Bbbk}{\otimes} \Bbbk[M]$$
$$m \mapsto m \otimes m, \quad m \in M$$

and $\epsilon : \Bbbk[M] \to \Bbbk$ by mapping every $m \in M$ to 1. Let us verify that these data give a $\Bbbk$-bialgebra.

First, $(\Bbbk[M], \Delta, \epsilon)$ is a coalgebra, since for every $m \in M$,

$$(\mathrm{id} \otimes \epsilon)(\Delta(m)) = m = (\epsilon \otimes \mathrm{id})(\Delta(m)),$$
$$(\Delta \otimes \mathrm{id})(\Delta(m)) = m \otimes m \otimes m = (\mathrm{id} \otimes \Delta)(\Delta(m)).$$

Note that $\Bbbk[M]$ is always cocommutative, whereas $\Bbbk[M]$ is commutative if and only if M is commutative.

Second, check that ϵ and Δ are both homomorphisms of $\Bbbk$-algebras. It is enough to verify, for every $m, m' \in M$, that

$$\epsilon(mm') = 1 = \epsilon(m)\epsilon(m'), \quad \epsilon(1) = 1,$$
$$\Delta(mm') = mm' \otimes mm' = (m \otimes m)(m' \otimes m') = \Delta(m)\Delta(m'),$$
$$\Delta(1) = 1 \otimes 1.$$

For an algebra A over $\mathcal{V}$, Definition 7.1.2 describes left A-modules. If A is a bialgebra, we can go further as follows.

1. On the unit object $\mathbf{1}$, define a left A-module structure by taking $\mu_{\mathbf{1}} : A \otimes \mathbf{1} \to \mathbf{1}$ to be the composite $A \otimes \mathbf{1} \xrightarrow{\sim} A \xrightarrow{\epsilon} \mathbf{1}$.

2. For left A-modules M_i, denote the corresponding scalar multiplication morphisms by μ_i $(i = 1, 2)$. Define the morphism $\mu_{M_1 \otimes M_2} : A \otimes (M_1 \otimes M_2) \to M_1 \otimes M_2$ to be the composite

$$A \otimes M_1 \otimes M_2 \xrightarrow{\Delta \otimes \mathrm{id}_{M_1 \otimes M_2}} A \otimes A \otimes M_1 \otimes M_2$$
$$\xrightarrow{\mathrm{id}_A \otimes c(A, M_1) \otimes \mathrm{id}_{M_2}} A \otimes M_1 \otimes A \otimes M_2 \xrightarrow{\mu_1 \otimes \mu_2} M_1 \otimes M_2.$$

Here, to simplify notation, parentheses in $\otimes$ operations and the associativity constraints have been suppressed; in other words, $\mathcal{V}$ is treated as a strict monoidal category.

These definitions are intended to give the objects $\mathbf{1}$ and $M_1 \otimes M_2$ of $\mathcal{V}$ left A-module structures. The case of right A-modules is, of course, similar.

Proposition 7.5.6 Let A be a bialgebra over a braided monoidal category $\mathcal{V}$. The definitions above make the category of left A-modules A-Mod a monoidal category with unit $(\mathbf{1}, \mu_{\mathbf{1}})$. The same holds for the category of right A-modules Mod-A.

Dually, the multiplication and unit of A make the category of left A-comodules A-Comod and the category of right A-comodules Comod-A into monoidal categories.

For all four categories above, the forgetful functor to $\mathcal{V}$ is always a monoidal functor.

Proof By duality, it is enough to prove the first assertion. The verification is lengthy but straightforward; we outline it. First, one must check that scalar multiplication satisfies the commutative diagrams in Definition 7.1.2. For $(\mathbf{1}, \mu_{\mathbf{1}})$, these properties follow from the fact that $\epsilon : A \to \mathbf{1}$ is an algebra morphism. For the module structure on $M_1 \otimes M_2$, the diagram for scalar multiplication by the unit follows from the fact that $\Delta : A \to A \otimes A$ is an algebra morphism, in particular from its compatibility with η and $\eta \otimes \eta$.

For associativity of scalar multiplication, after expanding the definition, the issue is to prove that two maps from $A \otimes A \otimes M_1 \otimes M_2$ to $M_1 \otimes M_2$ are equal:

(a) first apply $\Delta \otimes \Delta \otimes \mathrm{id}_{M_1} \otimes \mathrm{id}_{M_2}$, then, taking the braiding into account, successively let the first and third (or second and fourth) copies of A in $A \otimes A \otimes (A \otimes A \otimes M_1 \otimes M_2)$ act from the left on M_1 (or M_2);

(b) first apply $\mu : A \otimes A \to A$ in the first two positions, then apply $\mu_{M_1 \otimes M_2}$. Since

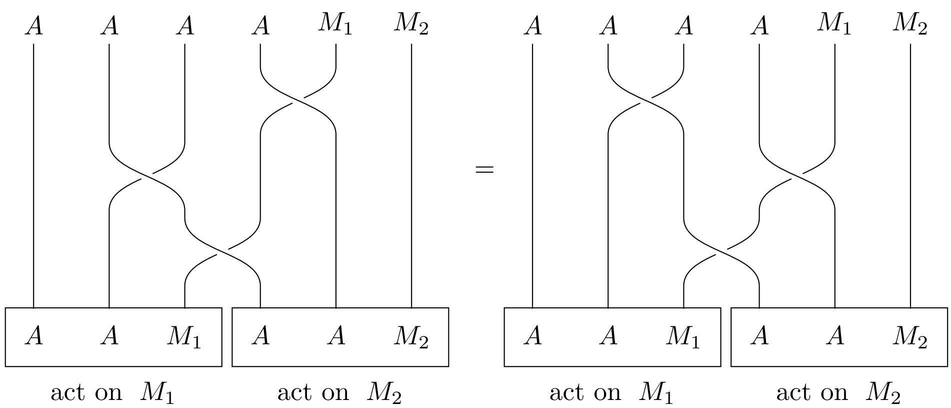
$$\Delta\mu = (\mu \otimes \mu)(\mathrm{id} \otimes c(A, A) \otimes \mathrm{id})(\Delta \otimes \Delta),$$

this map is the composite

$$\begin{aligned} A\otimes A\otimes M_1\otimes M_2 &\xrightarrow{\Delta\otimes\Delta\otimes\mathrm{id}_{M_1}\otimes\mathrm{id}_{M_2}} A\otimes A\otimes A\otimes A\otimes M_1\otimes M_2\\ &\xrightarrow{\mathrm{id}_A\otimes c(A,A)\otimes\mathrm{id}_A\otimes\mathrm{id}_{M_1}\otimes\mathrm{id}_{M_2}} \boxed{A\otimes A\otimes A\otimes A}\otimes M_1\otimes M_2\\ &\xrightarrow{\mu\otimes\mu\otimes\mathrm{id}_{M_1}\otimes\mathrm{id}_{M_2}} A\otimes A\otimes M_1\otimes M_2\\ &\xrightarrow{\mathrm{id}_A\otimes c(A,M_1)\otimes\mathrm{id}_{M_2}} A\otimes M_1\otimes A\otimes M_2\xrightarrow{\mu_1\otimes\mu_2} M_1\otimes M_2. \end{aligned}$$

The scalar multiplications μ_1 and μ_2 are each associative and c is functorial. Thus, taking the braiding into account, the last three arrows in this composite may also be regarded as first letting the first and second (or third and fourth) copies of A in the boxed part act successively from the left on M_1 (or M_2).

Braid diagrams help explain this, as at the end of [51, §7.4]. To prove that (a) equals (b), it is enough to verify the following equality[6)]



which is immediately apparent.

To show that A-Mod is a monoidal category, one must still prove that the associativity and unit constraints for $\otimes$ are all morphisms of left A-modules. The first assertion is straightforward. The second requires the coalgebra property $(\mathrm{id}\otimes\epsilon)\Delta = \mathrm{id}_A = (\epsilon\otimes\mathrm{id})\Delta$, after identifying $A\otimes\mathbf{1}\simeq A\simeq\mathbf{1}\otimes A$. $\square$

For a bialgebra A, one can still ask whether the monoidal category A-Mod has a braiding and whether that braiding is symmetric. A natural first idea is to use the braiding $c(M_1, M_2)$ of $\mathcal{V}$, but in general this is not a morphism in A-Mod. Even in the special case $\mathcal{V} = \Bbbk$-Mod, the answer already involves the concept of a **quasitriangular bialgebra**, which cannot be explained briefly. We give only the following example, which is so simple as to be somewhat misleading.

6) There is one harmless minor difference: in the cited reference, morphisms are composed from bottom to top, whereas here they are composed in the opposite direction.

Proposition 7.5.7 Let $\mathcal{V}$ be a symmetric monoidal category.

(i) If A is a cocommutative bialgebra over $\mathcal{V}$, then the monoidal categories A-Mod and Mod-A become symmetric monoidal categories with respect to $c(M_1, M_2) : M_1 \otimes M_2 \xrightarrow{\sim} M_2 \otimes M_1$.

(ii) Dually, if A is a commutative bialgebra over $\mathcal{V}$, then A-Comod and Comod-A are symmetric monoidal categories.

Proof It is enough to treat the cocommutative case. Using the preceding notation, the assertion that $c(M_1, M_2)$ is a morphism in A-Mod is equivalent to proving that the outer boundary of the following diagram commutes:

$$\begin{tikzcd}
A \otimes M_1 \otimes M_2 \arrow[r, "\Delta\otimes\mathrm{id}\otimes\mathrm{id}"] \arrow[d, "\mathrm{id}\otimes c(M_1,M_2)"] & A \otimes A \otimes M_1 \otimes M_2 \arrow[r, "\mathrm{id}\otimes c(A,M_1)\otimes\mathrm{id}"] \arrow[d, "c(A,A)\otimes c(M_1,M_2)"] & A \otimes M_1 \otimes A \otimes M_2 \arrow[r, "\mu_1\otimes\mu_2"] \arrow[d, "c(A\otimes M_1, A\otimes M_2)"] & M_1 \otimes M_2 \arrow[d, "c(M_1,M_2)"] \\
A \otimes M_2 \otimes M_1 \arrow[r, "\Delta\otimes\mathrm{id}\otimes\mathrm{id}"'] & A \otimes A \otimes M_2 \otimes M_2 \arrow[r, "\mathrm{id}\otimes c(A,M_2)\otimes\mathrm{id}"'] & A \otimes M_2 \otimes A \otimes M_1 \arrow[r, "\mu_2\otimes\mu_1"'] & M_2 \otimes M_1.
\end{tikzcd}$$

The square on the left commutes because A is cocommutative, while the square on the right commutes by functoriality of c. Drawing the braid diagram shows that

$$\begin{aligned} &c(A \otimes M_1, A \otimes M_2) = \\ &\qquad (\mathrm{id}_A \otimes c(A, M_2) \otimes \mathrm{id}_{M_1})(c(A, A) \otimes c(M_1, M_2))(\mathrm{id}_A \otimes c(A, M_1) \otimes \mathrm{id}_{M_2}), \end{aligned}$$

and since $\mathcal{V}$ is a symmetric monoidal category, the middle square also commutes. All other properties of the symmetric monoidal structure on A-Mod reduce to verifications in $\mathcal{V}$. □

We now introduce the concept of a Hopf algebra. Let $\mathcal{V}$ be a braided monoidal category.

Definition 7.5.8 Let the data $(A, \mu, \eta, \Delta, \epsilon)$ form a bialgebra A over $\mathcal{V}$. If an isomorphism $S : A \xrightarrow{\sim} A$ in $\mathcal{V}$ makes the following diagram commute,

$$\begin{tikzcd}
A \otimes A \arrow[d, "\mu"'] & A \otimes A \arrow[l, "\mathrm{id}_A\otimes S"'] \arrow[r, "S\otimes\mathrm{id}_A"] & A \otimes A \arrow[d, "\mu"] \\
A & A \arrow[l, "\eta\epsilon"] \arrow[r, "\eta\epsilon"'] \arrow[u, "\Delta"] & A,
\end{tikzcd}$$

that is,

$$\mu(\mathrm{id}_A \otimes S)\Delta = \eta\epsilon = \mu(S \otimes \mathrm{id}_A)\Delta,$$

then S is called an **antipode** of A. A bialgebra equipped with an antipode is called a **Hopf algebra**. Morphisms between Hopf algebras are defined to be morphisms between their underlying bialgebras.

For example, $\mathbf{1}$ is a Hopf algebra with antipode $\mathrm{id}_{\mathbf{1}}$.

Interpreted using the convolution in Remark 7.5.3, an antipode S is precisely a two-sided inverse of $\mathrm{id}_A \in \mathrm{End}(A)$ with respect to $\star$. The antipode in the definition of a Hopf algebra, if it exists, is unique. This follows immediately by applying the next result to $f = \mathrm{id}_A$; the result also shows that morphisms always preserve antipodes.

Proposition 7.5.9 Let A and A' be Hopf algebras over $\mathcal{V}$ with antipodes S and S', respectively. For every Hopf algebra morphism $f : A \to A'$, one has $S'f = fS$.

Proof By the functoriality of convolution $\star$ in (7.5.2), both maps

$$\mathrm{Hom}(A', A') \xrightarrow{g \mapsto gf} \mathrm{Hom}(A, A') \ , \qquad \mathrm{Hom}(A, A) \xrightarrow{h \mapsto fh} \mathrm{Hom}(A, A')$$

are homomorphisms of convolution monoids. Thus $S'f$ and fS are both convolution inverses of $f \in \mathrm{Hom}(A, A')$. □

Because the braiding on $\mathcal{V}$ gives a braiding on $\mathcal{V}^{\mathrm{op}}$, the concept of a bialgebra is plainly self-dual. The next result shows that the definition of an antipode is self-dual as well.

Proposition 7.5.10 If $(A, \mu, \eta, \Delta, \epsilon)$ is a bialgebra over $\mathcal{V}$, then $(A, \eta, \mu, \epsilon, \Delta)$ is a bialgebra over $\mathcal{V}^{\mathrm{op}}$. If the Hopf algebra $(A, \mu, \eta, \Delta, \epsilon)$ has antipode S, then S^{-1} is an antipode of $(A, \eta, \mu, \epsilon, \Delta)$ as a bialgebra over $\mathcal{V}^{\mathrm{op}}$; in particular, the latter object is a Hopf algebra over $\mathcal{V}^{\mathrm{op}}$.

Proof The diagram in Definition 7.5.8 is self-dual, but the direction of S is reversed.□

Proposition 7.5.11 For a bialgebra A, in accordance with Definition 7.1.4,

- ⋄ reversing the multiplication of A using the braiding $c(A, A)$ produces an algebra A^{op};
- ⋄ reversing the comultiplication of A using $c(A, A)^{-1}$ produces a coalgebra A_{cop}.

Together, these two operations produce a bialgebra $A^{\mathrm{op}}_{\mathrm{cop}}$. If A has an antipode S, then S is also an antipode of $A^{\mathrm{op}}_{\mathrm{cop}}$, and in this case $S : A \xrightarrow{\sim} A^{\mathrm{op}}_{\mathrm{cop}}$ is an isomorphism of bialgebras.

Proof A routine verification shows that $A^{\mathrm{op}}_{\mathrm{cop}}$ is indeed a bialgebra with antipode S; we omit the details. The remaining assertion has two parts. First, S preserves the unit and counit. This is easy: in Proposition 7.5.9, take $A = \mathbf{1}$ and $A' = \mathbf{1}$, respectively.

Second, S preserves the multiplication and comultiplication. The proof in general is more involved; see [22, §9, Proposition 2] or [2, Proposition 1.22]. Here we only sketch the special case $\mathcal{V} = \Bbbk\text{-Mod}$, $\otimes = \otimes_{\Bbbk}$ with its standard braiding. This is the only case that will be needed later.

Recall from §7.1 that $A \otimes A$ is naturally a coalgebra. Equip $\mathrm{Hom}(A \otimes A, A)$ with the convolution from Remark 7.5.3, denoted by $*$, and consider the elements

$$f := \mu, \quad g := \mu c(A, A)(S \otimes S), \quad h := S\mu.$$

It is enough to prove that $h * f = \eta_A \epsilon_{A \otimes A} = f * g$. This equality will show that f is invertible in the monoid $(\mathrm{Hom}(A \otimes A, A), *)$, and hence that $h = g$. Let $a, b \in A$ and write the expansions

$$\Delta(a) = \sum_i a_i^{(1)} \otimes a_i^{(2)}, \quad \Delta(b) = \sum_j b_j^{(1)} \otimes b_j^{(2)}.$$

Write μ simply as multiplication. From the standard braiding on $\Bbbk$-**Mod**, one obtains

$$\begin{aligned}
(h * f)(a \otimes b) &= \sum_{i,j} h\left(a_i^{(1)} \otimes b_j^{(1)}\right) f\left(a_i^{(2)} \otimes b_j^{(2)}\right) \\
&= \sum_{i,j} S\left(a_i^{(1)} b_j^{(1)}\right) a_i^{(2)} b_j^{(2)} \\
&= (\underbrace{S \star \mathrm{id}_A}_{\text{convolution on } \mathrm{End}(A)})(ab) = \eta_A \epsilon_A(ab), \\
(f * g)(a \otimes b) &= \cdots = \sum_{i,j} a_i^{(1)} b_j^{(1)} S\left(b_j^{(2)}\right) S\left(a_i^{(2)}\right) \\
&= \sum_i a_i^{(1)} \eta_A \epsilon_A(b) S\left(a_i^{(2)}\right) \\
&= \eta_A \epsilon_A(a) \cdot \eta_A \epsilon_A(b) = \eta_A \epsilon_A(ab), \\
\eta_A \epsilon_{A \otimes A}(a \otimes b) &= \eta_A\left(\epsilon_A(a) \epsilon_A(b)\right) = \eta_A \epsilon_A(ab),
\end{aligned}$$

as claimed. Thus S preserves the multiplication of the bialgebra. A similar argument shows that S preserves the comultiplication. □

In particular, if $\mathcal{V}$ is a symmetric monoidal category, then S^2 is an automorphism of A.

Corollary 7.5.12 Let the Hopf algebra A have antipode S. If A is commutative or cocommutative, then $S^2 = \mathrm{id}$.

Proof With respect to the convolution $\star$ on $\mathrm{End}(A)$ (Remark 7.5.3), we have

$$S \star S^2 = \mu(S \otimes S^2)\Delta = \mu(S \otimes S)(\mathrm{id} \otimes S)\Delta.$$

If A is commutative, then $\mu c(A, A) = \mu$, and Proposition 7.5.11 transforms the right-hand side above into

$$\mu c(A, A)(S \otimes S)(\mathrm{id} \otimes S)\Delta = S\mu(\mathrm{id} \otimes S)\Delta = S(\mathrm{id} \star S).$$

The definition of an antipode says that $\mathrm{id} \star S = \eta\epsilon$, while the same proposition gives $S\eta = \eta$; the result is $\eta\epsilon$.

If A is cocommutative, then $c(A, A)\Delta = \Delta$, and functoriality of the braiding transforms $S \star S^2$ into

$$\mu(S \otimes S)(\mathrm{id} \otimes S)c(A, A)\Delta = \mu c(A, A)(S \otimes S)(S \otimes \mathrm{id})\Delta,$$

which the same argument then transforms into $S(S \star \mathrm{id}) = \eta\epsilon$.

Thus S^2 is a convolution inverse of S, so $S^2 = \mathrm{id}$. □

All the following examples take $\mathcal{V} = \Bbbk\text{-}\mathsf{Mod}$ and $\otimes = \otimes_{\Bbbk}$, where $\Bbbk$ is a commutative ring.

Example 7.5.13 Let G be a group. Form the group $\Bbbk$-algebra $\Bbbk[G]$. Example 7.5.5 gives $\Bbbk[G]$ a bialgebra structure. Define the $\Bbbk$-module automorphism $S : \Bbbk[G] \to \Bbbk[G]$ by

$$S(g) = g^{-1}, \quad g \in G.$$

It is easy to verify that S satisfies the commutative diagram required of an antipode. Thus $\Bbbk[G]$ is a Hopf algebra.

Example 7.5.14 (H. Hopf) Write $\{\mathrm{pt}\}$ for a one-point set. Let a topological space X be equipped with continuous maps $m : X \times X \to X$ (multiplication) and $e : \{\mathrm{pt}\} \to X$ (equivalently, a choice of an element of X, the unit, also denoted by e), such that the two maps satisfy, up to homotopy, the properties required for associativity of multiplication and for a unit. Then (X, m, e) is called an **H-space**. If a continuous map $i : X \to X$ (taking inverses) is also chosen and satisfies, up to homotopy, the properties required of inverses, then (X, m, e, i) is called an **H-group**.

A topological monoid (or topological group) is of course an H-space (or H-group), but the H-space and H-group conditions are much weaker. A natural example is a loop space. For a topological space M with a chosen base point $x_0 \in M$, a continuous map $\gamma : [0, 1] \to M$ satisfying $\gamma(0) = x_0 = \gamma(1)$ is called a **loop** in M. All loops naturally form a topological space $\Omega(M, x_0)$. If m is taken to be end-to-start concatenation of two loops, e the constant map at x_0, and i reversal of the direction of a loop, then intuitively $\Omega(M, x_0)$ forms an H-group. The key point is that associativity and the inverse properties hold only up to homotopy.

Next, consider the full subcategory S of Top consisting of all CW complexes[7]; it contains spaces commonly encountered in geometry, such as manifolds. Let $\Bbbk$ be a field. The cohomology functor with coefficients in $\Bbbk$, namely $\mathrm{H}^\bullet$, is a functor from

[7] This is a topological term, somewhat different from the complexes defined in linear algebra.

S^{op} to the category of graded $\Bbbk$-vector spaces $\mathsf{Vect}(\Bbbk)^{\mathbb{Z}_{\geq 0}}$. Equip $\mathsf{Vect}(\Bbbk)^{\mathbb{Z}_{\geq 0}}$ with the braiding determined by the Koszul sign rule (7.1.3) (in the notation there, take $\epsilon(a) = a \bmod 2$). The cup product $\mu := \cup$ on cohomology makes the functor $\mathrm{H}^\bullet$ factor through $\mathsf{CAlg}\left(\mathsf{Vect}(\Bbbk)^{\mathbb{Z}_{\geq 0}}\right)$; in short, cohomology is a commutative algebra over $\mathsf{Vect}(\Bbbk)^{\mathbb{Z}_{\geq 0}}$.

On the other hand, give S the monoidal structure determined by the product of spaces, with $\{\mathrm{pt}\}$ as unit. The Künneth formula in topology shows that $\mathrm{H}^\bullet$ is a monoidal functor. Moreover, homotopic maps induce the same map on cohomology. Consequently, if $X \in \mathrm{Ob}(\mathsf{S})$ is an H-space, then $\mathrm{H}^\bullet(X)$ is also a coalgebra over $\mathsf{Vect}(\Bbbk)^{\mathbb{Z}_{\geq 0}}$.

Readers familiar with algebraic topology will readily conclude that the algebra and coalgebra structures on $\mathrm{H}^\bullet(X)$ are compatible and hence give a bialgebra over $\mathsf{Vect}(\Bbbk)^{\mathbb{Z}_{\geq 0}}$. If, in addition, X is an H-group and the automorphism induced by the map i is denoted by $S \in \mathrm{Aut}(\mathrm{H}^\bullet(X))$, then $\mathrm{H}^\bullet(X)$ becomes a Hopf algebra.

In fact, the multiplication (cup product) on $\mathrm{H}^\bullet(X)$ is precisely the image of the diagonal embedding $\mathrm{diag} : X \to X \times X$ under the Künneth formula, while its unit is the image of $X \to \{\mathrm{pt}\}$. Thus all the properties required of the bialgebra and antipode can be verified at the level of topological spaces, up to homotopy. For example, the antipode identity $\mu(S \otimes \mathrm{id})\Delta = \eta\epsilon$ comes from

$$\left[X \xrightarrow{\mathrm{diag}} X \times X \xrightarrow{(i,\mathrm{id})} X \times X \xrightarrow{m} X\right] \text{ being homotopy equivalent to } \left[X \to \{\mathrm{pt}\} \xrightarrow{e} X\right].$$

Thus the cohomology of an H-group is naturally a commutative Hopf algebra with a rich structure. This was Hopf's original motivation for studying such algebras.

The exercises will provide more examples of Hopf algebras. Most are commutative or cocommutative, but there are also many Hopf algebras having neither property. The most important class among them is that of **quantum groups**. The relevant discussion requires a foundation in Lie theory and is not treated in this book.

7.6 Beck's Monadicity Theorem

Let $\mathcal{C}$ be a category. Consider the functor category $\mathcal{C}^{\mathcal{C}}$, whose objects are all endofunctors $T : \mathcal{C} \to \mathcal{C}$ and whose morphisms are morphisms between endofunctors. This category is naturally monoidal: the multiplication $\otimes$ is composition of functors, and the unit object $\mathbf{1}$ is the identity functor. Since composition of functors is strictly associative, this is also a strict monoidal category.

Definition 7.6.1 (Monad and comonad) Let $\mathcal{C}$ be a category. An algebra in the monoidal category $\mathcal{C}^{\mathcal{C}}$ (Definition 7.1.1) is called a **monad** on $\mathcal{C}$. Dually, a monad on $\mathcal{C}^{\mathrm{op}}$ is called a **comonad** on $\mathcal{C}$.

Expanding Definition 7.1.1 in the category of endofunctors $\mathcal{C}^{\mathcal{C}}$, we find that a monad is data (T, μ, η), where $T : \mathcal{C} \to \mathcal{C}$ is a functor and $\mu : T^2 \to T$ and $\eta : \mathrm{id}_{\mathcal{C}} \to T$ are morphisms such that the following diagrams commute:

$$\begin{tikzcd} T \arrow[r, "\eta T"] \arrow[dr, "\mathrm{id}"'] & T^2 \arrow[d, "\mu"] & T \arrow[l, "T\eta"'] \arrow[dl, "\mathrm{id}"] \\ & T & \end{tikzcd} \qquad \begin{tikzcd} T^3 \arrow[r, "\mu T"] \arrow[d, "T\mu"'] & T^2 \arrow[d, "\mu"] \\ T^2 \arrow[r, "\mu"'] & T. \end{tikzcd}$$

Compared with the original definition, associativity constraints are no longer needed here, while $\eta \otimes \mathrm{id}$ (or $\mathrm{id} \otimes \eta$) becomes ηT (or $T\eta$), and so on.

Dually, a comonad (L, δ, ϵ) consists of a functor $L : \mathcal{C} \to \mathcal{C}$ and morphisms $\delta : L \to L^2$ and $\epsilon : L \to \mathrm{id}_{\mathcal{C}}$, subject to the commutativity of the following diagrams.

$$\begin{tikzcd} L & L^2 \arrow[l, "\epsilon L"'] \arrow[r, "L\epsilon"] & L \\ & L \arrow[u, "\delta"] \arrow[ul, "\mathrm{id}"] \arrow[ur, "\mathrm{id}"'] & \end{tikzcd} \qquad \begin{tikzcd} L^3 & L^2 \arrow[l, "\delta L"'] \\ L^2 \arrow[u, "L\delta"] & L \arrow[l, "\delta"] \arrow[u, "\delta"'] \end{tikzcd}$$

The data (T, μ, η) (or (L, δ, ϵ)) are customarily abbreviated simply as T (or L). The principal source of monads and comonads is adjoint pairs.

Example 7.6.2 (An adjoint pair determines a monad) Consider a pair of adjoint functors

$$F : \mathcal{C} \rightleftarrows \mathcal{D} : G,$$

with the corresponding unit $\eta : \mathrm{id}_{\mathcal{C}} \to GF$ and counit $\varepsilon : FG \to \mathrm{id}_{\mathcal{D}}$. Define a monad (T, μ, η) on $\mathcal{C}$ and a comonad (L, δ, ε) on $\mathcal{D}$ as follows.

$T := GF$	$L := FG$
$\mu := \left[GFGF \xrightarrow{G\varepsilon F} GF\right]$	$\delta := \left[FG \xrightarrow{F\eta G} FGFG\right]$
$\eta : \mathrm{id}_{\mathcal{C}} \to GF$	$\varepsilon : FG \to \mathrm{id}_{\mathcal{D}}$

By duality, it is enough to check the monad axioms in the case (T, μ, η). First, the unit diagram is

$$\begin{tikzcd} GF \arrow[r, "\eta GF"] \arrow[dr, equal] & GFGF \arrow[d, "G\varepsilon F"] & GF \arrow[l, "GF\eta"'] \arrow[dl, equal] \\ & GF & \end{tikzcd}$$

which commutes by the triangle identities $(G\varepsilon)(\eta G) = \mathrm{id}_G$ and $(\varepsilon F)(F\eta) = \mathrm{id}_F$. The associativity diagram is

$$\begin{tikzcd}
GFGFGF \arrow[r, "GFG\varepsilon F"] \arrow[d, "G\varepsilon FGF"'] & GFGF \arrow[d, "G\varepsilon F"] \\
GFGF \arrow[r, "G\varepsilon F"'] & GF
\end{tikzcd}$$

and it commutes: both composites are depicted by

$$\begin{tikzcd}
\mathcal{C} \arrow[r, "F"] & \mathcal{D} \arrow[r, "G"] \arrow[rr, bend right, "\mathrm{id}"'] & \mathcal{C} \arrow[r, "F"] \arrow[d, phantom, "\Downarrow\varepsilon"] & \mathcal{D} \arrow[r, "G"] \arrow[rr, bend right, "\mathrm{id}"'] & \mathcal{C} \arrow[r, "F"] \arrow[d, phantom, "\Downarrow\varepsilon"] & \mathcal{D} \arrow[r, "G"] & \mathcal{C} \\
 & & {} & & {} & &
\end{tikzcd}$$

with only the order of composition of the two curved regions differing; the result is the same. This is a special case of the interchange law for vertical and horizontal composition of morphisms [51, Lemma 2.2.7].

Since endofunctors act on $\mathcal{C}$, we wish to discuss "modules" in $\mathcal{C}$ under the action of a monad T. For historical reasons, such a structure is also called a T-algebra.

Definition 7.6.3 (S. Eilenberg, J. C. Moore) Let (T, μ, η) be a monad on $\mathcal{C}$. A ***T*-module** is data (M, a), where $M \in \mathrm{Ob}(\mathcal{C})$ and a is a morphism $T(M) \to M$ in $\mathcal{C}$, such that the following diagrams commute:

$$\begin{tikzcd}
T^2(M) \arrow[r, "Ta"] \arrow[d, "\mu_M"'] & T(M) \arrow[d, "a"] \\
T(M) \arrow[r, "a"'] & M
\end{tikzcd}
\qquad
\begin{tikzcd}
M \arrow[r, "\eta_M"] \arrow[dr, "\mathrm{id}_M"'] & T(M) \arrow[d, "a"] \\
 & M.
\end{tikzcd}$$

All T-modules form a category $\mathcal{C}^T$. A morphism from (M, a) to (M', a') is defined to be a morphism $f : M \to M'$ in $\mathcal{C}$ that makes the following diagram commute:

$$\begin{tikzcd}
T(M) \arrow[r, "a"] \arrow[d, "Tf"'] & M \arrow[d, "f"] \\
T(M') \arrow[r, "a'"'] & M'.
\end{tikzcd}$$

Dually, let (L, δ, ϵ) be a comonad on $\mathcal{C}$. An L-comodule is data (N, b), where $b : N \to L(N)$ is a morphism, subject to the commutative diagrams dual to those above. The category of L-comodules is denoted by $\mathcal{C}^L$.

The two commutative diagrams for a T-module should be viewed as the associativity and unit laws for the action of T, respectively. The examples and properties below are stated mainly for monads and modules over them; the versions for comonads and comodules are entirely dual.

Example 7.6.4 Write Mon for the category of monoids, conventionally taken to be realized on small sets, and consider the adjoint pair

$$\mathbf{M} : \mathsf{Set} \rightleftarrows \mathsf{Mon} : U,$$

where $\mathbf{M}$ maps a set X to the free monoid $\mathbf{M}(X)$, whose definition and construction are given in [51, Definition 4.8.1, Lemma 4.8.4], and U is the forgetful functor. Thus,

$$T := U\mathbf{M} : \mathsf{Set} \to \mathsf{Set}, \quad T(X) = \bigsqcup_{n \geq 0} X^n.$$

In other words, an element of $T(X)$ is a "word" $(x_1, \ldots, x_n)$ with $n \in \mathbb{Z}_{\geq 0}$. The unit of the adjoint pair $\eta_X : X \to U\mathbf{M}(X)$ maps $x \in X$ to the word of length 1, (x). For a monoid A, the counit $\varepsilon_A : \mathbf{M}U(A) \to A$ maps the word $(a_1, \ldots, a_n)$ to the product $a_1 \cdots a_n \in A$. Since multiplication in $\mathbf{M}(X)$ is concatenation of words, the map $\mu_X = U\varepsilon\mathbf{M} : T^2(X) \to T(X)$ corresponding to the monad T (or, viewed as a map $\mathbf{M}(\mathbf{M}(X)) \to \mathbf{M}(X)$) flattens a "word of words," that is, it concatenates a sequence of lists, as in

$$((x_1, \ldots, x_n), (y_1, \ldots, y_m), \ldots) \mapsto (x_1, \ldots, x_n, y_1, \ldots, y_m, \ldots).$$

Specifying a T-module (M, a) for $T = U\mathbf{M}$ is equivalent to specifying $m_1 \cdots m_n := a(m_1, \ldots, m_n)$ for every word $(m_1, \ldots, m_n)$ in the set M, such that $a(m) = m$ and the associativity law

$$a\left(a(x_1, \ldots, x_n), a(y_1, \ldots, y_m), \ldots\right) = a\left(x_1, \ldots, x_n, y_1, \ldots, y_m, \ldots\right)$$

holds. In short, a T-module is precisely a monoid, with the image under a of the empty word ($n = 0$) as its unit. It is easy to check that morphisms of T-modules are precisely monoid homomorphisms.

The case of the category of groups Grp is entirely similar; the key is to replace $\mathbf{M}$ by the free-group functor $\mathbf{F}$. The corresponding T-modules are precisely groups.

Example 7.6.5 Let R be a ring and consider the adjoint pair

$$\mathbf{F} : \mathsf{Set} \rightleftarrows R\text{-}\mathsf{Mod} : U,$$

where $\mathbf{F}$ maps X to the free left R-module $R^{\oplus X}$ and U is the forgetful functor. This gives a monad (T, μ, η) on Set. As in Example 7.6.4, $T = U\mathbf{F}$ maps a set X to $R^{\oplus X}$ regarded as a set; its elements are finite formal R-linear combinations of elements of X, "$r_1x_1 + \cdots + r_nx_n$". The unit η_X maps $x \in X$ to "x", while $\mu_X : R^{\oplus(R^{\oplus}X)} \to R^{\oplus X}$ flattens a two-level linear combination, namely,

$$\begin{aligned}&\text{“}\alpha(\text{“}r_1x_1 + \cdots + r_nx_n\text{”}) + \beta(\text{“}s_1y_1 + \cdots + s_my_m\text{”}) + \cdots\text{”}\\ &\qquad\qquad \mapsto \text{“}(\alpha r_1)x_1 + \cdots + (\alpha r_n)x_n + (\beta s_1)y_1 + \cdots + (\beta s_m)y_m + \cdots\text{”}.\end{aligned}$$

Specifying a T-module (M, a) is equivalent to specifying a set M together with a rule that maps formal linear combinations to elements of M, maps "x" to x, is associative with respect to addition and scalar multiplication, and makes scalar multiplication by $1 \in R$ the identity map. In short, T-modules are precisely left R-modules. It is clear that morphisms of T-modules are precisely module homomorphisms.

Return to the general theory. There is a forgetful functor $U^T : \mathcal{C}^T \to \mathcal{C}$ mapping an object (M, a) to M. We now give its left adjoint: the free construction.

Definition–Proposition 7.6.6 (Free T-module) Let (T, μ, η) be a monad on $\mathcal{C}$. Define a functor

$$\mathrm{Free}^T : \mathcal{C} \to \mathcal{C}^T, \qquad \begin{aligned} (\text{object } M) &\mapsto (TM, \mu_M : T^2M \to TM), \\ (\text{morphism } f : M \to M') &\mapsto Tf : TM \to TM'. \end{aligned}$$

This gives an adjoint pair

$$\mathrm{Free}^T : \mathcal{C} \rightleftarrows \mathcal{C}^T : U^T,$$

and the monad on $\mathcal{C}$ determined by this adjoint pair is precisely (T, μ, η).

Proof To check that (TM, μ_M) is a T-module, one must show that the following two diagrams commute:

$$\begin{tikzcd} T^3(M) \arrow[r, "T\mu_M"] \arrow[d, "\mu_{T(M)}"'] & T^2(M) \arrow[d, "\mu_M"] \\ T^2(M) \arrow[r, "\mu_M"'] & T(M) \end{tikzcd} \qquad \begin{tikzcd} T(M) \arrow[r, "\eta_{T(M)}"] \arrow[dr, "\mathrm{id}"'] & T^2(M) \arrow[d, "\mu_M"] \\ & T(M). \end{tikzcd}$$

But these diagrams are obtained by evaluating the functor diagrams in Definition 7.6.1 at the object M. Functoriality is clear.

Next, note that $U^T \mathrm{Free}^T = T$. Take $\eta : \mathrm{id}_{\mathcal{C}} \to T$ from the data of T, and define

$$\epsilon : \mathrm{Free}^T U^T \to \mathrm{id}_{\mathcal{C}^T}, \quad \epsilon_{(M,a)} : (TM, \mu_M) \xrightarrow{a} (M, a) \in \mathrm{Ob}(\mathcal{C}^T).$$

The left diagram in Definition 7.6.3 shows that $\epsilon_{(M,a)}$ is a morphism in $\mathcal{C}^T$. Its functoriality is also clear.

A routine verification shows that (Free^T, U^T) is an adjoint pair with unit η and counit ϵ; we omit the details. Moreover,

$$\begin{aligned} \left(U^T \epsilon \mathrm{Free}^T\right)_M &= U^T \epsilon_{(TM, \mu_M)} \\ &= \left[\mu_M : T^2(M) \to T(M)\right], \end{aligned}$$

which shows that the monad on $\mathcal{C}$ determined by the adjoint pair (Free^T, U^T) is (T, μ, η). $\square$

For the monads in Examples 7.6.4 and 7.6.5, the functor Free^T corresponds, respectively, to the constructions of the free monoid and the free module.

Dually, a comonad L on a category $\mathcal{D}$ also gives a free–forgetful adjoint pair; there is no need to discuss it further.

Lemma 7.6.7 Consider the monad T and comonad L arising from an adjoint pair (F, G). Let $N \in \mathrm{Ob}(\mathcal{D})$. The data

$$M := G(N) \in \mathrm{Ob}(\mathcal{C}), \quad a : T(M) = GFG(N) \xrightarrow{G\varepsilon_N} G(N) = M$$

give $(M, a) \in \mathrm{Ob}(\mathcal{C}^T)$. We thereby obtain a functor $\mathbb{K} : \mathcal{D} \to \mathcal{C}^T$ satisfying $G = U^T\mathbb{K}$.

Dually, F also has a canonical factorization $\mathcal{C} \xrightarrow{\mathbb{K}} \mathcal{D}^L \to \mathcal{D}$, which will henceforth be written $F = U^L\mathbb{K}$, where U^L is the forgetful functor[8)].

Proof It is enough to treat the case of T. In this case, the two diagrams in Definition 7.6.3 become

$$\begin{tikzcd} GFGFG(N) \arrow[r, "GFG\varepsilon"] \arrow[d, "G\varepsilon FG"'] & GFG(N) \arrow[d, "G\varepsilon"] \\ GFG(N) \arrow[r, "G\varepsilon"'] & G(N) \end{tikzcd} \qquad \begin{tikzcd} G(N) \arrow[r, "\eta G"] \arrow[dr, equal] & GFG(N) \arrow[d, "G\varepsilon"] \\ & G(N). \end{tikzcd}$$

Their commutativity is verified exactly as in Example 7.6.2, so it need not be repeated; the functoriality of $N \mapsto (M, a)$ is also immediate. ☐

For a given adjoint pair (F, G), applying a functor is a process that forgets structure. Lemma 7.6.7 factors G (or F) as $U^T\mathbb{K}$ (or $U^L\mathbb{K}$). We want to know whether structure is lost only at the second step, when the forgetful functor U^T is applied. If so, the information lost in the process can be reconstructed with the aid of the monad T (or comonad L). This consideration motivates the following concept.

Definition 7.6.8 Consider a pair of adjoint functors $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$, which determines a monad T on $\mathcal{C}$ and a comonad L on $\mathcal{D}$.

- ◇ If the functor $\mathbb{K} : \mathcal{D} \to \mathcal{C}^T$ in Lemma 7.6.7 is an equivalence of categories, the adjoint pair is called **monadic**.
- ◇ If its dual version $\mathbb{K} : \mathcal{C} \to \mathcal{D}^L$ is an equivalence of categories, the adjoint pair is called **comonadic**.

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8) Using the same symbol $\mathbb{K}$ will cause little confusion, since this book will not discuss monads and comonads simultaneously.

For example, the free–forgetful adjoint pairs in Examples 7.6.4 and 7.6.5 are both monadic; a direct check even shows that $\mathbb{K}$ is an isomorphism of categories in both examples.

Beck's theorem below, also called the monadicity theorem or the Barr–Beck theorem, characterizes monadicity. We need some preparation.

Definition–Proposition 7.6.9 Consider a diagram in a category $\mathcal{C}$

$$\begin{tikzcd} A \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & B \arrow[l, dashed, bend left, "t"] \arrow[r, "h"] & Z \arrow[l, dashed, bend left, "s"] \end{tikzcd}$$

such that the solid part commutes, that is, $hu = hv$, and h and v have sections (that is, right inverses) s and t, respectively, as indicated by the dashed arrows, with $sh = ut \in \mathrm{End}(B)$. The solid part of a diagram with these properties is called a **split fork**. The diagram automatically gives a coequalizer of u and v in $\mathcal{C}$.

Proof Consider the following diagram:

$$\begin{tikzcd} & & W \\ A \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & B \arrow[ur, "k"] \arrow[r, "h"'] & Z \end{tikzcd} \qquad \begin{array}{l} W \in \mathrm{Ob}(\mathcal{C}) \\ ku = kv. \end{array}$$

If there is a $\varphi : Z \to W$ with $\varphi h = k$, then $\varphi = \varphi hs = ks$. Conversely, $\varphi := ks$ does satisfy $\varphi h = ksh = kut = kvt = k$. This proves the required universal property. □

Unlike coequalizers in general, split forks are "absolute": their images under arbitrary functors are still split forks.

Example 7.6.10 Let (T, μ, η) be a monad on $\mathcal{C}$. Every $(M, a) \in \mathrm{Ob}(\mathcal{C}^T)$ gives a split fork in $\mathcal{C}$:

$$\begin{tikzcd} T^2(M) \arrow[r, shift left, "Ta"] \arrow[r, shift right, "\mu_M"'] & T(M) \arrow[l, dashed, bend left, "\eta_{TM}"] \arrow[r, "a"] & M \arrow[l, dashed, bend left, "\eta_M"] \end{tikzcd} \qquad \begin{array}{ll} a(Ta) = a\mu_M, & \eta_M a = (Ta)\eta_{TM}, \\ a\eta_M = \mathrm{id}_M, & \mu_M \eta_{TM} = \mathrm{id}_{TM}. \end{array}$$

The equalities on the right are part of Definitions 7.6.1 and 7.6.3; for example, $\eta_M a = (Ta)\eta_{TM}$ follows from the naturality of $\eta : \mathrm{id}_{\mathcal{C}} \to T$. In particular, the diagram above realizes M as the coequalizer of Ta and μ_M.

Definition 7.6.11 Consider a functor $G : \mathcal{D} \to \mathcal{C}$ and a pair of morphisms $f, g : X \rightrightarrows Y$ in $\mathcal{D}$. If there is a morphism $h : G(Y) \to Z$ such that the diagram

$$\begin{tikzcd} G(X) \arrow[r, shift left, "Gf"] \arrow[r, shift right, "Gg"'] & G(Y) \arrow[r, "h"] & Z \end{tikzcd}$$

is a split fork in $\mathcal{C}$, then (f, g) is called a ***G*-split pair** in $\mathcal{D}$.

Let (f, g) be a G-split pair. Since (Gf, Gg) has a coequalizer, it is natural to ask whether that coequalizer can be lifted to a coequalizer of (f, g) in $\mathcal{D}$, and whether the lift is unique. A coequalizer is a special case of a $\varinjlim$; the concepts of a functor creating or preserving $\varinjlim$ in Definition 1.5.2 provide the right terminology for this question.

Note that the forgetful functor $U^T : \mathcal{C}^T \to \mathcal{C}$ is conservative in the sense of Definition 1.5.4.

Lemma 7.6.12 Let (T, μ, η) be a monad on $\mathcal{C}$. The functor U^T creates the coequalizer of every U^T-split pair.

Proof Suppose we are given a pair of morphisms in $\mathcal{C}^T$, $u, v : (M, a) \rightrightarrows (M', a')$, and a split fork in $\mathcal{C}$:

$$\begin{tikzcd} M \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & M' \arrow[l, dashed, bend left, "t"] \arrow[r, "h"] & M'' \arrow[l, dashed, bend left, "s"] \end{tikzcd}.$$

Since this diagram gives a coequalizer in $\mathcal{C}$, it is enough to extend M'' uniquely to an object $(M'', a'') \in \mathrm{Ob}(\mathcal{C}^T)$ such that h is a morphism in $\mathcal{C}^T$, and then to show that this construction gives the coequalizer of u and v in $\mathcal{C}^T$.

The image of a split fork under any functor remains a split fork. Hence, in the following diagram, whose solid part is commutative,

$$\begin{tikzcd} T(M) \arrow[r, shift left, "Tu"] \arrow[r, shift right, "Tv"'] \arrow[d, "a"'] & T(M') \arrow[r, "Th"] \arrow[d, "a'"] & T(M'') \arrow[d, dashed, "a''"] \\ M \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & M' \arrow[r, "h"'] & M'' \end{tikzcd}$$

both rows are coequalizers in $\mathcal{C}$. The universal property therefore gives exactly one a'' that makes the whole diagram commute. If (M'', a'') is shown to be a T-algebra, then commutativity of the right-hand part of this diagram says precisely that h is a morphism in $\mathcal{C}^T$.

For this, it remains to verify that the following two diagrams commute.[9)]

$$\begin{tikzcd} M'' \arrow[r, "\eta_{M''}"] \arrow[dr, equal] & T(M'') \arrow[d, "a''"] \\ & M'' \end{tikzcd} \qquad \begin{tikzcd} T^2(M'') \arrow[r, "\mu_{M''}"] \arrow[d, "Ta''"'] & T(M'') \arrow[d, "a''"] \\ T(M'') \arrow[r, "a''"'] & M''. \end{tikzcd}$$

The coequalizer diagrams show that h and Th are epimorphisms (as is T^2h). The commutativity of the two diagrams above therefore follows immediately from that of the corresponding diagrams for M and M'.

9) Translator's note: the lower-right node of the first diagram is printed as $T(M'')$ in the source. The unit law for a T-algebra requires that node to be M''; the target is corrected here.

Finally, we show that h gives the coequalizer of u and v in $\mathcal{C}^T$. Suppose there is a morphism in $\mathcal{C}^T$ $k : (M', a') \to (W, b)$ with $ku = kv$.[10] In $\mathcal{C}$, there is a unique morphism $\varphi : M'' \to W$ with $\varphi h = k$. It remains to prove that φ is actually a morphism in $\mathcal{C}^T$. Consider the diagram

$$\begin{tikzcd} T(M') \arrow[r, "Th"] \arrow[d, "a'"'] & T(M'') \arrow[r, "T\varphi"] \arrow[d, "a''"] & T(W) \arrow[d, "b"] \\ M' \arrow[r, "h"'] & M'' \arrow[r, "\varphi"'] & W. \end{tikzcd}$$

The composites of the top and bottom rows are Tk and k, respectively. Since h and k are morphisms in $\mathcal{C}^T$, the outer boundary and the left square commute. Thus the right square commutes after composition with Th. Since Th is an epimorphism, the right square itself commutes. This shows that φ is a morphism in $\mathcal{C}^T$. □

Lemma 7.6.13 Let $G : \mathcal{D} \to \mathcal{C}$ be a functor with left adjoint F, and suppose G creates the corresponding coequalizers for all G-split pairs $f, g : X \rightrightarrows Y$ (Definition 7.6.11). Then the following diagram is a coequalizer:

$$FGFG(N) \overset{FG\varepsilon_N}{\underset{\varepsilon_{FG(N)}}{\rightrightarrows}} FG(N) \xrightarrow{\varepsilon_N} N,$$

where $N \in \mathrm{Ob}(\mathcal{D})$.

Proof Applying G to this diagram gives the split fork of Example 7.6.10 for $(M, a) := (G(N), G\varepsilon_N) = \mathbb{K}(N)$. Since G creates the corresponding coequalizer, the original diagram is a coequalizer. □

Theorem 7.6.14 (J. M. Beck) Let $G : \mathcal{D} \to \mathcal{C}$ be a functor with left adjoint F. The following statements are equivalent:

(i) the adjoint pair (F, G) is monadic (Definition 7.6.8);

(ii) G creates the corresponding coequalizer for every G-split pair $f, g : X \rightrightarrows Y$;

(iii) G is conservative (Definition 1.5.4), every G-split pair $f, g : X \rightrightarrows Y$ has a coequalizer $X \rightrightarrows Y \to Z$ in $\mathcal{D}$, and the image of that coequalizer under G is a coequalizer of (Gf, Gg).

If any of the conditions above holds, a quasi-inverse functor $\mathbb{L}$ to $\mathbb{K} : \mathcal{D} \to \mathcal{C}^T$ can be described as follows. For $(M, a) \in \mathrm{Ob}(\mathcal{C}^T)$, there is a coequalizer diagram

$$FGF(M) \overset{Fa}{\underset{\varepsilon_{FM}}{\rightrightarrows}} F(M) \longrightarrow \mathbb{L}(M, a). \tag{7.6.1}$$

[10] Translator's note: the source prints the domain of k as (M'', a''), which makes ku, kv, and $\varphi h = k$ undefined. The correct domain is (M', a').

For the comonad determined by (F, G), the criterion is entirely dual.

Proof First, we prove (i) $\implies$ (ii). Suppose $\mathbb{K} : \mathcal{D} \to \mathcal{C}^T$ has a quasi-inverse functor $\mathbb{L}$. The properties of being a split pair, being a split fork, and creating $\varinjlim$ are invariant under equivalences of categories. Since $G = U^T\mathbb{K}$, it is enough to prove that $U^T : \mathcal{C}^T \to \mathcal{C}$ creates the corresponding coequalizer for every U^T-split pair. This is precisely the content of Lemma 7.6.12.

Next, (i) $\implies$ (iii). Since (i) $\implies$ (ii) is already known, it remains only to show that G is conservative when G is monadic. But $G = U^T\mathbb{K}$, where $\mathbb{K}$ is an equivalence of categories and U^T is plainly conservative. Hence G is conservative as well.

We now prove (ii) $\implies$ (i) by constructing a quasi-inverse functor $\mathbb{L}$ to $\mathbb{K}$. For $(M, a) \in \mathrm{Ob}(\mathcal{C}^T)$, consider the pair of morphisms in $\mathcal{D}$ $FGF(M) \underset{\varepsilon_{FM}}{\overset{Fa}{\rightrightarrows}} F(M)$. Its image under G is $T^2(M) \underset{\mu_M}{\overset{Ta}{\rightrightarrows}} T(M)$, which can be extended to a split fork by Example 7.6.10. Thus (Fa, ε_{FM}) is a G-split pair, and the original diagram can be extended to a coequalizer diagram in $\mathcal{D}$, namely (7.6.1).

By (7.6.1) and the universal property of coequalizers, a morphism $(M, a) \xrightarrow{\varphi} (M', a')$ in $\mathcal{C}^T$ gives a unique morphism $\mathbb{L}(M, a) \to \mathbb{L}(M', a')$ making the following diagram commute:

$$\begin{tikzcd}
FGF(M) \arrow[r, shift left, "Fa"] \arrow[r, shift right, "\varepsilon_{FM}"'] \arrow[d, "FGF\varphi"'] & F(M) \arrow[r] \arrow[d, "F\varphi"] & \mathbb{L}(M,a) \arrow[d] \\
FGF(M') \arrow[r, shift left, "Fa'"] \arrow[r, shift right, "\varepsilon_{FM'}"'] & F(M') \arrow[r] & \mathbb{L}(M',a').
\end{tikzcd}$$

Thus $(M, a) \mapsto \mathbb{L}(M, a)$ becomes a functor $\mathbb{L} : \mathcal{C}^T \to \mathcal{D}$.

We next prove $\mathbb{LK} \simeq \mathrm{id}_{\mathcal{D}}$.[11)] For every $N \in \mathrm{Ob}(\mathcal{D})$, the construction above gives a coequalizer diagram

$$FGFG(N) \underset{\varepsilon_{FG(N)}}{\overset{FG\varepsilon_N}{\rightrightarrows}} FG(N) \longrightarrow \mathbb{LK}(N).$$

Comparing this with the coequalizer diagram in Lemma 7.6.13 immediately gives the canonical isomorphism $\mathbb{LK} \simeq \mathrm{id}_{\mathcal{D}}$.

Next, we verify $\mathbb{KL} \simeq \mathrm{id}_{\mathcal{C}^T}$. Clearly $\mathbb{K}F = \mathrm{Free}^T$; both send an object M to $(GF(M), G\varepsilon_{FM})$. By definition and $\mu := G\varepsilon F$, applying $\mathbb{K}$ to (7.6.1) gives

$$\begin{tikzcd}
(T^2(M), \ldots) \arrow[r, shift left, "Ta"] \arrow[r, shift right, "\mu_M"'] \arrow[d, equal] & (T(M), \ldots) \arrow[r] \arrow[d, equal] & \mathbb{KL}(M,a). \\
\mathrm{Free}^T(T(M)) & \mathrm{Free}^T(M) &
\end{tikzcd} \tag{7.6.2}$$

11) Translator's note: the source prints $\mathrm{id}_{\mathcal{C}}$ here and in the comparison sentence below. Since $\mathbb{LK}$ is an endofunctor of $\mathcal{D}$, the correct subscript is $\mathcal{D}$.

Applying the forgetful functor U^T to the two terms on the left of (7.6.2) gives $T^2(M) \overset{Ta}{\underset{\mu_M}{\rightrightarrows}} T(M)$. Example 7.6.10 extends this diagram to a split fork with coequalizer M. On the other hand, U^T creates this coequalizer (Lemma 7.6.12). Recalling Definition 1.5.2, it follows that (7.6.2) is also a coequalizer diagram.

On the other hand, apply Lemma 7.6.13 to the adjoint pair (Free^T, U^T) and the object $N = (M, a)$. Carefully expanding the description of this adjoint pair gives the coequalizer diagram

$$\mathrm{Free}^T(T(M)) \overset{Ta}{\underset{\mu_M}{\rightrightarrows}} \mathrm{Free}^T(M) \longrightarrow (M, a).$$

Comparison with the preceding paragraph immediately gives $\mathbb{KL} \simeq \mathrm{id}_{\mathcal{C}^T}$.

Finally, we prove (iii) $\implies$ (ii). The second part of (iii) says precisely that all G-split pairs have coequalizers and that G preserves those coequalizers. Since G is conservative, Remark 1.5.5 shows that (ii) holds.

Reversing all morphisms while retaining the direction of the functors, that is, replacing $\mathcal{C}$ and $\mathcal{D}$ by $\mathcal{C}^{\mathrm{op}}$ and $\mathcal{D}^{\mathrm{op}}$, gives the characterization of comonadicity. We omit the details. □

Corollary 7.6.15 Consider a pair of adjoint functors $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$ with corresponding unit η and counit ε. If this adjoint pair is monadic, then the diagram

$$FGFG(N) \overset{FG\varepsilon_N}{\underset{\varepsilon_{FG(N)}}{\rightrightarrows}} FG(N) \xrightarrow{\varepsilon_N} N$$

is a coequalizer for every $N \in \mathrm{Ob}(\mathcal{D})$.

Proof Combine Lemma 7.6.13 with Theorem 7.6.14 (ii). □

In practice, a linear version of Beck's theorem is often needed. More precisely, let $\Bbbk$ be a commutative ring and (F, G) an adjoint pair between $\Bbbk$-**Mod**-categories, with both functors $\Bbbk$-linear; see Definition 1.4.1 and Proposition 1.4.3. It is easy to see that $\mathcal{C}^T$ naturally becomes a $\Bbbk$-**Mod**-category and that, by definition, both functors in $\mathcal{D} \xrightarrow{\mathbb{K}} \mathcal{C}^T \xrightarrow{U^T} \mathcal{C}$ are $\Bbbk$-linear. If condition (ii) or (iii) of Beck's Theorem 7.6.14 is satisfied, the description by the diagram (7.6.1) immediately shows that the quasi-inverse functor $\mathbb{L}$ to $\mathbb{K}$ is also $\Bbbk$-linear. We therefore obtain a $\Bbbk$-linear equivalence $\mathbb{K} : \mathcal{D} \to \mathcal{C}^T$.

7.7 Morita Theory

Throughout this section, fix a Grothendieck universe $\mathcal{U}$ and a commutative ring $\Bbbk$. Relative to $\mathcal{U}$, all rings are understood to be small by default. Functors between $\Bbbk$-linear categories are also understood to be $\Bbbk$-linear unless otherwise stated, as in Definitions 1.4.1 and 1.4.4. When $\Bbbk = \mathbb{Z}$, $\Bbbk$-linearity is the same as additivity.

For $\Bbbk$-linear categories $\mathcal{C}$ and $\mathcal{D}$, the notation $\mathrm{Fct}(\mathcal{C}, \mathcal{D})$ or $\mathcal{D}^{\mathcal{C}}$ denotes the category of all functors $F : \mathcal{C} \to \mathcal{D}$. Although $\mathrm{Fct}(\mathcal{C}, \mathcal{D})$ need not be a $\mathcal{U}$-category, this causes no difficulty below because we shall not form limits or similar constructions within it.

Definition 7.7.1 Let $\mathcal{C}$ and $\mathcal{D}$ be $\Bbbk$-linear categories.

- If $\mathcal{C}$ and $\mathcal{D}$ are cocomplete, define the full subcategory $\mathrm{Fct}^c(\mathcal{C}, \mathcal{D})$ of $\mathrm{Fct}(\mathcal{C}, \mathcal{D})$ to consist of all functors that preserve small $\varinjlim$.
- If $\mathcal{C}$ and $\mathcal{D}$ are complete, define the full subcategory $\mathsf{Fct}_c(\mathcal{C}, \mathcal{D})$ of $\mathrm{Fct}(\mathcal{C}, \mathcal{D})$ to consist of all functors that preserve small $\varprojlim$.

In some of the literature, objects of Fct_c (or Fct^c) are called continuous (or cocontinuous) functors.

We use the notation of §4.12. For arbitrary $\Bbbk$-algebras A and B, define

$$\begin{aligned}
(A, B)\text{-}\mathsf{Mod} &:= \text{the category of } (A, B)\text{-bimodules},\\
A\text{-}\mathsf{Mod} &:= \text{the category of left } A\text{-modules} \simeq (A, \Bbbk)\text{-}\mathsf{Mod},\\
\mathsf{Mod}\text{-}B &:= \text{the category of right } B\text{-modules} \simeq (\Bbbk, B)\text{-}\mathsf{Mod}.
\end{aligned}$$

These are elementary examples of $\Bbbk$-linear categories.

Recall the following basic facts from module theory. Every (A, B)-bimodule P canonically induces functors preserving small $\varinjlim$,

$$P \underset{B}{\otimes} (\cdot) : B\text{-}\mathsf{Mod} \to A\text{-}\mathsf{Mod}, \quad (\cdot) \underset{A}{\otimes} P : \mathsf{Mod}\text{-}A \to \mathsf{Mod}\text{-}B,$$

while every (B, A)-bimodule Q canonically induces functors preserving small $\varprojlim$,

$$\mathrm{Hom}_{\mathsf{Mod}\text{-}B}(Q, \cdot) : B\text{-}\mathsf{Mod} \to A\text{-}\mathsf{Mod}, \quad \mathrm{Hom}_{\mathsf{Mod}\text{-}A}(Q, \cdot) : \mathsf{Mod}\text{-}A \to \mathsf{Mod}\text{-}B.$$

If instead one takes the functor $\mathrm{Hom}(\cdot, R)$, where R is a (B, A)-bimodule, its domain must be replaced by an opposite category such as $(B\text{-}\mathsf{Mod})^{\mathrm{op}}$ in order for the functor still to preserve small $\varprojlim$.

Theorem 7.7.2 (S. Eilenberg, C. E. Watts) For arbitrary $\Bbbk$-algebras A and B, there are equivalences

$$\begin{array}{ccccc}
\mathrm{Fct}^c\,(B\text{-}\mathsf{Mod}, A\text{-}\mathsf{Mod}) & \xleftarrow{\sim} & (A,B)\text{-}\mathsf{Mod} & \xrightarrow{\sim} & \mathrm{Fct}^c\,(\mathsf{Mod}\text{-}A, \mathsf{Mod}\text{-}B) \\
P \underset{B}{\otimes} (\cdot) & \longleftarrow\!\shortmid & P & \longmapsto & (\cdot) \underset{A}{\otimes} P \\
\mathrm{Fct}_c\,(B\text{-}\mathsf{Mod}, A\text{-}\mathsf{Mod}) & \xleftarrow{\sim} & (B,A)\text{-}\mathsf{Mod} & \xrightarrow{\sim} & \mathrm{Fct}_c\,(\mathsf{Mod}\text{-}A, \mathsf{Mod}\text{-}B) \\
\mathrm{Hom}_{B\text{-}\mathsf{Mod}}(Q,\cdot) & \longleftarrow\!\shortmid & Q & \longmapsto & \mathrm{Hom}_{\mathsf{Mod}\text{-}A}(Q,\cdot) \\
\mathrm{Fct}_c\,((B\text{-}\mathsf{Mod})^{\mathrm{op}}, \mathsf{Mod}\text{-}A) & \xleftarrow{\sim} & (B,A)\text{-}\mathsf{Mod} & \xrightarrow{\sim} & \mathrm{Fct}_c\,((\mathsf{Mod}\text{-}A)^{\mathrm{op}}, B\text{-}\mathsf{Mod}) \\
\mathrm{Hom}_{B\text{-}\mathsf{Mod}}(\cdot,R) & \longleftarrow\!\shortmid & R & \longmapsto & \mathrm{Hom}_{\mathsf{Mod}\text{-}A}(\cdot,R).
\end{array}$$

For the first and second equivalences, composition of functors corresponds to tensor product of bimodules; when $A = B$, the identity functor comes from A as an (A, A)-bimodule, up to isomorphism.

Proof The arguments for the various equivalences are similar. We discuss only

$$(A,B)\text{-}\mathsf{Mod} \to \mathrm{Fct}^c(B\text{-}\mathsf{Mod}, A\text{-}\mathsf{Mod}).$$

The discussion preceding the theorem has shown that $P \underset{B}{\otimes} (\cdot)$ is indeed an object of $\mathrm{Fct}^c(B\text{-}\mathsf{Mod}, A\text{-}\mathsf{Mod})$, and a bimodule homomorphism $P \to P'$ plainly induces a morphism of functors $P \underset{B}{\otimes} (\cdot) \to P' \underset{B}{\otimes} (\cdot)$. We now construct a quasi-inverse functor.

For an object F of $\mathrm{Fct}^c(B\text{-}\mathsf{Mod}, A\text{-}\mathsf{Mod})$, regard B as a left B-module and define $P := F(B)$. Since B also acts on itself on the right by multiplication and F is a $\Bbbk$-linear functor, P naturally acquires the structure of an (A, B)-bimodule. The canonical isomorphism $P \underset{B}{\otimes} B \simeq P$ shows that

$$(A,B)\text{-}\mathsf{Mod} \to \mathrm{Fct}^c\,(B\text{-}\mathsf{Mod}, A\text{-}\mathsf{Mod}) \xrightarrow{F \mapsto P} (A,B)\text{-}\mathsf{Mod} \text{ has composite } \simeq \mathrm{id}.$$

For the composite in the other direction, take a functor F as above and set $P := F(B)$. For any left B-module M, there is a canonical surjective homomorphism

$$\bigoplus_{\varphi \in \mathrm{Hom}(B,M)} B \twoheadrightarrow M.$$

Notice that $\bigoplus_{\varphi}$ is "small." Repeating the construction for the kernel of this surjective homomorphism gives a canonical exact sequence for M,

$$\bigoplus_{\psi} B \to \bigoplus_{\varphi} B \to M \to 0, \tag{7.7.1}$$

whose first map may be regarded as right multiplication by an infinite matrix with entries in $B = \mathrm{End}_{B\text{-}\mathsf{Mod}}(B)$. Since F and $P \underset{B}{\otimes} (\cdot)$ both preserve small $\varinjlim$, applying them to (7.7.1) gives exact sequences in A-Mod,

$$\bigoplus_{\psi} P \to \bigoplus_{\varphi} P \to F(M) \to 0,$$
$$\bigoplus_{\psi} P \to \bigoplus_{\varphi} P \to P \underset{B}{\otimes} M \to 0.$$

In both rows, the map $\bigoplus_{\psi} P \to \bigoplus_{\varphi} P$ comes from the same matrix (with entries in B), and hence there is a canonical isomorphism $F(M) \simeq P \underset{B}{\otimes} M$.

By the associativity constraint for tensor products, the first equivalence (A, B)-$\mathsf{Mod} \to \mathrm{Fct}^c(B\text{-}\mathsf{Mod}, A\text{-}\mathsf{Mod})$ sends tensor products of bimodules to composition of functors, up to isomorphism. For the second equivalence, the corresponding statement follows from the adjunction between tensor product and Hom; see [51, Theorem 6.6.5]. In both cases, it is easy to see that the (A, A)-bimodule A corresponds to the identity functor, up to isomorphism. □

The sizes of these functor categories are now easy to control.

Corollary 7.7.3 For arbitrary $\Bbbk$-algebras A and B, all the functor categories $\mathrm{Fct}^c(\cdots)$ and $\mathrm{Fct}_c(\cdots)$ appearing in Theorem 7.7.2 are $\mathcal{U}$-categories.

Proof Indeed, (A, B)-Mod and (B, A)-Mod are both $\mathcal{U}$-categories. □

Definition–Proposition 7.7.4 For arbitrary $\Bbbk$-algebras A and B, the following statements are equivalent.

(i) A-Mod and B-Mod are equivalent.

(ii) There are an (A, B)-bimodule P and a (B, A)-bimodule Q together with

- ⋄ an isomorphism of (A, A)-bimodules $P \underset{B}{\otimes} Q \simeq A$;
- ⋄ an isomorphism of (B, B)-bimodules $Q \underset{A}{\otimes} P \simeq B$.

(iii) Mod-A and Mod-B are equivalent.

When one of these conditions holds, A and B are called **Morita equivalent**.

Proof Equivalences necessarily preserve $\varinjlim$ and $\varprojlim$. Theorem 7.7.2 gives (i) $\iff$ (ii) and (iii) $\iff$ (ii). □

Condition (ii) is concise but not yet sufficiently explicit. It will be refined by Theorem 7.7.16 at the end of this section. Before that, we introduce several examples and related results.

Example 7.7.5 For commutative $\Bbbk$-algebras, Morita equivalence is the same as isomorphism of algebras. This rests on the canonical isomorphism $Z(A\text{-}\mathsf{Mod}) \simeq Z(A)$, where the left-hand side is the center of the abelian category and the right-hand side is the center of the $\Bbbk$-algebra.

Example 7.7.6 Let $n \in \mathbb{Z}_{\geq 1}$. Every $\Bbbk$-algebra A is Morita equivalent to the $n \times n$ matrix algebra $\mathrm{M}_n(A)$. To see this, use matrix operations to define

$$\begin{aligned} P &:= \text{the } (A, \mathrm{M}_{n\times n}(A))\text{-bimodule } A^n \quad \text{(row vectors)}, \\ Q &:= \text{the } (\mathrm{M}_{n\times n}(A), A)\text{-bimodule } A^n \quad \text{(column vectors)}. \end{aligned}$$

Matrix multiplication gives the required isomorphisms,

$$P \underset{\mathrm{M}_{n\times n}(A)}{\otimes} Q \xrightarrow{\sim} A \quad \text{(as an } (A, A)\text{-bimodule)},$$
$$Q \underset{A}{\otimes} P \xrightarrow{\sim} \mathrm{M}_{n\times n}(A) \quad \text{(as an } (\mathrm{M}_{n\times n}(A), \mathrm{M}_{n\times n}(A))\text{-bimodule)}.$$

Return to the general setting. For any (A, B)-bimodule P, a basic fact of module theory [51, Theorem 6.6.5] gives an adjoint pair

$$(\cdot) \underset{A}{\otimes} P : \mathsf{Mod}\text{-}A \rightleftarrows \mathsf{Mod}\text{-}B : \mathrm{Hom}_{\mathsf{Mod}\text{-}B}(P, \cdot). \tag{7.7.2}$$

Theorem 7.7.2 shows that, up to isomorphism, the functors on the left and right exhaust the objects of $\mathrm{Fct}^c(\mathsf{Mod}\text{-}A, \mathsf{Mod}\text{-}B)$ and $\mathrm{Fct}_c(\mathsf{Mod}\text{-}B, \mathsf{Mod}\text{-}A)$, respectively.

By Example 7.6.2, the adjoint pair (7.7.2) determines a monad T on $\mathsf{Mod}\text{-}A$ and a comonad L on $\mathsf{Mod}\text{-}B$. Their general description is as follows. Consider a right A-module N and a right B-module M:

unit η_N	counit ϵ_M
$N \to \mathrm{Hom}_{\mathsf{Mod}\text{-}B}\left(P, N \underset{A}{\otimes} P\right)$	$\mathrm{Hom}_{\mathsf{Mod}\text{-}B}(P, M) \underset{A}{\otimes} P \to M$
$x \mapsto [p \mapsto x \otimes p]$	$\varphi \otimes p \mapsto \varphi(p)$

(7.7.3)

Recall the definition of an exact functor in Definition 2.8.3.

Proposition 7.7.7 Let P be an (A, B)-bimodule. If $(\cdot) \underset{A}{\otimes} P$ (or $\mathrm{Hom}_{\mathsf{Mod}\text{-}B}(P, \cdot)$) is a faithful exact functor, then the adjoint pair (7.7.2) is comonadic (or monadic); see Definition 7.6.8 and its dual version.

Proof We give the proof for $(\cdot) \underset{A}{\otimes} P$; the argument for the other side is exactly the same.

We have already shown that $(\cdot) \underset{A}{\otimes} P$ has a right adjoint. It remains to verify the dual of condition (iii) in Theorem 7.6.14. First, we prove that $(\cdot) \underset{A}{\otimes} P$ is conservative.

For a homomorphism of right A-modules $f : N \to N'$, the required conservativity follows from

$$
\begin{aligned}
&f \text{ is an isomorphism} \iff 0 \to N \xrightarrow{f} N' \to 0 \text{ is exact} \\
&\overset{\text{Proposition 2.8.9 (iii)}}{\iff} 0 \to N \underset{A}{\otimes} P \xrightarrow{\mathrm{id} \otimes f} N' \underset{A}{\otimes} P \to 0 \text{ is exact} \iff \mathrm{id} \otimes f \text{ is an isomorphism.}
\end{aligned}
$$

Next, every pair of morphisms in $\mathsf{Mod}\text{-}A$ and $\mathsf{Mod}\text{-}B$ has an equalizer, and the faithful exactness hypothesis ensures that $(\cdot) \underset{A}{\otimes} P$ preserves equalizers. Thus all the conditions of Theorem 7.6.14 (iii) are satisfied. $\square$

This is only an abstract result. To apply it, the corresponding monad T and comonad L must be described explicitly in certain special cases. We begin with change of rings.

Example 7.7.8 (Change of rings) Fix a homomorphism of $\Bbbk$-algebras $f : A \to B$. In the preceding framework, take $P := B$, regarded through f as an (A, B)-bimodule. In this case, the right adjoint $\mathrm{Hom}_{\mathsf{Mod}\text{-}B}(B, \cdot)$ of $(\cdot) \underset{A}{\otimes} B$ is isomorphic, via $\varphi \mapsto \varphi(1)$, to the forgetful functor $\mathcal{F}_{B|A} : \mathsf{Mod}\text{-}B \to \mathsf{Mod}\text{-}A$, and

unit morphism η_N	counit morphism ϵ_M
$N \to \mathcal{F}_{B\|A}(N \underset{A}{\otimes} B)$ $x \mapsto x \otimes 1$	$\mathcal{F}_{B\|A}(M) \underset{A}{\otimes} B \to M$ $y \otimes b \mapsto yb$

N : a right A-module,
M : a right B-module.

In discussions of change of rings, $\mathcal{F}_{B|A}$ is usually omitted from the notation for simplicity. Thus the monad T and comonad L may respectively be written as

T	$\mathrm{id} \to T$	$T^2 \to T$
$N \mapsto N \underset{A}{\otimes} B$	$N \to N \underset{A}{\otimes} B$ $x \mapsto x \otimes 1$	$(N \underset{A}{\otimes} B) \underset{A}{\otimes} B \to N \underset{A}{\otimes} B$ $(x \otimes b) \otimes b' \mapsto x \otimes bb'$

L	$L \to \mathrm{id}$	$L \to L^2$
$M \mapsto M \underset{A}{\otimes} B$	$M \underset{A}{\otimes} B \to M$ $y \otimes b \mapsto yb$	$M \underset{A}{\otimes} B \to (M \underset{A}{\otimes} B) \underset{A}{\otimes} B$ $y \otimes b \mapsto (y \otimes 1) \otimes b$

The required verifications are routine.

We next consider a slightly more general situation. First, for arbitrary right B-modules P and X, regard B as a (B, B)-bimodule in order to define the left B-module

$$P^{\vee} := \mathrm{Hom}_{\mathsf{Mod}\text{-}B}(P, B) \tag{7.7.4}$$

and the canonical homomorphism of $\Bbbk$-modules

$$\begin{aligned} X \underset{B}{\otimes} P^\vee &\to \mathrm{Hom}_{\mathsf{Mod}\text{-}B}(P, X) \\ x \otimes \lambda &\mapsto x\lambda(\cdot). \end{aligned} \tag{7.7.5}$$

If P is an (A, B)-bimodule, then $P^\vee$ has the structure of a (B, A)-bimodule, $\mathrm{Hom}_{\mathsf{Mod}\text{-}B}(P, X)$ has the structure of a right A-module, and (7.7.5) is a homomorphism of right A-modules.

The same construction naturally has a version with left and right interchanged. Consequently, $P^{\vee\vee}$ is defined, and there is also a canonical homomorphism of right B-modules $P \to P^{\vee\vee}$.

Lemma 7.7.9 Let P be an (A, B)-bimodule which, as a right B-module, is finitely generated and projective. Then $P^\vee$, as a left B-module, is also finitely generated and projective, and in this situation:

- for every X, (7.7.5) gives a canonical isomorphism of right A-modules;
- $P \to P^{\vee\vee}$ is an isomorphism.

Proof As a right B-module, write P as a direct summand of $B^{\oplus n}$ for some $n \in \mathbb{Z}_{\geq 0}$. Then $P^\vee$ is naturally a direct summand of $B^{\oplus n}$. It is clear that replacing P by $B^{\oplus n}$ in (7.7.5) gives an isomorphism of $\Bbbk$-modules. Restricting to the direct summand P therefore still gives an isomorphism of $\Bbbk$-modules, hence an isomorphism of right A-modules. The assertion $P \xrightarrow{\sim} P^{\vee\vee}$ is proved in the same way: it is enough to check it for $B^{\oplus n}$. □

Under the preceding hypothesis, the adjoint pair (7.7.2) can therefore be rewritten as

$$(\cdot) \underset{A}{\otimes} P : \mathsf{Mod}\text{-}A \rightleftarrows \mathsf{Mod}\text{-}B : (\cdot) \underset{B}{\otimes} P^\vee. \tag{7.7.6}$$

Definition 7.7.10 Let P be an (A, B)-bimodule which, as a right B-module, is finitely generated and projective. Define the homomorphism of (B, B)-bimodules

$$\begin{aligned} \mathrm{ev} : \quad P^\vee \underset{A}{\otimes} P &\longrightarrow B \\ \lambda \otimes p &\longmapsto \lambda(p) \end{aligned}$$

and the homomorphism of (A, A)-bimodules

$$\begin{aligned} \mathrm{coev} : \quad A \longrightarrow P \underset{B}{\otimes} P^\vee &\xrightarrow[\sim]{\text{Lemma 7.7.9}} \mathrm{End}_{\mathsf{Mod}\text{-}B}(P) \\ a \longmapsto\quad\quad &\quad [p \mapsto ap]. \end{aligned}$$

It is easy to check that both maps are well defined. Combining (7.7.3) with Lemma 7.7.9, for every right A-module N and right B-module M, the unit morphism η_N and counit morphism ϵ_M of the adjoint pair (7.7.6) satisfy

$$\begin{array}{ccccc} N & \xrightarrow{\eta_N} & \mathrm{Hom}_{\mathsf{Mod}\text{-}B}(P, N \underset{A}{\otimes} P) & \xleftarrow{\sim} & N \underset{A}{\otimes} P \underset{B}{\otimes} P^\vee \\ x & \longmapsto & [p \mapsto x \otimes p] & \longleftarrow\!\shortmid & x \otimes \mathrm{coev}(1) \end{array}$$

$$\begin{array}{ccccc} M \underset{B}{\otimes} P^\vee \underset{A}{\otimes} P & \xrightarrow{\sim} & \mathrm{Hom}_{\mathsf{Mod}\text{-}B}(P, M) \underset{A}{\otimes} P & \xrightarrow{\epsilon_M} & M \\ x \otimes \lambda \otimes p & \longmapsto & x\lambda(\cdot) \otimes p & \longmapsto & x\lambda(p) \\ & & & & \shortparallel \\ & & & & (\mathrm{id}_M \otimes \mathrm{ev})(x \otimes \lambda \otimes p). \end{array}$$

Thus we may identify

$$\eta_N = \mathrm{id}_N \otimes \mathrm{coev} : N \to N \underset{A}{\otimes} P \underset{B}{\otimes} P^\vee,$$
$$\epsilon_M = \mathrm{id}_M \otimes \mathrm{ev} : M \underset{B}{\otimes} P^\vee \underset{A}{\otimes} P \to M.$$

The preceding constructions take place at the level of the bimodules P and $P^\vee$ and do not depend on M or N at all.

Definition 7.7.11 For an arbitrary $\Bbbk$-algebra A, the category (A, A)-Mod is monoidal under $\otimes := \underset{A}{\otimes}$. Hence §7.1 and §7.5 define what is meant by an algebra E (or coalgebra C) in (A, A)-Mod, as well as by left and right E-modules (or left and right C-comodules). Since the definition of a module (or comodule) involves $\otimes$ on only one side, the following slight generalization is needed.

- A left E-module means the following data: an $M \in \mathrm{Ob}(A\text{-}\mathsf{Mod})$ together with a morphism in A-Mod, $\mu_M : E \otimes M \to M$, such that the diagrams in Definition 7.1.2 commute in A-Mod.
- A left C-comodule means the following data: an $M \in \mathrm{Ob}(A\text{-}\mathsf{Mod})$ together with a morphism in A-Mod, $\rho_M : M \to C \otimes M$, such that the diagrams in Definition 7.5.2 commute in A-Mod.

Right E-modules and right C-comodules are defined similarly, and morphisms between modules or comodules are defined in the standard way.

Applying the preceding sequence of results to describe the monad and comonad determined by the adjoint pair (7.7.6) immediately gives the following result.

Proposition 7.7.12 Let P be an (A, B)-bimodule which, as a right B-module, is finitely generated and projective, and define the (B, A)-bimodule $P^\vee$ as above. Then:

- $P \underset{B}{\otimes} P^\vee$ is an algebra in the monoidal category (A, A)-Mod. Its multiplication is given by
$$\mathrm{id}_P \otimes \mathrm{ev} \otimes \mathrm{id}_{P^\vee} : P \underset{B}{\otimes} P^\vee \underset{A}{\otimes} P \underset{B}{\otimes} P^\vee \to P \underset{B}{\otimes} P^\vee$$
and its unit comes from $\mathrm{coev} : A \to P \underset{B}{\otimes} P^\vee$;
- $P^\vee \underset{A}{\otimes} P$ is a coalgebra in the monoidal category (B, B)-Mod. Its comultiplication is given by
$$\mathrm{id}_{P^\vee} \otimes \mathrm{coev} \otimes \mathrm{id}_P : P^\vee \underset{A}{\otimes} P \to P^\vee \underset{A}{\otimes} P \underset{B}{\otimes} P^\vee \underset{A}{\otimes} P$$
and its counit comes from $\mathrm{ev} : P^\vee \underset{A}{\otimes} P \to B$.

Next, consider the monad T on Mod-A determined by the adjoint pair (7.7.6), and the comonad L on Mod-B determined by it. Use Definition 7.7.11 when discussing modules and comodules.

- Specifying a T-module is equivalent to specifying a right A-module N together with a homomorphism satisfying the usual properties such as associativity and unitality,
$$N \underset{A}{\otimes} (P \underset{B}{\otimes} P^\vee) \to N;$$
in other words, it is equivalent to making N a $(P \underset{B}{\otimes} P^\vee)$-module.
- Specifying an L-comodule is equivalent to specifying a right B-module M together with a homomorphism satisfying the usual properties such as coassociativity and counitality,
$$M \to M \underset{B}{\otimes} (P^\vee \underset{A}{\otimes} P);$$
in other words, it is equivalent to making M a $(P^\vee \underset{A}{\otimes} P)$-comodule.

Proof The proof is easier than the statement. □

As a ring, $P \underset{B}{\otimes} P^\vee$ is simply $\mathrm{End}_{B\text{-}\mathsf{Mod}}(P)$. By contrast, the coalgebra structure on $P^\vee \underset{A}{\otimes} P$ looks cumbersome, but it is often more convenient in applications. Nevertheless, comodules do not have an absolute advantage: for example, the coalgebra must be flat for its comodule category to be abelian. We give only an outline of a simple proof; the exercises at the end of the chapter provide more.

Proposition 7.7.13 Let C be a coalgebra in (B, B)-Mod. Write the category of right (or left) C-comodules as Comod-C (or C-Comod), and write its forgetful functor to Mod-B (or B-Mod) as U. If C, as a left (or right) B-module, is flat [51, Definition 6.9.4], then Comod-C (or C-Comod) is an abelian category and U is faithful and exact.

Proof It is enough to discuss the version for $\mathsf{Comod}\text{-}C$. Let $f : M \to N$ be a homomorphism of right C-comodules and denote its cokernel at the level of B-modules by $\operatorname{coker}(Uf)$. Then there is a commutative diagram in $\mathsf{Mod}\text{-}B$ with exact rows,

$$\begin{tikzcd} M \arrow[r, "Uf"] \arrow[d] & N \arrow[r] \arrow[d] & \operatorname{coker}(Uf) \arrow[r] \arrow[d, dashed] & 0 \\ M \underset{B}{\otimes} C \arrow[r, "Uf\otimes\mathrm{id}"'] & N \underset{B}{\otimes} C \arrow[r] & \operatorname{coker}(Uf) \underset{B}{\otimes} C \arrow[r] & 0, \end{tikzcd}$$

where the dashed arrow is uniquely determined by functoriality of the cokernel. It is easy to prove that this dashed arrow gives $\operatorname{coker}(Uf)$ a comodule structure; we denote this object by $\operatorname{coker}(f)$, and it is the cokernel of f. The reader may check the details if desired.

We wish to lift $\ker(Uf)$ to a comodule in the same way. The key is to ensure that

$$0 \to \ker(Uf) \underset{B}{\otimes} C \to M \underset{B}{\otimes} C \xrightarrow{Uf\otimes\mathrm{id}} N \underset{B}{\otimes} C$$

is exact in $\mathsf{Mod}\text{-}B$. This is where the flatness of C as a left B-module is needed; the remaining routine verification is similar to the cokernel case.

To prove that $\mathsf{Comod}\text{-}C$ is an abelian category, it remains to show that the canonical morphism

$$\operatorname{coim}(f) = \operatorname{coker}[\ker(f) \hookrightarrow M] \to \ker[N \twoheadrightarrow \operatorname{coker}(f)] = \operatorname{im}(f)$$

is an isomorphism, but this need only be checked at the level of $\mathsf{Mod}\text{-}B$.

By construction, U preserves kernels and cokernels and is therefore exact; it is plainly faithful as well. $\square$

Returning to the main discussion, we continue the study of the functor $(\cdot) \underset{A}{\otimes} P$.

Proposition 7.7.14 Let P be an (A, B)-bimodule which, as a right B-module, is finitely generated and projective, and suppose that $(\cdot) \underset{A}{\otimes} P$ is a faithful exact functor. A right A-module N is finitely generated if and only if the right B-module $N \underset{A}{\otimes} P$ is finitely generated.

Proof For the "only if" direction, suppose there are $n \in \mathbb{Z}_{\geq 0}$ and a surjective homomorphism $A^{\oplus n} \twoheadrightarrow N$. Applying the functor gives a surjective homomorphism $P^{\oplus n} \twoheadrightarrow N \underset{A}{\otimes} P$, while P, as a right B-module, is finitely generated.

For the "if" direction, take a finitely generated submodule N' of N such that the image of $N' \underset{A}{\otimes} P$ contains a set of generators of $N \underset{A}{\otimes} P$ as a B-module. Since $(\cdot) \underset{A}{\otimes} P$ is faithful and exact, we in fact have $N' \underset{A}{\otimes} P \xrightarrow{\sim} N \underset{A}{\otimes} P$, and hence $N' = N$. $\square$

In fact, the property that a module is finitely generated admits an entirely categorical characterization, as follows.

Lemma 7.7.15 Let R be a ring and N a right R-module. Then N is finitely generated if and only if the following property holds: for every family $(N_i)_{i\in I}$ of subobjects of N, if $\sum_{i\in I} N_i = N$, then there is a finite subset $I_0 \subset I$ such that $\sum_{i\in I_0} N_i = N$.

Proof For the "only if" direction, let $x_1, \ldots, x_n$ be a set of generators for N. Each x_j belongs to some $\sum_{i\in I_j} N_i$, where $I_j \subset I$ is finite; it is enough to take $I_0 = \bigcup_{j=1}^n I_j$. For the "if" direction, consider $N = \sum_{x\in N} xR$. ☐

We now return to the Morita equivalences introduced in Definition–Proposition 7.7.4. We discuss only the case of right modules.

Theorem 7.7.16 (Kiiti Morita) Let A and B be $\Bbbk$-algebras. Let $\mathrm{Equiv}(\mathsf{Mod}\text{-}A, \mathsf{Mod}\text{-}B)$ be the category whose objects are all equivalences $F : \mathsf{Mod}\text{-}A \to \mathsf{Mod}\text{-}B$ and whose morphisms are isomorphisms between these equivalences. On the other hand, define a category $\mathcal{P}(A, B)$ whose objects are the (A, B)-bimodules P satisfying the following conditions:

- as a right B-module, P is a finitely generated projective generator of $\mathsf{Mod}\text{-}B$;
- left multiplication induces an isomorphism $A \xrightarrow{\sim} \mathrm{End}_{\mathsf{Mod}\text{-}B}(P)$.

Its morphisms are defined to be bimodule isomorphisms. There are quasi-inverse functors

$$\begin{array}{ccc} \mathrm{Equiv}(\mathsf{Mod}\text{-}A, \mathsf{Mod}\text{-}B) & \rightleftarrows & \mathcal{P}(A, B) \\ F & \longmapsto & F(A) \\ (\cdot) \underset{A}{\otimes} P & \longleftarrow\!\shortmid & P. \end{array}$$

Proof First, we show that $F \mapsto F(A)$ is well defined. Observe that A is a finitely generated projective generator of $\mathsf{Mod}\text{-}A$. Both the property of being a projective generator and the property of being finitely generated have categorical characterizations (Lemma 7.7.15), so the same holds for $F(A)$. Moreover, abstract considerations give

$$F : \mathrm{End}_{\mathsf{Mod}\text{-}A}(A) \xrightarrow{\sim} \mathrm{End}_{\mathsf{Mod}\text{-}B}(F(A)).$$

It is easy to check that this is exactly the homomorphism $A \to \mathrm{End}_{\mathsf{Mod}\text{-}B}(F(A))$ given by left multiplication by A on $F(A)$.

If the functor from right to left is also well defined, then Theorem 7.7.2 implies that the two functors are quasi-inverse. It therefore suffices to prove the following assertion: if $P \in \mathrm{Ob}(\mathcal{P}(A, B))$, then $F := (\cdot) \underset{A}{\otimes} P$ is an equivalence of categories.

Take P as above. In (7.7.6), we showed that the right adjoint of $(\cdot)\underset{A}{\otimes}P$ is $(\cdot)\underset{B}{\otimes}P^\vee$, where $P^\vee := \mathrm{Hom}_{\mathsf{Mod}\text{-}B}(P,B)$. We seek to show that these functors are quasi-inverse. Taking $Q = P^\vee$ in Definition–Proposition 7.7.4 (ii), it is enough to prove that the bimodule homomorphisms in Definition 7.7.10,

$$\mathrm{ev} : P^\vee \underset{A}{\otimes} P \to B, \qquad \mathrm{coev} : A \to P \underset{B}{\otimes} P^\vee \simeq \mathrm{End}_{\mathsf{Mod}\text{-}B}(P),$$

are both isomorphisms.

First consider ev. Since P is a generator of $\mathsf{Mod}\text{-}B$, there is a surjective homomorphism $P^{\oplus I} \twoheadrightarrow B$. In particular, there are $q_1,\ldots,q_n \in P^\vee$ and $p_1,\ldots,p_n \in P$ such that $\sum_{i=1}^n q_i(p_i) = 1_B$, or equivalently $\mathrm{ev}(\sum_i q_i \otimes p_i) = 1_B$. Thus surjectivity is proved.

Next, define an operation $P \times P^\vee \to A$, written as multiplication. It is characterized by the following associativity-like identity in P, for all $p, p' \in P$ and $q \in P^\vee$:

$$\underbrace{(pq)}_{\in A}\, p' = p\, \underbrace{(qp')}_{\in B}.$$

Here $qp' := q(p')$. Indeed, for fixed p and q, the right-hand side is an element of $\mathrm{End}_{\mathsf{Mod}\text{-}B}(P)$, while left multiplication induces $A \xrightarrow{\sim} \mathrm{End}_{\mathsf{Mod}\text{-}B}(P)$; thus $pq \in A$ is uniquely defined. The identity above also gives a second form of associativity, namely the following identity in $P^\vee$:

$$\underbrace{(qp)}_{\in B}\, q' = q\, \underbrace{(pq')}_{\in A}.$$

It is enough to check that the two sides take the same value on every $p' \in P$; this direct verification is left to the reader.

We can now prove the injectivity of ev. Choose p_i and q_i as in the preceding step. If $\mathrm{ev}(\sum_{j=1}^m q'_j \otimes p'_j) = 0$, then the two forms of associativity above give

$$\begin{aligned} \sum_j q'_j \otimes p'_j &= \sum_{i,j} (q_i p_i) q'_j \otimes p'_j = \sum_{i,j} q_i \underbrace{(p_i q'_j)}_{\in A} \otimes p'_j \\ &= \sum_{i,j} q_i \otimes (p_i q'_j) p'_j = \sum_{i,j} q_i \otimes p_i \underbrace{(q'_j p'_j)}_{\in B} = \sum_i q_i \otimes p_i \sum_j q'_j p'_j = 0. \end{aligned}$$

Finally, the assertion that coev is an isomorphism is precisely the second condition in the definition of $\mathcal{P}(A,B)$. □

Consequently, A and B are Morita equivalent if and only if $\mathcal{P}(A,B)$ is nonempty, while studying the category $\mathcal{P}(A,B)$ is an entirely concrete problem in module theory. Conversely, Morita equivalence may also be used to rewrite the definition of $\mathcal{P}(A,B)$ in a more symmetric form.

Corollary 7.7.17 An (A, B)-bimodule P is an object of the category $\mathcal{P}(A, B)$ in Theorem 7.7.16 if and only if the following conditions hold:

- ◇ as a left A-module, P is a finitely generated projective generator of $A\text{-}\mathsf{Mod}$, and as a right B-module, P is a finitely generated projective generator of $\mathsf{Mod}\text{-}B$;
- ◇ left multiplication induces an isomorphism $A \xrightarrow{\sim} \mathrm{End}_{\mathsf{Mod}\text{-}B}(P)$, while right multiplication induces an isomorphism $B^{\mathrm{op}} \xrightarrow{\sim} \mathrm{End}_{A\text{-}\mathsf{Mod}}(P)$.

Proof It is enough to prove the "only if" direction. Suppose $P \in \mathrm{Ob}(\mathcal{P}(A, B))$. The bimodule isomorphisms already proved,

$$P \underset{B}{\otimes} P^{\vee} \simeq A, \qquad P^{\vee} \underset{A}{\otimes} P \simeq B,$$

not only show that $(\cdot) \underset{A}{\otimes} P : \mathsf{Mod}\text{-}A \to \mathsf{Mod}\text{-}B$ is an equivalence, but also imply that $P \underset{B}{\otimes} (\cdot) : B\text{-}\mathsf{Mod} \to A\text{-}\mathsf{Mod}$ is an equivalence. Therefore the left-module version of Theorem 7.7.16 shows that P, as a left A-module, is a finitely generated projective generator, and that right multiplication induces $B^{\mathrm{op}} \xrightarrow{\sim} \mathrm{End}_{A\text{-}\mathsf{Mod}}(P)$. In fact, it is enough to repeat the first paragraph of the proof of Theorem 7.7.16 for left modules. □

7.8 Recognizing Module Categories

Throughout this section, continue to fix a commutative ring $\Bbbk$. All abelian categories and functors are understood to be $\Bbbk$-linear unless otherwise stated.

Categories of the form $R\text{-}\mathsf{Mod}$ or $\mathsf{Mod}\text{-}R$, where R is a $\Bbbk$-algebra, are collectively called module categories. In §7.7, we specifically studied module categories and functors between them. It is also natural to ask which categories are equivalent to module categories and how such equivalences can be described concretely. This section presents several concise, practical results about these questions.

Let $\mathcal{A}$ be an abelian category. Our approach is to consider a functor of the form $\mathrm{Hom}_{\mathcal{A}}(s, \cdot)$, which naturally lifts to a functor $\mathcal{A} \to \mathsf{Mod}\text{-}R$ with $R := \mathrm{End}_{\mathcal{A}}(s)$. The question is when $\mathrm{Hom}_{\mathcal{A}}(s, \cdot)$ is an equivalence.

Recall Definition A.2.2 of a compact object. Suppose $\mathcal{A}$ has all small filtered $\varinjlim$ and $X \in \mathrm{Ob}(\mathcal{A})$. The object X is called **compact** if, for every small filtered category I and functor $\alpha : I \to \mathcal{A}$, the following canonical morphism is an isomorphism:

$$\varinjlim_{i} \mathrm{Hom}\left(X, \alpha(i)\right) \to \mathrm{Hom}\left(X, \varinjlim \alpha\right).$$

Lemma 7.8.1 Let X be a projective object in a cocomplete abelian category $\mathcal{A}$. Then X is compact if and only if $\mathrm{Hom}_{\mathcal{A}}(X, \cdot)$ preserves all small direct sums; that is, for every small set I and family of objects $(Y_i)_{i \in I}$, $\bigoplus_{i \in I} \mathrm{Hom}_{\mathcal{A}}(X, Y_i) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{A}}(X, \bigoplus_{i \in I} Y_i)$. When this condition holds, $\mathrm{Hom}_{\mathcal{A}}(X, \cdot)$ in fact preserves all small $\varinjlim$.

Proof Suppose X is compact. A direct sum indexed by a small set I is the filtered $\varinjlim$ of the finite direct sums (take all finite subsets of I, ordered by inclusion). Since $\mathrm{Hom}_{\mathcal{A}}(X, \cdot)$ is known to preserve finite direct sums, it also preserves small direct sums.

Conversely, projectivity is equivalent to $\mathrm{Hom}_{\mathcal{A}}(X, \cdot)$ preserving cokernels, while a general small $\varinjlim$ can be constructed from small direct sums and cokernels. Hence $\mathrm{Hom}_{\mathcal{A}}(X, \cdot)$ preserves all small $\varinjlim$. □

Example 7.8.2 Let R be a $\Bbbk$-algebra and M a projective right R-module. Then M is compact if and only if M is finitely generated.

- Necessity: there are a small set I and homomorphisms $i : M \to R^{\oplus I}$ and $p : R^{\oplus I} \to M$ with $pi = \mathrm{id}_M$. Compactness implies that the image of i is contained in $R^{\oplus I_0}$ for some finite subset $I_0 \subset I$. Take $p' := p|_{R^{\oplus I_0}}$; then $p'i = \mathrm{id}_M$ still holds. Thus M can be realized as a direct summand of a finite-rank free module.

- Sufficiency: apply the criterion of Lemma 7.8.1. Every homomorphism $M \to \bigoplus_{i \in I} Y_i$ must land in $\bigoplus_{i \in I_0} Y_i$ for some finite subset $I_0 \subset I$ (it is enough to check the images of the generators). Hence M is compact.

Proposition 7.8.3 Let $\mathcal{A}$ be a cocomplete abelian category. Then $s \in \mathrm{Ob}(\mathcal{A})$ is a compact projective generator if and only if $\mathrm{Hom}_{\mathcal{A}}(s, \cdot) : \mathcal{A} \to \Bbbk\text{-}\mathsf{Mod}$ is a faithful exact functor preserving all small $\varinjlim$.

Proof By Proposition 2.8.15, being a projective generator is equivalent to $\mathrm{Hom}_{\mathcal{A}}(s, \cdot)$ being faithful and exact. By Lemma 7.8.1, under the projectivity assumption, compactness is equivalent to $\mathrm{Hom}_{\mathcal{A}}(s, \cdot)$ preserving all small $\varinjlim$. □

Theorem 7.8.4 (P. Gabriel) Let $s \in \mathrm{Ob}(\mathcal{A})$ and set $R := \mathrm{End}_{\mathcal{A}}(s)$. Then

$$\mathrm{Hom}_{\mathcal{A}}(s, \cdot) : \mathcal{A} \to \mathsf{Mod}\text{-}R$$

is an equivalence if and only if s is a compact projective generator of $\mathcal{A}$.

Proof For brevity, write $G := \mathrm{Hom}_{\mathcal{A}}(s, \cdot) : \mathcal{A} \to \mathsf{Mod}\text{-}R$. First prove necessity. Since G is an equivalence and $G(s) = R$, it is enough to check that R is a compact projective generator of $\mathsf{Mod}\text{-}R$, which is immediate.

We now prove sufficiency. Suppose s is a compact projective generator. Proposition 7.8.3 implies that G is a faithful exact functor preserving all small $\varinjlim$, because its composite with the forgetful functor $\mathsf{Mod}\text{-}R \to \Bbbk\text{-}\mathsf{Mod}$ has these properties. Since faithfulness is already known, it is enough to prove that:

- for all $X, Y \in \mathrm{Ob}(\mathcal{A})$, the corresponding map $\mathrm{Hom}_{\mathcal{A}}(X,Y) \to \mathrm{Hom}_R(GX, GY)$ is surjective;
- G is essentially surjective.

First, for every $X \in \mathrm{Ob}(\mathcal{A})$, define a morphism $\phi_X : s^{\oplus \mathrm{Hom}_{\mathcal{A}}(s,X)} \to X$ whose restriction to the direct summand corresponding to $f : s \to X$ is f. We show that ϕ_X is surjective. Otherwise, suppose that $\mathrm{coker}(\phi_X) \neq 0$. The generator property gives a nonzero morphism $s \to \mathrm{coker}(\phi_X)$ (Proposition 2.8.15), while projectivity makes this morphism factor as $s \xrightarrow{g} X \to \mathrm{coker}(\phi_X)$. It follows that the composite $s^{\oplus \mathrm{Hom}_{\mathcal{A}}(s,X)} \xrightarrow{\phi_X} X \to \mathrm{coker}(\phi_X)$ is nonzero on the direct summand corresponding to g, a contradiction.

Apply the same construction to the kernel $\ker(\phi_X)$. Thus every X fits into an exact sequence

$$s^{\oplus J} \to s^{\oplus I} \to X \to 0, \quad I = I_X,\ J = J_X : \text{small sets.}$$

Since G preserves all small $\varinjlim$, this gives an exact sequence of right R-modules

$$R^{\oplus J} \to R^{\oplus I} \to GX \to 0.$$

The first morphism in each sequence may be regarded as an $I \times J$ matrix over R acting by left multiplication; the two are in fact the same matrix.

Now consider a homomorphism of right R-modules $f : GX \to GY$. Choose exact sequences of the preceding form for X and Y. It is easy to see that f lifts to a commutative diagram with exact rows

$$\begin{tikzcd}
R^{\oplus J_X} \arrow[r] \arrow[d] & R^{\oplus I_X} \arrow[r] \arrow[d] & GX \arrow[r] \arrow[d, "f"] & 0 \\
R^{\oplus J_Y} \arrow[r] & R^{\oplus I_Y} \arrow[r] & GY \arrow[r] & 0.
\end{tikzcd}$$

The four arrows in the left square, however, can all be expressed as matrices over R. Those four matrices give the solid part of a commutative diagram with exact rows

$$\begin{tikzcd}
s^{\oplus J_X} \arrow[r] \arrow[d] & s^{\oplus I_X} \arrow[r] \arrow[d] & X \arrow[r] \arrow[d, dashed] & 0 \\
s^{\oplus J_Y} \arrow[r] & s^{\oplus I_Y} \arrow[r] & Y \arrow[r] & 0,
\end{tikzcd}$$

and the dashed part is uniquely determined by functoriality of the cokernel. Since G is exact, necessarily $G(X \dashrightarrow Y) = f$.

A similar argument shows that G is essentially surjective. For a given right R-module M, there is an exact sequence

$$R^{\oplus L} \to R^{\oplus K} \to M \to 0,$$

where the first morphism is regarded as a $K \times L$ matrix over R. The same matrix gives a cokernel exact sequence in $\mathcal{A}$,

$$s^{\oplus L} \to s^{\oplus K} \to X \to 0.$$

Since G preserves all small $\varinjlim$, necessarily $GX \simeq M$. This completes the proof. □

Another related problem is to recognize categories of finitely generated modules. Write $\mathsf{Mod}_{\mathrm{fg}}\text{-}R$ for the category of finitely generated right R-modules. If R is a right Noetherian ring, $\mathsf{Mod}_{\mathrm{fg}}\text{-}R$ remains an abelian category; see [51, Proposition 6.10.3, Lemma 6.10.4].

Theorem 7.8.5 Let $\mathcal{A}$ be an abelian category and $s \in \mathrm{Ob}(\mathcal{A})$. Suppose that:

- ◇ every $X \in \mathrm{Ob}(\mathcal{A})$ is a Noetherian object (Definition 2.4.12);
- ◇ $R := \mathrm{End}_{\mathcal{A}}(s)$ is a right Noetherian ring;
- ◇ s is a projective generator of $\mathcal{A}$.

Then the functor $\mathrm{Hom}_{\mathcal{A}}(s, \cdot) : \mathcal{A} \to \mathsf{Mod}\text{-}R$ factors as

$$\mathcal{A} \xrightarrow{\text{equivalence}} \mathsf{Mod}_{\mathrm{fg}}\text{-}R \subset \mathsf{Mod}\text{-}R.$$

Proof First show that every $X \in \mathrm{Ob}(\mathcal{A})$ can be placed in an exact sequence

$$s^{\oplus m} \to s^{\oplus n} \to X \to 0, \quad n, m \in \mathbb{Z}_{\geq 0}. \tag{7.8.1}$$

Indeed, since s is a generator, for every $X \neq 0$ there is a nonzero morphism $\phi_1 : s \to X$. If ϕ_1 is not surjective, there is a nonzero morphism $s \to \mathrm{coker}(\phi_1)$; the projectivity of s ensures that this morphism factors as $s \xrightarrow{\psi_1} X \to \mathrm{coker}(\phi_1)$. We thereby obtain $\phi_2 := (\phi_1, \psi_1) : s^{\oplus 2} \to X$ with $\mathrm{im}(\phi_2) \supsetneq \mathrm{im}(\phi_1)$. If ϕ_2 is not yet surjective, repeat the same procedure. Since X is a Noetherian object, this process must terminate after finitely many steps and yield a surjective morphism $\phi_X : s^{\oplus n} \twoheadrightarrow X$. Apply the same procedure to $\ker(\phi_X)$ to obtain (7.8.1).

Recall that s is a projective generator if and only if $G := \mathrm{Hom}_{\mathcal{A}}(s, \cdot)$ is faithful and exact. Applying G to (7.8.1) gives an exact sequence $R^{\oplus m} \to R^{\oplus n} \to GX \to 0$, showing that the image of G lies in $\mathsf{Mod}_{\mathrm{fg}}\text{-}R$.

On the other hand, since R is right Noetherian, every $M \in \mathrm{Ob}(\mathsf{Mod}_{\mathrm{fg}}\text{-}R)$ can also be placed in an exact sequence

$$R^{\oplus q} \to R^{\oplus p} \to M \to 0, \quad p, q \in \mathbb{Z}_{\geq 0}.$$

The rest of the proof follows the same idea as the proof of sufficiency in Theorem 7.8.4. In that argument, the compactness of s was used only to handle infinite direct sums; here only finite direct sums occur, and every additive functor preserves them. The remainder of the argument is identical and is not repeated. $\square$

7.9 Application: Descent of Modules

In geometry, given an adjoint pair $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$, the process of reconstructing information on $\mathcal{C}$ from comodules on $\mathcal{D}$ under the coaction of the corresponding comonad is usually called **descent**. This section explains the case of module theory.

Consider a homomorphism of commutative rings $K \to L$. The corresponding forgetful functor $\mathcal{F}_{L|K} : L\textsf{-Mod} \to K\textsf{-Mod}$ fits into the adjoint pair

$$L \underset{K}{\otimes} (\cdot) : K\textsf{-Mod} \rightleftarrows L\textsf{-Mod} : \mathcal{F}_{L|K},$$

We apply the results from the second half of §7.7 to study the corresponding comonad on L-$\textsf{Mod}$. More precisely, take $\Bbbk = K$ and apply the "change of rings" construction of Example 7.7.8. Apart from notation, we use left rather than right modules here. Since only commutative rings are considered, this makes no substantive difference.

Following the description in Example 7.7.8, denote the comonad determined on L-$\textsf{Mod}$ by the adjoint pair by $(\mathbf{L}, \delta, \epsilon)$, where

$$\mathbf{L} = L \underset{K}{\otimes} \mathcal{F}_{L|K}(\cdot), \quad \delta : \mathbf{L} \to \mathbf{L}^2, \quad \epsilon : \mathbf{L} \to \mathrm{id}.$$

From now on, as usual, the forgetful functor $\mathcal{F}_{L|K}$ will be omitted from formulas to simplify the notation.

A comodule over the comonad $\mathbf{L}$ is a datum (M, a), where M is an L-module and $a : M \to L \underset{K}{\otimes} M$ is an L-module homomorphism, subject to the requirement that the following diagrams commute:

$$\begin{tikzcd}
\ell \otimes 1 \otimes m & \in & L \underset{K}{\otimes} L \underset{K}{\otimes} M & L \underset{K}{\otimes} M \arrow[l, "\mathbf{L}a = \mathrm{id}_L \otimes a"'] \\
\ell \otimes m \arrow[u, mapsto] & \in & L \underset{K}{\otimes} M \arrow[u, "\delta_M"] & M \arrow[l, "a"] \arrow[u, "a"']
\end{tikzcd}
\qquad
\begin{tikzcd}
\ell m & \ell \otimes m \arrow[l, mapsto] \\
\rotatebox{90}{$\in$} & \rotatebox{90}{$\in$} \\
M & L \underset{K}{\otimes} M \arrow[l, "\epsilon_M"'] \\
& M. \arrow[ul, "\mathrm{id}_M"] \arrow[u, "a"']
\end{tikzcd}$$

All comodules (M, a) over the comonad $\mathbf{L}$ form a category L-$\textsf{Mod}^{\mathbf{L}}$. The dual of Lemma 7.6.7 gives the factorization

$$L \underset{K}{\otimes} (\cdot) = \left[K\textsf{-Mod} \xrightarrow{\mathbb{K}} L\textsf{-Mod}^{\mathbf{L}} \xrightarrow{U^{\mathbf{L}}} L\textsf{-Mod} \right] \quad \text{as a composite.}$$

The functor $U^{\mathbf{L}}$ forgets the coaction, $(M, a) \mapsto M$, while $\mathbb{K}$ maps an object N to $(L \underset{K}{\otimes} N, a)$, where

$$\begin{aligned} a : L \underset{K}{\otimes} N &\to L \underset{K}{\otimes} (L \underset{K}{\otimes} N) \quad (L\text{-module homomorphism}) \\ \ell \otimes x &\mapsto \ell \otimes 1 \otimes x. \end{aligned} \tag{7.9.1}$$

The central question is whether the adjoint pair is comonadic; equivalently, is $\mathbb{K}$ an equivalence?

Definition 7.9.1 Given a homomorphism of commutative rings $K \to L$, if $L \underset{K}{\otimes} (\cdot) : K\text{-}\mathsf{Mod} \to L\text{-}\mathsf{Mod}$ is a faithful exact functor, then L, as a commutative K-algebra, is called **faithfully flat**.

Theorem 7.9.2 (Flat descent) Let L be a faithfully flat commutative K-algebra. Then the adjoint pair $\left(L \underset{K}{\otimes} (\cdot), \mathcal{F}_{L|K} \right)$ is comonadic. In this case, a quasi-inverse $\mathbb{L}$ to $K\text{-}\mathsf{Mod} \xrightarrow{\mathbb{K}} L\text{-}\mathsf{Mod}^{\mathbf{L}}$ is described by the following equalizer diagram in $K\text{-}\mathsf{Mod}$:

$$\mathbb{L}(M, a) \longrightarrow M \underset{m \mapsto 1 \otimes m}{\overset{a}{\rightrightarrows}} L \underset{K}{\otimes} M, \qquad (M, a) \in \mathrm{Ob}\left(L\text{-}\mathsf{Mod}^{\mathbf{L}} \right).$$

Proof Proposition 7.7.7 establishes comonadicity, while the dual of Theorem 7.6.14 already contains the description of the quasi-inverse functor. $\square$

Remark 7.9.3 It must be emphasized that at the level of the derived categories $\mathsf{D}(K\text{-}\mathsf{Mod})$ and $\mathsf{D}(L\text{-}\mathsf{Mod})$, although the adjoint pair in question also has a derived version as in Example 4.12.18 and Example 4.12.10, there is no corresponding descent theorem.

The proof of flat descent is surprisingly simple. In practice, however, the comonad or the descent data must be described in a usable form; this is the theme of §7.10. In the remainder of this section we recast the definition of $L\text{-}\mathsf{Mod}^{\mathbf{L}}$ in another form for easy comparison with the literature. In essence, this is a concrete rewriting of Proposition 7.7.12 (in its left-module version, with $P = L = P^{\vee}$), though the procedure is somewhat circuitous.

For $i = 1, 2$, write $\iota_i : L \to L \underset{K}{\otimes} L$ for the homomorphism that inserts an element into the ith tensor slot. For every L-module M there is a canonical isomorphism of L-modules

$$\begin{aligned} L \underset{K}{\otimes} M &\xrightarrow{\sim} (L \underset{K}{\otimes} L) \underset{L, \iota_2}{\otimes} M \\ \ell \otimes m &\mapsto \ell \otimes 1 \otimes m, \end{aligned}$$

where the right-hand side is made an L-module through ι_1. By the standard adjunction in module theory, specifying an L-module homomorphism $a : M \to L \underset{K}{\otimes} M$ is therefore equivalent to specifying an $L \underset{K}{\otimes} L$-module homomorphism

$$\begin{aligned} \tilde{a} : (L \underset{K}{\otimes} L) \underset{L,\iota_1}{\otimes} M &\to (L \underset{K}{\otimes} L) \underset{L,\iota_2}{\otimes} M \\ 1 \otimes 1 \otimes m &\mapsto \sum_{i=1}^{n} \ell_i \otimes 1 \otimes m_i; \end{aligned} \tag{7.9.2}$$

here $a(m) = \sum_{i=1}^{n} \ell_i \otimes m_i$. Since $\iota_1|_K = \iota_2|_K$, when M comes from a K-module N, careful substitution from (7.9.1) shows that $\tilde{a}$ becomes $\mathrm{id} : L \underset{K}{\otimes} L \underset{K}{\otimes} N \to L \underset{K}{\otimes} L \underset{K}{\otimes} N$.

For brevity, write $L^{\otimes n} := L \underset{K}{\otimes} \cdots \underset{K}{\otimes} L$ (with n factors). Temporarily forget $L\text{-}\mathsf{Mod}^{\mathbf{L}}$ and consider an arbitrary $\tilde{a} : L^{\otimes 2} \underset{L,\iota_1}{\otimes} M \to L^{\otimes 2} \underset{L,\iota_2}{\otimes} M$. By the universal property of the tensor product [51, Proposition 7.3.4], specifying a pair of homomorphisms of commutative K-algebras $f, g : L \to E$ is equivalent to specifying a homomorphism $f \otimes g : L \underset{K}{\otimes} L \to E$. Thus $\tilde{a}$ defines the following E-module homomorphism:

$$\tilde{a}_{gf} := E \underset{f\otimes g}{\otimes} \tilde{a} : E \underset{L,f}{\otimes} M \to E \underset{L,g}{\otimes} M.$$

As (E, f, g) varies, we want to analyze the compatibility among these homomorphisms. The formula above shows that $(E, f, g) := (L \underset{K}{\otimes} L, \iota_1, \iota_2)$ is "universal" for this problem and corresponds to $\tilde{a}$, while a general (E, f, g) is obtained by applying $E \underset{f\otimes g}{\otimes} (\cdot)$ to it. To analyze the "universal" case, for $1 \leq i \neq j \leq 3$ write:

$\iota^3_{ij} : L^{\otimes 2} \to L^{\otimes 3}$	insertion into tensor slots (i, j)
$\iota^3_i : L \to L^{\otimes 3}$	insertion into the ith tensor slot

Using (7.9.2), define the $L^{\otimes 3}$-module homomorphism

$$\tilde{a}_{ji} := L^{\otimes 3} \underset{L,\iota^3_{ij}}{\otimes} \tilde{a} : \; L^{\otimes 3} \underset{L,\iota^3_i}{\otimes} M \to L^{\otimes 3} \underset{L,\iota^3_j}{\otimes} M.$$

Here we use $\iota^3_{ij}\iota_1 = \iota^3_i$ and $\iota^3_{ij}\iota_2 = \iota^3_j$. When M comes from a K-module, we may identify

$$\tilde{a} \text{ with } \mathrm{id}_{L^{\otimes 2} \underset{K}{\otimes} N}, \quad \tilde{a}_{ji} \text{ with } \mathrm{id}_{L^{\otimes 3} \underset{K}{\otimes} N}.$$

Then $\tilde{a}_{31} = \tilde{a}_{32}\tilde{a}_{21}$ is clear. This is the desired compatibility condition. Its important consequence is

$$\tilde{a}_{31} = \tilde{a}_{32}\tilde{a}_{21} \implies \begin{array}{l} \forall f, g, h : L \xrightarrow{K\text{-algebra}} E, \\ \tilde{a}_{hf} = \tilde{a}_{hg}\tilde{a}_{gf}; \end{array} \tag{7.9.3}$$

the left-hand side is the "universal version" of the right-hand side.

Definition 7.9.4 (Descent datum) Define $\mathrm{Desc}_{K\to L}$ to be the following category. Its objects are data $(M, \tilde{a})$, where M is an L-module and $\tilde{a} : L^{\otimes 2} \underset{L,\iota_1}{\otimes} M \to L^{\otimes 2} \underset{L,\iota_2}{\otimes} M$ satisfies

▷ Isomorphism condition $\tilde{a}$ is an isomorphism of $L^{\otimes 2}$-modules;

▷ Cocycle condition $\tilde{a}_{31} = \tilde{a}_{32}\tilde{a}_{21}$.

A morphism from $(M, \tilde{a})$ to $(M', \tilde{a}')$ is an L-module homomorphism $f : M \to M'$ satisfying $(\mathrm{id}^{\otimes 2} \otimes f)\tilde{a} = \tilde{a}'(\mathrm{id}^{\otimes 2} \otimes f)$. Such data $(M, \tilde{a})$ are called **descent data**.

The discussion before (7.9.3) shows that $L \underset{K}{\otimes} (\cdot)$ gives a functor $K\text{-}\mathsf{Mod} \to \mathrm{Desc}_{K\to L}$. We now relate descent data to $L\text{-}\mathsf{Mod}^{\mathbf{L}}$.

Lemma 7.9.5 There is an isomorphism of categories from $L\text{-}\mathsf{Mod}^{\mathbf{L}}$ to $\mathrm{Desc}_{K\to L}$, given on objects by $(M, a) \leftrightarrow (M, \tilde{a})$ and defined on morphisms in the standard way.

Moreover, the isomorphism condition in the definition of $\mathrm{Desc}_{K\to L}$ may be replaced by the following condition: define $m : L^{\otimes 2} \to L$ by $m(\ell \otimes \ell') = \ell\ell'$. The endomorphism $M \to M$ given by $L \underset{L^{\otimes 2},m}{\otimes} \tilde{a}$ is id_M.

Proof Specifying $\tilde{a}$ is known to be equivalent to specifying $a : M \to L \underset{K}{\otimes} M$. The issue is to match the conditions on the descent datum $\tilde{a}$ with the conditions on a that make (M, a) a comodule. This is where the second assertion will be used.

The comodule conditions are the coassociativity and counit laws. For the first, the equivalence of $\tilde{a}_{31} = \tilde{a}_{32}\tilde{a}_{21}$ with coassociativity is a mechanical verification left to the reader. For the second, write φ for the composite $M \xrightarrow{a} L \underset{K}{\otimes} M \xrightarrow{\epsilon_M} M$. The counit law is equivalent to $\varphi = \mathrm{id}$. Recalling that $\epsilon_M(\ell \otimes m) = \ell m$, a similar direct verification gives

$$\varphi = L \underset{L^{\otimes 2},m}{\otimes} \tilde{a}.$$

We next show that, under the assumption $\tilde{a}_{31} = \tilde{a}_{32}\tilde{a}_{21}$, the map $\tilde{a}$ is an isomorphism if and only if $\varphi = \mathrm{id}_M$. This will complete the proof.

Suppose $\tilde{a}$ is an isomorphism. Then φ is also an isomorphism. In (7.9.3), take $E = L$ and $f = g = h = \mathrm{id}_L$. The right-hand side readily becomes $\varphi = \varphi^2$, and hence $\varphi = \mathrm{id}_M$.

Conversely, suppose $\varphi = \mathrm{id}_M$. For every homomorphism of K-algebras $f : L \to E$, change scalars on $\tilde{a}$ along the two paths of the commutative diagram

$$\begin{tikzcd} L^{\otimes 2} \arrow[r, "m"] \arrow[dr, "f\otimes f"'] & L \arrow[d, "f"] \\ & E. \end{tikzcd}$$

This shows that $\tilde{a}_{ff} = \mathrm{id}$. Together with (7.9.3), it follows that $\tilde{a}_{fg}^{-1} = \tilde{a}_{gf}$ for every (E, f, g). In particular, the "universal version" $\tilde{a}$ is also an isomorphism. This proves the lemma. $\square$

Theorem 7.9.6 (Second form of flat descent) Let L be a faithfully flat commutative K-algebra. Then

$$L \underset{K}{\otimes} (\cdot) : K\text{-}\mathsf{Mod} \to \mathrm{Desc}_{K \to L}$$

is an equivalence of categories. One of its quasi-inverses, $\tilde{\mathbb{L}}$, may be taken to be the equalizer

$$0 \longrightarrow \tilde{\mathbb{L}}(M, \tilde{a}) \longrightarrow M \underset{m \mapsto 1 \otimes m}{\overset{a}{\rightrightarrows}} L \underset{K}{\otimes} M,$$

where a is the map corresponding to $\tilde{a}$ under (7.9.2).

Proof By Lemma 7.9.5, this is merely a reformulation of the flat descent Theorem 7.9.2. $\square$

Theorem 7.9.6 is the usual formulation of flat descent in algebraic geometry. There a faithfully flat homomorphism $K \to L$ corresponds, in a certain sense, to a covering of a geometric object.

The datum $\tilde{a}$, or the corresponding a, is sometimes also expressed as a homomorphism of left $L \underset{K}{\otimes} L$-modules, that is, of (L, L)-bimodules,

$$\begin{aligned} \bar{a} : M \underset{K}{\otimes} L &\to L \underset{K}{\otimes} M \\ m \otimes \ell &\mapsto \sum_{i=1}^{n} \ell_i \otimes \ell m_i, \end{aligned}$$

still assuming $a(m) = \sum_{i=1}^{n} \ell_i \otimes m_i$. This is obtained by contracting the tensor products in (7.9.2) in the appropriate way and has the advantage of a more compact form. A contracted version for $\tilde{a}_{ji}$ is obtained similarly.

Remark 7.9.7 (Flat descent for algebras and modules) We can go on to discuss how L-algebras, or modules over those algebras, descend to K. Concretely, suppose $(M, a), (M', a') \in \mathrm{Ob}(L\text{-}\mathsf{Mod}^{\mathbf{L}})$ are the images of $N, N' \in \mathrm{Ob}(K\text{-}\mathsf{Mod})$. Recall that $L \underset{K}{\otimes} (\cdot) : K\text{-}\mathsf{Mod} \to L\text{-}\mathsf{Mod}$ is a monoidal functor [51, Proposition 6.6.10]; that is, there are canonical isomorphisms such as

$$\begin{aligned} (L \underset{K}{\otimes} N_1) \underset{L}{\otimes} (L \underset{K}{\otimes} N_2) &\xrightarrow{\sim} L \underset{K}{\otimes} (N_1 \underset{K}{\otimes} N_2) \\ (\ell_1 \otimes x) \otimes (\ell_2 \otimes y) &\mapsto \ell_1 \ell_2 \otimes (x \otimes y). \end{aligned} \tag{7.9.4}$$

Use this to give $M \underset{L}{\otimes} M'$ the $\mathbf{L}$-comodule structure coming from $N \underset{K}{\otimes} N'$. Now take $N' = N$. Giving N a K-algebra structure is equivalent to giving M an L-algebra structure such that the multiplication $M \underset{L}{\otimes} M \to M$ and unit $L \to M$ are both morphisms in $L\text{-}\mathsf{Mod}^{\mathbf{L}}$; all properties such as associativity required of a K-algebra come from M. The descent of M-modules to N-modules should be understood in the same way. For this purpose, descent data of the form (7.9.2) may be more convenient than comodules over $\mathbf{L}$, because all the functors involved are monoidal on M, whereas $\mathbf{L}$ itself is not a monoidal functor.

7.10 Application: Galois Descent

In what follows, $L|K$ is a Galois extension of fields. This is a useful special case of flat descent from §7.9, called **Galois descent**. Our strategy is to describe the corresponding comonad directly. The only tools required are the form of flat descent in Theorem 7.9.2 and its preliminaries; the deeper discussion in the second half of §7.9 is not needed.

From now on, assume that K and L are fields and that $L|K$ is a Galois extension. The modules considered above become vector spaces. Write

$$\Gamma := \mathrm{Gal}(L|K), \quad \text{equipped with the Krull topology.}$$

Background on the Krull topology can be found in [51, §4.10, §9.2]; when $L|K$ is finite, this is just the discrete topology. In fact, Γ is a profinite group as introduced in §B.1.

A map $f : \Gamma \to X$ is called **smooth** if, for every $\sigma \in \Gamma$, there is an open subgroup Γ_0 such that $\sigma_0 \in \Gamma_0 \implies f(\sigma\sigma_0) = f(\sigma)$. Write

$$\mathrm{Maps}(\Gamma, X)^\infty := \{\text{smooth maps } f : \Gamma \to X\}.$$

Lemma 7.10.1 If $f \in \mathrm{Maps}(\Gamma, X)^\infty$, there is an open normal subgroup Γ' such that f factors through Γ/Γ'. In particular, f takes only finitely many values.

Proof For every $\sigma \in \Gamma$, there is an open subgroup Γ_0, depending on σ, such that f is constant on $\sigma\Gamma_0$. These cosets form an open cover of Γ. Since Γ is compact, there is a finite subcover $\sigma_1\Gamma_{0,1}, \ldots, \sigma_r\Gamma_{0,r}$. Now choose an open normal subgroup Γ' contained in $\bigcap_{i=1}^r \Gamma_{0,i}$. $\square$

We use this to describe concretely the comonad $\mathbf{L} : M \mapsto L \underset{K}{\otimes} M$ determined by $L|K$ on $\mathsf{Vect}(L)$.

Lemma 7.10.2 For every L-vector space M, give $\mathrm{Maps}(\Gamma, M)^\infty$ the following L-vector space structure: if $\ell \in L$, then $(\ell f)(\sigma) = \sigma(\ell)f(\sigma)$ for every $\sigma \in \Gamma$. There is an isomorphism of L-vector spaces

$$\begin{aligned} \Phi(M) : L \underset{K}{\otimes} M &\xrightarrow{\sim} \mathrm{Maps}(\Gamma, M)^\infty \\ \ell \otimes m &\longmapsto [f : \sigma \mapsto \sigma(\ell)m]. \end{aligned} \tag{7.10.1}$$

Proof It is clear that $\Phi(M)$ is well-defined ($\mathrm{Stab}_\Gamma(\ell)$ is always open, so f is smooth), and it is L-linear by definition. It also commutes with arbitrary direct sums: $\Phi(\bigoplus_i M_i) = \bigoplus_i \Phi(M_i)$ (every f takes only finitely many values, so $\mathrm{Maps}(\Gamma, \cdot)^\infty$ commutes with arbitrary direct sums). Choosing a basis of M, the verification of (7.10.1) reduces to the special case $M = L$,

$$\begin{aligned} \Phi(L) : L \underset{K}{\otimes} L &\xrightarrow{\sim} \mathrm{Maps}(\Gamma, L)^\infty \\ \ell \otimes \ell' &\longmapsto [f : \sigma \mapsto \sigma(\ell)\ell']. \end{aligned}$$

For every finite Galois subextension $L'|K$ of $L|K$, write $\Gamma' := \mathrm{Gal}(L|L')$. We claim that the same map as above gives

$$L' \underset{K}{\otimes} L' \xrightarrow{\sim} \mathrm{Maps}(\Gamma/\Gamma', L') := \{\text{maps } f : \Gamma/\Gamma' \to L'\}\,. \tag{7.10.2}$$

Once this is granted, since $\mathrm{Maps}(\Gamma, L)^\infty = \bigcup_{L'|K} \mathrm{Maps}(\Gamma/\Gamma', L')$ (apply Lemma 7.10.1; enlarging L' amounts to shrinking Γ'), taking the union of both sides of (7.10.2) over all $L'|K$ shows that $\Phi(L)$ is an isomorphism.

Choose α such that $L' = K(\alpha)$. Its minimal polynomial $p \in K[X]$ splits over L' as $\prod_{i=1}^d (X - \alpha_i)$, with $\alpha_1 = \alpha$ and no repeated roots. The group $\Gamma/\Gamma' = \mathrm{Gal}(L'|K)$ is in bijection with $\alpha_1, \dots, \alpha_d$ by $\sigma \mapsto \sigma(\alpha)$. Write $\ell \in L'$ as $g(\alpha)$ with $g \in K[X]$. Then

$$\begin{array}{ccccccccc} L' \underset{K}{\otimes} L' & \xleftarrow{\sim} & \dfrac{K[X]}{(p)} \underset{K}{\otimes} L' & \xrightarrow{\sim} & \dfrac{L'[X]}{\prod_{i=1}^{d}(X-\alpha_i)} & \xrightarrow{\sim} & \prod_{i=1}^{d} \underbrace{\dfrac{L'[X]}{(X-\alpha_i)}}_{\text{identified with } L'} & \xrightarrow{\sim} & (L')^{\Gamma/\Gamma'} \\ \ell \otimes \ell' & \longleftarrow & (g + (p)) \otimes \ell' & \longmapsto & g\ell' + (p) & \longmapsto & (g(\alpha_i)\ell')_{i=1}^d & \longmapsto & (\sigma_i(\ell)\ell')_{i=1}^d\,. \end{array}$$

This proves (7.10.2). $\square$

Lemma 7.10.3 Under the canonical isomorphism of Lemma 7.10.2, the comonad $(\mathbf{L}, \delta, \epsilon)$ is described by the following commutative diagram, where M is any L-vector space:

$$\begin{tikzcd}
M \arrow[d, equal] & L \underset{K}{\otimes} M \arrow[l, "\epsilon_M"'] \arrow[r, "\delta_M"] \arrow[d, "\wr"] & L \underset{K}{\otimes} L \underset{K}{\otimes} M \arrow[d, "\wr"] \\
M & \mathrm{Maps}(\Gamma, M)^\infty \arrow[l, "\epsilon'_M"'] \arrow[r, "\delta'_M"] & \mathrm{Maps}\left(\Gamma, (\mathrm{Maps}(\Gamma, M)^\infty)\right)^\infty \\[-20pt]
\rotatebox{90}{$\in$} & \rotatebox{90}{$\in$} & \rotatebox{90}{$\in$} \\[-20pt]
f(1) & f \arrow[l, mapsto] \arrow[r, mapsto] & [\sigma \mapsto [\tau \mapsto f(\tau\sigma)]]
\end{tikzcd}$$

where $1 := 1_\Gamma$.

Proof The counit $\epsilon_M : L \underset{K}{\otimes} M \to M$ is $\ell \otimes m \mapsto \ell m$, which clearly corresponds under (7.10.1) to $f \mapsto f(1)$. The comultiplication $\delta_M : L \underset{K}{\otimes} M \to L \underset{K}{\otimes} L \underset{K}{\otimes} M$ requires a little work. If $f \in \mathrm{Maps}(\Gamma, M)^\infty$ corresponds to $\ell \otimes m$, then the image of the latter under δ_M, namely $\ell \otimes (1 \otimes m)$, corresponds to the map

$$\Gamma \ni \sigma \mapsto \sigma(\ell) \cdot \underbrace{[\tau \mapsto m]}_{\in \mathrm{Maps}(\Gamma, M)^\infty} = \left[\tau \mapsto \tau(\sigma(\ell))m \overset{(7.10.1)}{=\!=\!=} f(\tau\sigma)\right].$$

This proves the claim. $\square$

Definition 7.10.4 Suppose that the group $\Gamma = \mathrm{Gal}(L|K)$ acts on the left on an L-vector space M. The action is called **semilinear** if $\sigma(m + m') = \sigma(m) + \sigma(m')$ and $\sigma(tm) = \sigma(t)\sigma(m)$ for all $\sigma \in \Gamma$, $t \in L$, and $m, m' \in M$. Define the category $\mathsf{Vect}^{\Gamma,\infty}(L)$ as follows.

- ◇ Objects: L-vector spaces M with a smooth semilinear Γ-action. Smoothness means that $\mathrm{Stab}_\Gamma(m)$ is an open subgroup of Γ for every $m \in M$.
- ◇ Morphisms: L-linear maps φ satisfying $\varphi(\sigma m) = \sigma(\varphi(m))$, also called Γ-equivariant linear maps.

Return to the comonad $\mathbf{L}$ determined by the Galois extension $L|K$. Its category of comodules is denoted $\mathsf{Vect}(L)^{\mathbf{L}}$.

Lemma 7.10.5 With the notation above, there is an isomorphism of categories

$$\mathsf{Vect}(L)^{\mathbf{L}} \simeq \mathsf{Vect}^{\Gamma,\infty}(L).$$

If M is an object of $\mathsf{Vect}^{\Gamma,\infty}(L)$, the corresponding object in $\mathsf{Vect}(L)^{\mathbf{L}}$ is (M, a), where a is characterized by the fact that the composite

$$M \xrightarrow{a} L \underset{K}{\otimes} M \xrightarrow[\sim]{\Phi(M)} \mathrm{Maps}(\Gamma, M)^\infty$$

maps $m \in M$ to the map $[\sigma \mapsto \sigma m]$ from Γ to M.

Proof First observe that $\mathrm{Maps}(\Gamma,\cdot)^\infty$ is a functor: every map $\varphi : M_1 \to M_2$ induces $\mathrm{Maps}(\Gamma, M_1)^\infty \to \mathrm{Maps}(\Gamma, M_2)^\infty$, mapping f to φf.

Next, use Lemma 7.10.3 to rewrite and combine the two commutative diagrams defining a comodule into

$$\begin{tikzcd}
\mathrm{Maps}\left(\Gamma, \mathrm{Maps}(\Gamma, M)^\infty\right)^\infty & \mathrm{Maps}(\Gamma, M)^\infty \arrow[l, "\mathrm{Maps}(\Gamma,a')^\infty"'] \arrow[r, "\epsilon'_M"] & M \\
\mathrm{Maps}(\Gamma, M)^\infty \arrow[u, "\delta'_M"] & M \arrow[l, "a'"] \arrow[u, "a'"'] \arrow[ur, equal] &
\end{tikzcd}$$

where a' corresponds to a. Specifying an L-linear map $a' : M \to \mathrm{Maps}(\Gamma, M)^\infty$ is equivalent to specifying a map $\beta : \Gamma \times M \to M$ such that

- $\diamond$ $\beta(\sigma, m + m') = \beta(\sigma, m) + \beta(\sigma, m')$ and $\beta(\sigma, tm) = \sigma(t)\beta(\sigma, m)$, with $t \in L$;
- $\diamond$ for every $m \in M$ and $\sigma \in \Gamma$, there is an open subgroup Γ_0 such that $\sigma_0 \in \Gamma_0 \implies \beta(\sigma\sigma_0, m) = \beta(\sigma, m)$.

The concrete correspondence is, of course, $\beta(\sigma, m) = a'(m)(\sigma)$. The commutative diagram above then translates into

$$\beta(\tau\sigma, m) = \beta(\tau, \beta(\sigma, m)), \quad \beta(1, m) = m.$$

At the level of objects, these conditions are precisely those of a smooth semilinear action of Γ. The verification on morphisms is immediate. $\square$

Lemma 7.10.6 If $(M, a) \in \mathrm{Ob}(\mathsf{Vect}(L)^{\mathbf{L}})$ comes from a K-vector space N, then the corresponding object of $\mathsf{Vect}^{\Gamma,\infty}(L)$ is $L \underset{K}{\otimes} N$ with the smooth semilinear action

$$\sigma(\ell \otimes x) = \sigma(\ell) \otimes x, \quad \ell \in L,\ x \in N, \quad \sigma \in \Gamma.$$

Proof We have $M = L \underset{K}{\otimes} N$. By (7.9.1), the homomorphism a maps $\ell \otimes x$ to $\ell \otimes 1 \otimes x$. The characterization of the semilinear action therefore gives

$$\sigma(\ell \otimes x) = \Phi(M)(\ell \otimes (1 \otimes x))(\sigma) = \sigma(\ell)(1 \otimes x) = \sigma(\ell) \otimes x,$$

for arbitrary $\ell \in L$ and $x \in N$. This proves the lemma. $\square$

Take $(M, a) \in \mathrm{Ob}(\mathsf{Vect}(L)^{\mathbf{L}})$. Since $\Phi(M)(1 \otimes m)(\sigma) = \sigma(1)m = m$, the description of a in Lemma 7.10.5 extends to the following commutative diagram in $\mathsf{Vect}(K)$:

$$\begin{tikzcd}
M \arrow[r, "a"] \arrow[dr, "m\mapsto[\sigma\mapsto\sigma m]"'] & L \underset{K}{\otimes} M \arrow[d, "\Phi(M)"] & M. \arrow[l, "1\otimes m \leftarrow\!\shortmid m"'] \arrow[dl, "{[m\leftarrow\!\shortmid\sigma]\leftarrow\!\shortmid m}"] \\
& \mathrm{Maps}(\Gamma, M)^\infty &
\end{tikzcd}$$

Thus the equalizer in Theorem 7.9.2 is the operation of taking Γ-invariants,

$$M^\Gamma := \{m \in M : \forall \sigma \in \Gamma,\ \sigma m = m\}.$$

Notice that $M \mapsto M^\Gamma$ gives a K-linear functor between K-linear categories,

$$(\cdot)^\Gamma : \mathsf{Vect}^{\Gamma,\infty}(L) \to \mathsf{Vect}(K).$$

Theorem 7.10.7 (Galois descent) Let $L|K$ be a Galois extension of fields and let $\Gamma = \mathrm{Gal}(L|K)$. Then the functor

$$\begin{aligned} \mathsf{Vect}(K) &\longrightarrow \mathsf{Vect}^{\Gamma,\infty}(L)\\ \text{object } N &\longmapsto L \underset{K}{\otimes} N\\ \text{morphism } \varphi &\longmapsto \mathrm{id} \otimes \varphi \end{aligned}$$

is an equivalence of categories. A quasi-inverse may be taken to be $(\cdot)^\Gamma : \mathsf{Vect}^{\Gamma,\infty}(L) \to \mathsf{Vect}(K)$.

Proof Clearly L is a faithfully flat K-algebra. Lemma 7.10.5 and the preceding description of the equalizer reduce the assertion to the flat descent Theorem 7.9.2. $\square$

Let us view Theorem 7.10.7 from another angle. For every $M \in \mathrm{Ob}(\mathsf{Vect}^{\Gamma,\infty}(L))$, define

$$\iota_M : L \underset{K}{\otimes} (M^\Gamma) \to M, \quad \ell \otimes x \mapsto \ell x. \tag{7.10.3}$$

This is clearly a Γ-equivariant linear map. There is a simple adjoint pair

$$\begin{gathered} L \underset{K}{\otimes} (\cdot) : \mathsf{Vect}(K) \rightleftarrows \mathsf{Vect}^{\Gamma,\infty}(L) : (\cdot)^\Gamma,\\ \mathrm{Hom}_{\mathsf{Vect}^{\Gamma,\infty}(L)}(L \underset{K}{\otimes} N, M) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{Vect}(K)}(N, M^\Gamma)\\ \varphi \longmapsto \varphi|_{1\otimes N}\\ \iota_M \circ (\mathrm{id}_L \otimes \psi) \longleftarrow\!\shortmid\ \psi. \end{gathered} \tag{7.10.4}$$

- ◇ It is easy to see that the unit of the adjunction $N \to (L \underset{K}{\otimes} N)^\Gamma$ maps x to $1 \otimes x$. This is an isomorphism: choose a basis B of N. An element $\sum_{b\in B} \ell_b \otimes b$ (a finite sum) of $L \underset{K}{\otimes} N$ is Γ-invariant if and only if each ℓ_b is invariant; equivalently, $\sum_b \ell_b \otimes b = 1 \otimes \sum_b \ell_b b \in N$.
- ◇ The counit morphism is precisely the previously defined ι_M.

  If $M = L \underset{K}{\otimes} N$, the preceding item gives $L \underset{K}{\otimes} (M^\Gamma) = L \underset{K}{\otimes} (1 \otimes N) \xrightarrow{\sim} L \underset{K}{\otimes} N = M$, so ι_M is an isomorphism in this case.

Once Theorem 7.10.7 is accepted, every M comes from some N, so the discussion above shows that $(L \underset{K}{\otimes} (\cdot), (\cdot)^{\Gamma})$ is an adjoint equivalence. Conversely, if ι_M can be shown to be an isomorphism for every M, the Galois descent Theorem 7.10.7 follows anew. Although the proof by flat descent has a natural conceptual flow, it is admittedly indirect. The exercises will give a direct method for proving that ι_M is an isomorphism; the technique is also useful elsewhere.

Remark 7.10.8 (Galois descent for algebras and modules) Let $N_1, N_2 \in \mathrm{Ob}(\mathsf{Vect}(K))$. Under (7.9.4), the natural semilinear Γ-action on $L \underset{K}{\otimes} (N_1 \underset{K}{\otimes} N_2)$ becomes the "diagonal action" on $(L \underset{K}{\otimes} N_1) \underset{L}{\otimes} (L \underset{K}{\otimes} N_2)$, namely the simultaneous action on both tensor slots. Specifying a K-algebra N is equivalent to specifying an L-algebra structure on $M := L \underset{K}{\otimes} N$ such that its algebra operations are compatible with the semilinear Γ-action:

$$\sigma(1_M) = 1_M, \quad \sigma(xy) = \sigma(x)\sigma(y).$$

Modules over K-algebras should be understood in the same way. This is analogous to the idea of Remark 7.9.7.

7.11 The Relation between Galois Descent and H^1

Descent is a powerful method for classifying various algebraic or geometric objects; its general formulation requires the language of algebraic geometry. For ease of exposition, we make two simplifying assumptions in this section.

1. We use only Galois descent (Theorem 7.10.7 and the discussion following it), rather than flat descent or more general techniques.

2. As far as possible, this book uses algebraic language, especially that of linear algebra, to explain the examples. We must therefore narrow the scope of the classification problem. In fact, we shall consider only a certain class of tensors in finite-dimensional vector spaces.

This viewpoint naturally leads to the nonabelian cohomology H^1 of the Galois group. The discussion below draws on the tools of §6.13 and §7.10. Since we need to allow infinite Galois extensions, we shall also continue to use the language of profinite groups; see §B.1. We first introduce the necessary notation.

Convention 7.11.1 Throughout this section, fix a field K and write $\otimes := \otimes_K$. For a finite-dimensional K-vector space V, write $V^{\vee}$ for its dual space. For each $(p, q) \in \mathbb{Z}^2_{\geq 0}$,

introduce the notation
$$T^{p,q}V := V^{\otimes p} \otimes (V^\vee)^{\otimes q}.$$
Following the convention of differential geometry, the elements of $T^{p,q}V$ are called **tensors of type (p,q) on V**.

Every isomorphism $g : V \to V'$ between finite-dimensional K-vector spaces naturally induces an isomorphism $T^{p,q}V \xrightarrow{\sim} T^{p,q}V'$; for convenience, this induced isomorphism is still denoted by g. Moreover, finite dimensionality gives a canonical isomorphism $(V_1 \otimes V_2)^\vee \simeq V_1^\vee \otimes V_2^\vee$.

In the classification problems considered in this section, the principal objects are data of the following form, defined over the field K:

$$\mathbf{t} = (V, I, (t_i, p_i, q_i)_{i\in I}), \tag{7.11.1}$$

where

- ◇ V is a finite-dimensional K-vector space;
- ◇ I is a set (which may be empty);
- ◇ (p_i, q_i) is a family of pairs of nonnegative integers, indexed by $i \in I$;
- ◇ $t_i \in T^{p_i,q_i}V$ is a family of tensors, indexed by $i \in I$.

In other words, we study a family of tensors of specified types in a finite-dimensional vector space. Given another datum

$$\mathbf{t}' = (V', I, (t_i', p_i, q_i)_{i\in I}),$$

if there is an isomorphism of K-vector spaces $g : V \xrightarrow{\sim} V'$ such that $gt_i = t_i'$ for every i, then g is called an isomorphism between the two data, and we write

$$g\mathbf{t} = \mathbf{t}' \quad \text{or} \quad g : \mathbf{t} \xrightarrow{\sim} \mathbf{t}'.$$

We wish to classify such data $\mathbf{t}$ up to isomorphism, or more specifically to classify those $\mathbf{t}$ satisfying additional conditions. These conditions must be algebraically expressible and preserved by isomorphisms. Although the theoretical framework is confined to linear algebra, its range of applications is already quite broad. Here are several special cases.

1. Consider data (V, b), where $b \in (V^\vee)^{\otimes 2}$ is a tensor of type $(0,2)$. Classifying these data is equivalent to classifying bilinear forms on V. We may also impose conditions such as symmetry, antisymmetry, or nondegeneracy on b; all of these are "algebraic" conditions. Multilinear forms may be treated similarly.

2. We may also classify $(V, \varphi_1, \ldots, \varphi_n)$, where $\varphi_i \in V \otimes V^\vee$ is a tensor of type $(1,1)$. Since $V \otimes V^\vee \simeq \mathrm{End}_K(V)$, this amounts to classifying V together with a family of linear endomorphisms. Various algebraic relations may be imposed, such as commutativity among the endomorphisms.

3. Next consider data (V, μ), where $\mu \in V \otimes (V^\vee)^{\otimes 2}$ is a tensor of type $(1,2)$, which may also be regarded as an element of $\mathrm{Hom}_K(V \otimes V, V)$. This amounts to classifying V together with a bilinear "multiplication" $V \times V \to V$. If the operation is required to be associative and to have a unit (the latter may be regarded as a tensor of type $(1,0)$), these conditions make V into a K-algebra. Other conditions may be imposed on μ in order to study structures such as Lie algebras and Jordan algebras.

More generally, (7.11.1) can be extended in a natural way to treat the classification of a family of vector subspaces, or at least lines, in $T^{p,q}V$. We shall not pursue this because of space constraints.

Definition 7.11.2 For a finite-dimensional K-vector space V, write $\mathrm{GL}(V)$ for its group of linear automorphisms. For data $\mathbf{t}$ as in (7.11.1), define the subgroup of $\mathrm{GL}(V)$

$$G(\mathbf{t}) := \{g \in \mathrm{GL}(V) : g\mathbf{t} = \mathbf{t}\}.$$

Choose a basis for V and write the conditions $gt_i = t_i$ in coordinates for every $i \in I$. Thus $G(\mathbf{t})$ is cut out inside $\mathrm{GL}(V)$ by a family of polynomial equations. In the examples above, if $\mathbf{t}$ corresponds to a nondegenerate symmetric (respectively, antisymmetric) bilinear form $b : V \times V \to K$, then the corresponding $G(\mathbf{t})$ is the orthogonal group $\mathrm{O}(V, b)$ (respectively, the symplectic group $\mathrm{Sp}(V, b)$). If $\mathbf{t}$ is the tensor of type $(0, \dim V)$ given by the determinant, then the corresponding group is the special linear group $\mathrm{SL}(V) := \{g : \det g = 1\}$.

Definition 7.11.3 For every field extension $L|K$ and K-vector space V, write $V_L := L \underset{K}{\otimes} V$, regarded as an L-vector space. Through the canonical isomorphism of L-vector spaces

$$(T^{p,q}V)_L \simeq T^{p,q}(V_L),$$

every datum $\mathbf{t}$ on V of the form (7.11.1) induces a datum $\mathbf{t}_L$ on V_L, defined over the extension field L.

Field extension does not change the numerical part of the data, $(\dim V, I, (p_i, q_i)_i)$.

Recall that when $L|K$ is a Galois extension, $\mathrm{Gal}(L|K)$ acts smoothly and semilinearly on V_L (Lemma 7.10.6); we write the action on the left. In fact, it acts smoothly and semilinearly on every $T^{p,q}(V_L)$ (Definition 7.10.4), "diagonally" on tensor products and by $\check{v} \mapsto \check{v}\sigma^{-1}$ on the dual space $V^\vee$ (the contragredient action, or "inverse step"). Relative to the isomorphism above, its invariant subspace is

$$T^{p,q}(V_L)^{\mathrm{Gal}(L|K)} = T^{p,q}V.$$

This is precisely Galois descent at the level of objects. At the level of morphisms, it is expressed by the following result.

Lemma 7.11.4 Let V and W be finite-dimensional K-vector spaces. For every $h \in \mathrm{Hom}_L(V_L, W_L)$ and $\sigma \in \mathrm{Gal}(L|K)$, define

$${}^\sigma h := \sigma h \sigma^{-1} : V_L \to W_L.$$

This map is again L-linear. The formula gives a smooth semilinear action of $\mathrm{Gal}(L|K)$ on the L-vector space $\mathrm{Hom}_L(V_L, W_L)$. This action is compatible with composition of linear maps, and

$$\mathrm{Hom}_L(V_L, W_L)^{\mathrm{Gal}(L|K)} = \mathrm{Hom}_K(V, W).$$

Here $\mathrm{Hom}_K(V, W)$ is embedded in $\mathrm{Hom}_L(V_L, W_L)$ by $f \mapsto f_L := \mathrm{id}_L \otimes f$.

Proof The reader may check directly that ${}^\sigma h$ is indeed L-linear. For smoothness, choose bases of V and W and identify the elements of $\mathrm{Hom}_L(V_L, W_L)$ with matrices. Then $\mathrm{Gal}(L|K)$ acts in the standard way on every matrix entry. Since h involves only finitely many matrix entries, they are all contained in some finite subextension $E|K$ of $L|K$. Hence the open subgroup $\mathrm{Gal}(L|E)$ fixes h. The remaining assertions about the action $(\sigma, h) \mapsto {}^\sigma h$ are readily verified.

The equality $\mathrm{Hom}_L(V_L, W_L)^{\mathrm{Gal}(L|K)} = \mathrm{Hom}_K(V, W)$ follows directly from Galois descent and is also easy to see directly from matrices. □

Taking the special case $V = W$, we see that $\mathrm{Gal}(L|K)$ acts on $\mathrm{GL}(V_L)$ by $(\sigma, g) \mapsto {}^\sigma g$, and that $\mathrm{GL}(V_L)^{\mathrm{Gal}(L|K)} = \mathrm{GL}(V)$.

Lemma 7.11.5 For data $\mathbf{t}$ of the form (7.11.1), the smooth action of $\mathrm{Gal}(L|K)$ on $\mathrm{GL}(V_L)$ preserves $G(\mathbf{t}_L)$. Thus $G(\mathbf{t}_L)$ becomes a smooth $\mathrm{Gal}(L|K)$-group in the sense of Remark 6.13.11; its invariant subgroup satisfies

$$G(\mathbf{t}_L)^{\mathrm{Gal}(L|K)} = G(\mathbf{t}).$$

Proof Let $g \in G(\mathbf{t}_L)$ and $\sigma \in \mathrm{Gal}(L|K)$. Then

$$({}^\sigma g)\mathbf{t}_L = ({}^\sigma g)(\sigma \mathbf{t}_L) = \sigma g \sigma^{-1}(\sigma \mathbf{t}_L) = \sigma(g\mathbf{t}_L) = \sigma(\mathbf{t}_L) = \mathbf{t}_L.$$

Thus the action of $\mathrm{Gal}(L|K)$ preserves $G(\mathbf{t}_L)$. The remaining assertions follow immediately. □

In practice, the classification of data (7.11.1) usually becomes simpler after extending scalars to a sufficiently large field L. For example, the isomorphism class of a nondegenerate quadratic form over an algebraically closed field is completely determined by its dimension, whereas the situation is more complicated over a general field such as $\mathbb{R}$, $\mathbb{Q}_p$, or $\mathbb{Q}$.

Conditions imposed in the classification problems above, such as nondegeneracy or associativity of multiplication, can usually be checked after any field extension. The problem therefore splits into two parts:

1. Over a sufficiently large field L, classify data of the form (7.11.1) satisfying the desired conditions; a typical choice is to take L to be a separably closed field.

2. Choose data $\mathbf{t}_0$ defined over K, take a sufficiently large Galois extension $L|K$, and study the data $\mathbf{t}$ satisfying $\mathbf{t}_L \simeq \mathbf{t}_{0,L}$.

The first part may be called geometric, while the second is arithmetic. This section focuses on the arithmetic part, which motivates the following definition.

Definition 7.11.6 Let $\mathbf{t}_0$ be data of the form (7.11.1) defined over the field K, and let $L|K$ be a Galois extension. Set

$$\mathcal{T}(\mathbf{t}_0) := \left\{ \begin{array}{r} \mathbf{t}: \text{ data of the form (7.11.1)} \\ \text{defined over the field } K \end{array} \middle| \text{ there exists an isomorphism } h : \mathbf{t}_{0,L} \xrightarrow{\sim} \mathbf{t}_L \right\}.$$

Also write $\mathscr{T}(\mathbf{t}_0) := \mathcal{T}(\mathbf{t}_0)/\simeq$.

This definition requires only the existence of an isomorphism h, without specifying a choice; any two choices differ by right multiplication by an element of $G(\mathbf{t}_{0,L})$. Write

$$\Gamma := \mathrm{Gal}(L|K), \qquad \text{equipped with the Krull topology.}$$

If $\sigma \in \Gamma$ and $h : \mathbf{t}_{0,L} \xrightarrow{\sim} \mathbf{t}_L$ (that is, $h(\mathbf{t}_{0,L}) = \mathbf{t}_L$), then

$${}^{\sigma}h(\mathbf{t}_{0,L}) = {}^{\sigma}h(\sigma\mathbf{t}_{0,L}) = \sigma(h(\mathbf{t}_{0,L})) = \sigma(\mathbf{t}_L) = \mathbf{t}_L.$$

Consequently, there is a unique map $c : \Gamma \to G(\mathbf{t}_{0,L})$ such that

$${}^{\sigma}h = hc(\sigma), \qquad \sigma \in \mathrm{Gal}(L|K).$$

From ${}^{\sigma\tau}h = {}^{\sigma}(hc(\tau)) = {}^{\sigma}h \cdot {}^{\sigma}c(\tau) = h \cdot c(\sigma) \cdot {}^{\sigma}c(\tau)$, we obtain

$$c(\sigma\tau) = c(\sigma) \cdot {}^{\sigma}c(\tau), \qquad \sigma, \tau \in \Gamma.$$

Without loss of generality, suppose that $\mathbf{t}_0$ and $\mathbf{t}$ are realized on spaces V and W, respectively, so that $h \in \mathrm{Hom}_L(V_L, W_L)$. Since the action of Γ is smooth, there is an open subgroup $\Gamma_0 \subset \Gamma$ that fixes h; consequently, c factors through Γ/Γ_0.

If the choice of h is changed to $h' = ha$, where $a \in G(\mathbf{t}_{0,L})$, then the corresponding c' satisfies

$$c'(\sigma) = (h')^{-1} \cdot {}^{\sigma}h' = a^{-1} \cdot h^{-1} \cdot {}^{\sigma}h \cdot {}^{\sigma}a = a^{-1} \cdot c(\sigma) \cdot {}^{\sigma}a.$$

Comparing these observations one by one with Definition 6.13.4 of nonabelian H^1, and using Remark 6.13.11 for the profinite version, we obtain a map

$$\begin{aligned} \mathcal{T}(\mathbf{t}_0) &\longrightarrow \mathrm{H}^1\left(\Gamma, G(\mathbf{t}_{0,L})\right), \\ \mathbf{t} &\longmapsto [c] := \text{the equivalence class of } c. \end{aligned} \tag{7.11.2}$$

This map is canonical and independent of all auxiliary choices.

Recall also that $\mathrm{H}^1\left(\Gamma, G(\mathbf{t}_{0,L})\right)$ is generally not a group, but a pointed set whose base point is represented by the constant map $\Gamma \to G(\mathbf{t}_{0,L})$ with value the identity automorphism 1. On the other hand, $\mathcal{T}(\mathbf{t}_0)$ has the obvious base point $\mathbf{t}_0$, as does its quotient $\mathscr{T}(\mathbf{t}_0)$. The map (7.11.2) clearly preserves base points, as one sees by taking $h = \mathrm{id}$.

Theorem 7.11.7 For data $\mathbf{t}_0$ and a Galois extension $L|K$ as above, with $\Gamma := \mathrm{Gal}(L|K)$, the map (7.11.2) induces a base-point-preserving bijection

$$\mathscr{T}(\mathbf{t}_0) \xrightarrow{1:1} \mathrm{H}^1\left(\Gamma, G(\mathbf{t}_{0,L})\right).$$

Proof First we show that (7.11.2) factors through $\mathscr{T}(\mathbf{t}_0)$. Let $f : \mathbf{t} \xrightarrow{\sim} \mathbf{t}'$, so that $f_L : \mathbf{t}_L \xrightarrow{\sim} \mathbf{t}'_L$. From $h : \mathbf{t}_{0,L} \xrightarrow{\sim} \mathbf{t}_L$ we obtain $f_L h : \mathbf{t}_{0,L} \xrightarrow{\sim} \mathbf{t}'_L$. Since ${}^{\sigma}f_L = f_L$, we have $(f_L h)^{-1} \cdot {}^{\sigma}(f_L h) = h^{-1} \cdot {}^{\sigma}h$, and hence the corresponding c is unchanged.

Next we show that $\mathscr{T}(\mathbf{t}_0) \to \mathrm{H}^1\left(\Gamma, G(\mathbf{t}_{0,L})\right)$ is injective. Suppose that data $\mathbf{t}$ and $\mathbf{t}'$ and isomorphisms

$$\mathbf{t}'_L \underset{\sim}{\xleftarrow{h'}} \mathbf{t}_{0,L} \underset{\sim}{\xrightarrow{h}} \mathbf{t}_L,$$

are given, and that there is an $a \in G(\mathbf{t}_{0,L})$ satisfying

$$(h')^{-1} \cdot {}^{\sigma}h' = a^{-1} \cdot h^{-1} \cdot {}^{\sigma}h \cdot {}^{\sigma}a, \qquad \sigma \in \Gamma.$$

Consequently,

$$ha(h')^{-1} = {}^{\sigma}h \cdot {}^{\sigma}a \cdot {}^{\sigma}(h')^{-1}, \qquad \sigma \in \Gamma.$$

In other words, $ha(h')^{-1} : \mathbf{t}'_L \xrightarrow{\sim} \mathbf{t}_L$ is invariant under the Γ-action and therefore descends to an isomorphism $\mathbf{t}' \xrightarrow{\sim} \mathbf{t}$.

Finally, we show that $\mathscr{T}(\mathbf{t}_0) \to \mathrm{H}^1\left(\Gamma, G(\mathbf{t}_{0,L})\right)$ is surjective. Take $c \in Z^1(\Gamma, G(\mathbf{t}_{0,L}))$. Since $c(\sigma) \in \mathrm{GL}(V_L)$, we may use it to define a new Γ-action, denoted by $\odot$, on V_L:

$$\sigma \odot v := c(\sigma)(\sigma v), \qquad \sigma \in \Gamma, \quad v \in V_L.$$

From $c(\mathrm{id}) = 1$ and

$$\sigma \odot (\tau \odot v) = c(\sigma)\sigma(c(\tau)(\tau v)) = c(\sigma)\big({}^{\sigma}c(\tau)(\sigma\tau v)\big) = (\sigma\tau) \odot v,$$

it follows that $\odot$ is indeed a Γ-action, while the smoothness of c ensures that $\odot$ is still a smooth semilinear action. Define the K-vector space $W := (V_L)^{\Gamma,\ \odot\text{-action}}$. By Galois descent, we obtain an isomorphism of L-vector spaces that preserves the semilinear Γ-actions,

$$h : (V_L,\ \odot\text{-action}) \xrightarrow{\sim} (W_L,\ \text{natural action}),$$
$$w \in W \implies h(w) = 1 \otimes w.$$

On the other hand, Γ also acts semilinearly through $\odot$ on each $T^{p,q}(V_L)$ (see the discussion following Definition 7.11.3), and because $c(\sigma) \in G(\mathbf{t}_{0,L})$, we have

$$\sigma \odot \mathbf{t}_{0,L} = c(\sigma)(\sigma\mathbf{t}_{0,L}) = c(\sigma)\mathbf{t}_{0,L} = \mathbf{t}_{0,L}.$$

Thus, relative to $\odot$, the datum $\mathbf{t}_{0,L}$ descends to a datum $\mathbf{t}$ on W, characterized by $h(\mathbf{t}_{0,L}) = \mathbf{t}_L$.

It follows that $\mathbf{t} \in \mathcal{T}(\mathbf{t}_0)$. The corresponding isomorphism h satisfies $\sigma(h(v)) = h(\sigma \odot v) = (hc(\sigma)\sigma)(v)$, that is, the identity $\sigma h = hc(\sigma)\sigma$. Rewriting this as $h^{-1} \cdot {}^{\sigma}h = c(\sigma)$ shows that $\mathbf{t}$ maps to $[c]$ under (7.11.2). This proves the assertion. $\square$

The simplest and most direct application is the following. It follows from the general fact that every finite-dimensional vector space has a basis, so that dimension determines the isomorphism class of a finite-dimensional vector space.

Theorem 7.11.8 (Hilbert's Theorem 90: the GL version) Let V be a finite-dimensional K-vector space and let $L|K$ be a Galois extension. Thus $\Gamma := \mathrm{Gal}(L|K)$ acts smoothly on the group $\mathrm{GL}(V_L)$. Then

$$\mathrm{H}^1(\Gamma, \mathrm{GL}(V_L)) = 1.$$

Proof Apply Theorem 7.11.7, but take data $\mathbf{t}_0$ with $I = \varnothing$. In other words, the data specify only a finite-dimensional K-vector space V, with no tensors. In this case, the elements of $\mathscr{T}(\mathbf{t}_0)$ correspond bijectively to the isomorphism classes of K-vector spaces W satisfying $W_L \simeq V_L$, equivalently $\dim W = \dim V$. Clearly, there is only one such isomorphism class. $\square$

Similarly, over a field of characteristic $\neq 2$, the general fact that every symplectic space (V, ω) has a symplectic basis leads to the following result.

Theorem 7.11.9 Let $\mathrm{char}(K) \neq 2$, let V be a K-vector space, and let $\omega : V \times V \to K$ be a nondegenerate antisymmetric bilinear form (also called a **symplectic form**, while the datum (V, ω) is called a **symplectic space**). The corresponding symplectic group is defined by

$$\begin{aligned}\mathrm{Sp}(V) &= \mathrm{Sp}(V, \omega) \\ &:= \{g \in \mathrm{GL}(V) : \forall v_1, v_2 \in V,\ \omega(gv_1, gv_2) = \omega(v_1, v_2)\}.\end{aligned}$$

Let $L|K$ be a Galois extension. Define the group $\mathrm{Sp}(V_L)$ and the action of $\Gamma := \mathrm{Gal}(L|K)$ on it similarly; this action is smooth. Then

$$\mathrm{H}^1(\Gamma, \mathrm{Sp}(V_L)) = 1.$$

Proof The data $\mathbf{t}_0$ under consideration consist of the space V and the tensor ω of type $(0, 2)$. The nondegeneracy and antisymmetry of a bilinear form can be checked after any field extension. Therefore the elements of $\mathscr{T}(\mathbf{t}_0)$ correspond bijectively to the isomorphism classes of symplectic spaces (W, η) over K satisfying $(W_L, \eta_L) \simeq (V_L, \omega_L)$. By the structure theorem for symplectic spaces—namely, that their dimension determines their isomorphism class—such data (W, η) form a single isomorphism class. $\square$

Example 7.11.10 Take $\dim V = 1$ in Theorem 7.11.8. Since $\mathrm{GL}(V_L) \simeq L^\times$ with the standard Γ-action, we obtain

$$\mathrm{H}^1(\Gamma, L^\times) = 0.$$

Since $L^\times$ is abelian, additive notation is used in this equation. If $L|K$ is finite, it can be written in terms of the Tate cohomology of Definition 6.11.1 as $\hat{\mathrm{H}}^1(\Gamma, L^\times) = 0$.

If $L|K$ is further assumed to be cyclic, the periodicity stated in Example 6.11.7 gives

$$\hat{\mathrm{H}}^{-1}(\Gamma, L^\times) \simeq \hat{\mathrm{H}}^1(\Gamma, L^\times) = 0.$$

Thus we naturally recover the multiplicative version of Hilbert's Theorem 90; see [51, Theorem 9.6.9].

In general, directly computing H^1 in order to classify mathematical objects may be rather difficult. Its value lies in the new perspective that it provides, as well as in the availability of cohomological tools such as the long exact sequence (Theorem 6.13.7) for studying classification problems.

Exercises

1. Complete the argument omitted from Definition 7.1.4, and prove that for an algebra A over a symmetric monoidal category $\mathcal{V}$, a left A-module is the same thing as a right A^{op}-module.

2. Let $\mathcal{V}$ be a braided monoidal category and let (A, μ_A, η_A) be an algebra over it. Prove that if there are morphisms $\mu' : A \otimes A \to A$ and $\eta' : \mathbf{1} \to A$ in $\mathsf{Alg}(\mathcal{V})$ that make A into an algebra over $\mathsf{Alg}(\mathcal{V})$, then necessarily $\eta' = \eta_A$ and $\mu' = \mu_A$. In particular, Lemma 7.1.7 then shows that A is commutative. Deduce (7.1.1).

Hint〉 It is easy to show that $\eta' = \eta_A$. For $\mu' = \mu_A$, write $\mu_{A \otimes A}$ for the multiplication on $A \otimes A$. The assumption gives the commutative diagram

$$\begin{tikzcd}
(A \otimes A) \otimes (A \otimes A) \arrow[r, "\mu' \otimes \mu'"] \arrow[d, "\mu_{A \otimes A}"'] & A \otimes A \arrow[d, "\mu_A"] \\
A \otimes A \arrow[r, "\mu'"'] & A.
\end{tikzcd}$$

If the four copies of A are placed in a 2×2 array, this commutative diagram can be visualized as the interchange law between horizontal and vertical multiplication:

$$\begin{array}{|cc|}\hline A & A \\ \hline A & A \\ \hline \end{array} = \begin{array}{|c|c|}\hline A & A \\ A & A \\ \hline \end{array} \qquad \begin{array}{rl} \text{Horizontal:} & \text{multiply using } \mu' \\[1em] \text{Vertical:} & \text{multiply using } \mu_A \end{array}$$

The argument for $\mu' = \mu_A$ can now be pictured as

$$\begin{array}{|cc|}\hline A & A \\ \hline \end{array} = \begin{array}{|c|c|}\hline A & \mathbf{1} \\ \mathbf{1} & A \\ \hline \end{array} = \begin{array}{|cc|}\hline A & \mathbf{1} \\ \hline \mathbf{1} & A \\ \hline \end{array} = \begin{array}{|c|}\hline A \\ A \\ \hline \end{array}$$

Try also to translate this picture into the corresponding commutative diagram or equation.

3. Consider a dg-algebra A over $\mathcal{A}$ (Definition 7.2.1). Verify that when $A = \mathbf{1}$ (the unit object), both left and right A-modules reduce to complexes over $\mathcal{A}$.

4. Fix a commutative ring $\Bbbk$. Let A be a dg-algebra over $\Bbbk$-Mod and let M be a complex. Prove that giving M the structure of a left A-module is equivalent to specifying a homomorphism of dg-algebras $A \to \mathrm{End}^\bullet(M) := \mathrm{Hom}^\bullet(M, M)$.

5. Let $\Bbbk$ be a commutative ring and R a dg-algebra over $\Bbbk$-Mod. For a right R-module X and a left R-module Y, identify R and X, Y, respectively, with a $\Bbbk$-algebra and $\Bbbk$-modules equipped with direct-sum decompositions

$$R = \bigoplus_n R^n, \quad X = \bigoplus_n X^n, \quad Y = \bigoplus_n Y^n$$

and degree-1 endomorphisms d satisfying $d^2 = 0$. Thus $X \underset{R}{\otimes} Y$ is already defined as a $\Bbbk$-module.

(i) Recall that $X \otimes Y = \bigoplus_n (\bigoplus_{p+q=n} X^p \otimes Y^q)$ is both a $\Bbbk$-module and a complex. For the evident $\Bbbk$-module homomorphism $\pi : X \otimes Y \twoheadrightarrow X \underset{R}{\otimes} Y$, verify that $\ker(\pi)$ is a subcomplex, so that $X \underset{R}{\otimes} Y$ has a canonical complex structure.

(ii) Prove that if X is an (A,R)-bimodule and Y is an (R,B)-bimodule, where A and B are also dg-algebras, then $X \underset{R}{\otimes} Y$ is naturally an (A,B)-bimodule. Use this to extend the adjunction of Proposition 4.12.6 to dg-modules. Here $\mathrm{Hom}^\bullet(Y_B, Z_B)$ and $\mathrm{Hom}^\bullet({}_AX, {}_AZ)$ are defined as complexes as in Example 7.2.6, but the signs must be handled carefully when assigning the (A,R)- and (R,B)-bimodule structures.

6. Fix a dg-algebra A over $\mathcal{A}$. It is called **non-positive** if $p > 0 \implies A^p = 0$. For a left (or right) A-module M and $n \in \mathbb{Z}$, take the truncation subcomplex $\tau^{\leq n}M$ as in Definition 3.9.3. Prove that if A is non-positive, then:

(i) every $d_M^p : M^p \to M^{p+1}$ is A^0-linear;

(ii) there is a short exact sequence of A-modules $0 \to \tau^{\leq n}M \to M \to \tilde{\tau}^{\geq n+1}M \to 0$.

7. Fix a dg-algebra A over $\mathcal{A}$. Recall that all left A-modules form the dg-category $A\text{-}\mathsf{dgMod}$ (Example 7.2.6); Remark 7.2.8 then gives the homotopy category $\mathrm{h}(A\text{-}\mathsf{dgMod})$. A left A-module M is called acyclic if it is acyclic as a complex; a morphism of A-modules $f : M \to N$ is called a quasi-isomorphism if it is a quasi-isomorphism of complexes.

(i) Let M be a left A-module whose structure is determined by a family of morphisms in $\mathcal{A}$, $\alpha_{A,M}^{p,q} : A^p \otimes M^q \to M^{p+q}$ $(p,q \in \mathbb{Z})$. Show that the formula

$$\alpha_{A,M[1]}^{p,q} := (-1)^p \alpha_{A,M}^{p,q+1}.$$

gives the complex $M[1]$ a left A-module structure. Then show that $M \mapsto M[1]$ defines a dg-functor from $A\text{-}\mathsf{dgMod}$ to itself.

(ii) For every morphism $f : M \to N$ in $A\text{-}\mathsf{dgMod}$, show that the mapping cone $\mathrm{Cone}(f)$ has a canonical left A-module structure, given in the notation above by

$$\alpha_{A,\mathrm{Cone}(f)}^{p,q} = \alpha_{A,M[1]}^{p,q} \oplus \alpha_{A,N}^{p,q}.$$

(iii) Following the construction of §4.4, use mapping cones to make $\mathrm{h}(A\text{-}\mathsf{dgMod})$ a triangulated category. Next perform Verdier localization at the acyclic A-modules, or equivalently adjoin inverses to the quasi-isomorphisms, to obtain the derived category $\mathsf{D}(A\text{-}\mathsf{dgMod})$. Define the corresponding $\mathsf{D}^\star(A\text{-}\mathsf{dgMod})$ for $\star \in \{+,-,\mathrm{b}\}$.

(iv) Assuming that A is non-positive, extend Proposition 4.4.9 to $\mathsf{D}(A\text{-}\mathsf{dgMod})$ and $\mathsf{D}^\star(A\text{-}\mathsf{dgMod})$ for $\star \in \{+,-,\mathrm{b}\}$. Hint Use the truncation functors.

(v) Following §3.15, define K-injective and K-projective A-modules and use them to describe the morphisms in $\mathsf{D}(A\text{-}\mathsf{dgMod})$. Place this discussion in the framework of §4.6 to study their relation to derived functors.

(vi) For a right A-module M, show that the complex $M[1]$ has the right A-module structure $\alpha_{M[1],A}^{p,q} = \alpha_{M,A}^{p,q+1}$ and that all the preceding properties remain valid.

8. Let $\mathcal{C}$ be a dg-category over $\Bbbk$. Define $\mathcal{C}^{\mathrm{op}}$ by $\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{C})$ and, for all $X, Y \in \mathrm{Ob}(\mathcal{C}^{\mathrm{op}})$, $\mathrm{Hom}^{\bullet}_{\mathcal{C}^{\mathrm{op}}}(X,Y) := \mathrm{Hom}^{\bullet}_{\mathcal{C}}(Y,X)$. Define composition by

$$\begin{aligned}\mathrm{Hom}^{p}_{\mathcal{C}^{\mathrm{op}}}(Y,Z) \underset{\Bbbk}{\otimes} \mathrm{Hom}^{q}_{\mathcal{C}^{\mathrm{op}}}(X,Y) &\to \mathrm{Hom}^{p+q}_{\mathcal{C}^{\mathrm{op}}}(X,Z) \quad p,q \in \mathbb{Z}\\ f \otimes g &\mapsto (-1)^{pq} \underbrace{f \circ g}_{\text{composition in } \mathrm{Hom}^{\bullet}_{\mathcal{C}}}.\end{aligned}$$

Verify that this indeed defines a dg-category $\mathcal{C}^{\mathrm{op}}$, called the opposite dg-category of $\mathcal{C}$.

9. Fix a commutative ring $\Bbbk$, a dg-algebra R over $\Bbbk$-Mod, a right R-module X, and a left R-module Y. Prove that the previously introduced functors

$$X \underset{R}{\otimes} (\cdot) : R\text{-}\mathsf{dgMod} \to \mathsf{C}(\Bbbk), \quad (\cdot) \underset{R}{\otimes} Y : \mathsf{dgMod}\text{-}R \to \mathsf{C}(\Bbbk),$$

upgrade naturally to dg-functors. Outline the corresponding statements when X or Y also has a bimodule structure.

Hint〉 The key is to define the morphisms between the Hom complexes. For $X \underset{R}{\otimes} (\cdot)$, let M and N be left R-modules. The Koszul sign rule requires the image of $f \in \mathrm{Hom}^{n}_{R}(M,N)$, namely $\mathrm{id}_X \otimes f \in \mathrm{Hom}^{n}(X \underset{R}{\otimes} M, X \underset{R}{\otimes} N)$, to be defined by

$$(\mathrm{id}_X \otimes f)(x \otimes m) = (-1)^{pn} x \otimes f(m), \quad x \in X^p,\ m \in M^q.$$

One must verify that $\mathrm{id}_X \otimes d_{\mathrm{Hom}^{\bullet}}(f) = d_{\mathrm{Hom}^{\bullet}}(\mathrm{id}_X \otimes f)$.

The case $(\cdot) \underset{R}{\otimes} Y$ is simpler: take $(f \otimes \mathrm{id}_Y)(m \otimes y) = f(m) \otimes y$.

10. Let $\mathcal{C}$ be a dg-category over a commutative ring $\Bbbk$ and let $M \in \mathrm{Ob}(\mathcal{C})$.

(i) Upgrade $\mathrm{Hom}^{\bullet}(M, \cdot) : \mathcal{C} \to \mathsf{C}(\Bbbk)$ to a dg-functor.

Hint〉 Let $X, Y \in \mathrm{Ob}(\mathcal{C})$ and $f \in \mathrm{Hom}^{n}(X,Y)$. Set $\mathcal{H}_X = \mathrm{Hom}^{\bullet}(M,X)$. The image of f should be

$$\left[f_* : g \xmapsto{\text{morphism of degree } n} fg \right] \in \mathrm{Hom}^{n}(\mathcal{H}_X, \mathcal{H}_Y).$$

Use the definition of the Hom complex to verify that $(d_{\mathrm{Hom}^{\bullet}(X,Y)} f)_* = d_{\mathrm{Hom}^{\bullet}(\mathcal{H}_X,\mathcal{H}_Y)}(f_*)$.

(ii) Upgrade $\mathrm{Hom}^{\bullet}(\cdot, M) : \mathcal{C} \to \mathsf{C}(\Bbbk)^{\mathrm{op}}$ to a dg-functor.

Hint〉 Set $\mathcal{K}_X = \mathrm{Hom}^{\bullet}(X,M)$. The Koszul sign rule requires the image of $f \in \mathrm{Hom}^{n}(X,Y)$ to be

$$\left[\begin{array}{c} f^* : g \xmapsto{\text{morphism of degree } n} (-1)^{np} gf \\ \forall p \in \mathbb{Z},\ \forall\, g \in \mathrm{Hom}^{p}(Y,M) \end{array} \right] \in \mathrm{Hom}^{n}(\mathcal{K}_Y, \mathcal{K}_X).$$

Verify carefully that

- ◇ $(d_{\mathrm{Hom}^{\bullet}(X,Y)} f)^* = d_{\mathrm{Hom}^{\bullet}(\mathcal{K}_Y,\mathcal{K}_X)}(f^*)$;
- ◇ $(fh)^*$ equals the composite of h^* and f^* in $\mathrm{Hom}^{\bullet}_{\mathsf{C}(\Bbbk)}$.

11. Prove that the canonical morphisms $[Y,Z]\otimes[X,Y]\to[X,Z]$ and $\mathbf{1}\to[X,X]$ defined for a closed monoidal category in §7.3 satisfy associativity and unit laws. Thus a closed monoidal category is enriched over itself with respect to its internal Hom. Hint Verify this directly, or consult [24, Section 1.6].

12. Consider the polynomial algebra $\Bbbk[t]$ over a commutative ring $\Bbbk$. Define Δ, ϵ, and S, respectively, by

$$\Delta(t^n)=\sum_{k=0}^{n}\binom{n}{k}t^k\otimes t^{n-k},\quad \epsilon(t^n)=\begin{cases}1, & n=0\\ 0, & n\neq 0\end{cases},\quad S(t^n)=(-1)^n t^n,$$

where $n\in\mathbb{Z}_{\geq 0}$. Verify that these data define a commutative and cocommutative Hopf $\Bbbk$-algebra.

13. Let A be a Hopf algebra over a commutative ring. An ideal I of A is called a **Hopf ideal** if $\Delta(I)\subset I\otimes A+A\otimes I$, $\epsilon(I)=0$, and $S(I)\subset I$. Show that quotienting by a Hopf ideal gives a Hopf algebra A/I.

14. Let $\Bbbk$ be a commutative ring and V a $\Bbbk$-module. On the tensor algebra $T(V)$ (see [51, Definition 7.5.1]) one can define $\Bbbk$-algebra homomorphisms $\Delta: T(V)\to T(V)\otimes T(V)$ and $\epsilon: T(V)\to\Bbbk$ such that, for every $v\in V$,

$$\Delta(v)=v\otimes 1+1\otimes v,\quad \epsilon(v)=0;$$

here the algebra structure on $T(V)\otimes T(V)$ is defined by the trivial braiding $x\otimes y\mapsto y\otimes x$.

(i) Verify that this makes $T(V)$ a bialgebra.

(ii) Show how to define a $\Bbbk$-module automorphism $S: T(V)\to T(V)$ such that $S(v)=-v$ and S is a bialgebra antiautomorphism (Proposition 7.5.11); more explicitly,

$$S(v_1\cdots v_n)=(-1)^n v_n\cdots v_1,\quad n\in\mathbb{Z}_{\geq 0},\quad v_1,\ldots,v_n\in V.$$

Verify that this makes $T(V)$ a Hopf algebra.

(iii) Show that the symmetric algebra $\mathrm{Sym}(V)$ has the analogous properties; in fact it is a quotient of $T(V)$ by a Hopf ideal.

(iv) Treat the version for the exterior algebra $\bigwedge(V)$, but define the multiplication on $\bigwedge(V)\otimes\bigwedge(V)$ using the Koszul braiding (7.1.3) (take $\epsilon(a)=a \bmod 2$ there), so that homogeneous elements $x,y,z,w\in\bigwedge(V)$ satisfy $(x\otimes y)(z\otimes w)=(-1)^{\deg y\deg z}xz\otimes yw$. This ensures that $(1\otimes v+v\otimes 1)^2$ is the zero element of $\bigwedge(V)\otimes\bigwedge(V)$ for every $v\in V$.

15. (E. Taft) Let q be a primitive nth root of unity in a field $\Bbbk$, where $n\geq 2$. Consider the $\Bbbk$-algebra H defined by generators g, x, and relations

$$g^n=1,\quad x^n=0,\quad gxg^{-1}=qx.$$

Show that one can define $\Delta : H \to H \underset{\Bbbk}{\otimes} H$ and $\epsilon : H \to \Bbbk$ satisfying

$$\Delta(g) = g \otimes g, \quad \Delta(x) = 1 \otimes x + x \otimes g,$$
$$\epsilon(g) = 1, \quad \epsilon(x) = 0,$$

and making H a Hopf $\Bbbk$-algebra with antipode

$$S : H \to H, \quad S(g) = g^{-1}, \quad S(x) = -g^{-1}x.$$

Show that H is neither commutative nor cocommutative. When $n = 2$, the corresponding Hopf algebra is also called Sweedler's Hopf algebra.

16. Let (A, Δ, ϵ) be a coalgebra over a commutative ring $\Bbbk$. If $x \in A$ satisfies $\Delta(x) = 1 \otimes x + x \otimes 1$, then x is called **primitive**.

(i) Prove that $\epsilon(x) = 0$ for every primitive element x.

(ii) Let $(A, \mu, \eta, \Delta, \epsilon)$ be a bialgebra and write μ as multiplication, so $xy = \mu(x \otimes y)$. Prove that if $x, y \in A$ are primitive, then $xy - yx$ is primitive as well.

(iii) Still taking A to be a bialgebra, let $x_1, \ldots, x_n \in A$ be primitive elements and let V be the free $\Bbbk$-module with basis $x_1, \ldots, x_n$. This gives a $\Bbbk$-algebra homomorphism $\phi : T(V) \to A$. Prove that ϕ is also a bialgebra homomorphism.

17. Let (A, Δ, ϵ) be a coalgebra over a commutative ring $\Bbbk$, with $A \neq \{0\}$ and A torsion-free as a $\Bbbk$-module. A nonzero element $x \in A$ satisfying $\Delta(x) = x \otimes x$ is called **group-like**. Denote the set of all group-like elements by $\mathcal{G}(A)$.

(i) Prove that $x \in \mathcal{G}(A)$ implies $\epsilon(x) = 1$.

(ii) Prove that if $(A, \mu, \eta, \Delta, \epsilon)$ is a bialgebra, then $\mathcal{G}(A)$ is a monoid under the multiplication μ of A, with unit $1_A := \eta(1)$.

(iii) For a monoid M and the corresponding bialgebra $\Bbbk[M]$ (Example 7.5.5), prove that $\mathcal{G}(\Bbbk[M]) = M$.

(iv) Let A be a Hopf algebra with antipode S. Prove that $\mathcal{G}(A)$ is a group and that the inverse of $x \in \mathcal{G}(A)$ is $S(x)$.

18. Let (A, Δ, ϵ) be a nonzero coalgebra over a field $\Bbbk$. Prove that $\mathcal{G}(A)$ is a linearly independent subset of A.

Hint Suppose otherwise, and choose a shortest nontrivial linear relation among these elements. Without loss of generality, write $x_1 = \sum_{i=2}^{n} a_i x_i$, where $x_1, \ldots, x_n \in \mathcal{G}(A)$ are pairwise distinct and $a_i \in \Bbbk$. Then $x_2, \ldots, x_n$ are linearly independent, while $x_1 \otimes x_1 = \sum_{i=2}^{n} a_i x_i \otimes x_i$.

19. Write $\mathsf{TopCMon}$ for the category of commutative topological monoids, $\mathsf{Top}_\bullet$ for the category of pointed topological spaces, and $U : \mathsf{TopCMon} \to \mathsf{Top}_\bullet$ for the forgetful functor (the base point corresponds to the unit). Construct a left adjoint $\mathrm{Sym} : \mathsf{Top}_\bullet \to \mathsf{TopCMon}$ to U. For a pointed topological space (X, x), the object $\mathrm{Sym}(X, x)$ is also called the **infinite symmetric product** of (X, x); explain this terminology. The Dold–Thom theorem in algebraic topology is based on this construction.

20. Let the (A, B)-bimodule P be finitely generated projective as a right B-module, and suppose that $(\cdot) \underset{A}{\otimes} P$ is a faithful exact functor. Prove that a right A-module N is finitely presented (Example A.2.5) if and only if $N \underset{A}{\otimes} P$ is finitely presented as a right B-module. This complements Proposition 7.7.14.

21. Consider a commutative ring $\Bbbk$ and a homomorphism of $\Bbbk$-algebras $f : A \to B$. We have an adjoint pair

$$\mathcal{F}_{B|A} : \mathsf{Mod}\text{-}B \rightleftarrows \mathsf{Mod}\text{-}A : \mathrm{Hom}_A(B, \cdot),$$

where, for every right A-module N, $\mathrm{Hom}_A(B, N)$ is given the right B-module structure $(\phi b)(x) = \phi(bx)$. More precisely,

- ◇ the unit morphism $M \to \mathrm{Hom}_A(B, M)$ maps m to $[b \mapsto mb]$;
- ◇ the counit morphism $\mathrm{Hom}_A(B, N) \to N$ maps ϕ to $\phi(1)$.

See [51, Corollary 6.6.8]. As a complement to Example 7.7.8, determine explicitly the corresponding monad and comonad.

22. Let R be a commutative ring. Write R-Alg for the category of R-algebras. Consider the adjoint pair $T(\cdot) : R\text{-}\mathsf{Mod} \rightleftarrows R\text{-}\mathsf{Alg} : U$, where $T(\cdot)$ is the tensor-algebra functor and U is the forgetful functor.

(i) Prove directly that modules under the corresponding monad on R-Mod are R-algebras, and hence that the adjoint pair is monadic.

(ii) Replace $T(\cdot)$ by $\mathrm{Sym}(\cdot)$ and $\bigwedge(\cdot)$, then state and prove the corresponding results; see [51, §7.6].

23. Complete the argument of Proposition 7.7.13, and then prove the following stronger version. Let C be a coalgebra in (B, B)-Mod. Prove that:

(i) the forgetful functor $U : \mathsf{Comod}\text{-}C \to \mathsf{Mod}\text{-}B$ has a right adjoint;

Hint〉 Write Δ for the comultiplication of C and ϵ for its counit. For a right B-module N, make $N \underset{B}{\otimes} C$ a right C-comodule via $\mathrm{id}_N \otimes \Delta$. For every right C-comodule M, verify that

$$\begin{aligned} \mathrm{Hom}_{\mathsf{Mod}\text{-}B}(M, N) &\rightleftarrows \mathrm{Hom}_{\mathsf{Comod}\text{-}C}\left(M, N \underset{B}{\otimes} C\right) \\ \varphi &\longmapsto (\varphi \otimes \mathrm{id}_C)\rho \\ (\mathrm{id}_N \otimes \epsilon)\psi &\longleftarrow\!\shortmid \psi \end{aligned}$$

are mutually inverse, where $\rho : M \to M \underset{B}{\otimes} C$ specifies the comodule structure of M.

(ii) the forgetful functor U creates all small $\varinjlim$; hence $\mathsf{Comod}\text{-}C$ is cocomplete;

(iii) the category $\mathsf{Comod}\text{-}C$ has a cogenerator (Definition 1.11.8);

Hint〉 Take a cogenerator of the Grothendieck category $\mathsf{Mod}\text{-}B$, then take its image under the right adjoint of U.

(iv) if $\mathsf{Comod}\text{-}C$ is an abelian category and U is exact, then C is flat as a left B-module.

Hint〉 Let $g : M \to N$ be a monomorphism of right B-modules. Use the exactness of U and the fact that a right adjoint preserves kernels to show that $g \otimes \mathrm{id}_C : M \underset{B}{\otimes} C \to N \underset{B}{\otimes} C$ remains injective.

24. As in the preceding exercise, continue to let C be a coalgebra in (B, B)-Mod and let M be a right C-comodule. Prove that if a quotient B-module M'' of M (or a B-submodule M' of M) has a comodule structure making $M \twoheadrightarrow M''$ (or $M' \hookrightarrow M$) a comodule homomorphism, then that comodule structure is unique. In the submodule case, additionally assume that C is flat as a left B-module.

25. Following the explanation at the end of §7.10, give a direct proof of the Galois descent Theorem 7.10.7 by the following steps. Use the notation from that section: let $M \in \mathrm{Ob}(\mathsf{Vect}^{\Gamma,\infty}(L))$ and define the canonical map ι_M by (7.10.3).

(i) Let M' be a Γ-invariant subspace of $L \underset{K}{\otimes} (M^\Gamma)$, that is, $\Gamma M' \subset M'$. Prove that if $M' \cap (1 \otimes M^\Gamma) = \{0\}$, then $M' = \{0\}$.

Hint〉 Choose a basis $(y_i)_{i \in I}$ of M^Γ. If $M' \neq \{0\}$, choose in it a nonzero expression with the fewest possible terms, $\sum_{i \in I} \ell_i \otimes y_i$ (a finite sum). After a suitable normalization, it may be assumed to have the form

$$m' = 1 \otimes y_{i_0} + \ell_1 \otimes y_{i_1} + \cdots, \quad \ell_1 \notin K.$$

Choose $\sigma \in \Gamma$ such that $\sigma(\ell_1) \neq \ell_1$, and consider $m' - \sigma(m')$ to obtain a contradiction.

(ii) Prove that $\iota_M : L \underset{K}{\otimes} (M^\Gamma) \to M$ is injective. Hint〉 Consider $M' := \ker(\iota_M)$.

(iii) Prove that ι_M is surjective.

Hint〉 Consider a finite Galois subextension $E|K$ of $L|K$. Take $a_1, \ldots, a_n$ as a basis of E as a vector space over K, and choose representatives $\mathrm{id} = \sigma_1, \ldots, \sigma_n \in \Gamma$ for the elements of $\mathrm{Gal}(E|K)$. Field theory shows that $(\sigma_i(a_j))_{1 \leq i,j \leq n}$ is an invertible matrix over E; write its inverse as $(b_{ij})_{1 \leq i,j \leq n}$. For $m \in M^{\mathrm{Gal}(L|E)}$, derive

$$m = \sum_{i=1}^{n} b_{i1} \sum_{j=1}^{n} \sigma_j(a_i m) \in \mathrm{im}(\iota_M).$$

(iv) Use the preceding results to show that (7.10.4) is an adjoint equivalence, thus giving a direct proof of Theorem 7.10.7.

26. Let G be a group and $L|K$ a Galois extension of fields. Write G-Mod (or G-Mod_L) for the category of G-modules with coefficients in K (or L); see Definition 6.1.1. Use the Galois descent methods of §§7.10—7.11 to describe G-Mod in terms of objects of G-Mod_L equipped with an appropriate $\mathrm{Gal}(L|K)$-action.

27. Use the notation of §7.11. Let $c \in Z^1(\Gamma, G(\mathbf{t}_{0,L}))$. Recall from the proof of Theorem 7.11.7 that, through Galois descent, c defines a K-vector space W with data $\mathbf{t}$ on it, as well as an isomorphism $h : \mathbf{t}_{0,L} \xrightarrow{\sim} \mathbf{t}_L$.

(i) Show that $g \mapsto h^{-1}gh$ gives a group isomorphism $\nu : G(\mathbf{t}_L) \xrightarrow{\sim} G(\mathbf{t}_{0,L})$. Show that $f \xmapsto{\sigma\in\Gamma} c(\sigma)({}^\sigma f)c(\sigma)^{-1}$ gives a new Γ-action on $G(\mathbf{t}_{0,L})$. Denote this structure by ${}^cG(\mathbf{t}_{0,L})$, and show that $\nu : G(\mathbf{t}_L) \xrightarrow{\sim} {}^cG(\mathbf{t}_{0,L})$ is Γ-equivariant.

(ii) Show that the following formula gives a well-defined map:

$$\begin{aligned} Z^1(\Gamma, G(\mathbf{t}_L)) &\to Z^1(\Gamma, G(\mathbf{t}_{0,L})) \\ \tilde{c} &\mapsto [\sigma \mapsto \nu(\tilde{c}(\sigma))c(\sigma)]. \end{aligned}$$

It induces a bijection $\mathrm{H}^1(\Gamma, G(\mathbf{t}_L)) \to \mathrm{H}^1(\Gamma, G(\mathbf{t}_{0,L}))$ that maps the base point to $[c]$. Hint Compare the "twisting" construction in the exercises for Chapter 6.

(iii) Show that, under the correspondence in Theorem 7.11.7, this bijection corresponds to the equality $\mathscr{T}(\mathbf{t}) = \mathscr{T}(\mathbf{t}_0)$.

28. Let K be a field with $\mathrm{char}(K) \neq 2$. A nondegenerate symmetric bilinear form $q : V \times V \to K$ on a finite-dimensional K-vector space V will simply be called a nondegenerate quadratic form. In the orthogonal group $\mathrm{O}(q) = \mathrm{O}(V, q)$, write $\mathrm{SO}(q) = \mathrm{SO}(V, q)$ for the subgroup cut out by $\det = 1$. For every Galois extension $L|K$, there is a nondegenerate quadratic form (V_L, q_L) and the corresponding groups, all equipped with a smooth action of $\Gamma := \mathrm{Gal}(L|K)$.

(i) Write $\mu_2 := \{\pm 1\} \subset K^\times$. Show that $\mathrm{O}(q)/\,\mathrm{SO}(q) \simeq \mu_2$. For the version $\mathrm{O}(q_L)/\,\mathrm{SO}(q_L)$, explain the relation between this isomorphism and the Γ-action.

(ii) Using the tools of §7.11, show that the elements of $\mathrm{H}^1(\Gamma, \mathrm{O}(q_L))$ correspond bijectively to the isomorphism classes of nondegenerate quadratic forms (V', q') over K with $\dim V' = \dim V$, where the base point corresponds to (V, q).

(iii) Apply Theorem 6.13.7 to obtain the exact sequence of pointed sets

$$\mathrm{O}(q) \xrightarrow{\det} \mu_2 \to \mathrm{H}^1(\Gamma, \mathrm{SO}(q_L)) \xrightarrow{\varphi} \mathrm{H}^1(\Gamma, \mathrm{O}(q_L)) \xrightarrow{\psi} \mathrm{H}^1(\Gamma, \mu_2).$$

Try to show that φ is injective. Hint Surjectivity of det only shows that $\varphi^{-1}(\text{base point}) = \{\text{base point}\}$. To prove injectivity, one must vary (V, q) and apply the twisting technique from the preceding exercise.

(iv) Show that $\mathrm{H}^1(\Gamma, \mu_2) \simeq K^\times / K^{\times, 2}$. Also show that ψ is the map

$$\big[(V', q') : \text{ nondegenerate quadratic form}\big] \mapsto \mathrm{disc}(q')/\mathrm{disc}(q),$$

where one chooses any basis of V', identifies q' with a symmetric matrix, and defines

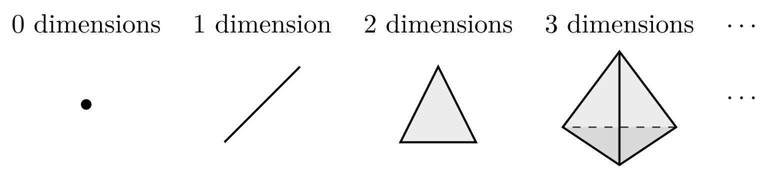
$$\mathrm{disc}(q') := (-1)^n \det(q') \bmod K^{\times 2}, \quad \dim V = 2n \text{ or } 2n+1.$$

(v) Show that the elements of $\mathrm{H}^1(\Gamma, \mathrm{SO}(q_L))$ correspond bijectively to the isomorphism classes of nondegenerate quadratic forms (V', q') over K satisfying $\dim V' = \dim V$ and $\mathrm{disc}(q') = \mathrm{disc}(q)$, with the base point corresponding to (V, q).

Chapter 8 Simplicial Methods

Let $\mathbf{\Delta}$ be the category of nonempty finite ordinals. By definition, a simplicial object in a category $\mathcal{C}$ is a functor $X : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{C}$. Equivalently, it can be described as a sequence of objects $X_0, X_1, \ldots$ together with morphisms $d_i : X_n \to X_{n-1}$ (faces) and $s_j : X_n \to X_{n+1}$ (degeneracies), for $0 \le i, j \le n$, satisfying the family of identities (8.1.2). Such functors form the category $\mathsf{s}\mathcal{C}$. In the special case $\mathcal{C} = \mathsf{Set}$, these objects are also called simplicial sets. Their basic theory is discussed in §§8.1–8.2. Write FinOrd for the category of finite ordinals. If the functor underlying a simplicial object X extends to a functor $\mathsf{FinOrd} \to \mathcal{C}$, then X is called an augmented simplicial object. This is equivalent to adding a term X_{-1} in degree -1 to the data $(X_n, d_i, s_j)_{n,i,j}$.

Simplicial sets originate in topology. The basic idea is to decompose a space as a gluing of simplices; the intuitive picture of simplices (a point, a line segment, a triangle, a tetrahedron, and so forth) is



Such a decomposition gives a concrete, combinatorial way to understand homology, homotopy, and other questions about spaces. The rigorous formulation used here was introduced by S. Eilenberg and J. A. Zilber in 1950. The language of simplicial sets has a broad reach and is not confined to classical topological questions. For example, it can describe a category through the construction called its "nerve", introduced in §8.3. Category theory concerns objects in diagrams (shown as zero-dimensional points), morphisms (shown as one-dimensional arrows), and commutativity (associated with two-dimensional pieces such as triangles or squares). It is therefore unsurprising that categories and spaces can be united within the same mathematical structure.

After introducing nerves and classifying spaces as basic examples of simplicial sets, §8.4 returns to the geometric intuition and defines the geometric realization $|X|$ of a simplicial set X. Theorem 8.4.7 shows that geometric realization and the singular-set functor Sing familiar from topology form an adjoint pair

$$|\cdot| : \mathsf{sSet} \rightleftarrows \mathsf{Top} : \mathrm{Sing}.$$

If Top is replaced by the category CGHaus of compactly generated Hausdorff spaces, customary in homotopy theory, geometric realization also satisfies $|X \times Y| \simeq |X| \times |Y|$. The product on the left is formed degreewise from the two simplicial sets. The proof is related to triangulating products and involves some nontrivial combinatorial details; since this is not a topology textbook, we shall not pursue them.

For the subject of this book, simplicial objects in an abelian category $\mathcal{A}$ are more important. In this situation, $\mathsf{s}\mathcal{A}$ is also an abelian category; for example, simplicial abelian groups form the abelian category sAb. The fundamental result is the Dold–Kan correspondence in §8.5 (Theorem 8.5.9), which, for an abelian category $\mathcal{A}$, gives an adjoint pair of equivalences

$$\Gamma : \mathsf{s}\mathcal{A} \rightleftarrows \mathsf{Ch}_{\geq 0}(\mathcal{A}) : \mathrm{N},$$

where $\mathsf{Ch}_{\geq 0}(\mathcal{A})$ denotes the abelian category of nonnegatively graded chain complexes over $\mathcal{A}$. The functor N maps a simplicial object X to the corresponding normalized chain complex $\mathrm{N}X$. More precisely, define the unnormalized chain complex of $X \in \mathrm{Ob}(\mathsf{s}\mathcal{A})$ by

$$\mathrm{C}X := (X_n, \partial_n)_{n\geq 0}, \quad \partial_n := \sum_{i=0}^{n} (-1)^i d_i : X_n \to X_{n-1}.$$

Then $\mathrm{N}X$ may be taken as the chain subcomplex of $\mathrm{C}X$ given by $(\mathrm{N}X)_n := \bigcap_{i=1}^{n} \ker(d_i)$, or regarded as the quotient of $\mathrm{C}X$ by its degenerate part. The isomorphism between these two forms is the content of Proposition 8.5.11. Moreover, Theorem 8.5.12 shows that the inclusion morphism $u : \mathrm{N}X \to \mathrm{C}X$ is a quasi-isomorphism of chain complexes.

The Dold–Kan correspondence makes simplicial theory one of the sources of homological algebra, not only historically but also conceptually and technically. Simplicial methods can explain and enrich the basic operations on complexes or chain complexes; for example, homotopy and contractibility have simplicial versions and thereby acquire topological interpretations. This is discussed in §8.6.

Another application connects simplicial methods with the theory of monads from §7.6. In this way, an object of a category can be canonically expanded into an augmented simplicial object or an augmented cosimplicial object. Because these techniques are related to the operation of "adding a bar" in Hochschild homology, they are collectively called bar constructions and form the subject of §8.7. Through the Dold–Kan correspondence and some observations about contractibility, bar constructions explain the standard resolutions in Hochschild theory (see §3.8) and group cohomology (see §6.1); they all turn out to arise from a single idea at a higher level. The Exercises contain another extension of the bar construction.

By definition, the bisimplicial objects discussed in §8.8 are simply functors $\mathbf{\Delta}^{\mathrm{op}} \times \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{C}$; they form the category $\mathsf{s}^2\mathcal{C}$. If $\mathcal{A}$ is an abelian category, the Eilenberg–Zilber Theorem 8.8.9, which directly concerns $\mathsf{s}^2\mathcal{A}$, is a fundamental construction in algebraic topology. From the viewpoint of pure category theory, if $\mathcal{A}$ is a monoidal category (for example, $\mathcal{A} = \mathsf{Ab}$), the theorem gives two canonical monoidal structures, in opposite directions, on the functor $\mathrm{N} : \mathsf{s}\mathcal{A} \to \mathsf{Ch}_{\geq 0}(\mathcal{A})$. The applications of bisimplicial objects are not limited to this, however.

The last two sections of this chapter, §8.9 and §8.10, discuss Hom objects and mapping cones for simplicial sets, respectively. On the one hand, they correspond to mapping spaces and cones in the topological intuition. On the other hand, in the case of abelian groups, the Dold–Kan correspondence gives the familiar Hom chain complexes and mapping cones. Thus the related constructions on complexes or chain complexes receive a transparent explanation, while for topologists all of this is already familiar.

阅读提示

Sections §§8.1–8.4 form the backbone of the simplicial methods in this chapter; §§8.5–8.7 are directly connected with chain complexes; and although §§8.8–8.10 accord with the contents of the preceding books, these sections are more advanced. Unlike the material of the other chapters, simplicial methods are not logically necessary. Their ideas and techniques are nevertheless part of a mathematician's basic toolkit and provide a foundation for entering higher algebra. Some arguments about simplicial objects occasionally involve lengthy routine verifications (as in §8.5); this is a consequence of the definitions, and readers may decide for themselves how far they wish to follow such details.

8.1 Simplicial Objects

In Remark 7.1.17, we recalled monotone maps between partially ordered sets and used them to define the category FinOrd of finite ordinals. This chapter focuses on its full subcategory $\mathbf{\Delta}$ consisting of the nonempty ordinals.

Definition 8.1.1 Let $\mathbf{\Delta}$ be the category of all nonempty finite ordinals and monotone maps between them. In other words, its objects are the totally ordered sets $[n] := \{0, \dots, n\}$ for $n \in \mathbb{Z}_{\geq 0}$, and its morphisms are monotone maps.

Note that $[n]$ should not be confused with $\mathbf{n}$, which in this book denotes the totally ordered set $\{0, \dots, n-1\}$ or the corresponding category.

Every morphism $[l] \to [n]$ in $\mathbf{\Delta}$ has a unique factorization as a surjective monotone map followed by an injective monotone map,

$$[l] \twoheadrightarrow [m] \hookrightarrow [n], \quad n \geq m \leq l.$$

An injective (respectively, surjective) map of this kind can be decomposed further into pieces that at each step omit exactly one element (respectively, identify exactly two elements). In other words, every morphism can be factored as a composite of the following two kinds of morphisms.

Coface	$\mathrm{d}^i = \mathrm{d}^{n,i} : [n-1] \hookrightarrow [n]$, where $0 \leq i \leq n$; the monotone injection omits only $i \in [n]$.
Codegeneracy	$\mathrm{s}^j = \mathrm{s}^{n,j} : [n+1] \twoheadrightarrow [n]$, where $0 \leq j \leq n$; the monotone surjection takes the value $j \in [n]$ twice.

An injective (respectively, surjective) monotone map can be decomposed into cofaces (respectively, codegeneracies) in several ways, depending on the order in which the elements are omitted (respectively, identified). The following **cosimplicial identities** are easy to verify:

$$\begin{aligned}
&\mathrm{d}^j\mathrm{d}^i = \mathrm{d}^i\mathrm{d}^{j-1}, && i < j,\\
&\mathrm{s}^j\mathrm{d}^i = \mathrm{d}^i\mathrm{s}^{j-1} && i < j,\\
&\mathrm{s}^j\mathrm{d}^j = \mathrm{id} = \mathrm{s}^j\mathrm{d}^{j+1}, && \forall j,\\
&\mathrm{s}^j\mathrm{d}^i = \mathrm{d}^{i-1}\mathrm{s}^j, && i > j+1,\\
&\mathrm{s}^j\mathrm{s}^i = \mathrm{s}^i\mathrm{s}^{j+1}, && i \leq j.
\end{aligned} \tag{8.1.1}$$

Examining how one passes between the various decompositions of an injective (respectively, surjective) monotone map shows that the cofaces, codegeneracies, and (8.1.1) give a complete system of generators and relations for the morphisms of $\mathbf{\Delta}$.

Viewed in $\mathbf{\Delta}^{\mathrm{op}}$, these give the corresponding morphisms

Face	$\mathrm{d}_i = \mathrm{d}_i^n : [n] \xrightarrow{\mathrm{op}} [n-1]$	$0 \le i \le n$
Degeneracy	$\mathrm{s}_j = \mathrm{s}_j^n : [n] \xrightarrow{\mathrm{op}} [n+1]$	$0 \le j \le n$

The symbol $\xrightarrow{\mathrm{op}}$ merely reminds us that the arrow lies in the opposite category $\mathbf{\Delta}^{\mathrm{op}}$. Reversing (8.1.1) gives the **simplicial identities**

$$\begin{aligned}
\mathrm{d}_i\mathrm{d}_j &= \mathrm{d}_{j-1}\mathrm{d}_i, && i < j\\
\mathrm{d}_i\mathrm{s}_j &= \mathrm{s}_{j-1}\mathrm{d}_i && i < j\\
\mathrm{d}_j\mathrm{s}_j &= \mathrm{id} = \mathrm{d}_{j+1}\mathrm{s}_j, && \forall j\\
\mathrm{d}_i\mathrm{s}_j &= \mathrm{s}_j\mathrm{d}_{i-1}, && i > j+1\\
\mathrm{s}_i\mathrm{s}_j &= \mathrm{s}_{j+1}\mathrm{s}_i, && i \le j.
\end{aligned} \tag{8.1.2}$$

Definition 8.1.2 For a category $\mathcal{C}$,

- a **simplicial object** in it means a functor $\mathbf{\Delta}^{\mathrm{op}} \to \mathcal{C}$; all simplicial objects form the category $\mathsf{s}\mathcal{C} := \mathcal{C}^{\mathbf{\Delta}^{\mathrm{op}}}$;
- a **cosimplicial object** in it means a functor $\mathbf{\Delta} \to \mathcal{C}$; all cosimplicial objects form the category $\mathsf{cs}\mathcal{C} := \mathcal{C}^{\mathbf{\Delta}}$.

Consequently, $(\mathsf{cs}\mathcal{C})^{\mathrm{op}} \simeq \mathsf{s}(\mathcal{C}^{\mathrm{op}})$. If the functor underlying a simplicial (respectively, cosimplicial) object has an extension to $\mathsf{FinOrd}^{\mathrm{op}}$ (respectively, FinOrd), then the object is called **augmented**.

By the preceding discussion, giving a simplicial object X is equivalent to giving a family of objects $(X_n)_{n\ge 0}$ in $\mathcal{C}$ together with a family of morphisms satisfying (8.1.2),

$$d_i = d_i^n : X_n \to X_{n-1} \text{ (face)}, \quad s_j = s_j^n : X_n \to X_{n+1} \text{ (degeneracy)}, \quad 0 \le i, j \le n.$$

The object X_n is called the degree-n term of X. A morphism from a simplicial object X to Y is equivalently a family of morphisms $(f_n : X_n \to Y_n)_{n\ge 0}$ in $\mathcal{C}$ satisfying, for every $n \ge 0$, the compatibility conditions

$$d_i^{n+1} f_{n+1} = f_n d_i^{n+1}, \quad s_j^n f_n = f_{n+1} s_j^n.$$

Dually, giving a cosimplicial object X is equivalent to giving a family of objects $(X^n)_{n\geq 0}$ in $\mathcal{C}$ together with a family of morphisms satisfying (8.1.1),

$$d^i = d^{n,i} : X^{n-1} \to X^n \text{ (coface)}, \quad s^j = s^{n,j} : X^{n+1} \to X^n \text{ (codegeneracy)}, \quad 0 \leq i, j \leq n.$$

Its morphisms are families of morphisms $f^n : X^n \to Y^n$ compatible with all d^i and s^j.

An augmentation of a simplicial (respectively, cosimplicial) object X is equivalently the addition to the data of a morphism $X_0 \to X_{-1}$ (respectively, $X^{-1} \to X^0$) and a commutative diagram $X_1 \underset{d_1}{\overset{d_0}{\rightrightarrows}} X_0 \xrightarrow{\epsilon} X_{-1}$ (respectively, its dual). This morphism is called the augmentation morphism.

Convention 8.1.3 For a morphism $\phi : [m] \to [n]$ in $\boldsymbol{\Delta}$ and a simplicial (respectively, cosimplicial) object X in a category $\mathcal{C}$, denote the corresponding morphism by $\phi^* : X_n \to X_m$ (respectively, $\phi_* : X^m \to X^n$).

Remark 8.1.4 All nonempty ordinals and the injective monotone maps between them form a category $\boldsymbol{\Delta}_+$. A functor of the form $\boldsymbol{\Delta}_+^{\mathrm{op}} \to \mathcal{C}$ is called a **semisimplicial object** in $\mathcal{C}$. This is equivalent to deleting the degeneracy morphisms s_j and the associated conditions from the definition of a simplicial object. Every simplicial object gives a corresponding semisimplicial object. Augmented semisimplicial objects may be defined similarly. The case of cosemisimplicial objects is entirely dual.

Example 8.1.5 Let $C \in \mathrm{Ob}(\mathcal{C})$. The corresponding **constant simplicial object** $\mathrm{const}(C)$ is determined by $\mathrm{const}(C)_n = C$ and $s_j = \mathrm{id}_C = d_i$ for all n, i, j. As a simple exercise, check that

$$\{\text{augmentations of } X\} \overset{1:1}{\longleftrightarrow} \{(C, \varphi) : C \in \mathrm{Ob}(\mathcal{C}),\ \varphi : X \to \mathrm{const}(C)\}.$$

Concretely, for a given augmentation of X, take $C = X_{-1}$ and let φ_n be the composite $X_n \xrightarrow{\iota_k^*} X_0 \to X_{-1}$, where $\iota_k : [0] \to [n]$ sends 0 to any k with $0 \leq k \leq n$. The case of constant cosimplicial objects is entirely similar.

Example 8.1.6 Let $B \to A$ be a morphism in a category $\mathcal{C}$. If the following fiber products exist,

$$X_n := \underbrace{B \underset{A}{\times} \cdots \underset{A}{\times} B}_{n+1 \text{ factors}}, \quad n \geq 0,$$

each morphism $f : [m] \to [n]$ induces a morphism $X_n \to X_m$ whose ith component is the projection $B \underset{A}{\times} \cdots \underset{A}{\times} B \xrightarrow{\mathrm{pr}_{f(i)}} B$. This gives a simplicial object X in $\mathcal{C}$. It is augmented by taking $B \to A$ as the morphism $X_0 \to X_{-1}$.

Remark 8.1.7 (Order-reversal duality) For each $n \in \mathbb{Z}_{\geq 0}$, define a bijection w_n of $\{0, \dots, n\}$ with itself by $w_n(i) = n - i$. Define an autoisomorphism w of the category $\boldsymbol{\Delta}$ that preserves the objects $[n]$ and maps a morphism $f : [n] \to [m]$ to $wf := w_m f w_n$. Note that

$$w^2 = \mathrm{id}_{\boldsymbol{\Delta}}, \quad w\mathrm{d}^{n,i} = \mathrm{d}^{n,n-i}, \quad w\mathrm{s}^{n,j} = \mathrm{s}^{n,n-j}.$$

A more intrinsic viewpoint is that w maps $[n]$ to the poset with the reversed order $[n]^{\mathrm{op}}$, or, when regarded as a category, to its opposite category. This category is still uniquely isomorphic to $[n]$. The action of w on morphisms is expressed by the commutative diagram

$$\begin{tikzcd} {[m]^{\mathrm{op}}} \arrow[r, "f^{\mathrm{op}}", "\text{monotone}"'] & {[n]^{\mathrm{op}}} \\ {[m]} \arrow[u, "\wr"] \arrow[r, "\text{monotone}", "wf"'] & {[n]} \arrow[u, "\wr"] \end{tikzcd} \qquad f^{\mathrm{op}} \xlongequal{\text{as a map}} f.$$

For every category $\mathcal{C}$, this gives an autoisomorphism $X \mapsto X \circ w$ of $\mathsf{s}\mathcal{C}$ (respectively, $\mathsf{cs}\mathcal{C}$).

Finally, we outline the relation between monoidal structures and simplicial objects.

Definition 8.1.8 Every functor $F : \mathcal{C} \to \mathcal{D}$ induces a functor $\mathsf{s}\mathcal{C} \to \mathsf{s}\mathcal{D}$ that maps the data $(X_n, d_i, s_j)_{n,i,j}$ to $(FX_n, Fd_i, Fs_j)_{n,i,j}$. Moreover, there is a direct relation $\mathsf{s}(\mathcal{C}_1 \times \mathcal{C}_2) \simeq \mathsf{s}\mathcal{C}_1 \times \mathsf{s}\mathcal{C}_2$.

Apply this observation to a monoidal category $\mathcal{C}$ and the bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$. For every $X, Y \in \mathrm{Ob}(\mathsf{s}\mathcal{C})$, define $X \otimes Y \in \mathrm{Ob}(\mathsf{s}\mathcal{C})$ so that its degree-n term is $X_n \otimes Y_n$, while its face and degeneracy morphisms have the forms $d_i \otimes d_i$ and $s_j \otimes s_j$, respectively.

Thus, if $\mathcal{C}$ is a monoidal category, then $\mathsf{s}\mathcal{C}$ also has a corresponding monoidal structure, with the constant simplicial object $\mathrm{const}(\mathbf{1})$ as its unit. The analogous statement holds for cosimplicial objects.

8.2 Simplicial Sets

The simplicial objects of Definition 8.1.2 in the case $\mathcal{C} = \mathsf{Set}$ are called simplicial sets. Relative to a fixed Grothendieck universe, the precise meaning of Set in this book is the category of all small sets; these details do not affect the content of this section.

Definition 8.2.1 A simplicial object in the category of sets Set is called a **simplicial set**; a cosimplicial object in it is called a **cosimplicial set**.

By Definition 8.1.2, all simplicial sets (or cosimplicial sets) form the category $\mathsf{sSet} := \mathsf{Set}^{\mathbf{\Delta}^{\mathrm{op}}}$ (or $\mathsf{csSet} := \mathsf{Set}^{\mathbf{\Delta}}$).

Definition 8.2.2 For every $n \in \mathbb{Z}_{\geq 0}$, write Δ^n for the simplicial set determined by $\mathrm{Hom}_{\mathbf{\Delta}}(\cdot, [n]) : \mathbf{\Delta}^{\mathrm{op}} \to \mathsf{Set}$. It is called the **standard n-simplex**.

Thus $(\Delta^n)_m = \mathrm{Hom}_{\mathbf{\Delta}}([m], [n])$ is the set of all order-preserving maps $[m] \to [n]$. The map $(\Delta^n)_{m'} \to (\Delta^n)_m$ induced by an order-preserving map $\phi : [m] \to [m']$ is precisely pullback of maps, $\phi^* : f \mapsto f\phi$.

Example 8.2.3 We introduce two common subobjects of the standard n-simplex Δ^n.

▷ Boundary Define the subfunctor $\partial\Delta^n$ of Δ^n by

$$(\partial\Delta^n)_m := \{f : [m] \to [n] \text{ order-preserving}, \mathrm{im}(f) \neq [n]\}, \quad m \in \mathbb{Z}_{\geq 0}.$$

If $m < n$, then $(\partial\Delta^n)_m = (\Delta^n)_m$; if $m \geq n$, the elements of $(\partial\Delta^n)_m$ are obtained by repeated degeneracies from $(\Delta^n)_h$ for $0 \leq h < n$.

▷ Horn For $0 \leq k \leq n$, define the subfunctor Λ^n_k of Δ^n by

$$(\Lambda^n_k)_m := \{f : [m] \to [n] \text{ order-preserving}, \mathrm{im}(f) \not\supset [n] \smallsetminus \{k\}\}, \quad m \in \mathbb{Z}_{\geq 0}.$$

We call k the vertex of the horn.

Every pullback $\phi^* : f \mapsto f\phi$ clearly preserves the conditions above, so both are subfunctors. The exercises in this chapter will show how Λ^n_i (or $\partial\Delta^n$) can be expressed precisely as a gluing of n (or $n+1$) copies of Δ^{n-1}.

The constant simplicial set given by the empty set, $\mathrm{const}(\varnothing)$, is abbreviated to $\varnothing$. Thus $\partial\Delta^0 = \varnothing$.

In §8.4 we will give an intuitive interpretation of simplicial sets. In particular, we will depict the geometric images of $\Delta^n \supset \Lambda^n_i \supset \partial\Delta^n$.

The Yoneda lemma (Theorem A.1.1) immediately gives the following canonical isomorphism, where $n \in \mathbb{Z}_{\geq 0}$ and $X \in \mathrm{Ob}(\mathsf{sSet})$:

$$\begin{aligned} \mathrm{Hom}_{\mathsf{sSet}}(\Delta^n, X) &\xrightarrow{\sim} X_n \\ \phi &\longmapsto \phi([n])(\mathrm{id}_{[n]}) \\ \phi([n]) : \mathrm{End}([n]) = \Delta^n([n]) &\to X([n]) =: X_n. \end{aligned} \tag{8.2.1}$$

Definition 8.2.4 An element of X_n is called an **n-simplex** of the simplicial set X. If $x \in X_n$ lies in the image of some s_j, it is called **degenerate**; otherwise it is called **non-degenerate**.

As a simple exercise, verify that the non-degenerate m-simplices of Δ^n correspond to the order-preserving injections $[m] \hookrightarrow [n]$; there are $\binom{n+1}{m+1}$ of them.

Furthermore, the Yoneda lemma ensures that $h_{\mathbf{\Delta}} : [n] \mapsto \Delta^n$ defines a fully faithful functor $\mathbf{\Delta} \to \mathsf{sSet}$. Every morphism $f : [m] \to [n]$ in $\mathbf{\Delta}$ induces $f : \Delta^m \to \Delta^n$. Specifying an $(m+1)$-element subset $\{k_0, \ldots, k_m\} \subset [n]$ is equivalent to specifying an order-preserving injection $[m] \hookrightarrow [n]$. The corresponding morphism between standard simplices is also written

$$\Delta^m \xrightarrow{\{k_0,\ldots,k_m\}} \Delta^n.$$

The intuitive meaning of simplicial sets will be explained by geometric realization in §8.4. Before that, we introduce a series of abstract constructions, beginning with the join and the left and right cones of simplicial sets.

Write FinLin_+ for the full subcategory of the category of finite totally ordered sets (also called finite linearly ordered sets) FinLin obtained by removing the empty set; its morphisms are order-preserving maps. Since $\mathbf{\Delta}$ is a skeleton of FinLin_+, a simplicial set can equivalently be viewed as a functor $\mathsf{FinLin}_+^{\mathrm{op}} \to \mathsf{Set}$.

Convention 8.2.5 Let J be an object of FinLin_+. Every subset $I \subset J$ is automatically a finite totally ordered set. If it also satisfies

$$\forall (i,j) \in I \times J, \quad j \leq i \implies j \in I,$$

then I is called an **initial segment** of J, written $I \sqsubset J$.

Definition 8.2.6 The **join** $X \star X'$ of simplicial sets X and X' is defined as the following functor $\mathsf{FinLin}_+^{\mathrm{op}} \to \mathsf{Set}$:

$$(X \star X')(J) = \bigsqcup_{I \sqsubset J} X(I) \times X'(J \smallsetminus I).$$

Here (and only here), $X(\varnothing)$ and $X'(\varnothing)$ are defined to be one-point sets. For every order-preserving map $f : J_1 \to J$ and $I \sqsubset J$, set $I_1 := f^{-1}(I)$. Then $I_1 \sqsubset J_1$ and there is a map

$$X(I) \times X'(J \smallsetminus I) \to X(I_1) \times X'(J_1 \smallsetminus I_1).$$

This yields an induced morphism $f^* : (X \star X')(J) \to (X \star X')(J_1)$, which determines the simplicial structure on $X \star X'$.

There are canonical morphisms $X \to X \star Y \leftarrow Y$. The definition can also be written in the earlier notation as

$$(X \star Y)_n = X_n \sqcup Y_n \sqcup \bigsqcup_{j+k=n-1} X_j \times Y_k.$$

The morphisms d_i and s_i are obtained directly from the corresponding morphisms on X and Y. For example, for $d_i : (X \star Y)_n \to (X \star Y)_{n-1}$, its restrictions to the subsets X_n and Y_n are the original d_i, while on $X_j \times Y_k$ it is given by

$$\begin{gathered} d_i(x, y) = \begin{cases} (d_i x, y), & i \leq j,\ j \neq 0 \\ (x, d_{i-j-1} y), & i > j,\ k \neq 0, \end{cases} \\ j = 0 \implies d_0(x, y) = y \in Y_{n-1}, \\ k = 0 \implies d_n(x, y) = x \in X_{n-1}. \end{gathered} \tag{8.2.2}$$

Proposition 8.2.7 Write $X_n^{\mathrm{nd}} \subset X_n$ for the subset of non-degenerate n-simplices of the simplicial set X (Definition 8.2.4). For all X and Y,

$$(X \star Y)_n^{\mathrm{nd}} = X_n^{\mathrm{nd}} \sqcup Y_n^{\mathrm{nd}} \sqcup \bigsqcup_{j+k=n-1} X_j^{\mathrm{nd}} \times Y_k^{\mathrm{nd}}.$$

Proof Let Z be a simplicial set. For every nonempty finite totally ordered set J, an element $x \in Z(J)$ is degenerate if and only if there is an order-preserving surjection that is not injective, $f : J \twoheadrightarrow J_0$, such that $x \in \mathrm{im}[f^* : Z(J_0) \to Z(J)]$. The rest follows immediately from this observation. $\square$

Definition 8.2.8 The **left cone** with base a simplicial set X is $X^{\triangleleft} := \Delta^0 \star X$, while the **right cone** is $X^{\triangleright} := X \star \Delta^0$.

An intuitive interpretation of cones is given in Example 8.4.5. The exercises will examine further properties of the join operation.

8.3 Example: Nerves of Categories

This section introduces the nerves of categories and, through them, the classifying spaces of groups. Both are important examples of simplicial sets.

Example 8.3.1 (Nerve of a category) Let $\mathcal{C}$ be a small category. Define the functor

$$\mathbf{\Delta}^{\mathrm{op}} \to \mathsf{Set}, \quad [n] \mapsto \{\text{all functors } [n] \to \mathcal{C}\}.$$

As a simplicial set it is denoted by $\mathrm{N}\mathcal{C}$ and called the **nerve** of $\mathcal{C}$. Specifying an element of $\mathrm{N}\mathcal{C}_n$ is equivalent to specifying a functor $[n] \to \mathcal{C}$, that is, a chain of morphisms in $\mathcal{C}$

$$(f_1, \ldots, f_n) : C_0 \xrightarrow{f_1} C_1 \xrightarrow{f_2} \cdots \xrightarrow{f_n} C_n.$$

Thus $\mathrm{N}\mathcal{C}_0$ may be identified with $\mathrm{Ob}(\mathcal{C})$, and $\mathrm{N}\mathcal{C}_1$ with $\mathrm{Mor}(\mathcal{C})$.

The face morphisms $d_i : \mathrm{N}\mathcal{C}_n \to \mathrm{N}\mathcal{C}_{n-1}$ and degeneracy morphisms $s_j : \mathrm{N}\mathcal{C}_n \to \mathrm{N}\mathcal{C}_{n+1}$ are given by

$$d_i(f_1, \ldots, f_n) = \begin{cases} (f_2, \cdots, f_n), & i = 0 \\ (\ldots, f_{i+1}f_i, \ldots), & 0 < i < n, \\ (f_1, \ldots, f_{n-1}), & i = n, \end{cases}$$

$$s_j(f_1, \ldots, f_n) = \begin{cases} (\mathrm{id}_{C_0}, f_1, \ldots, f_n), & j = 0 \\ (\ldots, f_j, \mathrm{id}_{C_j}, f_{j+1}, \ldots), & 0 < j < n \\ (f_1, \ldots, f_n, \mathrm{id}_{C_n}), & j = n. \end{cases}$$

Remark 8.3.2 Unfolding the definitions shows that the simplicial sets $\mathrm{N}(\mathcal{C}^{\mathrm{op}})$ and $\mathrm{N}\mathcal{C}$ are related by the order-reversing duality of Remark 8.1.7. The details are left to the reader.

The nerve realizes a category concretely through combinatorial/topological data and contains all the information of the original category. This is the content of Proposition 8.3.4, which will soon be proved. We first characterize which simplicial sets are nerves; the argument is instructive.

Proposition 8.3.3 (Characterization of nerves) Let X be a simplicial set. The following statements are equivalent.

(i) There is a small category $\mathcal{C}$ such that $\mathrm{N}\mathcal{C} \simeq X$.

(ii) X has the unique inner horn-filling property: for every integer $0 < i < n$ and every morphism $\sigma' : \Lambda^n_i \to X$ in sSet (such a Λ^n_i is called an inner horn), there is a unique $\sigma : \Delta^n \to X$ extending σ'.

Proof First prove (i) $\implies$ (ii). Let $X = \mathrm{N}\mathcal{C}$ and $0 < i < n$; we wish to extend $\sigma' : \Lambda^n_i \to X$ to Δ^n. For every $0 \leq k \leq n$ (or $0 < k \leq n$), the morphism $\Delta^0 \xrightarrow{\{k\}} \Delta^n$ (or $\Delta^1 \xrightarrow{\{k-1,k\}} \Delta^n$) factors through Λ^n_i. Denote its image under σ' by $C_k \in X_0 = \mathrm{Ob}(\mathcal{C})$ (or $[g_k : C_{k-1} \to C_k] \in X_1 = \mathrm{Mor}(\mathcal{C})$). This gives an element of X_n,

$$C_0 \xrightarrow{g_1} C_1 \xrightarrow{g_2} \cdots \xrightarrow{g_n} C_n.$$

Write σ for the corresponding morphism $\Delta^n \to X$. The construction also shows that σ is the only possible extension of σ'. Since $\Lambda^n_i = \bigcup_{j \neq i} \mathrm{im}[\mathrm{d}^j : \Delta^{n-1} \to \Delta^n]$, it remains to verify

$$\sigma \circ \mathrm{d}^j = \sigma' \circ \mathrm{d}^j : \Delta^{n-1} \to X, \quad j \neq i. \tag{8.3.1}$$

By the definition of the nerve, proving (8.3.1) reduces to the following verification. For each $j \neq i$ and each pair of adjacent elements $h < k$ in the sequence $0, \ldots, \widehat{j}, \ldots, n$ (where $\widehat{j}$ means that j is omitted), the pullbacks of σ and σ' along $\Delta^1 \xrightarrow{\{h,k\}} \Lambda^n_i \subset \Delta^n$ must agree.

- ◇ If $k = h+1$, this is guaranteed by the construction of σ. In particular, (8.3.1) always holds when $j \in \{0, n\}$.

- ◇ Suppose $(h,k) = (j-1, j+1)$. If $n = 2$, this case cannot occur. If $n > 2$, then either $j - 1 > 0$ or $j + 1 < n$. When $j - 1 > 0$, $\{j-1, j+1\}$ factors as
$$\Delta^1 \xrightarrow{\{j-2,j\}} \Delta^{n-1} \xrightarrow{\mathrm{d}^0=\{1,\ldots,n\}} \Lambda^n_i \subset \Delta^n.$$
By the already known $j = 0$ case of (8.3.1), the two pullbacks agree. Similarly, when $j + 1 < n$, the issue reduces to the known case $j = n$.

Now prove (ii) $\implies$ (i). Define a category $\mathcal{C}$ by $\mathrm{Ob}(\mathcal{C}) := X_0$ and, for all $C, C' \in X_0$,
$$\mathrm{Hom}_{\mathcal{C}}(C, C') := \{f \in X_1 : d_1(f) = C,\ d_0(f) = C'\}.$$
Use the degeneracy map $s_0 : X_0 \to X_1$ to define the identity morphism id_C as $s_0(C)$. We also depict a morphism f as $0 \xrightarrow{f} 1$ to emphasize that it corresponds to a 1-simplex $[1] = \{0,1\} \to X$.

If $d_0(f) = d_1(g)$, define the composite by $gf := d_1(\sigma)$, where $\sigma : \Delta^2 \to X$ is the unique extension of
$$\begin{tikzcd} & 1 \arrow[dr, "g"] & \\ 0 \arrow[ur, "f"] & & 2 \end{tikzcd} : \Lambda^2_1 \to X \quad \text{that is,} \quad d^0(\sigma) = g, \quad d^2(\sigma) = f.$$

The laws $f \circ \mathrm{id}_C = f$ and $\mathrm{id}_{C'} \circ f = f$ are witnessed, respectively, by the following elements of X_2.
$$\begin{tikzcd} & 1 \arrow[dr, bend left, "\mathrm{id}_{C'}=d_0 s_1(f)"] & \\ 0 \arrow[ur, bend left, "f=d_2 s_1(f)"] \arrow[rr, "f=d_1 s_1(f)"'] \arrow[rr, phantom, bend left=40, "s_1(f)"] & & 2 \end{tikzcd} \qquad \begin{tikzcd} & 1 \arrow[dr, bend left, "f=d_0 s_0(f)"] & \\ 0 \arrow[ur, bend left, "\mathrm{id}_C=d_2 s_0(f)"] \arrow[rr, "f=d_1 s_0(f)"'] \arrow[rr, phantom, bend left=40, "s_0(f)"] & & 2 \end{tikzcd}$$

For associativity $h(gf) = (hg)f$, construct $\sigma' : \Lambda^3_2 \to X$ whose three faces are, respectively,
$$\begin{tikzcd} & 1 \arrow[dr, "g"] & \\ 0 \arrow[ur, "f"] \arrow[rr, "gf"'] & & 2 \end{tikzcd} \quad \begin{tikzcd} & 2 \arrow[dr, "h"] & \\ 1 \arrow[ur, "g"] \arrow[rr, "hg"'] & & 3 \end{tikzcd} \quad \begin{tikzcd} & 2 \arrow[dr, "h"] & \\ 0 \arrow[ur, "gf"] \arrow[rr, "h(gf)"'] & & 3. \end{tikzcd}$$
This horn has a unique extension $\sigma : \Delta^3 \to X$ such that $d^2(\sigma) \in X_2$ has the form
$$\begin{tikzcd} & 1 \arrow[dr, "hg"] & \\ 0 \arrow[ur, "f"] \arrow[rr, "h(gf)"'] & & 3. \end{tikzcd}$$

By construction, however, this 2-simplex also determines the composite of f and hg, proving $(hg)f = h(gf)$.

Thus $\mathcal{C}$ is a category. There is a canonical morphism $X \to \mathrm{N}\mathcal{C}$: from a given $\Delta^n \to X$, pull back along each $\Delta^1 \xrightarrow{\{j,j+1\}} \Delta^n$ to obtain a chain of morphisms. By construction, $X_n \to \mathrm{N}\mathcal{C}_n$ is clearly a bijection for $n = 0, 1$. If $n \geq 2$, choose $0 < i < n$ and consider the commutative diagram

$$\begin{array}{ccc} \mathrm{Hom}(\Delta^n, X) & \longrightarrow & \mathrm{Hom}(\Delta^n, \mathrm{N}\mathcal{C}) \\ \downarrow & & \downarrow \\ \mathrm{Hom}(\Lambda^n_i, X) & \longrightarrow & \mathrm{Hom}(\Lambda^n_i, \mathrm{N}\mathcal{C}). \end{array}$$

By the assumption and the unique inner horn-filling property, both vertical arrows are bijections. Moreover, Λ^n_i can be expressed as the $\varinjlim$ of a family of Δ^{n-1} and Δ^{n-2} (that is, the gluing of n faces), so induction shows that the arrow in the second row is also a bijection. This proves the result. $\square$

Notice also that in the argument (ii) $\implies$ (i), if there is a small category $\mathcal{C}_0$ with $X = \mathrm{N}(\mathcal{C}_0)$, then the category $\mathcal{C}$ constructed in the proof is exactly $\mathcal{C}_0$.

Write Cat for the category of all small categories, with functors as morphisms. Every functor $F : \mathcal{C} \to \mathcal{C}'$ induces a morphism $\mathrm{N}F : \mathrm{N}\mathcal{C} \to \mathrm{N}\mathcal{C}'$ in sSet, mapping $(f_1, \ldots, f_n)$ to $(Ff_1, \ldots, Ff_n)$. This gives the nerve functor $\mathrm{N} : \mathsf{Cat} \to \mathsf{sSet}$.

Proposition 8.3.4 The nerve functor $\mathrm{N} : \mathsf{Cat} \to \mathsf{sSet}$ is fully faithful.

Proof The proof of Proposition 8.3.3 has already shown how to reconstruct morphisms and their composition from the nerve of a category. Hence a morphism $\mathrm{N}(\mathcal{C}) \to \mathrm{N}(\mathcal{C}')$ naturally induces a functor $\mathcal{C} \to \mathcal{C}'$. It is easy to see that this construction and the construction induced by N are mutually inverse. $\square$

Example 8.3.5 (Classifying space) Let Γ be a monoid. Define the categories $\mathcal{B}\Gamma$ and $\mathcal{E}\Gamma$ as follows:

Category	Object set	Morphisms	Composition of morphisms
$\mathcal{B}\Gamma$	$\{\star\}$	$\mathrm{End}_{\mathcal{B}\Gamma}(\star) = \Gamma$	multiplication in Γ
$\mathcal{E}\Gamma$	Γ	$\mathrm{Hom}_{\mathcal{E}\Gamma}(g, g') = \{h \in \Gamma : hg = g'\}$	multiplication in Γ

Define the nerves $\mathrm{B}\Gamma := \mathrm{N}((\mathcal{B}\Gamma)^{\mathrm{op}})$ and $\mathrm{E}\Gamma := \mathrm{N}((\mathcal{E}\Gamma)^{\mathrm{op}})$. The simplicial set $\mathrm{B}\Gamma$ is called the classifying space of Γ. Taking $(\cdots)^{\mathrm{op}}$ amounts to reversing the arrows in the definition of the nerve without changing their labels. Thus

Set	Element	Corresponding chain of morphisms in $\mathcal{B}\Gamma$ or $\mathcal{E}\Gamma$
$(\mathrm{B}\Gamma)_n = \Gamma^n$	$(g_1, \ldots, g_n)$	$\star \xleftarrow{g_1} \cdots \leftarrow \star \xleftarrow{g_n} \star$
$(\mathrm{E}\Gamma)_n = \Gamma^{n+1}$	$(g_1, \ldots, g_{n+1})$	$g_1 \cdots g_{n+1} \xleftarrow{g_1} \cdots \leftarrow g_n g_{n+1} \xleftarrow{g_n} g_{n+1}$

Substitution into the definitions gives, for every $n \geq 0$,

$$\mathrm{B}\Gamma: \quad \begin{aligned} d_i(g_1, \ldots, g_n) &= \begin{cases} (g_2, \ldots, g_n), & i = 0 \\ (\ldots, g_i g_{i+1}, \ldots), & 0 < i < n \\ (g_1, \ldots, g_{n-1}), & i = n, \end{cases} \\ s_j(g_1, \ldots, g_n) &= \begin{cases} (1, g_1, \ldots, g_n), & j = 0 \\ (\ldots, g_j, 1, g_{j+1}, \ldots), & 0 < j < n \\ (g_1, \ldots, g_n, 1), & j = n, \end{cases} \end{aligned}$$

and

$$\mathrm{E}\Gamma: \quad \begin{aligned} d_i(g_1, \ldots, g_{n+1}) &= \begin{cases} (g_2, \ldots, g_{n+1}), & i = 0 \\ (g_1, \ldots, g_i g_{i+1}, \ldots), & 0 < i \leq n, \end{cases} \\ s_j(g_1, \ldots, g_{n+1}) &= \begin{cases} (1, g_1, \ldots, g_{n+1}), & j = 0 \\ (\ldots, g_j, 1, g_{j+1}, \ldots), & 0 < j \leq n. \end{cases} \end{aligned}$$

There is a natural functor $(\mathcal{E}\Gamma)^{\mathrm{op}} \to (\mathcal{B}\Gamma)^{\mathrm{op}}$ mapping all objects to $\star$ and every morphism $h \in \Gamma$ to h. The induced morphism $\mathrm{E}\Gamma \to \mathrm{B}\Gamma$ simply maps $(g_1, \ldots, g_{n+1})$ to $(g_1, \ldots, g_n)$. The right multiplication action of Γ on $(\mathrm{E}\Gamma)_n$ acts along its fibers, or in other words changes the datum g_{n+1}:

$$(g_1, \ldots, g_n, g_{n+1})g = (g_1, \ldots, g_n, g_{n+1}g).$$

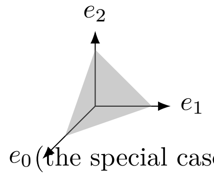
These actions clearly commute with d_i and s_j. If Γ is a group, the action is also free.

In topology, $\mathrm{B}\Gamma$ (or its geometric realization $|\mathrm{B}\Gamma|$; see §8.4) serves to classify Γ-torsors over spaces, while $\mathrm{E}\Gamma \to \mathrm{B}\Gamma$ gives the universal torsor. Example 8.6.9 will show that $\mathrm{E}\Gamma$ is “right-contractible.”

8.4 The Geometric-Realization Functor

We now begin to build a geometric picture of the simplicial sets introduced in §8.2. We start with the standard n-simplex, where $n \in \mathbb{Z}_{\geq 0}$. Write the standard ordered basis of $\mathbb{R}^{n+1}$ as $e_0, \dots, e_n$. Define

$$|\Delta^n| := \left\{ (x_0, \dots, x_n) \in (\mathbb{R}_{\geq 0})^{n+1} : \sum_{i=0}^{n} x_i = 1 \right\}$$



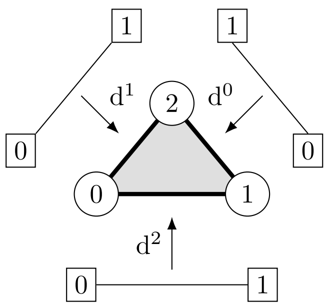
(the special case $n = 2$)

Its vertices are labeled $0, \dots, n$ by $i \leftrightarrow e_i$. Its interior is given the standard orientation for which $e_1 - e_0, \dots, e_n - e_{n-1}$ is a positively oriented basis at every point.

Every monotone map $f : [m] \to [n]$ induces a map

$$\begin{aligned} |f| : |\Delta^m| &\longrightarrow |\Delta^n| \\ \textstyle\sum_{i=0}^{m} y_i e_i &\longmapsto \textstyle\sum_{i=0}^{m} y_i e_{f(i)}. \end{aligned}$$

If f is not surjective, the image of $|f|$ lies in the boundary. The embedding $|\mathrm{d}^i| : |\Delta^{n-1}| \to |\Delta^n|$ may be used to compare the orientation on the interior of $|\Delta^{n-1}|$ with the orientation induced on the boundary by $|\Delta^n|$. A standard determinant calculation shows that they differ by a factor of $(-1)^i$. The following figure illustrates the case $n = 2$.



Orientation: $0 \longrightarrow 1$

$$\begin{tikzcd} & 2 \arrow[dl] & \\ 0 \arrow[rr] & & 1 \arrow[ul] \end{tikzcd}$$

Summarizing the discussion above, we obtain a functor

$$|\cdot| : \mathbf{\Delta} \to \mathsf{Top} \quad \begin{cases} \text{objects} & [n] \mapsto |\Delta^n| \\ \text{morphisms} & f \mapsto |f|. \end{cases}$$

We wish to extend it to a functor $\mathsf{sSet} \to \mathsf{Top}$. Recall that Top is cocomplete.

Definition 8.4.1 (Geometric-realization functor) The functor $|\cdot| : \mathsf{sSet} \to \mathsf{Top}$ is defined as follows. If X is a simplicial set, then

$$|X| := \varinjlim_{n,x} |\Delta^n|,$$

where $n \in \mathbb{Z}_{\geq 0}$ and $x \in X_n$, or equivalently, where x is a morphism $\Delta^n \to X$. Such pairs (n, x) form a category. A morphism from (n, x) to (m, y) is precisely an $f \in \mathrm{Hom}_{\mathbf{\Delta}}([n], [m])$ for which the diagram

$$\begin{tikzcd} \Delta^n \arrow[rr, "f"] \arrow[dr, "x"'] & & \Delta^m \arrow[dl, "y"] \\ & X & \end{tikzcd}$$

commutes; equivalently, $f^* : X_m \to X_n$ must satisfy $f^*(y) = x$.

For fixed $m \geq 0$, the category of data $(n, x : \Delta^n \to \Delta^m)$ has the terminal object $(m, \mathrm{id}_{\Delta^m})$. Therefore the geometric realization of Δ^m is canonically identified with the previously defined $|\Delta^m|$. Thus the notation is consistent.

Remark 8.4.2 For every simplicial set X, there is a canonical isomorphism $\varinjlim_{n,x} \Delta^n \xrightarrow{\sim} X$. This follows directly by applying the Density Theorem A.1.3 to the category $\mathsf{sSet} = \mathbf{\Delta}^\wedge$; the reader is invited to check it.

Remark 8.4.3 By Theorem 1.8.1, geometric realization is the left Kan extension of the functor $|\cdot| : \mathbf{\Delta} \to \mathsf{Top}$ along $\mathbf{\Delta} \to \mathsf{sSet}$; see Definition 1.7.1. Thus the construction is entirely natural.

By the construction above, for each $n \in \mathbb{Z}_{\geq 0}$ and $x \in X_n$, the universal property of $\varinjlim$ gives a canonical map $i_{n,x} : |\Delta^n| \to |X|$, compatible as (n, x) varies. Taking $\varinjlim$ in Top is simply the operation of gluing topological spaces. More concretely, the geometric realization can be written as the quotient space

$$|X| = \frac{\bigsqcup_{n \geq 0} X_n \times |\Delta^n|}{(f^*(y), t) \sim (y, |f|(t))}, \qquad \begin{array}{ccccc} (f^*(y), t) & \in & X_n & \times & |\Delta^n| & & [n] \\ & & {\scriptstyle f^*}\uparrow & & \downarrow{\scriptstyle |f|} & & \downarrow{\scriptstyle f} \\ (y, |f|(t)) & \in & X_m & \times & |\Delta^m| & & [m]. \end{array} \tag{8.4.1}$$

In keeping with the geometric intuition, a simplicial set X may be viewed as a model of a space. The sets $X_0, X_1, \ldots$ are the collections of building blocks of the model: each element of X_n corresponds to, and labels, one copy of $|\Delta^n|$. The various maps $f^* : X_m \to X_n$ serve as assembly instructions that determine how these copies of $|\Delta^n|$ are glued together.

Since the morphisms in $\mathbf{\Delta}$ have already been described explicitly, the equivalence relation in (8.4.1) can be generated by two special kinds of f.

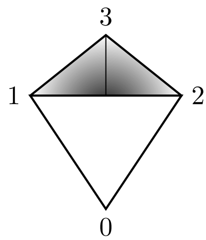
◇ Take $\mathrm{d}^i : [n-1] \hookrightarrow [n]$. Then $|\mathrm{d}^i|$ embeds $\{d_i(x)\} \times |\Delta^{n-1}|$ into $\{x\} \times |\Delta^n|$. Thus the face map $d_i : X_n \to X_{n-1}$ labels the ith face of the n-dimensional component corresponding to x by $d_i(x)$, and the gluing is then performed according to these labels.

- Take $\mathrm{s}^j : [n+1] \twoheadrightarrow [n]$. Then $|\mathrm{s}^j|$ "collapses" $\{s_j(x)\} \times |\Delta^{n+1}|$ onto $\{x\} \times |\Delta^n|$. Thus the degeneracy map $s_j : X_n \to X_{n+1}$ determines how an n-dimensional component corresponding to x gives rise to a "degenerate" $(n+1)$-dimensional component, labeled $s_j(x)$, which is collapsed onto its jth face during the gluing process.

Consequently, degenerate n-simplices (Definition 8.2.4) may be omitted from the gluing process, but the degeneracy data still cannot be ignored. For example, a face map $d_i : X_n \to X_{n-1}$ may well map a nondegenerate simplex to a degenerate simplex.

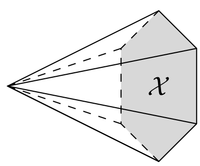
Example 8.4.4 As an illustration, $\partial\Delta^n$ in Example 8.2.3 realizes the boundary of the standard n-simplex in the intuitive sense, while the horn Λ^n_k is obtained by removing from $\partial\Delta^n$ the face opposite its kth vertex. Here is one illustration:

$|\Lambda^3_0| =$



Example 8.4.5 For every topological space $\mathcal{X}$, define the cone over it, $\mathrm{Cone}_{\mathrm{top}}(\mathcal{X})$, to be the space obtained by contracting the closed subset $\mathcal{X} \times \{0\}$ of $\mathcal{X} \times [0,1]$ to a single point. Visually,

$\mathrm{Cone}_{\mathrm{top}}(\mathcal{X}) =$



As a simple exercise, check that $|X^{\triangleleft}| \simeq \mathrm{Cone}_{\mathrm{top}}(|X|) \simeq |X^{\triangleright}|$. This shows that the cones in Definition 8.2.8 are aptly named; the only difference between the left and right cones is their orientation.

By presenting a triangulation of a space as a simplicial set, one can extract many important topological properties. This approach is aptly called "combinatorial topology". Further discussion and examples may be found in [49, Chapter VI].

Definition 8.4.6 (Singular-set functor) For every topological space E, define the simplicial set $\mathrm{Sing}(E)$ by

$$\mathrm{Sing}(E)_n := \mathrm{Hom}_{\mathsf{Top}}(|\Delta^n|, E), \quad n \in \mathbb{Z}_{\geq 0},$$

where $d_i : \mathrm{Sing}(E)_n \to \mathrm{Sing}(E)_{n-1}$ (respectively, $s_j : \mathrm{Sing}(E)_n \to \mathrm{Sing}(E)_{n+1}$) is given by pullback along $|\mathrm{d}^i| : |\Delta^{n-1}| \to |\Delta^n|$ (respectively, $|\mathrm{s}^j| : |\Delta^{n+1}| \to |\Delta^n|$). This is called the **(simplicial) singular set** associated with E. The construction gives a functor

$$\mathrm{Sing} : \mathsf{Top} \to \mathsf{sSet}.$$

Theorem 8.4.7 The geometric-realization functor is left adjoint to the singular-set functor. In other words, there is a natural bijection

$$\begin{array}{ccc}
\mathrm{Hom}_{\mathsf{Top}}(|X|, E) & \xleftrightarrow{\quad 1:1 \quad} & \mathrm{Hom}_{\mathsf{sSet}}(X, \mathrm{Sing}(E)) \\
\varphi & \longmapsto & \left[X_n \xrightarrow{x \mapsto \varphi \circ i_{n,x}} \mathrm{Hom}_{\mathsf{Top}}(|\Delta^n|, E)\right] \\
\varinjlim_{(n,x)} \left(|\Delta^n| \xrightarrow{\psi_n(x)} E\right) & \longleftarrow\!\!| & \psi = \left[\psi_n : X_n \to \mathrm{Hom}_{\mathsf{Top}}(|\Delta^n|, E)\right]_{n \geq 0},
\end{array}$$

where $X \in \mathrm{Ob}(\mathsf{sSet})$ and $E \in \mathrm{Ob}(\mathsf{Top})$.

Proof Recall that $|X| = \varinjlim_{n,x} |\Delta^n|$. There are canonical isomorphisms

$$\begin{aligned}
\mathrm{Hom}_{\mathsf{Top}}\left(\varinjlim_{n,x} |\Delta^n|, E\right) &\xrightarrow{\sim} \varprojlim_{n,x} \mathrm{Hom}_{\mathsf{Top}}(|\Delta^n|, E) = \varprojlim_{n,x} \mathrm{Sing}(E)_n \\
&\xrightarrow{\sim} \varprojlim_{n,x} \mathrm{Hom}_{\mathsf{sSet}}\left(\Delta^n, \mathrm{Sing}(E)\right) \\
\varphi &\mapsto \left(\psi_n(x) := \varphi \circ i_{n,x}\right)_{n,x}.
\end{aligned}$$

Every family of morphisms $(\psi_n(x))_{n,x}$ determined by a morphism $\psi : X \to \mathrm{Sing}(E)$ satisfies the compatibility conditions, and thus is an element of the inverse limit on the right-hand side above. To complete the proof, it remains to show that giving a morphism ψ is equivalent to giving a compatible family of morphisms; this is exactly what Remark 8.4.2 guarantees. □

Remark 8.4.8 For homotopy theory, the category CGHaus mentioned in §7.3 may be more convenient than Top. Since the inclusion functor $\mathsf{CGHaus} \to \mathsf{Top}$ has a right adjoint k, it preserves $\varinjlim$. Moreover, $|\Delta^n|$ is a compactly generated Hausdorff space. Consequently, the gluing (respectively, the formation of Hom) in geometric realization (respectively, the singular set) may be carried out within CGHaus[1)]. Thus the adjoint pair of Theorem 8.4.7 factors into two adjunctions, written with the same notation as

$$\mathsf{sSet} \underset{\mathrm{Sing}}{\overset{|\cdot|}{\rightleftarrows}} \mathsf{CGHaus} \underset{k}{\overset{\text{inclusion functor}}{\rightleftarrows}} \mathsf{Top}.$$

[1)] It must be explained that the $\varinjlim$ in question does indeed exist in CGHaus. This is a topological issue; see [14, III.1.8, III.2.1].

For any two simplicial sets X and Y, their product is defined degreewise:

$$\begin{aligned}(X \times Y)_n &= X_n \times Y_n,\\ d_i(x, y) &= (d_i(x), d_i(y)),\\ s_j(x, y) &= (s_j(x), s_j(y)).\end{aligned}$$

This is both the categorical product in $\mathsf{sSet} = \mathsf{Set}^{\boldsymbol{\Delta}^{\mathrm{op}}}$ and the construction of Definition 8.1.8 applied to the monoidal category $(\mathsf{Set}, \times)$. Although both definitions arise from category theory, the following fundamental result shows that they also have geometric meaning.

Theorem 8.4.9 For simplicial sets X and Y, there is a canonical isomorphism in CGHaus

$$|X \times Y| \simeq |X| \times |Y|.$$

More generally, the CGHaus-valued version of $|\cdot|$ preserves finite limits. If either X or Y has only finitely many nondegenerate simplices, then $|X \times Y| \simeq |X| \times |Y|$ also holds in Top.

The proof in the general case is rather involved; see [14, Chapter III] or [12]. The isomorphism does not hold in general for the Top-valued version of geometric realization. Notice also that the condition stated in the theorem for the Top-valued version is not optimal.

Corollary 8.4.10 Write Grp for the category of groups. If a simplicial set X can be promoted to an object of sGrp, then $|X|$ naturally has the structure of a topological group. Analogous results hold for other algebraic structures.

In view of Theorem 8.4.9, it is natural to define a homotopy from g to f, for morphisms $f, g : X \rightrightarrows Y$ in sSet, as a morphism $H : X \times \Delta^1 \to Y$ making the following diagram commute:

$$\begin{tikzcd} X \times \Delta^0 \arrow[r, "\sim"] \arrow[d, "\mathrm{id}_X \times \mathrm{d}^0"'] & X \arrow[d, "f"] \\ X \times \Delta^1 \arrow[r, dashed, "H"] & Y \\ X \times \Delta^0 \arrow[r, "\sim"'] \arrow[u, "\mathrm{id}_X \times \mathrm{d}^1"] & X \arrow[u, "g"'] \end{tikzcd} \tag{8.4.2}$$

As a special case, the composite $X \times \Delta^1 \xrightarrow{\text{projection}} X \xrightarrow{f} Y$ gives a constant homotopy from f to itself. Homotopy as defined here is not, however, an equivalence relation: a simple example already shows that it need not be symmetric. In homotopy theory one usually imposes an additional condition on Y, for example that it be a Kan complex. The homotopy relation then has the expected properties, allowing homotopy groups, weak equivalences, and related notions to be studied combinatorially.

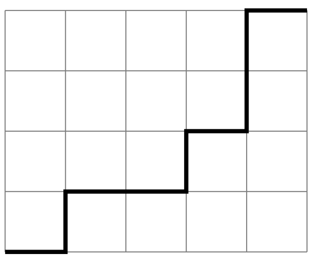
Although this book does not prove Theorem 8.4.9, it is useful to say a little more about the simple special case $X = \Delta^p$ and $Y = \Delta^q$. For example, how can the nondegenerate simplices of $\Delta^p \times \Delta^q$ be classified? The answer not only helps explain the geometric realization of a product; the related construction will also be needed later.

First, the product of two partially ordered sets S_1 and S_2 becomes a partially ordered set under $(a_1, a_2) \leq (b_1, b_2)$ if and only if $a_1 \leq b_1$ and $a_2 \leq b_2$. This is also the product of the corresponding categories.

Definition 8.4.11 Let $p, q \in \mathbb{Z}_{\geq 0}$. A ***(p, q)*-shuffle** is an injective monotone map $\sigma : [p+q] \to [p] \times [q]$.

For every $n \in \mathbb{Z}_{\geq 0}$, specifying a monotone map $\sigma : [n] \to [p] \times [q]$ is equivalent to specifying a pair of monotone maps $\sigma_- : [n] \to [p]$ and $\sigma_+ : [n] \to [q]$. It is also equivalent to specifying an n-simplex of $\Delta^p \times \Delta^q$. The condition that σ be injective says precisely that the path $i \mapsto (\sigma_-(i), \sigma_+(i))$ never remains stationary, for $i = 0, \ldots, n$. When $n = p + q$, the situation may be pictured as follows:

(p, q)



$i = 0$

exactly $p + q$ steps. (8.4.3)

Thus, for a (p, q)-shuffle σ, define

$$\begin{aligned} I_\pm &:= \{1 \leq i \leq p + q : \sigma_\pm(i-1) < \sigma_\pm(i)\} \\ &= \{1 \leq i \leq p + q : \sigma_\pm(i-1) = \sigma_\pm(i) - 1\} \\ &= \{1 \leq i \leq p + q : \sigma_\mp(i-1) = \sigma_\mp(i)\}\,, \end{aligned}$$

so that $I_+ \sqcup I_- = \{1, \ldots, p+q\}$. The subsets I_+ and I_- correspond, respectively, to the upward and rightward portions of the path in (8.4.3). Thus a (p, q)-shuffle may also be viewed as an ordering of p symbols $\to$ and q symbols $\uparrow$; there are $\binom{p+q}{p}$ such orderings. These observations also show that σ_+ and σ_- are surjective monotone maps for every (p, q)-shuffle.

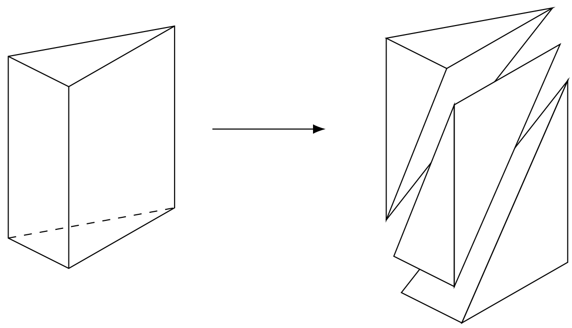
Proposition 8.4.12 Let $p, q \in \mathbb{Z}_{\geq 0}$. Consider an n-simplex of $\Delta^p \times \Delta^q$, that is, a monotone map $\sigma : [n] \to [p] \times [q]$. Set $(p_i, q_i) := \sigma(i)$ and $(p', q') := (p_n - p_0, q_n - q_0)$. Then σ is nondegenerate if and only if the following conditions hold:

◇ $p' + q' = n$;

◇ σ factors as a (p', q')-shuffle $\sigma' : [n] \to [p'] \times [q']$ followed by a monotone injection $[p'] \times [q'] \hookrightarrow [p] \times [q]$ of the form $f \times g$.

Proof Identify σ with the pair of monotone maps (σ_-, σ_+). Consider the path $i \mapsto (p_i, q_i)$ as in (8.4.3). The condition that σ be nondegenerate is equivalent to the condition that this path never remains stationary. The rest is clear. □

Using this description of the nondegenerate simplices, the reader may visualize the isomorphism $|\Delta^p \times \Delta^q| \simeq |\Delta^p| \times |\Delta^q|$ in the cases $(p, q) = (1, 1)$ and $(2, 1)$. For example, the following figure shows the triangulation of $|\Delta^2| \times |\Delta^1|$ into three tetrahedra, corresponding to the $3 = \binom{3}{2}$ $(2, 1)$-shuffles.



Definition 8.4.13 Let σ be a (p, q)-shuffle. Its sign is defined by

$$\operatorname{sgn}(\sigma) := (-1)^{|I_\sigma|}, \quad I_\sigma := \{(i, j) \in I_- \times I_+ : i > j\}.$$

If σ is viewed as an ordering of p symbols $\to$ and q symbols $\uparrow$, then I_σ is precisely the set of pairs (j, i), with $j < i$, that occur in the reversed order $(\uparrow, \to)$. A simple combinatorial exercise shows that there is a unique $\tau \in \mathfrak{S}_{p+q}$ that rearranges the word into the form $\to \cdots \to \uparrow \cdots \uparrow$ without changing the internal order within either I_+ or I_-. Thus Definition 8.4.13 is equivalent to $\operatorname{sgn}(\sigma) = \operatorname{sgn}(\tau)$.

For a (p, q)-shuffle σ, exchanging the roles of σ_- and σ_+ gives a (q, p)-shuffle σ'. The interpretation above and a basic combinatorial argument give

$$\operatorname{sgn}(\sigma) = (-1)^{pq} \operatorname{sgn}(\sigma'). \tag{8.4.4}$$

In the same spirit, define a (p, q, r)-shuffle to be an injective monotone map $\sigma = (\sigma_-, \sigma_0, \sigma_+) : [p + q + r] \to [p] \times [q] \times [r]$, or, equivalently, regard it as an ordering of the symbols $\to$, $\nearrow$, and $\uparrow$. Define $\operatorname{sgn}(\sigma) := (-1)^{|I_\sigma|}$, where I_σ consists of all triples of indices $k > j > i$ corresponding to odd permutations of $(\rightarrow, \nearrow, \uparrow)$. The same standard combinatorial exercise shows that if the (p,q,r)-shuffle σ has a factorization

$$[p+q+r] \xrightarrow{\sigma_1} [p+q] \times [r] \xrightarrow{\sigma_2 \times \mathrm{id}_{[r]}} [p] \times [q] \times [r],$$

then σ_1 is a $(p+q,r)$-shuffle, σ_2 is a (p,q)-shuffle, and

$$\mathrm{sgn}(\sigma) = \mathrm{sgn}(\sigma_1)\,\mathrm{sgn}(\sigma_2). \tag{8.4.5}$$

There is, of course, a corresponding statement for a factorization $[p+q+r] \to [p] \times [q+r] \to [p] \times [q] \times [r]$.

Equations (8.4.4) and (8.4.5) are entirely combinatorial in nature; their detailed verification is left to the exercises for this chapter. These observations will be used in §8.8.

8.5 The Dold–Kan Correspondence

Following the conventions of algebraic topology, for an additive category $\mathcal{A}$ we consider the category of chain complexes over it (Definition 2.2.4), denoted by $\mathsf{Ch}(\mathcal{A})$, together with its subcategory

$$\mathsf{Ch}_{\geq 0}(\mathcal{A}) := \{A = (A_n, \partial_n)_{n\in\mathbb{Z}} : \text{chain complex},\ \forall n < 0,\ A_n = 0\},$$

where $\partial_n = \partial_n^A : A_n \to A_{n-1}$. For an object of $\mathsf{Ch}_{\geq 0}(\mathcal{A})$, it plainly suffices to specify the data $(A_n)_{n\geq 0}$ and $(\partial_n)_{n\geq 1}$. The category $\mathsf{Ch}_{\geq m}(\mathcal{A})$ is defined similarly; compare the cochain complex version in Definition 3.9.1.

By enlarging the chosen Grothendieck universe if necessary, we may assume that $\mathcal{A}$ is a small additive category; see [51, Assumption 1.5.2]. From $\mathcal{A}$ one can define the category $\mathsf{s}\mathcal{A}$ of simplicial objects; the constant simplicial object corresponding to 0 is still denoted by $0 \in \mathrm{Ob}(\mathsf{s}\mathcal{A})$. This section aims to give a first account of the relationships among these categories. More precisely, we shall define three functors

$$\begin{tikzcd} \mathsf{Ch}_{\geq 0}(\mathcal{A}) \arrow[r, "\Gamma"] & \mathsf{s}\mathcal{A} \arrow[l, bend right, "\mathrm{C}"'] \arrow[l, bend left, "\mathrm{N}"] \end{tikzcd}$$

where N is defined only when $\mathcal{A}$ is an abelian category. Both C and N extend to the semisimplicial objects of Remark 8.1.4; in other words, neither of them involves the degeneracy morphisms.

Our treatment follows [32, §1.2]. We begin by making the relevant definitions explicit.

Definition 8.5.1 (Unnormalized chain complex) For a semisimplicial object X in $\mathcal{A}$ and every $n \in \mathbb{Z}_{\geq 1}$, define

$$\partial_n := \sum_{i=0}^{n} (-1)^i d_i : X_n \to X_{n-1}.$$

The objects $(X_n)_{n \geq 0}$ together with $(\partial_n)_n$ form an object $\mathrm{C}X$ of $\mathsf{Ch}_{\geq 0}(\mathcal{A})$, called the **unnormalized chain complex**, or **Moore chain complex**, given by X. Thus $(\mathrm{C}X)_n = X_n$.

We must show that $\partial_{n-1}\partial_n = 0$ when $n > 1$. Indeed, the identity $i < j \implies d_i d_j = d_{j-1} d_i$ in (8.1.2) gives

$$\begin{aligned}\partial_{n-1}\partial_n &= \sum_{i=0}^{n-1}\sum_{j=0}^{n}(-1)^{i+j} d_i d_j \\ &= \sum_{0 \leq i < j \leq n} (-1)^{i+j} d_{j-1} d_i + \sum_{n-1 \geq i \geq j \geq 0} (-1)^{i+j} d_i d_j = 0.\end{aligned}$$

This construction is plainly canonical in X and gives the functor C.

Example 8.5.2 Let $\Bbbk$ be a commutative ring and $\mathcal{A} = \Bbbk\text{-}\mathsf{Mod}$. For a group Γ, Example 8.3.5 gives the corresponding simplicial set $\mathrm{E}\Gamma$. Define a simplicial object $\Bbbk\mathrm{E}\Gamma$ in $\Bbbk\text{-}\mathsf{Mod}$ whose term in degree n is the free $\Bbbk$-module with basis $(\mathrm{E}\Gamma)_n$, and whose d_i and s_i are extended linearly. We claim that $\mathrm{C}(\Bbbk\mathrm{E}\Gamma)$ is precisely the chain complex $\mathsf{L} := (\mathsf{L}_n, \partial'_n)_n$ introduced in Definition 6.1.10. First, L_n is by definition the free $\Bbbk$-module with basis Γ^{n+1}. Take the map from $(\mathrm{E}\Gamma)_n = \Gamma^{n+1}$ to L_n

$$(g_1, \dots, g_{n+1}) \mapsto (g_1 \cdots g_{n+1},\ \dots,\ g_n g_{n+1},\ g_{n+1}).$$

Extend it linearly to $(\Bbbk\mathrm{E}\Gamma)_n$. The reader may verify that $\sum_i (-1)^i d_i$ thereby corresponds to $\partial'_n : \mathsf{L}_n \to \mathsf{L}_{n-1}$, giving the asserted isomorphism.

Moreover, Γ acts on the right on each $(\mathrm{E}\Gamma)_n$ by $(g_1, \dots, g_{n+1}) \overset{g}{\mapsto} (g_1, \dots, g_n, g_{n+1}g)$. In the basis coordinates of L_n, this action becomes $(g_0, \dots, g_n) \overset{g}{\mapsto} (g_0 g, \dots, g_n g)$. Thus we recover the right Γ-action on L from Definition 6.1.10. Consequently, the complexes obtained by taking degreewise Γ-coinvariants of the chain complexes $\mathrm{C}(\Bbbk\mathrm{E}\Gamma)$ and L are also isomorphic. On the $\mathrm{C}(\Bbbk\mathrm{E}\Gamma)$ side, this amounts to taking the degreewise quotient of $\mathrm{E}\Gamma$ and then the free $\Bbbk$-module. It follows that

$$\mathrm{C}(\Bbbk\mathrm{B}\Gamma) \simeq \mathsf{L} \underset{\Bbbk[\Gamma]}{\otimes} \Bbbk,$$

where the tensor product involves the augmentation homomorphism $\Bbbk[\Gamma] \to \Bbbk$. As a consequence, group homology (Definition 6.2.5) is interpreted as

$$\mathrm{H}_n(\mathrm{B}\Gamma; \Bbbk) := \mathrm{H}_n\, \mathrm{C}(\Bbbk\mathrm{B}\Gamma) \simeq \mathrm{H}_n(\Gamma, \Bbbk), \quad n \in \mathbb{Z}_{\geq 0}.$$

The left-hand side features the homology of a simplicial set, a general construction to be introduced in Definition 8.6.3. The exercises in this chapter will give a similar isomorphism for the normalized version $\overline{\mathsf{L}}$ of L, corresponding to the normalized chain complex $\mathrm{N}(\Bbbk\mathrm{E}\Gamma)$ to be discussed in Definition 8.5.4.

Definition 8.5.3 Let $(A_n, \partial_n)_{n\geq 0}$ be an object of $\mathsf{Ch}_{\geq 0}(\mathcal{A})$. Define an object $\Gamma(A)$ of $\mathsf{s}\mathcal{A}$ as follows:

$$(\Gamma A)_n := \bigoplus_{t:[n]\twoheadrightarrow[k]} A_k.$$

For every morphism $f : [m] \to [n]$ in $\mathbf{\Delta}$, the corresponding morphism $f^* : (\Gamma A)_n \to (\Gamma A)_m$ is represented, with respect to the direct-sum decomposition above, by the matrix

$$(\Phi_{u,t} : A_k \to A_l)_{\substack{u:[m]\twoheadrightarrow[l] \\ t:[n]\twoheadrightarrow[k]}}.$$

This matrix is determined as follows. Given $t : [n] \twoheadrightarrow [k]$, take the surjection–injection factorization of tf,

$$\begin{tikzcd} & {[n]} \arrow[dr, two heads, "t"] & \\ {[m]} \arrow[ur, hook, "f"] \arrow[dr, two heads, "u"'] & & {[k]} \\ & {[l]} \arrow[ur, hook, "v"'] & \end{tikzcd} \tag{8.5.1}$$

- ◇ if $l = k$, set $\Phi_{u,t} = \mathrm{id}$;
- ◇ if $l = k - 1$ and $v = \mathrm{d}^0$ (the order-preserving injection omitting 0), set $\Phi_{u,t} = \partial_k : A_k \to A_{k-1}$;
- ◇ in all other cases, set $\Phi_{u,t} = 0$.

Notice that once f and t are given, there is at most one u for which $\Phi_{u,t} \neq 0$.

The verification that $\Gamma A \in \mathrm{Ob}(\mathsf{s}\mathcal{A})$ is somewhat tedious and is omitted. The heart of the construction lies in the direct summand A_n of $(\Gamma A)_n$ and its behavior under order-preserving injections. All lower-degree direct summands A_k arise from degeneracies induced by surjections $[n] \twoheadrightarrow [k]$; the definition of f^* in the general case merely implements this idea.

The construction above is canonical and gives a functor $\Gamma : \mathsf{Ch}_{\geq 0}(\mathcal{A}) \to \mathsf{s}\mathcal{A}$.

Definition 8.5.4 Let $\mathcal{A}$ be an abelian category. For a semisimplicial object X in $\mathcal{A}$ and every $n \in \mathbb{Z}_{\geq 0}$, define

$$\begin{aligned} (\mathrm{N}X)_n &:= \bigcap_{i=1}^{n} \ker\left[d_i : X_n \to X_{n-1}\right], \quad n \geq 1, \\ (\mathrm{N}X)_0 &:= X_0. \end{aligned}$$

The morphisms $\mathrm{N}X_n \to \mathrm{N}X_{n-1}$ induced by $X_n \xrightarrow{d_0} X_{n-1}$ (verify this using (8.1.2)) are henceforth denoted by ∂_n. Together with these objects, they form an object $\mathrm{N}X$ of $\mathsf{Ch}_{\geq 0}(\mathcal{A})$, called the **normalized chain complex** corresponding to X.

Proposition 8.5.5 For X as above and every $n \geq 0$, write $u_n : (\mathrm{N}X)_n \hookrightarrow X_n$ for the natural inclusion. Then $(u_n)_{n\geq 0}$ forms a monomorphism $u : \mathrm{N}X \to \mathrm{C}X$ in $\mathsf{Ch}_{\geq 0}(\mathcal{A})$; this construction is functorial in X.

Proof This follows directly from the definitions of $\mathrm{N}X$ and $\mathrm{C}X$. $\square$

The next few results are preparations for the Dold–Kan theorem, Theorem 8.5.9.

Lemma 8.5.6 Let $\mathcal{A}$ be an abelian category. For every $A \in \mathrm{Ob}(\mathsf{Ch}_{\geq 0}(\mathcal{A}))$ and $n \geq 0$, we have $\mathrm{N}\Gamma(A)_n = A_n$. These identifications give an isomorphism of functors $\eta : \mathrm{id}_{\mathsf{Ch}_{\geq 0}(\mathcal{A})} \xrightarrow{\sim} \mathrm{N}\Gamma$.

Proof Given $t : [n] \twoheadrightarrow [k]$ with $k < n$, one can always choose $1 \leq i \leq n$ such that $t^{-1}(t(i))$ has at least two elements. Consequently td^i is surjective, and the following diagram commutes:

$$\begin{tikzcd}
 & {[n]} \arrow[dr, two heads, "t"] & \\
{[n-1]} \arrow[ur, "\mathrm{d}^i"] \arrow[dr, two heads, "t\mathrm{d}^i"'] & & {[k]} \\
 & {[k]} \arrow[ur, "\mathrm{id}"'] &
\end{tikzcd}$$

Definition 8.5.3 then shows that $\mathrm{N}\Gamma(A)_n$ is contained in the direct summand A_n corresponding to $\mathrm{id} : [n] \to [n]$. The same definition readily gives $\mathrm{N}\Gamma(A)_n = A_n$ and shows that $\mathrm{N}\Gamma(A)_{n+1} \to \mathrm{N}\Gamma(A)_n$ is precisely $\partial_{n+1} : A_{n+1} \to A_n$. This proves the result. $\square$

Lemma 8.5.7 Let $\mathcal{A}$ be an abelian category. The functors above form an adjunction

$$\Gamma : \mathsf{Ch}_{\geq 0}(\mathcal{A}) \rightleftarrows \mathsf{s}\mathcal{A} : \mathrm{N}.$$

- ⋄ Its unit morphism is the isomorphism η of Lemma 8.5.6.
- ⋄ Its counit morphism $\epsilon : \Gamma\mathrm{N} \to \mathrm{id}_{\mathsf{s}\mathcal{A}}$ is described as follows. For $t : [n] \twoheadrightarrow [k]$, on the corresponding direct summand $(\mathrm{N}X)_k$ of $\Gamma\mathrm{N}(X)_n$, the map $\epsilon_{X,n}$ is the composite $(\mathrm{N}X)_k \subset X_k \xrightarrow{t^*} X_n$.

Proof For $A \in \mathrm{Ob}(\mathsf{Ch}_{\geq 0}(\mathcal{A}))$ and $X \in \mathrm{Ob}(\mathsf{s}\mathcal{A})$, we first prove that the composite

$$\mathrm{Hom}_{\mathsf{s}\mathcal{A}}(\Gamma(A), X) \xrightarrow{\mathrm{N}} \mathrm{Hom}_{\mathsf{Ch}_{\geq 0}(\mathcal{A})}(\mathrm{N}\Gamma(A), \mathrm{N}(X)) \xrightarrow{\eta_A^*} \mathrm{Hom}_{\mathsf{Ch}_{\geq 0}(\mathcal{A})}(A, \mathrm{N}(X))$$

is a bijection. Its inverse is defined concretely as follows. Given $\phi = (\phi_n)_{n\geq 0} : A \to \mathrm{N}(X)$, define a morphism

$$\Phi_n : (\Gamma A)_n = \bigoplus_{t:[n]\twoheadrightarrow[k]} A_k \to X_n$$

whose restriction to the direct summand corresponding to $t : [n] \twoheadrightarrow [k]$ is the composite

$$A_k \xrightarrow{\phi_k} (\mathrm{N}X)_k \subset X_k \xrightarrow{t^*} X_n.$$

For such a t and an arbitrary $f : [m] \to [n]$, take the surjection–injection factorization of tf as in (8.5.1), and consider the following diagram in $\mathcal{A}$ (see Definition 8.5.3):

$$\begin{tikzcd}
A_k \arrow[r, "\phi_k"] \arrow[d, "\Phi_{u,t}"'] & (\mathrm{N}X)_k \arrow[r, hook] \arrow[d, "v^*|_{(\mathrm{N}X)_k}"] & X_k \arrow[r, "t^*"] \arrow[d, "v^*"] & X_n \arrow[d, "f^*"] \\
A_l \arrow[r, "\phi_l"'] & (\mathrm{N}X)_l \arrow[r, hook] & X_l \arrow[r, "u^*"'] & X_m.
\end{tikzcd}$$

The left square commutes by the definition of the normalized chain complex $\mathrm{N}X$, the middle square plainly commutes, and the right square commutes because $tf = vu$. Thus $\Phi = (\Phi_n)_n$ is a morphism in $\mathsf{s}\mathcal{A}$.

The verification that $\eta^*_A\mathrm{N}$ and $\phi \mapsto \Phi$ are mutually inverse is routine. In the description of $\phi \mapsto \Phi$, take $\phi = \mathrm{id}_{\mathrm{N}X}$; the result is exactly the asserted ϵ. $\square$

Lemma 8.5.8 Take $\mathcal{A} = \mathsf{Ab}$. Then Γ is an equivalence; more precisely, the adjunction $\Gamma : \mathsf{Ch}_{\geq 0}(\mathsf{Ab}) \rightleftarrows \mathsf{sAb} : \mathrm{N}$ is an adjoint equivalence in the sense of [51, Theorem 2.6.12].

Proof It suffices to prove that the counit morphism ϵ of Lemma 8.5.7 is an isomorphism. Let $X \in \mathrm{Ob}(\mathsf{s}\mathcal{A})$. We claim that $\epsilon_{X,n} : \Gamma(\mathrm{N}X)_n \to X_n$ is injective for every $n \geq 0$. Let $x = (x_t)_t$ be an element of its domain, where t ranges over all order-preserving surjections $[n] \twoheadrightarrow [k]$. For a given t, define the order-preserving injection

$$\begin{aligned} s = s_t : [k] &\hookrightarrow [n] \\ i &\mapsto \min t^{-1}(i), \end{aligned}$$

which satisfies $ts = \mathrm{id}_{[k]}$. Suppose that $x \neq 0$, and write

$$S := \{t : x_t \neq 0\} \neq \varnothing.$$

Choose the smallest k for which there is a $t : [n] \twoheadrightarrow [k]$ in S, then choose such a t minimizing $\sum_{i=0}^k \min t^{-1}(i)$, and construct s. We shall prove that $s^*(\epsilon_{X,n}(x)) = x_t \in X_k$, which shows that $\epsilon_{X,n}(x) \neq 0$.

By the concrete description of $\epsilon_{X,n}$, it is enough to show that for every $t' : [n] \twoheadrightarrow [k']$, $s^*(t')^*(x_{t'}) \neq 0$ implies $t = t'$. Set $u := t's : [k] \to [k']$. If $t' \notin S$, then $x_{t'} = 0$. Thus it remains to consider $t' \in S$ satisfying $s^*(t')^*(x_{t'}) = u^*(x_{t'}) \neq 0$.

Since t is order-preserving and surjective, $\min t^{-1}(0) = 0$, and the same holds for t', so $u(0) = 0$. Moreover, $x_{t'} \in (\mathrm{N}X)_{k'}$ implies that $u^*(x_{t'}) \neq 0$ is possible only if $\mathrm{im}(u) \supset \{1, \ldots, k'\}$. Hence we may assume that u is surjective. The minimality of k then forces $k' = k$ and $u = \mathrm{id}$.

Consequently, $t'(\min t^{-1}(i)) = i$ for $i = 0, \ldots, k$, and hence $\min(t')^{-1}(i) \leq \min t^{-1}(i)$. By the choice of t, we obtain $\min(t')^{-1}(i) = \min t^{-1}(i)$ for every $0 \leq i \leq k$. An order-preserving surjection is determined by the initial points of its fibers, so this is equivalent to $t = t'$. Thus $\epsilon_{X,n}$ is injective.

We now prove by induction on n that $\epsilon_{X,n}$ is surjective. We claim that, for every $0 \leq i \leq n$,

$$\mathrm{im}\,(\epsilon_{X,n}) \supset X(i)_n := \bigcap_{i<j\leq n} \ker(d_j) \subset X_n.$$

For $i = 0$, we have $X(i)_n = (\mathrm{N}X)_n$, and the inclusion above follows readily from the concrete description of $\epsilon_{X,n}$. Our goal is the case $i = n$. Let $n \geq i \geq 1$ and $y \in X(i)_n$. The induction hypothesis for $n-1$ and the description of $\epsilon_{X,n}$ give

$$s_{i-1}d_i(y) \in s_{i-1}\left(\mathrm{im}\,(\epsilon_{X,n-1})\right) \subset \mathrm{im}\,(\epsilon_{X,n})\,.$$

On the other hand, the simplicial identities (8.1.2) give

$$\begin{aligned} d_i s_{i-1} d_i &= d_i, \\ j > i \implies d_j s_{i-1} d_i &= s_{i-1} d_{j-1} d_i = s_{i-1} d_i d_j. \end{aligned}$$

Thus $y - s_{i-1}d_i(y) \in X(i-1)_n$, and the induction hypothesis for $i-1$ shows that this element lies in $\mathrm{im}(\epsilon_{X,n})$. Therefore $y \in \mathrm{im}(\epsilon_{X,n})$, as required. □

Theorem 8.5.9 (A. Dold, D. Kan) For every additive category $\mathcal{A}$, the functor $\Gamma : \mathsf{Ch}_{\geq 0}(\mathcal{A}) \to \mathsf{s}\mathcal{A}$ is fully faithful.

- ◇ If $\mathcal{A}$ is also a Karoubi category as mentioned in §2.5, meaning that every idempotent has a kernel, then Γ is an equivalence of categories.
- ◇ If, in addition, $\mathcal{A}$ is an abelian category (and hence also a Karoubi category), then N is a quasi-inverse functor to Γ, and $\Gamma : \mathsf{Ch}_{\geq 0}(\mathcal{A}) \rightleftarrows \mathsf{s}\mathcal{A} : \mathrm{N}$ is an adjoint equivalence. The unit and counit morphisms of the adjunction are described in Lemma 8.5.7.

Proof Consider the functor

$$\tilde{h}_{\mathcal{A}} : \mathcal{A} \to \tilde{\mathcal{A}}^{\wedge} := \mathsf{Ab}^{(\mathcal{A}^{\mathrm{op}})}, \quad Y \mapsto \mathrm{Hom}_{\mathcal{A}}(\cdot, Y).$$

Its composite with the forgetful functor $\mathsf{Ab} \to \mathsf{Set}$ is the Yoneda embedding $h_{\mathcal{A}} : \mathcal{A} \to \mathcal{A}^\wedge$ recalled in §A.1. The functor $\tilde{h}_{\mathcal{A}}$ is fully faithful. This is simply the Ab-enriched version of the Yoneda lemma, but it can also be deduced from the original version. For every $Y_1, Y_2 \in \mathrm{Ob}(\mathcal{A})$, the composite map

$$\mathrm{Hom}_{\mathcal{A}}(Y_1, Y_2) \to \mathrm{Hom}_{\tilde{\mathcal{A}}^\wedge}\left(\tilde{h}_{\mathcal{A}}(Y_1), \tilde{h}_{\mathcal{A}}(Y_2)\right) \to \mathrm{Hom}_{\mathcal{A}^\wedge}\left(h_{\mathcal{A}}(Y_1), h_{\mathcal{A}}(Y_2)\right)$$

is known to be bijective, while the second map is plainly injective. Therefore both maps are bijective.

Proposition 2.1.4 shows that $\tilde{\mathcal{A}}^\wedge$ is naturally an abelian category. Use the same symbol $\tilde{h}_{\mathcal{A}}$ for the induced functors on chain complexes and simplicial objects, and consider the diagram

$$\begin{tikzcd}
\mathsf{Ch}_{\geq 0}(\mathcal{A}) \arrow[r, "\Gamma"] \arrow[d, "\tilde{h}_{\mathcal{A}}"'] & \mathsf{s}\mathcal{A} \arrow[d, "\tilde{h}_{\mathcal{A}}"] \\
\mathsf{Ch}_{\geq 0}(\tilde{\mathcal{A}}^\wedge) \arrow[r, "\Gamma"'] & \mathsf{s}\tilde{\mathcal{A}}^\wedge.
\end{tikzcd}$$

This diagram commutes up to a canonical isomorphism. The observation in the preceding paragraph shows that both vertical arrows are fully faithful, while applying Lemma 8.5.8 objectwise shows that the second row is an equivalence of categories. Hence Γ in the first row is fully faithful.

For any functor $F : \mathcal{C} \to \mathcal{C}'$, an object $X' \in \mathrm{Ob}(\mathcal{C}')$ is said to lie in the **essential image** of F if there is an $X \in \mathrm{Ob}(\mathcal{C})$ such that $X' \simeq FX$. Together with the description in Lemma 8.5.8 of the quasi-inverse to $\mathsf{Ch}_{\geq 0}(\mathsf{Ab}) \to \mathsf{sAb}$, this yields the following conclusion: $X \in \mathrm{Ob}(\mathsf{s}\mathcal{A})$ lies in the essential image of Γ if and only if $(\mathrm{N}\tilde{h}_{\mathcal{A}}(X))_n$ lies in the essential image of $\tilde{h}_{\mathcal{A}} : \mathcal{A} \to \tilde{\mathcal{A}}^\wedge$ for every $n \geq 0$.

Notice that $\tilde{h}_{\mathcal{A}} : \mathcal{A} \to \tilde{\mathcal{A}}^\wedge$ preserves finite direct sums. If $\mathcal{A}$ is a Karoubi category, the essential image of $\mathcal{A} \to \tilde{\mathcal{A}}^\wedge$ is therefore closed under taking direct summands. Since $(\mathrm{N}\tilde{h}_{\mathcal{A}}(X))_n$ is a direct summand of $(\Gamma\mathrm{N}\tilde{h}_{\mathcal{A}}(X))_n \simeq \tilde{h}_{\mathcal{A}}(X)_n$,[2)] in this case Γ is fully faithful and essentially surjective, hence an equivalence.

Finally, the adjoint equivalence theorem [51, Theorem 2.6.12] ensures that a quasi-inverse to $\Gamma : \mathsf{Ch}_{\geq 0}(\mathcal{A}) \to \mathsf{s}\mathcal{A}$ can always be made into its right adjoint. If $\mathcal{A}$ is an abelian category, Lemma 8.5.7 has already shown that N is the right adjoint of Γ, and hence is also its quasi-inverse. This proves the result. □

Convention 8.5.10 In view of this result, if $\mathcal{A}$ is a Karoubi category, we may choose a quasi-inverse functor to $\Gamma : \mathsf{Ch}_{\geq 0}(\mathcal{A}) \to \mathsf{s}\mathcal{A}$ and naturally denote it by N.

[2)] Translator's note: the final term is printed as $h_{\mathcal{A}}(X)_n$ in the source. The other objects in this isomorphism are Ab-valued presheaves, so the correctly typed term is $\tilde{h}_{\mathcal{A}}(X)_n$.

Although NX was initially defined as a subobject of CX, it can also be understood as a quotient object; the latter viewpoint is often more convenient. We now explain this.

Let $\mathcal{A}$ be a Karoubi category and $X \in \mathrm{Ob}(\mathsf{s}\mathcal{A})$. For every $n \geq 0$, define the morphism Ψ_n in $\mathcal{A}$ as the one characterized by the commutativity of the following diagram for every $0 \leq j < n$:

$$\begin{tikzcd} \bigoplus_{0 \leq i < n} X_{n-1} \arrow[r, "\Psi_n"] & X_n \\ X_j \arrow[u, hook, "\text{direct summand}"] \arrow[ur, "s_j"'] & \end{tikzcd}$$

Thus taking $\mathrm{coker}(\Psi_n)$ (that is, the coequalizer of Ψ_n and 0) intuitively removes every degenerate part from X_n, at least when $\mathcal{A}$ is an abelian category. The next result not only guarantees that $\mathrm{coker}(\Psi_n)$ exists, but also identifies it with $(\mathrm{N}X)_n$ via the natural embedding $u : \mathrm{N}X \to \mathrm{C}X$ in Proposition 8.5.5. This gives another perspective on NX.

Proposition 8.5.11 In the situation above, $\mathrm{coker}(\Psi_n)$ exists and is canonically isomorphic to $(\mathrm{N}X)_n$. Consider the corresponding composite $v_n : X_n \twoheadrightarrow \mathrm{coker}(\Psi_n) \xrightarrow{\sim} (\mathrm{N}X)_n$. Then $(v_n)_n$ gives a morphism $v : \mathrm{C}X \to \mathrm{N}X$ satisfying $vu = \mathrm{id}_{\mathrm{N}X}$ for the natural embedding $u : \mathrm{N}X \hookrightarrow \mathrm{C}X$; this construction is functorial in X.

Proof By Theorem 8.5.9, we may assume that $X = \Gamma A$, where $A \in \mathrm{Ob}(\mathsf{Ch}_{\geq 0}(\mathcal{A}))$. In this case, Ψ_n can be written as

$$\bigoplus_{0 \leq i < n} \bigoplus_{t:[n-1] \twoheadrightarrow [k]} A_k \to \bigoplus_{t':[n] \twoheadrightarrow [k]} A_k.$$

From Definition 8.5.3 of the functor Γ, we see that Ψ_n acts on the (i,t)-summand on the left as follows. Form the composite $[n] \xrightarrow{\mathrm{s}^i} [n-1] \xrightarrow{t} [k]$, which is still an order-preserving surjection, and denote it by t'; then A_k is mapped identically to the A_k-summand on the right corresponding to t'.

Observe that $t' : [n] \twoheadrightarrow [k]$ arises from such a pair (i,t) if and only if $k < n$. Using the η of Lemma 8.5.6, define v_n to be the composite $(\Gamma A)_n \xrightarrow{\text{projection}} A_n \underset{\sim}{\xrightarrow{\eta}} (\mathrm{N}\Gamma A)_n$. The discussion above shows that

$$[\text{quotient} : (\Gamma A)_n \to \mathrm{coker}\, \Psi_n] \simeq [v_n : (\Gamma A)_n \twoheadrightarrow (\mathrm{N}\Gamma A)_n].$$

Expanding the definitions immediately gives $v_n u_n = \mathrm{id}_{(\mathrm{N}\Gamma A)_n}$. We next show that $(v_n)_n$ gives a morphism v in $\mathsf{Ch}_{\geq 0}(\mathcal{A})$. This is equivalent to saying that the outer rectangle of the following diagram commutes:

$$\begin{tikzcd}
(\Gamma A)_n \arrow[r, "\sum_i (-1)^i d_i^{\Gamma A}"] \arrow[d, "\text{projection}"'] & (\Gamma A)_{n-1} \arrow[d, "\text{projection}"] \\
A_n \arrow[r, "\partial_n^A"] \arrow[d, "\eta"'] & A_{n-1} \arrow[d, "\eta"] \\
(\mathrm{N}\Gamma A)_n \arrow[r, "\partial_n^{\mathrm{N}\Gamma A}"'] & (\mathrm{N}\Gamma A)_{n-1}.
\end{tikzcd}$$

The lower part commutes because η is a morphism, while the commutativity of the upper part follows by directly applying the definition of $d_i^{\Gamma A}$ (take $f = \mathrm{d}^i$ in Definition 8.5.3). Finally, the functoriality of v in X is clear. $\square$

Our next goal is to compare C and N more precisely in the case of an abelian category. There is a notion of homotopy for morphisms between chain complexes; see Definition 3.2.6 and Remark 3.2.10. A morphism of chain complexes is called a **quasi-isomorphism** if it induces isomorphisms on homology.

Theorem 8.5.12 Let $\mathcal{A}$ be an abelian category and $X \in \mathrm{Ob}(\mathsf{s}\mathcal{A})$. Then $u : \mathrm{N}X \to \mathrm{C}X$ from Lemma 8.5.5 and $v : \mathrm{C}X \to \mathrm{N}X$ from Proposition 8.5.11 are both quasi-isomorphisms.

Proof We know that $vu = \mathrm{id}_{\mathrm{N}X}$, so it is enough to prove that v is a quasi-isomorphism. Since v has a right inverse, it is an epimorphism. The long exact sequence of chain complexes (Proposition 3.6.4) reduces the problem to proving that $\ker(v)$ is acyclic. By Theorem 8.5.9, we may henceforth assume that $X = \Gamma A$, where $A \in \mathrm{Ob}(\mathsf{Ch}_{\geq 0}(\mathcal{A}))$. Therefore

$$(\mathrm{C}X)_n = \bigoplus_{t:[n] \twoheadrightarrow [k]} A_k,$$
$$\partial_n : (\mathrm{C}X)_n \to (\mathrm{C}X)_{n-1}.$$

For $n \geq 0$ and $i \in \mathbb{Z}$, let $(\mathrm{C}^{\leq i} X)_n$ denote the subobject of $(\mathrm{C}X)_n$ cut out by the direct summands satisfying

$$\exists\, j, \quad \max\{n-i, 1\} \leq j \leq n, \quad t(j) = t(j-1).$$

We have $i \leq i' \implies (\mathrm{C}^{\leq i} X)_n \subset (\mathrm{C}^{\leq i'} X)_n$.

Since v_n is simply the projection onto the summand $t = \mathrm{id}_{[n]}$, if $i \geq n-1$ then $(\mathrm{C}^{\leq i} X)_n = \ker(v_n)$. We now show that

$$\big((\mathrm{C}^{\leq i} X)_n\big)_{n \geq 0} \text{ forms a chain subcomplex of } \mathrm{C}X \text{ denoted by } \mathrm{C}^{\leq i} X. \tag{8.5.2}$$

To prove this, fix a map $t : [n] \twoheadrightarrow [k]$ and an index $\max\{n-i, 1\} \leq j \leq n$ satisfying the condition above. From now on write $d_i = d_i^X$ and $s_j = s_j^X$. For every $0 \leq h \leq n$, take $m = n-1$ and $f = \mathrm{d}^h : [n-1] \hookrightarrow [n]$ in the diagram (8.5.1).

- ◇ Suppose $h \notin \{j-1, j\}$ and set $j' := (\mathrm{d}^h)^{-1}(j)$. Then $u : [n-1] \twoheadrightarrow [l]$ satisfies $u(j') = u(j'-1)$, and it is easy to see that $n-1-i \leq j' \leq n-1$. In this case, $(-1)^h d_h : X_n \to X_{n-1}$ maps A_k to the direct summand of $(\mathrm{C}^{\leq i}X)_{n-1}$ determined by u.
- ◇ In the situation of the preceding item, if also $h \geq n-i-1$, the range of j' can be sharpened to $n-i \leq j' \leq n-1$. In other words, the image of A_k then lies in $(\mathrm{C}^{\leq i-1}X)_{n-1}$.
- ◇ For $h \in \{j-1, j\}$, verify that $t\mathrm{d}^{j-1} = t\mathrm{d}^j$. It follows that $d_{j-1}, d_j : X_n \rightrightarrows X_{n-1}$ agree on the direct summand A_k corresponding to t, so $(-1)^{j-1}d_{j-1}$ and $(-1)^j d_j$ cancel on that summand.

This proves (8.5.2). The second and third items also give the congruence

$$\sum_{h=0}^{n}(-1)^h d_h \equiv \sum_{h=0}^{n-i-2}(-1)^h d_h \pmod{\mathrm{C}^{\leq i-1}X}. \tag{8.5.3}$$

Thus it is enough to prove that $\mathrm{C}^{\leq i}X$ is acyclic. Set

$$D := \mathrm{C}^{\leq i}X/\mathrm{C}^{\leq i-1}X.$$

Notice that $\mathrm{C}^{<-1}X = 0$. The long exact sequence of chain complexes then reduces the problem to proving inductively that D is acyclic, for $i \geq 0$.

To prove that D is acyclic, consider the family of morphisms

$$(-1)^{n-i-1}s_{n-i-1} : X_n \to X_{n+1}, \quad n \geq i+1.$$

A direct verification shows that these morphisms preserve both $(\mathrm{C}^{\leq i-1}X)_\bullet$ and $(\mathrm{C}^{\leq i}X)_\bullet$ (this choice of bounds is optimal), and therefore induce

$$h_n : D_n \to D_{n+1}.$$

Extend the definition of s_k by zero to every $k \in \mathbb{Z}$, and thereby extend the definition of h_n by zero to the case $n \leq i$. Notice that $n \leq i \implies D_n = 0$, so this is the only reasonable choice. We claim that

$$\partial_{n+1}^D h_n + h_{n-1}\partial_n^D = \mathrm{id}_{D_n}, \quad n \in \mathbb{Z}.$$

This will show that D is acyclic. First, applying (8.5.3) gives

$$\partial_n^D = \sum_{j=0}^{n-i-2} (-1)^j d_j \mod \mathrm{C}^{\leq i-1}(X)_{n-1}.$$

Together with (8.1.2), this gives, for $n \geq i+1$,

$$\begin{aligned}
\partial_{n+1}^D h_n &= (-1)^{n-i-1} \sum_{j=0}^{n-i-1} (-1)^j d_j s_{n-i-1} \mod \mathrm{C}^{\leq i-1}(X)_n \\
&= (-1)^{n-i-1} \sum_{j=0}^{n-i-2} (-1)^j s_{n-i-2} d_j + \mathrm{id}_{\mathrm{C}^{\leq i}(X)_n} \mod \mathrm{C}^{\leq i-1}(X)_n \\
&= -h_{n-1}\partial_n^D + \mathrm{id}_{D_n}.
\end{aligned}$$

The identity also holds trivially for $n \leq i$. This proves the claim. □

Remark 8.5.13 Naturally, one may ask for corresponding versions of the various theorems in this section, particularly the Dold–Kan correspondence, for the category of complexes $\mathsf{C}(\mathcal{A})$ or its subcategories $\mathsf{C}_{\leq 0}(\mathcal{A})$ and $\mathsf{C}_{\geq 0}(\mathcal{A})$. There are at least two simple approaches: reflection, and reversal or duality.

1. As explained in Remark 2.2.6, $\mathsf{C}(\mathcal{A})$ and $\mathsf{Ch}(\mathcal{A})$ correspond by $X^n = X_{-n}$ and $d^n = \partial_{-n}$. All the results in this section can be transported to complexes through this correspondence. For example, the reflected version of Theorem 8.5.9 is an adjoint equivalence between $\mathsf{C}_{\leq 0}(\mathcal{A})$ and $\mathsf{s}\mathcal{A}$.

2. Another approach is to consider the category of cosimplicial objects $\mathsf{cs}\mathcal{A}$. There are equivalences of categories

$$\mathsf{cs}\mathcal{A} \simeq \mathsf{s}\left(\mathcal{A}^{\mathrm{op}}\right)^{\mathrm{op}} \simeq \mathsf{Ch}_{\geq 0}(\mathcal{A}^{\mathrm{op}})^{\mathrm{op}} \simeq \mathsf{C}_{\geq 0}(\mathcal{A}),$$

where the final step comes from reversing the arrows, as in Remark 2.2.6.

Operations such as taking the normalized complex can likewise be translated into the category of complexes.

8.6 Homology Computation

A classical application of the functors C and N is the definition of the homology of a simplicial set, which is closely connected to the study of algebraic topology.

Definition 8.6.1 For every $K \in \mathrm{Ob}(\mathsf{sSet})$, define $\mathbb{Z}K \in \mathrm{Ob}(\mathsf{sAb})$ by taking $(\mathbb{Z}K)_n$ to be the free $\mathbb{Z}$-module with basis K_n, with the maps d_i, s_j induced by the corresponding maps on K. This construction gives a functor $\mathbb{Z}(\cdot) : \mathsf{sSet} \to \mathsf{sAb}$.

Proposition 8.6.2 There is a natural adjunction $\mathbb{Z}(\cdot) : \mathsf{sSet} \rightleftarrows \mathsf{sAb} : \text{forgetful}$.

Proof Both $\mathbb{Z}(\cdot)$ and the forgetful functor are defined degreewise. The entire assertion follows directly from the free–forgetful adjunction $\mathsf{Set} \leftrightarrows \mathsf{Ab}$. □

Consequently, for every $X \in \mathrm{Ob}(\mathsf{sAb})$, specifying a morphism $\mathbb{Z}\Delta^n \to X$ is equivalent to specifying an element of X_n.

Moreover, $\mathbb{Z}(X \times Y) \simeq \mathbb{Z}X \otimes \mathbb{Z}Y$, where the tensor product on the right is formed degreewise, with n-th term $\mathbb{Z}X_n \underset{\mathbb{Z}}{\otimes} \mathbb{Z}Y_n$.

Definition 8.6.3 For a simplicial set K and every $n \in \mathbb{Z}_{\geq 0}$, write $\mathrm{H}_n(K;\mathbb{Z}) := \mathrm{H}_n\big(\mathrm{C}(\mathbb{Z}K)\big)$; this group is called the ***n*-th homology group of *K***. The construction gives a family of functors $\mathrm{H}_n(\cdot;\mathbb{Z}) : \mathsf{sSet} \to \mathsf{Ab}$.

- ◇ More generally, for a $\mathbb{Z}$-module M, homology with coefficients in M is defined by $\mathrm{H}_n(K;M) := \mathrm{H}_n\big(\mathrm{C}(\mathbb{Z}K) \otimes M\big)$, where $\cdot \otimes M$ means taking $\cdot \underset{\mathbb{Z}}{\otimes} M$ in every term of the chain complex.

- ◇ Cohomology with coefficients in M is defined by $\mathrm{H}^n(K;M) := \mathrm{H}^n\big(\mathrm{Hom}_{\mathbb{Z}}(\mathrm{C}(\mathbb{Z}K), M)\big)$, with the expression in parentheses regarded as a complex. By the universal property of $\mathbb{Z}(\cdot)$, the n-th term of this complex can also be identified with

$$\mathrm{Maps}(K_n, M) := \{\text{maps } K_n \to M\}.$$

  This set becomes a $\mathbb{Z}$-module under pointwise operations, and its differential homomorphism is identified with $\sum_i (-1)^i (d_i)^*$.

If $\mathrm{C}(\mathbb{Z}K)$ in the definition above is replaced by $\mathrm{N}(\mathbb{Z}K)$, the resulting homology (or cohomology) groups are canonically isomorphic; this is a consequence of Theorem 8.5.12. Using $\mathrm{N}(\mathbb{Z}K)$ is sometimes simpler, because Proposition 8.5.11 gives

$$\mathrm{N}(\mathbb{Z}K)_n = \bigoplus_{x \in K_n^{\mathrm{nd}}} \mathbb{Z}x.$$

Here $K_n^{\mathrm{nd}} \subset K_n$ is the subset of nondegenerate n-simplices, while the map $\mathrm{N}(\mathbb{Z}K)_n \to \mathrm{N}(\mathbb{Z}K)_{n-1}$ is obtained by first taking $\sum_{i=0}^n (-1)^i d_i$ and then projecting to the K_{n-1}^{nd} part. For example,

$$\begin{aligned} \mathrm{N}(\mathbb{Z}\Delta^0) &= \mathbb{Z} \text{ placed in degree } 0, \\ \mathrm{C}(\mathbb{Z}\Delta^0) &= \left[\cdots \xrightarrow{1} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{1} \mathbb{Z} \xrightarrow{0} \mathbb{Z}\right]. \end{aligned}$$

Example 8.6.4 As a useful special case, let $(Q, \leq)$ be a nonempty small poset with a least element e. This gives order-preserving maps

$$[0] \xrightarrow{0 \mapsto e} Q \longrightarrow [0].$$

Regard Q as a category and take the nerve $S := \mathrm{N}(Q)$ from Example 8.3.1. By the original definition of the nerve, we also have $\mathrm{N}([n]) = \Delta^n$. We thus obtain morphisms in **sSet**

$$\Delta^0 \xrightarrow{i} S \xrightarrow{q} \Delta^0, \qquad qi = \mathrm{id}_{\Delta^0}.$$

It follows that we have

$$\mathbb{Z} = \mathrm{N}(\mathbb{Z}\Delta^0) \xrightarrow{\mathrm{N}i} \mathrm{N}(\mathbb{Z}S) \xrightarrow{\mathrm{N}q} \mathbb{Z},$$

where $\mathbb{Z}$ is regarded as a chain complex placed in degree 0. We claim that $\mathrm{N}i$ and $\mathrm{N}q$ are mutually inverse in the homotopy category of chain complexes. In particular, $\mathrm{H}_0(Q; M) \simeq M$, while $\mathrm{H}_n(Q; M) = \{0\}$ for $n \neq 0$.

Since $qi = \mathrm{id}$, it remains only to construct a homotopy from $\mathrm{N}i\mathrm{N}q$ to id. For every $n \geq 0$, observe that the elements of S_n^{nd} are the chains in Q of the form $q_0 < \cdots < q_n$. Define

$$\begin{aligned} h_n : S_n^{\mathrm{nd}} &\longrightarrow S_{n+1}^{\mathrm{nd}} \cup \{0\}, \qquad n \geq 0, \\ q_0 < \cdots < q_n &\longmapsto \begin{cases} e < q_0 < \cdots < q_n, & q_0 \neq e, \\ 0, & q_0 = e. \end{cases} \end{aligned}$$

Extend this map linearly to $\mathbb{Z}S_n^{\mathrm{nd}} \simeq \mathrm{N}(\mathbb{Z}S)_n$. It readily satisfies the conditions for a homotopy; the details are left to the reader as an exercise.

The computation of homology is inseparable from homotopy. Under the Dold–Kan correspondence, homotopies of chain complexes reflect homotopies of simplicial objects, and the latter can be defined combinatorially.

Definition 8.6.5 (Simplicial homotopy) Let X and Y be simplicial objects in a category $\mathcal{C}$, and let $f, g : X \rightrightarrows Y$ be a pair of morphisms. A **simplicial homotopy from g to f** is a family of morphisms

$$h_i = h_i^n : X_n \to Y_{n+1}, \qquad 0 \leq i \leq n,$$

briefly denoted by h, satisfying

$$\begin{aligned} d_0 h_0 &= f_n, \\ d_{n+1} h_n &= g_n, \\ d_i h_j &= \begin{cases} h_{j-1} d_i, & 0 \leq i < j, \\ d_i h_{i-1}, & 1 \leq i = j, \\ h_j d_{i-1}, & i > j+1, \end{cases} \\ s_i h_j &= \begin{cases} h_{j+1} s_i, & i \leq j, \\ h_j s_{i-1}, & i > j. \end{cases} \end{aligned}$$

The definition above is combinatorial. When $\mathcal{C} = \mathsf{Set}$, however, it is the same as (8.4.2), which is based on topological intuition. Why is this so? For every $n \in \mathbb{Z}_{\geq 0}$, write $(\Delta^1)_n = \{\alpha_{-1}, \ldots, \alpha_n\}$ explicitly, where $\alpha_i : [n] \to [1]$ maps the numbers $\leq i$ to 0 and all the others to 1. Then specifying $H_n : (X \times \Delta^1)_n \to Y_n$ is equivalent to specifying $H^n_{-1}, \ldots, H^n_n : X_n \to Y_n$.

- Given the data $(h^n_i)_{i,n}$ of a simplicial homotopy, define $H^n_{-1} = g$, $H^n_n = f$, and $H^n_i = d_{n+1} h^n_i$ for $0 \leq i \leq n-1$.
- Given the data H of the commutative diagram (8.4.2), define $h^n_i := H^{n+1}_i s_i : X_n \to Y_{n+1}$.

A routine verification shows that the two maps are well defined and mutually inverse.

For the case $\mathcal{C} = \mathsf{Ab}$, simply replace $X \times \Delta^1$ in (8.4.2) by $X \otimes \mathbb{Z}\Delta^1$ and work in sAb; the same argument applies without change. The following result is therefore entirely unsurprising.

Proposition 8.6.6 Consider a pair of morphisms $f, g : X \rightrightarrows Y$ in $\mathsf{s}\mathcal{A}$. If h is a simplicial homotopy from g to f, then $\left(\sum_{j=0}^n (-1)^j h^n_j\right)_{n \geq 0}$ defines a homotopy between $\mathrm{C}f, \mathrm{C}g : \mathrm{C}X \rightrightarrows \mathrm{C}Y$.

Proof As usual, denote all face morphisms of X and Y uniformly by d_i, and omit the superscripts on h_j; this creates no ambiguity. We have the following equality in $\mathrm{Hom}_{\mathcal{A}}(X_n, Y_n)$:

$$\begin{aligned} &\sum_{i=0}^{n+1} (-1)^i d_i \sum_{j=0}^{n} (-1)^j h_j + \sum_{j=0}^{n-1} (-1)^j h_j \sum_{i=0}^{n} (-1)^i d_i \\ &\qquad = f_n - g_n + \sum_{\substack{0 \leq i \leq n+1 \\ 0 \leq j \leq n \\ (i,j) \neq (0,0), (n+1,n)}} (-1)^{i+j} d_i h_j + \sum_{\substack{0 \leq i \leq n \\ 0 \leq j \leq n-1}} (-1)^{i+j} h_j d_i. \end{aligned}$$

The first sum on the last line can be rewritten as

$$\sum_{\substack{0\le i\le n+1\\ i<j\le n}} (-1)^{i+j} h_{j-1} d_i + \sum_{i=1}^{n} d_i h_i - \sum_{i=1}^{n} d_i h_{i-1} + \sum_{\substack{1\le i\le n+1\\ 0\le j<i-1}} (-1)^{i+j} h_j d_{i-1}$$
$$= - \sum_{\substack{0\le i\le n\\ i\le j\le n-1}} (-1)^{i+j} h_j d_i - \sum_{\substack{0\le i\le n\\ 0\le j<i}} (-1)^{i+j} h_j d_i.$$

Here we used $1 \le i \le n \implies d_i h_i = d_i h_{i-1}$. This expression cancels the second sum and leaves only $f_n - g_n$. □

We next consider augmented semisimplicial objects (Remark 8.1.4) and their corresponding chain complexes; these results will be used in §8.7. Unless stated otherwise, the augmentation morphism of an augmented semisimplicial object X will always be denoted by $\epsilon : X_0 \to X_{-1}$.

Definition 8.6.7 For an augmented semisimplicial object X in an additive category $\mathcal{A}$, the corresponding complex $\mathrm{C}X$ can be extended to the chain complex

$$\mathrm{C}^{\mathrm{aug}}X := \left[\cdots \xrightarrow{\partial_3} X_2 \xrightarrow{\partial_2} X_1 \xrightarrow{\partial_1} X_0 \xrightarrow{\partial_0 := \epsilon} X_{-1} \to 0 \to \cdots\right],$$
$$\partial_n := \sum_{i=0}^{n} (-1)^i d_i : X_n \to X_{n-1}, \qquad n \ge 1.$$

The equality $\epsilon d_0 = \epsilon d_1$ shows that $\mathrm{C}^{\mathrm{aug}}X$ is indeed an object of $\mathsf{Ch}_{\ge -1}(\mathcal{A})$. It can also be regarded as the following morphism in $\mathsf{Ch}_{\ge 0}(\mathcal{A})$:

$$\epsilon : \mathrm{C}X \to \underbrace{X_{-1}}_{\text{placed in degree } 0}, \qquad \epsilon \text{ in degree 0, and 0 in every other degree.}$$

Definition 8.6.8 Let X be an augmented semisimplicial object in an arbitrary category $\mathcal{C}$. If there is a family of morphisms $k_n : X_n \to X_{n+1}$, with $n \in \mathbb{Z}_{\ge -1}$, call X **left contractible** or **right contractible**, according to which column of conditions below is satisfied:

Left contractible	Right contractible
$\epsilon k_{-1} = \mathrm{id}_{X_{-1}}$	$\epsilon k_{-1} = \mathrm{id}_{X_{-1}}$
$d_0 k_n = \mathrm{id}_{X_n}$	$d_{n+1} k_n = \mathrm{id}_{X_n}$
$d_i k_n = k_{n-1} d_{i-1}$ $(1 \le i \le n+1,\ n \ge 1)$	$d_i k_n = k_{n-1} d_i$ $(0 \le i \le n,\ n \ge 1)$
$d_1 k_0 = k_{-1}\epsilon$	$d_0 k_0 = k_{-1}\epsilon$

The reader may verify that left and right contractibility are order-reversal duals in the sense of Remark 8.1.7. Moreover, both are "absolute" properties: if X is left (or right) contractible, with corresponding data $(k_n)_{n\geq -1}$, then for every functor $F: \mathcal{C} \to \mathcal{D}$ the data $(Fk_n)_{n\geq -1}$ make $F(X)$ left (or right) contractible.

Example 8.6.9 For any monoid Γ, make the $\mathrm{E}\Gamma$ of Example 8.3.5 into an augmented simplicial set by taking $(\mathrm{E}\Gamma)_{-1} = \{1\}$ and the augmentation morphism $\epsilon : (\mathrm{E}\Gamma)_0 \to (\mathrm{E}\Gamma)_{-1}$ to be the constant map. This object is right contractible: for $n \geq -1$, define $k_n : (\mathrm{E}\Gamma)_n \to (\mathrm{E}\Gamma)_{n+1}$ by $(g_1, \dots, g_{n+1}) \mapsto (g_1, \dots, g_{n+1}, 1)$. This map plainly satisfies $\epsilon k_{-1} = \mathrm{id}$; after expanding the definitions, the verifications that $d_{n+1}k_n = \mathrm{id}$, $d_i k_n = k_{n-1}d_i$ for $0 \leq i \leq n$, $n \geq 1$, and $d_0 k_0 = k_{-1}\epsilon$ are also immediate.

If $\Bbbk$ is a commutative ring, identify the chain complex $\mathrm{C}(\Bbbk\mathrm{E}\Gamma)$ with the chain complex $\mathsf{L} = (\mathsf{L}_n, \partial'_n)_n$ from Definition 6.1.10, as in Example 8.5.2. The augmentation $\mathsf{L} \to \Bbbk$ is precisely the augmentation of the resolution in Proposition 6.1.13. In view of the next proposition, the contractibility of $\mathrm{E}\Gamma$ (and hence of $\Bbbk\mathrm{E}\Gamma$) gives a topological explanation for the fact that $\mathsf{L} \to \Bbbk$ is a quasi-isomorphism.

Proposition 8.6.10 Let $\mathcal{A}$ be an additive category and let X be a left or right contractible augmented semisimplicial object in $\mathcal{A}$. Then the identity morphism from $\mathrm{C}^{\mathrm{aug}}X$ to itself is null-homotopic.

Proof First consider the left-contractible case, and take the family of morphisms $k_n : X_n \to X_{n+1}$ from its definition. Define $k_n = 0$ for $n < -1$ (or $\partial_n = 0$ for $n < 0$), and also define $\partial_0 = \epsilon$. We want to use $(k_n)_n$ as a witness to the null homotopy, that is, to prove

$$\partial_{n+1}k_n + k_{n-1}\partial_n = \mathrm{id}_{X_n}.$$

For $n < -1$, both sides equal 0. For $n = -1$, the left-hand side is $\mathrm{id}_{X_{-1}}$. For $n = 0$, the left-hand side is $d_0k_0 - d_1k_0 + k_{-1}\epsilon = \mathrm{id}_{X_0}$. For $n \geq 1$, computing the left-hand side gives

$$\begin{aligned}&\sum_{i=0}^{n+1}(-1)^i d_i k_n + \sum_{i=0}^{n}(-1)^i k_{n-1}d_i \\ &\qquad = \mathrm{id}_{X_n} + \sum_{i=1}^{n+1}(-1)^i k_{n-1}d_{i-1} + \sum_{i=0}^{n}(-1)^i k_{n-1}d_i = \mathrm{id}_{X_n}.\end{aligned}$$

For the right-contractible case, use $\big((-1)^{n+1}k_n\big)_n$ instead. □

8.7 The Bar Construction

Definition 7.6.1 explains what is meant by a monad on a category $\mathcal{C}$, as well as its dual notion, a comonad. The starting point of this section is a way to produce an augmented cosimplicial (respectively, simplicial) object in $\mathrm{End}(\mathcal{C})$ from a monad (respectively, a comonad). In this way, many basic constructions in homological algebra can be explained naturally, including Hochschild homology and cohomology, which will be revisited at the end of the section (Example 8.7.7).

Example 8.7.1 Let (T, μ, η) be a monad on a category $\mathcal{C}$, that is, an algebra in the category of endofunctors $\mathrm{End}(\mathcal{C}) = \mathcal{C}^{\mathcal{C}}$. The "walking algebra" construction of Remark 7.1.17 gives the corresponding monoidal functor

$$\mathsf{FinOrd} \to \mathrm{End}(\mathcal{C}), \quad \mathbf{n} \mapsto T^n,$$

that is, an augmented cosimplicial object in $\mathrm{End}(\mathcal{C})$. Dually, a comonad (L, δ, ϵ) on a category $\mathcal{D}$ gives an augmented simplicial object in $\mathrm{End}(\mathcal{D})$; here $\delta : L \to L^2$ and $\epsilon : L \to \mathrm{id}_{\mathcal{D}}$.

- ◇ In this section we shall mainly consider the augmented simplicial object on $\mathcal{D}$ given by the comonad. Explicitly, its term in degree n is L^{n+1}, while its face and degeneracy morphisms are readily written as

$$\begin{aligned} d_i &:= L^i \epsilon L^{n-i} : L^{n+1} \to L^n, \\ s_j &:= L^j \delta L^{n-j} : L^{n+1} \to L^{n+2}, \quad 0 \leq i, j \leq n; \end{aligned}$$

  the term in degree -1 of the augmented object is the identity functor. Notice that the formula for d_0 still makes sense when $n = 0$ and gives the augmentation morphism $\epsilon : L \to \mathrm{id}_{\mathcal{D}}$.

- ◇ Correspondingly, the augmented cosimplicial object on $\mathcal{C}$ has T^{n+1} as its term in degree n, and

$$\begin{aligned} d^i &:= T^i \eta T^{n-i} : T^n \to T^{n+1}, \\ s^j &:= T^j \mu T^{n-j} : T^{n+2} \to T^{n+1}, \quad 0 \leq i, j \leq n; \end{aligned}$$

  when $n = 0$, d^0 becomes the augmentation morphism $\eta : \mathrm{id}_{\mathcal{C}} \to T$.

An important source of monads and comonads is adjoint pairs of functors. In view of their origins in classical theory (see §3.8), constructions of this kind are collectively called **bar constructions**.

Example 8.7.2 (Bar construction: an adjoint pair) Suppose we are given an adjoint pair

$$F : \mathcal{C} \rightleftarrows \mathcal{D} : G, \qquad \begin{array}{ll} \eta : \mathrm{id}_{\mathcal{C}} \to GF & \text{unit} \\ \epsilon : FG \to \mathrm{id}_{\mathcal{D}} & \text{counit} \end{array}$$

By Example 7.6.2, this gives

- the monad $(T, \mu, \eta) := (GF, G\epsilon F, \eta)$ on $\mathcal{C}$, whose corresponding augmented cosimplicial object is denoted by

$$\mathrm{Cobar}(F, G) : \mathsf{FinOrd} \to \mathrm{End}(\mathcal{C});$$

- the comonad $(L, \delta, \varepsilon) := (FG, F\eta G, \epsilon)$ on $\mathcal{D}$, whose corresponding augmented simplicial object is denoted by

$$\mathrm{Bar}(F, G) : \mathsf{FinOrd}^{\mathrm{op}} \to \mathrm{End}(\mathcal{D}).$$

The corresponding face and degeneracy morphisms (and their dual versions) have been described in Example 8.7.1.

Since $\mathrm{Bar}(F, G)$ (respectively, $\mathrm{Cobar}(F, G)$) takes values in the functor category $\mathrm{End}(\mathcal{D})$ (respectively, $\mathrm{End}(\mathcal{C})$), we can compose each of its terms with G on the left (respectively, on the right). The former composition produces an augmented simplicial object $G\mathrm{Bar}(F, G) : \mathsf{FinOrd}^{\mathrm{op}} \to \mathcal{C}^{\mathcal{D}}$, whose general term is the functor

$$\begin{aligned} G\mathrm{Bar}(F, G)_n &= GL^{n+1} = (GF)^{n+1}G \\ &= T^{n+1}G : \mathcal{D} \to \mathcal{C}, \quad n \geq -1. \end{aligned}$$

The face morphism $d_i : T^{n+1}G \to T^nG$ (understood as the augmentation morphism when $i = n = 0$) is

$$d_i = \begin{cases} GL^i\epsilon L^{n-i} & = T^iG\epsilon FT^{n-i-1}G \\ & = T^i\mu T^{n-i-1}G, \qquad \text{if } 0 \leq i < n, \\ GL^n\epsilon & = T^nG\epsilon, \qquad \text{if } i = n. \end{cases}$$

The degeneracy morphism with source the term in degree $n \geq 0$, $s_j : T^{n+1}G \to T^{n+2}G$, is given by

$$\begin{aligned} s_j = GL^j\delta L^{n-j} &= GL^jF\eta GL^{n-j} \\ &= T^{j+1}\eta T^{n-j}G, \quad 0 \leq j \leq n. \end{aligned}$$

The augmented cosimplicial object $\mathrm{Cobar}(F,G)G$ has a corresponding description: its term in degree n is again $T^{n+1}G = GL^{n+1}$, while

$$\left[d^i : GL^n \to GL^{n+1}\right] = \begin{cases} GL^{i-1}\delta L^{n-i}, & 0 \le i < n \\ \eta GL^n, & i = n, \end{cases}$$
$$\left[s^j : GL^{n+2} \to GL^{n+1}\right] = GL^j \epsilon L^{n-j+1}, \quad 0 \le j \le n.$$

[3)] Likewise, when $i = n = 0$, d^0 is to be understood as the augmentation morphism $\eta G : G \to GFG = GL$.

Readers new to the subject are advised to pause here. For a homomorphism $f : A \to B$ of $\Bbbk$-algebras and the adjoint pair

$$B \underset{A}{\otimes} (\cdot) : A\text{-}\mathsf{Mod} \rightleftarrows B\text{-}\mathsf{Mod} : \text{forgetful}$$

describe the value of Bar (respectively, Cobar) at a left B-module M (respectively, a left A-module N), and then use the functor C of Definition 8.5.1 to write down the corresponding chain complex. For the right-module version, the corresponding comonad (respectively, monad) was described completely in Example 7.7.8.

Example 8.7.3 (Bar construction: free and forgetful) Let (T, μ, η) be a monad on a category $\mathcal{C}$. Definition–Proposition 7.6.6 gives an adjoint pair

$$\mathrm{Free}^T : \mathcal{C} \underset{\text{forgetful}}{\overset{\text{free}}{\rightleftarrows}} \mathcal{C}^T : U^T,$$

and its induced monad is precisely (T, μ, η). Substituting this into Example 8.7.2 produces the augmented simplicial object

$$\mathrm{Bar}(T) := \mathrm{Bar}(\mathrm{Free}^T, U^T) : \mathsf{FinOrd}^{\mathrm{op}} \to \mathrm{End}(\mathcal{C}^T).$$

Evaluation at $(M, a) \in \mathrm{Ob}(\mathcal{C}^T)$ is a functor from $\mathrm{End}(\mathcal{C}^T)$ to $\mathcal{C}^T$. Evaluating every term of $\mathrm{Bar}(T)$ gives the augmented simplicial object

$$\mathrm{Bar}(M, a) : \mathsf{FinOrd}^{\mathrm{op}} \to \mathcal{C}^T.$$

Now consider $U^T\mathrm{Bar}(M, a) : \mathsf{FinOrd}^{\mathrm{op}} \to \mathcal{C}$; this is also the result of evaluating $U^T\mathrm{Bar}(T)$ at (M, a).

◇ Substitution into Example 8.7.2 shows that the general term of $U^T\mathrm{Bar}(M, a)$ is

$$U^T\mathrm{Bar}(M, a)_n = T^{n+1}U^T(M, a) = T^{n+1}M, \quad n \ge -1.$$

[3)] Translator's note: in the printed case distinction for d^i, the boundary conditions are interchanged. Consistently with $d^i = T^i\eta T^{n-i}$, the first line applies for $1 \le i \le n$, while the second applies for $i = 0$.

- Again denote the counit of (Free^T, U^T) by ϵ. The proof of Definition–Proposition 7.6.6 has shown that

$$\left[\epsilon_{(M,a)} : \mathrm{Free}^T U^T(M,a) = (TM, \mu_M) \to (M,a)\right] = a,$$

so the face morphisms of $U^T\mathrm{Bar}(M,a)$ are

$$\left[d_i : T^{n+1}M \to T^n M\right] = \begin{cases} T^i \mu_{T^{n-i-1}M}, & 0 \le i < n \\ T^n a, & i = n. \end{cases}$$

This formula also makes sense when $i = n = 0$ and gives the augmentation morphism, namely $a : TM \to M$.

- Its degeneracy morphisms are

$$\left[s_j : T^{n+1}M \to T^{n+2}M\right] = T^{j+1}\eta_{T^{n-j}M}, \quad 0 \le j \le n.$$

Thus $U^T\mathrm{Bar}(M,a)$ can be written concisely as

$$\cdots \overset{\mu_{T^2M}}{\underset{T^3a}{\rightrightarrows\!\!\rightrightarrows}} T^3M \overset{\mu_{TM}}{\underset{T^2a}{\overset{-T\mu_M\to}{\rightrightarrows}}} T^2M \overset{\mu_M}{\underset{Ta}{\rightrightarrows}} TM \xrightarrow{\quad a \quad} M. \tag{8.7.1}$$

We have already explained abstractly that these arrows satisfy the simplicial identities (8.1.2), although a direct verification also presents no essential difficulty. Moreover, Example 7.6.10 shows that the right end of the diagram realizes $a : TM \to M$ as a coequalizer.

The augmented simplicial object in (8.7.1) given by the bar construction can be viewed heuristically as a replacement for M. To clarify this idea, consider the following question. Let Γ be a group, and equip the one-point set $\{\mathrm{pt}\}$ with the trivial right Γ-action. How should we understand

$$\text{the “quotient space” } \{\mathrm{pt}\}/\Gamma?$$

With the naive definition of a quotient set or topological space, the answer is of course uninteresting; in general, taking the quotient by a nonfree action erases information from the original space. Various needs arising in practice, however, prompt us to seek a more refined definition of a quotient by a nonfree action.

To this end, write $\mathsf{Set}\text{-}\Gamma$ for the category of all small right Γ-sets, and consider the free–forgetful adjunction $\mathsf{Set} \leftrightarrows \mathsf{Set}\text{-}\Gamma$. From the one-point right Γ-set $\{\mathrm{pt}\}$, the bar construction produces a simplicial object in $\mathsf{Set}\text{-}\Gamma$; this is precisely the simplicial set $\mathrm{E}\Gamma$ introduced in Example 8.3.5. The detailed verification is left as an exercise in this chapter. Once this fact is understood and the bar construction is accepted as a replacement for $\{\mathrm{pt}\}$ in this setting, it is reasonable to define $\{\mathrm{pt}\}/\Gamma$ as $(\mathrm{E}\Gamma)/\Gamma \simeq \mathrm{B}\Gamma$. This is a quotient by a free action and gives the classifying space of Γ; it is more natural, and usually more useful, than the naive quotient set.

In what sense can the augmented simplicial objects in Examples 8.7.2 and 8.7.3 be said to provide a kind of replacement, or "resolution", of the original object? The idea comes from topology, and its precise meaning involves the contractibility of Definition 8.6.8. One related result is the following.

Proposition 8.7.4 In the setting of Example 8.7.2, suppose we are given an adjoint pair

$$F : \mathcal{C} \rightleftarrows \mathcal{D} : G, \qquad \begin{aligned} &\eta : \mathrm{id}_{\mathcal{C}} \to GF \quad \text{unit} \\ &\epsilon : FG \to \mathrm{id}_{\mathcal{D}} \quad \text{counit} \end{aligned}$$

and form the augmented simplicial object $\mathrm{Bar}(F,G)$ in $\mathrm{End}(\mathcal{D})$. Then $G\mathrm{Bar}(F,G)$ is left contractible.

Proof Use the notation of Example 8.7.2, such as $T = GF$ and $\mu : T^2 \to T$. Recall that $G\mathrm{Bar}(F,G)_n = T^{n+1}G$. For every $n \geq -1$, define

$$k_n := \eta T^{n+1}G : T^{n+1}G \to T^{n+2}G.$$

Recall also that the face morphism $d_i : T^{n+1}G \to T^nG$ of $G\mathrm{Bar}(F,G)$ is $T^i\mu T^{n-i-1}G$ if $0 \leq i < n$, and $T^nG\epsilon$ if $i = n$; its augmentation morphism is $G\epsilon : TG \to G$ (note that this notation differs from that in Definition 8.6.8). We now verify the conditions for left contractibility one by one.

First, the triangle identity for the adjunction gives $(G\epsilon)k_{-1} = (G\epsilon)(\eta G) = \mathrm{id}_G$.

Second, (T,μ,η) is a monad, so $\mu(\eta T) = \mathrm{id}_T$. If $n \geq 0$, composing both sides on the right with T^nG gives $d_0k_n = \mu T^nG \cdot \eta T^{n+1}G = \mathrm{id}_{T^{n+1}G}$.

Finally, by the naturality of η, the following diagram commutes when $1 \leq i \leq n+1$:

$$\begin{tikzcd} T^{n+1}G \arrow[r, "k_n = \eta T^{n+1}G"] \arrow[d, "d_{i-1} = \varphi"'] & T^{n+2}G \arrow[d, "d_i = T\varphi"] \\ T^nG \arrow[r, "k_{n-1} = \eta T^nG"'] & T^{n+1}G \end{tikzcd} \qquad \varphi := \begin{cases} T^{i-1}\mu T^{n-i}G, & 1 \leq i \leq n \\ T^nG\epsilon, & i = n+1. \end{cases}$$

This gives $d_1k_0 = k_{-1}(G\epsilon)$ when $i = 1$ and $n = 0$, and $d_ik_n = k_{n-1}d_{i-1}$ when $n \geq 1$. The result follows. $\square$

Although the contractibility of $G\mathrm{Bar}(F,G)$ does not necessarily imply that $\mathrm{Bar}(F,G)$ is contractible, Proposition 8.7.4 is already sufficient for the applications below. The exercises in this chapter will introduce further results concerning contractibility.

For additive categories, the meaning of "resolution" can also be carried to the level of chain complexes and thus understood through linear algebra. First, by Definition 8.6.7, an augmented simplicial object X in an additive category gives an augmentation $\mathrm{C}^{\mathrm{aug}}X$ of the chain complex $\mathrm{C}X$. If the identity morphism of a chain complex is null-homotopic, the chain complex itself is called **null-homotopic**.

Theorem 8.7.5 Let $\mathcal{A}$ be an additive category, suppose an adjoint pair is given,

$$F : \mathcal{A} \rightleftarrows \mathcal{B} : G,$$

and form the augmented simplicial object $\mathrm{Bar}(F, G)$ in $\mathrm{End}(\mathcal{B})$ (respectively, $G\mathrm{Bar}(F, G)$ in $\mathcal{A}^{\mathcal{B}}$).

- ◇ Notice that $\mathcal{A}^{\mathcal{B}}$ is an additive category (see the first part of §2.1). The chain complex $\mathrm{C}^{\mathrm{aug}}\big(G\mathrm{Bar}(F, G)\big)$ from Definition 8.6.7 is null-homotopic.
- ◇ For every $Y \in \mathrm{Ob}(\mathcal{B})$, evaluate $\mathrm{Bar}(F, G)$ (respectively, $G\mathrm{Bar}(F, G)$) at Y degreewise. This produces an augmented simplicial object $\mathrm{Bar}(Y)$ in $\mathcal{B}$ (respectively, $G\mathrm{Bar}(Y)$ in $\mathcal{A}$), and the chain complex $\mathrm{C}^{\mathrm{aug}}\big(G\mathrm{Bar}(Y)\big)$ in $\mathcal{A}$ is null-homotopic.

Proof Proposition 8.7.4 shows that $G\mathrm{Bar}(F, G)$ is left contractible. Since contractibility is an absolute condition, $G\mathrm{Bar}(Y)$ obtained through the evaluation functor $\mathrm{ev}_Y : \mathcal{A}^{\mathcal{B}} \to \mathcal{A}$ is naturally left contractible as well. Apply Proposition 8.6.10 to both objects. □

Now suppose that $\mathcal{A}$ and $\mathcal{B}$ are both additive categories. For an adjoint pair as in Theorem 8.7.5 and $Y \in \mathrm{Ob}(\mathcal{B})$, since $\mathrm{Bar}(Y)_{-1} = Y$, flattening $\mathrm{C}^{\mathrm{aug}}(\mathrm{Bar}(Y))$ gives a morphism in $\mathsf{Ch}_{\geq 0}(\mathcal{B})$

$$\epsilon_Y : \mathrm{C}(\mathrm{Bar}(Y)) \to Y; \tag{8.7.2}$$

it is canonical in Y, and its degree-zero part is precisely the augmentation morphism $\epsilon_Y : FG(Y) \to Y$ of $\mathrm{Bar}(Y)$, so the notation is appropriate.

Corollary 8.7.6 Suppose an adjoint pair of abelian categories is given[4] $F : \mathcal{A} \rightleftarrows \mathcal{B} : G$ and let $Y \in \mathrm{Ob}(\mathcal{B})$. The morphism obtained by applying G degreewise to (8.7.2), namely

$$G\epsilon_Y : \mathrm{C}(G\mathrm{Bar}(Y)) \to GY,$$

is a quasi-isomorphism in $\mathsf{Ch}(\mathcal{A})$.

[4] Here F and G are necessarily additive functors; see Corollary 1.3.6.

Proof Theorem 8.7.5 implies that $\mathrm{C}^{\mathrm{aug}}(G\mathrm{Bar}(Y))$ is acyclic, and the result follows immediately. □

Example 8.7.7 (The bar resolution of a module and Hochschild homology/cohomology) Let $\Bbbk$ be a commutative ring and R a $\Bbbk$-algebra. Write $\otimes := \otimes_{\Bbbk}$. By Example 7.7.8, the comonad determined by the adjoint pair

$$R \otimes (\cdot) : \Bbbk\text{-}\mathsf{Mod} \rightleftarrows R\text{-}\mathsf{Mod} : \text{forgetful}$$

is the endofunctor $L : M \mapsto R \otimes M$ on R-Mod. There is a corresponding version for right R-modules, which we shall not repeat.

Now apply the bar construction. For every left R-module M, we have a canonical morphism in $\mathsf{Ch}_{\geq 0}(R\text{-}\mathsf{Mod})$

$$\epsilon_M : \mathrm{C}(\mathrm{Bar}(M)) \to M;$$

this is a quasi-isomorphism because Corollary 8.7.6 ensures that ϵ_M is a quasi-isomorphism at the level of $\Bbbk$-Mod. Explicitly,

$$\mathrm{C}(\mathrm{Bar}(M))_n = L^{n+1}(M) = R^{\otimes(n+1)} \otimes M.$$

According to Example 8.7.1 and the description of $L \to \mathrm{id}$ in Example 7.7.8 (namely, "multiplying in"), the map $L^{n+1}(M) \to L^n(M)$ is

$$\partial_n : r_0 \otimes \cdots \otimes r_n \otimes x \mapsto \sum_{k=0}^{n-1} (-1)^k \cdots \otimes r_k r_{k+1} \otimes \cdots \otimes x + (-1)^n r_0 \otimes \cdots \otimes r_{n-1} \otimes r_n x,$$

where $x \in M$ and $r_i \in R$. This formula still makes sense when $n = 0$ ($r_0 \otimes x \mapsto r_0 x$), and thus gives the augmented version $\epsilon : \mathrm{C}(\mathrm{Bar}(M)) \to M$, namely $\mathrm{C}^{\mathrm{aug}}(\mathrm{Bar}(M))$.

Now consider the special case $M = R$. In this case

$$\mathrm{C}(\mathrm{Bar}(R))_n = R^{\otimes(n+2)} = R \otimes R^{\otimes n} \otimes R,$$

$$\partial_n(r_0 \otimes \cdots \otimes r_{n+1}) = \sum_{k=0}^{n} (-1)^k \cdots \otimes r_k r_{k+1} \otimes \cdots .$$

Although $\mathrm{C}(\mathrm{Bar}(R))$ and its augmentation $\mathrm{C}^{\mathrm{aug}}(\mathrm{Bar}(R))$ are, by definition, chain complexes of left R-modules, the description above upgrades both to chain complexes of (R, R)-bimodules: R acts by left multiplication on the first factor and by right multiplication on the last factor of $R \otimes R^{\otimes n} \otimes R$. This upgrade of structure is no coincidence. Since $\mathrm{C}^{\mathrm{aug}}(\mathrm{Bar}(\cdot))$ is a functor, every endomorphism of R as a left R-module (for example, right multiplication) lifts naturally to the entire bar construction.

Comparison with Definition 3.8.1 shows immediately that these are precisely the chain complexes used to define Hochschild homology and cohomology:

$$\mathrm{C}(\mathrm{Bar}(R)) = \mathsf{B}R, \quad \mathrm{C}^{\mathrm{aug}}(\mathrm{Bar}(R)) = \mathsf{B}'R.$$

Notice also that Lemma 3.8.2 is just a special case of Theorem 8.7.5.

Write $R^e := R \otimes R^{\mathrm{op}}$ and, according to Convention 3.8.3, identify left R^e-modules, (R, R)-bimodules, and right R^e-modules. From this viewpoint, the Hochschild homology $\mathrm{HH}_\bullet(M)$ (respectively, cohomology $\mathrm{HH}^\bullet(M)$) of a bimodule M is a derived version of $M \underset{R^e}{\otimes} R$ (respectively, $\mathrm{Hom}_{R^e}(R, M)$): the bimodule R is replaced by its bar resolution $\mathrm{C}(\mathrm{Bar}(R))$, after which homology (respectively, cohomology) is computed.

This idea of replacement pervades the study of derived functors and derived categories, but the situation here is slightly different. In the framework of Chapter 3 and Chapter 4, R should be replaced by its flat resolution (respectively, projective resolution) as a bimodule. This yields $\mathrm{Tor}_\bullet^{R^e}(M, R)$ (respectively, $\mathrm{Ext}^\bullet_{R^e}(R, M)$), or the corresponding object in the derived category. Example 3.14.11 (respectively, Example 3.14.6) shows that if R is flat (respectively, projective) as a $\Bbbk$-module, these objects are naturally isomorphic to $\mathrm{HH}_\bullet(M)$ (respectively, $\mathrm{HH}^\bullet(M)$); this need not hold in general.

Example 8.7.8 (The bar resolution and group homology) Let Γ be a group. Continue the discussion of Example 8.7.7 for the group algebra $R := \Bbbk[\Gamma]$; in this case, M is the same as a left Γ-module in the sense of §6.1. Recall that $(\cdot)_\Gamma : \Gamma\text{-}\mathsf{Mod} \to \Bbbk\text{-}\mathsf{Mod}$ is the coinvariants functor. We claim that $\epsilon_M : \mathrm{C}(\mathrm{Bar}(M)) \to M$ is a $(\cdot)_\Gamma$-acyclic resolution in Γ-Mod (Convention 3.12.8). Its quasi-isomorphism property is already known. To see that every $\mathrm{Bar}(M)_n = \Bbbk[\Gamma]^{\otimes(n+1)} \otimes M$ is $(\cdot)_\Gamma$-acyclic, first use Shapiro's lemma (Theorem 6.4.3), which gives, for every $\Bbbk$-module P,

$$\mathrm{H}_k(\Gamma, \Bbbk[\Gamma] \otimes P) = \mathrm{H}_k\left(\Gamma, \mathrm{ind}_{\{1\}}^{\Gamma}(P)\right) \simeq \mathrm{H}_k(\{1\}, P),$$

and the right-hand side is always 0 when $k > 0$. Apply this with $P = \Bbbk[\Gamma]^{\otimes n} \otimes M$ to obtain the claim.

Consequently, Corollary 3.12.10 implies $\mathrm{H}_n(\Gamma, M) \simeq \mathrm{H}_n\big(\mathrm{C}((\mathrm{Bar}(M)_\Gamma))\big)$; here the functor $(\cdot)_\Gamma$ may be applied either to the simplicial object or, equivalently, to the corresponding chain complex. This interprets group homology as comonad homology in the sense of the exercises in this chapter. More precisely, taking Γ-coinvariants of $\mathrm{Bar}(M)_n$ amounts to removing the first tensor factor in $\Bbbk[\Gamma]^{\otimes(n+1)} \otimes M$ via $\Bbbk[\Gamma] \twoheadrightarrow \Bbbk$, giving $\Bbbk[\Gamma]^{\otimes n} \otimes M$. Under this identification, $\mathrm{Bar}(M)_{n,\Gamma} \to \mathrm{Bar}(M)_{n-1,\Gamma}$ is given by

$$g_1 \otimes \cdots \otimes g_n \otimes x \mapsto g_2 \otimes \cdots \otimes g_n \otimes x + \sum_{k=1}^{n-1} (-1)^k \cdots \otimes g_k g_{k+1} \otimes \cdots \otimes x \\ + (-1)^n g_1 \otimes \cdots \otimes g_{n-1} \otimes g_n x,$$

where $g_i \in \Gamma$ and $x \in M$. Comparison with Proposition 6.2.9 shows immediately that $\mathrm{C}(\mathrm{Bar}(M))_\Gamma$ is precisely the standard chain complex computing $\mathrm{H}_n(\Gamma, M)$.

8.8 Bisimplicial Objects

We begin with some general definitions. For objects $[m_1], \ldots, [m_n]$ of $\mathbf{\Delta}$, from now on we shall also write the object $([m_1], \ldots, [m_n])$ of $(\mathbf{\Delta})^n$ as $[m_1] \times \cdots \times [m_n]$ for typographical convenience.

Definition 8.8.1 Let $\mathcal{C}$ be an arbitrary category and $n \in \mathbb{Z}_{\geq 1}$. A functor of the form $X : (\mathbf{\Delta}^{\mathrm{op}})^n \to \mathcal{C}$ (respectively, $\mathbf{\Delta}^n \to \mathcal{C}$) is called an ***n*-fold simplicial object** (respectively, an ***n*-fold cosimplicial object**) in $\mathcal{C}$; its morphisms are morphisms of such functors. When $n = 2$, the corresponding objects are called bisimplicial (respectively, bicosimplicial) objects. The value of an n-fold simplicial (respectively, cosimplicial) object X at $[m_1] \times \cdots \times [m_n]$ is denoted by $X_{m_1,\ldots,m_n}$ (respectively, $X^{m_1,\ldots,m_n}$).

Thus an n-fold simplicial object is determined by a family of objects $X_{m_1,\ldots,m_n}$, where $m_1, \ldots, m_n \in \mathbb{Z}_{\geq 0}$, together with its face morphisms

$${}^k d_i : X_{m_1,\ldots,m_n} \to X_{\ldots,m_k-1,\ldots}, \quad 1 \leq k \leq n, \quad 0 \leq i \leq m_k$$

and its degeneracy morphisms

$${}^k s_j : X_{m_1,\ldots,m_n} \to X_{\ldots,m_k+1,\ldots}, \quad 1 \leq k \leq n, \quad 0 \leq j \leq m_k.$$

These morphisms are required to satisfy (8.1.2) for each k, and morphisms corresponding to different values of k must commute. This is simply the multivariable version of Definition 8.1.2.

More generally, for every morphism in $\mathbf{\Delta}^n$

$$f : [m_1] \times \cdots \times [m_n] \to [m_1'] \times \cdots \times [m_n'],$$

there is a corresponding pullback $f^* : X_{m_1',\ldots,m_n'} \to X_{m_1,\ldots,m_n}$. The case of n-fold cosimplicial objects is dual.

Example 8.8.2 Take $\mathcal{C} = \mathsf{Set}$. As in Definition 8.2.2, we can define the standard n-fold simplicial set

$$\Delta^{p_1,\dots,p_n} := \mathrm{Hom}_{\mathbf{\Delta}^n}\left(\cdot, [p_1] \times \cdots \times [p_n]\right).$$

Next take $\mathcal{C} = \mathsf{Ab}$. Imitating Definition 8.6.1, for every n-fold simplicial set K we can define an n-fold simplicial object $\mathbb{Z}K$ in Ab. Specifying a morphism $\mathbb{Z}\Delta^{p_1,\dots,p_n} \to X$ is equivalent to specifying an element of $X_{p_1,\dots,p_n}$.

If $\mathcal{C}$ is a small category, the category of all n-fold simplicial objects is denoted by $\mathsf{s}^n\mathcal{C}$. Thus $\mathsf{s}^1\mathcal{C} = \mathsf{s}\mathcal{C}$, while, if $n > 1$,

$$\mathsf{s}^n\mathcal{C} = \mathsf{s}\big(\mathsf{s}^{n-1}\mathcal{C}\big) = \mathsf{s}^{n-1}\big(\mathsf{s}\mathcal{C}\big),$$

and so on. The case of n-fold cosimplicial objects is analogous.

The semisimplicial objects of Remark 8.1.4 plainly have an n-fold version, obtained by removing the degeneracy morphisms and the related conditions from the definition above.

Definition 8.8.3 With the notation above, the **diagonal functor** $d : \mathsf{s}^n\mathcal{C} \to \mathsf{s}\mathcal{C}$ maps an n-fold simplicial object X to the simplicial object

$$d(X)_n := X_{n,\dots,n},$$

whose face and degeneracy morphisms are defined respectively by $d_i = \prod_k {}^k d_i$ and $s_j = \prod_k {}^k s_j$. On morphisms, the functor is defined in the evident way. Equivalently, $d(X)$ is the composite of X with the diagonal embedding $\mathbf{\Delta}^{\mathrm{op}} \to (\mathbf{\Delta}^{\mathrm{op}})^n$. The cosimplicial case is entirely dual.

Example 8.8.4 Let $(\mathcal{C}, \otimes)$ be a monoidal category, for example Set with the product $\times$. For $X_1, \dots, X_n \in \mathrm{Ob}(\mathsf{s}\mathcal{C})$, we obtain an object of $\mathsf{s}^n\mathcal{C}$ whose $(m_1, \dots, m_n)$-term is $X_{1,m_1} \otimes \cdots \otimes X_{n,m_n}$. This n-fold simplicial object is denoted by

$$X_1 \boxtimes \cdots \boxtimes X_n \in \mathrm{Ob}(\mathsf{s}^n\mathcal{C}).$$

This should not be confused with $X_1 \otimes \cdots \otimes X_n \in \mathrm{Ob}(\mathsf{s}\mathcal{C})$ from Definition 8.1.8; the two are related by

$$X_1 \otimes \cdots \otimes X_n = d(X_1 \boxtimes \cdots \boxtimes X_n).$$

A basic example is obtained from the monoidal category $(\mathsf{Set}, \times)$. In this case $\Delta^{p_1} \boxtimes \cdots \boxtimes \Delta^{p_n} = \Delta^{p_1,\dots,p_n}$, while $\Delta^{p_1} \otimes \cdots \otimes \Delta^{p_n} = \Delta^{p_1} \times \cdots \times \Delta^{p_n}$, the simplicial set obtained by taking the product degreewise.

General n-fold simplicial objects play an indispensable role in homotopy theory and its applications. This section is mainly concerned with abelian categories, focusing on the special case $n = 2$. The first step is to introduce a variant of Definition 8.5.1.

Recall the definition of a double complex in Definition 3.5.1, or more precisely its chain-complex version. The category of double chain complexes is denoted by $\mathsf{Ch}^2(\cdots)$, and $\mathsf{Ch}^2_{\geq 0}(\cdots)$ is defined similarly. For a bisemisimplicial object, we also introduce the notation

$$ {}^{\triangleright}d_i := {}^1 d_i, \quad {}^{\vartriangle}d_i := {}^2 d_i, $$

for the face morphisms in the "horizontal" and "vertical" directions. For a bisimplicial object, the degeneracy morphisms are similarly denoted by ${}^{\triangleright}s_j = {}^1 s_j$ and ${}^{\vartriangle}s_j = {}^2 s_j$.

Definition 8.8.5 Let $\mathcal{A}$ be an additive category and X a bisemisimplicial object in $\mathcal{A}$. The corresponding **unnormalized double chain complex** (or **Moore double chain complex**) consists of $\left(X_{p,q}\right)_{p,q\in\mathbb{Z}_{\geq 0}}$ together with the data

$$ {}^{\triangleright}\partial_{p,q} := \sum_{i=0}^{p} (-1)^{i\,\triangleright} d_i : X_{p,q} \to X_{p-1,q}, $$
$$ {}^{\vartriangle}\partial_{p,q} := \sum_{i=0}^{q} (-1)^{i\,\vartriangle} d_i : X_{p,q} \to X_{p,q-1}. $$

By convention, $X_{p,q} := 0$ when $(p, q) \notin \mathbb{Z}^2_{\geq 0}$. The resulting double chain complex is denoted by $\mathrm{C}^2 X \in \mathrm{Ob}(\mathsf{Ch}^2(\mathcal{A}))$.

To show that this really is a double chain complex, we must verify ${}^{\triangleright}\partial_{p-1,q}{}^{\triangleright}\partial_{p,q} = 0$, ${}^{\vartriangle}\partial_{p,q-1}{}^{\vartriangle}\partial_{p,q} = 0$, and ${}^{\vartriangle}\partial_{p-1,q}{}^{\triangleright}\partial_{p,q} = {}^{\triangleright}\partial_{p,q-1}{}^{\vartriangle}\partial_{p,q}$. The first two identities are exactly as in Definition 8.5.1; the last follows directly from the commutativity of ${}^{\triangleright}d_i$ and ${}^{\vartriangle}d_j$. We thus obtain an additive functor

$$ \mathrm{C}^2 : \mathsf{s}^2\mathcal{A} \to \mathsf{Ch}^2(\mathcal{A}). $$

Since $(p, q) \notin \mathbb{Z}^2_{\geq 0}$ implies $X_{p,q} = 0$, we may now naturally define its total chain complex by

$$ (\mathrm{tot}\, X)_n := \bigoplus_{p+q=n} X_{p,q}, \qquad \begin{tikzcd} (\mathrm{tot}\, X)_n \arrow[r, "\partial_n"] & (\mathrm{tot}\, X)_{n-1} \\ X_{p,q} \arrow[u, hook] \arrow[r, "({}^{\triangleright}\partial_{p,q},\, (-1)^{p\,\vartriangle}\partial_{p,q})"'] & X_{p-1,q} \oplus X_{p,q-1}. \arrow[u, hook] \end{tikzcd} $$

The same argument as in Definition 3.5.4 shows that $\mathrm{tot}\, X$ really is a chain complex. Thus the composite functor $\mathrm{tot} \circ \mathrm{C}^2 : \mathsf{s}^2\mathcal{A} \to \mathsf{Ch}_{\geq 0}(\mathcal{A})$ is defined.

The same construction can be applied to n-fold simplicial objects over $\mathcal{A}$ to obtain a functor $\mathrm{C}^n : \mathsf{s}^n\mathcal{A} \to \mathsf{Ch}^n(\mathcal{A})$ and the corresponding total chain complex; see Remark 3.5.7.

Example 8.8.6 Suppose $\mathcal{A}$ has the structure of a monoidal category such that the bifunctor $\otimes : \mathcal{A} \times \mathcal{A} \to \mathcal{A}$ is additive in each variable. If $C_1, C_2 \in \mathrm{Ob}(\mathsf{Ch}_{\geq 0}(\mathcal{A}))$, we naturally obtain a double chain complex $C_1 \boxtimes C_2$ whose (p,q)-term is $C_{1,p} \otimes C_{2,q}$ and whose ${}^{\triangleright}\partial$ and ${}^{\vartriangle}\partial$ come from C_1 and C_2, respectively. If $A, B \in \mathrm{Ob}(\mathsf{s}\mathcal{A})$, there is a simple relation between the double chain complex $\mathrm{C}A \boxtimes \mathrm{C}B$ and the bisimplicial object $A \boxtimes B$ of Example 8.8.4:

$$\mathrm{C}^2(A \boxtimes B) = \mathrm{C}A \boxtimes \mathrm{C}B.$$

Moreover, $\mathsf{Ch}_{\geq 0}(\mathcal{A})$ becomes a monoidal category with

$$C_1 \otimes C_2 := \mathrm{tot}(C_1 \boxtimes C_2),$$

and its unit is plainly $\mathbf{1}$ placed in degree zero.

Definition 8.8.3 gives an additive functor $d : \mathsf{s}^2\mathcal{A} \to \mathsf{s}\mathcal{A}$. The aim of this section is to explain, under the assumption that $\mathcal{A}$ is an abelian category, the relationship between the functors

$$\mathrm{tot} \circ \mathrm{C}^2,\ \mathrm{C} \circ d :\ \mathsf{s}^2\mathcal{A} \rightrightarrows \mathsf{Ch}_{\geq 0}(\mathcal{A}).$$

This relationship involves the (p,q)-shuffle σ of Definition 8.4.11 and its sign $\mathrm{sgn}(\sigma)$ from Definition 8.4.13.

Definition 8.8.7 Let $\mathcal{A}$ be an additive category and X a bisimplicial object in $\mathcal{A}$. For a fixed $(p,q) \in \mathbb{Z}^2_{\geq 0}$, write $n := p + q$.

(i) Define $\overline{\mathrm{EZ}}_{p,q} : X_{p,q} \to X_{n,n}$ by

$$\overline{\mathrm{EZ}}_{p,q} := \sum_{\sigma:(p,q)\text{-shuffle}} \mathrm{sgn}(\sigma)\, (\sigma_- \times \sigma_+ : [n] \times [n] \to [p] \times [q])^* \,.$$

A lengthy but straightforward argument, omitted here, shows that these maps combine to form a morphism of chain complexes called the **shuffle morphism**, or **Eilenberg–Zilber morphism**:

$$\overline{\mathrm{EZ}} : \mathrm{tot}(\mathrm{C}^2 X) \to \mathrm{C}(dX).$$

(ii) Define $\overline{\mathrm{AW}}_n := \sum_{p+q=n} \overline{\mathrm{AW}}_{p,q} : \mathrm{C}(dX)_n \to \mathrm{tot}(\mathrm{C}^2X)_n$, where $\overline{\mathrm{AW}}_{p,q} : X_{n,n} \to X_{p,q}$ is defined as $(\lambda^n_p \times \rho^n_q)^*$, with

$$\lambda^n_p : [p] \hookrightarrow [n], \quad i \mapsto i,$$
$$\rho^n_q : [q] \hookrightarrow [n], \quad i \mapsto n - q + i.$$

Similarly, these maps combine to form a morphism of chain complexes $\overline{\mathrm{AW}}$: $\mathrm{C}(dX) \to \mathrm{tot}(\mathrm{C}^2X)$, called the **Alexander–Whitney morphism**.

These morphisms are functorial not only in X, but also in the category $\mathcal{A}$ and in functors between such categories. They were originally introduced in topology in connection with the homology of product spaces.

We next introduce the normalized versions of $\overline{\mathrm{EZ}}$ and $\overline{\mathrm{AW}}$.

Let $\mathcal{A}$ be a monoidal abelian category such that $\otimes : \mathcal{A} \times \mathcal{A} \to \mathcal{A}$ is additive in each variable. Take $A, B \in \mathrm{Ob}(\mathsf{s}\mathcal{A})$. Since $\mathrm{C}^2(A \boxtimes B) = \mathrm{C}A \boxtimes \mathrm{C}B$, we have

$$\mathrm{C}A \otimes \mathrm{C}B := \mathrm{tot}\,(\mathrm{C}A \boxtimes \mathrm{C}B) = \mathrm{tot}\;\mathrm{C}^2(A \boxtimes B).$$

We also use the simplified notation

$$A \otimes B := d(A \boxtimes B),$$
$$\mathrm{N}A \otimes \mathrm{N}B := \mathrm{tot}\,(\mathrm{N}A \boxtimes \mathrm{N}B)\,.$$

Lemma 8.8.8 Consider the monoidal abelian category $\mathcal{A}$ above. Recall that, for every $Y \in \mathrm{Ob}(\mathsf{s}(\mathcal{A}))$, Proposition 8.5.11 identifies $\mathrm{N}Y$ with $\mathrm{C}Y/\text{degenerate part}$. Under this identification, the morphisms $\overline{\mathrm{AW}}$ and $\overline{\mathrm{EZ}}$ each factor uniquely through the following commutative diagram:

$$\begin{tikzcd}
\mathrm{C}A \otimes \mathrm{C}B \arrow[r, "\overline{\mathrm{EZ}}"] \arrow[d, two heads] & \mathrm{C}(A \otimes B) \arrow[r, "\overline{\mathrm{AW}}"] \arrow[d, two heads] & \mathrm{C}A \otimes \mathrm{C}B \arrow[d, two heads] \\
\mathrm{N}A \otimes \mathrm{N}B \arrow[r, "\mathrm{EZ}"'] & \mathrm{N}(A \otimes B) \arrow[r, "\mathrm{AW}"'] & \mathrm{N}A \otimes \mathrm{N}B.
\end{tikzcd}$$

Proof For AW, it is enough to verify, for all $p, q \geq 0$ and $n := p + q$, the following equalities of morphisms in $\mathbf{\Delta}$:

$$\lambda^n_p \circ \mathrm{s}^j = \mathrm{s}^j \circ \lambda^{n+1}_{p+1},$$
$$\rho^n_q \circ \mathrm{s}^j = \mathrm{s}^{j+n-q} \circ \rho^{n+1}_{q+1},$$

where the respective conditions on j are $0 \leq j \leq p$ and $0 \leq j \leq q$. This verification is immediate.

For EZ, let σ be a (p, q)-shuffle and let $0 \leq j \leq p - 1$. Consider the restriction of $\overline{\mathrm{EZ}}_{p,q}$ to $\mathrm{im}(s_j) \otimes B_q$. The earlier discussion of (p, q)-shuffles shows that there is a unique $0 \leq k < n := p + q$ such that $\sigma_-(k) = j$ and $\sigma_-(k + 1) = j + 1$; at the same time, $\sigma_+(k + 1) = \sigma_+(k)$. This gives the equalities in $\mathbf{\Delta}$

$$\mathrm{s}^j\sigma_- \underbrace{\mathrm{d}^k\mathrm{s}^k}_{[n]\leftarrow[n]} = \mathrm{s}^j\sigma_- : [n] \to [p-1],$$
$$\sigma_+\mathrm{d}^k\mathrm{s}^k = \sigma_+ : [n] \to [q].$$

Pulling back along these equalities, we see that $\overline{\mathrm{EZ}}_{p,q}|_{\mathrm{im}(s_j)\otimes B_q}$ is contained in the part degenerate through s_k. The case $0 \leq j \leq q-1$ and $\overline{\mathrm{EZ}}_{p,q}|_{A_p\otimes\mathrm{im}(s_j)}$ is handled in the same way. $\square$

The following result is also called the generalized Eilenberg–Zilber theorem.

Theorem 8.8.9 (S. Eilenberg, J. A. Zilber, P. Cartier) Let $\mathcal{A}$ be an abelian category and $X \in \mathrm{Ob}(\mathsf{s}^2\mathcal{A})$. The canonical morphisms defined above, $\mathrm{tot}\left(\mathrm{C}^2(X)\right) \underset{\overline{\mathrm{AW}}}{\overset{\overline{\mathrm{EZ}}}{\rightleftarrows}} \mathrm{C}(d(X))$ are mutually inverse on homology; in particular, both are quasi-isomorphisms.

Proof Since $\mathsf{s}^2\mathcal{A} \simeq \mathsf{s}(\mathsf{s}\mathcal{A})$ and $\mathsf{Ch}^2_{\geq 0}(\mathcal{A}) \simeq \mathsf{Ch}_{\geq 0}(\mathsf{Ch}_{\geq 0}(\mathcal{A}))$, the Dold–Kan correspondence has a double version, which we continue to denote by the equivalence of categories $\Gamma : \mathsf{Ch}^2_{\geq 0}(\mathcal{A}) \to \mathsf{s}^2\mathcal{A}$. We may therefore assume that $X = \Gamma A$ for $A \in \mathrm{Ob}(\mathsf{Ch}^2_{\geq 0}(\mathcal{A}))$. If $p + q > n$ implies $A_{p,q} = 0$, we say that A has degree $\leq n$.

First suppose that $\mathcal{A} = \mathsf{Ab}$, with the monoidal structure from the tensor product $\otimes = \otimes_{\mathbb{Z}}$. Observe that tot, C^2, C, and d are all right-exact additive functors and preserve arbitrary direct sums. The canonical morphisms $\overline{\mathrm{EZ}}$ and $\overline{\mathrm{AW}}$ are likewise compatible with arbitrary direct sums. Write A as a filtered union of bounded subobjects. Since the filtered colimit $\varinjlim$ preserves homology, the problem reduces to the case where A has degree $\leq n$ for a fixed $n \geq 0$. We argue by induction on n.

Brutal truncation of A gives a short exact sequence $0 \to A' \to A \to A'' \to 0$, where

- ◇ A' has degree $\leq n-1$;
- ◇ all nonzero terms of A'' lie in the part $p + q = n$;
- ◇ the short exact sequence splits in every bidegree (p, q).

The description of the functor Γ shows that $0 \to \Gamma A' \to \Gamma A \to \Gamma A'' \to 0$ is likewise a degreewise split short exact sequence in $\mathsf{s}^2\mathsf{Ab}$. Strictly speaking, the double version of Γ is needed here, but its form is analogous.

We therefore obtain a commutative diagram of chain complexes with exact rows:

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{tot}\left(\mathrm{C}^2(\Gamma A')\right) \arrow[r] \arrow[d, shift left, "\overline{\mathrm{EZ}}"] & \mathrm{tot}\left(\mathrm{C}^2(\Gamma A)\right) \arrow[r] \arrow[d, shift left] & \mathrm{tot}\left(\mathrm{C}^2(\Gamma A'')\right) \arrow[r] \arrow[d, shift left] & 0 \\
0 \arrow[r] & \mathrm{C}(d\Gamma A') \arrow[r] \arrow[u, shift left, "\overline{\mathrm{AW}}"] & \mathrm{C}(d\Gamma A) \arrow[r] \arrow[u, shift left] & \mathrm{C}(d\Gamma A'') \arrow[r] \arrow[u, shift left] & 0.
\end{tikzcd}$$

This diagram inductively reduces the problem to the case of A''. In other words, under the induction hypothesis, the original problem depends only on $(A_{p,q})_{p+q=n}$.

It therefore suffices to handle the case in which A is a direct sum of several copies of $C \otimes (\mathrm{N}(\mathbb{Z}\Delta^p) \boxtimes \mathrm{N}(\mathbb{Z}\Delta^q))$, with $p + q = n$ and $C \in \mathrm{Ob}(\mathsf{Ab})$. We can further reduce to a single copy for one pair (p, q), and then factor id_C out of all the maps to reduce to the special case $C = \mathbb{Z}$. Notice that the part of $\mathrm{N}(\mathbb{Z}\Delta^p)$ in degrees below p does not affect the problem; the same holds with q in place of p. Correspondingly, $X = \Gamma A \simeq \mathbb{Z}(\Delta^p \times \Delta^q)$. By Theorem 8.5.12 and Lemma 8.8.8, it suffices to prove that

$$\mathrm{N}(\mathbb{Z}\Delta^p) \otimes \mathrm{N}(\mathbb{Z}\Delta^q) \underset{\mathrm{AW}}{\overset{\mathrm{EZ}}{\rightleftarrows}} \mathrm{N}(\mathbb{Z}(\Delta^p \times \Delta^q)) \quad \text{are mutually inverse on homology.} \tag{8.8.1}$$

The posets $[p]$, $[q]$, and $[p] \times [q]$ each have a least element, 0 or $(0, 0)$, so the homotopy equivalences $\mathrm{N}i$ and $\mathrm{N}q$ from Example 8.6.4 apply. It is easy to verify that the following diagram commutes:

$$\begin{tikzcd}
\mathbb{Z} \otimes \mathbb{Z} \arrow[r, "\sim", "\text{standard}"'] \arrow[d, "\mathrm{N}i \otimes \mathrm{N}i"'] & \mathbb{Z} \arrow[r, "\sim", "\text{standard}"'] \arrow[d, "\mathrm{N}i"] & \mathbb{Z} \otimes \mathbb{Z} \arrow[d, "\mathrm{N}i \otimes \mathrm{N}i"'] \\
\mathrm{N}(\mathbb{Z}\Delta^p) \otimes \mathrm{N}(\mathbb{Z}\Delta^q) \arrow[r, "\mathrm{EZ}"'] & \mathrm{N}(\mathbb{Z}(\Delta^p \times \Delta^q)) \arrow[r, "\mathrm{AW}"'] & \mathrm{N}(\mathbb{Z}\Delta^p) \otimes \mathrm{N}(\mathbb{Z}\Delta^q).
\end{tikzcd}$$

The first row is concentrated in degree zero, and the standard isomorphism $\mathbb{Z} \otimes \mathbb{Z} \simeq \mathbb{Z}$ maps $1 \otimes 1$ to 1. The verification of commutativity involves only the degree-zero terms of EZ and AW, and is therefore immediate. This proves (8.8.1).

We now return to a general category $\mathcal{A}$. Suppose there is a faithful exact functor (Proposition 2.8.9) $F : \mathcal{A} \to \mathsf{Ab}$. By definition, tot, C^2, C, and d commute with F, while $\overline{\mathrm{EZ}}$ and $\overline{\mathrm{AW}}$ are plainly compatible with F. Faithfulness and exactness then reduce the problem to the known case $\mathcal{A} = \mathsf{Ab}$.

If $\mathcal{A}$ has a projective generator, for example $\mathcal{A} = \mathsf{Mod}\text{-}R$ for a ring R, there is always a faithful exact functor to Ab. For a general abelian category $\mathcal{A}$, the existence of F is guaranteed by the Freyd–Mitchell theorem B.6.3; for this purpose, the Grothendieck universe may be enlarged as necessary to make $\mathcal{A}$ a small category. ☐

Remark 8.8.10 (Monoidal structures) Let $\mathcal{A}$ be a monoidal category and suppose that $\otimes$ is additive in each variable. Lemma 8.8.8 has already given canonical morphisms of the form

$$\mathrm{C}A \otimes \mathrm{C}B \xrightarrow{\overline{\mathrm{EZ}}} \mathrm{C}(A \otimes B) \xrightarrow{\overline{\mathrm{AW}}} \mathrm{C}A \otimes \mathrm{C}B,$$

$$\mathrm{N}A \otimes \mathrm{N}B \xrightarrow[\mathrm{EZ}]{} \mathrm{N}(A \otimes B) \xrightarrow[\mathrm{AW}]{} \mathrm{N}A \otimes \mathrm{N}B$$

for $A, B \in \mathrm{Ob}(\mathsf{s}\mathcal{A})$. The special case of the generalized Eilenberg–Zilber theorem with $X = A \boxtimes B$ states that these morphisms are mutually inverse on homology. Recall that $\mathsf{Ch}_{\geq 0}(\mathcal{A})$ is also a monoidal category (Example 8.8.6), and there are the evident morphisms

$$\mathbf{1} \text{ in degree zero} \xrightarrow{\sim} \mathrm{N}(\mathrm{const}(\mathbf{1})) \underset{\text{quotient by the degenerate part}}{\overset{\text{inclusion}}{\rightleftarrows}} \mathrm{C}(\mathrm{const}(\mathbf{1})).$$

From an abstract viewpoint, the shuffle morphism (respectively, the Alexander–Whitney morphism) equips C and N with the structure of right-lax (respectively, left-lax) monoidal functors; see [51, Remark 3.1.8]. For the shuffle morphism, this roughly means that $\overline{\mathrm{EZ}}_{p,q}$ and $\mathrm{EZ}_{p,q}$ are associative and compatible with the unit; the case of $\overline{\mathrm{AW}}$ and AW is dual. These assertions can also be verified directly, though at length. For example, associativity of the shuffle morphism involves (8.4.5), while the case of the Alexander–Whitney morphism is simpler.

If $\mathcal{A}$ is a symmetric monoidal category, $\overline{\mathrm{EZ}}$ and EZ also preserve the Koszul braiding introduced in (7.1.3). For $\overline{\mathrm{EZ}}$, this is equivalent to the commutativity of the following diagram in $\mathcal{A}$:

$$\begin{tikzcd} (\mathrm{C}A)_p \otimes (\mathrm{C}B)_q \arrow[r, "\overline{\mathrm{EZ}}_{p,q}"] \arrow[d, "(-1)^{pq}\mathrm{swap}"'] & \mathrm{C}(A \otimes B)_{p+q} \arrow[d, "\mathrm{C(swap)}"] \\ (\mathrm{C}B)_q \otimes (\mathrm{C}A)_p \arrow[r, "\overline{\mathrm{EZ}}_{q,p}"'] & \mathrm{C}(B \otimes A)_{q+p}. \end{tikzcd}$$

Here $p, q \in \mathbb{Z}_{\geq 0}$. The maps swap in the two columns come from the braiding $c(X,Y) : X \otimes Y \xrightarrow{\sim} Y \otimes X$ on $\mathcal{A}$ and its induced degreewise action on simplicial objects; the maps themselves involve no sign. The proof follows directly from the definition and (8.4.4). By contrast, $\overline{\mathrm{AW}}$ and AW need not be compatible with the braiding.

Broadly speaking, the Dold–Kan correspondence functor N of Theorem 8.5.9, together with its unnormalized version C, is monoidal in both the right-lax and left-lax senses. The required data are supplied respectively by the shuffle morphism and the Alexander–Whitney morphism. If $\mathcal{A}$ is symmetric monoidal, the right-lax monoidal functor structure is also symmetric. The concept of a bilax monoidal functor [2, Chapters 3, 5] gives a more precise formulation of these structures.

8.9 Closed Structures

The aim of this section is to explain the closed structures on sSet and sAb. For sSet, this is the Cartesian closed structure of Definition 7.3.2; for sAb, it is the monoidal structure given by the tensor product that is closed. In other words, we seek to promote the Hom sets to suitable simplicial objects. We first discuss the category sSet of simplicial sets, then its version for sAb and its comparison, via the Dold–Kan correspondence, with the Hom complex. The last step requires the results of §8.8.

Definition 8.9.1 For $X, Y \in \mathrm{Ob}(\mathsf{sSet})$, define $\mathcal{H}om(X,Y) = \mathcal{H}om_{\mathsf{sSet}}(X,Y) \in \mathrm{Ob}(\mathsf{sSet})$ as follows:

$$\mathcal{H}om(X,Y)_n := \mathrm{Hom}_{\mathsf{sSet}}(X \times \Delta^n, Y), \qquad n \in \mathbb{Z}_{\geq 0}.$$

For every morphism $f : [m] \to [n]$ in $\mathbf{\Delta}$, define $f^* : \mathcal{H}\mathrm{om}(X, Y)_n \to \mathcal{H}\mathrm{om}(X, Y)_m$ by pullback, that is, by precomposition, along

$$\mathrm{id}_X \times f : X \times \Delta^m \to X \times \Delta^n.$$

Also define the evaluation morphism

$$\mathrm{ev}_{X,Y} : \mathcal{H}\mathrm{om}(X, Y) \times X \to Y$$

as follows: the image of $(\varphi, x) \in \mathrm{Hom}_{\mathsf{sSet}}(X \times \Delta^n, Y) \times X_n$ is $\mathrm{ev}_{X,Y,n}(\varphi, x) := \varphi(x, \mathrm{id}_{[n]}) \in Y_n$. Recall that $(\Delta^n)_n = \mathrm{End}_{\mathbf{\Delta}}([n])$.

We must check that $\mathrm{ev}_{X,Y}$ is indeed a morphism in sSet. This is immediate: for $f : [m] \to [n]$ and $(\varphi, x) \in \mathcal{H}\mathrm{om}(X, Y)_n \times X_n$, we have

$$\begin{aligned} \mathrm{ev}_{X,Y,m}\big(f^*(\varphi), f^*(x)\big) &= ((\mathrm{id}_X \times f)^*\varphi)\big(f^*(x), \mathrm{id}_{[m]}\big) \\ &= \varphi(\underbrace{f^*(x), f}_{\in X_m \times (\Delta^n)_m}) = \varphi\big(f^*(x), f^*\mathrm{id}_{[n]}\big) \\ &= f^*\big(\varphi(x, \mathrm{id}_{[n]})\big) = f^*\big(\mathrm{ev}_{X,Y,n}(\varphi, x)\big). \end{aligned}$$

As X, Y vary, this construction gives a functor $\mathcal{H}\mathrm{om} : \mathsf{sSet}^{\mathrm{op}} \times \mathsf{sSet} \to \mathsf{sSet}$, while ev is natural in X and Y.

Proposition 8.9.2 For every $X, Y, Z \in \mathrm{Ob}(\mathsf{sSet})$, there is a canonical bijection

$$\mathrm{Hom}_{\mathsf{sSet}}\big(X, \mathcal{H}\mathrm{om}(Y, Z)\big) \xrightarrow{1:1} \mathrm{Hom}_{\mathsf{sSet}}(X \times Y, Z).$$

It sends $g : X \to \mathcal{H}\mathrm{om}(Y, Z)$ to the composite

$$X \times Y \xrightarrow{g \times \mathrm{id}_Y} \mathcal{H}\mathrm{om}(Y, Z) \times Y \xrightarrow{\mathrm{ev}_{Y,Z}} Z.$$

Conversely, it sends $h : X \times Y \to Z$ to the following morphism. For $n \in \mathbb{Z}_{\geq 0}$ and $x \in X_n$, let $\iota_x : \Delta^n \to X$ be the morphism corresponding to x. The image of x is the composite

$$Y \times \Delta^n \xrightarrow{\mathrm{id}_Y \times \iota_x} Y \times X \xrightarrow[\sim]{\mathrm{swap}} X \times Y \xrightarrow{h} Z.$$

Proof A routine verification. □

Corollary 8.9.3 The bijection in Proposition 8.9.2 makes sSet a Cartesian closed category in the sense of Definition 7.3.2, with bifunctor $\mathcal{H}\mathrm{om}$.

Proof Recall that the product in sSet is simply the product $X \times Y$ of simplicial sets, and similarly for the other products. It remains to compare (7.3.1) with the canonical bijection in Proposition 8.9.2. □

The general theory of closed monoidal categories (see §7.3) gives a canonical morphism $\mathcal{H}\mathrm{om}(X,Y)\times X\to Y$: it is the image of id under

$$\mathrm{End}_{\mathsf{sSet}}\big(\mathcal{H}\mathrm{om}(X,Y)\big)\xrightarrow{1:1}\mathrm{Hom}_{\mathsf{sSet}}\big(\mathcal{H}\mathrm{om}(X,Y)\times X,Y\big).$$

Unwinding the definitions shows that this morphism is precisely the $\mathrm{ev}_{X,Y}$ defined above.

Example 8.9.4 Proposition 8.3.4 says that the nerve functor N embeds Cat fully faithfully into sSet. How is $\mathcal{H}\mathrm{om}$ reflected at the level of categories? The answer is simple: for any small categories $\mathcal{C}$ and $\mathcal{D}$, there is a canonical isomorphism

$$\mathcal{H}\mathrm{om}(\mathrm{N}\mathcal{C},\mathrm{N}\mathcal{D})\simeq\mathrm{N}(\mathcal{D}^{\mathcal{C}}).$$

Indeed, N takes products of categories to products of simplicial sets, and $\mathrm{N}([n])=\Delta^n$. Since Cat is also Cartesian closed, we obtain

$$\begin{aligned}\mathrm{Hom}_{\mathsf{sSet}}(\mathrm{N}\mathcal{C}\times\Delta^n,\mathrm{N}\mathcal{D})&=\mathrm{Hom}_{\mathsf{sSet}}(\mathrm{N}(\mathcal{C}\times[n]),\mathrm{N}\mathcal{D})\\&\simeq\mathrm{Hom}_{\mathsf{Cat}}(\mathcal{C}\times[n],\mathcal{D})\simeq\mathrm{Hom}_{\mathsf{Cat}}([n],\mathcal{D}^{\mathcal{C}})=\mathrm{N}(\mathcal{D}^{\mathcal{C}})_n.\end{aligned}$$

It is easy to check that the pullbacks on both sides correspond for every morphism $f:[m]\to[n]$ in $\mathbf{\Delta}$.

Now consider sAb. Recall that it has the symmetric monoidal structure arising from $\otimes:=\otimes_{\mathbb{Z}}$, namely the degreewise tensor product. For $X\in\mathrm{Ob}(\mathsf{sAb})$ and $K\in\mathrm{Ob}(\mathsf{sSet})$, define

$$X\otimes K:=X\otimes\mathbb{Z}K\in\mathrm{Ob}(\mathsf{sAb}).$$

In particular, $X\otimes\Delta^n$ is defined. This construction gives a bifunctor $\mathsf{sAb}\times\mathsf{sSet}\to\mathsf{sAb}$.

Definition 8.9.5 For $X,Y\in\mathrm{Ob}(\mathsf{sAb})$, define $\mathcal{H}\mathrm{om}(X,Y)=\mathcal{H}\mathrm{om}_{\mathsf{sAb}}(X,Y)\in\mathrm{Ob}(\mathsf{sAb})$ by $\mathcal{H}\mathrm{om}(X,Y)_n:=\mathrm{Hom}_{\mathsf{sAb}}(X\otimes\Delta^n,Y)$. The pullback morphisms are defined as in Definition 8.9.1.

Proposition 8.9.6 For every $X,Y,Z\in\mathrm{Ob}(\mathsf{sAb})$, there is a canonical commutative diagram

$$\begin{array}{ccc}\mathrm{Hom}_{\mathsf{sSet}}\big(X,\mathcal{H}\mathrm{om}_{\mathsf{sSet}}(Y,Z)\big)&\xrightarrow{1:1}&\mathrm{Hom}_{\mathsf{sSet}}(X\times Y,Z)\\ \cup&&\uparrow\\ \mathrm{Hom}_{\mathsf{sAb}}\big(X,\mathcal{H}\mathrm{om}_{\mathsf{sAb}}(Y,Z)\big)&\xrightarrow[1:1]{}&\mathrm{Hom}_{\mathsf{sAb}}(X\otimes Y,Z)\end{array}$$

Proof The first row is Proposition 8.9.2. The vertical arrow on the right embeds $\mathrm{Hom}_{\mathsf{sAb}}(X\otimes Y,Z)$ as the bilinear part of $\mathrm{Hom}_{\mathsf{sSet}}(X\times Y,Z)$. The image of the embedding on the left consists of all $\varphi:X\to\mathcal{H}\mathrm{om}_{\mathsf{sSet}}(Y,Z)$ for which the family of morphisms

$$\varphi_n : X_n \to \mathrm{Hom}_{\mathsf{sSet}}(Y \times \Delta^n, Z)$$

satisfies, for every $n \in \mathbb{Z}_{\geq 0}$,

- $\varphi_n(x) : Y \times \Delta^n \to Z$ is additive in the Y variable for every $x \in X_n$, or equivalently factors through $\mathrm{Hom}_{\mathsf{sAb}}(Y \otimes \Delta^n, Z)$;
- the resulting map $\varphi_n : X_n \to \mathrm{Hom}_{\mathsf{sAb}}(Y \otimes \Delta^n, Z)$ is additive.

Unwinding the definitions immediately shows that these two descriptions correspond.□

In fact, sAb is an additive category, and the bijection $\mathrm{Hom}_{\mathsf{sAb}}\big(X, \mathcal{H}\mathrm{om}_{\mathsf{sAb}}(Y, Z)\big) \simeq \mathrm{Hom}_{\mathsf{sAb}}(X \otimes Y, Z)$ is an isomorphism of $\mathbb{Z}$-modules.

Corollary 8.9.7 Let $X, Y \in \mathrm{Ob}(\mathsf{sAb})$ and $K \in \mathrm{Ob}(\mathsf{sSet})$. There are canonical isomorphisms of $\mathbb{Z}$-modules

$$\begin{aligned}\mathrm{Hom}_{\mathsf{sAb}}(X \otimes K, Y) &\simeq \mathrm{Hom}_{\mathsf{sSet}}\big(K, \mathcal{H}\mathrm{om}_{\mathsf{sAb}}(X, Y)\big)\\ &\simeq \mathrm{Hom}_{\mathsf{sAb}}\big(X, \mathcal{H}\mathrm{om}_{\mathsf{sSet}}(K, Y)\big).\end{aligned}$$

By the universal property, the underlying simplicial set of $\mathcal{H}\mathrm{om}_{\mathsf{sSet}}(K, Y)$ in the last term is naturally identified with the underlying simplicial set of $\mathcal{H}\mathrm{om}_{\mathsf{sAb}}(\mathbb{Z}K, Y)$, and thus acquires the structure of an object of sAb.

Proof Proposition 8.9.6 gives

$$\begin{aligned}\mathrm{Hom}_{\mathsf{sAb}}(X \otimes K, Y) &\simeq \mathrm{Hom}_{\mathsf{sAb}}(\mathbb{Z}K \otimes X, Y)\\ &\simeq \mathrm{Hom}_{\mathsf{sAb}}\big(\mathbb{Z}K, \mathcal{H}\mathrm{om}_{\mathsf{sAb}}(X, Y)\big).\end{aligned}$$

The adjunction property of $\mathbb{Z}(\cdot)$ shows that the last term is $\mathrm{Hom}_{\mathsf{sSet}}(K, \mathcal{H}\mathrm{om}_{\mathsf{sAb}}(X, Y))$. Similarly, $\mathrm{Hom}_{\mathsf{sAb}}(X \otimes K, Y) \simeq \mathrm{Hom}_{\mathsf{sAb}}(X, \mathcal{H}\mathrm{om}_{\mathsf{sAb}}(\mathbb{Z}K, Y))$. □

Corollary 8.9.8 The canonical bijection in Proposition 8.9.6 makes $(\mathsf{sAb}, \otimes)$ a closed monoidal category in the sense of Definition 7.3.1, with bifunctor $\mathcal{H}\mathrm{om}$.

As for simplicial sets, the general theory of closed monoidal categories gives a canonical evaluation morphism

$$\mathrm{ev}_{X,Y} : \mathcal{H}\mathrm{om}_{\mathsf{sAb}}(X, Y) \otimes X \to Y.$$

This morphism has a concrete description similar to Definition 8.9.1, obtained simply by unwinding the definitions.

We now study the relationship between $\mathcal{H}\mathrm{om}_{\mathsf{sAb}}$ and the Hom complex from the perspective of the Dold–Kan correspondence (Theorem 8.5.9). They are the internal

Hom objects of the closed monoidal categories sAb and $\mathsf{Ch}_{\geq 0}(\mathsf{Ab})$, respectively. Although the normalized chain complex functor $\mathrm{N} : \mathsf{sAb} \to \mathsf{Ch}_{\geq 0}(\mathsf{Ab})$ is an equivalence, it is not a monoidal functor. We therefore cannot directly deduce an isomorphism between $\mathrm{N}\mathcal{H}\mathrm{om}_{\mathsf{sAb}}(X, Y)$ and $\mathrm{Hom}^{\bullet}(\mathrm{N}X, \mathrm{N}Y)$. Nevertheless, the results of §8.8 still provide a pair of canonical morphisms

$$\mathrm{N}\mathcal{H}\mathrm{om}_{\mathsf{sAb}}(X, Y) \rightleftarrows \mathrm{Hom}^{\bullet}(\mathrm{N}X, \mathrm{N}Y). \tag{8.9.1}$$

- By the closed monoidal structure on $\mathsf{Ch}_{\geq 0}(\mathsf{Ab})$, giving a morphism from left to right is equivalent to giving a morphism

  $$\mathrm{N}\mathcal{H}\mathrm{om}_{\mathsf{sAb}}(X, Y) \otimes \mathrm{N}X \longrightarrow \mathrm{N}Y.$$

  Take this morphism to be the composite

  $$\mathrm{N}\mathcal{H}\mathrm{om}_{\mathsf{sAb}}(X, Y) \otimes \mathrm{N}X \xrightarrow{\mathrm{EZ}} \mathrm{N}\big(\mathcal{H}\mathrm{om}_{\mathsf{sAb}}(X, Y) \otimes X\big) \xrightarrow{\mathrm{N}(\mathrm{ev}_{X,Y})} \mathrm{N}Y.$$

- Apply the left adjoint Γ of N, which is also its quasi-inverse. Giving a morphism from right to left is equivalent to giving a morphism

  $$\Gamma\,\mathrm{Hom}^{\bullet}(\mathrm{N}X, \mathrm{N}Y) \longrightarrow \mathcal{H}\mathrm{om}_{\mathsf{sAb}}(X, Y).$$

  By the closed monoidal structure on sAb, this is also equivalent to giving a morphism $\Gamma\,\mathrm{Hom}^{\bullet}(\mathrm{N}X, \mathrm{N}Y) \otimes X \to Y$. Concretely, choose this morphism so that its image under N is the composite

  $$\begin{aligned}\mathrm{N}\big(\Gamma\,\mathrm{Hom}^{\bullet}(\mathrm{N}X, \mathrm{N}Y) \otimes X\big) &\xrightarrow{\mathrm{AW}} \mathrm{N}\Gamma\,\mathrm{Hom}^{\bullet}(\mathrm{N}X, \mathrm{N}Y) \otimes \mathrm{N}X \\ &\simeq \mathrm{Hom}^{\bullet}(\mathrm{N}X, \mathrm{N}Y) \otimes \mathrm{N}X \xrightarrow{\text{evaluation}} \mathrm{N}Y.\end{aligned}$$

The point of the construction above is that N is both a right lax and a left lax monoidal functor. If N were monoidal, or equivalently if EZ and AW were both isomorphisms, an abstract argument would show that the two morphisms in (8.9.1) are mutually inverse; in reality, they are not. However, the general Eilenberg–Zilber theorem, Theorem 8.8.9, states that EZ and AW are mutually inverse on homology. This gives the following relationship between the Hom complex and $\mathcal{H}\mathrm{om}_{\mathsf{sAb}}$.

Proposition 8.9.9 For every $X, Y \in \mathrm{Ob}(\mathsf{sAb})$, the pair of morphisms in (8.9.1) are mutually inverse on homology.

If $\Bbbk$ is a commutative ring, an analogous result holds after replacing Ab by the symmetric monoidal category $\Bbbk\text{-}\mathsf{Mod}$ and considering simplicial objects in it.

Remark 8.9.10 The definition of $\mathcal{H}\mathrm{om}$ can be extended to an arbitrary additive category $\mathcal{A}$. Indeed, in the description of $\mathcal{H}\mathrm{om}_{\mathsf{sAb}}(X,Y)_n$, we have

$$(X \otimes \mathbb{Z}\Delta^n)_k = X_k \otimes \left(\bigoplus_{u:[k]\to[n]} \mathbb{Z} \right) \simeq \bigoplus_{u:[k]\to[n]} X_k.$$

The last term involves only finite direct sums and no elements; the maps d_i, s_j, and so forth also have simple descriptions in this direct-sum presentation. We can therefore define a bifunctor

$$\mathcal{H}\mathrm{om}_{\mathcal{A}} : (\mathsf{s}\mathcal{A})^{\mathrm{op}} \times \mathsf{s}\mathcal{A} \to \mathsf{s}\mathcal{A},$$

which is additive in each variable.

8.10 The Mapping Cone Revisited

As we saw in §3.3 and §4.4, the mapping cone $\mathrm{Cone}(f)$ plays an important role in the theory of complexes and in the study of derived categories. The aim of this section is to interpret it through the Dold–Kan correspondence and to explain why a mapping cone really is a "cone." More precisely, we shall study the chain complex version of the mapping cone; see Remark 3.3.13.

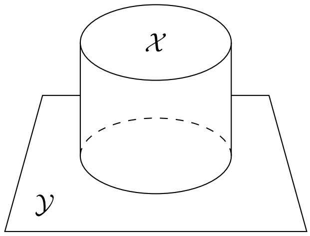
We begin with mapping cones and mapping cylinders in topology. Consider a continuous map $F : \mathcal{X} \to \mathcal{Y}$ with $\mathcal{X} \neq \varnothing$. Its mapping cylinder is the quotient space

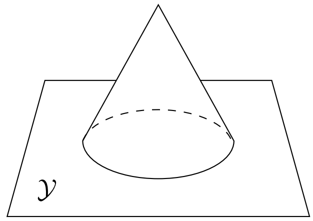
$$\mathrm{Cyl}_{\mathrm{top}}(F) := \frac{(\mathcal{X} \times [0,1]) \sqcup \mathcal{Y}}{\forall x \in \mathcal{X},\ (x,1) \sim F(x)},$$

where $\times$ (respectively, $\sqcup$) denotes the product (respectively, disjoint union) of topological spaces. The mapping cone is defined by

$$\begin{aligned} \mathrm{Cone}_{\mathrm{top}}(F) &:= \frac{(\mathcal{X} \times [0,1]) \sqcup \mathcal{Y}}{\forall x \in \mathcal{X},\ (x,1) \sim F(x),\ (x,0) \sim (\star, 0)} \\ &\simeq \frac{\mathrm{Cyl}_{\mathrm{top}}(F)}{\forall x \in \mathcal{X},\ (x,0) \sim (\star,0)}, \end{aligned}$$

where $\star \in \mathcal{X}$ is an arbitrary point. Intuitively, $\mathcal{X} \times [0,1]$ may be pictured as a tube with cross-section $\mathcal{X}$. The cylinder $\mathrm{Cyl}_{\mathrm{top}}(F)$ is obtained by gluing the end $\mathcal{X} \times \{1\}$ of the tube to $\mathcal{Y}$ via F, while $\mathrm{Cone}_{\mathrm{top}}(F)$ further contracts the end $\mathcal{X} \times \{0\}$ to a single point. Schematically:

$\mathrm{Cyl}_{\mathrm{top}}(F) =$ 

$\mathrm{Cone}_{\mathrm{top}}(F) =$ 

Simplicial sets provide a concrete way to describe topological spaces. More precisely, Theorem 8.4.7 gives an adjoint pair

$$|\cdot| : \mathsf{sSet} \rightleftarrows \mathsf{Top} : \mathrm{Sing},$$

and the two categories model equivalent homotopy theories in a precise sense. The same construction works with Top replaced by CGHaus. We now translate the preceding discussion into sSet. We may assume that

$$\mathcal{X} = |X|, \quad \mathcal{Y} = |Y|, \quad F = |f| : |X| \to |Y|,$$

where $f : X \to Y$ is a morphism in sSet. From the viewpoint of simplicial sets, the mapping cone is easier to describe than the mapping cylinder. We begin with the cone on $\mathcal{X}$, $\mathrm{Cone}_{\mathrm{top}}(\mathcal{X}) := \mathrm{Cone}_{\mathrm{top}}(\mathrm{id}_{\mathcal{X}})$. If $\mathcal{X} = |X|$, then $\mathrm{Cone}_{\mathrm{top}}(\mathcal{X}) \simeq |X^{\triangleleft}| \simeq |X^{\triangleright}|$ (Example 8.4.5). Below we focus on the left cone $X^{\triangleleft} = \Delta^0 \star X$ and its corresponding chain complex.

For every simplicial set Z, write Z_n^{nd} for the set of all nondegenerate n-simplices in it, and write $\{\mathrm{pt}\} = (\Delta^0)_0 = (\Delta^0)_0^{\mathrm{nd}}$. Proposition 8.2.7, or topological intuition, gives

$$(\Delta^0 \star X)_n^{\mathrm{nd}} = \begin{cases} X_n^{\mathrm{nd}} \sqcup \left(\{\mathrm{pt}\} \times X_{n-1}^{\mathrm{nd}}\right), & n > 0 \\ \{\mathrm{pt}\} \sqcup X_0, & n = 0. \end{cases} \tag{8.10.1}$$

Substituting (8.2.2) and the general formulas discussed above it shows that, for $n \geq 1$, the restrictions of $d_i : (\Delta^0 \star X)_n \to (\Delta^0 \star X)_{n-1}$ to $\{\mathrm{pt}\} \times X_{n-1}$ and X_n are as follows:

$$\begin{array}{cc} 1 \leq i \leq n & i = 0 \\ \boxed{\begin{tikzcd} \{\mathrm{pt}\} \times X_{n-1} \arrow[d, "\mathrm{id} \times d_{i-1}"'] & X_n \arrow[d, "d_i"] \\ \{\mathrm{pt}\} \times X_{n-2} & X_{n-1} \end{tikzcd}} & \boxed{\begin{tikzcd} \{\mathrm{pt}\} \times X_{n-1} \arrow[dr, "\mathrm{pr}_2"] & X_n \arrow[d, "d_0"] \\ \{\mathrm{pt}\} \times X_{n-2} & X_{n-1} \end{tikzcd}} \end{array} \tag{8.10.2}$$

If $n = 1$, the term $\{\mathrm{pt}\} \times X_{n-2}$ in this formula is to be understood as $\{\mathrm{pt}\}$.

Since our goal is a chain complex, the next step is to linearize the problem using the functor $\mathbb{Z}(\cdot)$ of Definition 8.6.1, and then pass to chain complexes via the Dold–Kan correspondence:

$$\mathrm{N}(\mathbb{Z}X^{\lhd}) \simeq \mathrm{C}(\mathbb{Z}X^{\lhd})/\text{degenerate part} \quad (\text{Proposition 8.5.11}).$$

To simplify notation in the rest of this section, write $\mathrm{N}X := \mathrm{N}(\mathbb{Z}X)$ and $\mathrm{N}f := \mathrm{N}(\mathbb{Z}f)$. The chain-complex structure on $\mathrm{C}(\mathbb{Z}X^{\lhd})$ comes from $\partial_n := \sum_{i=0}^{n}(-1)^i d_i$. Combining (8.10.1), (8.10.2), and the description of $\mathrm{Cone}(\mathrm{id}_{\mathrm{N}X})$ in Remark 3.3.13, we obtain a short exact sequence of chain complexes

$$0 \to \underbrace{\mathbb{Z}\{\mathrm{pt}\}}_{\text{degree zero}} \to \mathrm{N}(\mathbb{Z}X^{\lhd}) \to \mathrm{Cone}(\mathrm{id}_{\mathrm{N}X}) \to 0.$$

Next, the functors $|\cdot|$ and $\mathrm{N}\mathbb{Z}(\cdot)$ preserve pushout diagrams: the former is a left adjoint, while the latter is the composite of a left adjoint with an equivalence; in topology, a pushout diagram means gluing. Therefore, for $f : X \to Y$ define

$$\mathrm{Cone_s}(f) := X^{\lhd} \underset{X,f}{\sqcup} Y.$$

We then obtain

$$\begin{tikzcd}
\mathcal{X} \arrow[r, "\text{base}"] \arrow[d, "F"'] \arrow[dr, phantom, "\boxplus"] & \mathrm{Cone_{top}}(\mathcal{X}) \arrow[d] & & X \arrow[r, "\text{canonical}"] \arrow[d, "f"'] \arrow[dr, phantom, "\boxplus"] & X^{\lhd} \arrow[d] & & \mathrm{N}X \arrow[r] \arrow[d, "\mathrm{N}f"'] \arrow[dr, phantom, "\boxplus"] & \mathrm{N}X^{\lhd} \arrow[d] \\
\mathcal{Y} \arrow[r] & \mathrm{Cone_{top}}(F) & \arrow[l, phantom, "\overset{|\cdot|}{\longleftarrow}"] & Y \arrow[r] & \mathrm{Cone_s}(f) & \arrow[l, phantom, "\overset{\mathrm{N}}{\longrightarrow}"] & \mathrm{N}Y \arrow[r] & \mathrm{N}\,\mathrm{Cone_s}(f)
\end{tikzcd}$$

This chain of constructions shows that the linear-algebraic counterpart of $\mathrm{Cone_{top}}(F)$ should be $\mathrm{N}\,\mathrm{Cone_s}(f)$.

Proposition 8.10.1 With the notation above, there is a canonical short exact sequence of chain complexes

$$0 \to \underbrace{\mathbb{Z}\{\mathrm{pt}\}}_{\text{degree zero}} \to \mathrm{N}\,\mathrm{Cone_s}(f) \to \mathrm{Cone}(\mathrm{N}f) \to 0.$$

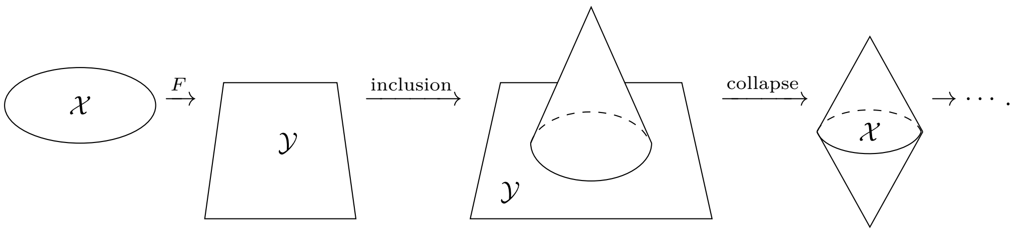
Proof The pushout does not affect the chain subcomplex $\mathbb{Z}\{\mathrm{pt}\}$ of $\mathrm{N}X^{\lhd}$ corresponding to the cone point. In degree n, it replaces every X_n by Y_n and replaces the projection pr_2 in (8.10.2) by $f_{n-1}\mathrm{pr}_2$. Thus, on the quotient $\mathrm{Cone}(\mathrm{id}_{\mathrm{N}X})$, the resulting pushout is precisely $\mathrm{Cone}(\mathrm{N}f)$; see Remark 3.3.13. □

Remark 8.10.2 The mapping cylinder $\mathrm{Cyl_{top}}(F)$ and its chain-complex version can be compared similarly; here there is no need to quotient out the contribution of the cone point. However, the mapping cylinder involves products of simplicial sets, whereas N is not a monoidal functor. Topological accounts therefore usually use CW complexes rather than simplicial sets, since products are easier to handle in the world of CW complexes.

In short, the mapping cone of chain complexes is the quotient of the mapping cone arising from simplicial sets (or topology) by a copy of $\mathbb{Z}$ coming from the cone point pt. In the study of chain complexes, the main role of the mapping cone is that every map $\phi : A \to B$ extends to a sequence

$$A \xrightarrow{\phi} B \to \mathrm{Cone}(\phi) \to A[-1] \xrightarrow{-\phi[-1]} B[-1] \to \cdots . \tag{8.10.3}$$

In the topological setting, the map corresponding to $\mathrm{Cone}(\phi) \to A[-1]$ is obtained by contracting the base $\mathcal{Y}$ of $\mathrm{Cone}_{\mathrm{top}}(F)$ to a point. The result is denoted by $S\mathcal{X}$; it is independent of $\mathcal{Y}$ (see the figure below) and defines a functor S. The corresponding sequence of maps is



This sequence is called the **cofiber sequence** generated by f (more precisely, the unpointed version). If we work in sSet and take the chain complexes of the corresponding sequence, the result is roughly (8.10.3). The qualification "roughly" is needed because:

- ◇ The map $S\mathcal{X} \to S\mathcal{Y}$ cannot simply be defined as $S(F)$; the vertical coordinate must be reversed appropriately. At the level of linear algebra, this corresponds to $-\phi[-1]$ in (8.10.3); see a topology textbook for details.
- ◇ As Proposition 8.10.1 shows, the various chain complexes contain one extra copy of $\mathbb{Z}$ corresponding to the cone point of spaces such as $\mathrm{Cone}_{\mathrm{top}}(F)$ and $S\mathcal{X}$. In topology, this issue is avoided by using pointed spaces and, in the topological cone construction, contracting the part associated with the base point.

If we set all details aside for the moment, the theory of derived categories constructed in §4.4, with $\mathcal{A} = \mathsf{Ab}$, may also be viewed in broad outline as a somewhat coarse "linearization" of homotopy theory[5)].

As the related questions increasingly involve topological ideas and methods, we stop here so as not to stray too far afield.

5) More precisely, stable homotopy theory, since chain complexes allow terms in negative degrees.

Exercises

1. Prove that, for every category $\mathcal{C}$, the construction $C \mapsto \mathrm{const}(C)$ of Example 8.1.5 gives a fully faithful functor $\mathcal{C} \to \mathsf{s}\mathcal{C}$.

2. Let X be a simplicial set and $x \in X_n$. Prove that there is a sequence of indices $j_1 < \cdots < j_h$ and a nondegenerate simplex y such that $x = s_{j_h} \cdots s_{j_1}(y)$, and that y is unique.

Hint The difficulty lies in uniqueness. Suppose that $Sy = x = S'y'$, where S and S' are both of the form in the statement. We can choose a composite D of a sequence of face morphisms such that $y = DSy = DS'y'$. Rewrite DS' in the form $\widetilde{S'}\widetilde{D}$ (the meaning of the notation is clear). Since y is nondegenerate, we must have $\widetilde{S'} = \mathrm{id}$, and hence $y = \widetilde{D}y'$. Use symmetry to conclude that $y = y'$ and $\widetilde{D} = \mathrm{id}$.

3. Show that the following diagram exists in sSet, with both the upper and lower parts commutative, for $0 \leq i < j \leq n$:

$$\begin{tikzcd}
\Delta^{n-2} \arrow[r, "\mathrm{d}^{j-1}"] \arrow[d, "\iota_{i<j}"'] & \Delta^{n-1} \arrow[d, "\iota_i"] \arrow[dr, "\mathrm{d}^i"] & \\
\coprod_{0 \leq i' < j' \leq n} \Delta^{n-2} \arrow[r, shift left] \arrow[r, shift right] & \coprod_{i'=0}^{n} \Delta^{n-1} \arrow[r] & \partial\Delta^n \\
\Delta^{n-2} \arrow[r, "\mathrm{d}^i"'] \arrow[u, "\iota_{i<j}"] & \Delta^{n-1} \arrow[u, "\iota_j"'] \arrow[ur, "\mathrm{d}^j"'] &
\end{tikzcd}$$

Here $\coprod$ is taken degreewise in $\mathsf{sSet} = \mathsf{Set}^{\mathbf{\Delta}^{\mathrm{op}}}$, while $\iota_{i<j}$ (respectively, ι_i) denotes the embedding into the $i < j$ term (respectively, the ith term). All arrows in the middle row are characterized by this commutativity. Show, moreover, that the middle row is a coequalizer.

Similarly, show that for every $0 \leq k \leq n$ there is also a diagram

$$\begin{tikzcd}
\Delta^{n-2} \arrow[r, "\mathrm{d}^{j-1}"] \arrow[d, "\iota_{i<j}"'] & \Delta^{n-1} \arrow[d, "\iota_i"] \arrow[dr, "\mathrm{d}^i"] & \\
\coprod_{\substack{0 \leq i' < j' \leq n \\ i', j' \neq k}} \Delta^{n-2} \arrow[r, shift left] \arrow[r, shift right] & \coprod_{\substack{0 \leq i' \leq n \\ i' \neq k}} \Delta^{n-1} \arrow[r] & \Lambda^n_k \\
\Delta^{n-2} \arrow[r, "\mathrm{d}^i"'] \arrow[u, "\iota_{i<j}"] & \Delta^{n-1} \arrow[u, "\iota_j"'] \arrow[ur, "\mathrm{d}^j"'] &
\end{tikzcd}$$

whose middle row is a coequalizer and where $i, j \neq k$. Try to explain the topological meaning of these coequalizers.

4. Verify the following properties of the join operation $\star$ in Definition 8.2.6.

(i) Simplicial sets naturally form a monoidal category under $\star$, with the constant empty simplicial set $\varnothing$ as unit object.

(ii) The order-reversal duality of Remark 8.1.7 gives an automorphism of sSet, provisionally denoted by $X \mapsto \hat{X}$. Show that $(X \star Y)^\wedge \simeq \hat{Y} \star \hat{X}$.

(iii) Verify that $\Delta^p \star \Delta^q \simeq \Delta^{p+q+1}$.

5. Show that taking the order-reversal dual of a simplicial set does not change its geometric realization, up to canonical homeomorphism. Hint Begin with the case of Δ^n.

6. Verify the assertion of Example 8.4.5.

7. Let $\mathcal{C}$ and $\mathcal{C}'$ be categories. Define the category $\mathcal{C} \star \mathcal{C}'$ as follows:

$$\mathrm{Ob}(\mathcal{C} \star \mathcal{C}') := \mathrm{Ob}(\mathcal{C}) \sqcup \mathrm{Ob}(\mathcal{C}'),$$

$$\mathrm{Hom}_{\mathcal{C} \star \mathcal{C}'}(X, Y) := \begin{cases} \mathrm{Hom}_{\mathcal{C}}(X, Y), & X, Y \in \mathrm{Ob}(\mathcal{C}) \\ \mathrm{Hom}_{\mathcal{C}'}(X, Y), & X, Y \in \mathrm{Ob}(\mathcal{C}'), \\ \text{one-point set}, & X \in \mathrm{Ob}(\mathcal{C}),\ Y \in \mathrm{Ob}(\mathcal{C}'), \\ \varnothing, & X \in \mathrm{Ob}(\mathcal{C}'),\ Y \in \mathrm{Ob}(\mathcal{C}). \end{cases}$$

Composition and identity morphisms are defined in the evident way. Prove that the join operation $\star$ on simplicial sets and the nerve functor of Example 8.3.1 are related by

$$\mathrm{N}(\mathcal{C}) \star \mathrm{N}(\mathcal{C}') \simeq \mathrm{N}(\mathcal{C} \star \mathcal{C}').$$

8. Give a more detailed proof of Equations (8.4.4) and (8.4.5).

9. For a simplicial object X in an arbitrary category, define the simplicial objects called the decalages $\mathrm{Déc}_0 X$ and $\mathrm{Déc}^0 X$ by $(\mathrm{Déc}_0 X)_n = (\mathrm{Déc}^0 X)_n = X_{n+1}$ and

$$d_i^{\mathrm{Déc}_0 X, n} = d_i^{X, n+1}, \quad s_j^{\mathrm{Déc}_0 X, n} = s_j^{X, n+1},$$
$$d_i^{\mathrm{Déc}^0 X, n} = d_{i+1}^{X, n+1}, \quad s_j^{\mathrm{Déc}^0 X, n} = s_{j+1}^{X, n+1}.$$

Show that both are well-defined and are related by the order-reversal duality (Remark 8.1.7). Also show that for an additive category, $\mathrm{C}(\mathrm{Déc}^0 X) = \mathrm{C}(X)[1]$; see Remark 3.1.9.

10. Continuing the preceding exercise, show that $d_0 : X_1 \to X_0$ (respectively, $d_1 : X_1 \to X_0$) gives an augmentation of $\mathrm{Déc}_0 X$ (respectively, $\mathrm{Déc}^0 X$), with X_0 as the term of degree -1. Show further that, after augmentation, $\mathrm{Déc}_0 X$ is right contractible and $\mathrm{Déc}^0 X$ is left contractible; see Definition 8.6.8. Hint It suffices to treat $\mathrm{Déc}_0 X$. Take $k_{n-1} := s_n : X_n \to X_{n+1}$.

11. Prove that a small category $\mathcal{C}$ is a groupoid (that is, all its morphisms are invertible) if and only if $\mathrm{N}\mathcal{C}$ satisfies the following condition: for every $n \in \mathbb{Z}_{\geq 1}$ and $0 \leq i \leq n$, every morphism $\Lambda_i^n \to \mathrm{N}\mathcal{C}$ can be extended to $\Delta^n \to \mathrm{N}\mathcal{C}$, without requiring uniqueness. A simplicial set satisfying this extension condition is called a Kan complex.

12. Continue the discussion of Example 8.5.2. Show that the normalized complex $\overline{\mathsf{L}}$, after being reversed into a chain complex, is isomorphic to $\mathrm{N}(\Bbbk E\Gamma)$. Hint First use Proposition 8.5.11 to identify $\mathrm{N}(X)$ with a quotient of $\mathrm{C}(X)$.

13. Let Γ be a monoid. Write $\mathsf{Set}\text{-}\Gamma$ for the category of all small sets with a right Γ-action.

(i) Describe explicitly the free–forgetful adjoint pair

$$\mathrm{F} : \mathsf{Set} \rightleftarrows \mathsf{Set}\text{-}\Gamma : U$$

together with its unit and counit morphisms. Show that the corresponding monad T gives an equivalence $\mathsf{Set}\text{-}\Gamma \simeq \mathsf{Set}^T$.

Hint⟩ The functor F sends a set X to $X \times \Gamma$, with Γ acting by right multiplication on the second component. The unit morphism is $x \mapsto (x, 1)$, while the counit morphism is the action map.

(ii) For every right Γ-set X, describe the corresponding simplicial object $\mathrm{Bar}(X)$ in $\mathsf{Set}\text{-}\Gamma$; see Examples 8.7.2 and 8.7.3.

(iii) Take the one-point set $X = \{\mathrm{pt}\}$ with trivial right Γ-action. Show that the corresponding simplicial object is isomorphic to $\mathrm{E}\Gamma$ of Example 8.3.5; it becomes a simplicial object in $\mathsf{Set}\text{-}\Gamma$ via the right Γ-action on each $(\mathrm{E}\Gamma)_n$.

14. Fix a group Γ. In Example 8.7.7, take the group algebra $R = \Bbbk[\Gamma]$ and take M to be the trivial Γ-module $\Bbbk$; see Example 6.1.2. Show that the corresponding $\epsilon_{\Bbbk} : \mathrm{C}(\mathrm{Bar}(\Bbbk)) \to \Bbbk$ is isomorphic to the resolution $\mathsf{L} = (\mathsf{L}_n, \partial'_n)_n \to \Bbbk$ of Proposition 6.1.13. Hint⟩ Use (6.1.4).

15. Let $\Bbbk$ be a commutative ring and $\mathcal{A} = \Bbbk\text{-}\mathsf{Mod}$, with $\otimes = \otimes_{\Bbbk}$. Let Γ be a group. Consider the simplicial object $\Bbbk\mathrm{E}\Gamma$ of Example 8.6.9. Apply Definition 8.8.7 of the Alexander–Whitney morphism to $X := \Bbbk\mathrm{E}\Gamma \boxtimes \Bbbk\mathrm{E}\Gamma$, and then consider the morphism

$$\mathrm{C}(\Bbbk\mathrm{E}\Gamma) \to \mathrm{C}(\underbrace{\Bbbk\mathrm{E}\Gamma \otimes \Bbbk\mathrm{E}\Gamma}_{\simeq \Bbbk(\mathrm{E}\Gamma\times\mathrm{E}\Gamma)}) \xrightarrow{\overline{\mathrm{AW}}} \mathrm{C}(\Bbbk\mathrm{E}\Gamma) \otimes \mathrm{C}(\Bbbk\mathrm{E}\Gamma).$$

The first arrow is induced by the diagonal embedding of simplicial sets $\mathrm{E}\Gamma \to \mathrm{E}\Gamma \times \mathrm{E}\Gamma$. In view of Example 8.5.2, the composite can also be regarded as a morphism of chain complexes

$$\mathsf{L} \to \mathsf{L} \otimes \mathsf{L}.$$

Use this to interpret the morphism $\Delta : \mathsf{L} \to \mathsf{L} \otimes \mathsf{L}$ used in the definition of the cup product in group cohomology; see Lemma 6.10.8 (ii).

Hint⟩ Substitution of the definition of $\overline{\mathrm{AW}}$ shows that the (p, q) component of the composite morphism differs from that in Lemma 6.10.8 (ii) by a factor $(-1)^{pq}$. This difference arises because the discussion there uses $(\mathsf{L}_n, \partial_n)_n$, whereas Example 8.5.2 uses $(\mathsf{L}_n, \partial'_n)_n$. At the level of the simplicial set $\mathrm{E}\Gamma$, the two differ by order reversal (Remark 8.1.7). Observe that order-reversal duality interchanges the roles of λ^n_p and ρ^n_q in $\overline{\mathrm{AW}}$, or equivalently interchanges the two copies of $\mathrm{C}(\Bbbk\mathrm{E}\Gamma)$.

16. Consider a comonad (L, δ, ϵ) on a category $\mathcal{D}$. For every $\mathcal{M} \in \mathrm{Ob}(\mathcal{D})$, if $\epsilon_{\mathcal{M}} : L\mathcal{M} \to \mathcal{M}$ has a right inverse $f : \mathcal{M} \to L\mathcal{M}$, then $\mathcal{M}$ is called an ***L*-projective object**. Dually, from a monad (T, μ, η) on a category $\mathcal{C}$, we define a ***T*-injective object** in $\mathcal{C}$.

By duality, the following discussion focuses mainly on the concept of L-projectivity. By the method of Example 8.7.1, we obtain in $\mathrm{End}(\mathcal{D})$ an augmented simplicial object $(L^{n+1})_{n\geq -1}$; the data d_i, s_j, and so forth are omitted from the notation. For an L-projective object $\mathcal{M}$ and a corresponding f, define

$$k_n := L^{n+1}f : L^{n+1}\mathcal{M} \to L^{n+2}\mathcal{M}, \quad n \geq -1.$$

Prove that this makes the augmented simplicial object $(L^{n+1}\mathcal{M})_{n\geq -1}$ right contractible (Definition 8.6.8).

Hint We know that $\epsilon_{\mathcal{M}} k_{-1} = \epsilon_{\mathcal{M}} f = \mathrm{id}_{\mathcal{M}}$. Applying L^{n+1} to both sides gives $d_{n+1}k_n = \mathrm{id}_{L^{n+1}\mathcal{M}}$, while naturality ensures that

$$\begin{tikzcd} L\mathcal{M} \arrow[r, "Lf"] \arrow[d, "\epsilon_{\mathcal{M}}"'] & L^2\mathcal{M} \arrow[d, "\epsilon_{L\mathcal{M}}"] \\ \mathcal{M} \arrow[r, "f"'] & L\mathcal{M} \end{tikzcd}$$ commutes, that is, $d_0k_0 = f\epsilon_{\mathcal{M}} = k_{-1}\epsilon_{\mathcal{M}}$.

Replace f by $L^{n-i}f$, and then apply L^i to the corresponding diagram. The same method gives $d_ik_n = k_{n-1}d_i$ for $0 \leq i \leq n$.

17. For a ring R, the free–forgetful adjoint pair $\mathsf{Set} \leftrightarrows R\text{-}\mathsf{Mod}$ determines a comonad L on $R\text{-}\mathsf{Mod}$. Prove that a left R-module is L-projective if and only if it is a projective module.

18. Consider an adjoint pair $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$ and the corresponding comonad $(L, \delta, \epsilon) = (FG, F\eta G, \epsilon)$ on $\mathcal{D}$ (Example 7.6.2). Prove that every object of the form $F\mathcal{N}$, for $\mathcal{N} \in \mathrm{Ob}(\mathcal{C})$, is L-projective. Consequently, for every $\mathcal{M} \in \mathrm{Ob}(\mathcal{D})$, every term of the simplicial object $(L^{n+1}\mathcal{M})_{n\geq 0}$ is L-projective.

Hint Take $f = F\eta_{\mathcal{N}} : F(\mathcal{N}) \to FGF(\mathcal{N}) = L(F(\mathcal{N}))$. Then $\epsilon_{F(\mathcal{N})}f = \mathrm{id}_{F(\mathcal{N})}$ is precisely one of the triangle identities of the adjunction.

19. Continue to consider the comonad L arising from the adjoint pair (F, G). Show that $\mathcal{P} \in \mathrm{Ob}(\mathcal{D})$ is L-projective if and only if it has the following lifting property. For every pair of morphisms $\alpha : \mathcal{M}_1 \to \mathcal{M}_2$ and $\phi : \mathcal{P} \to \mathcal{M}_2$ in $\mathcal{D}$, if $G\alpha$ has a right inverse, then there is a morphism $\beta : \mathcal{P} \to \mathcal{M}_1$ such that $\phi = \alpha\beta$.

Hint For the "if" direction, apply the lifting property to $\alpha := \epsilon_{\mathcal{P}} : FG(\mathcal{P}) \to \mathcal{P}$ and $\phi := \mathrm{id}$. For the "only if" direction, first explain why $\mathcal{P}$ in the lifting property can be replaced by $FG(\mathcal{P})$, and then use the adjunction to show that

$$\alpha_* : \mathrm{Hom}(FG(\mathcal{P}), \mathcal{M}_1) \to \mathrm{Hom}(FG(\mathcal{P}), \mathcal{M}_2)$$

is surjective.

20. For an arbitrary commutative ring $\Bbbk$ and a homomorphism of $\Bbbk$-algebras $S \to R$, consider the adjoint pairs

$$R \underset{S}{\otimes} (\cdot) : S\text{-}\mathsf{Mod} \rightleftarrows R\text{-}\mathsf{Mod} : \text{forgetful},$$

$$\text{forgetful} : R\text{-}\mathsf{Mod} \rightleftarrows S\text{-}\mathsf{Mod} : \mathrm{Hom}_S(R, \cdot)$$

which give a comonad L and a monad T, respectively, on R-Mod. Use the lifting property above, or its dual, to determine necessary and sufficient conditions for a left R-module M to be an L-projective (respectively, T-injective) object, and compare these conditions with projectivity (respectively, injectivity) of modules.

On this basis one can construct a theory called **relative homological algebra**; see [19]. Its distinguishing feature is that it considers only those exact sequences of R-modules that are split exact over S. The next exercise examines relative versions of the Tor and Ext functors.

21. Consider an adjoint pair between abelian categories $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$. Prove for every $\mathcal{M} \in \mathrm{Ob}(\mathcal{D})$ that:

(i) There is an object $\mathcal{P}_\bullet$ of $\mathsf{Ch}_{\geq 0}(\mathcal{D})$ together with a morphism $\epsilon : \mathcal{P}_\bullet \to \mathcal{M}$ such that

- ◇ each $\mathcal{P}_n$ is L-projective for $n \geq 0$;
- ◇ the identity morphism on $\cdots \to G\mathcal{P}_1 \to G\mathcal{P}_0 \to G\mathcal{M} \to 0$ is null-homotopic; in other words, this is a split exact complex.

Hint〉 Apply Proposition 8.7.5.

(ii) For any two morphisms $\epsilon : \mathcal{P}_\bullet \to \mathcal{M}$ and $\epsilon' : \mathcal{P}'_\bullet \to \mathcal{M}$ satisfying the conditions above, there is a homotopy equivalence $f : \mathcal{P} \to \mathcal{P}'$ such that $\epsilon' f = \epsilon$. Hint〉 Use the lifting property introduced above.

22. (M. Barr, J. M. Beck) Let L be a comonad on a category $\mathcal{D}$. Write $\mathrm{Bar}(\mathcal{M})$ for the simplicial object obtained by evaluating the construction of Example 8.7.1 at $\mathcal{M} \in \mathrm{Ob}(\mathcal{D})$. For every abelian category $\mathcal{A}$ and functor $E : \mathcal{D} \to \mathcal{A}$, define the **comonad homology** of (L, E) to be the family of objects in $\mathcal{A}$

$$\mathrm{H}_n^L(E; \mathcal{M}) := \mathrm{H}_n\left(\mathrm{C}(E\mathrm{Bar}(\mathcal{M}))\right), \quad n \in \mathbb{Z}_{\geq 0},$$

where $E\mathrm{Bar}(\mathcal{M})$ means that E is applied to every term of $\mathrm{Bar}(\mathcal{M})$. Dually, if T is a monad on a category $\mathcal{C}$ and $F : \mathcal{C} \to \mathcal{A}$ is a functor to an abelian category, one can similarly define the value of the **monad cohomology** (T, F) at $\mathcal{N} \in \mathrm{Ob}(\mathcal{C})$; we omit the details.

(i) Write down the natural morphism $\mathrm{H}_0^L(E; \mathcal{M}) \to E\mathcal{M}$ and show that it is an isomorphism if $\mathcal{M}$ is L-projective.

(ii) Suppose that L arises from an adjoint pair (F, G) between abelian categories and that E is an additive functor. Prove that if $\mathcal{P}_\bullet \to \mathcal{M}$ is a morphism in $\mathsf{Ch}_{\geq 0}(\mathcal{D})$ satisfying

- ◇ every $\mathcal{P}_n$ is L-projective for $n \geq 0$;
- ◇ the identity morphism on $\cdots \to G\mathcal{P}_1 \to G\mathcal{P}_0 \to G\mathcal{M} \to 0$ is null-homotopic,

then $\mathrm{H}_n^L(E; \mathcal{M}) \simeq \mathrm{H}_n(E\mathcal{P}_\bullet)$. Thus comonad homology can be computed using such L-projective resolutions.

(iii) Continue under the assumptions above. A pair of morphisms $\mathcal{M}' \to \mathcal{M} \to \mathcal{M}''$ in $\mathcal{D}$ is called a ***G*-split** short exact sequence if, after applying G, it becomes a split short exact sequence in $\mathcal{C}$. Prove that a G-split short exact sequence naturally induces a long exact sequence for $\mathrm{H}^L_n(E; \cdot)$.

23. (G. Hochschild [19]) For an arbitrary commutative ring $\Bbbk$ and a homomorphism of $\Bbbk$-algebras $S \to R$, consider the comonad L on $R\text{-}\mathsf{Mod}$ determined by the adjoint pair

$$R \underset{S}{\otimes} (\cdot) : S\text{-}\mathsf{Mod} \rightleftarrows R\text{-}\mathsf{Mod} : \text{forgetful}$$

For a right R-module A and left R-modules B, N, define the $\Bbbk$-modules

$$\mathrm{Tor}_n^{R|S}(A, N) := \mathrm{H}^L_n\left(A \underset{R}{\otimes} (\cdot); N\right),$$
$$\mathrm{Ext}^n_{R|S}(N, B) := \mathrm{H}^L_n\left(\mathrm{Hom}_R(\cdot, B); N\right),$$

where $\mathrm{Hom}_R(\cdot, B)$ is regarded as a functor $R\text{-}\mathsf{Mod} \to S\text{-}\mathsf{Mod}^{\mathrm{op}}$. These are called the relative Tor and relative Ext functors.

(i) Show that both are functorial in every variable. Describe the corresponding chain complexes explicitly, and show that

$$\mathrm{Tor}_0^{R|S}(A, N) \simeq A \underset{R}{\otimes} N, \quad \mathrm{Ext}^0_{R|S}(N, B) \simeq \mathrm{Hom}_R(N, B).$$

(ii) Show that the bifunctor $\mathrm{Tor}_n^{R|S}$ has a "balanced" property analogous to Theorem 3.14.2.[6)]

(iii) Prove that if I is a two-sided ideal of S and $R = S/I$, then for $n > 0$, $\mathrm{Tor}_n^{R|S} = 0 = \mathrm{Ext}^n_{R|S}$. Thus relative Tor and relative Ext differ from their absolute versions in §3.14. Hint In this case $L = \mathrm{id}_{R\text{-}\mathsf{Mod}}$.

(iv) Write $R^e = R \underset{\Bbbk}{\otimes} R^{\mathrm{op}}$. Prove that $\mathrm{HH}_n(M) \simeq \mathrm{Tor}_n^{R^e|\Bbbk}(M, R)$ and $\mathrm{HH}^n(M) \simeq \mathrm{Ext}^n_{R^e|\Bbbk}(R, M)$; see Example 8.7.7. Hint Reduce the problem to the assertion that $\mathrm{C}(\mathrm{Bar}(R))_n = R \otimes R^{\otimes n} \otimes R$ is an L-projective R^e-module.

[6)] The $\mathrm{Ext}^n_{R|S}$ case is not discussed because it involves the notion of monad cohomology, which we have not developed here; see the references for details.

Chapter 9 Duality

The notion of duality originates in the theory of vector spaces. Let $\Bbbk$ be a field, and denote the dual space of a $\Bbbk$-vector space V by $V^\vee := \mathrm{Hom}_\Bbbk(V, \Bbbk)$. There are

- ◇ the evaluation map $\mathrm{ev} : V \otimes V^\vee \to \Bbbk$, which sends $v \otimes \lambda$ to $\lambda(v)$;
- ◇ if V is finite-dimensional, also the coevaluation map $\mathrm{coev} : \Bbbk \to V^\vee \otimes V$, which sends 1 to $\sum_{i=1}^n \check{v}_i \otimes v_i$, where $(v_i)_{i=1}^n$ is any basis of V and $(\check{v}_i)_{i=1}^n$ its dual basis.

In the finite-dimensional case, they satisfy the fundamental identities

$$(\mathrm{id}_{V^\vee} \otimes \mathrm{ev})(\mathrm{coev} \otimes \mathrm{id}_{V^\vee}) = \mathrm{id}_{V^\vee}, \quad (\mathrm{ev} \otimes \mathrm{id}_V)(\mathrm{id}_V \otimes \mathrm{coev}) = \mathrm{id}_V.$$

In a general monoidal category, the duality data defined in §9.1 directly generalize the situation above: they consist of a pair of objects (L, R) together with morphisms $\mathrm{ev} : L \otimes R \to \mathbf{1}$ and $\mathrm{coev} : \mathbf{1} \to R \otimes L$ satisfying the preceding fundamental identities. In this situation, R is called a right dual of L, while L is called a left dual of R. A monoidal category in which every object has a left dual (respectively, a right dual) is called left rigid (respectively, right rigid).

For braided monoidal categories, §9.2 will explain how the two kinds of dual are related. Thus, if an object X has a dual, we can define the trace $\mathrm{Tr}(f)$ of $f \in \mathrm{End}(X)$ and subsequently its dimension $\dim X := \mathrm{Tr}(\mathrm{id}_X)$. Both are elements of $\mathrm{End}(\mathbf{1})$. Strictly speaking, each has a left and a right version, but the two need not be distinguished in a symmetric monoidal category. This book mainly discusses abelian categories equipped with symmetric monoidal structures, such as categories of modules over commutative rings (see Proposition 9.2.2). As long as the reader understands the usefulness of dual vector spaces, the value of this generalization will be self-evident. Duality becomes still more powerful in the setting of 2-categories, but that lies beyond the scope of this book.

Examples of duality are the subject of §9.3. The examples closest to algebra are the category $\mathsf{Mod}_{\mathrm{f}}\text{-}A$ of finite-dimensional right A-modules for a $\Bbbk$-Hopf algebra A, and its comodule counterpart $\mathsf{Comod}_{\mathrm{f}}\text{-}A$ (Proposition 9.3.5). In both cases, duality reflects properties of the antipode $S : A \to A$. Examples involving Hopf algebras are important in the study of quantum groups and also prepare the way for the theory of Tannakian categories.

The material in §§9.4—9.6 of this chapter concerns the coalgebra reconstruction problem. For every coalgebra L, the abelian category $\mathsf{Comod}_{\mathrm{f}}\text{-}L$ is locally finite (Definition A.4.2) and comes equipped with a faithful exact forgetful functor $U : \mathsf{Comod}_{\mathrm{f}}\text{-}L \to \mathsf{Vect}_{\mathrm{f}}(\Bbbk)$. The reconstruction problem asks for a converse:

> How can one construct a coalgebra L from a locally finite $\Bbbk$-linear abelian category $\mathcal{A}$ and a faithful exact functor $\omega : \mathcal{A} \to \mathsf{Vect}_{\mathrm{f}}(\Bbbk)$ in such a way that $(\mathcal{A}, \omega)$ is naturally equivalent to $(\mathsf{Comod}_{\mathrm{f}}\text{-}L, U)$?

The approach taken in this chapter uses the coendomorphism coalgebra $\mathrm{coE}(\omega) = \mathrm{coE}_{\Bbbk}(\omega)$ of the functor ω (Definition–Proposition 9.4.3). This coalgebra coacts on each $\omega(X)$ and is "universal" with respect to these coactions.

The proof of the coalgebra reconstruction theorem, Theorem 9.5.5, involves the theory of comonads from §7.6, basic Morita theory from §7.7, a technical result about locally finite abelian categories from §A.4, and the Ind-ization introduced in §B.4. If $(\mathcal{A}, \omega)$ is equipped with a monoidal structure, then $\mathrm{coE}(\omega)$ also becomes a bialgebra. If $\mathcal{A}$ is both left and right rigid, then $\mathrm{coE}(\omega)$ is even a Hopf algebra (Theorem 9.6.5). The presentation in this part mainly follows [10].

After all these preparations, §9.7 will introduce the precise definition of a Tannakian category. These are symmetric monoidal categories $\mathcal{T}$ with special properties. The formal definition is due to N. Saavedra Rivano and P. Deligne, while its motivation goes back to work of Tadao Tannaka and his successors. Among the conditions in Definition 9.7.1, the most prominent is the existence of a nonzero commutative $\Bbbk$-algebra B and a right exact monoidal functor $\omega : \mathcal{T} \to \mathsf{Mod}\text{-}B$ compatible with the braiding; this is called a fiber functor for $\mathcal{T}$ over B.

The setting of algebraic geometry requires that B be allowed to be general, but many elementary applications involve only neutral fiber functors. In that case, ω is a faithful exact functor with values in $\mathsf{Vect}_{\mathrm{f}}(\Bbbk)$. The corresponding neutral Tannakian reconstruction theorem, Theorem 9.7.3, is simply an application of the preceding results: it states that $\mathrm{coE}(\omega)$ is a commutative Hopf algebra and identifies $(\mathcal{T}, \omega)$ with $(\mathsf{Comod}_{\mathrm{f}}\text{-}\,\mathrm{coE}(\omega), U)$. [1)] Many properties of $(\mathcal{T}, \omega)$ are thereby translated into properties of this Hopf algebra.

For a general fiber functor, Corollary 9.7.8 relates the commutative Hopf algebra $\mathrm{coE}_B(\omega)$ over B to the group $\mathrm{Aut}^{\otimes}(\omega)$ of $\otimes$-automorphisms of ω. All the information about $\mathrm{coE}_B(\omega)$ is contained in $\mathrm{Aut}^{\otimes}$, provided the latter is understood as a group functor $B\text{-}\mathsf{CAlg} \to \mathsf{Grp}$ rather than merely as a single group. This viewpoint is entirely natural in algebraic geometry.

Finally, §9.8 returns to the historical origins of Tannakian categories. Using the preceding theory, it explains how to reconstruct a finite group G from the category $G\text{-}\mathsf{Mod}_{\mathrm{f}}$ of finite-dimensional G-modules (Definition 6.1.1), together with the forgetful functor ω to $\mathsf{Vect}_{\mathrm{f}}(\Bbbk)$. Since in abstract harmonic analysis $G\text{-}\mathsf{Mod}_{\mathrm{f}}$ is in some sense a "dual" of G, this result is also called the finite-group version of the Tannaka–Krein duality theorem. In fact, once it is known that G-modules are the same as right comodules over the function space $C(G)$, the desired isomorphism $G \xrightarrow{\sim} \mathrm{Aut}^{\otimes}(\omega)$ has a shorter proof. The detour through the reconstruction theorem is taken only to explain the origin of Tannakian categories.

阅读提示

The sections on Hopf algebras, as well as the proof of the reconstruction theorem in §9.5, are based on some of the material in Chapter 7. In addition, the proof of the reconstruction theorem uses §A.4 and §B.4 from the appendices; neither needs to be studied deeply on a first reading. The final section, §9.8, uses the basic notion of a G-module explained in the first half of §6.1; the discussion of resolutions in the second half of that section is not relevant to this chapter.

[1)] Translator's note: the source prints the second pair as $(\mathrm{coE}(\omega), U)$. Since the first component of that pair must be a category carrying the forgetful functor U, the well-typed expression is $\mathsf{Comod}_{\mathrm{f}}\text{-}\,\mathrm{coE}(\omega)$.

9.1 Duality in Monoidal Categories

We first introduce a notion that applies to all monoidal categories and then refine it step by step.

Definition 9.1.1 Let $\mathcal{C}$ be a monoidal category and $L, R \in \mathrm{Ob}(\mathcal{C})$. If there are morphisms

$$\mathrm{ev} : L \otimes R \to \mathbf{1}, \quad \mathrm{coev} : \mathbf{1} \to R \otimes L$$

such that the following two composites are id_R and id_L, respectively, then L is called a **left dual** of R, while R is called a **right dual** of L:

$$R \simeq \mathbf{1} \otimes R \xrightarrow{\mathrm{coev} \otimes \mathrm{id}_R} (R \otimes L) \otimes R \simeq R \otimes (L \otimes R) \xrightarrow{\mathrm{id}_R \otimes \mathrm{ev}} R \otimes \mathbf{1} \simeq R,$$
$$L \simeq L \otimes \mathbf{1} \xrightarrow{\mathrm{id}_L \otimes \mathrm{coev}} L \otimes (R \otimes L) \simeq (L \otimes R) \otimes L \xrightarrow{\mathrm{ev} \otimes \mathrm{id}_L} \mathbf{1} \otimes L \simeq L.$$

The quadruple $(L, R, \mathrm{ev}, \mathrm{coev})$ is called **duality data**.

The morphisms ev and coev in duality data should be understood as "evaluation" and its dual. The unit constraints, associativity constraints, and placement of parentheses involved in the definition cause no difficulty and will often be suppressed from now on.

Definition 9.1.1 may remind the reader of the unit and counit morphisms of adjoint functors. Duality does indeed encompass the theory of adjoint functors, but to see this one must work in 2-categories or more general structures; that lies beyond the scope of this book.

Proposition 9.1.2 Let $F : \mathcal{C} \to \mathcal{D}$ be a monoidal functor between monoidal categories. Every duality datum $(L, R, \mathrm{ev}, \mathrm{coev})$ in $\mathcal{C}$ canonically induces the duality datum $(FL, FR, F\mathrm{ev}, F\mathrm{coev})$ in $\mathcal{D}$.

Proof To interpret $(FL, FR, F\mathrm{ev}, F\mathrm{coev})$ as duality data, it suffices to consider the commutative diagram

$$\begin{tikzcd}
FL \otimes FR \arrow[r] & \mathbf{1}_{\mathcal{D}} \arrow[r] & FR \otimes FL \\
F(L \otimes R) \arrow[u, "\wr"] \arrow[r, "F\mathrm{ev}"'] & F(\mathbf{1}_{\mathcal{C}}) \arrow[u, "\wr"] \arrow[r, "F\mathrm{coev}"'] & F(R \otimes L) \arrow[u, "\wr"']
\end{tikzcd}$$

The vertical arrows come from the monoidal-functor structure. The required property is therefore reduced to the corresponding property in $\mathcal{C}$. $\square$

Definition–Proposition 9.1.3 Let $\mathcal{C}$ be a monoidal category and $X \in \mathrm{Ob}(\mathcal{C})$. If X has a right dual (respectively, a left dual), write its duality data as $(X, X^*, \mathrm{ev}, \mathrm{coev})$ (respectively, $({}^*X, X, \mathrm{ev}, \mathrm{coev})$). Then these data are unique up to unique isomorphism.

Thus, without ambiguity, we may speak of the right dual X^* (respectively, the left dual ${}^*X$) of X.

Proof Reversing the order of $\otimes$ turns left duals into right duals and vice versa, so it suffices to treat the case of X^*. Suppose that the data $(X_i^*, \mathrm{ev}_i, \mathrm{coev}_i)$ give right duals of X, for $i = 1, 2$. First we prove uniqueness of the isomorphism. If $\alpha : X_1^* \xrightarrow{\sim} X_2^*$ is compatible with the duality morphisms ev_i and coev_i, then the following diagram commutes:

$$\begin{tikzcd}
X_1^* \arrow[r, "\mathrm{coev}_2\otimes\mathrm{id}"] \arrow[dd, "\alpha"'] & X_2^*\otimes X\otimes X_1^* \arrow[dr, "\mathrm{id}\otimes\mathrm{ev}_1"] \arrow[dd, "\mathrm{id}\otimes\alpha"] & \\
 & & X_2^* \\
X_2^* \arrow[r, "\mathrm{coev}_2\otimes\mathrm{id}"'] & X_2^*\otimes X\otimes X_2^* \arrow[ur, "\mathrm{id}\otimes\mathrm{ev}_2"'] &
\end{tikzcd}$$

By the definition of a right dual, the composite along the lower path is $(\mathrm{id}\otimes\mathrm{ev}_2)(\mathrm{coev}_2\otimes\mathrm{id}) = \mathrm{id}_{X_2^*}$. The corresponding statement also holds for the inverse morphism $\beta : X_2^* \xrightarrow{\sim} X_1^*$. Hence the desired morphisms $\alpha : X_1^* \to X_2^*$ and $\beta : X_2^* \to X_1^*$ can only be the composites

$$X_1^* \xrightarrow{\mathrm{coev}_2\otimes\mathrm{id}_{X_1^*}} X_2^*\otimes X\otimes X_1^* \xrightarrow{\mathrm{id}_{X_2^*}\otimes\mathrm{ev}_1} X_2^*,$$
$$X_2^* \xrightarrow{\mathrm{coev}_1\otimes\mathrm{id}_{X_2^*}} X_1^*\otimes X\otimes X_2^* \xrightarrow{\mathrm{id}_{X_1^*}\otimes\mathrm{ev}_2} X_1^*.$$

We now show that the α and β defined above are indeed compatible with the duality morphisms. By symmetry, it suffices to consider α. The definition of a right dual gives the commutative diagrams

$$\begin{tikzcd}
X\otimes X_1^* \arrow[d, "\mathrm{id}\otimes\mathrm{coev}_2\otimes\mathrm{id}"'] \arrow[dr, "\mathrm{id}"] & \\
X\otimes X_2^*\otimes X\otimes X_1^* \arrow[r, "\mathrm{ev}_2\otimes\mathrm{id}"'] \arrow[d, "\mathrm{id}\otimes\mathrm{ev}_1"'] & X\otimes X_1^* \arrow[d, "\mathrm{ev}_1"] \\
X\otimes X_2^* \arrow[r, "\mathrm{ev}_2"'] & \mathbf{1}
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbf{1} \arrow[r, "\mathrm{coev}_1"] \arrow[d, "\mathrm{coev}_2"'] & X_1^*\otimes X \arrow[d, "\mathrm{coev}_2\otimes\mathrm{id}"] \\
X_2^*\otimes X \arrow[r, "\mathrm{id}\otimes\mathrm{coev}_1"] \arrow[dr, "\mathrm{id}"'] & X_2^*\otimes X\otimes X_1^*\otimes X \arrow[d, "\mathrm{id}\otimes\mathrm{ev}_1\otimes\mathrm{id}"] \\
 & X_2^*\otimes X
\end{tikzcd}$$

Their vertical composites are $\mathrm{id}\otimes\alpha$ and $\alpha\otimes\mathrm{id}$, respectively, proving the required compatibility.

Next we prove that α and β are mutually inverse. Again, it suffices to prove

$\beta\alpha = \mathrm{id}$. Consider the diagram

$$\begin{tikzcd}
X_1^* \arrow[r, "\mathrm{coev}_1 \otimes \mathrm{id}"] \arrow[d, "\mathrm{coev}_2 \otimes \mathrm{id}"'] & X_1^* \otimes X \otimes X_1^* \arrow[d, "\mathrm{id} \otimes \mathrm{id} \otimes \mathrm{coev}_2 \otimes \mathrm{id}"'] \arrow[dr, "\mathrm{id}"] & \\
X_2^* \otimes X \otimes X_1^* \arrow[r, "\mathrm{coev}_1 \otimes \mathrm{id}"'] \arrow[d, "\mathrm{id} \otimes \mathrm{ev}_1"'] & X_1^* \otimes X \otimes X_2^* \otimes X \otimes X_1^* \arrow[r, "\mathrm{id} \otimes \mathrm{ev}_2 \otimes \mathrm{id} \otimes \mathrm{id}"'] \arrow[d, "\mathrm{id} \otimes \mathrm{ev}_1"'] & X_1^* \otimes X \otimes X_1^* \arrow[d, "\mathrm{id} \otimes \mathrm{ev}_1"] \\
X_2^* \arrow[r, "\mathrm{coev}_1 \otimes \mathrm{id}"'] & X_1^* \otimes X \otimes X_2^* \arrow[r, "\mathrm{id} \otimes \mathrm{ev}_2"'] & X_1^*
\end{tikzcd}$$

The three squares plainly commute, while the triangle commutes by the definition of a right dual. Hence the entire diagram commutes. The composite along is $\beta\alpha$, while the composite along is $\mathrm{id}_{X_1^*}$. This completes the proof. □

The relationship between left and right duals can be written as ${}^*(X^*) = X$ and $({}^*X)^* = X$. Although duality data are often chosen in an argument or exposition, duality is fundamentally a property of an object, not additional structure.

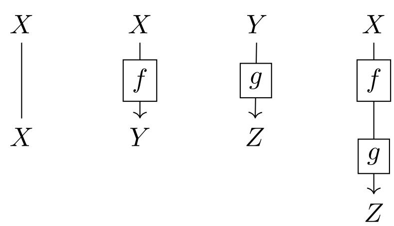
Example 9.1.4 In every monoidal category,

$$\mathbf{1}^* = \mathbf{1} = {}^*\mathbf{1},$$

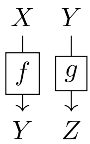
and the required morphisms ev and coev both arise from the unit constraint $\mathbf{1} \otimes \mathbf{1} \simeq \mathbf{1}$.

Definition 9.1.5 (N. Saavedra Rivano) If every object in a monoidal category $\mathcal{C}$ has a left dual (respectively, a right dual), then $\mathcal{C}$ is called a **left rigid** (respectively, **right rigid**) category.

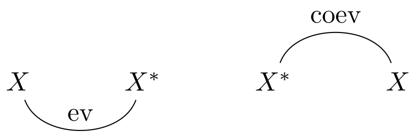
The idea behind the proof of Definition–Proposition 9.1.3 is actually simple, but its execution produces many formulas and commutative diagrams. In arguments involving duality, a visualization technique called a **string diagram** is very useful; compare §7.1 or [51, Proof of Theorem 2.6.12]. More precisely, objects are drawn as vertices and morphisms as arrows, with composition read from top to bottom. For example, id_X, $f : X \to Y$, $g : Y \to Z$, and gf are drawn, respectively, as



Thus, pre- or postcomposing a morphism with id amounts to extending its arrow. The $\otimes$-product of morphisms is represented by placing their arrows side by side; for example, $f \otimes g$ is drawn as



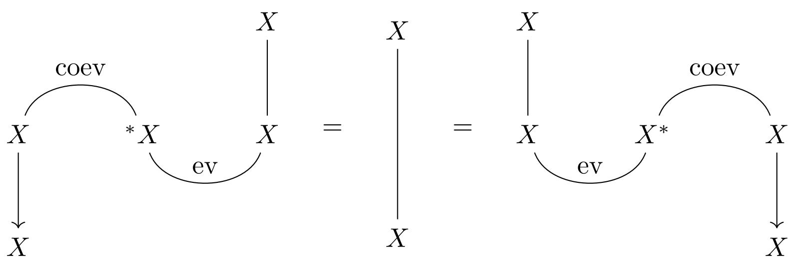
Since $X \otimes \mathbf{1} \simeq X \simeq \mathbf{1} \otimes X$, the unit $\mathbf{1}$ can be omitted from a string diagram; in other words, the string has no end at that point. If the dual exists, the morphisms $\mathrm{ev} : X \otimes X^* \to \mathbf{1}$ and $\mathrm{coev} : \mathbf{1} \to X^* \otimes X$ are therefore drawn, respectively, as



Under the same assumptions, the identities in Definition 9.1.1,

$$(\mathrm{id}_X \otimes \mathrm{ev})(\mathrm{coev} \otimes \mathrm{id}_X) = \mathrm{id}_X = (\mathrm{ev} \otimes \mathrm{id}_X)(\mathrm{id}_X \otimes \mathrm{coev})$$

are represented by



(9.1.1)

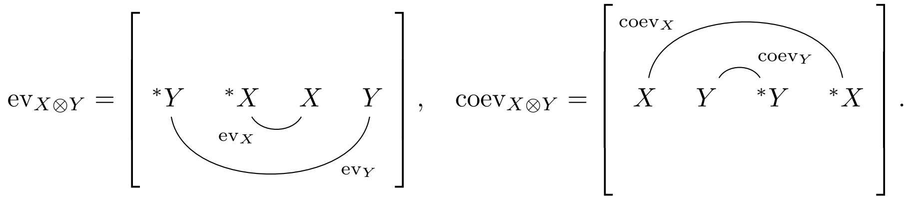
provided, of course, that the duals in question exist. These diagrams give Definition 9.1.1 the operational flavor of "straightening a string." In some of the literature, the diagrams above are drawn after a rotation by $\frac{\pi}{2}$ and are therefore also called the **Z identities**.

By the correspondence between algebraic identities and operations on string diagrams, basic algebraic identities involving duality can be found simply by following the diagrams, or even proved directly from them. As an exercise, the reader may try to rewrite the proof of Definition–Proposition 9.1.3 as a simple diagram.

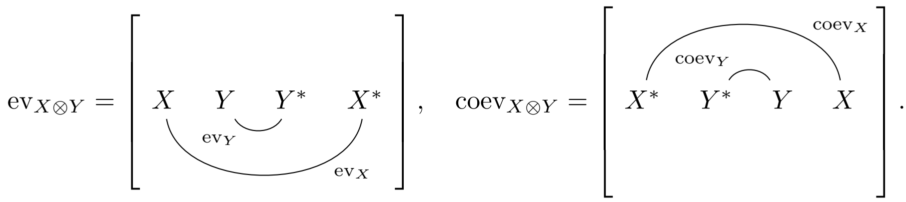
The following result shows that duality naturally reverses the order of $\otimes$.

Proposition 9.1.6 Let $\mathcal{C}$ be a monoidal category and $X, Y \in \mathrm{Ob}(\mathcal{C})$.

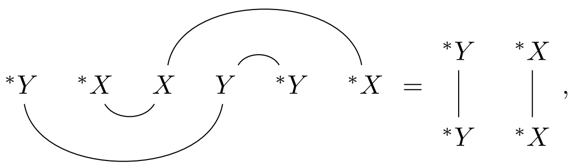
(i) If X and Y have left duals ${}^*X$ and ${}^*Y$, respectively, then ${}^*Y \otimes {}^*X$ is a left dual of $X \otimes Y$. The corresponding data can be depicted in terms of the duality data of X and Y as

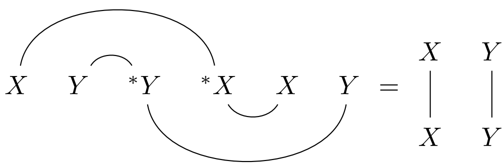


(ii) If X and Y have right duals X^* and Y^*, respectively, then $Y^* \otimes X^*$ is a right dual of $X \otimes Y$. The corresponding data can be depicted in terms of the duality data of X and Y as



Proof We must verify the identities in Definition 9.1.1, namely the Z identities (9.1.1). Equivalently, we must check that



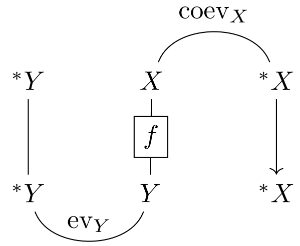


Strictly speaking, the diagrams on the left should be stretched in the vertical direction as in (9.1.1): insert several copies of id, shift them horizontally as appropriate, and then straighten them. The identities above are then clear. The corresponding formal verification is based on the various naturality properties of the unit constraints; the details are left to the interested reader. □

Definition 9.1.7 Let $f : X \to Y$ be a morphism in a monoidal category $\mathcal{C}$. Suppose that X and Y both have left duals ${}^*X$ and ${}^*Y$ (respectively, right duals X^* and Y^*). The left dual morphism ${}^*f : {}^*Y \to {}^*X$ (respectively, the right dual morphism

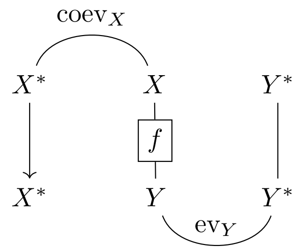
$f^* : Y^* \to X^*$) of f is defined as follows:

$$ {}^*f := (\mathrm{ev}_Y \otimes \mathrm{id})(\mathrm{id} \otimes f \otimes \mathrm{id})(\mathrm{id} \otimes \mathrm{coev}_X) $$



,

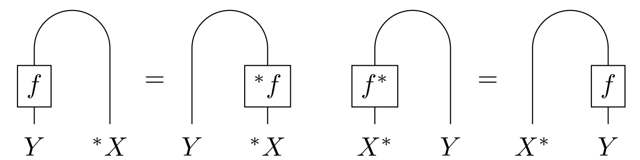
$$ f^* := (\mathrm{id} \otimes \mathrm{ev}_Y)(\mathrm{id} \otimes f \otimes \mathrm{id})(\mathrm{coev}_X \otimes \mathrm{id}) $$



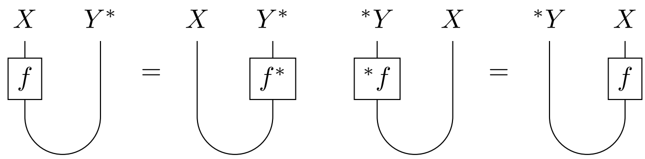
.

From the relationship between left and right duals and Definition–Proposition 9.1.1, one checks that ${}^*(f^*) = f = ({}^*f)^*$; this is immediately apparent upon drawing a "diagram within a diagram". Moreover, ${}^*(\mathrm{id}_X) = \mathrm{id}_{{}^*X}$ and $(\mathrm{id}_X)^* = \mathrm{id}_{X^*}$ follow directly from the definitions.

Here are two useful sets of identities concerning the left and right duals of $f : X \to Y$; once again, the diagrams provide the proof.



(9.1.2)



(9.1.3)

Proposition 9.1.8 Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms in a monoidal category $\mathcal{C}$. If all these objects have left duals (respectively, right duals), then ${}^*(gf) = {}^*f\ {}^*g$ (respectively, $(gf)^* = f^*g^*$).

Proof Apply the Zorro identity (9.1.1); one picture is worth a thousand words. $\square$

Corollary 9.1.9 Let $\mathcal{C}$ be a left rigid (respectively, right rigid) category, and equip $\mathcal{C}^{\mathrm{op}}$ with the monoidal structure that reverses both the arrows and the order of the arguments of $\otimes$. Then there is a monoidal functor $\mathcal{C} \to \mathcal{C}^{\mathrm{op}}$ sending an object X to ${}^*X$ (respectively, X^*) and a morphism f to ${}^*f$ (respectively, f^*).

Somewhat surprisingly, every morphism between monoidal functors whose source is a left or right rigid category must be an isomorphism. This helps explain the meaning of the term “rigid”.

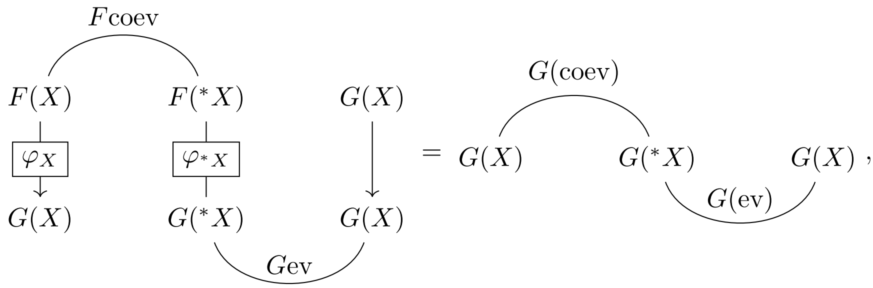
Proposition 9.1.10 Let $F, G : \mathcal{C} \rightrightarrows \mathcal{D}$ be monoidal functors between monoidal categories. If $\mathcal{C}$ is left rigid (respectively, right rigid), then every morphism $\varphi : F \to G$ of monoidal functors is an isomorphism. In fact, $(\varphi_X)^{-1} = (\varphi_{{}^*X})^*$ (respectively, ${}^*(\varphi_{X^*})$).

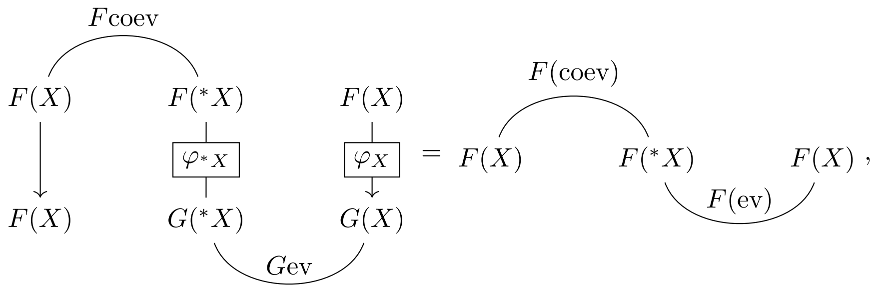
Proof It suffices to consider the left rigid case. Fix $X \in \mathrm{Ob}(\mathcal{C})$, together with its left dual ${}^*X$ and the data ev and coev. We have the commutative diagrams

$$\begin{tikzcd}
\mathbf{1} \arrow[r, "\sim"] \arrow[dr, "\sim"'] & F(\mathbf{1}) \arrow[r, "F\mathrm{coev}"] \arrow[d, "\varphi_{\mathbf{1}}"] & F(X \otimes {}^*X) \arrow[r, "\sim"] \arrow[d, "\varphi_{X \otimes {}^*X}"] & F(X) \otimes F({}^*X) \arrow[d, "\varphi_X \otimes \varphi_{{}^*X}"] \\
& G(\mathbf{1}) \arrow[r, "G\mathrm{coev}"'] & G(X \otimes {}^*X) \arrow[r, "\sim"'] & G(X) \otimes G({}^*X)
\end{tikzcd}$$

$$\begin{tikzcd}
F({}^*X) \otimes F(X) \arrow[r, "\sim"] \arrow[d, "\varphi_{{}^*X} \otimes \varphi_X"'] & F({}^*X \otimes X) \arrow[r, "F\mathrm{ev}"] \arrow[d, "\varphi_{{}^*X \otimes X}"] & F(\mathbf{1}) \arrow[r, "\sim"] \arrow[d, "\varphi_{\mathbf{1}}"] & \mathbf{1} \\
G({}^*X) \otimes G(X) \arrow[r, "\sim"'] & G({}^*X \otimes X) \arrow[r, "G\mathrm{ev}"'] & G(\mathbf{1}) \arrow[ur, "\sim"'] &
\end{tikzcd}$$

Moreover, Proposition 9.1.2 shows that composition along the two rows makes $F({}^*X)$ and $G({}^*X)$ into left duals of $F(X)$ and $G(X)$, respectively. Thus $(\varphi_{{}^*X})^* : G(X) \to F(X)$ is defined. We thereby obtain the following diagrammatic identities:

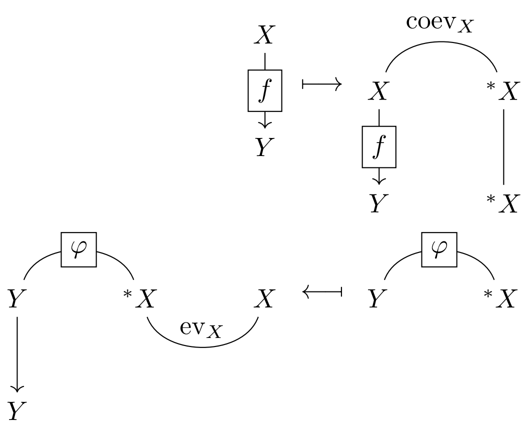




By (9.1.1), the two right-hand sides are id_{GX} and id_{FX}, respectively. This proves that $(\varphi_{{}^*X})^*$ is indeed the inverse of φ_X. $\square$

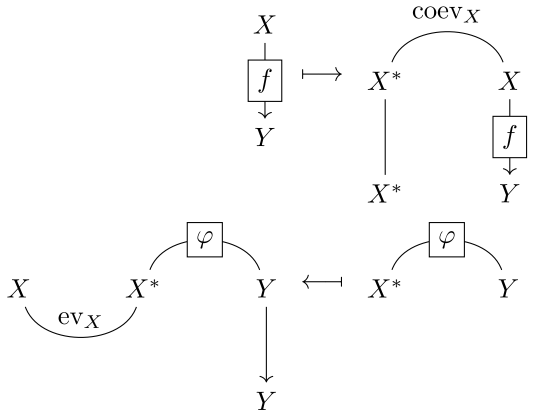
Proposition 9.1.11 Let $\mathcal{C}$ be a monoidal category and let $X, Y \in \mathrm{Ob}(\mathcal{C})$.

(i) If X has a left dual ${}^*X$, there are mutually inverse bijections $\mathrm{Hom}_{\mathcal{C}}(X, Y) \overset{1:1}{\longleftrightarrow} \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Y \otimes {}^*X)$ as follows.



(ii) If X has a right dual X^*, there are mutually inverse bijections $\mathrm{Hom}_{\mathcal{C}}(X, Y) \overset{1:1}{\longleftrightarrow}$

$\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X^* \otimes Y)$ as follows.



(iii) More generally, whenever the duals in question exist, there are canonical bijections

$$\begin{aligned} \mathrm{Hom}_{\mathcal{C}}(W \otimes X, Y) &\xrightarrow{1:1} \mathrm{Hom}_{\mathcal{C}}(W, Y \otimes {}^*X) \\ f &\longmapsto (f \otimes \mathrm{id}_{{}^*X})(\mathrm{id}_W \otimes \mathrm{coev}_X) \end{aligned}$$

and

$$\begin{aligned} \mathrm{Hom}_{\mathcal{C}}(X \otimes W, Y) &\xrightarrow{1:1} \mathrm{Hom}_{\mathcal{C}}(W, X^* \otimes Y) \\ f &\longmapsto (\mathrm{id}_{X^*} \otimes f)(\mathrm{coev}_X \otimes \mathrm{id}_W). \end{aligned}$$

Proof For (i) and (ii), checking that the displayed maps are mutually inverse is merely an application of (9.1.1). The same idea, expressed by a slightly more awkward diagram, also proves (iii). □

In the settings of (i) and (ii), it is easy to describe composition of morphisms on the element φ on the right-hand side of the bijection (hint: apply the morphism ev). These canonical operations lead to the following result.

Corollary 9.1.12 Let $\mathcal{C}$ be a left rigid (respectively, right rigid) category in the sense of Definition 9.1.5. Defining the internal Hom by $\mathcal{H}\mathrm{om}_{\mathrm{left}}(X, Y) := Y \otimes {}^*X$ (respectively, $\mathcal{H}\mathrm{om}_{\mathrm{right}}(X, Y) := X^* \otimes Y$) makes $\mathcal{C}$ a right closed (respectively, left closed) monoidal category in the sense of Definition 7.3.1.

Proof Apply Proposition 9.1.11 (iii) to Definition 7.3.1; the remaining verifications are routine. □

Corollary 9.1.13 Let $X \in \mathrm{Ob}(\mathcal{C})$.

- ◇ If X has a left dual ${}^*X$, then $(\cdot)\otimes X$ preserves all small $\varinjlim$, while $X\otimes(\cdot)$ preserves all small $\varprojlim$.
- ◇ If X has a right dual X^*, then $(\cdot)\otimes X$ preserves all small $\varprojlim$, while $X\otimes(\cdot)$ preserves all small $\varinjlim$.

Proof Consider the case in which X has a left dual ${}^*X$. Proposition 9.1.11 (iii) implies that the functor $(\cdot)\otimes X$ has right adjoint $(\cdot)\otimes {}^*X$, whereas the functor $X\otimes(\cdot)\simeq({}^*X)^*\otimes(\cdot)$ has left adjoint ${}^*X\otimes(\cdot)$.

Reversing the order of $\otimes$ in the preceding argument gives the case in which X has a right dual X^*. □

9.2 Duality: Trace and Dimension

In §9.1 we introduced several concepts related to duality. Each comes in left and right versions, depending on the position of the object in $\otimes$. In many applications the monoidal category under consideration is symmetric, so there is no need to distinguish the two. We begin with the case of a braided monoidal category. As usual, its braiding is written in the form

$$c(X,Y) : X\otimes Y \xrightarrow{\sim} Y\otimes X$$

The braiding is symmetric precisely when $c(X,Y)^{-1}=c(Y,X)$. Braid diagrams may be used together with the string-diagram techniques introduced in §9.1.

Lemma 9.2.1 Let $\mathcal{C}$ be a braided monoidal category. If X^* is a right dual of $X\in\mathrm{Ob}(\mathcal{C})$, then X^* is also a left dual of X. More precisely, define

$$\begin{aligned}\mathrm{ev}' &:= \mathrm{ev}\circ c(X,X^*)^{-1} : X^*\otimes X\to\mathbf{1},\\ \mathrm{coev}' &:= c(X^*,X)\circ\mathrm{coev} : \mathbf{1}\to X\otimes X^*,\end{aligned}$$

Then $(X^*,X,\mathrm{ev}',\mathrm{coev}')$ is duality data. The case of a left dual ${}^*X$ is similar. If $\mathcal{C}$ is symmetric monoidal, then $\mathrm{ev}''=\mathrm{ev}$ and $\mathrm{coev}''=\mathrm{coev}$ as well.

Proof This is an application of the axioms of a braiding. For example, $(\mathrm{id}\otimes\mathrm{ev}')(\mathrm{coev}'\otimes\mathrm{id})=\mathrm{id}$ reduces to the commutativity of the following diagram:

$$\begin{tikzcd}
X\otimes\mathbf{1} \arrow[r, "\mathrm{id}\otimes\mathrm{coev}"] \arrow[d, "c(X{,}\mathbf{1})"] & X\otimes X^*\otimes X \arrow[rrr, "\mathrm{ev}\otimes\mathrm{id}"] \arrow[d, "c(X{,}X^*\otimes X)"] & & & \mathbf{1}\otimes X \arrow[d, "c(\mathbf{1}{,}X)"'] \\
\mathbf{1}\otimes X \arrow[r, "\mathrm{coev}\otimes\mathrm{id}"'] & X^*\otimes X\otimes X \arrow[r, "c(X^*{,}X)\otimes\mathrm{id}"'] & X\otimes X^*\otimes X \arrow[r, "\mathrm{id}\otimes c(X{,}X^*)^{-1}"'] & X\otimes X\otimes X^* \arrow[r, "\mathrm{id}\otimes\mathrm{ev}"'] & X\otimes\mathbf{1}
\end{tikzcd}$$

The small square commutes by functoriality of the braiding, whereas the large square on the right is a standard braid-diagram argument; compare §7.1. □

Thus an object X in a braided monoidal category has a left dual if and only if it has a right dual. In a symmetric monoidal category we may identify the two versions and uniformly denote dual objects and dual morphisms by X^* and f^*.

The most typical example is, of course, the category of modules over a commutative ring. With the tensor product of modules it is a symmetric monoidal category, and duality in it has a simple characterization.

Proposition 9.2.2 Let R be a commutative ring, and make R-Mod into a symmetric monoidal category via $\otimes_R$. An R-module M has a dual M^* if and only if M is finitely generated and projective. In that case one may take $M^* := \mathrm{Hom}_R(M, R)$, with ev_M and coev_M as in Definition 7.7.10 (take $A = B = \Bbbk = R$ and $P = M$; the present M^* is denoted there by $M^\vee$), up to possibly reversing the order of the tensor factors.

Proof The "if" direction is easier, since the choice of $(M^*, \mathrm{ev}_M, \mathrm{coev}_M)$ is explicit. In the special case $M = R$, we may identify M^* with R, and then $\mathrm{ev}_M(x \otimes y) = xy$ and $\mathrm{coev}_M(1) = 1 \otimes 1$; all required identities are immediate.

The construction of $(M^*, \mathrm{ev}_M, \mathrm{coev}_M)$ is plainly compatible with direct sums, giving the case $M = R^{\oplus n}$. Every finitely generated projective module M is a direct summand of some $R^{\oplus n}$; hence M has a dual.

Now consider the "only if" direction. Suppose the R-module M has a dual M^*. By Proposition 9.1.11 (ii), for every R-module M' there is an isomorphism of R-modules

$$\begin{array}{ccccc} \mathrm{Hom}_R(M, M') & \xleftarrow{\sim} & \mathrm{Hom}_R(R, M^* \underset{R}{\otimes} M') & \xrightarrow{\sim} & M^* \underset{R}{\otimes} M' \\ (\mathrm{ev}_M \otimes \mathrm{id}_{M'})(\mathrm{id}_M \otimes \varphi) & \longleftarrow\!\shortmid & \varphi & \longmapsto & \varphi(1). \end{array}$$

Taking $M' = R$, we obtain $\mathrm{Hom}_R(M, R) \simeq \mathrm{Hom}_R(R, M^*) \simeq M^*$. This isomorphism sends $\lambda \in M^*$ to the following homomorphism:

$$\begin{array}{ccccccc} M & \xrightarrow{\sim} & M \underset{R}{\otimes} R & \xrightarrow{\mathrm{id}_M \otimes \lambda} & M \underset{R}{\otimes} M^* & \xrightarrow{\mathrm{ev}_M} & R \\ m & \longmapsto & m \otimes 1 & \longmapsto & m \otimes \lambda & \longmapsto & \mathrm{ev}_M(m \otimes \lambda). \end{array}$$

Thus M^* may be identified with $\mathrm{Hom}_R(M, R)$ and ev_M with evaluation.

Consider the identity $(\mathrm{ev}_M \otimes \mathrm{id}_M)(\mathrm{id}_M \otimes \mathrm{coev}_M) = \mathrm{id}_M$. Write $\mathrm{coev}_M(1) = \sum_{i=1}^n \lambda_i \otimes m_i$. This identity is equivalent to $\sum_i \lambda_i(m) m_i = m$ for every m, that is, id_M factors as

$$M \xrightarrow{(\lambda_1, \dots, \lambda_n)} R^{\oplus n} \xrightarrow{(m_1, \dots, m_n)} M,$$

This shows that M is a direct summand of $R^{\oplus n}$. □

Definition 9.2.3 For a braided monoidal category, left rigidity and right rigidity in Definition 9.1.5 are equivalent. A braided monoidal category satisfying either condition is called a **rigid category**.

Once an abelian category structure is added, the unit object of a rigid braided monoidal category displays a special property. We first prove that $\otimes$ is automatically additive in each variable.

Proposition 9.2.4 Let $\mathcal{C}$ be a rigid braided monoidal category. If $\mathcal{C}$ is also an additive category, then $\otimes$ is an additive bifunctor.

Proof Fix $X \in \mathrm{Ob}(\mathcal{C})$. Proposition 9.1.11 (iii) implies that the functor $X \otimes (\cdot)$ has right adjoint $X^* \otimes (\cdot)$. Thus Corollary 1.3.6 (v) ensures that $X \otimes (\cdot)$ is additive. By the braiding, the same holds for $(\cdot) \otimes X$. □

Proposition 9.2.5 Let $\mathcal{C}$ be both a rigid braided monoidal category and an abelian category. If $\iota : U \hookrightarrow \mathbf{1}$ is a subobject, then $\mathbf{1} = U \oplus \ker(\iota^*)$. Consequently:

(i) the object $\mathbf{1}$ is split (Definition 2.7.6);

(ii) if $\mathrm{End}_{\mathcal{C}}(\mathbf{1})$ is a field, then $\mathbf{1}$ is a simple object.

Proof Set $V := \mathrm{coker}(\iota)$. We know that $\otimes$ is an exact additive functor in each variable (Corollary 9.1.13 and Proposition 9.2.4). We therefore obtain the following commutative diagram, whose solid-line rows are exact:

$$\begin{tikzcd}
0 \arrow[r] & U \arrow[r, "\iota"] & \mathbf{1} \arrow[r] & V \arrow[r] & 0 \\
0 \arrow[r] & U \otimes U \arrow[r, "\mathrm{id}\otimes\iota"'] \arrow[u, hook, "\iota\otimes\mathrm{id}"] & U \arrow[r] \arrow[u, hook, "\iota"] \arrow[ur, dashed] & U \otimes V \arrow[r] \arrow[u, hook, "\iota\otimes\mathrm{id}"'] & 0
\end{tikzcd}$$

The composite represented by the dashed arrow is 0. Hence $U \otimes V = 0$ and $U \otimes U \simeq U$.

For every object T, the same argument gives a monomorphism $\iota \otimes \mathrm{id}_T : U \otimes T \hookrightarrow T$. Thus $U \otimes T = 0$ if and only if $\iota \otimes \mathrm{id}_T = 0$. In turn, this is equivalent to the vanishing of the corresponding morphism $T \to U^* \otimes T$ under Proposition 9.1.11 (iii). It is not hard to show that the latter morphism is precisely $\iota^* \otimes \mathrm{id}_T$ (exercise in this chapter). Therefore, for every object X, the largest subobject $T \subset X$ satisfying $U \otimes T = 0$ is the largest subobject for which $T \to U^* \otimes T \hookrightarrow U^* \otimes X$ vanishes, and this is also

$$\ker\left[\iota^* \otimes \mathrm{id}_X : X \to U^* \otimes X\right] \simeq \ker(\iota^*) \otimes X.$$

- Apply this to $X = V$ and use $U \otimes V = 0$; we obtain $\ker(\iota^*) \otimes V \simeq V$.
- Apply this to $X = U$. Since every subobject T of U is also a subobject of $\mathbf{1}$, we have $U \otimes T \supset T \otimes T \simeq T$. Consequently, $\ker(\iota^*) \otimes U = 0$.

Substituting these two results into the short exact sequence

$$0 \to \ker(\iota^*) \otimes U \to \ker(\iota^*) \to \ker(\iota^*) \otimes V \to 0,$$

immediately gives $\mathbf{1} \supset \ker(\iota^*) \xrightarrow{\sim} V$. This splits the sequence $0 \to U \to \mathbf{1} \to V \to 0$. Since U was an arbitrary subobject, this is precisely the assertion that $\mathbf{1}$ is split.

Finally, Corollary 2.5.5 implies that if $\mathrm{End}_{\mathcal{C}}(\mathbf{1})$ is a field, then $\mathbf{1}$ is indecomposable; hence $\mathbf{1}$ is simple. □

In any monoidal category $\mathcal{C}$, $\mathrm{End}(\mathbf{1})$ is a monoid, and it is easy to prove that it is commutative; see [51, Chapter 3 exercises]. The unit constraint $X \simeq \mathbf{1} \otimes X$ makes every element of $\mathrm{End}(\mathbf{1})$ act on each X by an endomorphism. These endomorphisms determine a monoid homomorphism $\mathrm{End}(\mathbf{1}) \to Z(\mathcal{C})$, where $Z(\mathcal{C})$ denotes the center of the category $\mathcal{C}$. By the general properties of the center, composition of morphisms is bilinear with respect to the action of $\mathrm{End}(\mathbf{1})$:

$$(zf)g = z(fg) = f(zg), \quad X \xrightarrow{g} Y \xrightarrow{f} Z, \quad z \in \mathrm{End}(\mathbf{1}).$$

Moreover, the axioms of $\otimes$ (see [51, §3.1]) readily imply that $\otimes$ is bilinear as well:

$$(zf_1) \otimes f_2 = z(f_1 \otimes f_2) = f_1 \otimes (zf_2), \quad f_i \in \mathrm{Hom}(X_i, Y_i), \quad z \in \mathrm{End}(\mathbf{1}).$$

Thus $\mathrm{End}(\mathbf{1})$ may be regarded as providing a kind of “scalars” for the monoidal category $\mathcal{C}$.

Definition 9.2.6 Let $f \in \mathrm{End}(X)$ be an endomorphism in a braided monoidal category $\mathcal{C}$, and suppose X has a dual. Define the left trace of f (abbreviated simply to **trace**) by

$$\mathrm{Tr}(f) = \mathrm{Tr}_{\mathrm{left}}(f) := \mathrm{ev}_X\,(f \otimes \mathrm{id}_{X^*})\,\mathrm{coev}'_X \ \in \mathrm{End}(\mathbf{1}),$$

and define its right trace by

$$\mathrm{Tr}_{\mathrm{right}}(f) := \mathrm{ev}'_X\,(\mathrm{id}_{X^*} \otimes f)\,\mathrm{coev}_X,$$

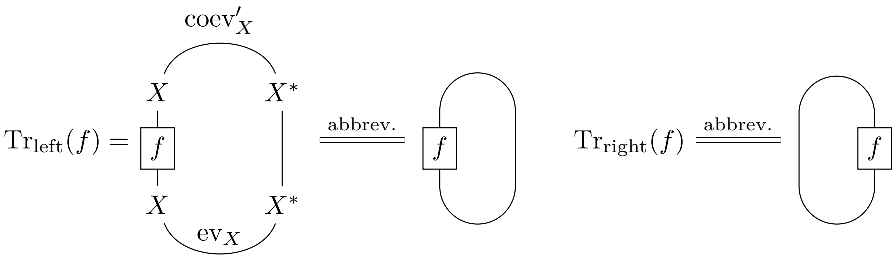
where coev'_X and ev'_X are as in Lemma 9.2.1. By Definition–Proposition 9.1.3, neither trace depends on the choice of duality data.

Definition 9.2.7 Let X be an object of a braided monoidal category $\mathcal{C}$, and suppose X has a dual. Its left dimension (abbreviated simply to **dimension**) is defined by

$$\dim X = \dim_{\mathrm{left}} X := \mathrm{Tr}(\mathrm{id}_X) \ \in \mathrm{End}(\mathbf{1}),$$

and its right dimension is defined by $\dim_{\mathrm{right}} X := \mathrm{Tr}_{\mathrm{right}}(\mathrm{id}_X)$.

The left and right traces are depicted, respectively, as follows:



If $\mathcal{C}$ is a symmetric monoidal category, the left and right versions of both trace and dimension always agree. This is the principal setting for later applications.

When the role of the category $\mathcal{C}$ needs to be emphasized, we use notation such as $\mathrm{Tr}_{\mathcal{C}}$ and $\dim_{\mathcal{C}}$. Some authors also call these the quantum trace and quantum dimension. Example 9.3.1 will explain their relation to the classical versions.

Proposition 9.2.8 Let $F : \mathcal{C} \to \mathcal{D}$ be a monoidal functor between braided monoidal categories that preserves the braiding. For every endomorphism $f \in \mathrm{End}(X)$ in $\mathcal{C}$, provided X has a dual, we have $F\,(\mathrm{Tr}_{\mathcal{C}}(f)) = \mathrm{Tr}_{\mathcal{D}}(Ff)$; the same holds for the right trace. In particular, $F\,(\dim_{\mathcal{C}} X) = \dim_{\mathcal{D}}(FX)$.

Proof The proof is exactly the same as that of Proposition 9.1.2. □

The trace defined above has properties analogous to those of the trace of a linear map.

Proposition 9.2.9 Let $\mathcal{C}$ be a braided monoidal category. In the following statements, whenever an endomorphism $f \in \mathrm{End}(X)$ is considered, it is understood that X has a dual.

(i) We have $\mathrm{Tr}_{\mathrm{left}}(f) = \mathrm{Tr}_{\mathrm{right}}({}^*f)$ and $\mathrm{Tr}_{\mathrm{right}}(f) = \mathrm{Tr}_{\mathrm{left}}(f^*)$.

(ii) Suppose X and Y both have duals. For every $X \underset{g}{\overset{f}{\rightleftarrows}} Y$, we have $\mathrm{Tr}_{\mathrm{left}}(gf) = \mathrm{Tr}_{\mathrm{left}}(({}^{**}f)g)$ and $\mathrm{Tr}_{\mathrm{right}}(gf) = \mathrm{Tr}_{\mathrm{right}}((f^{**})g)$. Observe that if $\mathcal{C}$ is symmetric monoidal, then ${}^{**}f = f = f^{**}$.

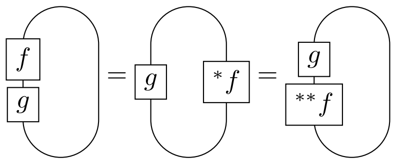
(iii) For every $f \in \mathrm{End}(X)$ and $g \in \mathrm{End}(Y)$, we have $\mathrm{Tr}(f \otimes g) = \mathrm{Tr}(f)\,\mathrm{Tr}(g)$.

(iv) If $a \in \mathrm{End}(\mathbf{1})$, then $\mathrm{Tr}(af) = a\,\mathrm{Tr}(f)$.

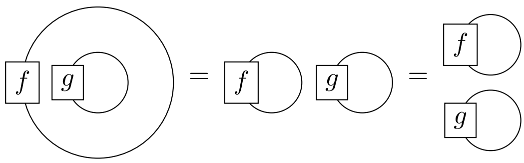
(v) Suppose $\mathcal{C}$ is an **Ab**-category and $\otimes$ is additive in the first (respectively, second) variable. Then for every $f, g \in \mathrm{End}(X)$ we have $\mathrm{Tr}_{\mathrm{left}}(f+g) = \mathrm{Tr}_{\mathrm{left}}(f)+\mathrm{Tr}_{\mathrm{left}}(g)$ (respectively, $\mathrm{Tr}_{\mathrm{right}}(f + g) = \mathrm{Tr}_{\mathrm{right}}(f) + \mathrm{Tr}_{\mathrm{right}}(g)$).

In (iii) and (iv), Tr may mean either the left trace or the right trace.

Proof As usual, the proof relies chiefly on pictures. By symmetry it suffices to consider the left trace. First, in the trace diagrams, closing off the bottom of every term in (9.1.2) immediately gives (i). Similarly, (ii) is depicted using (9.1.3) as follows.



For (iii), the key is the intuitive identity



Its formal proof is based on the various naturality properties of the unit object.

Assertion (iv) follows from the bilinearity of morphism composition and $\otimes$ with respect to $\mathrm{End}(\mathbf{1})$. Assertion (v) follows directly from the written definitions of the left and right traces, without recourse to diagrams. $\square$

Proposition 9.2.9 (iv) and (v) may be understood as the linearity properties of the trace. As a special case of (iii), we also have $\dim(X \otimes Y) = \dim X \, \dim Y$.

9.3 Examples of Duality

We next examine some basic examples of duality theory.

Example 9.3.1 Consider the category $\mathsf{Vect}(\Bbbk)$ of vector spaces over a field $\Bbbk$, with its standard symmetric monoidal structure and unit $\mathbf{1} := \Bbbk$. This is the basic model for duality. Proposition 9.2.2 implies that for every $\Bbbk$-vector space V,

$$V \text{ has a dual} \iff V \text{ is finite-dimensional.}$$

More concretely, for an n-dimensional $\Bbbk$-vector space, its dual (with no need to distinguish left from right) is taken to be

$$V^* := \mathrm{Hom}_{\Bbbk}(V, \Bbbk),$$

with the corresponding morphisms

$$\begin{array}{ccccccc} V \otimes V^* & \xrightarrow{\text{ev}} & \Bbbk & \qquad & \Bbbk & \xrightarrow{\text{coev}} & V^* \otimes V \\ v \otimes \check{v} & \longmapsto & \check{v}(v) & \qquad & 1 & \longmapsto & \sum_{i=1}^{n} \check{v}_i \otimes v_i. \end{array}$$

Here $v_1, \ldots, v_n$ is any basis of V, while $\check{v}_1, \ldots, \check{v}_n$ is its dual basis. These definitions are a special case of Proposition 9.2.2, and can also be verified directly without difficulty.

Finite-dimensional $\Bbbk$-vector spaces form a rigid category under $\otimes$, denoted by $\mathsf{Vect}_{\mathsf{f}}(\Bbbk)$. All operations involving duals reduce to the following elementary facts.

- ◇ The isomorphism $V^* \otimes W \simeq \mathrm{Hom}(\Bbbk, V^* \otimes W) \xrightarrow{\sim} \mathrm{Hom}(V, W)$ sends $\sum_i \check{v}_i \otimes w_i$ to the linear map $\sum_i \check{v}_i(\cdot)w_i$.
- ◇ The dual $f^* : W^* \to V^*$ of a morphism $f : V \to W$ is precisely the transpose of the linear map; the diagrammatic identity (9.1.3) says it all.
- ◇ Trace and dimension take values in $\mathrm{End}(\Bbbk) \simeq \Bbbk$. Since $\mathsf{Vect}_{\mathsf{f}}(\Bbbk)$ is symmetric, there is no need to distinguish left from right.
- ◇ Upon identifying $\mathrm{End}(V)$ with $V^* \otimes V$,

$$\mathrm{Tr}\left(\sum_i \check{v}_i \otimes v_i\right) = \sum_i \check{v}_i(v_i)$$

  is precisely the classical trace. Likewise, $\dim V$ is the image of the classical dimension under the homomorphism $\mathbb{Z} \to \Bbbk$.

Example 9.3.2 Consider the symmetric monoidal category $\mathsf{Vect}^-_{\mathbb{Z}/2\mathbb{Z}}(\Bbbk)$ from Example 7.1.13. Its objects and morphisms are written, respectively, as $V = V_0 \oplus V_1$ and $f = (f_0, f_1)$, with $\Bbbk = \Bbbk \oplus \{0\}$ as the unit. As in Example 9.3.1, an object V has a dual if and only if it is finite-dimensional as a $\Bbbk$-vector space; all such objects form the full subcategory $\mathsf{Vect}^-_{\mathbb{Z}/2\mathbb{Z},f}(\Bbbk)$. If V is finite-dimensional, its dual may be taken to be

$$V^* = V_0^* \oplus V_1^*, \quad V_i^* := \mathrm{Hom}_{\Bbbk}(V_i, \Bbbk),$$

with

$$\begin{aligned} V \otimes V^* &\xrightarrow{\quad\mathrm{ev}\quad} \Bbbk \\ (v_0, v_1) \otimes (\check{v}_0, \check{v}_1) &\longmapsto \check{v}_0(v_0) - \check{v}_1(v_1), \\ \Bbbk &\xrightarrow{\quad\mathrm{coev}\quad} V^* \otimes V \\ 1 &\longmapsto \textstyle\sum_{i=1}^{n_0} \check{v}_{0,i} \otimes v_{0,i} - \sum_{i=1}^{n_1} \check{v}_{1,i} \otimes v_{1,i}. \end{aligned}$$

Here $v_{0,1}, \ldots, v_{0,n_0}$ (respectively, $v_{1,1}, \ldots, v_{1,n_1}$) is any basis of V_0 (respectively, V_1), while $\check{v}_{0,1}, \ldots, \check{v}_{0,n_0}$ (respectively, $\check{v}_{1,1}, \ldots, \check{v}_{1,n_1}$) is the dual basis. The conditions of Definition 9.1.1 can be checked directly.

- ◇ The isomorphism $(V^* \otimes W)_0 \simeq \mathrm{Hom}(\Bbbk, V^* \otimes W) \xrightarrow{\sim} \mathrm{Hom}(V, W)$ sends $\sum_i \check{v}_i \otimes w_i$ to the linear map $\sum_i \check{v}_i(\cdot)w_i$, where either $\check{v}_i \in V_0^*$ and $w_i \in W_0$, or $\check{v}_i \in V_1^*$ and $w_i \in W_1$.

- The dual $f^* : W^* \to V^*$ of a morphism $f : V \to W$ is again the transpose of the linear map, applied separately to each direct summand.

- Trace and dimension again take values in $\mathrm{End}(\Bbbk) \simeq \Bbbk$, with no distinction between left and right. The trace of an endomorphism $f = (f_0, f_1) \in \mathrm{End}(V)$ is $\mathrm{Tr}(f_0) - \mathrm{Tr}(f_1)$ and is called the **supertrace**. Correspondingly, the **superdimension** $\dim V$ is the image of $\dim V_0 - \dim V_1$ under $\mathbb{Z} \to \Bbbk$.

A similar description applies to the more general category $\mathsf{Vect}_I^{\epsilon}(\Bbbk)$ of graded $\Bbbk$-vector spaces, where I is a commutative monoid and $\epsilon : I \to \mathbb{Z}/2\mathbb{Z}$; see Example 7.1.11. Objects and morphisms of this symmetric monoidal category still have the forms $\bigoplus_{i\in I} V_i$ and $(f_i)_{i\in I}$, while the signs depend on $\epsilon(i)$.

Observe that $\mathsf{Vect}(\Bbbk)$ embeds into $\mathsf{Vect}^{-}_{\mathbb{Z}/2\mathbb{Z}}(\Bbbk)$ via $V \mapsto (V_0, V_1) := (V, \{0\})$. Thus Example 9.3.2 subsumes the classical theory of Example 9.3.1. Furthermore, when $\Bbbk \in \{\mathbb{R}, \mathbb{C}\}$, $\Bbbk$-super vector spaces may be generalized to super vector bundles over a topological manifold M. Supertrace and superdimension extend accordingly to super vector bundles; the sole difference is that they take values in the endomorphism monoid of the trivial super vector bundle $M \times (\Bbbk \oplus \{0\})$, namely in $\{$continuous functions $M \to \Bbbk\}$.

Next come two nonlinear examples.

Example 9.3.3 (Correspondences) Let Corr be the following category. Its objects are all small sets, and a morphism from X to Y is defined to be an isomorphism class of diagrams $[X \xleftarrow{u} A \xrightarrow{v} Y]$, where A is any small set and u, v are arbitrary maps; isomorphism is understood in the evident way. Such a diagram is called a "correspondence" from X to Y and generalizes the notion of a map.[2)] Composition of $[Y \leftarrow B \to Z]$ with $[X \leftarrow A \to Y]$ is defined by the fiber product:

$$\begin{tikzcd}
 & & A \underset{Y}{\times} B \arrow[dl] \arrow[dr] & & \\
 & A \arrow[dl] \arrow[dr] & & B \arrow[dl] \arrow[dr] & \\
X & & Y & & Z.
\end{tikzcd}$$

The identity morphism of an object X is $X \xleftarrow{\mathrm{id}} X \xrightarrow{\mathrm{id}} X$, or any diagram isomorphic to it. Special cases of composition are

$$[Y = Y \xrightarrow{w} Z] \circ [X \xleftarrow{u} A \xrightarrow{v} Y] = [X \xleftarrow{u} A \xrightarrow{wv} Z],$$
$$[Y \xleftarrow{z} B \xrightarrow{w} Z] \circ [X \xleftarrow{u} Y = Y] = [X \xleftarrow{uz} B \xrightarrow{w} Z].$$

2) If $f : X \to Y$ is a map, take A to be the graph of f.

Abstract sets are used here purely for pedagogical reasons. In applications, X, Y, and so forth are usually taken to be suitable geometric objects, such as topological spaces or schemes.

Now equip Corr with a monoidal structure: define $X \otimes Y := X \times Y$, with the one-point set pt as the unit $\mathbf{1}$. This category is naturally symmetric monoidal and is also rigid: for every X, take $X^* = X$ together with

$$\mathrm{coev}_X := \left[\ \mathrm{pt} \longleftarrow X \xrightarrow{\text{diagonal}} X \times X\ \right], \quad \mathrm{ev}_X := \left[\ X \times X \xleftarrow{\text{diagonal}} X \longrightarrow \mathrm{pt}.\ \right]$$

The reader is invited to verify carefully the conditions of Definition 9.1.1.

- Once the definitions are expanded, the isomorphism $\mathrm{Hom}_{\mathrm{Corr}}(X, Y) \simeq \mathrm{Hom}_{\mathrm{Corr}}(\mathbf{1}, X^* \otimes Y)$ becomes a tautology.

- The dual of the morphism $f = [X \xleftarrow{u} A \xrightarrow{v} Y]$ is $f^* = [Y \xleftarrow{v} A \xrightarrow{u} X]$. This follows immediately by working through the definitions; the details are left to the reader.

- Trace and dimension take values in $\mathrm{End}_{\mathrm{Corr}}(\mathbf{1})$. The elements of this set are isomorphism classes of diagrams $[\mathrm{pt} \leftarrow A \rightarrow \mathrm{pt}]$ and are completely determined by the cardinality $|A|$.

- For an arbitrary map $u : A \to X$, write $\Gamma_u := \{(a, u(a)) : a \in A\} \subset A \times X$ for its graph. The trace of the endomorphism $f = [X \xleftarrow{u} A \xrightarrow{v} X]$ is described by the diagram

$$\begin{tikzcd}
 & & & \Gamma_u \cap \Gamma_v \arrow[dl, hook'] \arrow[dr, hook] & & & \\
 & & \Gamma_u \arrow[dl, "\text{projection}"'] \arrow[dr, hook] & & \Gamma_v \arrow[dl, hook'] \arrow[dr, "\text{projection}"] & & \\
 & X \arrow[dl] \arrow[dr, "\text{diagonal}"'] & & A \times X \arrow[dl, "u \times \mathrm{id}"] \arrow[dr, "v \times \mathrm{id}"'] & & X \arrow[dl, "\text{diagonal}"] \arrow[dr] & \\
\mathrm{pt} & & X \times X & & X \times X & & \mathrm{pt}.
\end{tikzcd}$$

  Each diamond is a pullback square. Observe that $\Gamma_u \cap \Gamma_v \simeq \{a \in A : u(a) = v(a)\}$.

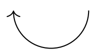
  Write $\Delta := \Gamma_{\mathrm{id}_X} \subset X \times X$ for the diagonal subset. In the special case $A = X$ and $u = \mathrm{id}_X$, the vertex of the diagram is the fixed-point set $\Gamma_v \cap \Delta = \{x \in X : v(x) = x\}$. Thus $\mathrm{Tr}(f)$ does nothing but count fixed points.

- In the special case $u = v = \mathrm{id}_X$, we see that $\dim X$ is the self-intersection $\Delta \cap \Delta$ of the diagonal. In the present setting of sets, $\Delta \cap \Delta$ is simply X, so

$\dim X \in \mathrm{End}_{\mathrm{Corr}}(\mathbf{1})$ merely records $|X|$. In more refined geometric or topological settings, together with a suitable "derived" framework, the self-intersection of the diagonal can yield more subtle and important information.

Example 9.3.4 (Cobordisms) Consider the monoidal category n-Cob from [51, Example 3.1.4]. Its objects are compact closed oriented $(n-1)$-manifolds, where $n \in \mathbb{Z}_{\geq 1}$, while its morphisms are equivalence classes of "cobordisms" represented by oriented n-manifolds with boundary; equivalence means a diffeomorphism preserving the boundary. The monoidal structure comes from disjoint union of oriented manifolds, with the empty manifold $\varnothing$ as unit. Concrete depictions of cobordisms, commonly called "pairs of pants", may be found in the cited reference. To respect the string-diagram convention of §9.1, however, cobordisms should be drawn from top to bottom, rather than from left to right as in that reference.

Diffeomorphic closed manifolds are also isomorphic in n-Cob, so the monoidal structure is symmetric. This category is also rigid: for every object X, reversing orientation gives X^*. More precisely, the morphisms $\mathrm{ev}_X : X \sqcup X^* \to \varnothing$ and $\mathrm{coev}_X : \varnothing \to X \sqcup X^*$ make the manifold with boundary $X \times [0,1]$ into a cobordism in two different ways, which may be understood as products of oriented manifolds:

$\mathrm{ev}_X := X \times$  , $\mathrm{coev}_X := X \times$  .

- As in the reference above, the definition of n-Cob already incorporates the bijection

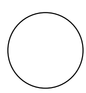
$$\begin{aligned}\mathrm{Hom}_{n\text{-}\mathsf{Cob}}(X,Y) &\overset{1:1}{\longleftrightarrow} \mathrm{Hom}_{n\text{-}\mathsf{Cob}}(\varnothing, X^* \sqcup Y)\\ &= \left\{W : \text{oriented with boundary, together with data } \partial W \xrightarrow{\sim} X^* \sqcup Y\right\}/\sim .\end{aligned}$$

- The dual of a morphism, that is, of a cobordism, simply reverses $\partial W \xrightarrow{\sim} X^* \sqcup Y$ to $\partial W \xrightarrow{\sim} (Y^*)^* \sqcup X^*$.

- Trace and dimension take values in

$$\mathrm{End}_{n\text{-}\mathsf{Cob}}(\varnothing) = \{\text{compact closed manifolds of dimension } n\}/\sim .$$

- By the description of ev_X and coev_X above, taking the trace of $f \in \mathrm{End}_{n\text{-}\mathsf{Cob}}(X)$ amounts to gluing together the two X-ends of the cobordism f. Since the identity morphism id_X corresponds to the trivial cobordism $X \times [0,1]$, we obtain

$\dim X = X \times$  .

For n-Cob, the various diagrams introduced in §9.1 acquire literal meaning. Here the method and the objects share the same geometric substance.

Finally, we discuss how duality appears for modules and comodules over Hopf algebras; see §7.5. Let $\Bbbk$ be a field and let $(A, \mu, \eta, \Delta, \epsilon)$ be a $\Bbbk$-bialgebra, that is, a bialgebra in $\mathsf{Vect}(\Bbbk)$. Proposition 7.5.6 explained how the algebra and coalgebra structures are used together to equip the categories

$$A\text{-}\mathsf{Mod}, \quad \mathsf{Mod}\text{-}A, \quad A\text{-}\mathsf{Comod}, \quad \mathsf{Comod}\text{-}A$$

with natural monoidal structures. We wish to characterize duality in them. Only the cases of right modules and right comodules are described below.

Since the forgetful functor from $\mathsf{Mod}\text{-}A$ (respectively, $\mathsf{Comod}\text{-}A$) to $\mathsf{Vect}(\Bbbk)$ is monoidal, if a right A-module (respectively, right A-comodule) M has a left or right dual, then:

- ◇ $\dim_{\Bbbk} M < \infty$;
- ◇ both the left and right dual must be realized on the dual vector space $M^\vee := \mathrm{Hom}_{\Bbbk}(M, \Bbbk)$;
- ◇ at the level of $\mathsf{Vect}_{\mathrm{f}}(\Bbbk)$, the morphisms ev and coev are determined, up to isomorphism, as in Example 9.3.1.

An A-module (respectively, A-comodule) is called finite-dimensional if it is finite-dimensional as a $\Bbbk$-vector space. We henceforth focus on finite-dimensional M; the corresponding categories are denoted by $\mathsf{Mod}_{\mathrm{f}}\text{-}A$, and so forth.[3)] Readers familiar with commutative ring theory will have no difficulty generalizing the following results to an arbitrary commutative ring $\Bbbk$, provided M is required to be a finitely generated projective $\Bbbk$-module.

The key is to equip $M^\vee$ with a suitable right A-module (respectively, right A-comodule) structure. Write $\otimes := \otimes_{\Bbbk}$. The dual functor in the finite-dimensional case extends to

$$(\cdot)^\vee := \mathrm{Hom}_{\Bbbk}(\cdot, \Bbbk) : \mathsf{Vect}(\Bbbk)^{\mathrm{op}} \to \mathsf{Vect}(\Bbbk).$$

This functor is only right lax monoidal: there is a canonical morphism $N^\vee \otimes M^\vee \to (M \otimes N)^\vee$. We also have the family of evaluation morphisms

$$\mathrm{ev}_M : M \otimes M^\vee \to \Bbbk.$$

[3)] Do not confuse this with $\mathsf{Mod}_{\mathrm{fg}}\text{-}A$ in Theorem 7.8.5, the category of finitely generated right A-modules.

First suppose M is a right A-module, with scalar multiplication given by $a : M \otimes A \to M$. Since $\mathsf{Vect}(\Bbbk)$ is a concrete category, the language of elements and maps is more direct. Take the following transpose ${}^t a : A \otimes M^\vee \to M^\vee$, which makes $M^\vee$ a left A-module:

$$\mathrm{ev}\left(m, {}^t a(t \otimes \lambda)\right) = \mathrm{ev}\left(a(m \otimes t), \lambda\right),$$

where $m \in M$, $\lambda \in M^\vee$, and $t \in A$.

For a right A-comodule M, choose a basis $v_1, \ldots, v_n$ of M and the dual basis $\check{v}_1, \ldots, \check{v}_n$ of $M^\vee$. Its comodule structure may be written as

$$\rho(v_i) = \sum_{j=1}^{n} v_j \otimes t_{ji}, \quad t_{ji} \in A, \quad 1 \leq i \leq n. \tag{9.3.1}$$

The preceding transpose operation has a dual version for ρ: define

$${}^t\rho : M^\vee \to A \otimes M^\vee, \quad \check{v}_i \mapsto \sum_{j=1}^{n} t_{ij} \otimes \check{v}_j.$$

It follows from (7.5.1) and its left-comodule version that ${}^t\rho$ makes $M^\vee$ a left A-comodule; one can also check that this structure is independent of the choice of basis.

The question is how to return to right A-modules and right A-comodules. This can be done canonically when A is a Hopf algebra, as follows.

Proposition 9.3.5 Let $\Bbbk$ be a field and let A be a Hopf $\Bbbk$-algebra with antipode S (Definition 7.5.8), with data $(A, \mu, \eta, \Delta, \epsilon)$. If M is a finite-dimensional right A-module (respectively, right A-comodule), then M has both a left dual and a right dual, both realized on the dual vector space $M^\vee$, as follows.

- Suppose the right A-module structure on M is given by $a : M \otimes A \to M$. Then:

Left dual	Right dual
${}^\vee a : M^\vee \otimes A \to M^\vee$	$a^\vee : M^\vee \otimes A \to M^\vee$
${}^\vee a(\lambda \otimes t) = {}^t a\left(S^{-1}t \otimes \lambda\right)$	$a^\vee(\lambda \otimes t) = {}^t a\left(St \otimes \lambda\right)$

  where ${}^t a$ is the transpose defined above, $\lambda \in M^\vee$, and $t \in A$.

- Suppose the right A-comodule structure on M is given by $\rho : M \to M \otimes A$ and, after choosing a basis, is described as in (9.3.1). Then:

Left dual	Right dual
${}^\vee\rho : M^\vee \to M^\vee \otimes A$	$\rho^\vee : M^\vee \to M^\vee \otimes A$
${}^\vee\rho(\check{v}_i) = \sum_j \check{v}_j \otimes St_{ij}$	$\rho^\vee(\check{v}_i) = \sum_j \check{v}_j \otimes S^{-1}t_{ij}$

Consequently, the category $\mathsf{Mod}_{\mathrm{f}}\text{-}A$ of finite-dimensional right A-modules (respectively, the category $\mathsf{Comod}_{\mathrm{f}}\text{-}A$ of finite-dimensional right A-comodules) is both left and right rigid. The cases of left A-modules or comodules are entirely similar, with left and right interchanged.

Proof First consider a right A-module M. With the notation above, since S gives a homomorphism $A \to A^{\mathrm{op}}_{\mathrm{cop}}$ (Proposition 7.5.11) and the braiding of $\mathsf{Vect}(\Bbbk)$ is symmetric, both constructions in the table make $M^\vee$ a right A-module. We claim that, for the right A-module structure on $M^\vee$ given by $a^\vee$, $\mathrm{ev} : M \otimes M^\vee \to \Bbbk$ is a homomorphism of right A-modules.

Write a and $a^\vee$ directly as right multiplication, as is customary in ring theory; also write $\mu : A \otimes A \to A$ as multiplication and omit $\eta : \Bbbk \to A$ from the notation. Let $\Delta(t) = \sum_i t_i^{(1)} \otimes t_i^{(2)}$. Right-multiplying $m \otimes \lambda \in M \otimes M^\vee$ by t and then evaluating gives

$$\mathrm{ev}\left(\sum_i m t_i^{(1)}, \lambda t_i^{(2)}\right) = \mathrm{ev}\left(\sum_i m t_i^{(1)} S\left(t_i^{(2)}\right), \lambda\right).$$

The definition of the antipode gives $\sum_i t_i^{(1)} S(t_i^{(2)}) = \epsilon(t)$, so this expression becomes $\epsilon(t)\mathrm{ev}(m \otimes \lambda)$. Since the A-module structure on $\Bbbk$ is induced by ϵ, this proves that ev is a homomorphism of right A-modules.

Next, check that $\mathrm{coev} : \Bbbk \to M^\vee \otimes M$ is a homomorphism of right A-modules, still using the structure $a^\vee$ on $M^\vee$. Identify $M^\vee \otimes M$ with $\mathrm{End}_\Bbbk(M)$; then $\mathrm{coev}(1) = \mathrm{id}_M$. It suffices to check that for every $t \in A$,

$$\epsilon(t) \cdot \mathrm{id}_M = \sum_i \left(t_i^{(2)} \text{ acting on the right}\right) \circ \mathrm{id}_M \circ \left(S t_i^{(1)} \text{ acting on the right}\right).$$

This is equivalent to proving $\epsilon(t) = \sum_i S(t_i^{(1)}) t_i^{(2)}$, which again follows directly from the definition of the antipode.

Thus $M^\vee$, with the structure $a^\vee$, gives the right dual of M and is denoted by M^*. The left dual can be handled similarly, or reduced to the following simple observation. Equip $M^\vee$ with the right A-module structure ${}^\vee a$ and denote the result by ${}^*M$. It is easy to see that $({}^*M)^*$, as given by the preceding construction, returns to M; hence ${}^*M$ is a left dual of M.

For a finite-dimensional right A-comodule M, we use the method of undetermined coefficients. Suppose $M^\vee$ has a right comodule structure determined by an $n \times n$ matrix over A, $U = (u_{ij})_{i,j}$, via

$$\check{v}_i \mapsto \sum_{j=1}^n \check{v}_j \otimes u_{ji}, \quad 1 \le i \le n.$$

The necessary and sufficient conditions for U to define a right comodule structure were listed in (7.5.1).

Recall that $\mathrm{ev}(v_i, \check{v}_j) = \delta_{i,j}$, where δ is the Kronecker symbol, and $\mathrm{coev}(1) = \sum_i \check{v}_i \otimes v_i$. The condition that ev and coev be homomorphisms of right A-comodules can then be written as

$$\sum_k t_{ki} u_{kj} = \delta_{i,j} = \sum_k u_{ik} t_{jk}. \tag{9.3.2}$$

Write $T = (t_{ij})_{i,j}$ and let ${}^tT$ denote its transpose. In matrix language the preceding condition becomes

$$({}^tT)U = 1_{n\times n} = U({}^tT).$$

Thus finding a right dual of M amounts to finding $({}^tT)^{-1}$. Similarly, finding a left dual of M amounts to finding ${}^t(T^{-1})$. The assertions about the right and left duals reduce, respectively, to checking

$$\sum_k t_{ki} S^{-1}(t_{jk}) = \delta_{i,j} = \sum_k S^{-1}(t_{ki}) t_{jk},$$
$$\sum_k t_{ik} S(t_{kj}) = \delta_{i,j} = \sum_k S(t_{ik}) t_{kj}.$$

This is immediate. Since $\Delta(t_{ij}) = \sum_k t_{ik} \otimes t_{kj}$, the definition of the antipode shows that both sides in the second line equal $\epsilon(t_{ij}) = \delta_{i,j}$. Apply S^{-1} to both equalities in the second line and use the fact that S preserves the unit and reverses the order of multiplication; rearranging yields the first line. This proves the assertion. □

If A is noncommutative, ${}^t(T^{-1})$ and $({}^tT)^{-1}$ need not be equal. Thus the argument above also shows that left and right duals do not in general coincide.

In fact, Corollary 9.6.6 below will show conversely how the rigidity of the category of finite-dimensional comodules determines a Hopf structure on a bialgebra. A sequence of theoretical preparations will be needed first.

9.4 The Endomorphism Coalgebra

Beginning with this section, we lay a series of foundations for the theory of Tannakian categories (see §9.7); some of the tools and ideas involved have broader applications. The principal reference is [10].

Let $\Bbbk$ be a commutative ring. For any category $\mathcal{A}$ and functor $\xi : \mathcal{A} \to \Bbbk\text{-}\mathsf{Mod}$, we may consider the $\Bbbk$-algebra $\mathrm{End}(\xi)$ of endomorphisms of the functor. It gives a family of $\Bbbk$-linear maps

$$\mathrm{End}(\xi) \underset{\Bbbk}{\otimes} \xi(X) \to \xi(X), \quad X \in \mathrm{Ob}(\mathcal{A}),$$

functorial in X, or briefly $\mathrm{End}(\xi) \underset{\Bbbk}{\otimes} \xi \to \xi$. Moreover, $\mathrm{End}(\xi)$ is universal for such actions: tautologically, giving an action of a $\Bbbk$-algebra E on ξ is equivalent to giving an algebra homomorphism $E \to \mathrm{End}(\xi)$. The multiplication on $\mathrm{End}(\xi)$ can be derived abstractly from this universal property.

For the problems we shall address, the dual version of this construction is both more useful and simpler. We must, however, consider functors taking values in $\mathsf{Mod}\text{-}B$, where B is a $\Bbbk$-algebra, assumed nonzero by convention. This construction is closely related to the abstract categorical construction called a **coend**; an exercise in this chapter gives further details.

Definition 9.4.1 Let $\Bbbk$ be a commutative ring, B a $\Bbbk$-algebra, and $\omega_1, \omega_2 : \mathcal{A} \to \mathsf{Mod}\text{-}B$ functors. Consider a (B, B)-bimodule L together with a natural transformation $a : \omega_2 \to \omega_1 \underset{B}{\otimes} L$, that is, a family of right B-module homomorphisms

$$a_X : \omega_2(X) \to \omega_1(X) \underset{B}{\otimes} L, \quad X \in \mathrm{Ob}(\mathcal{A}),$$

such that the following diagram commutes for every $f \in \mathrm{Hom}_{\mathcal{A}}(X, Y)$:

$$\begin{tikzcd} \omega_2(X) \arrow[r, "a_X"] \arrow[d, "\omega_2(f)"'] & \omega_1(X) \underset{B}{\otimes} L \arrow[d, "\omega_1(f)\otimes \mathrm{id}"] \\ \omega_2(Y) \arrow[r, "a_Y"'] & \omega_1(Y) \underset{B}{\otimes} L. \end{tikzcd}$$

A morphism from the data (L, a) to (L', a') is defined to be a bimodule homomorphism $t : L \to L'$ that makes the following diagram commute for all X:

$$\begin{tikzcd} \omega_2(X) \arrow[r, "a_X"] \arrow[dr, "a'_X"'] & \omega_1(X) \underset{B}{\otimes} L \arrow[d, "\mathrm{id}\otimes t"] \\ & \omega_1(X) \underset{B}{\otimes} L'. \end{tikzcd}$$

Conversely, data (L, a) and a bimodule homomorphism $t : L \to L'$ determine a' by this diagram, together with a morphism $t : (L, a) \to (L', a')$.

If the category of all such data (L, a) and morphisms has an initial object, denote the corresponding (B, B)-bimodule by

$$\mathrm{coH}(\omega_1, \omega_2) = \mathrm{coH}_{\Bbbk}(\omega_1, \omega_2),$$

and denote the corresponding morphism by $\lambda : \omega_2 \to \omega_1 \underset{B}{\otimes} L$. We also write

$$\mathrm{coE}(\omega) = \mathrm{coE}_{\Bbbk}(\omega) := \mathrm{coH}(\omega, \omega).$$

The symbols coH and coE stand for coHom and coEnd, respectively; they should be viewed as dual to Hom and End, and the abbreviations are merely typographical. Note that the notion of a (B,B)-bimodule depends not only on the ring structure of B, but also on $\Bbbk$.

Remark 9.4.2 The equality $\mathrm{coH}(\omega_1,\omega_2)=0$ means that every morphism $\omega_2 \to \omega_1 \underset{B}{\otimes} L$ is zero; the definition does not exclude this possibility. By contrast, if $\mathrm{coE}(\omega)$ exists and ω is not the constant zero functor, then $\mathrm{coE}(\omega)\neq 0$, since $\omega \xrightarrow{\sim} \omega \underset{B}{\otimes} B$.

If $\mathrm{coE}(\omega)$ exists, applying the universal property to $\omega \xrightarrow{\sim} \omega \underset{B}{\otimes} B$ gives a bimodule homomorphism

$$\epsilon : \mathrm{coE}(\omega) \to B.$$

On the other hand, assuming the relevant coH objects exist, successive application to $\omega_1,\omega_2,\omega_3$ yields the morphism

$$\omega_3 \to \omega_2 \underset{B}{\otimes} \mathrm{coH}(\omega_2,\omega_3) \to \omega_1 \underset{B}{\otimes} \mathrm{coH}(\omega_1,\omega_2) \underset{B}{\otimes} \mathrm{coH}(\omega_2,\omega_3).$$

The universal property then gives a canonical bimodule homomorphism

$$\mathrm{coH}(\omega_1,\omega_3) \to \mathrm{coH}(\omega_1,\omega_2) \underset{B}{\otimes} \mathrm{coH}(\omega_2,\omega_3). \tag{9.4.1}$$

A routine check with the functors $\omega_1,\ldots,\omega_4$ shows that (9.4.1) satisfies coassociativity. Thus $\mathrm{coE}(\omega)$ becomes a coalgebra in $(B,B)\text{-}\mathsf{Mod}$, with $\epsilon : \mathrm{coE}(\omega) \to B$ as counit, while every $\omega(X)$ becomes a $\mathrm{coE}(\omega)$-comodule via the canonical morphism λ. This discussion is summarized as follows.

Definition–Proposition 9.4.3 (Endomorphism coalgebra) Let $\omega : \mathcal{A} \to \mathsf{Mod}\text{-}B$ be a functor and suppose $\mathrm{coE}(\omega)$ exists.

- ◇ As an object of the monoidal category $(B,B)\text{-}\mathsf{Mod}$, $\mathrm{coE}(\omega)$ has a canonical coalgebra structure such that every $\omega(X)$ becomes a $\mathrm{coE}(\omega)$-comodule via the canonical morphism, and morphisms in $\mathcal{A}$ induce comodule homomorphisms. We call $\mathrm{coE}(\omega)$ the **endomorphism coalgebra** of ω.[4)]
- ◇ For every coalgebra L in $(B,B)\text{-}\mathsf{Mod}$, compatibly giving all $\omega(X)$ an L-comodule structure is equivalent to giving a coalgebra homomorphism $\mathrm{coE}(\omega) \to L$.

Remark 9.4.4 Given a functor $T : \mathcal{A}' \to \mathcal{A}$ and $\omega_1,\omega_2 : \mathcal{A} \to \mathsf{Mod}\text{-}B$, assuming the relevant coH objects exist, the canonical morphism $\omega_2(TX) \to \omega_1(TX) \underset{B}{\otimes} \mathrm{coH}(\omega_1,\omega_2)$ together with the universal property naturally induces a homomorphism

$$\mathrm{coH}(\omega_1 T, \omega_2 T) \to \mathrm{coH}(\omega_1,\omega_2),$$

4) Perhaps “coendomorphism coalgebra” would be a more accurate name.

compatible with all the structures above. In particular, taking $\omega_1 = \omega = \omega_2$ gives a coalgebra homomorphism $\mathrm{coE}(\omega T) \to \mathrm{coE}(\omega)$.

Proposition 9.4.5 Let $\omega_1, \omega_2 : \mathcal{A} \to \mathsf{Mod}\text{-}B$ be functors. Suppose $\mathcal{A}$ is the union of a family of subcategories, each denoted by $\mathcal{A}'$, and write $\omega_i' := \omega_i|_{\mathcal{A}'}$. If $\mathrm{coH}(\omega_1', \omega_2')$ exists for every $\mathcal{A}'$, then $\mathrm{coH}(\omega_1, \omega_2)$ exists and there is a canonical isomorphism

$$\varinjlim_{\mathcal{A}'} \mathrm{coH}(\omega_1', \omega_2') \xrightarrow{\sim} \mathrm{coH}(\omega_1, \omega_2).$$

The colimit on the left is ordered by inclusion; its transition homomorphisms are given as in Remark 9.4.4. The isomorphism is compatible with (9.4.1).

Proof For every (B, B)-bimodule L, giving a morphism $\omega_2 \to \omega_1 \underset{B}{\otimes} L$ is equivalent to compatibly giving $\omega_2' \to \omega_1' \underset{B}{\otimes} L$ for every $\mathcal{A}'$; this in turn is equivalent to giving a compatible family of homomorphisms $\mathrm{coH}(\omega_1', \omega_2') \to L$. $\square$

The issue is how to ensure the existence of $\mathrm{coH}(\omega_1, \omega_2)$, which is not automatic. We must first control the size of the category. Fix a Grothendieck universe $\mathcal{U}$, relative to which we may speak of small sets and small categories. A category is called **essentially small** if the isomorphism classes of its objects form a small set (Definition A.4.1).

Convention 9.4.6 Let $\Bbbk$ be a commutative ring and B a $\Bbbk$-algebra. We shall need the following notation.

- ◇ Consider the full subcategories

$$\underbrace{\mathsf{Mod}_{\mathrm{pfg}}\text{-}B}_{\text{projective + finitely generated}} \subset \underbrace{\mathsf{Mod}_{\mathrm{fg}}\text{-}B}_{\text{finitely generated}} \subset \mathsf{Mod}\text{-}B.$$

- ◇ For every right B-module P, define the left B-module $P^\vee$ according to (7.7.4). Every homomorphism $\varphi : P \to Q$ induces the dual homomorphism $\varphi^\vee : Q^\vee \to P^\vee$.

Following this book's convention for algebraic structures, $\Bbbk$ and B are assumed to be realized on small sets. Finite generation ensures that $\mathsf{Mod}_{\mathrm{fg}}\text{-}B$ and $\mathsf{Vect}_{\mathrm{f}}(\Bbbk)$ are essentially small categories; they are also $\Bbbk$-linear. The essentially small category $\mathcal{A}$ considered in this section, however, need not have a $\Bbbk$-linear structure.

For every $P_1, P_2 \in \mathrm{Ob}(\mathsf{Mod}_{\mathrm{pfg}}\text{-}B)$, there is a canonical isomorphism

$$\begin{aligned}\mathrm{Hom}_{\mathsf{Mod}\text{-}B}\left(P_2, P_1 \underset{B}{\otimes} L\right) &\xrightarrow{\sim} \mathrm{Hom}_{(B,B)\text{-}\mathsf{Mod}}\left(P_1^\vee \underset{\Bbbk}{\otimes} P_2, L\right)\\ \varphi &\mapsto [\check{p}_1 \otimes p_2 \mapsto (\check{p}_1 \otimes \mathrm{id}_L)(\varphi(p_2))].\end{aligned}$$

This is easily reduced to the case in which P_1 and P_2 are finite-rank free modules. Therefore, when $\omega_1, \omega_2 : \mathcal{A} \to \mathsf{Mod}\text{-}B$ take values in $\mathsf{Mod}_{\mathrm{pfg}}\text{-}B$, giving data $a = (a_X)_X$ as in Definition 9.4.1 is equivalent to giving a family of homomorphisms

$$\alpha_X : \omega_1(X)^\vee \underset{\Bbbk}{\otimes} \omega_2(X) \to L, \quad X \in \mathrm{Ob}(\mathcal{A}),$$

whose functoriality in X translates into commutativity of the following diagram for every $f \in \mathrm{Hom}_{\mathcal{A}}(X, Y)$:

$$\begin{tikzcd}
\omega_1(Y)^\vee \underset{\Bbbk}{\otimes} \omega_2(X) \arrow[r, "\omega_1(f)^\vee \otimes \mathrm{id}"] \arrow[d, "\mathrm{id}\otimes\omega_2(f)"'] & \omega_1(X)^\vee \underset{\Bbbk}{\otimes} \omega_2(X) \arrow[d, "\alpha_X"] \\
\omega_1(Y)^\vee \underset{\Bbbk}{\otimes} \omega_2(Y) \arrow[r, "\alpha_Y"'] & L.
\end{tikzcd} \tag{9.4.2}$$

Proposition 9.4.7 Let $\mathcal{A}$ be an essentially small category and let $\omega_1, \omega_2 : \mathcal{A} \to \mathsf{Mod}_{\mathrm{pfg}}\text{-}B$ be arbitrary functors. Then $\mathrm{coH}(\omega_1, \omega_2)$ of Definition 9.4.1 exists.

Proof The only point to prove is the existence of an initial object. Use the commutative diagram (9.4.2). Let L_0 be the direct sum of all $\omega_1(X)^\vee \underset{\Bbbk}{\otimes} \omega_2(X)$, where X ranges over a set of representatives for $\mathrm{Ob}(\mathcal{A})/\simeq$; this is where essential smallness is needed. For every $f : X \to Y$, define

$$\delta_f := \omega_1(f)^\vee \otimes \mathrm{id} - \mathrm{id} \otimes \omega_2(f) : \omega_1(Y)^\vee \underset{\Bbbk}{\otimes} \omega_2(X) \to L_0.$$

Then $\mathrm{coH}(\omega_1, \omega_2) := L_0 / \sum_f \mathrm{im}(\delta_f)$ is the desired object. □

Convention 9.4.8 The following concrete notation will be useful. For $t \in \omega_1(X)^\vee \underset{\Bbbk}{\otimes} \omega_2(X)$, write $[t]$ for its image in $\mathrm{coH}(\omega_1, \omega_2)$; these images generate $\mathrm{coH}(\omega_1, \omega_2)$. In the universal property of Definition 9.4.3, the image in L of $[\phi \otimes x] \in \mathrm{coE}(\omega)$ is obtained by first mapping $x \in \omega(X)$ through $\omega(X) \to \omega(X) \underset{B}{\otimes} L$ and then contracting with $\phi \in \omega(X)^\vee$.

For a functor $\omega : \mathcal{A} \to \mathsf{Mod}_{\mathrm{pfg}}\text{-}B$, the evaluation homomorphism $\mathrm{ev}_{\omega(X)} : \omega(X)^\vee \underset{\Bbbk}{\otimes} \omega(X) \to B$ is a homomorphism of (B, B)-bimodules. The canonical homomorphism that it determines is precisely the previously defined homomorphism

$$\begin{aligned} \epsilon : \mathrm{coE}(\omega) &\to B \\ [t] &\mapsto \mathrm{ev}_{\omega(X)}(t). \end{aligned} \tag{9.4.3}$$

If $\omega_1(X)$ is free, choose a B-basis $(v_i)_{i=1}^n$ and its dual basis $(\check{v}_i)_{i=1}^n$. Then $\lambda_X : \omega_2(X) \to \omega_1(X) \underset{B}{\otimes} \mathrm{coH}(\omega_1, \omega_2)$ may be written as

$$w \mapsto \sum_{i=1}^n v_i \otimes [\check{v}_i \otimes w]; \tag{9.4.4}$$

It follows that for arbitrary $\omega_1, \omega_2, \omega_3$, if $\omega_1(X)$ and $\omega_2(X)$ are free, $\lambda \in \omega_1(X)^\vee$, and $w \in \omega_3(X)$, then

$$\begin{aligned} \mathrm{coH}(\omega_1, \omega_3) &\to \mathrm{coH}(\omega_1, \omega_2) \underset{B}{\otimes} \mathrm{coH}(\omega_2, \omega_3) \\ [\lambda \otimes w] &\mapsto \sum_i [\lambda \otimes v_i] \otimes [\check{v}_i \otimes w]. \end{aligned} \tag{9.4.5}$$

The next step is to introduce a monoidal structure. For this, B must be commutative and we must take $\Bbbk = B$ in the preceding construction. Then Mod-B and $\mathsf{Mod}_{\mathrm{pfg}}$-$B$ are both symmetric monoidal categories under $\otimes_B$, with B as unit. To emphasize this setting, we denote the corresponding coH and coE by coH_B and coE_B; in this case, a (B, B)-bimodule is the same thing as a B-module.

Let $\mathcal{A}$, $\mathcal{A}'$, and so forth be arbitrary categories. First take functors

$$\omega_i : \mathcal{A} \to \mathsf{Mod}\text{-}B, \quad \omega_i' : \mathcal{A}' \to \mathsf{Mod}\text{-}B, \quad i = 1, 2.$$

Define $\omega_i \boxtimes \omega_i' : \mathcal{A} \times \mathcal{A}' \to \mathsf{Mod}\text{-}B$ by sending an object (X, X') to $\omega_i(X) \underset{B}{\otimes} \omega_i'(X')$.

Assuming that the relevant coH objects exist, apply the universal property to the family of morphisms

$$\begin{aligned} \omega_2(X) \underset{B}{\otimes} \omega_2'(X') &\to \omega_1(X) \underset{B}{\otimes} \mathrm{coH}_B(\omega_1, \omega_2) \underset{B}{\otimes} \omega_1'(X') \underset{B}{\otimes} \mathrm{coH}_B(\omega_1', \omega_2') \\ &\simeq \omega_1(X) \underset{B}{\otimes} \omega_1'(X') \underset{B}{\otimes} \mathrm{coH}_B(\omega_1, \omega_2) \underset{B}{\otimes} \mathrm{coH}_B(\omega_1', \omega_2') \end{aligned}$$

to obtain a canonical morphism

$$\nu : \mathrm{coH}_B(\omega_1 \boxtimes \omega_1', \omega_2 \boxtimes \omega_2') \to \mathrm{coH}_B(\omega_1, \omega_2) \underset{B}{\otimes} \mathrm{coH}_B(\omega_1', \omega_2'),$$

that makes the following diagram commute for every $(X, X') \in \mathrm{Ob}(\mathcal{A} \times \mathcal{A}')$:

$$\begin{tikzcd} \omega_2(X) \underset{B}{\otimes} \omega_2'(X') \arrow[r, "\text{canonical}"] \arrow[dr, "\text{canonical}"'] & \omega_1(X) \underset{B}{\otimes} \omega_1'(X') \underset{B}{\otimes} \mathrm{coH}_B(\omega_1 \boxtimes \omega_1', \omega_2 \boxtimes \omega_2') \arrow[d, "\mathrm{id} \otimes \mathrm{id} \otimes \nu"] \\ & \omega_1(X) \underset{B}{\otimes} \omega_1'(X') \underset{B}{\otimes} \mathrm{coH}_B(\omega_1, \omega_2) \underset{B}{\otimes} \mathrm{coH}_B(\omega_1', \omega_2'). \end{tikzcd}$$

Lemma 9.4.9 With the notation above, suppose in addition that all ω_i and ω_i' take values in $\mathsf{Mod}_{\mathrm{pfg}}$-$B$. Then $\omega_i \boxtimes \omega_i'$ also takes values there and, in this setting, ν is an isomorphism. Following Convention 9.4.8, it can be described explicitly by

$$\nu : [(\phi \otimes \phi') \otimes (x \otimes x')] \mapsto [\phi \otimes x] \otimes [\phi' \otimes x'].$$

Proof Since $\mathsf{Mod}_{\mathrm{pfg}}$-$B$ is closed under $\otimes_B$, the functor $\omega \boxtimes \omega'$ takes values there. Returning to the construction in (9.4.2) and Proposition 9.4.7 immediately gives the explicit description of ν.

Next we show that ν is an isomorphism. By the construction in Proposition 9.4.7, $\mathrm{coH}_B(\omega_1, \omega_2)$ may suitably be expressed as $\varinjlim_{X,Y} \omega_1(Y)^\vee \underset{B}{\otimes} \omega_2(X)$, with the relevant morphisms coming from (9.4.2); the same is of course true for the other coH_B objects involved in ν. Thus the assertion that ν is an isomorphism is equivalent to saying that the evident morphism

$$\begin{aligned} &\varinjlim_{X,X',Y,Y'} \omega_1(Y)^\vee \underset{B}{\otimes} \omega_1'(Y')^\vee \underset{B}{\otimes} \omega_2(X) \underset{B}{\otimes} \omega_2'(X') \\ &\qquad \to \varinjlim_{X,Y} \left(\omega_1(Y)^\vee \underset{B}{\otimes} \omega_2(X) \right) \underset{B}{\otimes} \varinjlim_{X',Y'} \left(\omega_1'(Y')^\vee \underset{B}{\otimes} \omega_2'(X') \right) \end{aligned}$$

is an isomorphism. This in turn reduces to the standard fact from module theory that $\underset{B}{\otimes}$ preserves all small $\varinjlim$ [51, Proposition 6.9.2]. $\square$

We continue the preceding discussion. Consider the functors

$$\otimes : \mathcal{A} \times \mathcal{A}' \to \mathcal{A}'',$$
$$\omega_i : \mathcal{A} \to \mathsf{Mod}_{\mathrm{pfg}}\text{-}B, \quad \omega_i' : \mathcal{A}' \to \mathsf{Mod}_{\mathrm{pfg}}\text{-}B, \quad \omega_i'' : \mathcal{A}'' \to \mathsf{Mod}_{\mathrm{pfg}}\text{-}B \quad i = 1, 2,$$

together with isomorphisms $\theta_i : \omega_i \boxtimes \omega_i' \xrightarrow{\sim} \omega_i'' \circ \otimes$, whose dual versions are denoted by $\theta_i^\vee$. The preceding preparations give a B-linear homomorphism

$$\begin{tikzcd}[column sep=small] \mathrm{coH}_B(\omega_1, \omega_2) \underset{B}{\otimes} \mathrm{coH}_B(\omega_1', \omega_2') \arrow[r, "\nu^{-1}", "\theta_1, \theta_2"'] & \mathrm{coH}_B(\omega_1'' \circ \otimes,\ \omega_2'' \circ \otimes) \arrow[r, "\text{Remark 9.4.4}"] & \mathrm{coH}_B(\omega_1'', \omega_2'') \\ {[\phi \otimes x] \otimes [\phi' \otimes x']} \arrow[rr, mapsto] & & {[\theta_1^\vee(\phi \otimes \phi') \otimes \theta_2(x \otimes x')].} \end{tikzcd} \tag{9.4.6}$$

The preceding commutative diagram for ν implies that the following diagram commutes; the details are left to the reader.

$$\begin{tikzcd} \omega_2''(X \otimes X') \arrow[rr, "\text{canonical}"] \arrow[d, "\theta_2^{-1}", "\wr"'] & & \omega_1''(X \otimes X') \underset{B}{\otimes} \mathrm{coH}_B(\omega_1'', \omega_2'') \\ \omega_2(X) \underset{B}{\otimes} \omega_2'(X') \arrow[d, "\text{canonical}"] & & \\ \omega_1(X) \underset{B}{\otimes} \omega_1'(X') \underset{B}{\otimes} \mathrm{coH}_B(\omega_1, \omega_2) \underset{B}{\otimes} \mathrm{coH}_B(\omega_1', \omega_2') \arrow[rr, "(9.4.6)"'] & & \omega_1(X) \underset{B}{\otimes} \omega_1'(X') \underset{B}{\otimes} \mathrm{coH}_B(\omega_1'', \omega_2''). \arrow[uu, "\theta_1 \otimes \mathrm{id}"', "\wr"] \end{tikzcd} \tag{9.4.7}$$

Now suppose further that $\mathcal{A} = \mathcal{A}' = \mathcal{A}''$ is a monoidal category, $\otimes$ is its product, and $\omega_i'' = \omega_i' = \omega_i$ are monoidal functors ($i = 1, 2$). Write $\mathcal{A}_0$ for the category with just one object and one morphism (the identity), and define $\iota : \mathcal{A}_0 \to \mathcal{A}$ by sending its unique object to $\mathbf{1}$. A direct manipulation of the definitions gives

$$\mathrm{coH}_B(\omega_1 \iota, \omega_2 \iota) \simeq B \underset{B}{\otimes} B \simeq B.$$

Functoriality in Remark 9.4.4 then gives the canonical map

$$B \simeq \mathrm{coH}_B(\omega_1\iota, \omega_2\iota) \to \mathrm{coH}_B(\omega_1, \omega_2), \quad b \mapsto [b \otimes 1] = [1 \otimes b]. \tag{9.4.8}$$

Proposition 9.4.10 In the setting above, suppose $\mathcal{A} = \mathcal{A}' = \mathcal{A}''$ is a monoidal category, ω_1 and ω_2 are monoidal functors, and $\omega_i'' = \omega_i' = \omega_i$.

(i) The corresponding homomorphism $\mathrm{coH}_B(\omega_1, \omega_2) \underset{B}{\otimes} \mathrm{coH}_B(\omega_1, \omega_2) \to \mathrm{coH}_B(\omega_1, \omega_2)$ makes $\mathrm{coH}_B(\omega_1, \omega_2)$ a B-algebra, with (9.4.8) as its unit.

(ii) In the special case $\omega_1 = \omega = \omega_2$, this structure makes $\mathrm{coE}_B(\omega)$ a bialgebra.

(iii) If $\mathcal{A}$ is a symmetric monoidal category and ω_1, ω_2 are compatible with the braiding, then $\mathrm{coH}_B(\omega_1, \omega_2)$ is a commutative B-algebra.

Proof Since the multiplication on $\mathrm{coH}_B(\omega_1, \omega_2)$ has a natural characterization through (9.4.6) and (9.4.7), all the remaining assertions merely reflect the monoidal structure and require only formal arguments; the details are omitted. □

9.5 The Reconstruction Theorem

Throughout this section, $\Bbbk$ is a fixed commutative ring. Unless stated otherwise, all categories and functors considered below are $\Bbbk$-linear; in particular, the bifunctor $\otimes$ of a monoidal category is understood to be $\Bbbk$-linear in each variable. For a $\Bbbk$-algebra B, Convention 9.4.6 defined the essentially small categories $\mathsf{Mod}_{\mathrm{pfg}}\text{-}B \subset \mathsf{Mod}_{\mathrm{fg}}\text{-}B$. If C is a coalgebra in the monoidal category $(B, B)\text{-}\mathsf{Mod}$, write $\mathsf{Comod}\text{-}C$ for the category of right C-comodules; these comodule structures are superimposed on structures in $\mathsf{Mod}\text{-}B$, as explained after Definition 7.7.11. Imposing projectivity and finiteness conditions as right B-modules gives the full subcategories

$$\underbrace{\mathsf{Comod}_{\mathrm{pf}}\text{-}C}_{\text{projective + finitely generated}} \subset \underbrace{\mathsf{Comod}_{\mathrm{f}}\text{-}C}_{\text{finitely generated}} \subset \mathsf{Comod}\text{-}C.$$

Finite generation ensures that $\mathsf{Comod}_{\mathrm{f}}\text{-}C$ is also an essentially small category in the sense of Definition A.4.1.

To gain a more concrete understanding of $\mathrm{coE}(\omega) = \mathrm{coE}_{\Bbbk}(\omega)$ from §9.4, consider the setting of Proposition 7.7.12. Take $\Bbbk$-algebras A, B, not necessarily commutative, and an (A, B)-bimodule P. Suppose P is finitely generated and projective as a right B-module. Let $\mathcal{A}$ be a full subcategory of $\mathsf{Mod}\text{-}A$ satisfying

- ◇ $A \in \mathrm{Ob}(\mathcal{A})$;

◇ $\omega := (\cdot) \underset{A}{\otimes} P$ defines a functor $\mathcal{A} \to \mathsf{Mod}_{\mathrm{pfg}}\text{-}B$.

Proposition 7.7.12 asserts that the (B, B)-bimodule $P^\vee \underset{A}{\otimes} P$ is the coalgebra corresponding to the comonad determined by the adjoint pair

$$(\cdot) \underset{A}{\otimes} P : \mathsf{Mod}\text{-}A \rightleftarrows \mathsf{Mod}\text{-}B : (\cdot) \underset{B}{\otimes} P^\vee.$$

The coalgebra $P^\vee \underset{A}{\otimes} P$ naturally coacts on every $\omega(N) = N \underset{A}{\otimes} P$. Explicitly, the coaction is

$$N \underset{A}{\otimes} P \xrightarrow{\mathrm{id}_N \otimes \mathrm{coev} \otimes \mathrm{id}_P} N \underset{A}{\otimes} P \underset{B}{\otimes} P^\vee \underset{A}{\otimes} P, \quad N \in \mathrm{Ob}(\mathsf{Mod}\text{-}A).$$

Definition–Proposition 9.4.3 then gives a canonical coalgebra homomorphism

$$\mathrm{coE}(\omega) \to P^\vee \underset{A}{\otimes} P.$$

Proposition 9.5.1 In the setting above, the homomorphism $\mathrm{coE}(\omega) \to P^\vee \underset{A}{\otimes} P$ is an isomorphism.

Proof Let $\mathcal{A}^\flat$ be the full subcategory of $\mathsf{Mod}\text{-}A$ with $\mathrm{Ob}(\mathcal{A}^\flat) = \{A\}$, and write i for the inclusion functor. Remark 9.4.4 gives the commutative diagram

$$\begin{tikzcd} \mathrm{coE}(\omega i) \arrow[rr] \arrow[dr] & & \mathrm{coE}(\omega), \arrow[dl] \\ & P^\vee \underset{A}{\otimes} P & \end{tikzcd}$$

The reader can check directly that $\mathrm{coE}(\omega i) \xrightarrow{\sim} P^\vee \underset{A}{\otimes} P$. Hence $\mathrm{coE}(\omega i) \to \mathrm{coE}(\omega)$ has a left inverse. If this morphism is surjective, then $\mathrm{coE}(\omega i) \xrightarrow{\sim} \mathrm{coE}(\omega)$, and consequently $\mathrm{coE}(\omega) \xrightarrow{\sim} P^\vee \underset{A}{\otimes} P$.

By the concrete constructions of $\mathrm{coE}(\omega)$ and $\mathrm{coE}(\omega i)$, it remains to prove, for every $Y \in \mathrm{Ob}(\mathcal{A})$, that

$$\mathrm{im}\left[\omega(Y)^\vee \underset{\Bbbk}{\otimes} \omega(Y) \to \mathrm{coE}(\omega)\right] \subset \mathrm{im}\left[\omega(A)^\vee \underset{\Bbbk}{\otimes} \omega(A) \to \mathrm{coE}(\omega)\right].$$

Write $t \in \omega(Y)^\vee \underset{\Bbbk}{\otimes} \omega(Y)$ as a finite sum $\sum_i \lambda_i \otimes o_i$. By the concrete form of ω, each $o_i \in \omega(Y)$ is a finite linear combination of elements of the form $\omega(f)(x)$, where $x \in \omega(A)$ and $f \in \mathrm{Hom}_{\mathcal{A}}(A, Y)$. Thus the problem further reduces to the case $t = \lambda \otimes \omega(f)(x)$. This follows by considering the image of $\lambda \otimes x \in \omega(Y)^\vee \underset{\Bbbk}{\otimes} \omega(A)$ in the commutative diagram

$$\begin{tikzcd}
\omega(Y)^\vee \underset{\Bbbk}{\otimes} \omega(A) \arrow[r, "\omega(f)^\vee\otimes\mathrm{id}"] \arrow[d, "\mathrm{id}\otimes\omega(f)"'] & \omega(A)^\vee \underset{\Bbbk}{\otimes} \omega(A) \arrow[d] \\
\omega(Y)^\vee \underset{\Bbbk}{\otimes} \omega(Y) \arrow[r] & \mathrm{coE}(\omega).
\end{tikzcd}$$

□

For any essentially small category $\mathcal{A}$ and any functor $\omega : \mathcal{A} \to \mathsf{Mod}_{\mathrm{pfg}}\text{-}B$, Definition–Proposition 9.4.3 canonically factors ω as

$$\mathcal{A} \xrightarrow{\overline{\omega}} \mathsf{Comod}_{\mathrm{pf}}\text{-}\,\mathrm{coE}(\omega) \xrightarrow{U} \mathsf{Mod}_{\mathrm{pfg}}\text{-}B, \tag{9.5.1}$$

where U is the forgetful functor. We wish to determine when $\overline{\omega}$ is an equivalence.

The following lemma underlies the various reconstruction theorems below. The core of its proof uses Beck's Monadicity Theorem 7.6.14. We will also need the characterization of faithful exact functors from Proposition 2.8.9, the notion of a locally finite abelian category from Definition A.4.2, and some Ind-completion techniques from Appendix B.

Lemma 9.5.2 Let $\Bbbk$ be a field, let $\mathcal{A}$ be a locally finite abelian category, and let s be a projective generator of $\mathcal{A}$. Consider a faithful exact functor

$$\omega : \mathcal{A} \to \mathsf{Mod}\text{-}B,$$

and suppose that it takes values in $\mathsf{Mod}_{\mathrm{pfg}}\text{-}B$.

(i) The coalgebra $\mathrm{coE}(\omega)$ can be written concretely as $P^\vee \underset{A}{\otimes} P$, where $A = \mathrm{End}_{\mathcal{A}}(s)$ and $P = \omega(s)$.

(ii) The functor $\overline{\omega}$ in the canonical factorization (9.5.1) is an equivalence of categories.

(iii) We have $\mathsf{Comod}_{\mathrm{pf}}\text{-}\,\mathrm{coE}(\omega) = \mathsf{Comod}_{\mathrm{f}}\text{-}\,\mathrm{coE}(\omega)$.

Proof Observe that A is a finite-dimensional $\Bbbk$-algebra. All the hypotheses of Theorem 7.8.5 hold, and $\mathrm{Hom}_{\mathcal{A}}(s, \cdot)$ gives an equivalence $\mathcal{A} \to \mathsf{Mod}_{\mathrm{fg}}\text{-}A$. Thus the problem reduces to the case $\mathcal{A} = \mathsf{Mod}_{\mathrm{fg}}\text{-}A$ and $s = A$. The first observation is that ω now extends canonically to a functor

$$\Omega : \mathsf{Mod}\text{-}A \to \mathsf{Mod}\text{-}B,$$

by the following construction.

▷ On objects Take the small filtered colimit

$$\Omega(N) = \varinjlim_{\substack{N' \subset N \\ \text{finitely generated}}} \omega(N'). \tag{9.5.2}$$

▷ On morphisms For a homomorphism of right A-modules $f : N_1 \to N_2$, take

$$\begin{aligned} \Omega(f) = \varprojlim_{N_1' \subset N_1} \varinjlim_{\substack{N_2' \subset N_2 \\ N_2' \supset \mathrm{im}(f|_{N_1'})}} \left[\omega(f|_{N_1'}) : \omega(N_1') \to \omega(N_2')\right] \\ \in \varprojlim_{N_1' \subset N_1} \varinjlim_{N_2' \subset N_2} \mathrm{Hom}_{\mathsf{Mod}\text{-}B}\left(\omega(N_1'), \omega(N_2')\right). \end{aligned}$$

The right-hand side determines a homomorphism $\Omega(N_1) \to \Omega(N_2)$ in the evident way.

For readers using the theory of Ind-completion from §§B.2–B.3, this construction merely identifies $\mathsf{Mod}\text{-}A$ with $\mathrm{Ind}(\mathsf{Mod}_{\mathrm{fg}}\text{-}A)$ and then uses the cocompleteness of $\mathsf{Mod}\text{-}B$ to extend ω to $\mathrm{Ind}(\mathsf{Mod}_{\mathrm{fg}}\text{-}A)$ by filtered colimits; see Definition–Proposition B.4.4.

The following properties are general facts about Ind-completion and can also be checked directly from the construction of Ω.

- ◇ The restriction of Ω to $\mathsf{Mod}_{\mathrm{fg}}\text{-}A$ is isomorphic to ω.
- ◇ Ω is an exact $\Bbbk$-linear functor; see Proposition B.5.5(ii) and the preceding discussion, which use the exactness of ω and the fact that $\mathsf{Mod}\text{-}B$ is a Grothendieck category.
- ◇ Ω preserves all small colimits; see Proposition B.5.5(iii), which uses the fact that $\mathsf{Mod}_{\mathrm{fg}}\text{-}A$ is essentially small.

Moreover, since ω is left exact, for all finitely generated submodules $N' \subset N''$ of N, the transition homomorphism $\omega(N') \to \omega(N'')$ in (9.5.2) is monic. The elementary construction of filtered colimits in the category of modules then gives

$$\Omega(N) = 0 \iff \forall N',\ \omega(N') = 0 \overset{\because\ \omega \text{ faithful}}{\iff} \forall N',\ N' = 0 \iff N = 0.$$

Thus Ω is faithful and exact.

We may now apply Theorem 7.7.2 to Ω to obtain

$$\Omega \simeq (\cdot) \underset{A}{\otimes} P, \quad P := \Omega(A) = \omega(A) \in \mathrm{Ob}(\mathsf{Mod}_{\mathrm{pfg}}\text{-}B).$$

Its restriction to $\mathsf{Mod}_{\mathrm{fg}}\text{-}A$ gives the original functor ω. Together with Proposition 9.5.1, this also gives the description of $\mathrm{coE}(\omega)$ in (i).

Since Ω is faithful and exact, Proposition 7.7.7 shows that the adjoint pair

$$(\cdot) \underset{A}{\otimes} P : \mathsf{Mod}\text{-}A \rightleftarrows \mathsf{Mod}\text{-}B : (\cdot) \underset{B}{\otimes} P^\vee$$

is comonadic, with corresponding coalgebra $P^\vee \underset{A}{\otimes} P$. Thus Ω factors canonically as an equivalence followed by the forgetful functor:

$$\mathsf{Mod}\text{-}A \xrightarrow[\sim]{\overline{\Omega}} \mathsf{Comod}\text{-}\left(P^\vee \underset{A}{\otimes} P\right) \xrightarrow{U} \mathsf{Mod}\text{-}B.$$

By Proposition 7.7.14, this factorization restricts to the commutative diagram

$$\begin{tikzcd} \mathsf{Mod}_{\mathrm{fg}}\text{-}A \arrow[r, "\sim"'] \arrow[dr, bend right] & \mathsf{Comod}_{\mathrm{f}}\text{-}\left(P^\vee \underset{A}{\otimes} P\right) \arrow[r, "U"] \arrow[d, "\mathrm{(i)}"', "\wr"] & \mathsf{Mod}_{\mathrm{fg}}\text{-}B. \\ & \mathsf{Comod}_{\mathrm{f}}\text{-}\,\mathrm{coE}(\omega) & \end{tikzcd}$$

The composite of the first row is ω, which is known to take values in $\mathsf{Mod}_{\mathrm{pfg}}\text{-}B$; this proves (iii). The curved arrow in the diagram is therefore $\overline{\omega}$, and (ii) follows. □

The relation between the preceding result and abelian subcategories is also easy to describe.

Lemma 9.5.3 Suppose that $\mathcal{A}$ and ω satisfy the hypotheses of Lemma 9.5.2. Every abelian subcategory $\mathcal{A}'$, together with $\omega|_{\mathcal{A}'}$, also satisfies these hypotheses. Moreover, the canonical homomorphism $\mathrm{coE}(\omega|_{\mathcal{A}'}) \to \mathrm{coE}(\omega)$ from Remark 9.4.4 is monic.

Proof It is clear that $\mathcal{A}'$ is locally finite. Without loss of generality, we may suppose that $\mathrm{Ob}(\mathcal{A}') \subset \mathrm{Ob}(\mathcal{A})$ is closed under isomorphisms. We use the following general properties; their proofs are not difficult and can be found in the exercises of Appendix A.

- ◇ Choose a projective generator s of $\mathcal{A}$. It has a largest quotient object lying in $\mathcal{A}'$, denoted by s', and s' is a projective generator of $\mathcal{A}'$.
- ◇ Write $A := \mathrm{End}(s)$ and $A' := \mathrm{End}(s') \simeq \mathrm{Hom}(s, s')$. There is a two-sided ideal $\mathfrak{a}$ such that $A \to A'$ induces $A/\mathfrak{a} \xrightarrow{\sim} A'$. More explicitly, if $t := \ker[s \twoheadrightarrow s'] \subset s$, then $\mathfrak{a} = \mathrm{Hom}(s, t) \subset A$.
- ◇ After identifying $\mathcal{A}$ with $\mathsf{Mod}_{\mathrm{fg}}\text{-}A$, the category $\mathcal{A}'$ is identified with the full subcategory consisting of modules annihilated by $\mathfrak{a}$.

Choose a basis $\phi_1, \ldots, \phi_m$ of the $\Bbbk$-vector space $\mathrm{Hom}(s, t)$. The generator property gives an exact sequence

$$s^{\oplus m} \xrightarrow{(\phi_i)_i} s \to s' \to 0.$$

Write $P := \omega(s)$ and $P' := \omega(s')$. The algebra A acts on P from the left through ω, and the corresponding exact sequence

$$P^{\oplus m} \xrightarrow{(\omega(\phi_i))_i} P \to P' \to 0$$

shows that $P' \simeq P/\mathfrak{a}P$.

The proof of Lemma 9.5.2 showed that $(\cdot) \underset{A}{\otimes} P : \mathsf{Mod}\text{-}A \to \mathsf{Mod}\text{-}B$ is exact. Together with the description above, this gives

$$\begin{aligned}(P')^\vee \underset{A/\mathfrak{a}}{\otimes} P' &\simeq (A/\mathfrak{a} \underset{A}{\otimes} P)^\vee \underset{A/\mathfrak{a}}{\otimes} (A/\mathfrak{a} \underset{A}{\otimes} P) \\ &\simeq (A/\mathfrak{a} \underset{A}{\otimes} P)^\vee \underset{A}{\otimes} P \\ &\hookrightarrow P^\vee \underset{A}{\otimes} P.\end{aligned}$$

It is not difficult to check that this is precisely the canonical homomorphism $\mathrm{coE}(\omega|_{\mathcal{A}'}) \to \mathrm{coE}(\omega)$. $\square$

Example 9.5.4 Here are some initial applications of Lemma 9.5.2 for $B = \Bbbk$; recall that $\Bbbk$ is assumed to be a field. For a nonempty set I, define $\mathsf{Vect}_I(\Bbbk)$ to be the category of I-graded $\Bbbk$-vector spaces (Example 7.1.11, with trivial ϵ). The objects $(M^i)_{i\in I}$ satisfying $\sum_i \dim_{\Bbbk} M^i < \infty$ form the full subcategory $\mathcal{A} := \mathsf{Vect}_{I,\mathrm{f}}(\Bbbk)$. Write $\Bbbk[I]$ for the $\Bbbk$-vector space with basis I; the following structure makes it a cocommutative coalgebra:

$$\epsilon(i) = 1, \quad \Delta(i) = i \otimes i, \quad i \in I.$$

Define a functor $\omega_I : \mathcal{A} \to \mathsf{Vect}_{\mathrm{f}}(\Bbbk)$ by sending $(M^i)_{i\in I}$ to $\bigoplus_i M^i$.

1. The simplest case $|I| = 1$ amounts to taking $\mathcal{A} = \mathsf{Vect}_{\mathrm{f}}(\Bbbk)$ and $\omega_I = \mathrm{id}$; the projective generator s may be taken to be $\Bbbk$. A direct calculation from the definition, or Lemma 9.5.2(i), gives

$$\mathrm{coE}(\omega_I) \simeq \Bbbk, \quad \epsilon(1) = 1, \quad \Delta(1) = 1 \otimes 1.$$

Thus, when $|I| = 1$, we have $\mathrm{coE}(\omega_I) \simeq \Bbbk \simeq \Bbbk[I]$.

2. Next, if I is finite, then $\mathrm{coE}(\omega_I) \simeq \Bbbk[I]$. This can be deduced from the preceding case or understood by applying Lemma 9.5.2(i), taking as projective generator s the object defined by $M^i = \Bbbk$ for every i.

   Observe that if $I \subset J$, the embedding $\mathsf{Vect}_{I,\mathrm{f}}(\Bbbk) \subset \mathsf{Vect}_{J,\mathrm{f}}(\Bbbk)$ induces $\Bbbk[I] \to \Bbbk[J]$, namely the coalgebra embedding induced by $I \hookrightarrow J$.

3. For every nonempty set I, we likewise have $\mathrm{coE}(\omega_I) \simeq \Bbbk[I]$. More precisely, if $x \in M^i$ and $\phi \in (M^i)^\vee$ satisfy $\phi(x) = 1$, then $[\phi \otimes x] \in \mathrm{coE}(\omega_I)$ corresponds to $i \in \Bbbk[I]$.

   The set I is the filtered union of its finite subsets, and both the category $\mathsf{Vect}_{I,\mathrm{f}}(\Bbbk)$ and the coalgebra $\Bbbk[I]$ reduce to the finite case in the same way. Since the case of finite I is known, Proposition 9.4.5 immediately gives

$$\mathrm{coE}(\omega_I) \simeq \varinjlim_{J\subset I, |J|<\infty} \mathrm{coE}(\omega_J).$$

Taking colimits is a key technique that will reappear in the next proof.

The following theorem is the main intermediate result so far, and its proof depends on the technical results of §A.4. Recall that if $(L, \mu, \eta, \Delta, \epsilon)$ is a B-bialgebra, Proposition 7.5.6 equips $\mathsf{Comod}\text{-}L$ with a natural monoidal structure: the operation $\otimes$ comes from μ, and the unit is B with the coaction given by $\eta : B \to B \underset{B}{\otimes} L \simeq L$. Observe that $\mathsf{Comod}_{\mathrm{pf}}\text{-}L$ is a monoidal subcategory.

Theorem 9.5.5 Let $\Bbbk$ be a field, let B be a $\Bbbk$-algebra, let $\mathcal{A}$ be a locally finite abelian category, and let $\omega : \mathcal{A} \to \mathsf{Mod}\text{-}B$ be a faithful exact functor taking values in $\mathsf{Mod}_{\mathrm{pfg}}\text{-}B$.

(i) The functor $\overline{\omega}$ in the canonical factorization (9.5.1) is an equivalence of categories. Moreover, there is a family of abelian subcategories $\mathcal{A}'$ of $\mathcal{A}$ such that

- ◇ $\bigcup_{\mathcal{A}'} \mathcal{A}' = \mathcal{A}$ is a filtered union of subcategories;
- ◇ every $\mathcal{A}'$ has a projective generator,

and for this family there is an isomorphism of coalgebras

$$\mathrm{coE}(\omega) \simeq \varinjlim_{\mathcal{A}'} \mathrm{coE}(\omega'), \quad \omega' := \omega|_{\mathcal{A}'}.$$

The transition homomorphisms on the right come from Remark 9.4.4, and all of them are coalgebra embeddings. In this situation,

$$\mathsf{Comod}_{\mathrm{f}}\text{-}\,\mathrm{coE}(\omega) = \mathsf{Comod}_{\mathrm{pf}}\text{-}\,\mathrm{coE}(\omega).$$

(ii) If $\mathcal{A}$ has a monoidal-category structure, $B = \Bbbk$, and ω is a monoidal functor, then with respect to the $\Bbbk$-bialgebra structure on $\mathrm{coE}(\omega)$ from Proposition 9.4.10, both $\overline{\omega}$ and U are monoidal functors.

(iii) Under the same assumptions, if in addition $\mathcal{A}$ is a symmetric monoidal category and ω is compatible with the braidings, then $\mathrm{coE}(\omega)$ is commutative as an algebra, while $\overline{\omega}$ and U are also compatible with the braidings.

Proof Proposition A.4.4 writes $\mathcal{A}$ as the filtered union of abelian subcategories $\mathcal{A}'$ having projective generators; for example, we may take $\mathcal{A}' = \langle X \rangle$ for $X \in \mathrm{Ob}(\mathcal{A})$. Proposition 9.4.5 shows that $\mathrm{coE}(\omega) \simeq \varinjlim_{\mathcal{A}'} \mathrm{coE}(\omega')$, and Lemma 9.5.3 shows that all the transition homomorphisms are coalgebra embeddings. The canonical factorization (9.5.1) is obtained by taking the filtered union of

$$\mathcal{A}' \xrightarrow{\overline{\omega'}} \mathsf{Comod}_{\mathrm{pf}}\text{-}\,\mathrm{coE}(\omega') \xrightarrow{U} \mathsf{Mod}_{\mathrm{pfg}}\text{-}B.$$

Lemma 9.5.2 states that every $\overline{\omega'}$ is an equivalence, so $\overline{\omega}$ is also an equivalence. Moreover, $\mathsf{Comod}_{\mathrm{pf}}\text{-}\operatorname{coE}(\omega') = \mathsf{Comod}_{\mathrm{f}}\text{-}\operatorname{coE}(\omega')$; the same process simultaneously gives this equality for $\operatorname{coE}(\omega)$. This proves (i).

If $\mathcal{A}$ is monoidal and $B = \Bbbk$, the forgetful functor U is always monoidal for the structure on $\mathsf{Comod}\text{-}\operatorname{coE}(\omega)$ coming from the bialgebra $\operatorname{coE}(\omega)$. The commutative diagram (9.4.7), with all of $\omega_i, \omega'_i, \omega''_i$ replaced by ω, shows that $\overline{\omega}$ is also monoidal. This proves (ii), and a similar argument proves (iii). □

Example 9.5.6 Let Γ be a monoid and let $B = \Bbbk$ be a field, so that $\mathsf{Mod}_{\mathrm{pfg}}\text{-}B = \mathsf{Vect}_{\mathrm{f}}(\Bbbk)$. Example 9.5.4 shows that the coalgebra corresponding to

$$\omega_\Gamma : \mathsf{Vect}_{\Gamma,\mathrm{f}}(\Bbbk) \to \mathsf{Vect}_{\mathrm{f}}(\Bbbk), \quad (M^\gamma)_{\gamma\in\Gamma} \to \bigoplus_\gamma M^\gamma$$

is $\Bbbk[\Gamma]$. Moreover, $\Bbbk[\Gamma]$ carries a bialgebra structure (Example 7.5.5). On the other hand, $\mathsf{Vect}_{\Gamma,\mathrm{f}}(\Bbbk)$ is a monoidal category and ω_Γ is a monoidal functor (Example 7.1.11); this also equips $\Bbbk[\Gamma]$ with a bialgebra structure. The two structures coincide, as one checks by comparing the description of multiplication in (9.4.6) with Example 9.5.4; the details are left to the reader.

9.6 Reconstruction of Hopf Algebras

This section continues the discussion of §9.5. Theorem 9.5.5 constructs the coalgebra $\operatorname{coE}(\omega)$ from the data $(\mathcal{A}, \omega)$ and identifies $\mathcal{A}$ with $\mathsf{Comod}_{\mathrm{f}}\text{-}\operatorname{coE}(\omega)$. The most basic situation is when $B = \Bbbk$ is a field. For a $\Bbbk$-coalgebra L, take

$$\mathcal{A} := \mathsf{Comod}_{\mathrm{f}}\text{-}L,$$
$$\omega := U : \mathsf{Comod}_{\mathrm{f}}\text{-}L \xrightarrow{\text{forget}} \mathsf{Vect}_{\mathrm{f}}(\Bbbk).$$

Since $\Bbbk$ is a field, Proposition 7.7.13 with $B = \Bbbk$ shows that $\mathsf{Comod}\text{-}L$ is an abelian category, while $\mathsf{Comod}_{\mathrm{f}}\text{-}L$, formed by the finite-dimensional comodules, is clearly a locally finite abelian subcategory. It is now natural to ask: does the corresponding $\operatorname{coE}(\omega)$ reconstruct the original datum L, or does it instead produce a new coalgebra? As expected, the answer is “reconstruction”, but a little argument is required.

The first step is to make clear how to compare $\operatorname{coE}(\omega)$ and L. Since L coacts from the right on every $\omega(X)$, the universal property induces a coalgebra homomorphism

$$u : \operatorname{coE}(\omega) \to L. \tag{9.6.1}$$

The original question is equivalent to asking whether u is an isomorphism. We first prove a simple result.

Lemma 9.6.1 Let $\Bbbk$ be a field and let L be a $\Bbbk$-coalgebra, with the comultiplication homomorphism denoted by Δ and the counit homomorphism by ϵ. Every right L-comodule V is a union of finite-dimensional subcomodules.

Proof Suppose that the comodule structure is given by $\rho : V \to V \underset{\Bbbk}{\otimes} L$. For each $x \in V \underset{\Bbbk}{\otimes} L$, define the subspace

$$V(x) := \langle (\mathrm{id} \otimes \lambda)(x) : \lambda \in \mathrm{Hom}_{\Bbbk}(L, \Bbbk) \rangle \ \subset V;$$

if x is written concretely as $\sum_{i=1}^{n} v_i \otimes \ell_i$, with $\ell_1, \dots, \ell_n$ linearly independent, then clearly $V(x) = \langle v_1, \dots, v_n \rangle$, and hence $x \in V(x) \underset{\Bbbk}{\otimes} L$.

Take $v \in V$. Setting $x = \rho(v)$ and $\lambda = \epsilon$ gives $v \in V(\rho(v))$. We next show that $\rho(V(\rho(v))) \subset V(\rho(v)) \underset{\Bbbk}{\otimes} L$. Write again $\rho(v) = \sum_i v_i \otimes \ell_i$; the equation $(\rho \otimes \mathrm{id})\rho(v) = (\mathrm{id} \otimes \Delta)\rho(v)$ in the definition of a comodule can be written as the following equation in $V \underset{\Bbbk}{\otimes} L \underset{\Bbbk}{\otimes} L$:

$$\sum_i \rho(v_i) \otimes \ell_i = \sum_i v_i \otimes \Delta(\ell_i).$$

Extend $\ell_1, \dots, \ell_n$ to a basis and contract with the dual basis. This gives $\rho(v_i) \in V(\rho(v)) \underset{\Bbbk}{\otimes} L$ for $i = 1, \dots, n$.

Thus every finite-dimensional subspace $\langle v_1, \dots, v_m \rangle$ of V is contained in the subcomodule $\sum_{i=1}^{m} V(\rho(v_i))$, which is clearly finite-dimensional. □

Lemma 9.6.2 With the notation above, $u : \mathrm{coE}(\omega) \to L$ in (9.6.1) is an isomorphism of coalgebras.

Proof We show that all right L-comodules V, together with all homomorphisms between them, lift naturally to right $\mathrm{coE}(\omega)$-comodules. For $\dim_{\Bbbk} V < \infty$, this is by definition—apply Definition–Proposition 9.4.3 to $\mathcal{A} = \mathsf{Comod}_{\mathrm{f}}\text{-}L$ and $\omega = U$; the general case then follows from Lemma 9.6.1.

Now take $V = L$ with the right L-comodule structure determined by Δ. The lifted $\mathrm{coE}(\omega)$-comodule structure corresponds to $\tilde{\Delta} : L \to L \underset{\Bbbk}{\otimes} \mathrm{coE}(\omega)$. This is the same as saying that the triangular part of the following diagram commutes:

$$\begin{tikzcd}
L \arrow[r, "\tilde{\Delta}"] \arrow[dr, "\Delta"'] & L \underset{\Bbbk}{\otimes} \mathrm{coE}(\omega) \arrow[r, "\epsilon\otimes\mathrm{id}"] \arrow[d, "\mathrm{id}\otimes u"] & \mathrm{coE}(\omega) \arrow[d, "u"] \\
& L \underset{\Bbbk}{\otimes} L \arrow[r, "\epsilon\otimes\mathrm{id}"'] & L,
\end{tikzcd}$$

and its square part clearly commutes as well. Denote the composite of the first row by γ; then $u\gamma = \mathrm{id}_L$. It remains only to prove that γ is surjective.

Let $\rho : V \to V \underset{\Bbbk}{\otimes} L$ be a finite-dimensional right L-comodule. The commutative diagram in the definition

$$\begin{tikzcd} V \arrow[r, "\rho"] \arrow[d, "\rho"'] & V \underset{\Bbbk}{\otimes} L \arrow[d, "\mathrm{id}\otimes\Delta"] \\ V \underset{\Bbbk}{\otimes} L \arrow[r, "\rho\otimes\mathrm{id}"'] & V \underset{\Bbbk}{\otimes} L \underset{\Bbbk}{\otimes} L \end{tikzcd}$$

is also equivalent to saying that ρ is a homomorphism of L-comodules, provided that $V \underset{\Bbbk}{\otimes} L$ is equipped with the comodule structure coming from L and Δ. Now lift V naturally to a $\mathrm{coE}(\omega)$-comodule, corresponding to $\tilde{\rho} : V \to V \underset{\Bbbk}{\otimes} \mathrm{coE}(\omega)$; if the comodule $V \underset{\Bbbk}{\otimes} L$ is lifted similarly, its structure corresponds to $\mathrm{id} \otimes \tilde{\Delta} : V \underset{\Bbbk}{\otimes} L \to V \underset{\Bbbk}{\otimes} L \underset{\Bbbk}{\otimes} \mathrm{coE}(\omega)$. The comodule homomorphism ρ also lifts to the $\mathrm{coE}(\omega)$ level, which is the same as saying that the square in the following diagram commutes:

$$\begin{tikzcd} V \arrow[r, "\rho"] \arrow[d, "\tilde{\rho}"'] & V \underset{\Bbbk}{\otimes} L \arrow[d, "\mathrm{id}\otimes\tilde{\Delta}"] & \\ V \underset{\Bbbk}{\otimes} \mathrm{coE}(\omega) \arrow[r, "\rho\otimes\mathrm{id}"'] & V \underset{\Bbbk}{\otimes} L \underset{\Bbbk}{\otimes} \mathrm{coE}(\omega) \arrow[r, "\mathrm{id}\otimes\epsilon\otimes\mathrm{id}"'] & V \underset{\Bbbk}{\otimes} \mathrm{coE}(\omega) \end{tikzcd}$$

Since the composite of the second row is id, we obtain $\tilde{\rho} = (\mathrm{id}_V \otimes \gamma)\rho$. One checks that the map $\alpha_V : V^\vee \underset{\Bbbk}{\otimes} V \to \mathrm{coE}(\omega)$ induced by $\tilde{\rho}$ therefore factors as

$$V^\vee \underset{\Bbbk}{\otimes} V \xrightarrow{\text{induced by } \rho} L \xrightarrow{\gamma} \mathrm{coE}(\omega).$$

Recalling the construction of $\mathrm{coE}(\omega)$, as V varies, the images $\mathrm{im}(\alpha_V)$ generate all of $\mathrm{coE}(\omega)$. Thus γ is surjective. □

As one might expect, if L is a bialgebra, then (9.6.1) is also an isomorphism of bialgebras. The required argument is entirely routine and need not be spelled out.

Definition 9.6.3 For the chosen field $\Bbbk$, define the category $\mathsf{Fib}(\Bbbk)$ as follows.

- Its objects are data $(\mathcal{A}, \omega)$, where $\mathcal{A}$ is a locally finite abelian category (and hence essentially small), and $\omega : \mathcal{A} \to \mathsf{Vect}_{\mathrm{f}}(\Bbbk)$ is a faithful exact functor.
- A morphism from $(\mathcal{A}, \omega)$ to $(\mathcal{A}', \omega')$ is a datum (F, α), where $F : \mathcal{A} \to \mathcal{A}'$ is a functor and $\alpha : \omega \xrightarrow{\sim} \omega' F$ is an isomorphism of functors. Identities and composition are defined in the evident way.

Similarly, define the category $\mathsf{Fib}^{\otimes}(\Bbbk)$, with the following additional conditions:

- ◇ $\mathcal{A}$ in an object $(\mathcal{A}, \omega)$ must be equipped with a monoidal structure, and ω must be a monoidal functor;

- ◇ F in a morphism must also be a monoidal functor, and α must be an isomorphism of monoidal functors.

Observe that if $(\mathcal{A}, \omega)$ is an object of $\mathsf{Fib}^{\otimes}(\Bbbk)$, faithfulness and exactness imply that $\mathrm{End}_{\mathcal{A}}(\mathbf{1}) \to \Bbbk$ is a monomorphism of $\Bbbk$-algebras; thus in this case $\mathrm{End}_{\mathcal{A}}(\mathbf{1}) \xrightarrow{\sim} \Bbbk$.

On the other hand, denote the category of $\Bbbk$-coalgebras (respectively, $\Bbbk$-bialgebras) and their homomorphisms by $\Bbbk\text{-}\mathsf{coAlg}$ (respectively, $\Bbbk\text{-}\mathsf{biAlg}$).

- ◇ There are evident functors $\mathsf{Fib}(\Bbbk) \to \Bbbk\text{-}\mathsf{coAlg}$ and $\mathsf{Fib}^{\otimes}(\Bbbk) \to \Bbbk\text{-}\mathsf{biAlg}$. Both send an object $(\mathcal{A}, \omega)$ to $\mathrm{coE}(\omega)$, while a morphism $(\mathcal{A}, \omega) \to (\mathcal{A}', \omega')$ induces a coalgebra or bialgebra homomorphism $\mathrm{coE}(\omega) \simeq \mathrm{coE}(\omega' F) \to \mathrm{coE}(\omega')$.

- ◇ There are also functors in the opposite direction: for a coalgebra or bialgebra L, take the corresponding datum $(\mathsf{Comod}_{\mathrm{f}}\text{-}L, U)$; on morphisms, for a homomorphism $\phi : L \to L'$, a right L-comodule (M, ρ) becomes a right L'-comodule through $M \xrightarrow{(\mathrm{id}\otimes\phi)\rho} M \otimes L'$. This gives $(\mathsf{Comod}_{\mathrm{f}}\text{-}L, U) \to (\mathsf{Comod}_{\mathrm{f}}\text{-}L', U')$.

Theorem 9.6.4 Let $\Bbbk$ be a field. The constructions above give two pairs of mutually quasi-inverse functors

$$\mathsf{Fib}(\Bbbk) \rightleftarrows \Bbbk\text{-}\mathsf{coAlg},$$

$$\mathsf{Fib}^{\otimes}(\Bbbk) \rightleftarrows \Bbbk\text{-}\mathsf{biAlg}.$$

Proof Write the functors in the two directions as $Z : \star \leftrightarrows \star : Y$. For the first pair, the isomorphism u in Lemma 9.6.2 gives $ZY \xrightarrow{\sim} \mathrm{id}$, while the equivalence $\overline{\omega}$ in Theorem 9.5.5 (i) gives $\mathrm{id} \xrightarrow{\sim} YZ$.

For the second pair, since it is already known that when L is a bialgebra, u in (9.6.1) is also an isomorphism of bialgebras, we still have $ZY \xrightarrow{\sim} \mathrm{id}$ in the monoidal setting; meanwhile, $\mathrm{id} \xrightarrow{\sim} YZ$ is now the content of Theorem 9.5.5 (ii). □

We now add the duality introduced in §9.1 to the framework of Reconstruction Theorem 9.5.5. As usual, we consider only the simple situation in which $B = \Bbbk$ is a field.

Theorem 9.6.5 Under the assumptions of Theorem 9.6.4, suppose further that $(\mathcal{A}, \omega)$ is an object of $\mathsf{Fib}^{\otimes}(\Bbbk)$ and that $\mathcal{A}$ is both left and right rigid. Then $\mathrm{coE}(\omega)$ is a Hopf algebra.

Proof It suffices to show that the bialgebra $\mathrm{coE}(\omega)$ has an antipode. Use the left rigidity of $\mathcal{A}$ to define the functor $X \mapsto {}^*X$. For each $X \in \mathrm{Ob}(\mathcal{A})$, since monoidal functors preserve duals, we obtain $\omega({}^*X) \simeq \omega(X)^\vee$; since the monoidal structure on $\mathsf{Vect}_{\mathrm{f}}(\Bbbk)$ is symmetric, we also obtain $\omega(X) \simeq \omega({}^*X)^\vee$. Now consider

$$\phi \in \omega(X)^\vee \simeq \omega({}^*X), \quad x \in \omega(X) \simeq \omega({}^*X)^\vee,$$

where $X \in \mathrm{Ob}(\mathcal{A})$ is arbitrary. In the notation of Convention 9.4.8, both $[\phi \otimes x]$ and $[x \otimes \phi]$ are therefore elements of $\mathrm{coE}(\omega)$. We claim that

$$\text{there is a linear map } S : \mathrm{coE}(\omega) \to \mathrm{coE}(\omega), \text{ characterized by } S[\phi \otimes x] = [x \otimes \phi]. \tag{9.6.2}$$

For this, observe that for every morphism $f : X \to Y$ in $\mathcal{A}$, there are commutative diagrams

$$\begin{tikzcd} \omega(X) \arrow[r, "\omega(f)"] \arrow[d, "\wr"'] & \omega(Y) \arrow[d, "\wr"] \\ \omega({}^*X)^\vee \arrow[r, "\omega({}^*f)^\vee"'] & \omega({}^*Y)^\vee \end{tikzcd} \qquad \begin{tikzcd} \omega(Y)^\vee \arrow[r, "\omega(f)^\vee"] \arrow[d, "\wr"'] & \omega(X)^\vee \arrow[d, "\wr"] \\ \omega({}^*Y) \arrow[r, "\omega({}^*f)"'] & \omega({}^*X), \end{tikzcd}$$

where all the vertical arrows come from the isomorphisms in the preceding paragraph. For $\psi \in \omega(Y)^\vee$ and $x \in \omega(X)$, there are corresponding elements $\psi' \in \omega({}^*Y)$ and $x' \in \omega({}^*X)^\vee$ such that, respectively,

$$\omega(f)^\vee(\psi) \leftrightarrow \omega({}^*f)(\psi'), \quad \omega(f)(x) \leftrightarrow \omega({}^*f)^\vee(x').$$

Substituting this into the construction and description of $\mathrm{coE}(\omega)$ in the proof of Proposition 9.4.7, we see that δ_f there corresponds to $\delta_{{}^*f}$. Thus $[\phi \otimes x] \mapsto [x \otimes \phi]$ indeed defines a unique linear map S, proving (9.6.2).

Our goal is to prove that S is an antipode. The construction in the preceding paragraph can also be carried out with X^* in place of ${}^*X$, yielding a linear map $T : \mathrm{coE}(\omega) \to \mathrm{coE}(\omega)$; this map is likewise written in the form $[\phi \otimes x] \mapsto [x \otimes \phi]$. From ${}^*(X^*) = X = ({}^*X)^*$, it is easy to see that $ST = \mathrm{id} = TS$; in particular, S is an isomorphism. This is the first requirement for an antipode.

Next, denote the multiplication, comultiplication, unit, and counit homomorphisms of $\mathrm{coE}(\omega)$ by μ, Δ, η, and ϵ, respectively. Choose an object X and a basis $v_1, \dots, v_n$ of the $\Bbbk$-vector space $\omega(X)$. Denote the dual basis of $\omega(X)^\vee$ by $\check{v}_1, \dots, \check{v}_n$. Denote the morphisms for the left dual of X by

$$\mathrm{ev}_X : {}^*X \otimes X \to \mathbf{1}, \quad \mathrm{coev}_X : \mathbf{1} \to X \otimes {}^*X.$$

Their images under ω realize the duality between $\omega(X)$ and $\omega({}^*X)$. Recall that $(\cdot)^\vee$ is the dual in the symmetric monoidal category $\mathsf{Vect}_{\mathrm{f}}(\Bbbk)$, with no distinction between left and right; henceforth we may identify

$$\omega(\mathbf{1}) \simeq \omega(\mathbf{1})^\vee, \quad \omega(X \otimes {}^*X) \simeq \omega({}^*X \otimes X)^\vee,$$

and correspondingly identify $\omega(\mathrm{coev}_X)$ with $\omega(\mathrm{ev}_X)^\vee$.

For every $\phi \in \omega(X)^\vee$ and $x \in \omega(X)$, the description of Δ (respectively, μ) in (9.4.5) (respectively, (9.4.6)) gives

$$\begin{aligned}
\Delta([\phi \otimes x]) &= \sum_{i=1}^{n} [\phi \otimes v_i] \otimes [\check{v}_i \otimes x], \\
\mu(S \otimes \mathrm{id})\Delta([\phi \otimes x]) &= \sum_{i=1}^{n} \mu\left([v_i \otimes \phi] \otimes [\check{v}_i \otimes x]\right) \\
&= \sum_{i=1}^{n} \left[(v_i \otimes \check{v}_i) \otimes (\phi \otimes x)\right] \\
&= \left[\mathrm{coev}_{\omega(X)}(1) \otimes (\phi \otimes x)\right] \\
&= \left[\omega(\mathrm{coev}_X)(1) \otimes (\phi \otimes x)\right].
\end{aligned}$$

However, (9.4.2) implies that the following diagram commutes:

$$\begin{tikzcd}
\omega(\mathbf{1}) \otimes \omega({}^*X \otimes X) \arrow[r, "\omega(\mathrm{coev}_X)\otimes \mathrm{id}"] \arrow[d, "\mathrm{id}\otimes\omega(\mathrm{ev}_X)"'] & \omega(X \otimes {}^*X) \otimes \omega(X \otimes {}^*X), \arrow[d, "{[\cdot]}"] \\
\omega(\mathbf{1}) \otimes \omega(\mathbf{1}) \arrow[r, "{[\cdot]}"] \arrow[d, "\wr"'] & \mathrm{coE}(\omega) \\
\Bbbk \arrow[ur, "\eta"'] &
\end{tikzcd}$$

Therefore,

$$[\omega(\mathrm{coev}_X)(1) \otimes (\phi \otimes x)] = \eta\left(\omega(\mathrm{ev}_X)(\phi \otimes x)\right) \overset{(9.4.3)}{=\!=\!=} \eta\epsilon\left(\phi \otimes x\right),$$

which proves $\mu(S \otimes \mathrm{id})\Delta = \eta\epsilon$. Similarly, $\mu(\mathrm{id} \otimes S)\Delta = \eta\epsilon$. Thus S is an antipode. □

Corollary 9.6.6 Under the bijection given by Theorem 9.6.4,

$$\mathrm{Ob}(\mathsf{Fib}^{\otimes}(\Bbbk))/\simeq \;\overset{1:1}{\rightleftarrows}\; \mathrm{Ob}(\Bbbk\text{-}\mathsf{biAlg})/\simeq,$$

the category $\mathcal{A}$ in an object $(\mathcal{A}, \omega)$ on the left is both left and right rigid if and only if the object L on the right is a Hopf algebra.

Proof The "only if" direction follows from Theorem 9.6.5, whereas the "if" direction was explained in Proposition 9.3.5. □

In the proof of Theorem 9.6.5, the left rigidity of $\mathcal{A}$ plays the principal role; right rigidity is used only to ensure that S is an isomorphism. If the definition of a Hopf algebra allows a noninvertible S, then the category of right comodules $\mathsf{Comod}_{\mathrm{f}}\text{-}L$ is only left rigid; see the formulas in Proposition 9.3.5.

9.7 Tannakian Categories

The theory of Tannakian categories is a thorough reformulation, due to Saavedra Rivano and Deligne [39, 10], of the work of Tadao Tannaka [44] and M. Krein [26] on representations of compact groups; its guiding ideas originate with Grothendieck. Related ideas were later developed far more broadly in Lurie and Bhatt's work on derived algebraic geometry. We shall broadly follow Deligne's treatment.

Unless stated otherwise, throughout this section we retain the assumptions of §9.5, with a fixed field $\Bbbk$.

Definition 9.7.1 Let $\mathcal{T}$ be an essentially small abelian category (Definition A.4.1) equipped with a symmetric monoidal structure for which $\otimes$ is $\Bbbk$-linear in each variable. The category $\mathcal{T}$ is called a **Tannakian category over the field** $\Bbbk$ if it has the following properties.

- ◇ If $\mathbf{1}$ denotes the unit of $\mathcal{T}$, then $\mathrm{End}_{\mathcal{T}}(\mathbf{1}) = \Bbbk$.
- ◇ As a symmetric monoidal category, $\mathcal{T}$ is rigid (Definition 9.2.3).
- ◇ There exist a commutative $\Bbbk$-algebra B, taken by convention to be nonzero, and a right-exact monoidal functor compatible with the braiding

$$\omega : \mathcal{T} \to \mathsf{Mod}\text{-}B.$$

Such a functor is called a **fiber functor** for $\mathcal{T}$ over B.[5)]

A fiber functor over $\Bbbk$ is called **neutral**. If a Tannakian category $\mathcal{T}$ has a neutral fiber functor, then $\mathcal{T}$ is called a **neutral Tannakian category**.

By the definition of the dual of a morphism in Definition 9.1.7, the duality functor $(\cdot)^*$ of a Tannakian category is necessarily $\Bbbk$-linear.

The preceding definition is quite flexible. If ω is a fiber functor over B and $B \to B'$ is a homomorphism of commutative $\Bbbk$-algebras, then $\omega \underset{B}{\otimes} B'$ is a fiber functor over B'. These axioms have several nontrivial consequences, in which duality plays a key role.

5) The term comes from the Grothendieck school and is rooted in an analogy with the theory of covering spaces.

Lemma 9.7.2 Let $\mathcal{T}$ be a Tannakian category.

(i) For every commutative $\Bbbk$-algebra B, every fiber functor ω over B takes values in $\mathsf{Mod}_{\mathrm{pfg}}\text{-}B$ and is automatically faithful and exact.

(ii) There exist a field extension $\Bbbk'|\Bbbk$ and a fiber functor over $\Bbbk'$.

(iii) If ω is a fiber functor over B, the canonical map

$$\mathrm{Hom}_{\mathcal{T}}(X,Y) \underset{\Bbbk}{\otimes} B \to \mathrm{Hom}_{\mathsf{Mod}\text{-}B}(\omega(X), \omega(Y))$$

is injective.

(iv) The category $\mathcal{T}$ is locally finite (Definition A.4.2).

Proof For (i), since ω is monoidal and $\mathcal{T}$ is rigid, Proposition 9.2.2 shows that ω takes values in $\mathsf{Mod}_{\mathrm{pfg}}\text{-}B$. We next prove that ω is left exact. Consider any exact sequence in $\mathcal{T}$

$$0 \to X \to Y \to Z.$$

Its dual is also exact, so applying ω gives an exact sequence in $\mathsf{Mod}\text{-}B$

$$\omega(Z)^\vee \to \omega(Y)^\vee \to \omega(X)^\vee \to 0.$$

Apply $(\cdot)^\vee = \mathrm{Hom}_{\mathsf{Mod}\text{-}B}(\cdot, B)$ once more. Together with Proposition 2.8.11 and $\omega(\cdot)^{\vee\vee} \simeq \omega(\cdot)$, this yields the exact sequence

$$0 \to \omega(X) \to \omega(Y) \to \omega(Z).$$

For faithfulness, the definition of duality shows that $X \neq 0$ if and only if $\mathrm{ev}_X : X \otimes X^* \to \mathbf{1}$ is nonzero. Meanwhile, Proposition 9.2.5 and $\mathrm{End}_{\mathcal{T}}(\mathbf{1}) = \Bbbk$ show that this is equivalent to ev_X being an epimorphism. Since ω is both monoidal and exact, $X \neq 0$ therefore gives an epimorphism $\omega(X) \underset{B}{\otimes} \omega(X)^\vee \twoheadrightarrow B$, and hence $\omega(X) \neq 0$.

For (ii), choose a fiber functor ω over B and a maximal ideal $\mathfrak{m}$ of B, and put $\Bbbk' := B/\mathfrak{m}$. Then $\omega \underset{B}{\otimes} \Bbbk'$ is a fiber functor over $\Bbbk'$.

For (iii), take linearly independent elements $f_1, \ldots, f_n \in \mathrm{Hom}_{\mathcal{T}}(X,Y)$. In the notation of Corollary 9.1.12, they define a morphism $\Phi : \mathbf{1}^{\oplus n} \to \mathcal{H}\mathrm{om}(X,Y)$. We claim that Φ is a monomorphism. Indeed, $\mathbf{1}$ is simple (Proposition 9.2.5), so $\ker(\Phi)$ has finite length. If $\ker(\Phi) \neq 0$, all its composition factors are $\mathbf{1}$; see §2.7. In particular there is a monomorphism $\mathbf{1} \hookrightarrow \ker(\Phi) \hookrightarrow \mathbf{1}^{\oplus n}$. Since $\mathrm{End}_{\mathcal{T}}(\mathbf{1}) = \Bbbk$, composing with Φ gives a morphism $\mathbf{1} \to \mathcal{H}\mathrm{om}(X,Y)$ that is zero on the one hand, but on the other hand corresponds to $\sum_{i=1}^n a_i f_i \in \mathrm{Hom}_{\mathcal{T}}(X,Y)$ for some $a_1, \ldots, a_n \in \Bbbk$ not all zero. This contradicts linear independence.

Applying the exact monoidal functor ω to Φ gives a monomorphism of B-modules $B^{\oplus n} \hookrightarrow \mathrm{Hom}_{\mathsf{Mod}\text{-}B}(\omega(X), \omega(Y))$. This is precisely the restriction of the canonical map under consideration to $\bigoplus_{i=1}^{n} \Bbbk f_i \underset{\Bbbk}{\otimes} B$.

For (iv), first use (ii) to obtain a fiber functor ω over a field extension $\Bbbk'$. The faithful exactness in (i) already implies that every $X \in \mathrm{Ob}(\mathcal{T})$ has finite length. Next, (iii) gives a $\Bbbk'$-linear embedding

$$\mathrm{Hom}_{\mathcal{T}}(X, Y) \underset{\Bbbk}{\otimes} \Bbbk' \hookrightarrow \mathrm{Hom}_{\mathsf{Vect}_{\mathrm{f}}(\Bbbk')}(\omega(X), \omega(Y)).$$

The right-hand side is finite-dimensional, and hence $\mathrm{Hom}_{\mathcal{T}}(X, Y)$ is a finite-dimensional $\Bbbk$-vector space. □

By Lemma 9.7.2, the theoretical framework of §9.5 applies to every Tannakian category $\mathcal{T}$ and every fiber functor $\omega : \mathcal{T} \to \mathsf{Mod}_{\mathrm{pfg}}\text{-}B$ over a commutative $\Bbbk$-algebra B. From the viewpoint of algebraic geometry, $\mathsf{Mod}_{\mathrm{pfg}}\text{-}B$ corresponds to the category of vector bundles on the affine scheme Spec B. It is therefore necessary to consider general B, although doing so also requires additional theoretical tools. The reconstruction theorem below treats only the case $B = \Bbbk$.

Theorem 9.7.3 Let $\mathcal{T}$ be a neutral Tannakian category and let $\omega : \mathcal{T} \to \mathsf{Vect}_{\mathrm{f}}(\Bbbk)$ be a neutral fiber functor. Then ω factors canonically as

$$\mathcal{T} \xrightarrow{\overline{\omega}} \mathsf{Comod}_{\mathrm{f}}\text{-}\,\mathrm{coE}(\omega) \xrightarrow{U} \mathsf{Vect}_{\mathrm{f}}(\Bbbk),$$

where $\mathrm{coE}(\omega)$ carries a canonical commutative Hopf algebra structure, and $\overline{\omega}$ is an equivalence of symmetric monoidal categories.

Proof Since $\mathcal{T}$ is now known to be locally finite, the assertion follows immediately from Theorems 9.5.5 and 9.6.5. □

Example 9.7.4 Let H be a commutative Hopf $\Bbbk$-algebra. Proposition 9.3.5 shows that the category of finite-dimensional comodules $\mathsf{Comod}_{\mathrm{f}}\text{-}H$ is naturally a neutral Tannakian category, with the forgetful functor as its standard neutral fiber functor:

$$\omega : \mathsf{Comod}_{\mathrm{f}}\text{-}H \to \mathsf{Vect}_{\mathrm{f}}(\Bbbk), \quad M \mapsto (M \text{ as a vector space}).$$

Theorem 9.7.3 says that, up to equivalence, this family of examples exhausts all neutral Tannakian categories.

Example 9.7.5 Let $(\Gamma, +)$ be an abelian group and put $\mathcal{T} = \mathsf{Vect}_{\Gamma,\mathrm{f}}(\Bbbk)$, in the notation of Examples 9.5.4 and 9.5.6. Equip $\mathcal{T}$ with the standard braiding

$$\begin{aligned} c(M, N) : M \otimes N &\xrightarrow{\sim} N \otimes M \\ x \otimes y &\mapsto y \otimes x, \quad x \in M^{\gamma},\ y \in N^{\eta}. \end{aligned}$$

This makes $\mathcal{T}$ a neutral Tannakian category, with $\omega_\Gamma : (M^\gamma)_{\gamma\in\Gamma} \mapsto \bigoplus_\gamma M^\gamma$ as neutral fiber functor. The key point is rigidity, which is precisely the multivariable version of Example 9.3.1. The corresponding endomorphism coalgebra is already known: it is the Hopf algebra $\Bbbk[\Gamma]$.

Although $\Gamma = \mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$ are both allowed, the braiding here does not involve the Koszul sign rule (7.1.3); if it did, ω_Γ would not preserve the braiding. The theory of Tannakian categories therefore cannot be applied directly to categories such as the category of super vector spaces $\mathsf{Vect}^-_{\mathbb{Z}/2\mathbb{Z},\mathrm{f}}(\Bbbk)$, because it has no fiber functor to $\mathsf{Vect}_{\mathrm{f}}(\Bbbk)$.

Remark 9.7.6 Some assertions of linear algebra—that is, assertions about fiber functors—can be translated into the language of Hopf algebras by Theorem 9.7.3. As a simple and useful example, for a neutral fiber functor ω and an abelian group $(\Gamma, +)$, consider the following three kinds of data.

(i) Equip each $\omega(X)$ with a Γ-grading; that is, specify a family of isomorphisms

$$\alpha_X : \omega(X) \xrightarrow{\sim} \omega_\Gamma(FX), \quad FX \in \mathrm{Ob}\,(\mathsf{Vect}_{\Gamma,\mathrm{f}}(\Bbbk))\,,$$

with FX and α_X functorial in X, and such that $\omega(\mathbf{1}) \simeq \Bbbk$ and $\omega(X \otimes X') \simeq \omega(X) \otimes \omega(X')$ both lift to the level of Γ-graded vector spaces.

(ii) Specify a monoidal functor $F : \mathcal{T} \to \mathsf{Vect}_{\Gamma,\mathrm{f}}(\Bbbk)$ together with an isomorphism of monoidal functors $\alpha : \omega \xrightarrow{\sim} \omega_\Gamma F$.

(iii) Specify a homomorphism of $\Bbbk$-bialgebras $\mathrm{coE}(\omega) \to \Bbbk[\Gamma]$.

Data (i) and (ii) differ only in wording. In the language of Definition 9.6.3, (ii) is the same as specifying the morphism in $\mathsf{Fib}^{\otimes}(\Bbbk)$ $(F, \alpha) : (\mathcal{T}, \omega) \to (\mathsf{Vect}_{\Gamma,\mathrm{f}}(\Bbbk), \omega_\Gamma)$. Hence the Reconstruction Theorem 9.6.4 ensures that (ii) and (iii) are equivalent.

For example, for the most common grading $\Gamma = \mathbb{Z}$, data (iii) amount to giving a homomorphism of Hopf algebras $\mathrm{coE}(\omega) \to \Bbbk[T, T^{-1}]$, where T is an indeterminate.

Finally, we briefly discuss the relationship between fiber functors. Unlike the preceding discussion, this part does not involve the reconstruction theorem. Let $\mathcal{T}$ be a Tannakian category, and let ω_1, ω_2 be fiber functors over a commutative $\Bbbk$-algebra B. For every commutative B-algebra B', define the set

$$\mathcal{H}\mathrm{om}^{\otimes}_B(\omega_2, \omega_1)(B') := \mathrm{Hom}_{\text{monoidal functors}}\left(\omega_2 \underset{B}{\otimes} B', \omega_1 \underset{B}{\otimes} B'\right).$$

As B' varies, this defines a functor $B\text{-}\mathsf{CAlg} \to \mathsf{Set}$. Recall that Proposition 9.4.10 (iii) equips $\mathrm{coH}_B(\omega_1, \omega_2)$ with the structure of a commutative B-algebra.

Proposition 9.7.7 For a Tannakian category $\mathcal{T}$ and fiber functors $\omega_1, \omega_2 : \mathcal{T} \to \mathsf{Mod}\text{-}B$, there is an isomorphism of functors

$$\mathcal{H}\mathrm{om}_B^{\otimes}(\omega_2, \omega_1) \simeq \mathrm{Hom}_{B\text{-}\mathsf{CAlg}}(\mathrm{coH}_B(\omega_1, \omega_2), \,\cdot\,).$$

Composition of homomorphisms on the right is induced by (9.4.1).

Proof To simplify notation, write $\mathrm{coH}_B(\omega_1, \omega_2)$ as coH from now on. Specifying a morphism, without regard to the monoidal structure,

$$\tilde{\varphi} : \omega_2 \underset{B}{\otimes} B' \to \omega_1 \underset{B}{\otimes} B' \quad \text{(both valued in } \mathsf{Mod}\text{-}B')$$

is equivalent to specifying a morphism

$$\varphi : \omega_2 \to \omega_1 \underset{B}{\otimes} B' \quad \text{(both valued in } \mathsf{Mod}\text{-}B),$$

which, by the universal property of coH, is in turn equivalent to specifying a homomorphism of B-modules

$$\psi : \mathrm{coH} \to B'.$$

It remains to prove that $\tilde{\varphi}$ preserves the monoidal structure if and only if ψ is a homomorphism of B-algebras. The argument is formal, although spelling it out is somewhat lengthy.

Write the multiplication morphisms as $m : B' \underset{B}{\otimes} B' \to B'$ and $\mu : \mathrm{coH} \underset{B}{\otimes} \mathrm{coH} \to \mathrm{coH}$. It is immediate that $\tilde{\varphi}$ preserves the unit, respectively $\otimes$, if and only if the left-hand, respectively right-hand, diagram below always commutes:

$$\begin{tikzcd}
\omega_2(\mathbf{1}) \arrow[r, "\varphi_{\mathbf{1}}"] \arrow[dd, "\wr"] & \omega_1(\mathbf{1}) \underset{B}{\otimes} B' \arrow[d, "\wr"] \\
& B \underset{B}{\otimes} B' \arrow[d, "\wr"] \\
B \arrow[r] & B'
\end{tikzcd}
\qquad
\begin{tikzcd}
\omega_2(X \otimes Y) \arrow[r, "\varphi_{X \otimes Y}"] \arrow[dd, "\wr"] & \omega_1(X \otimes Y) \underset{B}{\otimes} B' \arrow[d, "\wr"] \\
& \omega_1(X) \underset{B}{\otimes} \omega_1(Y) \underset{B}{\otimes} B' \\
\omega_2(X) \underset{B}{\otimes} \omega_2(Y) \arrow[r, "\varphi_X \otimes \varphi_Y"'] & \omega_1(X) \underset{B}{\otimes} B' \underset{B}{\otimes} \omega_1(Y) \underset{B}{\otimes} B'. \arrow[u, "\text{induced by } m"']
\end{tikzcd}$$

First consider the left-hand diagram. Its first row factors as $\omega_2(\mathbf{1}) \to \omega_1(\mathbf{1}) \underset{B}{\otimes} \mathrm{coH} \xrightarrow{\mathrm{id} \otimes \psi} \omega_1(\mathbf{1}) \underset{B}{\otimes} B'$. After identifying every $\omega_i(\mathbf{1})$ with B, commutativity of the left-hand diagram is equivalent to commutativity of $\begin{tikzcd} B \arrow[r] \arrow[d] & B' \\ \mathrm{coH} \arrow[ur, "\psi"'] & \end{tikzcd}$, which in turn is equivalent to ψ preserving the unit.

We now show that commutativity of the right-hand diagram is equivalent to ψ preserving multiplication. Consider

$$\begin{tikzcd}
\omega_2(X \otimes Y) \arrow[r] \arrow[dd, "\wr"'] \arrow[rr, bend left, "\varphi_{X\otimes Y}"] & \omega_1(X \otimes Y) \underset{B}{\otimes} \mathrm{coH} \arrow[r, "\mathrm{id}\otimes\psi"] & \omega_1(X \otimes Y) \underset{B}{\otimes} B' \arrow[d, "\wr"] \\
 & \omega_1(X \otimes Y) \underset{B}{\otimes} \mathrm{coH} \underset{B}{\otimes} \mathrm{coH} \arrow[u, "\mathrm{id}\otimes\mu"] & \omega_1(X) \underset{B}{\otimes} \omega_1(Y) \underset{B}{\otimes} B' \\
\omega_2(X) \underset{B}{\otimes} \omega_2(Y) \arrow[r] \arrow[rr, bend right, "\varphi_X\otimes\varphi_Y"'] & \omega_1(X) \underset{B}{\otimes} \mathrm{coH} \underset{B}{\otimes} \omega_1(Y) \underset{B}{\otimes} \mathrm{coH} \arrow[u, "\wr"] \arrow[r, "(\mathrm{id}\otimes\psi)^{\otimes 2}"] & \omega_1(X) \underset{B}{\otimes} B' \underset{B}{\otimes} \omega_1(Y) \underset{B}{\otimes} B', \arrow[u, "\text{induced by } m"']
\end{tikzcd} \tag{9.7.1}$$

The upper and lower curved regions always commute, while commutativity of the square on the left reduces to the characterization of μ in (9.4.7).

Define $\eta := \omega_1^\vee \underset{B}{\otimes} \omega_2$, which is a monoidal functor $\mathcal{T} \to \mathsf{Mod}\text{-}B$. The square portion of (9.7.1) can be rearranged as

$$\begin{tikzcd}
\eta(X \otimes Y) \arrow[r] \arrow[d, "\wr"'] & \mathrm{coH} \arrow[r, "\psi"] & B' \\
\eta(X) \underset{B}{\otimes} \eta(Y) \arrow[r] & \mathrm{coH} \underset{B}{\otimes} \mathrm{coH} \arrow[u, "\mu"] \arrow[r, "\psi\otimes\psi"'] & B' \underset{B}{\otimes} B', \arrow[u, "m"']
\end{tikzcd} \tag{9.7.2}$$

The left-hand square still commutes, while ψ preserves multiplication if and only if the right-hand square commutes.

If $\tilde{\varphi}$ preserves $\otimes$, then the large outer frame in (9.7.1) commutes, and hence so does the outer frame of its square portion. Thus the outer frame of (9.7.2) commutes. This shows that the right-hand square commutes after precomposition with $\eta(X) \underset{B}{\otimes} \eta(Y) \to \mathrm{coH} \underset{B}{\otimes} \mathrm{coH}$. Since X and Y are arbitrary, the universal property of coH, or its concrete construction in Proposition 9.4.7, suffices to show that the right-hand square in (9.7.2) commutes.

Conversely, suppose the right-hand square in (9.7.2) commutes. Then the entire diagram (9.7.2) commutes, so the large outer frame in (9.7.1) commutes; hence $\tilde{\varphi}$ preserves $\otimes$. This completes the verification. $\square$

The rigidity of $\mathcal{T}$ implies that every element of $\mathcal{H}\mathrm{om}^{\otimes}_B(\omega_2, \omega_1)(B')$ is an isomorphism; this is an application of Proposition 9.1.10. The next result is therefore immediate.

Corollary 9.7.8 For a Tannakian category $\mathcal{T}$, a commutative $\Bbbk$-algebra B, and a fiber functor $\omega : \mathcal{T} \to \mathsf{Mod}\text{-}B$, write $\mathrm{Aut}^{\otimes}(\omega)$ for the group of monoidal automorphisms of ω. Then, for every commutative B-algebra B', there is a canonical isomorphism

$$\mathrm{Aut}^{\otimes}\left(\omega \underset{B}{\otimes} B'\right) \simeq \mathrm{Hom}_{B\text{-}\mathsf{CAlg}}\left(\mathrm{coE}_B(\omega), B'\right),$$

where multiplication in the group on the right is reflected by the comultiplication of $\mathrm{coE}_B(\omega)$.

One can further show that the identity element and inversion in the automorphism group arise respectively from the counit morphism and the antipode of $\mathrm{coE}_B(\omega)$. The verification is simple and formal, so it is left as an exercise for this chapter. Considering the Yoneda embedding of B-CAlg gives the following conclusion:

$$\begin{gathered}\text{knowing the functor } \mathrm{Aut}^{\otimes}\left(\omega \underset{B}{\otimes} (\cdot)\right) : B\text{-}\mathsf{CAlg} \to \mathsf{Grp} \\ \Updownarrow \\ \text{knowing the Hopf } B\text{-algebra } \mathrm{coE}_B(\omega).\end{gathered}$$

Thus, once $\mathrm{Aut}^{\otimes}$ is viewed functorially, the entire long story of the endomorphism coalgebra ultimately returns to the group of $\otimes$-automorphisms of the fiber functor.

In the situation of Proposition 9.7.7, Deligne proved in [10, 1.11–1.13] that $\mathrm{coH}_B(\omega_1, \omega_2)$ is faithfully flat as a B-module; in particular, $\mathrm{coH}_B(\omega_1, \omega_2) \neq 0$. We shall not give the detailed proof, but its importance in algebraic geometry is worth mentioning. It guarantees the existence of a faithfully flat B-algebra B' and an isomorphism $\omega_1 \underset{B}{\otimes} B' \simeq \omega_2 \underset{B}{\otimes} B'$; for example, one may take $B' = \mathrm{coH}_B(\omega_1, \omega_2)$. Thus fiber functors are unique up to faithfully flat change of rings, while properties over the original ring B can in principle be reconstructed from descent data as in §7.9. This is the principal benefit of faithful flatness.

9.8 The Tannaka–Krein Theorem for Finite Groups

Classical Tannaka–Krein theory [44, 26] is chiefly concerned with the following question: given a compact topological group G, how can G be reconstructed from its category of representations? The answer is closely tied to the tensor-product structure and duality; see [54, §4.3]. This section gives a concise interpretation of the Tannaka–Krein reconstruction theorem based on the preceding results, but treats only finite groups rather than compact groups.

Let G be a group and $\Bbbk$ a field. Henceforth write $\otimes := \otimes_{\Bbbk}$, and denote the identity element of G by 1_G.

The principal objects of interest are the G-modules introduced in Definition 6.1.1, which in many contexts are also called representations of G. In brief, they are $\Bbbk$-vector spaces V equipped with a linear left G-action $G \times V \to V$, written as left multiplication $(g, v) \mapsto gv$. All G-modules form a $\Bbbk$-linear Abelian category $G\text{-}\mathsf{Mod}$, whose Hom is denoted by Hom_G. The forgetful functor

$$\omega : G\text{-}\mathsf{Mod} \to \mathsf{Vect}(\Bbbk)$$

of course sends a G-module V to its underlying $\Bbbk$-vector space V.

Write the group algebra of G as $\Bbbk[G]$; by Example 7.5.13, it is a Hopf algebra. As stated in Proposition 6.1.4, $G\text{-}\mathsf{Mod}$ is isomorphic to $\Bbbk[G]\text{-}\mathsf{Mod}$, and we shall switch between these two viewpoints without further comment.

Convention 9.8.1 The dimension of a G-module means its dimension as a $\Bbbk$-vector space. Write $G\text{-}\mathsf{Mod}_{\mathrm{f}}$ for the full subcategory of $G\text{-}\mathsf{Mod}$ consisting of finite-dimensional G-modules.

The field $\Bbbk$ endowed with the trivial G-action is called the trivial G-module (Example 6.1.2). For G-modules V and W, the spaces $V \otimes W$ and $\mathrm{Hom}(V, W) := \mathrm{Hom}_{\Bbbk}(V, W)$ carry canonical G-module structures. The special case $\mathrm{Hom}(V, \Bbbk)$ of the latter is called the contragredient G-module of V, and is also denoted by V^* to distinguish it from the vector-space dual. These operations were described in detail after Definition 6.1.5.

Proposition 9.8.2 The tensor-product operation above, $(V, W) \mapsto V \otimes W$, gives $G\text{-}\mathsf{Mod}$ a symmetric monoidal structure whose unit is the trivial G-module $\Bbbk$, and makes the forgetful functor $G\text{-}\mathsf{Mod} \to \mathsf{Vect}(\Bbbk)$ monoidal. The full subcategory $G\text{-}\mathsf{Mod}_{\mathrm{f}}$ is then a neutral Tannakian category, and the forgetful functor $\omega : G\text{-}\mathsf{Mod}_{\mathrm{f}} \to \mathsf{Vect}_{\mathrm{f}}(\Bbbk)$ is its fiber functor.

Proof The symmetric monoidal structure on G-Mod can be checked directly from the definition of the tensor product. Readers who prefer a big-picture account may instead reason as follows.

- ◇ Since $\Bbbk[G]$ is a Hopf algebra, Proposition 7.5.6 equips $\Bbbk[G]$-Mod with a monoidal structure for which the forgetful functor is monoidal. Because the comultiplication on $\Bbbk[G]$ is $g \mapsto g \otimes g$, carefully expanding the definitions shows that this is precisely the tensor-product operation defined above.
- ◇ Since $\Bbbk[G]$ is cocommutative, Proposition 7.5.7 (i) further implies that $\Bbbk[G]$-Mod is a symmetric monoidal category.

Clearly G-$\mathsf{Mod}_{\mathrm{f}}$ is a monoidal subcategory of G-Mod and is also an Abelian category. For every $n \in \mathbb{Z}_{\geq 0}$, all G-module structures that can be placed on $\Bbbk^n$ form a small set; hence G-$\mathsf{Mod}_{\mathrm{f}}$ is essentially small. Plainly,

$$\mathrm{End}_G\left(\Bbbk : \text{ the trivial } G\text{-module}\right) = \Bbbk.$$

Checking the definition of duality in §9.1.1 step by step shows that the contragredient operation $V \mapsto V^*$ makes G-$\mathsf{Mod}_{\mathrm{f}}$ rigid. Its duality data come from the familiar morphisms

$$\begin{aligned} V \otimes V^* &\xrightarrow{\mathrm{ev}} \Bbbk & \qquad \Bbbk &\xrightarrow{\mathrm{coev}} V^* \otimes V \\ v \otimes \lambda &\longmapsto \lambda(v), & 1 &\longmapsto \textstyle\sum_i \check{v}_i \otimes v_i, \end{aligned}$$

where $(v_i)_i$ is any basis of V as a vector space and $(\check{v}_i)_i$ is its dual basis; both are morphisms in G-Mod. From the viewpoint of the Hopf algebra $\Bbbk[G]$, rigidity also follows from Proposition 9.3.5 and the fact that the antipode of $\Bbbk[G]$ is $g \mapsto g^{-1}$.

The constructions above also show that the forgetful functor $\omega : G\text{-}\mathsf{Mod}_{\mathrm{f}} \to \mathsf{Vect}_{\mathrm{f}}(\Bbbk)$ is monoidal, compatible with the symmetric monoidal structures on both sides, and plainly exact. Thus ω is a neutral fiber functor on the neutral Tannakian category G-$\mathsf{Mod}_{\mathrm{f}}$. □

From now on suppose that G is finite. The central question of this section is:

How can the group G be reconstructed from the data $(G\text{-}\mathsf{Mod}_{\mathrm{f}}, \omega)$?

The results obtained in §9.7 already bring us close to an answer, but the machinery still requires some adjustment. First we must interpret G-$\mathsf{Mod}_{\mathrm{f}}$ as a comodule category.

Definition 9.8.3 Continue to assume that G is finite. Define the $\Bbbk$-vector space

$$C(G) := \{\text{maps } f : G \to \Bbbk\},$$

with vector-space structure given by pointwise operations. There is moreover a canonical isomorphism

$$\begin{aligned} C(G) \otimes C(G) &\xrightarrow{\ \sim\ } C(G \times G) \\ f_1 \otimes f_2 &\longmapsto [(g_1, g_2) \mapsto f_1(g_1) f_2(g_2)]. \end{aligned}$$

Identifying $C(G) \otimes C(G)$ with $C(G \times G)$, the space $C(G)$ has the following structure:

structure map	formula
Multiplication	Pointwise multiplication
Comultiplication Δ	$f \mapsto [(g_1, g_2) \mapsto f(g_1 g_2)]$
Unit	Constant function 1
Counit ϵ	$f \mapsto f(1_G)$
Antipode S	$(Sf)(g) = f(g^{-1})$

The reader is invited to give a brief verification that this is a commutative Hopf algebra. Consequently, Proposition 7.5.7 (ii) ensures that Comod-$C(G)$ is a symmetric monoidal category, while Proposition 9.3.5 makes $\mathsf{Comod_f}$-$C(G)$ a rigid symmetric monoidal category; compare the big-picture part of the preceding proof.

For every $g \in G$, let $\mathbf{1}_g \in C(G)$ be the element equal to 1 at g and 0 elsewhere. These elements form a basis of $C(G)$, and multiplication and comultiplication are respectively determined by $\mathbf{1}_{g_1}\mathbf{1}_{g_2} = \delta_{g_1,g_2}\mathbf{1}_{g_1}$ (with δ the Kronecker symbol) and $\Delta(\mathbf{1}_g) = \sum_{xy=g} \mathbf{1}_x \otimes \mathbf{1}_y$.

Proposition 9.8.4 There is an isomorphism of categories G-$\mathsf{Mod} \simeq \mathsf{Comod}$-$C(G)$. The right $C(G)$-comodule corresponding to a G-module V is V equipped with the linear map

$$\begin{aligned} \rho : V &\longrightarrow V \otimes C(G) \xrightarrow{\ \sim\ } \{\text{maps } G \to V\} \\ v &\longmapsto \sum_{g \in G} gv \otimes \mathbf{1}_g \longmapsto [g \mapsto gv], \end{aligned}$$

while the G-module corresponding to a right $C(G)$-comodule (V, ρ) is V equipped with the map

$$\begin{aligned} G \times V &\to V \\ (g, v) &\mapsto \underbrace{\rho(v)}_{\text{map } G \to V}(g). \end{aligned}$$

Furthermore, the symmetric monoidal structures on G-Mod and Comod-$C(G)$ correspond under this isomorphism. All these isomorphisms preserve the forgetful functors to $\mathsf{Vect}(\Bbbk)$, as expressed by the commutative diagram

$$\begin{tikzcd}
G\text{-}\mathsf{Mod} \arrow[rr, "\sim"] \arrow[dr, "\omega"'] & & \mathsf{Comod}\text{-}C(G) \arrow[dl, "\omega_{C(G)}"] \\
& \mathsf{Vect}(\Bbbk). &
\end{tikzcd}$$

In particular, G-$\mathsf{Mod}_{\mathsf{f}}$ and $\mathsf{Comod}_{\mathsf{f}}$-$C(G)$ are likewise isomorphic, and the isomorphism preserves their forgetful functors to $\mathsf{Vect}_{\mathsf{f}}(\Bbbk)$.

Proof Unwind the definitions. $\square$

Lemma 9.8.5 There is a bijection $G \xrightarrow{1:1} \mathrm{Hom}_{\Bbbk\text{-}\mathsf{CAlg}}(C(G), \Bbbk)$ sending g to the evaluation homomorphism $f \mapsto f(g)$. Endow the right-hand side with the binary operation induced by $\Delta : C(G) \to C(G \times G) \simeq C(G) \otimes C(G)$, which sends (φ_1, φ_2) to $(\varphi_1 \otimes \varphi_2) \circ \Delta$. Then this bijection is an isomorphism of groups.

Proof Every $\Bbbk$-algebra homomorphism $\varphi : C(G) \to \Bbbk$ is surjective, so $\ker(\varphi)$ is a maximal ideal. As a $\Bbbk$-algebra, however, $C(G)$ is the direct product of G copies of $\Bbbk$, via the map $f \mapsto (f(g))_{g \in G}$. It is easy to show that the maximal ideals of $\prod_{g \in G} \Bbbk$ correspond bijectively to the elements of G through

$$g \in G \quad \leftrightarrow \quad \Bbbk \times \cdots \times \underbrace{\{0\}}_{g\text{-component}} \times \cdots \times \Bbbk$$

See [51, Lemma 8.9.7], or prove this directly. The required bijection follows at once.

To see how group multiplication is reflected on $\mathrm{Hom}_{\Bbbk\text{-}\mathsf{CAlg}}(C(G), \Bbbk)$, it suffices to verify that

$$\begin{tikzcd}
(g_1, g_2) \arrow[d, mapsto] & \in & G \times G \arrow[r, "1:1"] \arrow[d] & \mathrm{Hom}_{\Bbbk\text{-}\mathsf{CAlg}}(C(G \times G), \Bbbk) \arrow[d] & \ni & {[F \mapsto F(g_1, g_2)]} \arrow[d, mapsto] \\
g_1 g_2 & \in & G \arrow[r, "1:1"'] & \mathrm{Hom}_{\Bbbk\text{-}\mathsf{CAlg}}(C(G), \Bbbk) & \ni & {[f \mapsto (\Delta f)(g_1, g_2)]}
\end{tikzcd}$$

is commutative. This presents no essential difficulty. $\square$

We can now answer the reconstruction question posed above. The answer involves the group $\mathrm{Aut}^{\otimes}(\omega)$ introduced in Corollary 9.7.8; the point is to describe the isomorphism explicitly.

Theorem 9.8.6 (T. Tannaka, M. Krein) Let $\Bbbk$ be a field and G a finite group. Consider the forgetful functor $\omega : G$-$\mathsf{Mod}_{\mathsf{f}} \to \mathsf{Vect}_{\mathsf{f}}(\Bbbk)$. There is an isomorphism of groups

$$A : G \xrightarrow{\sim} \mathrm{Aut}^{\otimes}(\omega),$$

described explicitly as follows: for $g \in G$ and a finite-dimensional G-module V, the map $A(g)_V : V \to V$ is the $\Bbbk$-linear map $v \mapsto gv$.

Proof We know that ω is a neutral fiber functor on the Tannakian category G-$\mathsf{Mod}_{\mathrm{f}}$ (Proposition 9.8.2), and therefore it has an associated commutative Hopf algebra $\mathrm{coE}(\omega)$. Taking $B = B' = \Bbbk$ in Corollary 9.7.8 gives an isomorphism of groups

$$\mathrm{Aut}^{\otimes}(\omega) \simeq \mathrm{Hom}_{\Bbbk\text{-}\mathsf{CAlg}}\left(\mathrm{coE}(\omega), \Bbbk\right).$$

The commutative diagram in Proposition 9.8.4 implies

$$\mathrm{coE}(\omega) \simeq \mathrm{coE}\left(\omega_{C(G)} : \mathsf{Comod}_{\mathrm{f}}\text{-}C(G) \to \mathsf{Vect}_{\mathrm{f}}(\Bbbk)\right),$$

while Lemma 9.6.2, together with the explanation following it, further gives

$$\mathrm{coE}\left(\omega_{C(G)}\right) \simeq C(G);$$

all of these are canonical isomorphisms of bialgebras. Substituting Lemma 9.8.5, we obtain isomorphisms of groups

$$\begin{array}{ccccc} G & \xrightarrow{\sim} & \mathrm{Hom}_{\Bbbk\text{-}\mathsf{CAlg}}\left(C(G), \Bbbk\right) & \xrightarrow{\sim} & \mathrm{Aut}^{\otimes}(\omega) \\ g & \longmapsto & B(g) & \longmapsto & A(g). \end{array}$$

The image $B(g)$ under the first isomorphism is the evaluation homomorphism $f \mapsto f(g)$. On the other hand, by the universal property of $\mathrm{coE}(\omega) = \mathrm{coH}(\omega, \omega) \simeq C(G)$ in Definition 9.4.1 (with $L = \Bbbk$), the second isomorphism in fact extends to

$$\mathrm{Hom}_{\Bbbk}(C(G), \Bbbk) \xrightarrow{\sim} \mathrm{Hom}_{\Bbbk}(\omega, \omega),$$

and that universal property shows that the correspondence between A and B is characterized by the following commutative diagram in $\mathsf{Vect}_{\mathrm{f}}(\Bbbk)$:

$$\begin{tikzcd} & \omega(V) \arrow[dl, "\rho"'] \arrow[dr, "A(g)_V"] & \\ \omega(V) \otimes C(G) \arrow[r, "\mathrm{id} \otimes B(g)"'] & \omega(V) \otimes \Bbbk \arrow[r, "\sim"'] & \omega(V), \end{tikzcd}$$

where V ranges over the finite-dimensional G-modules and ρ is defined as in Proposition 9.8.4. Substituting $B(g)(f) = f(g)$ immediately gives $A(g)_V(v) = gv$, as required. $\square$

Under suitable conditions on $\Bbbk$, one can go further and characterize those commutative Hopf algebras isomorphic to $C(G)$ for some finite group G. Such questions belong to elementary algebraic geometry or commutative ring theory, and we shall not pursue them here.

Finally, there is another method that does not depend on a fiber functor and reconstructs a compact group solely from its category of G-modules. It is called the Doplicher–Roberts reconstruction theorem and has its origins in quantum field theory. Interested readers may consult the survey [35].

Exercises

1. Let $\Bbbk$ be a field and let V, W be $\Bbbk$-vector spaces; as usual, set $V^\vee := \mathrm{Hom}_\Bbbk(V, \Bbbk)$, and so on. Write down the canonical linear map $V^\vee \underset{\Bbbk}{\otimes} W^\vee \to (V \underset{\Bbbk}{\otimes} W)^\vee$. Show that it is always injective, but is not surjective when both V and W are infinite-dimensional.

2. Let T be any object of a monoidal category $\mathcal{C}$, and let $\iota : U \to \mathbf{1}$ be a morphism. Assuming that the left dual of U (respectively the right dual U^*) exists, prove that the bijection in Proposition 9.1.11(iii)

$$\mathrm{Hom}(T \otimes U, T) \to \mathrm{Hom}(T, T \otimes {}^*U)$$

(respectively $\mathrm{Hom}(U \otimes T, T) \to \mathrm{Hom}(T, U^* \otimes T)$) sends $\mathrm{id}_T \otimes \iota$ to $\mathrm{id}_T \otimes {}^*\iota$ (respectively sends $\iota \otimes \mathrm{id}_T$ to $\iota^* \otimes \mathrm{id}_T$).

3. Suppose that the monoidal category $\mathcal{C}$ is also additive, with zero object 0, and that $\otimes$ is an additive bifunctor. Show that $X \otimes 0 = 0 = 0 \otimes X$ for every X.

4. Let $\mathcal{C}$ be a rigid braided monoidal category that is also Abelian, and suppose that $\mathrm{End}_{\mathcal{C}}(\mathbf{1})$ is a field. Given $X \in \mathrm{Ob}(\mathcal{C})$ and its right dual X^*, prove that $X \neq 0$ is equivalent to $X \otimes X^* \neq 0$, and also to $X^* \otimes X \neq 0$. Hint Observe that $X \neq 0$ is equivalent to $\mathrm{ev}_X \neq 0$, and also to $\mathrm{coev}_X \neq 0$; apply Proposition 9.2.5.

5. Let $\mathcal{C}$ and $\mathcal{C}'$ be rigid symmetric monoidal categories, both of which are also Abelian, with $\mathrm{End}_{\mathcal{C}}(\mathbf{1})$ a field and $\mathbf{1}' \neq 0$. Prove that every exact monoidal functor compatible with the braidings, $F : \mathcal{C} \to \mathcal{C}'$, satisfies $X \neq 0 \iff F(X) \neq 0$. Hint Repeat the argument of Lemma 9.7.2(i), or use the method of the preceding exercise.

6. (Nobuo Yoneda) Let $\mathcal{C}, \mathcal{D}$ be categories and let $F : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{D}$ be a functor. A **wedge** for F means data $\big(W, (e_X)_{X \in \mathrm{Ob}(\mathcal{C})}\big)$, also written $e : W \xrightarrow{\bullet} F$, consisting of

- ◇ an object W of $\mathcal{D}$;
- ◇ a morphism $e_X : W \to F(X, X)$ in $\mathcal{D}$,

such that the following diagram commutes for every $f : X \to Y$ in $\mathcal{C}$:

$$\begin{tikzcd} W \arrow[r, "e_Y"] \arrow[d, "e_X"'] & F(Y,Y) \arrow[d, "F(f,Y)"] \\ F(X,X) \arrow[r, "F(X,f)"'] & F(X,Y). \end{tikzcd}$$

Dually, define a **cowedge** to be data $(M, (f_X)_X)$, written $f : F \xrightarrow{\bullet} M$, subject to the commutativity of

$$\begin{tikzcd} F(Y,X) \arrow[r, "F(f,X)"] \arrow[d, "F(Y,f)"'] & F(X,X) \arrow[d, "f_X"] \\ F(Y,Y) \arrow[r, "f_Y"'] & M. \end{tikzcd}$$

(i) Explain how all wedges (respectively cowedges) for F form a category. If the category of wedges (respectively cowedges) has a terminal (respectively initial) object, that object is called the **end** (respectively **coend**) of F, and is denoted by

$$\text{end} = \int_{X \in \mathcal{C}} F(X,X), \qquad \text{coend} = \int^{X \in \mathcal{C}} F(X,X).$$

Here the integral is only notation, but it is not unmotivated.

(ii) Interpret (9.4.2) and the construction of $\mathrm{coH}(\omega_1, \omega_2)$ in Proposition 9.4.7 as a special case of a coend.

(iii) Suppose that $\mathcal{C}$ is essentially small relative to the chosen Grothendieck universe (Definition A.4.1), and that $\mathcal{D}$ is complete (respectively cocomplete). Prove that every F has an end (respectively coend).

(iv) For two functors $F, G : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \rightrightarrows \mathcal{D}$, define a **dinatural transformation** $\alpha : F \xrightarrow{\bullet} G$ to be data $(\alpha_X)_{X \in \mathrm{Ob}(\mathcal{C})}$, where $\alpha_X : F(X,X) \to G(X,X)$, such that the following diagram commutes for every $f : X \to Y$ in $\mathcal{C}$:

$$\begin{tikzcd} & F(X,X) \arrow[r, "\alpha_X"] & G(X,X) \arrow[dr, "G(X,f)"] & \\ F(Y,X) \arrow[ur, "F(f,X)"] \arrow[dr, "F(Y,f)"'] & & & G(X,Y). \\ & F(Y,Y) \arrow[r, "\alpha_Y"'] & G(Y,Y) \arrow[ur, "G(f,Y)"'] & \end{tikzcd}$$

Show that the wedges and cowedges above are special cases of dinatural transformations. Also show that natural transformations, that is, morphisms between functors $\mathcal{C} \to \mathcal{D}$ or $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$, are special cases as well.

A detailed account of ends and coends can be found in the monograph [30].

7. Consider two functors $F, G : \mathcal{C} \to \mathcal{D}$, where $\mathcal{C}$ is essentially small. Prove that the set of morphisms between them can be interpreted as the end

$$\mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(F, G) \simeq \int_{X \in \mathcal{C}} \mathrm{Hom}_{\mathcal{D}}(FX, GX).$$

Use this to interpret the center $Z(\mathcal{C})$ of the category $\mathcal{C}$ as an end determined by $\mathrm{Hom}_{\mathcal{C}}$.

8. Let $\Bbbk$ be a field. Prove that the coend of the functor $\mathrm{Hom}_{\Bbbk} : \mathsf{Vect}_{\mathrm{f}}(\Bbbk)^{\mathrm{op}} \times \mathsf{Vect}_{\mathrm{f}}(\Bbbk) \to \mathsf{Vect}_{\mathrm{f}}(\Bbbk)$ can be identified with $\Bbbk$ together with the trace maps

$$\mathrm{Tr}_V : \mathrm{End}_{\Bbbk}(V) \to \Bbbk, \qquad V : \text{ a finite-dimensional } \Bbbk\text{-vector space.}$$

9. In general, an end resembles an operation of taking invariants, whereas a coend resembles gluing. For readers familiar with simplicial sets, interpret the geometric realization functor $|\cdot|$ discussed in §8.4 as the coend

$$|X| \simeq \int^{[n]\in\boldsymbol{\Delta}} X_n \times |\Delta^n|, \quad X \in \mathrm{Ob}(\mathsf{sSet}).$$

10. For a small category $\mathcal{C}$, define the functor category $\mathcal{C}^\wedge := \mathsf{Set}^{\mathcal{C}^{\mathrm{op}}}$. Prove that for every $\mathcal{F} \in \mathrm{Ob}(\mathcal{C}^\wedge)$ there is a canonical isomorphism in $\mathcal{C}^\wedge$

$$\int^{X\in\mathrm{Ob}(\mathcal{C})} \mathcal{F}(X) \times \mathrm{Hom}_{\mathcal{C}}(\cdot, X) \xrightarrow{\sim} \mathcal{F}.$$

This coend comes from the functor

$$\mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{C}^\wedge, \quad (X, Y) \mapsto \mathcal{F}(X) \times \mathrm{Hom}_{\mathcal{C}}(\cdot, Y).$$

Hint〉 See §A.1, especially the density theorem A.1.3.

11. Let $\mathcal{T}$ be a Tannakian category, let $\omega : \mathcal{T} \to \mathsf{Mod}\text{-}B$ be a fiber functor, and let B' be a commutative B-algebra. Show that, under the isomorphism of Corollary 9.7.8, the identity element and inversion in the group $\mathrm{Aut}^{\otimes}(\omega \underset{B}{\otimes} B')$ arise respectively from the counit and antipode of $\mathrm{coE}_B(\omega)$ through the functor $\mathrm{Hom}_{B\text{-}\mathsf{CAlg}}(\cdot, B')$. Hint〉 The identity and inverse in a group are uniquely determined by multiplication. Multiplication is already known to correspond to the comultiplication of $\mathrm{coE}(\omega)$.

12. Let G be a group, let G-Set denote the category of all small sets with a left G-action, and let $U : G\text{-}\mathsf{Set} \to \mathsf{Set}$ be the forgetful functor. Give a natural isomorphism of groups

$$G \simeq \mathrm{Aut}(U).$$

Explain how this may be viewed as a simple prototype of the Tannaka–Krein theorem 9.8.6. Hint〉 Apply the Yoneda lemma to $U \simeq \mathrm{Hom}_{G\text{-}\mathsf{Set}}(G, \cdot)$ to obtain an isomorphism of monoids $G \simeq \mathrm{End}(U)$.

Appendix A Further Material on Abelian Categories

This appendix is a collection of topics related, directly or indirectly, to Abelian categories; it is almost a miscellaneous stew. In principle, the prerequisites extend no further than the first half of Chapter 1 and a small part of Chapter 2.

Section A.1 begins with a brief review of the Yoneda embedding. Its density theorem, Theorem A.1.3, is standard knowledge in category theory, yet is often omitted from textbooks. These tools will be used frequently in §B.2.

Although the compact objects and presentable categories introduced in §A.2 are not absolutely necessary elsewhere in this book, they provide a convenient and common language. The discussion of regular cardinals there also paves the way for §2.10.

The next two sections are directly related to Abelian categories. The Gabriel–Popescu theorem in §A.3 explains the structure of Grothendieck categories, while the locally finite Abelian categories discussed in §A.4 are needed in §9.5. The focus of the latter section is Proposition A.4.4, due to O. Gabber, whose proof requires some ingenuity.

A.1 Density of the Yoneda Embedding

We first recall the theoretical framework surrounding the Yoneda embedding. For any category $\mathcal{C}$, define the functor categories

$$\mathcal{C}^\wedge := \mathsf{Set}^{\mathcal{C}^{\mathrm{op}}}, \quad \mathcal{C}^\vee := \left(\mathsf{Set}^{\mathrm{op}}\right)^{\mathcal{C}^{\mathrm{op}}} = \left(\mathsf{Set}^{\mathcal{C}}\right)^{\mathrm{op}}.$$

They are dual to each other: $(\mathcal{C}^\vee)^{\mathrm{op}} = (\mathcal{C}^{\mathrm{op}})^\wedge$. Relative to a previously chosen Grothendieck universe $\mathcal{U}$, the categories $\mathcal{C}^\wedge$ and $\mathcal{C}^\vee$ are generally large unless $\mathcal{C}$ is small.

The **Yoneda embeddings** are the following functors:

$$\begin{aligned} h_{\mathcal{C}} : \mathcal{C} &\longrightarrow \mathcal{C}^\wedge \\ S &\longmapsto \mathrm{Hom}_{\mathcal{C}}(\cdot, S), \end{aligned} \qquad\qquad \begin{aligned} k_{\mathcal{C}} : \mathcal{C} &\longrightarrow \mathcal{C}^\vee \\ S &\longmapsto \mathrm{Hom}_{\mathcal{C}}(S, \cdot). \end{aligned}$$

The following restates [51, Theorem 2.5.1].

Theorem A.1.1 (Nobuo Yoneda) For every $S \in \mathrm{Ob}(\mathcal{C})$, $A \in \mathrm{Ob}(\mathcal{C}^\wedge)$, and $B \in \mathrm{Ob}(\mathcal{C}^\vee)$, the canonical maps

$$\begin{aligned} \mathrm{Hom}_{\mathcal{C}^\wedge}(h_{\mathcal{C}}(S), A) &\longrightarrow A(S) \\ \left[\mathrm{Hom}_{\mathcal{C}}(\cdot, S) \xrightarrow{\phi} A(\cdot)\right] &\longmapsto \phi_S(\mathrm{id}_S) \end{aligned} \qquad \begin{aligned} &\mathrm{Hom}_{\mathcal{C}^\vee}(B, k_{\mathcal{C}}(S)) \\ &\qquad\qquad \shortparallel \\ &\mathrm{Hom}_{\mathsf{Set}^{\mathcal{C}}}(k_{\mathcal{C}}(S), B) \longrightarrow B(S) \\ &\left[\mathrm{Hom}_{\mathcal{C}}(S, \cdot) \xrightarrow{\psi} B(\cdot)\right] \longmapsto \psi_S(\mathrm{id}_S) \end{aligned}$$

are bijections. Consequently, $h_{\mathcal{C}}$ and $k_{\mathcal{C}}$ are both fully faithful functors.

Definition A.1.2 A functor $\mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$ (respectively $\mathcal{C} \to \mathsf{Set}$) that, up to isomorphism, lies in the image of $h_{\mathcal{C}}$ (respectively $k_{\mathcal{C}}$) is called a **representable functor**.

Although $\mathcal{C}^\wedge$ (respectively $\mathcal{C}^\vee$) is much larger than $\mathcal{C}$, each of its objects can be written canonically as a $\varinjlim$ (respectively $\varprojlim$) of representable functors. Before stating this density theorem, we need a few definitions.

- ⋄ For each $A \in \mathrm{Ob}(\mathcal{C}^\wedge)$, define the category $(h_{\mathcal{C}}/A)$ as in Definition 1.6.2. Its objects are data $\underline{S} := \left(S, h_{\mathcal{C}}(S) \xrightarrow{\phi_{\underline{S}}} A\right)$. A morphism from $\underline{S}$ to $\underline{S}'$ is an $f \in \mathrm{Hom}_{\mathcal{C}}(S, S')$ satisfying $\phi_{\underline{S}} = \phi_{\underline{S}'} h_{\mathcal{C}}(f)$.
- ⋄ Dually, for $B \in \mathrm{Ob}(\mathcal{C}^\vee)$ there is a category $(B/k_{\mathcal{C}})$. Its objects are data $\overline{S} := \left(S, B \xrightarrow{\psi_{\overline{S}}} k_{\mathcal{C}}(S)\right)$, where $\psi_{\overline{S}}$ is regarded as a morphism in $\mathcal{C}^\vee$; morphisms $\overline{S} \to \overline{S}'$ are defined similarly.

Theorem A.1.3 (Density) For every $A \in \mathrm{Ob}(\mathcal{C}^\wedge)$ and $B \in \mathrm{Ob}(\mathcal{C}^\vee)$, the families of morphisms $\phi_{\underline{S}}$ and $\psi_{\overline{S}}$ above respectively give canonical isomorphisms in $\mathcal{C}^\wedge$ and $\mathcal{C}^\vee$

$$\varinjlim_{\underline{S}} h_{\mathcal{C}}(S) \xrightarrow{\sim} A, \quad B \xrightarrow{\sim} \varprojlim_{\overline{S}} k_{\mathcal{C}}(S).$$

Proof It suffices to prove the first assertion. For any $A' \in \mathrm{Ob}(\mathcal{C}^\wedge)$, there is a map

$$\begin{aligned} \mathrm{Hom}_{\mathcal{C}^\wedge}(A, A') &\longrightarrow \left\{ \begin{array}{l} \text{compatible families of morphisms } a'_{\underline{S}} : h_{\mathcal{C}}(S) \to A', \\ \text{where } \underline{S} = (S, \phi_{\underline{S}}) \in \mathrm{Ob}((h_{\mathcal{C}}/A)) \end{array} \right\} \\ \varphi &\longmapsto \left(a'_{\underline{S}} := \varphi\phi_{\underline{S}} \right)_{\underline{S}}. \end{aligned} \tag{A.1.1}$$

Define the reverse map as follows. Given data $(a'_{\underline{S}})_{\underline{S}}$, for every $S \in \mathrm{Ob}(\mathcal{C})$ and $a_S \in A(S)$ take, by Theorem A.1.1, the corresponding morphism $\phi : h_{\mathcal{C}}(S) \to A$, thereby determining $\underline{S} := (S, \phi) \in \mathrm{Ob}((h_{\mathcal{C}}/A))$. The assignment $a_S \mapsto a'_{\underline{S}}$ then gives a map $A(S) \to A'(S)$. We claim:

- as S varies, the maps $A(S) \to A'(S)$ give a morphism $\varphi : A \to A'$ in $\mathcal{C}^\wedge$;
- the map $(a'_{\underline{S}})_{\underline{S}} \mapsto \varphi$ and the map in (A.1.1) are mutually inverse.

As with many results concerning the Yoneda embedding, the detailed verification is nearly tautological and is omitted.

Thus (A.1.1) is a bijection. In the special case $A' = A$, it sends id_A to $(a'_{\underline{S}} = \phi_{\underline{S}})_{\underline{S}}$. This verifies precisely that A, together with the morphisms $\phi_{\underline{S}} : h_{\mathcal{C}}(S) \to A$, has the universal property of the colimit. □

The reader may also consult the treatment in [33, pp. 76–77].

Remark A.1.4 The isomorphism involving B can also be restated in $\mathsf{Set}^{\mathcal{C}}$ as $\varinjlim_{\overline{S}} \mathrm{Hom}_{\mathcal{C}}(S, \cdot) \xrightarrow{\sim} B$, where $\overline{S}$ ranges over data $(S, \psi'_{\overline{S}})$ with $S \in \mathrm{Ob}(\mathcal{C})$ and $\psi'_{\overline{S}} : \mathrm{Hom}_{\mathcal{C}}(S, \cdot) \to B(\cdot)$ a morphism in $\mathsf{Set}^{\mathcal{C}}$.

One application of Theorem A.1.3 is the extension of functors defined on a small category. We state only the version for $\mathcal{C}^\wedge$. Let $\mathcal{C}$ be a small category, let $\mathcal{D}$ be a cocomplete category, possibly large but required to have all small colimits, and let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Define

$$\begin{aligned} \tilde{F} : \mathcal{C}^\wedge &\to \mathcal{D} \\ X &\mapsto \varinjlim_{\underline{S}} F(S), \end{aligned} \tag{A.1.2}$$

where $\underline{S} = (S, \phi_{\underline{S}} : h_{\mathcal{C}}(S) \to X)$ ranges over the small category $(h_{\mathcal{C}}/X)$. If X comes from an object T of $\mathcal{C}$, then $(h_{\mathcal{C}}/X)$ has the terminal object $(T, h_{\mathcal{C}}(T) \xrightarrow{\sim} X)$. Thus the meaning of extension is that the following diagram commutes up to canonical isomorphism:

$$\begin{tikzcd} \mathcal{C}^\wedge \arrow[r, "\tilde{F}"] & \mathcal{D}. \\ \mathcal{C} \arrow[u, "h_{\mathcal{C}}"] \arrow[ur, "F"'] & \end{tikzcd}$$

We now show that $\tilde{F}$ always has a right adjoint and therefore preserves all small colimits. This fact will be used in §B.3.

Proposition A.1.5 Let $\mathcal{C}$ be a small category and $\mathcal{D}$ a cocomplete category, possibly large. For a functor $F : \mathcal{C} \to \mathcal{D}$, take the canonical extension $\tilde{F} : \mathcal{C}^\wedge \to \mathcal{D}$ defined in (A.1.2). Then $\tilde{F}$ has the right adjoint

$$\mathcal{H}\mathrm{om}(F, \cdot) : \mathcal{D} \to \mathcal{C}^\wedge,$$

defined by:

- ◇ $\mathcal{H}\mathrm{om}(F, Y)(S) = \mathrm{Hom}_{\mathcal{D}}(FS, Y)$ for every $S \in \mathrm{Ob}(\mathcal{C})$ and $Y \in \mathrm{Ob}(\mathcal{D})$;
- ◇ a morphism $f : Y \to Y'$ induces a morphism $f_* : \mathrm{Hom}(F(\cdot), Y) \to \mathrm{Hom}(F(\cdot), Y')$ in $\mathcal{C}^\wedge$.

Consequently, $\tilde{F}$ preserves all small colimits.

Proof For $X \in \mathrm{Ob}(\mathcal{C}^\wedge)$ and $Y \in \mathrm{Ob}(\mathcal{D})$, the equality $X = \varinjlim_{S} h_{\mathcal{C}}(S)$ and the universal property of the colimit give canonical bijections

$$\begin{aligned} \mathrm{Hom}_{\mathcal{C}^\wedge}(X, \mathcal{H}\mathrm{om}(F, Y)) &\simeq \varprojlim_{S} \mathrm{Hom}_{\mathcal{C}^\wedge}(h_{\mathcal{C}}(S), \mathcal{H}\mathrm{om}(F, Y)) \simeq \varprojlim_{S} \mathrm{Hom}_{\mathcal{D}}(FS, Y) \\ &\simeq \mathrm{Hom}_{\mathcal{D}}(\varinjlim_{S} FS, Y) = \mathrm{Hom}_{\mathcal{D}}(\tilde{F}X, Y). \end{aligned}$$

Thus $\mathcal{H}\mathrm{om}(F, \cdot)$ is indeed right adjoint to $\tilde{F} : \mathcal{C}^\wedge \to \mathcal{D}$. □

A.2 Compact Objects and Presentable Categories

Compactness in category theory is a common refinement of many finiteness conditions in algebra, geometry, and algebraic geometry. Throughout this section we fix a category $\mathcal{C}$; unless stated otherwise, we always assume that $\mathcal{C}$ has all small filtered $\varinjlim$.

Consider a small filtered category I and a functor $\alpha : I \to \mathcal{C}$. For $X \in \mathrm{Ob}(\mathcal{C})$, the canonical family of morphisms $\iota_i : \alpha(i) \to \varinjlim \alpha$ induces a family $(\iota_i)_* : \mathrm{Hom}(X, \alpha(i)) \to$

$\mathrm{Hom}\left(X, \varinjlim \alpha\right)$, with $i \in \mathrm{Ob}(I)$, and hence a canonical map

$$\varinjlim_{i \in \mathrm{Ob}(I)} \mathrm{Hom}\left(X, \alpha(i)\right) \to \mathrm{Hom}\left(X, \varinjlim \alpha\right). \tag{A.2.1}$$

Definition A.2.1 Let $(P, \leq)$ be a nonempty partially ordered set and let κ be an infinite cardinal. If every subset P_0 satisfying $|P_0| < \kappa$ has an upper bound in P, then $(P, \leq)$ is called **κ-filtered**.

If $\kappa \leq \kappa'$, then κ'-filtered implies κ-filtered. Taking $\kappa := \aleph_0 = \omega$ recovers the filtered partially ordered sets introduced in Example 1.6.7.

Definition A.2.2 If $X \in \mathrm{Ob}(\mathcal{C})$ is such that (A.2.1) is an isomorphism for every small filtered category I and every $\alpha : I \to \mathcal{C}$, then X is called a **compact object** of $\mathcal{C}$.

Given a small cardinal κ, if I is further restricted to small κ-filtered partially ordered sets, then an X satisfying the corresponding condition is called a **κ-compact object**.

In the terminology of Definition 1.5.2, X is compact if and only if $\mathrm{Hom}(X, \cdot) : \mathcal{C} \to \mathsf{Set}$ preserves small filtered $\varinjlim$. If $\kappa \leq \kappa'$, then κ-compactness implies κ'-compactness.

Remark A.2.3 At first glance the definitions of compact and $\aleph_0$-compact objects appear different: the former involves all small filtered categories, whereas the latter permits only small filtered partially ordered sets. There is, however, a nontrivial fact [1, Theorem 1.5]:

> For every small filtered category I, there is a small filtered partially ordered set $(\mathcal{P}, \leq)$ together with a cofinal functor $\mathcal{P} \to I$.

It follows that compactness is equivalent to $\aleph_0$-compactness. A proof will be sketched in the exercises.

Using the definition of a filtered category and the concrete description of filtered $\varinjlim$ in Set (Proposition 1.6.11), we see that

- ◇ the map (A.2.1) is surjective $\iff$ every $f : X \to \varinjlim \alpha$ factors as $X \xrightarrow{f_i} \alpha(i) \xrightarrow{\iota_i} \varinjlim \alpha$;[1)]
- ◇ the map (A.2.1) is injective $\iff$ for every $i \in \mathrm{Ob}(I)$ and every $f, g \in \mathrm{Hom}(X, \alpha(i))$ satisfying $\iota_i f = \iota_i g$, there is a morphism $i \to j$ in I such that the images of f and g in $\mathrm{Hom}(X, \alpha(j))$ are equal.

[1)] Translator's note: the source prints the target of the last arrow as X. Since ι_i is the canonical morphism to the colimit, the type-correct target is $\varinjlim \alpha$.

Example A.2.4 For $\mathcal{C} = \mathsf{Set}$, an object X is compact if and only if X is a finite set. The if direction follows immediately from the preceding discussion and the concrete description of $\varinjlim \alpha$. For the only-if direction, express X as the filtered $\varinjlim$ of its finite subsets and consider id_X.

Example A.2.5 Let R be a ring. For $\mathcal{C} = R\text{-}\mathsf{Mod}$, an object X is compact if and only if it is finitely presented; in other words, there exist $a, b \in \mathbb{Z}_{\geq 0}$ and an exact sequence of R-modules

$$R^{\oplus b} \to R^{\oplus a} \xrightarrow{p} X \to 0.$$

For filtered $\varinjlim$ in $R\text{-}\mathsf{Mod}$, see the explanation in [51, Remark 6.2.3]. The only-if direction is left as an exercise. We sketch the if direction. Take the images of the standard basis of $R^{\oplus a}$ to obtain generators $x_1, \ldots, x_a$ of X. First we prove that (A.2.1) is injective. Suppose that $f, g \in \mathrm{Hom}(X, \alpha(i))$ satisfy $\iota_i f = \iota_i g$. For every $x \in \{x_1, \ldots, x_a\}$, there is an $i \to j_x$ such that the images of $f(x)$ and $g(x)$ in $\alpha(j_x)$ are equal. The filteredness of I then gives the desired $i \to j$, for which the images of f and g in $\mathrm{Hom}(X, \alpha(j))$ are equal.

Next we prove that (A.2.1) is surjective. Consider $f : X \to \varinjlim \alpha$. For every $x \in \{x_1, \ldots, x_a\}$, there are j_x and $m_x \in \alpha(j_x)$ such that $f(x) = \iota_{j_x}(m_x)$. Again, filteredness ensures that the j_x may be chosen independently of x; denote the common choice by j. This determines a morphism $m : R^{\oplus a} \to \alpha(j)$ such that $fp = \iota_j m$. Since b is finite, by passing sufficiently far along a morphism $j \to i$ and then replacing j by i, we may also arrange for m to vanish identically on the image of $R^{\oplus b}$. This gives the required f_i.

For compact objects determined by various kinds of algebraic structure, [1, Examples 1.2; Theorem 3.12] gives a more complete discussion. For instance, the compact objects in the category of groups Grp are precisely the finitely presented groups.

Another important class of compact objects comes from the Yoneda embedding $h_{\mathcal{D}} : \mathcal{D} \to \mathcal{D}^{\wedge}$ recalled in §A.1.

Proposition A.2.6 Let $\mathcal{D}$ be any category and $X \in \mathrm{Ob}(\mathcal{D})$. Then $h_{\mathcal{D}}(X)$ is compact in $\mathcal{D}^{\wedge}$.

Proof Given a small filtered category I and a functor $\alpha : I \to \mathcal{D}^{\wedge}$, Theorem A.1.1 and the pointwise construction of $\varinjlim$ in $\mathcal{D}^{\wedge}$ (see [51, Proposition 2.7.8], where it is denoted by "$\varinjlim$") give

$$\begin{aligned}\varinjlim_i \mathrm{Hom}_{\mathcal{D}^\wedge}(h_{\mathcal{D}}(X), \alpha(i)) &\simeq \varinjlim_i (\alpha(i)(X)) \\ &= (\varinjlim \alpha)(X) \\ &\simeq \mathrm{Hom}_{\mathcal{D}^\wedge}(h_{\mathcal{D}}(X), \varinjlim \alpha).\end{aligned}$$

A routine verification shows that this is the canonical map in (A.2.1). Thus compactness follows. □

By further restricting the filtered category I in (A.2.1), one obtains various versions of compactness. We first focus on the regular cardinals introduced in [51, Definition 1.4.10]; a more careful treatment is useful here.[2)] First let $\alpha > 0$ be any infinite ordinal. Consider a limit ordinal $\theta > 0$ and a strictly increasing sequence of ordinals $(a_\beta)_{\substack{\beta:\text{ordinal} \\ \beta<\theta}}$. If $\sup_{\beta<\theta} a_\beta = \alpha$, the sequence is called **cofinal** in α. If α is a limit ordinal, define

$$\mathrm{cf}(\alpha) := \inf\{\theta > 0 : \text{limit ordinal}, \ \exists (a_\beta)_{\beta<\theta} \text{ as above, cofinal in } \alpha\};$$

this inf is well-defined and is attained by some θ of the indicated kind; for sup and inf of ordinals, see [51, Theorem 1.2.10]. Taking $a_\beta := \beta$ immediately gives $\mathrm{cf}(\alpha) \le \alpha$.

Note that every infinite cardinal, viewed as an ordinal, is necessarily a limit ordinal; see the discussion following [51, Lemma 1.4.6].

Definition A.2.7 An infinite cardinal κ is called a **regular cardinal** if, viewed as an ordinal, it satisfies $\mathrm{cf}(\kappa) = \kappa$.

Proposition A.2.8 Let $\alpha > 0$ be a limit ordinal.

(i) $\mathrm{cf}(\mathrm{cf}(\alpha)) = \mathrm{cf}(\alpha)$;

(ii) if a nonempty subset $S \subset \alpha$ satisfies $\sup S = \alpha$, then $|S| \ge \mathrm{cf}(\alpha)$ as ordinals;

(iii) $\mathrm{cf}(\alpha)$ is a regular cardinal.

Proof For (i), the key is to prove $\mathrm{cf}(\alpha) \le \mathrm{cf}(\mathrm{cf}(\alpha))$. Suppose that $(a_\beta)_{\beta<\theta}$ is cofinal in α and $(b(\gamma))_{\gamma<\psi}$ is cofinal in θ. Then $\left(a_{b(\gamma)}\right)_{\gamma<\psi}$ is cofinal in α. Hence $\mathrm{cf}(\alpha) \le \mathrm{cf}(\theta)$. Now take $\theta = \mathrm{cf}(\alpha)$.

For (ii), choose any bijection $f : |S| \to S$. Recall that $|S|$ is an ordinal, while $\alpha \notin S$. We claim that there are a limit ordinal θ and an order-preserving embedding $i : \theta \hookrightarrow |S|$ (hence $\theta \le |S|$) such that the sequence of ordinals $(f(i(\beta)))_{\beta<\theta}$ in S is strictly increasing and $\sup_{\beta<\theta} f(i(\beta)) = \sup S = \alpha$.

[2)] Readers uninterested in set theory may skip the following discussion.

Indeed, this is an application of transfinite recursion. More explicitly, take

$$\begin{aligned} i(0) &:= \inf |S| = 0, \\ i(n+1) &:= \inf \{t \in |S| : f(t) > f(i(n))\}, \quad n \in \omega := \{0, 1, 2, \ldots\}, \\ i(\omega) &:= \inf \{t \in |S| : \forall k < \omega,\ f(t) > f(i(k))\}, \end{aligned}$$

and continue transfinitely in this manner until stopping at θ. This proves the claim, and immediately yields $|S| \geq \theta \geq \mathrm{cf}(\alpha)$.

For (iii), in view of (i), it remains only to show that $\mathrm{cf}(\alpha)$ is a cardinal. If $\beta > 0$ is a limit ordinal, then (ii) implies $\beta \geq |\beta| \geq \mathrm{cf}(\beta)$. Substituting $\beta = \mathrm{cf}(\alpha)$ and using (i), we obtain $\mathrm{cf}(\alpha) = |\mathrm{cf}(\alpha)|$, so $\mathrm{cf}(\alpha)$ is a cardinal. □

Proposition A.2.9 (See [21, Theorem 5.10]) Let λ be an infinite cardinal. Then $\mathrm{cf}(2^\lambda) > \lambda$.

Proof This is a standard result in set theory. Let $0 < \alpha \leq \lambda$ be a limit ordinal and let $(a_\beta)_{\beta<\alpha}$ be a sequence of ordinals satisfying $a_\beta < 2^\lambda$. Write $\kappa_\beta := |a_\beta| < 2^\lambda$. By the definitions of the supremum of ordinals and cardinal arithmetic,

$$\left| \sup_{\beta<\alpha} a_\beta \right| = \left| \bigcup_{\beta<\alpha} a_\beta \right| \leq \sum_{\beta<\alpha} \kappa_\beta.$$

König's lemma for cardinal arithmetic (see [51, Exercise 1.6], or [21, Theorem 5.10]) gives

$$\begin{aligned} \sum_{\beta<\alpha} \kappa_\beta < \prod_{\beta<\alpha} 2^\lambda &= (2^\lambda)^{|\alpha|} \\ &= 2^{\lambda \cdot |\alpha|} \leq 2^{\lambda\cdot\lambda} = 2^\lambda; \end{aligned}$$

the last step uses [51, Theorem 1.4.8]. Thus $\lambda \geq \mathrm{cf}(2^\lambda)$ is impossible. □

Lemma A.2.10 Let T be a small set. Then there is a small regular cardinal μ such that $\mu > |T|$.

Proof Recall that a Grothendieck universe $\mathcal{U}$ has already been fixed. The question depends only on $|T|$, so we may assume without loss of generality that $T \in \mathcal{U}$ is infinite. Its power set $P(T)$ then belongs to $\mathcal{U}$. Write $\lambda := |T|$. Proposition A.2.9 says that $\mathrm{cf}(2^\lambda) > \lambda$. Since $\mathrm{cf}(2^\lambda)$ is both a regular cardinal (Proposition A.2.8 (iii)) and embeddable as a subset of $2^\lambda = |P(T)|$, the choice $\mu := \mathrm{cf}(2^\lambda)$ has the required properties. □

We now return to the main line of category theory. The following notion originates in [15]; a textbook reference is [1, Chapters 1, 2].

Definition A.2.11 (P. Gabriel, F. Ulmer) Let $\mathcal{C}$ be a category as above and let κ be a small regular cardinal.

1. The category $\mathcal{C}$ is called **κ-accessible** if it satisfies the following conditions:
   - ◇ $\mathcal{C}$ has all small κ-filtered $\varinjlim$ (Definition A.2.1);
   - ◇ there is a small subset $S \subset \mathrm{Ob}(\mathcal{C})$ consisting of κ-compact objects (Definition A.2.2) such that every $X \in \mathrm{Ob}(\mathcal{C})$ can be expressed as a small κ-filtered $\varinjlim$ of elements of S.
2. A functor between κ-accessible categories is called κ-accessible if it preserves small κ-filtered $\varinjlim$.
3. A cocomplete κ-accessible category is called a **κ-presentable category**.[3)]

When κ' is sufficiently large relative to κ, the corresponding conditions are weaker than those for κ. A category that is κ-accessible (respectively κ-presentable) for some small regular cardinal κ is called accessible (respectively presentable). Similarly, a functor between presentable categories is called accessible if it is κ-accessible for some small regular cardinal κ.

One can verify that the categories Set and $R\text{-}\mathsf{Mod}$ discussed in Examples A.2.4 and A.2.5 are both presentable; in fact, they are $\aleph_0$-presentable. See also the exercises.

For the following important result we can give only part of the proof here.

Theorem A.2.12 (P. Gabriel, F. Ulmer) Let κ be a small regular cardinal and let $F : \mathcal{C} \to \mathcal{D}$ be a functor between presentable categories. Then:

(i) F has a right adjoint if and only if it preserves small $\varinjlim$;

(ii) F has a left adjoint if and only if it is accessible and preserves small $\varprojlim$.

Proof The only-if direction of (i) is well known. The converse follows from the special adjoint functor theorem 1.11.13 and the following facts. First, Lemma 1.11.9 implies that the set S in Definition A.2.11 is a generating family for $\mathcal{C}$; second, every presentable category is co-well-powered, by [1, Theorem 1.58].

Assertion (ii) follows from Freyd's adjoint functor theorem 1.11.6; see [1, Theorem 1.66] for details. □

3) Gabriel once called these algebraic categories; the literature [1, 15] calls them locally presentable categories, while here the terminology is changed following [31, A.1.1].

A.3 The Gabriel–Popescu Theorem

This section continues and supplements §2.10. Fix a Grothendieck category $\mathcal{A}$, and abbreviate $\mathrm{Hom}_{\mathcal{A}}$ to Hom. Write $X^{\oplus I}$ for the direct sum in $\mathcal{A}$ of I copies of an object X, where I is a small set.

The Gabriel–Popescu theorem will show that $\mathcal{A}$ can always be realized as a reflective localization (Remark 1.9.20) of a category of right modules $\mathsf{Mod}\text{-}R$ for some ring R. The ring R and the functors involved depend on a generator s. We follow L. Kuhn's proof [27]; one of its steps requires the classical theory of derived functors, for which see §3.12.

Choose $s \in \mathrm{Ob}(\mathcal{A})$, put $R := \mathrm{End}(s)$, and consider the functor

$$\begin{aligned} G : \mathcal{A} &\to \mathsf{Mod}\text{-}R \\ X &\mapsto \mathrm{Hom}(s, X), \end{aligned}$$

where R acts on the right of $\mathrm{Hom}(s, X)$ by composition of morphisms. It is easy to see that G is an additive functor between Abelian categories and preserves all small $\varprojlim$. Henceforth write $\mathrm{Hom}_R := \mathrm{Hom}_{\mathsf{Mod}\text{-}R}$ to distinguish it from $\mathrm{Hom} := \mathrm{Hom}_{\mathcal{A}}$.

For every $r \in R$, define $\lambda_r : x \mapsto rx$ in $\mathrm{End}_R(R)$. Notice that $G(s) = R$, and that $\lambda_r : R \to R$ is precisely the image under G of the morphism $r : s \to s$.

Lemma A.3.1 The functor $G : \mathcal{A} \to \mathsf{Mod}\text{-}R$ above has a left adjoint $P : \mathsf{Mod}\text{-}R \to \mathcal{A}$. The counit $\varepsilon : PG \to \mathrm{id}$ of this adjunction makes $\varepsilon_s : PG(s) = P(R) \to s$ an isomorphism, and the following diagram commutes:

$$\begin{tikzcd}
\lambda_r & \in & \mathrm{End}_R(R) \arrow[r, "P"] & \mathrm{End}(P(R)) \arrow[d, "\wr"] & \ni & \varphi \arrow[d, mapsto] \\
r \arrow[u, mapsto] & \in & R \arrow[u, "\wr"] \arrow[r, equal] & \mathrm{End}(s) & \ni & \varepsilon_s \varphi \varepsilon_s^{-1}.
\end{tikzcd}$$

Proof The existence of the left adjoint P follows from the special adjoint functor theorem 1.11.13 and Corollary 2.10.8. We next claim that, for every $Y \in \mathrm{Ob}(\mathcal{A})$, the following diagram commutes:

$$\begin{tikzcd}
& \mathrm{Hom}(PG(s), Y) \arrow[d, "\wr"] & & \\
\mathrm{Hom}(s, Y) \arrow[ur, "\varepsilon_s^*"] \arrow[r, "G"] \arrow[dr, "\mathrm{id}"'] & \mathrm{Hom}_R(G(s), G(Y)) \arrow[d, "\wr"] & \ni & \psi \arrow[d, mapsto] \\
& G(Y) & \ni & \psi(1_R).
\end{tikzcd}$$

- ◇ Commutativity of the upper part is a general property of the adjunction (P, G). Given $f \in \operatorname{Hom}(s, Y)$, the morphism $f\varepsilon_s : PGs \to Y$ corresponds to $G(f\varepsilon_s) \circ \eta_{Gs} : Gs \to GY$; see [51, (2.5)]. By the triangle identity for the adjunction, the latter equals $(Gf) \circ ((G\varepsilon_s)\eta_{Gs}) = Gf$.

- ◇ Commutativity of the lower part follows directly from the definition of G.

Thus ε_s^* is an isomorphism. Taking Y to be the injective cogenerator supplied by Corollary 2.10.15 then shows that ε_s is an isomorphism. Finally, the assertion about the first diagram is equivalent to the commutativity of

$$\begin{tikzcd} P(R) \arrow[r, "P(\lambda_r)"] \arrow[d, "\varepsilon_s"'] & P(R) \arrow[d, "\varepsilon_s"] \\ s \arrow[r, "r"'] & s \end{tikzcd} \qquad (r \in R)$$

But $R = Gs$ and $\lambda_r = Gr$, so this follows from the naturality of ε. □

How can P be described? For every right R-module M, there are small sets I, J and an exact sequence

$$R^{\oplus J} \to R^{\oplus I} \to M \to 0;$$

Since P preserves small $\varinjlim$, it follows that $PM \simeq \operatorname{coker}\left[s^{\oplus J} \to s^{\oplus I}\right]$.

The following result is stated in terms of the Serre quotient of an Abelian category; see Theorem 2.9.6 for details. A Serre quotient is a special case of a localization.

Theorem A.3.2 (P. Gabriel, N. Popescu, L. Kuhn) Let s be a generator of the Grothendieck category $\mathcal{A}$. Then:

- ◇ the functor $G = \operatorname{Hom}(s, \cdot) : \mathcal{A} \to \mathsf{Mod}\text{-}R$ has an exact left adjoint P;

- ◇ G is fully faithful, and the counit of the adjunction gives an isomorphism $\varepsilon : PG \xrightarrow{\sim} \mathrm{id}_{\mathcal{A}}$;[4)]

- ◇ P induces an equivalence of categories $\mathsf{Mod}\text{-}R/\ker(P) \xrightarrow{\sim} \mathcal{A}$.

Proof Lemma A.3.1 has shown that G has a left adjoint P. We claim that if $u : M \to GX$ is a monomorphism in $\mathsf{Mod}\text{-}R$, then the corresponding morphism $v := \varepsilon_X \circ Pu : PM \to X$ (see [51, (2.5)]) is a monomorphism in $\mathcal{A}$. Express M as the filtered $\varinjlim$ of its finitely generated R-submodules. Since P preserves $\varinjlim$ and filtered $\varinjlim$ in $\mathcal{A}$ are exact, the problem reduces to the case where M is finitely generated.

[4)] Translator's note: the source prints $\mathrm{id}_{\mathsf{Mod}\text{-}R}$. Since PG is an endofunctor of $\mathcal{A}$, the type-correct target of the counit is $\mathrm{id}_{\mathcal{A}}$.

Identify s with $P(R)$ via ε_s. Choose a finite set I and a surjection $R^{\oplus I} \twoheadrightarrow M$. This gives $e : s^{\oplus I} \twoheadrightarrow PM$. To conclude that v is a monomorphism, it suffices to prove $\ker(ve) = \ker(e)$. By the generator property (for instance, Proposition 2.10.7), the problem further reduces to showing that any morphism $f : s \to s^{\oplus I}$ satisfying $vef = 0$ also satisfies $ef = 0$. Make two observations:

- ◇ By construction, e is the image under P of a morphism. Since I is finite, write f as an element of $R^{\oplus I}$ and inspect its components one by one; the commutative diagram in Lemma A.3.1 ensures that f is also in the image of P.
- ◇ If a morphism $\psi : N \to M$ in $\mathsf{Mod}\text{-}R$ satisfies $v \circ P\psi = 0$, then $\psi = 0$. This follows from the monicity of u and the commutative diagram in Ab

$$\begin{tikzcd} v \in \mathrm{Hom}(PM, X) \arrow[r, "\sim"] \arrow[d, "(P\psi)^*"'] & \mathrm{Hom}_R(M, GX) \ni u \arrow[d, "\psi^*"] \\ \mathrm{Hom}(PN, X) \arrow[r, "\sim"'] & \mathrm{Hom}_R(N, GX). \end{tikzcd}$$

Thus $ef : s \to PM$ is always in the image of P, while $vef = 0$ implies $ef = 0$. This proves the claim.

The counit ε of an adjunction is an isomorphism if and only if the right adjoint G is fully faithful; this general fact is an exercise in Chapter 1. We now prove that ε is an isomorphism. Let $X \in \mathrm{Ob}(\mathcal{A})$. Since $\varepsilon_X : PG(X) \hookrightarrow X$ corresponds to id_{GX}, taking $u = \mathrm{id}_{GX}$ in the preceding step shows that ε_X is a monomorphism. For surjectivity, Proposition 2.10.7 reduces the problem to proving that every $\alpha : s \to X$ factors through $\varepsilon_X : PG(X) \hookrightarrow X$, and the required factorization is supplied by the diagram

$$\begin{tikzcd} PG(s) \arrow[r, "\varepsilon_s", "\sim"'] \arrow[d, "PG(\alpha)"'] & s \arrow[d, "\alpha"] \\ PG(X) \arrow[r, "\varepsilon_X"'] & X \end{tikzcd}$$

Next we prove that P is exact. It is already right exact, so it suffices to show that the left derived functor $\mathrm{L}_1 P = 0$. For any right R-module M, take a short exact sequence

$$0 \to K \xrightarrow{u} F \to M \to 0, \quad F : \text{a free module.}$$

By dimension shifting (Proposition 3.12.9),

$$\mathrm{L}_1 P(M) \simeq \ker[Pu : P(K) \to P(F)].$$

The problem therefore reduces to proving that Pu is monic. Express F as the filtered $\varinjlim$ of its finite-rank free submodules F' (or as their increasing union), and correspondingly write $K = \varinjlim_{F'}(F' \cap K)$. Since P preserves $\varinjlim$ and filtered $\varinjlim$ in $\mathcal{A}$ are exact, the problem reduces to the case $F = R^{\oplus I} \simeq G(s^{\oplus I})$ for a finite set I. But at the start of the proof we showed that if $u : K \to G(s^{\oplus I})$ is monic, then $v = \varepsilon_{s^{\oplus I}} \circ Pu : P(K) \to s^{\oplus I}$ is monic; hence Pu is monic. Exactness follows.

Finally, applying the universal property in Theorem 2.9.6 to P factors it through a faithful additive functor $\overline{P} : \mathsf{Mod}\text{-}R/\ker(P) \to \mathcal{A}$. Let $\overline{G}$ be the composite $\mathcal{A} \xrightarrow{G} \mathsf{Mod}\text{-}R \to \mathsf{Mod}\text{-}R/\ker(P)$. Then $\overline{P} \circ \overline{G} \simeq P \circ G \simeq \mathrm{id}_{\mathcal{A}}$. It follows that $\overline{P}$ is also full and essentially surjective, with $\overline{G}$ as a quasi-inverse, as desired. □

Remark A.3.3 The Gabriel–Popescu theorem A.3.2 implies that a Grothendieck category is presentable in the sense of Definition A.2.11. Indeed, $\mathsf{Mod}\text{-}R \simeq R^{\mathrm{op}}\text{-}\mathsf{Mod}$ is known to be presentable, and [1, Corollary 1.40] reflects this property to $\mathcal{A}$.

A.4 Locally Finite Abelian Categories

The aim of this section is to introduce a finiteness condition on Abelian categories that supplies the technical support needed in §9.5. Readers are advised first to learn the statements of Definition A.4.1, Definition A.4.2, and Proposition A.4.4, while skipping the subsequent lemmas and proofs. We fix a Grothendieck universe $\mathcal{U}$ in order to discuss what is meant by small sets and small categories.

Definition A.4.1 A category $\mathcal{C}$ is called **essentially small** if it is equivalent to a small category, or equivalently if $\mathrm{Ob}(\mathcal{C})/\simeq$ is a small set.

Fix a field $\Bbbk$. All Abelian categories $\mathcal{A}$ considered below are understood to be $\Bbbk$-linear.

Definition A.4.2 Relative to the fixed field $\Bbbk$, an essentially small Abelian category $\mathcal{A}$ (Definition A.4.1) is called **locally finite** if it has the following properties:

$$\forall X, Y \in \mathrm{Ob}(\mathcal{A}) \quad \begin{cases} X \text{ is an object of finite length (Definition 2.4.12)}, \\ \dim_{\Bbbk} \mathrm{Hom}_{\mathcal{A}}(X, Y) < \infty. \end{cases}$$

Convention A.4.3 For an Abelian category $\mathcal{A}$ and $X \in \mathrm{Ob}(\mathcal{A})$, define the Abelian subcategory $\langle X \rangle$ of $\mathcal{A}$ by

$$Y \in \mathrm{Ob}(\langle X \rangle) \iff \exists n \in \mathbb{Z}_{\geq 0} \text{ such that } Y \text{ is a subquotient of } X^{\oplus n}.$$

It is easy to see that $\mathcal{A}$ is the union of all the subcategories $\langle X \rangle$. Moreover, $\langle X \rangle \subset \langle X \oplus Y \rangle \supset \langle Y \rangle$ shows that this union is filtered. We are interested in locally finite Abelian categories of the form $\langle X \rangle$.

Proposition A.4.4 (O. Gabber) Let $\mathcal{A}$ be a locally finite Abelian category, and suppose there is an $X \in \mathrm{Ob}(\mathcal{A})$ such that $\mathcal{A} = \langle X \rangle$. Then $\mathcal{A}$ has a projective generator.

The proof requires some preparation.

Definition A.4.5 In any Abelian category, let $\alpha : E \twoheadrightarrow Y$ be an epimorphism. If there is no proper subobject E' of E for which $\alpha|_{E'}$ is still epic, then α is called an **essential extension**.

Lemma A.4.6 Let S be a simple object in any Abelian category, and let $\alpha : E \twoheadrightarrow S$ be an essential extension. Then, for every simple object T, pullback along α gives an isomorphism $\alpha^* : \mathrm{Hom}(S, T) \to \mathrm{Hom}(E, T)$.

Proof Pullback along the epimorphism α is always injective; we prove surjectivity. Let $\phi \in \mathrm{Hom}(E, T) \smallsetminus \{0\}$. Since $\alpha|_{\ker(\phi)}$ is not epic, simplicity of S implies $\ker(\phi) \subset \ker(\alpha)$. The induced morphism $T \simeq E/\ker(\phi) \twoheadrightarrow E/\ker(\alpha) \simeq S$ must be an isomorphism, so $\ker(\phi) = \ker(\alpha)$. Hence there is a $\phi' \in \mathrm{Hom}(S, T)$ such that $\phi = \phi'\alpha = \alpha^*(\phi')$. □

For an object Y of finite length in an Abelian category, write $\mathrm{JH}(Y)$ for the multiset of its composition factors, counted with multiplicity (Definition–Theorem 2.7.4).

Lemma A.4.7 Let $\mathcal{A}$ be an Abelian category and X an object of finite length. Consider a simple object S of $\langle X \rangle$ and an essential extension $\alpha : E \twoheadrightarrow S$. For every $Y \in \mathrm{Ob}(\langle X \rangle)$, let $\ell_S(Y)$ be the number, counted with multiplicity, of composition factors in $\mathrm{JH}(Y)$ isomorphic to S. Then

$$\dim_{\Bbbk} \mathrm{Hom}(E, Y) \leq \ell_S(Y) \dim_{\Bbbk} \mathrm{End}(S); \tag{A.4.1}$$

if Y is simple, equality holds.

In addition, the following statements are equivalent:

(i) E is a projective object of $\langle X \rangle$;

(ii) equality holds in (A.4.1) for every $Y \in \mathrm{Ob}(\langle X \rangle)$;

(iii) equality holds in (A.4.1) for $Y = X$.

Proof As a function of Y, the right-hand side of (A.4.1) is additive on short exact sequences $0 \to Y' \to Y \to Y'' \to 0$. On the other hand, $\mathrm{Hom}(E, \cdot)$ is left exact, so the left-hand side of (A.4.1) is subadditive:

$$\dim_{\Bbbk} \mathrm{Hom}(E, Y) \leq \dim_{\Bbbk} \mathrm{Hom}(E, Y') + \dim_{\Bbbk} \mathrm{Hom}(E, Y'').$$

Equality in this formula holds for every short exact sequence if and only if E is projective in $\langle X\rangle$.

If Y is simple, Lemma A.4.6 implies that the left-hand side of (A.4.1) is $\dim_{\Bbbk} \operatorname{End}(S)$ when $Y \simeq S$, and is 0 otherwise. The right-hand side has the same property, so equality in (A.4.1) holds in this case. For general Y, consider a composition series and apply the first paragraph. This proves both the inequality (A.4.1) and (i) $\implies$ (ii). Conversely, equality in (A.4.1) makes $\dim_{\Bbbk} \operatorname{Hom}(E, \cdot)$ additive, proving (ii) $\implies$ (i).

Clearly (ii) $\implies$ (iii). We prove (iii) $\implies$ (ii). Since the inequality (A.4.1) has already been established, the first paragraph shows that if equality holds for Y, it also holds for Y' and Y''. By assumption it holds for X, hence for every $X^{\oplus n}$ and all its subquotients. This gives (ii). $\square$

We now return to the goal of this section.

Proof (Proposition A.4.4) We may assume that $X \neq 0$. First suppose that, for every element S of $\mathrm{JH}(X)$ without multiplicities, there is an epimorphism $P_S \twoheadrightarrow S$ such that P_S is projective in $\langle X\rangle$. We show that the direct sum P of all these P_S is a projective generator of $\langle X\rangle$.

Indeed, P is automatically projective in $\langle X\rangle$. By Proposition 2.8.15, it remains to show that $Y \neq 0 \implies \operatorname{Hom}(P, Y) \neq 0$ for every $Y \in \operatorname{Ob}(\langle X\rangle)$. Choose a simple quotient $Y \twoheadrightarrow S$. Projectivity of P_S factors $P_S \twoheadrightarrow S$ as $P_S \to Y \twoheadrightarrow S$, and this gives a nonzero morphism $P \xrightarrow{\text{projection}} P_S \to Y$.

Now fix a simple object S of $\langle X\rangle$ and construct $P_S \twoheadrightarrow S$. Choose a composition series of X

$$X = X_0 \supsetneq \cdots \supsetneq X_r = 0, \quad r \geq 1.$$

For $i = 1, \ldots, r$, we shall recursively construct a sequence of essential extensions $P_i \twoheadrightarrow S$ such that

$$\dim_{\Bbbk} \operatorname{Hom}(P_i, X/X_i) = \ell_S(X/X_i) \dim_{\Bbbk} \operatorname{End}(S).$$

Once this is proved, Lemma A.4.7 shows that P_r is projective in $\langle X\rangle$, and $P_S := P_r \twoheadrightarrow S$ is the desired morphism.

Take $P_1 = S$; Lemma A.4.7 shows that the equality holds. Now let $1 \leq i < r$ and suppose P_i has been constructed. Choose a $\Bbbk$-basis $\phi_1, \ldots, \phi_n$ of $\operatorname{Hom}(P_i, X/X_i)$. Form the fiber products

$$\begin{tikzcd} Q_j \arrow[r, "\psi_j"] \arrow[d] \arrow[dr, phantom, "\square"] & X/X_{i+1} \arrow[d, two heads] \\ P_i \arrow[r, "\phi_j"'] & X/X_i \end{tikzcd} \qquad 1 \leq j \leq n.$$

Notice that $Q_j \twoheadrightarrow P_i$ (Proposition 2.1.6). Let Q be the fiber product of $Q_1, \ldots, Q_n$ over P_i. Since fiber products may be formed one step at a time, the same reasoning shows that $Q \to P_i$ is epic. Choose a subobject P_{i+1} of Q such that the composite $P_{i+1} \hookrightarrow Q \to P_i$, denoted p_i^{i+1}, is epic, and such that P_{i+1} is minimal with this property. By construction, the composite $P_{i+1} \xrightarrow{p_i^{i+1}} P_i \twoheadrightarrow S$ is an essential extension. Also write t_j for the composite $P_{i+1} \hookrightarrow Q \xrightarrow{\text{projection}} Q_j$.

We claim that $(p_i^{i+1})^* : \operatorname{Hom}(P_i, X/X_i) \to \operatorname{Hom}(P_{i+1}, X/X_i)$ is an isomorphism. It is injective, and the dimension of the left-hand side is known to be $\ell_S(X/X_i) \dim_{\Bbbk} \operatorname{End}(S)$, while Lemma A.4.7 bounds the dimension of the right-hand side by the same number. This proves the claim.

Denote the map $\operatorname{Hom}(P_{i+1}, X/X_{i+1}) \to \operatorname{Hom}(P_{i+1}, X/X_i)$ by Φ_i. By the claim, we can define a reverse $\Bbbk$-linear map Ψ_i by sending $\phi_j p_i^{i+1}$ to $\psi_j t_j$, for $1 \le j \le n$. We verify that $\Phi_i \Psi_i = \mathrm{id}$. On each $\phi_j p_i^{i+1}$, this follows immediately from the commutative diagram

$$\begin{tikzcd}
P_{i+1} \arrow[r, "t_j"] \arrow[d, "p_i^{i+1}"'] & Q_j \arrow[r, "\psi_j"] \arrow[dl] & X/X_{i+1} \arrow[d, two heads] \\
P_i \arrow[rr, "\phi_j"'] & & X/X_i,
\end{tikzcd}$$

which says precisely that $\Phi_i(\psi_j t_j) = \phi_j p_i^{i+1}$.

We therefore have a split short exact sequence

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Hom}(P_{i+1}, \underbrace{X_i/X_{i+1}}_{\text{simple object}}) \arrow[r] & \operatorname{Hom}(P_{i+1}, X/X_{i+1}) \arrow[r, "\Phi_i"] & \operatorname{Hom}(P_{i+1}, X/X_i) \arrow[r] & 0. \\
& & & \operatorname{Hom}(P_i, X/X_i) \arrow[u, "\wr"] &
\end{tikzcd}$$

Lemma A.4.7 and the induction hypothesis show that the dimensions of the terms at the left and right are respectively

$$\ell_S(X_i/X_{i+1}) \dim_{\Bbbk} \operatorname{End}(S), \quad \ell_S(X/X_i) \dim_{\Bbbk} \operatorname{End}(S).$$

By additivity of $\ell_S(\cdot)$, the middle term therefore has the required dimension. This completes the proof. $\square$

Exercises

1. In the discussion of §A.1, fix a commutative ring $\Bbbk$, require $\mathcal{C}$ to be a $\Bbbk$-Mod-category (Definition 1.4.1), replace Set by $\Bbbk$-Mod in the definitions of $\mathcal{C}^\wedge$ and $\mathcal{C}^\vee$, and require the functors to be $\Bbbk$-linear. Formulate the corresponding $\Bbbk$-linear versions of Theorems A.1.1 and A.1.3.

2. Complete the only-if part of Example A.2.5. Hint〉 Every R-module can be written as a filtered $\varinjlim$ of finitely presented R-modules, not necessarily its submodules; for simplicity, write $X = \varinjlim \alpha$. Consider a preimage of id_X under (A.2.1). This gives $f_i \in \mathrm{Hom}(X, \alpha(i))$ such that $\iota_i f_i = \mathrm{id}_X$, and hence a decomposition $\alpha(i) \simeq X \oplus N$. Since N is isomorphic to a quotient of $\alpha(i)$, it is a finitely generated R-module. The problem reduces to the fact that the quotient of a finitely presented R-module by a finitely generated submodule is still finitely presented.

3. Explain the relation between the cofinal ordinal sequences introduced in §A.2 and the cofinal subcategories of Definition 1.6.4.

4. (J. Adámek, J. Rosický) Prove the fact mentioned in Remark A.2.3 as follows: for every small filtered category I, there is a small filtered partially ordered set $(\mathcal{P}, \leq)$ together with a cofinal functor $\mathcal{P} \to I$.

(i) Suppose that the small filtered category I has the following property: every finite subcategory $\mathcal{A}$ is contained in a finite subcategory $\mathcal{A}'$ having a unique terminal object. Note that a subcategory need not be full, and that a finite category means one with finite sets of objects and morphisms. Let $\mathcal{I}$ be the set of all subcategories having a unique terminal object, partially ordered by inclusion $\subset$. Show that one can define a functor

$$H : \mathcal{I} \to I,$$

sending an object $\mathcal{A}$ to its unique terminal object $H\mathcal{A}$ and an inclusion $\mathcal{A}_1 \subset \mathcal{A}_2$ to the unique morphism $H\mathcal{A}_1 \to H\mathcal{A}_2$ in $\mathcal{A}_2$.

(ii) In the notation above, show that $(\mathcal{I}, \subset)$ is a small filtered partially ordered set and that H is cofinal.

(iii) Now let I be any small filtered category. Show that the product category $I \times \omega$ has the property assumed in (i), and that the projection functor $I \times \omega \to I$ is cofinal. Here ω is the totally ordered set of nonnegative integers.

Hint〉 Clearly $I \times \omega$ is filtered. Given a finite subcategory $\mathcal{A}$ of $I \times \omega$, show that there are an object (i, n) of $I \times \omega$ and a cone whose base is the inclusion functor $\mathcal{A} \to I \times \omega$ and whose vertex is (i, n) (Convention 1.5.1). Adjoin to $\mathcal{A}$ the object $(i, n+1)$, its identity morphism, and all composites of the morphisms $(j, k) \to (i, n)$ in the cone with the canonical morphism $(i, n) \to (i, n+1)$. This produces a finite subcategory $\mathcal{A}'$ having the unique terminal object $(i, n+1)$.

(iv) Use (iii) to prove the existence of the required cofinal functor $\mathcal{P} \to I$.

5. Let $\mathcal{C}$ be an accessible category.

(i) Choose a small subset $S \subset \mathrm{Ob}(\mathcal{C})$ as in Definition A.2.11, and let $\mathcal{S}$ be the full subcategory it generates. Prove that the Yoneda embedding restricts to a fully faithful functor $\mathcal{C} \to \mathsf{Set}^{\mathcal{S}^{\mathrm{op}}}$.

(ii) Prove that $\mathcal{C}$ is well-powered (Definition 1.11.11). Hint For every object F of $\mathsf{Set}^{\mathcal{S}^{\mathrm{op}}}$ and every $X \in S$, the set FX and its power set are both small. Use this to show that Sub_F is a small set.

6. Fix a field $\Bbbk$. Let $\mathcal{A}$ be a locally finite $\Bbbk$-linear Abelian category, let $\mathcal{A}'$ be an Abelian subcategory, and write its inclusion functor as $i : \mathcal{A}' \to \mathcal{A}$. Assume that the subset $\mathrm{Ob}(\mathcal{A}') \subset \mathrm{Ob}(\mathcal{A})$ is closed under isomorphism.

(i) Show that for every $X \in \mathrm{Ob}(\mathcal{A})$, the intersection of the partially ordered set of quotient objects Quot_X (respectively the partially ordered set of subobjects Sub_X) with $\mathcal{A}'$ has a unique greatest element, denoted $i^*(X)$ (respectively $i^!(X)$), and that i^* (respectively $i^!$) gives a left (respectively right) adjoint to i.

(ii) Suppose that $\mathcal{A}$ has a projective generator s. Prove that $s' := i^*(s)$ is a projective generator of $\mathcal{A}'$.

(iii) For s, s' as above, put $A := \mathrm{End}(s)$ and $A' := \mathrm{End}(s')$. Prove that

$$A = \mathrm{Hom}(s, s) \to \mathrm{Hom}(s, s') \xleftarrow{\sim} \mathrm{Hom}(s', s') = A'$$

makes A' a quotient ring of A, written $A' = A/\mathfrak{a}$. Prove that $\mathcal{A}$ and s satisfy the hypotheses of Theorem 7.8.5, so that $\mathcal{A}$ is equivalent to $\mathsf{Mod}_{\mathrm{fg}}\text{-}A$, while $\mathcal{A}'$ is equivalent to the full subcategory consisting of modules annihilated by the two-sided ideal $\mathfrak{a}$.

Appendix B Introduction to Ind-Objects and Pro-Objects

Given a category $\mathcal{C}$, what is an ind-object over it? Through the Yoneda embedding $\mathcal{C} \to \mathcal{C}^\wedge$, an ind-object can be defined rigorously as an object of $\mathcal{C}^\wedge$ of the form $X = \text{“}\varinjlim\text{”} X_i$, where i ranges over a small filtered category I and $X_i \in \mathrm{Ob}(\mathcal{C})$. The symbol “$\varinjlim$” distinguishes $\varinjlim$ in $\mathcal{C}^\wedge$ from $\varinjlim$ in $\mathcal{C}$. One can prove that

$$\mathrm{Hom}(X, Y) = \varprojlim_i \varinjlim_j \mathrm{Hom}_{\mathcal{C}}(X_i, Y_j), \quad X = \text{“}\varinjlim\text{”} X_i, \; Y = \text{“}\varinjlim\text{”} Y_j.$$

Thus, roughly speaking, an ind-object X may be imagined as a filtered system represented by a family of objects X_i of $\mathcal{C}$ together with a compatible family of morphisms $X_i \to X_j$, as $i \to j$ ranges over $\mathrm{Mor}(I)$. The choice of $(X_i)_i$ representing X, however, is not unique. There is also a dual notion, called a pro-object over $\mathcal{C}$. The presentation here follows [23].

As a source of examples, §B.1 begins with profinite groups. These are both a special class of topological groups and, equivalently, filtered $\varprojlim$ of finite groups in the category of topological groups (Remark B.1.6). On this basis, Example B.2.5 will later show that profinite groups are precisely the pro-objects over the category of finite groups, and so can be handled by algebraic methods. This class of topological groups occurs especially often in mathematics and is also background knowledge for several chapters of this book.

Section B.2 gives the formal definitions of ind-objects and pro-objects, a description of their morphisms, and some preliminary examples. The subsequent discussion focuses on ind-objects. Section B.3 studies the category $\mathrm{Ind}\,\mathcal{C}$ of all ind-objects over $\mathcal{C}$, also called the Ind-completion of $\mathcal{C}$; it contains $\mathcal{C}$ as a full subcategory. We shall study how to recognize ind-objects inside $\mathcal{C}^\wedge$ (Proposition B.3.3), that is, how to characterize when a functor $\mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$ is ind-representable, and then discuss the interaction between $\mathcal{C} \to \mathrm{Ind}\,\mathcal{C}$ and limits. In particular, we shall show that $\mathrm{Ind}\,\mathcal{C}$ has all small filtered $\varinjlim$.

Dually, $\mathcal{C}$ also has a Pro-completion: all pro-objects form the category $\mathrm{Pro}\,\mathcal{C} = \mathrm{Ind}\,(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}}$.

The Ind-extension of functors is the subject of §B.4. The results include how to extend a functor $\mathcal{C} \to \mathcal{D}$ to $\mathrm{Ind}\,\mathcal{C}$ (Definition–Proposition B.4.4), and how to recognize whether a category $\mathcal{C}$ is the Ind-completion of a full subcategory $\mathcal{C}'$ (Proposition B.4.5).

In §B.5, these results are applied to an Abelian category $\mathcal{C}$ to show that both embeddings $\mathcal{C} \to \mathrm{Ind}\,\mathcal{C}$ and $\mathcal{C} \to \mathrm{Pro}\,\mathcal{C}$ are fully faithful exact functors between Abelian categories (Theorem B.5.1). That section also discusses exactness of functor extensions for Abelian categories (Proposition B.5.5). Furthermore, Theorem B.5.7 shows that the Ind-completion of a small Abelian category is a Grothendieck category.

Using the preceding results and the basic fact that Grothendieck categories have injective cogenerators, §B.6 proves the Freyd–Mitchell embedding theorem: every small Abelian category admits a fully faithful exact embedding into $\mathsf{Mod}\text{-}R$, where R is a ring realized on a small set. The route through Ind-completion is neither the most elementary nor the shortest; compared with the Freyd–Mitchell theorem itself, however, Ind-completion may be a more valuable technique.

B.1 Prelude: Profinite Groups

We first define profinite groups from the topological viewpoint: they are topological groups with specified properties. On the few occasions when set size must be distinguished, the topological groups under consideration are understood to be realized on small sets; otherwise we shall not mention this point again.

Definition B.1.1 A **profinite group** is a topological group G satisfying:

- G is a compact Hausdorff group;
- G has a neighborhood basis at the identity 1_G consisting of normal subgroups.

Let **TopGrp** denote the category of all topological groups, with continuous homomorphisms as morphisms. The profinite groups form a full subcategory of **TopGrp**.

If G is profinite, a neighborhood basis at 1_G may be taken to consist of all open normal subgroups $K \lhd G$. For a detailed discussion, see [54, §1.9] or [51, §4.10].

Remark B.1.2 Another characterization is that a topological group G is profinite if and only if it is a totally disconnected compact group; see [54, §1.7 and Proposition 1.9.3]. The following related fact is also useful: a Hausdorff topological space E is locally compact and totally disconnected if and only if every $e \in E$ has a neighborhood basis consisting of compact open subsets.

If H is an open subgroup of G, choose a representative g for each coset $\bar{g} \in G/H$. The decomposition $G = \bigsqcup_{\bar{g} \in G/H} gH$, together with compactness, implies that $(G : H)$ is finite. Moreover, $H = G \smallsetminus \bigcup_{\bar{g} \neq H} gH$ shows that H is closed.

We record a few more basic properties.

Lemma B.1.3 Let G be a topological group and H a subgroup. Then:

(i) the quotient map $\pi : G \to G/H$ is open with respect to the quotient topology on G/H;

(ii) G/H with the quotient topology is Hausdorff if and only if H is closed;

(iii) the quotient topology on G/H is discrete if and only if H is open.

The same assertions of course hold for $H\backslash G$.

Proof For (i), if U is an open subset of G,[1] then $\pi^{-1}(\pi(U)) = \bigcup_{h \in H} Uh$ is open. Hence $\pi(U)$ is open by the definition of the quotient topology.

The only-if direction of (ii) follows from $H = \pi^{-1}(1_G \cdot H)$. For the if direction, given distinct cosets $xH \neq yH$, choose an open subset $V \ni 1_G$ of G such that $Vx \cap yH = \varnothing$, and then choose an open subset $U \ni 1_G$ such that $U^{-1}U \subset V$. This ensures $UxH \cap UyH = \varnothing$, while (i) shows that $\pi(Ux)$ and $\pi(Uy)$ are both open.

The only-if direction of (iii) again follows from $H = \pi^{-1}(1_G \cdot H)$. Now suppose that H is open. By (i), $\{1_G \cdot H\} = \pi(H)$ is open; translation shows that every singleton in G/H is open. This proves the if direction. □

Proposition B.1.4 Let G be a profinite group.

(i) If H is a closed subgroup of G, then H is profinite.

[1] Translator's note: the source says that U is an open subset of G/H, but the expressions $\pi(U)$ and $\bigcup_{h \in H} Uh$ require $U \subset G$.

(ii) If H is a closed normal subgroup of G, then G/H is profinite with the quotient topology.

Proof For (i), the closed subgroup H is of course compact and Hausdorff. A neighborhood basis at 1_G is given by all $K \cap H$, where K ranges over the open normal subgroups of G.

For (ii), closedness of H implies that G/H is Hausdorff, and compactness of G implies that G/H is compact. A neighborhood basis of G/H at $1_G \cdot H$ is given by all KH/H, where K ranges over the open normal subgroups of G; each KH/H is an open normal subgroup of G/H. □

It is easy to prove that the product of a family of profinite groups $(G_i)_i$, indexed by a small set, is again profinite. For any small category I and functor $\beta : I^{\mathrm{op}} \to \mathsf{TopGrp}$, abbreviated as $(G_i)_{i \in \mathrm{Ob}(I)}$ with $G_i := \beta(i)$, give

$$\left(\varprojlim_i G_i := \varprojlim \beta \right) \hookrightarrow \prod_i G_i$$

the topology inherited from the product space on the right, making it a closed subgroup. This gives the $\varprojlim$ in TopGrp. Hence if every G_i is profinite, then $\varprojlim_i G_i$ is profinite. It follows that the category of profinite groups is complete.

Example B.1.5 The simplest example is a finite group G; in this case there is only one Hausdorff topology, namely the discrete topology. The following examples are also common.

- Let $G := \mathrm{Gal}(E|F)$ be the Galois group of a Galois extension $E|F$, endowed with the Krull topology [51, §9.2]. The corresponding neighborhood basis consists of the subgroups $\mathrm{Gal}(E|L)$, where $L|F$ ranges over the finite Galois subextensions of $E|F$.

- For a prime p, let $\mathbb{Z}_p$ be the ring of p-adic integers, and let $\mathrm{SL}(n, \mathbb{Z}_p)$ be the group of $n \times n$ matrices over it with determinant 1. Endowed with the topology inherited from the space $\mathrm{M}_n(\mathbb{Z}_p)$ of $n \times n$ matrices, this too is a profinite group.

- For any group G, its profinite completion is defined by

$$\hat{G} := \varprojlim_{\substack{N \lhd G \\ (G:N) < \infty}} G/N,$$

where each G/N is given the discrete topology. As the name suggests, this is a profinite group.

Remark B.1.6 The preceding discussion shows that if finite groups are given the discrete topology, then their small $\varprojlim$ are profinite. Conversely, every profinite group G can be written as $\varprojlim_i G_i$, where $(I, \leq)$ is a small filtered partially ordered set and every G_i is finite; see [51, Theorem 4.10.6]. Concretely, one may choose a neighborhood basis $(K_i)_{i\in I}$ at 1_G consisting of open normal subgroups, order the index set by reverse inclusion, and put $G_i := G/K_i$.

For a Galois-group example, $\mathrm{Gal}(E|F) = \varprojlim_{L|F} \mathrm{Gal}(L|F)$, where $L|F$ ranges over the finite Galois subextensions of $E|F$.

Consider profinite groups $G = \varprojlim_i G_i$ and $H = \varprojlim_j H_j$ presented as above. Since the subgroups $\ker[G \to G_i]$ form a neighborhood basis at 1_G, there are canonical isomorphisms

$$\begin{aligned}\mathrm{Hom}_{\mathsf{TopGrp}}(G, H) &= \mathrm{Hom}_{\mathsf{TopGrp}}\left(\varprojlim_i G_i, \varprojlim_j H_j\right)\\ &\simeq \varprojlim_j \mathrm{Hom}_{\mathsf{TopGrp}}\left(\varprojlim_i G_i, H_j\right) \quad \text{(by the universal property of inverse limits)}\\ &\simeq \varprojlim_j \varinjlim_i \mathrm{Hom}_{\mathsf{Grp}}(G_i, H_j) \quad \text{(every continuous homomorphism factors through some } G_i\text{);}\end{aligned}$$

Thus profinite groups can also be handled algebraically; see Example B.2.5 below.

Definition B.1.7 Let p be a prime. A profinite group G is called a **pro-p group** if, for every sufficiently small open normal subgroup $K \lhd G$, the quotient G/K is a p-group. Equivalently, G can be written as $\varprojlim_i G_i$, where $(I, \leq)$ is a filtered partially ordered set and every G_i is a p-group.

Such groups occur frequently in number theory. The exercises contain further discussion of pro-p groups.

The following result shows that quotient maps of profinite groups by closed subgroups have particularly simple topological behavior, quite unlike the usual situation for Lie groups.

Lemma B.1.8 Let H and H' be closed subgroups of a profinite group G, with $H' \subset H$. Then the quotient map $\pi : G/H' \twoheadrightarrow G/H$ has a continuous section $s : G/H \to G/H'$; in other words, s is continuous and satisfies $\pi s = \mathrm{id}_{G/H}$.[2)]

An analogous assertion holds for the quotient map $H'\backslash G \twoheadrightarrow H\backslash G$.

Proof First consider the case in which $(H : H')$ is finite. Then H is the disjoint union of H' and finitely many of its left translates, so H' is open in H. There is an open normal subgroup $U \lhd G$ such that $U \cap H \subset H'$. The restriction of π is therefore a bijection $UH'/H' \to UH/H$; by compactness it is a homeomorphism, and its inverse

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[2)] Translator's note: the source has the undefined subscript G/H_2 on this identity; the codomain of π makes G/H the type-correct subscript.

gives a continuous section on the open subset UH/H of G/H. Finally, translate this section to extend it over all of G/H.

In the general case, the reader is asked to reduce the problem to $H' = \{1_G\}$. In that case, consider all data (T, t), where $T \subset H$ is a closed subgroup and t is a continuous section of $G/T \twoheadrightarrow G/H$. Partially order these data by declaring $(T_1, t_1) \preceq (T_2, t_2)$ if $T_2 \subset T_1$ and t_1 factors as $G/H \xrightarrow{t_2} G/T_2 \twoheadrightarrow G/T_1$. Zorn's lemma ensures the existence of a maximal element. The key point is that if $((S_i, s_i))_i$ is a chain for $\preceq$ and $S' := \bigcap_i S_i$, then the canonical map $G/S' \to \varprojlim_i G/S_i$ is continuous and injective with dense image; compactness therefore makes it a homeomorphism. We may consequently define $s' := \varprojlim_i s_i : G/H \to G/S'$, obtaining an upper bound (S', s') for the chain.

Let (S, s) be a maximal element of this partially ordered set. If $S = \{1\}$, we are done. If $S \neq \{1\}$, choose an open normal subgroup U of G such that $S \cap U \neq S$. Since $(S : S \cap U)$ is finite, the first step of the proof gives a continuous section of $G/(S \cap U) \to G/S$. Composing it with s gives a continuous section of $G/(S \cap U) \to G/H$, contradicting the maximality of (S, s).

The version for $H'\backslash G \twoheadrightarrow H\backslash G$ is identical. □

The most important special case is $H' = \{1_G\}$. Then a continuous section s induces a homeomorphism of topological spaces $(G/H) \times H \xrightarrow{\sim} G$, sending (x, h) to $s(x)h$. The conclusion for $G \twoheadrightarrow H\backslash G$ is of course analogous.

B.2 Ind-Objects and Pro-Objects

From this section onward, fix a Grothendieck universe $\mathcal{U}$; the terms small category and small set refer to this choice. Unless stated otherwise, an unqualified category is understood to be a $\mathcal{U}$-category; otherwise it is called a large category. Recall that Set denotes the category of small sets.

Let $\mathcal{C}$ be a category. In §A.1 we recalled the two Yoneda embeddings

$$h_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}^{\wedge}, \quad k_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}^{\vee}.$$

They are of course dual to each other. Note that unless $\mathcal{C}$ is small, $\mathcal{C}^{\wedge}$ and $\mathcal{C}^{\vee}$ are generally large categories. When no confusion can arise, we shall often omit the functors $h_{\mathcal{C}}$ and $k_{\mathcal{C}}$ from the notation.

Recall the lessons of [51, Propositions 2.7.8, 2.7.9]:

- $\mathcal{C}^{\wedge}$ has all small $\varinjlim$ and small $\varprojlim$, constructed pointwise in Set;
- in general, the Yoneda embedding $\mathcal{C} \to \mathcal{C}^{\wedge}$ preserves $\varprojlim$ but not $\varinjlim$.

Following the convention there,[3] in this section the $\varinjlim$ of $\alpha : I \to \mathcal{C}^\wedge$ will be written separately as

$$“\varinjlim”\alpha \text{ or } “\varinjlim”\alpha(i),$$

where i ranges over $\mathrm{Ob}(I)$. This distinguishes it from $\varinjlim$ in $\mathcal{C}$.

Dually, we write “$\varprojlim$” for the $\varprojlim$ in $\mathcal{C}^\vee$ obtained by the pointwise construction, since in general $\mathcal{C} \to \mathcal{C}^\vee$ preserves $\varinjlim$ but not $\varprojlim$.

The density theorem A.1.3 says that every object of $\mathcal{C}^\wedge$ can be expressed as a “$\varinjlim$” of objects of $\mathcal{C}$. The ind-objects are those which can be so expressed using a filtered “$\varinjlim$”; the dual objects in $\mathcal{C}^\vee$ are called pro-objects.

Definition B.2.1 Let $\mathcal{C}$ be a category.

- If $X \in \mathrm{Ob}(\mathcal{C}^\wedge)$ can be written as “$\varinjlim$”X_i, where the index ranges over a small filtered category I and $X_i \in \mathrm{Ob}(\mathcal{C})$ (equivalently, the data define a functor $I \to \mathcal{C}$), then X is called an **ind-object** over $\mathcal{C}$.

- If $X \in \mathrm{Ob}(\mathcal{C}^\vee)$ can be written as “$\varprojlim$”X_i, where the index ranges over a small filtered category I and $X_i \in \mathrm{Ob}(\mathcal{C})$ (equivalently, the data define a functor $I^{\mathrm{op}} \to \mathcal{C}$), then X is called a **pro-object** over $\mathcal{C}$.

All ind-objects form a category $\mathrm{Ind}\,\mathcal{C}$ and all pro-objects form a category $\mathrm{Pro}\,\mathcal{C}$; they are respectively full subcategories of $\mathcal{C}^\wedge$ and $\mathcal{C}^\vee$.

We therefore have fully faithful functors $\mathcal{C} \to \mathrm{Ind}\,\mathcal{C}$ and $\mathcal{C} \to \mathrm{Pro}\,\mathcal{C}$. Corollary B.2.3 will control the sizes of $\mathrm{Ind}\,\mathcal{C}$ and $\mathrm{Pro}\,\mathcal{C}$ and show that they are not large categories.

It is immediate from the definition that $(\mathrm{Ind}\,\mathcal{C})^{\mathrm{op}} \simeq \mathrm{Pro}(\mathcal{C}^{\mathrm{op}})$. Henceforth we write ind-objects (respectively pro-objects) as “$\varinjlim$”X_i (respectively “$\varprojlim$”Y_i) without further comment; remember that such presentations are not unique.

Proposition B.2.2 Let $X =$ “$\varinjlim$”X_i and $Y =$ “$\varinjlim$”Y_j be ind-objects over $\mathcal{C}$. There is a canonical bijection

$$\mathrm{Hom}_{\mathrm{Ind}\,\mathcal{C}}(X, Y) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}_{\mathcal{C}}(X_i, Y_j).$$

Similarly, for pro-objects $X =$ “$\varprojlim$”X_i and $Y =$ “$\varprojlim$”Y_j, there is a canonical bijection

$$\mathrm{Hom}_{\mathrm{Pro}\,\mathcal{C}}(X, Y) \simeq \varprojlim_j \varinjlim_i \mathrm{Hom}_{\mathcal{C}}(X_i, Y_j).$$

The limits on the right are all understood to be taken in Set.

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[3] This notation was introduced by P. Deligne.

Proof For ind-objects,

$$\begin{aligned}
\mathrm{Hom}_{\mathcal{C}^\wedge}\left(\text{“}\varinjlim\text{”}X_i, \text{“}\varinjlim\text{”}Y_j\right) &= \varprojlim_i \mathrm{Hom}_{\mathcal{C}^\wedge}\left(X_i, \text{“}\varinjlim\text{”}Y_j\right) && (\because \varinjlim \text{ in } \mathcal{C}^\wedge) \\
&= \varprojlim_i \left[\left(\text{“}\varinjlim\text{”}Y_j\right)(X_i)\right] && (\because \text{ Theorem A.1.1}) \\
&= \varprojlim_i \varinjlim_j \mathrm{Hom}_{\mathcal{C}}(X_i, Y_j) && (\because \text{ the construction of “}\varinjlim\text{”}).
\end{aligned}$$

The last two equalities can also be viewed as an expression of the compactness of X_i in $\mathcal{C}^\wedge$ (Proposition A.2.6). The pro-object case is dual. □

Corollary B.2.3 The categories $\mathrm{Ind}\,\mathcal{C}$ and $\mathrm{Pro}\,\mathcal{C}$ constructed above are $\mathcal{U}$-categories, just as $\mathcal{C}$ is.

Proof Every $\mathrm{Hom}_{\mathcal{C}}(X_i, Y_j)$ in Proposition B.2.2 is a $\mathcal{U}$-small set, and the $\varprojlim$ and $\varinjlim$ are also taken over small categories. □

Notice that the definitions of $\mathrm{Ind}\,\mathcal{C}$ and $\mathrm{Pro}\,\mathcal{C}$ depend on the choice of universe $\mathcal{U}$. For their precise relation to $\mathcal{U}$, see [23, Proposition 6.1.21]; we shall not pursue this point here.

All algebraic structures in the following examples are understood to be realized on small sets.

Example B.2.4 Let $\Bbbk$ be a field. Denote the category of finite-dimensional $\Bbbk$-vector spaces by $\mathsf{Vect}_{\mathrm{f}}(\Bbbk)$ and the category of all $\Bbbk$-vector spaces by $\mathsf{Vect}(\Bbbk)$. We shall show that

$$\mathrm{Ind}\,\mathsf{Vect}_{\mathrm{f}}(\Bbbk) \text{ is equivalent to } \mathsf{Vect}(\Bbbk).$$

Define a functor $\mathrm{Ind}\,\mathsf{Vect}_{\mathrm{f}}(\Bbbk) \to \mathsf{Vect}(\Bbbk)$ by sending “$\varinjlim$”V_i to $V := \varinjlim_i V_i$ in $\mathsf{Vect}(\Bbbk)$. The key to showing that it is fully faithful is

$$\mathrm{Hom}_{\Bbbk}(V, W) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}_{\Bbbk}(V_i, W_j).$$

Indeed, specifying a linear map $f : V \to W$ is equivalent to specifying a compatible family of linear maps $f_i : V_i \to W$, while the concrete description of filtered $\varinjlim$ in $\mathsf{Vect}(\Bbbk)$ shows that each f_i factors through some W_j.

The functor $\mathrm{Ind}\,\mathsf{Vect}_{\mathrm{f}}(\Bbbk) \to \mathsf{Vect}(\Bbbk)$ is also essentially surjective. For any $\Bbbk$-vector space V, its finite-dimensional subspaces $V^\flat$ form a filtered partially ordered set under inclusion, and there is a canonical isomorphism $\varinjlim V^\flat \xrightarrow{\sim} V$ in $\mathsf{Vect}(\Bbbk)$.

A similar and even simpler argument shows that Set is equivalent to $\mathrm{Ind}\,\mathsf{FinSet}$, where FinSet denotes the category of finite small sets. Observe that FinSet and

$\mathsf{Vect}_{\mathrm{f}}(\Bbbk)$ are both essentially small: the isomorphism classes of their objects form a small set. Thus ind-objects are a construction that obtains the large from the small.

There is no shortage of such examples; Proposition B.4.6 will give a unified criterion for recognizing the Ind-completion of a category.

Example B.2.5 Let FinGrp denote the category of finite groups and let $\mathsf{proFinGrp}$ denote the category of profinite groups from Definition B.1.1. We shall show that $\mathsf{proFinGrp}$ is equivalent to $\mathrm{Pro}\,\mathsf{FinGrp}$.

Define a functor $\mathrm{Pro}\,\mathsf{FinGrp} \to \mathsf{proFinGrp}$ by sending "$\varprojlim$"G_i to $G := \varprojlim_i G_i$ in the category TopGrp of topological groups, giving each G_i the discrete topology. By definition, the data $(G_i)_i$ come from a functor $I^{\mathrm{op}} \to \mathsf{TopGrp}$, where I is a small filtered category; Remark A.2.3 shows, however, that I may also be taken to be a filtered partially ordered set. As in Example B.2.4, the equivalence reduces to proving that

- ◇ the group G above is profinite;
- ◇ $\mathrm{Hom}_{\mathsf{TopGrp}}(G, H) \simeq \varprojlim_j \varinjlim_i \mathrm{Hom}_{\mathsf{FinGrp}}(G_i, H_j)$;
- ◇ every profinite group can be expressed as a small filtered $\varprojlim$ of finite groups.

But these are precisely the contents of Remark B.1.6.

When $\mathcal{C}$ itself already has small filtered $\varinjlim$, there is also a relation in the reverse direction between $\mathrm{Ind}\,\mathcal{C}$ and $\mathcal{C}$. We give the dual statement for pro-objects at the same time.

Proposition B.2.6 Suppose that for every small filtered category I and every functor $\alpha : I \to \mathcal{C}$ (respectively $\beta : I^{\mathrm{op}} \to \mathcal{C}$), the object $\varinjlim \alpha$ (respectively $\varprojlim \beta$) always exists. Then the embedding $\iota : \mathcal{C} \to \mathrm{Ind}\,\mathcal{C}$ (respectively $\mathcal{C} \to \mathrm{Pro}\,\mathcal{C}$) has a left adjoint $\sigma : \mathrm{Ind}\,\mathcal{C} \to \mathcal{C}$ (respectively a right adjoint $\tau : \mathrm{Pro}\,\mathcal{C} \to \mathcal{C}$), with $\sigma\iota \simeq \mathrm{id}_{\mathcal{C}}$ (respectively $\tau\iota \simeq \mathrm{id}_{\mathcal{C}}$).

More concretely, if $X =$ "$\varinjlim$"X_i (respectively $X =$ "$\varprojlim$"X_i), then there is a canonical isomorphism $\sigma(X) \simeq \varinjlim_i X_i$ (respectively $\tau(X) \simeq \varprojlim_i X_i$).

Proof It suffices to treat the ind-version. By [51, Proposition 2.6.9], the existence of a left adjoint σ is equivalent to the representability, for every ind-object X, of the functor

$$\mathrm{Hom}_{\mathcal{C}^\wedge}(X, \cdot) : \mathcal{C} \to \mathsf{Set}$$

Write $X =$ "$\varinjlim$"X_i. This functor is isomorphic to

$$\varprojlim_i \mathrm{Hom}_{\mathcal{C}}(X_i, \cdot) \simeq \mathrm{Hom}_{\mathcal{C}}\left(\varinjlim_i X_i, \cdot \right).$$

This proves both the existence of the left adjoint σ and the asserted description. □

For theoretical purposes, we often wish to align morphisms in $\operatorname{Ind}\mathcal{C}$ or $\operatorname{Pro}\mathcal{C}$. The following result does this.

Lemma B.2.7 Let X and Y be ind-objects (respectively pro-objects) over $\mathcal{C}$.

(i) There are a small filtered category K and functors $\gamma, \delta : K \to \mathcal{C}$ (respectively $K^{\mathrm{op}} \to \mathcal{C}$) such that $X = \text{“}\varinjlim\text{”}\gamma$ and $Y = \text{“}\varinjlim\text{”}\delta$ (respectively $X = \text{“}\varprojlim\text{”}\gamma$ and $Y = \text{“}\varprojlim\text{”}\delta$).

(ii) Given a morphism $f : X \to Y$, the data in (i) may be chosen together with a morphism of functors $\varphi : \gamma \to \delta$ such that f equals $\text{“}\varinjlim\text{”}\varphi$ (respectively $\text{“}\varprojlim\text{”}\varphi$).

(iii) More generally, given two morphisms $f, g : X \rightrightarrows Y$, the data may be chosen so that they are represented by the $\text{“}\varinjlim\text{”}$ (respectively $\text{“}\varprojlim\text{”}$) of two morphisms $\varphi, \psi : \gamma \to \delta$.

Proof It suffices to treat the ind-version. For (i), first suppose that X and Y arise respectively from functors $\alpha : I \to \mathcal{C}$ and $\beta : J \to \mathcal{C}$. Define K to be the product category $I \times J$; recall that its objects are pairs $(i, j) \in \mathrm{Ob}(I) \times \mathrm{Ob}(J)$ and that $\mathrm{Hom}_K((i, j), (i', j')) := \mathrm{Hom}_I(i, i') \times \mathrm{Hom}_J(j, j')$. It is easy to see that K is small and filtered, and that both projection functors

$$I \leftarrow K \to J, \quad i \leftarrow\!\shortmid (i, j) \mapsto j$$

are cofinal (Definition 1.6.4); this follows by expanding the definitions. Define γ and δ by composing these projections with α and β, respectively. Proposition 1.6.5 then gives $\text{“}\varinjlim\text{”}\alpha = \text{“}\varinjlim\text{”}\gamma$ and $\text{“}\varinjlim\text{”}\beta = \text{“}\varinjlim\text{”}\delta$.

For (ii), modify the construction as follows. This time, let K be the category whose objects are triples (i, j, t) forming a commutative diagram

$$\begin{tikzcd} \alpha(i) \arrow[r, "t"] \arrow[d, "\text{canonical}"'] & \beta(j) \arrow[d, "\text{canonical}"] \\ X \arrow[r, "f"'] & Y, \end{tikzcd}$$

with morphisms defined by the evident commutative triangular diagrams. There are functors $I \leftarrow K \to J$ given on objects by $i \leftarrow\!\shortmid (i, j, t) \mapsto j$. Again define γ and δ by composing these functors with α and β. The reader is asked to verify that

- ◇ K is a small filtered category;
- ◇ both functors $I \leftarrow K \to J$ are cofinal;
- ◇ the data t define a morphism $\varphi : \gamma \to \delta$.

These facts give the required commutative diagram

$$\begin{tikzcd}
\text{“}\varinjlim\text{”}\gamma \arrow[r, "\text{“}\varinjlim\text{”}\varphi"] \arrow[d, "\wr"'] & \text{“}\varinjlim\text{”}\delta \arrow[d, "\wr"] \\
X \arrow[r, "f"'] & Y.
\end{tikzcd}$$

For the pair of morphisms $f, g : X \rightrightarrows Y$ in (iii), the argument is similar: define the objects of K to be quadruples (i, j, s, t) with $s, t : \alpha(i) \rightrightarrows \beta(j)$.[4)] The details are left to the reader. $\square$

The result above generalizes naturally to any finite collection of morphisms $f, g, \ldots \in \mathrm{Hom}_{\mathrm{Ind}\,\mathcal{C}}(X, Y)$.

B.3 Ind-Completion of Categories

Continuing the discussion of §B.2, we now focus on ind-objects. The passage from $\mathcal{C}$ to $\mathrm{Ind}\,\mathcal{C}$ is an important technique, often called the Ind-completion of $\mathcal{C}$.

Definition B.3.1 Let I be a category. If there is a small category J together with a cofinal functor $J \to I$ (Definition 1.6.4), then I is called **cofinally small**.

Lemma B.3.2 A category I is cofinally small if and only if it has a full subcategory I' such that the inclusion functor $I' \to I$ is cofinal and I' is small. If I is filtered, then I' is filtered as well.

Proof It suffices to explain the only-if direction and the final assertion. Take a cofinal functor $H : J \to I$ as in the definition. Let I' be the full subcategory consisting of all objects $H(j)$, for $j \in \mathrm{Ob}(J)$; it is plainly small. Factor H as

$$J \xrightarrow{H'} I' \xrightarrow{G:\text{inclusion functor}} I.$$

To prove that $G : I' \to I$ is cofinal is equivalent to proving that the comma category (i/G) is connected for every $i \in \mathrm{Ob}(I)$; see Definition 1.6.4. This follows easily from the connectedness of (i/H), and the verification is left to the reader.

For the filteredness of the cofinal full subcategory I', see Proposition 1.6.8. $\square$

Proposition B.3.3 (Recognizing ind-objects) Let $X \in \mathrm{Ob}(\mathcal{C}^\wedge)$. The following statements are equivalent:

4) Translator's note: the source prints $\alpha(j)$ as the target, but j indexes β and the maps represent morphisms to Y; the type-correct target is $\beta(j)$.

(i) X is an ind-object over $\mathcal{C}$;

(ii) the category $(h_{\mathcal{C}}/X)$ appearing in Theorem A.1.3 is filtered and cofinally small.

Proof We first prove (i) $\implies$ (ii). For filteredness, suppose that $X = \text{“}\varinjlim\text{”} X_i$, where i ranges over the objects of a small filtered category I and $X_i \in \mathrm{Ob}(\mathcal{C})$. Let $\phi : S \to X$ and $\phi' : S' \to X$ be any two objects of $(h_{\mathcal{C}}/X)$. Proposition A.2.6 shows that there is an i (respectively, an i') such that ϕ (respectively, ϕ') factors through X_i (respectively, $X_{i'}$). Since I is filtered, we may assume without loss of generality that $i = i'$. Thus both ϕ and ϕ' factor through the object $X_i \to X$ of $(h_{\mathcal{C}}/X)$. This is the first condition for a filtered category.

The same argument also shows that the family of canonical morphisms $(X_i \to X)_i$ defines a cofinal functor $I \to (h_{\mathcal{C}}/X)$. Hence $(h_{\mathcal{C}}/X)$ is cofinally small.

Next, consider two morphisms in $(h_{\mathcal{C}}/X)$ of the form

$$\begin{tikzcd} S \arrow[rr, shift left, "f"] \arrow[rr, shift right, "g"'] \arrow[dr, "\phi"'] & & S'. \arrow[dl, "\phi'"] \\ & X & \end{tikzcd}$$

We know that ϕ' factors through some $\phi'_i : S' \to X_i$. The equality $\phi' f = \phi' g$ implies that there is a morphism $\alpha : i \to j$ in I such that $X(\alpha)\phi'_i f = X(\alpha)\phi'_i g$, where $X(\alpha) : X_i \to X_j$. Equivalently, f and g are equalized by the following diagram:

$$\begin{tikzcd} S \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] \arrow[dr, "\phi"'] & S' \arrow[r, "X(\alpha)\phi'_i"] \arrow[d, "\phi'"] & X_j. \arrow[dl] \\ & X & \end{tikzcd}$$

This proves that $(h_{\mathcal{C}}/X)$ is filtered.

We now prove (ii) $\implies$ (i). Theorem A.1.3 shows that X is the “$\varinjlim$” of the functor $(h_{\mathcal{C}}/X) \to \mathcal{C}^\wedge$ that sends $\phi : S \to X$ to S. By Lemma B.3.2, choose a cofinal full subcategory I of $(h_{\mathcal{C}}/X)$ such that I is a small filtered category. We may therefore write X as the ind-object $\text{“}\varinjlim\text{”} X_i$. $\square$

A functor $X : \mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$ that lies in $\mathrm{Ind}\,\mathcal{C}$ is also called **ind-representable**. Proposition B.3.3 is equivalently a criterion for ind-representability. Dually, one may speak of **pro-representable** functors and give an analogous criterion.

The following results use the terminology of Definition 1.5.2.

Proposition B.3.4 The fully faithful functor $\mathrm{Ind}\,\mathcal{C} \to \mathcal{C}^\wedge$ creates small filtered $\varinjlim$. In particular, $\mathrm{Ind}\,\mathcal{C}$ has all small filtered $\varinjlim$.

Proof Take a small filtered category I and a functor $\alpha : I \to \mathrm{Ind}\,\mathcal{C}$, and write $X :=$ "$\varinjlim$"$\alpha \in \mathrm{Ob}(\mathcal{C}^\wedge)$. Our goal is to prove that X lies in $\mathrm{Ind}\,\mathcal{C}$. By Proposition B.3.3, it suffices to prove that $(h_\mathcal{C}/X)$ is filtered and cofinally small.

For the first condition of filteredness, take $\phi : S \to X$ and $\phi' : S' \to X$. Proposition A.2.6 implies that there are $i, i' \in \mathrm{Ob}(I)$ such that these morphisms factor respectively through $\phi_i : S \to X_i$ and $\phi'_{i'} : S' \to X_{i'}$;[5] without loss of generality, we may assume that $i = i'$. Since X_i is an ind-object, write it as "$\varinjlim$"X_{ij}. Applying Proposition A.2.6 once more, we may assume that both ϕ_i and ϕ'_i factor through some $X_{ij} \in \mathrm{Ob}(\mathcal{C})$; the canonical morphism $X_{ij} \to X$ makes it an object of $(h_\mathcal{C}/X)$.

The second condition, concerning the equalization of morphisms, is handled in the same way.

We now check that $(h_\mathcal{C}/X)$ is cofinally small. By Lemma B.3.2, for each i there is a small subset S_i of $\mathrm{Ob}((h_\mathcal{C}/X_i))$ such that every object of $(h_\mathcal{C}/X_i)$ has a morphism to an object of S_i. Write $F_i : (h_\mathcal{C}/X_i) \to (h_\mathcal{C}/X)$ for the evident functor, and set

$$S := \bigcup_i F_i(S_i) \ \subset \mathrm{Ob}((h_\mathcal{C}/X)).$$

Then S is a small set, and the preceding argument in fact shows that every object of $(h_\mathcal{C}/X)$ has a morphism to an object of S. Thus S determines a cofinal full subcategory of $(h_\mathcal{C}/X)$ (apply Proposition 1.6.8); consequently, $(h_\mathcal{C}/X)$ is cofinally small. □

Lemma B.3.5 Suppose that $\mathcal{C}$ has finite $\varprojlim$. Then the functor $\mathrm{Ind}\,\mathcal{C} \to \mathcal{C}^\wedge$ creates finite $\varprojlim$, while $\mathcal{C} \to \mathrm{Ind}\,\mathcal{C}$ preserves finite $\varprojlim$. In particular, $\mathrm{Ind}\,\mathcal{C}$ has finite $\varprojlim$.

Under these hypotheses, finite $\varprojlim$ in $\mathrm{Ind}\,\mathcal{C}$ commute with small filtered $\varinjlim$.

Proof It is known that the embedding $\mathcal{C} \to \mathcal{C}^\wedge$ preserves small $\varprojlim$. The assertion in the first paragraph therefore reduces to proving that $\mathrm{Ind}\,\mathcal{C}$ is closed under finite $\varprojlim$ in $\mathcal{C}^\wedge$.

Since $\mathcal{C} \to \mathcal{C}^\wedge$ preserves small $\varprojlim$, it sends a terminal object to a terminal object. The problem thus reduces further to proving that $\mathrm{Ind}\,\mathcal{C}$ is closed under finite products and equalizers in $\mathcal{C}^\wedge$. For the product of two ind-objects X and Y, use Lemma B.2.7 (i) to choose a filtered category K and present them as $X =$ "$\varinjlim$"X_k and $Y =$ "$\varinjlim$"Y_k. We claim that "$\varinjlim$"$(X_k \times Y_k)$ gives $X \times Y$ in $\mathcal{C}^\wedge$. Indeed, for every $S \in \mathrm{Ob}(\mathcal{C})$,

$$\begin{aligned}\left(\text{“}\varinjlim\text{”}(X_k \times Y_k)\right)(S) &= \varinjlim_k((X_k \times Y_k)(S)) = \varinjlim_k(X_k(S) \times Y_k(S)) \\ &\simeq \varinjlim_k X_k(S) \times \varinjlim_k Y_k(S) = X(S) \times Y(S);\end{aligned}$$

[5] Translator's note: the source prints the domain of $\phi'_{i'}$ as S. Since ϕ' has domain S', the type-correct domain is S'.

the canonical isomorphism on the second line uses the fact that small filtered $\varinjlim$ commute with finite $\varprojlim$ in Set (Proposition 1.6.12).

Next we handle equalizers. Consider morphisms $f, g : X \rightrightarrows Y$ between ind-objects. By Lemma B.2.7 (iii), we may assume that $X = \text{“}\varinjlim\text{”} X_i$, $Y = \text{“}\varinjlim\text{”} Y_i$, and that f, g arise respectively from two families of morphisms $f_i, g_i : X_i \rightrightarrows Y_i$. Take $\ker(f_i, g_i)$ in $\mathcal{C}$. For every $S \in \mathrm{Ob}(\mathcal{C})$ and every i, we obtain an equalizer diagram in Set:

$$\mathrm{Hom}_{\mathcal{C}}(S, \ker(f_i, g_i)) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(S, X_i) \underset{g_{i,*}}{\overset{f_{i,*}}{\rightrightarrows}} \mathrm{Hom}_{\mathcal{C}}(S, Y_i).$$

Take $\varinjlim$ over i and apply once again the fact that small filtered $\varinjlim$ commute with finite $\varprojlim$ in Set. This gives an equalizer diagram

$$\left(\text{“}\varinjlim\text{”} \ker(f_i, g_i)\right)(S) \longrightarrow X(S) \underset{g}{\overset{f}{\rightrightarrows}} Y(S).$$

As S varies, this shows that the ind-object “$\varinjlim$” $\ker(f_i, g_i)$ gives $\ker(f, g)$ in $\mathcal{C}^\wedge$.

In fact, the argument shows that small filtered $\varinjlim$ commute with finite $\varprojlim$ in $\mathcal{C}^\wedge$; this is checked by reducing to Set. Since $\mathrm{Ind}\,\mathcal{C} \to \mathcal{C}^\wedge$ creates small filtered $\varinjlim$ (Proposition B.3.4) and finite $\varprojlim$, the commutation property continues to hold in $\mathrm{Ind}\,\mathcal{C}$. □

Lemma B.3.6 Suppose that $\mathcal{C}$ has finite $\varinjlim$. Then $\mathrm{Ind}\,\mathcal{C}$ also has finite $\varinjlim$, and $\mathcal{C} \to \mathrm{Ind}\,\mathcal{C}$ preserves them.

Under these hypotheses, $\mathrm{Ind}\,\mathcal{C}$ is cocomplete.

Proof Decompose finite $\varinjlim$ into three cases: an initial object, coproducts of two objects, and coequalizers.

Suppose that X is an initial object of $\mathcal{C}$. Then, for every ind-object $Y = \text{“}\varinjlim\text{”} Y_j$, we have $\mathrm{Hom}_{\mathcal{C}^\wedge}(X, Y) \simeq \varinjlim_j \mathrm{Hom}_{\mathcal{C}}(X, Y_j)$, which is plainly a singleton. Thus X is an initial object of $\mathrm{Ind}\,\mathcal{C}$.

Next consider the coproduct of ind-objects X and Y. Use Lemma B.2.7 (i) to choose a filtered category K and present both objects as $X = \text{“}\varinjlim\text{”} X_k$ and $Y = \text{“}\varinjlim\text{”} Y_k$. We claim that their coproduct is given by the ind-object “$\varinjlim$”$(X_k \sqcup Y_k)$. For every ind-object $S = \text{“}\varinjlim\text{”} S_h$,

$$\begin{aligned}\mathrm{Hom}_{\mathcal{C}^\wedge}(\text{“}\varinjlim\text{”}(X_k \sqcup Y_k), S) &\simeq \varprojlim_k \varinjlim_h \mathrm{Hom}_{\mathcal{C}}(X_k \sqcup Y_k, S_h) \\ &\simeq \varprojlim_k \varinjlim_h \left(\mathrm{Hom}_{\mathcal{C}}(X_k, S_h) \times \mathrm{Hom}_{\mathcal{C}}(Y_k, S_h)\right) \\ &\simeq \varprojlim_k \varinjlim_h \mathrm{Hom}_{\mathcal{C}}(X_k, S_h) \times \varprojlim_k \varinjlim_h \mathrm{Hom}_{\mathcal{C}}(Y_k, S_h),\end{aligned}$$

where the last step uses the commutation of $\varprojlim$ with $\varprojlim$, as well as the commutation of small filtered $\varinjlim$ with finite $\varprojlim$ in Set. The final result is $\mathrm{Hom}_{\mathcal{C}^\wedge}(X,S) \times \mathrm{Hom}_{\mathcal{C}^\wedge}(Y,S)$, and all the isomorphisms are canonical.

Now consider the coequalizer case. Let $f, g : X \rightrightarrows Y$ be morphisms between ind-objects. By Lemma B.2.7 (iii), we may assume that $X = \text{“}\varinjlim\text{”} X_i$, $Y = \text{“}\varinjlim\text{”} Y_i$, and that f, g arise from two families of morphisms $f_i, g_i : X_i \rightrightarrows Y_i$. For every $S \in \mathrm{Ob}(\mathcal{C})$ and every i, we have an equalizer diagram in Set:

$$\mathrm{Hom}_{\mathcal{C}}(\mathrm{coker}(f_i, g_i), S) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(Y_i, S) \overset{f_i^*}{\underset{g_i^*}{\rightrightarrows}} \mathrm{Hom}_{\mathcal{C}}(X_i, S).$$

If, via the Yoneda embedding, $\mathrm{Hom}_{\mathcal{C}}$ is rewritten as $\mathrm{Hom}_{\mathcal{C}^\wedge}$ and the object $S \in \mathrm{Ob}(\mathcal{C})$ above is enlarged to an ind-object $S = \text{“}\varinjlim\text{”} S_j$, the diagram remains an equalizer. Indeed, by Proposition B.2.2, this merely amounts to replacing S by S_j in the formula above and then applying one additional $\varinjlim_j$; an equalizer is a finite $\varprojlim$, however, and hence is preserved by $\varinjlim_j$.

Take $\varprojlim_i$ of the resulting equalizer diagram, and then move $\varprojlim_i$ into the first argument of Hom, where it becomes “$\varinjlim$”. The result is an equalizer diagram in Set:

$$\mathrm{Hom}_{\mathcal{C}^\wedge}(\text{“}\varinjlim\text{”}\,\mathrm{coker}(f_i, g_i), S) \longrightarrow \mathrm{Hom}_{\mathcal{C}^\wedge}(Y, S) \overset{f^*}{\underset{g^*}{\rightrightarrows}} \mathrm{Hom}_{\mathcal{C}^\wedge}(X, S).$$

Thus we have verified that “$\varinjlim$” $\mathrm{coker}(f_i, g_i)$ has the universal property required of $\mathrm{coker}(f, g)$ in $\mathrm{Ind}\,\mathcal{C}$.

Finally, to prove that $\mathrm{Ind}\,\mathcal{C}$ is cocomplete, it suffices to show that it has small coproducts. But an arbitrary small coproduct can be expressed as a filtered $\varinjlim$ of finite coproducts. Apply Proposition B.3.4. This proves the assertion. □

B.4 Ind-Completion and Extension of Functors

We continue the discussion of the preceding section, but shift the focus to functors.

Definition–Proposition B.4.1 (Ind-completion of a functor) Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. There is a canonical functor $\mathrm{Ind}\,F : \mathrm{Ind}\,\mathcal{C} \to \mathrm{Ind}\,\mathcal{D}$ such that the following diagram commutes up to canonical isomorphism:

$$\begin{tikzcd} \mathrm{Ind}\,\mathcal{C} \arrow[r, "\mathrm{Ind}\,F"] & \mathrm{Ind}\,\mathcal{D} \\ \mathcal{C} \arrow[u] \arrow[r, "F"'] & \mathcal{D}. \arrow[u] \end{tikzcd}$$

For every ind-object $X = “\varinjlim” X_i$ over $\mathcal{C}$, there is a canonical isomorphism $(\operatorname{Ind} F)(X) \simeq “\varinjlim” F(X_i)$.

Proof The density theorem for the Yoneda embedding, Theorem A.1.3, canonically presents every $X \in \mathrm{Ob}(\mathcal{C}^\wedge)$ as

$$X \simeq “\varinjlim”_{\phi: S \to X} S,$$

where the data on the right are indexed by the category $(h_{\mathcal{C}}/X)$. For an ind-object X, we wish to define

$$(\operatorname{Ind} F)(X) := “\varinjlim”_{\phi: S \to X} F(S),$$

which is plainly canonical; the issue is to show that the right-hand side is an ind-object over $\mathcal{D}$. By Proposition B.3.3, $(h_{\mathcal{C}}/X)$ is a filtered, cofinally small category. By Lemma B.3.2, choose a cofinal full subcategory I that is small and filtered. Thus $(\operatorname{Ind} F)(X)$ is presented as a small filtered colimit indexed by I, and hence is indeed an ind-object.

For a given ind-object $X = “\varinjlim” X_i$, the proof of Proposition B.3.3 showed that the family of canonical morphisms $(X_i \to X)_i$ gives a cofinal functor $I \to (h_{\mathcal{C}}/X)$. It follows immediately that $(\operatorname{Ind} F)(X) \simeq “\varinjlim” F(X_i)$. Taking $X \in \mathrm{Ob}(\mathcal{C})$ and taking I to be the filtered poset with a single object gives the desired commutative diagram, up to canonical isomorphism. □

Lemma B.4.2 Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. If $\mathcal{C}$ and $\mathcal{D}$ have finite $\varinjlim$ (respectively, finite $\varprojlim$), and if F preserves them, then the functor $\operatorname{Ind} F : \operatorname{Ind} \mathcal{C} \to \operatorname{Ind} \mathcal{D}$ constructed in Definition–Proposition B.4.1 also preserves them.

Proof The existence of finite $\varinjlim$ (respectively, finite $\varprojlim$) in $\operatorname{Ind} \mathcal{C}$ and $\operatorname{Ind} \mathcal{D}$ is guaranteed by Lemma B.3.6 (respectively, Lemma B.3.5). Reviewing their proofs, one sees that all the $\varinjlim$ (respectively, $\varprojlim$) considered there are described “index by index,” once the ind-objects and morphisms in question have been suitably aligned; meanwhile, $\operatorname{Ind} F$ has the corresponding description. The issue therefore reduces to the assumption that $F : \mathcal{C} \to \mathcal{D}$ preserves finite $\varinjlim$ (respectively, finite $\varprojlim$). □

Lemma B.4.3 Let $F : \mathcal{C} \to \mathcal{D}$ be a functor, and suppose that $\mathcal{C}$ is small. Write $F^\wedge : \mathcal{C}^\wedge \to \mathcal{D}^\wedge$ for the functor obtained by applying (A.1.2) to the composite $\mathcal{C} \xrightarrow{F} \mathcal{D} \xrightarrow{h_{\mathcal{D}}} \mathcal{D}^\wedge$. Up to isomorphism, the following diagram commutes:

$$\begin{tikzcd} \operatorname{Ind} \mathcal{C} \arrow[r, "\operatorname{Ind} F"] \arrow[d] & \operatorname{Ind} \mathcal{D} \arrow[d] \\ \mathcal{C}^\wedge \arrow[r, "F^\wedge"'] & \mathcal{D}^\wedge, \end{tikzcd}$$

where both vertical functors are the evident embeddings.

Consequently, $\operatorname{Ind} F$ preserves small filtered $\varinjlim$.

Proof Recall that $\mathcal{D}^\wedge$ is cocomplete, so $F^\wedge$ is indeed defined. For the first assertion, take an ind-object $X = \text{“}\varinjlim\text{''} X_i$ over $\mathcal{C}$, where i ranges over a small filtered category I, or more generally over a cofinally small filtered category; the latter can canonically be taken to be $(h_{\mathcal{C}}/X)$. By the construction in (A.1.2), its image under $F^\wedge$ is “$\varinjlim$”$F(X_i)$ in $\mathcal{D}^\wedge$. By the proof of Definition–Proposition B.4.1, this is also precisely the image of $(\operatorname{Ind} F)(X)$ in $\mathcal{D}^\wedge$.

For the second assertion, Proposition A.1.5 states that $F^\wedge$ preserves small $\varinjlim$. Since the vertical functors in the diagram create small filtered $\varinjlim$ (Proposition B.3.4), it follows immediately that $\operatorname{Ind} F$ preserves small filtered $\varinjlim$. □

Definition–Proposition B.4.4 Let $F : \mathcal{C} \to \mathcal{D}$ be a functor, and suppose that $\mathcal{D}$ has small filtered $\varinjlim$. Define $\hat{F} : \operatorname{Ind} \mathcal{C} \to \mathcal{D}$ to be the composite functor

$$\operatorname{Ind} \mathcal{C} \xrightarrow{\operatorname{Ind} F} \operatorname{Ind} \mathcal{D} \xrightarrow[\text{Proposition B.2.6}]{\sigma} \mathcal{D};$$

if $X = \text{“}\varinjlim\text{''} X_i$ is an ind-object over $\mathcal{C}$, then there is a canonical isomorphism $\hat{F}(X) \simeq \varinjlim_i F(X_i)$. In particular, the composite of $\hat{F}$ with the embedding $\mathcal{C} \to \operatorname{Ind} \mathcal{C}$ is isomorphic to F.

If, in addition, $\mathcal{C}$ and $\mathcal{D}$ have finite $\varinjlim$, and F preserves them, then $\hat{F}$ preserves them as well.

Proof The first assertion merely combines the relevant statements in Proposition B.2.6 and Definition–Proposition B.4.1.

For the second assertion, Lemma B.4.2 shows that $\operatorname{Ind} F$ preserves finite $\varinjlim$. On the other hand, since σ has a right adjoint, it too preserves $\varinjlim$. □

When $\mathcal{C}$ is a small category, Definition–Proposition B.4.4 can be strengthened further.

Proposition B.4.5 Let $F : \mathcal{C} \to \mathcal{D}$ be a functor, suppose that $\mathcal{C}$ is small, and suppose that $\mathcal{D}$ has small filtered $\varinjlim$. Then $\hat{F} : \operatorname{Ind} \mathcal{C} \to \mathcal{D}$ preserves all small filtered $\varinjlim$.

Proof Lemma B.4.3 shows that $\operatorname{Ind} F$ preserves small filtered $\varinjlim$, while σ has a right adjoint. □

Proposition B.4.6 (Recognizing an Ind-completion) Suppose that $\mathcal{C}$ has small filtered $\varinjlim$ and that $\mathcal{C}'$ is a full subcategory of $\mathcal{C}$. The embedding functor $\iota : \mathcal{C}' \to \mathcal{C}$ induces a functor $\hat{\iota} : \operatorname{Ind}(\mathcal{C}') \to \mathcal{C}$ by the construction of Definition–Proposition B.4.4. Suppose that $\mathcal{C}'$ satisfies the following conditions:

- ◇ every object of $\mathcal{C}'$ is compact in $\mathcal{C}$ (Definition A.2.2);
- ◇ every object of $\mathcal{C}$ can be presented as a small filtered $\varinjlim$ of objects in $\mathcal{C}'$.

Then $\hat{\iota}$ is an equivalence.

Proof By construction, $\hat{\iota}$ sends an ind-object "$\varinjlim$"X_i over $\mathcal{C}'$ to $\varinjlim X_i$ in $\mathcal{C}$; here $I \to \mathcal{C}'$ is the given functor and I is small and filtered. By assumption, every object of $\mathcal{C}$ can be presented in this form, so $\hat{\iota}$ is essentially surjective.

We next prove that $\hat{\iota}$ is fully faithful. Consider ind-objects "$\varinjlim$"X_i and "$\varinjlim$"Y_j over $\mathcal{C}'$. By compactness of the objects of $\mathcal{C}'$ in $\mathcal{C}$, an argument similar to that of Proposition B.2.2 gives

$$\mathrm{Hom}_{\mathcal{C}}\left(\varinjlim_i X_i,\ \varinjlim_j Y_j\right) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}_{\mathcal{C}'}(X_i, Y_j).$$

This is a canonical bijection, proving the assertion. □

Conversely, if $\mathcal{C} = \mathrm{Ind}(\mathcal{C}')$, then $\mathcal{C}'$ embeds as a full subcategory of $\mathcal{C}$ and satisfies all the conditions above.

B.5 Ind-Completion of Abelian Categories

This section continues the discussion and standing assumptions of §§B.2—B.4.

Theorem B.5.1 If $\mathcal{C}$ is an Abelian category, then $\mathrm{Ind}\,\mathcal{C}$ is also Abelian and $\mathcal{C} \to \mathrm{Ind}\,\mathcal{C}$ is a fully faithful exact functor.

Dually, if $\mathcal{C}$ is an Abelian category, then $\mathcal{C} \to \mathrm{Pro}\,\mathcal{C}$ is likewise a fully faithful exact functor between Abelian categories.

Proof We first treat the assertion about $\mathrm{Ind}\,\mathcal{C}$. Below, write $\mathrm{Hom} := \mathrm{Hom}_{\mathrm{Ind}\,\mathcal{C}}$. Since $\mathcal{C} \to \mathrm{Ind}\,\mathcal{C}$ preserves finite $\varinjlim$ (Lemma B.3.6) and finite $\varprojlim$ (Lemma B.3.5), it sends a zero object to a zero object. This also shows that $\mathrm{Ind}\,\mathcal{C}$ has finite $\varinjlim$ and finite $\varprojlim$.

Next, the canonical morphism in (1.1.1),

$$\delta : Y \sqcup Y \to Y \times Y, \quad Y \in \mathrm{Ob}(\mathrm{Ind}\,\mathcal{C}),$$

is always an isomorphism: by the "index-by-index" construction of products and coproducts of ind-objects, verification reduces immediately to the case $X, Y \in \mathrm{Ob}(\mathcal{C})$. Following the method of Proposition 1.3.5, we may therefore define addition on $\mathrm{Hom}(X, Y)$ by taking $f + g$ to be the composite

$$X \to X \times X \xrightarrow{f \times g} Y \times Y \xrightarrow{\delta^{-1}} Y \sqcup Y \to Y.$$

The reader may check that under

$$\mathrm{Hom}(X, Y) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}_{\mathcal{C}}(X_i, Y_j),$$

this operation becomes the addition on $\mathrm{Hom}_{\mathcal{C}}$ characterized in the same way (recall that $\varinjlim_j$ is filtered). Thus this operation indeed makes $\mathrm{Ind}\,\mathcal{C}$ an Ab-category, then an additive category, and makes $\mathcal{C} \to \mathrm{Ind}\,\mathcal{C}$ an additive functor.

Consider any morphism $f : X \to Y$ in $\mathrm{Ind}\,\mathcal{C}$. We claim that the canonical morphism $\mathrm{coim}(f) \to \mathrm{im}(f)$ is always an isomorphism. By Lemma B.2.7 (ii), we may assume that f is induced by a family of morphisms $f_i : X_i \to Y_i$ in $\mathcal{C}$. In an additive category, images and coimages are described by kernels and cokernels (Proposition 1.3.12); recall also the properties established in the preceding proof, such as $\ker(f) = “\varinjlim” \ker(f_i)$. The assertion therefore reduces immediately to $\mathrm{coim}(f_i) \xrightarrow{\sim} \mathrm{im}(f_i)$. Hence $\mathrm{Ind}\,\mathcal{C}$ is Abelian.

Finally, exactness of $\mathcal{C} \to \mathrm{Ind}\,\mathcal{C}$ follows because this functor preserves finite $\varinjlim$ and finite $\varprojlim$.

The concept of an Abelian category is self-dual (Proposition 2.1.3), and $\mathrm{Pro}\,\mathcal{C} \simeq \mathrm{Ind}(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}}$. The dual assertion about $\mathrm{Pro}\,\mathcal{C}$ follows at once. □

Recall that an Abelian category is a special kind of additive category, and additivity is a property of a category, not additional structure; see the discussion following Corollary 1.3.6.

Remark B.5.2 If $\mathcal{C}$ is further assumed to be a $\Bbbk$-linear Abelian category, where $\Bbbk$ is a commutative ring, then the same is true of $\mathrm{Ind}\,\mathcal{C}$, and $\mathcal{C} \to \mathrm{Ind}\,\mathcal{C}$ is a $\Bbbk$-linear functor. The key is to define canonically a $\Bbbk$-module structure on every $\mathrm{Hom}(X, Y)$ such that, on the right-hand side of the bijection

$$\mathrm{Hom}(X, Y) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}_{\mathcal{C}}(X_i, Y_j),$$

it is reflected by the $\Bbbk$-module structure on each $\mathrm{Hom}_{\mathcal{C}}(X_i, Y_j)$. This condition can be taken as the definition, but one must show that it depends only on X and Y, not on the choices of $(X_i)_i$ and $(Y_j)_j$. The verification presents no essential difficulty; the details are left as an exercise.

Lemma B.5.3 Let $F : \mathcal{C} \to \mathcal{D}$ be a functor between Abelian categories. The functor $\mathrm{Ind}\,F : \mathrm{Ind}\,\mathcal{C} \to \mathrm{Ind}\,\mathcal{D}$ of Definition–Proposition B.4.1 is also additive.

Proof Consider ind-objects $X = “\varinjlim” X_i$ and $Y = “\varinjlim” Y_j$ over $\mathcal{C}$. By construction, the map induced by $\mathrm{Ind}\,F$ on morphisms is

$$\varprojlim_i \varinjlim_j \mathrm{Hom}_{\mathcal{C}}(X_i, Y_j) \xrightarrow{\text{induced by } F} \varprojlim_i \varinjlim_j \mathrm{Hom}_{\mathcal{D}}(FX_i, FY_j).$$

In view of the addition on the Hom-sets of $\mathrm{Ind}\,\mathcal{C}$ and $\mathrm{Ind}\,\mathcal{D}$ (see the proof of Theorem B.5.1), this map is a homomorphism of additive groups. □

Lemma B.5.4 If the Abelian category $\mathcal{C}$ has small filtered $\varinjlim$, then the functor $\sigma : \mathrm{Ind}\,\mathcal{C} \to \mathcal{C}$ of Proposition B.2.6 is additive.

Proof This can be verified directly as above, or deduced from the following fact: σ is left adjoint to $\mathrm{Ind}\,\mathcal{C} \to \mathcal{C}$, so Corollary 1.3.6 (v) implies its additivity. □

If $\mathcal{C}$ and $\mathcal{D}$ are $\Bbbk$-linear, where $\Bbbk$ is a commutative ring, and F is also $\Bbbk$-linear, the preceding results have evident generalizations.

The following result is used in §9.5.

Proposition B.5.5 Let $F : \mathcal{C} \to \mathcal{D}$ be a right-exact functor between Abelian categories. Suppose that $\mathcal{D}$ has small filtered $\varinjlim$.

(i) The extension $\hat{F} : \mathrm{Ind}\,\mathcal{C} \to \mathcal{D}$ given by Definition–Proposition B.4.4 is right exact.

(ii) If, in addition, F is exact and small filtered $\varinjlim$ are exact in $\mathcal{D}$, then $\hat{F}$ is exact.

(iii) Under the hypotheses of (i), if $\mathcal{C}$ is also assumed to be small, then $\hat{F}$ preserves all small $\varinjlim$.

Proof The preceding two lemmas show that $\hat{F}$ is additive, while Definition–Proposition B.4.4 shows that $\hat{F}$ preserves finite $\varinjlim$. This proves (i).

Now assume the hypotheses of (ii). It suffices to prove that $\hat{F}$ preserves ker. Let $f : X \to Y$ be a morphism in $\mathrm{Ind}\,\mathcal{C}$. By Lemma B.2.7 (ii), we may assume that $X = \text{“}\varinjlim\text{”} X_i$, $Y = \text{“}\varinjlim\text{”} Y_i$, and $f = \text{“}\varinjlim\text{”} f_i$, where the morphisms $f_i : X_i \to Y_i$ satisfy the compatibility conditions. First apply F and then $\varinjlim_i$ to the exact sequence

$$0 \to \ker(f_i) \to X_i \xrightarrow{f_i} Y_i.$$

The result is the exact sequence

$$0 \to \varinjlim_i F\ker(f_i) \to \varinjlim_i FX_i \to \varinjlim_i FY_i.$$

By the proof of Lemma B.3.5, $\text{“}\varinjlim\text{”}\ker(f_i) = \text{“}\varinjlim\text{”}\ker(f_i, 0)$ gives $\ker(f) = \ker(f, 0)$; hence the first term of this sequence is identified with $\hat{F}(\ker(f))$. On the other hand, the morphism in the second part is identified with $\hat{F}f : \hat{F}X \to \hat{F}Y$. This proves (ii).

Now consider (iii). Proposition B.4.5 states that $\hat{F}$ preserves all small filtered $\varinjlim$.[6] Together with right exactness, this implies that $\hat{F}$ preserves all small $\varinjlim$. □

[6] Translator's note: the source adds the pointer "(ii)," although Proposition B.4.5 is not divided into numbered parts.

We continue by focusing on the Ind-completion of a small category. We shall need the theory of Grothendieck categories from §2.10.

Lemma B.5.6 If $\mathcal{C}$ is a small category with finite $\varinjlim$, then $\operatorname{Ind}\mathcal{C}$ has a generator.

Proof We know that $\operatorname{Ind}\mathcal{C}$ is cocomplete (Lemma B.3.6) and that $\mathcal{C}$ is small. Thus, in $\operatorname{Ind}\mathcal{C}$ we may form

$$s := \coprod_{X \in \operatorname{Ob}(\mathcal{C})} X.$$

We show that s is a generator. For a pair of morphisms $f, g : X \rightrightarrows Y$ in $\operatorname{Ind}\mathcal{C}$, write $X = \text{“}\varinjlim\text{''} X_i$. By the construction of s, for each i one may choose ϵ_i such that the composite

$$X_i \xrightarrow{\text{canonical coproduct morphism}} s \xrightarrow{\epsilon_i} X$$

is the canonical morphism $X_i \to X$. If $f\epsilon = g\epsilon$ for every morphism $\epsilon : s \to X$, then taking $\epsilon = \epsilon_i$ shows that the restrictions of f and g to every X_i agree; hence $f = g$. □

Theorem B.5.7 If $\mathcal{C}$ is a small Abelian category, then $\operatorname{Ind}\mathcal{C}$ is a Grothendieck category.

Proof Theorem B.5.1 states that $\operatorname{Ind}\mathcal{C}$ is Abelian. We verify the conditions for a Grothendieck category one by one.

- ▷ Cocompleteness This is contained in Lemma B.3.6.
- ▷ Generator Its existence is guaranteed by Lemma B.5.6.
- ▷ Exactness of small filtered $\varinjlim$ Let I be a small filtered category, let $\alpha, \beta, \gamma : I \to \operatorname{Ind}\mathcal{C}$ be functors, and let $\alpha \to \beta \to \gamma$ be morphisms such that

  $$0 \to \alpha(i) \to \beta(i) \to \gamma(i) \to 0$$

  is a short exact sequence for every $i \in \operatorname{Ob}(I)$. We wish to prove that

  $$0 \to \varinjlim \alpha \to \varinjlim \beta \to \varinjlim \gamma \to 0$$

  is also exact. As explained in Example 2.8.6, $\varinjlim$ always preserves coker; the point is to prove that it preserves ker. But ker is a finite $\varprojlim$, so this follows from the second assertion of Lemma B.3.5.

This proves the theorem. □

B.6 The Freyd–Mitchell Embedding Theorem

We continue to fix a Grothendieck universe $\mathcal{U}$ in order to distinguish categories (which by default mean $\mathcal{U}$-categories) from small categories. By convention, all groups, rings, and modules mentioned below are by default realized on small sets (that is, $\mathcal{U}$-sets).

Lemma B.6.1 Let $\mathcal{C}$ be a cocomplete Abelian category with a projective generator, and let O be a small subset of $\mathrm{Ob}(\mathcal{C})$. Then $\mathcal{C}$ has a projective generator S such that every $X \in O$ is a quotient of S.

Proof Choose any projective generator s of $\mathcal{C}$. Since $\mathcal{C}$ is cocomplete, form the direct sum

$$S := \bigoplus_{X \in O} s^{\oplus \operatorname{Hom}(s,X)}.$$

This object remains projective. For each $X \in O$, define the canonical morphism $S \twoheadrightarrow X$ whose restriction to the direct-summand copy of s corresponding to $f \in \operatorname{Hom}(s, X)$ is f, and whose restriction to every other direct-summand copy is 0. The generator property readily shows that this is an epimorphism; see the proof of Theorem 7.8.4 for details. □

The following result is a simple variation on Theorem 7.8.4.

Lemma B.6.2 Let S be a projective generator of an Abelian category $\mathcal{C}$. Define $R := \operatorname{End}_{\mathcal{C}}(S)$, thereby obtaining a faithful exact functor (Proposition 2.8.15)

$$G := \operatorname{Hom}_{\mathcal{C}}(S, \cdot) : \mathcal{C} \to \mathsf{Mod}\text{-}R.$$

If $X \in \mathrm{Ob}(\mathcal{C})$ is a quotient of a direct sum of finitely many copies of S, then the map induced by G,

$$\operatorname{Hom}_{\mathcal{C}}(X, Y) \to \operatorname{Hom}_{\mathsf{Mod}\text{-}R}(GX, GY)$$

is bijective for every $Y \in \mathrm{Ob}(\mathcal{C})$.

Proof Take $m \in \mathbb{Z}_{\geq 0}$ and a short exact sequence in $\mathcal{C}$

$$0 \to X' \to S^{\oplus m} \to X \to 0.$$

Applying G gives a short exact sequence in $\mathsf{Mod}\text{-}R$. Consider the following commutative diagram with exact rows in Ab:

$$\begin{tikzcd} 0 \arrow[r] & \mathrm{Hom}_{\mathcal{C}}(X,Y) \arrow[r] \arrow[d, hook] & \mathrm{Hom}_{\mathcal{C}}(S^{\oplus m},Y) \arrow[r] \arrow[d, hook] & \mathrm{Hom}_{\mathcal{C}}(X',Y) \arrow[d, hook] \\ 0 \arrow[r] & \mathrm{Hom}_R(GX,GY) \arrow[r] & \mathrm{Hom}_R(G(S^{\oplus m}),GY) \arrow[r] & \mathrm{Hom}_R(GX',GY), \end{tikzcd} \tag{B.6.1}$$

where all vertical arrows arise from the faithful functor G and hence are injective. Observe that the composite

$$\mathrm{Hom}_{\mathcal{C}}(S,Y) \xrightarrow{\mathrm{id}} GY \xleftarrow[\psi(1_R) \leftarrow\!\shortmid \psi]{\sim} \mathrm{Hom}_R(R,GY) = \mathrm{Hom}_R(GS,GY)$$

is the same as the map on Hom induced by G. This is of course nearly a tautology, and was checked in the proof of Lemma A.3.1. Since G is additive, the middle vertical arrow in (B.6.1) is an isomorphism. A straightforward diagram chase in Ab shows that the left vertical arrow is also an isomorphism. □

Theorem B.6.3 (P. Freyd, B. Mitchell [13, Chapter 4]) Let $\mathcal{A}$ be a small Abelian category. Then there are a ring R and a fully faithful exact functor $F : \mathcal{A} \to \mathsf{Mod}\text{-}R$.

Proof The following argument is taken from [23, Theorem 9.6.10]. Since $\mathcal{A}^{\mathrm{op}}$ is again a small Abelian category, Theorem B.5.7 shows that $\mathrm{Ind}(\mathcal{A}^{\mathrm{op}})$ is a Grothendieck category. Corollary 2.10.15 therefore guarantees that it has an injective cogenerator. Recalling that

$$\mathrm{Pro}(\mathcal{A}) \simeq \mathrm{Ind}(\mathcal{A}^{\mathrm{op}})^{\mathrm{op}},$$

we conclude that $\mathrm{Pro}(\mathcal{A})$ has a projective generator. Moreover, $\mathcal{A} \to \mathrm{Pro}(\mathcal{A})$ is a fully faithful exact functor between Abelian categories (Theorem B.5.1).

Regard $\mathcal{A}$ as a full subcategory of $\mathrm{Pro}(\mathcal{A})$. Applying Lemma B.6.1, with $O = \mathrm{Ob}(\mathcal{A})$, gives a projective generator S of $\mathrm{Pro}(\mathcal{A})$ such that every object of $\mathcal{A}$ is a quotient of S. Now consider the functor

$$\mathcal{A} \to \mathrm{Pro}(\mathcal{A}) \xrightarrow{G} \mathsf{Mod}\text{-}R, \quad G := \mathrm{Hom}_{\mathrm{Pro}(\mathcal{A})}(S,\cdot),\ R := \mathrm{End}_{\mathrm{Pro}(\mathcal{A})}(S).$$

[7] Write the composite as F. Since each factor is exact, F is exact. Lemma B.6.2 shows that G is fully faithful; therefore F is fully faithful. This proves the theorem. □

Corollary B.6.4 Let $\mathcal{A}$ be a small Abelian category. Then there is a faithful exact functor $E : \mathcal{A} \to \mathsf{Ab}$.

[7] Translator's note: the source prints the endomorphism-ring subscript as $\mathrm{Pro}\,\mathcal{C}$. Here S is an object of $\mathrm{Pro}(\mathcal{A})$, so the type-correct subscript is $\mathrm{Pro}(\mathcal{A})$.

The Freyd–Mitchell theorem allows many commutative-diagram arguments about general Abelian categories to be reduced to the case of $\mathsf{Mod}\text{-}R$, or even of Ab. Since these two categories are concrete, their objects have elements, and diagram-chasing techniques [51, §6.8] greatly simplify many problems. Under suitable set-theoretic assumptions, the requirement that the category be small can be met by enlarging the Grothendieck universe.

Exercises

1. Let E be a Hausdorff topological group, let M be a finite normal subgroup of E, and suppose that E/M is a profinite group with respect to the quotient topology. Prove that E is also profinite.

2. Define $\hat{\mathbb{N}}$ to be the set of all maps $n : \{\text{primes}\} \to \mathbb{Z}_{\geq 0} \sqcup \{+\infty\}$; its elements may be written formally as products $\prod_{p:\text{prime}} p^{n_p}$. Via prime factorization, $\mathbb{Z}_{\geq 1}$ embeds as a subset of $\hat{\mathbb{N}}$. Multiplication, greatest common divisors, least common multiples, and the notion of being relatively prime are defined for these expressions in the evident way. Let H be a closed subgroup of a profinite group G. Define

$$(G : H) := \text{the least common multiple of all } (G/K : H/(H \cap K)) \text{ in } \hat{\mathbb{N}},$$

where K ranges over the open normal subgroups of G.

(i) Prove that $(G : H)$ is also the least common multiple of all $(G : L)$, where L ranges over the open subgroups containing H. Hint〉 Every such L must contain an open subgroup of the form HK, with K as above.

(ii) Prove that if $H_2 \subset H_1 \subset G$, then $(G : H_2) = (G : H_1)(H_1 : H_2)$.
Hint〉 Set $G_K = G/K$ and $H_{i,K} = H_i/H_i \cap K$. It is known that $(G_K : H_{2,K}) = (G_K : H_{1,K})(H_{1,K} : H_{2,K})$; take least common multiples on both sides of the equality.

(iii) Let $H_1 \supset H_2 \supset \cdots$ be a descending sequence of closed subgroups, and put $H := \bigcap_i H_i$. Prove that $(G : H)$ is the least common multiple of all $(G : H_i)$.

(iv) Prove that H is open if and only if $(G : H) \in \mathbb{Z}_{\geq 1}$.

3. Let p be a prime and let G be a profinite group. A closed subgroup P satisfying (a) P is a pro-p group, (b) $(G : P) \in \hat{\mathbb{N}}$ is relatively prime to p, is called a **pro-p Sylow subgroup** of G.

(i) Prove that G always has a pro-p Sylow subgroup. Hint〉 Every G/K has a Sylow p-subgroup P_K. Apply the fact that a filtered $\varprojlim$ of nonempty finite sets is nonempty (the next exercise) to choose P_K compatibly for every G/K, and then consider $\varprojlim_K P_K \hookrightarrow G$.

(ii) For the profinite completion $\hat{\mathbb{Z}} := \varprojlim_{n\geq 1} \mathbb{Z}/n\mathbb{Z}$ of $\mathbb{Z}$, verify that the additive group of the ring of p-adic integers $\mathbb{Z}_p$ is a pro-p Sylow subgroup of $\hat{\mathbb{Z}}$.

(iii) Prove that any two pro-p Sylow subgroups are conjugate. Hint〉 As in the argument for (i), reduce to the case of finite groups.

(iv) Prove that every pro-p subgroup of G is contained in a pro-p Sylow subgroup.[8)]

4. (N. Bourbaki [6, III.58 Théorème 1]) Let $(I, \leq)$ be a nonempty filtered partially ordered set, and let a functor $X : I^{\mathrm{op}} \to \mathsf{Set}$ be given, written as $(X_i)_{i\in \mathrm{Ob}(I)}$; denote the map corresponding to $i \leq j$ by $f_{ij} : X_j \to X_i$. By the following method, prove that if every X_i is a nonempty finite set, then $\varprojlim_i X_i \neq \varnothing$.

(i) Consider families of sets $(A_i)_{i\in \mathrm{Ob}(I)}$ satisfying

$$\varnothing \neq A_i \subset X_i, \quad i \leq j \implies f_{ij}(A_j) \subset A_i.$$

These families form a set Σ. Give it the partial order $\preceq$ defined as follows: $(A_i)_i \preceq (A'_i)_i$ means that $A'_i \subset A_i$ for every i. Use Zorn's lemma to show that $(\Sigma, \preceq)$ has a maximal element.

Hint〉 Clearly Σ is nonempty. For a chain in $(\Sigma, \preceq)$, check that taking intersections at each index i still gives an element of Σ, and hence gives an upper bound for the chain. Nonemptiness requires the hypothesis that the sets are finite.

(ii) Prove that if $(A_i)_i$ is a maximal element of $(\Sigma, \preceq)$, then $f_{ij}(A_j) = A_i$ for every $i \leq j$.

Hint〉 Set $A'_i := \bigcap_{i\leq j} f_{ij}(A_j) \subset A_i$. It suffices to prove that $(A'_i)_i \in \Sigma$. Nonemptiness follows from the finiteness hypothesis; the only other point to prove is that $f_{ij}(A'_j) \subset A'_i$. First observe that $f_{ij}(A'_j) \subset \bigcap_{j\leq k} f_{ik}(A_k)$. Then use filteredness to show that $\bigcap_{j\leq k} f_{ik}(A_k) = \bigcap_{i\leq h} f_{ih}(A_h) = A'_i$.

(iii) Continuing from the preceding parts, prove that every A_i in a maximal element $(A_i)_i$ is a singleton, and thereby prove that $\varprojlim_i X_i \neq \varnothing$.

Hint〉 Choose $i \in \mathrm{Ob}(I)$ and $x_i \in A_i$. For each j, define

$$B_j := \begin{cases} A_j \cap f_{ij}^{-1}(x_i), & i \leq j \\ A_j, & \text{otherwise.} \end{cases}$$

Prove that $(B_j)_j \in \Sigma$, so that $A_j = B_j$ always, including when $i = j$.

Observe that the case $(I, \leq) = (\mathbb{Z}_{\geq 0}, \leq)$ can be handled by Lemma 3.13.12.

5. Form the category $\mathsf{ext}_{\mathrm{alg}}(F)$ (respectively, $\mathsf{ext}_{\mathrm{f}}(F)$) of all algebraic extensions (respectively, finite extensions) of a field F. Show that $\mathsf{ext}_{\mathrm{alg}}(F)$ is equivalent to $\mathrm{Ind}\,(\mathsf{ext}_{\mathrm{f}}(F))$.

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8) Translator's note: in this item the source says "Sylow p-subgroup," whereas the object defined and discussed here is a pro-p Sylow subgroup.

6. Let A be a coalgebra over a field $\Bbbk$, that is, a coalgebra in the monoidal category $(\mathsf{Vect}(\Bbbk), \otimes_{\Bbbk})$ (Definition 7.5.1). Write $\mathsf{Comod}\text{-}A$ for the category of right A-comodules (Definition 7.5.2), and write $\mathsf{Comod}_{\mathrm{f}}\text{-}A$ for its full subcategory consisting of finite-dimensional right A-comodules. Prove that $\mathsf{Comod}\text{-}A$ is equivalent to $\mathrm{Ind}\,(\mathsf{Comod}_{\mathrm{f}}\text{-}A)$.

Hint〉 As in the case of vector spaces in Example B.2.4, apply Lemma 9.6.1.

7. Consider a full subcategory $\mathcal{C}' \subset \mathcal{C}$ and a functor $F' : \mathcal{C}' \to \mathcal{D}$. Assume that

- ◇ $\mathcal{C}'$ is small, and all its objects are compact in $\mathcal{C}$;
- ◇ every object of $\mathcal{C}$ can be written as a small filtered $\varinjlim$ of objects of $\mathcal{C}'$;
- ◇ $\mathcal{D}$ has small filtered $\varinjlim$.

Prove that under these hypotheses F' extends to a functor $F : \mathcal{C} \to \mathcal{D}$ that preserves small filtered $\varinjlim$; up to isomorphism, F is unique.

Hint〉 First extend F' to $\widehat{F'} : \mathrm{Ind}(\mathcal{C}') \to \mathcal{D}$ (Definition–Proposition B.4.4), and then obtain $F : \mathcal{C} \to \mathcal{D}$ via the equivalence in Proposition B.4.6. We also need Proposition B.4.5.

For uniqueness, observe that if an object X of $\mathcal{C}$ is written as $\varinjlim_i X_i$, where $I \to \mathcal{C}'$ is a functor and I is a small filtered category, then necessarily $F(X) \simeq \varinjlim_i F'(X_i)$.

8. Suppose that the category $\mathcal{C}$ has finite $\varinjlim$. Prove that $\mathrm{Ind}\,\mathcal{C}$ is an $\aleph_0$-presentable category in the sense of Definition A.2.11.

9. A compact Hausdorff totally disconnected topological space is called a **Stone space**; by convention, topological spaces are by default realized on small sets. Write Stone for the category of all Stone spaces, and FinSet for the category of all finite small sets.

(i) Prove that the Cantor set and $\mathbb{Z}_p$ are Stone spaces, for any prime p.

(ii) Prove that Stone is equivalent to $\mathrm{Pro}(\mathsf{FinSet})$; thus Stone spaces may be regarded as profinite sets.

(iii) Prove that profinite groups are precisely the group objects in Stone.

(iv) Prove that the category of locally compact Hausdorff totally disconnected topological spaces is equivalent to $\mathrm{Ind}\,\mathrm{Pro}(\mathsf{FinSet})$; prove that $\mathbb{Q}_p$ is such a space.

10. Check the details of Remark B.5.2.

11. Show that $\mathrm{Ind}\,\mathcal{C}$ can be characterized by the following property: for every category $\mathcal{D}$ with small filtered $\varinjlim$, the functor induced by $\mathcal{C} \to \mathrm{Ind}\,\mathcal{C}$,

$$\mathrm{Fct}^0(\mathrm{Ind}\,\mathcal{C}, \mathcal{D}) \to \mathrm{Fct}(\mathcal{C}, \mathcal{D})$$

is an equivalence. Here $\mathrm{Fct}(\cdots)$ denotes a functor category, while the superscript 0 denotes the full subcategory consisting of functors that preserve small filtered $\varinjlim$.

Mastery Bridges

Mastery Bridge 001: Diagram Chasing

Prerequisite map This unit can be read after mastering the following short chain of prerequisites.

- ◇ Kernels, cokernels, and their functoriality: §1.3, especially (1.3.4).
- ◇ Images, coimages, and epi–mono factorization: §1.2.
- ◇ Abelian categories and the stability of epimorphisms and monomorphisms under pullback and pushout: §2.1 and Proposition 2.1.6.
- ◇ Complexes, cohomology, and exactness: §2.2, especially Definition 2.2.5.
- ◇ The categorical construction of connecting morphisms and the Snake Lemma: §2.3, Remark 2.3.2, and Theorem 2.3.3.

Conceptually, the progression is

$$\text{kernels/cokernels} \Longrightarrow \text{images and exactness} \Longrightarrow \text{lift–map–descend} \Longrightarrow \text{connecting morphisms.}$$

Working method: start with the obstruction For element chasing in a module category, use the following discipline.

1. State precisely what must be proved: membership in a kernel or image, or equality of two cokernel classes.
2. Start with the element that measures the obstruction, usually a kernel element or a cokernel class.
3. Lift an element only along an arrow known to be surjective. Map an element forward along any available arrow.
4. Use commutativity to move an equation to the other side of the diagram, then use exactness to replace a statement of "being zero" by "coming from the preceding image."
5. If a choice is required, prove that the difference between any two choices vanishes in the final target.

In a general abelian category, an "element" can be replaced by a morphism from a test object. When a lift is not directly available, use an epimorphic cover as in Lemma 2.3.1. In this way, the element chases below are not merely heuristic: they are concrete models for arguments with kernels, cokernels, and factorizations.

As a worked example, consider the commutative diagram of R-modules

$$\begin{tikzcd} 0 \arrow[r] & A' \arrow[r, "i"] \arrow[d, "a'"'] & A \arrow[r, "p"] \arrow[d, "a"'] & A'' \arrow[r] \arrow[d, "a''"] & 0 \\ 0 \arrow[r] & B' \arrow[r, "j"'] & B \arrow[r, "q"'] & B'' \arrow[r] & 0 \end{tikzcd}$$

with both rows exact. Suppose that a' and a'' are injective. To prove that a is injective, start with the obstruction $x \in \ker(a)$. Commutativity gives $a''p(x) = qa(x) = 0$, so the injectivity of a'' gives $p(x) = 0$. Exactness of the top row gives $x = i(x')$ for some $x' \in A'$. Next,

$$0 = a(x) = ai(x') = ja'(x').$$

Because j and a' are injective, $x' = 0$, and hence $x = 0$. Notice the recurring pattern: start in a kernel, move right, use injectivity, return to the left by exactness, and then use injectivity once more.

Exercises and complete solutions

Exercise 1 — Functoriality of kernels, cokernels, and images Let

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "a"'] & B \arrow[d, "b"] \\ A' \arrow[r, "g"'] & B' \end{tikzcd} \qquad (ga = bf)$$

be a commutative square of R-modules.

(i) Construct the natural homomorphisms $\overline{a} : \ker(f) \to \ker(g)$ and $\overline{b} : \operatorname{coker}(f) \to \operatorname{coker}(g)$.

(ii) Prove that $b(\operatorname{im}(f)) \subset \operatorname{im}(g)$, with equality if a is surjective.

(iii) Prove that $\ker(f) = a^{-1}(\ker(g))$ if b is injective.

Solution. If $x \in \ker(f)$, then $g(a(x)) = b(f(x)) = 0$, so $a(x) \in \ker(g)$. Restricting a gives $\overline{a} : \ker(f) \to \ker(g)$. For cokernels, define

$$\overline{b}([y]) = [b(y)], \qquad [y] \in B/\operatorname{im}(f).$$

If y is replaced by $y + f(x)$, then $b(y + f(x)) = b(y) + g(a(x))$, which determines the same class modulo $\operatorname{im}(g)$. Thus $\overline{b}$ is well defined. These two constructions are precisely those in (1.3.4).

Next,

$$b(\operatorname{im}(f)) = b(f(A)) = g(a(A)) \subset g(A') = \operatorname{im}(g).$$

If a is surjective, then $a(A) = A'$, so the inclusion is an equality. Finally, $x \in \ker(f)$ always implies $a(x) \in \ker(g)$. Conversely, if $a(x) \in \ker(g)$, then $0 = g(a(x)) = b(f(x))$; injectivity of b gives $f(x) = 0$. Thus $x \in \ker(f)$, and consequently $\ker(f) = a^{-1}(\ker(g))$.

Exercise 2 — Intersections, sums, and the isomorphism theorem Let U and V be submodules of an R-module M. Define

$$r : U \cap V \longrightarrow U \oplus V, \quad r(w) = (w, -w), \qquad s : U \oplus V \longrightarrow U + V, \quad s(u, v) = u + v.$$

Prove that

$$0 \longrightarrow U \cap V \xrightarrow{r} U \oplus V \xrightarrow{s} U + V \longrightarrow 0$$

is exact. Use a kernel-and-image chase to obtain the canonical isomorphism

$$U/(U \cap V) \xrightarrow{\sim} (U + V)/V.$$

Solution. If $r(w) = 0$, then the first component gives $w = 0$, so r is injective. Moreover, $s(r(w)) = w - w = 0$, and hence $\operatorname{im}(r) \subset \ker(s)$. If $(u, v) \in \ker(s)$, then $u + v = 0$. Thus $u = -v$ belongs to both U and V, and

$$(u, v) = (u, -u) = r(u).$$

Consequently $\ker(s) = \operatorname{im}(r)$. Every element of $U + V$ has the form $u + v$, so s is surjective. The sequence is exact.

Now consider the homomorphism

$$\theta : U \longrightarrow (U + V)/V, \qquad u \longmapsto u + V.$$

This map is surjective because $(u + v) + V = u + V$. Its kernel consists of all $u \in U$ that also belong to V, namely $U \cap V$. Epi–mono factorization, or the Isomorphism Theorem 2.6.8, gives the canonical isomorphism $U/(U \cap V) \xrightarrow{\sim} (U + V)/V$.

Exercise 3 — Short Five Lemma In the commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & A' \arrow[r, "i"] \arrow[d, "a'"'] & A \arrow[r, "p"] \arrow[d, "a"'] & A'' \arrow[r] \arrow[d, "a''"] & 0 \\ 0 \arrow[r] & B' \arrow[r, "j"'] & B \arrow[r, "q"'] & B'' \arrow[r] & 0, \end{tikzcd}$$

suppose that both rows are exact.

(i) If a' and a'' are injective, prove that a is injective.

(ii) If a' and a'' are surjective, prove that a is surjective.

(iii) Conclude that if a' and a'' are isomorphisms, then a is also an isomorphism.

Solution. For (i), take $x \in A$ with $a(x) = 0$. Then $a''p(x) = qa(x) = 0$. Since a'' is injective, $p(x) = 0$; exactness of the top row gives $x = i(x')$. Next, $0 = ai(x') = ja'(x')$. Since j and a' are injective, $x' = 0$, and hence $x = 0$.

For (ii), take $y \in B$ and set $y'' = q(y)$. Surjectivity of a'' gives $x'' \in A''$ with $a''(x'') = y''$. Since p is surjective, choose $x \in A$ with $p(x) = x''$. Now

$$q(y - a(x)) = y'' - a''p(x) = 0.$$

Exactness of the bottom row gives $y - a(x) = j(y')$ for some $y' \in B'$. Surjectivity of a' gives $x' \in A'$ with $a'(x') = y'$. Therefore

$$y = a(x) + j(a'(x')) = a(x + i(x')).$$

Thus a is surjective. If the two outer morphisms are isomorphisms, (i) and (ii) show that a is both injective and surjective, and therefore an isomorphism.

Exercise 4 — Nine Lemma Consider the commutative diagram of R-modules

$$\begin{tikzcd} & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\ 0 \arrow[r] & A' \arrow[r, "f'"] \arrow[d, "\alpha"'] & B' \arrow[r, "g'"] \arrow[d, "\beta"'] & C' \arrow[r] \arrow[d, "\gamma"'] & 0 \\ 0 \arrow[r] & A \arrow[r, "f"] \arrow[d, "\overline{\alpha}"'] & B \arrow[r, "g"] \arrow[d, "\overline{\beta}"'] & C \arrow[r] \arrow[d, "\overline{\gamma}"'] & 0 \\ 0 \arrow[r] & A'' \arrow[r, "f''"] \arrow[d] & B'' \arrow[r, "g''"] \arrow[d] & C'' \arrow[r] \arrow[d] & 0 \\ & 0 & 0 & 0 & \end{tikzcd}$$

with all three columns exact. Suppose that the top and middle rows are exact. Prove that the bottom row is also exact.

Solution. First we prove that f'' is injective. Take $a'' \in A''$ with $f''(a'') = 0$, and lift a'' to $a \in A$. Since $\overline{\beta} f(a) = f''\overline{\alpha}(a) = 0$, there is a $b' \in B'$ with $\beta(b') = f(a)$. Applying g gives

$$\gamma(g'(b')) = g(\beta(b')) = g(f(a)) = 0.$$

The morphism γ is injective, so $g'(b') = 0$. Exactness of the top row gives $b' = f'(a')$ for some $a' \in A'$. Hence $f(a - \alpha(a')) = 0$. Since f is injective, $a = \alpha(a')$, and thus $a'' = \overline{\alpha}(a) = 0$.

Next, g'' is surjective. For $c'' \in C''$, choose $c \in C$ mapping to c''. Surjectivity of g gives $b \in B$ with $g(b) = c$; hence $g''(\overline{\beta}(b)) = \overline{\gamma}(g(b)) = c''$.

It remains to check exactness at B''. Commutativity immediately gives $g''f'' = 0$. Conversely, take $b'' \in B''$ with $g''(b'') = 0$ and lift b'' to $b \in B$. Since $\overline{\gamma}(g(b)) = g''(\overline{\beta}(b)) = 0$, there is a $c' \in C'$ with $\gamma(c') = g(b)$. Choose $b' \in B'$ with $g'(b') = c'$. Then

$$g(b - \beta(b')) = g(b) - \gamma(g'(b')) = 0.$$

Exactness of the middle row gives $a \in A$ with $f(a) = b - \beta(b')$. Mapping this equation to the bottom row yields

$$f''(\overline{\alpha}(a)) = \overline{\beta}(f(a)) = \overline{\beta}(b) = b''.$$

Thus $\ker(g'') = \operatorname{im}(f'')$, and the bottom row is exact.

Exercise 5 — Constructing the connecting morphism Return to the diagram of short exact sequences

$$\begin{tikzcd}
0 \arrow[r] & A' \arrow[r, "i"] \arrow[d, "\alpha'"'] & A \arrow[r, "p"] \arrow[d, "\alpha"'] & A'' \arrow[r] \arrow[d, "\alpha''"] & 0 \\
0 \arrow[r] & B' \arrow[r, "j"'] & B \arrow[r, "q"'] & B'' \arrow[r] & 0.
\end{tikzcd}$$

For $x'' \in \ker(\alpha'')$, choose $x \in A$ with $p(x) = x''$. Show that $\alpha(x) = j(y')$ for some $y' \in B'$, and then define

$$\delta(x'') := [y'] \in \operatorname{coker}(\alpha') = B'/\operatorname{im}(\alpha').$$

Prove that δ is well defined, is a homomorphism, and is natural with respect to morphisms between two such diagrams.

Solution. Since $x'' \in \ker(\alpha'')$,

$$q(\alpha(x)) = \alpha''(p(x)) = \alpha''(x'') = 0.$$

Exactness of the bottom row gives $\alpha(x) \in \operatorname{im}(j)$, so there is a $y' \in B'$ with $j(y') = \alpha(x)$. The element y' is unique because j is injective.

If x_1 is another lift of x'', then $x_1 - x \in \ker(p) = \operatorname{im}(i)$; write $x_1 - x = i(a')$. If $j(y_1') = \alpha(x_1)$, commutativity gives

$$j(y_1' - y' - \alpha'(a')) = \alpha(x_1) - \alpha(x) - \alpha i(a') = 0.$$

Injectivity of j gives $y_1' - y' = \alpha'(a')$. Thus $[y_1'] = [y']$ in $\operatorname{coker}(\alpha')$, and δ is independent of the choice of x. For x_1'' and x_2'', the sum of chosen lifts and the sum of the corresponding elements y' are valid data for $x_1'' + x_2''$; therefore δ is additive and R-linear.

For naturality, take a morphism from the diagram above to a similar diagram whose every square commutes. If x and y' are chosen for x'', their images in the second diagram are valid choices for the image of x''. Hence the two paths in the square

$$\begin{tikzcd}
\ker(\alpha'') \arrow[r, "\delta"] \arrow[d] & \operatorname{coker}(\alpha') \arrow[d] \\
\ker(\widetilde{\alpha}'') \arrow[r, "\widetilde{\delta}"'] & \operatorname{coker}(\widetilde{\alpha}')
\end{tikzcd}$$

send x'' to the same class of the image of y'. The square commutes, so δ is natural. This is the elementwise version of the canonical property in Remark 2.3.2.

Exercise 6 — Proving the Snake Lemma Using the diagram and δ from Exercise 5, prove that the sequence

$$0 \longrightarrow \ker(\alpha') \longrightarrow \ker(\alpha) \longrightarrow \ker(\alpha'') \xrightarrow{\delta} \operatorname{coker}(\alpha') \\ \longrightarrow \operatorname{coker}(\alpha) \longrightarrow \operatorname{coker}(\alpha'') \longrightarrow 0$$

is exact. State clearly where exactness of the two rows is used in the chase.

Solution. All arrows other than δ are induced by i, p, j, q. The map $\ker(\alpha') \to \ker(\alpha)$ is injective because i is injective.

At $\ker(\alpha)$, take $x \in \ker(\alpha)$ that maps to zero in $\ker(\alpha'')$. Then $p(x) = 0$, so exactness of the top row gives $x = i(x')$. The equation $0 = \alpha(x) = j\alpha'(x')$ and injectivity of j give $x' \in \ker(\alpha')$. The reverse inclusion follows immediately from $pi = 0$. Thus the sequence is exact at $\ker(\alpha)$.

At $\ker(\alpha'')$, an element coming from $x \in \ker(\alpha)$ has $y' = 0$ in the construction of Exercise 5, so it maps to zero under δ. Conversely, suppose that $x'' \in \ker(\alpha'')$ and

$\delta(x'') = 0$. Choose $x \in A$ and $y' \in B'$ as in the construction of δ. Since $[y'] = 0$ in $\operatorname{coker}(\alpha')$, there is an $a' \in A'$ with $y' = \alpha'(a')$. Therefore

$$\alpha(x - i(a')) = j(y') - j\alpha'(a') = 0, \qquad p(x - i(a')) = x''.$$

Thus x'' comes from $\ker(\alpha)$.

At $\operatorname{coker}(\alpha')$, the next arrow sends $[y']$ to $[j(y')]$. If $[y'] = \delta(x'')$ and x is the lift used, then $j(y') = \alpha(x)$, so $[j(y')] = 0$ in $\operatorname{coker}(\alpha)$. Conversely, if $[j(y')] = 0$, there is an $x \in A$ with $j(y') = \alpha(x)$. Commutativity gives

$$\alpha'' p(x) = q\alpha(x) = qj(y') = 0.$$

Thus $p(x) \in \ker(\alpha'')$, and the construction immediately gives $\delta(p(x)) = [y']$.

At $\operatorname{coker}(\alpha)$, the image of every $[y'] \in \operatorname{coker}(\alpha')$ maps to $[qj(y')] = 0$. Conversely, suppose that $[b] \in \operatorname{coker}(\alpha)$ maps to zero. This means that $q(b) = \alpha''(x'')$ for some $x'' \in A''$. Surjectivity of p gives $x \in A$ with $p(x) = x''$. Then

$$q(b - \alpha(x)) = q(b) - \alpha'' p(x) = 0.$$

Exactness of the bottom row gives $b - \alpha(x) = j(y')$ for some $y' \in B'$. Since $\alpha(x)$ vanishes in $\operatorname{coker}(\alpha)$, the class $[b]$ is the image of $[y']$.

Finally, $\operatorname{coker}(\alpha) \to \operatorname{coker}(\alpha'')$ is surjective because q itself is surjective: lift any representative in B'' to B. Exactness has now been checked at every term. This is the Snake Lemma 2.3.3 for two short exact sequences of modules.

Exercise 7 — Computing the snake explicitly Apply the Snake Lemma to the diagram of abelian groups

$$\begin{tikzcd}
0 \arrow[r] & \mathbb{Z} \arrow[r, "\times 6"] \arrow[d, "\times 4"'] & \mathbb{Z} \arrow[r, "\pi"] \arrow[d, "\times 4"'] & \mathbb{Z}/6\mathbb{Z} \arrow[r] \arrow[d, "\times 4"] & 0 \\
0 \arrow[r] & \mathbb{Z} \arrow[r, "\times 6"'] & \mathbb{Z} \arrow[r, "\pi"'] & \mathbb{Z}/6\mathbb{Z} \arrow[r] & 0.
\end{tikzcd}$$

Compute all kernels and cokernels, compute δ on a generator, and then write down and verify the resulting exact sequence.

Solution. The first two vertical maps are multiplication by 4 on $\mathbb{Z}$, so their kernels are zero and their cokernels are $\mathbb{Z}/4\mathbb{Z}$. In the third column,

$$\ker(\times 4 : \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}) = \{[0],[3]\} \simeq \mathbb{Z}/2\mathbb{Z}.$$

The image of multiplication by 4 on $\mathbb{Z}/6\mathbb{Z}$ is $\{[0],[2],[4]\}$, so its cokernel is isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

To compute $\delta([3])$, lift $[3]$ to $3 \in \mathbb{Z}$. The middle vertical map sends it to 12. Since $12 = 6 \cdot 2$, the construction of the connecting morphism gives

$$\delta([3]) = [2] \in \mathbb{Z}/4\mathbb{Z}.$$

Under the identification of generators $[3] \leftrightarrow [1] \in \mathbb{Z}/2\mathbb{Z}$, the map δ is $[1] \mapsto [2]$.

The map from the first cokernel to the second is induced by $\times 6$, and hence on $\mathbb{Z}/4\mathbb{Z}$ it equals $\times 2$. The next map is reduction modulo 2. The Snake Lemma sequence is therefore

$$0 \longrightarrow 0 \longrightarrow 0 \longrightarrow \mathbb{Z}/2\mathbb{Z} \xrightarrow{[1]\mapsto[2]} \mathbb{Z}/4\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}/4\mathbb{Z} \xrightarrow{\bmod 2} \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

The image of δ is $\{[0],[2]\} = \ker(\times 2)$, while the image of $\times 2$ is also $\{[0],[2]\} = \ker(\bmod 2)$. The last map is surjective, and δ is injective. Thus the sequence is indeed exact.

Exercise 8 — The connecting morphism in cohomology Let

$$0 \longrightarrow A^\bullet \xrightarrow{i} B^\bullet \xrightarrow{q} C^\bullet \longrightarrow 0$$

be a short exact sequence of cochain complexes of R-modules, exact in every degree. For a class $[c] \in \mathrm{H}^n(C^\bullet)$, choose a cocycle representative $c \in C^n$ and a lift $b \in B^n$ with $q(b) = c$.

(i) Prove that there is an $a \in A^{n+1}$ with $i(a) = d_B(b)$, and that a is a cocycle.

(ii) Prove that the formula $\partial^n[c] = [a]$ gives a well-defined homomorphism $\partial^n : \mathrm{H}^n(C^\bullet) \to \mathrm{H}^{n+1}(A^\bullet)$.

(iii) Prove exactness of the segment

$$\mathrm{H}^n(A^\bullet) \longrightarrow \mathrm{H}^n(B^\bullet) \longrightarrow \mathrm{H}^n(C^\bullet) \xrightarrow{\partial^n} \mathrm{H}^{n+1}(A^\bullet) \longrightarrow \mathrm{H}^{n+1}(B^\bullet).$$

(iv) Prove that ∂^n is natural with respect to morphisms of short exact sequences of complexes.

Solution. Since c is a cocycle,

$$q(d_B b) = d_C(qb) = d_C c = 0.$$

Exactness at B^{n+1} gives $d_B b \in \operatorname{im}(i^{n+1})$, so there is an $a \in A^{n+1}$ with $i(a) = d_B b$. Next,

$$i(d_A a) = d_B(i(a)) = d_B^2 b = 0.$$

Since i is injective in degree $n+2$, we obtain $d_A a = 0$, so a is a cocycle.

If the lift b is replaced by $b + i(u)$, the corresponding element changes from a to $a + d_A u$, so the class $[a]$ does not change. If the representative c is replaced by $c + d_C v$, choose a lift $w \in B^{n-1}$ of v and use $b + d_B w$ as a lift of the new representative. Since $d_B(b + d_B w) = d_B b$, the resulting class remains $[a]$. In particular, if c is a coboundary, we may choose a lift of the form $d_B w$ and obtain $a = 0$. Thus ∂^n depends only on the cohomology class. Taking sums of lifts shows that ∂^n is a homomorphism.

We check exactness at each term. At $\mathrm{H}^n(B^\bullet)$, the image of a class coming from A clearly vanishes after applying q. Conversely, if $b \in B^n$ is a cocycle and $[q(b)] = 0$, write $q(b) = d_C v$ and lift v to $w \in B^{n-1}$. Then $b - d_B w \in \ker(q) = \operatorname{im}(i)$; write $b - d_B w = i(a)$. Since $b - d_B w$ is a cocycle and i is injective, a is also a cocycle. Thus $[b]$ comes from $\mathrm{H}^n(A^\bullet)$.

At $\mathrm{H}^n(C^\bullet)$, the image of a cocycle $b \in B^n$ has $d_B b = 0$, so its class maps to zero under ∂^n. Conversely, if $\partial^n[c] = 0$, the construction gives $i(a) = d_B b$ with $a = d_A u$ for some $u \in A^n$. Hence

$$d_B(b - i(u)) = i(a) - i(d_A u) = 0, \qquad q(b - i(u)) = c.$$

Thus $[c]$ comes from $\mathrm{H}^n(B^\bullet)$.

At $\mathrm{H}^{n+1}(A^\bullet)$, the class $\partial^n[c] = [a]$ maps to $[i(a)] = [d_B b] = 0$. Conversely, suppose that $a \in A^{n+1}$ is a cocycle and $[i(a)] = 0$ in $\mathrm{H}^{n+1}(B^\bullet)$. Choose $b \in B^n$ with $i(a) = d_B b$. The element $c = q(b)$ is a cocycle because $d_C c = q(d_B b) = q(i(a)) = 0$, and the construction gives $\partial^n[c] = [a]$. The requested segment is therefore exact.

Finally, take a morphism between two short exact sequences of complexes. If c, b, a are the chase data above, their images in the second sequence still satisfy

$$q(b) = c, \qquad i(a) = d_B b.$$

Thus applying the morphism first and then forming the connecting morphism gives the same class as forming ∂^n first and then applying the induced map on cohomology. Therefore ∂^n is natural.

Carrying out this construction for every n joins the exact segments into the long exact sequence

$$\cdots \to \mathrm{H}^n(A^\bullet) \to \mathrm{H}^n(B^\bullet) \to \mathrm{H}^n(C^\bullet) \xrightarrow{\partial^n} \mathrm{H}^{n+1}(A^\bullet) \to \mathrm{H}^{n+1}(B^\bullet) \to \cdots.$$

The morphism ∂^n is conceptually the same connecting morphism as in the Snake Lemma: lift a cocycle, map its differential forward, and then descend the result, which now lies in a kernel, to the left-hand term.

Mastery Bridge 002: Derived Functors and Spectral Sequences

Prerequisite map The shortest prerequisite chain is as follows.

- ◇ Abelian categories, complexes, and cohomology: §2.1, §2.2, and Definition 2.2.5.
- ◇ Injective/projective objects and the existence of enough such objects: §2.8, Definition 2.8.12, and Definition 2.8.16.
- ◇ Resolutions and uniqueness up to homotopy: §3.11, Definition 3.11.1, and Theorem 3.11.5.
- ◇ Classical derived functors, long exact sequences, and dimension shifting: §3.12, Theorem 3.12.4, and Proposition 3.12.9.
- ◇ Spectral sequences, edge maps, and composition of functors: §5.2, Remark 5.2.13, and Theorem 5.6.6.
- ◇ Cohomology of cyclic groups and multiplicative structure: Definition 6.2.5, Proposition 6.7.5, §5.1, and §5.7.

The conceptual progression is

$$\begin{gathered}\text{replace by a resolution} \Longrightarrow \text{take cohomology},\\ \text{organize the total filtration} \Longrightarrow \text{read the pages and edges}.\end{gathered}$$

Working method: three ledgers To keep a computation organized, record its information in three separate "ledgers" that must not be conflated.

1. The **resolution ledger** records which object is replaced by a resolution, the direction of that resolution, and why its terms are injective, projective, or acyclic for the functor to be applied.

2. The **degree ledger** records cohomological indices, total degree $n = p + q$, the direction of the differential $d_r : E_r^{p,q} \to E_r^{p+r,q-r+1}$, and every vanishing term.

3. The **filtration ledger** records the abutment object H^n, the filtration $\mathrm{F}^p H^n$, the identification $\mathrm{gr}^p H^n \simeq E_\infty^{p,n-p}$, and multiplicative compatibility.

The most common mistake is to identify $E_2^{p,q}$ directly with a subobject of the abutment. What is canonically available is the graded term $E_\infty^{p,q}$; returning to the abutment still requires solving the filtration extension problem.

As a worked example, take a first-quadrant cohomological spectral sequence

$$E_2^{p,q} \Rightarrow H^{p+q}.$$

In total degree 1, no differential touches $E_2^{1,0}$, while the only differential that can leave $E_2^{0,1}$ is $d_2 : E_2^{0,1} \to E_2^{2,0}$. Therefore

$$E_\infty^{1,0} = E_2^{1,0}, \qquad E_\infty^{0,1} = \ker(d_2 : E_2^{0,1} \to E_2^{2,0}).$$

The filtration of H^1 has a subobject $\mathrm{F}^1 H^1 \simeq E_\infty^{1,0}$ and quotient $H^1/\mathrm{F}^1 H^1 \simeq E_\infty^{0,1}$. In total degree 2,

$$E_\infty^{2,0} \simeq E_2^{2,0}/\,\mathrm{im}(d_2 : E_2^{0,1} \to E_2^{2,0}) \hookrightarrow H^2.$$

These three observations immediately produce the five-term sequence

$$0 \to E_2^{1,0} \to H^1 \to E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \to H^2.$$

The method is always the same: mark the vanishing region, determine which differentials remain possible, compute E_∞, and only then use the abutment filtration.

Exercises and complete solutions

Exercise 1 — Comparing resolutions and independence of choices Let $\mathcal{A}$ be an abelian category with enough injectives, let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor, and let

$$0 \to X \to I^0 \to I^1 \to \cdots, \qquad 0 \to X \to J^0 \to J^1 \to \cdots$$

be two injective resolutions of X.

(i) Construct cochain maps $u : I^\bullet \to J^\bullet$ and $v : J^\bullet \to I^\bullet$ lifting id_X.

(ii) Prove by a cochain homotopy that $vu \simeq \mathrm{id}_I$ and $uv \simeq \mathrm{id}_J$.

(iii) Conclude that $\mathrm{H}^n(FI^\bullet)$ and $\mathrm{H}^n(FJ^\bullet)$ are canonically isomorphic for every n.

Solution. Because $X \hookrightarrow I^0$ is a monomorphism and J^0 is injective, the morphism $X \to J^0$ extends to $u^0 : I^0 \to J^0$. Suppose that $u^0, \dots, u^n$ have been constructed and satisfy the cochain-map condition. The morphism $d_J^n u^n : I^n \to J^{n+1}$ vanishes on $\ker(d_I^n) = \mathrm{im}(d_I^{n-1})$, and hence factors through $\mathrm{im}(d_I^n) \hookrightarrow I^{n+1}$. Since J^{n+1} is injective, this factorization extends to $u^{n+1} : I^{n+1} \to J^{n+1}$ with

$$u^{n+1} d_I^n = d_J^n u^n.$$

Induction produces u. Interchanging I and J produces v.

We also need the fact that any two lifts $r, s : I^\bullet \to J^\bullet$ of id_X are homotopic. In degree zero, $r^0 - s^0$ vanishes on $X = \ker(d_I^0)$, and hence factors through $\mathrm{im}(d_I^0) \hookrightarrow I^1$. Injectivity of J^0 gives $h^1 : I^1 \to J^0$ with $r^0 - s^0 = h^1 d_I^0$. Inductively, once $h^1, \dots, h^n$ have been constructed, the morphism

$$r^n - s^n - d_J^{n-1} h^n : I^n \longrightarrow J^n$$

vanishes on $\ker(d_I^n)$. It factors through $\mathrm{im}(d_I^n) \hookrightarrow I^{n+1}$ and, because J^n is injective, extends to $h^{n+1} : I^{n+1} \to J^n$. Thus

$$r - s = d_J h + h d_I,$$

that is, $r \simeq s$.

The composites vu and id_I both lift id_X, so they are homotopic; similarly, $uv \simeq \mathrm{id}_J$. Since F is additive, it carries the homotopy equations to homotopy equations. Therefore $\mathrm{H}^n(Fu)$ and $\mathrm{H}^n(Fv)$ are mutually inverse. If other comparison maps are chosen, they are homotopic to the first ones and induce the same maps on cohomology. The isomorphism is consequently canonical. This is the concrete content of independence of resolutions in Theorem 3.11.5.

Exercise 2 — Computing Ext from a short resolution Let $m \geq 1$ and let M be an abelian group. Use the projective resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \longrightarrow 0$$

to compute $\mathrm{Ext}^n_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, M)$ for all $n \geq 0$. Explain why the answer is independent of the chosen projective resolution.

Solution. Apply $\mathrm{Hom}_{\mathbb{Z}}(-, M)$ to the resolution. Since $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, M) \simeq M$, this gives the cochain complex

$$0 \longrightarrow \underbrace{M}_{\text{degree } 0} \xrightarrow{\times m} \underbrace{M}_{\text{degree } 1} \longrightarrow 0.$$

Thus

$$\mathrm{Ext}^n_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, M) \simeq \begin{cases} M[m] := \{x \in M : mx = 0\}, & n = 0, \\ M/mM, & n = 1, \\ 0, & n \geq 2. \end{cases}$$

For $n = 0$, this identification is the usual one $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, M) \simeq M[m]$. Two projective resolutions of $\mathbb{Z}/m\mathbb{Z}$ are connected by comparison maps unique up to homotopy, the dual version of Exercise 1. The functor $\mathrm{Hom}_{\mathbb{Z}}(-, M)$ carries these homotopies to homotopies, so the cohomology is unchanged. Hence the answer above is independent of the resolution used.

Exercise 3 — The long exact sequence of derived functors Let $F : \mathcal{A} \to \mathcal{B}$ be an additive left-exact functor, and suppose that $\mathcal{A}$ has enough injectives. For a short exact sequence

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0,$$

use compatible injective resolutions to construct the connecting morphisms

$$\delta^n : \mathrm{R}^n F(Z) \longrightarrow \mathrm{R}^{n+1} F(X).$$

Prove that these morphisms are well defined and natural, and that they produce a long exact sequence.

Solution. The injective version of the Horseshoe Lemma gives a diagram of resolutions whose rows are exact and which is split in every degree:

$$\begin{tikzcd} 0 \arrow[r] & X \arrow[r] \arrow[d] & Y \arrow[r] \arrow[d] & Z \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & I_X^\bullet \arrow[r] & I_Y^\bullet \arrow[r] & I_Z^\bullet \arrow[r] & 0. \end{tikzcd}$$

Because the bottom row is split in every degree, every additive functor preserves its exactness. Thus

$$0 \to F(I_X^\bullet) \to F(I_Y^\bullet) \to F(I_Z^\bullet) \to 0$$

is a short exact sequence of complexes.

Take a class $[z] \in \mathrm{H}^n(FI_Z^\bullet)$ with cocycle representative z. Lift z to $y \in F(I_Y^n)$. Since the image of $d(y)$ in $F(I_Z^{n+1})$ is $d(z) = 0$, there is an $x \in F(I_X^{n+1})$ whose image is $d(y)$. The cochain map on the left is injective, and $d^2 = 0$ then shows that $d(x) = 0$. Define

$$\delta^n[z] := [x].$$

If y is replaced by another lift, the difference comes from $F(I_X^n)$ and x changes by a coboundary. If z is replaced by $z + d(w)$, lift w and correct y by the differential of that lift; the class $[x]$ remains unchanged. Thus δ^n is well defined.

Exactness is checked by the same chase. For example, $[y] \in \mathrm{H}^n(FI_Y)$ maps to zero in $\mathrm{H}^n(FI_Z)$ if and only if, after correcting y by a coboundary, it comes from a cocycle in $F(I_X^n)$. Next, $[z]$ lies in the kernel of δ^n if and only if x is a coboundary; correcting the lift y by a preimage of that coboundary then produces a cocycle mapping to z. The analogous argument at the next term completes the exactness check and yields

$$\cdots \to \mathrm{R}^n F(X) \to \mathrm{R}^n F(Y) \to \mathrm{R}^n F(Z) \xrightarrow{\delta^n} \mathrm{R}^{n+1} F(X) \to \cdots .$$

For a morphism between two short exact sequences, comparison maps of resolutions carry the choices z, y, x to corresponding choices in the second diagram. Both orders of operations produce the same class, so δ^n is natural. This construction realizes Theorem 3.12.4.

Exercise 4 — Dimension shifting Let $F : \mathcal{A} \to \mathcal{B}$ be a left-exact functor, and suppose that

$$0 \longrightarrow X \longrightarrow A \longrightarrow B \longrightarrow 0$$

is exact, with A being F-acyclic.

(i) Prove the natural isomorphisms

$$\mathrm{R}^1 F(X) \simeq \operatorname{coker}[FA \to FB], \qquad \mathrm{R}^n F(X) \simeq \mathrm{R}^{n-1} F(B) \qquad (n \geq 2).$$

(ii) For an injective resolution $0 \to X \to I^0 \to I^1 \to \cdots$, define $C^0 = X$ and $C^{r+1} = \operatorname{coker}[C^r \to I^r]$. Apply (i) repeatedly to obtain a cohomological formula for $\mathrm{R}^n F(X)$.

(iii) If X has an injective resolution with $I^r = 0$ for $r > d$, prove that $\mathrm{R}^n F(X) = 0$ for $n > d$.

Solution. The long exact sequence from Exercise 3 contains

$$FA \longrightarrow FB \longrightarrow \mathrm{R}^1F(X) \longrightarrow \underbrace{\mathrm{R}^1F(A)}_{0}$$

and, for $n \geq 2$,

$$\underbrace{\mathrm{R}^{n-1}F(A)}_{0} \longrightarrow \mathrm{R}^{n-1}F(B) \longrightarrow \mathrm{R}^nF(X) \longrightarrow \underbrace{\mathrm{R}^nF(A)}_{0}.$$

Exactness gives both isomorphisms in (i), and their naturality follows from the naturality of the long exact sequence.

The injective resolution decomposes into short exact sequences

$$0 \to C^r \to I^r \to C^{r+1} \to 0, \qquad r \geq 0.$$

Because I^r is injective, it is F-acyclic. Therefore, for $n \geq 1$,

$$\mathrm{R}^nF(X) \simeq \mathrm{R}^{n-1}F(C^1) \simeq \cdots \simeq \mathrm{R}^1F(C^{n-1}) \simeq \operatorname{coker}[FI^{n-1} \to FC^n].$$

Left exactness of F identifies FC^n with $\ker[FI^n \to FI^{n+1}]$. Hence

$$\mathrm{R}^nF(X) \simeq \frac{\ker[FI^n \to FI^{n+1}]}{\operatorname{im}[FI^{n-1} \to FI^n]} = \mathrm{H}^n(FI^\bullet).$$

If the resolution ends at I^d, the complex $FI^\bullet$ has no terms in degrees greater than d, so its cohomology vanishes in degrees $n > d$. This also follows by shifting dimension until one reaches the final injective cosyzygy. It is the computational form of Proposition 3.12.9.

Exercise 5 — Setting up the Grothendieck spectral sequence Let

$$\mathcal{A} \xrightarrow{F} \mathcal{B} \xrightarrow{G} \mathcal{C}$$

be additive left-exact functors. Suppose that $\mathcal{A}$ and $\mathcal{B}$ have enough injectives, and that F maps every injective object to a G-acyclic object. For $X \in \mathrm{Ob}(\mathcal{A})$, prove the existence of a first-quadrant spectral sequence

$$E_2^{p,q} = \mathrm{R}^pG(\mathrm{R}^qF(X)) \Rightarrow \mathrm{R}^{p+q}(GF)(X).$$

Explain explicitly the role of the G-acyclicity hypothesis.

Solution. Choose an injective resolution $0 \to X \to I^\bullet$. The complex $F(I^\bullet)$ lies in nonnegative degrees. Choose a Cartan–Eilenberg resolution $F(I^\bullet) \to J^{\bullet,\bullet}$ in $\mathcal{B}$ and apply G to obtain a first-quadrant double complex $GJ^{p,q}$. The two filtrations of the total complex $\mathrm{tot}(GJ)$ give two spectral sequences converging to the same abutment.

For the first filtration, take cohomology in the direction that resolves each $F(I^p)$. This gives

$$\mathrm{H}^q(GJ^{p,\bullet}) \simeq \mathrm{R}^q G(F(I^p)).$$

The G-acyclicity hypothesis says precisely that this term vanishes for $q > 0$, while for $q = 0$ it is $GF(I^p)$. Thus this spectral sequence has only one nonzero row and degenerates. Its total cohomology is therefore

$$\mathrm{H}^n(GF(I^\bullet)) = \mathrm{R}^n(GF)(X).$$

For the second filtration, the defining property of a Cartan–Eilenberg resolution says that the complex obtained after taking cohomology in the $F(I^\bullet)$ direction is an injective resolution of

$$\mathrm{H}^q(F(I^\bullet)) = \mathrm{R}^q F(X).$$

After applying G and taking cohomology, one consequently obtains

$$E_2^{p,q} = \mathrm{R}^p G(\mathrm{R}^q F(X)).$$

Both filtrations compute the same total cohomology, which the first calculation identifies as $\mathrm{R}^{p+q}(GF)(X)$. Since the double complex lies in the first quadrant, the filtration in every total degree is finite and convergence follows. This is the construction in Theorem 5.6.6; without G-acyclicity, the first spectral sequence need not collapse, and this argument cannot identify the abutment with the derived functors of the composite.

Exercise 6 — Edge maps and the Grothendieck five-term sequence Use the spectral sequence from Exercise 5.

(i) Identify the two edge morphisms

$$\mathrm{R}^n G(FX) \longrightarrow \mathrm{R}^n(GF)(X), \qquad \mathrm{R}^n(GF)(X) \longrightarrow G(\mathrm{R}^n F(X)).$$

(ii) Derive the five-term exact sequence

$$\begin{aligned} 0 \to \mathrm{R}^1 G(FX) \to \mathrm{R}^1(GF)(X) &\to G(\mathrm{R}^1 F(X)) \\ &\xrightarrow{d_2} \mathrm{R}^2 G(FX) \to \mathrm{R}^2(GF)(X). \end{aligned}$$

(iii) Prove that the first edge morphism is an isomorphism for every n if $\mathrm{R}^q F(X) = 0$ for $q > 0$, and that the second edge morphism is an isomorphism for every n if $\mathrm{R}^p G(\mathrm{R}^q F(X)) = 0$ for $p > 0$.

Solution. Along the bottom edge, $E_2^{n,0} = \mathrm{R}^n G(FX)$ maps surjectively, page by page, to $E_\infty^{n,0}$. The latter term is the deepest filtration piece $\mathrm{F}^n\mathrm{R}^n(GF)(X)$, and its inclusion in the abutment gives the first edge morphism. Along the left edge, the quotient

$$\mathrm{R}^n(GF)(X)/\mathrm{F}^1 \simeq E_\infty^{0,n}$$

embeds in $E_2^{0,n} = G(\mathrm{R}^n F(X))$; their composite gives the second edge morphism. These directions agree with (5.2.2).

The computation in the working method above applies to every first-quadrant spectral sequence and gives

$$0 \to E_2^{1,0} \to H^1 \to E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \to H^2.$$

Substituting

$$E_2^{p,q} = \mathrm{R}^p G(\mathrm{R}^q F(X)), \qquad H^n = \mathrm{R}^n(GF)(X)$$

gives exactly the stated five-term sequence.

If $\mathrm{R}^q F(X) = 0$ for $q > 0$, only the row $q = 0$ is nonzero. There are no differentials or cross-row extension problems, so

$$E_2^{n,0} = E_\infty^{n,0} = \mathrm{R}^n(GF)(X),$$

and the first edge morphism is an isomorphism. If all terms with $p > 0$ vanish, only the column $p = 0$ remains; similarly,

$$\mathrm{R}^n(GF)(X) = E_\infty^{0,n} = E_2^{0,n} = G(\mathrm{R}^n F(X)),$$

and the second edge morphism is an isomorphism.

Exercise 7 — Cohomology of a cyclic group in concrete terms Let $C_m = \langle \sigma \rangle$ be the cyclic group of order m, and equip $A = \mathbb{Z}/r\mathbb{Z}$ with the trivial C_m-action. Write $d = \gcd(m, r)$. Use the periodic resolution

$$\cdots \xrightarrow{\nu} \mathbb{Z}[C_m] \xrightarrow{\sigma - 1} \mathbb{Z}[C_m] \xrightarrow{\nu} \mathbb{Z}[C_m] \xrightarrow{\sigma - 1} \mathbb{Z}[C_m] \to \mathbb{Z} \to 0, \qquad \nu = 1 + \sigma + \cdots + \sigma^{m-1},$$

to compute $\mathrm{H}^n(C_m, A)$ for all $n \geq 0$. As a check, also determine the answer for the trivial coefficients $A = \mathbb{Z}$.

Solution. Every $\mathrm{Hom}_{\mathbb{Z}[C_m]}(\mathbb{Z}[C_m], A)$ is canonically isomorphic to A. Because the action is trivial, $\sigma - 1$ acts as zero, while ν acts as multiplication by m. Applying $\mathrm{Hom}_{\mathbb{Z}[C_m]}(-, A)$ therefore gives the complex

$$0 \to \underset{0}{\underline{A}} \xrightarrow{0} \underset{1}{\underline{A}} \xrightarrow{\times m} A \xrightarrow{0} A \xrightarrow{\times m} A \to \cdots .$$

Consequently,

$$\mathrm{H}^n(C_m, A) \simeq \begin{cases} A, & n = 0, \\ A[m] := \ker(\times m : A \to A), & n > 0 \text{ odd}, \\ A/mA, & n > 0 \text{ even}. \end{cases}$$

For $A = \mathbb{Z}/r\mathbb{Z}$, the subgroup $A[m]$ is generated by the class of r/d and has order d, so $A[m] \simeq \mathbb{Z}/d\mathbb{Z}$. The subgroup mA has order r/d, so the quotient A/mA is also cyclic of order d. Thus

$$\mathrm{H}^n(C_m, \mathbb{Z}/r\mathbb{Z}) \simeq \begin{cases} \mathbb{Z}/r\mathbb{Z}, & n = 0, \\ \mathbb{Z}/d\mathbb{Z}, & n > 0. \end{cases}$$

For $A = \mathbb{Z}$, the kernel of multiplication by m is zero and its cokernel is $\mathbb{Z}/m\mathbb{Z}$. Hence

$$\mathrm{H}^n(C_m, \mathbb{Z}) \simeq \begin{cases} \mathbb{Z}, & n = 0, \\ 0, & n > 0 \text{ odd}, \\ \mathbb{Z}/m\mathbb{Z}, & n > 0 \text{ even}. \end{cases}$$

This is the direct specialization of Proposition 6.7.5.

Exercise 8 — Checking a multiplicative filtration Let $(A^\bullet, d, \mu)$ be a cohomological dg algebra over a commutative ring $\Bbbk$, equipped with a decreasing filtration that is exhaustive and bounded in every degree,

$$d(\mathrm{F}^p A) \subset \mathrm{F}^p A, \qquad \mathrm{F}^p A \cdot \mathrm{F}^{p'} A \subset \mathrm{F}^{p+p'} A.$$

(i) Prove that $E_0 = \mathrm{gr}_\mathrm{F} A$ has a graded multiplication and that d_0 satisfies the Leibniz rule.

(ii) If x and y represent classes in $E_r^{p,q}$ and $E_r^{p',q'}$, prove that xy represents a class in $E_r^{p+p',q+q'}$, that its product is independent of the chosen representatives, and that

$$d_r(xy) = d_r(x)y + (-1)^{p+q} x \, d_r(y).$$

(iii) For the induced filtration $\mathrm{F}^p \mathrm{H}^n(A) = \mathrm{im}[\mathrm{H}^n(\mathrm{F}^p A) \to \mathrm{H}^n(A)]$, prove that

$$\mathrm{F}^p \mathrm{H}^n(A) \cdot \mathrm{F}^{p'} \mathrm{H}^{n'}(A) \subset \mathrm{F}^{p+p'} \mathrm{H}^{n+n'}(A)$$

and that the convergence isomorphism $E_\infty \simeq \mathrm{gr}_\mathrm{F} \mathrm{H}(A)$ preserves multiplication.

Solution. For homogeneous classes $[x] \in \operatorname{gr}_{\mathrm{F}}^{p} A^{n}$ and $[y] \in \operatorname{gr}_{\mathrm{F}}^{p'} A^{n'}$, define

$$[x][y] := [xy] \in \operatorname{gr}_{\mathrm{F}}^{p+p'} A^{n+n'}.$$

If x or y is changed by an element at the next filtration level, multiplicative compatibility places the resulting change in xy inside $\mathrm{F}^{p+p'+1}$, so the multiplication is well defined. Since d preserves the filtration, it induces d_0, and the Leibniz rule in A gives

$$d_0([x][y]) = d_0([x])\,[y] + (-1)^n [x]\, d_0([y]).$$

On the E_r page, representatives can be chosen so that

$$x \in \mathrm{F}^p A^{p+q}, \quad dx \in \mathrm{F}^{p+r} A^{p+q+1},$$

and similarly $y \in \mathrm{F}^{p'} A^{p'+q'}$ with $dy \in \mathrm{F}^{p'+r} A^{p'+q'+1}$. Filtration compatibility and the Leibniz rule give

$$xy \in \mathrm{F}^{p+p'} A^{p+q+p'+q'},$$

and

$$d(xy) = dx\, y + (-1)^{p+q} x\, dy \in \mathrm{F}^{p+p'+r} A^{p+q+p'+q'+1}.$$

Thus xy indeed determines a class in $E_r^{p+p',q+q'}$, and the Leibniz formula for d_r follows immediately.

To check that the result is independent of the representatives, it is enough to consider changes by r-boundaries and by elements in a deeper filtration. If, for example, x is replaced by $x + du$, with $u \in \mathrm{F}^{p-r+1} A^{p+q-1}$, then

$$(du)y = d(uy) - (-1)^{p+q-1} u\, dy.$$

The first term is a boundary, while the second lies in

$$\mathrm{F}^{p-r+1+p'+r} A = \mathrm{F}^{p+p'+1} A,$$

so it does not change the class in question. A change of the representative y is handled symmetrically; changes in a deeper filtration are even more immediate. Thus multiplication is well defined on every page and passes to $E_{r+1} \simeq \mathrm{H}(E_r, d_r)$.

Finally, if $[x] \in \mathrm{F}^p\, \mathrm{H}^n(A)$ and $[y] \in \mathrm{F}^{p'}\, \mathrm{H}^{n'}(A)$, choose closed representatives $x \in \mathrm{F}^p A^n$ and $y \in \mathrm{F}^{p'} A^{n'}$. Their product xy is closed and lies in $\mathrm{F}^{p+p'} A^{n+n'}$, which proves the required filtration inclusion. The isomorphism $E_\infty^{p,q} \simeq \operatorname{gr}_{\mathrm{F}}^{p} \mathrm{H}^{p+q}(A)$ is built from the same filtered representatives on both sides. The product of two representatives maps to the class of their product, so this isomorphism preserves multiplication. This check is the argument represented in Proposition 5.7.6.

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Index of Symbols

Index of Terms

English terms are arranged alphabetically.